

Canonical GHZ configurations

species: electron

source configurations: 376

canonical configurations collected: 235

canonical display criterion: exactly 2 retained final-state components

retained-component cutoff: fraction $\geq 2.0\%$

normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

canonical configurations by polarization cluster:

P1: 72

P2: 56

P3: 53

P4: 54

all retained-component count distribution:

2 components: 235 configurations

3 components: 1 configurations

4 components: 32 configurations

5 components: 22 configurations

6 components: 72 configurations

7 components: 13 configurations

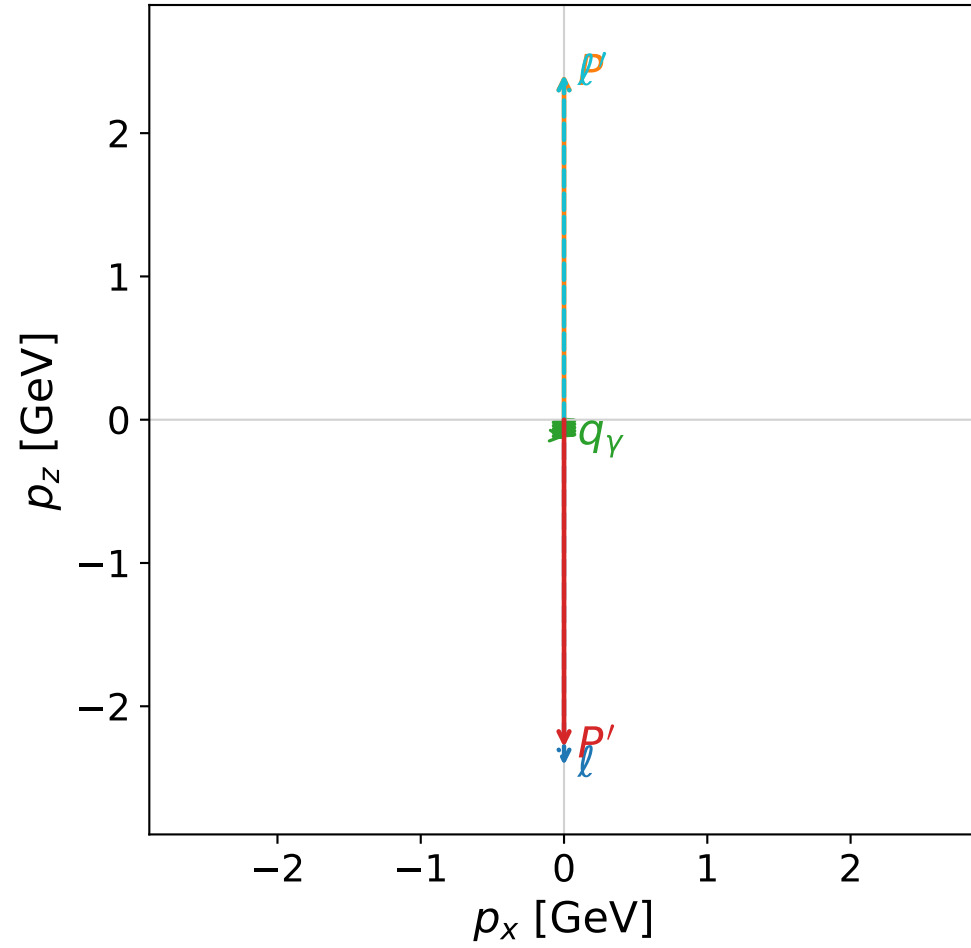
8 components: 1 configurations

CSV index: Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\canonical\canonical_dghz_configurations_electron.csv

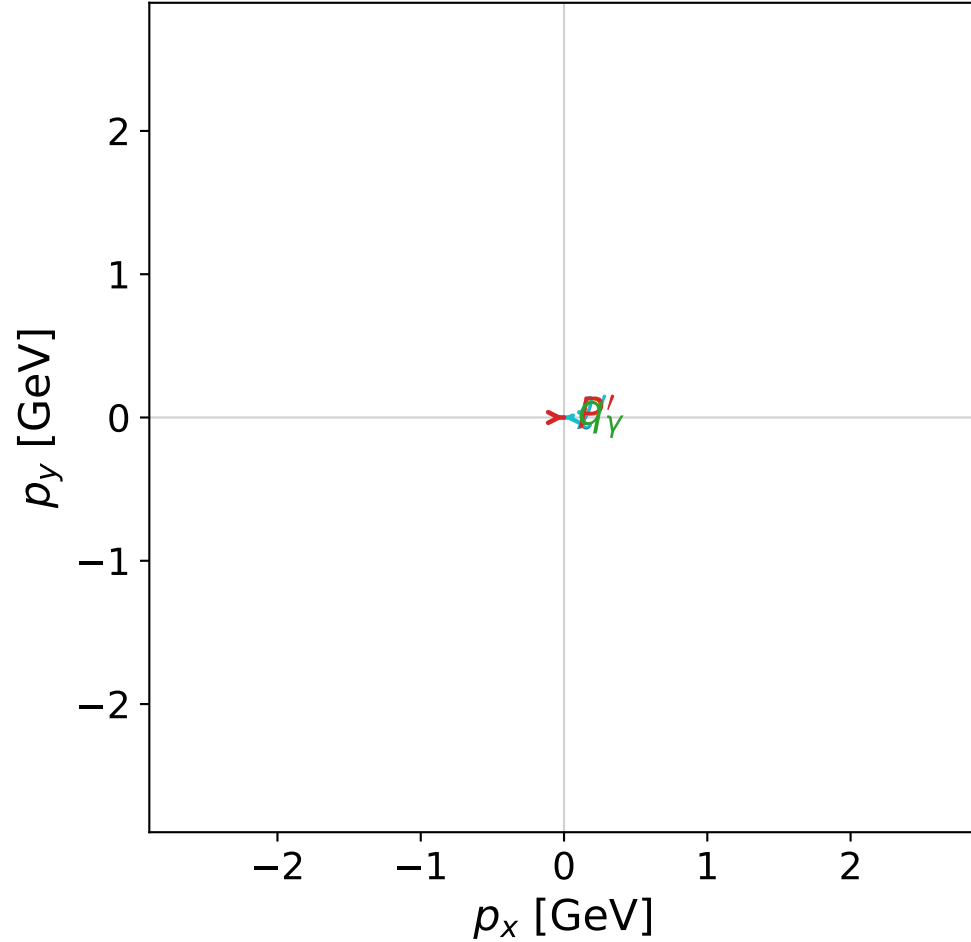
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



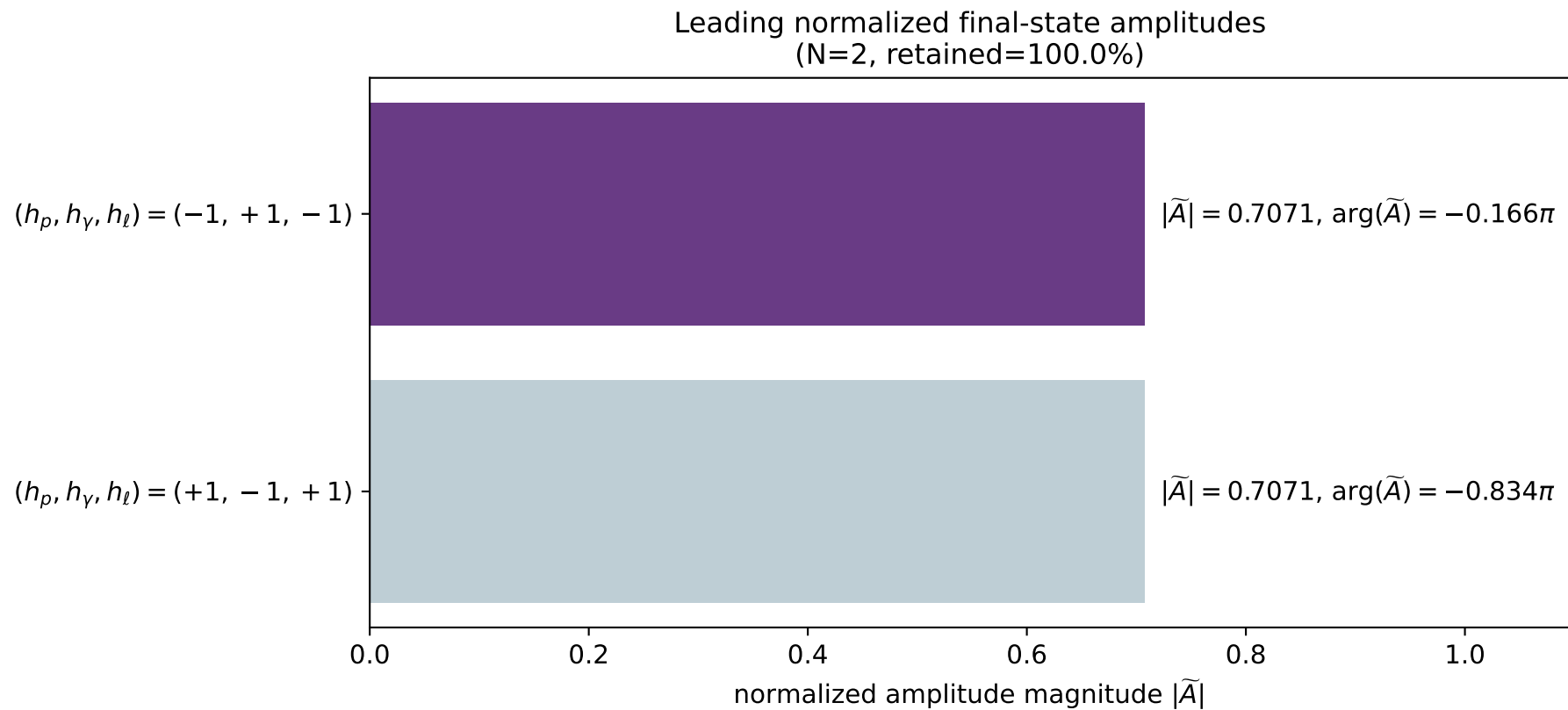
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_2
 $d_{\{\rm GHZ\}}=2.29125\text{e-}08$
 region: polarization_cluster_01_local_minimum_2
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0002
 lepton mixing angle: $\alpha_e=0.87696633$ rad
 incoming lepton: $0.639486|+\rangle + 0.768802|-\rangle$
 proton mixing angle: $\alpha_p=0.87697731$ rad
 incoming proton: $0.639478|+\rangle + 0.768809|-\rangle$

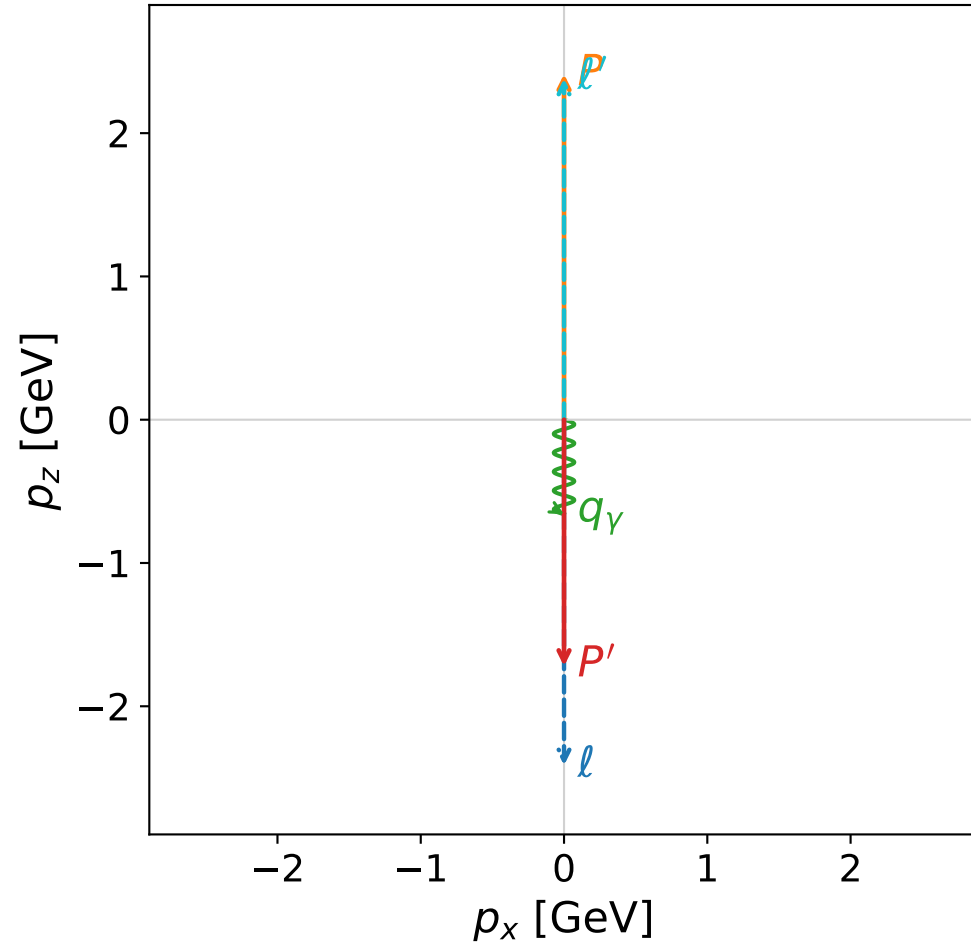
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28627$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.0335096$, $\phi_\gamma=0.522884$
 $E_\gamma=0.121256$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.19891$
 $Q^2=23.228$, $x_B=0.998073$, $t=-22.0603$, $W^2=0.924694$, $y=0.964872$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.4075$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.4075$
 P' $E= 2.4712$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2863$
 q_γ $E= 0.1213$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1213$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

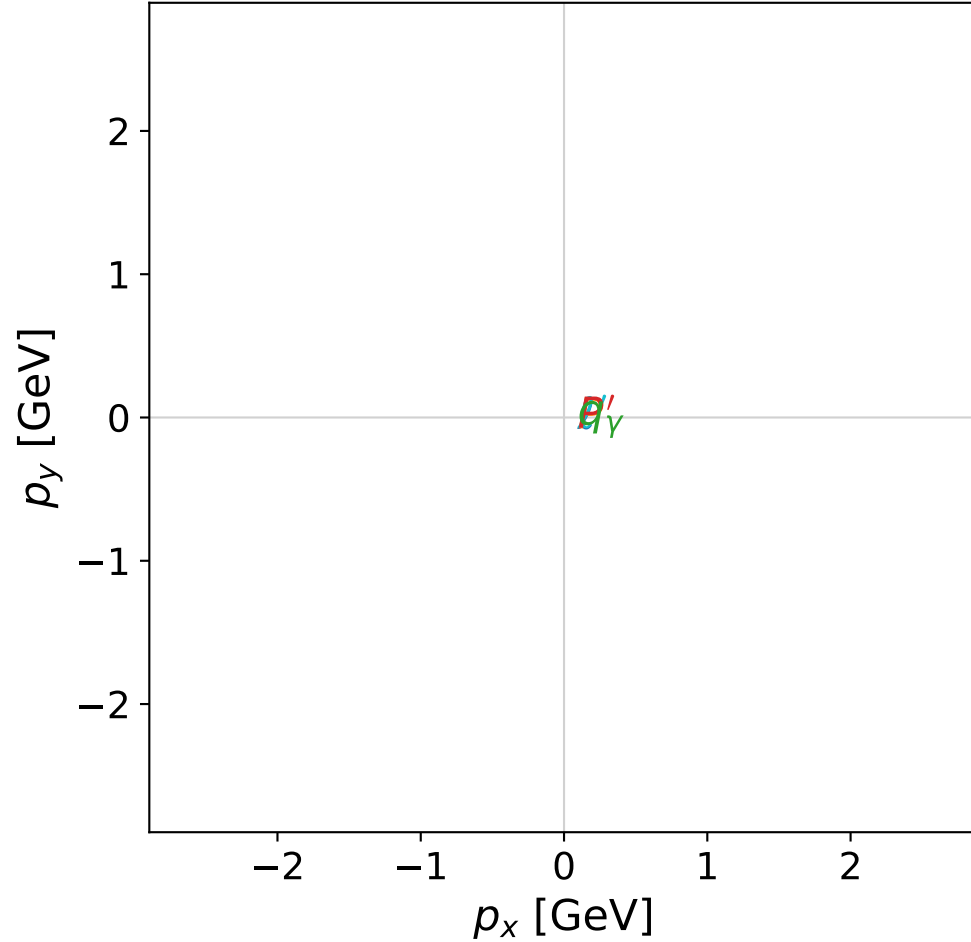


dghz_mixing_angles_polarization_01_local_minimum_11
 $d_{\{\rm GHZ\}}=2.44872\text{e-}08$
 region: polarization_cluster_01_local_minimum_11
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0011
 lepton mixing angle: $\alpha_e=0.59862965$ rad
 incoming lepton: $0.826109|+\rangle +0.563511|-\rangle$
 proton mixing angle: $\alpha_p=0.59861028$ rad
 incoming proton: $0.82612|+\rangle +0.563495|-\rangle$

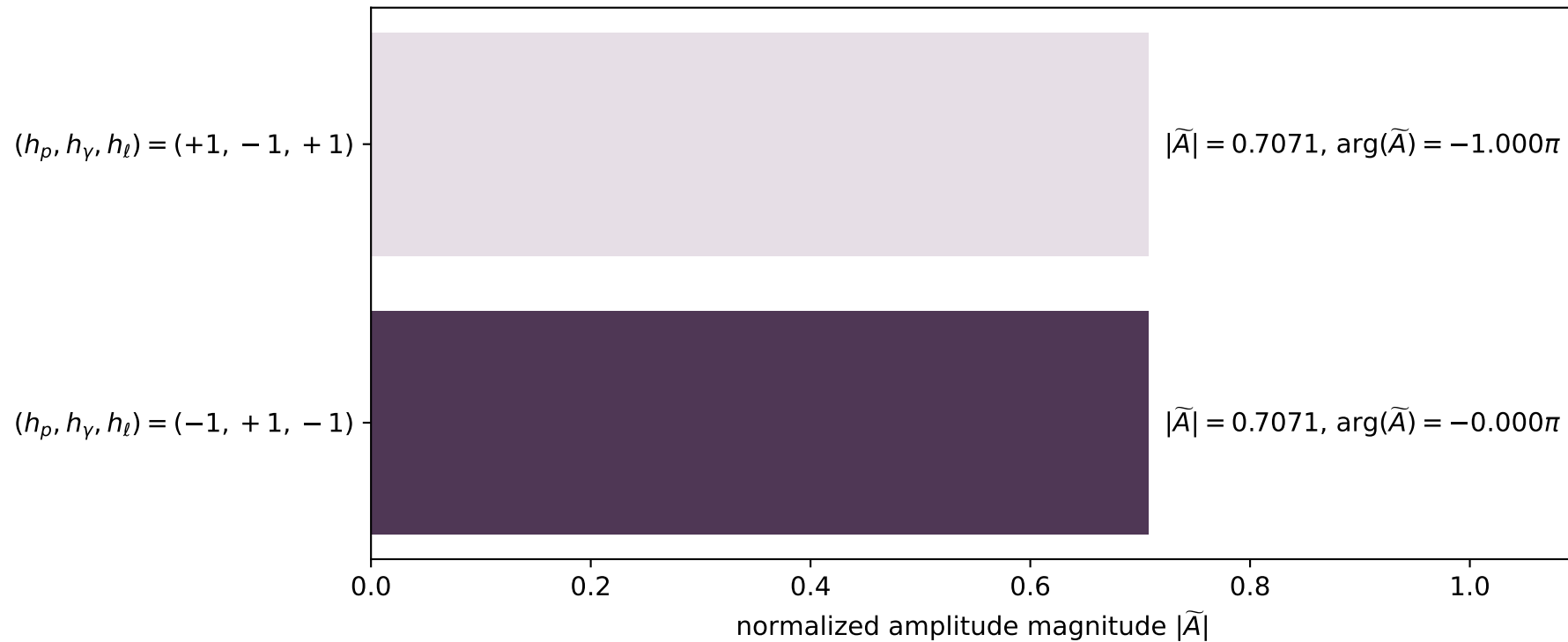
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.72071$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.57447$, $\phi_\gamma=1.17554\text{e-}06$
 $E_\gamma=0.659762$, $\theta_{l'}=0$, $\phi_{l'}=0.674751$
 $Q^2=22.9669$, $x_B=0.986451$, $t=-16.6848$, $W^2=1.19529$, $y=0.965266$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.3805$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3805$
 P' $E= 1.9598$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.7207$
 q_γ $E= 0.6598$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6598$

x--y plane

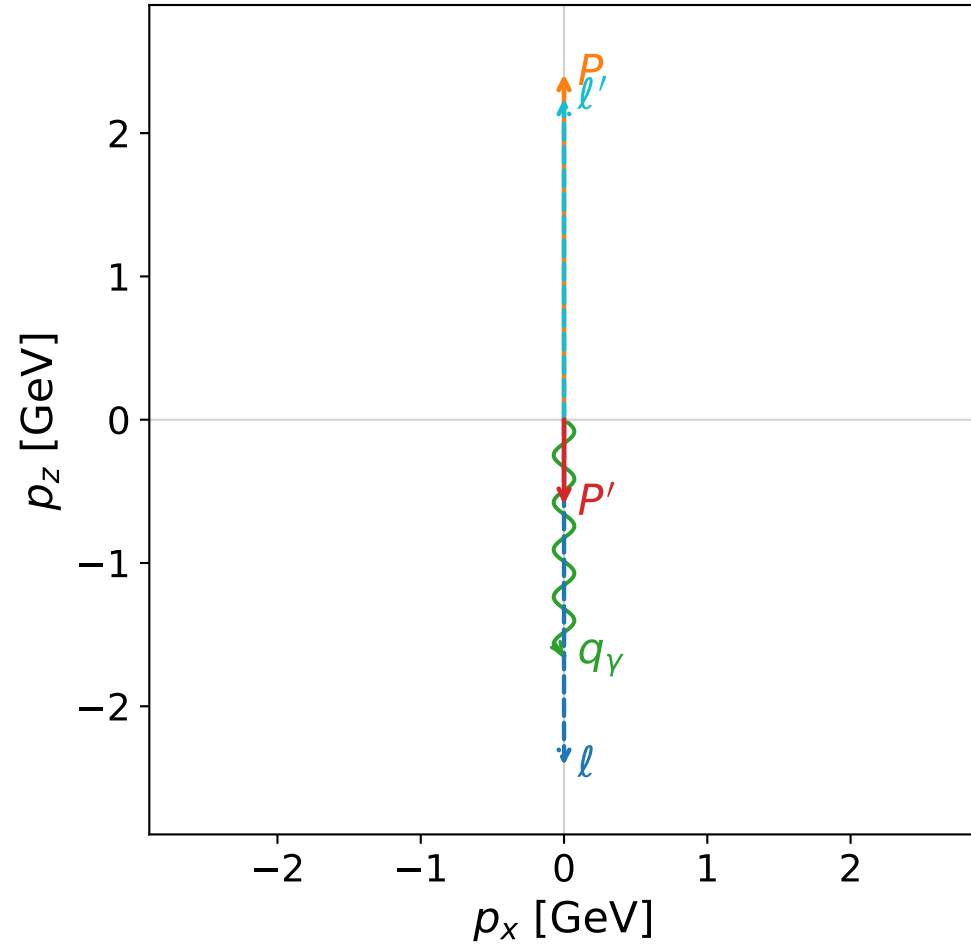


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

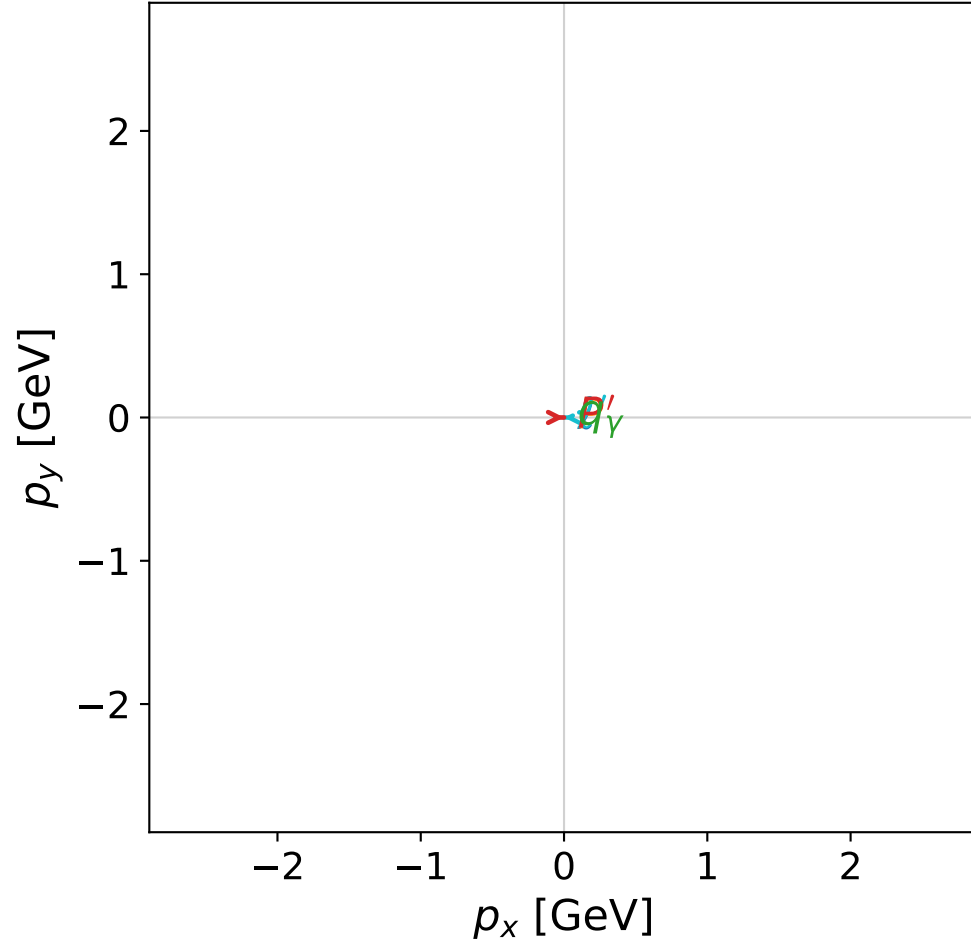


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



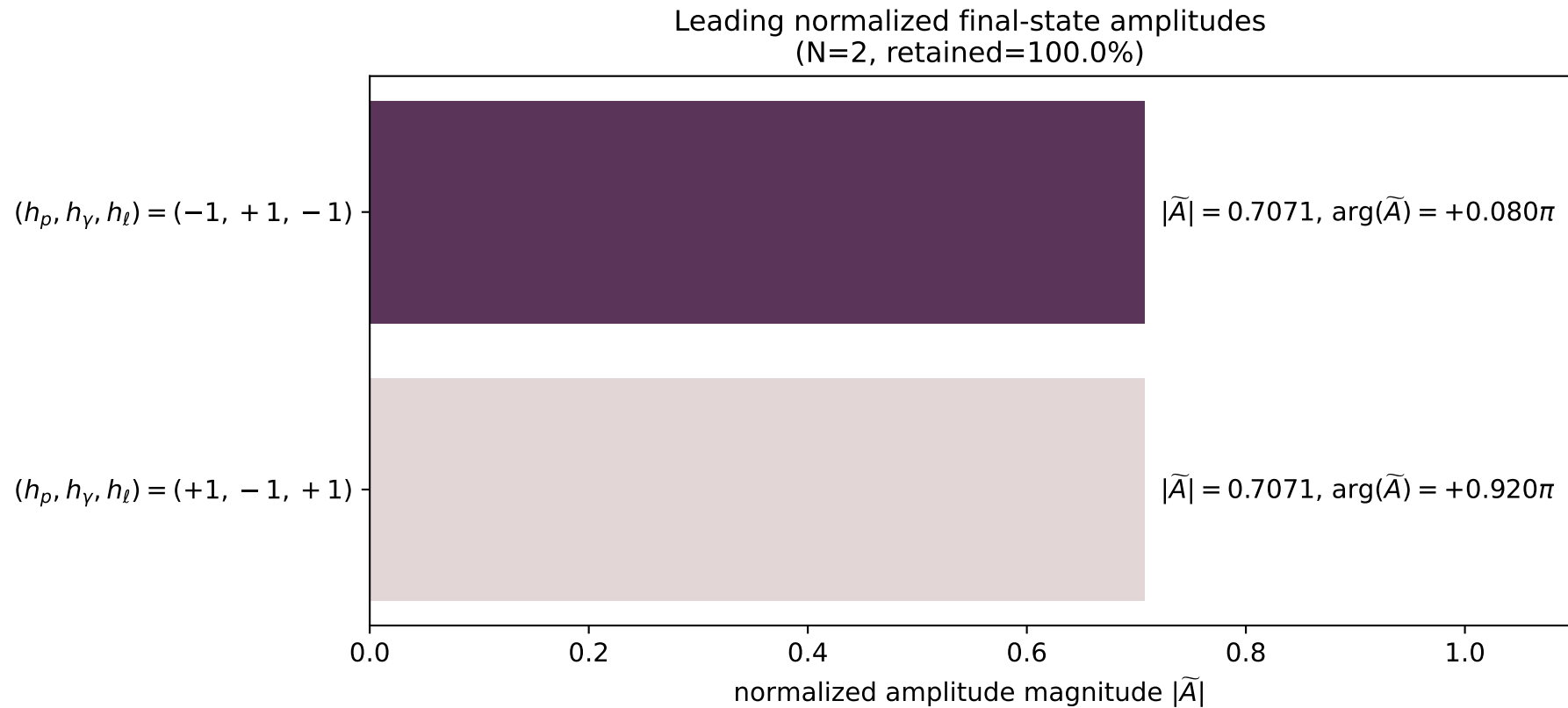
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_16
 $d_{\{\rm GHZ\}}=2.47608\text{e-}08$
 region: polarization_cluster_01_local_minimum_16
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0016
 lepton mixing angle: $\alpha_e=0.89433114$ rad
 incoming lepton: $0.626041|+\rangle + 0.779791|-\rangle$
 proton mixing angle: $\alpha_p=0.89433384$ rad
 incoming proton: $0.626038|+\rangle + 0.779792|-\rangle$

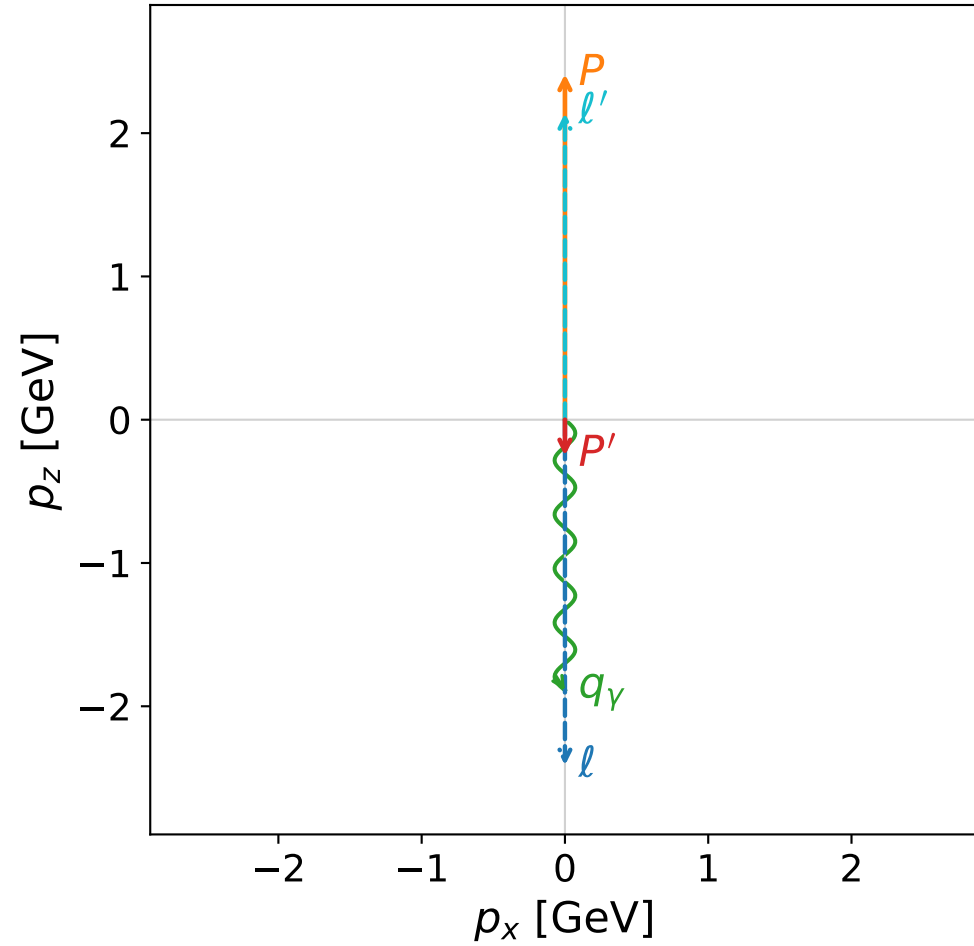
$s=24.9994$, $\sqrt{s}=4.99994$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.59265$
 $\theta_{p'}=3.1415$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.28079$, $\phi_\gamma=6.02949$
 $E_\gamma=1.64887$, $\theta_{l'}=2.53716\text{e-}05$, $\phi_{l'}=3.1392$
 $Q^2=21.6261$, $x_B=0.926939$, $t=-6.84211$, $W^2=2.58442$, $y=0.967293$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.2415$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.2415$
 P' $E= 1.1095$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.5927$
 q_γ $E= 1.6489$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.6489$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

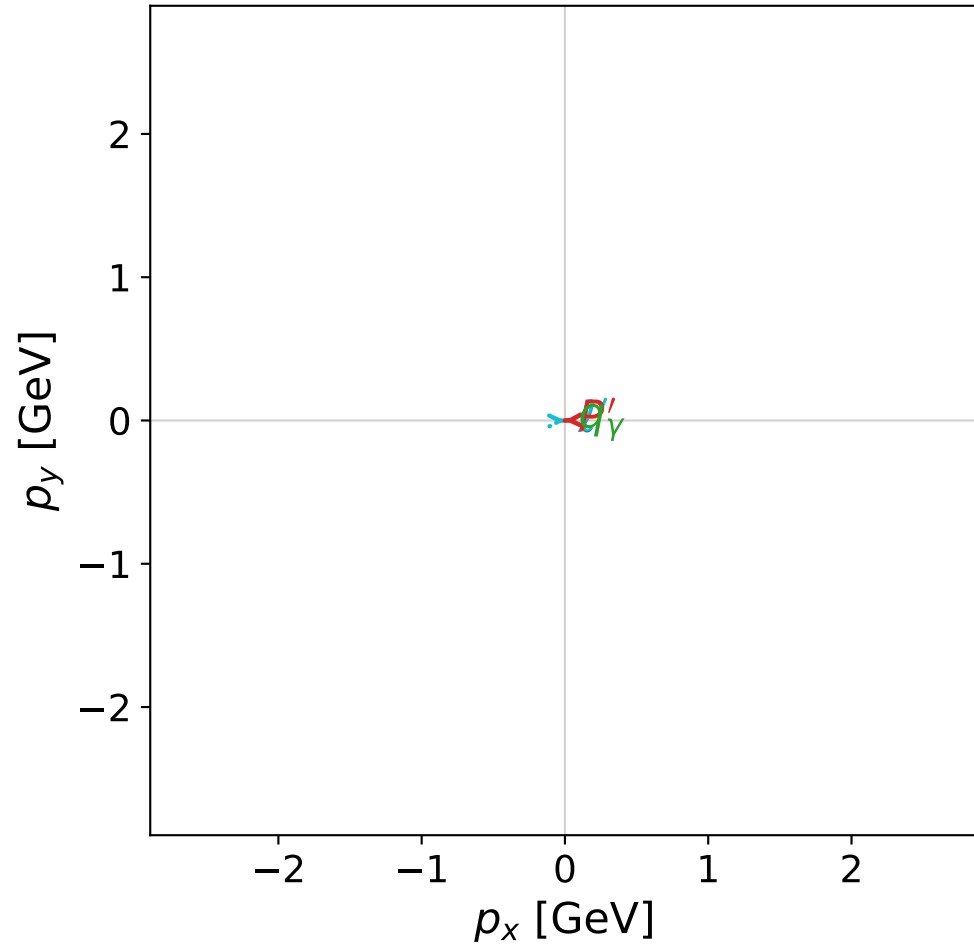


dghz_mixing_angles_polarization_01_local_minimum_19
 $d_{\text{GHZ}}=2.53539\text{e-}08$
 region: polarization_cluster_01_local_minimum_19
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0019
 lepton mixing angle: $\alpha_e=0.78519375$ rad
 incoming lepton: $0.707251|+\rangle + 0.706962|-\rangle$
 proton mixing angle: $\alpha_p=0.7851987$ rad
 incoming proton: $0.707248|+\rangle + 0.706966|-\rangle$

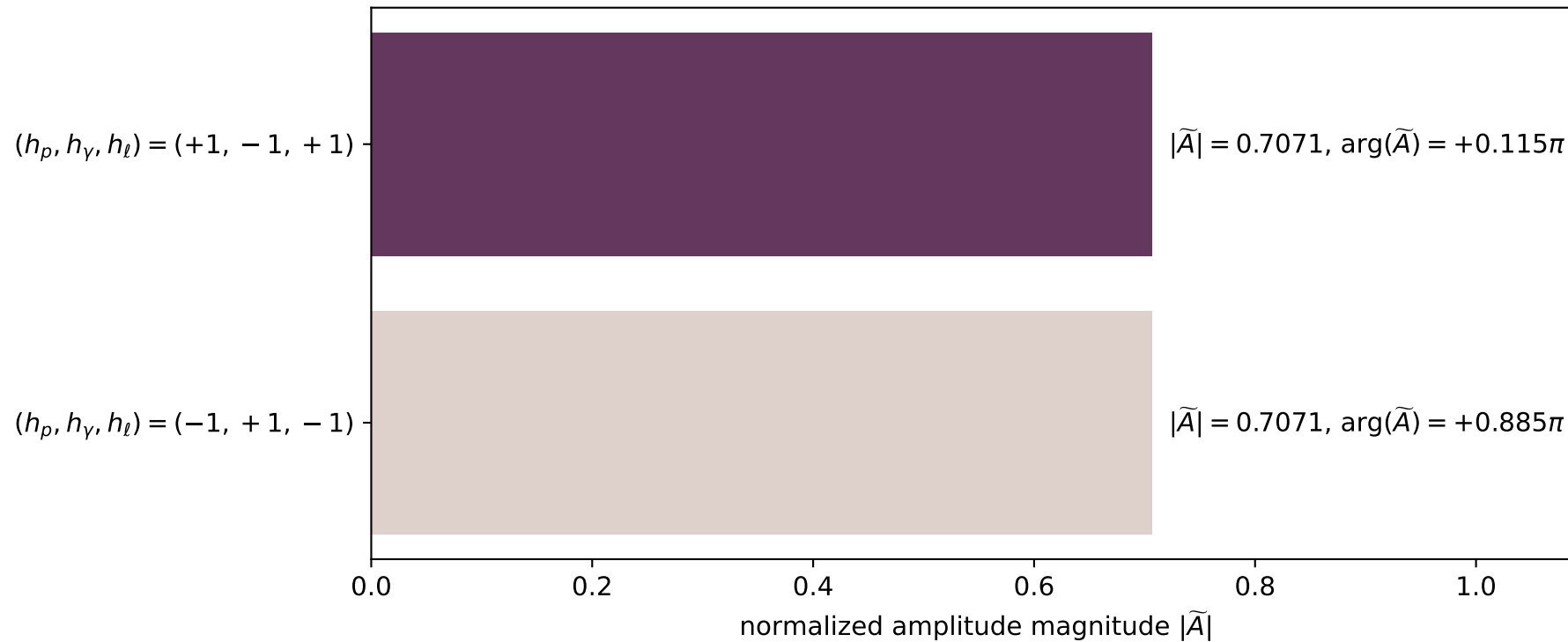
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.251318$
 $\theta_{P'}=3.14131$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.17031$, $\phi_\gamma=0.388985$
 $E_\gamma=1.8888$, $\theta_{l'}=3.33062\text{e-}05$, $\phi_{l'}=0.0287211$
 $Q^2=20.648$, $x_B=0.88364$, $t=-4.47898$, $W^2=3.59883$, $y=0.968774$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.1401$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.1401$
 P' $E= 0.9711$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.2513$
 q_γ $E= 1.8888$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8888$

x--y plane

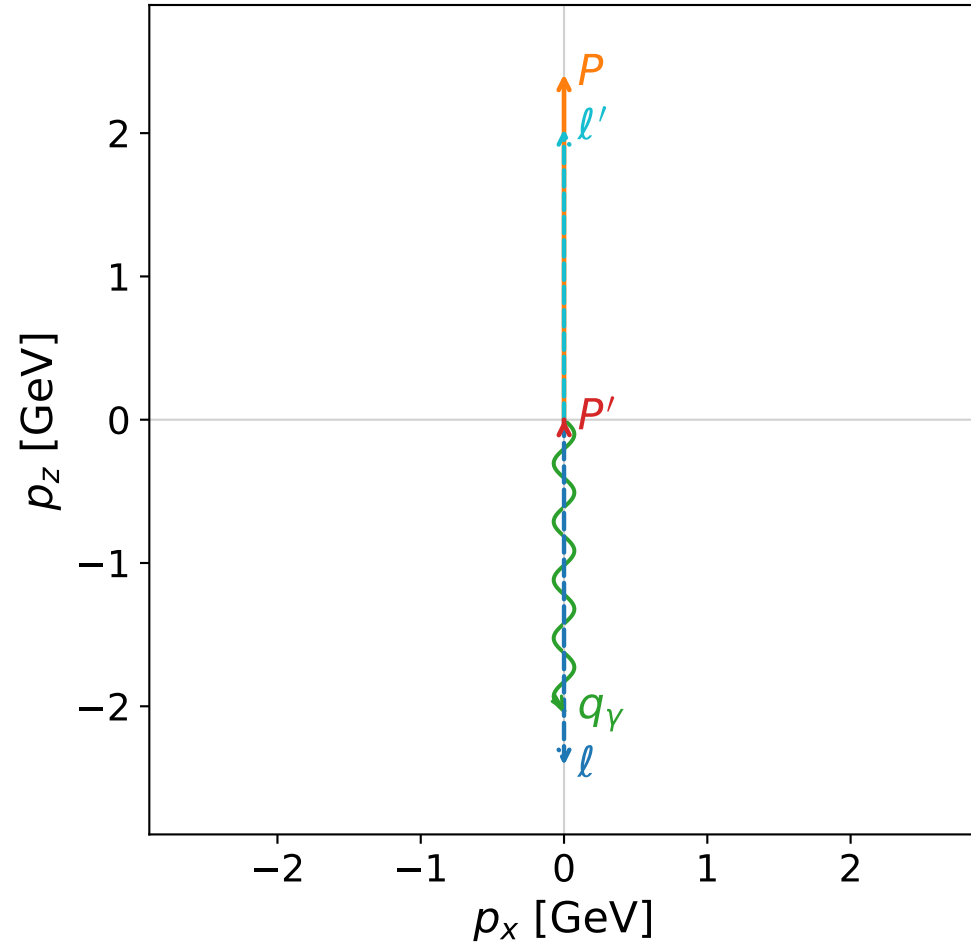


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

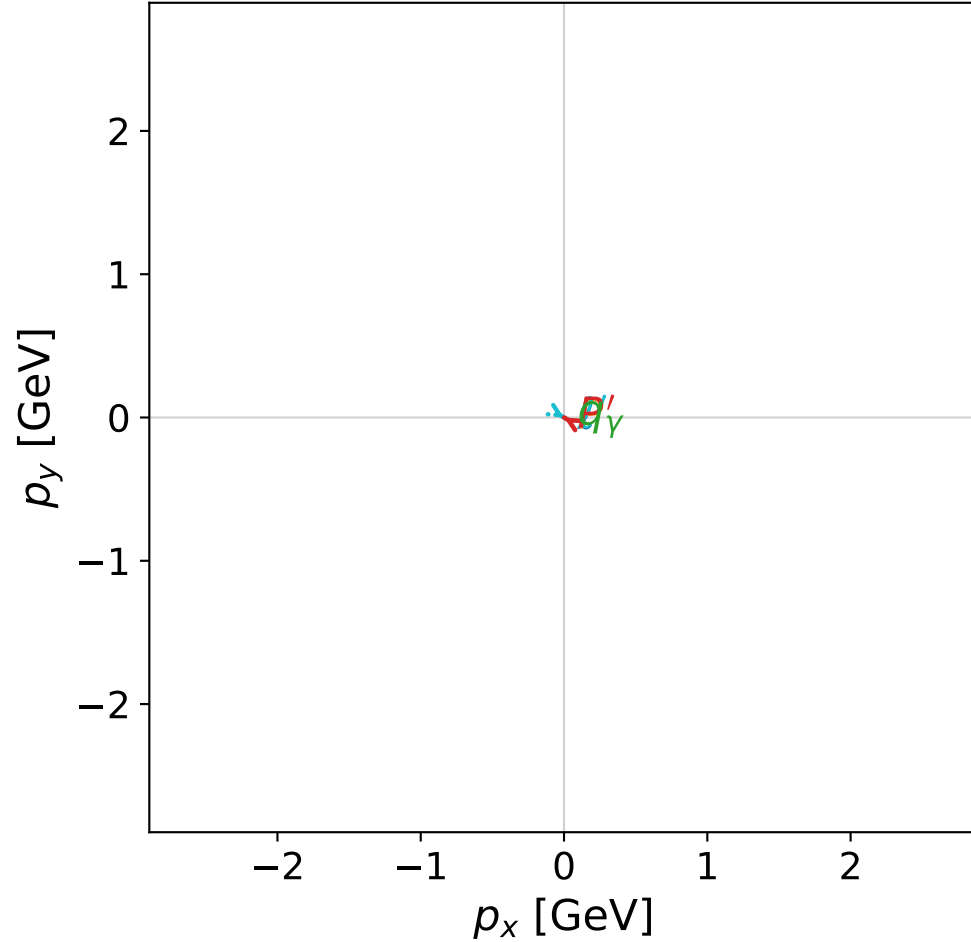


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



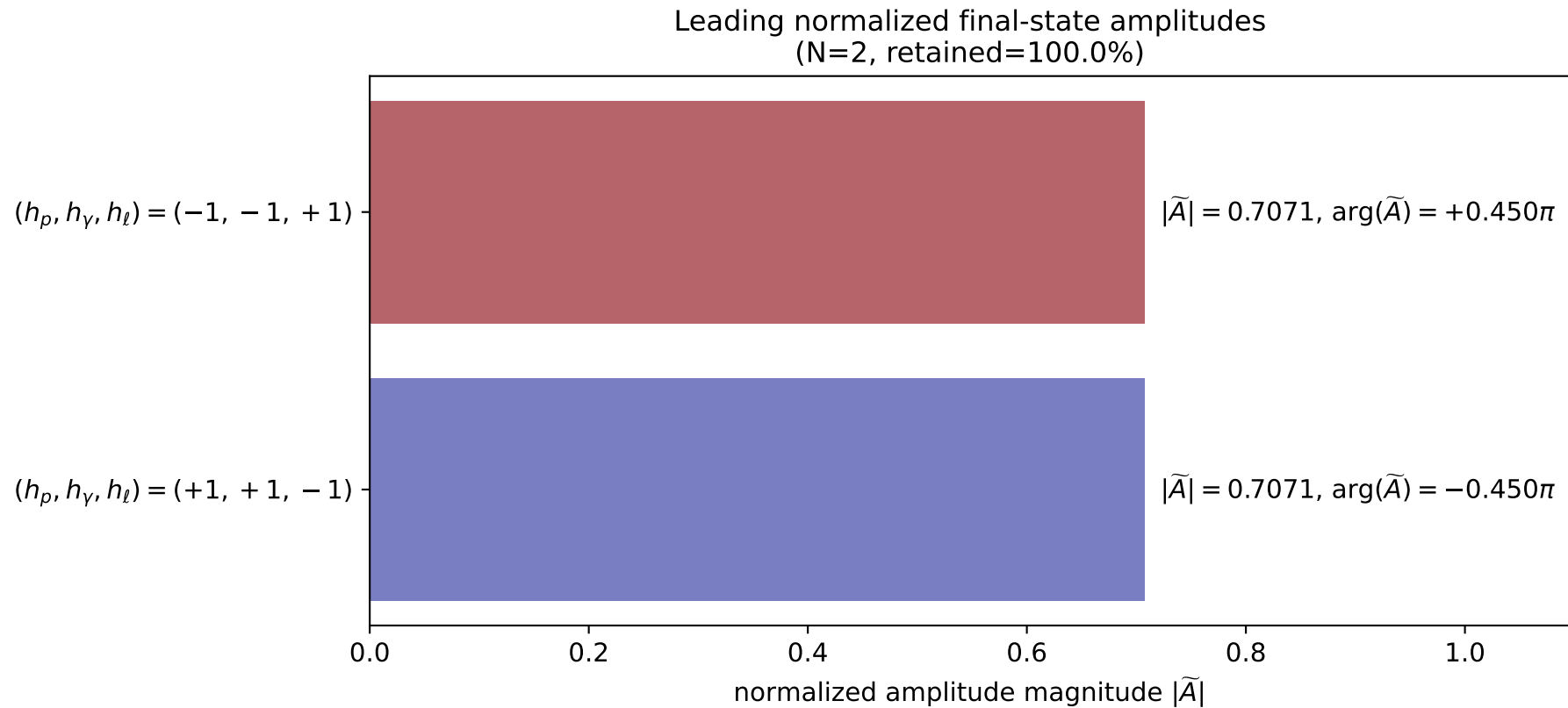
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_23
 $d_{\text{GHZ}}=2.57818\text{e-}08$
 region: polarization_cluster_01_local_minimum_23
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0023
 lepton mixing angle: $\alpha_e=0.77571838$ rad
 incoming lepton: $0.713918|+\rangle + 0.700229|-\rangle$
 proton mixing angle: $\alpha_p=0.77571056$ rad
 incoming proton: $0.713924|+\rangle + 0.700224|-\rangle$

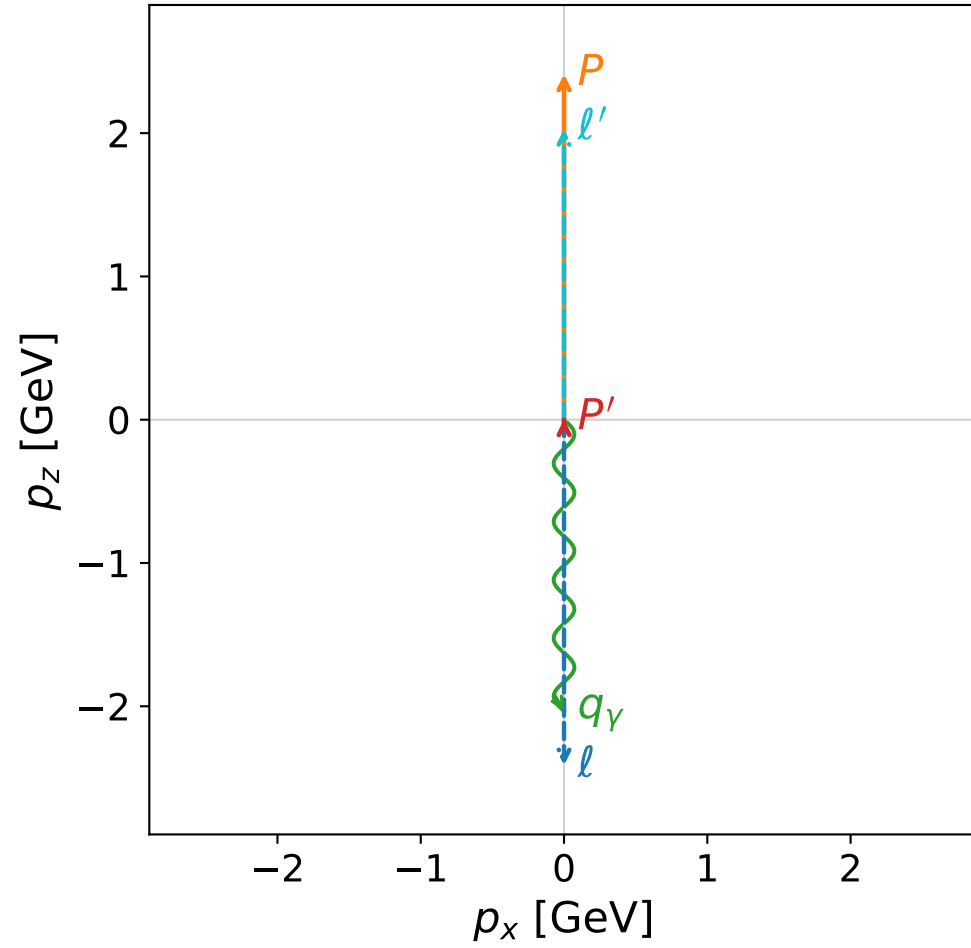
$s=24.9977$, $\sqrt{s}=4.99977$, $|\vec{P}|=2.41189$, $|\vec{P}'|=0.00408939$
 $\theta_{P'}=0.0106052$, $\theta_Y=3.14159$, $\phi_{P'}=2.61744$, $\phi_Y=4.55652$
 $E_Y=2.03292$, $\theta_{l'}=2.13758\text{e-}05$, $\phi_{l'}=5.75903$
 $Q^2=19.5733$, $x_B=0.836333$, $t=-3.07548$, $W^2=4.71027$, $y=0.970393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 l' $E= 2.0288$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0288$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0041$
 q_Y $E= 2.0329$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0329$

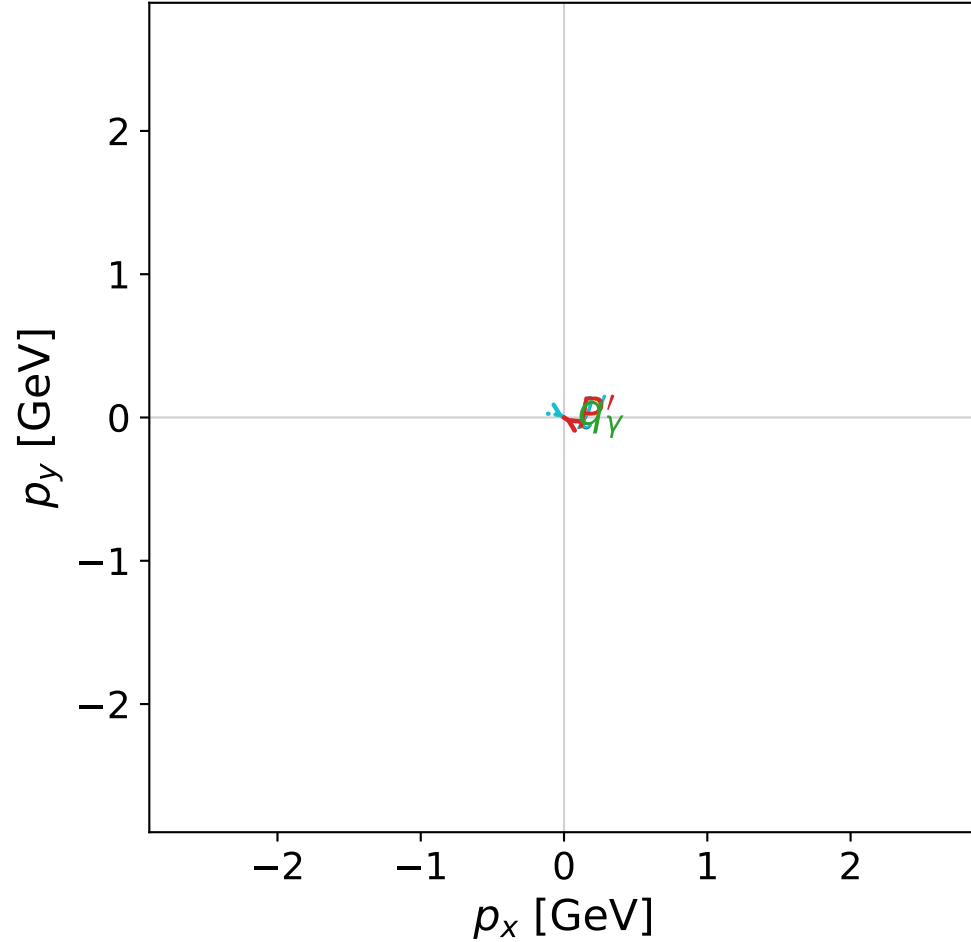


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



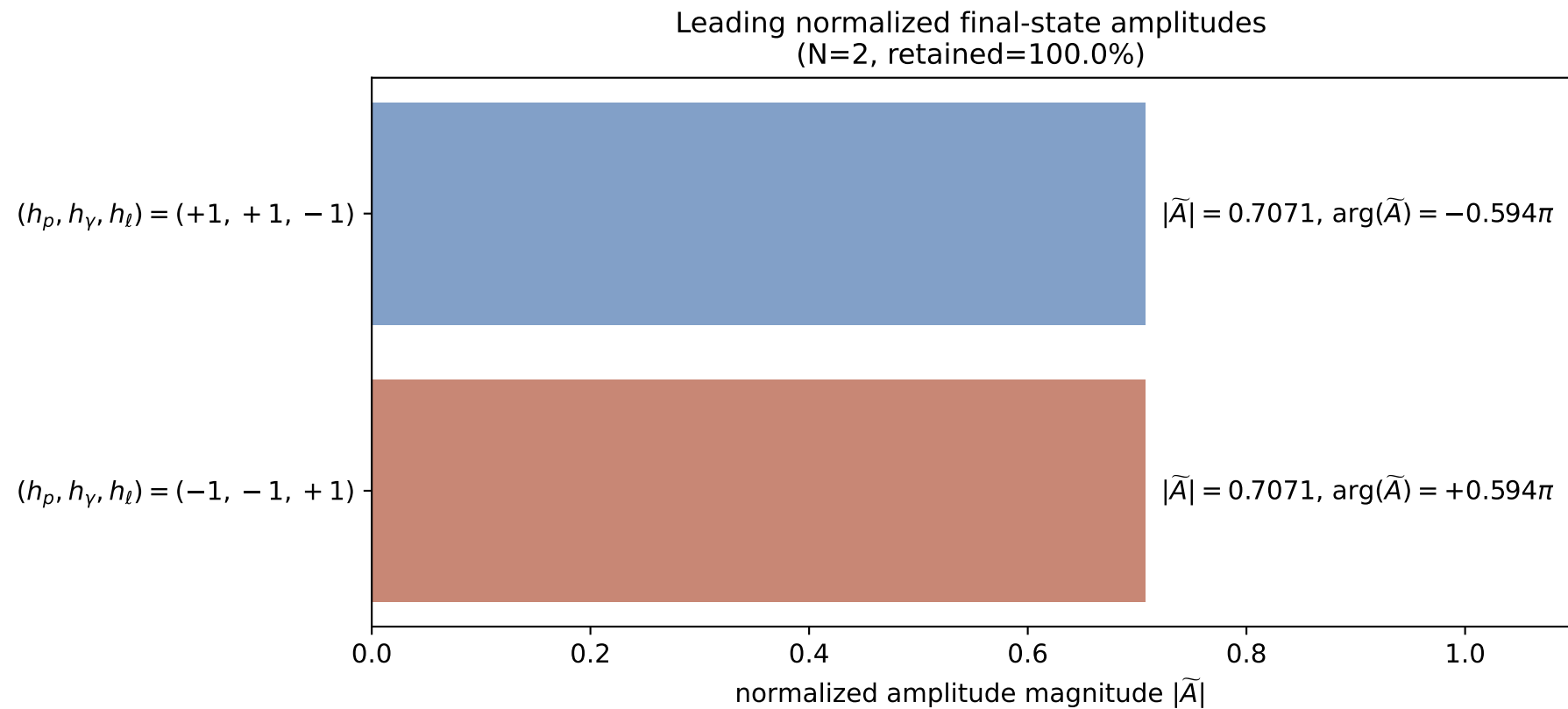
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_24
 $d_{\{\text{rm GHZ}\}}=2.58712\text{e-}08$
 region: polarization_cluster_01_local_minimum_24
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0024
 lepton mixing angle: $\alpha_e=0.742527$ rad
 incoming lepton: $0.736762|+\rangle + 0.676152|-\rangle$
 proton mixing angle: $\alpha_p=0.74253335$ rad
 incoming proton: $0.736758|+\rangle + 0.676157|-\rangle$

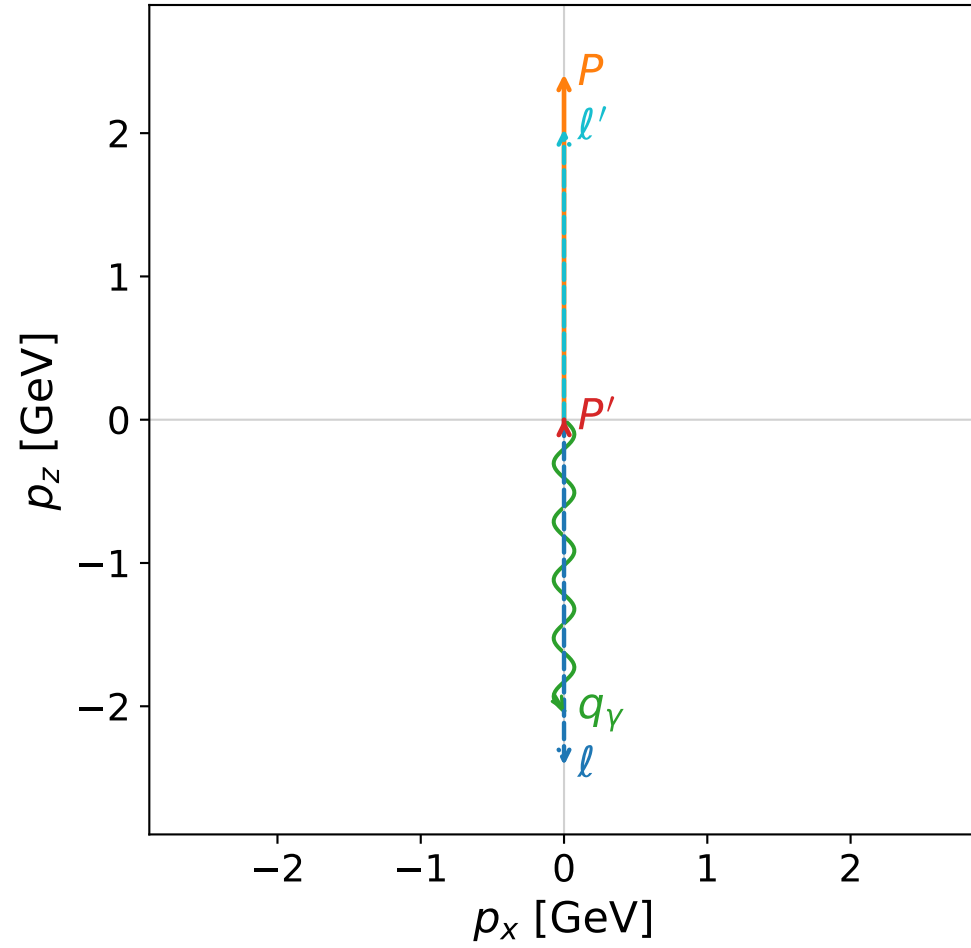
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0041485$
 $\theta_{p'}=0.00973758$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.58165$, $\phi_\gamma=5.00822$
 $E_\gamma=2.03307$, $\theta_{l'}=1.99099\text{e-}05$, $\phi_{l'}=5.72325$
 $Q^2=19.5752$, $x_B=0.836327$, $t=-3.07541$, $W^2=4.71079$, $y=0.970396$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0289$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0289$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0041$
 q_γ $E= 2.0331$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0331$

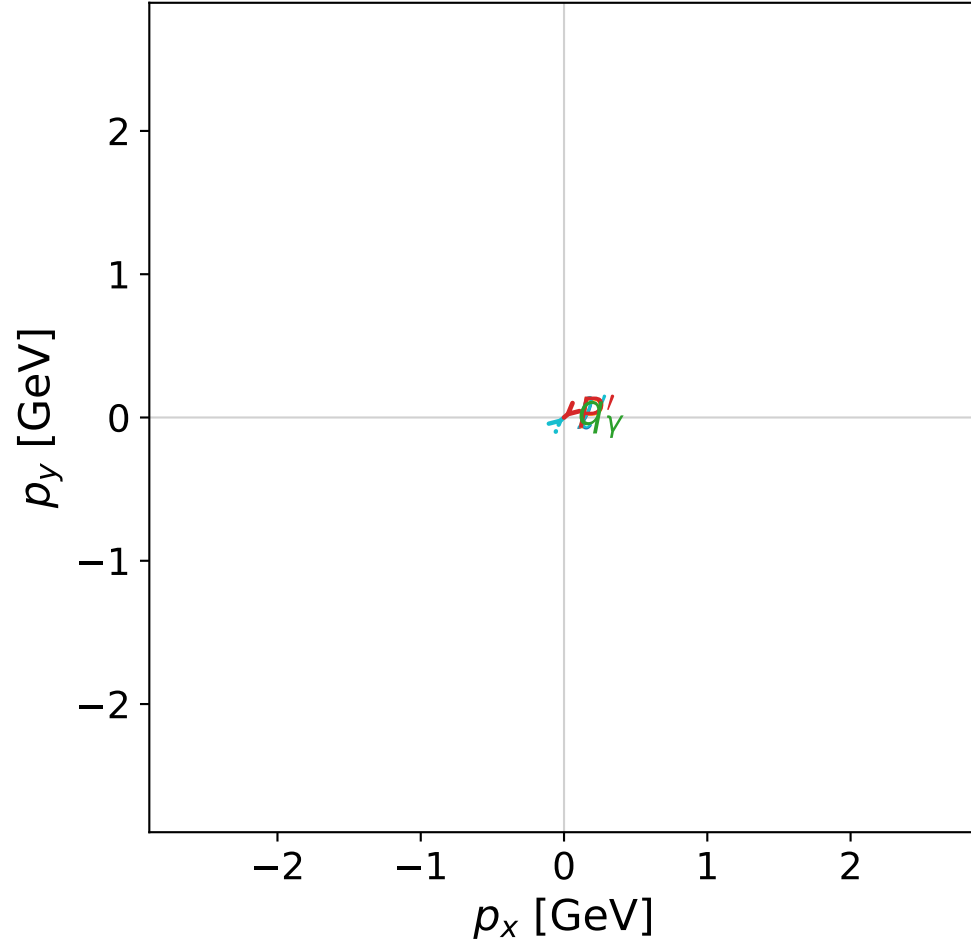


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



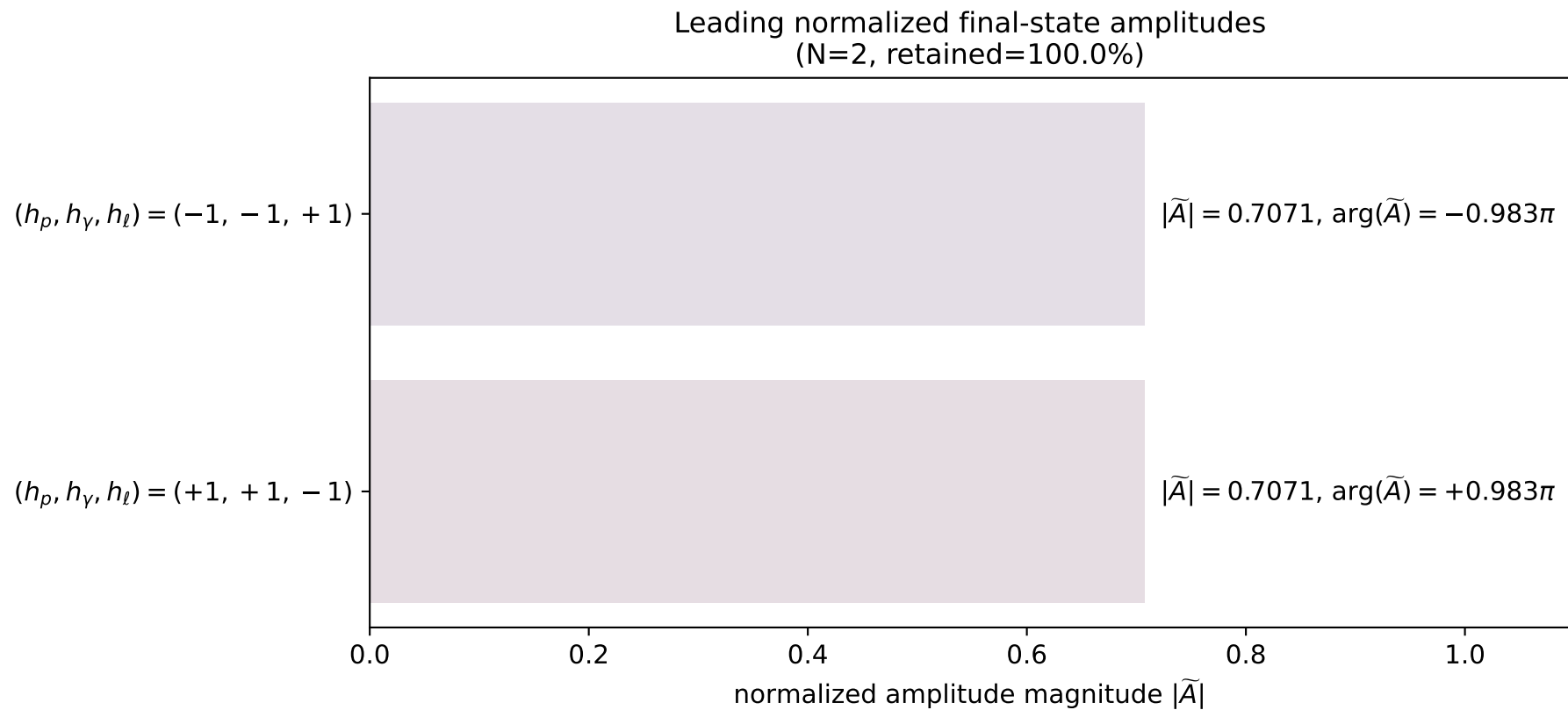
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_25
 $d_{\text{GHZ}}=2.59286\text{e-}08$
 region: polarization_cluster_01_local_minimum_25
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0025
 lepton mixing angle: $\alpha_e=0.768686$ rad
 incoming lepton: $0.718825|+\rangle + 0.695191|-\rangle$
 proton mixing angle: $\alpha_p=0.76868388$ rad
 incoming proton: $0.718826|+\rangle + 0.69519|-\rangle$

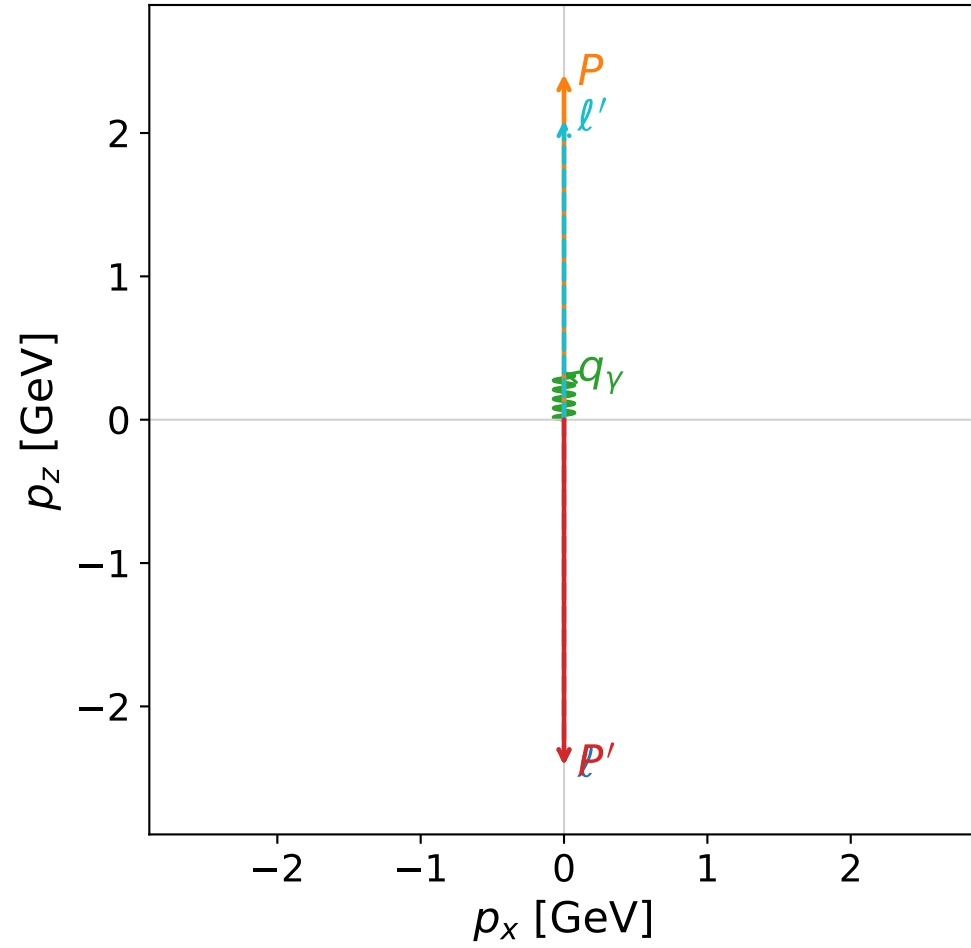
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00436041$
 $\theta_{P'}=0.00736311$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.8561$, $\phi_\gamma=0.0545742$
 $E_\gamma=2.03318$, $\theta_{\ell'}=1.5825\text{e-}05$, $\phi_{\ell'}=0.714507$
 $Q^2=19.5741$, $x_B=0.836282$, $t=-3.07439$, $W^2=4.71185$, $y=0.970398$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0288$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0288$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0044$
 q_γ $E= 2.0332$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0332$

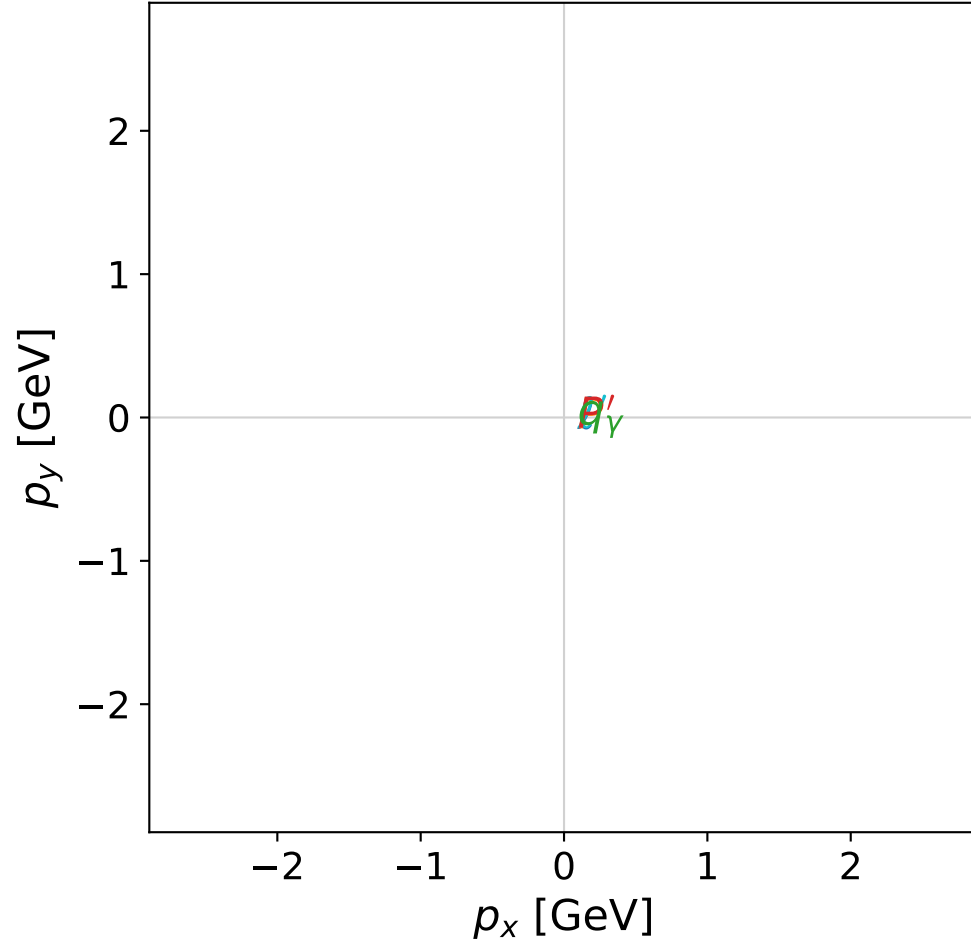


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



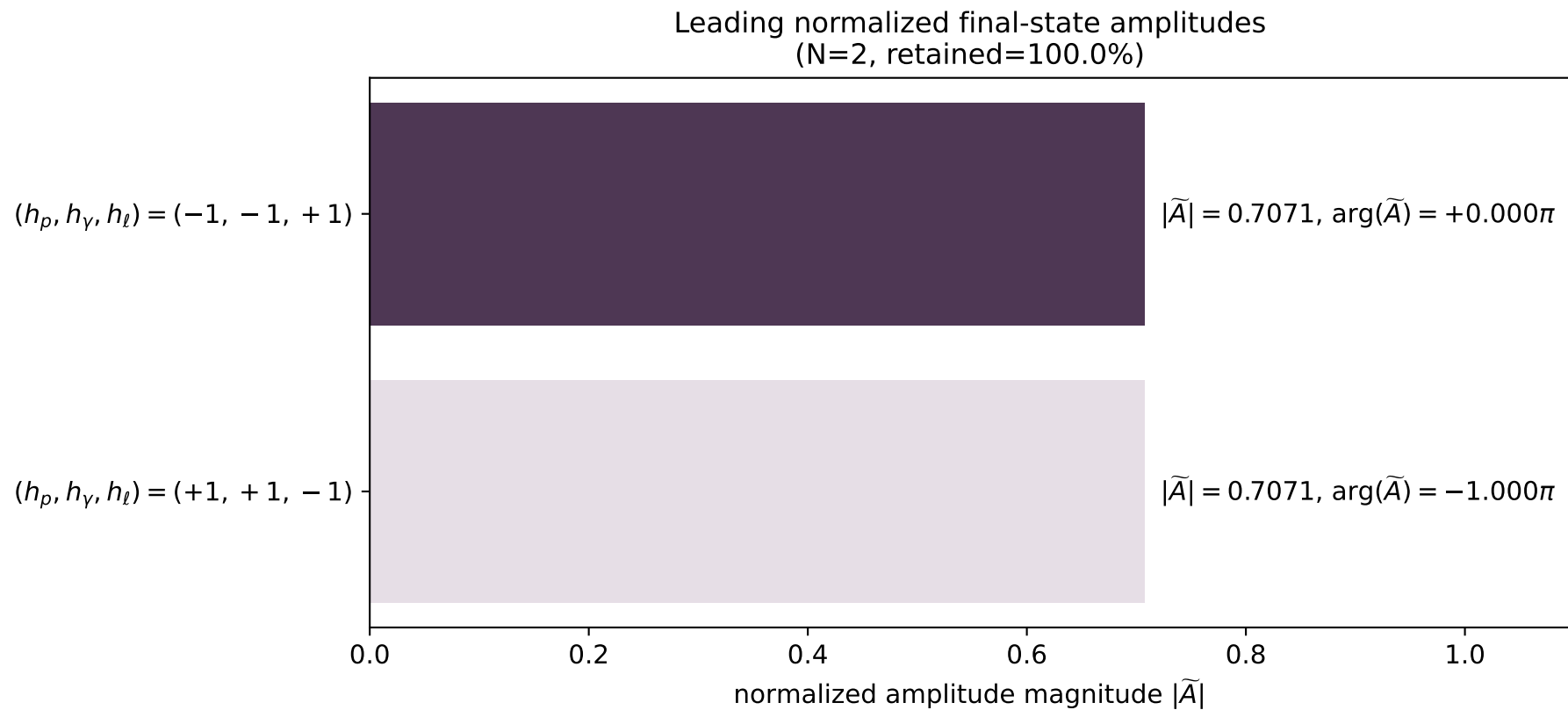
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_27
 $d_{\{\text{rm GHZ}\}}=2.59874\text{e-}08$
 region: polarization_cluster_01_local_minimum_27
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0027
 lepton mixing angle: $\alpha_e=0.75349531$ rad
 incoming lepton: $0.729302|+\rangle + 0.684192|-\rangle$
 proton mixing angle: $\alpha_p=0.81730854$ rad
 incoming proton: $0.684187|+\rangle + 0.729307|-\rangle$

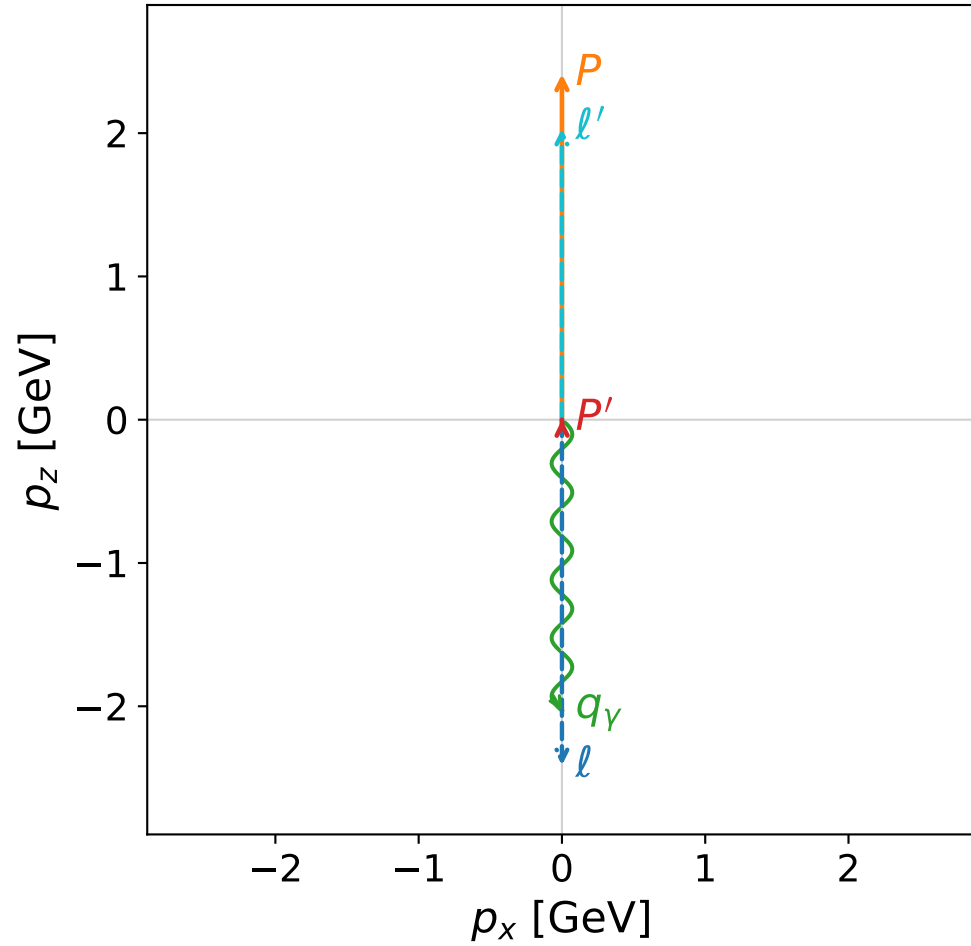
$s=24.9779$, $\sqrt{s}=4.99779$, $|\vec{P}|=2.41087$, $|\vec{P}'|=2.41087$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.0939447$, $\phi_\gamma=0.966166$
 $E_\gamma=0.322487$, $\theta_{l'}=0$, $\phi_{l'}=3.23554$
 $Q^2=20.1393$, $x_B=0.862026$, $t=-23.2492$, $W^2=4.10329$, $y=0.969487$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4109$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4109$
 P $E= 2.5869$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4109$
 ℓ' $E= 2.0884$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0884$
 P' $E= 2.5869$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4109$
 q_γ $E= 0.3225$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3225$

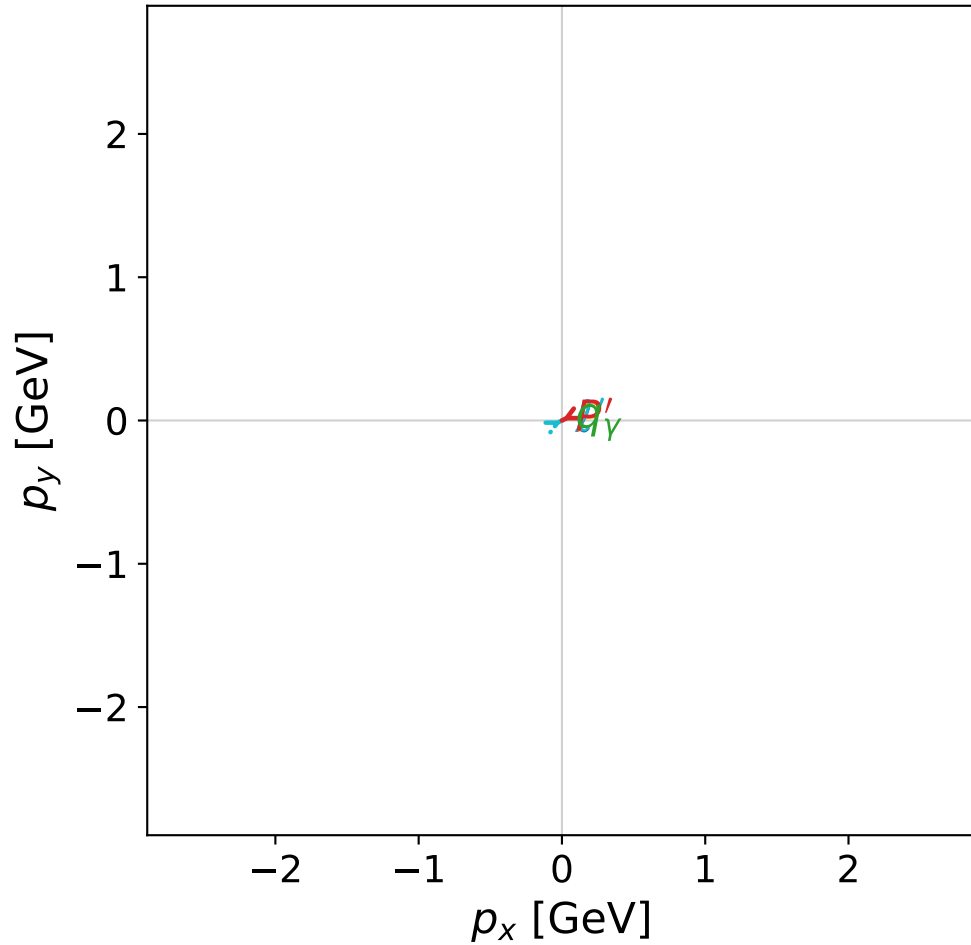


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

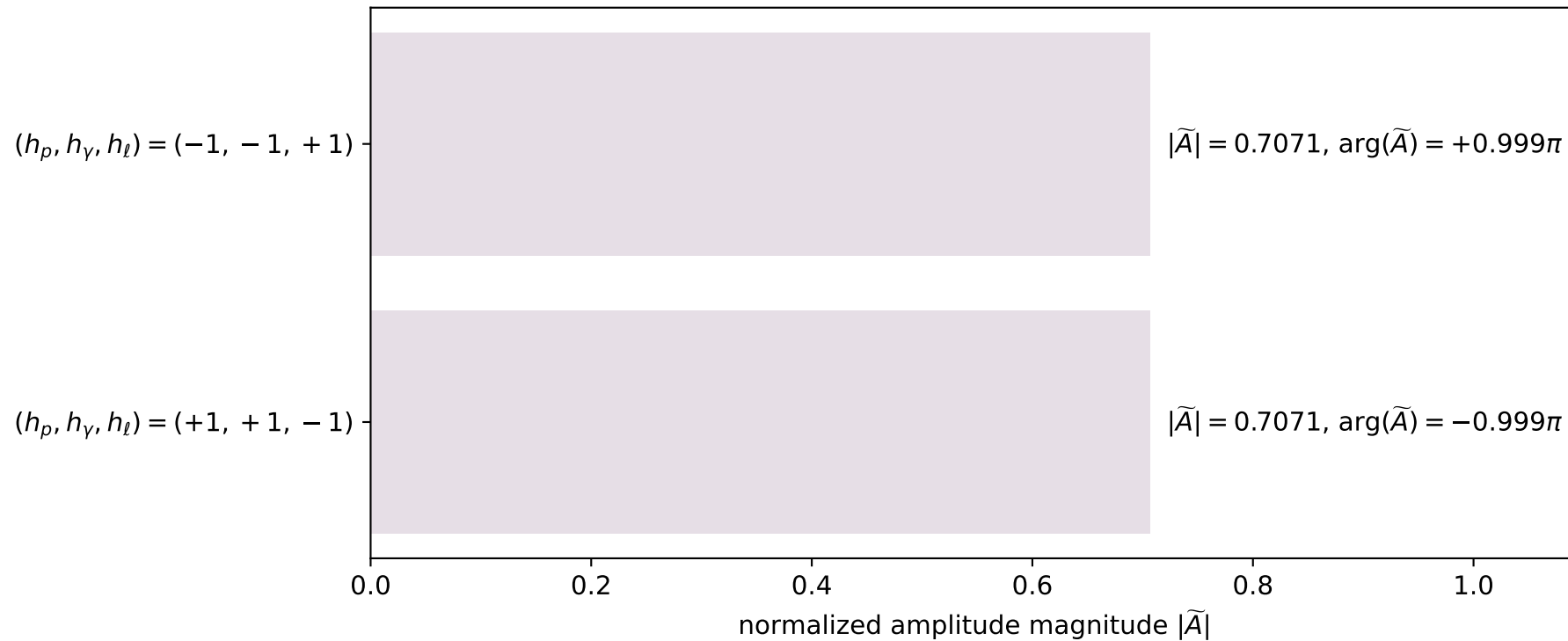


dghz_mixing_angles_polarization_01_local_minimum_28
 $d_{\text{GHZ}}=2.60293\text{e-}08$
 region: polarization_cluster_01_local_minimum_28
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0028
 lepton mixing angle: $\alpha_e=0.7050247$ rad
 incoming lepton: $0.761596|+\rangle + 0.648053|-\rangle$
 proton mixing angle: $\alpha_p=0.70501577$ rad
 incoming proton: $0.761601|+\rangle + 0.648046|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.000139487$
 $\theta_{P'}=0.0108069$, $\theta_Y=3.14159$, $\phi_{P'}=3.60549$, $\phi_Y=6.28094$
 $E_Y=2.03107$, $\theta_{\ell'}=7.42072\text{e-}07$, $\phi_{\ell'}=0.463893$
 $Q^2=19.5945$, $x_B=0.837181$, $t=-3.0947$, $W^2=4.6907$, $y=0.970367$

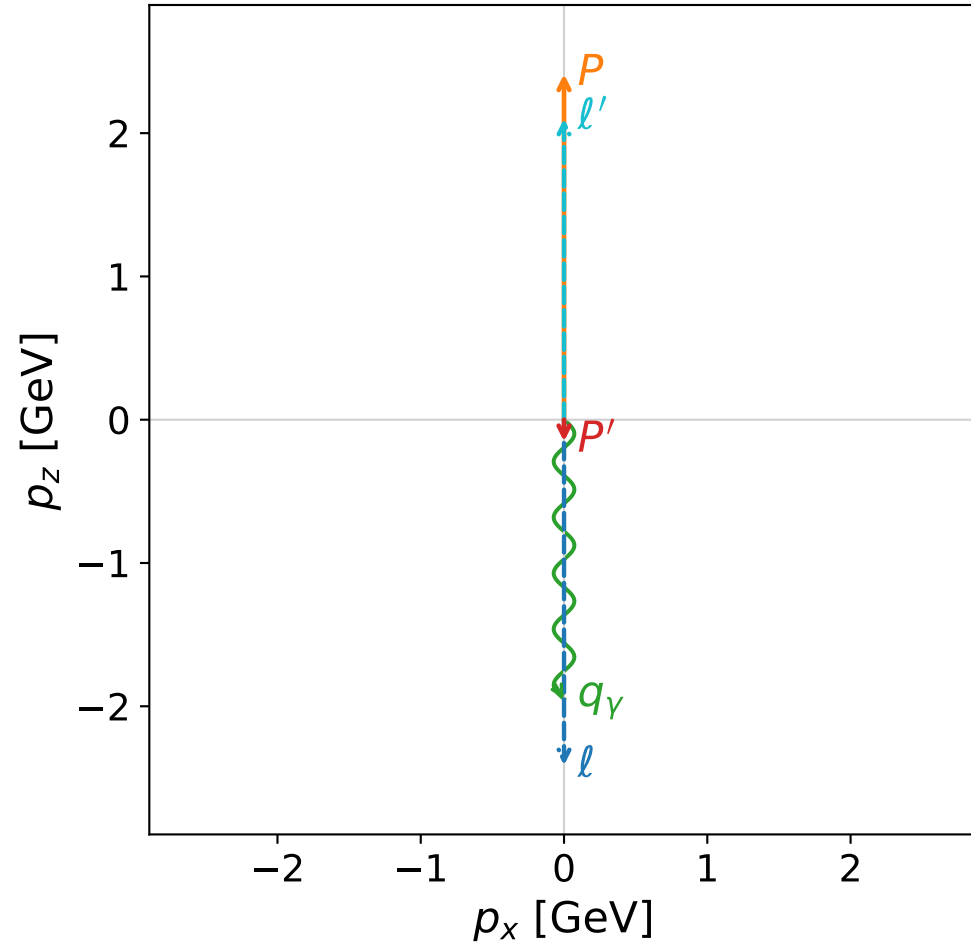
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0309$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0001$
 q_Y $E= 2.0311$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0311$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

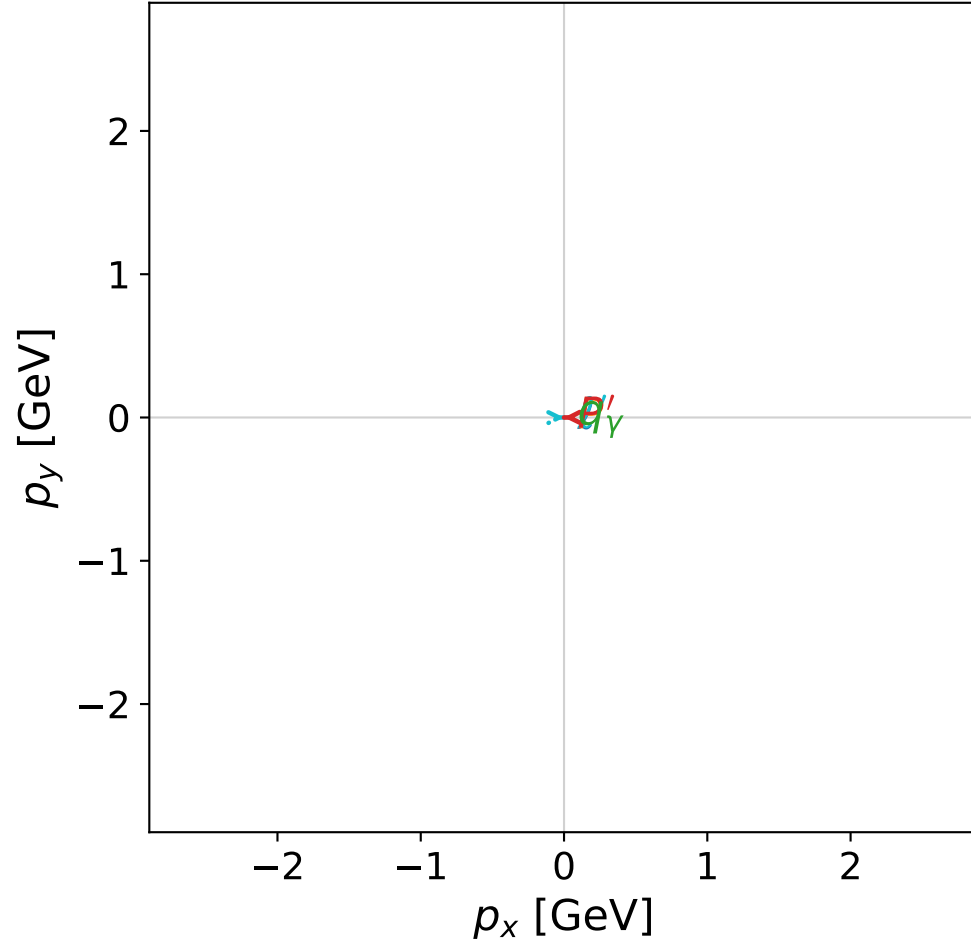


dghz_mixing_angles_polarization_01_local_minimum_32
 $d_{\{\rm GHZ\}}=2.62922\text{e-}08$
 region: polarization_cluster_01_local_minimum_32
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0032
 lepton mixing angle: $\alpha_e=0.6915138$ rad
 incoming lepton: $0.770282|+\rangle + 0.637704|-\rangle$
 proton mixing angle: $\alpha_p=0.69151265$ rad
 incoming proton: $0.770282|+\rangle + 0.637703|-\rangle$

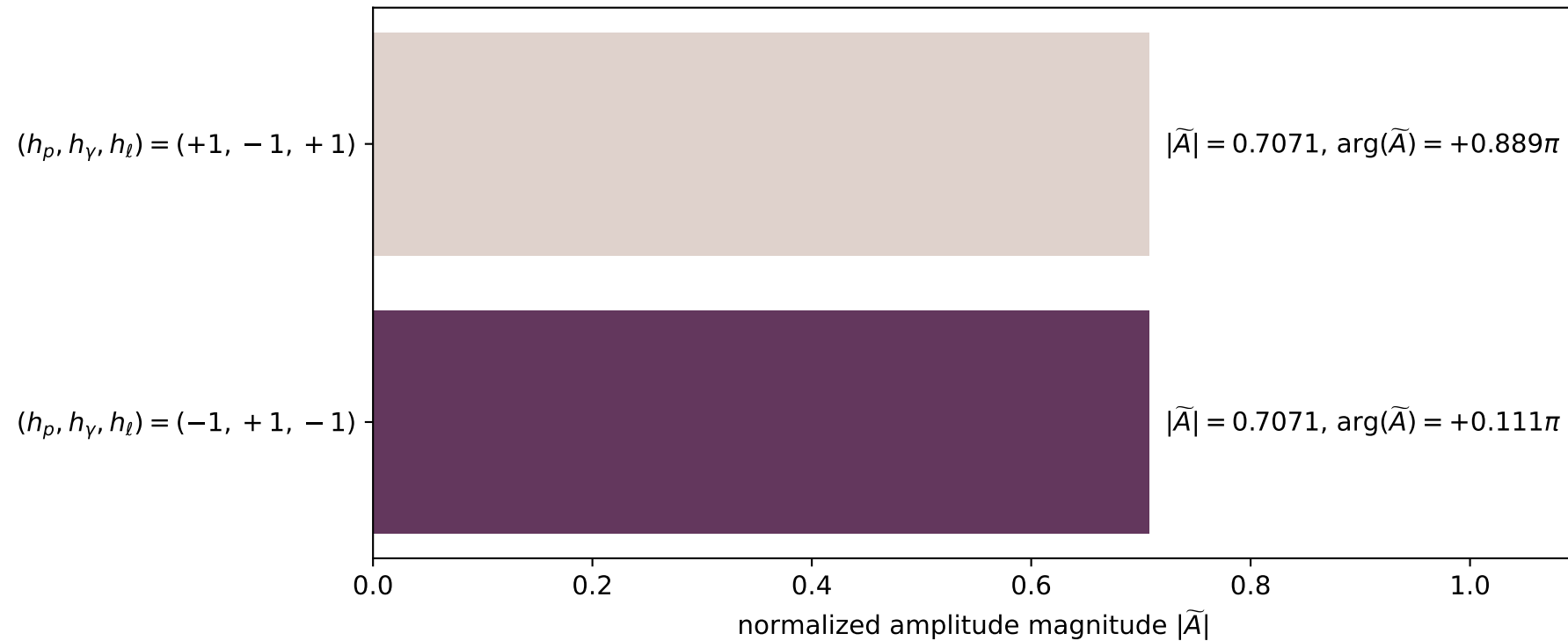
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.152285$
 $\theta_{p'}=3.14116$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.14913$, $\phi_\gamma=2.80196$
 $E_\gamma=1.94872$, $\theta_{l'}=3.10422\text{e-}05$, $\phi_{l'}=0.00753968$
 $Q^2=20.2706$, $x_B=0.866979$, $t=-3.89357$, $W^2=3.98998$, $y=0.969344$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.1010$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.1010$
 P' $E= 0.9503$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.1523$
 q_γ $E= 1.9487$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9487$

x--y plane

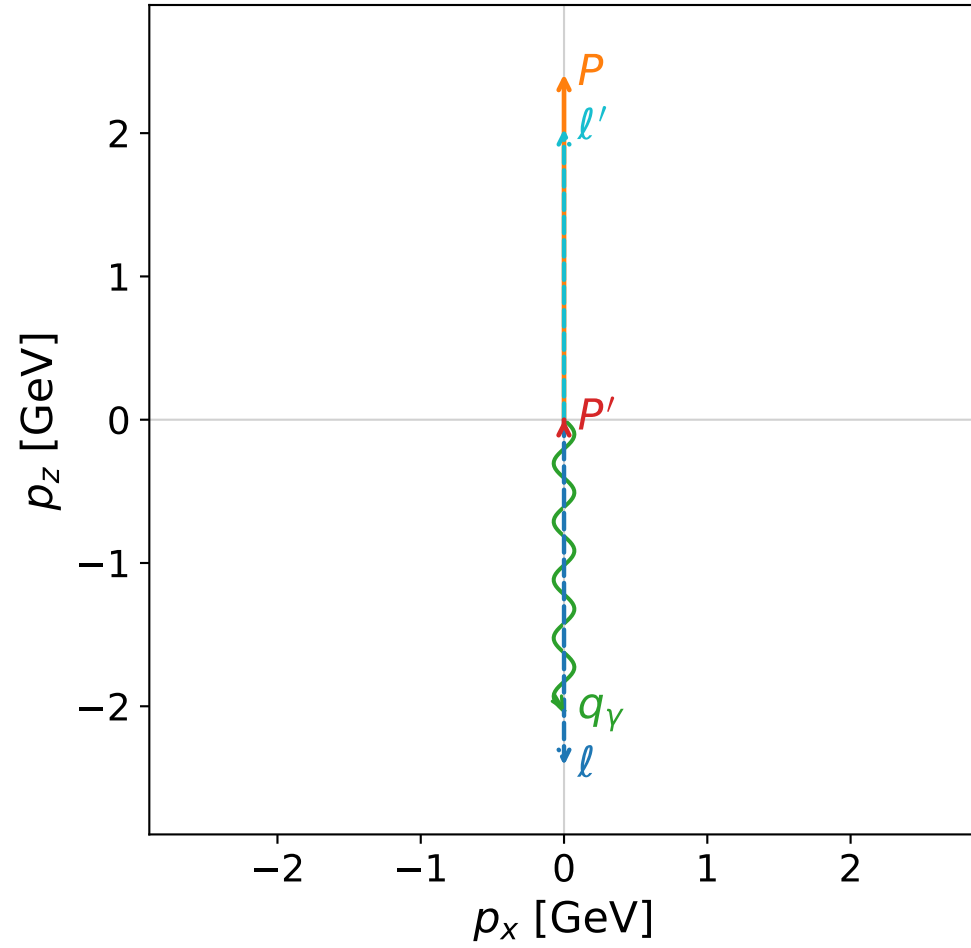


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

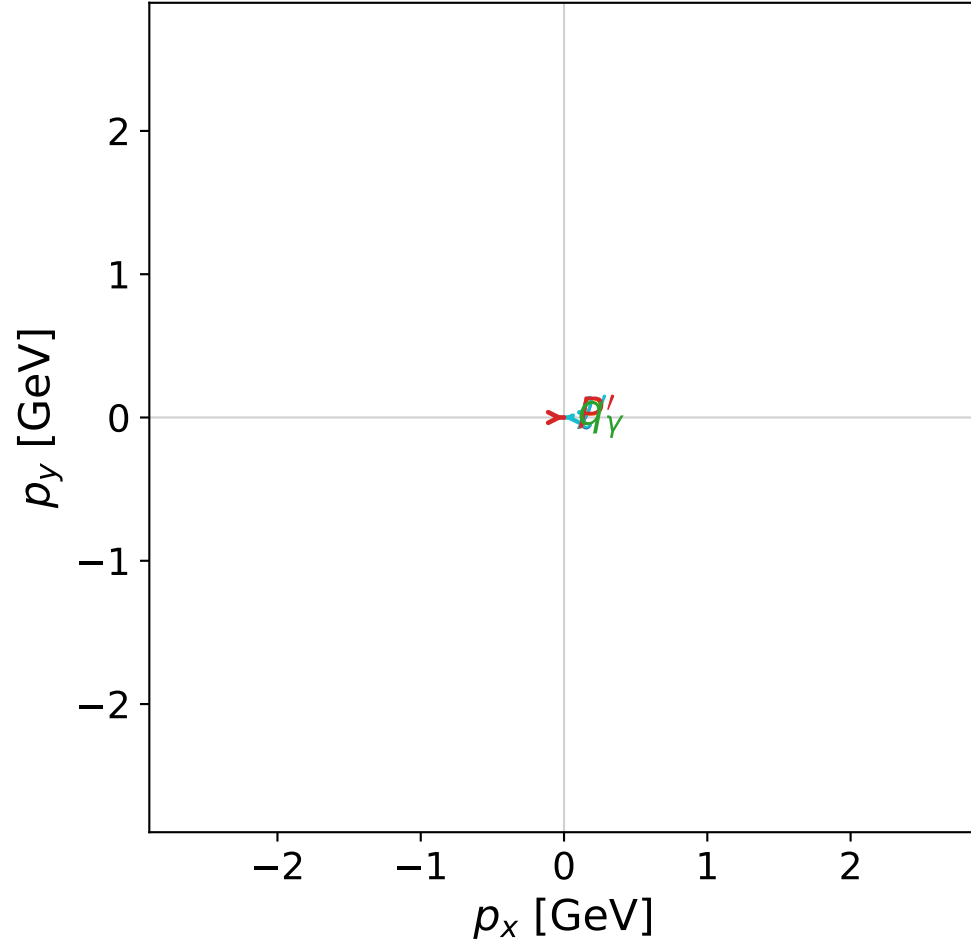


dghz_mixing_angles_polarization_01_local_minimum_35
 $d_{\{\rm GHZ\}}=2.63636\text{e-}08$
 region: polarization_cluster_01_local_minimum_35
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0035
 lepton mixing angle: $\alpha_e=0.86247052$ rad
 incoming lepton: $0.650563|+\rangle + 0.759452|-\rangle$
 proton mixing angle: $\alpha_p=0.86246408$ rad
 incoming proton: $0.650568|+\rangle + 0.759448|-\rangle$

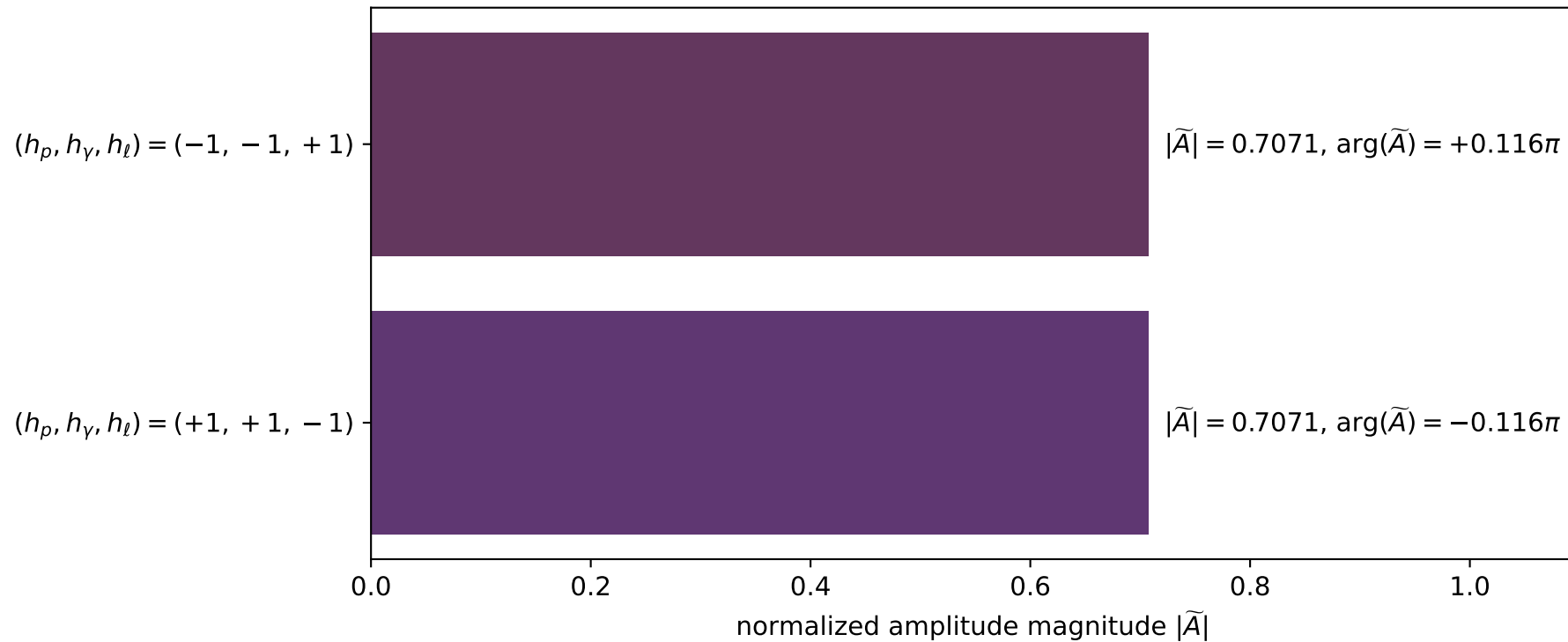
$s=24.999$, $\sqrt{s}=4.9999$, $|\vec{P}|=2.41196$, $|\vec{P}'|=0.00239249$
 $\theta_{P'}=0.00487673$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.28293$, $\phi_\gamma=3.50677$
 $E_\gamma=2.03215$, $\theta_{\ell'}=5.74822\text{e-}06$, $\phi_{\ell'}=3.14134$
 $Q^2=19.5828$, $x_B=0.836699$, $t=-3.08376$, $W^2=4.70189$, $y=0.970382$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0298$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0298$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0024$
 q_γ $E= 2.0321$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0321$

x--y plane

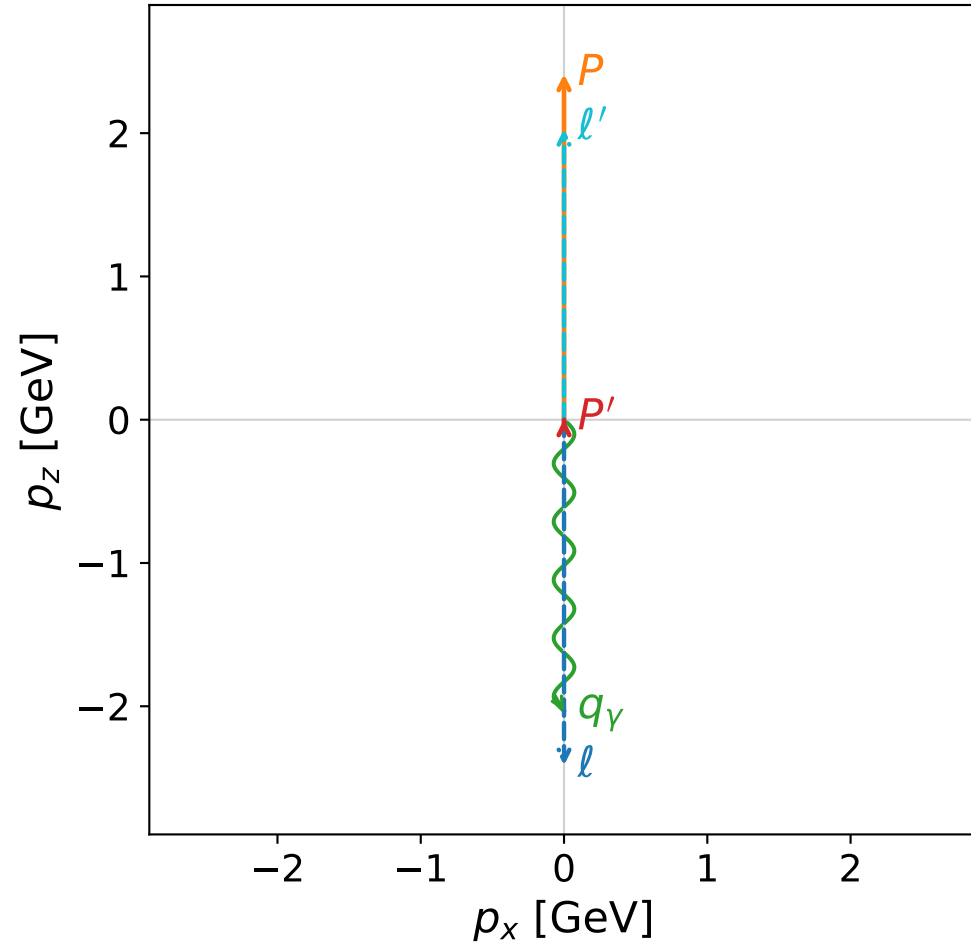


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

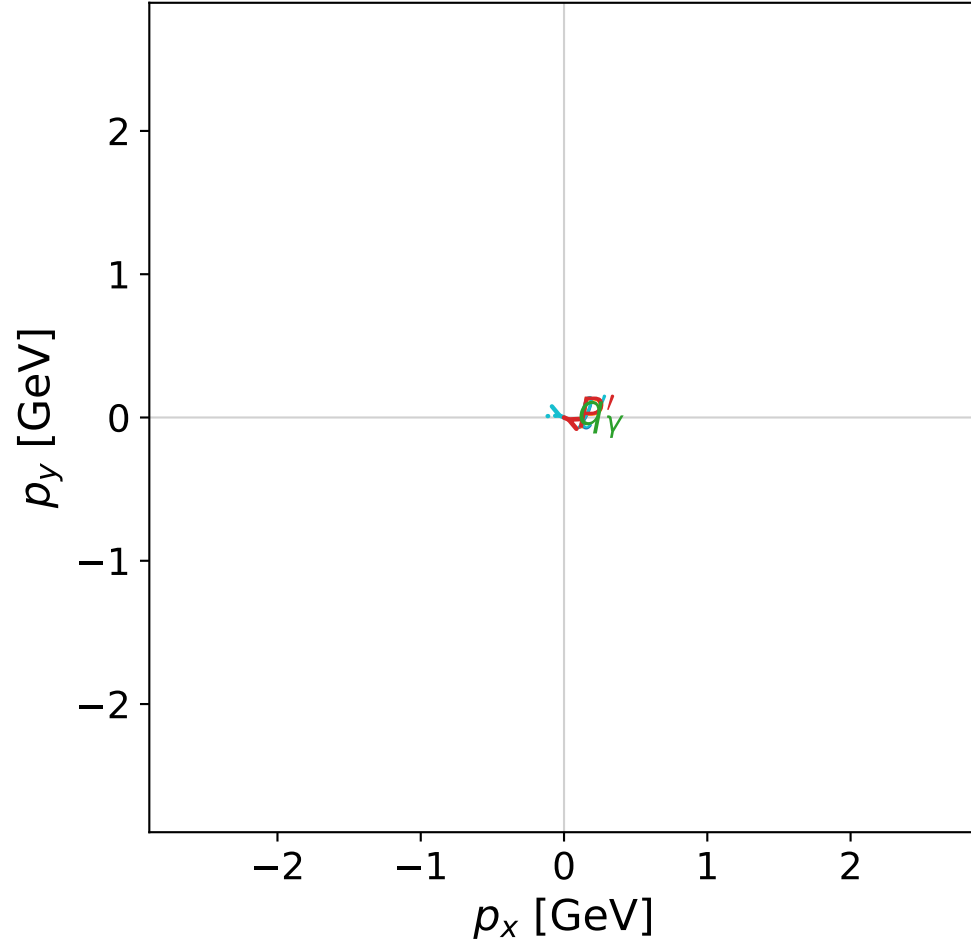


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



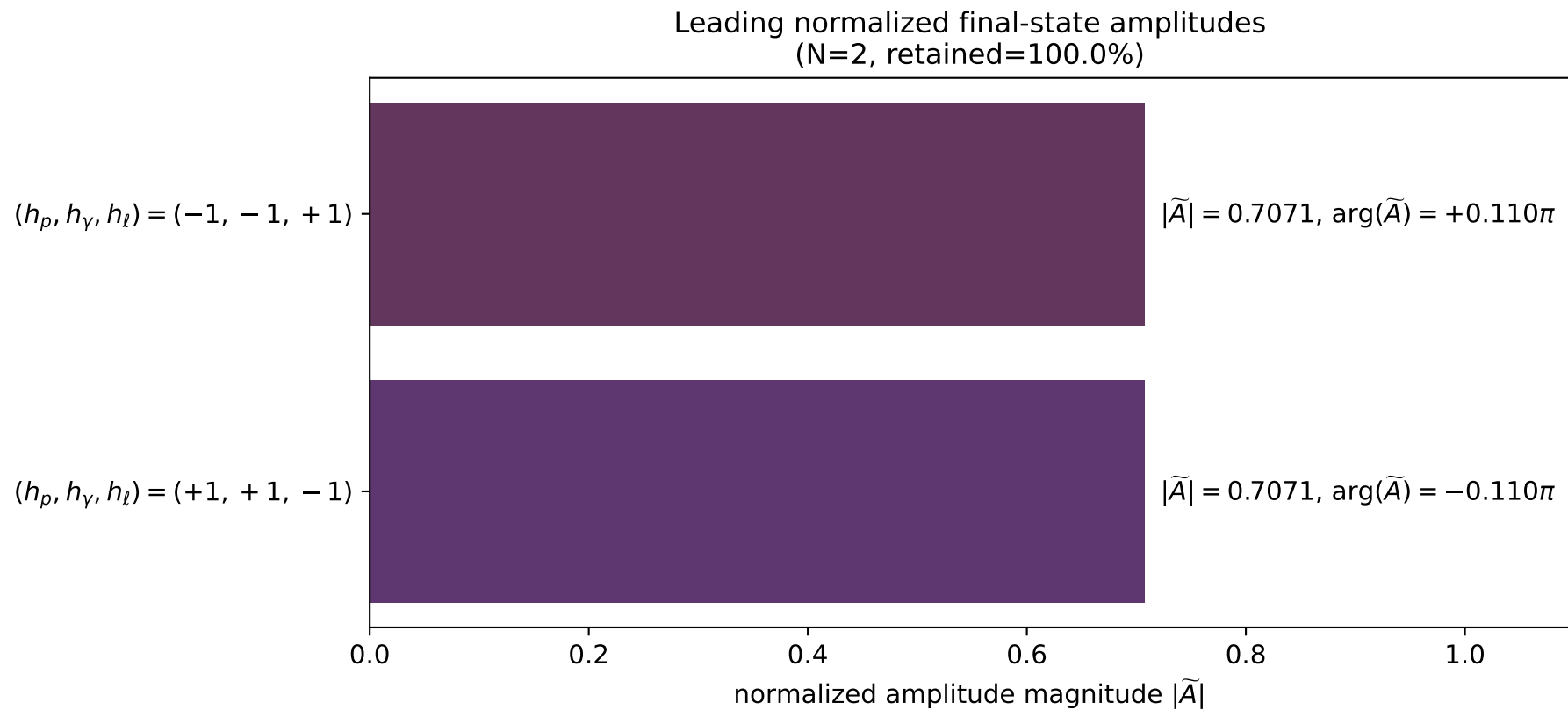
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_44
 $d_{\{\rm GHZ\}}=2.65428\text{e-}08$
 region: polarization_cluster_01_local_minimum_44
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0044
 lepton mixing angle: $\alpha_e=0.70477583$ rad
 incoming lepton: $0.761757|+\rangle +0.647863|-\rangle$
 proton mixing angle: $\alpha_p=0.70475776$ rad
 incoming proton: $0.761769|+\rangle +0.647849|-\rangle$

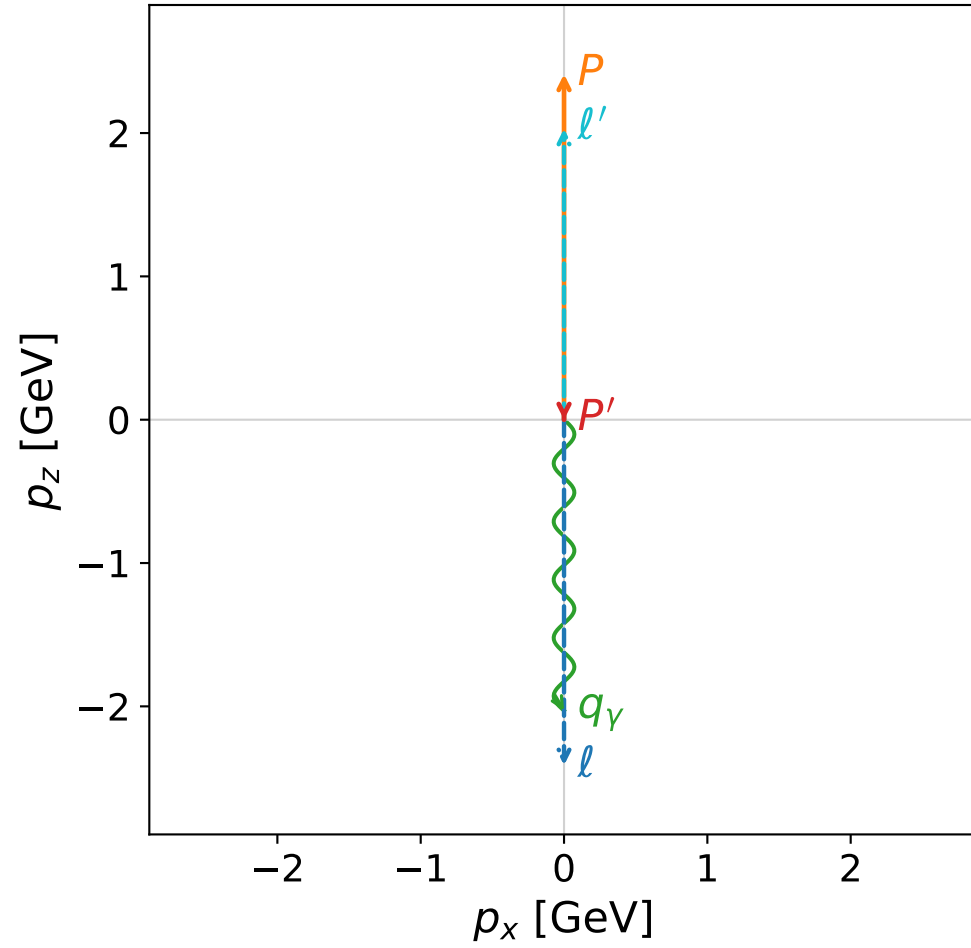
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00358496$
 $\theta_{p'}=2.38484\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.72855$, $\phi_\gamma=3.48858$
 $E_\gamma=2.03279$, $\theta_{l'}=4.21468\text{e-}08$, $\phi_{l'}=5.87014$
 $Q^2=19.5779$, $x_B=0.836447$, $t=-3.07811$, $W^2=4.70796$, $y=0.970392$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0292$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0292$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0036$
 q_γ $E= 2.0328$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0328$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

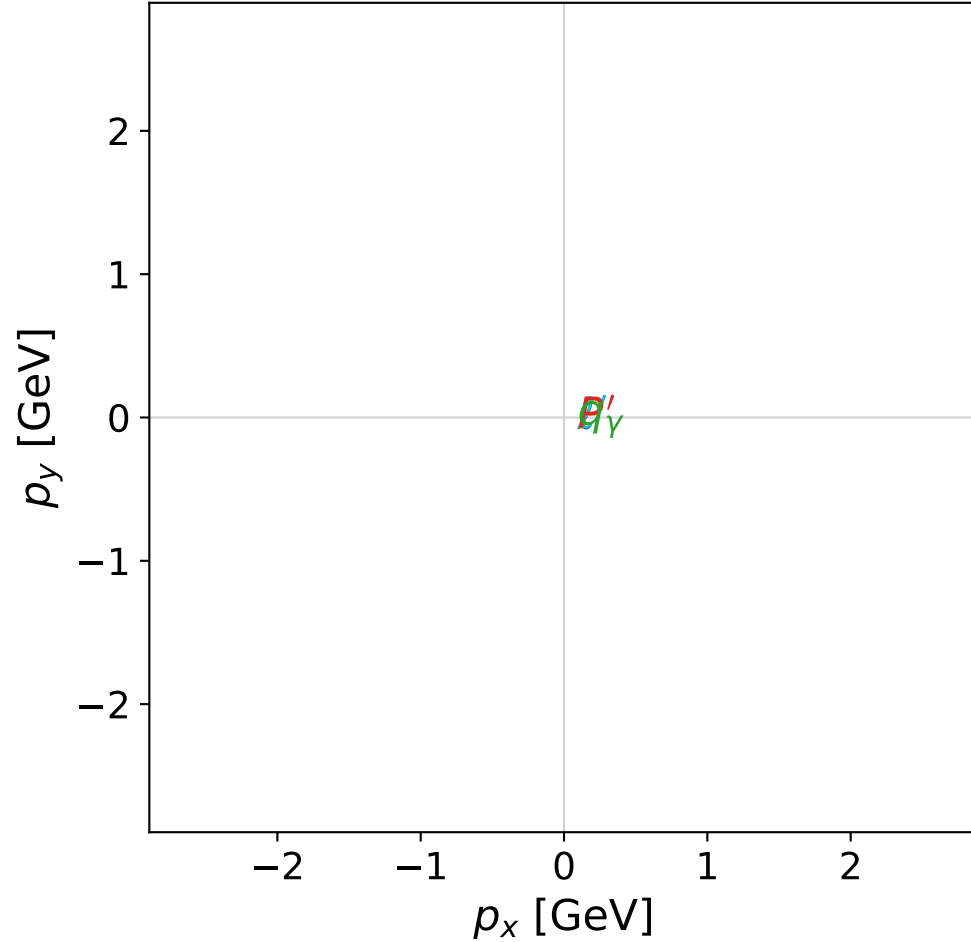


dghz_mixing_angles_polarization_01_local_minimum_46
 $d_{\{\rm GHZ\}}=2.66651\text{e-}08$
 region: polarization_cluster_01_local_minimum_46
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0046
 lepton mixing angle: $\alpha_e=0.78323852$ rad
 incoming lepton: $0.708632|+\rangle + 0.705578|-\rangle$
 proton mixing angle: $\alpha_p=0.78324505$ rad
 incoming proton: $0.708628|+\rangle + 0.705583|-\rangle$

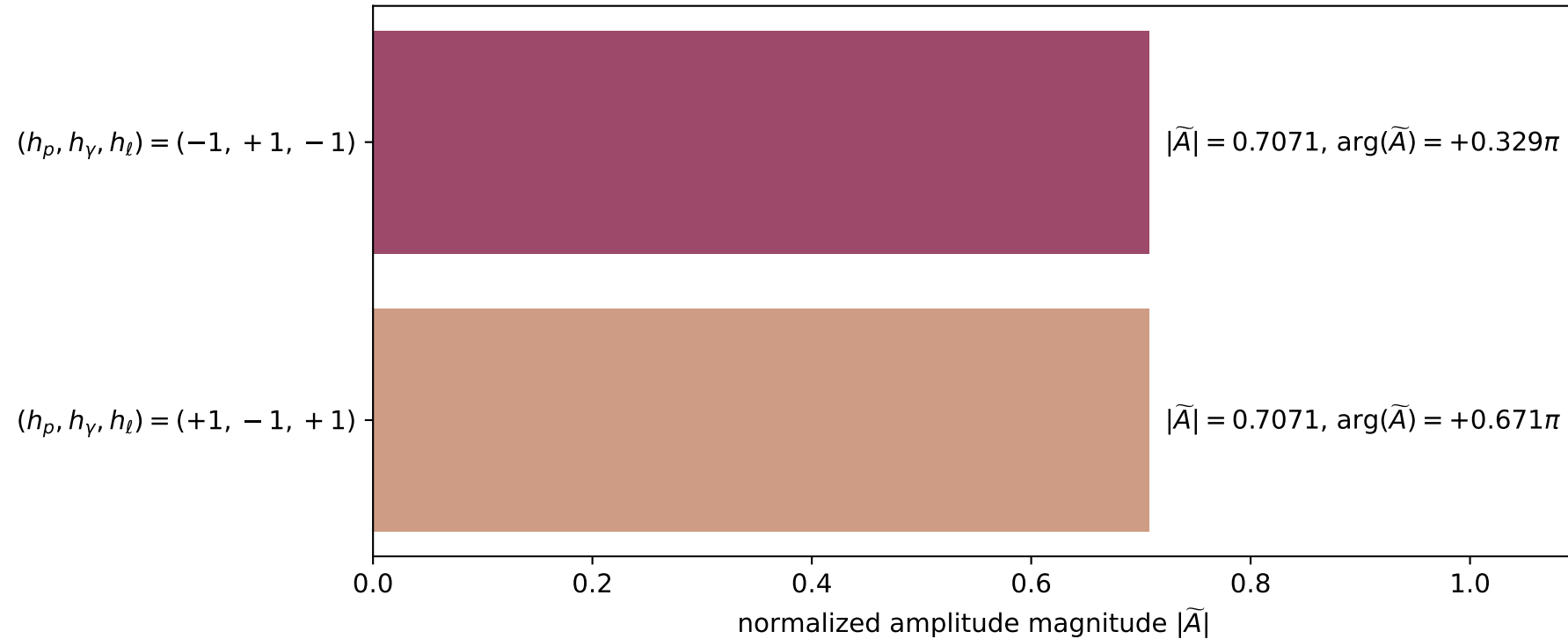
$s=24.9844$, $\sqrt{s}=4.99844$, $|\vec{P}|=2.41121$, $|\vec{P}'|=0.000949457$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.14724$, $\phi_\gamma=5.24835$
 $E_\gamma=2.02974$, $\theta_{l'}=0$, $\phi_{l'}=2.10635$
 $Q^2=19.5857$, $x_B=0.837366$, $t=-3.09854$, $W^2=4.68379$, $y=0.970342$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4112$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4112$
 P $E= 2.5872$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4112$
 l' $E= 2.0307$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0307$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0009$
 q_γ $E= 2.0297$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0297$

x--y plane

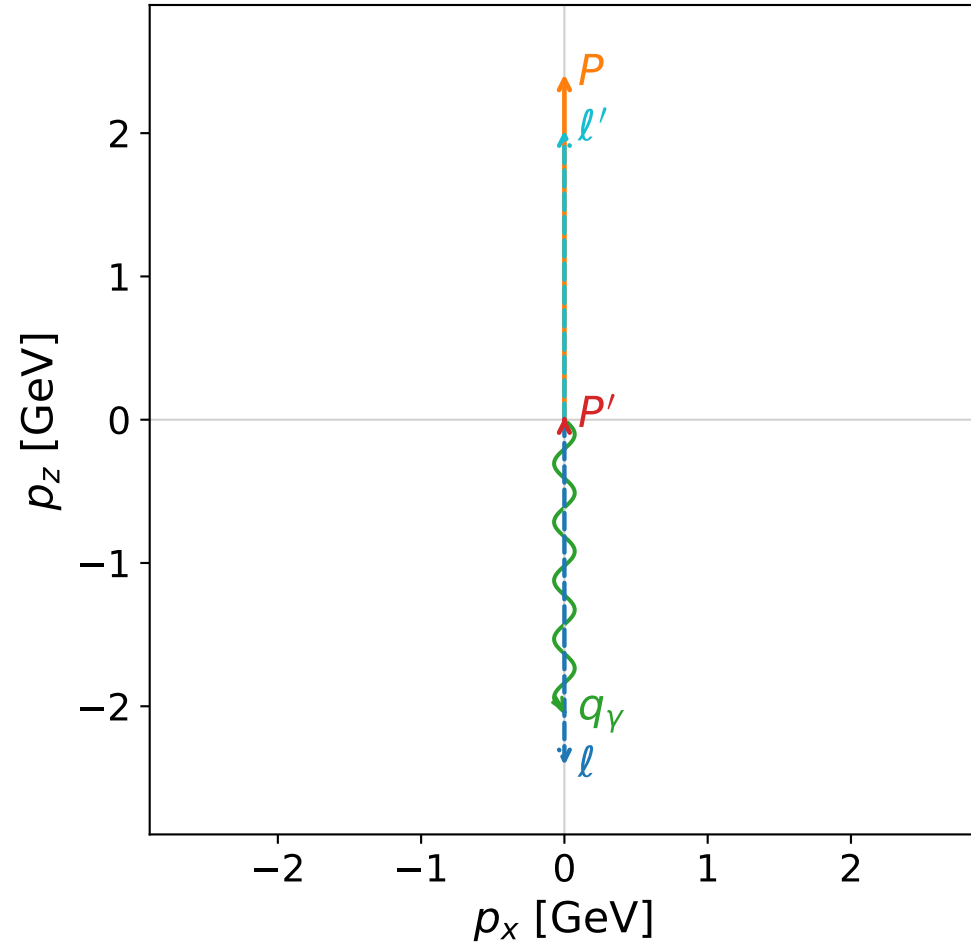


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

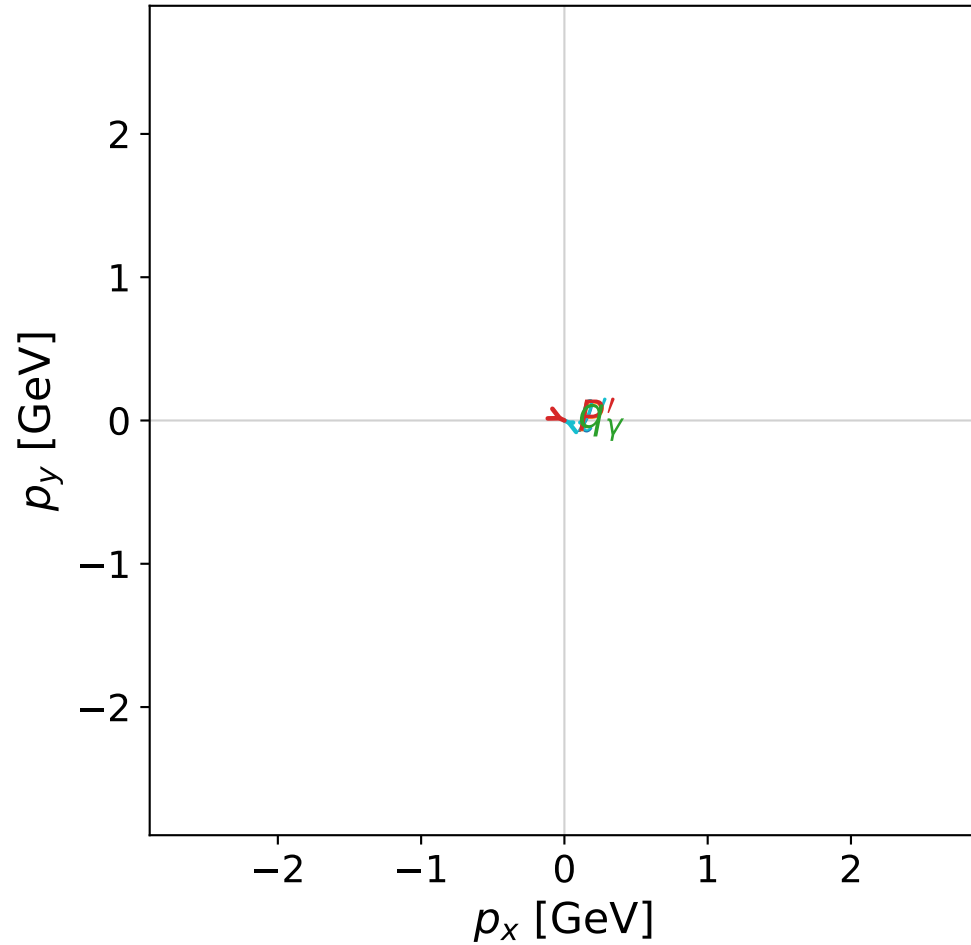


dghz_mixing_angles_polarization_01_local_minimum_47
 $d_{\text{GHZ}}=2.6669\text{e-}08$
 region: polarization_cluster_01_local_minimum_47
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0047
 lepton mixing angle: $\alpha_e=0.77596456$ rad
 incoming lepton: $0.713746|+\rangle + 0.700405|-\rangle$
 proton mixing angle: $\alpha_p=0.77595102$ rad
 incoming proton: $0.713755|+\rangle + 0.700395|-\rangle$

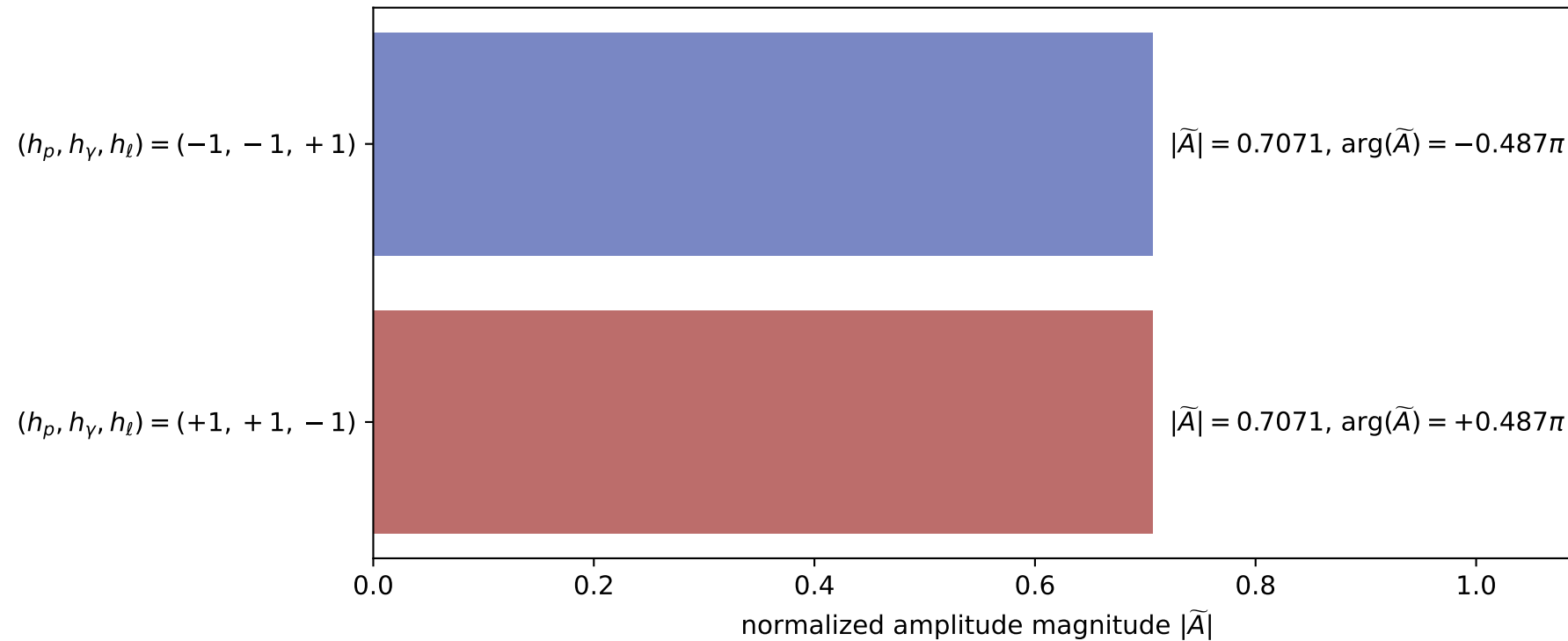
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0205162$
 $\theta_{P'}=0.000153398$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.83508$, $\phi_\gamma=1.61218$
 $E_\gamma=2.04115$, $\theta_{\ell'}=1.55751\text{e-}06$, $\phi_{\ell'}=2.69349$
 $Q^2=19.4952$, $x_B=0.832806$, $t=-2.99756$, $W^2=4.7937$, $y=0.970517$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0206$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0205$
 q_γ $E= 2.0411$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0411$

x--y plane

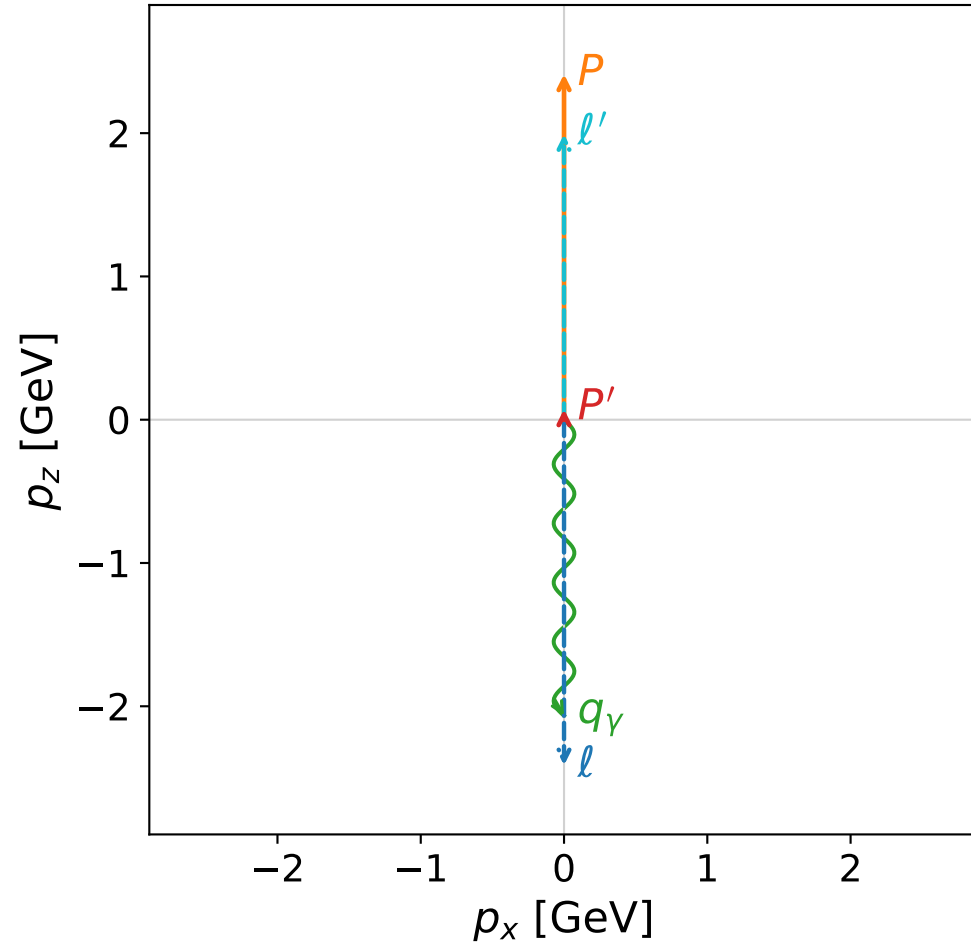


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

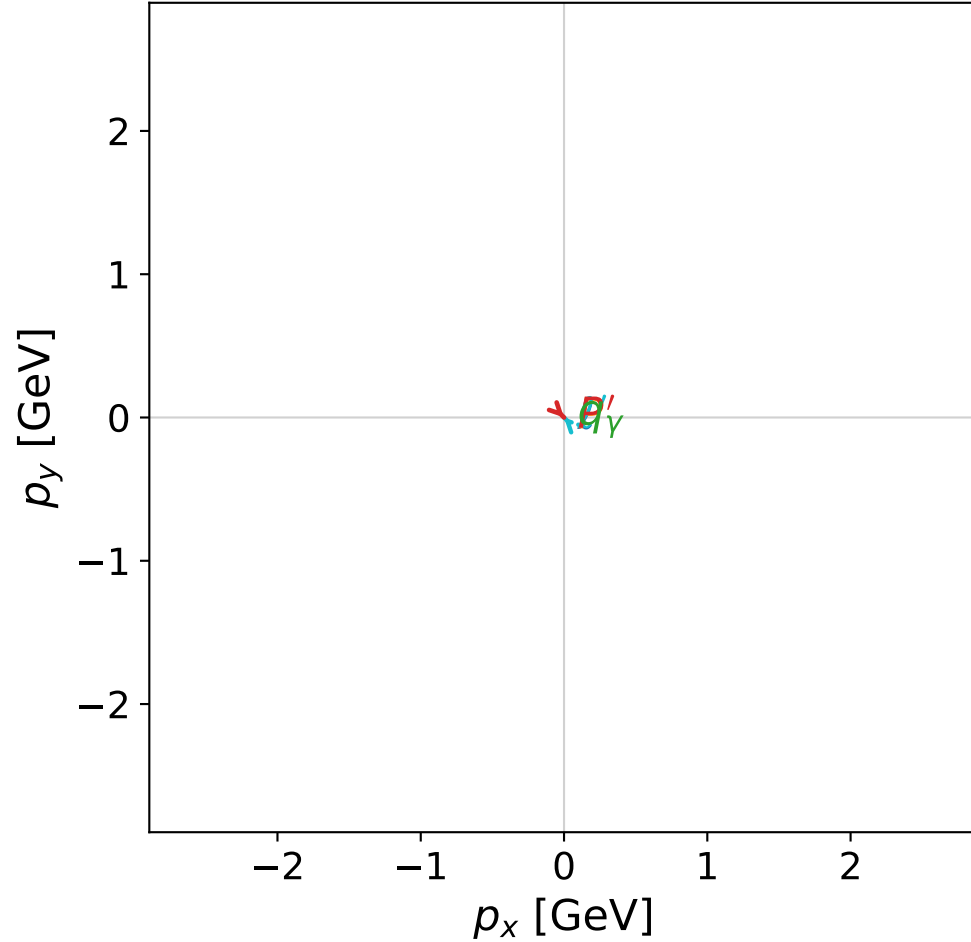


dghz_mixing_angles_polarization_01_local_minimum_57
 $d_{\text{GHZ}}=2.72176\text{e-}08$
 region: polarization_cluster_01_local_minimum_57
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0057
 lepton mixing angle: $\alpha_e=0.84620821$ rad
 incoming lepton: $0.662827|+\rangle + 0.748772|-\rangle$
 proton mixing angle: $\alpha_p=0.84620257$ rad
 incoming proton: $0.662831|+\rangle + 0.748769|-\rangle$

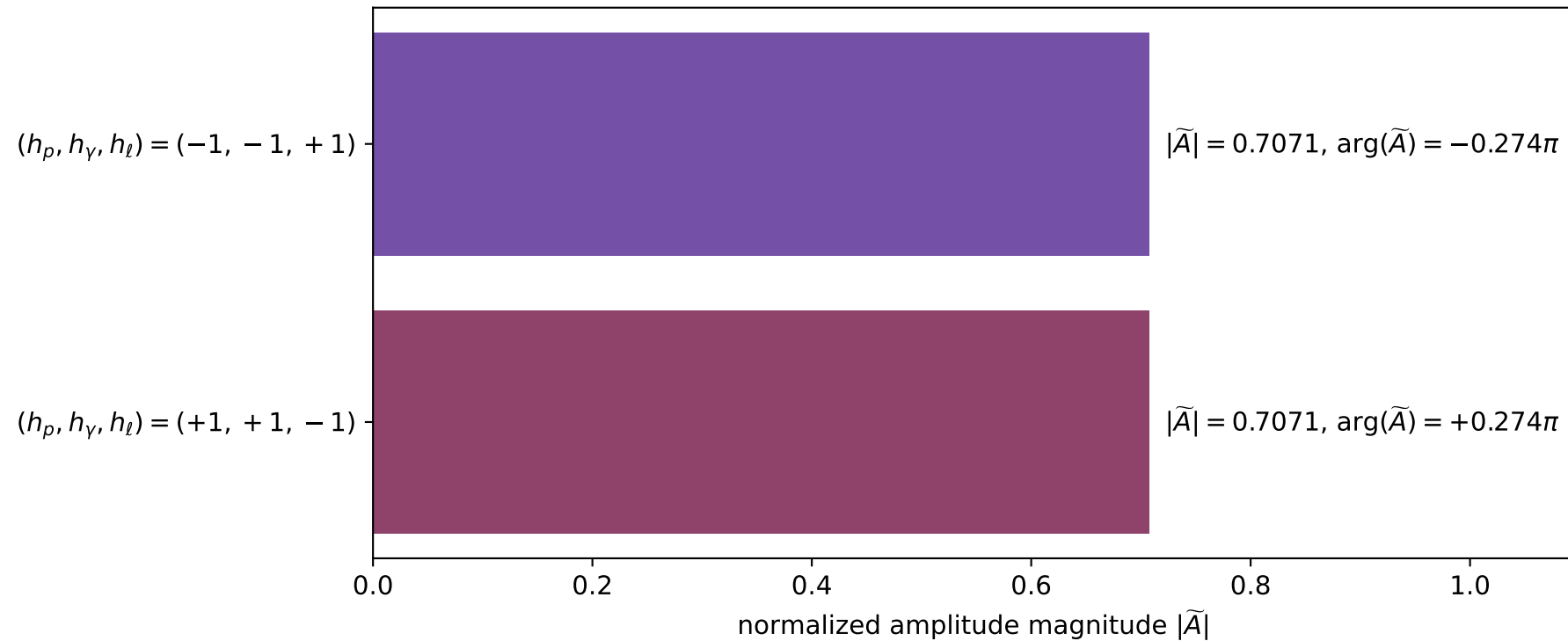
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0736114$
 $\theta_{P'}=0.000405073$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.49139$, $\phi_\gamma=2.28109$
 $E_\gamma=2.06636$, $\theta_{\ell'}=1.49632\text{e-}05$, $\phi_{\ell'}=2.34979$
 $Q^2=19.2262$, $x_B=0.820972$, $t=-2.75519$, $W^2=5.07248$, $y=0.970924$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9928$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9928$
 P' $E= 0.9409$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0736$
 q_γ $E= 2.0664$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0664$

x--y plane

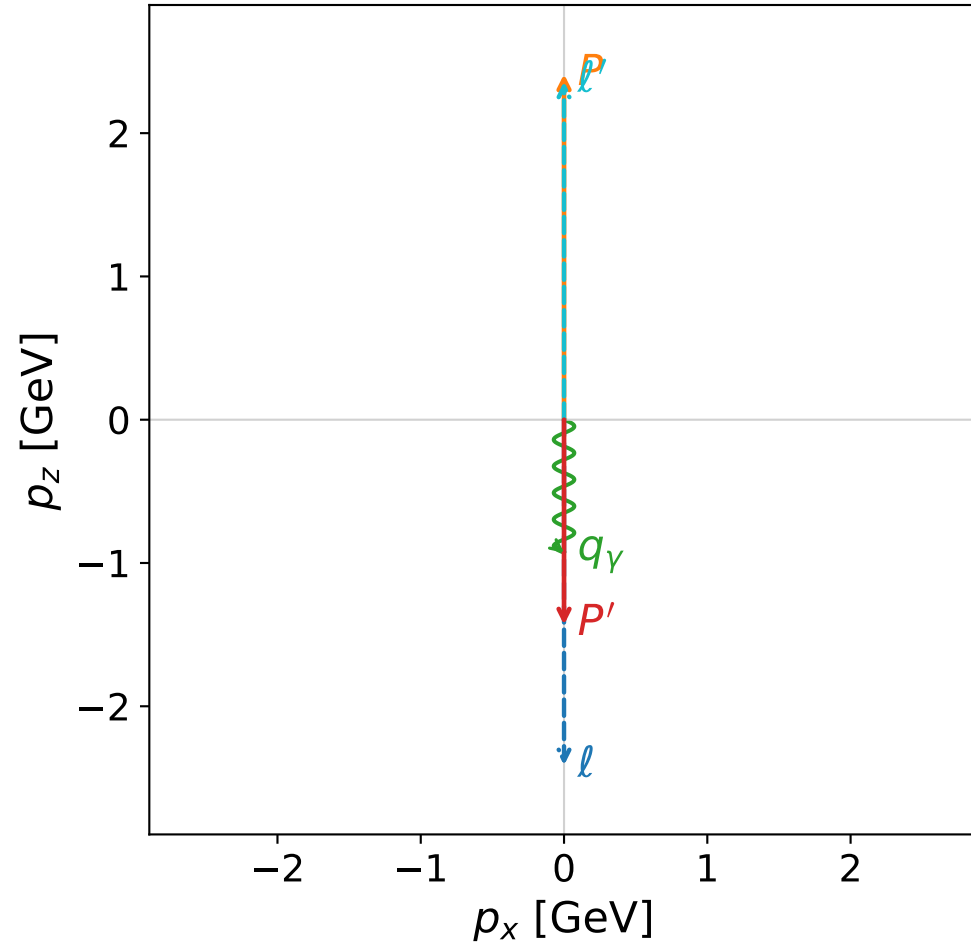


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

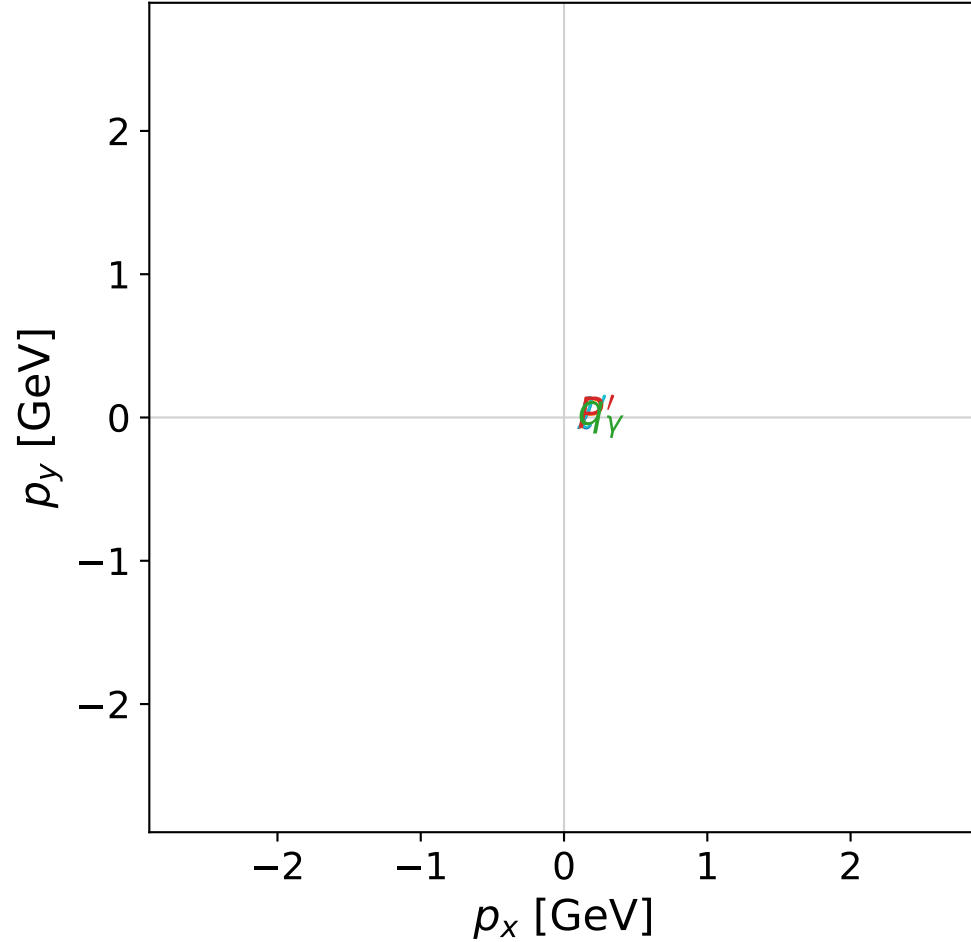


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



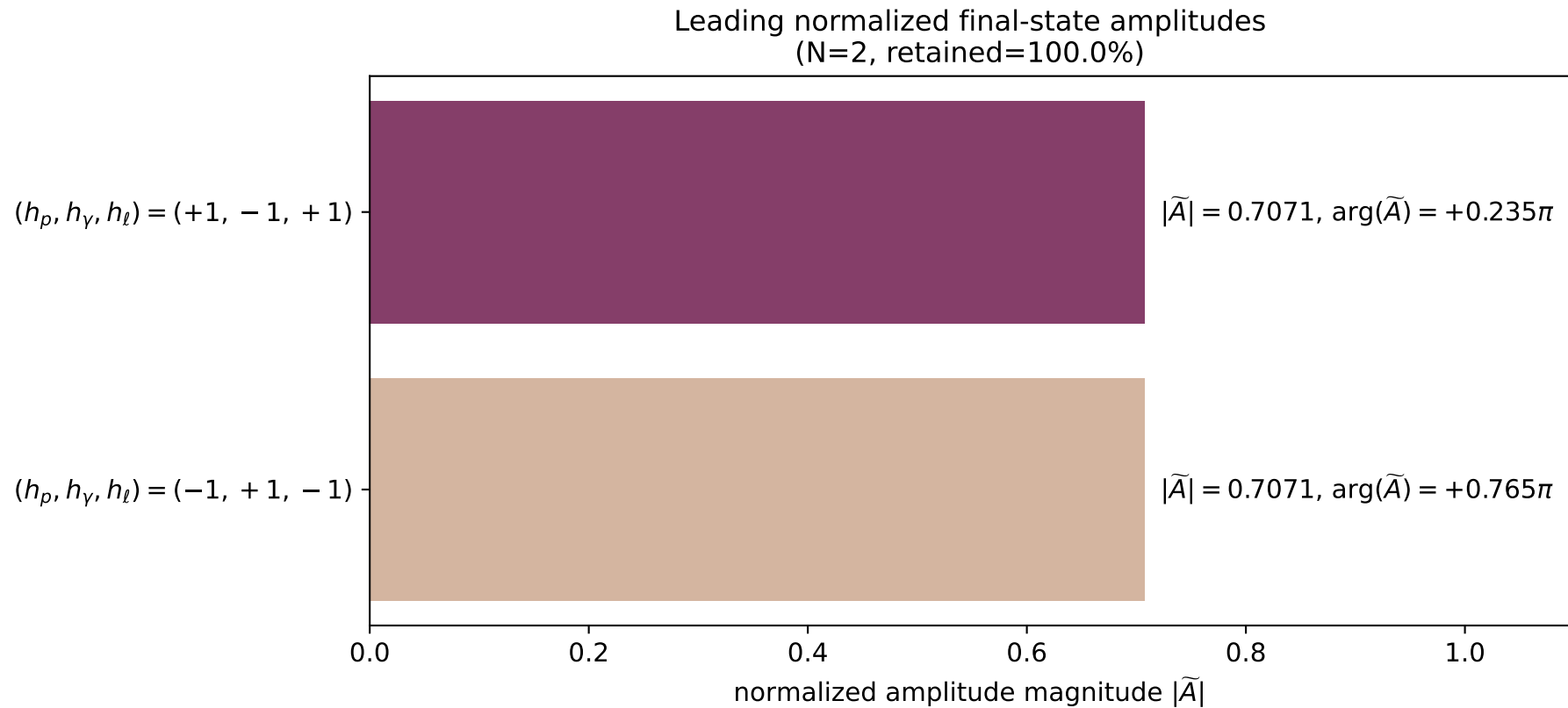
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_58
 $d_{\{\rm GHZ\}}=2.7258\text{e-}08$
 region: polarization_cluster_01_local_minimum_58
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0058
 lepton mixing angle: $\alpha_e=1.0707816$ rad
 incoming lepton: $0.479438|+\rangle + 0.877576|-\rangle$
 proton mixing angle: $\alpha_p=1.0707773$ rad
 incoming proton: $0.479442|+\rangle + 0.877573|-\rangle$

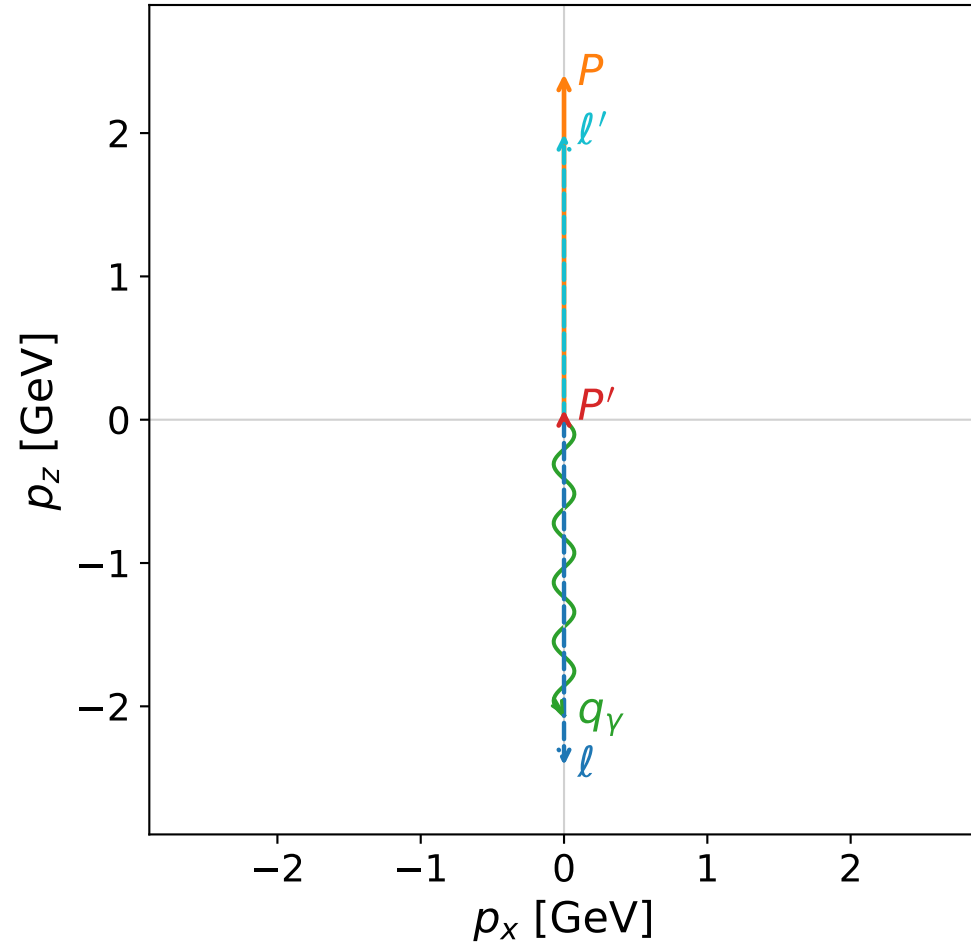
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.43163$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.31472$, $\phi_\gamma=3.87889$
 $E_\gamma=0.92841$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.393495$
 $Q^2=22.7698$, $x_B=0.977683$, $t=-14.0055$, $W^2=1.39961$, $y=0.965565$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3600$
 P' $E= 1.7116$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4316$
 q_γ $E= 0.9284$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.9284$

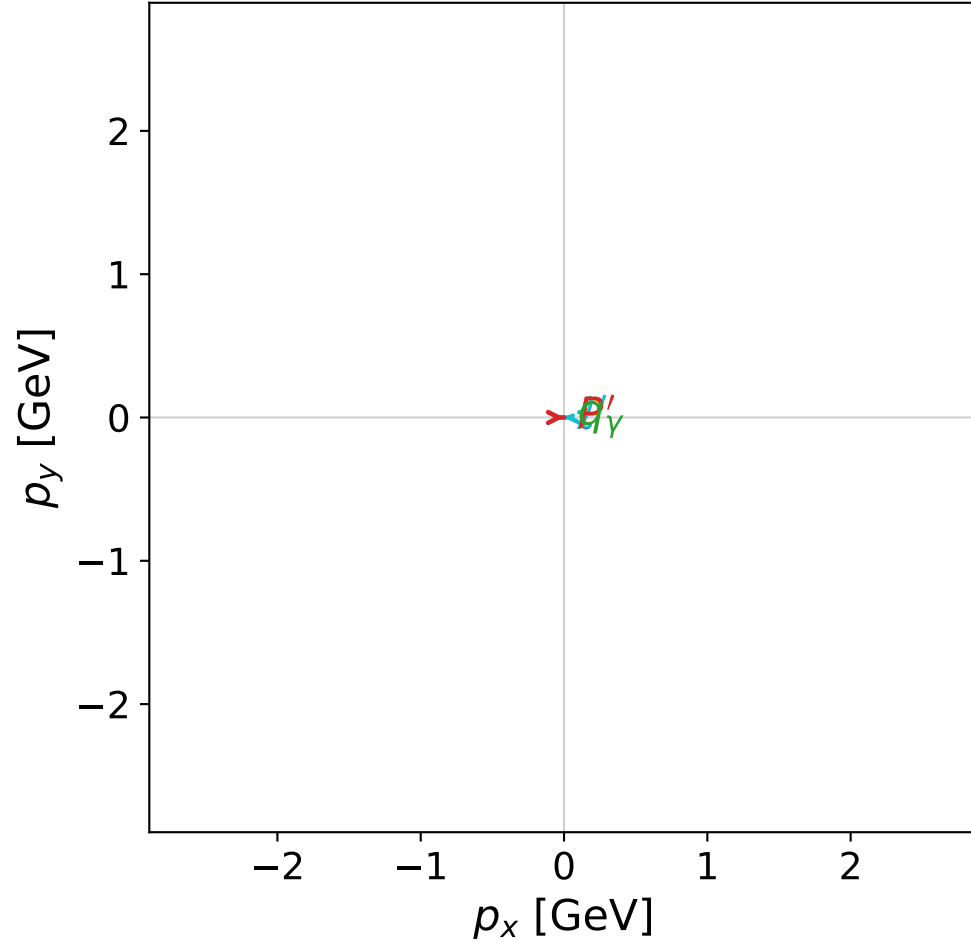


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



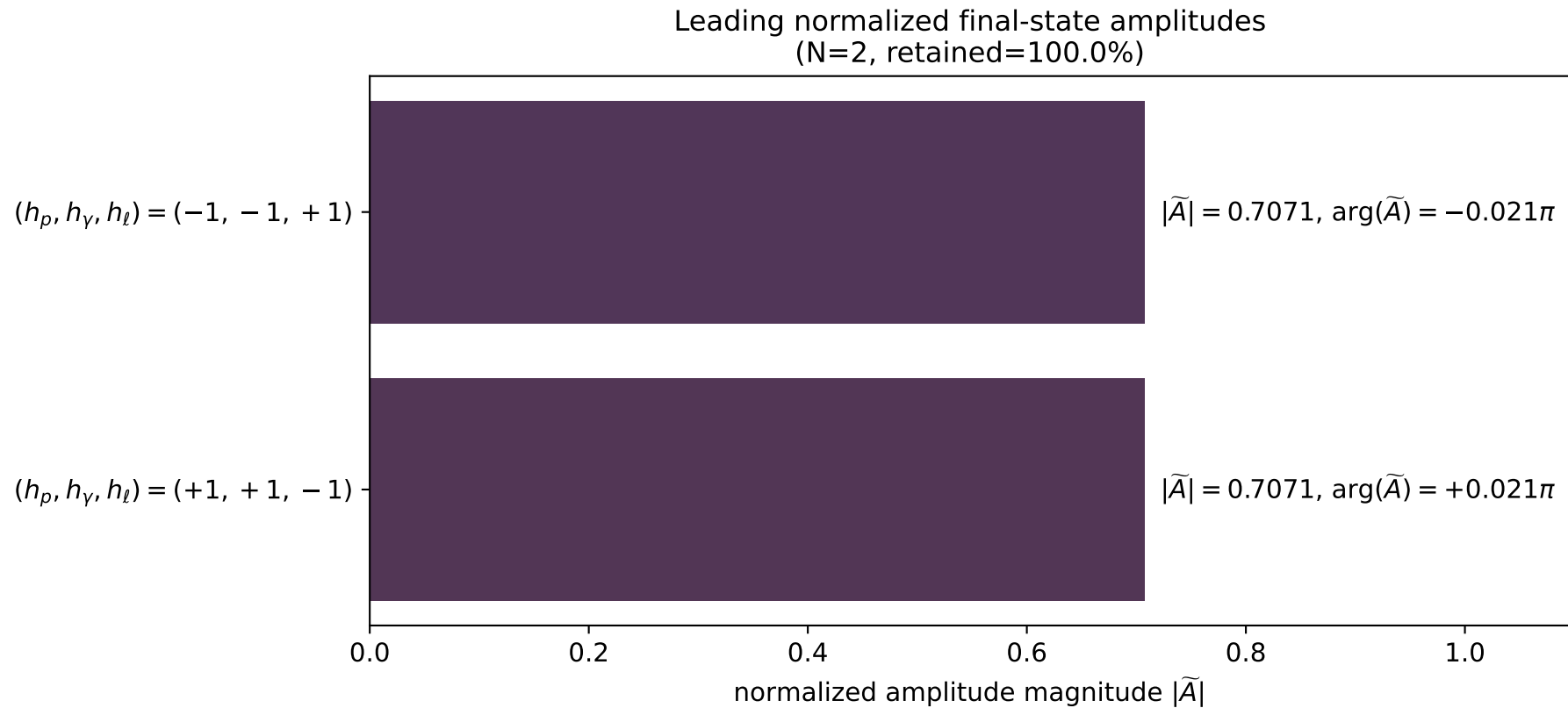
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_61
 $d_{\{\rm GHZ\}}=2.75163\text{e-}08$
 region: polarization_cluster_01_local_minimum_61
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0061
 lepton mixing angle: $\alpha_e=0.88093079$ rad
 incoming lepton: $0.636433|+\rangle +0.771332|-\rangle$
 proton mixing angle: $\alpha_p=0.88091145$ rad
 incoming proton: $0.636448|+\rangle +0.771319|-\rangle$

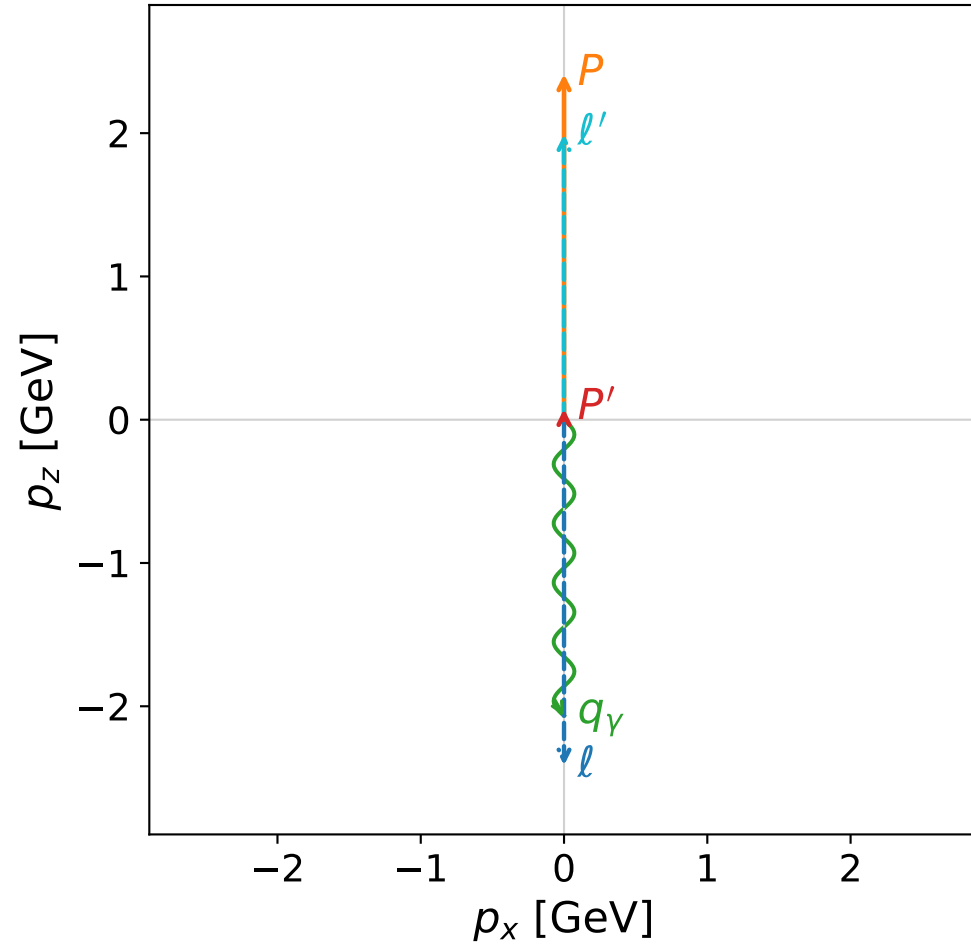
$s=24.9904$, $\sqrt{s}=4.99903$, $|\vec{P}|=2.41152$, $|\vec{P}'|=0.0704794$
 $\theta_{P'}=0.000153398$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.000314416$, $\phi_\gamma=3.07524$
 $E_\gamma=2.06444$, $\theta_\ell=5.4221\text{e-}06$, $\phi_\ell=3.14191$
 $Q^2=19.2338$, $x_B=0.821656$, $t=-2.76826$, $W^2=5.05464$, $y=0.970889$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4115$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4115$
 P $E= 2.5875$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4115$
 ℓ' $E= 1.9940$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9940$
 P' $E= 0.9406$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0705$
 q_γ $E= 2.0644$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0644$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

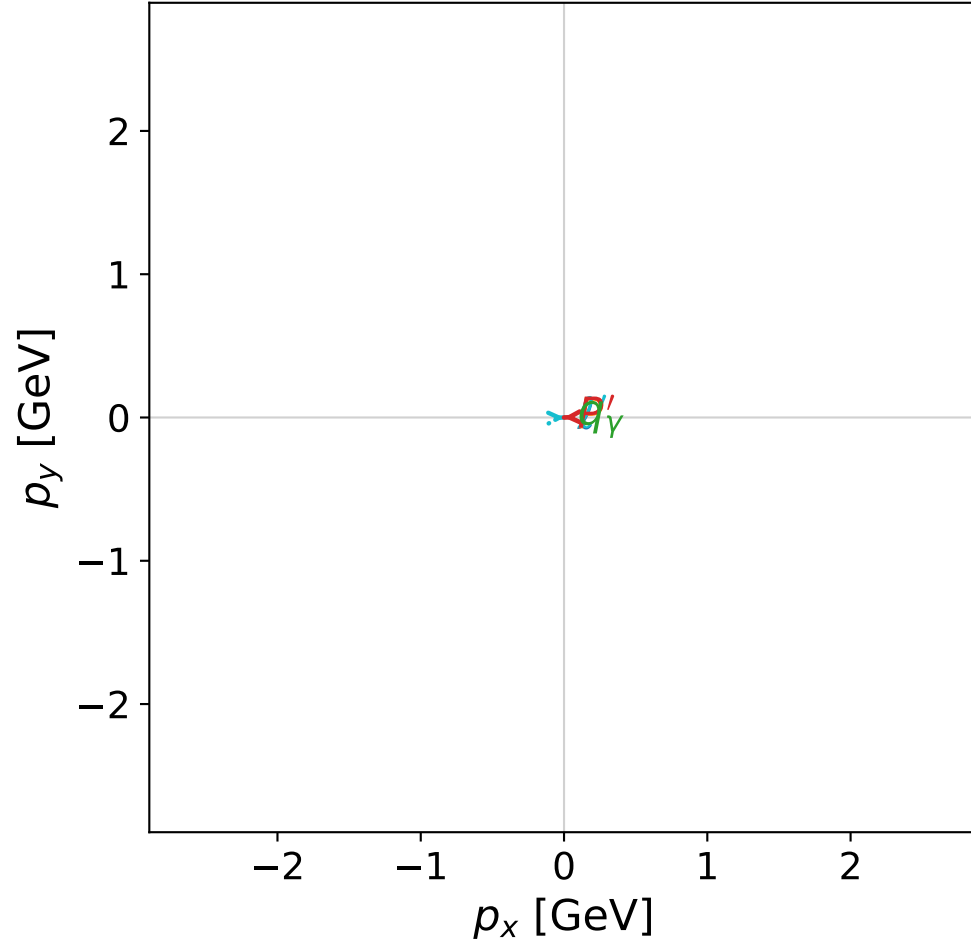


dghz_mixing_angles_polarization_01_local_minimum_65
 $d_{\{\rm GHZ\}}=2.78329\text{e-}08$
 region: polarization_cluster_01_local_minimum_65
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0065
 lepton mixing angle: $\alpha_e=0.5734809$ rad
 incoming lepton: $0.840017|+\rangle + 0.542559|-\rangle$
 proton mixing angle: $\alpha_p=0.57346771$ rad
 incoming proton: $0.840025|+\rangle + 0.542548|-\rangle$

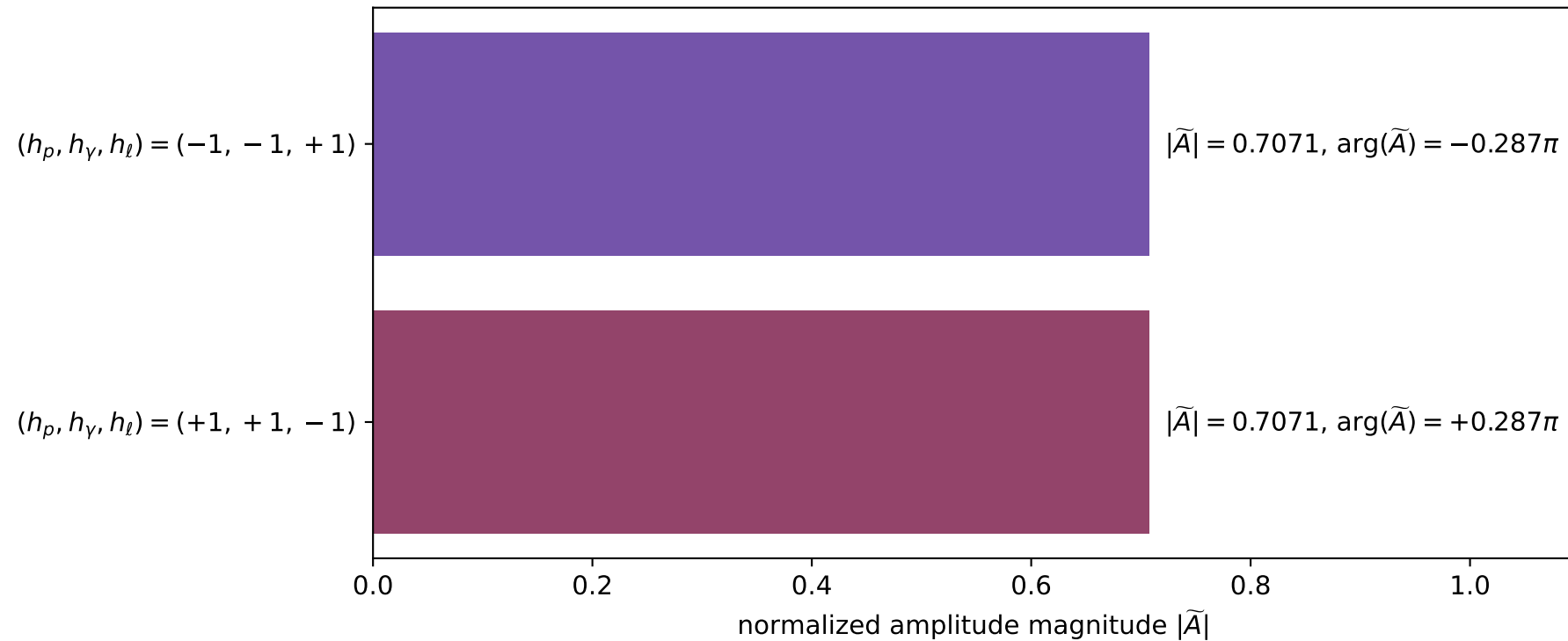
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0755206$
 $\theta_{p'}=2.17755\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.18$, $\phi_\gamma=2.24128$
 $E_\gamma=2.06724$, $\theta_\ell=8.1617\text{e-}08$, $\phi_\ell=0.0384033$
 $Q^2=19.2163$, $x_B=0.820535$, $t=-2.74677$, $W^2=5.08278$, $y=0.970939$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9917$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9917$
 P' $E= 0.9410$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0755$
 q_γ $E= 2.0672$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0672$

x--y plane

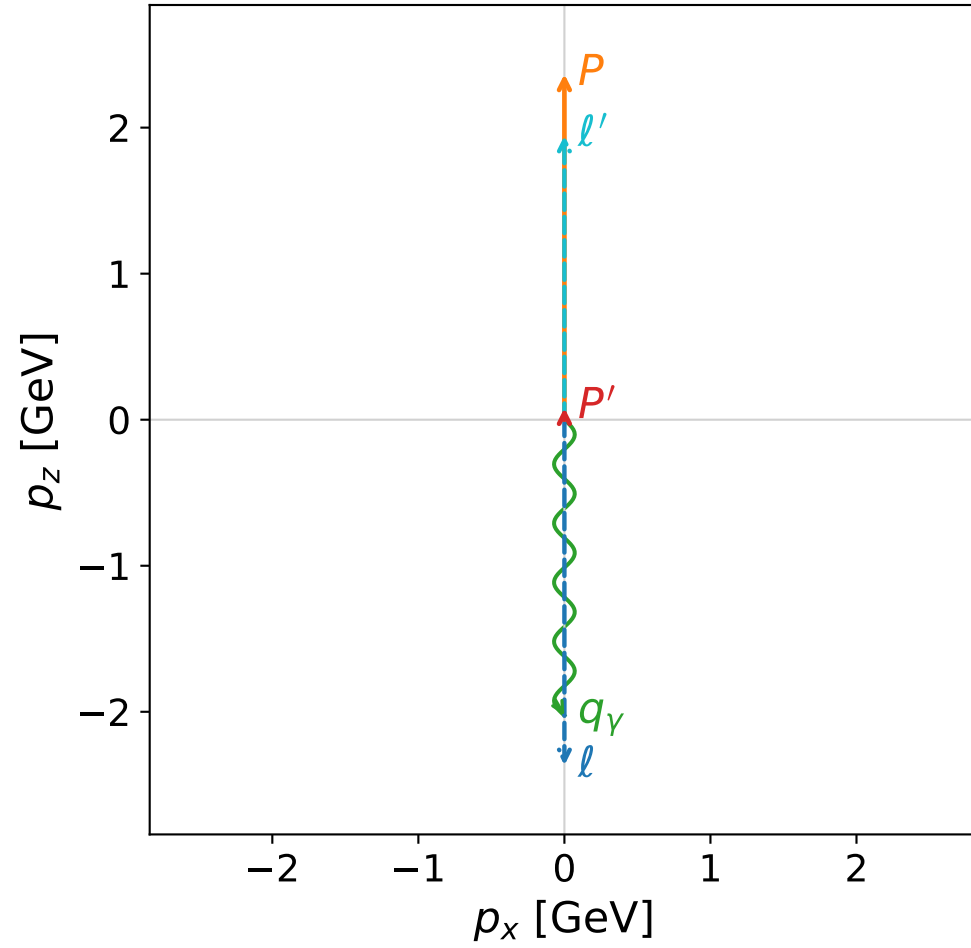


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

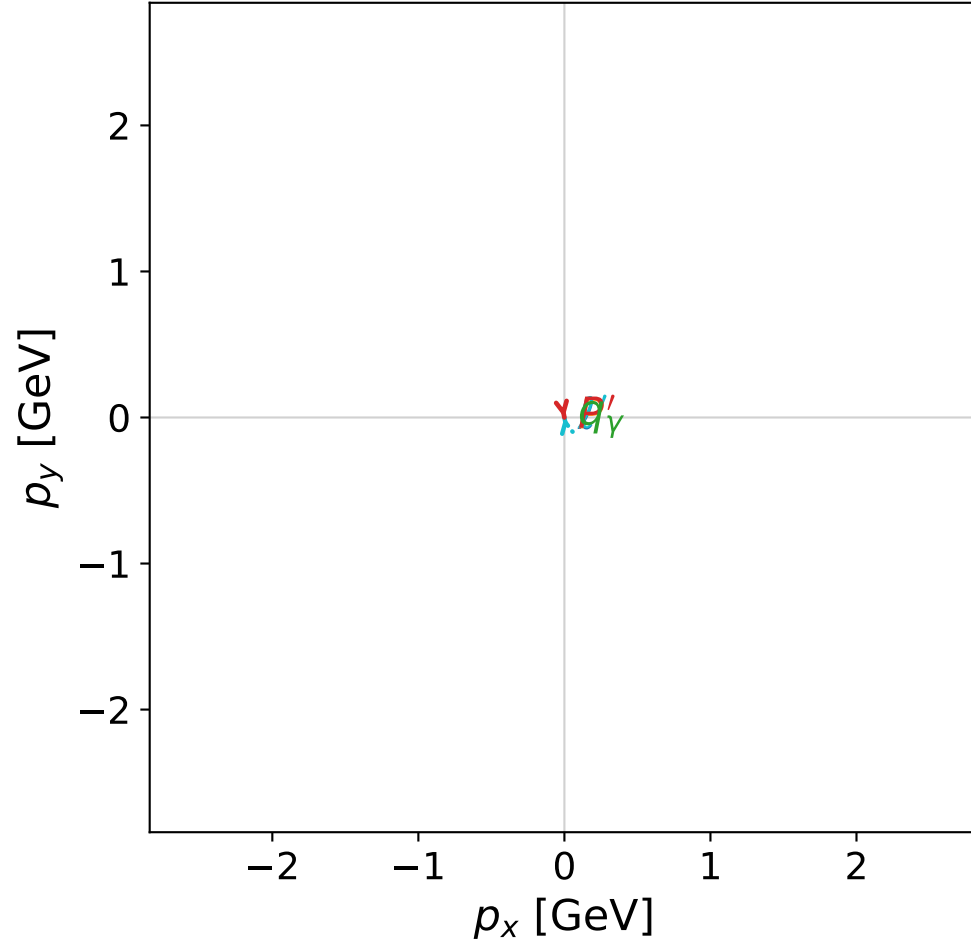


dghz_mixing_angles_polarization_01_local_minimum_69
 $d_{\text{GHZ}}=2.83483\text{e-}08$
 region: polarization_cluster_01_local_minimum_69
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0069
 lepton mixing angle: $\alpha_e=0.82158482$ rad
 incoming lepton: $0.681062|+\rangle + 0.732226|-\rangle$
 proton mixing angle: $\alpha_p=0.82156325$ rad
 incoming proton: $0.681077|+\rangle + 0.732211|-\rangle$

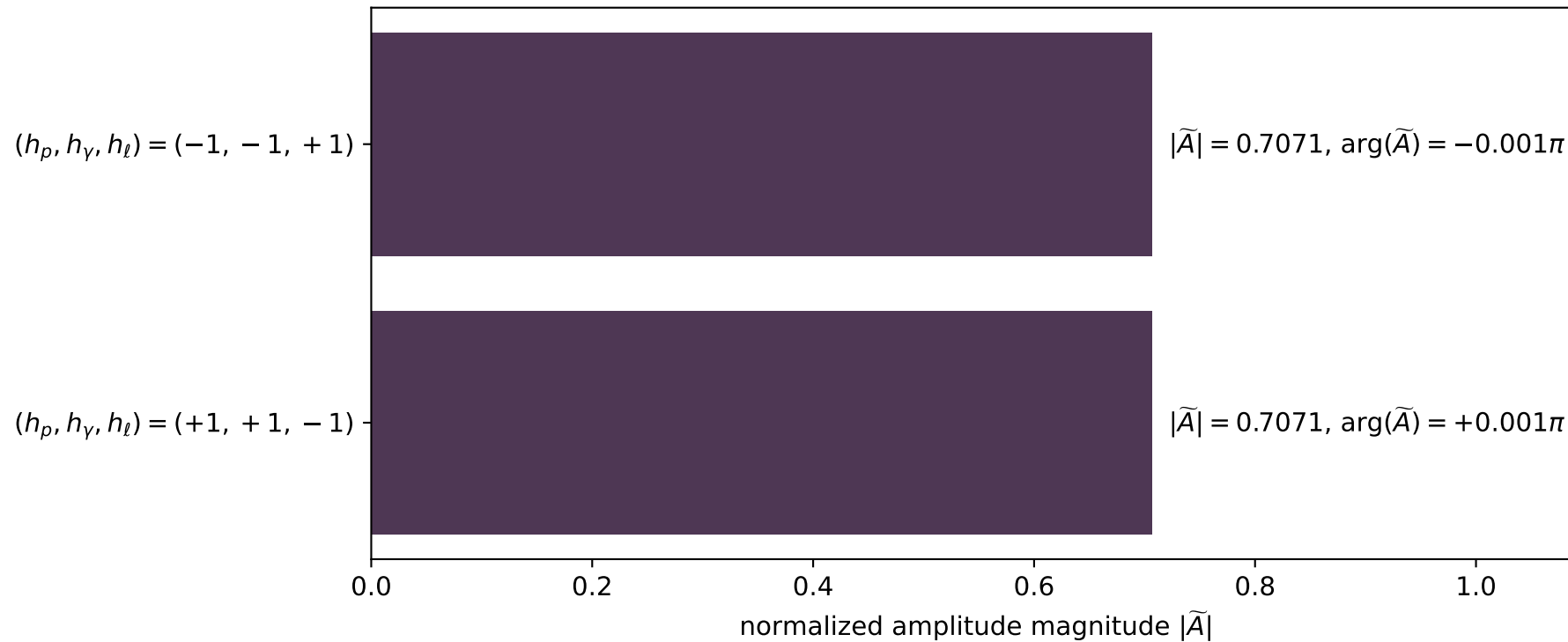
$s=24.1272$, $\sqrt{s}=4.91194$, $|\vec{P}|=2.36641$, $|\vec{P}'|=0.080731$
 $\theta_{P'}=0.00091855$, $\theta_\gamma=3.14159$, $\phi_{P'}=4.90382$, $\phi_\gamma=3.13744$
 $E_\gamma=2.0256$, $\theta_\ell=3.81287\text{e-}05$, $\phi_\ell=1.76223$
 $Q^2=18.4095$, $x_B=0.816362$, $t=-2.6513$, $W^2=5.02098$, $y=0.970029$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3664$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3664$
 P $E= 2.5455$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3664$
 ℓ' $E= 1.9449$ $p_x= -0.0000$ $p_y= 0.0001$ $p_z= 1.9449$
 P' $E= 0.9415$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= 0.0807$
 q_γ $E= 2.0256$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0256$

x--y plane

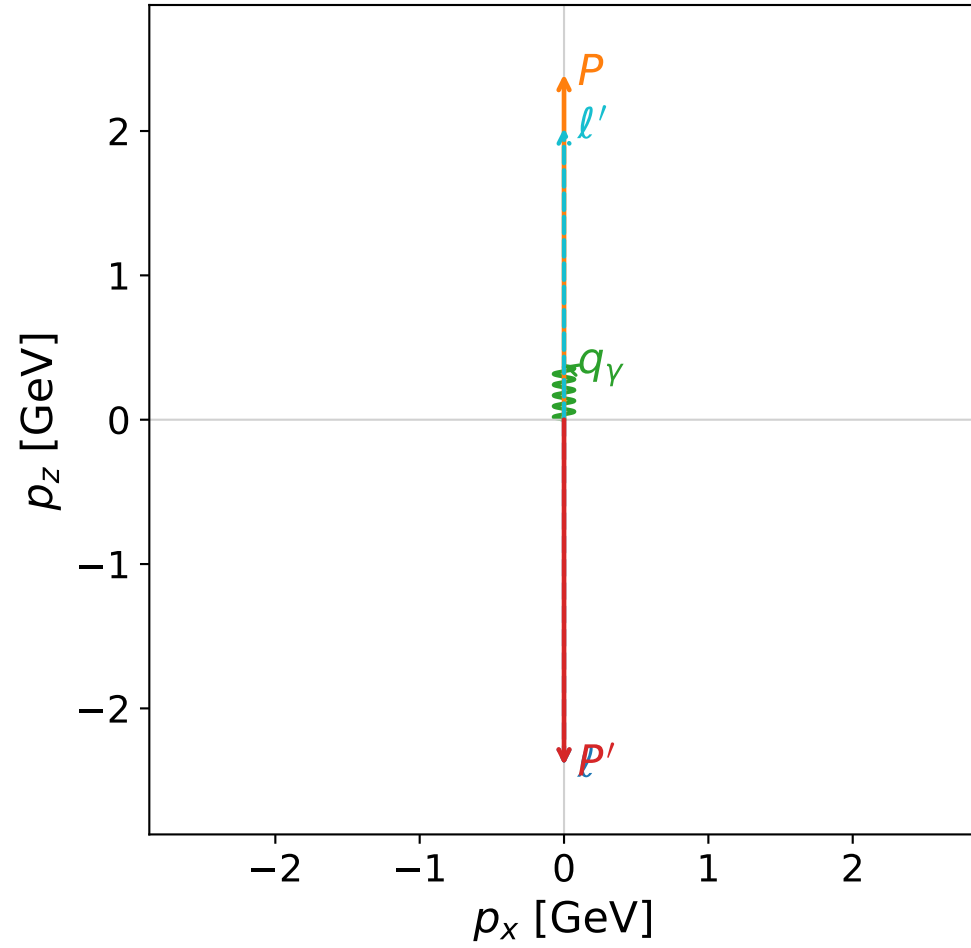


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

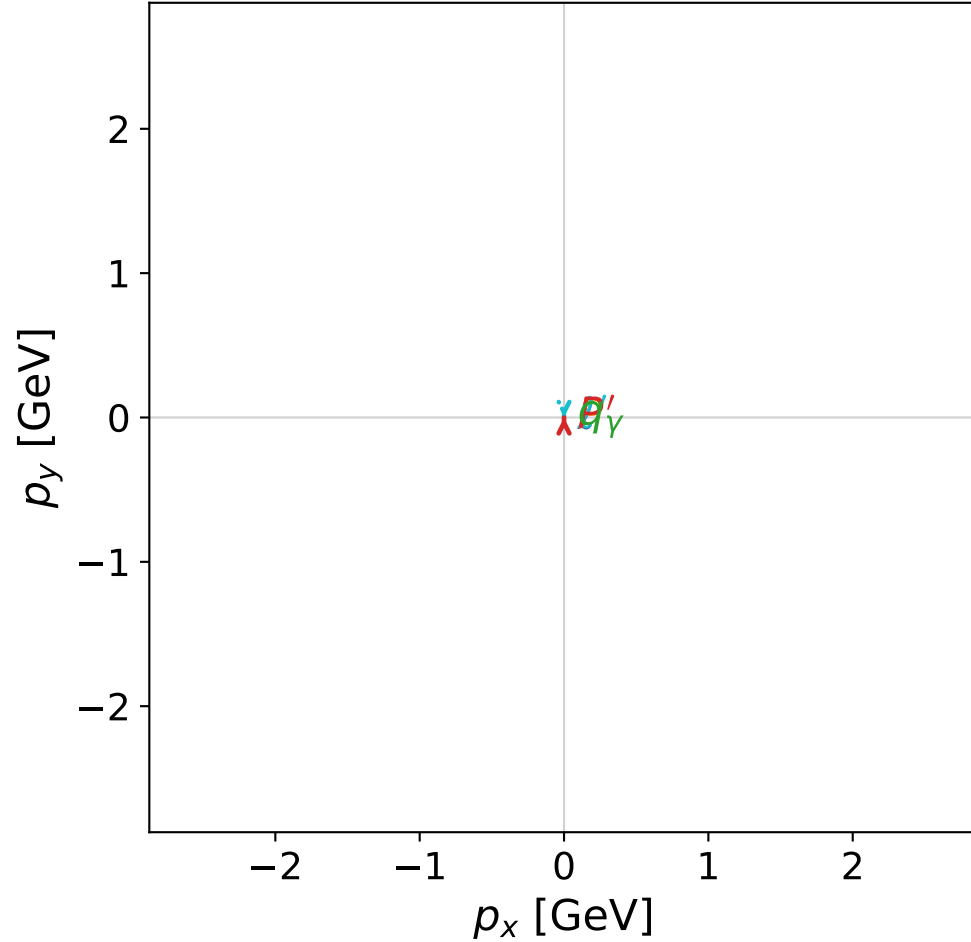


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



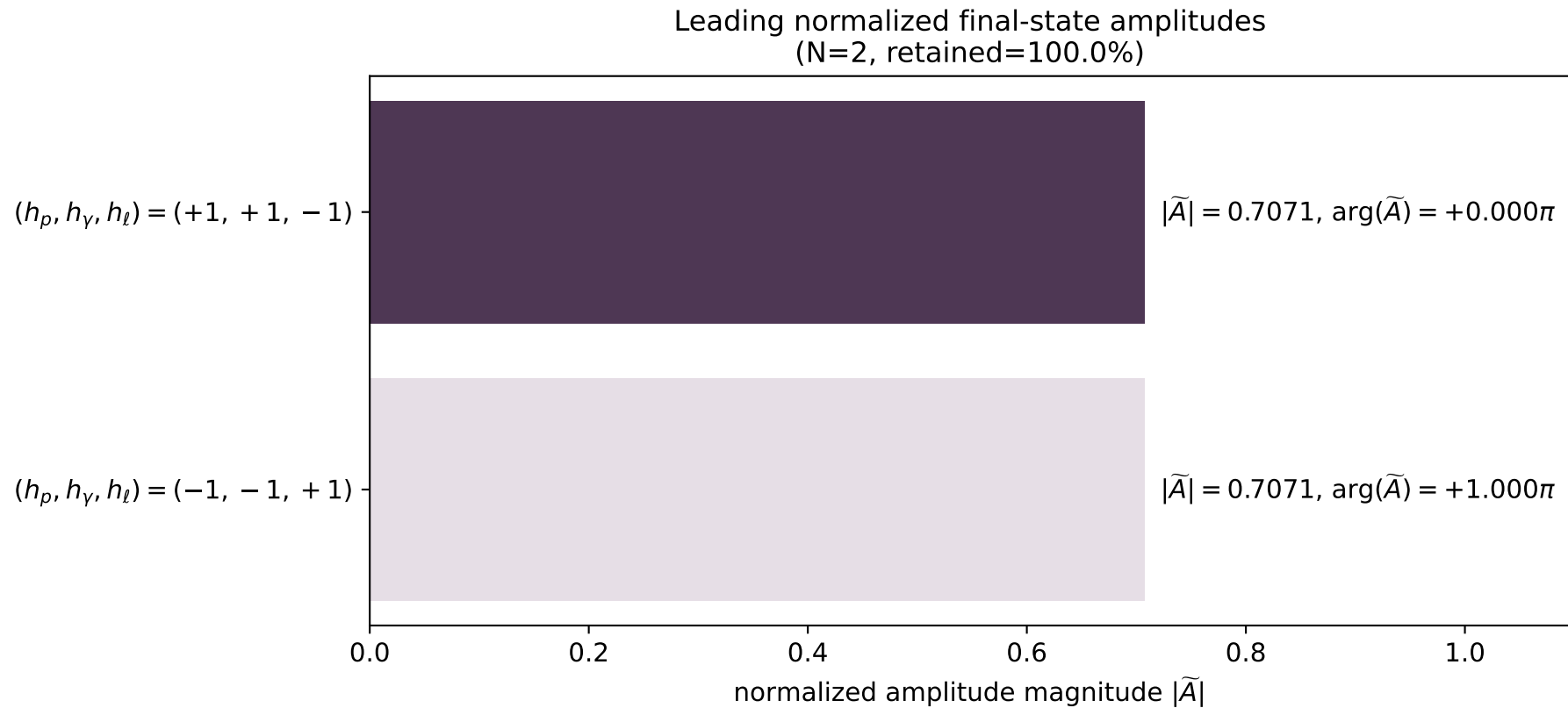
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_71
 $d_{\{\text{rm GHZ}\}}=2.87516\text{e-}08$
 region: polarization_cluster_01_local_minimum_71
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0071
 lepton mixing angle: $\alpha_e=0.61281827$ rad
 incoming lepton: $0.81803|+\rangle + 0.575175|-\rangle$
 proton mixing angle: $\alpha_p=0.95795616$ rad
 incoming proton: $0.575193|+\rangle + 0.818018|-\rangle$

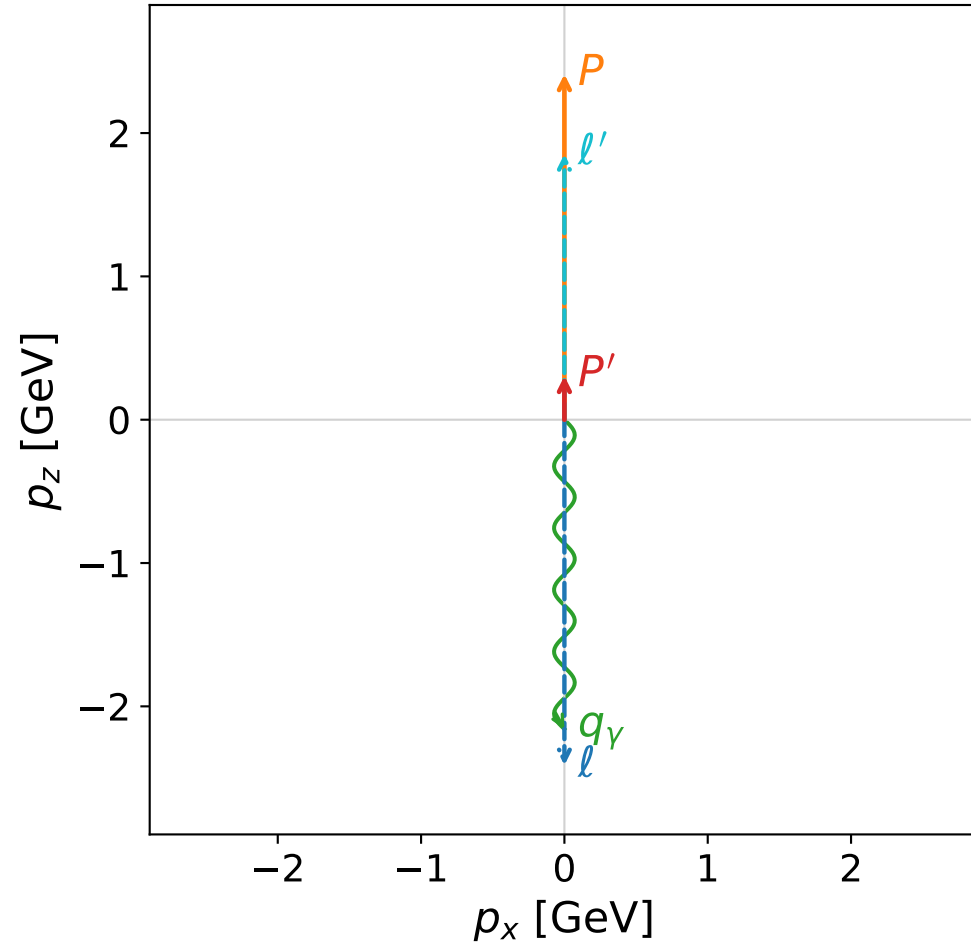
$s=24.6579$, $\sqrt{s}=4.96567$, $|\vec{P}|=2.39424$, $|\vec{P}'|=2.39424$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.69602$, $\phi_\gamma=3.17247$
 $E_\gamma=0.373172$, $\theta_{l'}=0$, $\phi_{l'}=4.83761$
 $Q^2=19.3557$, $x_B=0.839297$, $t=-22.9296$, $W^2=4.58594$, $y=0.969879$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3942$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3942$
 P $E= 2.5714$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3942$
 l' $E= 2.0211$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0211$
 P' $E= 2.5714$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.3942$
 q_γ $E= 0.3732$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.3732$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

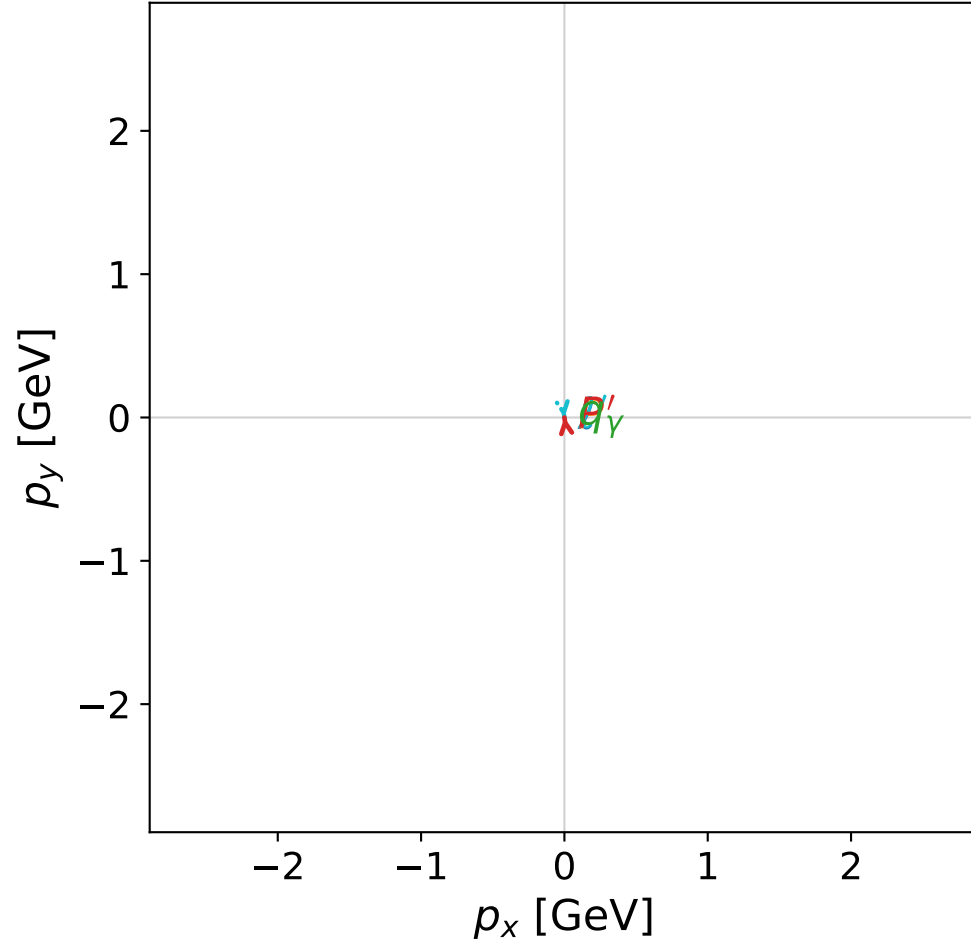


dghz_mixing_angles_polarization_01_local_minimum_72
 $d_{\text{GHZ}}=2.90848\text{e-}08$
 region: polarization_cluster_01_local_minimum_72
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0072
 lepton mixing angle: $\alpha_e=0.79108284$ rad
 incoming lepton: $0.703076|+\rangle + 0.711115|-\rangle$
 proton mixing angle: $\alpha_p=0.7910723$ rad
 incoming proton: $0.703083|+\rangle + 0.711108|-\rangle$

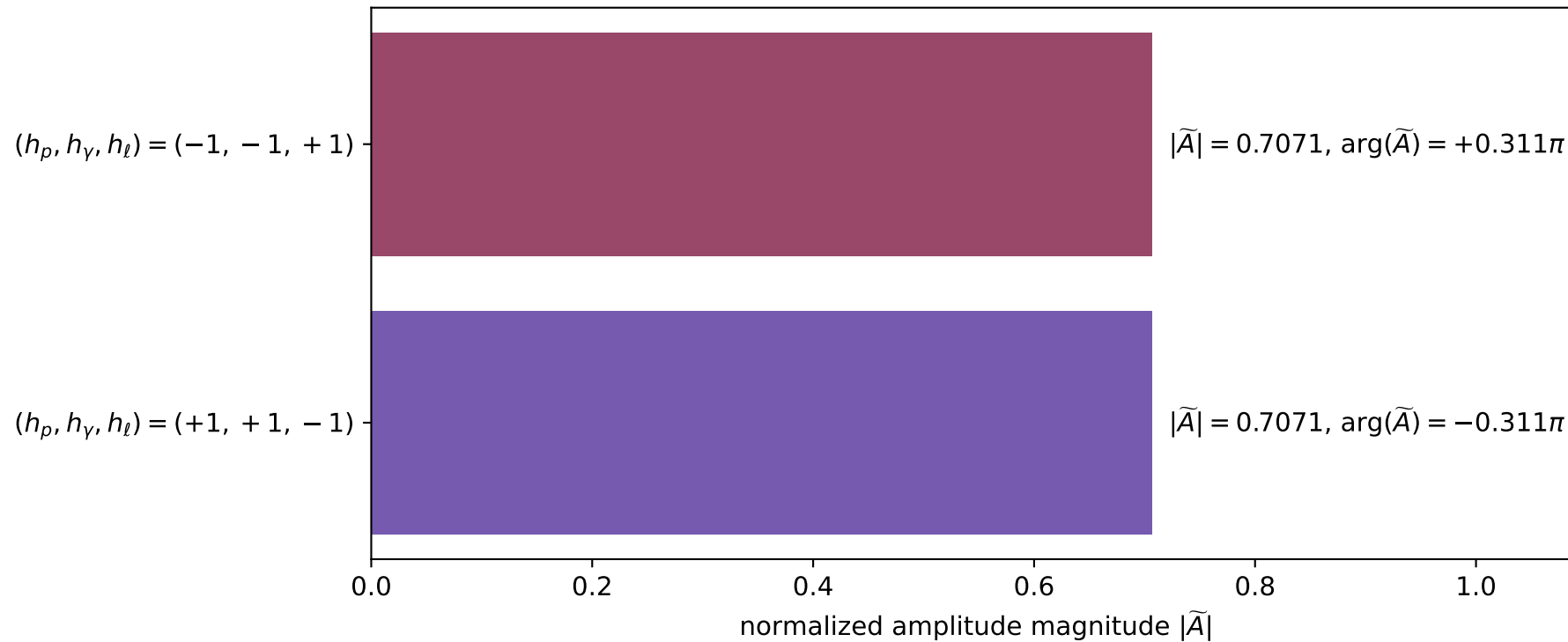
$s=24.9857$, $\sqrt{s}=4.99857$, $|\vec{P}|=2.41128$, $|\vec{P}'|=0.305039$
 $\theta_{P'}=0.000101732$, $\theta_Y=3.14159$, $\phi_{P'}=1.70875$, $\phi_Y=4.11801$
 $E_Y=2.15863$, $\theta_{\ell'}=1.67417\text{e-}05$, $\phi_{\ell'}=4.85035$
 $Q^2=17.8781$, $x_B=0.762282$, $t=-1.87322$, $W^2=6.45512$, $y=0.97293$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 1.8536$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.8536$
 P' $E= 0.9864$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3050$
 q_Y $E= 2.1586$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1586$

x--y plane

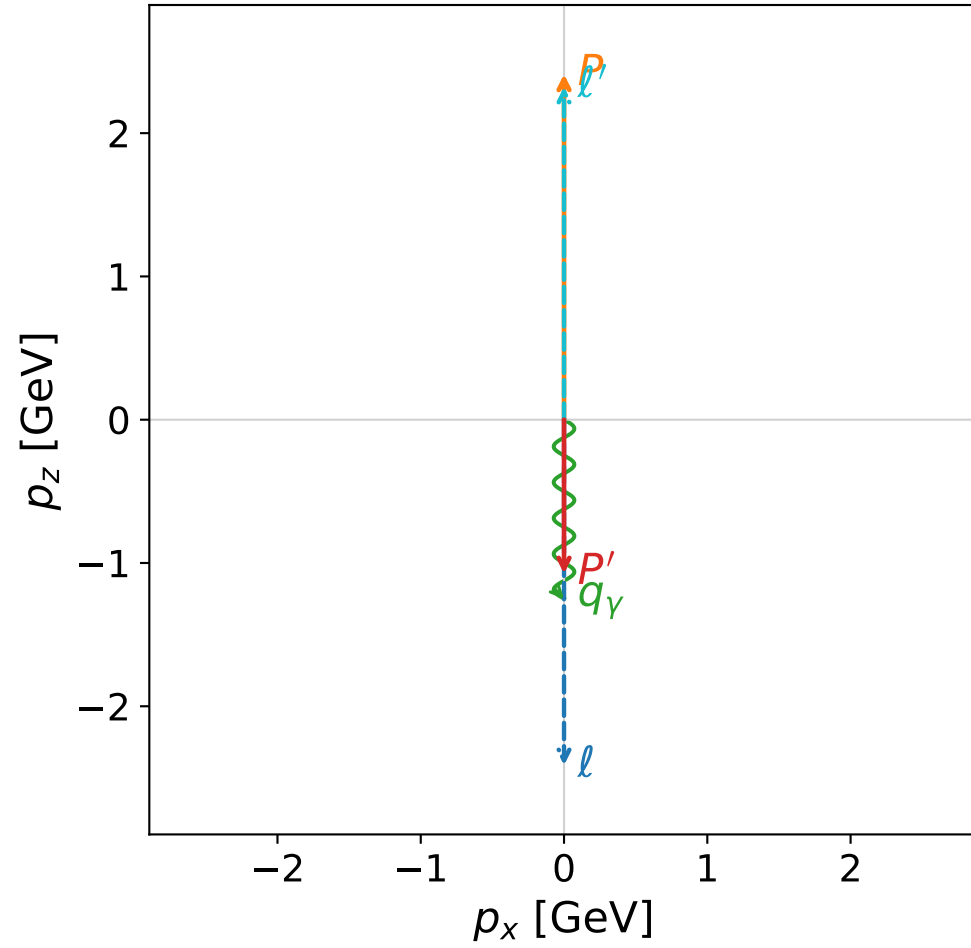


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

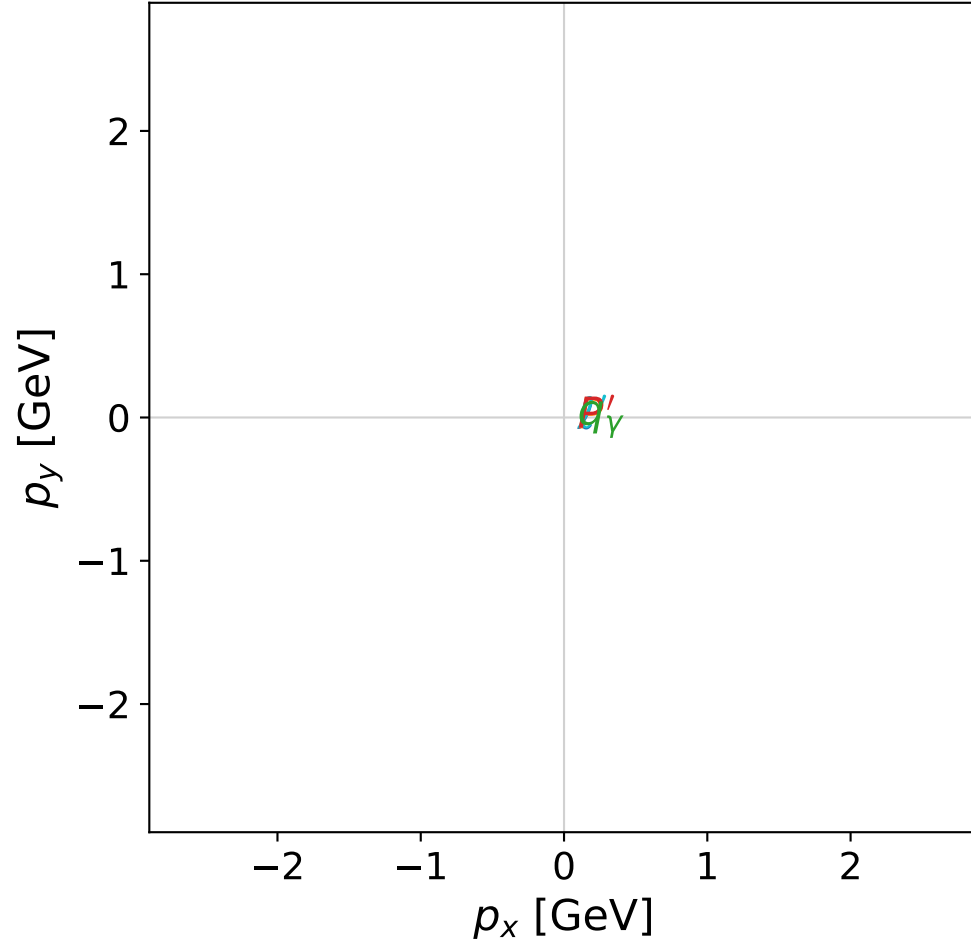


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



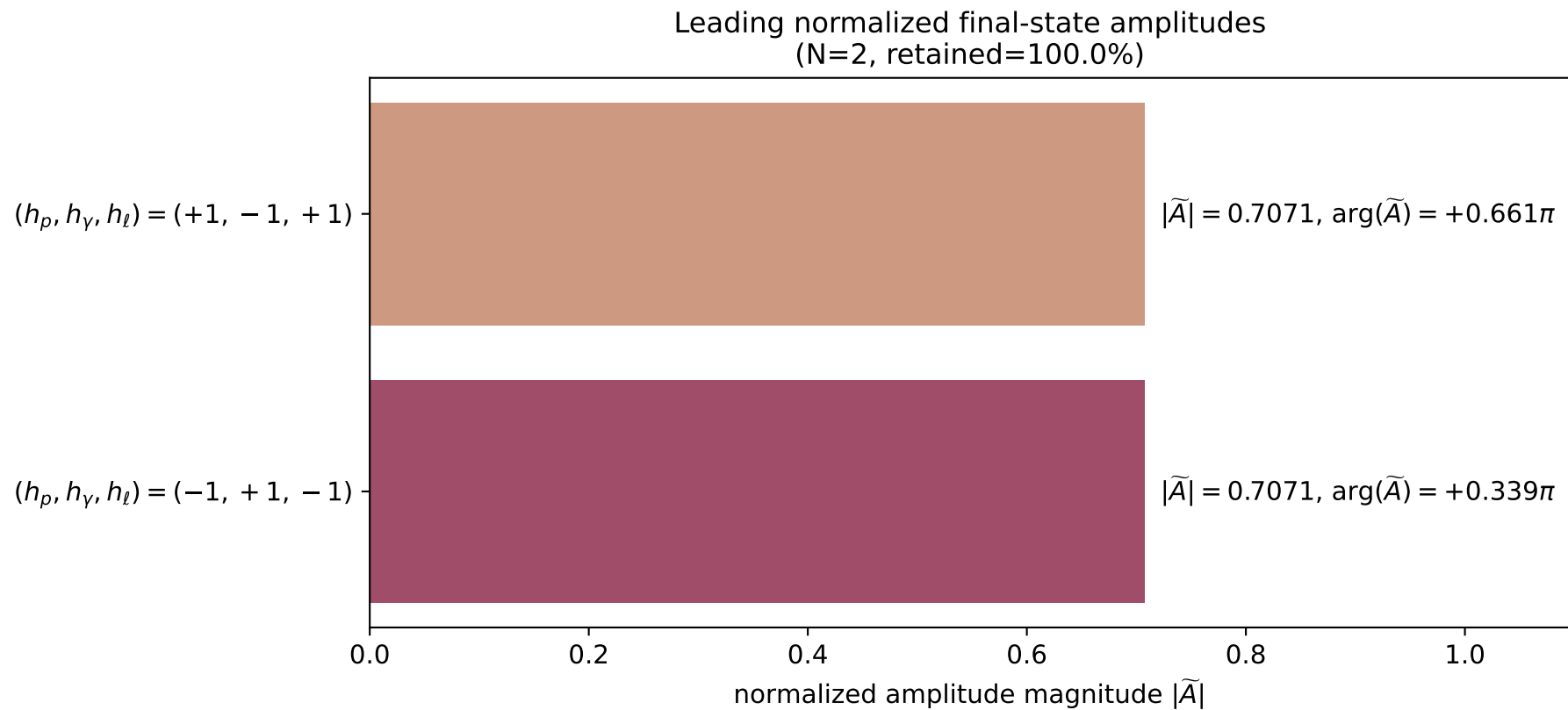
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_75
 $d_{\{\rm GHZ\}}=2.94743\text{e-}08$
 region: polarization_cluster_01_local_minimum_75
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0075
 lepton mixing angle: $\alpha_e=1.1150181$ rad
 incoming lepton: $0.440161|+\rangle + 0.897919|-\rangle$
 proton mixing angle: $\alpha_p=1.1150063$ rad
 incoming proton: $0.440172|+\rangle + 0.897914|-\rangle$

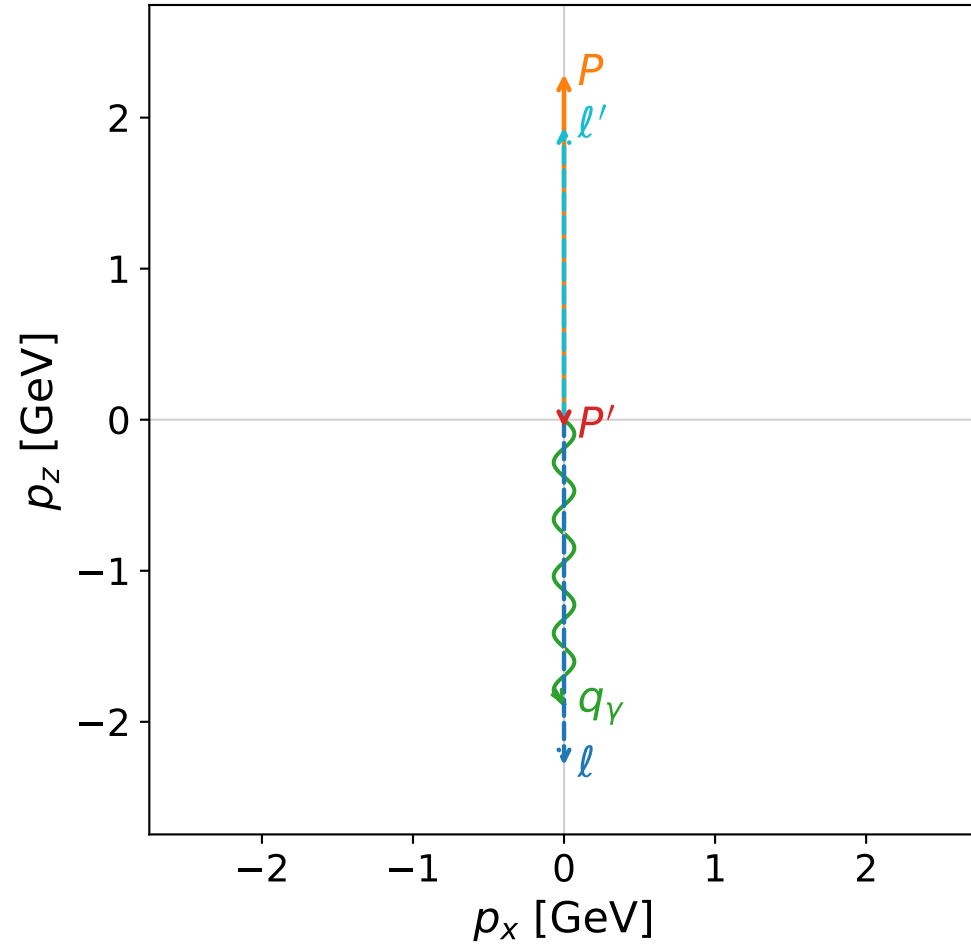
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.07529$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.49134$, $\phi_\gamma=5.21955$
 $E_\gamma=1.2489$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.11298$
 $Q^2=22.4239$, $x_B=0.962308$, $t=-10.8132$, $W^2=1.75814$, $y=0.966088$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3242$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.3242$
 P' $E= 1.4269$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.0753$
 q_γ $E= 1.2489$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.2489$

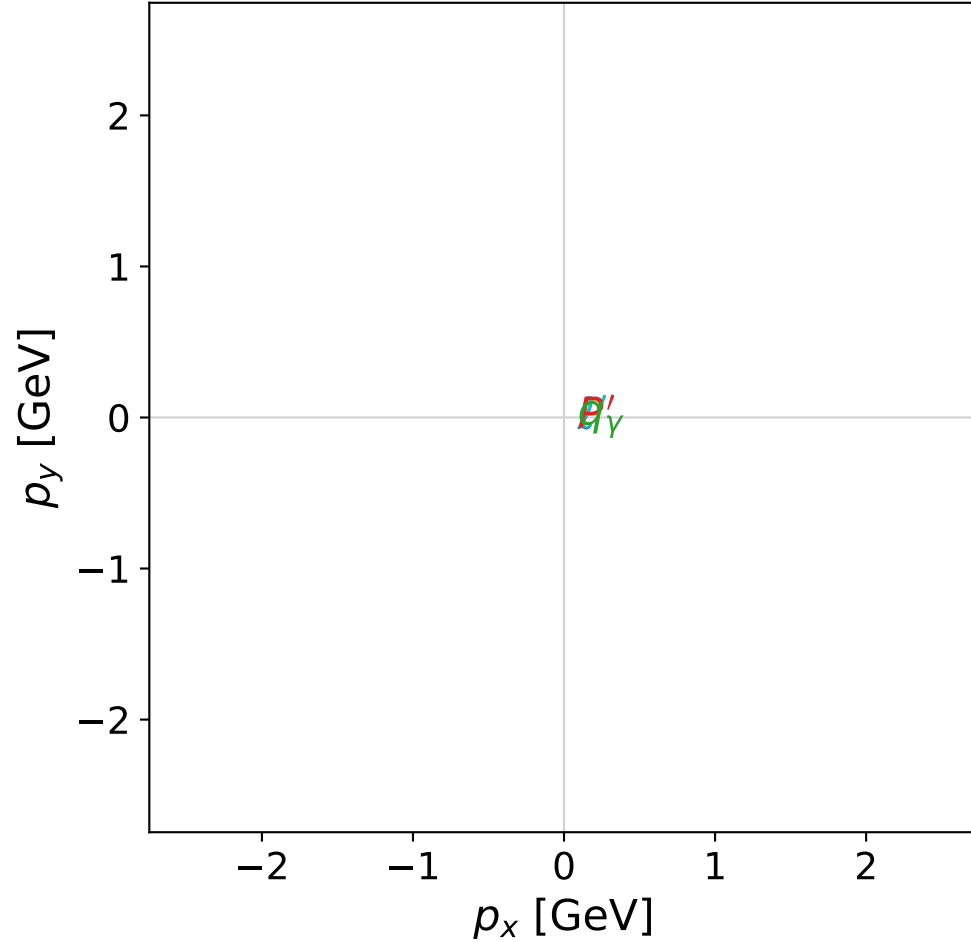


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



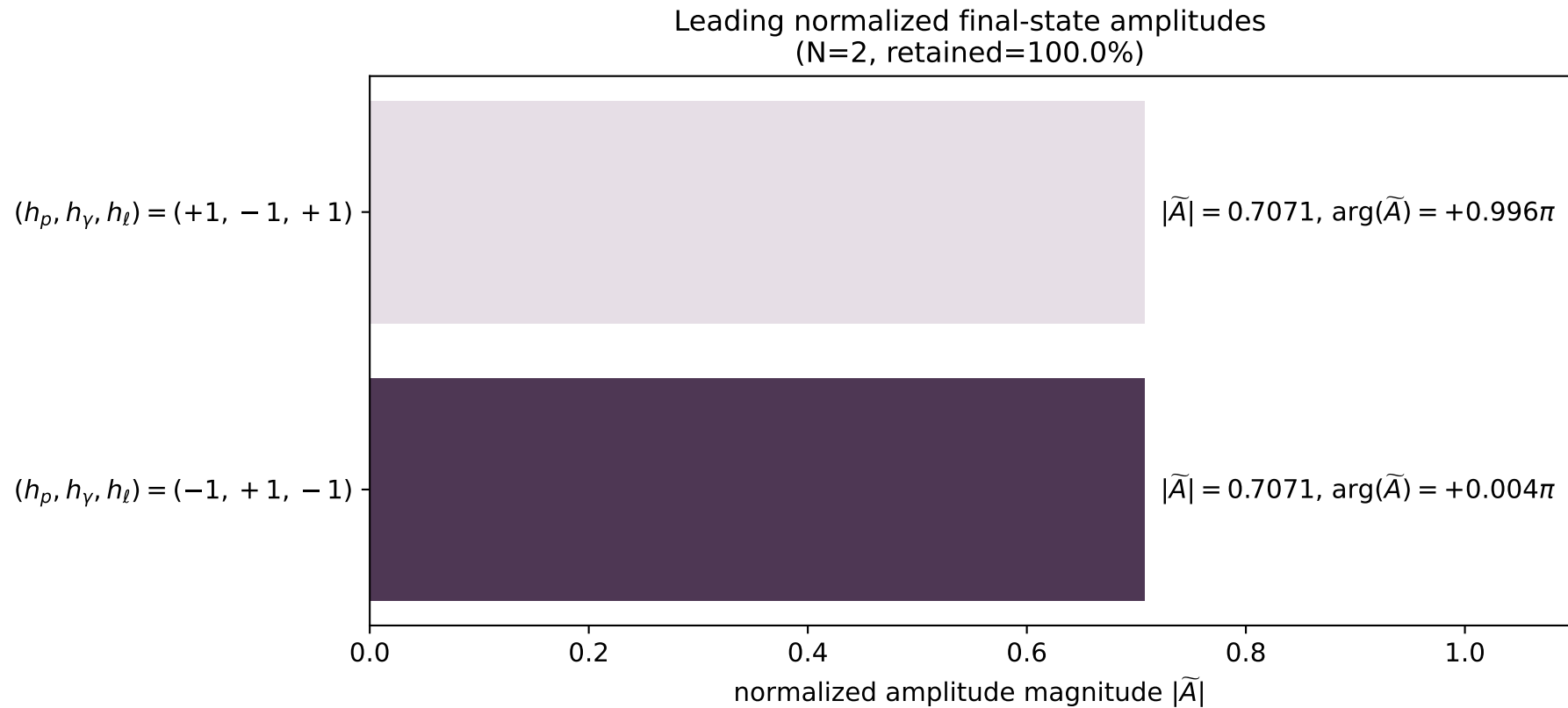
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_76
 $d_{\{\rm GHZ\}}=2.95168\text{e-}08$
 region: polarization_cluster_01_local_minimum_76
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0076
 lepton mixing angle: $\alpha_e=0.77617426$ rad
 incoming lepton: $0.713599|+\rangle +0.700555|-\rangle$
 proton mixing angle: $\alpha_p=0.77615415$ rad
 incoming proton: $0.713613|+\rangle +0.70054|-\rangle$

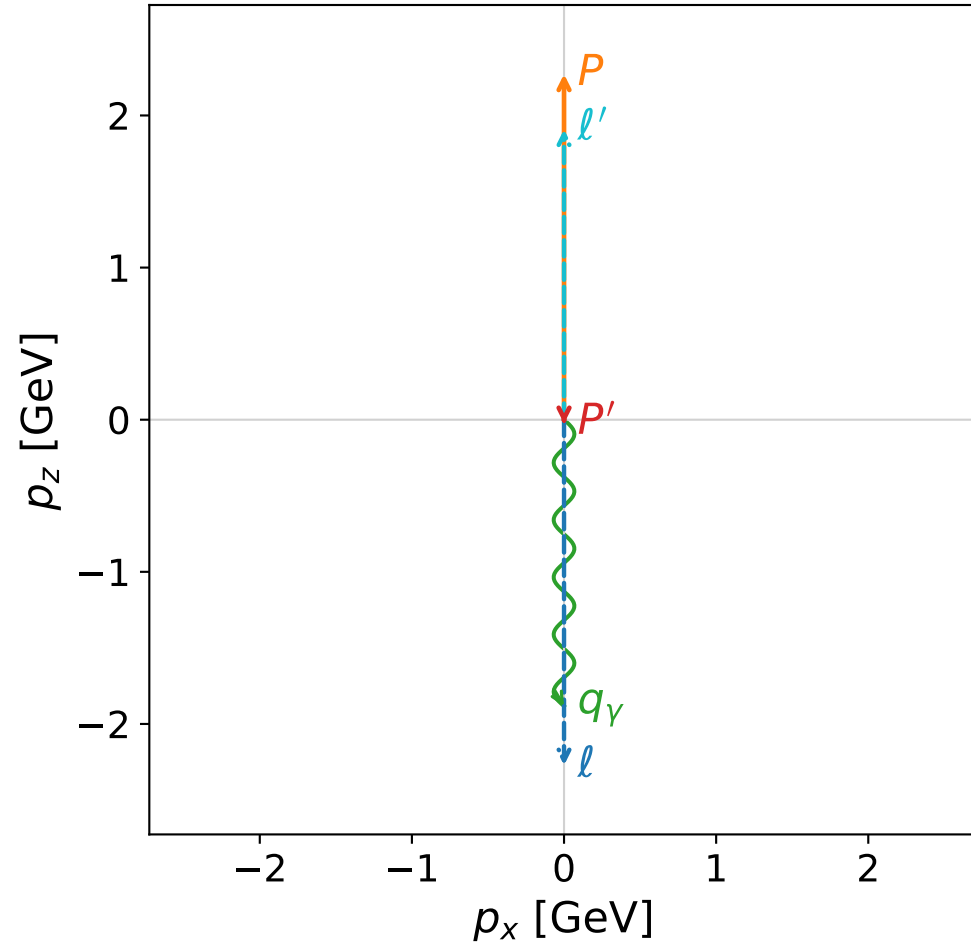
$s=22.6644$, $\sqrt{s}=4.76071$, $|\vec{P}|=2.28795$, $|\vec{P}'|=0.0517317$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.27601$, $\phi_\gamma=6.27171$
 $E_\gamma=1.88478$, $\theta_{l'}=0$, $\phi_{l'}=3.13024$
 $Q^2=17.7225$, $x_B=0.841177$, $t=-3.12298$, $W^2=4.22605$, $y=0.967142$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2879$
 P $E= 2.4728$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2879$
 l' $E= 1.9365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9365$
 P' $E= 0.9394$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0517$
 q_γ $E= 1.8848$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8848$

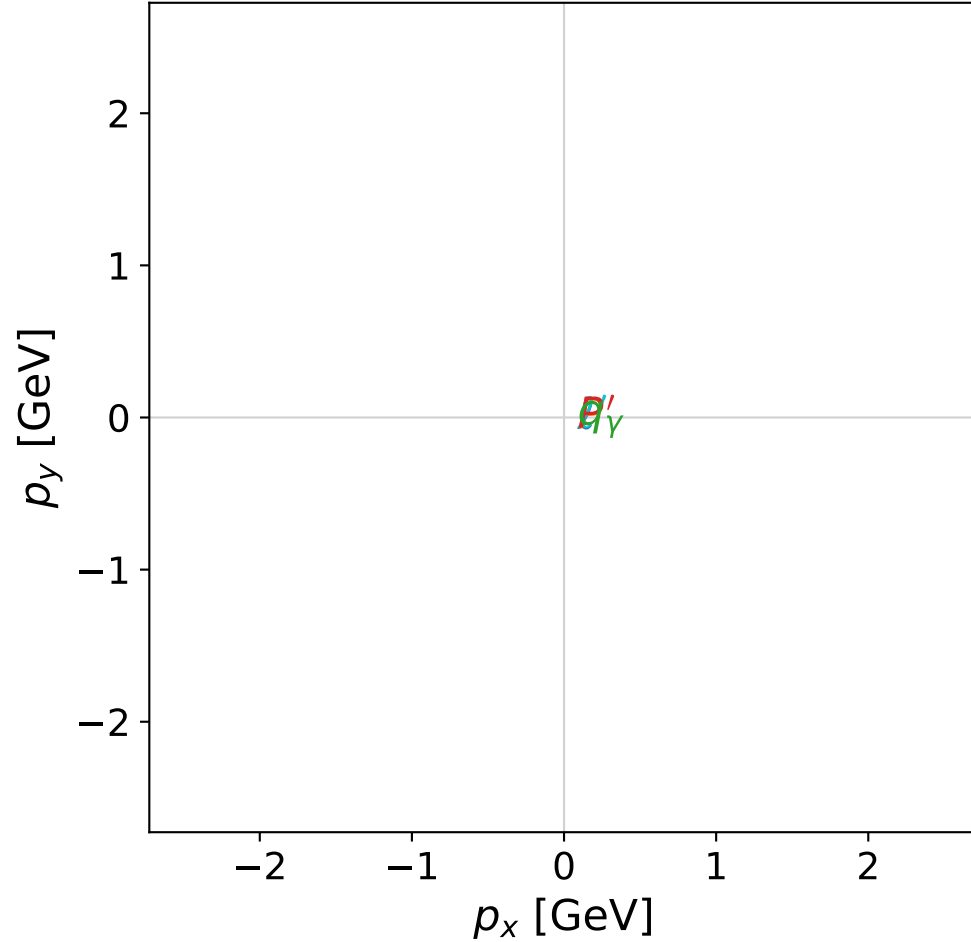


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



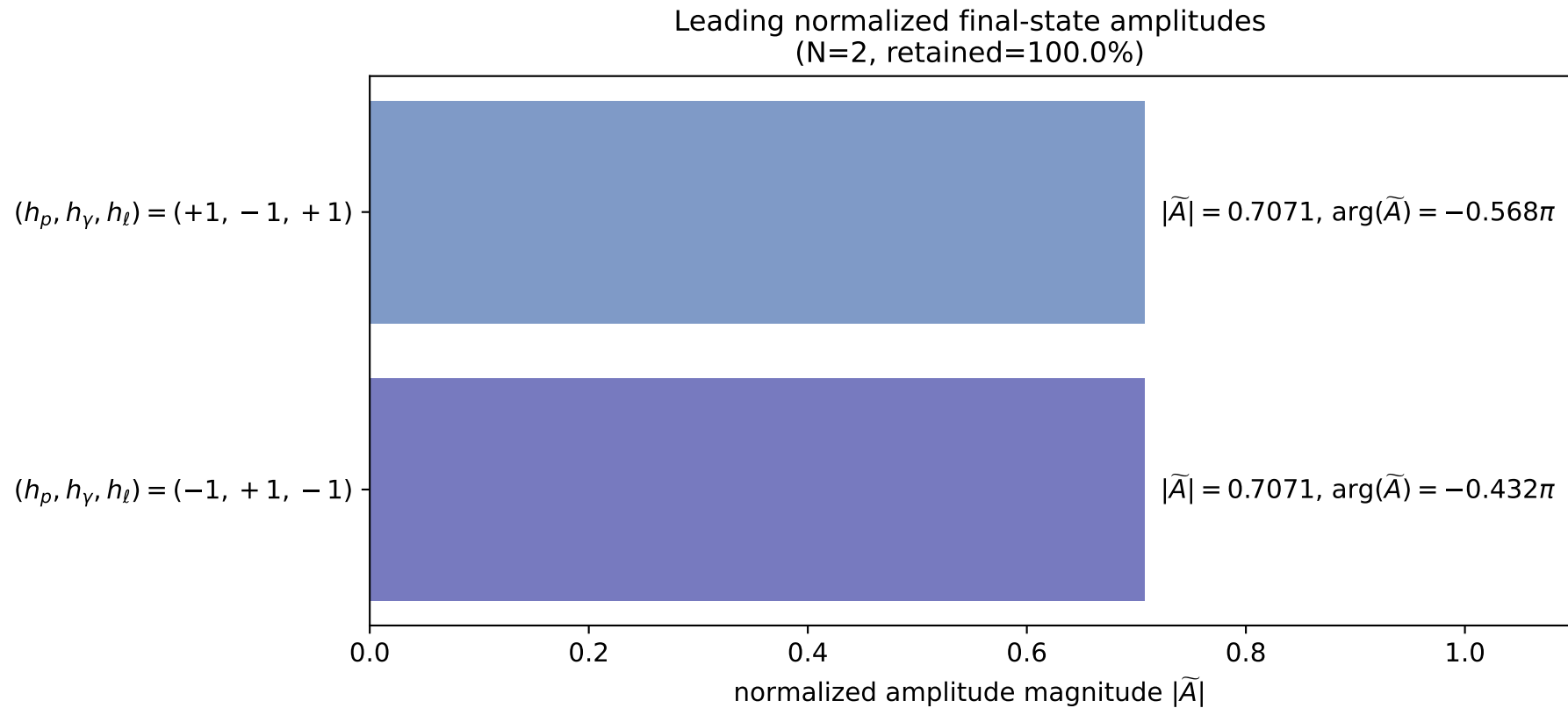
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_79
 $d_{\{\rm GHZ\}}=3.01399\text{e-}08$
 region: polarization_cluster_01_local_minimum_79
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0079
 lepton mixing angle: $\alpha_e=0.76405178$ rad
 incoming lepton: $0.722039|+\rangle +0.691853|-\rangle$
 proton mixing angle: $\alpha_p=0.76404422$ rad
 incoming proton: $0.722044|+\rangle +0.691847|-\rangle$

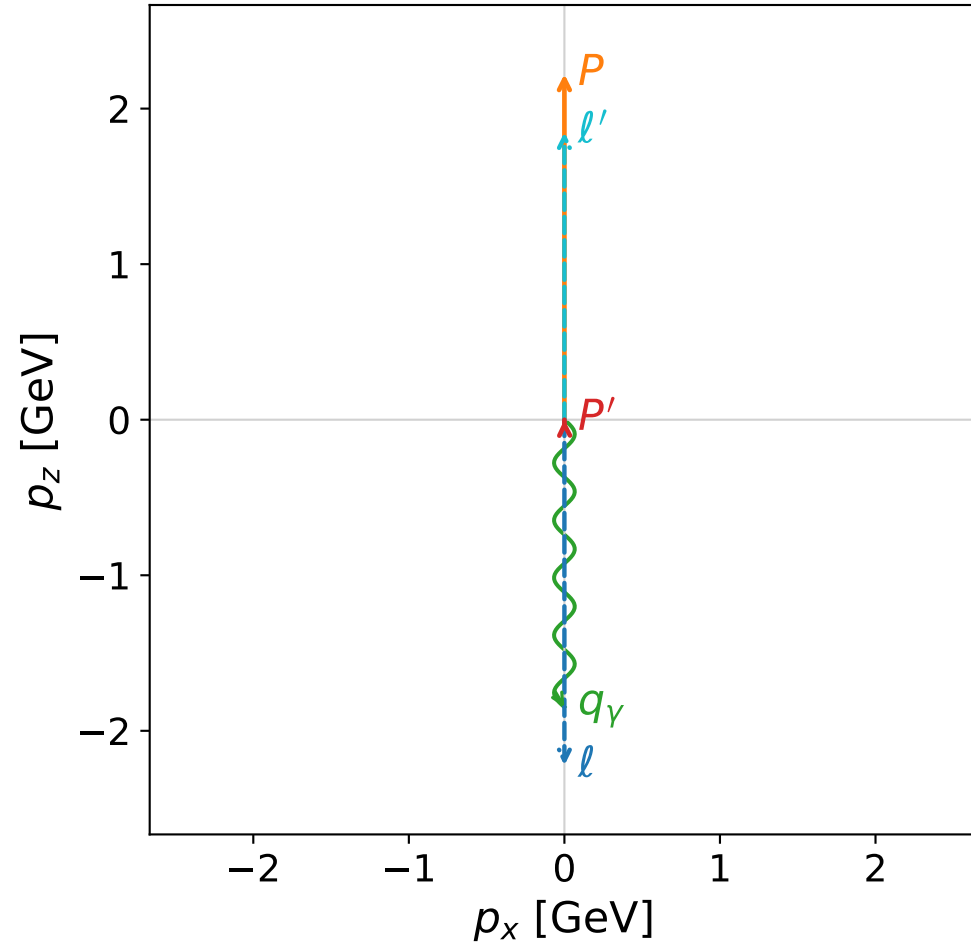
$s=22.3692$, $\sqrt{s}=4.72961$, $|\vec{P}|=2.27179$, $|\vec{P}'|=0.0259335$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.0568276$, $\phi_\gamma=1.35777$
 $E_\gamma=1.88266$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.48614$
 $Q^2=17.3437$, $x_B=0.834663$, $t=-2.97077$, $W^2=4.31541$, $y=0.966955$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2718$
 P $E= 2.4578$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2718$
 ℓ' $E= 1.9086$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9086$
 P' $E= 0.9384$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0259$
 q_γ $E= 1.8827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8827$

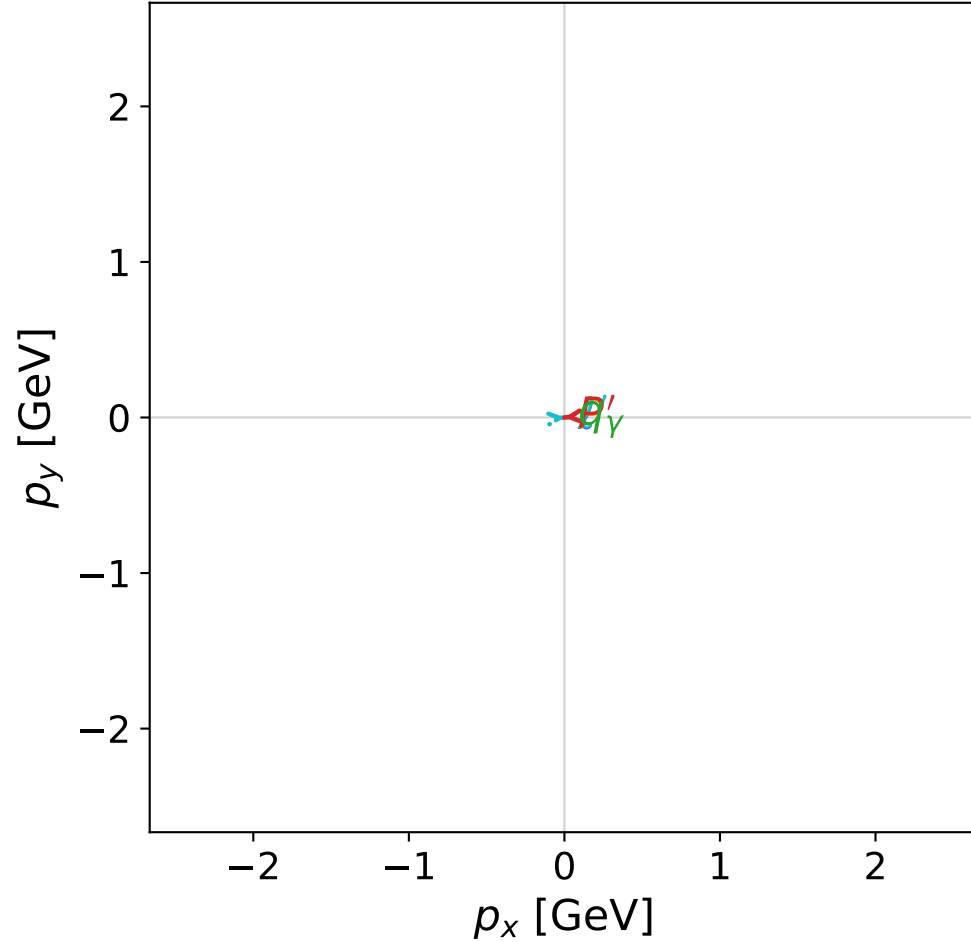


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



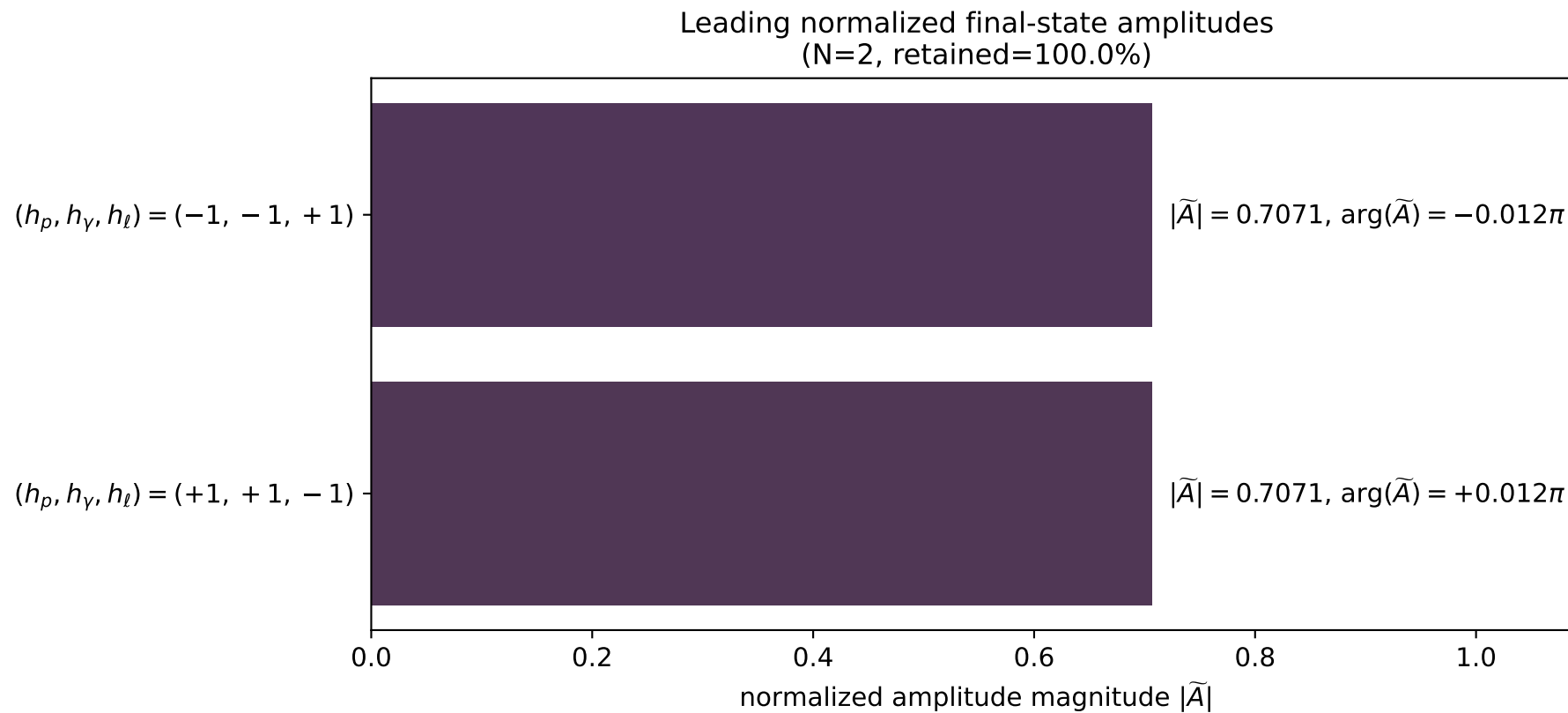
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_83
 $d_{\text{GHZ}}=3.05691\text{e-}08$
 region: polarization_cluster_01_local_minimum_83
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0083
 lepton mixing angle: $\alpha_e=0.74952575$ rad
 incoming lepton: $0.732012|+\rangle +0.681292|-\rangle$
 proton mixing angle: $\alpha_p=0.74952252$ rad
 incoming proton: $0.732014|+\rangle +0.681289|-\rangle$

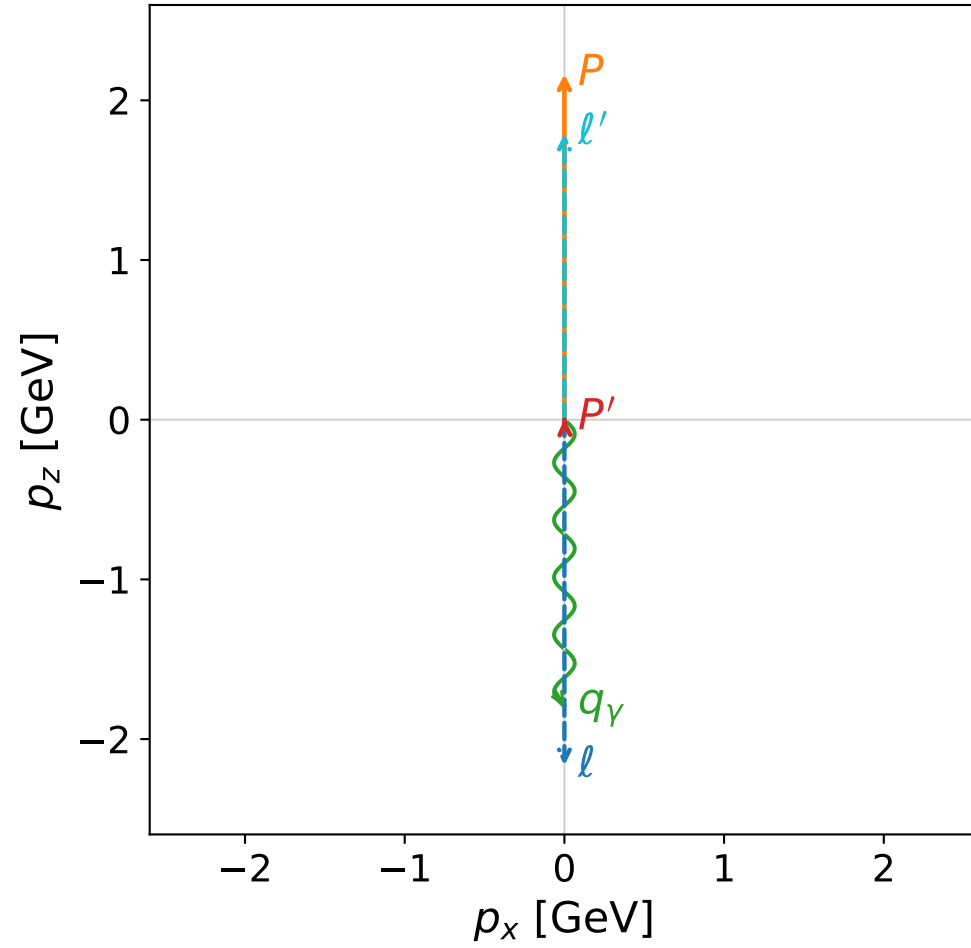
$s=21.4675$, $\sqrt{s}=4.63331$, $|\vec{P}|=2.22171$, $|\vec{P}'|=8.39622\text{e-}05$
 $\theta_{P'}=0.0161201$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.23565$, $\phi_\gamma=3.1028$
 $E_\gamma=1.8477$, $\theta_\ell=7.32586\text{e-}07$, $\phi_{\ell'}=0.0940553$
 $Q^2=16.4194$, $x_B=0.825677$, $t=-2.7641$, $W^2=4.34643$, $y=0.965916$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2217$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2217$
 P $E= 2.4116$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2217$
 ℓ' $E= 1.8476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8476$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0001$
 q_γ $E= 1.8477$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.8477$

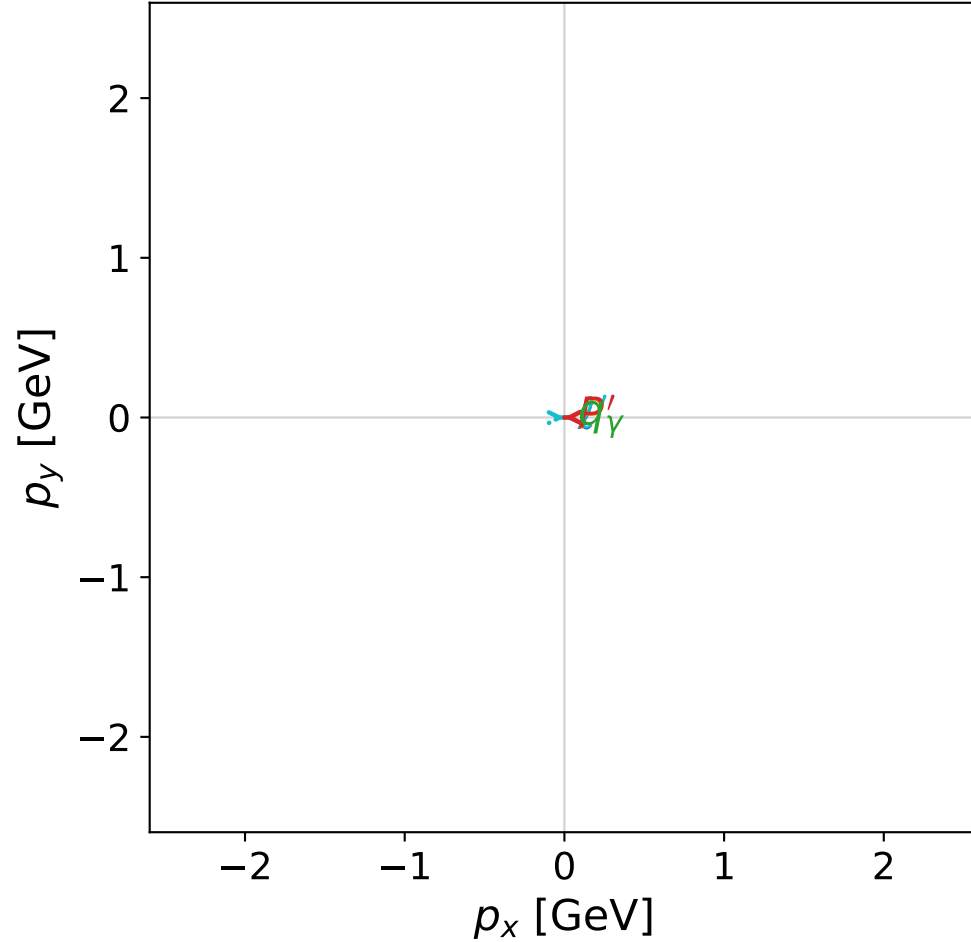


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



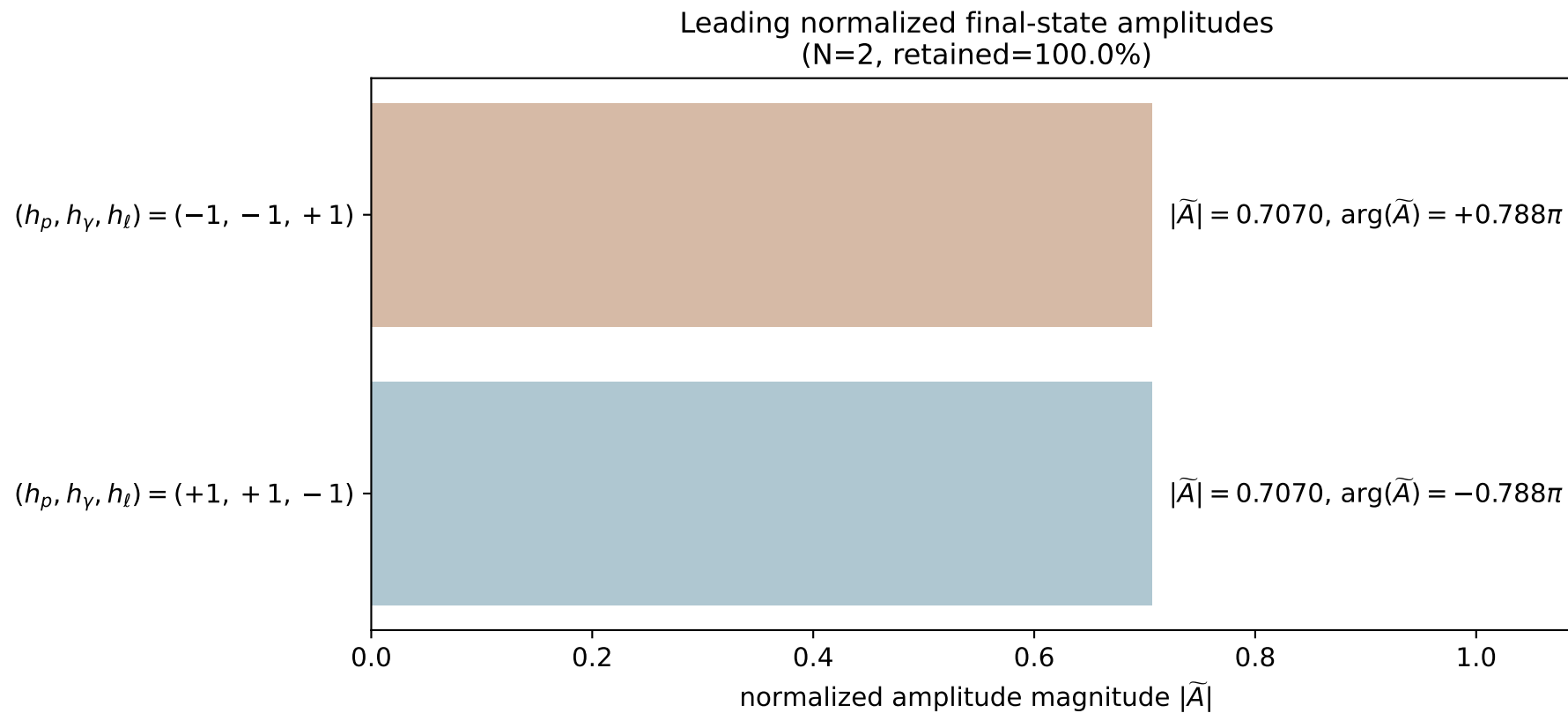
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_95
 $d_{\text{GHZ}}=3.40442\text{e-}08$
 region: polarization_cluster_01_local_minimum_95
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0095
 lepton mixing angle: $\alpha_e=0.83191558$ rad
 incoming lepton: $0.673461|+\rangle + 0.739223|-\rangle$
 proton mixing angle: $\alpha_p=0.83190695$ rad
 incoming proton: $0.673467|+\rangle + 0.739217|-\rangle$

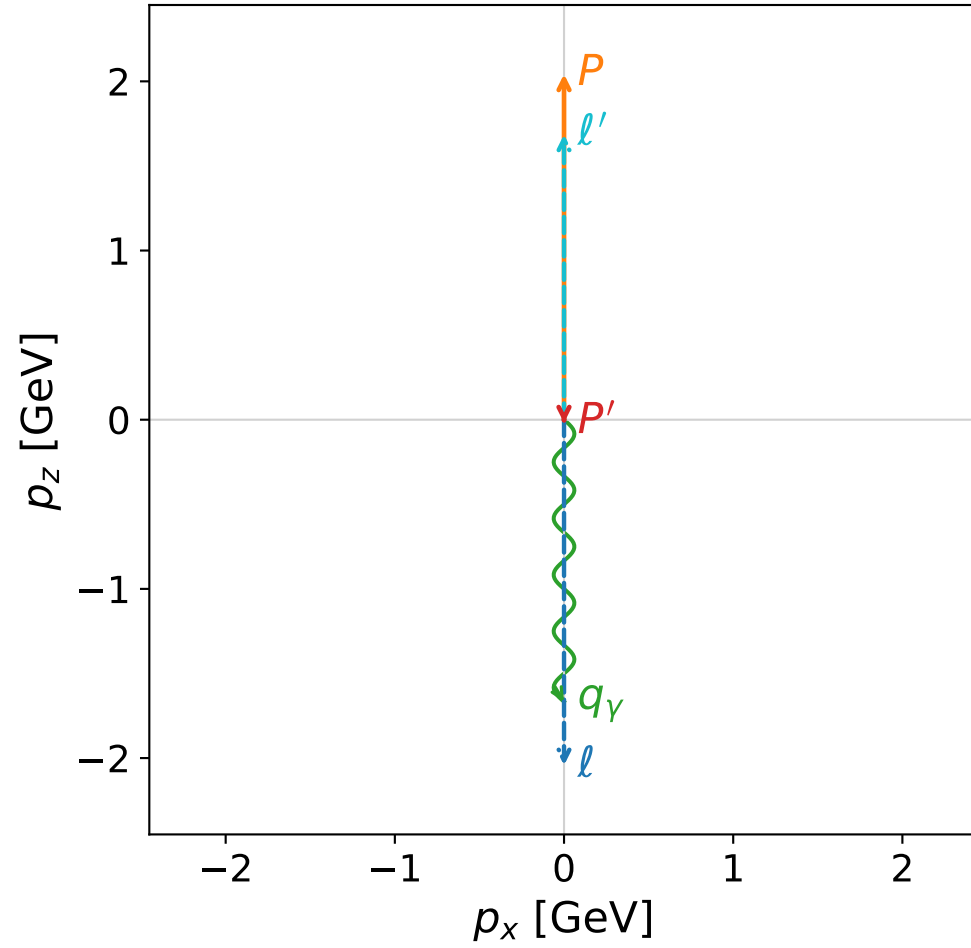
$s=20.4568$, $\sqrt{s}=4.52291$, $|\vec{P}|=2.16419$, $|\vec{P}'|=0.00379415$
 $\theta_{P'}=0.0374492$, $\theta_Y=3.14159$, $\phi_{P'}=3.15038$, $\phi_Y=5.61715$
 $E_Y=1.79435$, $\theta_{l'}=7.93354\text{e-}05$, $\phi_{l'}=0.00878615$
 $Q^2=15.5004$, $x_B=0.820986$, $t=-2.6489$, $W^2=4.25968$, $y=0.964415$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1642$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1642$
 P $E= 2.3587$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1642$
 ℓ' $E= 1.7906$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.7906$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 0.0038$
 q_Y $E= 1.7943$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.7943$

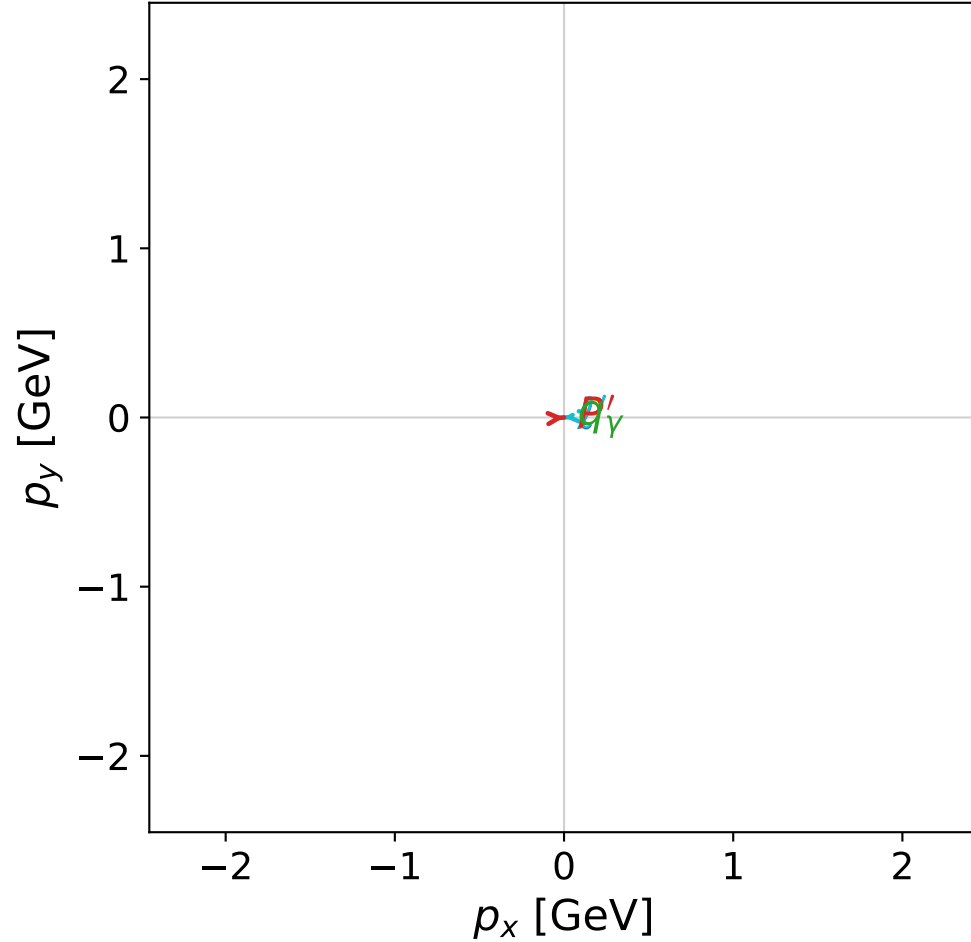


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



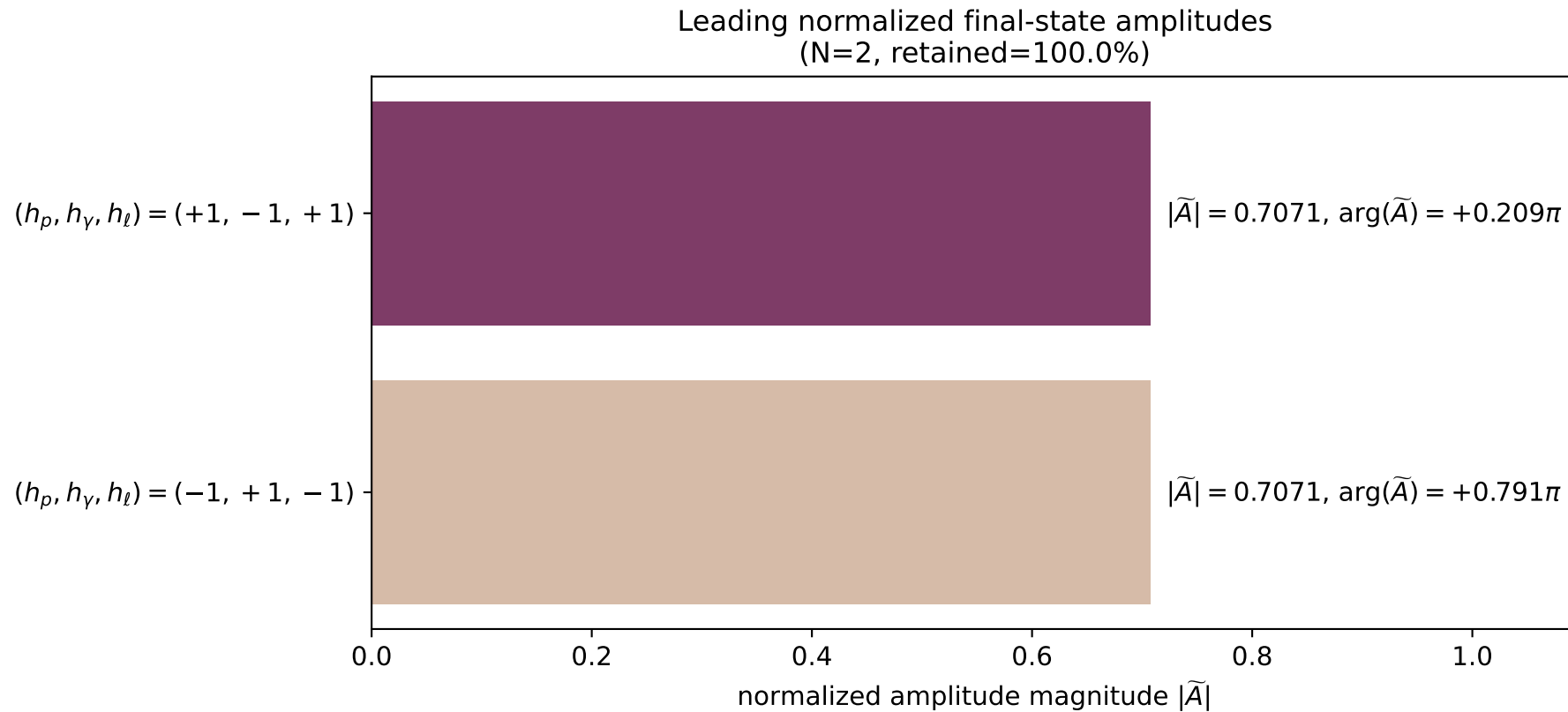
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_104
 $d_{\text{GHZ}}=3.80072\text{e-}08$
 region: polarization_cluster_01_local_minimum_104
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0104
 lepton mixing angle: $\alpha_e=0.79818758$ rad
 incoming lepton: $0.698006|+\rangle +0.716092|-\rangle$
 proton mixing angle: $\alpha_p=0.79817765$ rad
 incoming proton: $0.698013|+\rangle +0.716085|-\rangle$

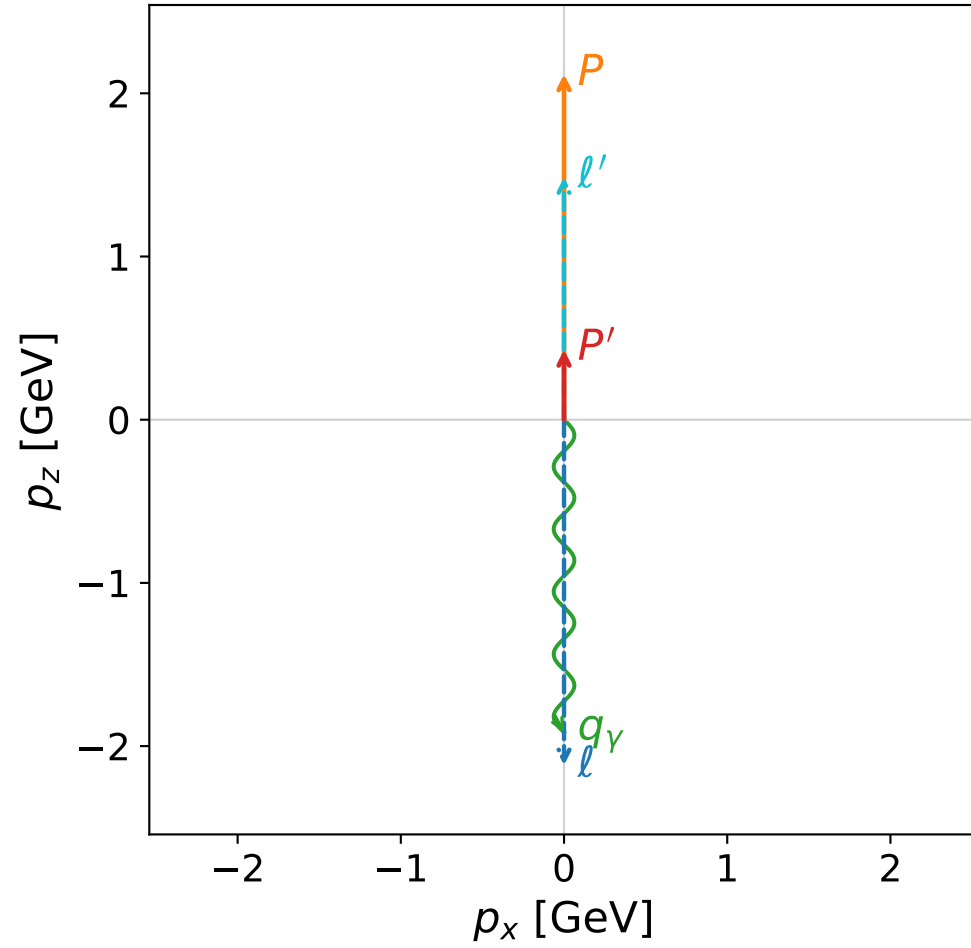
$s=18.4089$, $\sqrt{s}=4.29056$, $|\vec{P}|=2.04275$, $|\vec{P}'|=0.0189369$
 $\theta_{P'}=3.13802$, $\theta_Y=3.14159$, $\phi_{P'}=0.06771$, $\phi_Y=3.86678$
 $E_Y=1.66672$, $\theta_{\ell'}=4.01396\text{e-}05$, $\phi_{\ell'}=3.2093$
 $Q^2=13.7735$, $x_B=0.818012$, $t=-2.53544$, $W^2=3.94412$, $y=0.960561$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0427$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0427$
 P $E= 2.2478$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0427$
 ℓ' $E= 1.6857$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 1.6857$
 P' $E= 0.9382$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.0189$
 q_Y $E= 1.6667$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.6667$

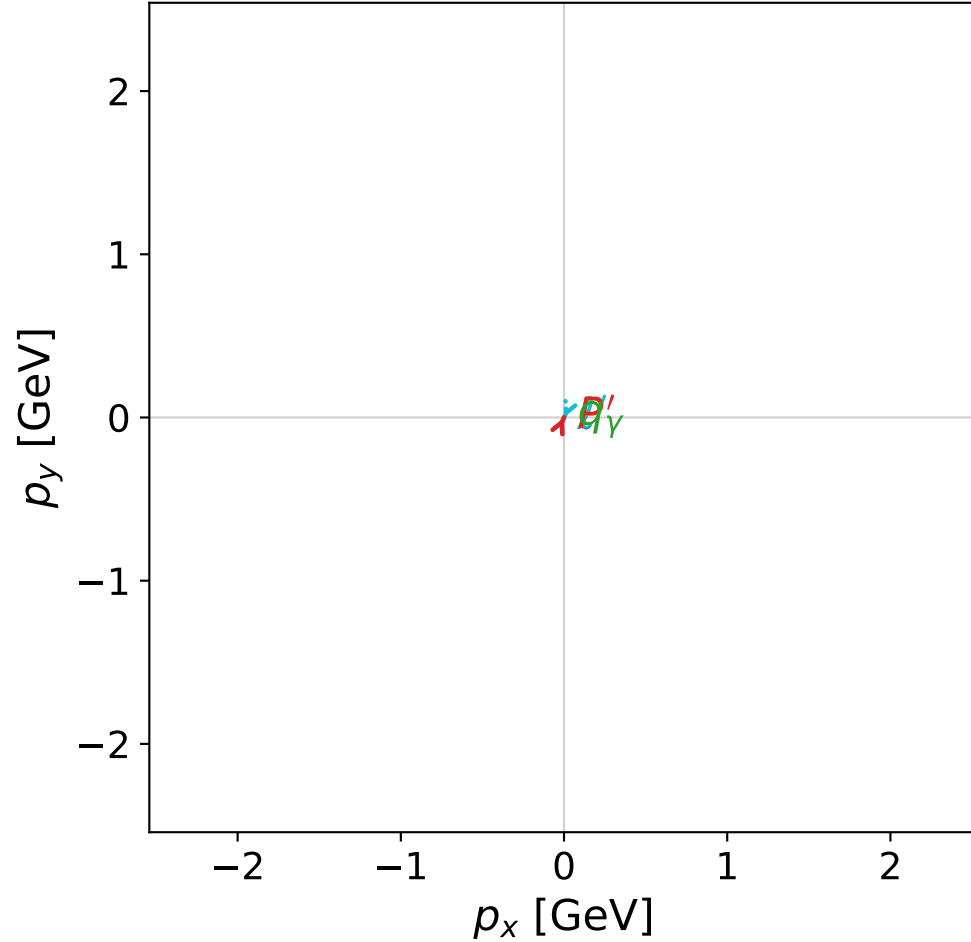


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



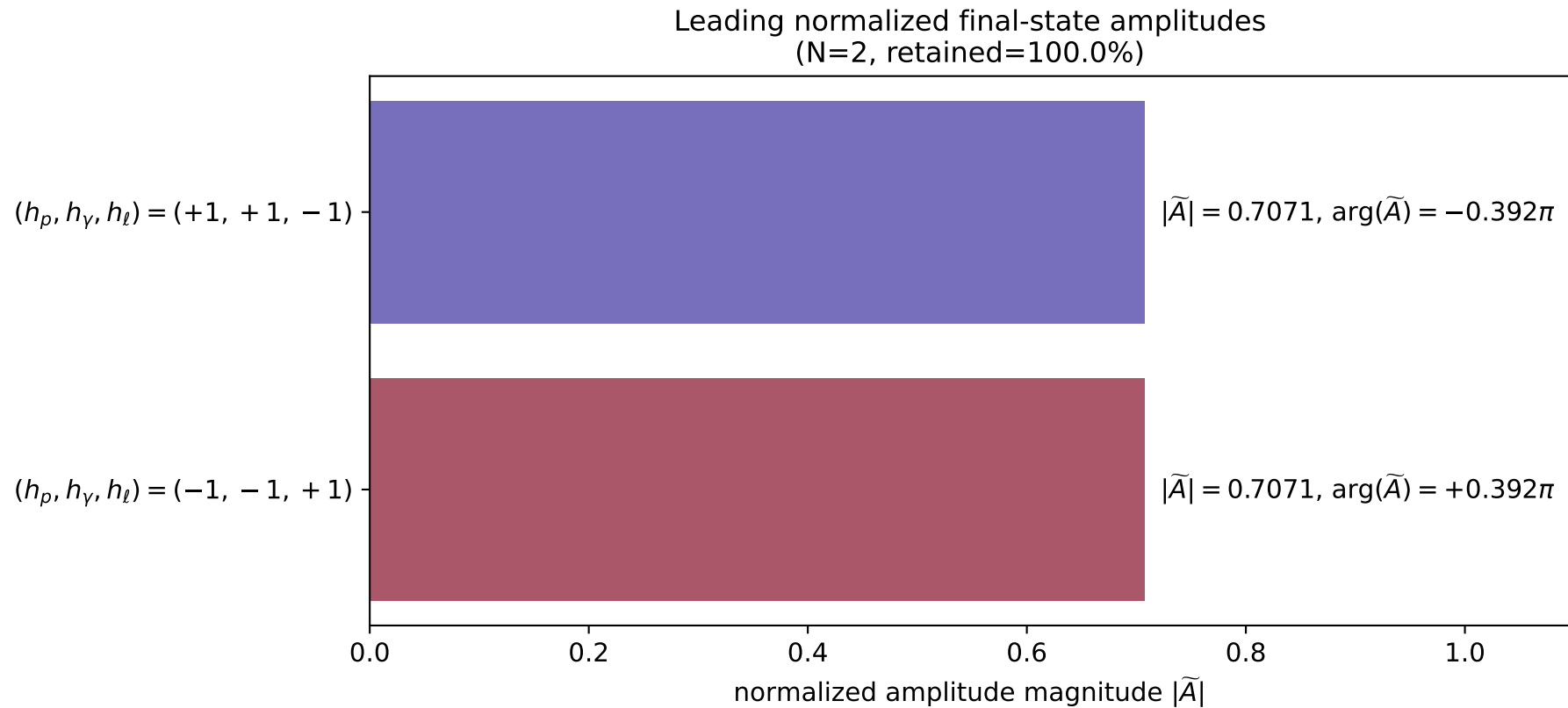
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_112
 $d_{\{\text{rm GHZ}\}}=4.14029\text{e-}08$
 region: polarization_cluster_01_local_minimum_112
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0112
 lepton mixing angle: $\alpha_e=0.7523389$ rad
 incoming lepton: $0.730093|+\rangle + 0.683348|-\rangle$
 proton mixing angle: $\alpha_p=0.75235965$ rad
 incoming proton: $0.730078|+\rangle + 0.683363|-\rangle$

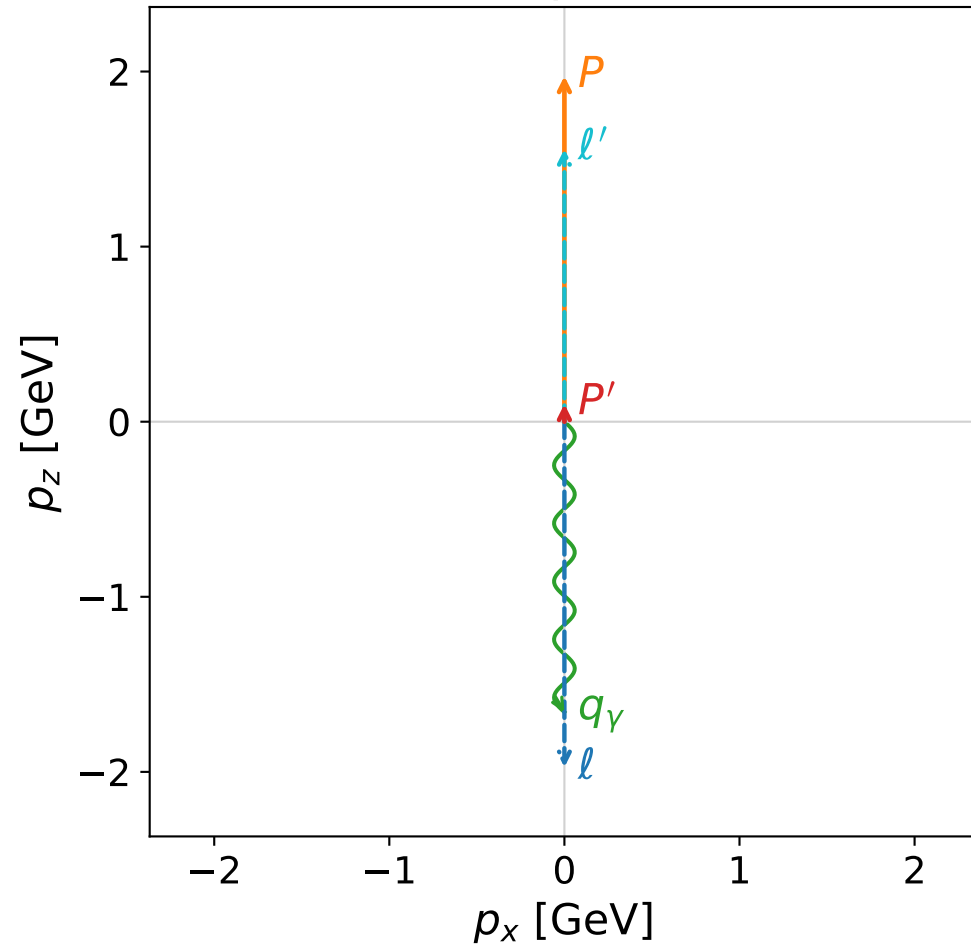
$s=19.6572$, $\sqrt{s}=4.43364$, $|\vec{P}|=2.1176$, $|\vec{P}'|=0.429483$
 $\theta_{p'}=7.6699\text{e-}05$, $\theta_Y=3.14159$, $\phi_{p'}=1.14894$, $\phi_Y=4.37324$
 $E_Y=1.91574$, $\theta_{\ell'}=2.21637\text{e-}05$, $\phi_{\ell'}=4.29053$
 $Q^2=12.5892$, $x_B=0.692189$, $t=-1.20006$, $W^2=6.47813$, $y=0.968585$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1176$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1176$
 P $E= 2.3160$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1176$
 ℓ' $E= 1.4863$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4863$
 P' $E= 1.0316$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4295$
 q_Y $E= 1.9157$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9157$

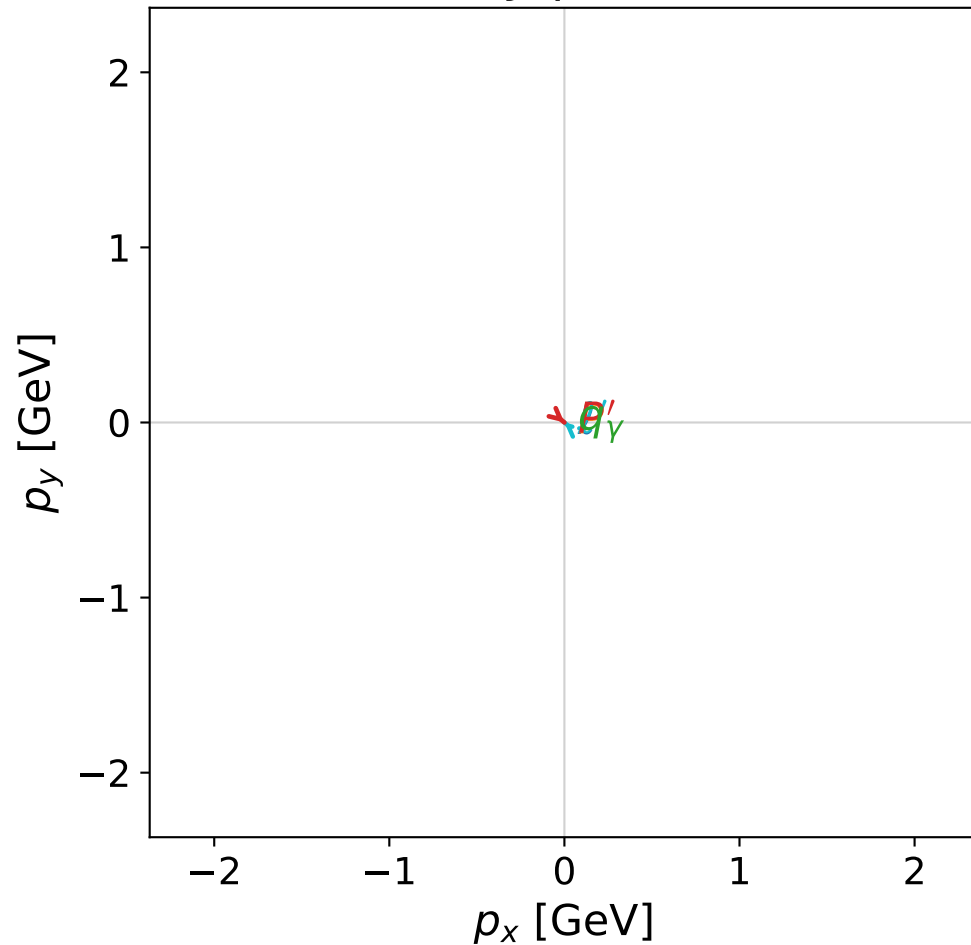


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



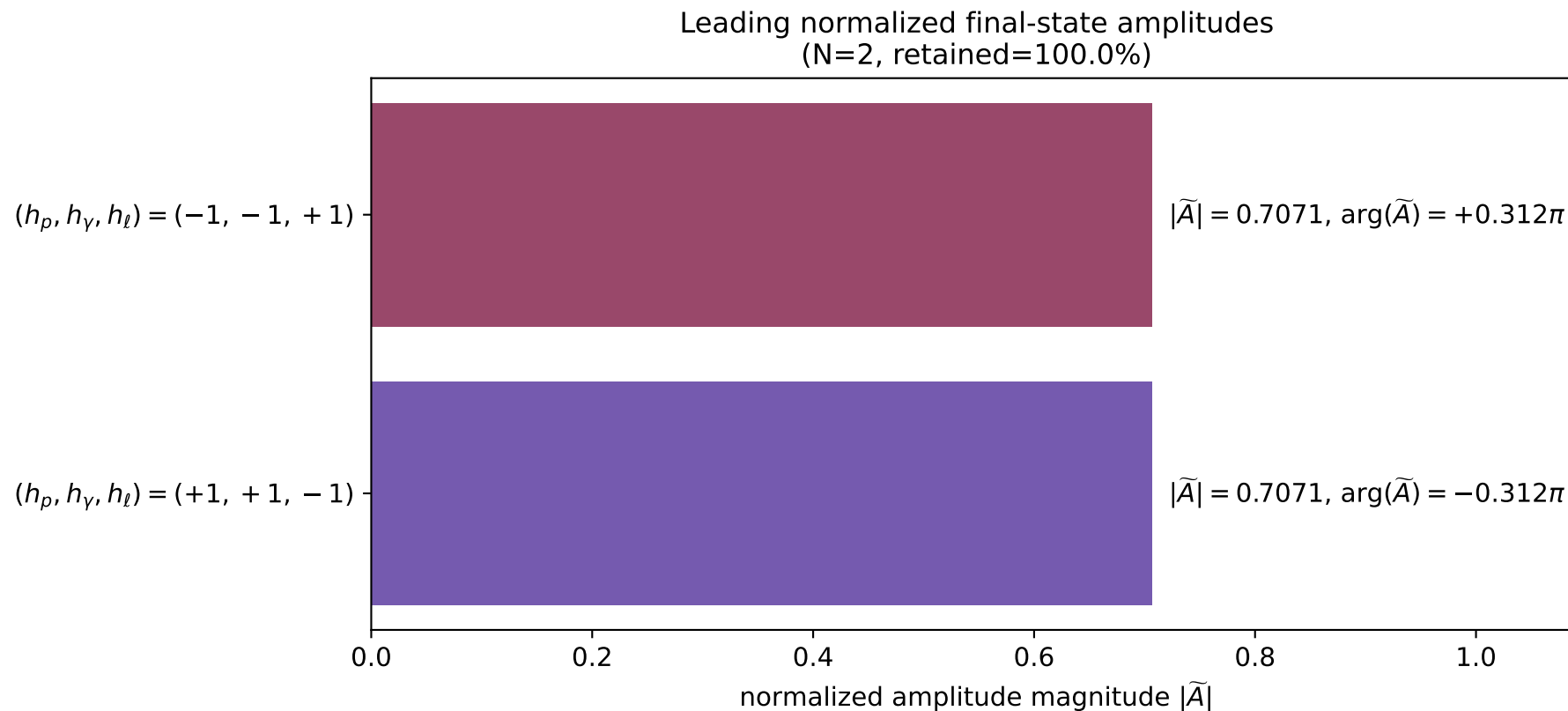
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_113
 $d_{\text{GHZ}}=4.24901\text{e-}08$
 region: polarization_cluster_01_local_minimum_113
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0113
 lepton mixing angle: $\alpha_e=0.83948165$ rad
 incoming lepton: $0.667849|+\rangle + 0.744297|-\rangle$
 proton mixing angle: $\alpha_p=0.8394633$ rad
 incoming proton: $0.667862|+\rangle + 0.744285|-\rangle$

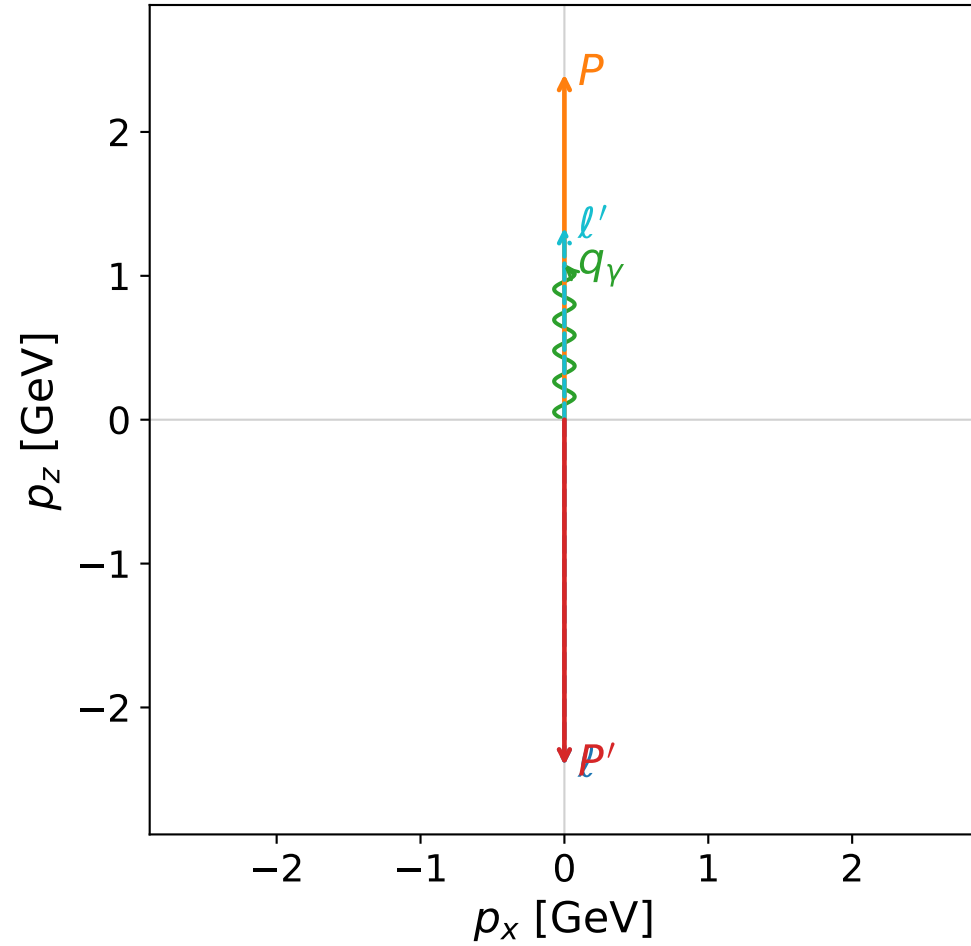
$s=17.2983$, $\sqrt{s}=4.15912$, $|\vec{P}|=1.97379$, $|\vec{P}'|=0.101055$
 $\theta_{P'}=0.000400551$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.58175$, $\phi_\gamma=4.12102$
 $E_\gamma=1.65837$, $\theta_{\ell'}=2.59918\text{e-}05$, $\phi_{\ell'}=2.44016$
 $Q^2=12.2953$, $x_B=0.780179$, $t=-1.9648$, $W^2=4.34413$, $y=0.959869$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9738$
 P $E= 2.1853$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9738$
 ℓ' $E= 1.5573$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5573$
 P' $E= 0.9434$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.1011$
 q_γ $E= 1.6584$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.6584$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

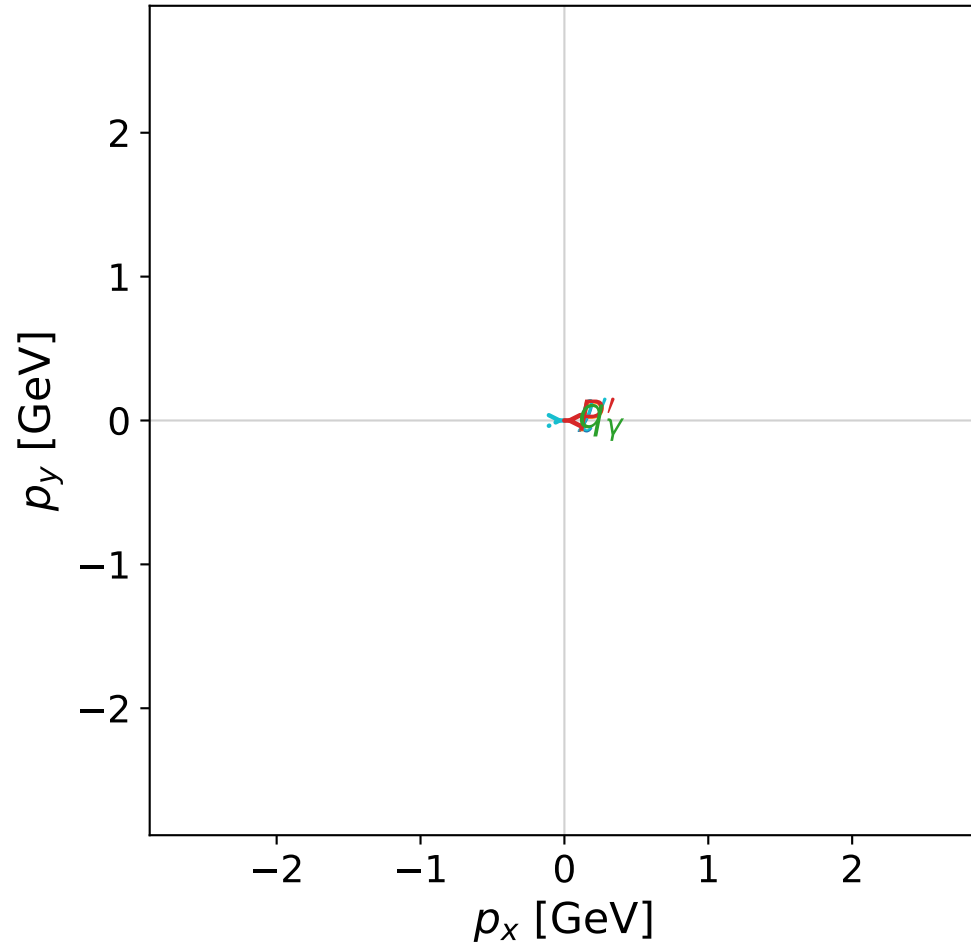


dghz_mixing_angles_polarization_01_local_minimum_114
 $d_{\text{GHZ}}=4.29606\text{e-}08$
 region: polarization_cluster_01_local_minimum_114
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0114
 lepton mixing angle: $\alpha_e=0.94445649$ rad
 incoming lepton: $0.586183|+\rangle + 0.810178|-\rangle$
 proton mixing angle: $\alpha_p=0.62636419$ rad
 incoming proton: $0.810164|+\rangle + 0.586203|-\rangle$

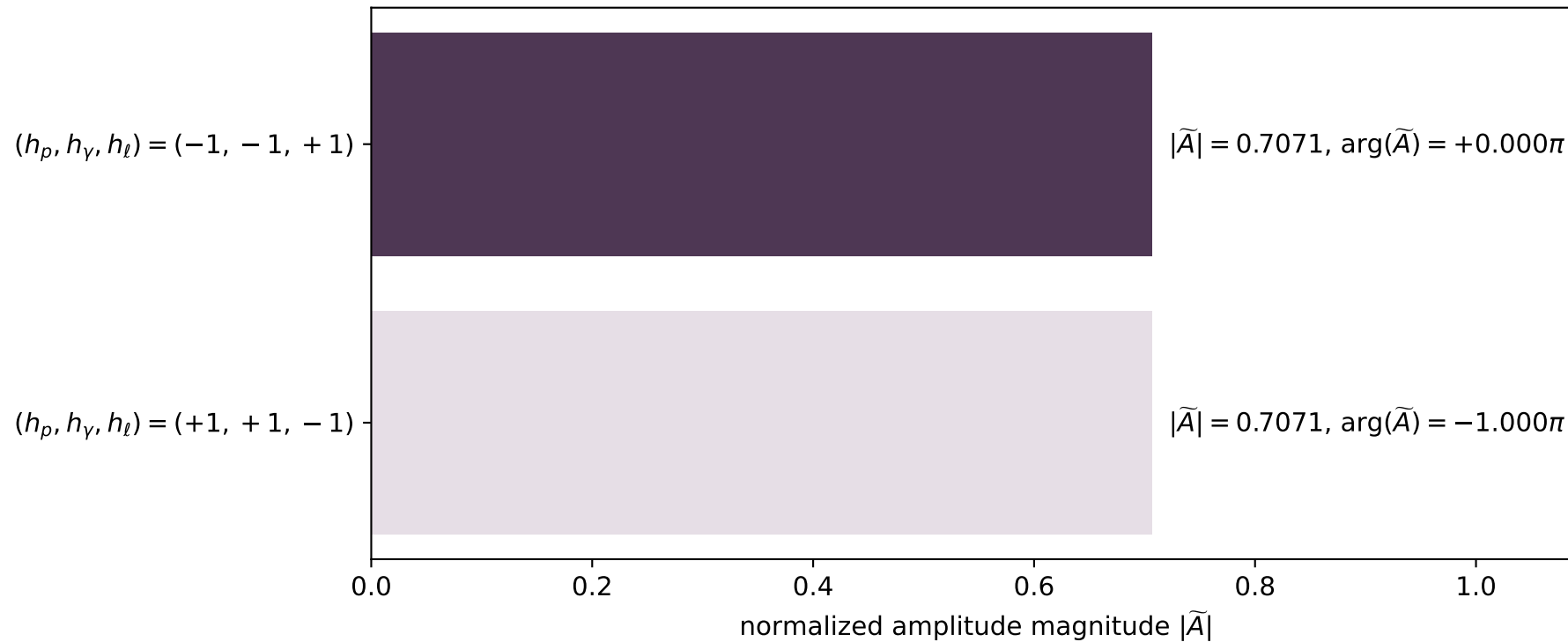
$s=24.8065$, $\sqrt{s}=4.98061$, $|\vec{P}|=2.40198$, $|\vec{P}'|=2.40198$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.70525$, $\phi_Y=5.75553$
 $E_Y=1.06741$, $\theta_{l'}=0$, $\phi_{l'}=5.84684$
 $Q^2=12.8224$, $x_B=0.546678$, $t=-23.078$, $W^2=11.5126$, $y=0.980293$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4020$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4020$
 P $E= 2.5786$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4020$
 ℓ' $E= 1.3346$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.3346$
 P' $E= 2.5786$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4020$
 q_Y $E= 1.0674$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0674$

x--y plane

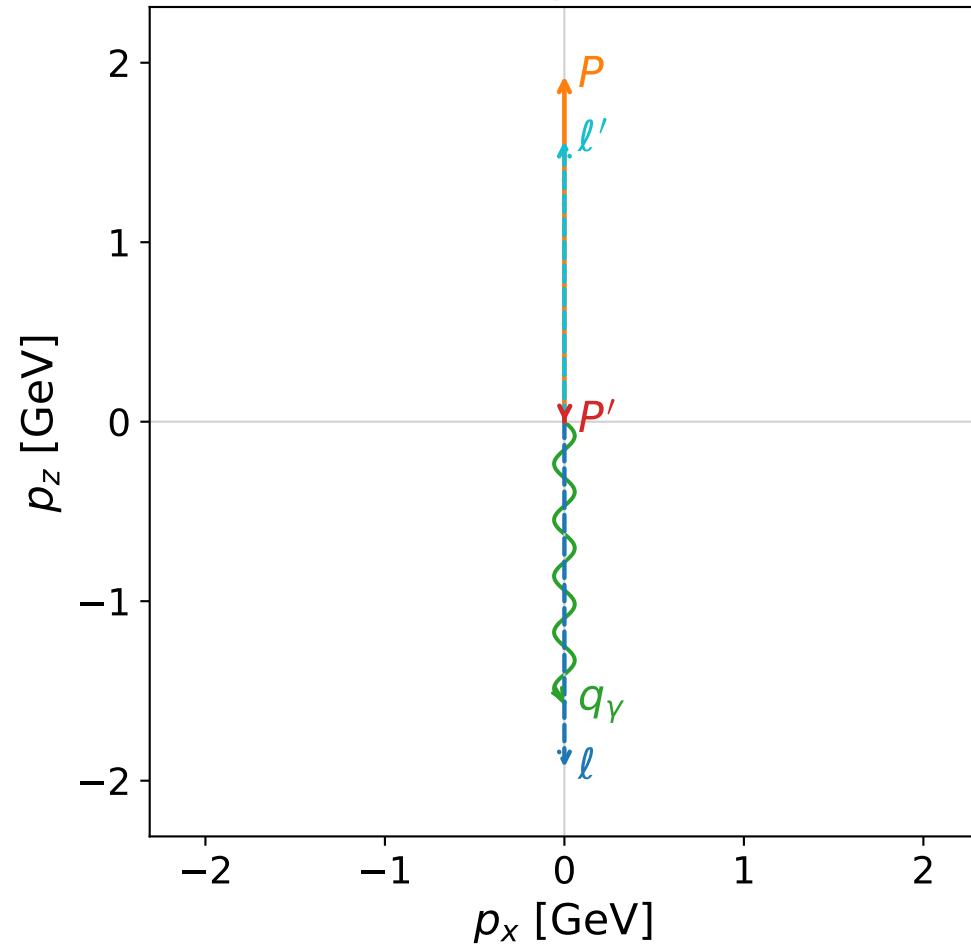


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

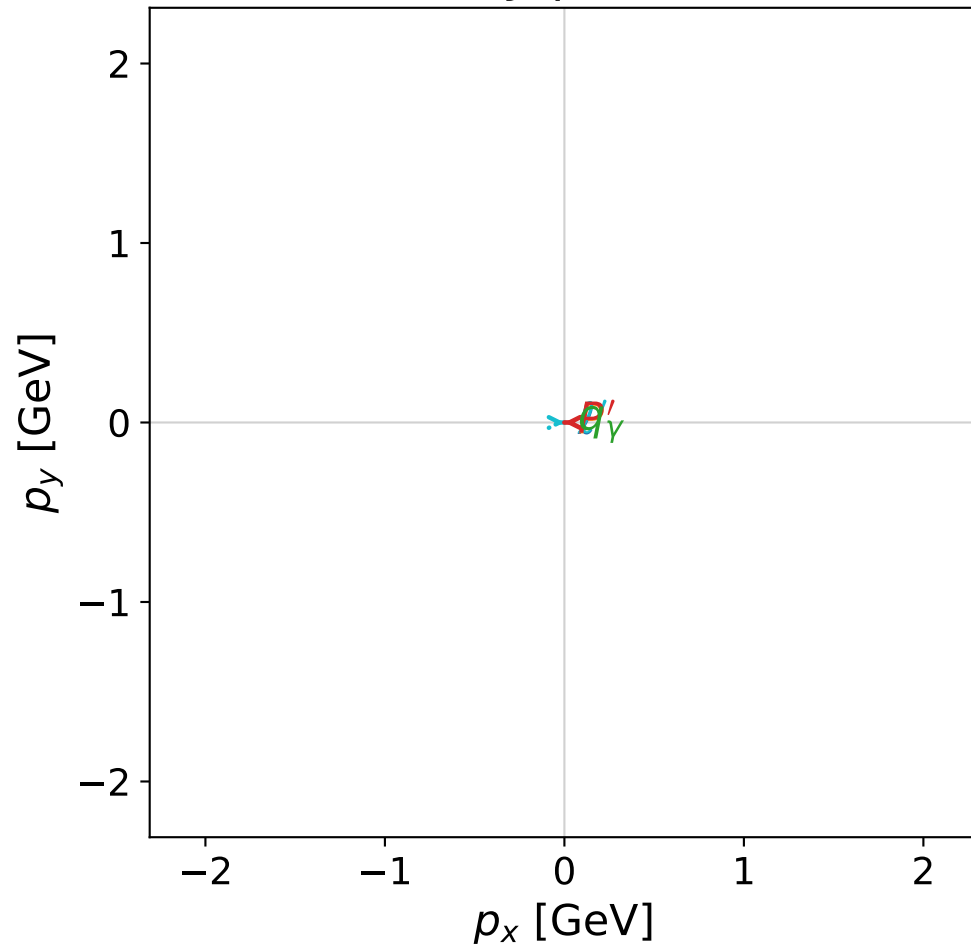


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



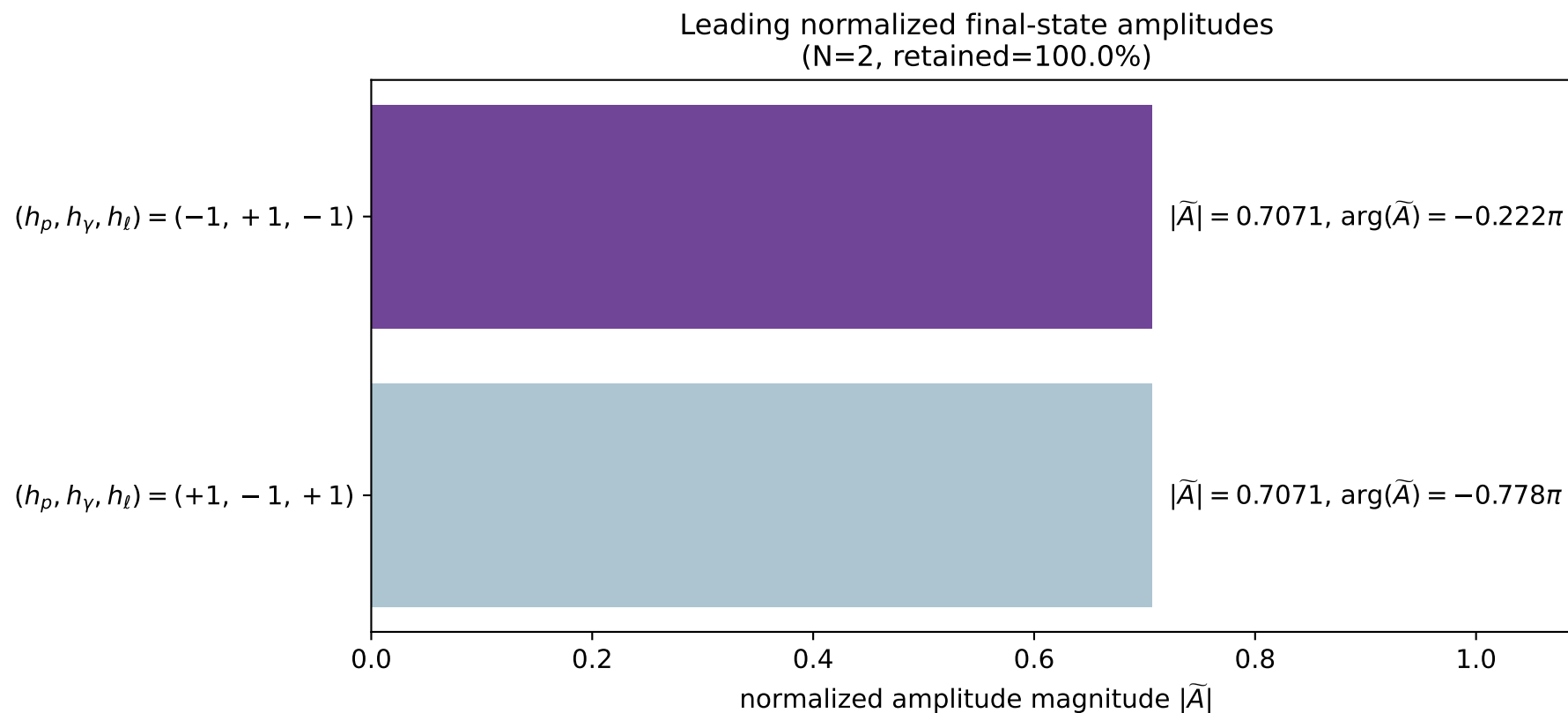
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_115
 $d_{\text{GHZ}}=4.30756\text{e-}08$
 region: polarization_cluster_01_local_minimum_115
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0115
 lepton mixing angle: $\alpha_e=0.76452518$ rad
 incoming lepton: $0.721711|+\rangle + 0.692194|-\rangle$
 proton mixing angle: $\alpha_p=0.76454044$ rad
 incoming proton: $0.721701|+\rangle + 0.692205|-\rangle$

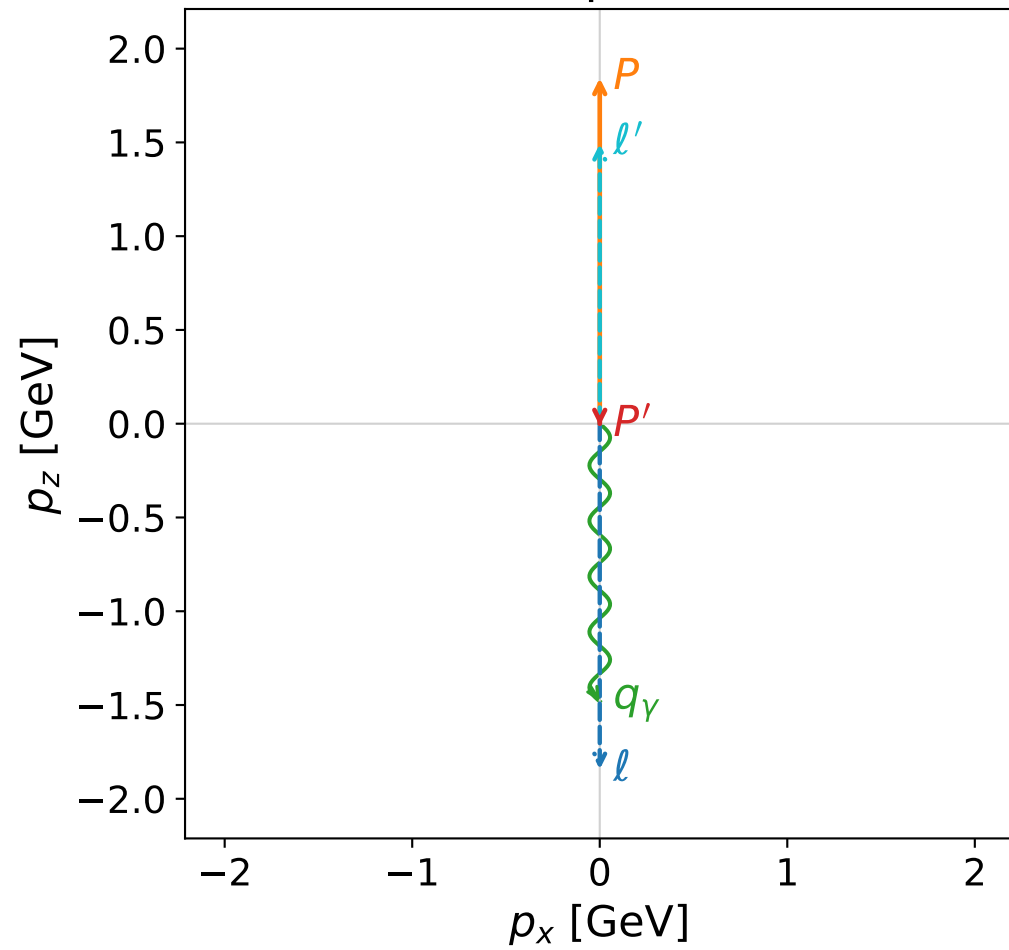
$s=16.5413$, $\sqrt{s}=4.0671$, $|\vec{P}|=1.92538$, $|\vec{P}'|=0.000570618$
 $\theta_{P'}=3.11934$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14223$, $\phi_\gamma=3.8405$
 $E_\gamma=1.56427$, $\theta_{\ell'}=8.11432\text{e-}06$, $\phi_{\ell'}=0.000639486$
 $Q^2=12.0516$, $x_B=0.804278$, $t=-2.26037$, $W^2=3.81262$, $y=0.95677$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9254$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9254$
 P $E= 2.1417$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9254$
 ℓ' $E= 1.5648$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5648$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0006$
 q_γ $E= 1.5643$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.5643$

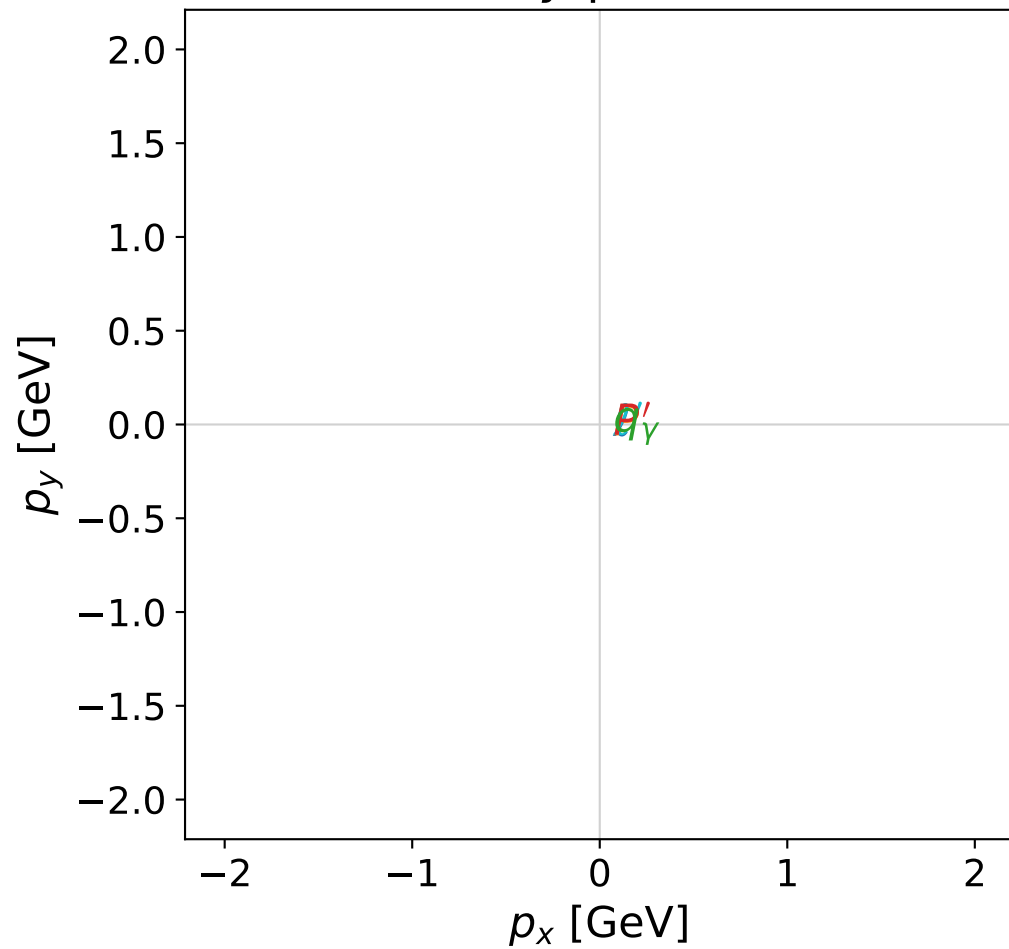


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



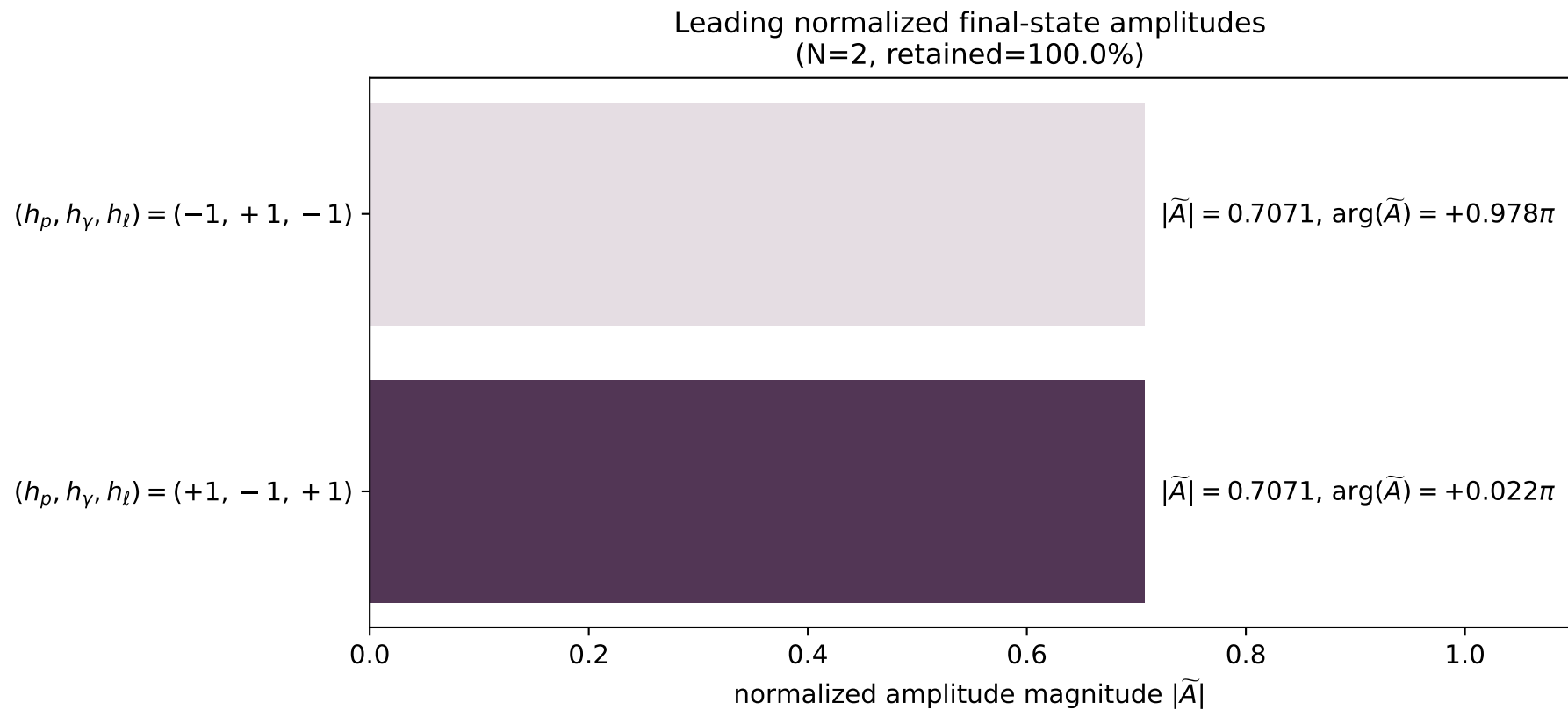
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_121
 $d_{\text{GHZ}}=4.75198\text{e-}08$
 region: polarization_cluster_01_local_minimum_121
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0121
 lepton mixing angle: $\alpha_e=0.80409735$ rad
 incoming lepton: $0.693762|+\rangle + 0.720205|-\rangle$
 proton mixing angle: $\alpha_p=0.80410559$ rad
 incoming proton: $0.693756|+\rangle + 0.72021|-\rangle$

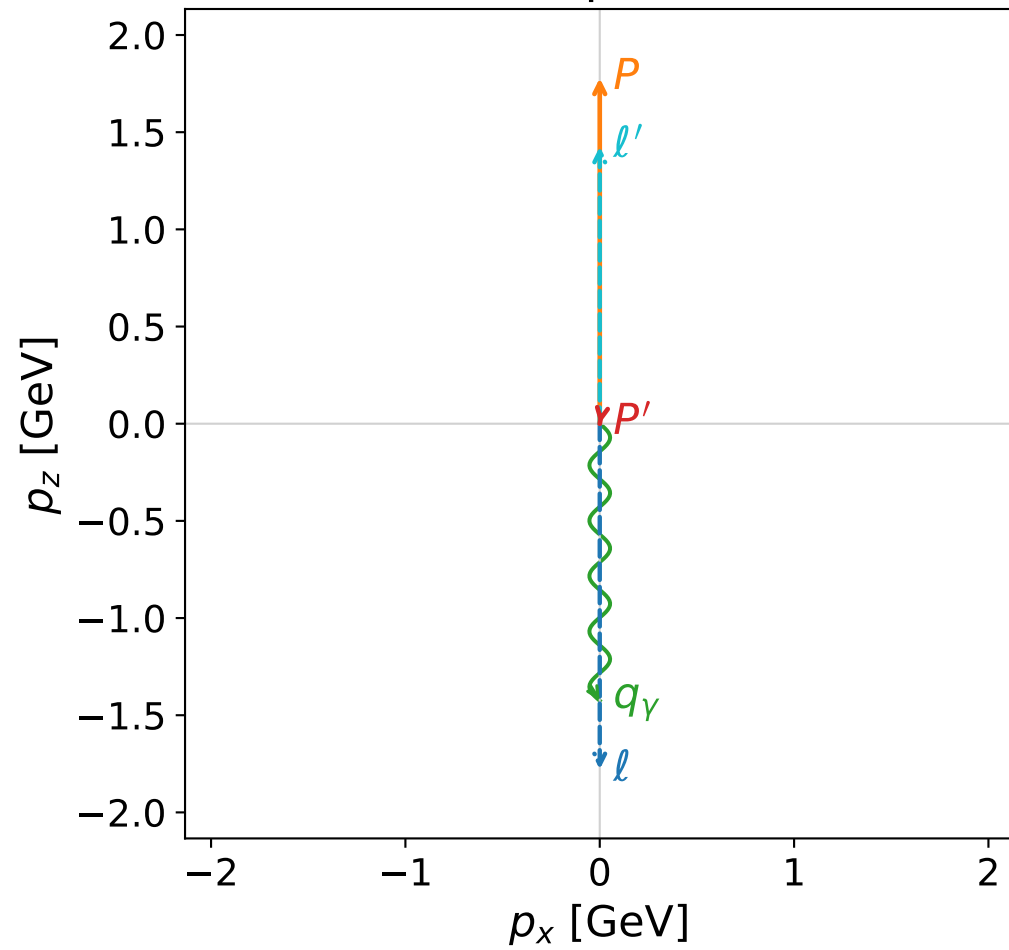
$s=15.2928$, $\sqrt{s}=3.9106$, $|\vec{P}|=1.8428$, $|\vec{P}'|=0.0114504$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.39183$, $\phi_\gamma=3.21079$
 $E_\gamma=1.48054$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.0705831$
 $Q^2=10.9978$, $x_B=0.800329$, $t=-2.16199$, $W^2=3.62364$, $y=0.953419$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8428$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8428$
 P $E= 2.0678$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8428$
 ℓ' $E= 1.4920$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4920$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0115$
 q_γ $E= 1.4805$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4805$

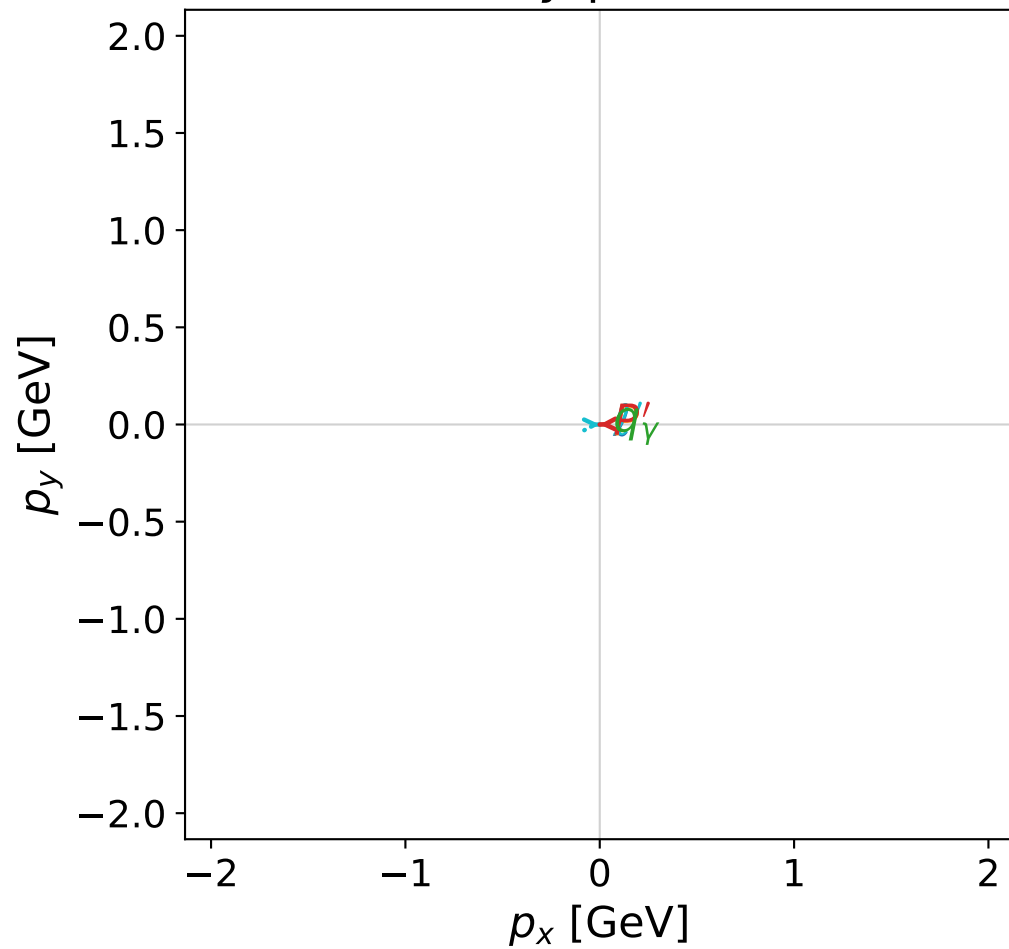


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



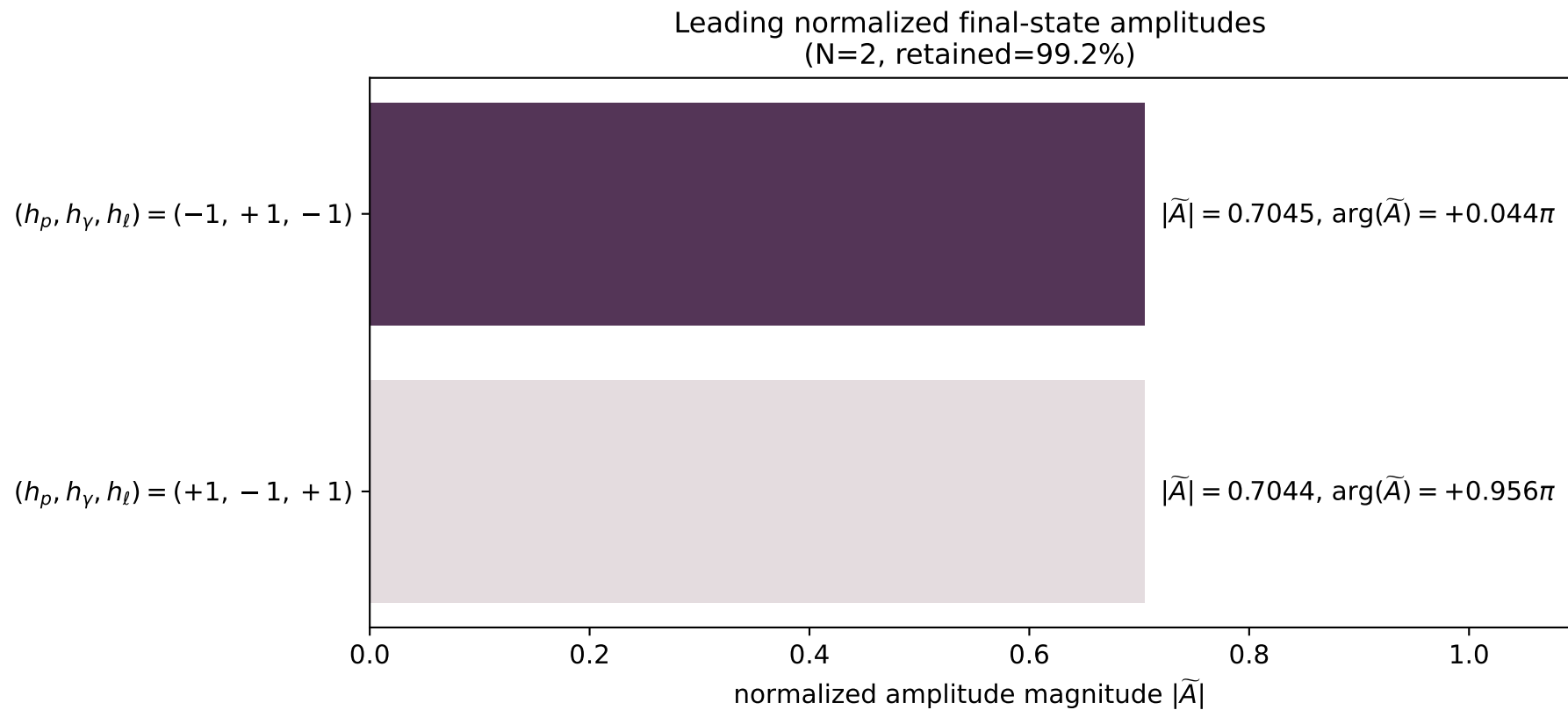
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_126
 $d_{\text{GHZ}}=4.95818\text{e-}08$
 region: polarization_cluster_01_local_minimum_126
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0126
 lepton mixing angle: $\alpha_e=0.79324484$ rad
 incoming lepton: $0.701537|+\rangle +0.712633|-\rangle$
 proton mixing angle: $\alpha_p=0.79327627$ rad
 incoming proton: $0.701514|+\rangle +0.712655|-\rangle$

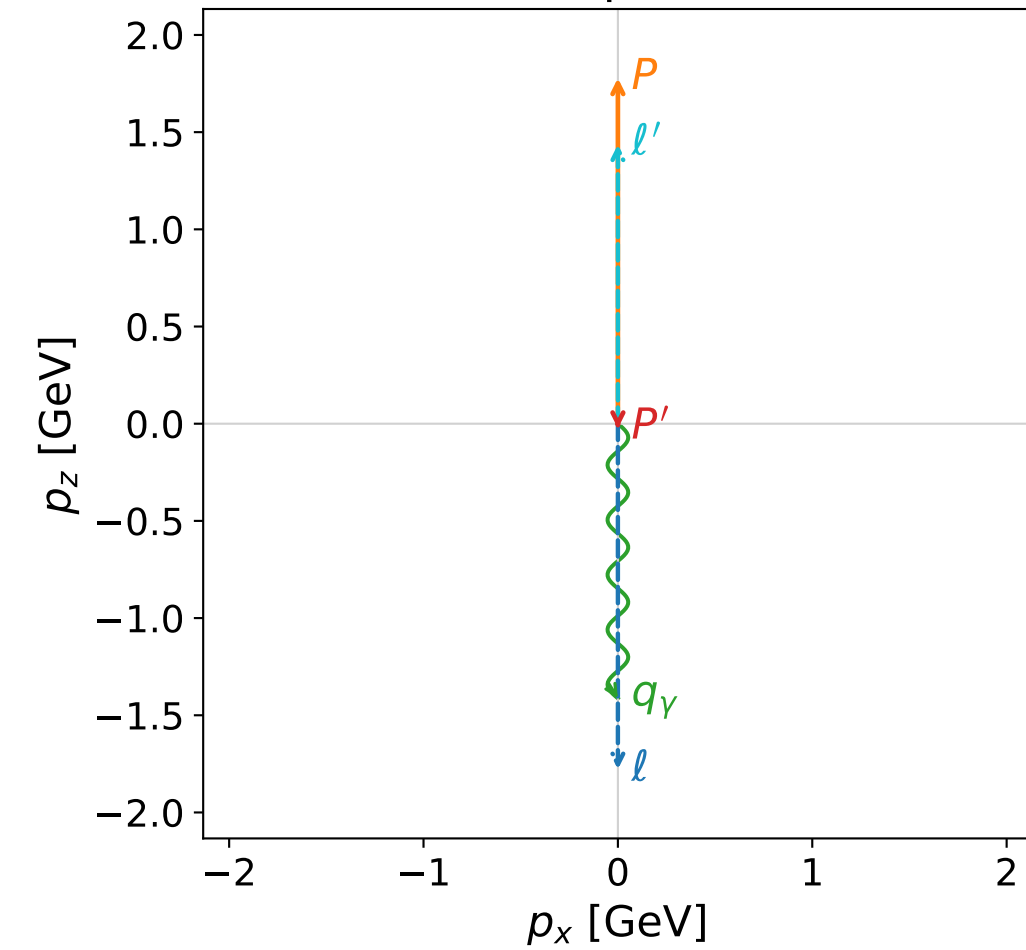
$s=14.3543$, $\sqrt{s}=3.78871$, $|\vec{P}|=1.77824$, $|\vec{P}'|=0.000403116$
 $\theta_{p'}=2.96794$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.16594$, $\phi_\gamma=3.0289$
 $E_\gamma=1.42516$, $\theta_{l'}=4.88583\text{e-}05$, $\phi_{l'}=0.0243458$
 $Q^2=10.1399$, $x_B=0.791416$, $t=-2.01337$, $W^2=3.55231$, $y=0.950862$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7782$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7782$
 P $E= 2.0105$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7782$
 l' $E= 1.4256$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.4256$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.0004$
 q_γ $E= 1.4252$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.4252$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

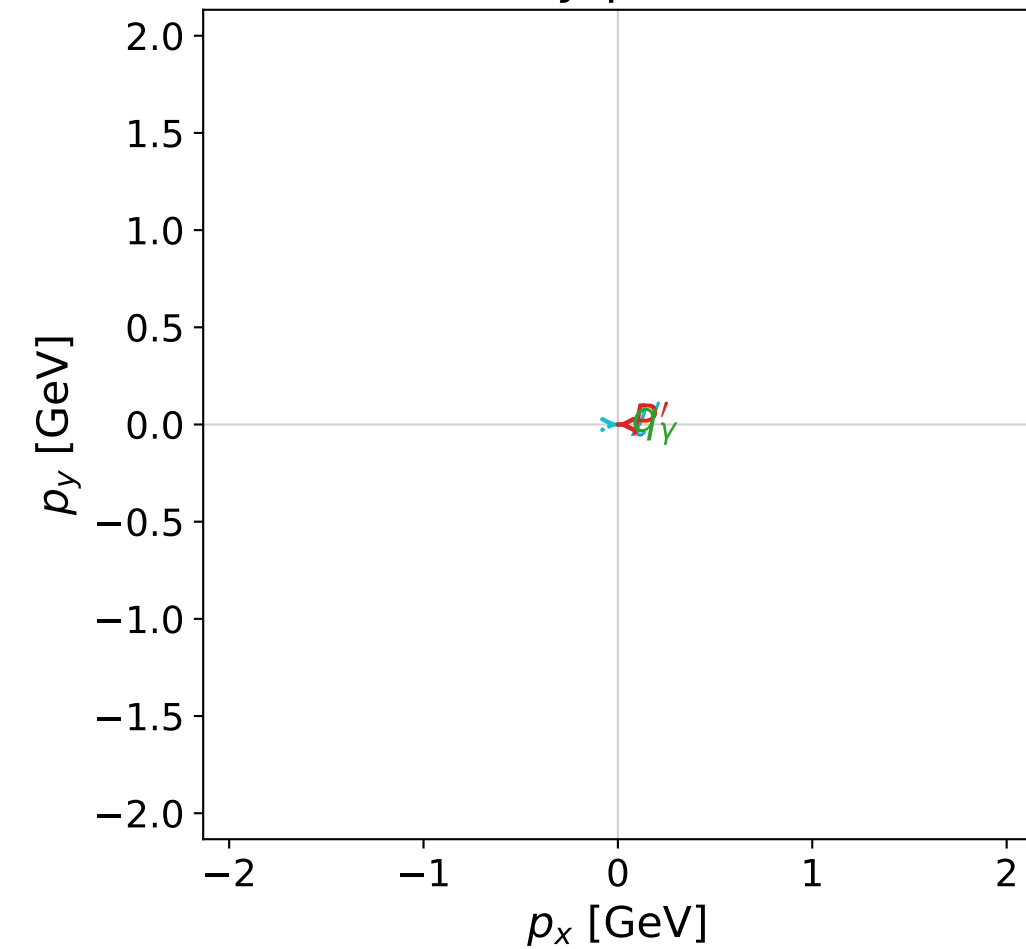


dghz_mixing_angles_polarization_01_local_minimum_129
 $d_{\text{GHZ}}=5.2277\text{e-}08$
 region: polarization_cluster_01_local_minimum_129
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0129
 lepton mixing angle: $\alpha_e=0.89040334$ rad
 incoming lepton: $0.629099|+\rangle + 0.777326|-\rangle$
 proton mixing angle: $\alpha_p=0.89039462$ rad
 incoming proton: $0.629105|+\rangle + 0.77732|-\rangle$

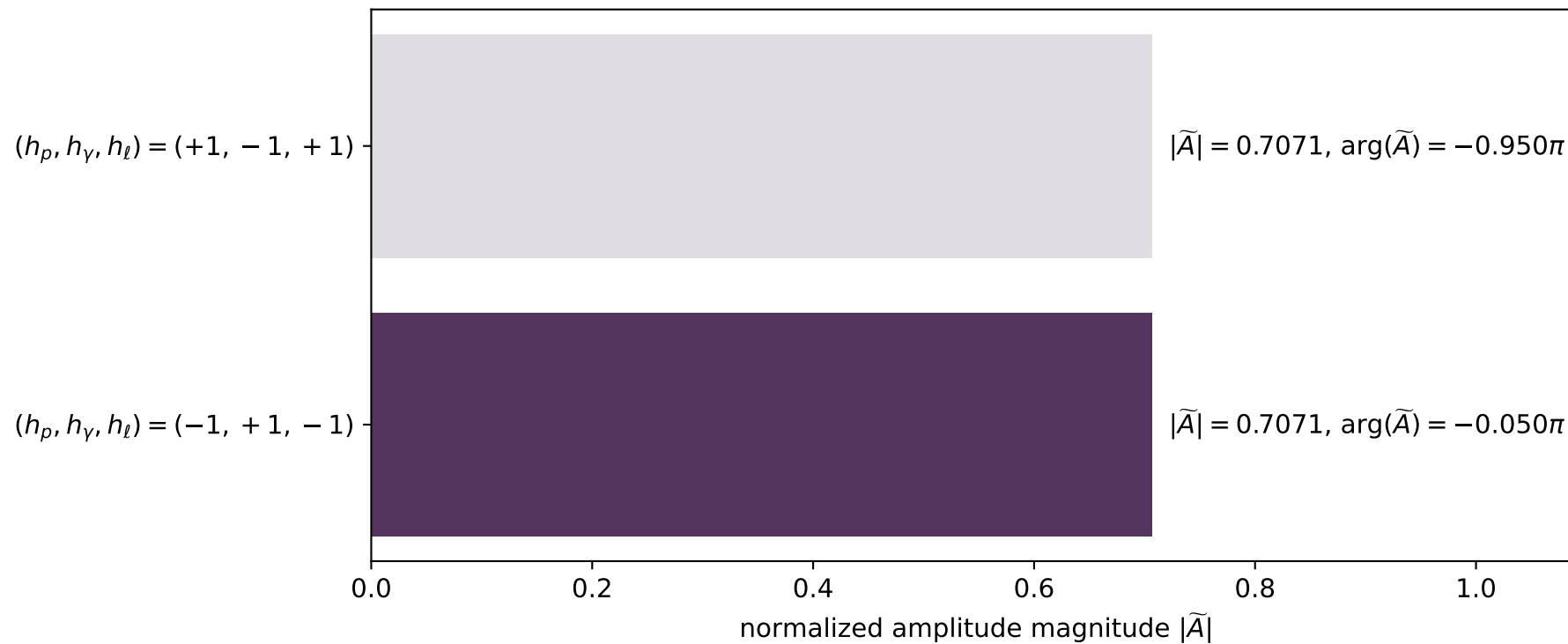
$s=14.3507$, $\sqrt{s}=3.78823$, $|\vec{P}|=1.77798$, $|\vec{P}'|=0.024118$
 $\theta_{P'}=3.13955$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14474$, $\phi_\gamma=3.30063$
 $E_\gamma=1.4129$, $\theta_{\ell'}=3.42918\text{e-}05$, $\phi_{\ell'}=0.00314395$
 $Q^2=10.22$, $x_B=0.79823$, $t=-2.09854$, $W^2=3.46317$, $y=0.950447$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7780$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7780$
 P $E= 2.0102$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7780$
 ℓ' $E= 1.4370$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4370$
 P' $E= 0.9383$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0241$
 q_γ $E= 1.4129$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4129$

x--y plane

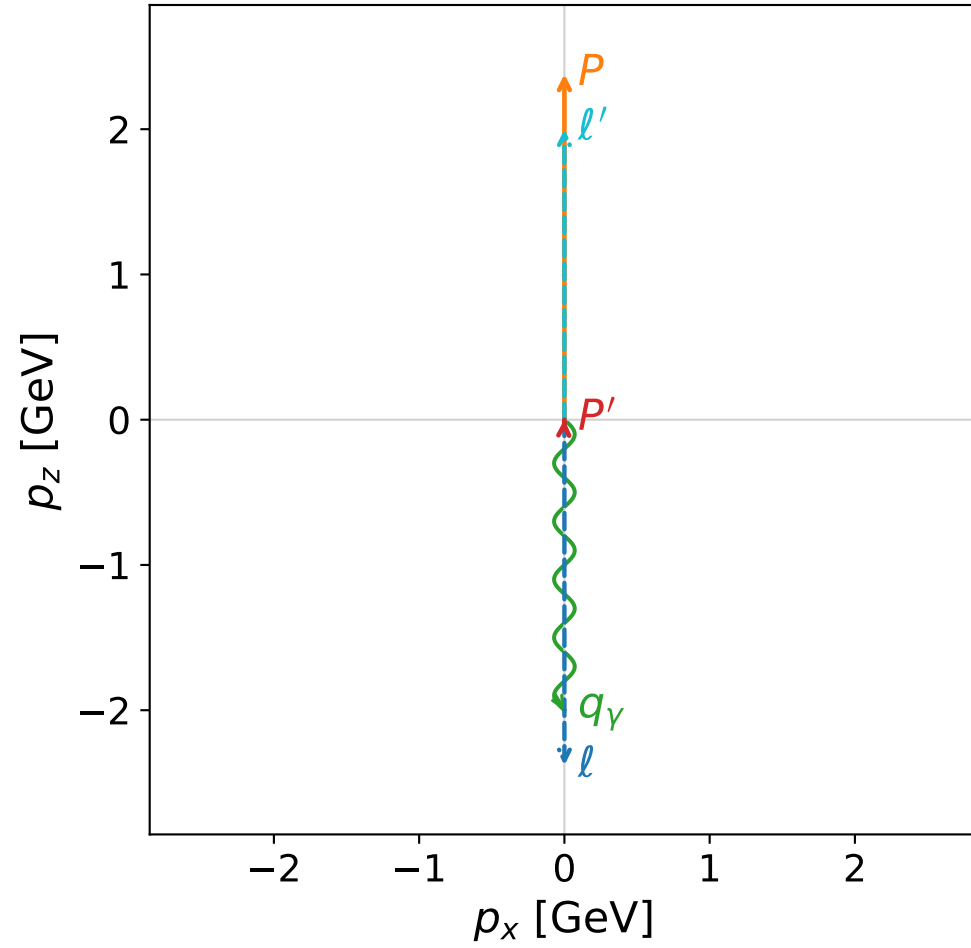


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

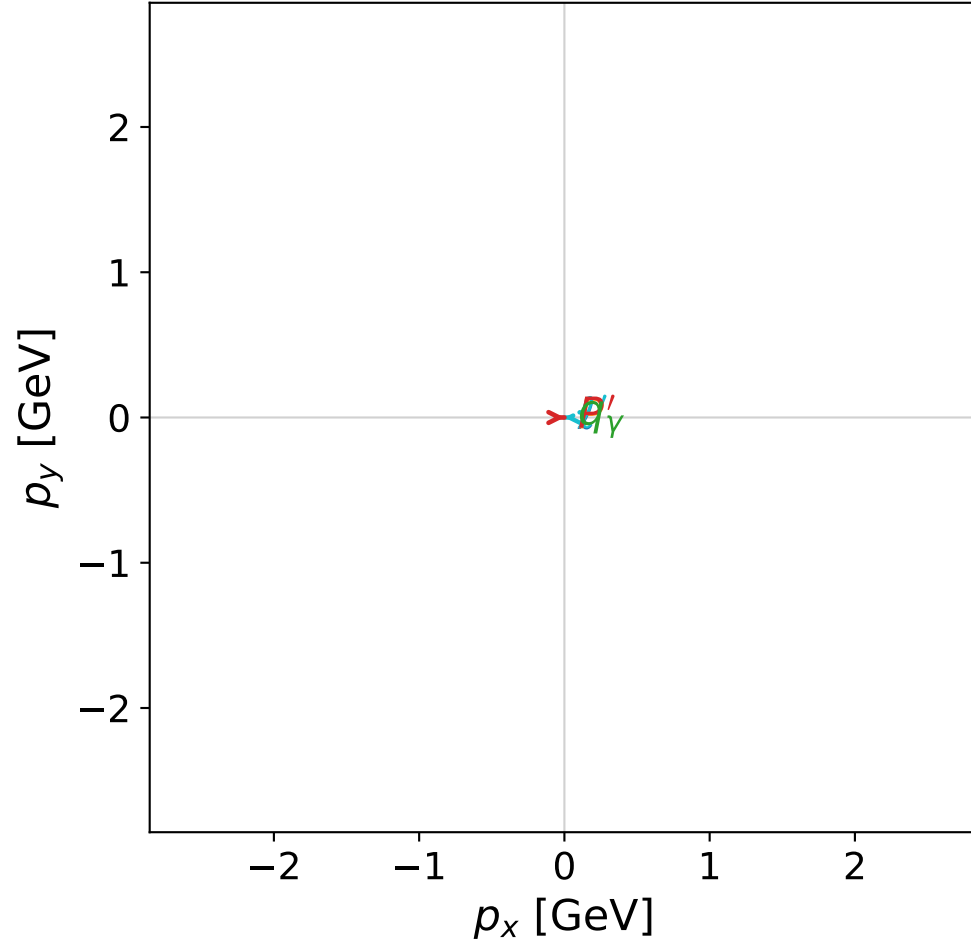


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



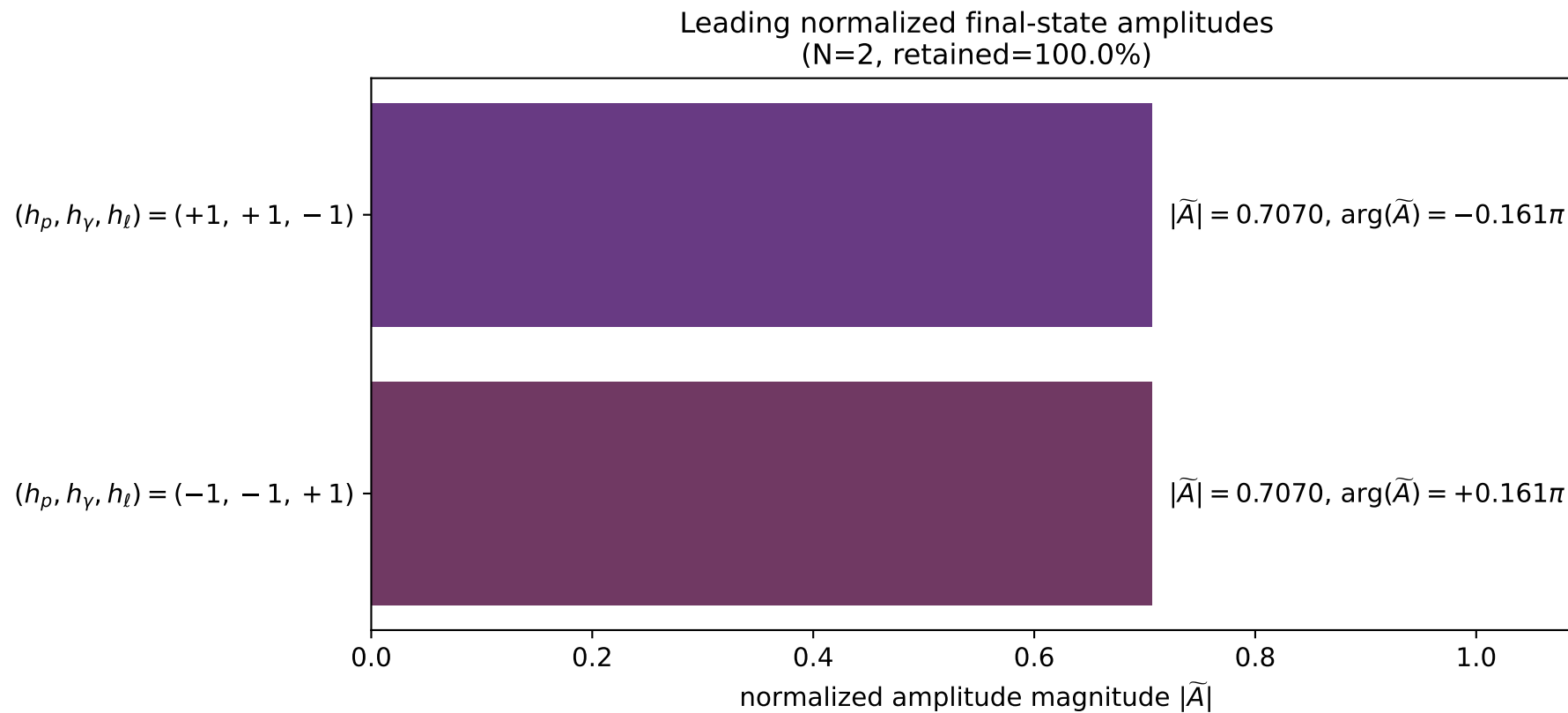
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_135
 $d_{\text{GHZ}}=5.61544\text{e-}08$
 region: polarization_cluster_01_local_minimum_135
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0135
 lepton mixing angle: $\alpha_e=1.3218649$ rad
 incoming lepton: $0.246369|+\rangle + 0.969176|-\rangle$
 proton mixing angle: $\alpha_p=1.3218678$ rad
 incoming proton: $0.246366|+\rangle + 0.969177|-\rangle$

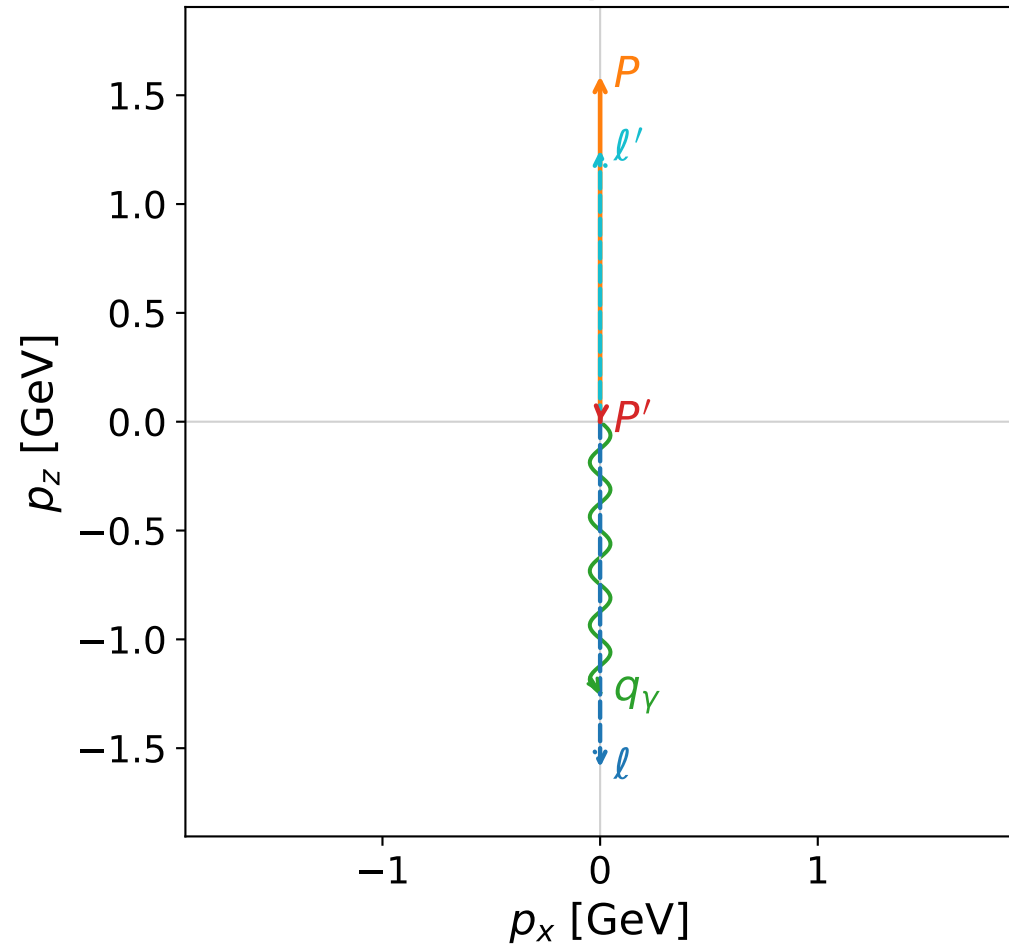
$s=24.3709$, $\sqrt{s}=4.93669$, $|\vec{P}|=2.37923$, $|\vec{P}'|=0.000777917$
 $\theta_{P'}=0.0397154$, $\theta_Y=3.14159$, $\phi_{P'}=0.000309928$, $\phi_Y=3.64684$
 $E_Y=1.99973$, $\theta_{\ell'}=1.54516\text{e-}05$, $\phi_{\ell'}=3.1419$
 $Q^2=19.0239$, $x_B=0.835169$, $t=-3.0344$, $W^2=4.63445$, $y=0.969668$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3792$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3792$
 P $E= 2.5575$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3792$
 ℓ' $E= 1.9990$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9990$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0008$
 q_Y $E= 1.9997$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9997$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



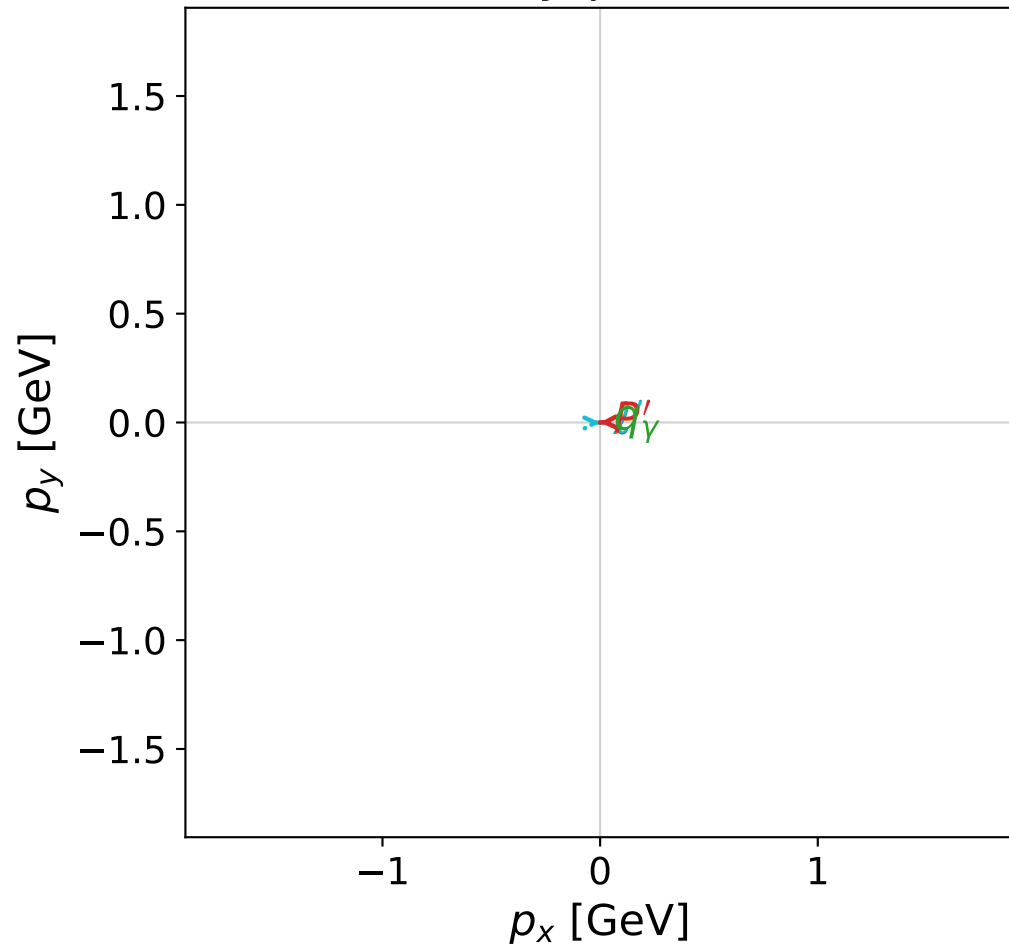
dghz_mixing_angles_polarization_01_local_minimum_138
 $d_{\text{GHZ}}=6.56334\text{e-}08$
 region: polarization_cluster_01_local_minimum_138
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0138
 lepton mixing angle: $\alpha_e=0.83474474$ rad
 incoming lepton: $0.671367|+\rangle + 0.741125|-\rangle$
 proton mixing angle: $\alpha_p=0.83473256$ rad
 incoming proton: $0.671376|+\rangle + 0.741117|-\rangle$

$s=11.7789$, $\sqrt{s}=3.43203$, $|\vec{P}|=1.58784$, $|\vec{P}'|=0.00100941$
 $\theta_{P'}=3.06084$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.17093$, $\phi_\gamma=2.78545$
 $E_\gamma=1.24651$, $\theta_{\ell'}=6.52663\text{e-}05$, $\phi_{\ell'}=0.0293406$
 $Q^2=7.92343$, $x_B=0.772311$, $t=-1.70322$, $W^2=3.2158$, $y=0.941313$

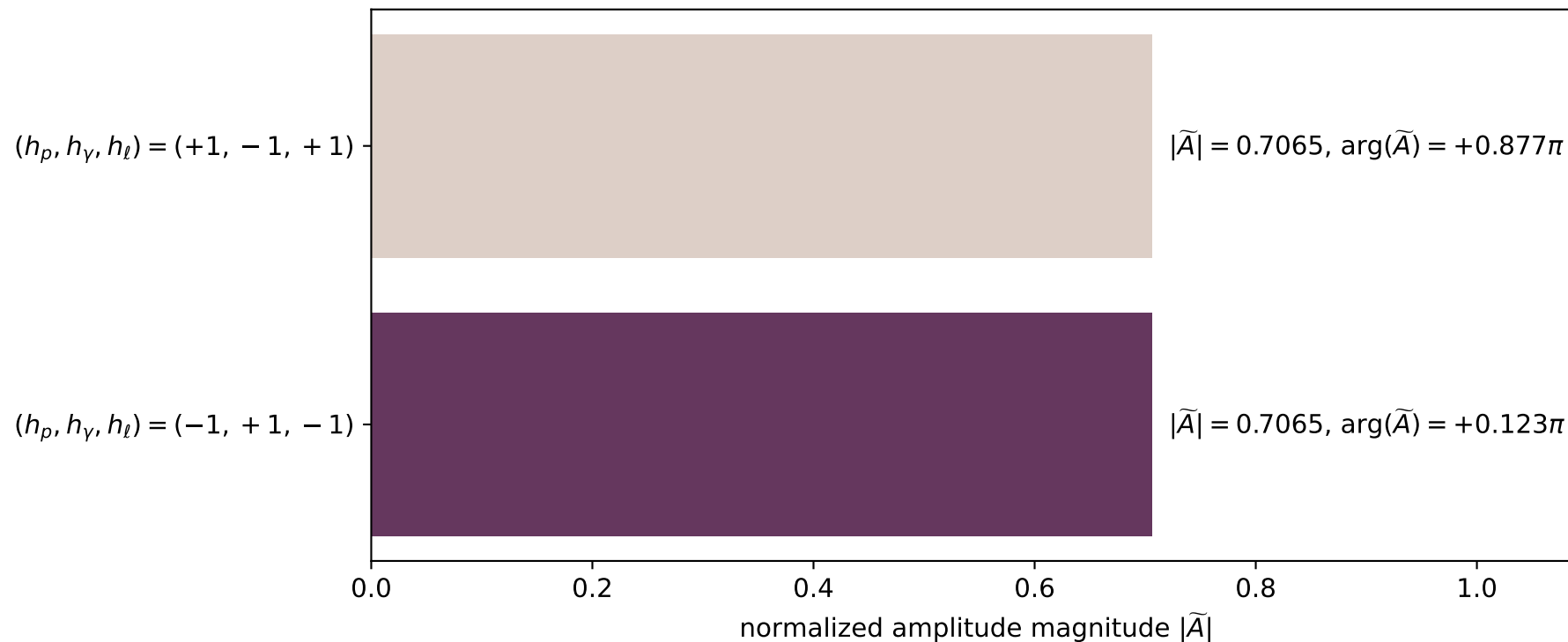
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	1.5878	0.0000	0.0000	-1.5878
P	1.8442	0.0000	0.0000	1.5878
ℓ'	1.2475	0.0001	0.0000	1.2475
P'	0.9380	-0.0001	-0.0000	-0.0010
q_γ	1.2465	-0.0000	0.0000	-1.2465

x--y plane

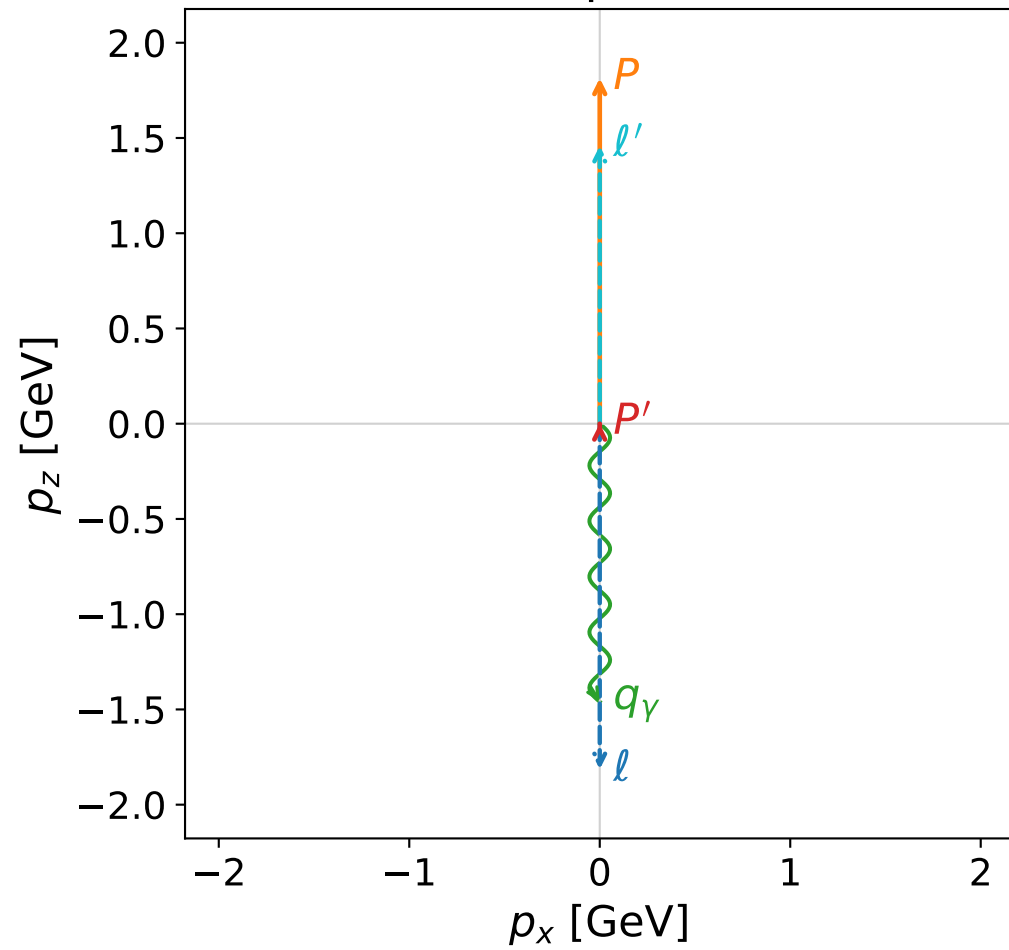


Leading normalized final-state amplitudes
(N=2, retained=99.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

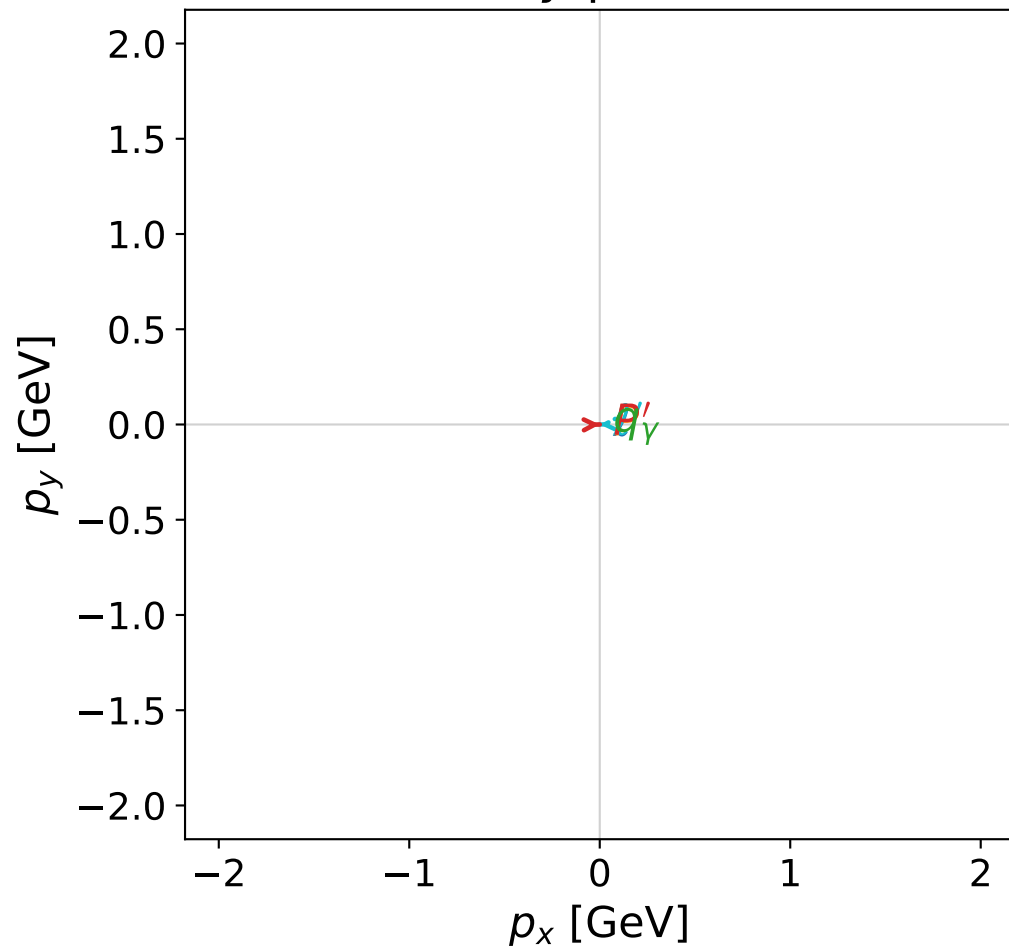


dghz_mixing_angles_polarization_01_local_minimum_139
 $d_{\text{GHZ}}=6.7347\text{e-}08$
 region: polarization_cluster_01_local_minimum_139
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0139
 lepton mixing angle: $\alpha_e=0.40504113$ rad
 incoming lepton: $0.919086|+\rangle + 0.394057|-\rangle$
 proton mixing angle: $\alpha_p=0.40503483$ rad
 incoming proton: $0.919089|+\rangle + 0.394051|-\rangle$

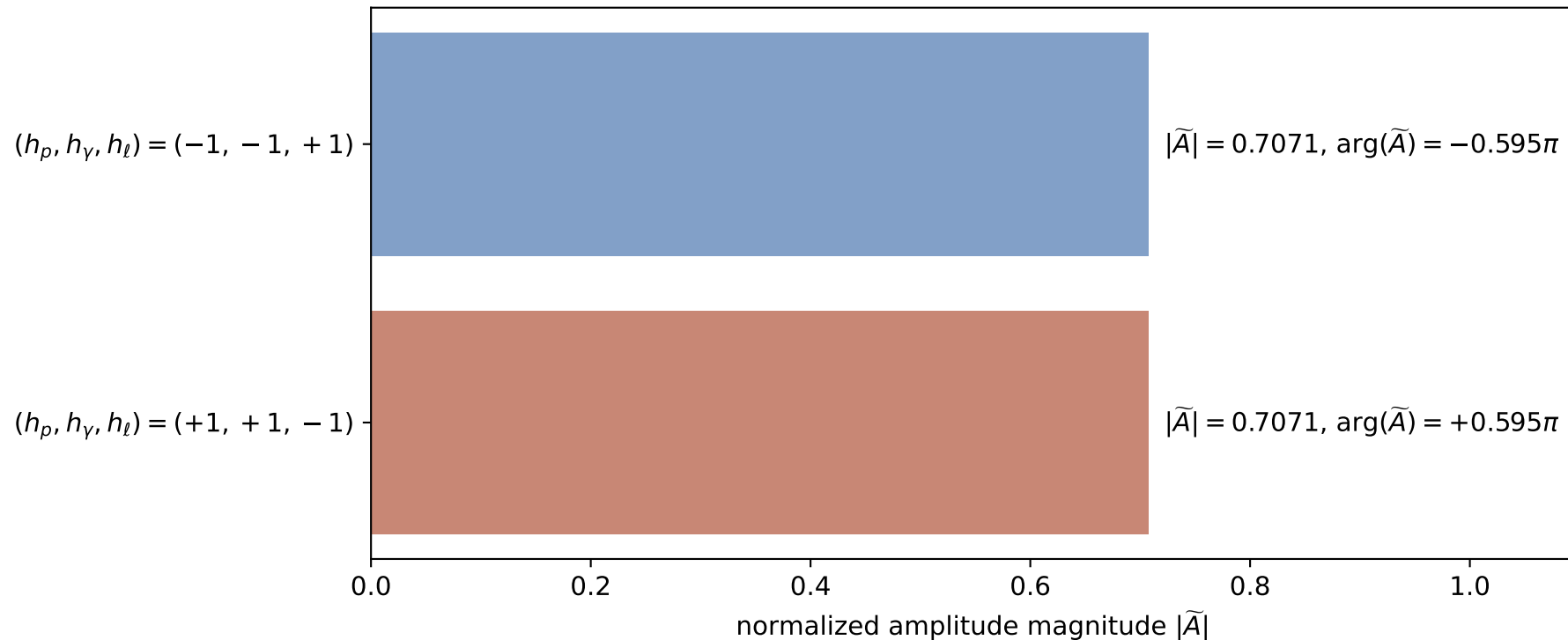
$s=14.8742$, $\sqrt{s}=3.85671$, $|\vec{P}|=1.81429$, $|\vec{P}'|=0.000584182$
 $\theta_{P'}=0.00490874$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0235349$, $\phi_\gamma=1.27295$
 $E_\gamma=1.45965$, $\theta_{\ell'}=1.96537\text{e-}06$, $\phi_{\ell'}=3.16513$
 $Q^2=10.5886$, $x_B=0.794427$, $t=-2.06977$, $W^2=3.61985$, $y=0.952429$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8143$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8143$
 P $E= 2.0424$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8143$
 ℓ' $E= 1.4591$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4591$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0006$
 q_γ $E= 1.4596$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4596$

x--y plane

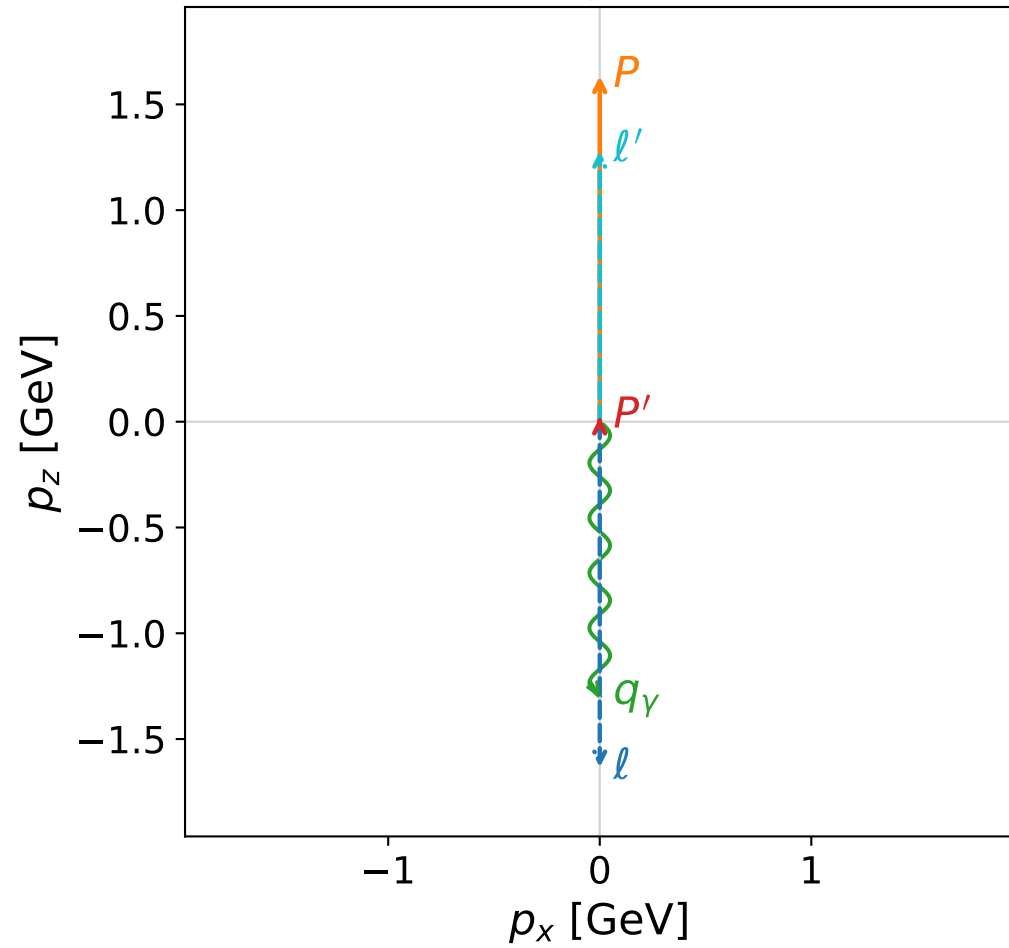


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

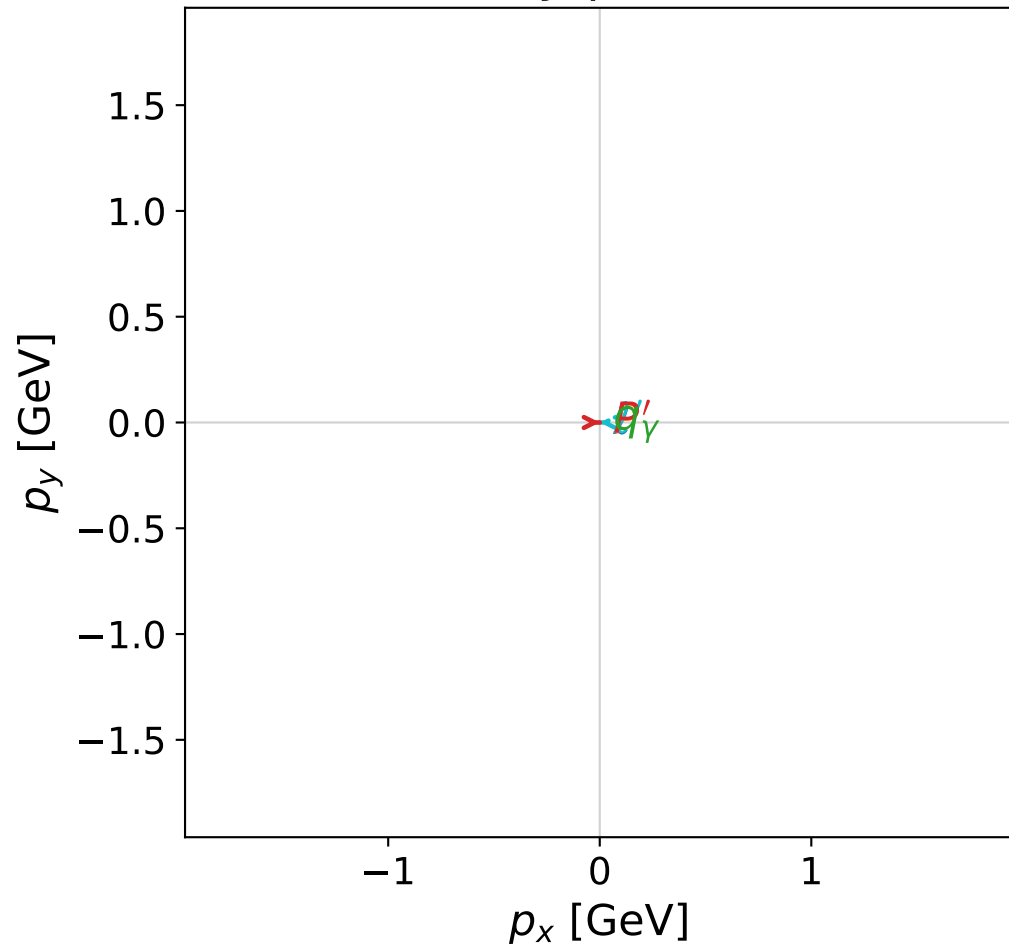


dghz_mixing_angles_polarization_01_local_minimum_140
 $d_{\text{GHZ}}=7.0516\text{e-}08$
 region: polarization_cluster_01_local_minimum_140
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0140
 lepton mixing angle: $\alpha_e=0.57080929$ rad
 incoming lepton: $0.841464|+\rangle + 0.540313|-\rangle$
 proton mixing angle: $\alpha_p=0.57079935$ rad
 incoming proton: $0.841469|+\rangle + 0.540305|-\rangle$

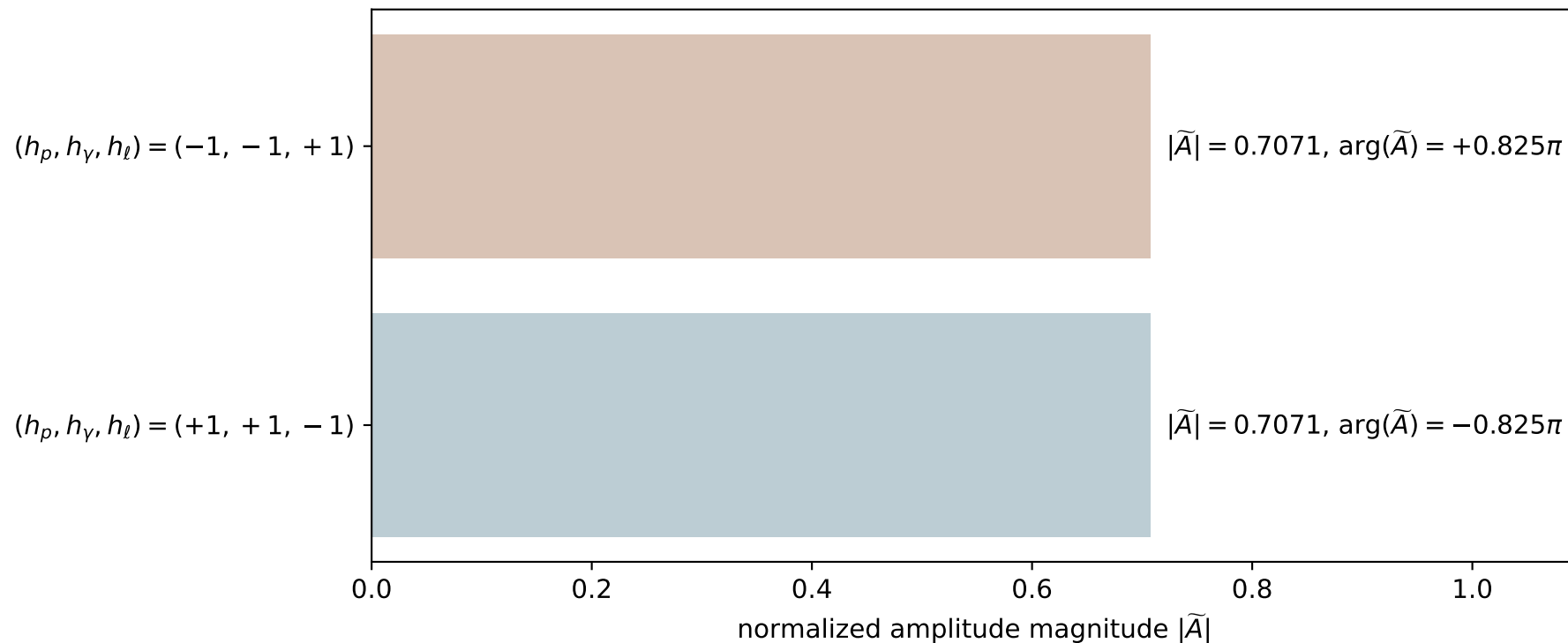
$s=12.3715$, $\sqrt{s}=3.51731$, $|\vec{P}|=1.63358$, $|\vec{P}'|=0.0205126$
 $\theta_{P'}=0.00635735$, $\theta_Y=3.14159$, $\phi_{P'}=6.28239$, $\phi_Y=5.7342$
 $E_Y=1.2998$, $\theta_{l'}=0.000101936$, $\phi_{l'}=3.1408$
 $Q^2=8.35928$, $x_B=0.770326$, $t=-1.70802$, $W^2=3.37217$, $y=0.944306$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.6336$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6336$
 P $E= 1.8837$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6336$
 l' $E= 1.2793$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 1.2793$
 P' $E= 0.9382$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0205$
 q_Y $E= 1.2998$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.2998$

x--y plane

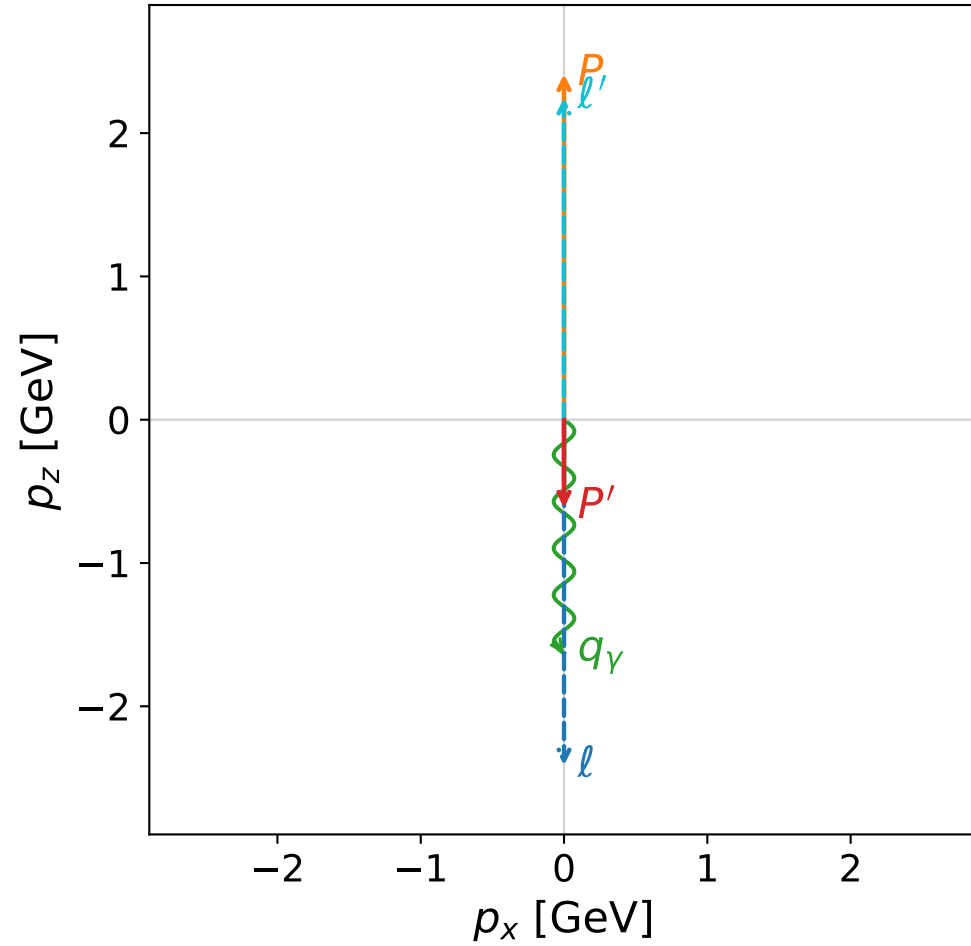


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

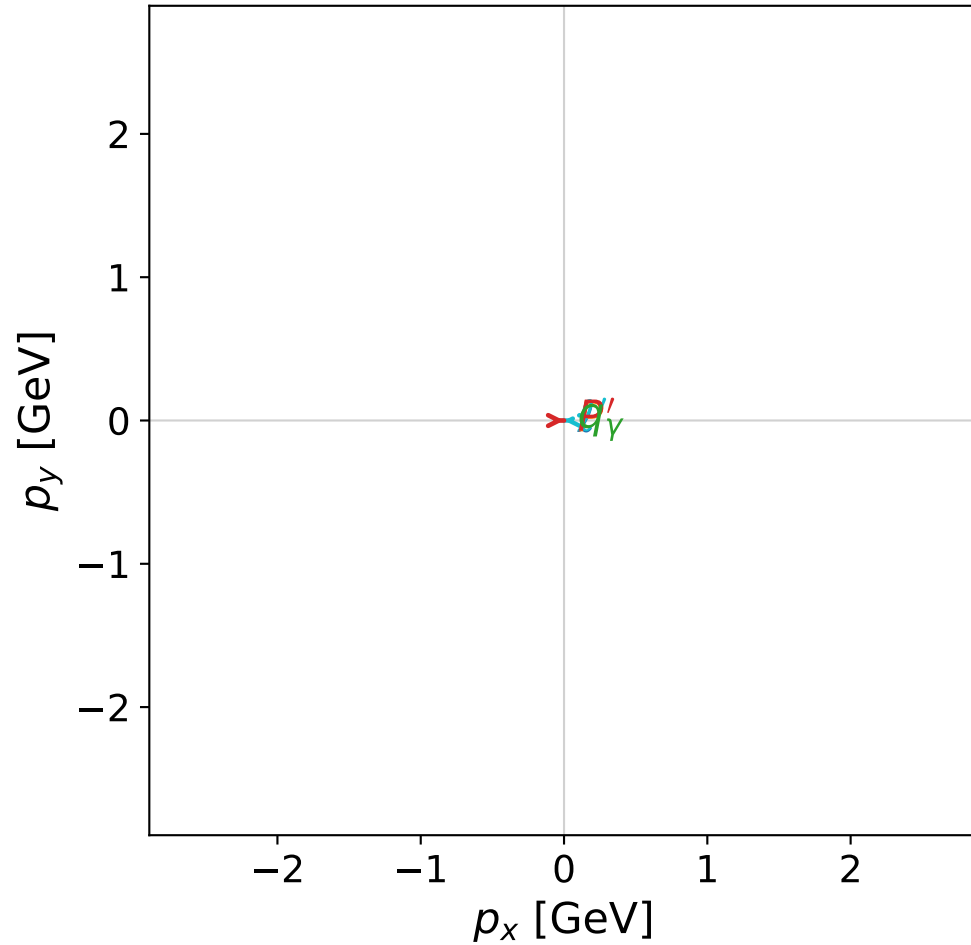


dghz_mixing_angles_polarization_01_local_minimum_146
 $d_{\{\text{rm GHZ}\}}=7.9425\text{e-}08$
 region: polarization_cluster_01_local_minimum_146
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0146
 lepton mixing angle: $\alpha_e=0.15440538$ rad
 incoming lepton: $0.988103|+\rangle + 0.153793|-\rangle$
 proton mixing angle: $\alpha_p=0.15439576$ rad
 incoming proton: $0.988105|+\rangle + 0.153783|-\rangle$

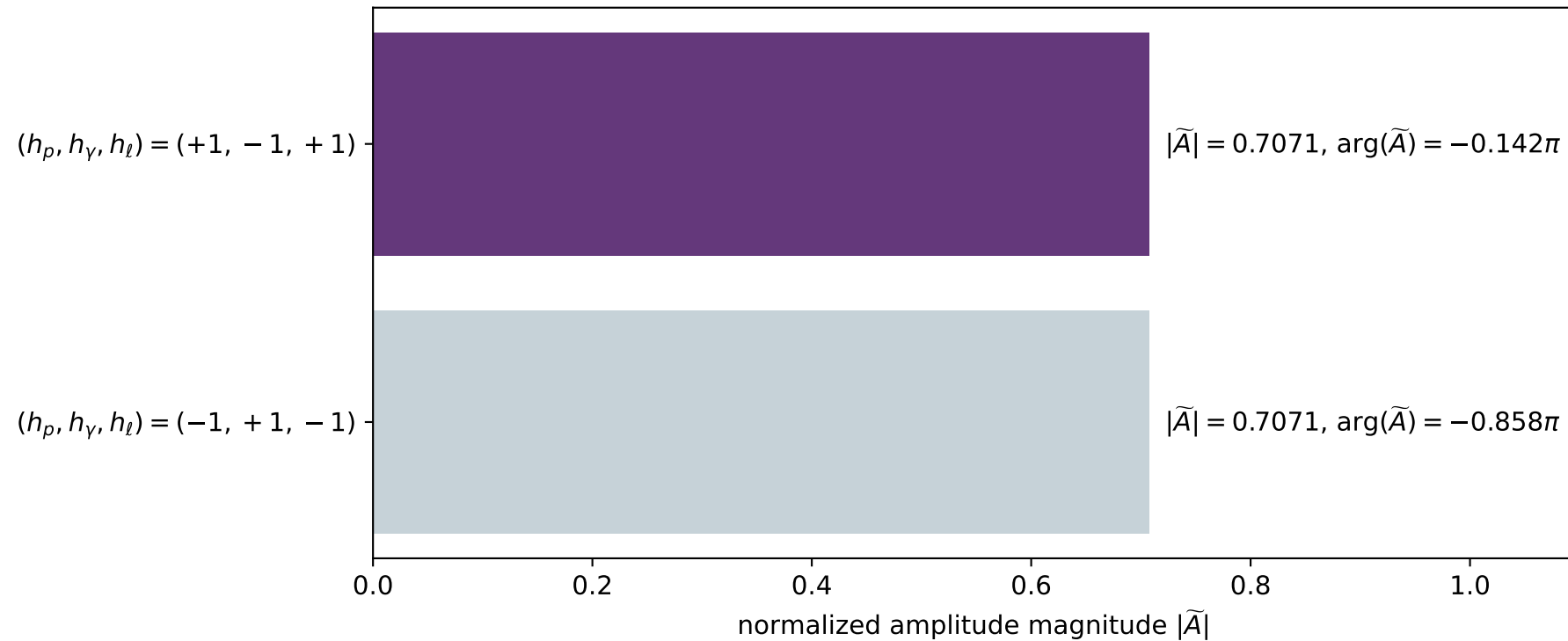
$s=24.9939$, $\sqrt{s}=4.99939$, $|\vec{P}|=2.4117$, $|\vec{P}'|=0.615551$
 $\theta_{p'}=3.14157$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.000416712$, $\phi_\gamma=2.69483$
 $E_\gamma=1.63095$, $\theta_{l'}=7.35473\text{e-}06$, $\phi_{l'}=3.14201$
 $Q^2=21.6715$, $x_B=0.929179$, $t=-7.01582$, $W^2=2.53163$, $y=0.967209$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4117$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4117$
 P $E= 2.5877$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4117$
 l' $E= 2.2465$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2465$
 P' $E= 1.1219$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6156$
 q_γ $E= 1.6309$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.6309$

x--y plane

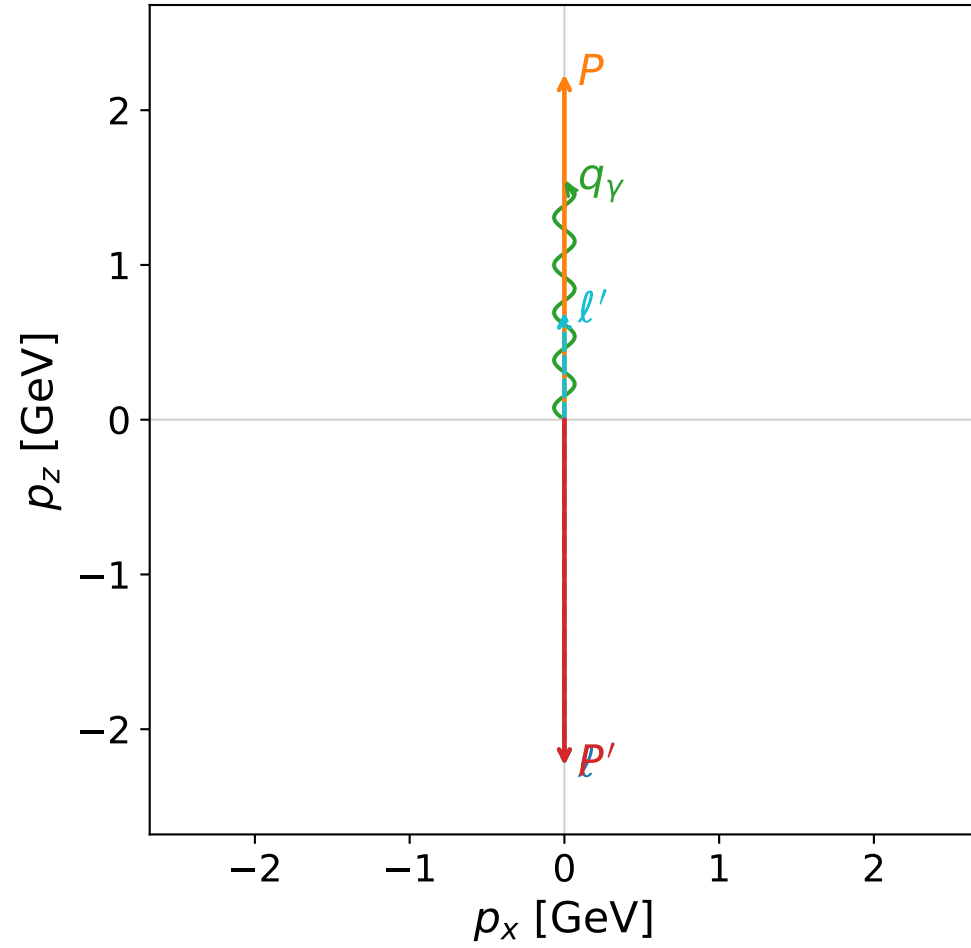


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

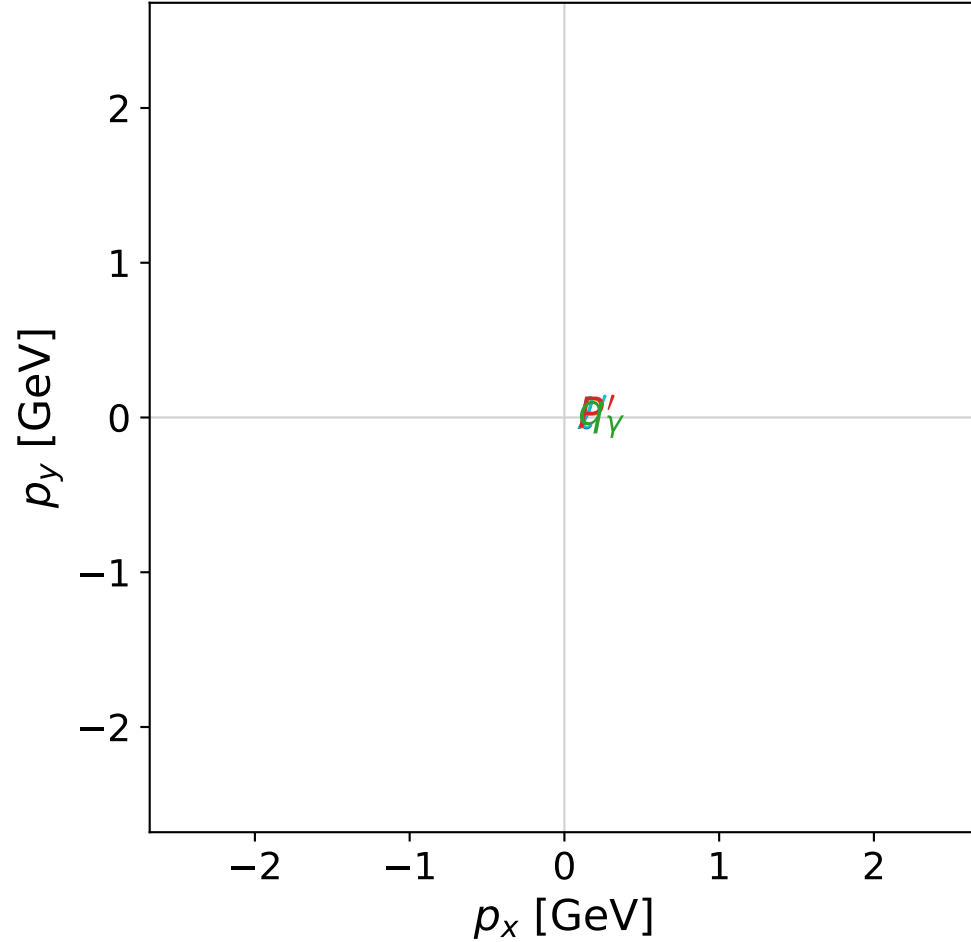


dghz_mixing_angles_polarization_01_local_minimum_149
 $d_{\text{GHZ}}=8.40053\text{e-}08$
 region: polarization_cluster_01_local_minimum_149
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0149
 lepton mixing angle: $\alpha_e=0.79062427$ rad
 incoming lepton: $0.703402|+\rangle + 0.710793|-\rangle$
 proton mixing angle: $\alpha_p=0.78016465$ rad
 incoming proton: $0.710798|+\rangle + 0.703396|-\rangle$

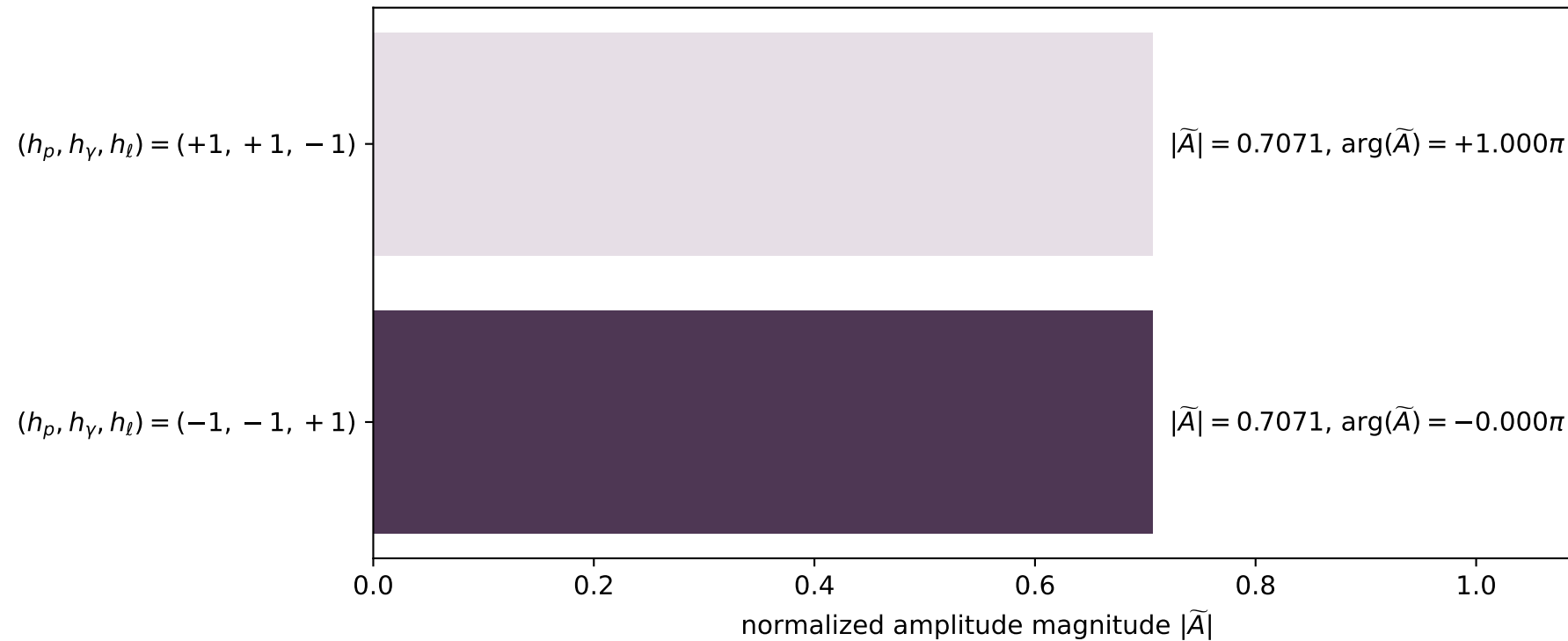
$s=21.6733$, $\sqrt{s}=4.65546$, $|\vec{P}|=2.23324$, $|\vec{P}'|=2.23324$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.62947$, $\phi_\gamma=0.0541682$
 $E_\gamma=1.53726$, $\theta_{l'}=0$, $\phi_{l'}=2.48787$
 $Q^2=6.2171$, $x_B=0.302824$, $t=-19.9494$, $W^2=15.1932$, $y=0.987349$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2332$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2332$
 P $E= 2.4222$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2332$
 l' $E= 0.6960$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6960$
 P' $E= 2.4222$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.2332$
 q_γ $E= 1.5373$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5373$

x--y plane

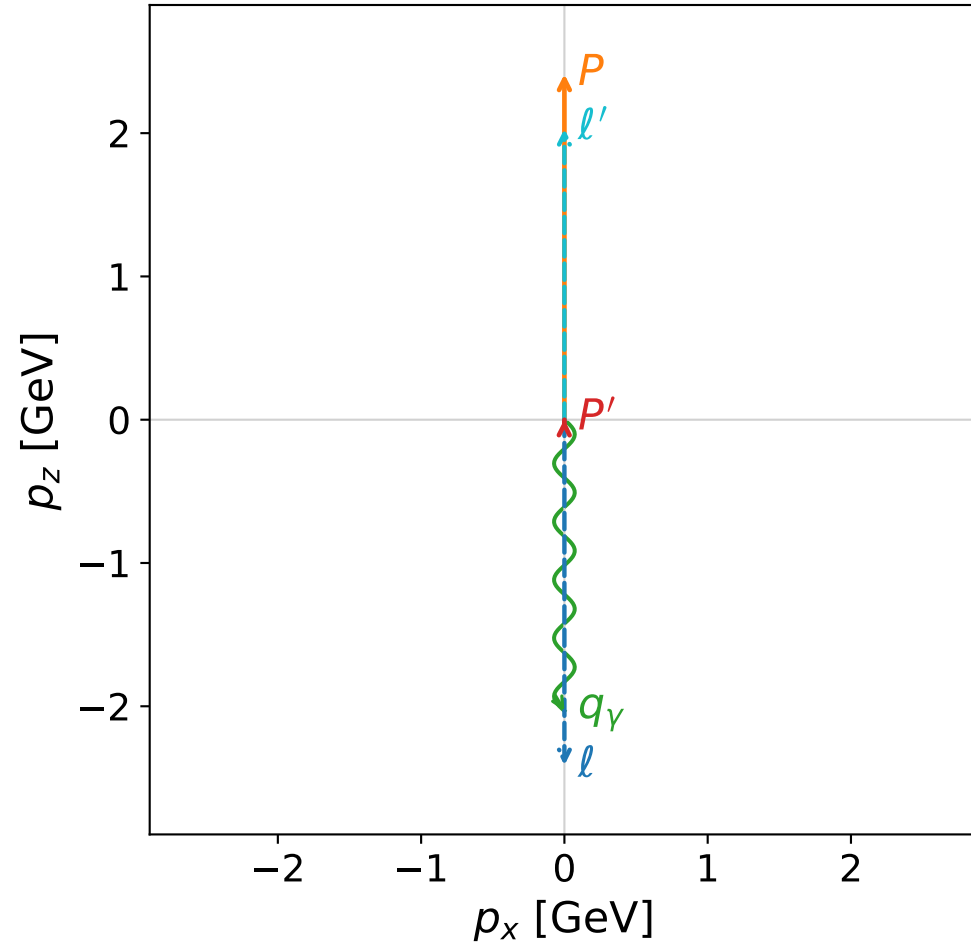


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

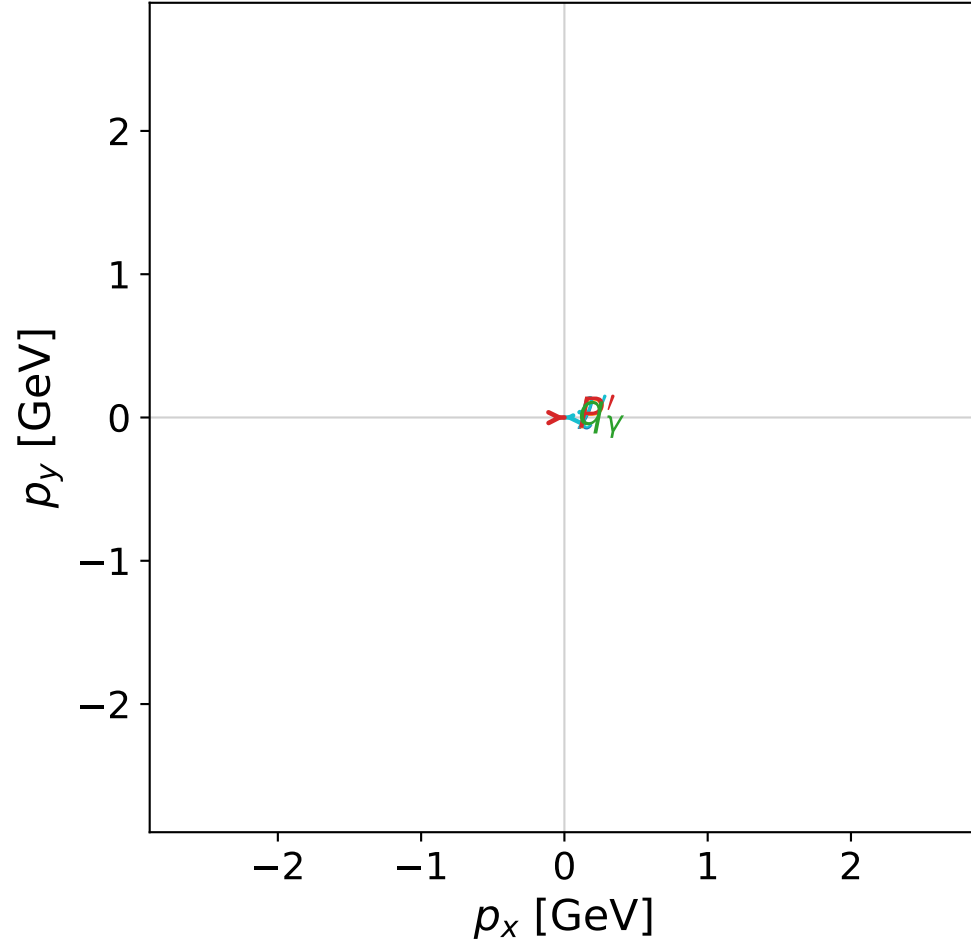


dghz_mixing_angles_polarization_01_local_minimum_151
 $d_{\text{GHZ}}=8.54757\text{e-}08$
 region: polarization_cluster_01_local_minimum_151
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0151
 lepton mixing angle: $\alpha_e=0.15707963$ rad
 incoming lepton: $0.987688|+\rangle + 0.156434|-\rangle$
 proton mixing angle: $\alpha_p=0.15707963$ rad
 incoming proton: $0.987688|+\rangle + 0.156434|-\rangle$

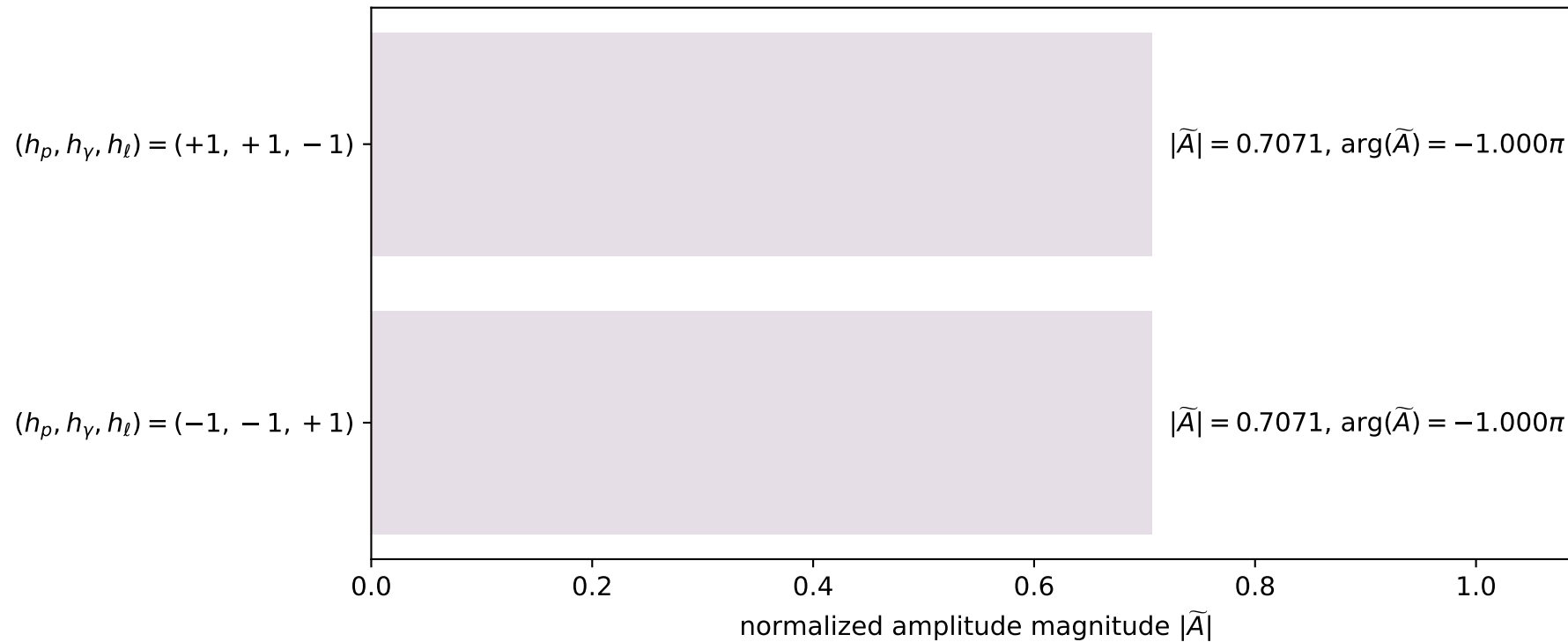
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00283891$
 $\theta_{P'}=0.00981748$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.00981748$, $\phi_\gamma=0$
 $E_\gamma=2.03242$, $\theta_{\ell'}=1.37321\text{e-}05$, $\phi_{\ell'}=3.15141$
 $Q^2=19.5815$, $x_B=0.836606$, $t=-3.0817$, $W^2=4.70422$, $y=0.970386$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0296$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0296$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0028$
 q_γ $E= 2.0324$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0324$

x--y plane

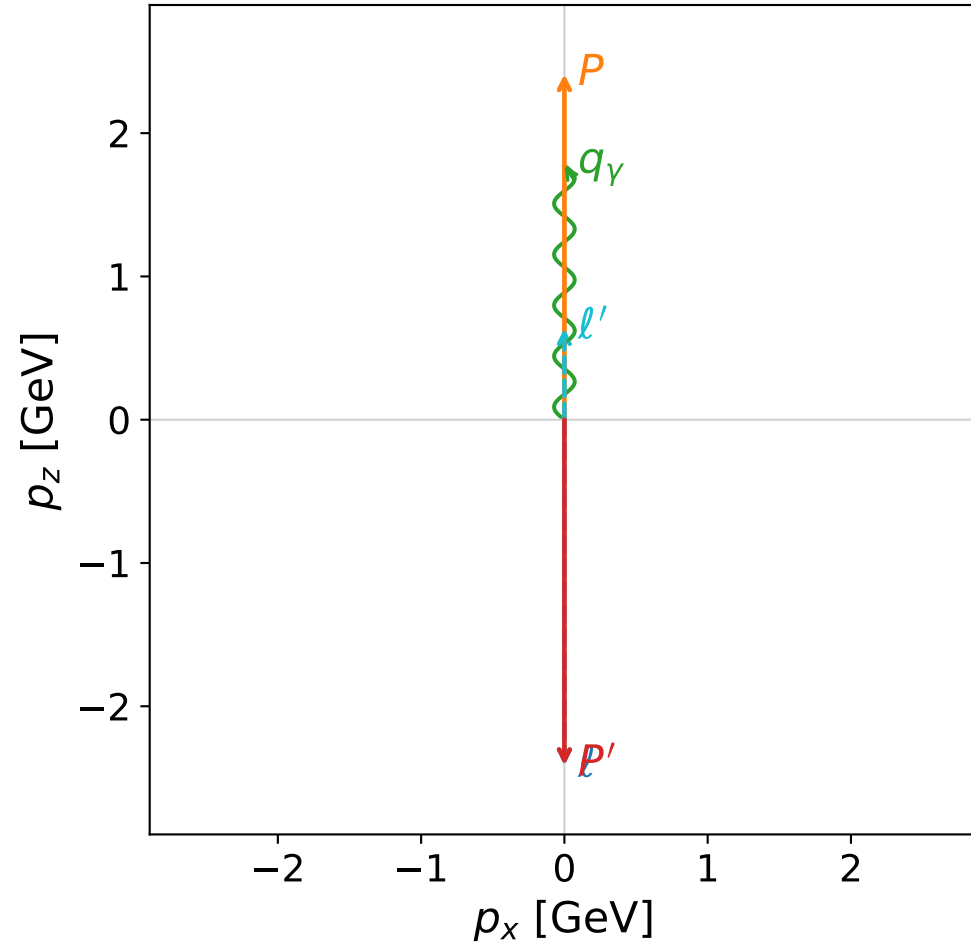


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

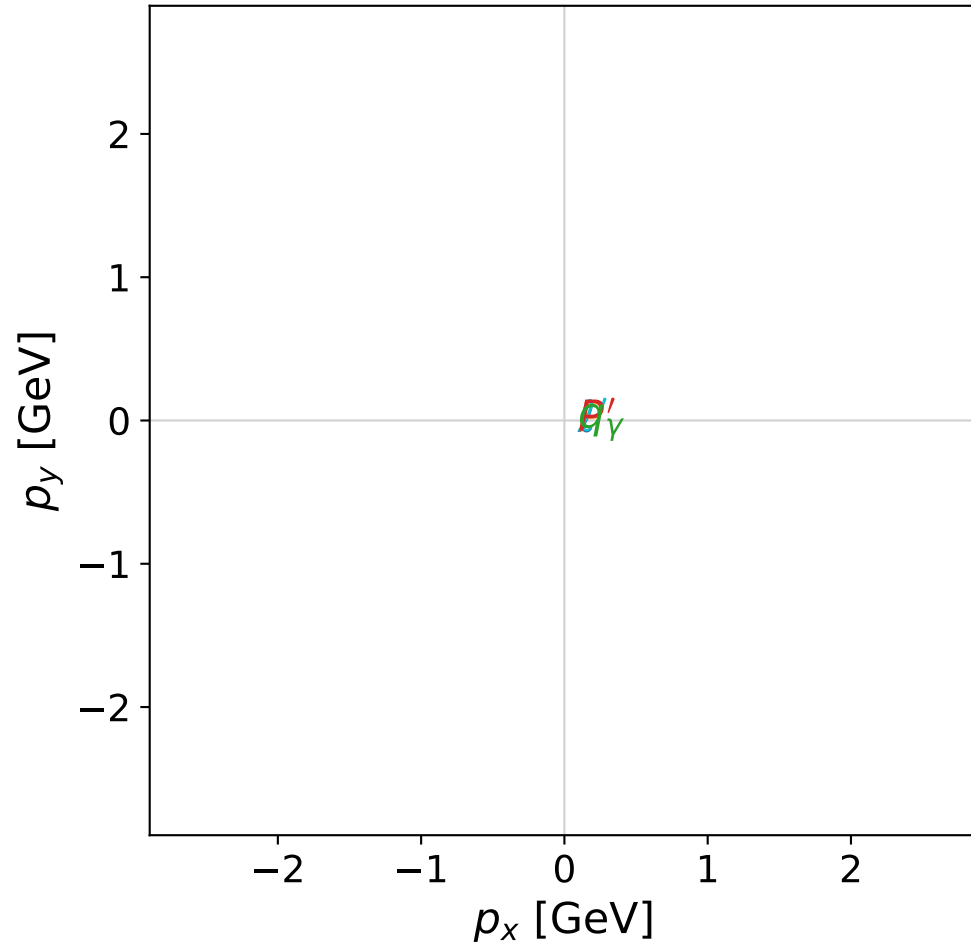


dghz_mixing_angles_polarization_01_local_minimum_152
 $d_{\text{GHZ}}=8.63787\text{e-}08$
 region: polarization_cluster_01_local_minimum_152
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0152
 lepton mixing angle: $\alpha_e=0.69075328$ rad
 incoming lepton: $0.770766|+\rangle +0.637118|-\rangle$
 proton mixing angle: $\alpha_p=0.88003373$ rad
 incoming proton: $0.637125|+\rangle +0.77076|-\rangle$

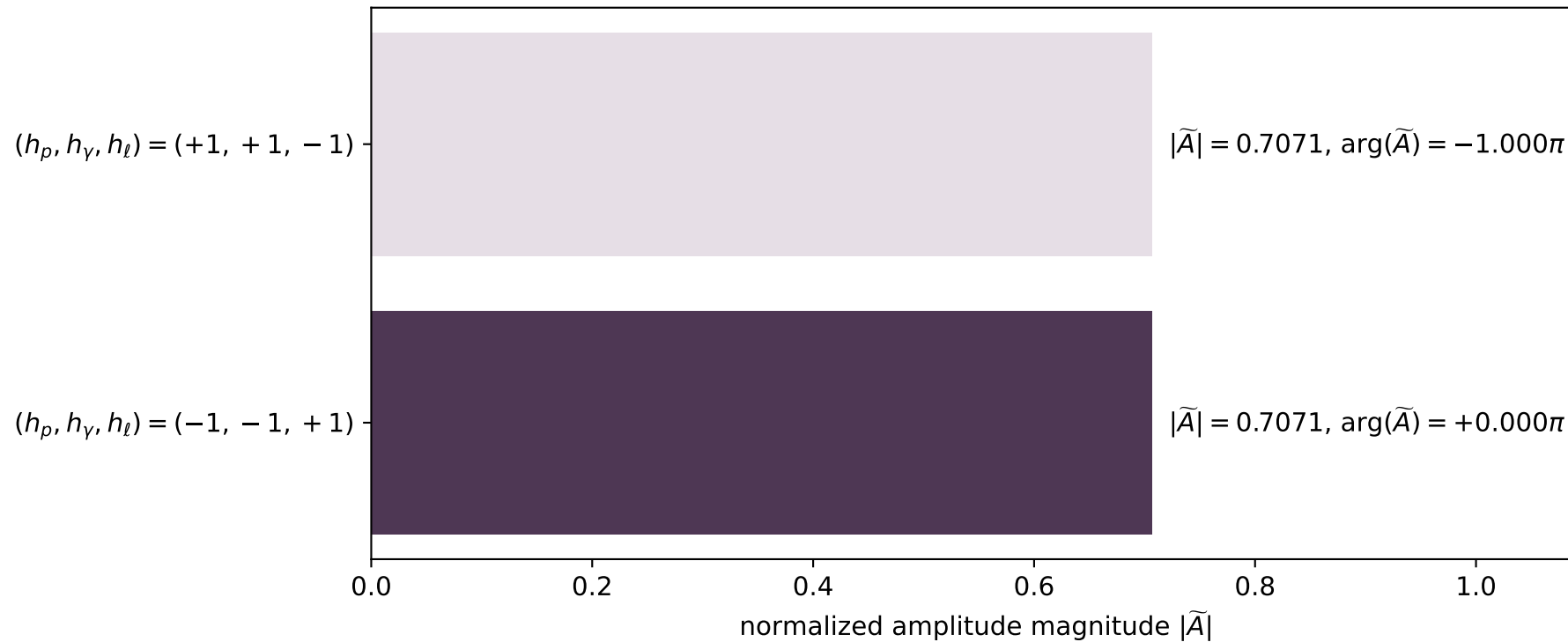
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.42356$, $\phi_\gamma=5.97423$
 $E_\gamma=1.77397$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.56515$
 $Q^2=6.15594$, $x_B=0.257618$, $t=-23.2713$, $W^2=18.6195$, $y=0.99069$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 0.6380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6380$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 1.7740$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7740$

x--y plane

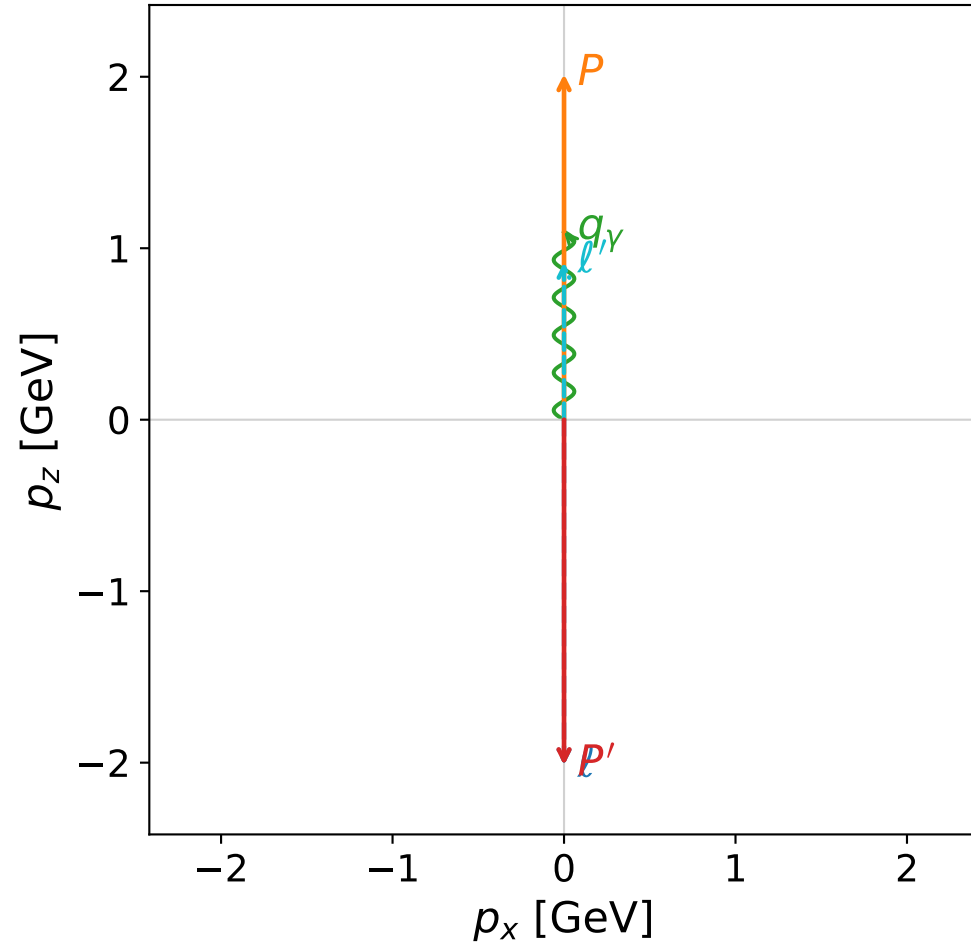


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

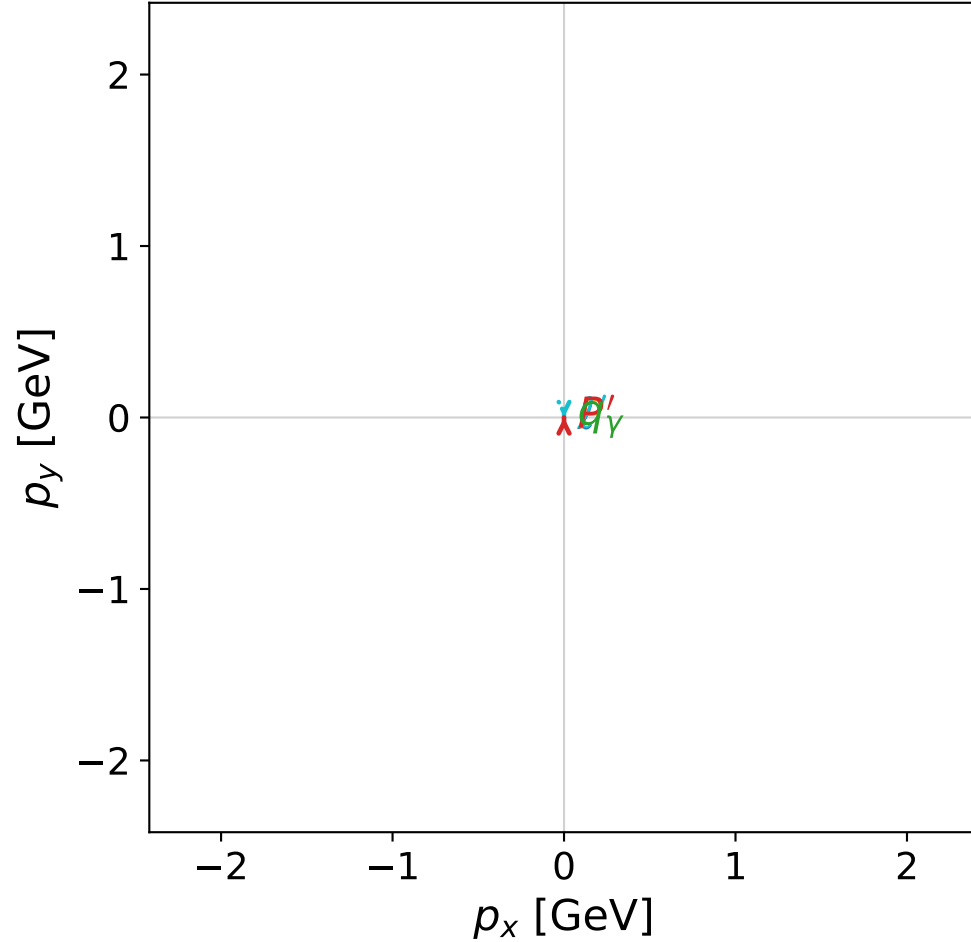


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



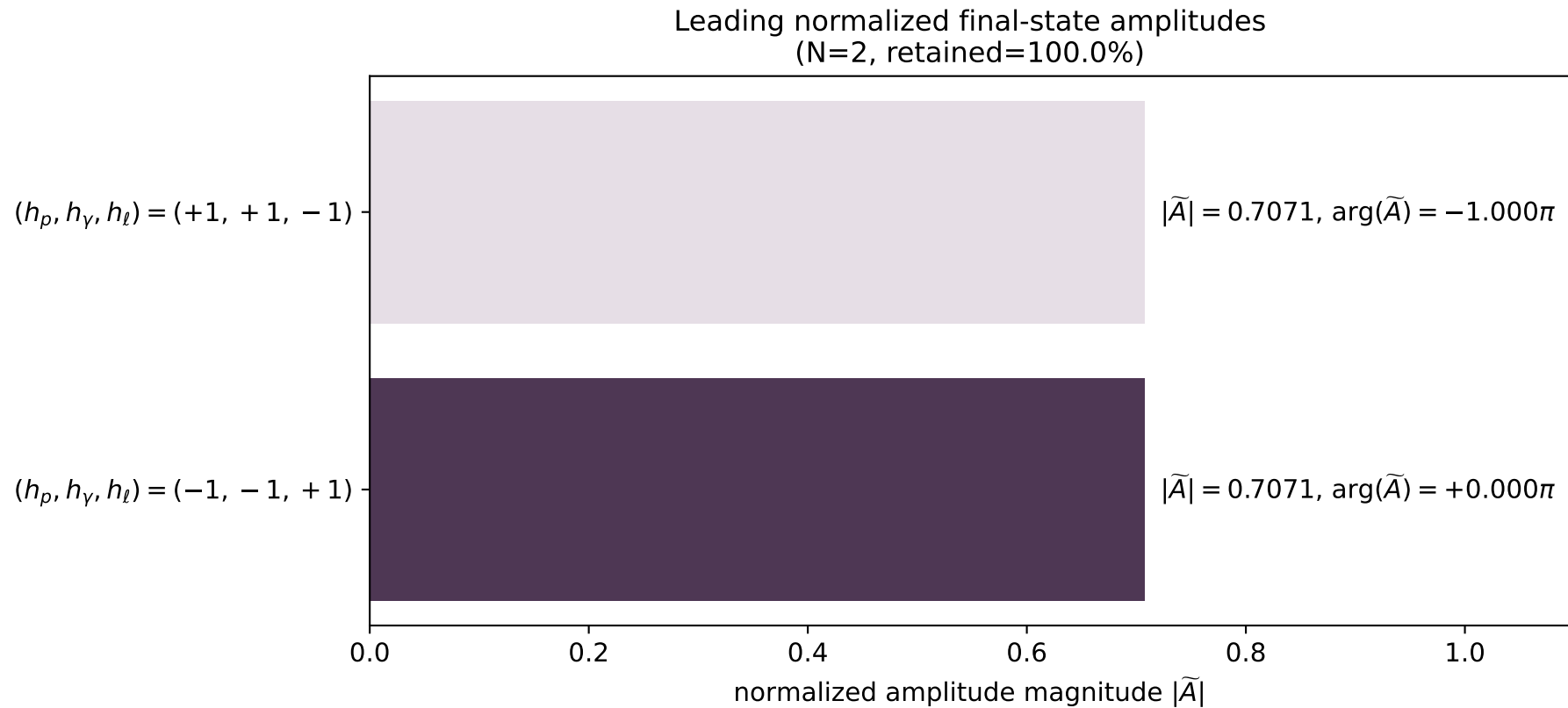
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_163
 $d_{\{\rm GHZ\}}=1.18968\text{e-}07$
 region: polarization_cluster_01_local_minimum_163
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0163
 lepton mixing angle: $\alpha_e=0.31723973$ rad
 incoming lepton: $0.9501|+\rangle +0.311945|-\rangle$
 proton mixing angle: $\alpha_p=1.2535406$ rad
 incoming proton: $0.31196|+\rangle +0.950095|-\rangle$

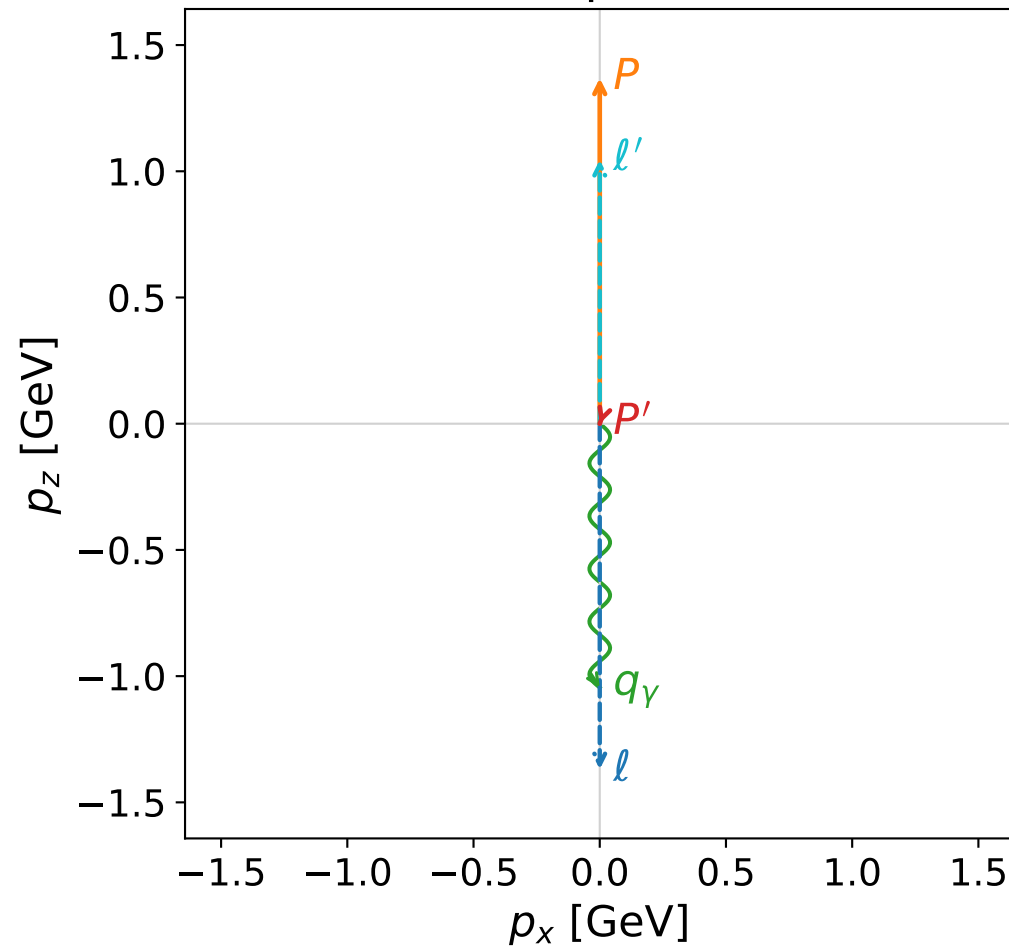
$s=17.9637$, $\sqrt{s}=4.23836$, $|\vec{P}|=2.01539$, $|\vec{P}'|=2.01539$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.26708$, $\phi_\gamma=1.45652$
 $E_\gamma=1.09647$, $\theta_{l'}=0$, $\phi_{l'}=4.40868$
 $Q^2=7.40786$, $x_B=0.443522$, $t=-16.2471$, $W^2=10.1743$, $y=0.977668$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0154$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0154$
 P $E= 2.2230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0154$
 l' $E= 0.9189$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.9189$
 P' $E= 2.2230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0154$
 q_γ $E= 1.0965$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0965$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

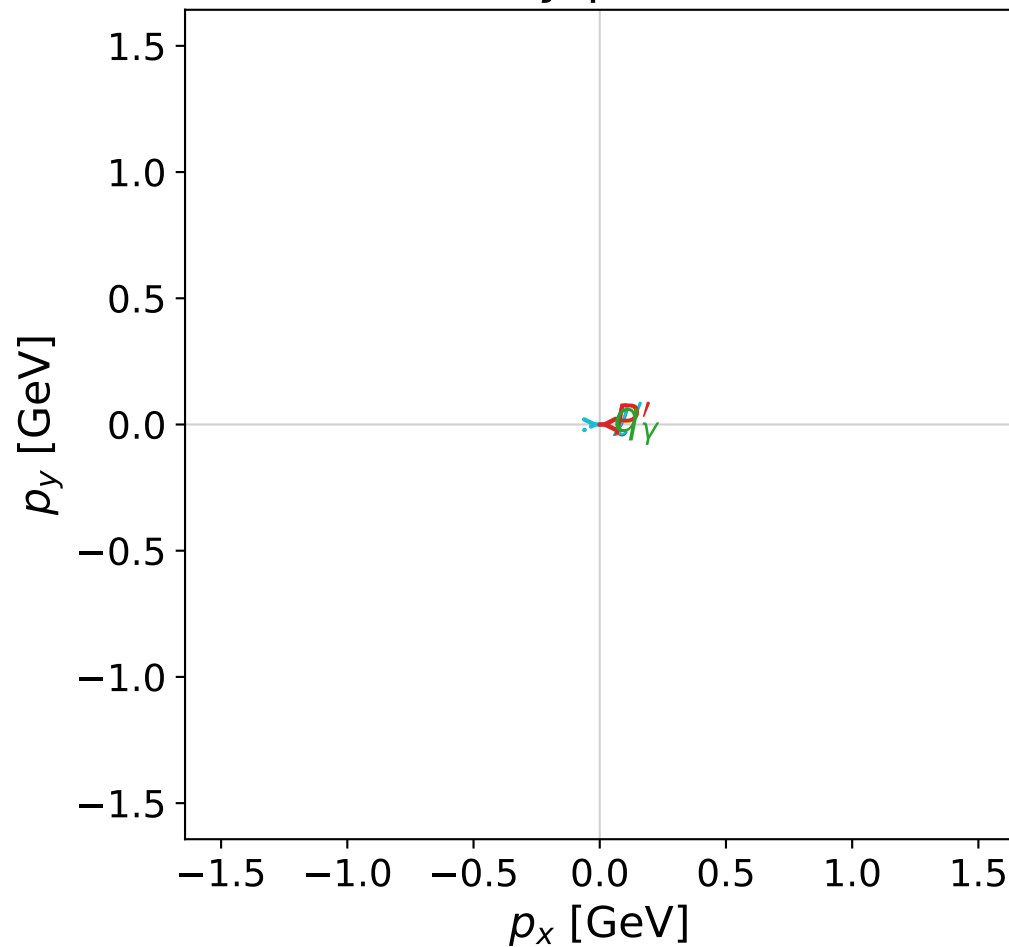


dghz_mixing_angles_polarization_01_local_minimum_165
 $d_{\{\rm GHZ\}}=1.20835\text{e-}07$
 region: polarization_cluster_01_local_minimum_165
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0165
 lepton mixing angle: $\alpha_e=0.40930724$ rad
 incoming lepton: $0.917397|+\rangle + 0.397974|-\rangle$
 proton mixing angle: $\alpha_p=0.40933059$ rad
 incoming proton: $0.917387|+\rangle + 0.397995|-\rangle$

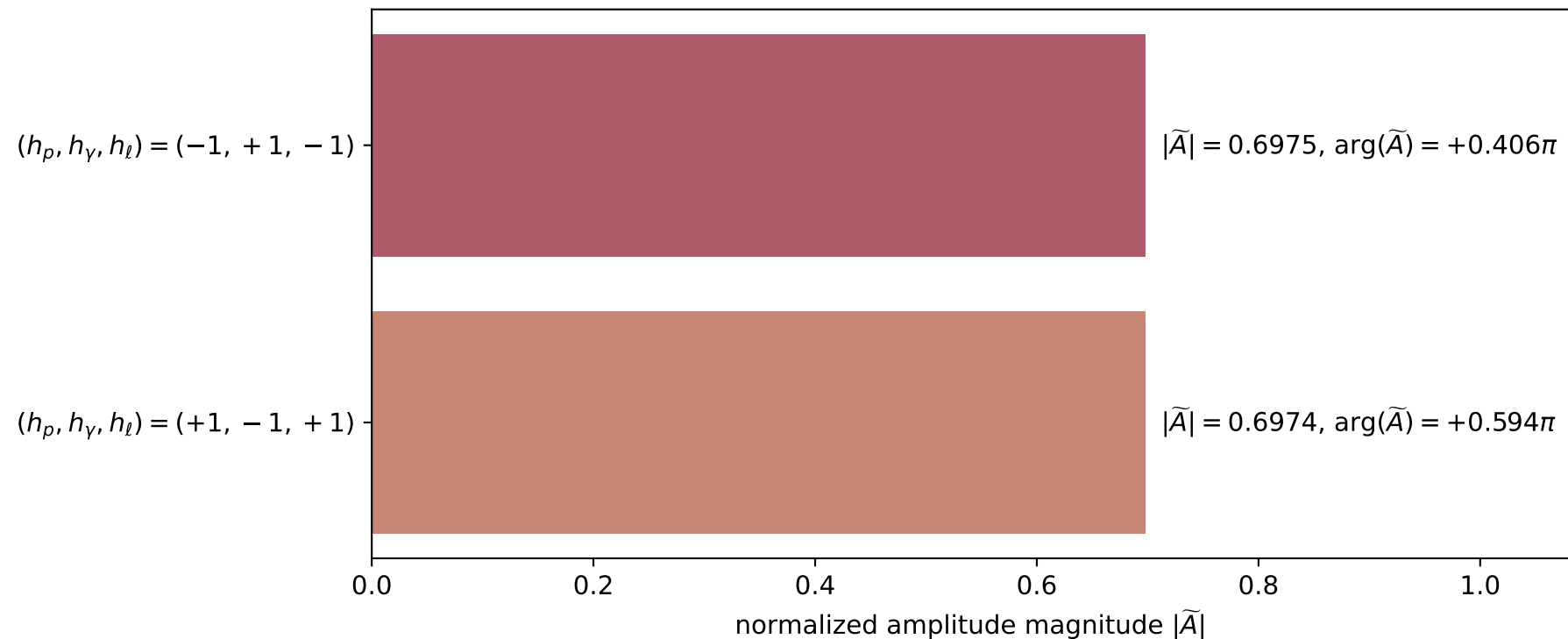
$s=9.17245$, $\sqrt{s}=3.0286$, $|\vec{P}|=1.36905$, $|\vec{P}'|=7.26171\text{e-}05$
 $\theta_{p'}=2.81094$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.1542$, $\phi_\gamma=1.87819$
 $E_\gamma=1.04527$, $\theta_{\ell'}=2.25534\text{e-}05$, $\phi_{\ell'}=0.0126103$
 $Q^2=5.72446$, $x_B=0.744864$, $t=-1.35383$, $W^2=2.84062$, $y=0.926758$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3690$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3690$
 P $E= 1.6596$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3690$
 ℓ' $E= 1.0453$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0453$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0001$
 q_γ $E= 1.0453$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.0453$

x--y plane

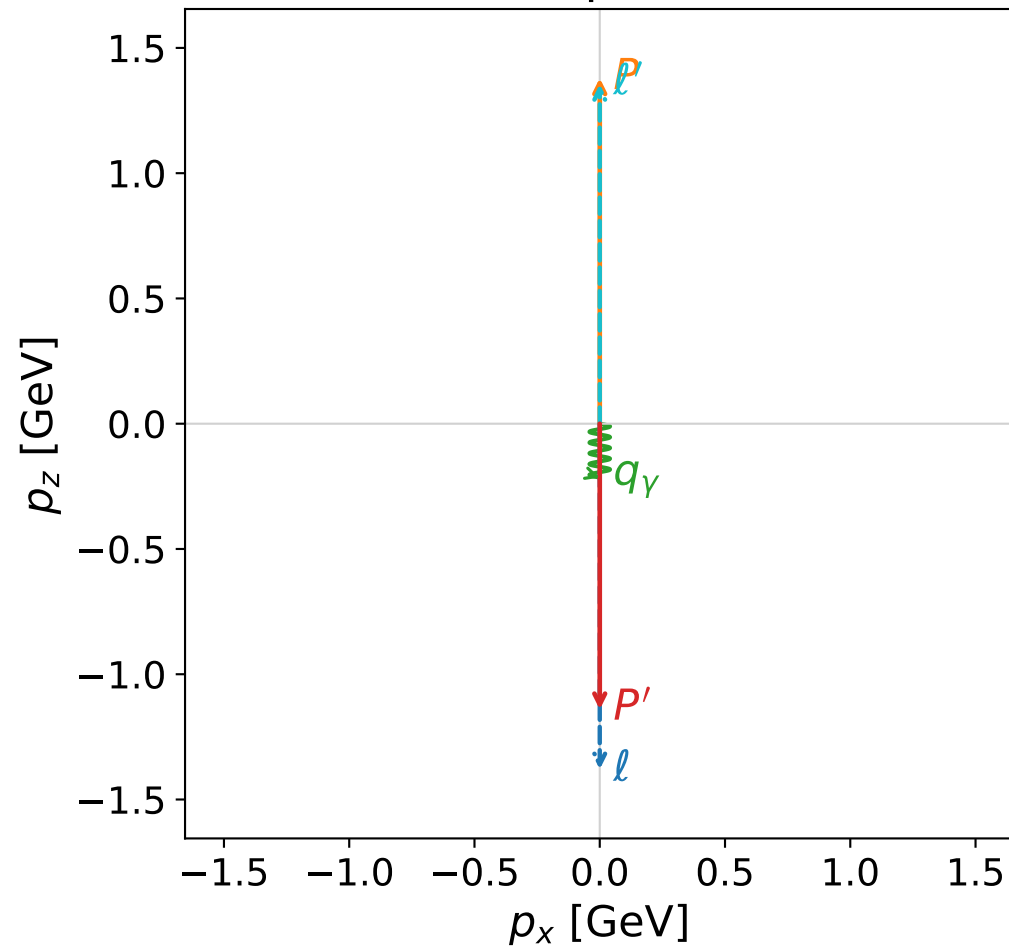


Leading normalized final-state amplitudes
 (N=2, retained=97.3%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

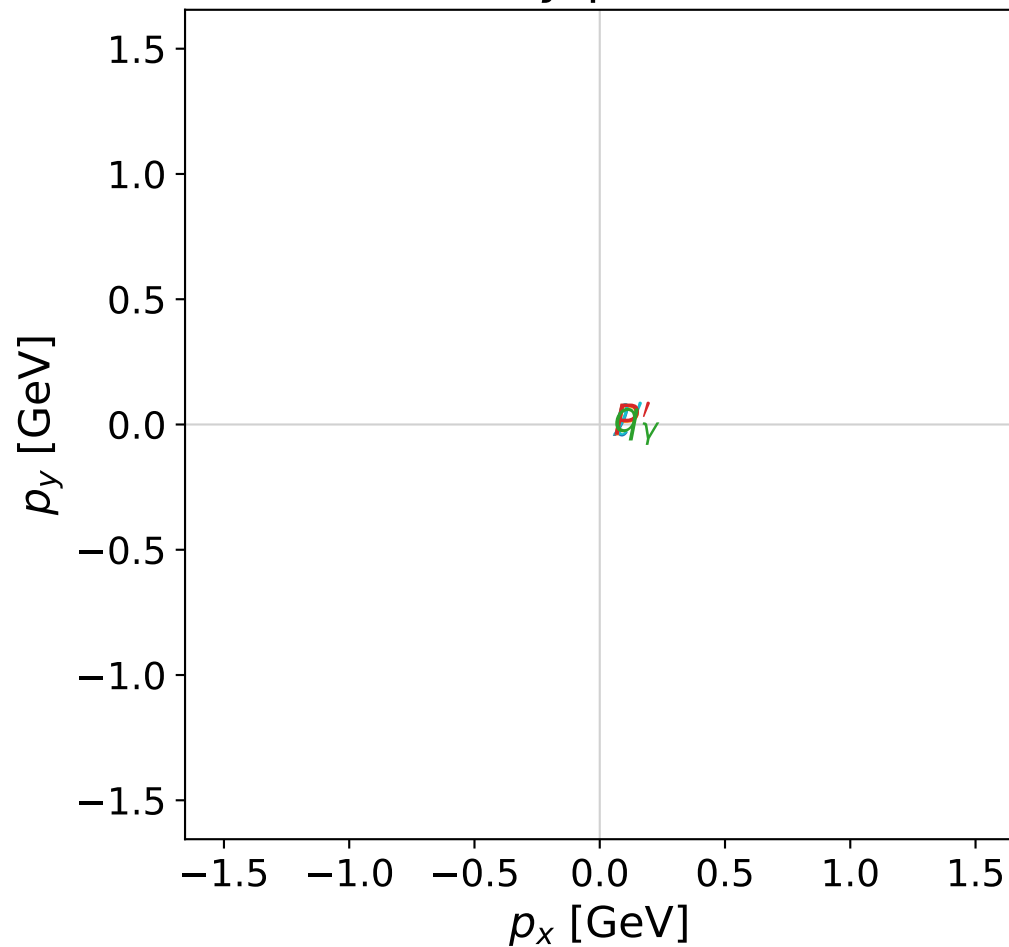


dghz_mixing_angles_polarization_01_local_minimum_170
 $d_{\{\rm GHZ\}}=1.94761\text{e-}07$
 region: polarization_cluster_01_local_minimum_170
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0170
 lepton mixing angle: $\alpha_e=0.18329382$ rad
 incoming lepton: $0.983249|+\rangle + 0.182269|-\rangle$
 proton mixing angle: $\alpha_p=0.18329532$ rad
 incoming proton: $0.983248|+\rangle + 0.182271|-\rangle$

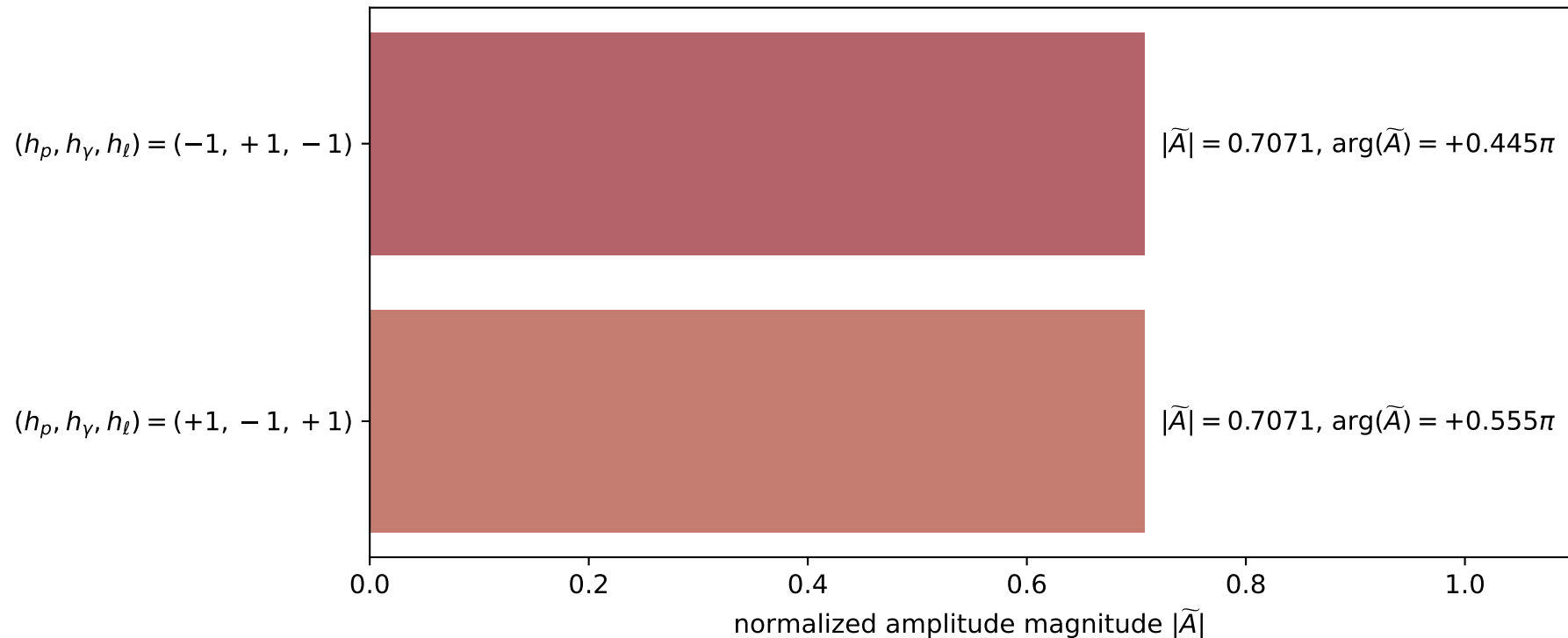
$s=9.28794$, $\sqrt{s}=3.04761$, $|\vec{P}|=1.37946$, $|\vec{P}'|=1.14077$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=5.38865$, $\phi_\gamma=4.88405$
 $E_\gamma=0.214975$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.16901$
 $Q^2=7.48078$, $x_B=0.981048$, $t=-6.31497$, $W^2=1.02436$, $y=0.906898$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3795$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3795$
 P $E= 1.6682$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3795$
 ℓ' $E= 1.3557$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.3557$
 P' $E= 1.4769$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.1408$
 q_γ $E= 0.2150$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.2150$

x--y plane

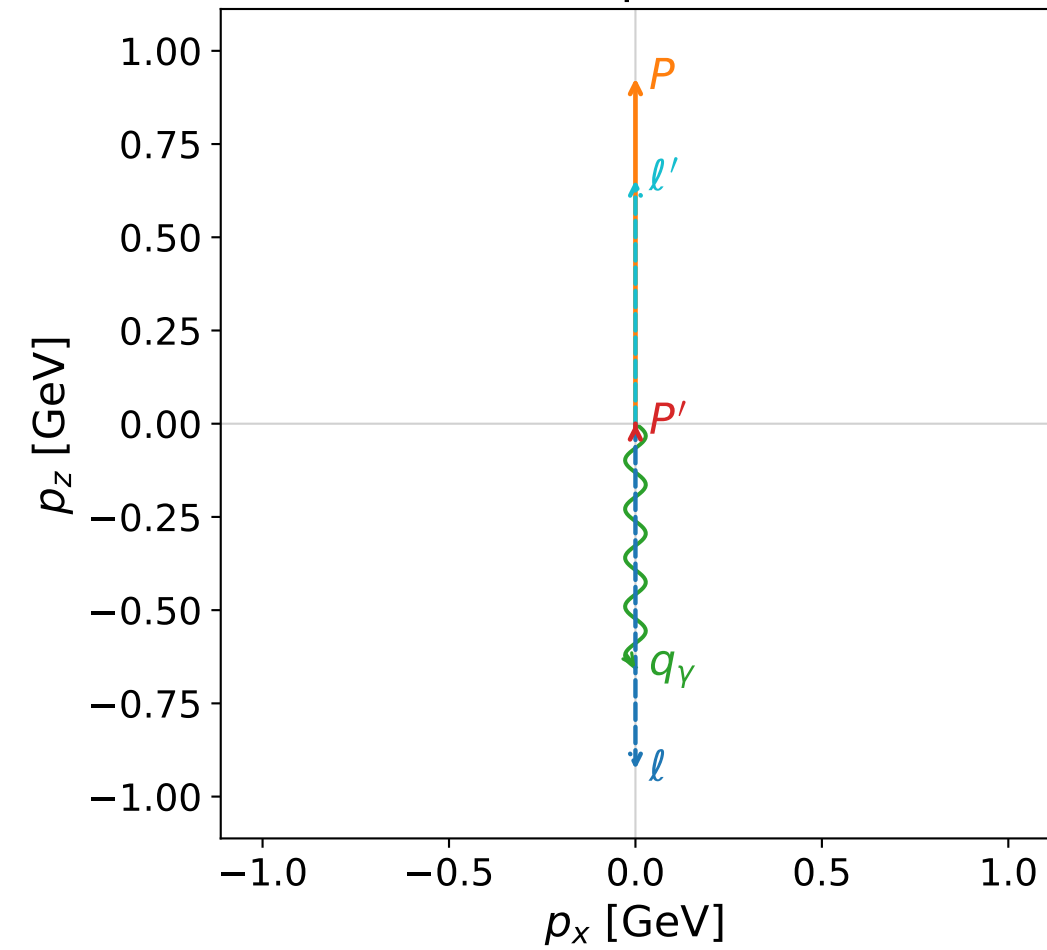


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

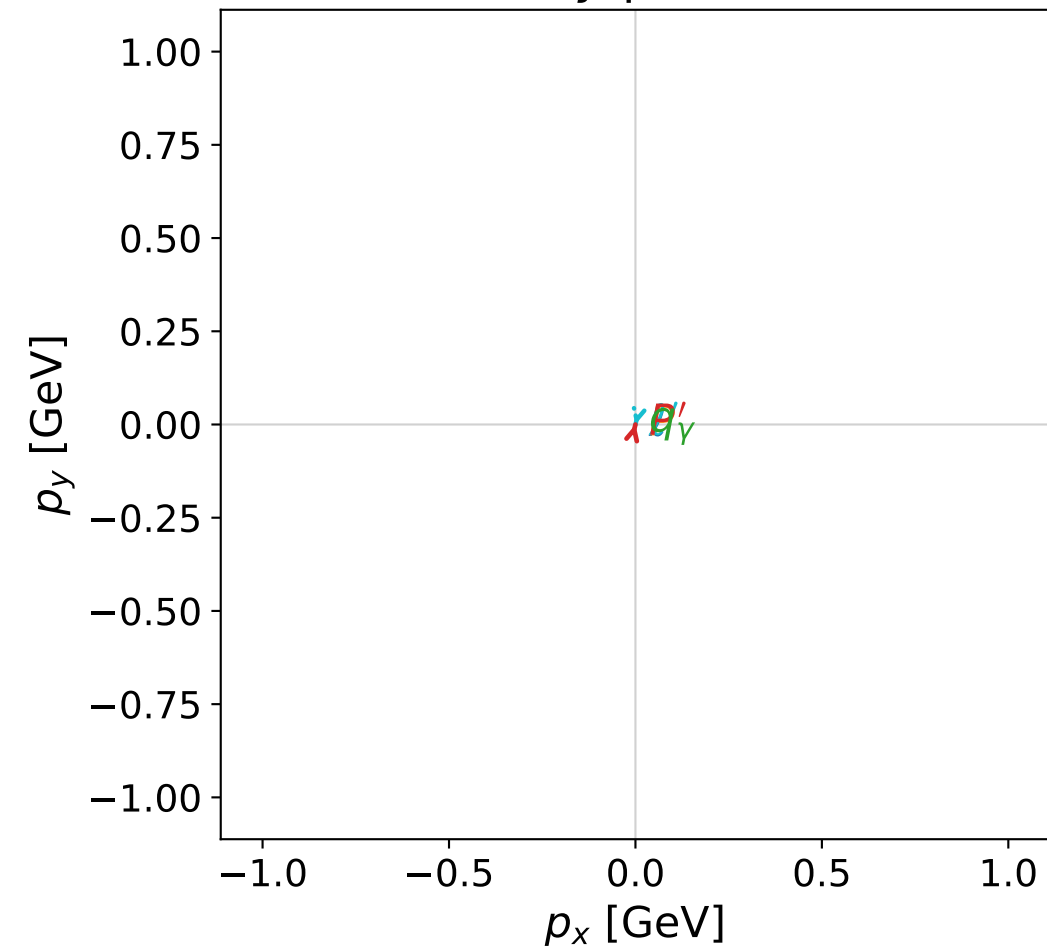


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

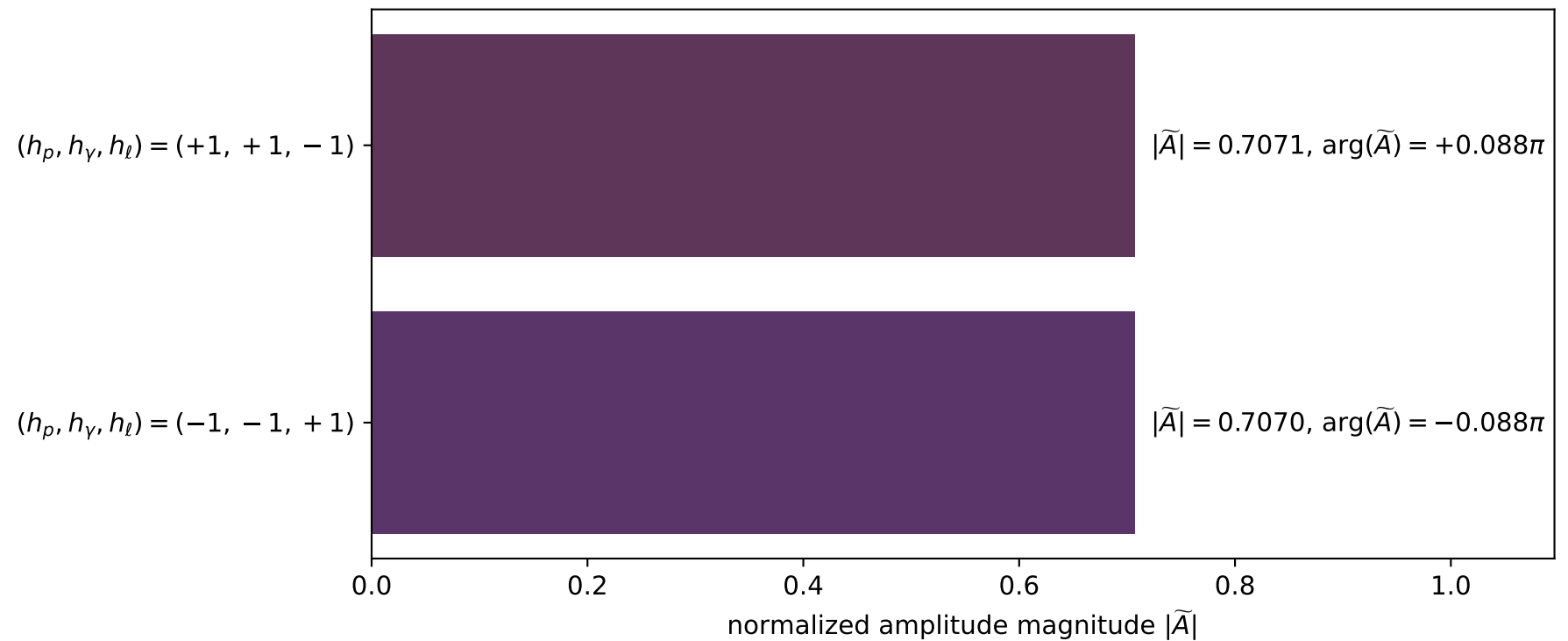


dghz_mixing_angles_polarization_01_local_minimum_172
 $d_{\text{GHZ}}=2.15034\text{e-}07$
 region: polarization_cluster_01_local_minimum_172
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0172
 lepton mixing angle: $\alpha_e=0.74567333$ rad
 incoming lepton: $0.734631|+\rangle +0.678467|-\rangle$
 proton mixing angle: $\alpha_p=0.74572493$ rad
 incoming proton: $0.734596|+\rangle +0.678505|-\rangle$

$s=5.04216$, $\sqrt{s}=2.24548$, $|\vec{P}|=0.926823$, $|\vec{P}'|=0.000424397$
 $\theta_{P'}=0.0244936$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.33005$, $\phi_\gamma=2.86593$
 $E_\gamma=0.65395$, $\theta_{l'}=1.59044\text{e-}05$, $\phi_{l'}=4.47164$
 $Q^2=2.42281$, $x_B=0.663752$, $t=-0.713318$, $W^2=2.10721$, $y=0.876957$

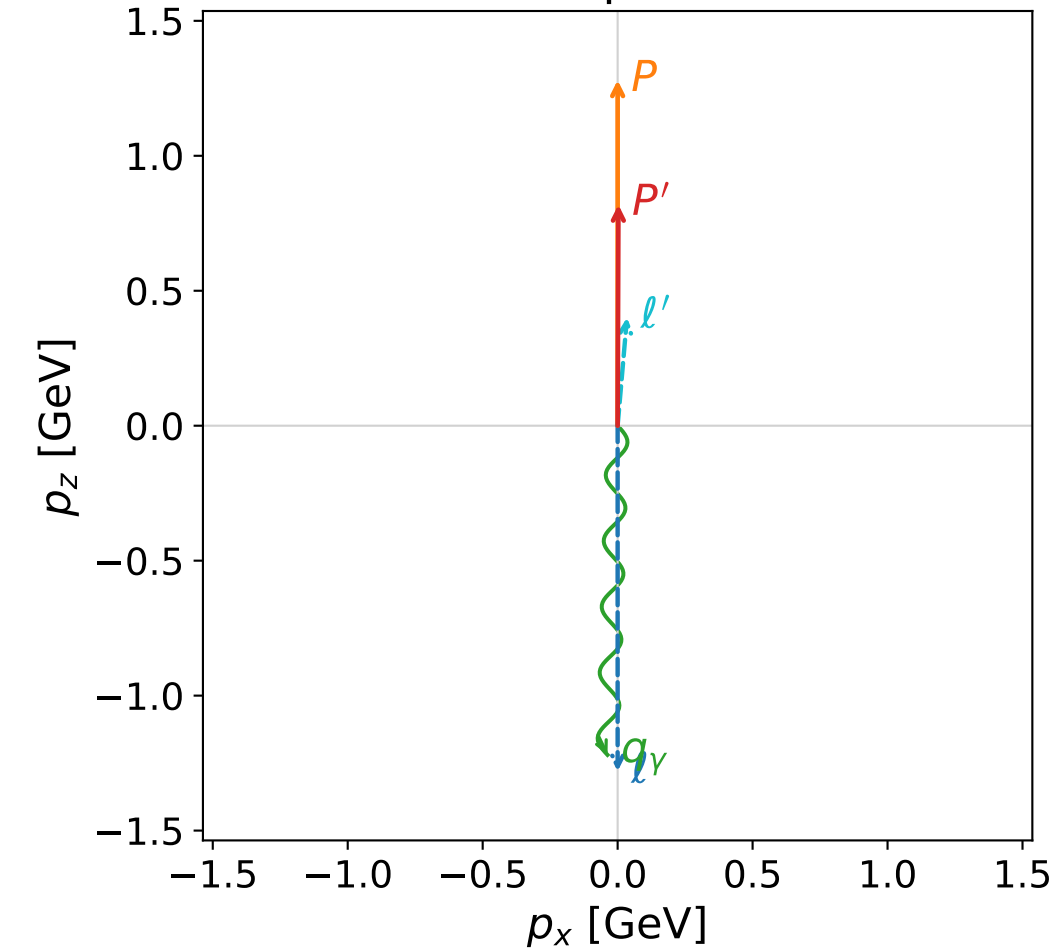
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.9268$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9268$
 P $E= 1.3187$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9268$
 l' $E= 0.6535$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.6535$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0004$
 q_γ $E= 0.6539$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.6539$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



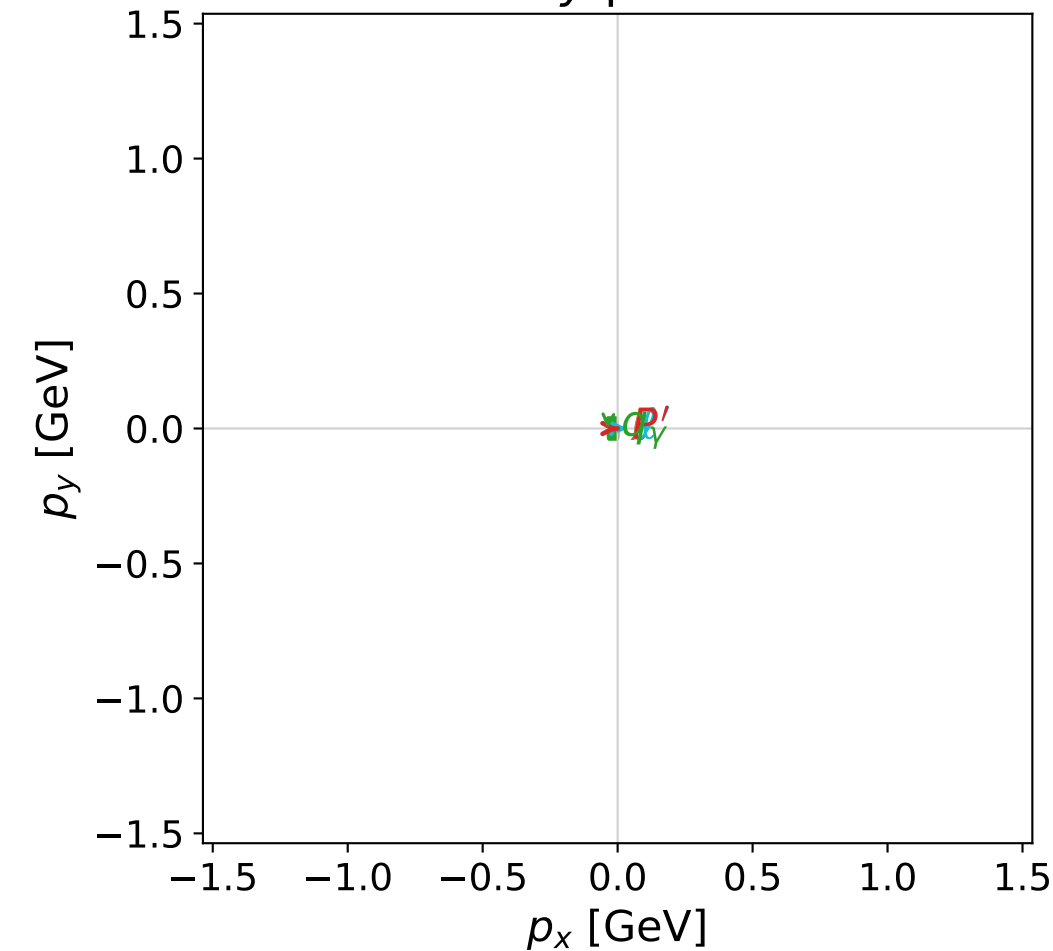
dghz_mixing_angles_polarization_01_local_minimum_193
 $d_{\text{GHZ}}=1.77496\text{e-}05$
 region: polarization_cluster_01_local_minimum_193
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0193
 lepton mixing angle: $\alpha_e=0.24273564$ rad
 incoming lepton: $0.970684|+\rangle +0.240359|-\rangle$
 proton mixing angle: $\alpha_p=1.3360225$ rad
 incoming proton: $0.232623|+\rangle +0.972567|-\rangle$

$s=8.22294$, $\sqrt{s}=2.86757$, $|\vec{P}|=1.28037$, $|\vec{P}'|=0.818827$
 $\theta_{P'}=0.00352053$, $\theta_Y=3.11084$, $\phi_{P'}=6.28317$, $\phi_Y=3.14158$
 $E_Y=1.22018$, $\theta_{l'}=0.086202$, $\phi_{l'}=6.28317$
 $Q^2=2.0564$, $x_B=0.289943$, $t=-0.0960184$, $W^2=5.91586$, $y=0.965862$

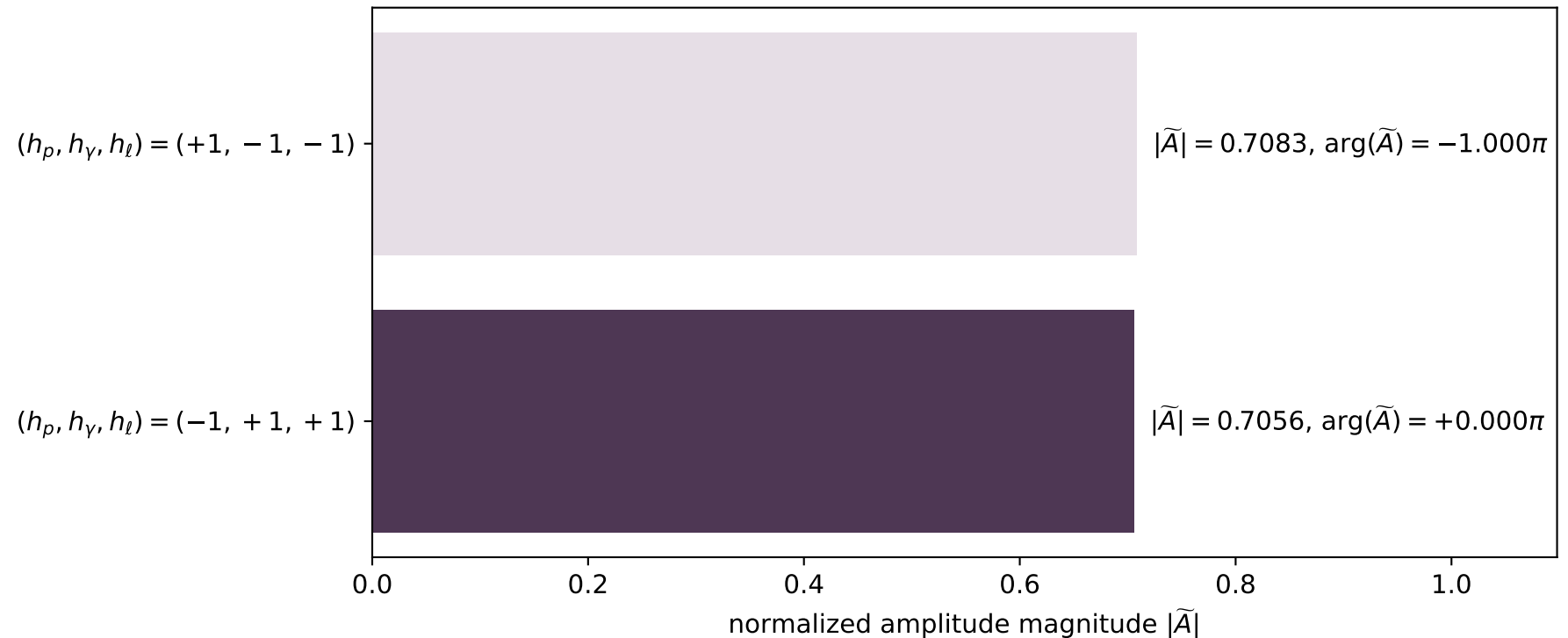
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	1.2804	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-1.2804
P	$E=$	1.5872	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	1.2804
ℓ'	$E=$	0.4023	$p_x=$	0.0346	$p_y=$	-0.0000	$p_z=$	0.4008
P'	$E=$	1.2451	$p_x=$	0.0029	$p_y=$	-0.0000	$p_z=$	0.8188
q_Y	$E=$	1.2202	$p_x=$	-0.0375	$p_y=$	0.0000	$p_z=$	-1.2196

x--y plane

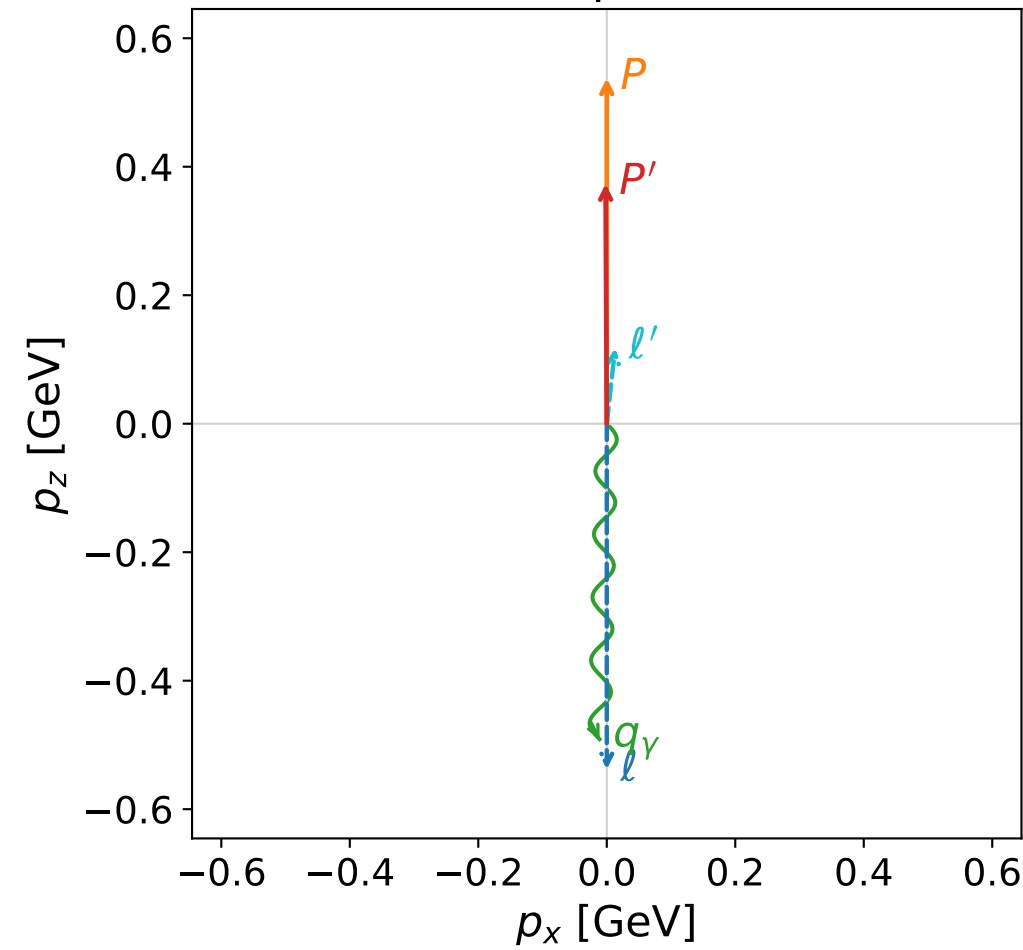


Leading normalized final-state amplitudes
(N=2, retained=99.9%)

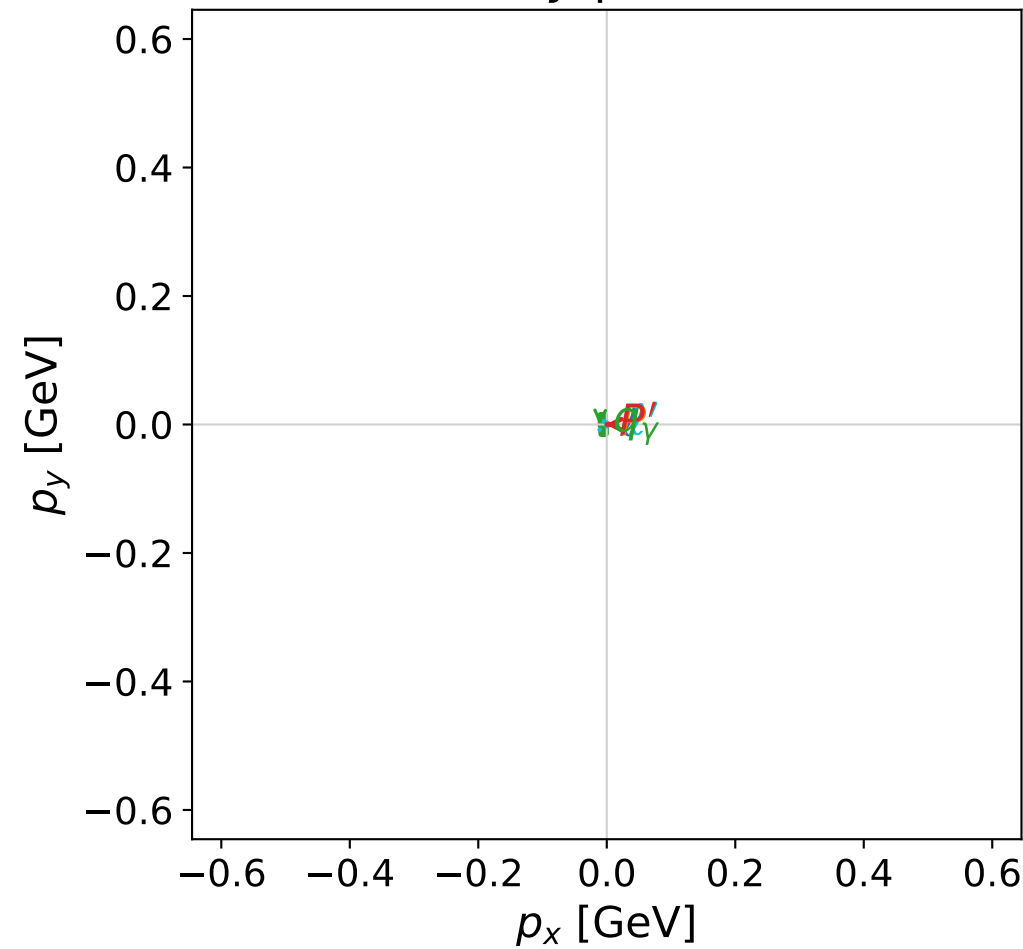


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

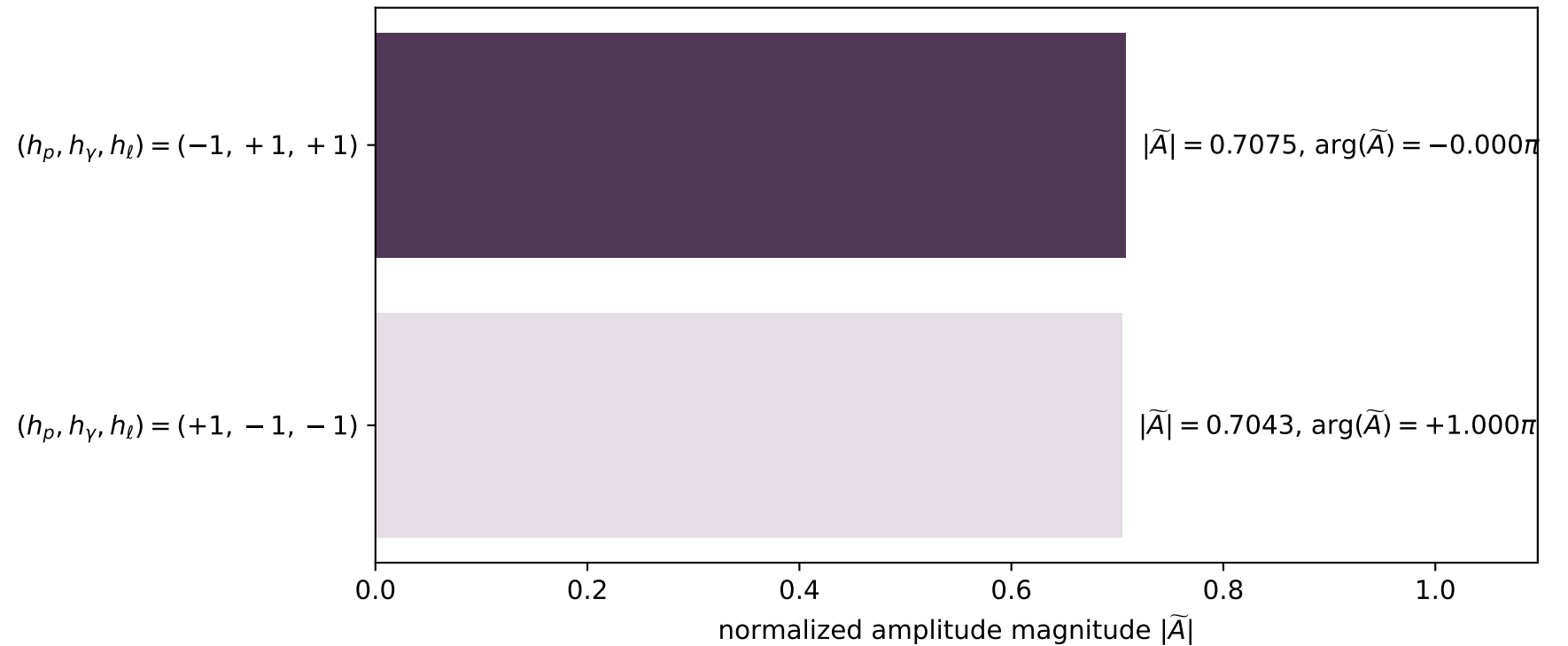


dghz_mixing_angles_polarization_01_local_minimum_198
 $d_{\text{GHZ}}=2.02825\text{e-}05$
 region: polarization_cluster_01_local_minimum_198
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0198
 lepton mixing angle: $\alpha_e=0.31842078$ rad
 incoming lepton: $0.949731|+\rangle +0.313067|-\rangle$
 proton mixing angle: $\alpha_p=1.2260476$ rad
 incoming proton: $0.33796|+\rangle +0.94116|-\rangle$

$s=2.62187$, $\sqrt{s}=1.61922$, $|\vec{P}|=0.537922$, $|\vec{P}'|=0.373454$
 $\theta_{P'}=0.00509094$, $\theta_\gamma=3.11889$, $\phi_{P'}=3.14157$, $\phi_\gamma=3.14159$
 $E_\gamma=0.491231$, $\theta_{l'}=0.110485$, $\phi_{l'}=6.28318$
 $Q^2=0.253934$, $x_B=0.157469$, $t=-0.0219156$, $W^2=2.23851$, $y=0.925704$

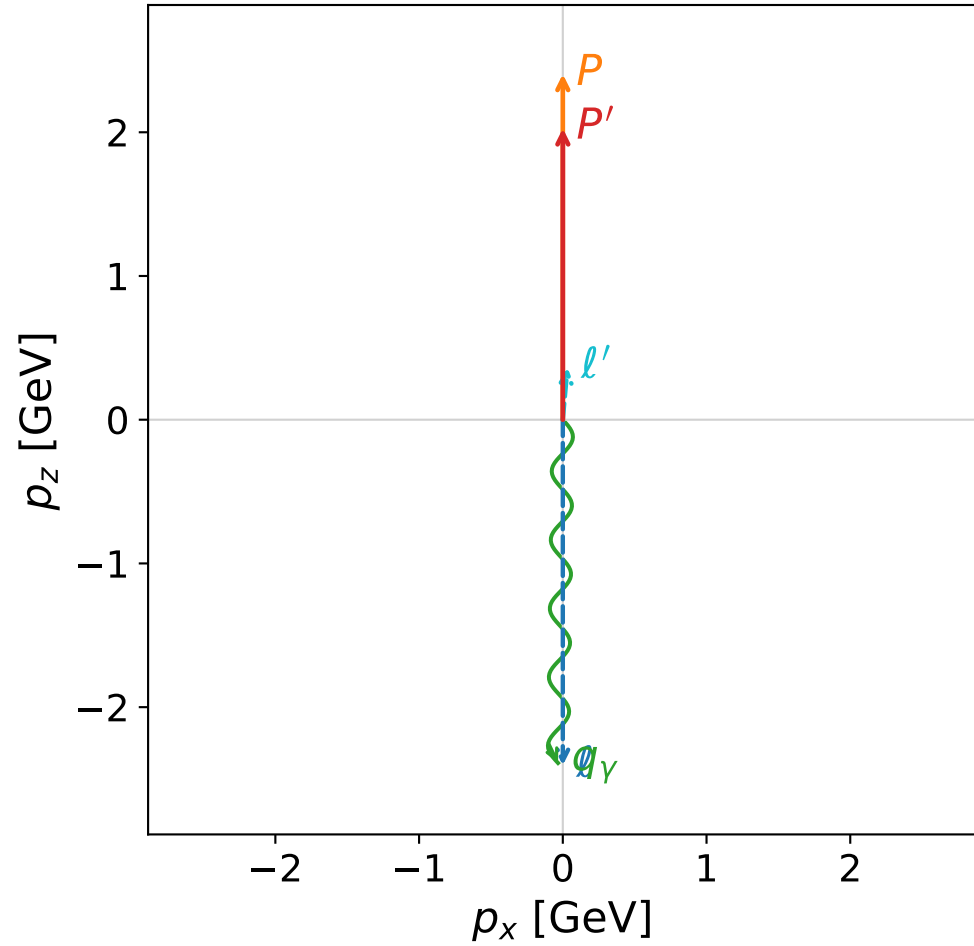
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.5379$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5379$
 P $E= 1.0813$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5379$
 l' $E= 0.1184$ $p_x= 0.0131$ $p_y= -0.0000$ $p_z= 0.1177$
 P' $E= 1.0096$ $p_x= -0.0019$ $p_y= 0.0000$ $p_z= 0.3734$
 q_γ $E= 0.4912$ $p_x= -0.0112$ $p_y= 0.0000$ $p_z= -0.4911$

Leading normalized final-state amplitudes
(N=2, retained=99.7%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

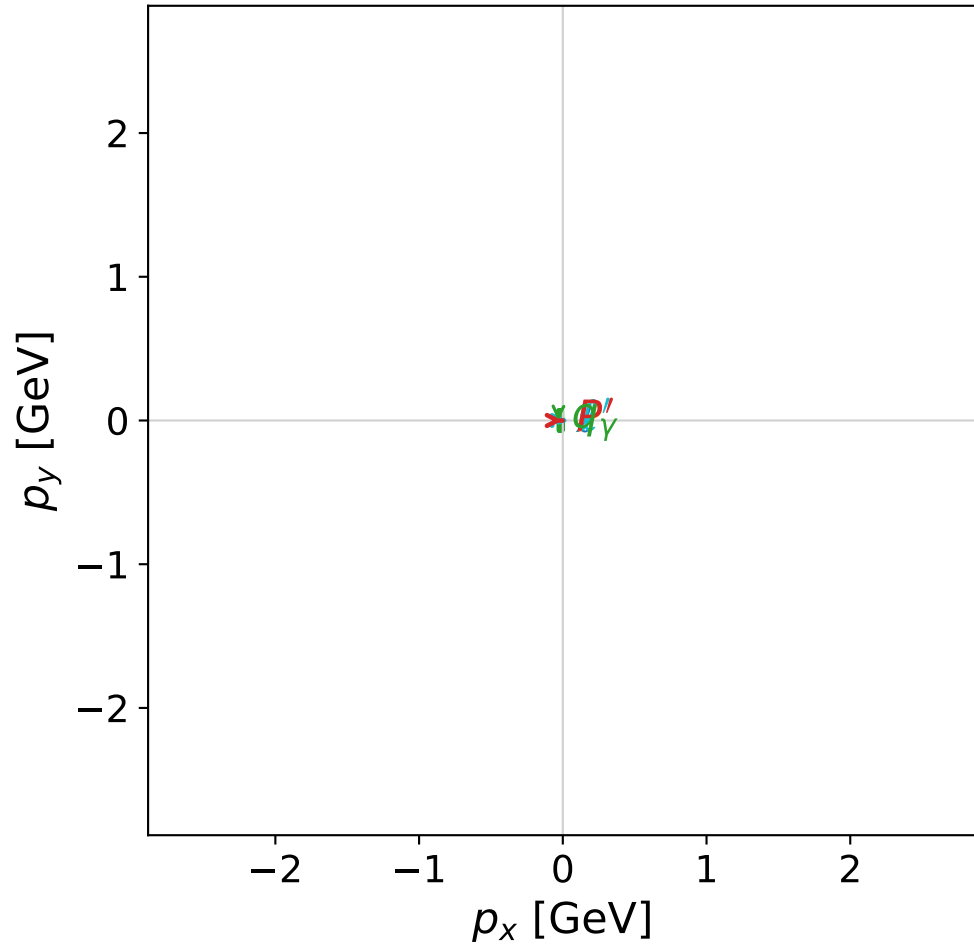


dghz_mixing_angles_polarization_01_local_minimum_203
 $d_{\text{GHZ}}=2.34128\text{e-}05$
 region: polarization_cluster_01_local_minimum_203
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0203
 lepton mixing angle: $\alpha_e=0.64267685$ rad
 incoming lepton: $0.800494|+\rangle + 0.59934|-\rangle$
 proton mixing angle: $\alpha_p=0.93529282$ rad
 incoming proton: $0.593583|+\rangle + 0.804773|-\rangle$

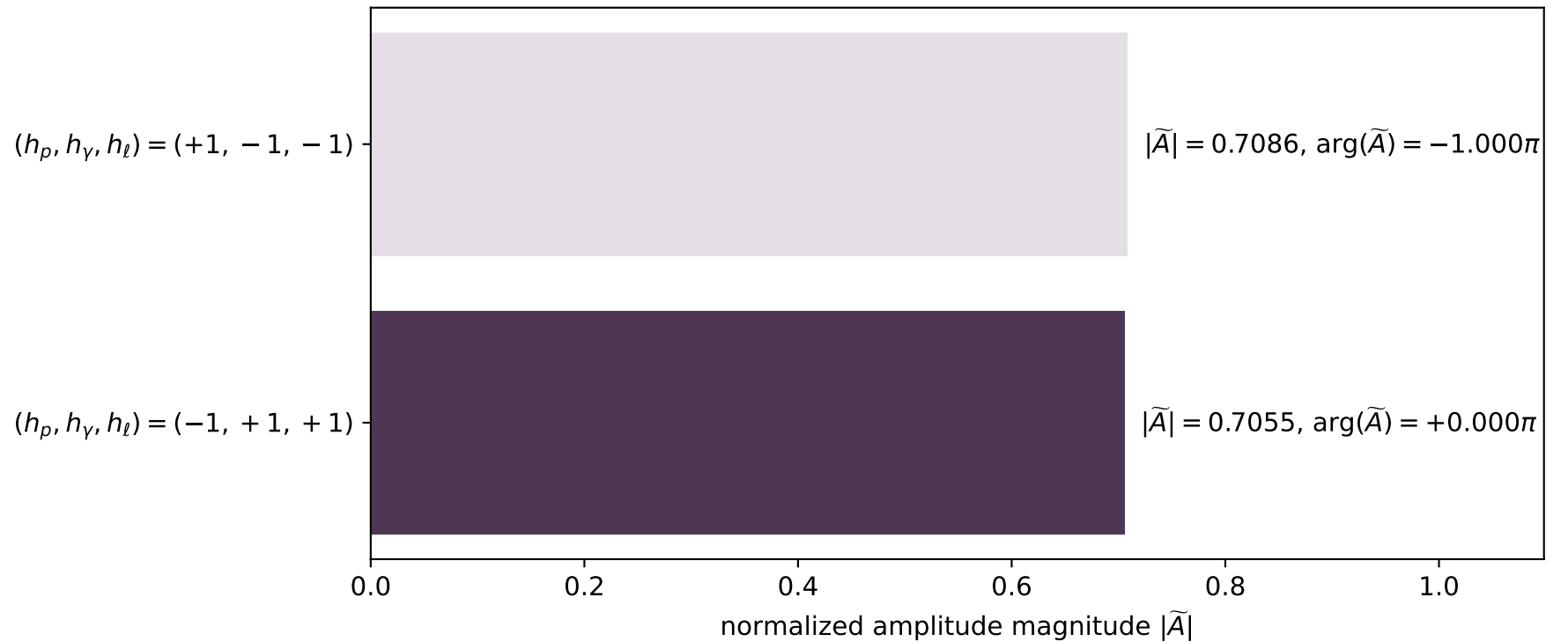
$s=24.8566$, $\sqrt{s}=4.98564$, $|\vec{P}|=2.40458$, $|\vec{P}'|=2.02645$
 $\theta_{p'}=0.000191071$, $\theta_\gamma=3.12775$, $\phi_{p'}=6.28238$, $\phi_\gamma=3.14161$
 $E_\gamma=2.38892$, $\theta_{l'}=0.0899866$, $\phi_{l'}=2.49864\text{e-}05$
 $Q^2=3.49121$, $x_B=0.146436$, $t=-0.0218485$, $W^2=21.2299$, $y=0.994351$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4046$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4046$
 P $E= 2.5811$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4046$
 l' $E= 0.3637$ $p_x= 0.0327$ $p_y= 0.0000$ $p_z= 0.3622$
 P' $E= 2.2330$ $p_x= 0.0004$ $p_y= -0.0000$ $p_z= 2.0264$
 q_γ $E= 2.3889$ $p_x= -0.0331$ $p_y= -0.0000$ $p_z= -2.3887$

x--y plane

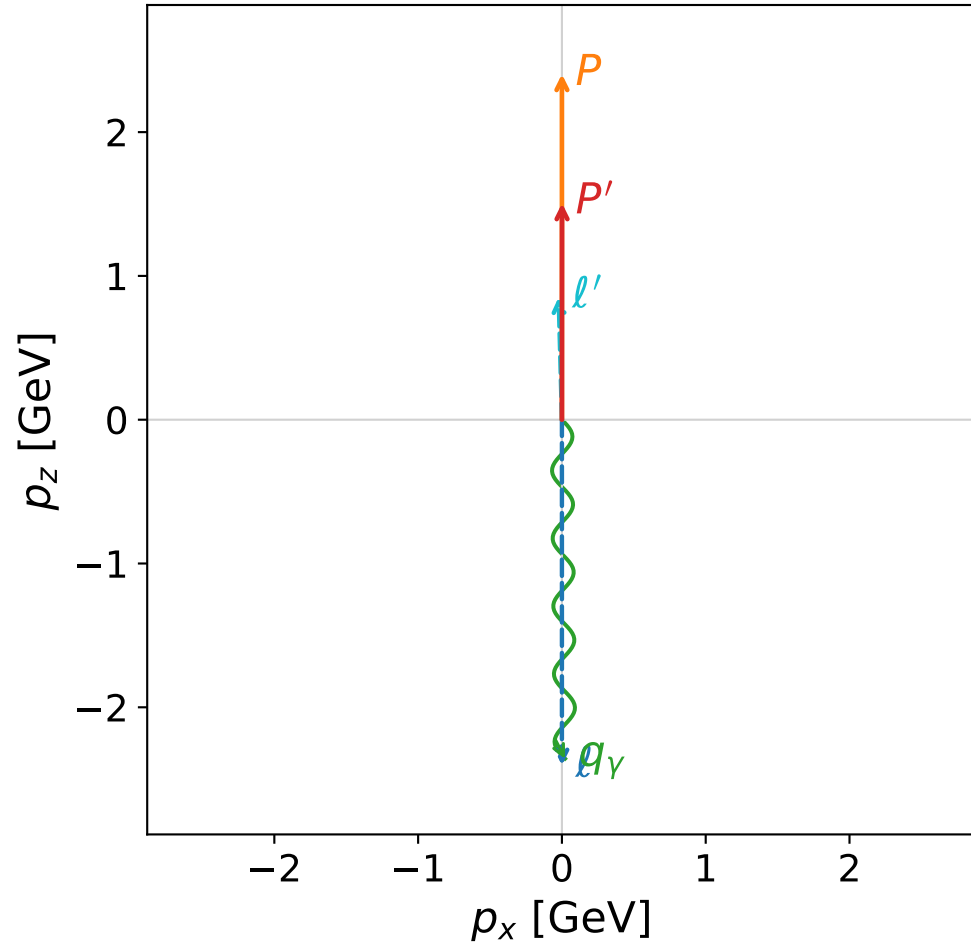


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

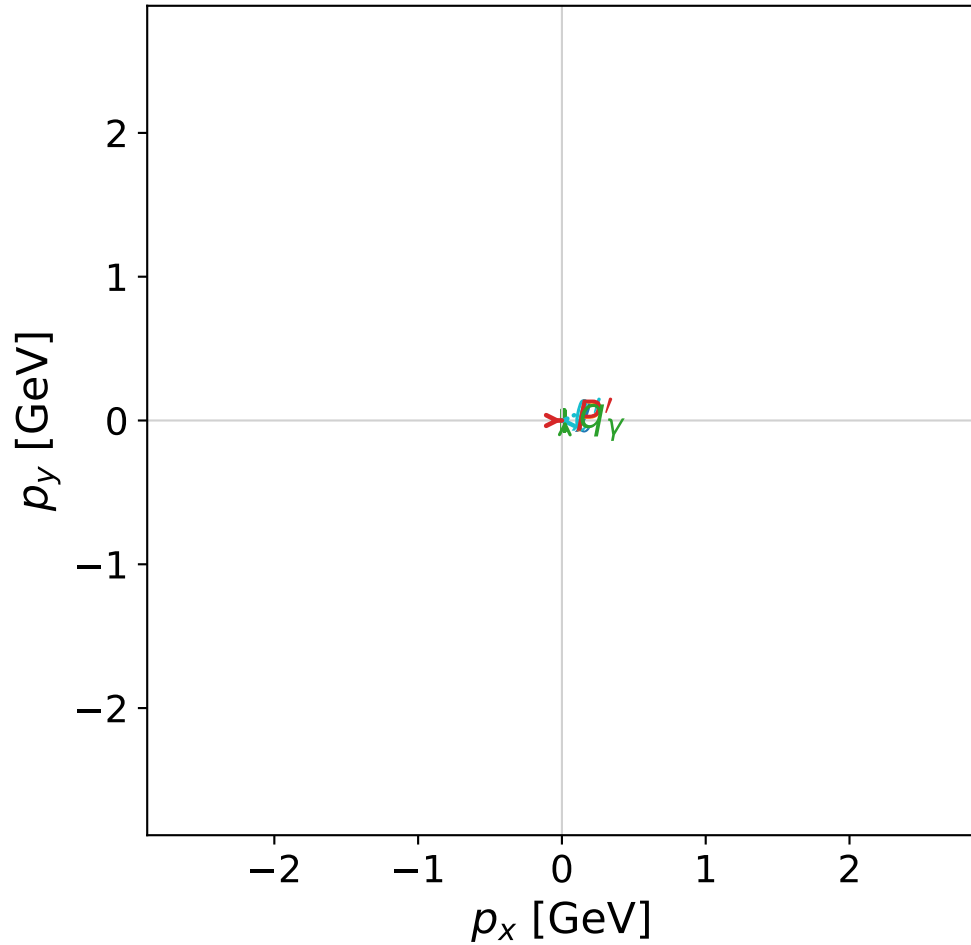


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

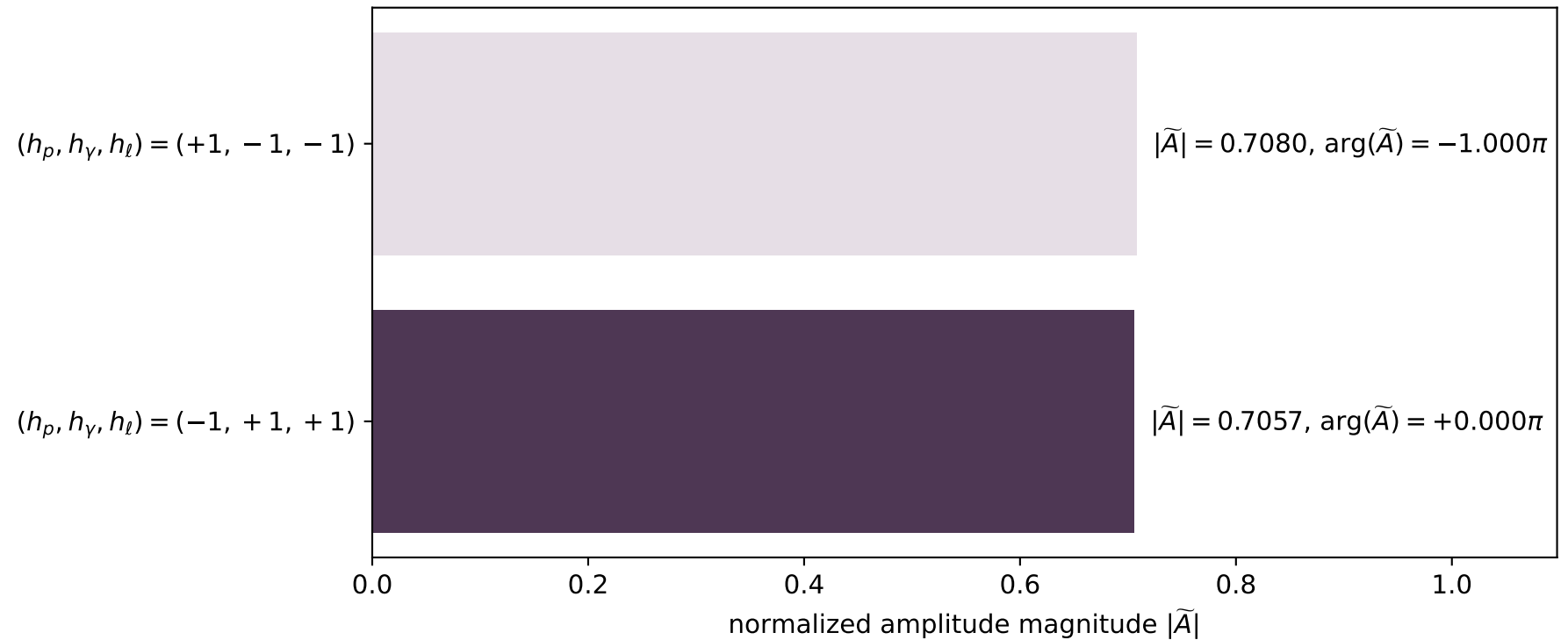


dghz_mixing_angles_polarization_01_local_minimum_206
 $d_{\{\rm GHZ\}}=2.59585\text{e-}05$
 region: polarization_cluster_01_local_minimum_206
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0206
 lepton mixing angle: $\alpha_e=1.1210285$ rad
 incoming lepton: $0.434756|+\rangle +0.900548|-\rangle$
 proton mixing angle: $\alpha_p=0.45230063$ rad
 incoming proton: $0.899444|+\rangle +0.437036|-\rangle$

$s=24.8343$, $\sqrt{s}=4.9834$, $|\vec{P}|=2.40342$, $|\vec{P}'|=1.50557$
 $\theta_{P'}=0.000624821$, $\theta_\gamma=3.13227$, $\phi_{P'}=6.28282$, $\phi_\gamma=0$
 $E_\gamma=2.35745$, $\theta_{\ell'}=0.0268985$, $\phi_{\ell'}=3.14158$
 $Q^2=8.19023$, $x_B=0.346279$, $t=-0.156318$, $W^2=16.3417$, $y=0.987378$

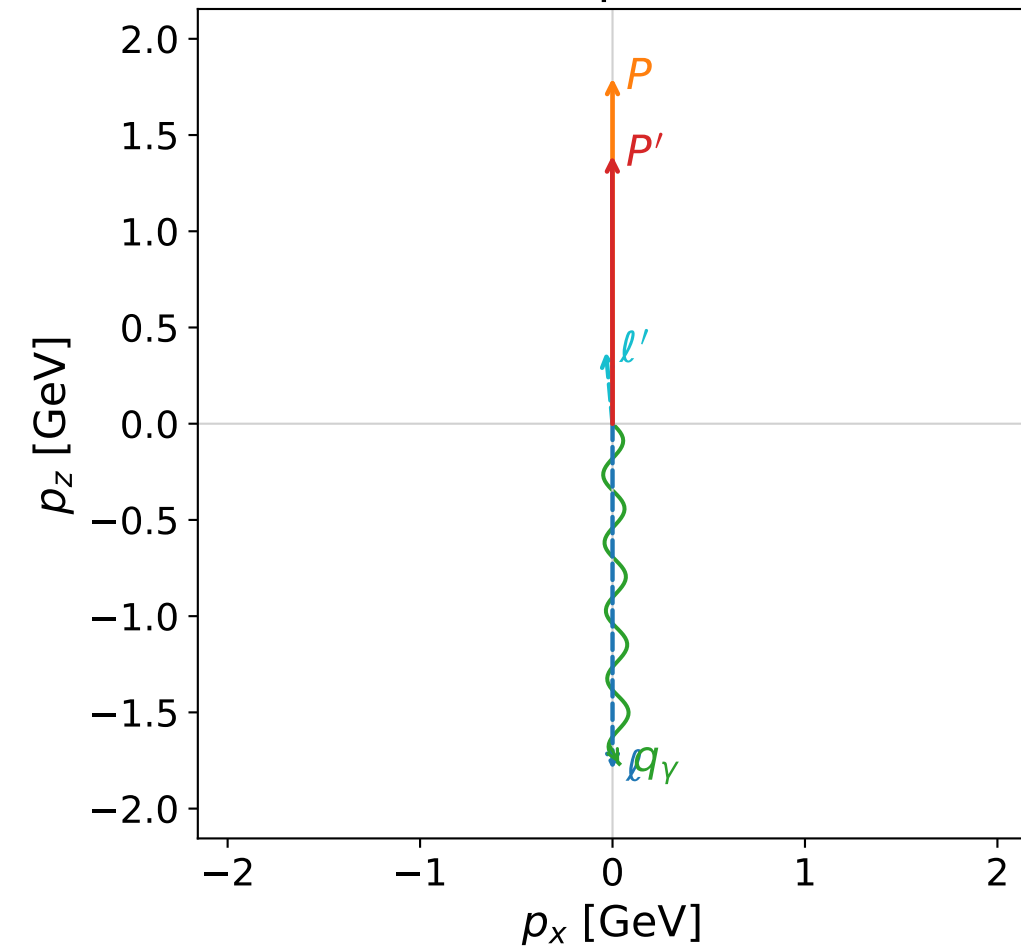
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4034$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4034$
 P $E= 2.5800$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4034$
 ℓ' $E= 0.8521$ $p_x= -0.0229$ $p_y= 0.0000$ $p_z= 0.8518$
 P' $E= 1.7739$ $p_x= 0.0009$ $p_y= -0.0000$ $p_z= 1.5056$
 q_γ $E= 2.3575$ $p_x= 0.0220$ $p_y= 0.0000$ $p_z= -2.3574$

Leading normalized final-state amplitudes
(N=2, retained=99.9%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_01_local_minimum_207

$d_{\{\rm GHZ\}}=2.62949\text{e-}05$

region: polarization_cluster_01_local_minimum_207

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0207

lepton mixing angle: $\alpha_e=1.0858098$ rad

incoming lepton: $0.466196|+\rangle + 0.884681|-\rangle$

proton mixing angle: $\alpha_p=0.47680636$ rad

incoming proton: $0.888465|+\rangle + 0.458944|-\rangle$

$s=14.6057$, $\sqrt{s}=3.82175$, $|\vec{P}|=1.79576$, $|\vec{P}'|=1.39318$

$\theta_{p'}=0.000700268$, $\theta_\gamma=3.12182$, $\phi_{p'}=3.14146$, $\phi_\gamma=1.35121\text{e-}05$

$E_\gamma=1.7671$, $\theta_{l'}=0.0906712$, $\phi_{l'}=3.14161$

$Q^2=2.68895$, $x_B=0.198481$, $t=-0.0420382$, $W^2=11.7385$, $y=0.987013$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 1.7958$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7958$

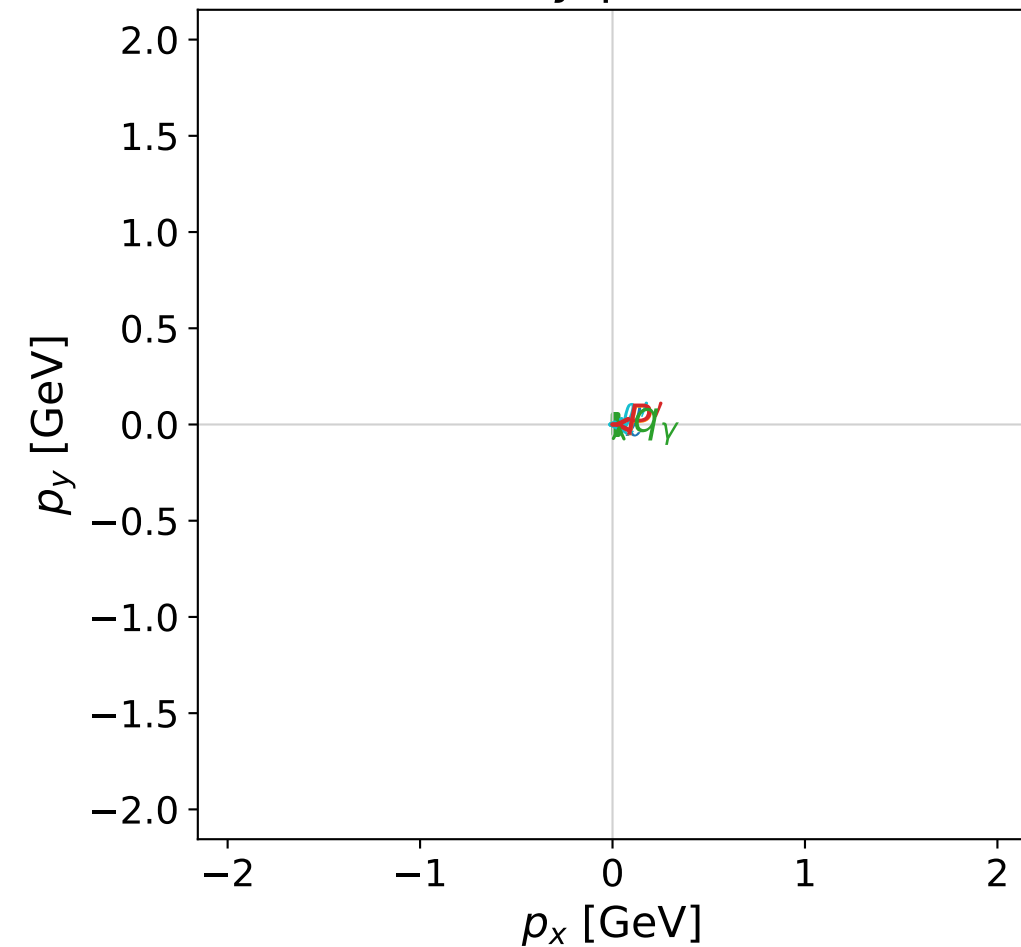
P $E= 2.0260$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7958$

l' $E= 0.3751$ $p_x= -0.0340$ $p_y= -0.0000$ $p_z= 0.3736$

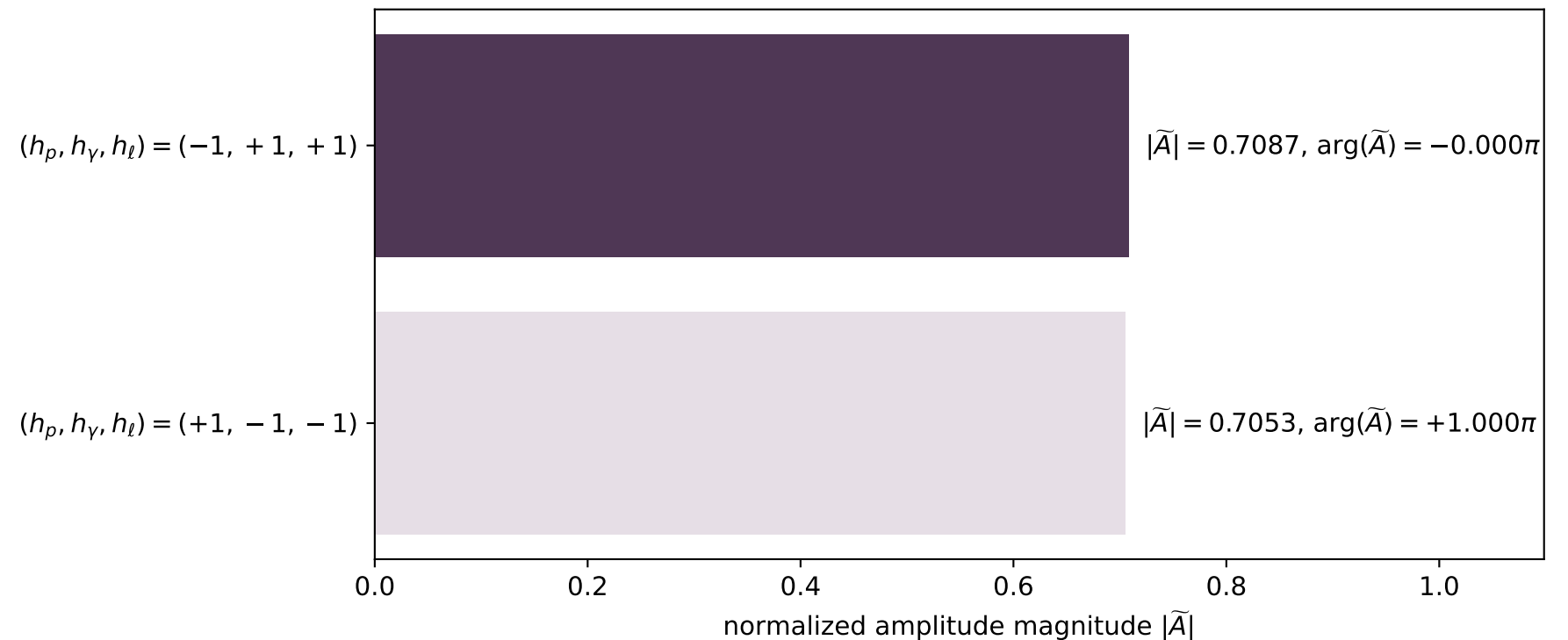
P' $E= 1.6795$ $p_x= -0.0010$ $p_y= 0.0000$ $p_z= 1.3932$

q_γ $E= 1.7671$ $p_x= 0.0349$ $p_y= 0.0000$ $p_z= -1.7668$

x--y plane

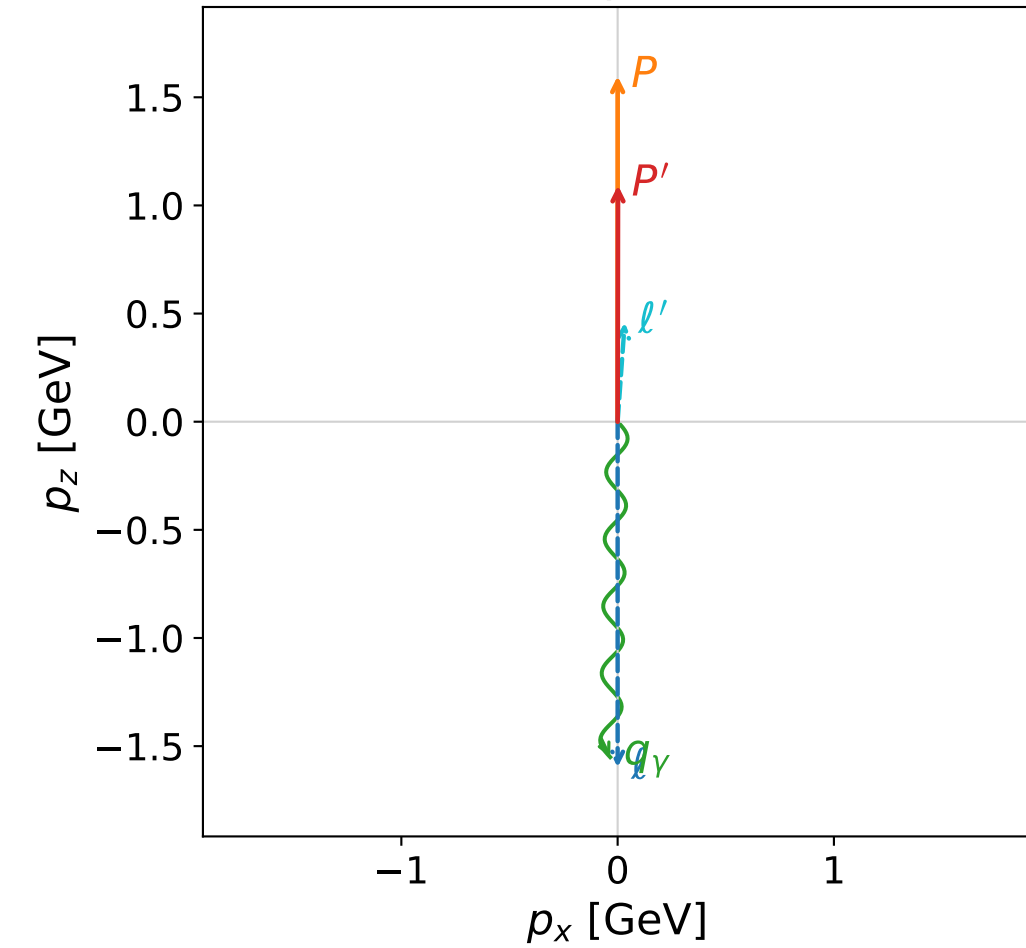


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

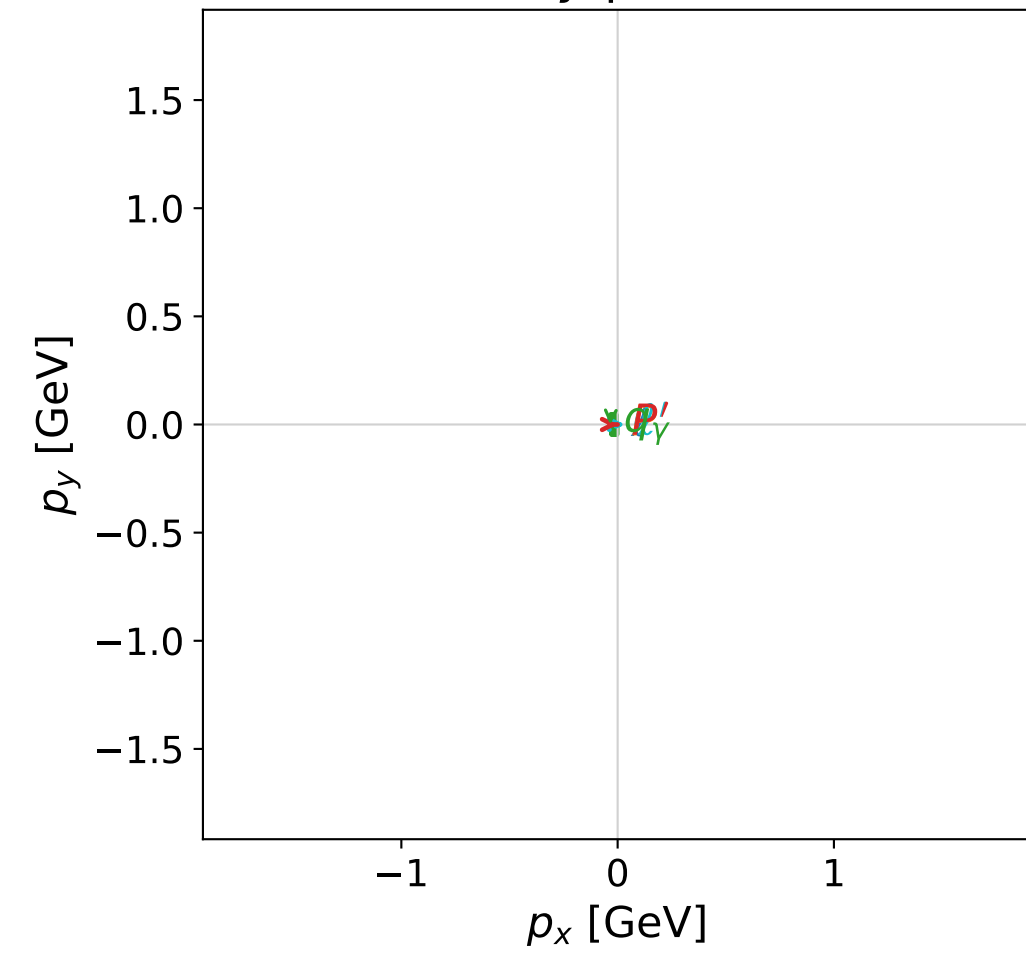


dghz_mixing_angles_polarization_01_local_minimum_216
 $d_{\text{GHZ}}=3.98745\text{e-}05$
 region: polarization_cluster_01_local_minimum_216
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0216
 lepton mixing angle: $\alpha_e=0.32329631$ rad
 incoming lepton: $0.948193|+\rangle +0.317694|-\rangle$
 proton mixing angle: $\alpha_p=1.2539312$ rad
 incoming proton: $0.311589|+\rangle +0.950217|-\rangle$

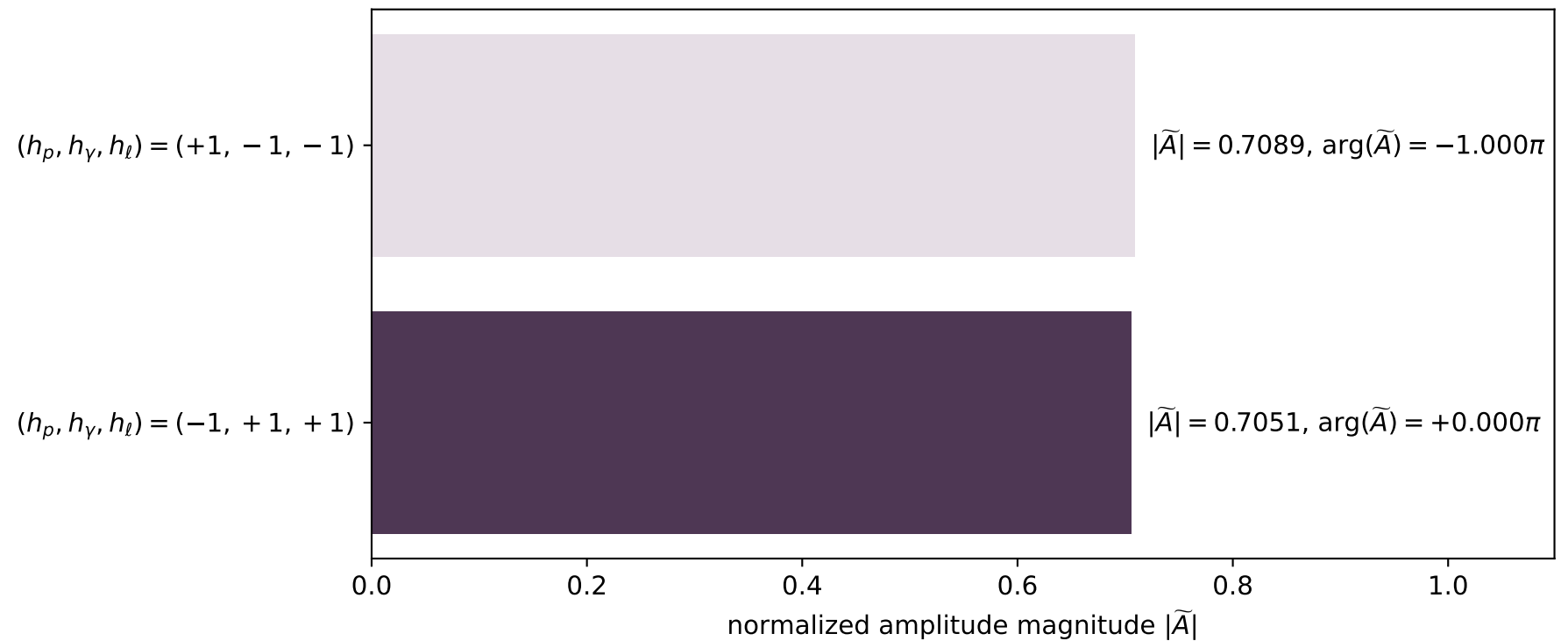
$s=11.9072$, $\sqrt{s}=3.45069$, $|\vec{P}|=1.59785$, $|\vec{P}'|=1.09272$
 $\theta_{p'}=0.00155077$, $\theta_\gamma=3.1194$, $\phi_{p'}=6.28293$, $\phi_\gamma=3.14161$
 $E_\gamma=1.55126$, $\theta_{l'}=0.0712991$, $\phi_{l'}=2.66731\text{e-}05$
 $Q^2=2.93205$, $x_B=0.271753$, $t=-0.0848181$, $W^2=8.73722$, $y=0.97842$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.5979$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5979$
 P $E= 1.8528$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5979$
 l' $E= 0.4593$ $p_x= 0.0327$ $p_y= 0.0000$ $p_z= 0.4582$
 P' $E= 1.4401$ $p_x= 0.0017$ $p_y= -0.0000$ $p_z= 1.0927$
 q_γ $E= 1.5513$ $p_x= -0.0344$ $p_y= -0.0000$ $p_z= -1.5509$

x--y plane

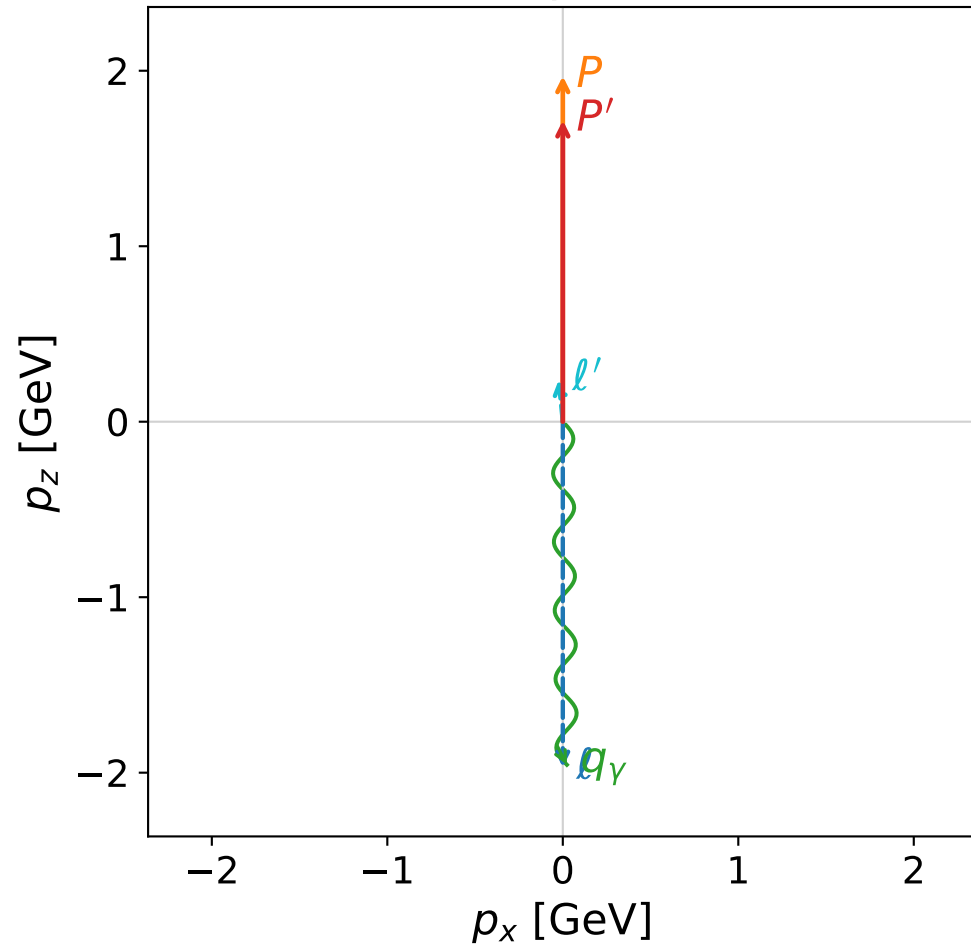


Leading normalized final-state amplitudes (N=2, retained=100.0%)

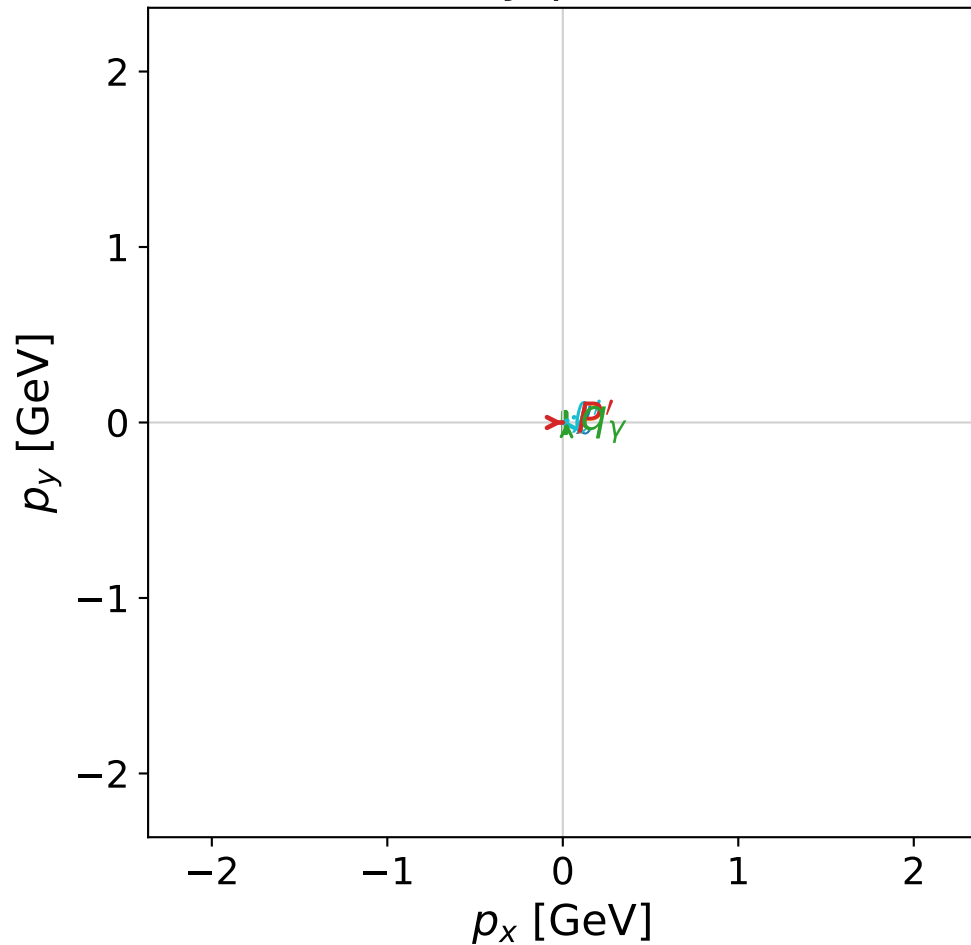


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_01_local_minimum_219

$d_{\text{GHZ}}=4.26109\text{e-}05$

region: polarization_cluster_01_local_minimum_219

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0219

lepton mixing angle: $\alpha_e=0.90729952$ rad

incoming lepton: $0.615876|+\rangle + 0.787843|-\rangle$

proton mixing angle: $\alpha_p=0.66718354$ rad

incoming proton: $0.785568|+\rangle + 0.618776|-\rangle$

$s=17.2279$, $\sqrt{s}=4.15065$, $|\vec{P}|=1.96934$, $|\vec{P}'|=1.71682$

$\theta_{P'}=0.00016275$, $\theta_Y=3.12967$, $\phi_{P'}=0.000585085$, $\phi_Y=1.82782\text{e-}05$

$E_Y=1.95504$, $\theta_{l'}=0.0987232$, $\phi_{l'}=3.14162$

$Q^2=1.88008$, $x_B=0.115754$, $t=-0.0131571$, $W^2=15.2418$, $y=0.993515$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 1.9693$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9693$

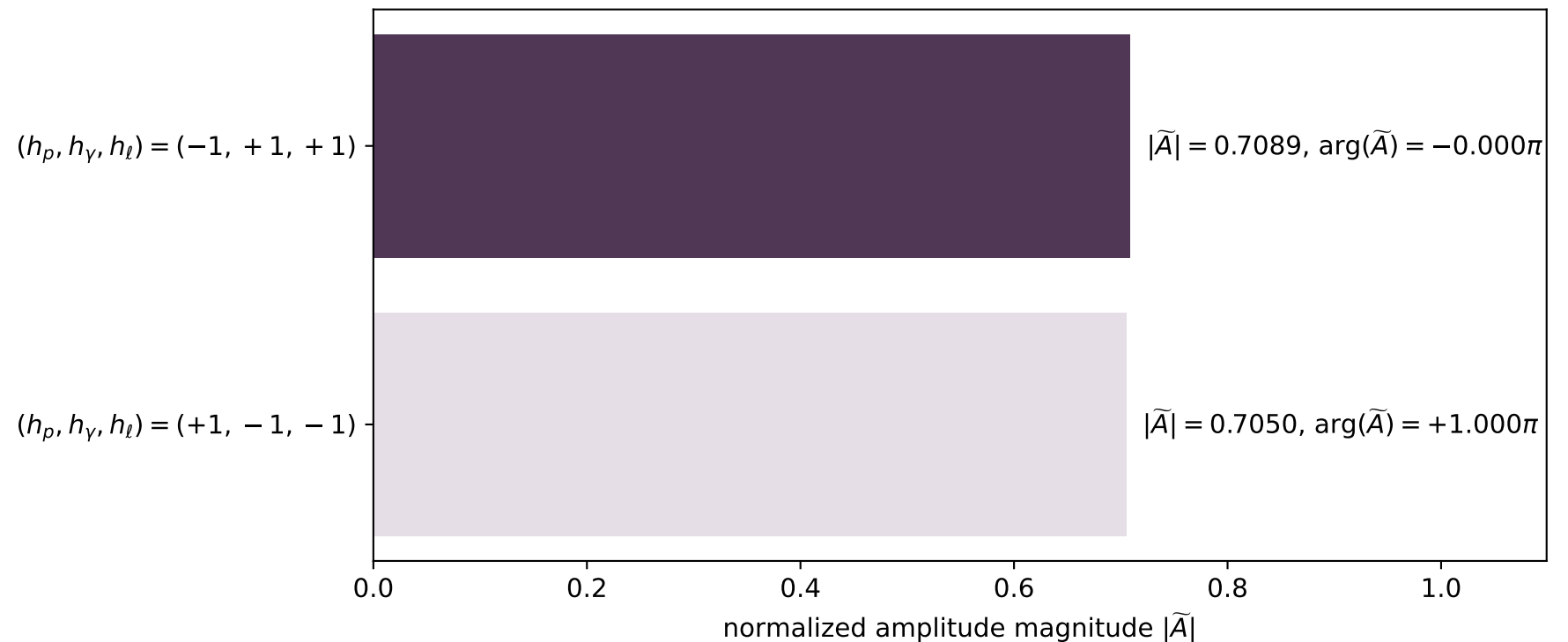
P $E= 2.1813$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9693$

ℓ' $E= 0.2393$ $p_x= -0.0236$ $p_y= -0.0000$ $p_z= 0.2381$

P' $E= 1.9564$ $p_x= 0.0003$ $p_y= 0.0000$ $p_z= 1.7168$

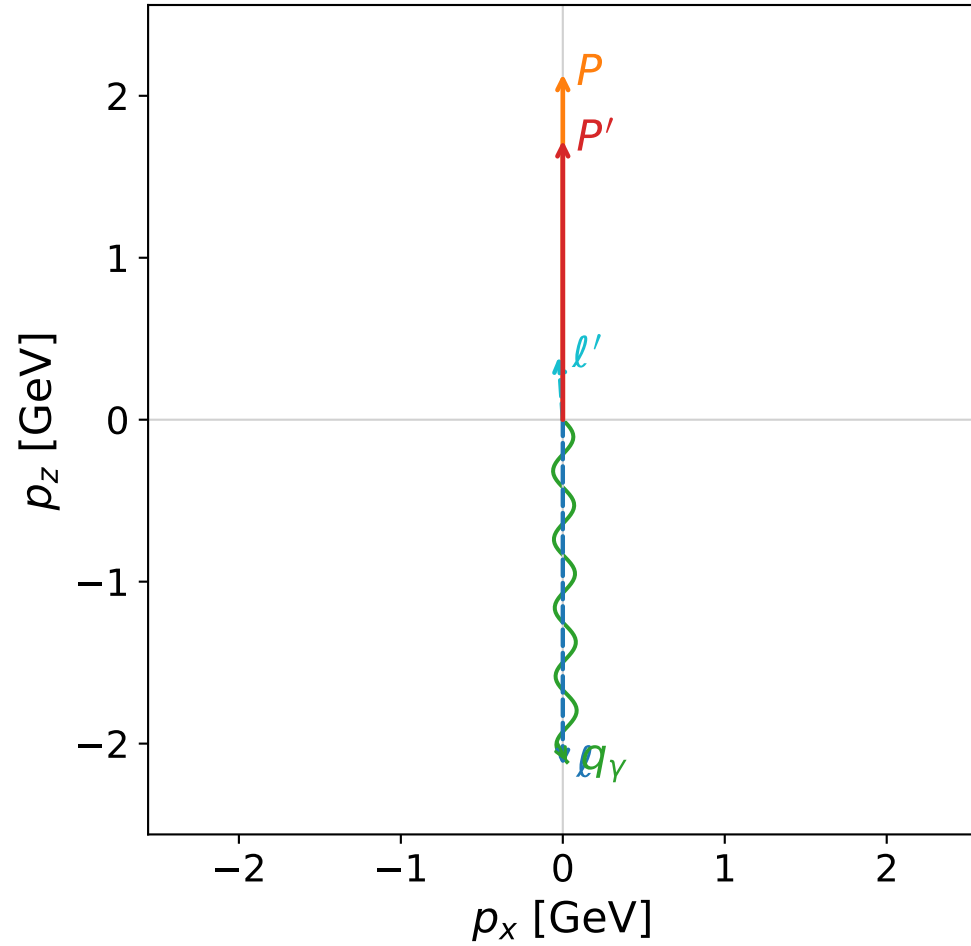
q_Y $E= 1.9550$ $p_x= 0.0233$ $p_y= 0.0000$ $p_z= -1.9549$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

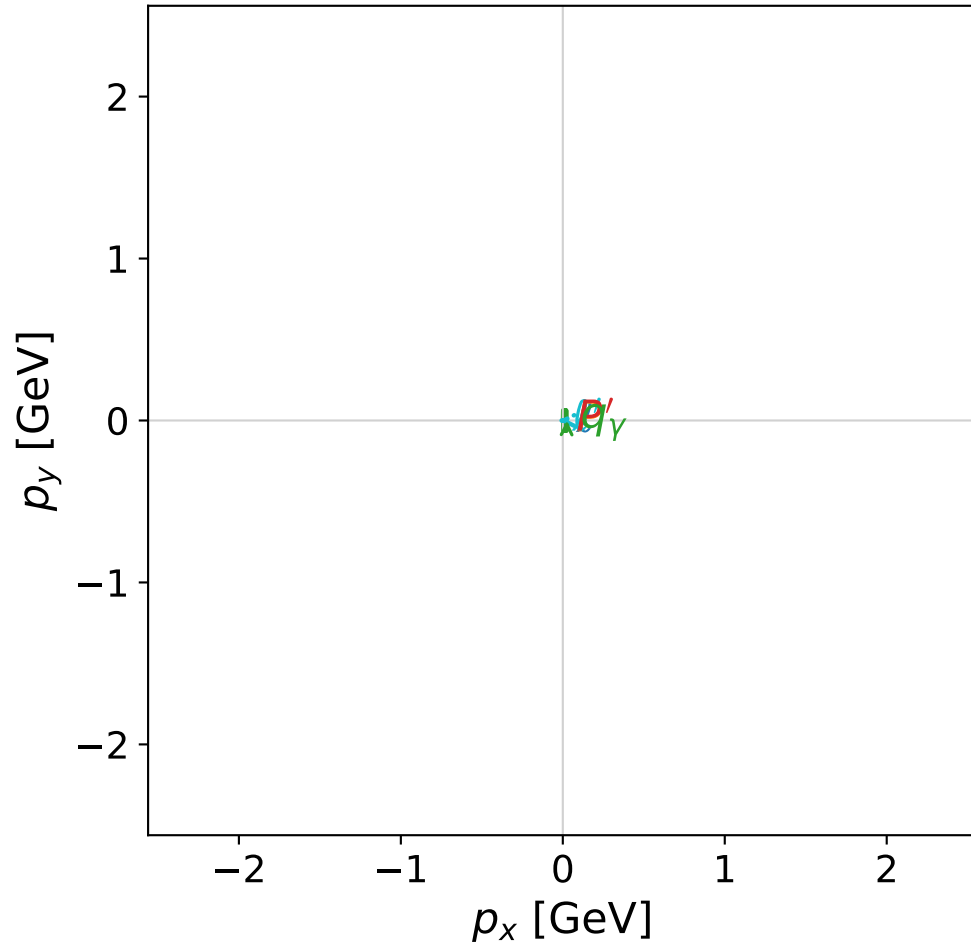


dghz_mixing_angles_polarization_01_local_minimum_231
 $d_{\{\rm GHZ\}}=5.7484\text{e-}05$
 region: polarization_cluster_01_local_minimum_231
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0231
 lepton mixing angle: $\alpha_e=0.97671817$ rad
 incoming lepton: $0.559745|+\rangle + 0.828665|-\rangle$
 proton mixing angle: $\alpha_p=0.59468799$ rad
 incoming proton: $0.828323|+\rangle + 0.56025|-\rangle$

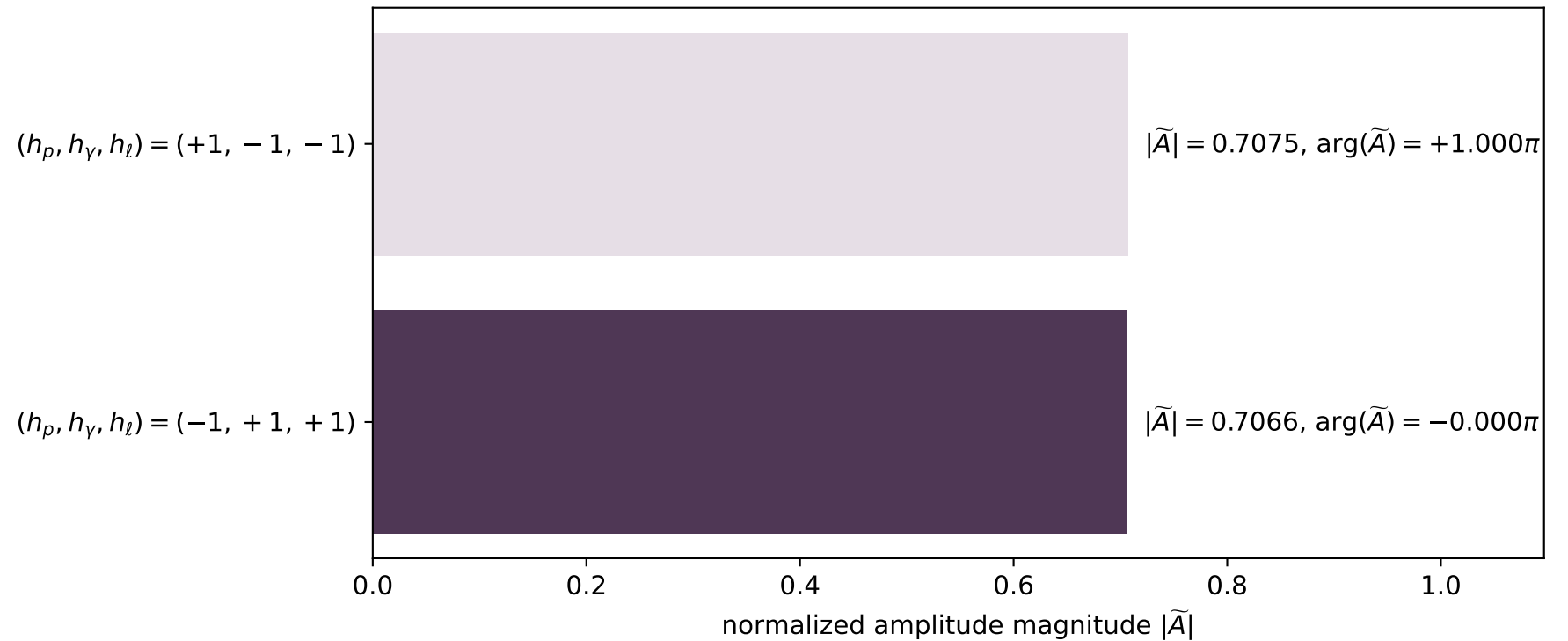
$s=19.9347$, $\sqrt{s}=4.46483$, $|\vec{P}|=2.13388$, $|\vec{P}'|=1.72462$
 $\theta_{p'}=0$, $\theta_\gamma=3.12944$, $\phi_{p'}=1.59257$, $\phi_\gamma=6.28318$
 $E_\gamma=2.11278$, $\theta_{l'}=0.066056$, $\phi_{l'}=3.14158$
 $Q^2=3.31544$, $x_B=0.175439$, $t=-0.0322616$, $W^2=16.4624$, $y=0.991767$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1339$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1339$
 P $E= 2.3309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1339$
 l' $E= 0.3889$ $p_x= -0.0257$ $p_y= 0.0000$ $p_z= 0.3880$
 P' $E= 1.9632$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7246$
 q_γ $E= 2.1128$ $p_x= 0.0257$ $p_y= -0.0000$ $p_z= -2.1126$

x--y plane

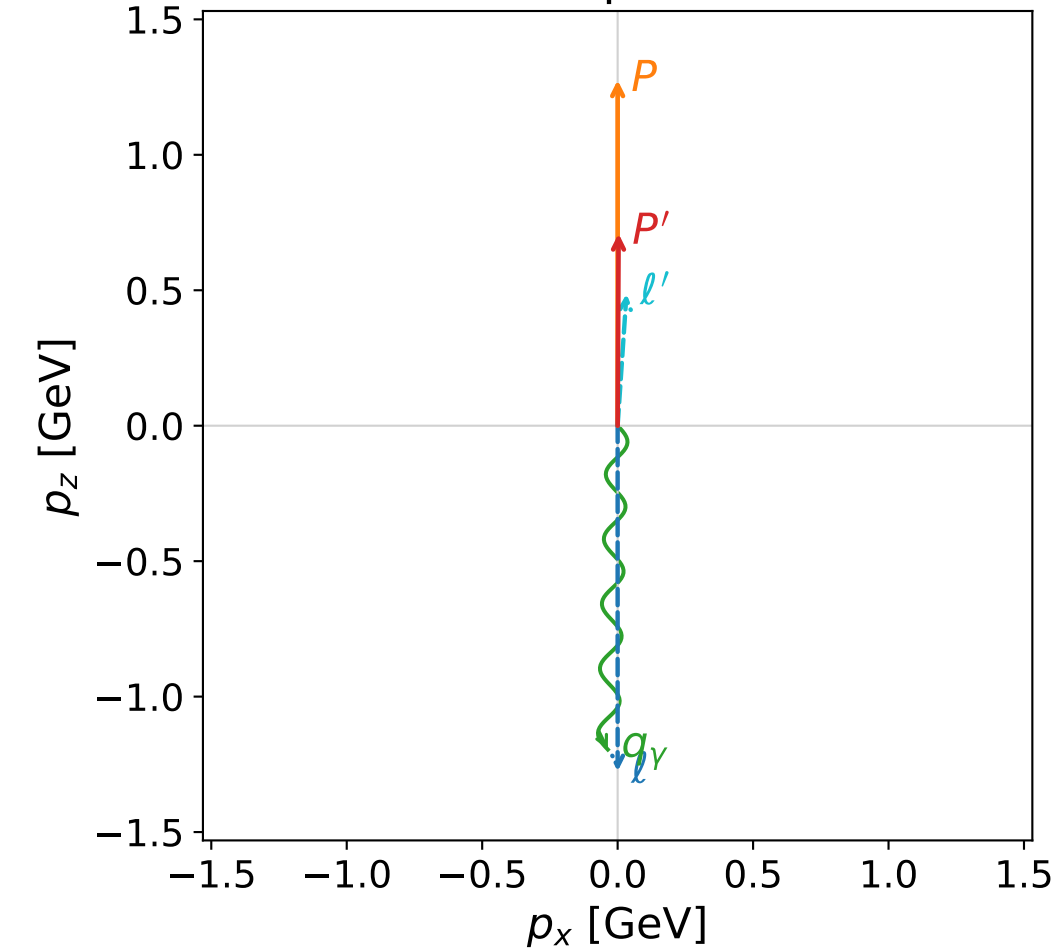


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



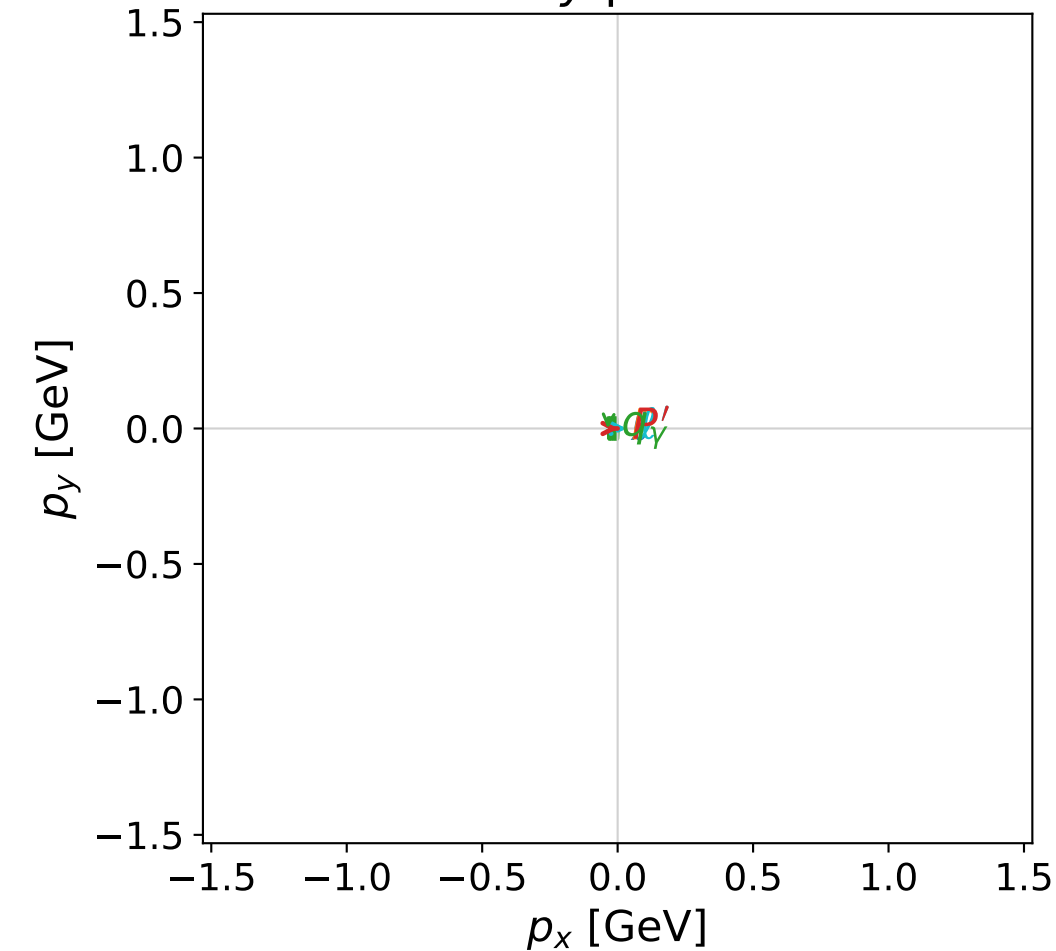
dghz_mixing_angles_polarization_01_local_minimum_236
 $d_{\text{GHZ}}=6.73836\text{e-}05$
 region: polarization_cluster_01_local_minimum_236
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0236
 lepton mixing angle: $\alpha_e=0.17076605$ rad
 incoming lepton: $0.985455|+\rangle +0.169937|-\rangle$
 proton mixing angle: $\alpha_p=1.4059166$ rad
 incoming proton: $0.164134|+\rangle +0.986438|-\rangle$

$s=8.17465$, $\sqrt{s}=2.85913$, $|\vec{P}|=1.2757$, $|\vec{P}'|=0.709145$
 $\theta_{P'}=0.00496652$, $\theta_\gamma=3.11136$, $\phi_{P'}=6.28315$, $\phi_\gamma=3.14159$
 $E_\gamma=1.19591$, $\theta_{l'}=0.0670059$, $\phi_{l'}=4.64873\text{e-}06$
 $Q^2=2.48393$, $x_B=0.355249$, $t=-0.154922$, $W^2=5.38799$, $y=0.958502$

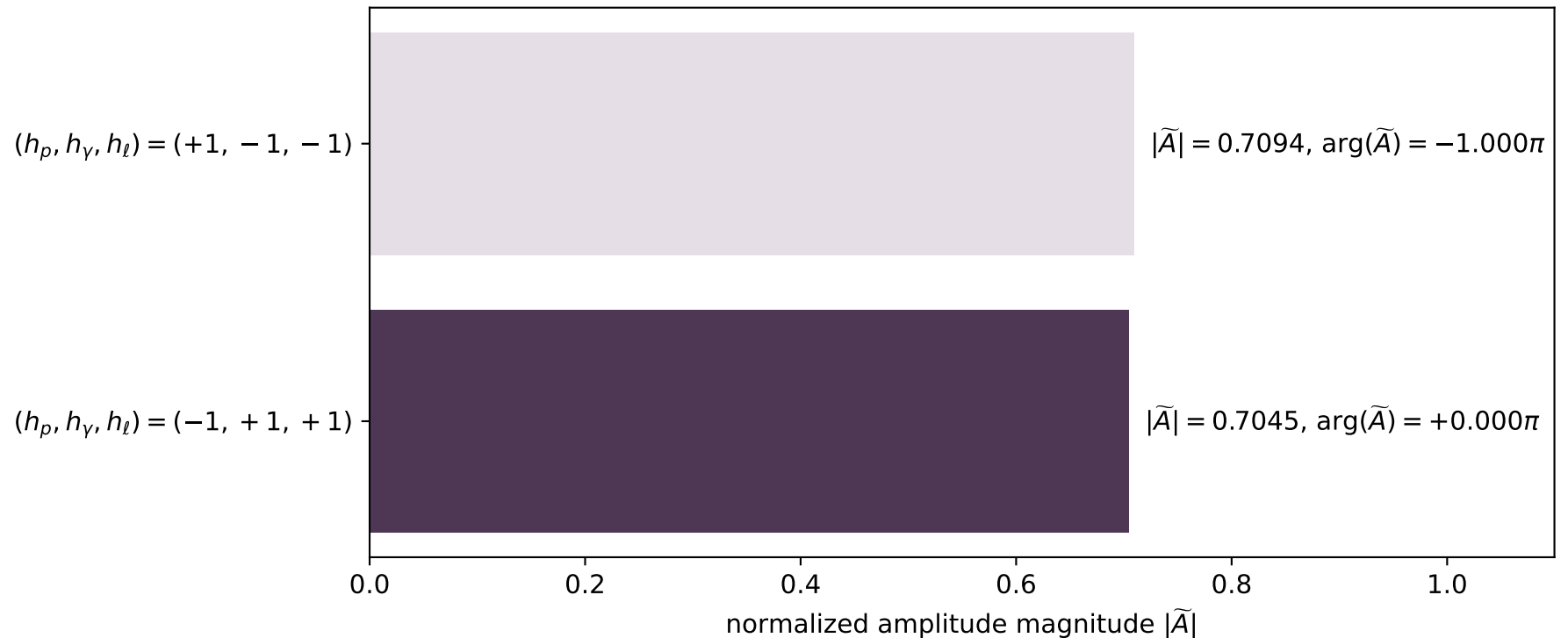
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l	$E=$	1.2757	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-1.2757
P	$E=$	1.5834	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	1.2757
l'	$E=$	0.4873	$p_x=$	0.0326	$p_y=$	0.0000	$p_z=$	0.4862
P'	$E=$	1.1759	$p_x=$	0.0035	$p_y=$	-0.0000	$p_z=$	0.7091
q_γ	$E=$	1.1959	$p_x=$	-0.0362	$p_y=$	-0.0000	$p_z=$	-1.1954

x--y plane

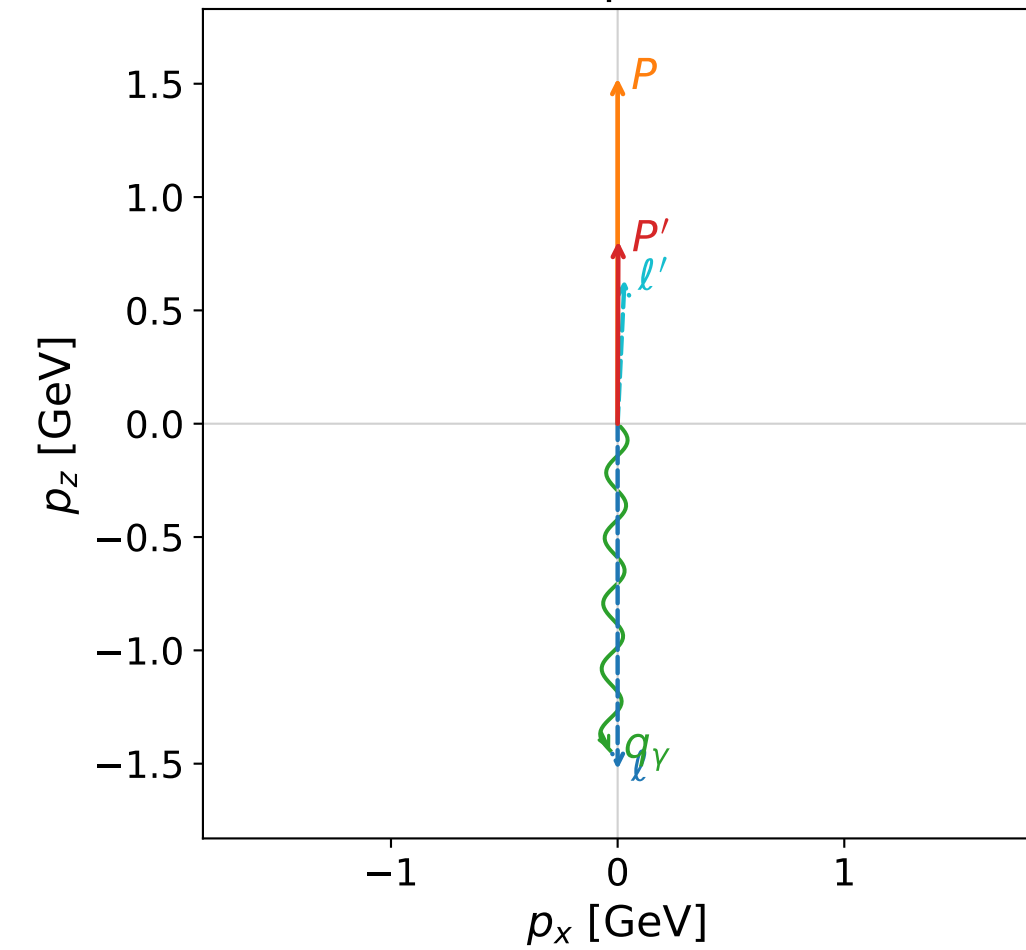


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



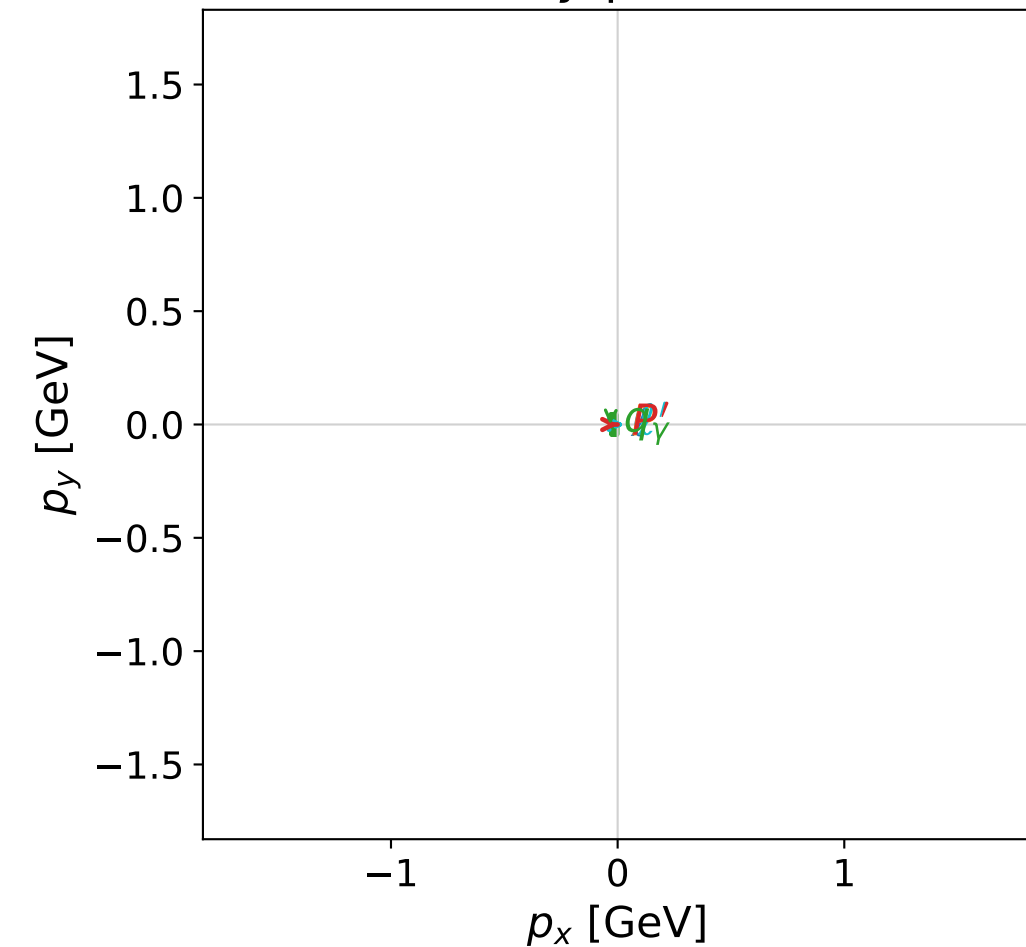
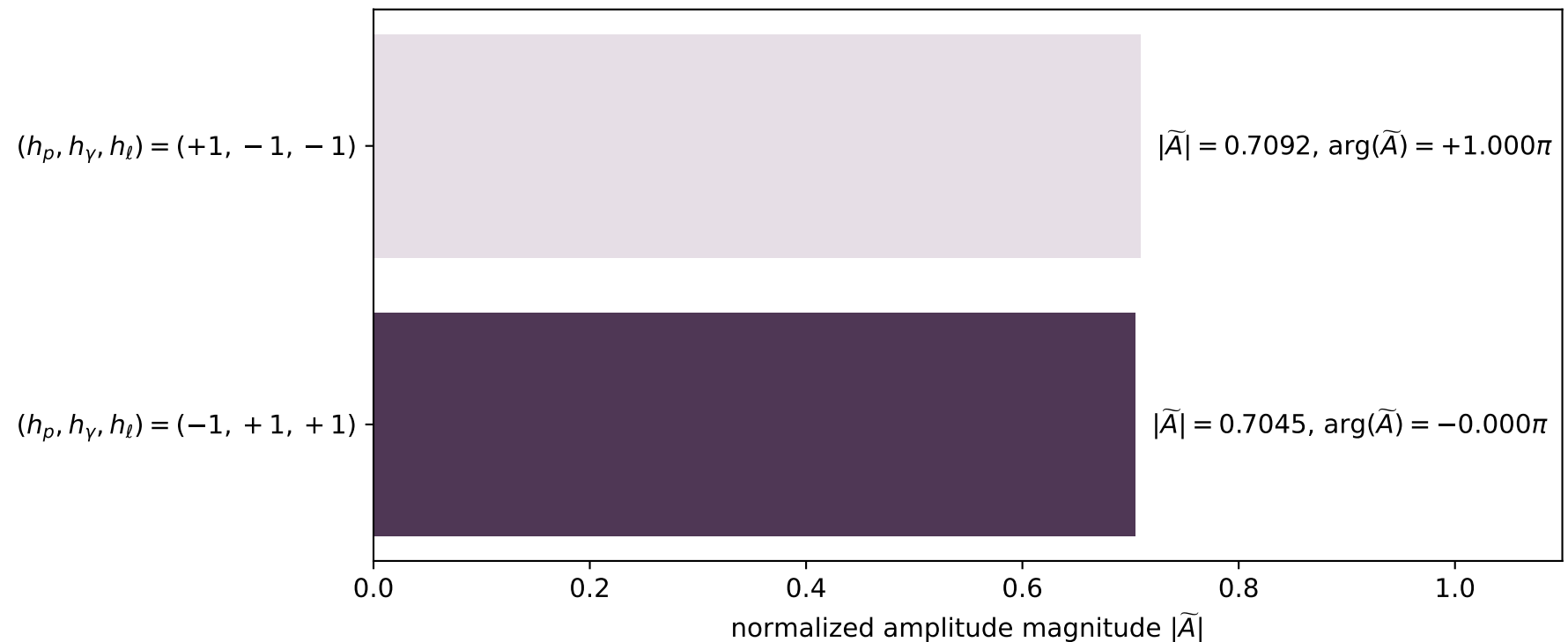
dghz_mixing_angles_polarization_01_local_minimum_241
 $d_{\text{GHZ}}=8.98184\text{e-}05$
 region: polarization_cluster_01_local_minimum_241
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0241
 lepton mixing angle: $\alpha_e=0.17233494$ rad
 incoming lepton: $0.985187|+\rangle +0.171483|-\rangle$
 proton mixing angle: $\alpha_p=1.4019813$ rad
 incoming proton: $0.168014|+\rangle +0.985785|-\rangle$

$s=10.9892$, $\sqrt{s}=3.31499$, $|\vec{P}|=1.52479$, $|\vec{P}'|=0.805866$
 $\theta_{p'}=0.0033848$, $\theta_\gamma=3.11869$, $\phi_{p'}=0.000438679$, $\phi_\gamma=3.14159$
 $E_\gamma=1.44194$, $\theta_{l'}=0.0476231$, $\phi_{l'}=6.28315$
 $Q^2=3.87942$, $x_B=0.397103$, $t=-0.210426$, $W^2=6.76972$, $y=0.966365$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

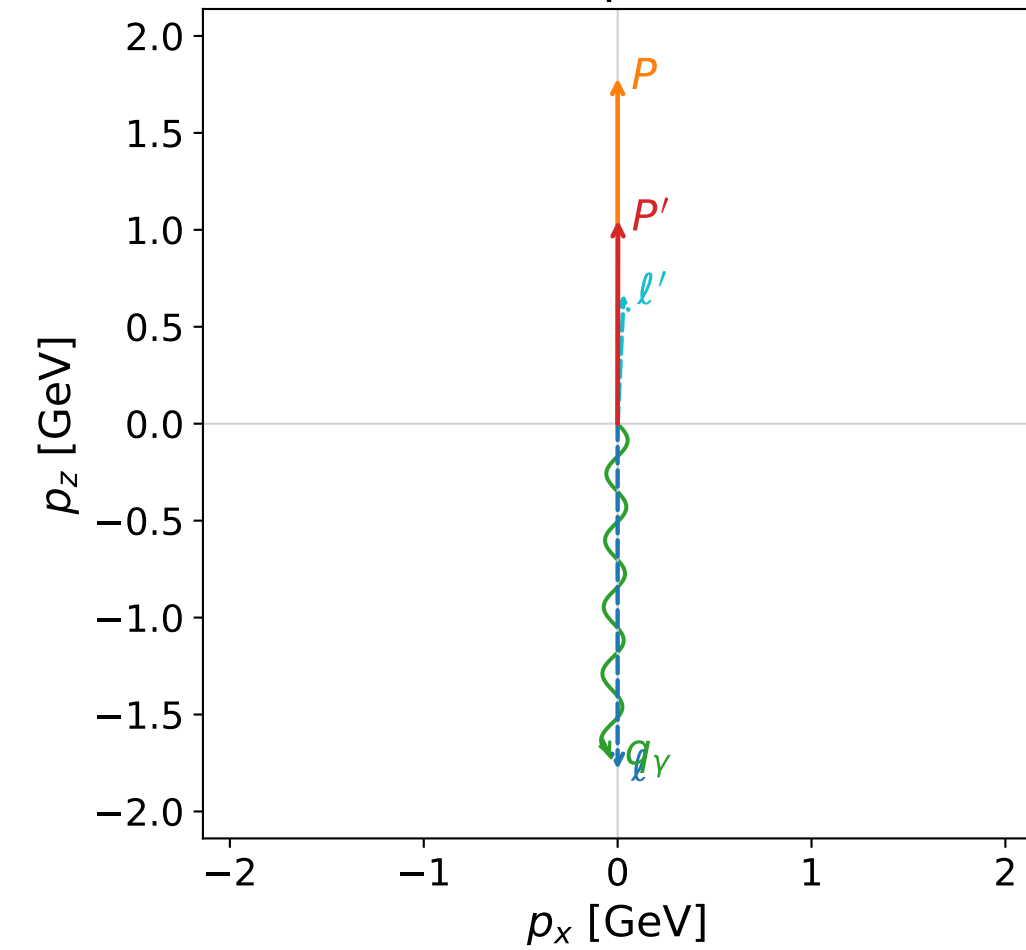
Particle	E	p_x	p_y	p_z
l	1.5248	0.0000	0.0000	-1.5248
P	1.7902	0.0000	0.0000	1.5248
l'	0.6364	0.0303	-0.0000	0.6357
P'	1.2366	0.0027	0.0000	0.8059
q_γ	1.4419	-0.0330	-0.0000	-1.4416

x--y plane

Leading normalized final-state amplitudes
(N=2, retained=99.9%)

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

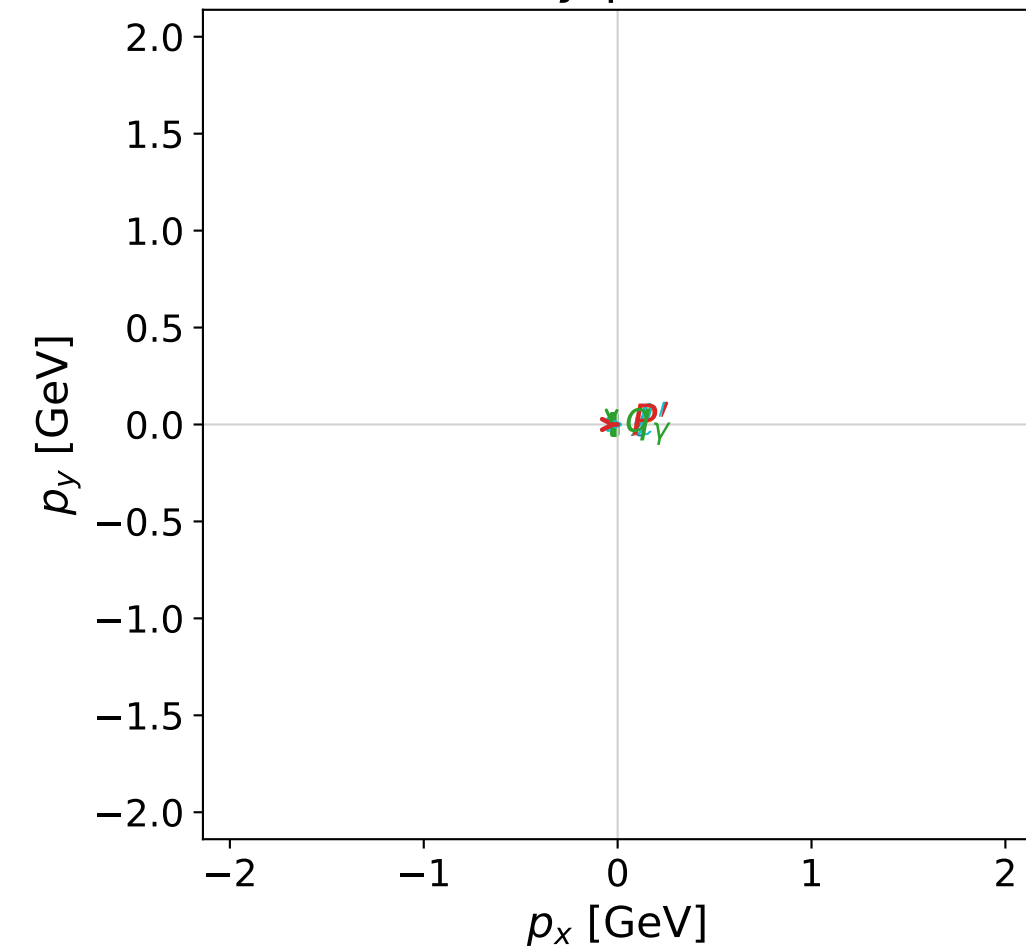


dghz_mixing_angles_polarization_01_local_minimum_242
 $d_{\text{GHZ}}=0.000102039$
 region: polarization_cluster_01_local_minimum_242
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0242
 lepton mixing angle: $\alpha_e=0.24367421$ rad
 incoming lepton: $0.970458|+\rangle +0.24127|-\rangle$
 proton mixing angle: $\alpha_p=1.3312341$ rad
 incoming proton: $0.237277|+\rangle +0.971442|-\rangle$

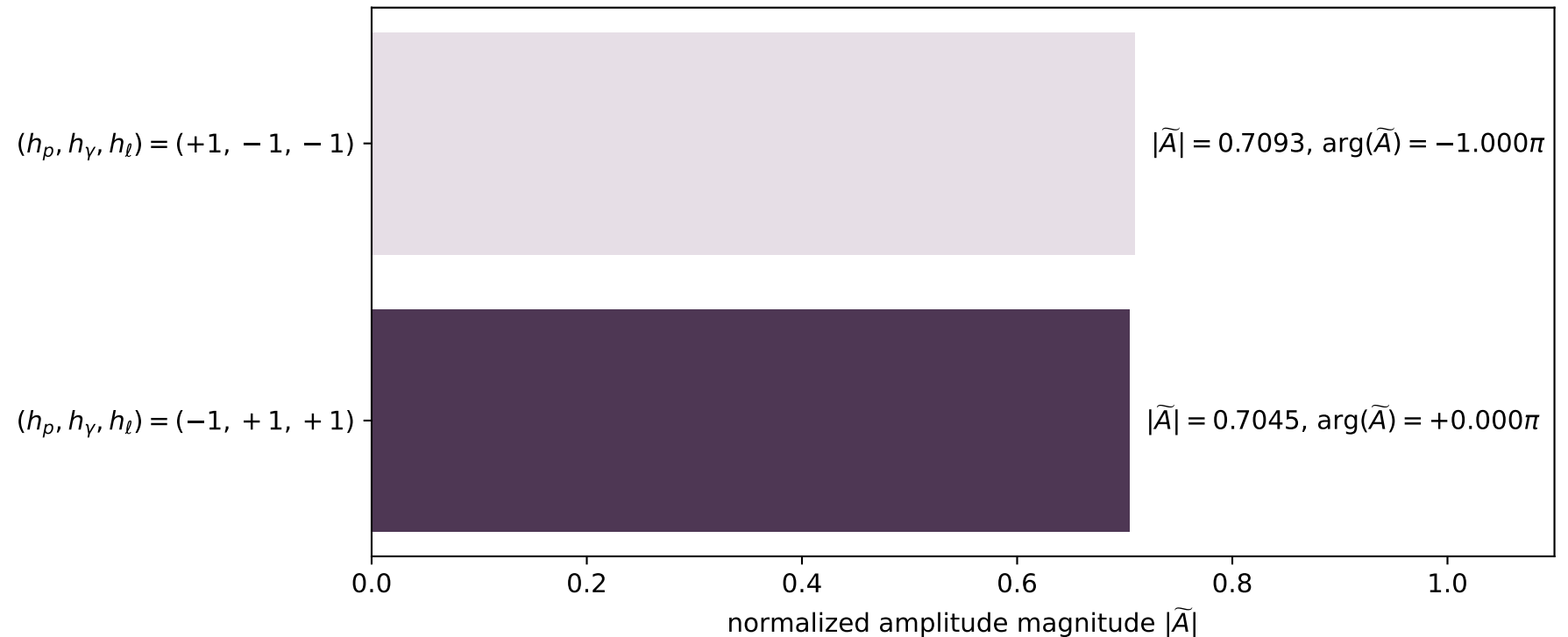
$s=14.4155$, $\sqrt{s}=3.79678$, $|\vec{P}|=1.78252$, $|\vec{P}'|=1.0492$
 $\theta_{P'}=0.00202721$, $\theta_\gamma=3.12202$, $\phi_{P'}=6.28087$, $\phi_\gamma=3.14159$
 $E_\gamma=1.7191$, $\theta_{l'}=0.0470478$, $\phi_{l'}=0.000148497$
 $Q^2=4.77679$, $x_B=0.361266$, $t=-0.169449$, $W^2=9.32542$, $y=0.976852$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7825$
 P $E= 2.0143$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7825$
 l' $E= 0.6703$ $p_x= 0.0315$ $p_y= 0.0000$ $p_z= 0.6696$
 P' $E= 1.4074$ $p_x= 0.0021$ $p_y= -0.0000$ $p_z= 1.0492$
 q_γ $E= 1.7191$ $p_x= -0.0337$ $p_y= 0.0000$ $p_z= -1.7188$

x--y plane

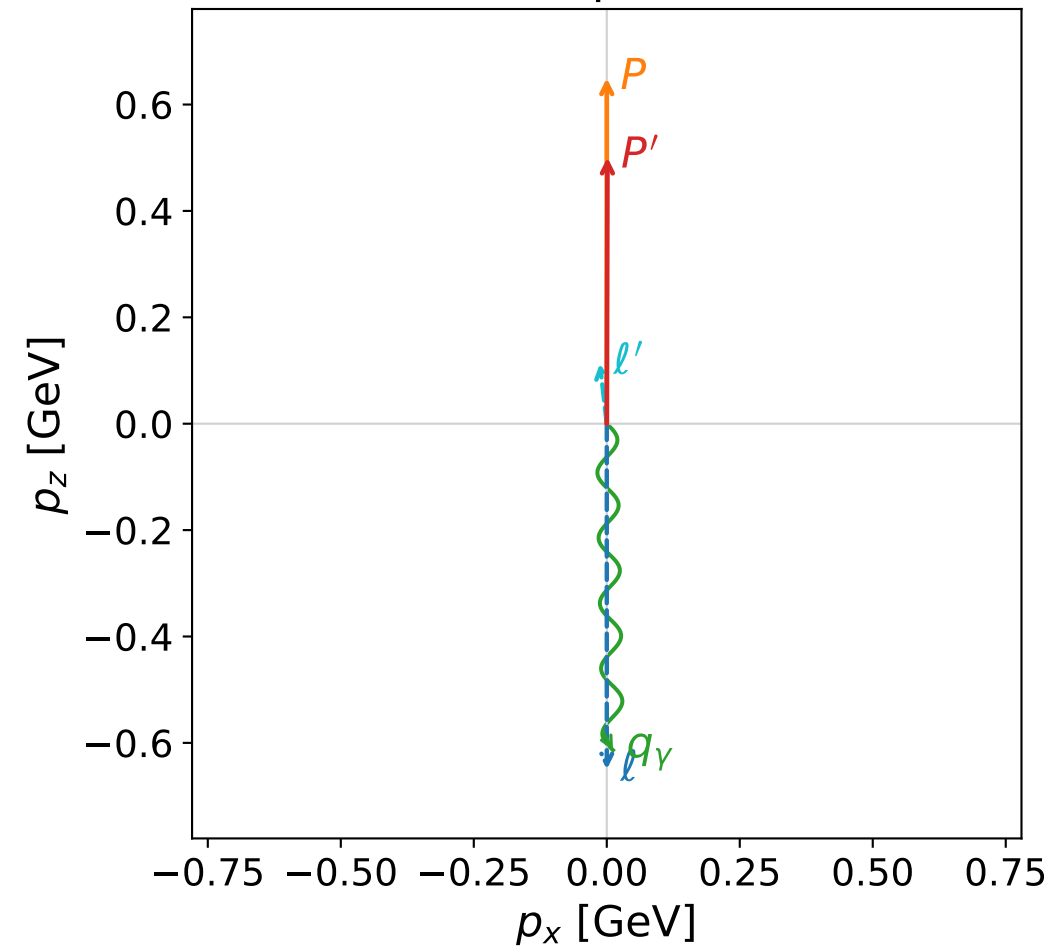


Leading normalized final-state amplitudes
 (N=2, retained=99.9%)

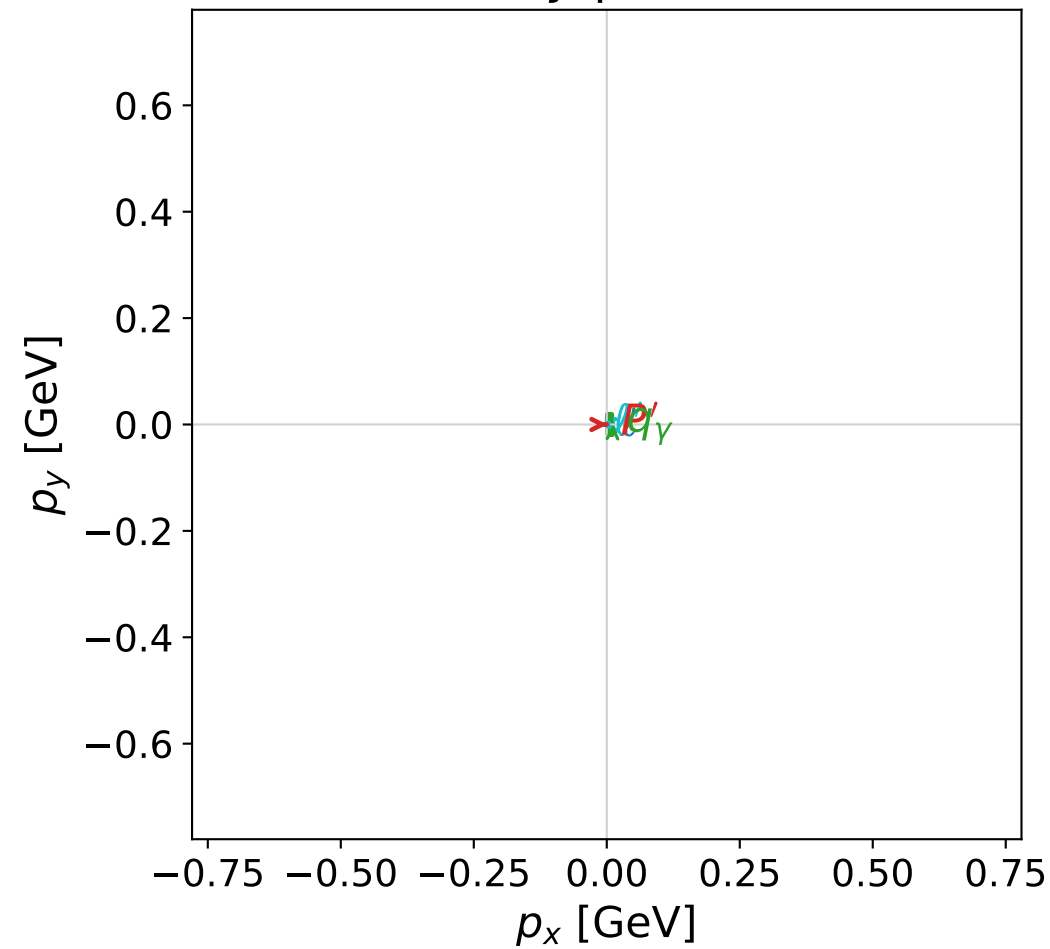


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

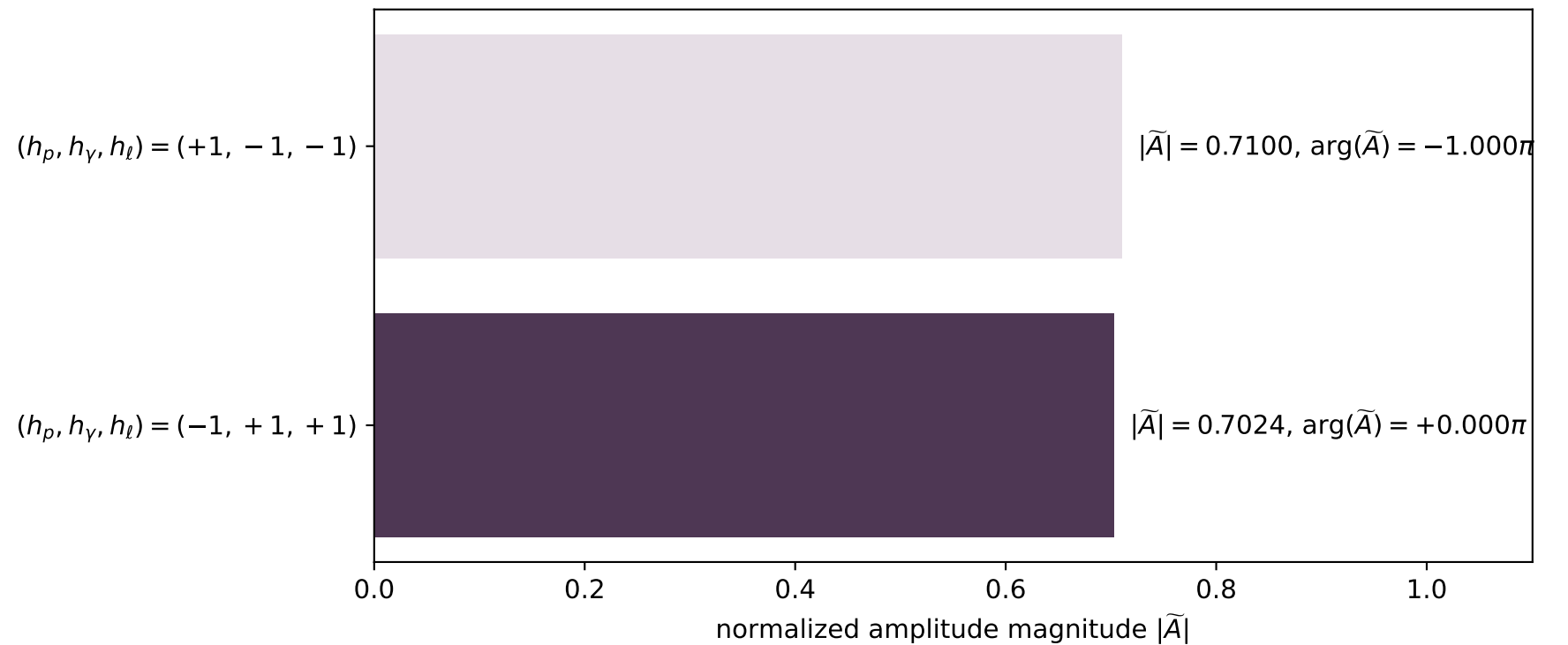


dghz_mixing_angles_polarization_01_local_minimum_249
 $d_{\text{GHZ}}=0.000132871$
 region: polarization_cluster_01_local_minimum_249
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0249
 lepton mixing angle: $\alpha_e=1.1524667$ rad
 incoming lepton: $0.406235|+\rangle +0.913769|-\rangle$
 proton mixing angle: $\alpha_p=0.44527808$ rad
 incoming proton: $0.902491|+\rangle +0.430709|-\rangle$

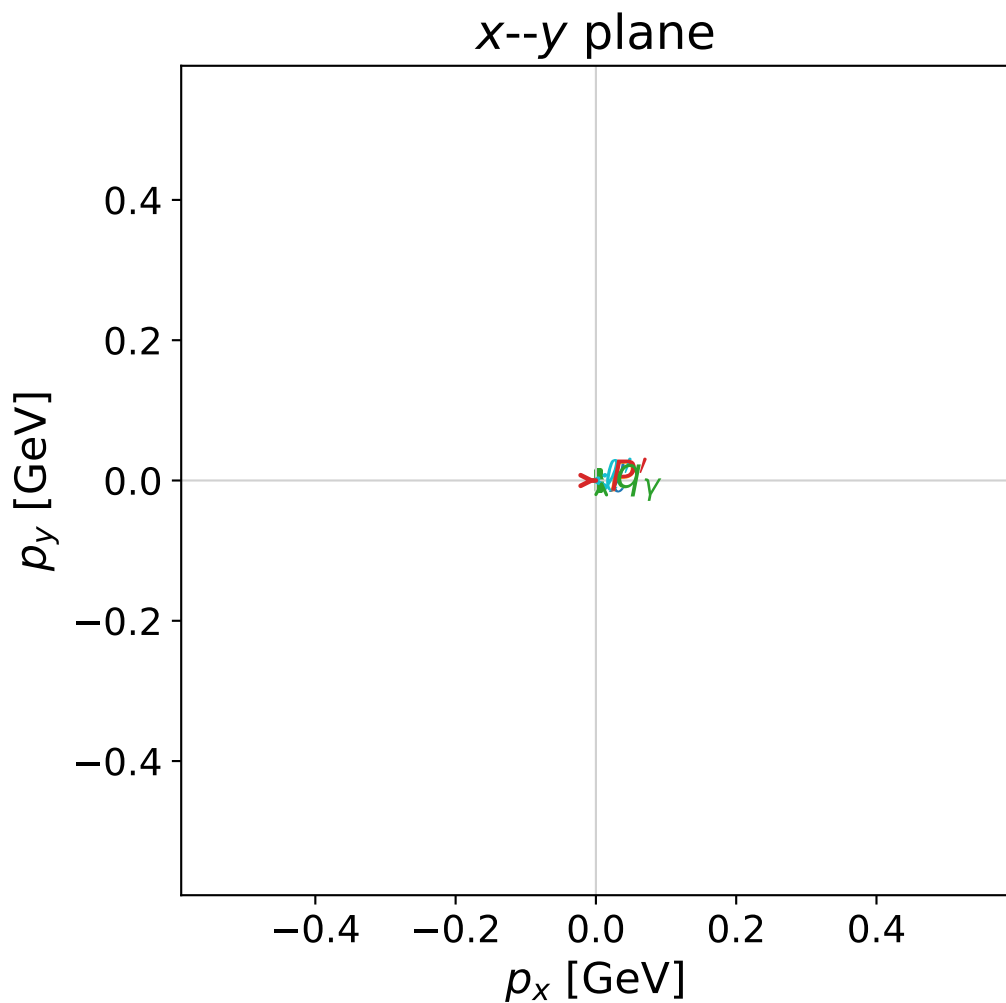
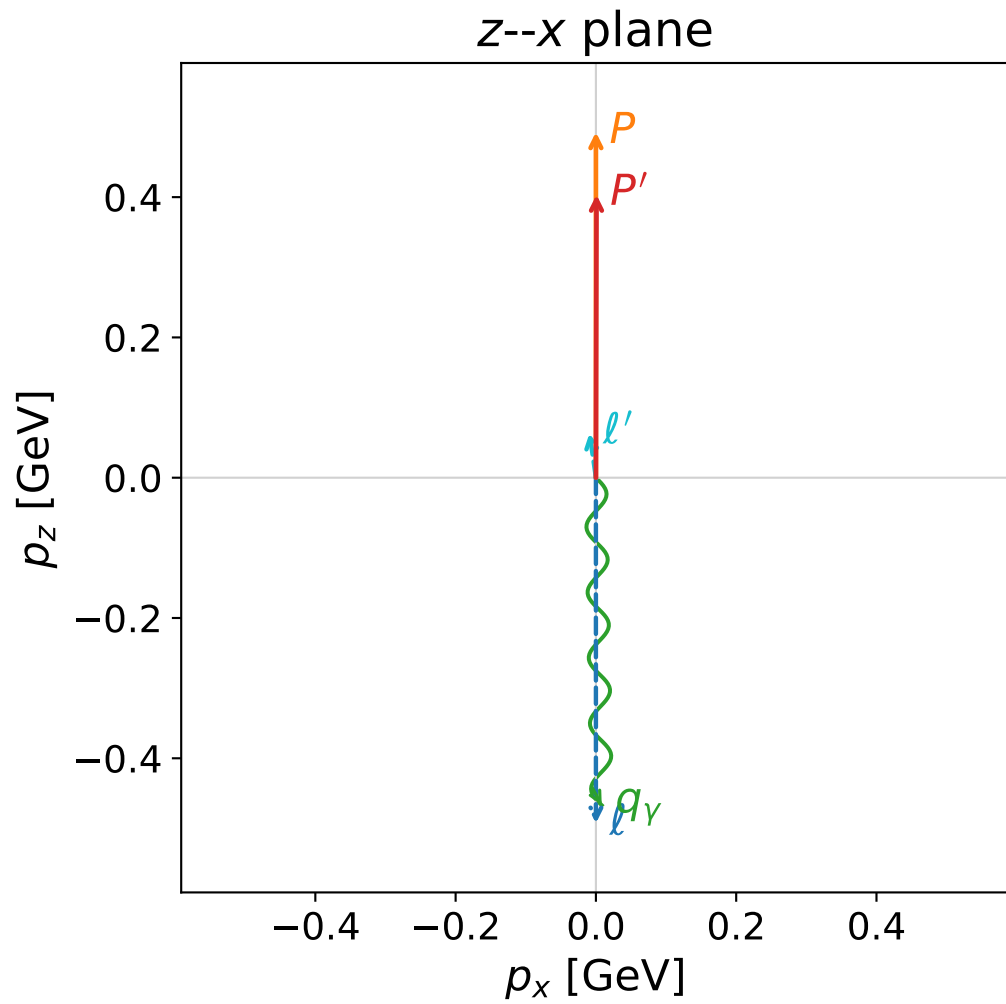
$s=3.20651$, $\sqrt{s}=1.79067$, $|\vec{P}|=0.649662$, $|\vec{P}'|=0.500629$
 $\theta_{P'}=0.0025126$, $\theta_Y=3.12259$, $\phi_{P'}=6.28317$, $\phi_Y=6.28316$
 $E_Y=0.613719$, $\theta_{\ell'}=0.113834$, $\phi_{\ell'}=3.14157$
 $Q^2=0.294553$, $x_B=0.133043$, $t=-0.0161642$, $W^2=2.79925$, $y=0.951558$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.6497$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.6497$
 P $E=1.1410$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.6497$
 ℓ' $E=0.1137$ $p_x=-0.0129$ $p_y=0.0000$ $p_z=0.1130$
 P' $E=1.0632$ $p_x=0.0013$ $p_y=-0.0000$ $p_z=0.5006$
 q_Y $E=0.6137$ $p_x=0.0117$ $p_y=-0.0000$ $p_z=-0.6136$

Leading normalized final-state amplitudes
(N=2, retained=99.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

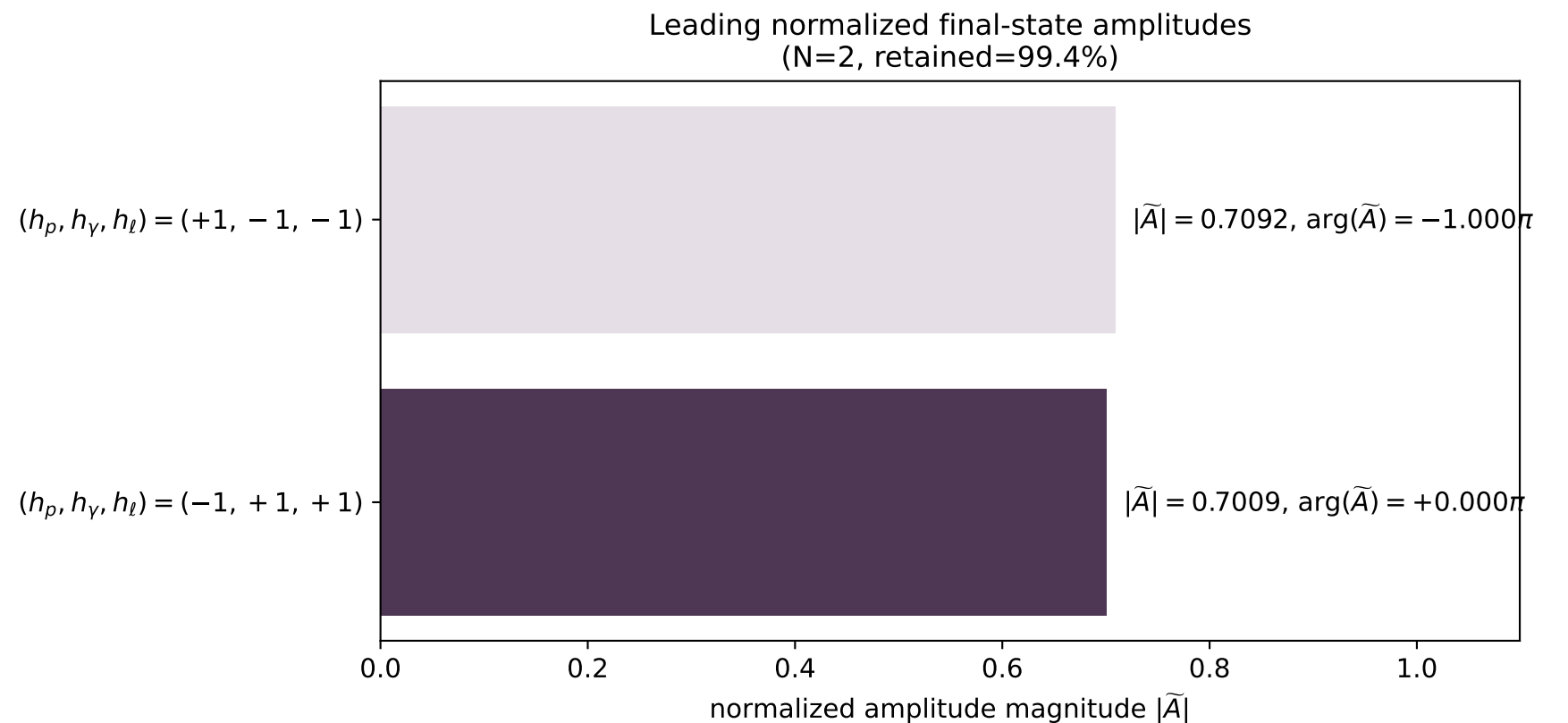


dghz_mixing_angles_polarization_01_local_minimum_253
 $d_{\{\rm GHZ\}}=0.000141867$
 region: polarization_cluster_01_local_minimum_253
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0253
 lepton mixing angle: $\alpha_e=1.0645582$ rad
 incoming lepton: $0.484891|+\rangle +0.874575|-\rangle$
 proton mixing angle: $\alpha_p=0.55250545$ rad
 incoming proton: $0.851212|+\rangle +0.524822|-\rangle$

$s=2.40938$, $\sqrt{s}=1.55222$, $|\vec{P}|=0.492693$, $|\vec{P}'|=0.403492$
 $\theta_{P'}=0.00213162$, $\theta_Y=3.12381$, $\phi_{P'}=6.28318$, $\phi_Y=2.67404e-05$
 $E_Y=0.46701$, $\theta_{\ell'}=0.143434$, $\phi_{\ell'}=3.14162$
 $Q^2=0.125686$, $x_B=0.0863101$, $t=-0.00648154$, $W^2=2.21037$, $y=0.952061$

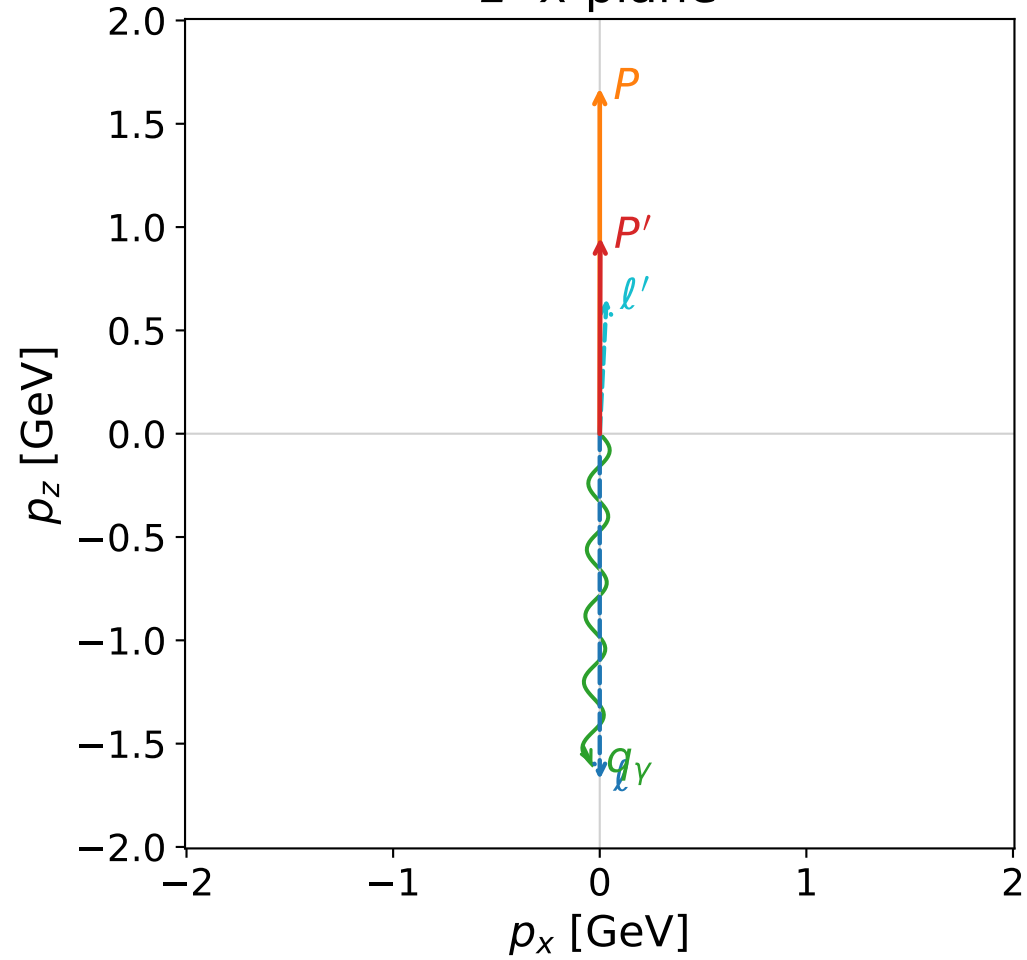
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.4927	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.4927
P	$E=$	1.0595	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.4927
ℓ'	$E=$	0.0641	$p_x=$	-0.0092	$p_y=$	-0.0000	$p_z=$	0.0634
P'	$E=$	1.0211	$p_x=$	0.0009	$p_y=$	-0.0000	$p_z=$	0.4035
q_Y	$E=$	0.4670	$p_x=$	0.0083	$p_y=$	0.0000	$p_z=$	-0.4669

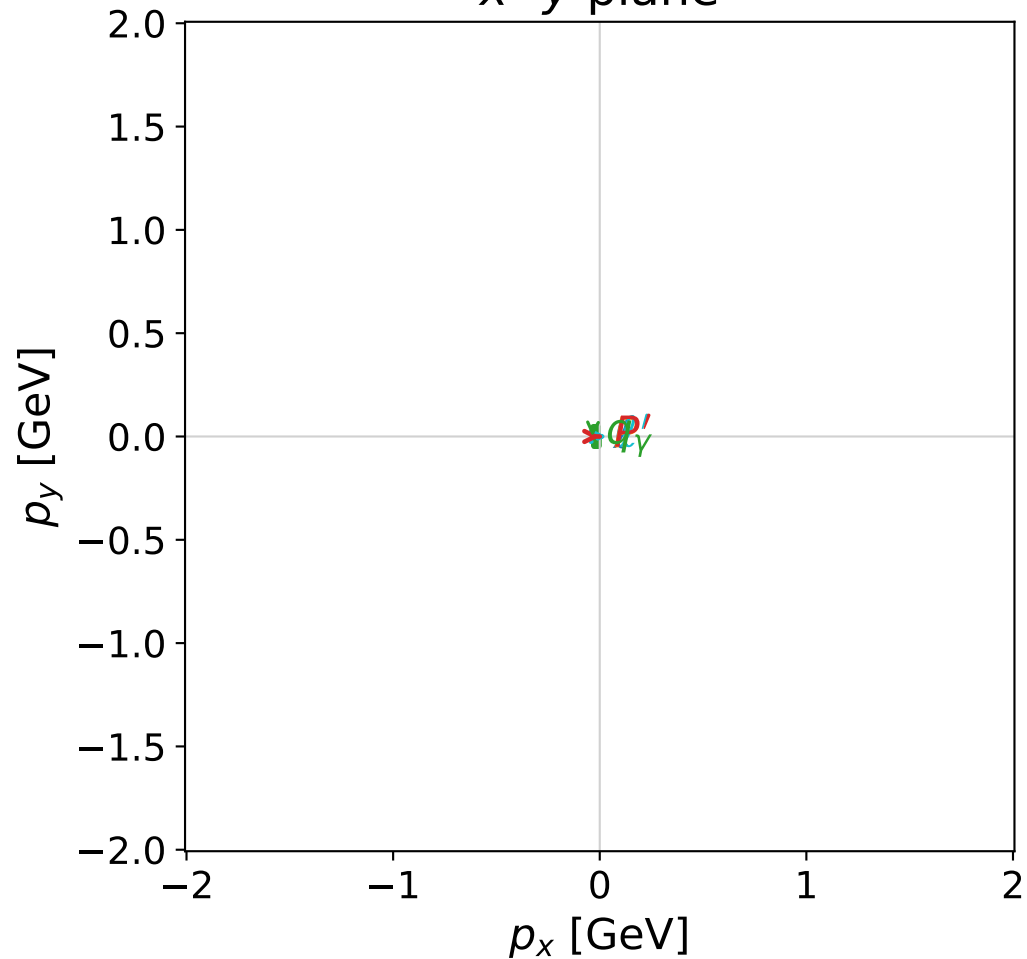


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



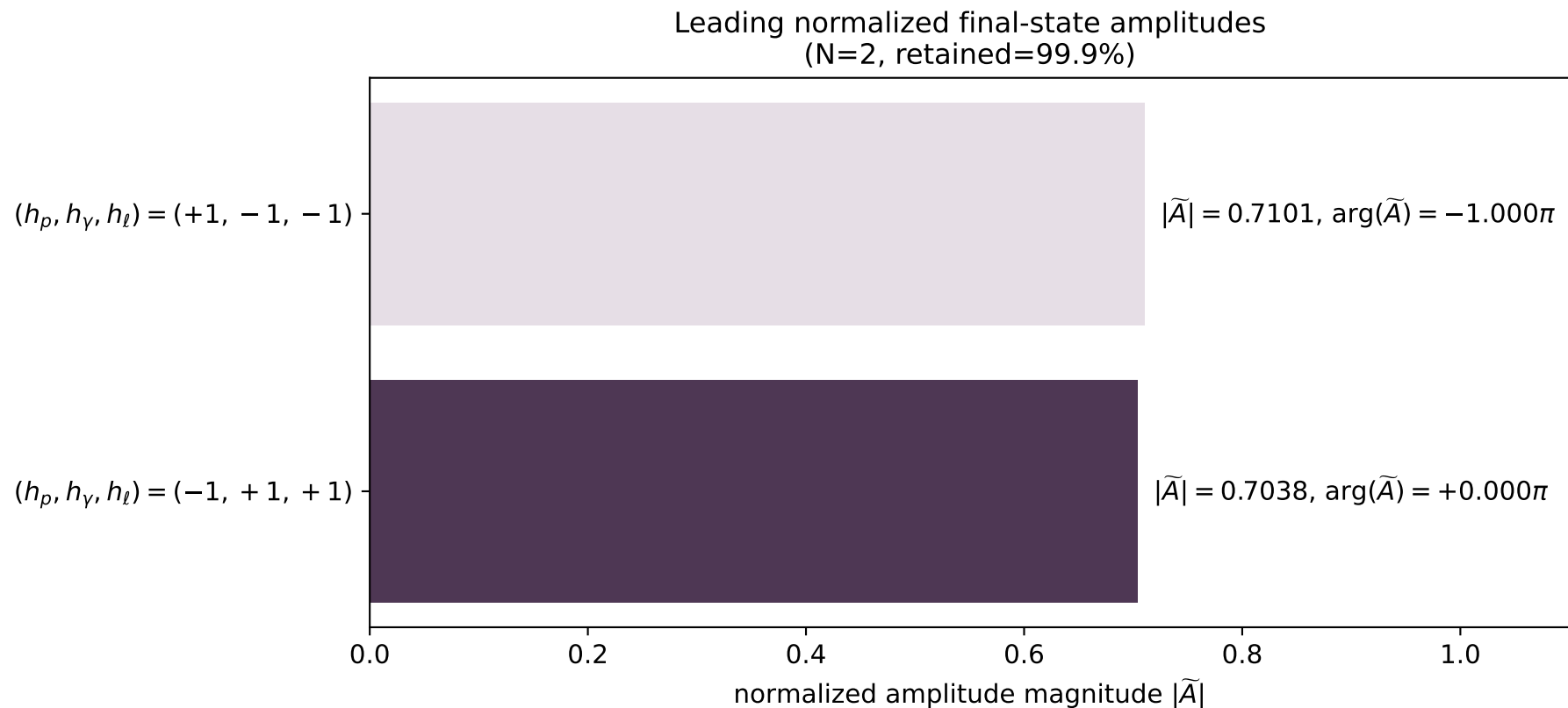
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_266
 $d_{\text{GHZ}}=0.000189044$
 region: polarization_cluster_01_local_minimum_266
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0266
 lepton mixing angle: $\alpha_e=0.20844456$ rad
 incoming lepton: $0.978354|+\rangle + 0.206938|-\rangle$
 proton mixing angle: $\alpha_p=1.3671115$ rad
 incoming proton: $0.202279|+\rangle + 0.979328|-\rangle$

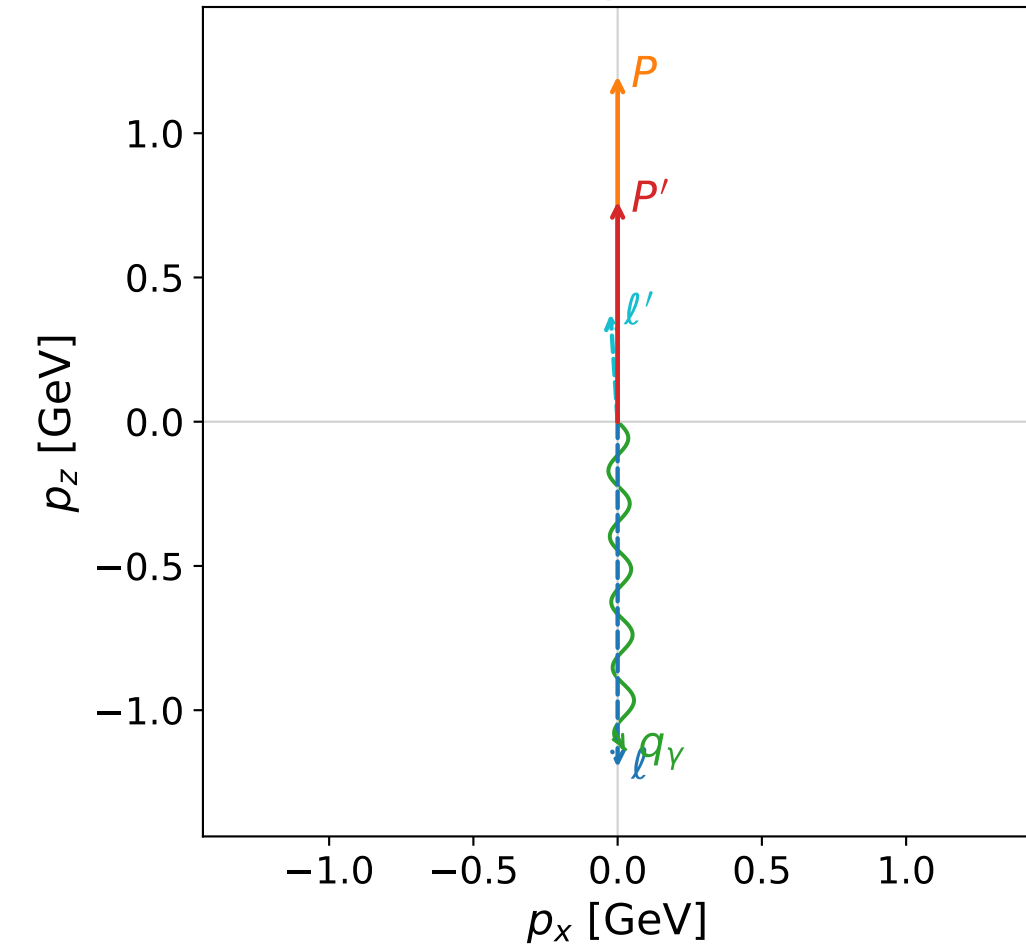
$s=12.8901$, $\sqrt{s}=3.59027$, $|\vec{P}|=1.67261$, $|\vec{P}'|=0.948936$
 $\theta_{P'}=0.00291456$, $\theta_Y=3.11931$, $\phi_{P'}=6.2819$, $\phi_Y=3.14164$
 $E_Y=1.60224$, $\theta_{\ell'}=0.0504001$, $\phi_{\ell'}=0.000160702$
 $Q^2=4.37105$, $x_B=0.374008$, $t=-0.183378$, $W^2=8.19583$, $y=0.97309$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.6726$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6726$
 P $E= 1.9177$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6726$
 ℓ' $E= 0.6537$ $p_x= 0.0329$ $p_y= 0.0000$ $p_z= 0.6529$
 P' $E= 1.3343$ $p_x= 0.0028$ $p_y= -0.0000$ $p_z= 0.9489$
 q_Y $E= 1.6022$ $p_x= -0.0357$ $p_y= -0.0000$ $p_z= -1.6018$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

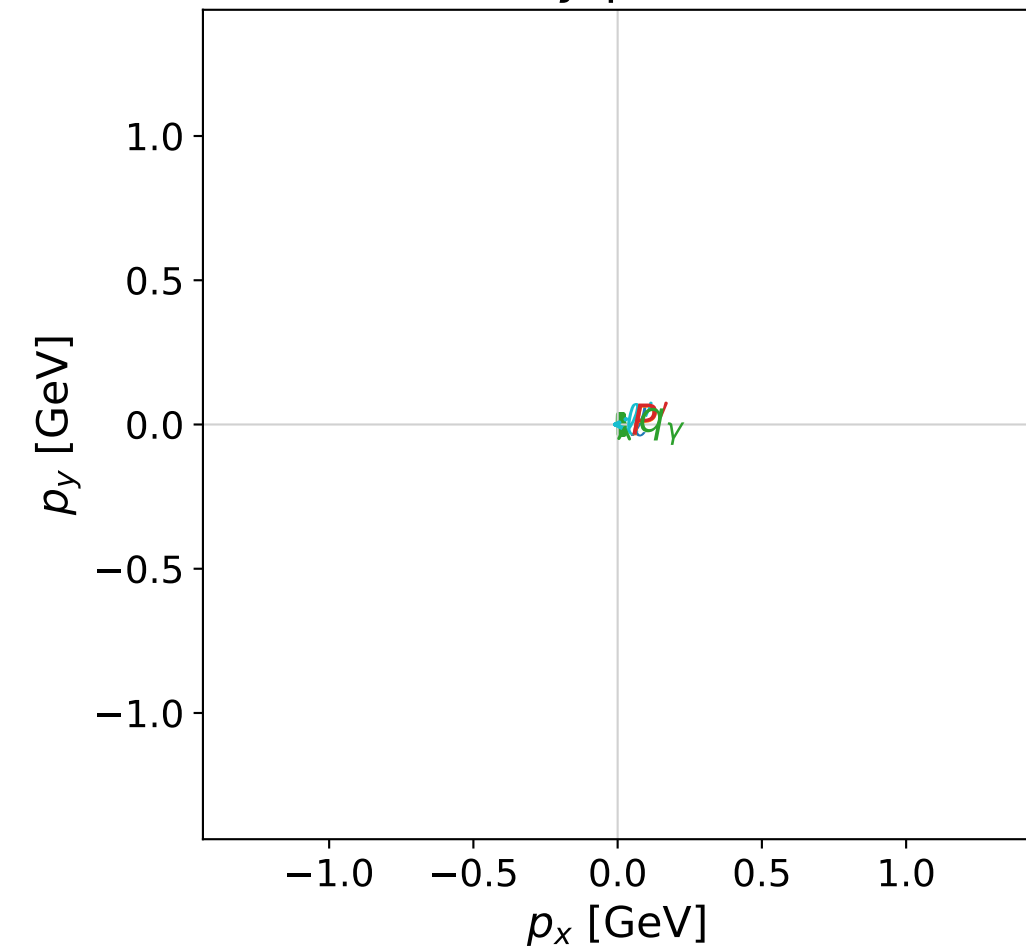


dghz_mixing_angles_polarization_01_local_minimum_314
 $d_{\text{GHZ}}=0.000387584$
 region: polarization_cluster_01_local_minimum_314
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0314
 lepton mixing angle: $\alpha_e=1.3236924$ rad
 incoming lepton: $0.244597|+\rangle +0.969625|-\rangle$
 proton mixing angle: $\alpha_p=0.24858017$ rad
 incoming proton: $0.969263|+\rangle +0.246028|-\rangle$

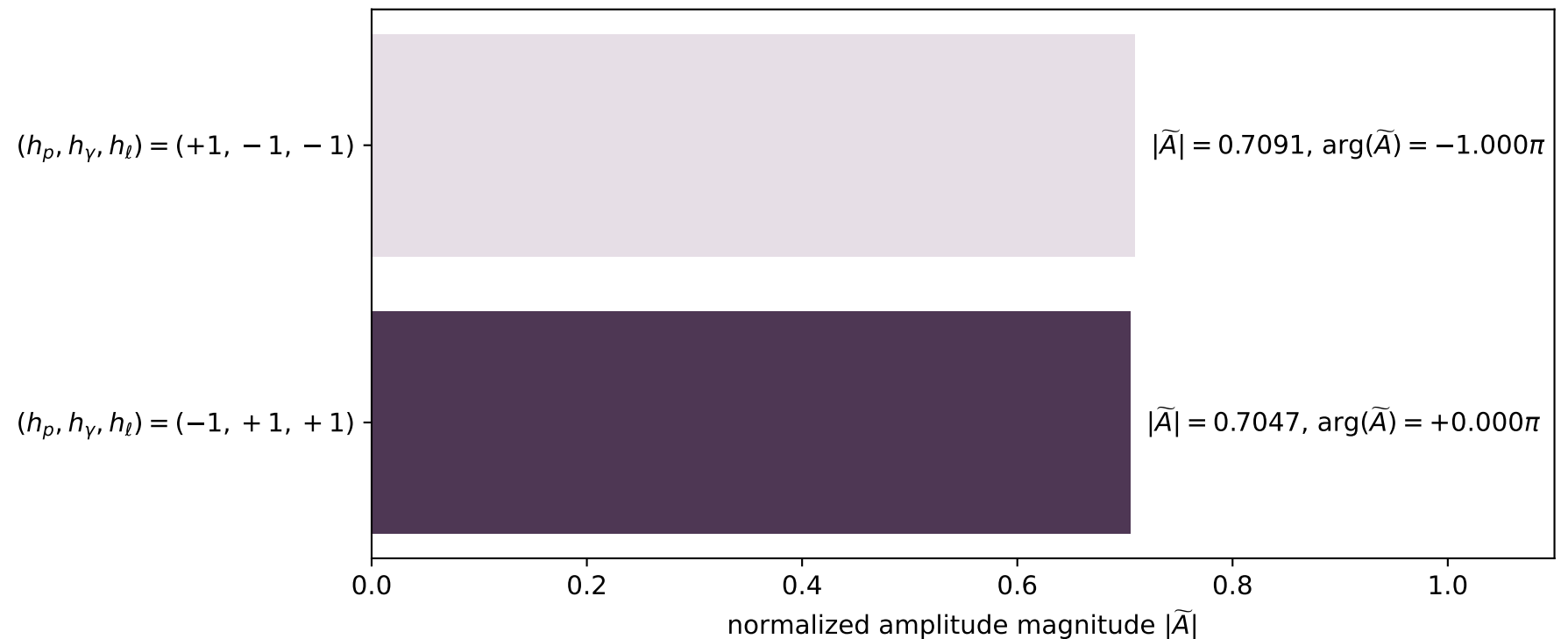
$s=7.3952$, $\sqrt{s}=2.71941$, $|\vec{P}|=1.19794$, $|\vec{P}'|=0.763049$
 $\theta_{P'}=0$, $\theta_Y=3.11984$, $\phi_{P'}=3.82165$, $\phi_Y=2.0169\text{e-}05$
 $E_Y=1.13637$, $\theta_{\ell'}=0.0661538$, $\phi_{\ell'}=3.14161$
 $Q^2=1.78954$, $x_B=0.285346$, $t=-0.091589$, $W^2=5.36178$, $y=0.962567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1979$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1979$
 P $E= 1.5215$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1979$
 ℓ' $E= 0.3739$ $p_x= -0.0247$ $p_y= -0.0000$ $p_z= 0.3731$
 P' $E= 1.2092$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7630$
 q_Y $E= 1.1364$ $p_x= 0.0247$ $p_y= 0.0000$ $p_z= -1.1361$

x--y plane

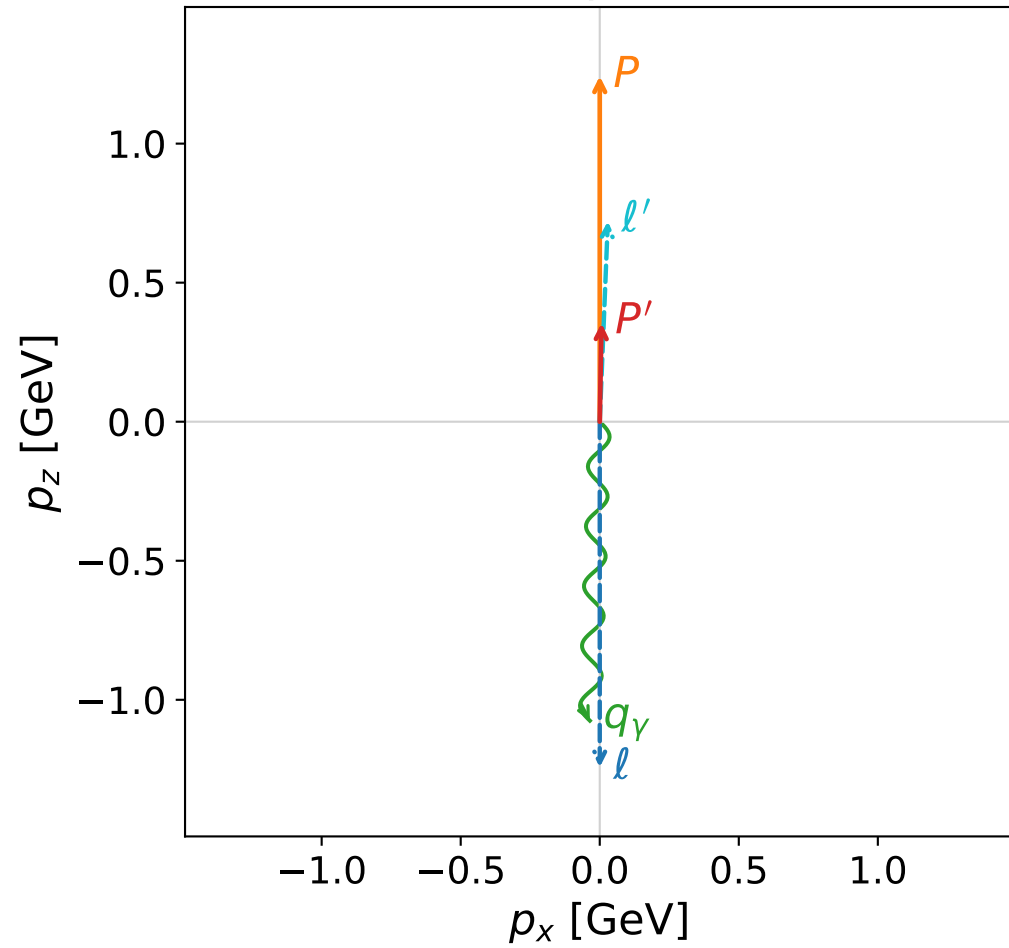


Leading normalized final-state amplitudes
 (N=2, retained=99.9%)

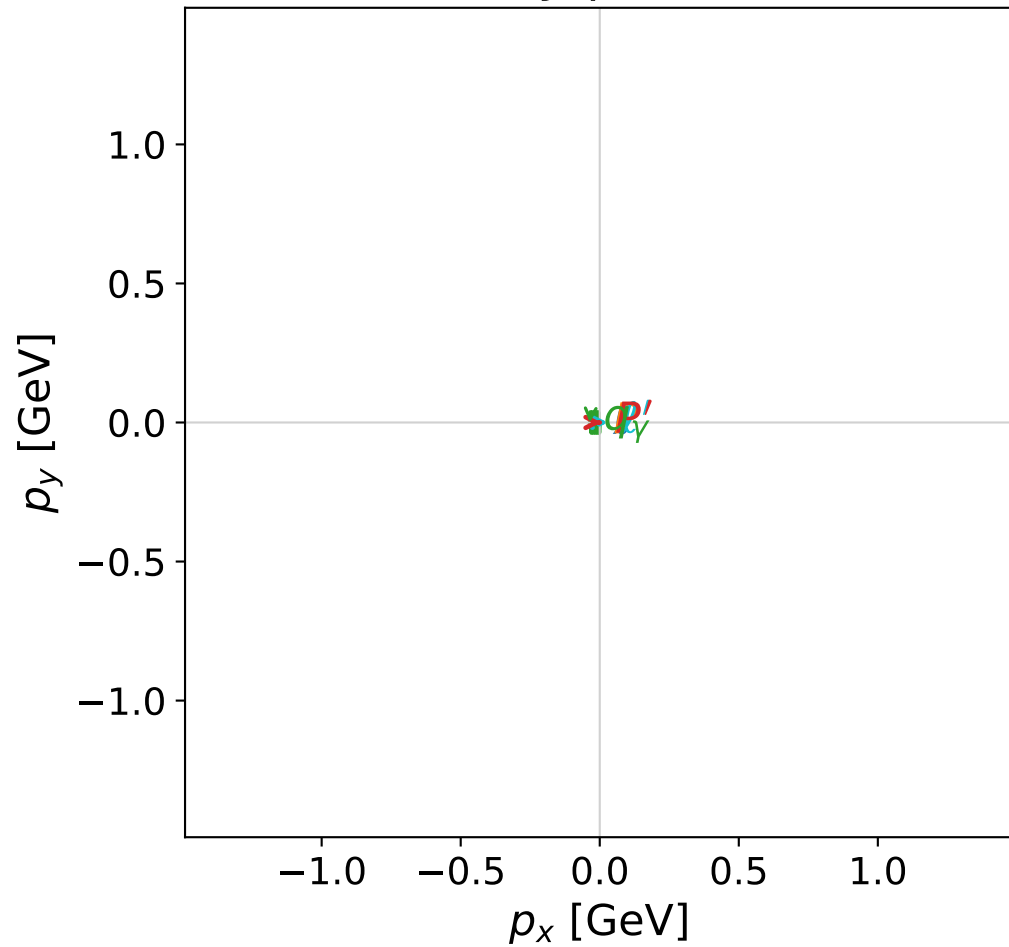


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

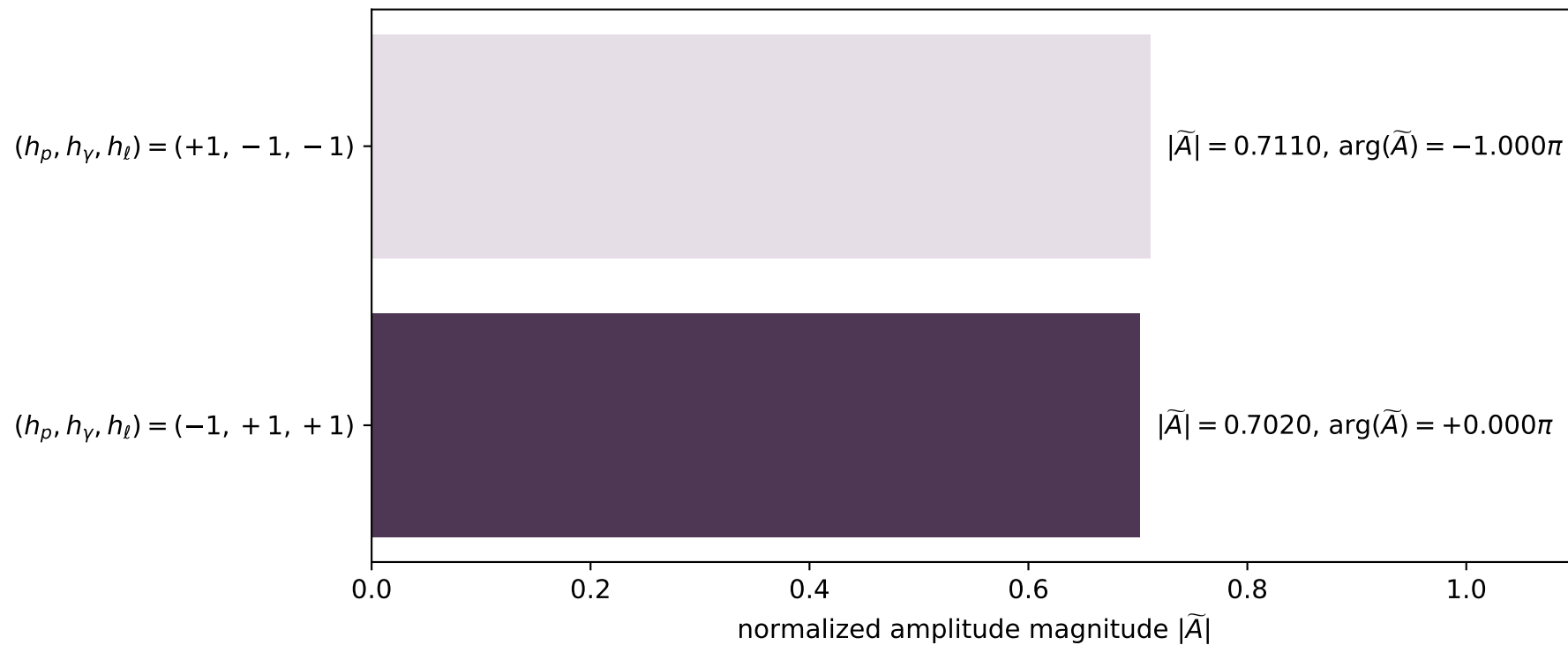


dghz_mixing_angles_polarization_01_local_minimum_331
 $d_{\text{GHZ}}=0.000498586$
 region: polarization_cluster_01_local_minimum_331
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0331
 lepton mixing angle: $\alpha_e=0.063309233$ rad
 incoming lepton: $0.997997|+\rangle + 0.063267|-\rangle$
 proton mixing angle: $\alpha_p=1.5095946$ rad
 incoming proton: $0.0611635|+\rangle + 0.998128|-\rangle$

$s=7.83845$, $\sqrt{s}=2.79972$, $|\vec{P}|=1.24273$, $|\vec{P}'|=0.354558$
 $\theta_{P'}=0.0179191$, $\theta_\gamma=3.1086$, $\phi_{P'}=6.28227$, $\phi_\gamma=3.14163$
 $E_\gamma=1.07572$, $\theta_\ell=0.0404006$, $\phi_\ell=0.000247116$
 $Q^2=3.5837$, $x_B=0.551013$, $t=-0.481835$, $W^2=3.79999$, $y=0.934646$

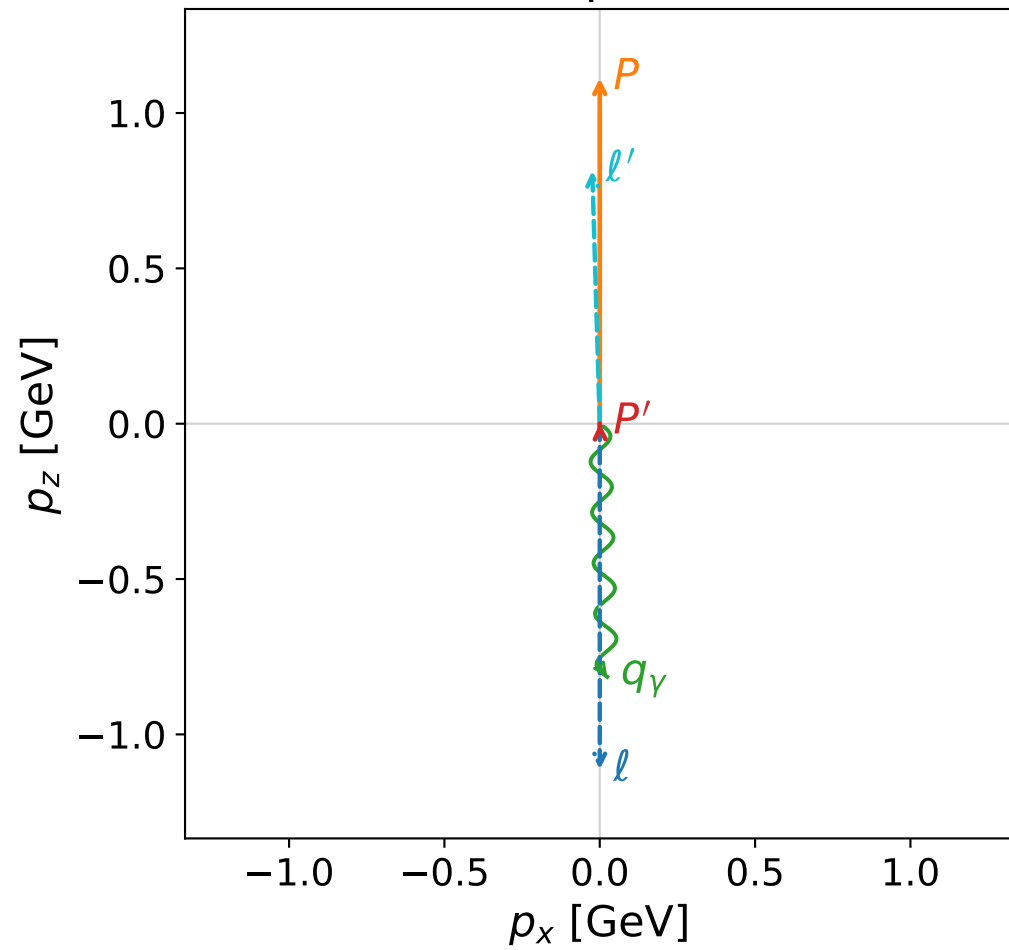
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2427$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2427$
 P $E= 1.5570$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2427$
 ℓ' $E= 0.7212$ $p_x= 0.0291$ $p_y= 0.0000$ $p_z= 0.7206$
 P' $E= 1.0028$ $p_x= 0.0064$ $p_y= -0.0000$ $p_z= 0.3545$
 q_γ $E= 1.0757$ $p_x= -0.0355$ $p_y= -0.0000$ $p_z= -1.0751$

Leading normalized final-state amplitudes
 (N=2, retained=99.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

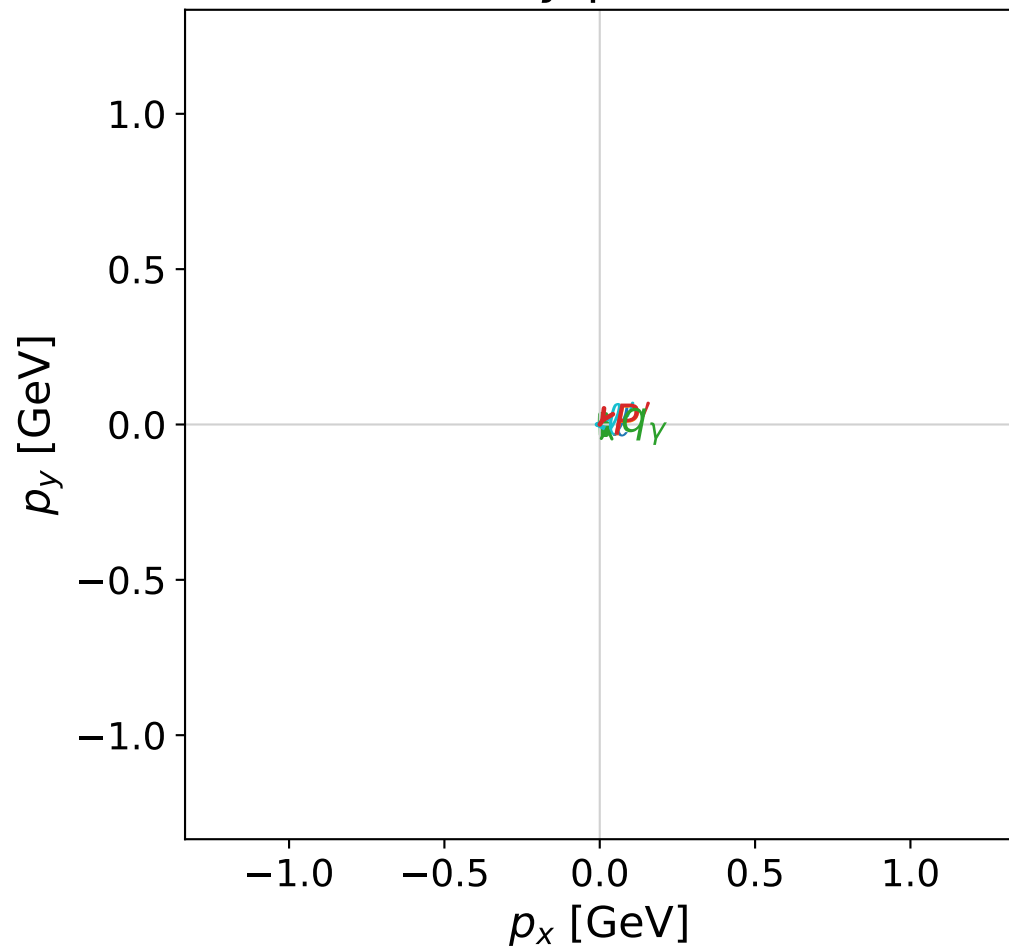


dghz_mixing_angles_polarization_01_local_minimum_333
 $d_{\text{GHZ}}=0.000507492$
 region: polarization_cluster_01_local_minimum_333
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0333
 lepton mixing angle: $\alpha_e=1.5422293$ rad
 incoming lepton: $0.0285631|+\rangle + 0.999592|-\rangle$
 proton mixing angle: $\alpha_p=0.028304256$ rad
 incoming proton: $0.999599|+\rangle + 0.0283005|-\rangle$

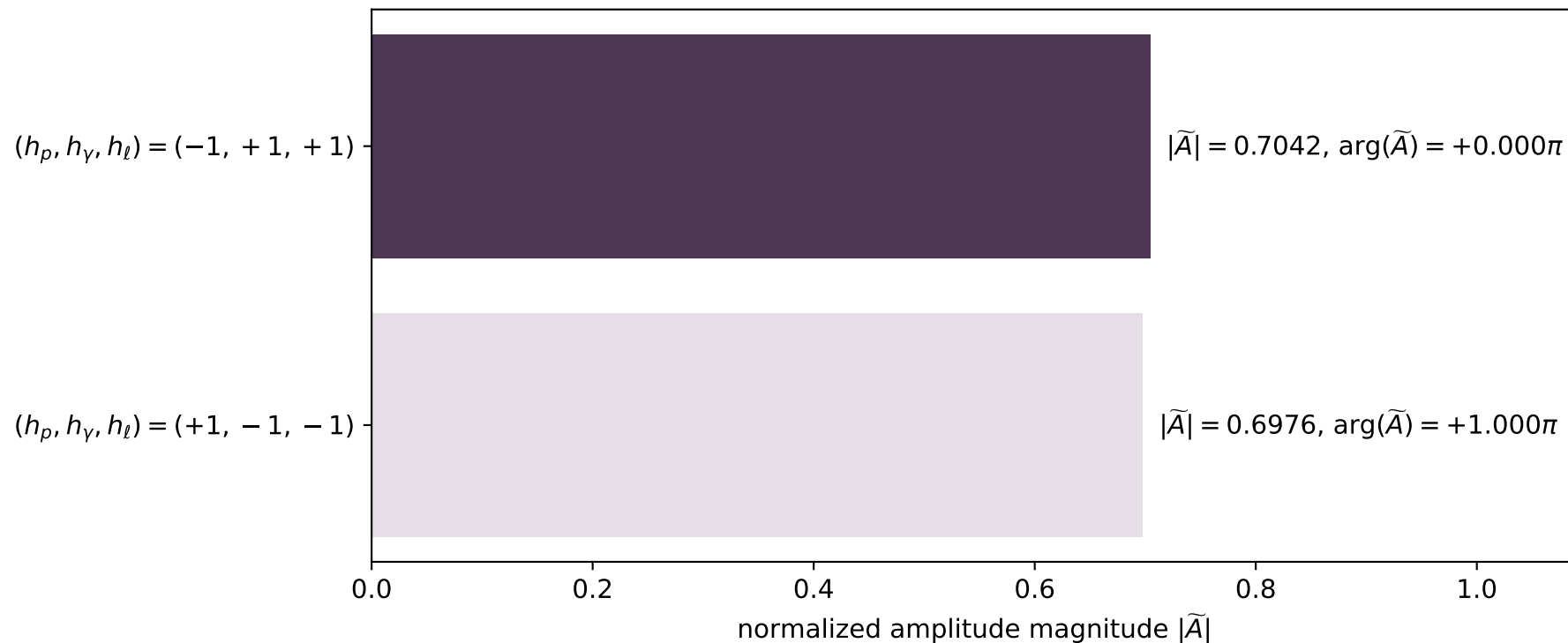
$s=6.59164$, $\sqrt{s}=2.56742$, $|\vec{P}|=1.11236$, $|\vec{P}'|=0.00112869$
 $\theta_{p'}=0.054477$, $\theta_\gamma=3.11194$, $\phi_{p'}=4.13085$, $\phi_\gamma=7.31348\text{e-}07$
 $E_\gamma=0.815273$, $\theta_{\ell'}=0.0296545$, $\phi_{\ell'}=3.13947$
 $Q^2=3.6217$, $x_B=0.702834$, $t=-0.967495$, $W^2=2.41114$, $y=0.902166$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1124$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1124$
 P $E= 1.4551$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1124$
 ℓ' $E= 0.8141$ $p_x= -0.0241$ $p_y= 0.0001$ $p_z= 0.8138$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 0.0011$
 q_γ $E= 0.8153$ $p_x= 0.0242$ $p_y= 0.0000$ $p_z= -0.8149$

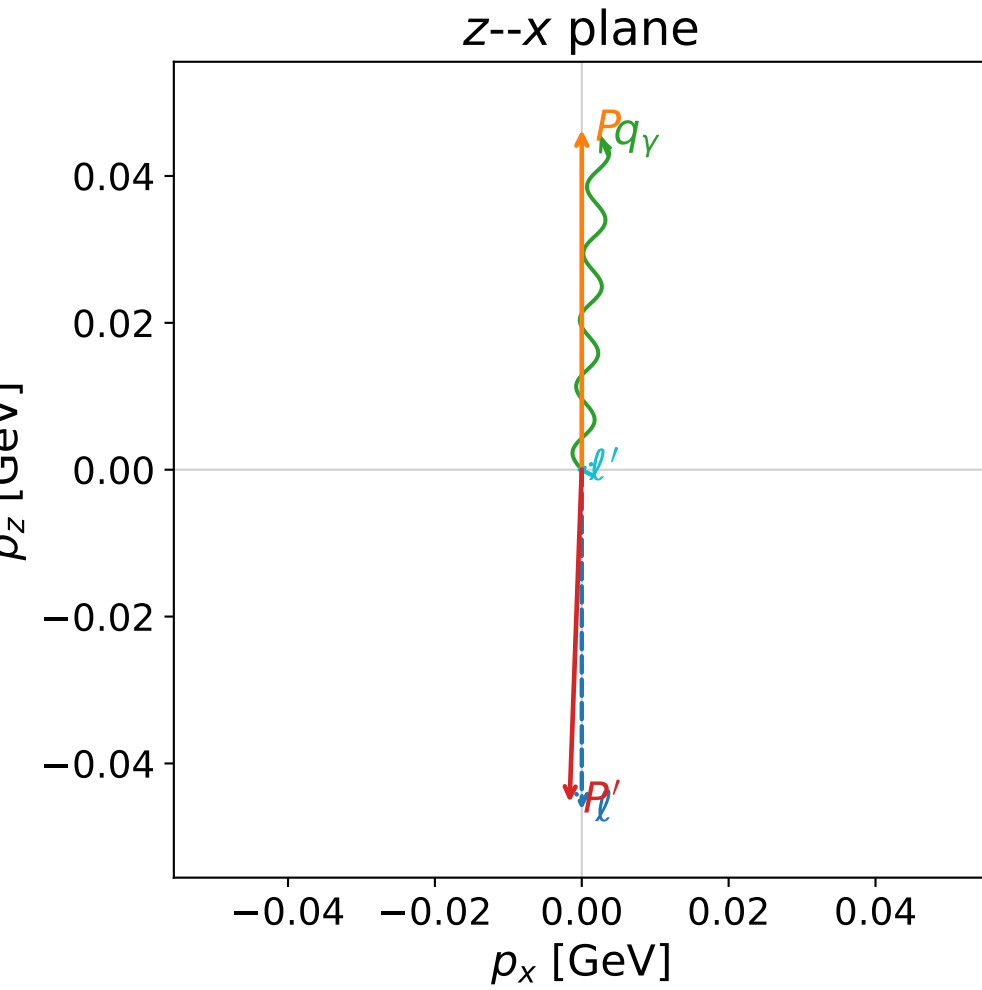
x--y plane



Leading normalized final-state amplitudes
 (N=2, retained=98.3%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

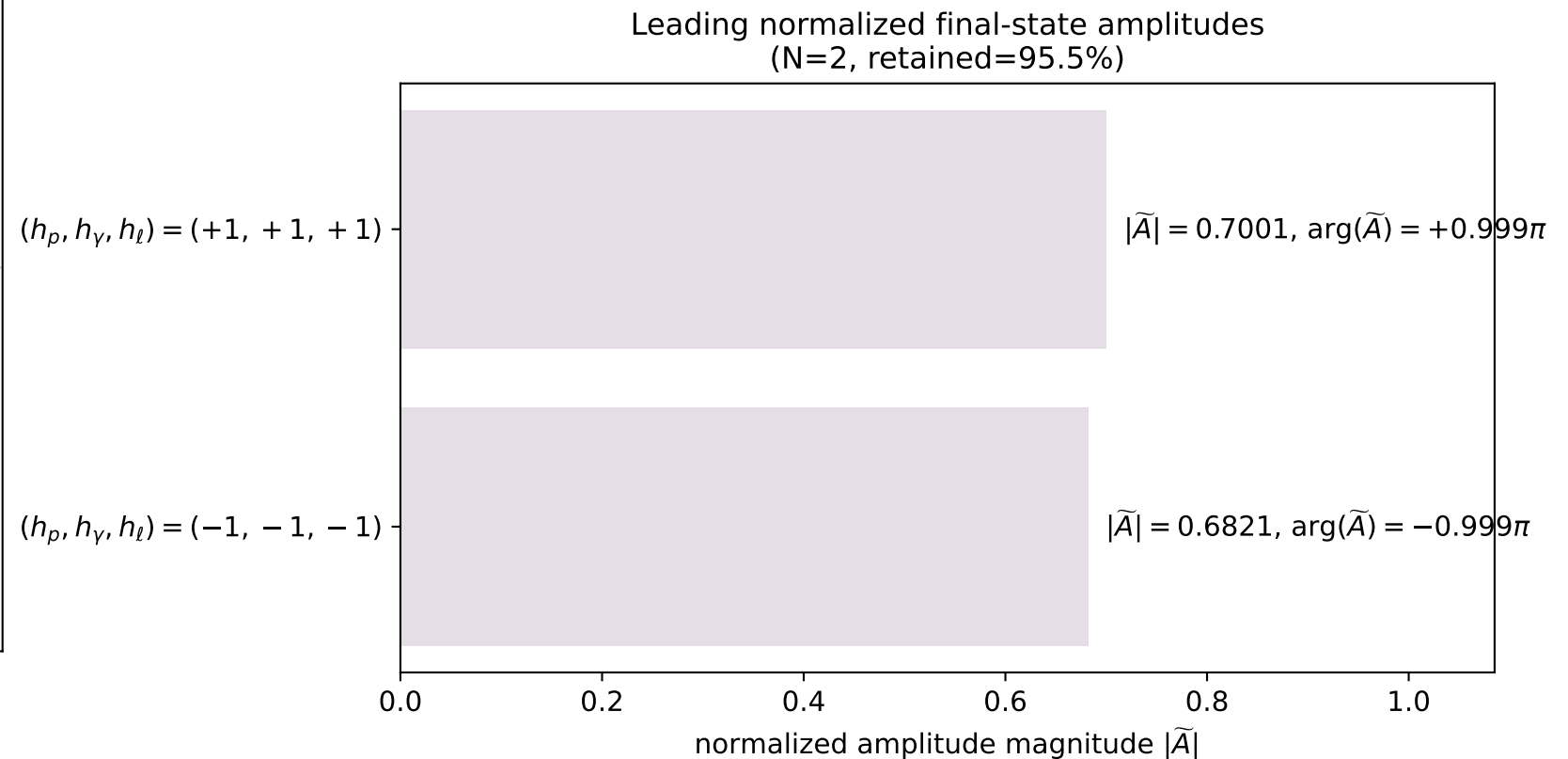
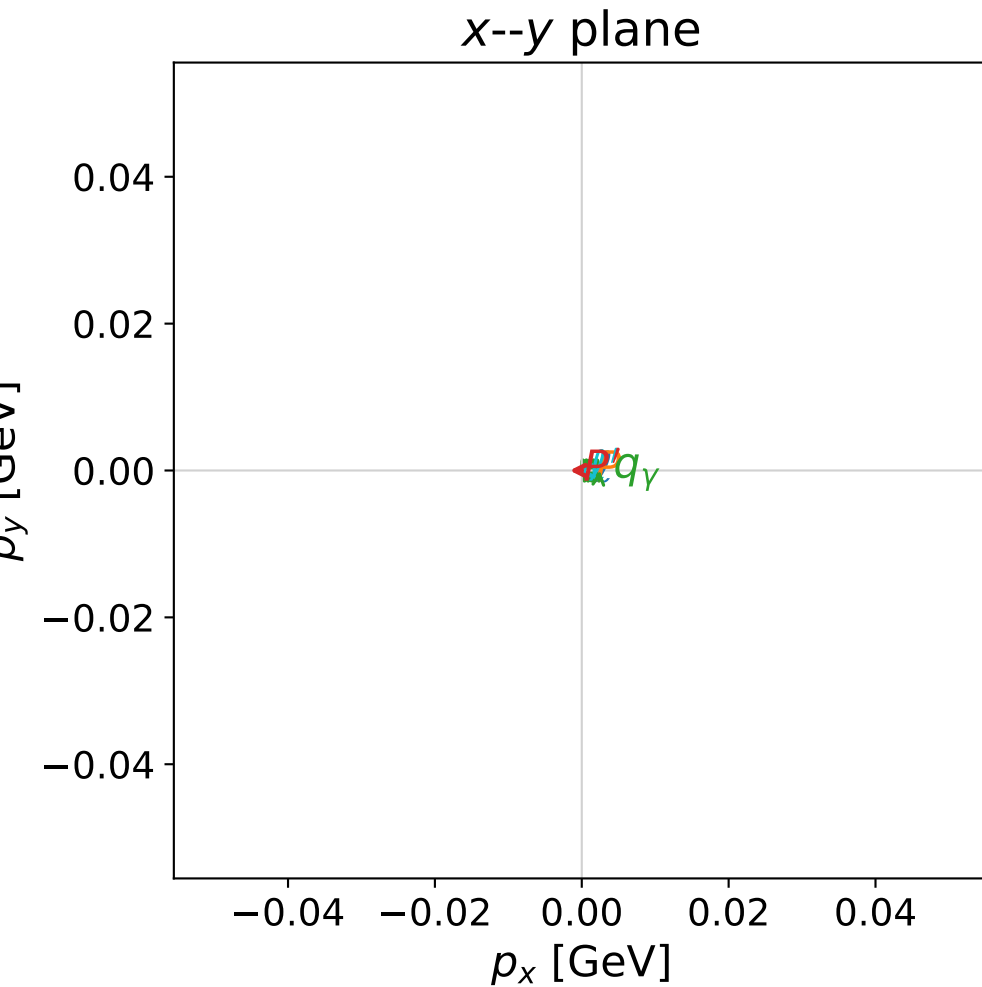


dghz_mixing_angles_polarization_01_local_minimum_351
 $d_{\text{GHZ}}=0.000879976$
 region: polarization_cluster_01_local_minimum_351
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0351
 lepton mixing angle: $\alpha_e=0.76448436$ rad
 incoming lepton: $0.721739|+\rangle +0.692165|-\rangle$
 proton mixing angle: $\alpha_p=0.73414056$ rad
 incoming proton: $0.742407|+\rangle +0.669949|-\rangle$

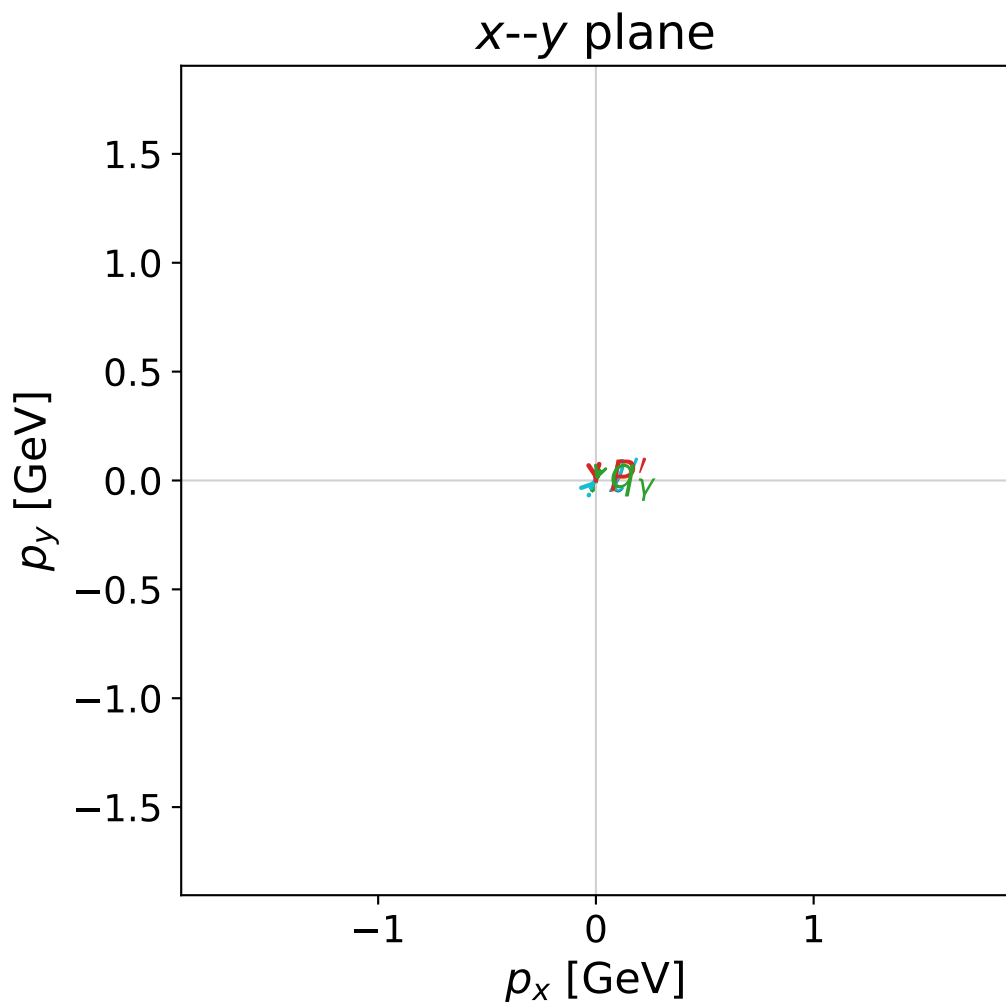
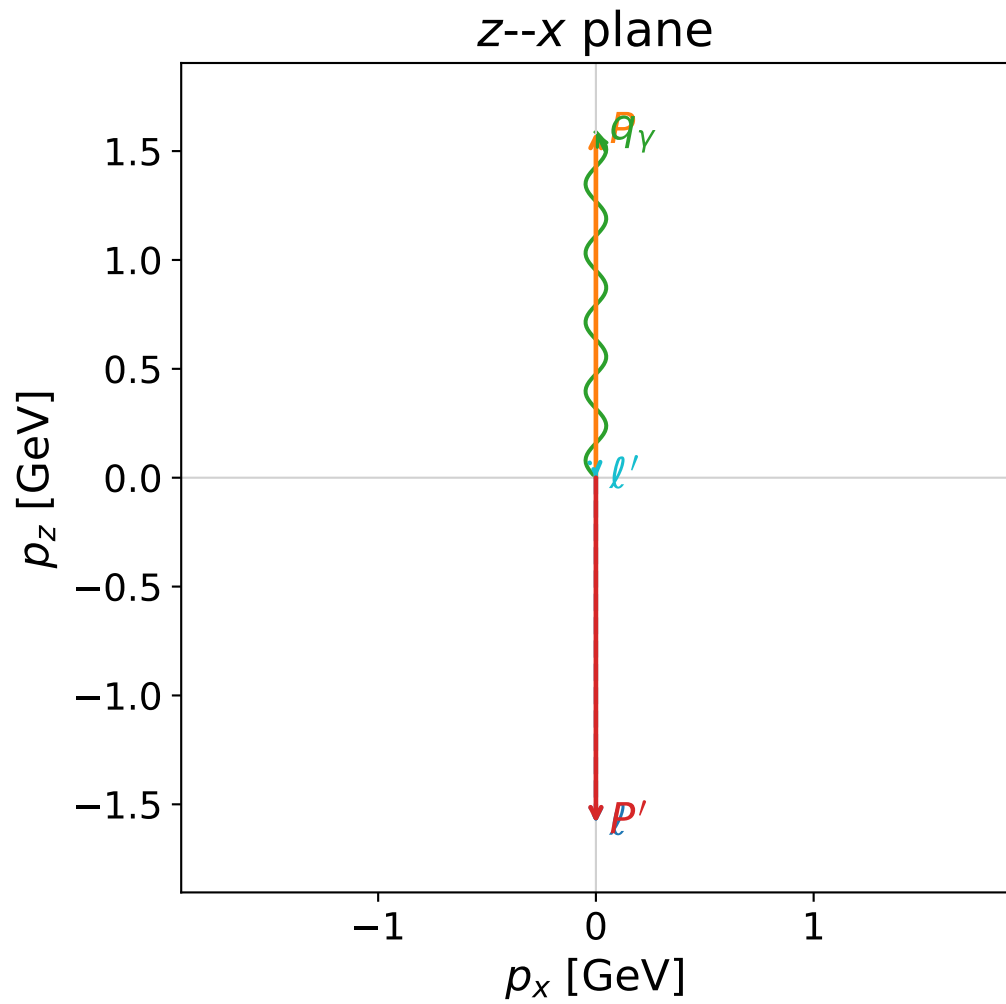
$s=0.971085$, $\sqrt{s}=0.985437$, $|\vec{P}|=0.0462921$, $|\vec{P}'|=0.0453374$
 $\theta_{p'}=3.10496$, $\theta_\gamma=0.0542554$, $\phi_{p'}=3.141$, $\phi_\gamma=0.000630486$
 $E_\gamma=0.045391$, $\theta_{l'}=1.59229$, $\phi_{l'}=3.14476$
 $Q^2=8.58888\text{e-}05$, $x_B=0.000960145$, $t=-0.00839316$, $W^2=0.969212$, $y=0.980410$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	0.0463	0.0000	0.0000	-0.0463
P	0.9391	0.0000	0.0000	0.0463
ℓ'	0.0010	-0.0008	-0.0000	-0.0000
P'	0.9391	-0.0017	0.0000	-0.0453
q_γ	0.0454	0.0025	0.0000	0.0453



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

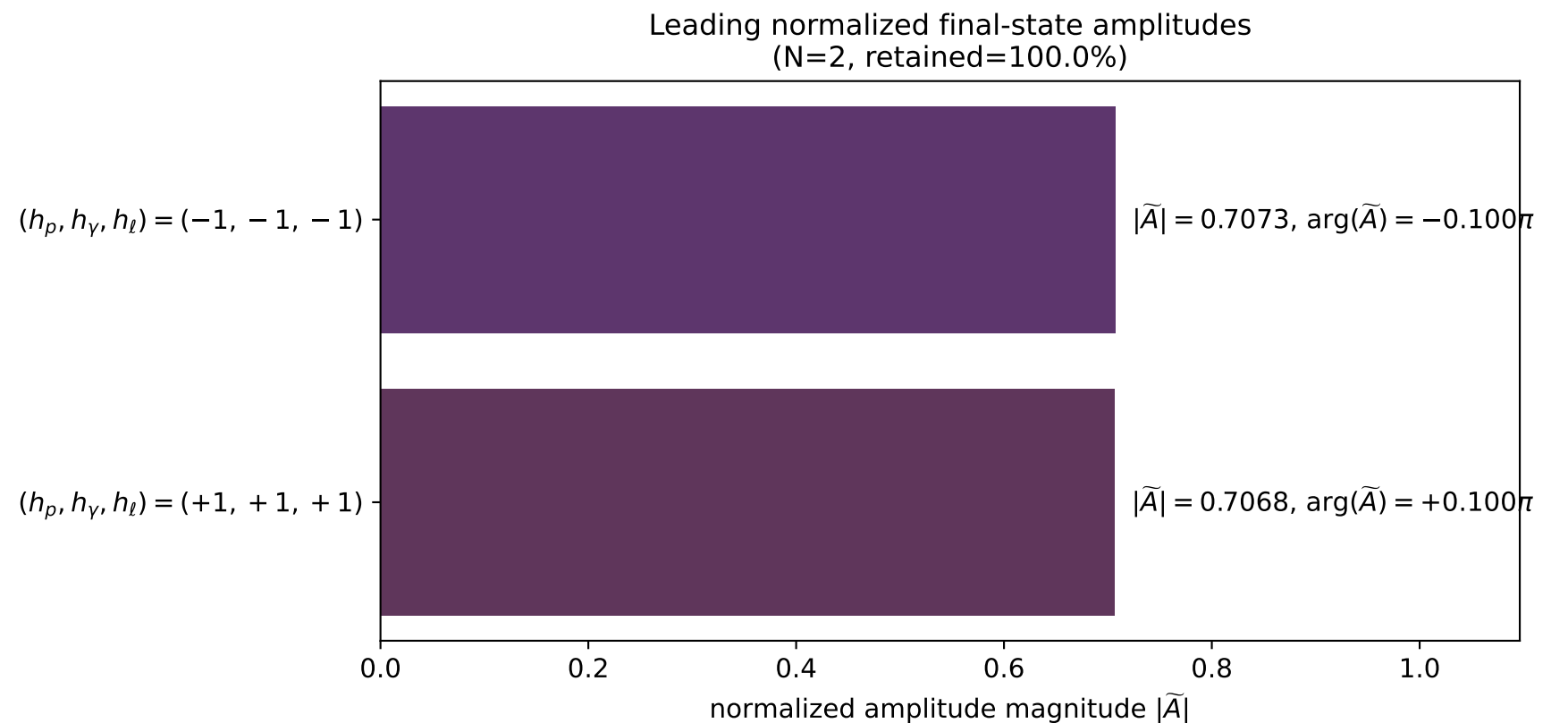


dghz_mixing_angles_polarization_01_local_minimum_356
 $d_{\text{GHZ}}=0.0011903$
 region: polarization_cluster_01_local_minimum_356
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0356
 lepton mixing angle: $\alpha_e=0.78547747$ rad
 incoming lepton: $0.707051|+\rangle +0.707163|-\rangle$
 proton mixing angle: $\alpha_p=0.78569818$ rad
 incoming proton: $0.706895|+\rangle +0.707319|-\rangle$

$s=11.7704$, $\sqrt{s}=3.4308$, $|\vec{P}|=1.58717$, $|\vec{P}'|=1.58616$
 $\theta_{p'}=3.14154$, $\theta_\gamma=4.27948\text{e-}05$, $\phi_{p'}=4.8388$, $\phi_\gamma=2.80317$
 $E_\gamma=1.58703$, $\theta_{\ell'}=3.05347$, $\phi_{\ell'}=0.792465$
 $Q^2=0.000448243$, $x_B=4.11835\text{e-}05$, $t=-10.07$, $W^2=11.7634$, $y=0.999402$

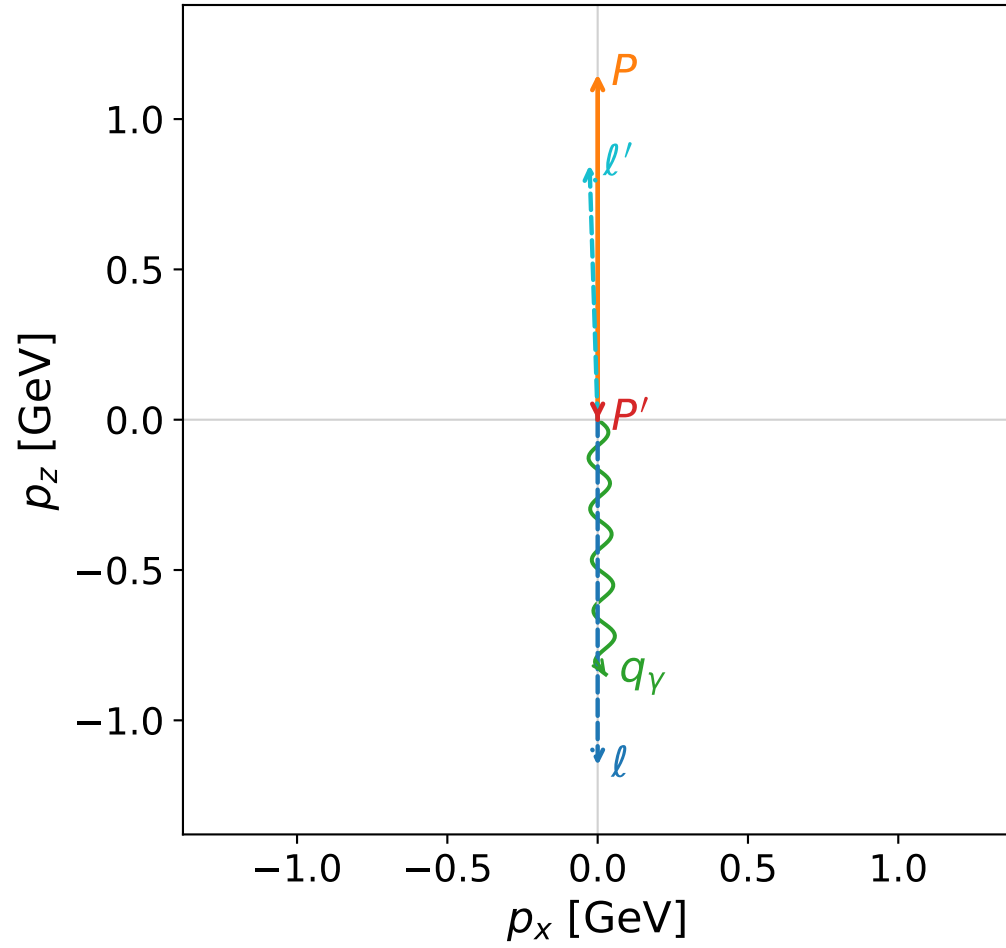
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	1.5872	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-1.5872
P	$E=$	1.8436	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	1.5872
ℓ'	$E=$	0.0010	$p_x=$	0.0001	$p_y=$	0.0001	$p_z=$	-0.0009
P'	$E=$	1.8428	$p_x=$	0.0000	$p_y=$	-0.0001	$p_z=$	-1.5862
q_γ	$E=$	1.5870	$p_x=$	-0.0001	$p_y=$	0.0000	$p_z=$	1.5870

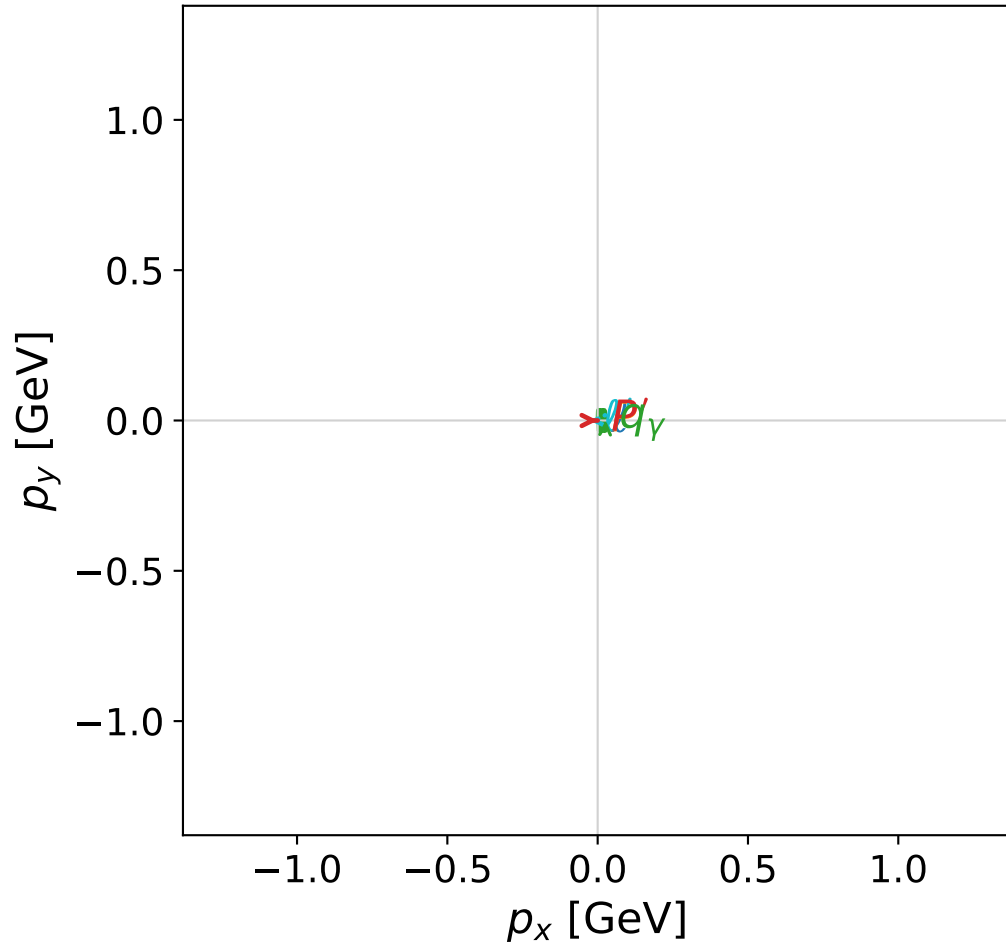


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



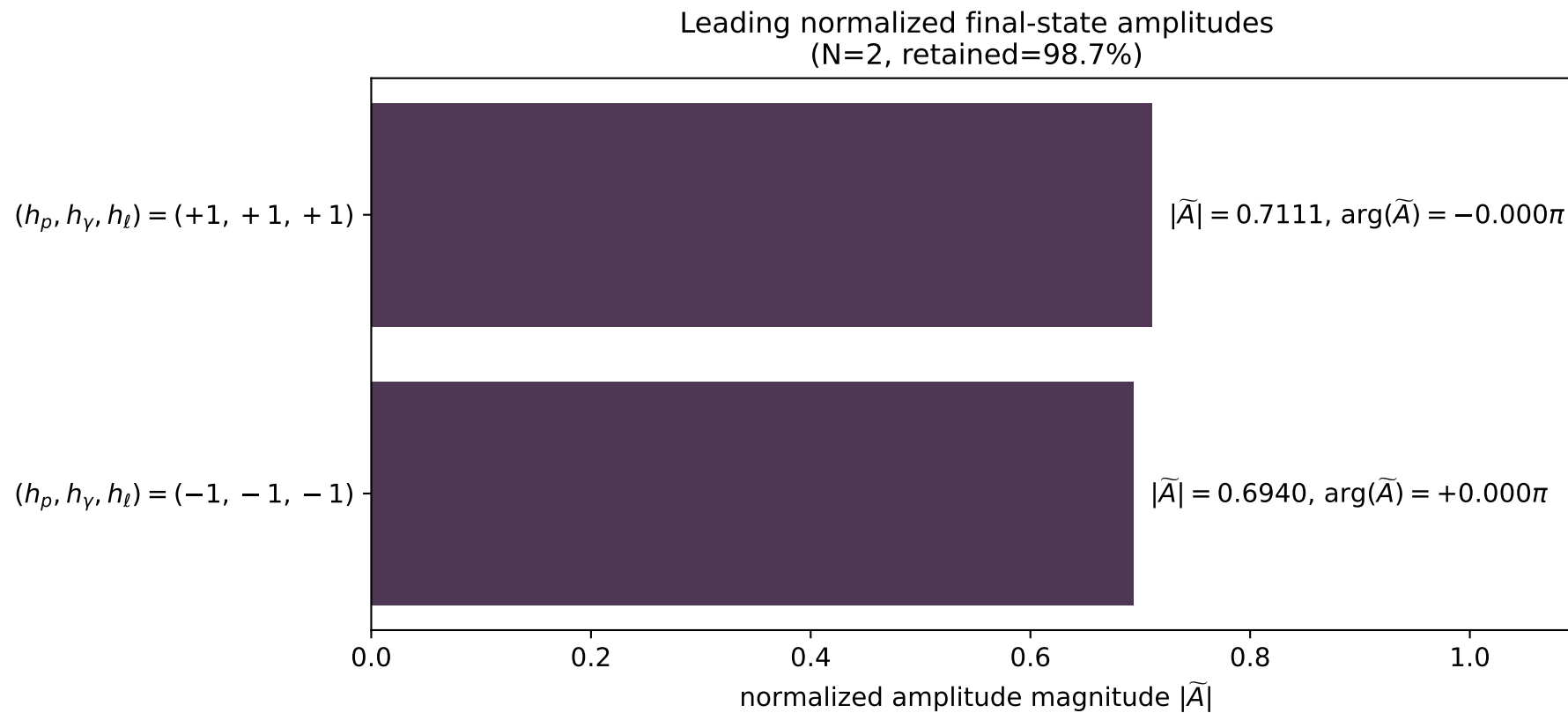
x--y plane



dghz_mixing_angles_polarization_01_local_minimum_359
 $d_{\text{GHZ}}=0.00157373$
 region: polarization_cluster_01_local_minimum_359
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0359
 lepton mixing angle: $\alpha_e=1.5434269$ rad
 incoming lepton: $0.027366|+\rangle + 0.999625|-\rangle$
 proton mixing angle: $\alpha_p=0.026716948$ rad
 incoming proton: $0.999643|+\rangle + 0.0267138|-\rangle$

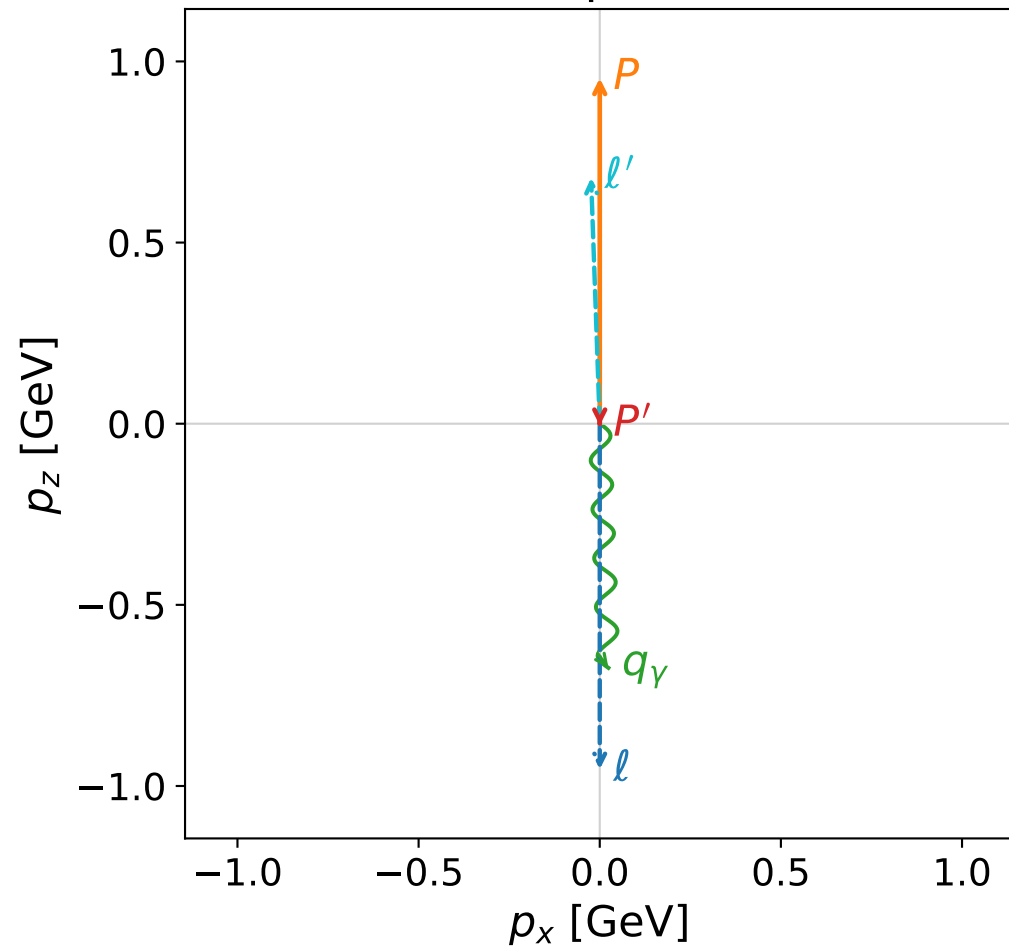
$s=6.9343$, $\sqrt{s}=2.63331$, $|\vec{P}|=1.14959$, $|\vec{P}'|=3.60035\text{e-}07$
 $\theta_{P'}=3.14082$, $\theta_\gamma=3.10976$, $\phi_{P'}=0.000770639$, $\phi_\gamma=6.28315$
 $E_\gamma=0.847653$, $\theta_{l'}=0.0318365$, $\phi_{l'}=3.14156$
 $Q^2=3.89684$, $x_B=0.71019$, $t=-1.02376$, $W^2=2.47004$, $y=0.90628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.1496$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1496$
 P $E= 1.4837$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1496$
 l' $E= 0.8477$ $p_x= -0.0270$ $p_y= 0.0000$ $p_z= 0.8472$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 0.8477$ $p_x= 0.0270$ $p_y= -0.0000$ $p_z= -0.8472$

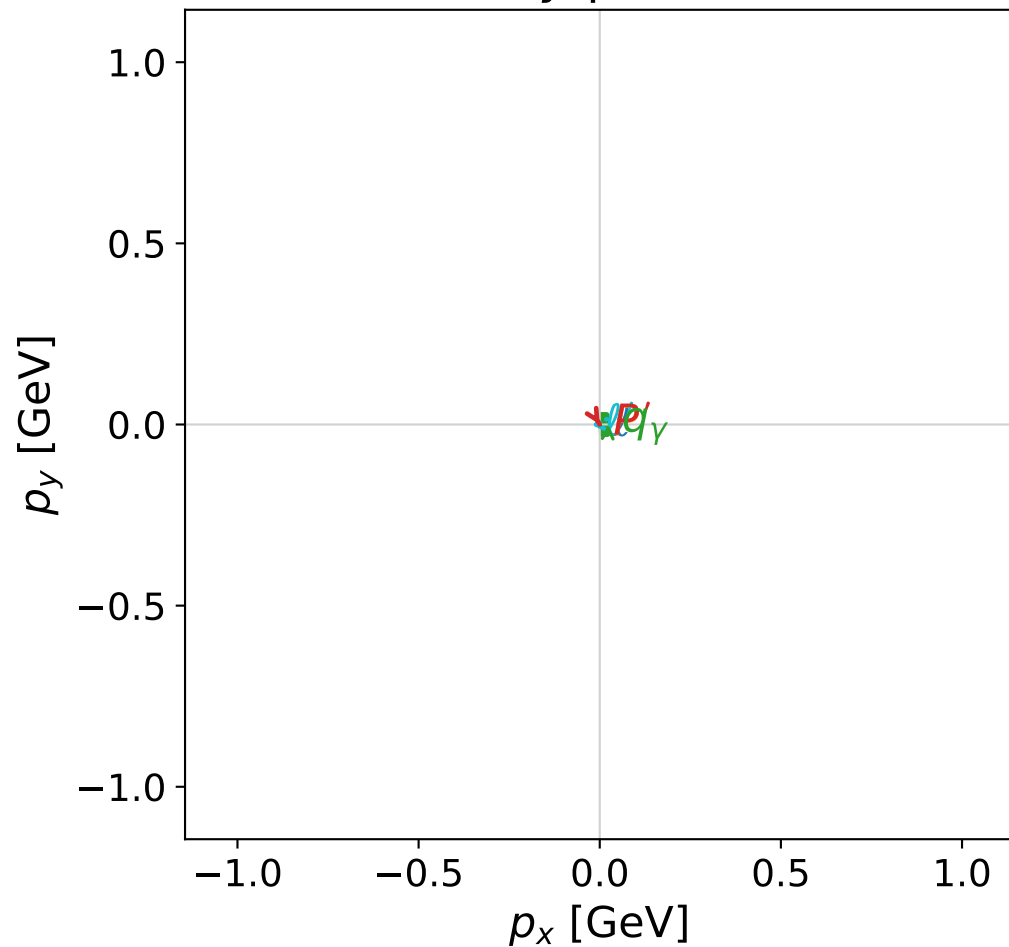


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



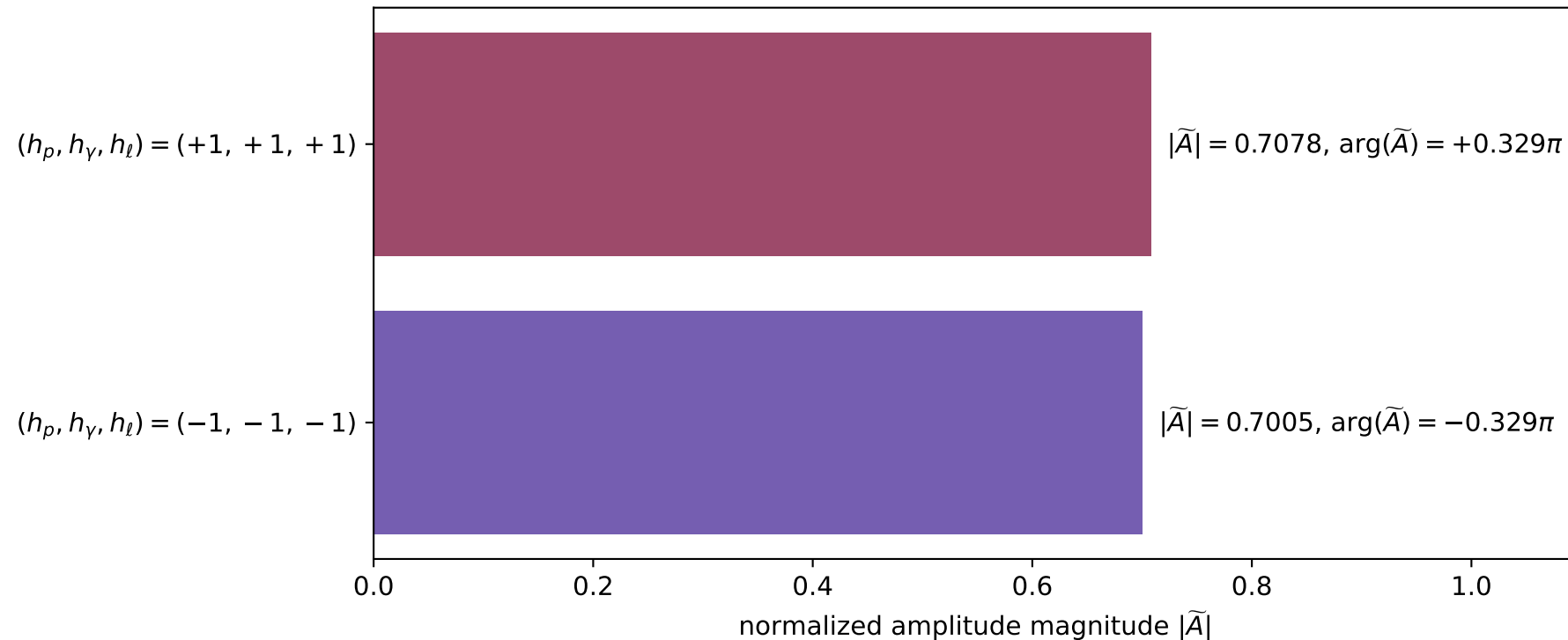
dghz_mixing_angles_polarization_01_local_minimum_360
 $d_{\text{GHZ}}=0.00187001$
 region: polarization_cluster_01_local_minimum_360
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0360
 lepton mixing angle: $\alpha_e=1.5430613$ rad
 incoming lepton: $0.0277314|+\rangle + 0.999615|-\rangle$
 proton mixing angle: $\alpha_p=0.027451861$ rad
 incoming proton: $0.999623|+\rangle + 0.0274484|-\rangle$

$s=5.25221$, $\sqrt{s}=2.29177$, $|\vec{P}|=0.953927$, $|\vec{P}'|=0.00502558$
 $\theta_{p'}=3.14128$, $\theta_\gamma=3.10607$, $\phi_{p'}=5.2495$, $\phi_\gamma=6.28316$
 $E_\gamma=0.674367$, $\theta_{\ell'}=0.0352562$, $\phi_{\ell'}=3.14151$
 $Q^2=2.59155$, $x_B=0.673146$, $t=-0.759728$, $W^2=2.1382$, $y=0.880508$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

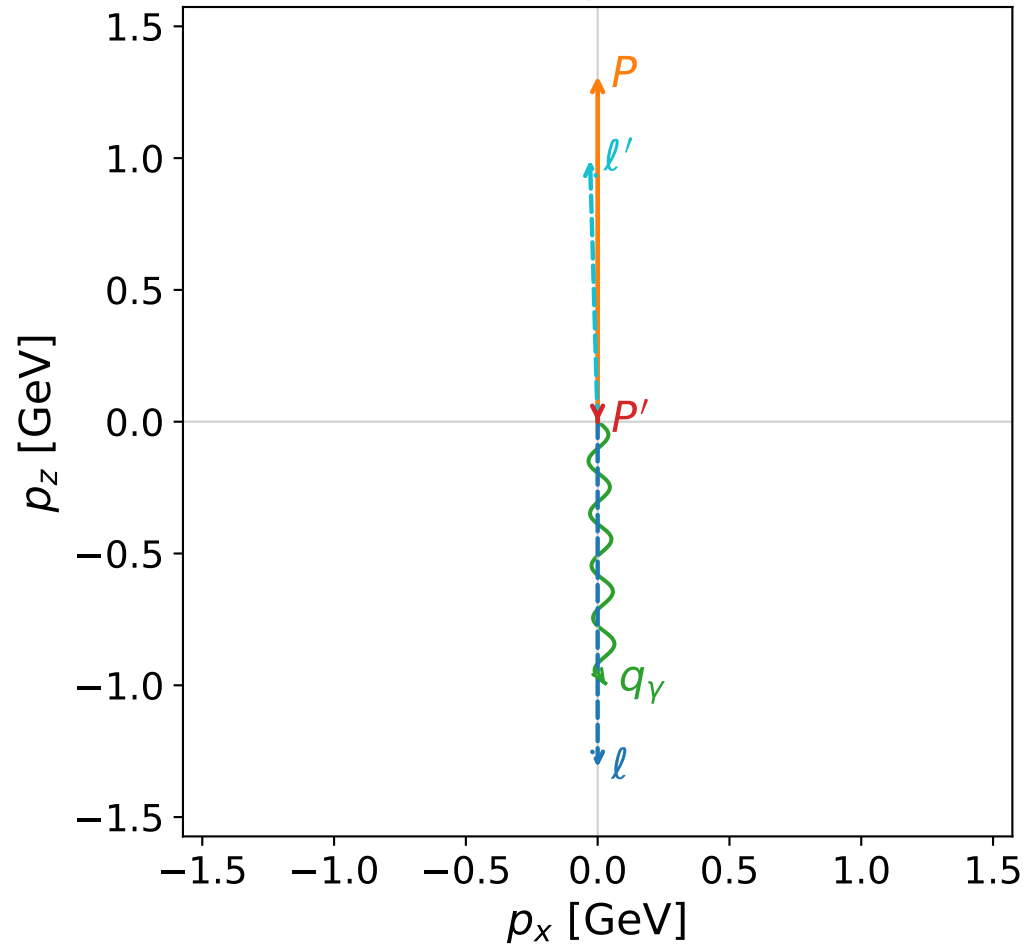
ℓ	$E=$	0.9539	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.9539
P	$E=$	1.3378	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.9539
ℓ'	$E=$	0.6794	$p_x=$	-0.0239	$p_y=$	0.0000	$p_z=$	0.6790
P'	$E=$	0.9380	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	-0.0050
q_γ	$E=$	0.6744	$p_x=$	0.0239	$p_y=$	-0.0000	$p_z=$	-0.6739

Leading normalized final-state amplitudes
(N=2, retained=99.2%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

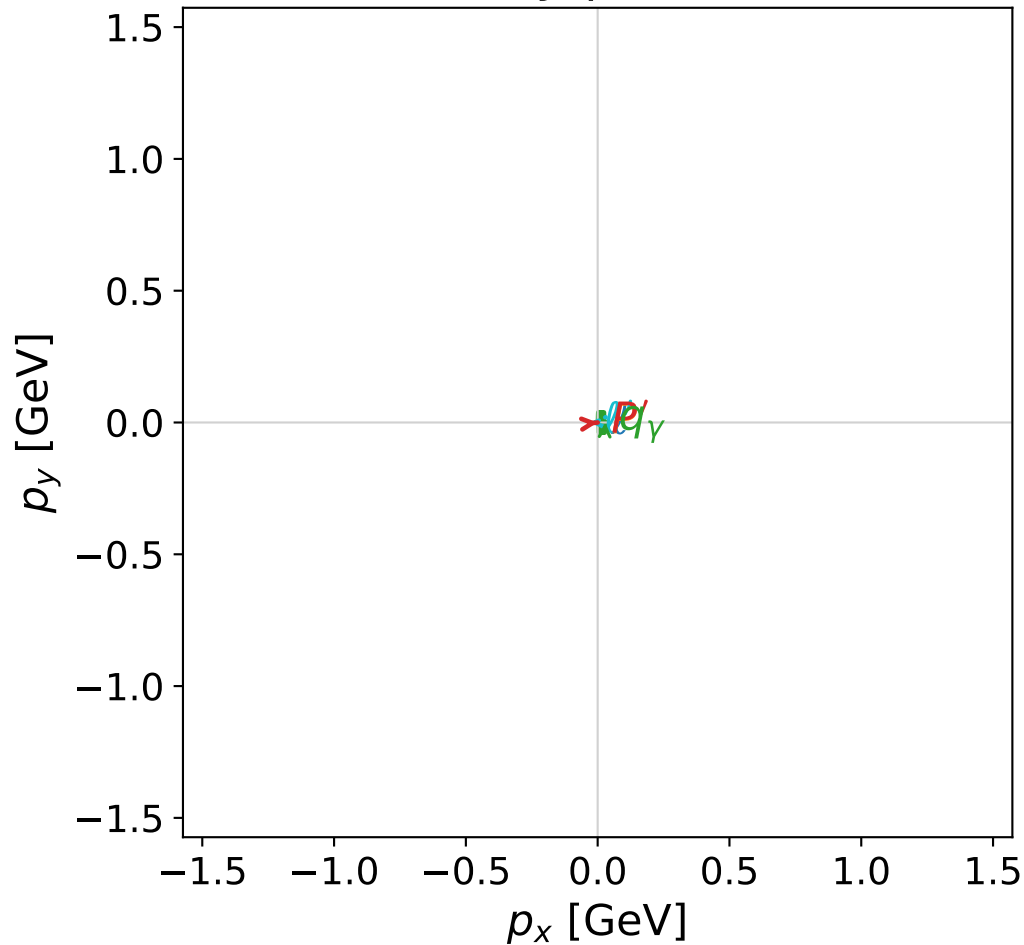


dghz_mixing_angles_polarization_01_local_minimum_361
 $d_{\text{GHZ}}=0.0018947$
 region: polarization_cluster_01_local_minimum_361
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0361
 lepton mixing angle: $\alpha_e=1.5434414$ rad
 incoming lepton: $0.0273516|+\rangle + 0.999626|-\rangle$
 proton mixing angle: $\alpha_p=0.026758462$ rad
 incoming proton: $0.999642|+\rangle + 0.0267553|-\rangle$

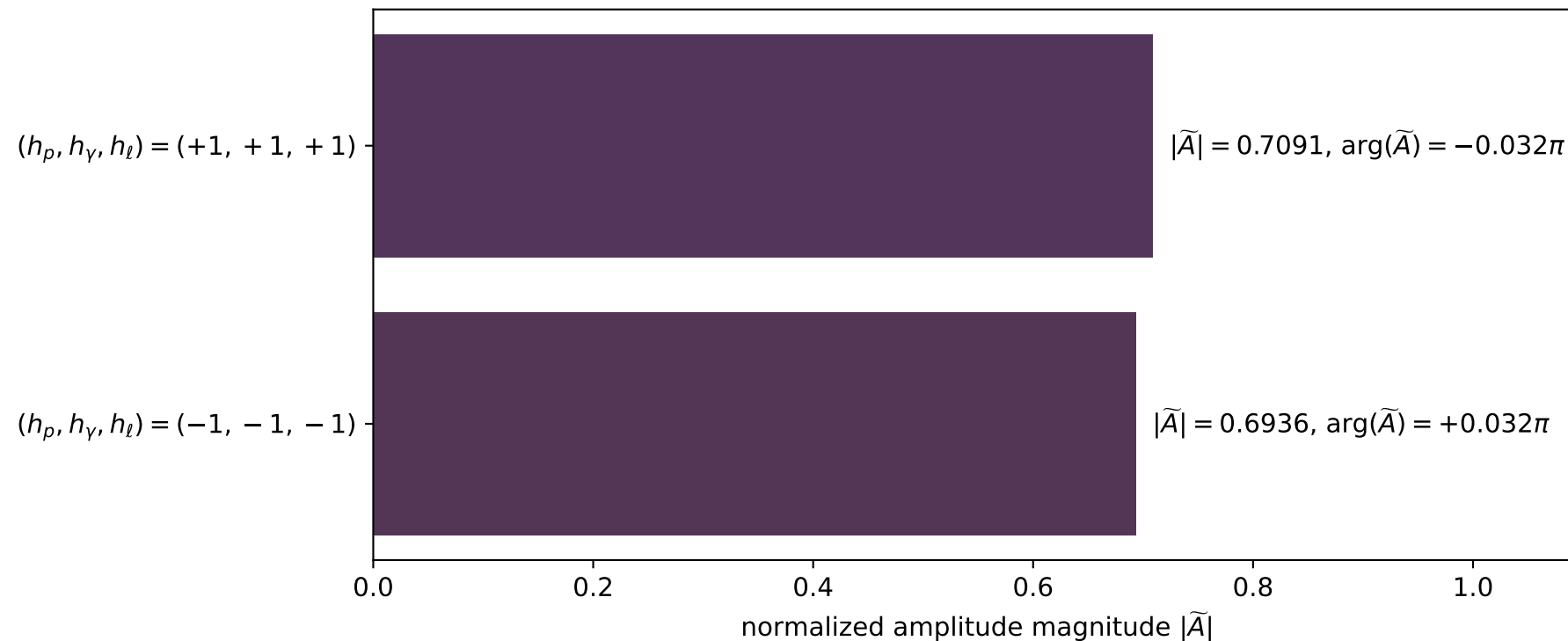
$s=8.54507$, $\sqrt{s}=2.9232$, $|\vec{P}|=1.3111$, $|\vec{P}'|=2.76355\text{e-}06$
 $\theta_{P'}=3.13818$, $\theta_\gamma=3.1125$, $\phi_{P'}=0.101143$, $\phi_\gamma=6.28318$
 $E_\gamma=0.992596$, $\theta_{l'}=0.0290944$, $\phi_{l'}=3.14159$
 $Q^2=5.2045$, $x_B=0.736492$, $t=-1.2646$, $W^2=2.74195$, $y=0.921904$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.3111$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3111$
 P $E= 1.6121$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3111$
 l' $E= 0.9926$ $p_x= -0.0289$ $p_y= 0.0000$ $p_z= 0.9922$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 0.9926$ $p_x= 0.0289$ $p_y= -0.0000$ $p_z= -0.9922$

x--y plane

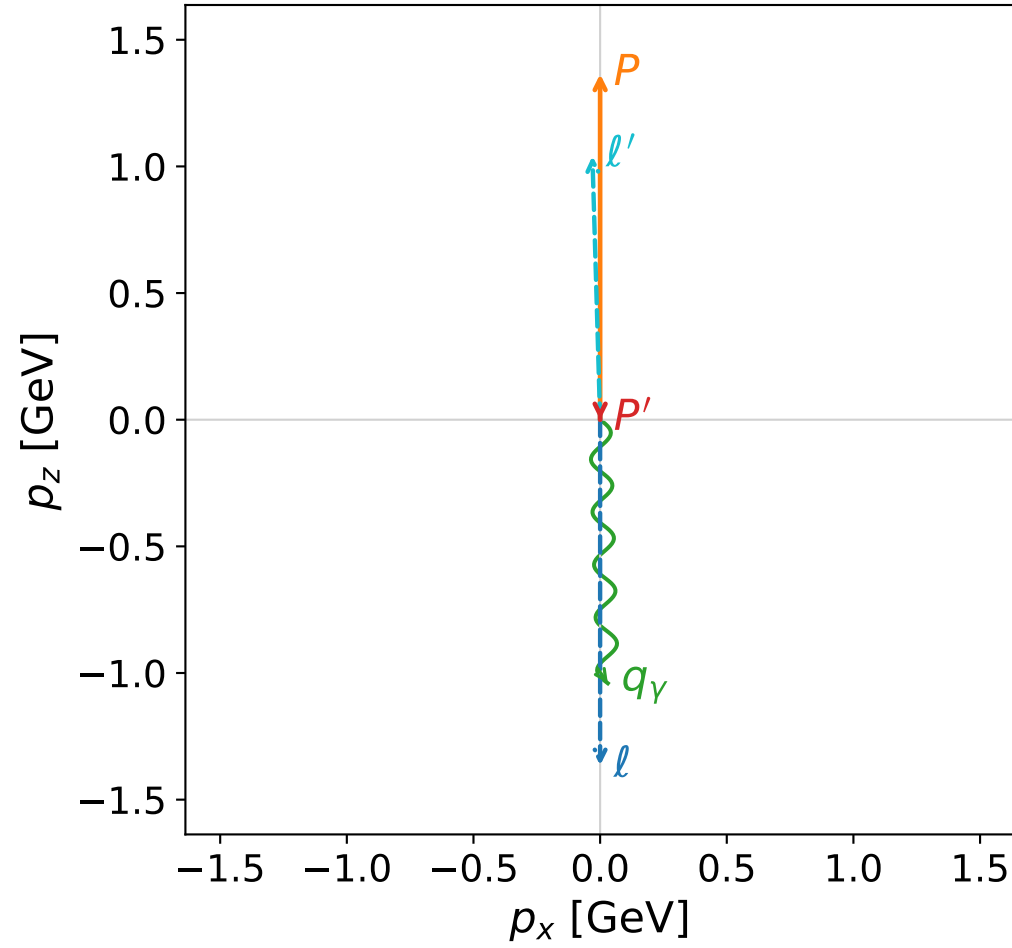


Leading normalized final-state amplitudes
(N=2, retained=98.4%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

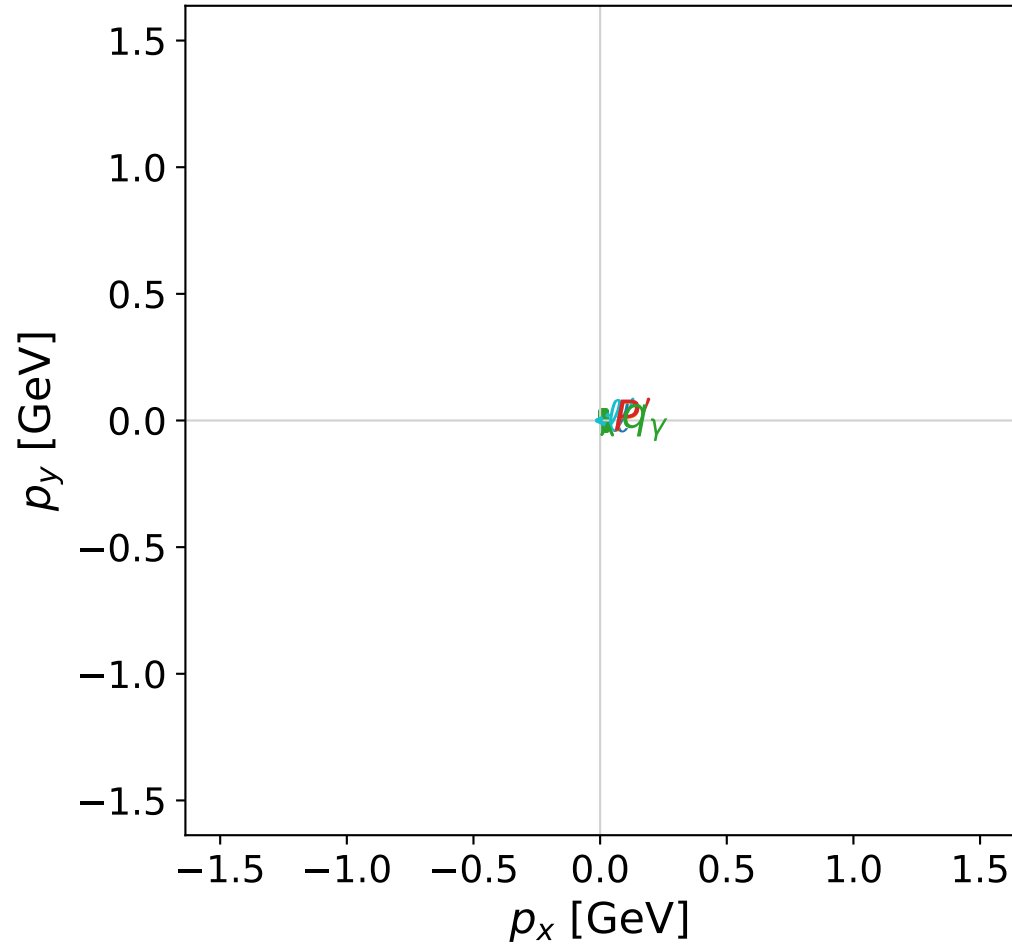


dghz_mixing_angles_polarization_01_local_minimum_364
 $d_{\text{GHZ}}=0.00240137$
 region: polarization_cluster_01_local_minimum_364
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0364
 lepton mixing angle: $\alpha_e=1.5438592$ rad
 incoming lepton: $0.0269338|+\rangle + 0.999637|-\rangle$
 proton mixing angle: $\alpha_p=0.026354725$ rad
 incoming proton: $0.999653|+\rangle + 0.0263517|-\rangle$

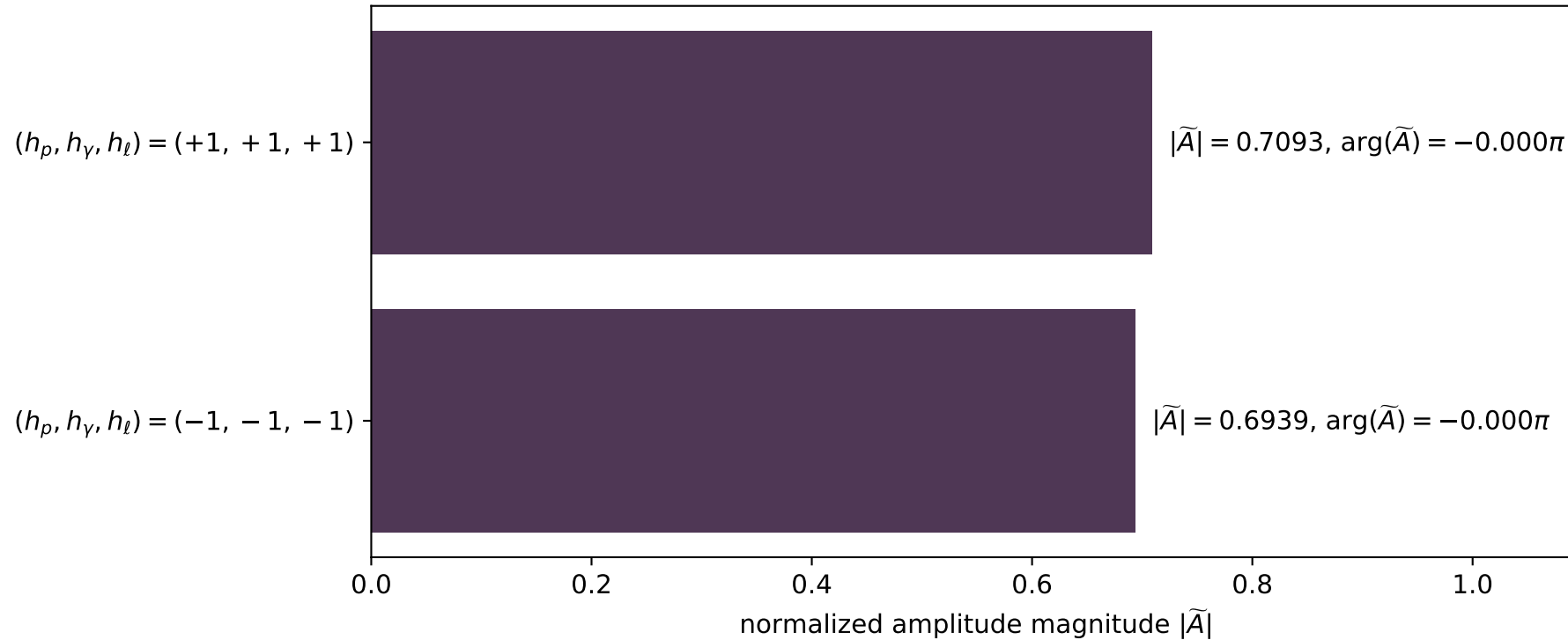
$s=9.12098$, $\sqrt{s}=3.0201$, $|\vec{P}|=1.36438$, $|\vec{P}'|=4.15695\text{e-}06$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.11231$, $\phi_{P'}=0.518887$, $\phi_\gamma=6.28317$
 $E_\gamma=1.04105$, $\theta_{\ell'}=0.0292827$, $\phi_{\ell'}=3.14158$
 $Q^2=5.68034$, $x_B=0.74415$, $t=-1.34644$, $W^2=2.83284$, $y=0.926248$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3644$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3644$
 P $E= 1.6557$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3644$
 ℓ' $E= 1.0410$ $p_x= -0.0305$ $p_y= 0.0000$ $p_z= 1.0406$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 1.0410$ $p_x= 0.0305$ $p_y= -0.0000$ $p_z= -1.0406$

x--y plane

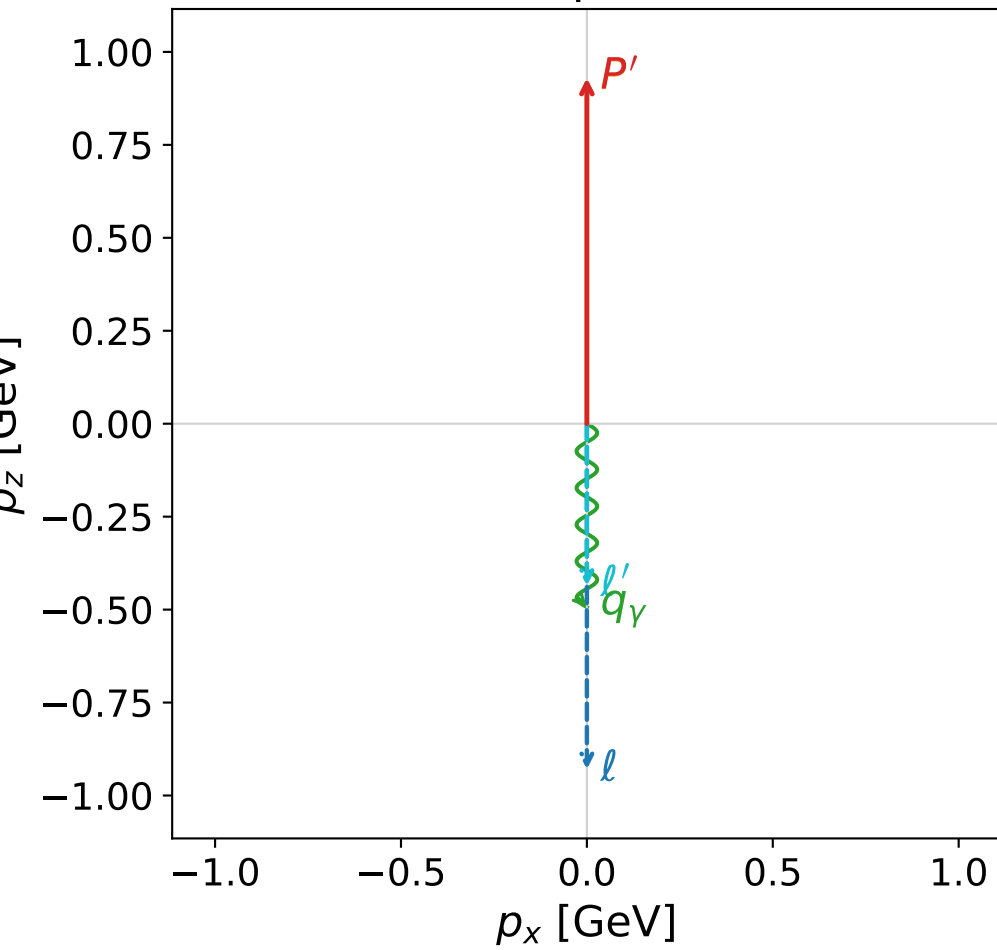


Leading normalized final-state amplitudes
(N=2, retained=98.5%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

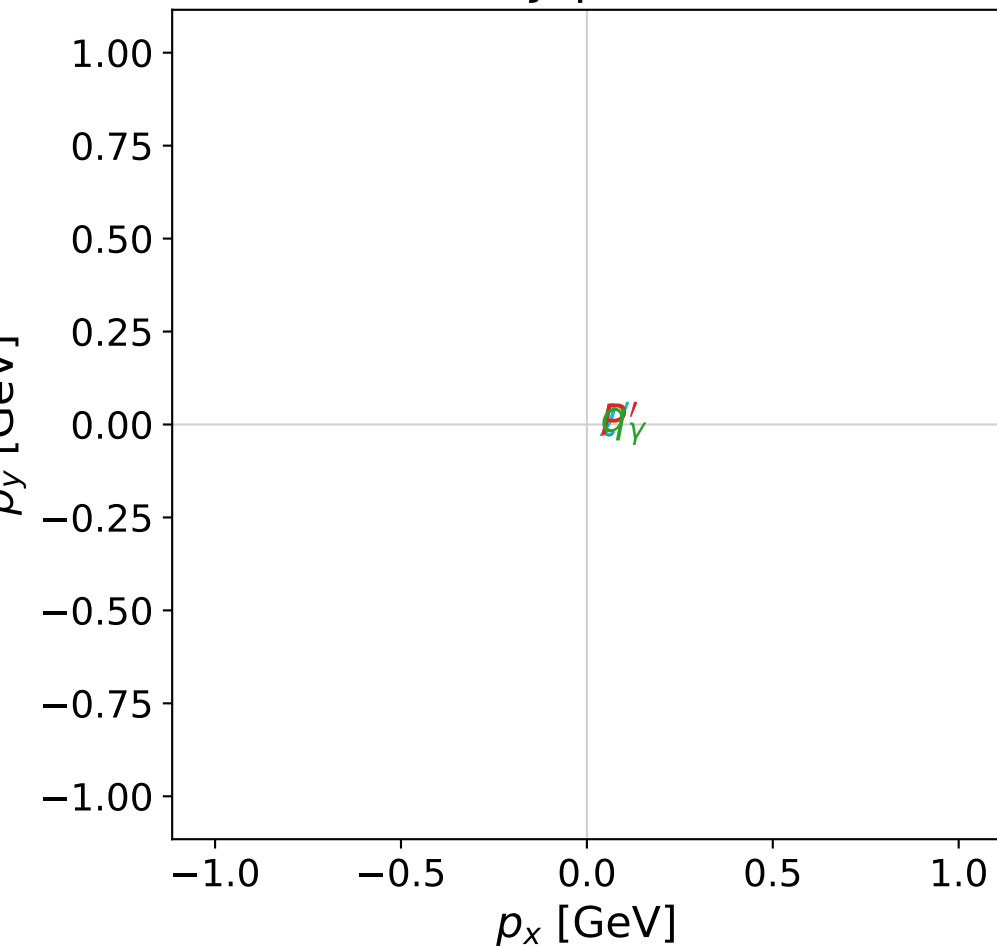


dghz_mixing_angles_polarization_01_local_minimum_367
 $d_{\{\rm GHZ\}}=0.0171798$
 region: polarization_cluster_01_local_minimum_367
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0367
 lepton mixing angle: $\alpha_e=6.1169479\text{e-}06$ rad
 incoming lepton: $1|+\rangle +6.11695\text{e-}06|-\rangle$
 proton mixing angle: $\alpha_p=0$ rad
 incoming proton: $1|+\rangle +0|-\rangle$

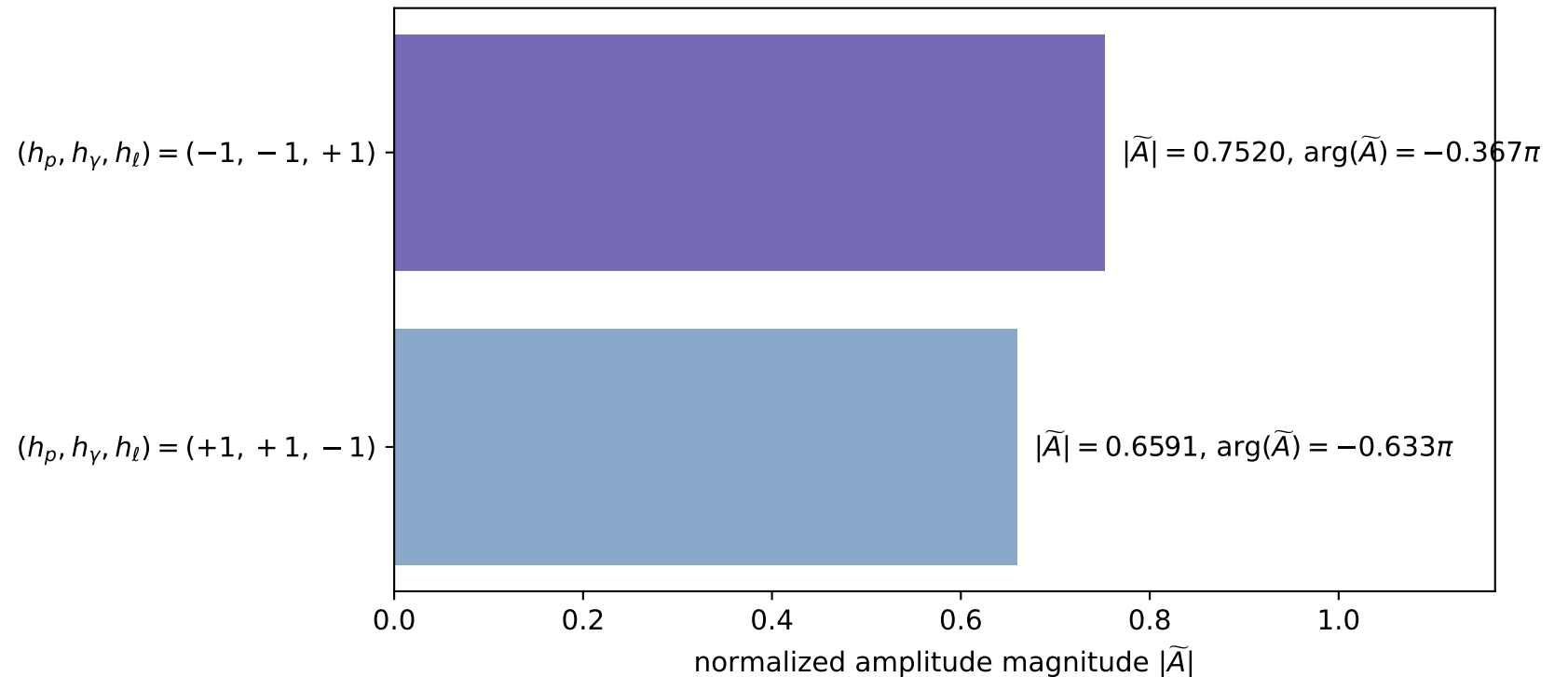
$s=5.06282$, $\sqrt{s}=2.25007$, $|\vec{P}|=0.929521$, $|\vec{P}'|=0.929521$
 $\theta_{p'}=0$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.9999$, $\phi_\gamma=5.13056$
 $E_\gamma=0.493368$, $\theta_{\ell'}=3.14159$, $\phi_{\ell'}=1.98897$
 $Q^2=1.56777\text{e-}07$, $x_B=7.06132\text{e-}08$, $t=-4.3871\text{e-}15$, $W^2=3.10007$, $y=0.530776$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.9295$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9295$
 P $E= 1.3206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9295$
 ℓ' $E= 0.4362$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.4362$
 P' $E= 1.3206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.9295$
 q_γ $E= 0.4934$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.4934$

x--y plane

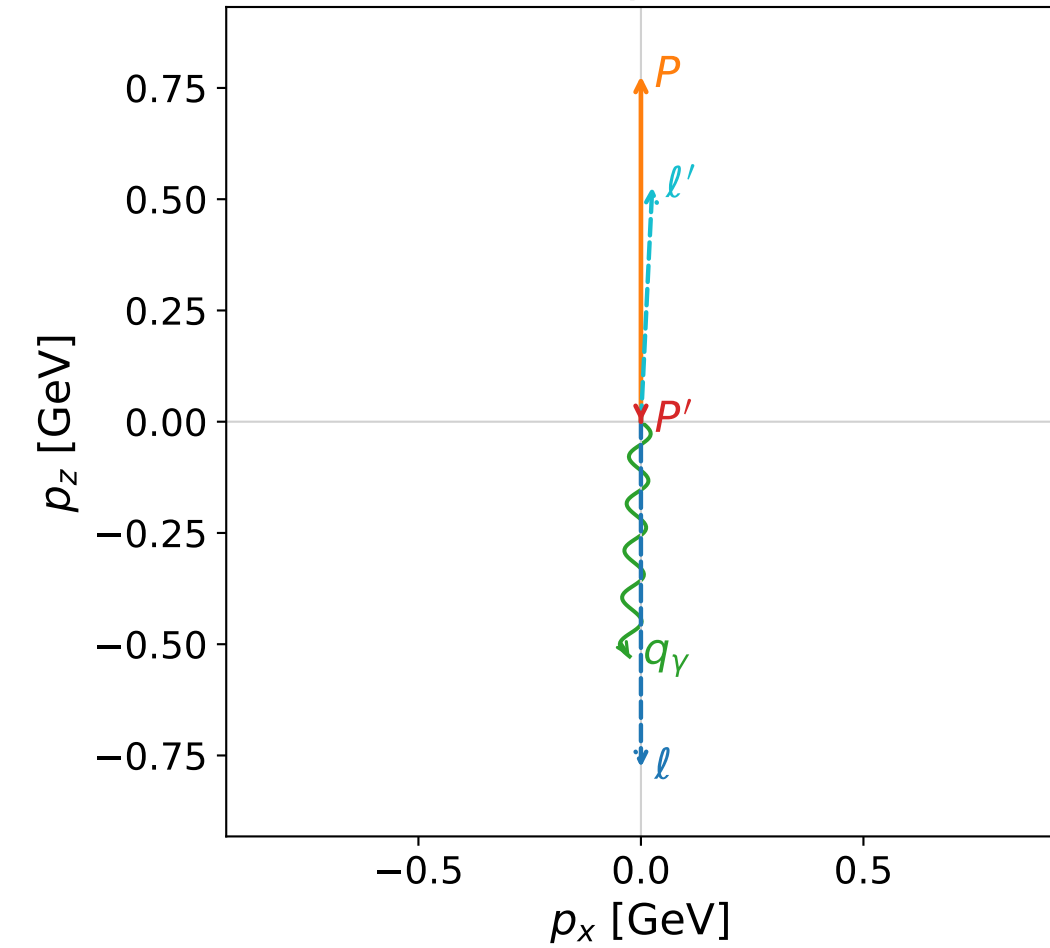


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

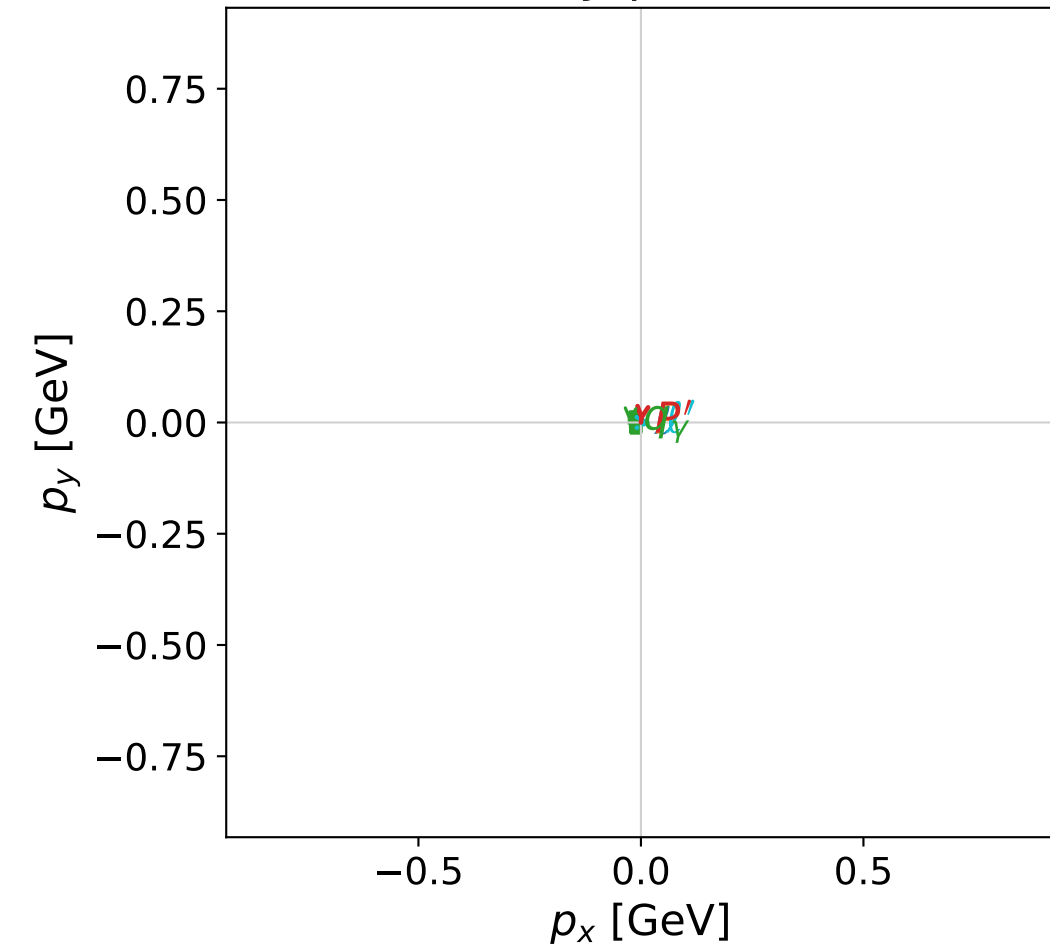


dghz_mixing_angles_polarization_01_local_minimum_369
 $d_{\text{GHZ}}=0.0208267$
 region: polarization_cluster_01_local_minimum_369
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0369
 lepton mixing angle: $\alpha_e=1.5707901$ rad
 incoming lepton: $6.21409\text{e-}06|+\rangle +1|-\rangle$
 proton mixing angle: $\alpha_p=3.1415775$ rad
 incoming proton: $-1|+\rangle +1.51327\text{e-}05|-\rangle$

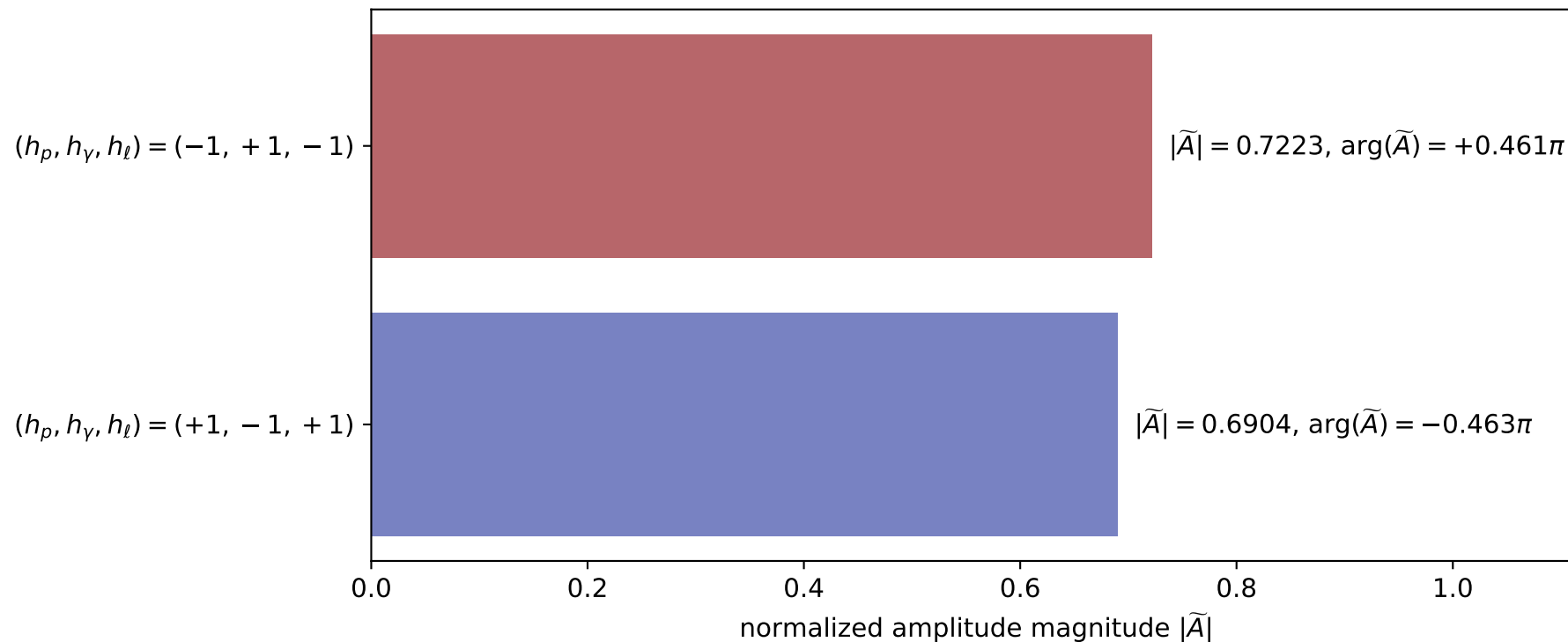
$s=3.97747$, $\sqrt{s}=1.99436$, $|\vec{P}|=0.776596$, $|\vec{P}'|=4.18979\text{e-}06$
 $\theta_{P'}=3.14157$, $\theta_Y=3.09338$, $\phi_{P'}=4.59353$, $\phi_Y=3.13875$
 $E_Y=0.528177$, $\theta_{\ell'}=0.0482106$, $\phi_{\ell'}=6.28034$
 $Q^2=1.63978$, $x_B=0.62334$, $t=-0.524841$, $W^2=1.8707$, $y=0.849244$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.7766$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7766$
 P $E= 1.2178$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7766$
 ℓ' $E= 0.5282$ $p_x= 0.0255$ $p_y= -0.0001$ $p_z= 0.5276$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_Y $E= 0.5282$ $p_x= -0.0255$ $p_y= 0.0001$ $p_z= -0.5276$

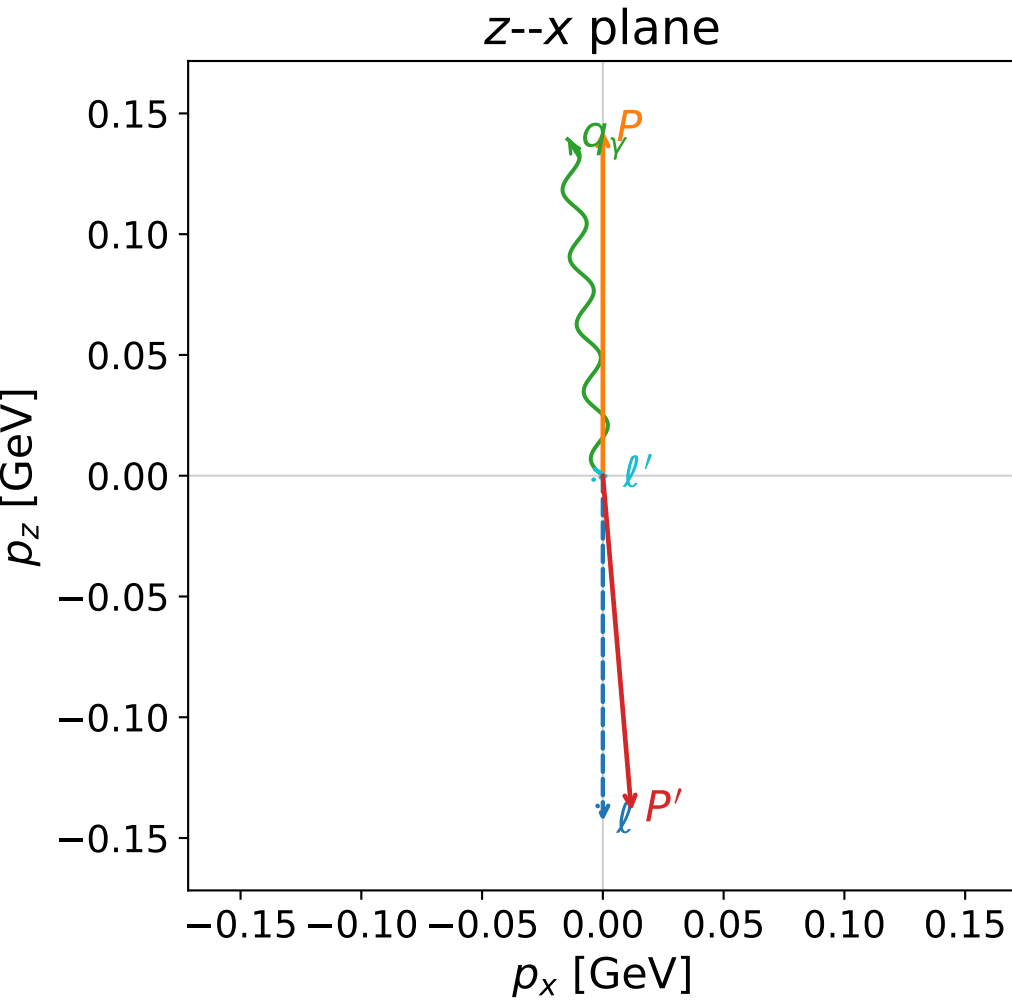
x--y plane



Leading normalized final-state amplitudes
 (N=2, retained=99.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

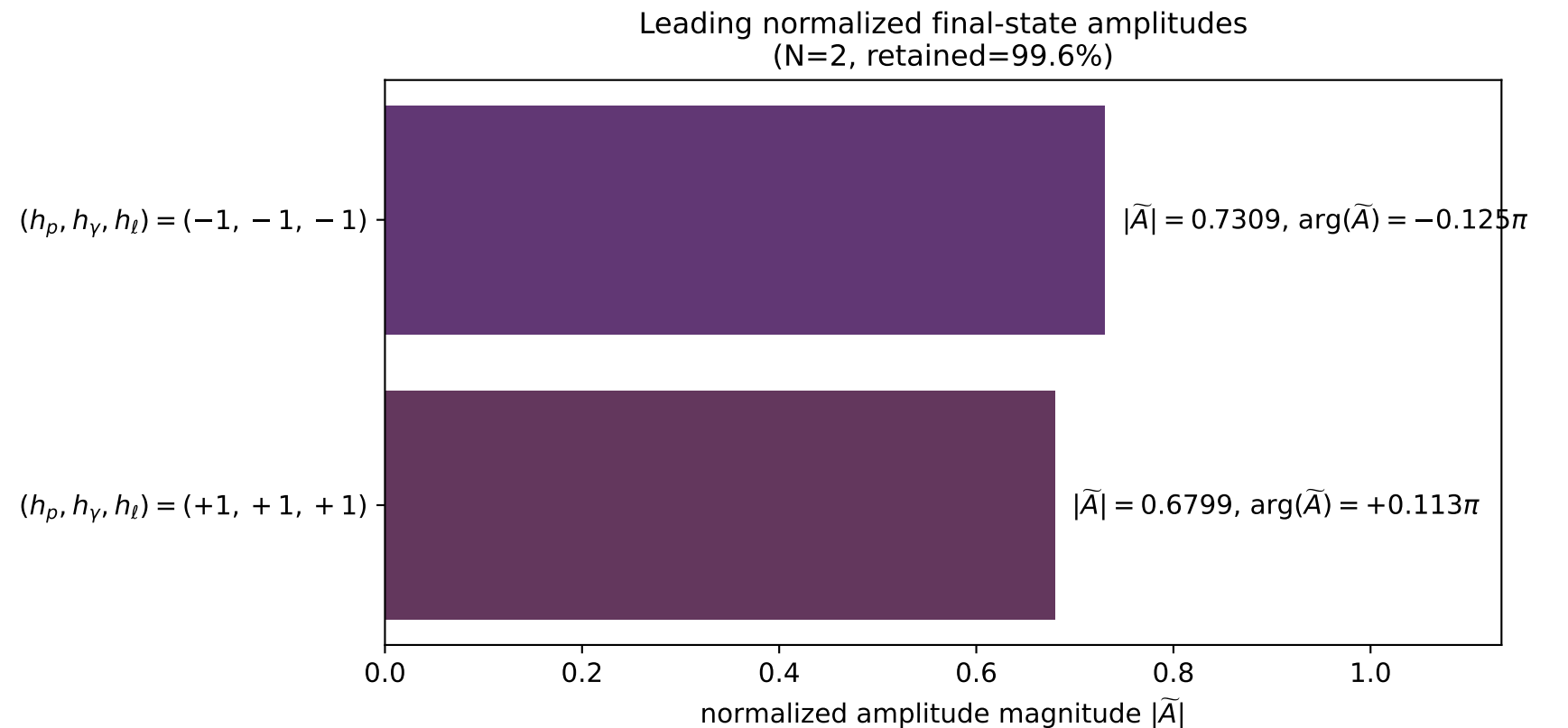
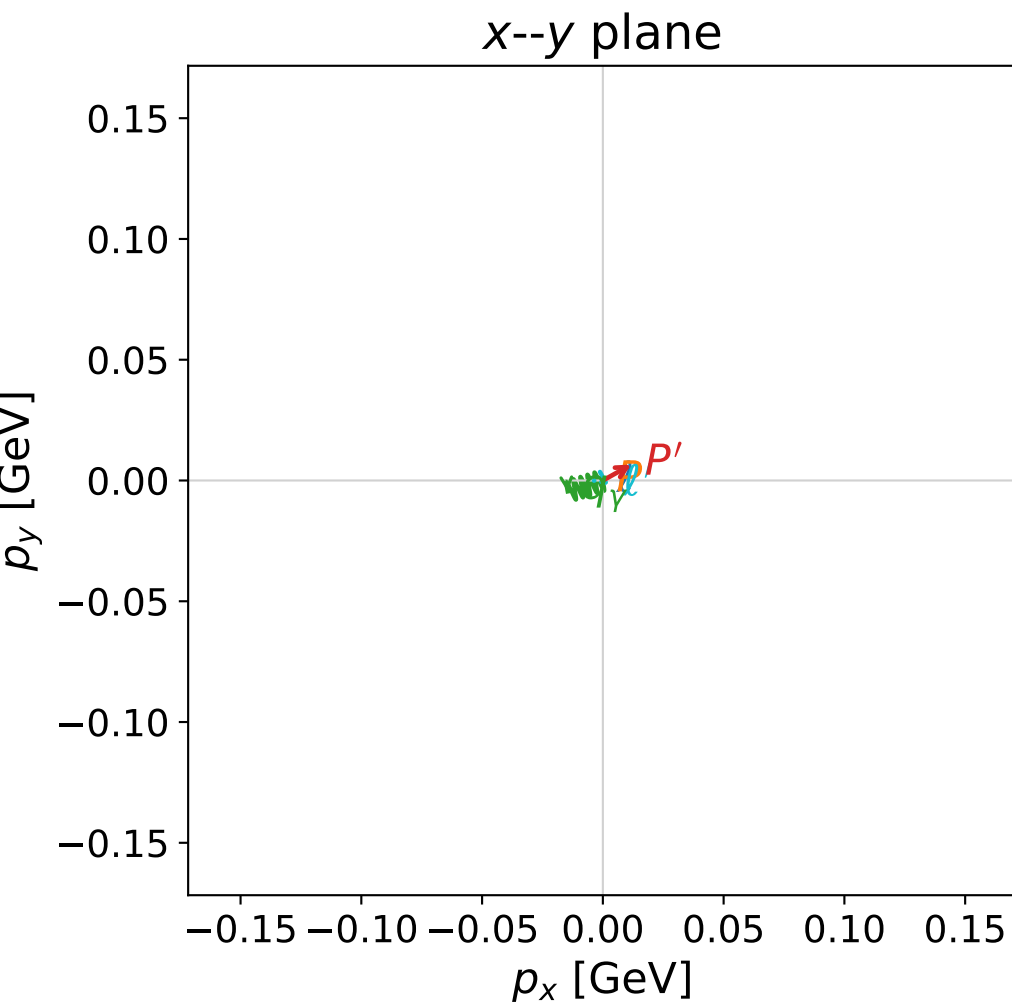


dghz_mixing_angles_polarization_01_local_minimum_373
 $d_{\{\rm GHZ\}}=0.0300609$
 region: polarization_cluster_01_local_minimum_373
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0373
 lepton mixing angle: $\alpha_e=0.74214409$ rad
 incoming lepton: $0.737021|+\rangle +0.67587|-\rangle$
 proton mixing angle: $\alpha_p=0.90015049$ rad
 incoming proton: $0.621492|+\rangle +0.78342|-\rangle$

$s=1.19232$, $\sqrt{s}=1.09193$, $|\vec{P}|=0.143083$, $|\vec{P}'|=0.139544$
 $\theta_{p'}=3.04448$, $\theta_\gamma=0.109503$, $\phi_{p'}=0.509518$, $\phi_\gamma=3.44722$
 $E_\gamma=0.140121$, $\theta_{\ell'}=1.68565$, $\phi_{\ell'}=5.66363$
 $Q^2=0.000885013$, $x_B=0.00289466$, $t=-0.0796895$, $W^2=1.1847$, $y=0.978442$

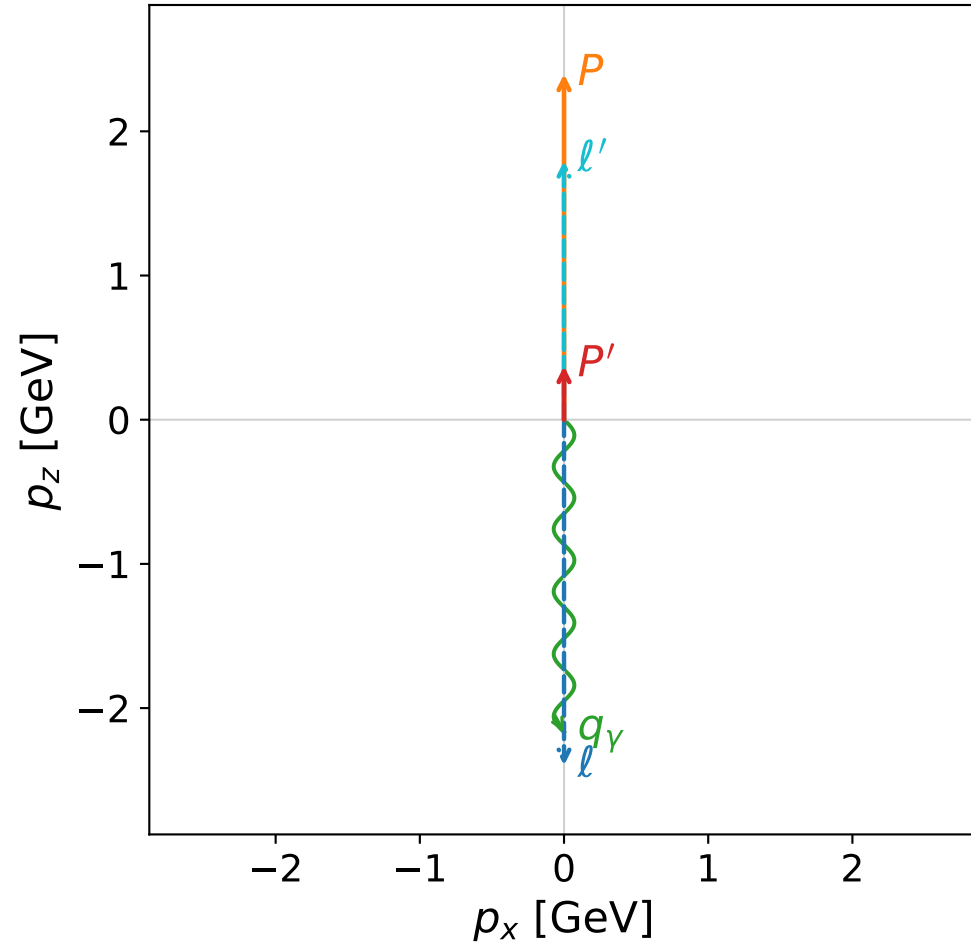
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1431	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1431
P	$E=$	0.9489	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1431
ℓ'	$E=$	0.0035	$p_x=$	0.0028	$p_y=$	-0.0020	$p_z=$	-0.0004
P'	$E=$	0.9483	$p_x=$	0.0118	$p_y=$	0.0066	$p_z=$	-0.1389
q_γ	$E=$	0.1401	$p_x=$	-0.0146	$p_y=$	-0.0046	$p_z=$	0.1393

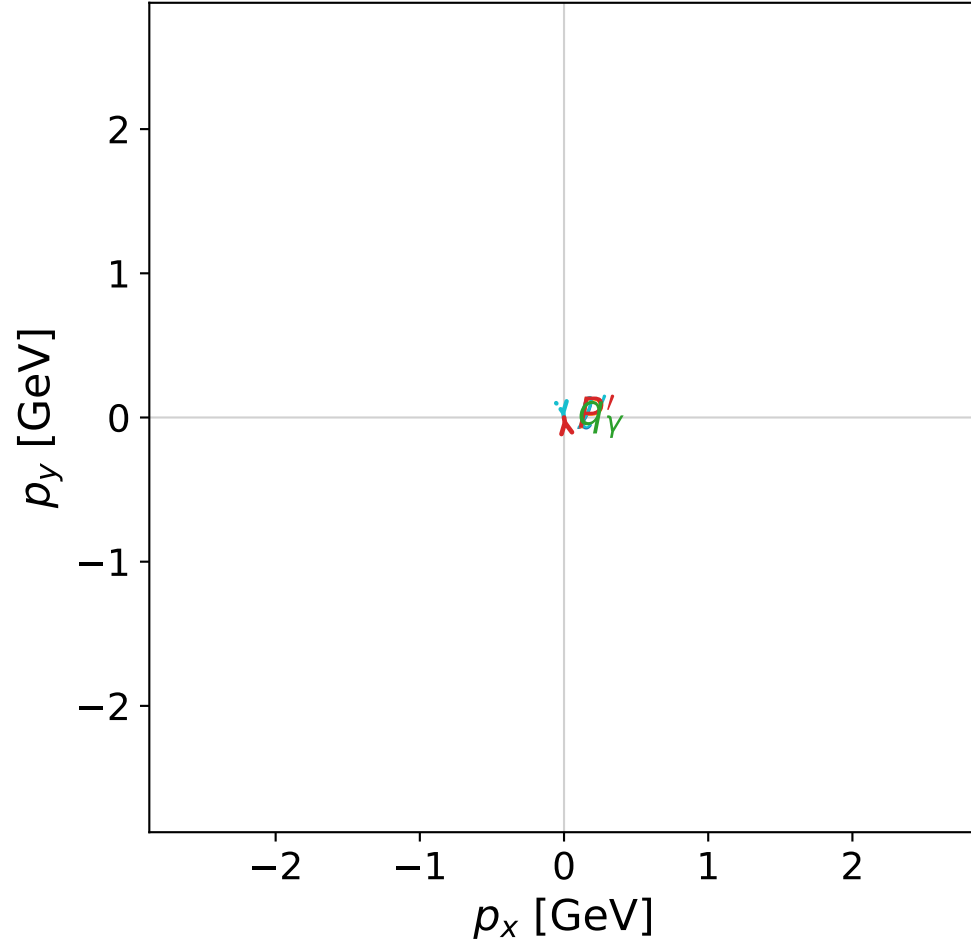


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

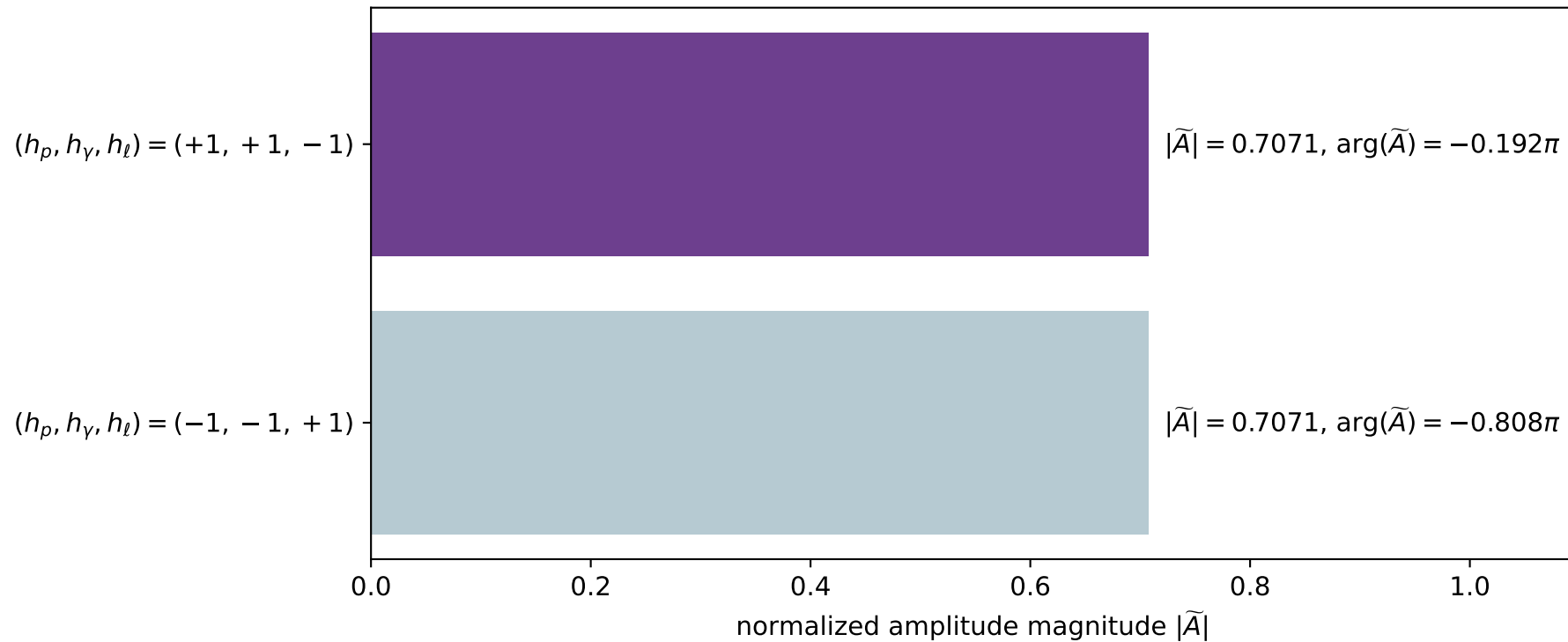


dghz_mixing_angles_polarization_02_local_minimum_0
 $d_{\text{GHZ}}=2.22835\text{e-}08$
 region: polarization_cluster_02_local_minimum_0
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0000
 lepton mixing angle: $\alpha_e=0.7579928$ rad
 incoming lepton: $0.726217|+\rangle + 0.687465|-\rangle$
 proton mixing angle: $\alpha_p=2.3835959$ rad
 incoming proton: $-0.726215|+\rangle + 0.687468|-\rangle$

$s=24.7062$, $\sqrt{s}=4.97053$, $|\vec{P}|=2.39676$, $|\vec{P}'|=0.371823$
 $\theta_{p'}=1.30111\text{e-}07$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.74218$, $\phi_\gamma=3.74627$
 $E_\gamma=2.16667$, $\theta_{l'}=2.58096\text{e-}08$, $\phi_{l'}=4.88378$
 $Q^2=17.2073$, $x_B=0.741984$, $t=-1.65188$, $W^2=6.86346$, $y=0.973331$

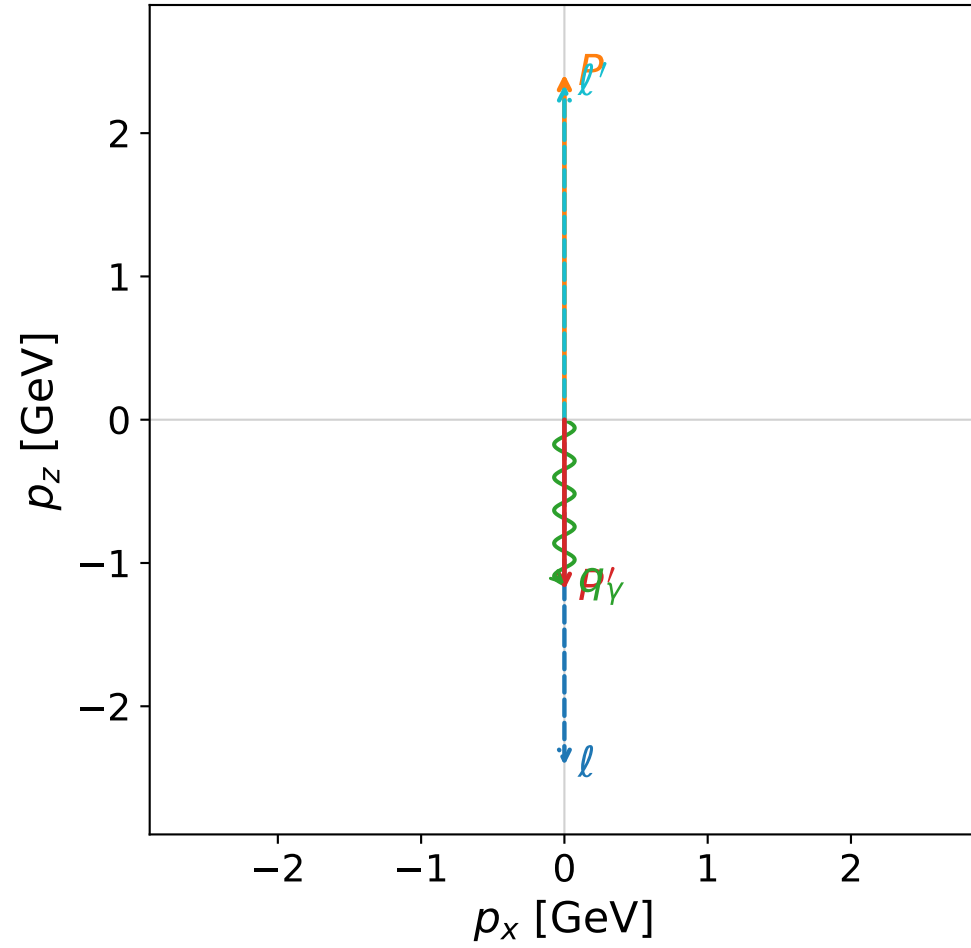
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3968$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3968$
 P $E= 2.5738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3968$
 l' $E= 1.7948$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7948$
 P' $E= 1.0090$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3718$
 q_γ $E= 2.1667$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1667$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

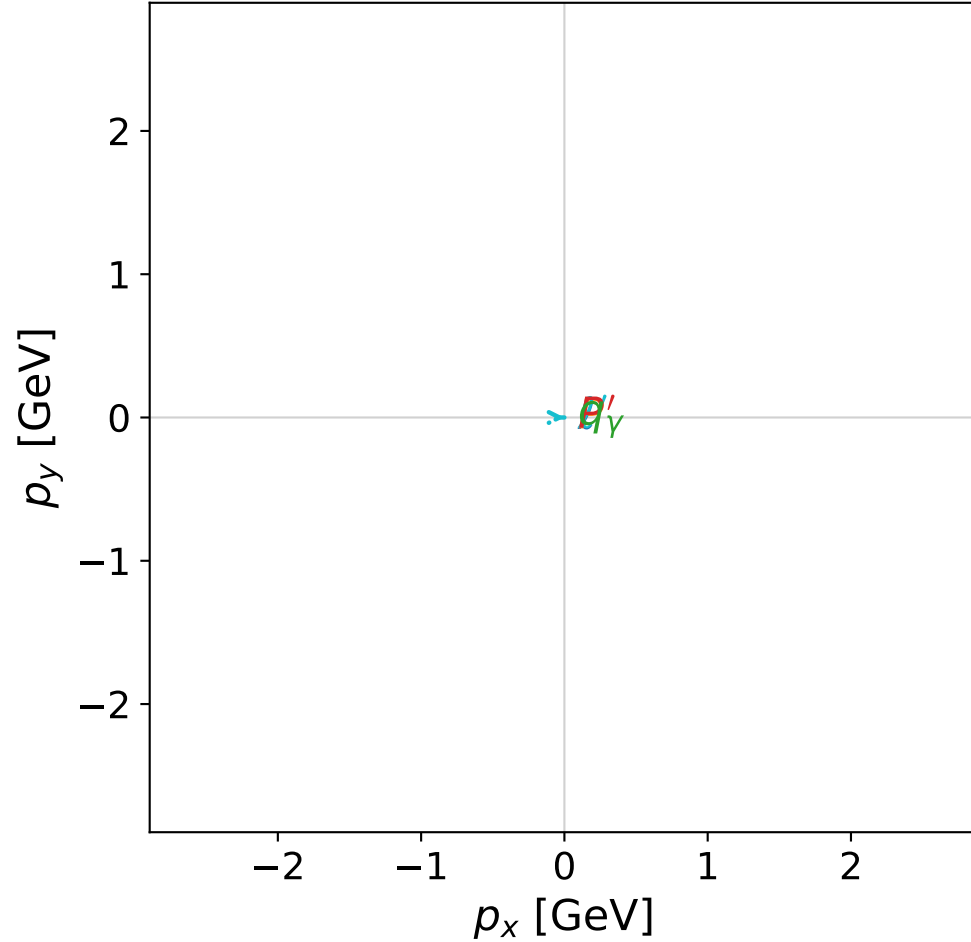


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



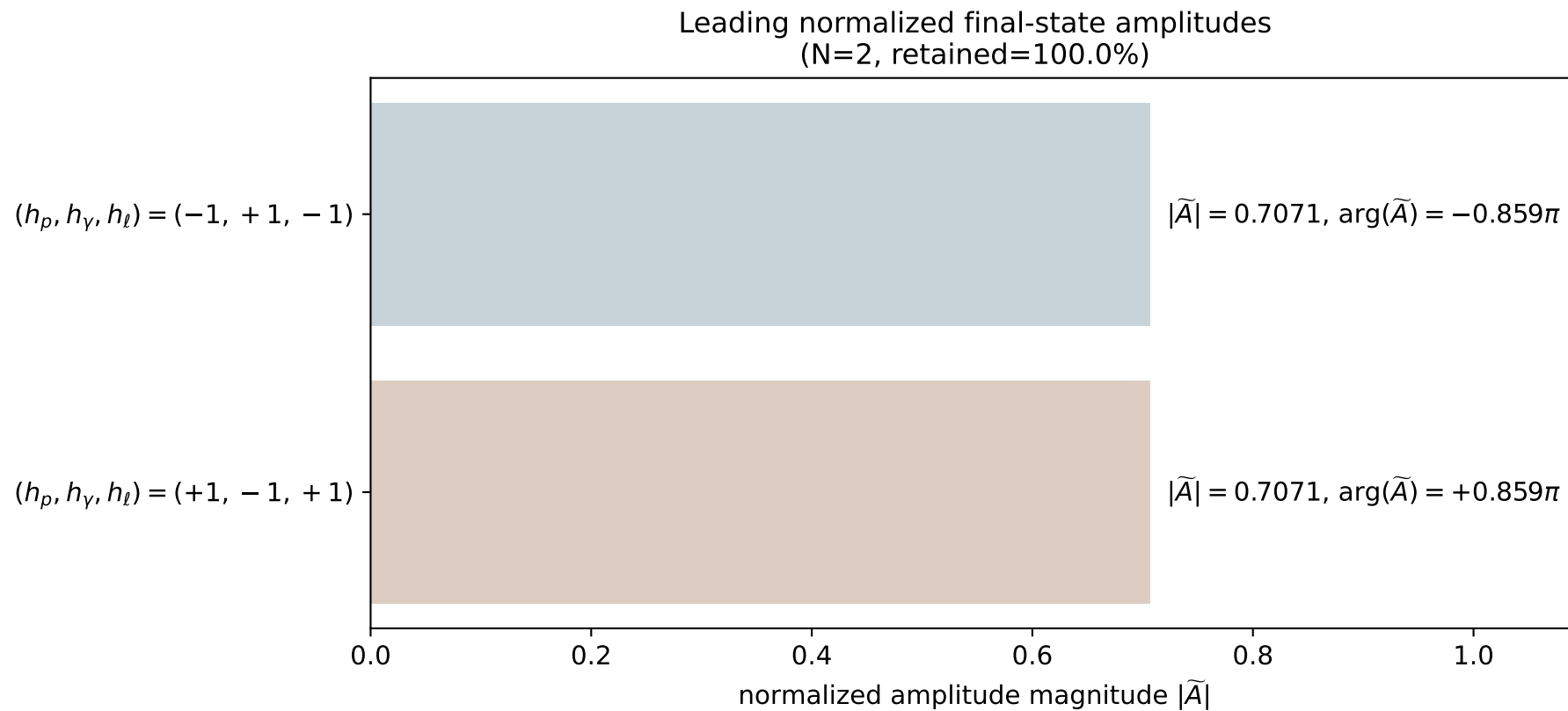
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_3
 $d_{\text{GHZ}}=2.32433\text{e-}08$
 region: polarization_cluster_02_local_minimum_3
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0003
 lepton mixing angle: $\alpha_e=0.82685667$ rad
 incoming lepton: $0.677192|+\rangle + 0.735806|-\rangle$
 proton mixing angle: $\alpha_p=2.3147292$ rad
 incoming proton: $-0.677187|+\rangle + 0.735811|-\rangle$

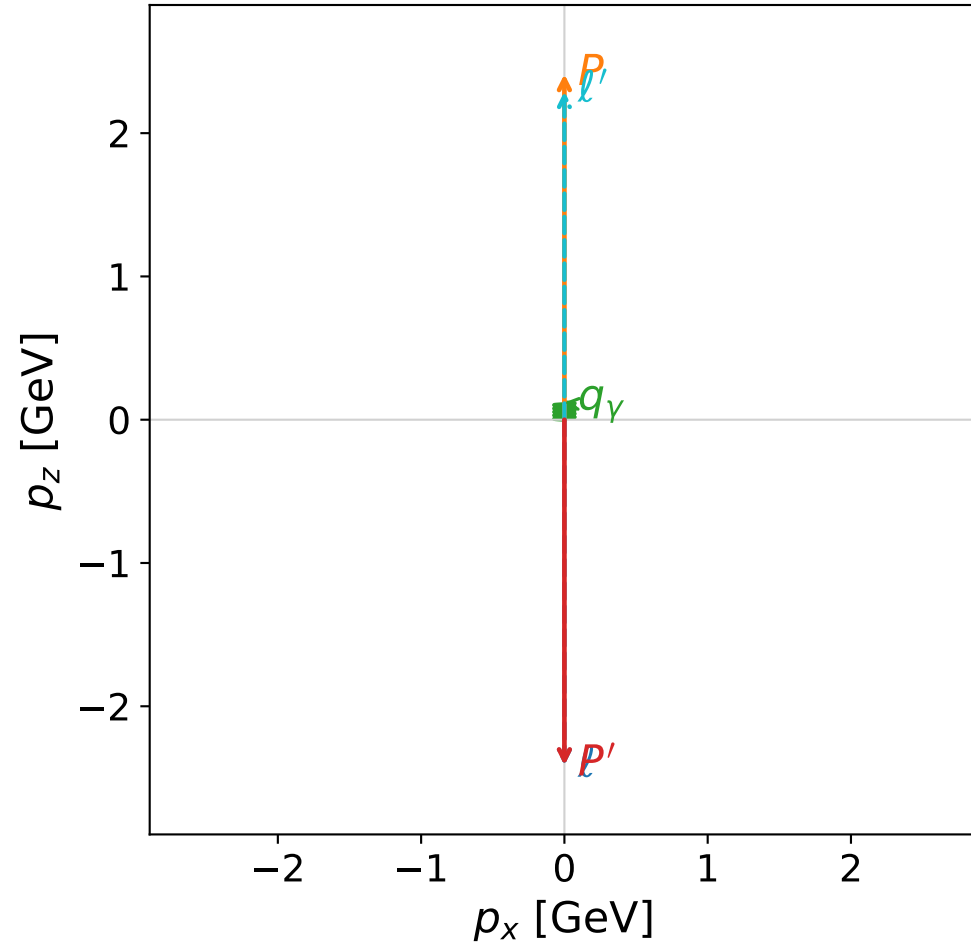
$s=24.9996$, $\sqrt{s}=4.99996$, $|\vec{P}|=2.412$, $|\vec{P}'|=1.18707$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.98106$, $\phi_\gamma=2.69995$
 $E_\gamma=1.14997$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.98435$
 $Q^2=22.5478$, $x_B=0.96783$, $t=-11.7976$, $W^2=1.62932$, $y=0.965899$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3370$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.3370$
 P' $E= 1.5129$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.1871$
 q_γ $E= 1.1500$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.1500$

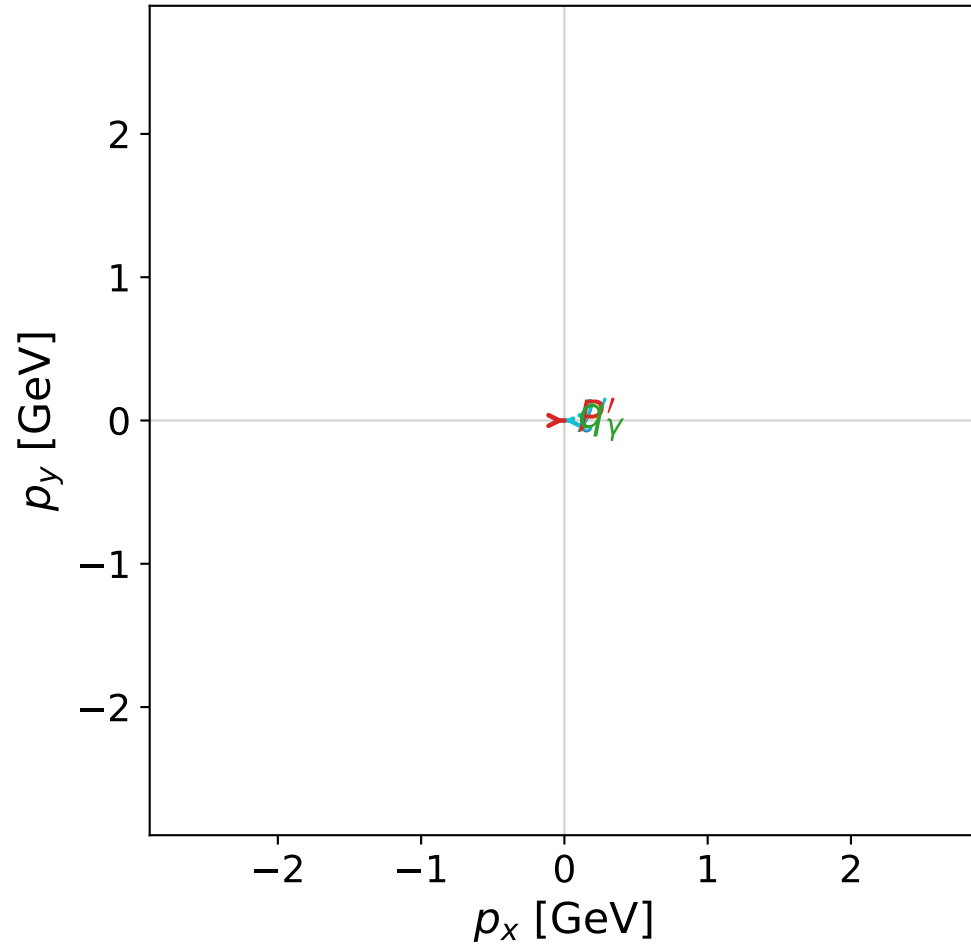


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



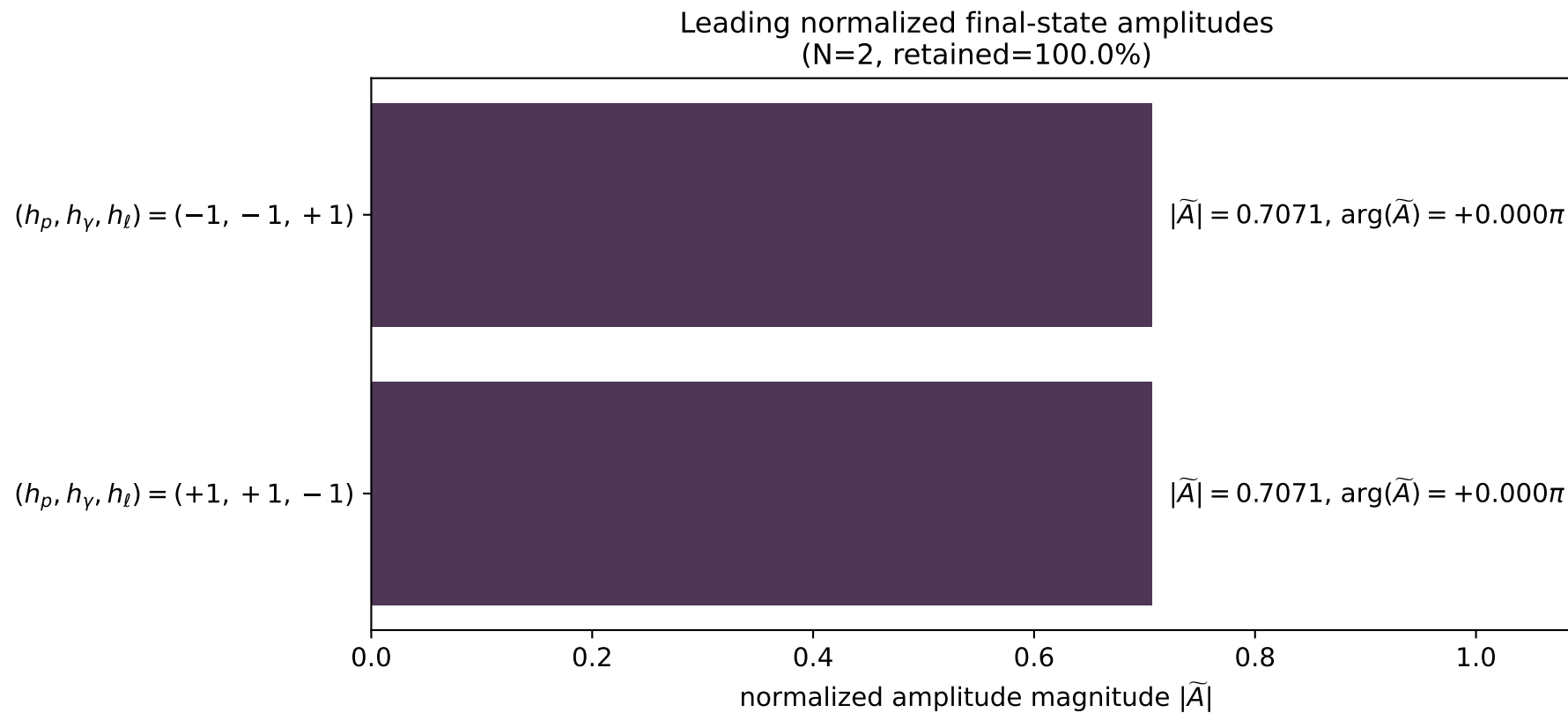
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_4
 $d_{\text{GHZ}}=2.36714\text{e-}08$
 region: polarization_cluster_02_local_minimum_4
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0004
 lepton mixing angle: $\alpha_e=0.81232036$ rad
 incoming lepton: $0.687816|+\rangle + 0.725885|-\rangle$
 proton mixing angle: $\alpha_p=2.3831047$ rad
 incoming proton: $-0.725877|+\rangle + 0.687825|-\rangle$

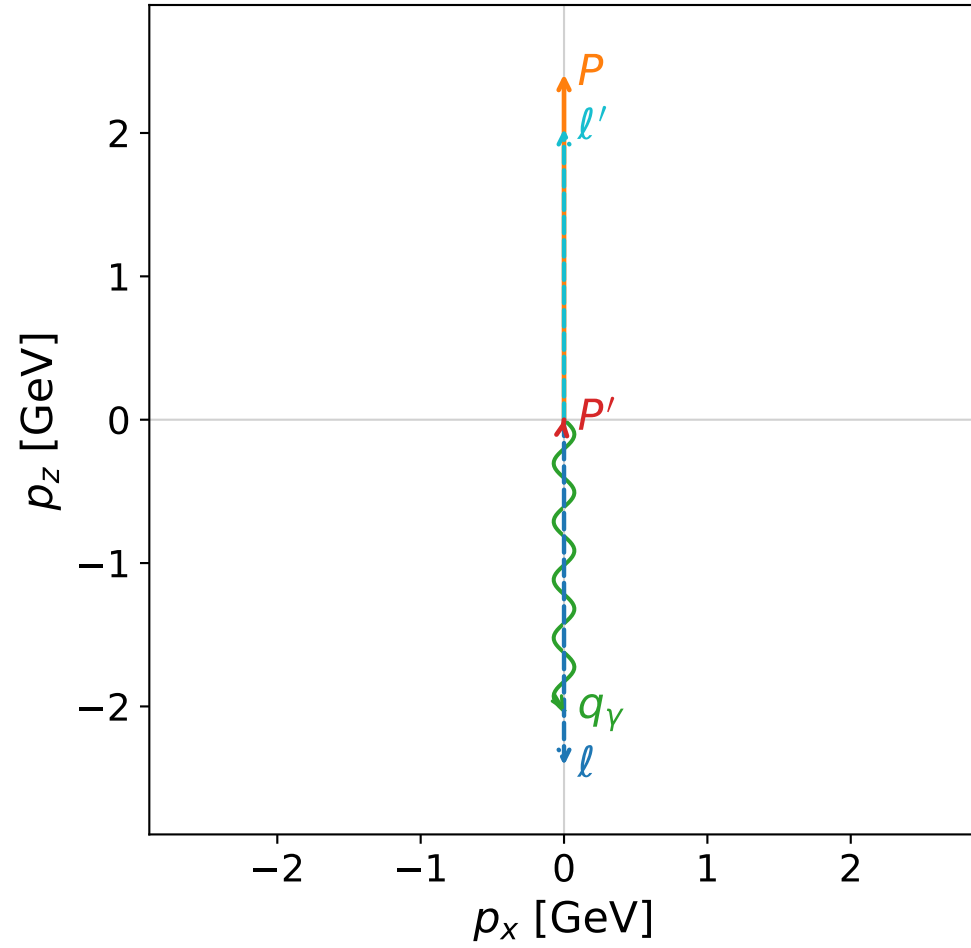
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0$, $\phi_\gamma=0.437635$
 $E_\gamma=0.120596$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

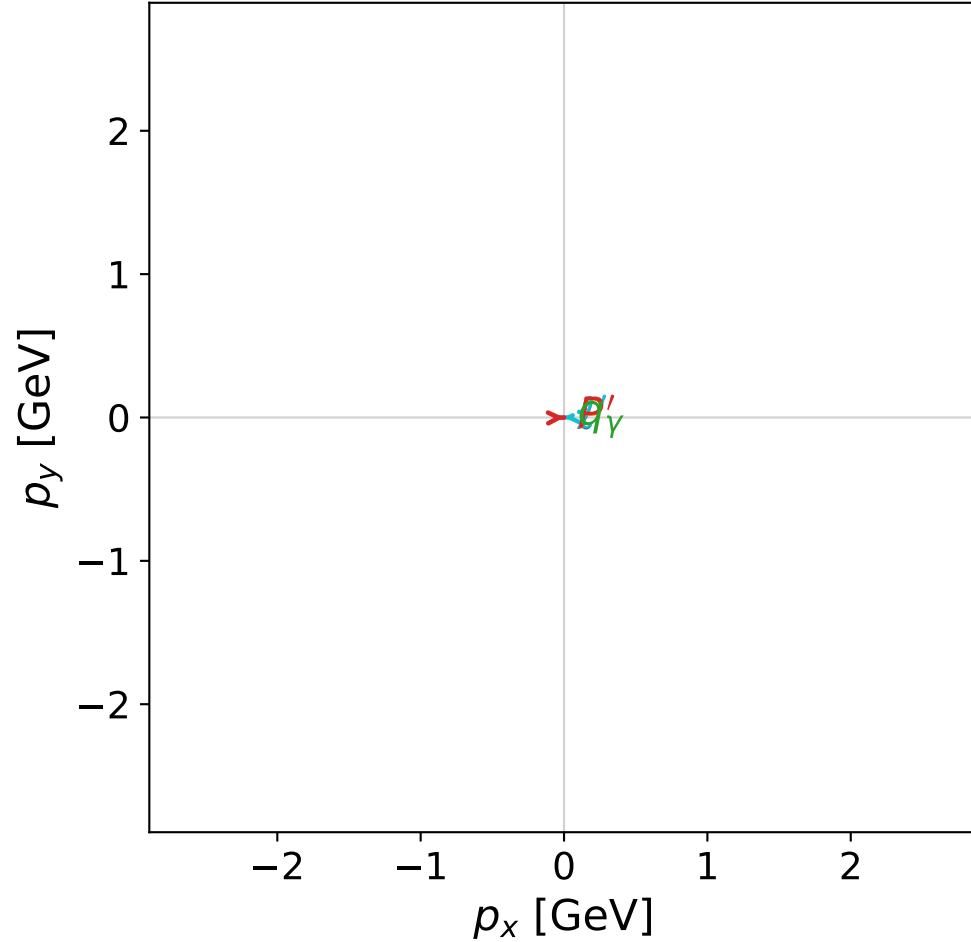


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



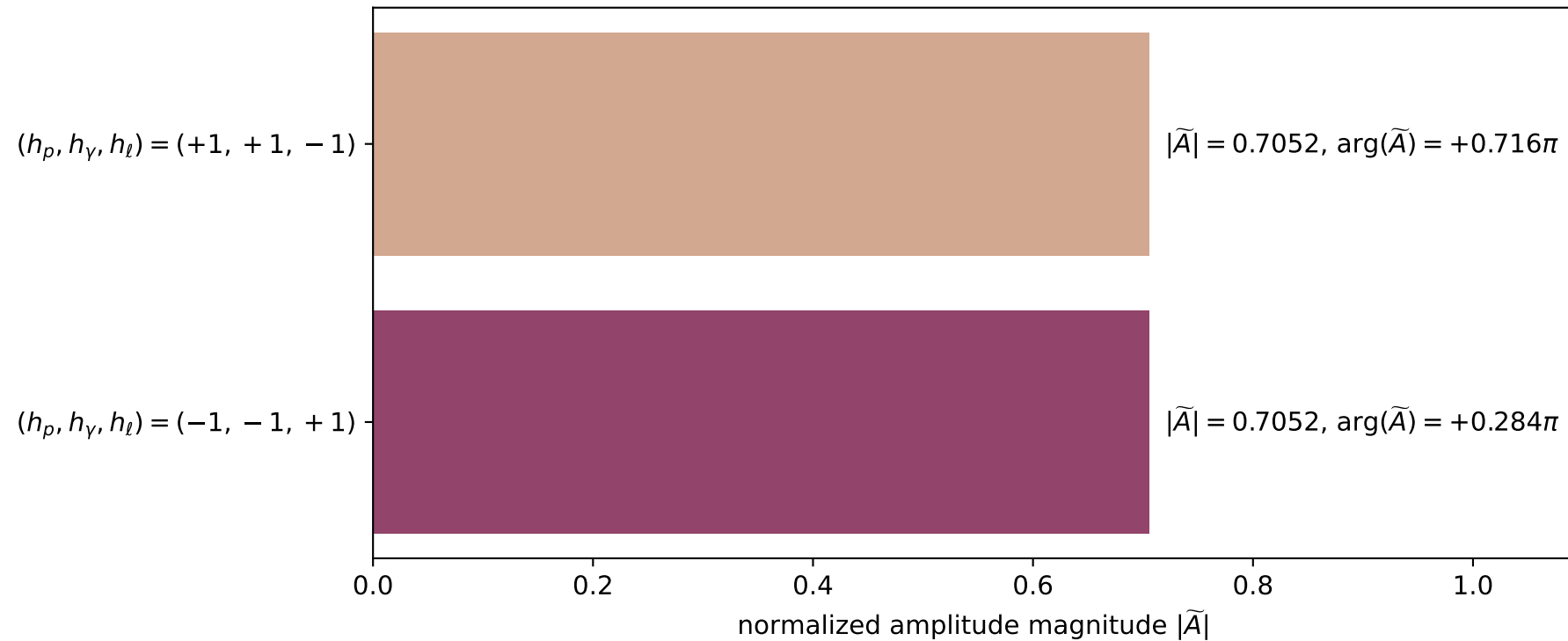
dghz_mixing_angles_polarization_02_local_minimum_6
 $d_{\{\text{rm GHZ}\}}=2.39329\text{e-}08$
 region: polarization_cluster_02_local_minimum_6
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0006
 lepton mixing angle: $\alpha_e=0.81040845$ rad
 incoming lepton: $0.689203|+\rangle + 0.724569|-\rangle$
 proton mixing angle: $\alpha_p=2.331173$ rad
 incoming proton: $-0.689194|+\rangle + 0.724576|-\rangle$

$s=24.9736$, $\sqrt{s}=4.99736$, $|\vec{P}|=2.41065$, $|\vec{P}'|=0.000556214$
 $\theta_{P'}=0.146926$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0285838$, $\phi_\gamma=0.891756$
 $E_\gamma=2.02996$, $\theta_{\ell'}=4.01243\text{e-}05$, $\phi_{\ell'}=3.17018$
 $Q^2=19.5688$, $x_B=0.837016$, $t=-3.09033$, $W^2=4.69028$, $y=0.970341$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

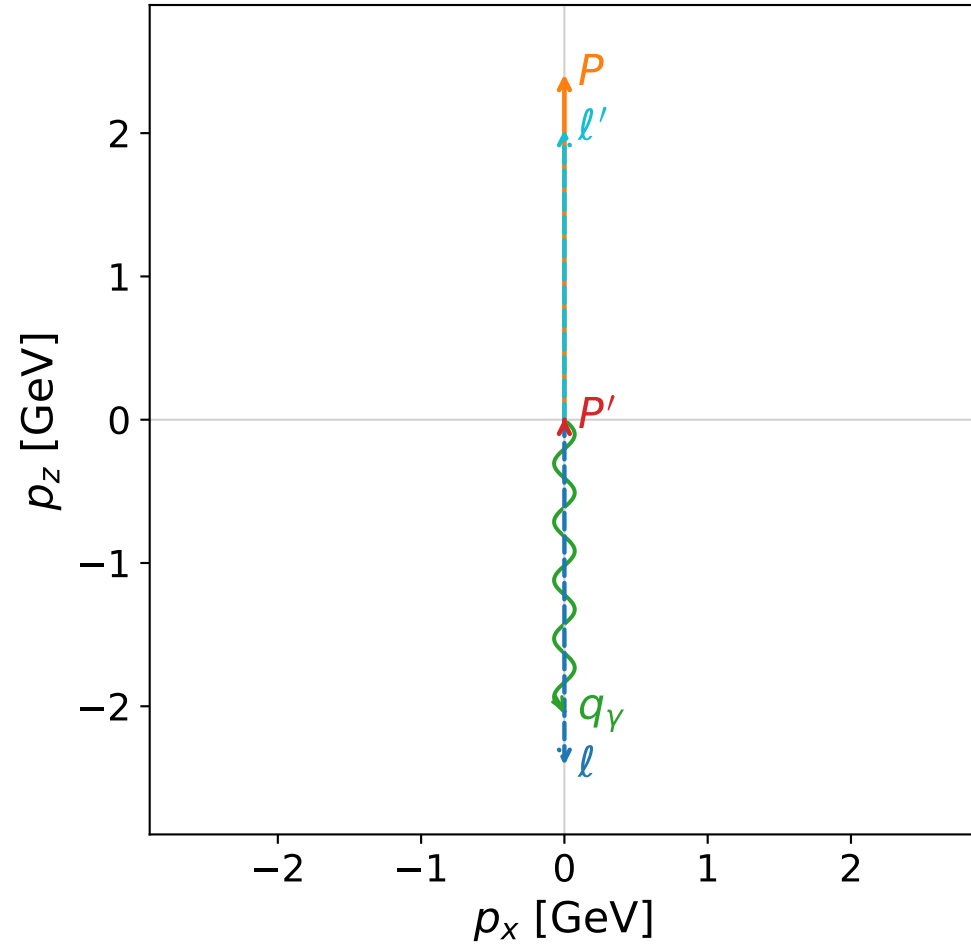
ℓ	$E=$	2.4107	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-2.4107
P	$E=$	2.5867	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	2.4107
ℓ'	$E=$	2.0294	$p_x=$	-0.0001	$p_y=$	-0.0000	$p_z=$	2.0294
P'	$E=$	0.9380	$p_x=$	0.0001	$p_y=$	0.0000	$p_z=$	0.0006
q_γ	$E=$	2.0300	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-2.0300

Leading normalized final-state amplitudes
(N=2, retained=99.5%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

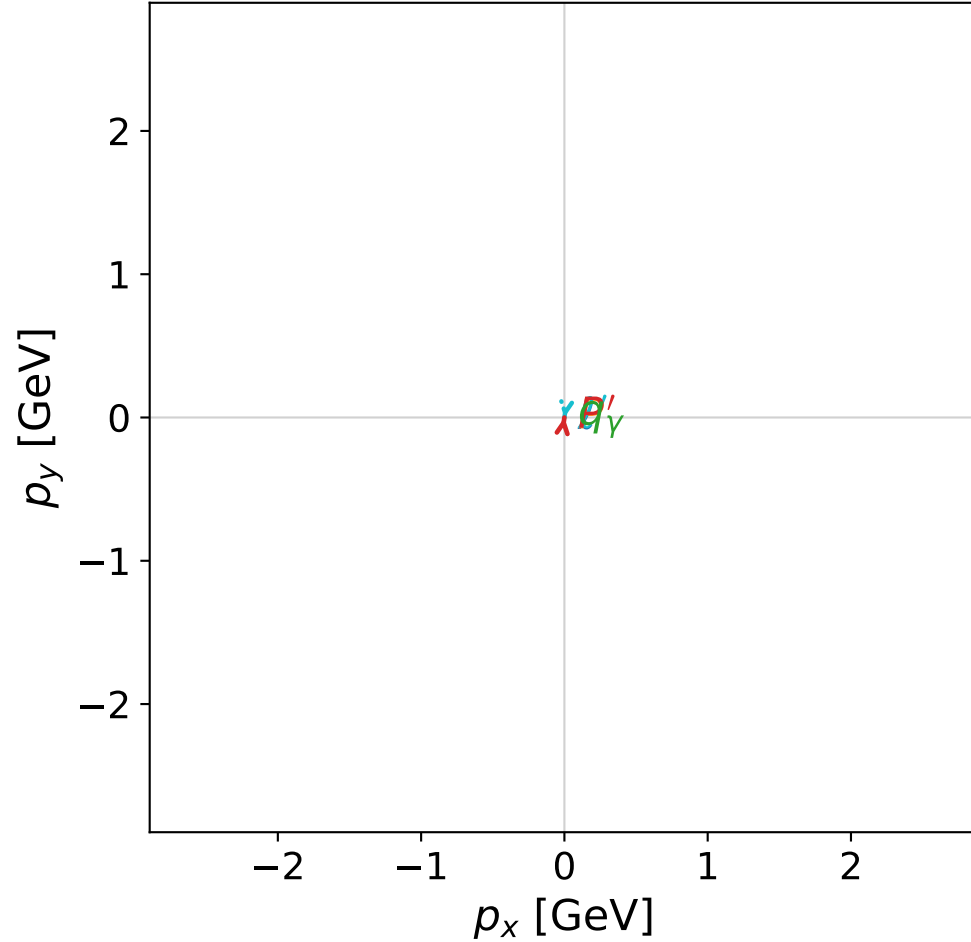


dghz_mixing_angles_polarization_02_local_minimum_7
 $d_{\text{GHZ}}=2.41149\text{e-}08$
 region: polarization_cluster_02_local_minimum_7
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0007
 lepton mixing angle: $\alpha_e=0.78796497$ rad
 incoming lepton: $0.705289|+\rangle +0.708919|-\rangle$
 proton mixing angle: $\alpha_p=2.3536584$ rad
 incoming proton: $-0.705311|+\rangle +0.708898|-\rangle$

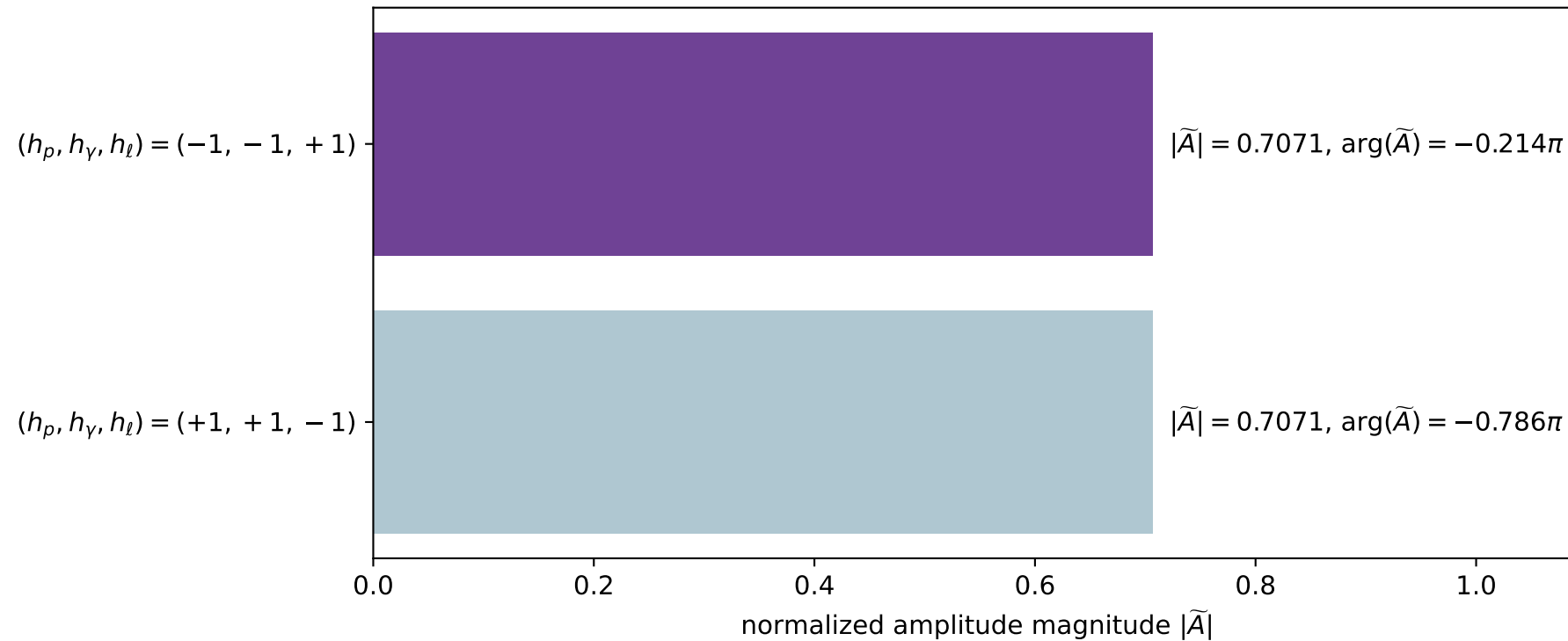
$s=24.9993$, $\sqrt{s}=4.99993$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.0126281$
 $\theta_{P'}=7.54042\text{e-}07$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.42848$, $\phi_\gamma=5.61149$
 $E_\gamma=2.03724$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.57008$
 $Q^2=19.5333$, $x_B=0.834508$, $t=-3.03483$, $W^2=4.7535$, $y=0.970458$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0246$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0246$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0126$
 q_γ $E= 2.0372$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0372$

x--y plane

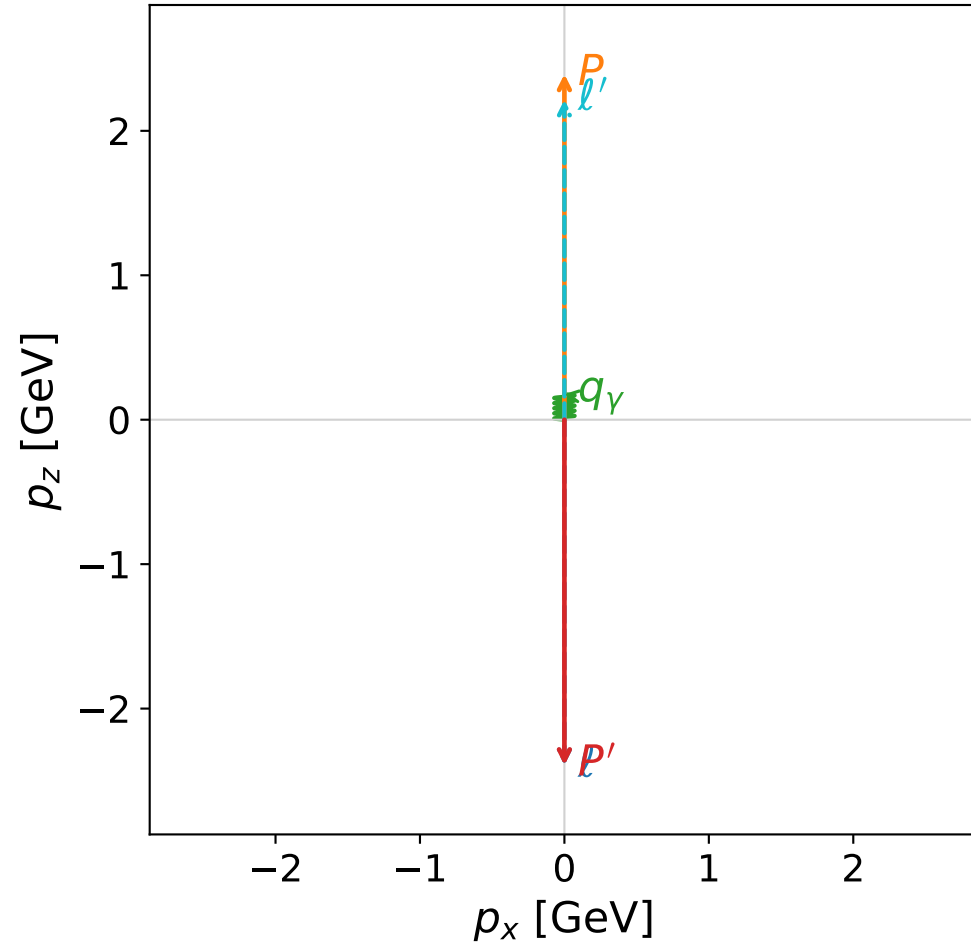


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

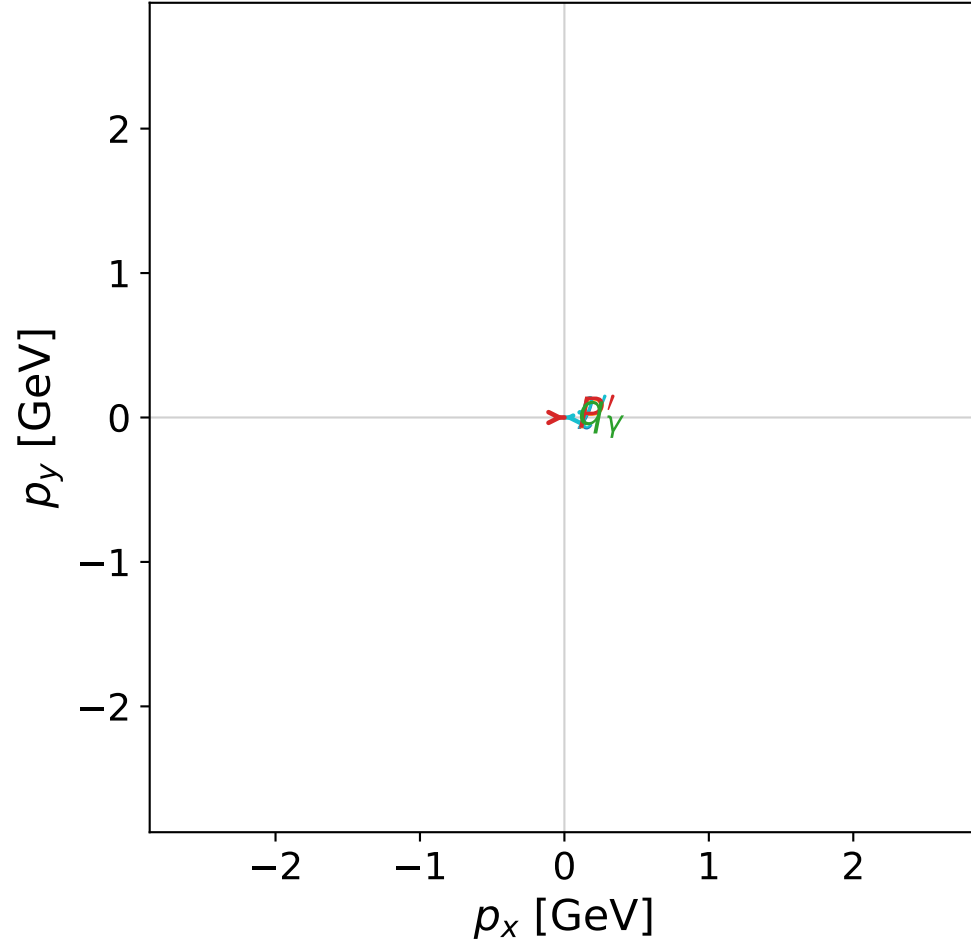


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

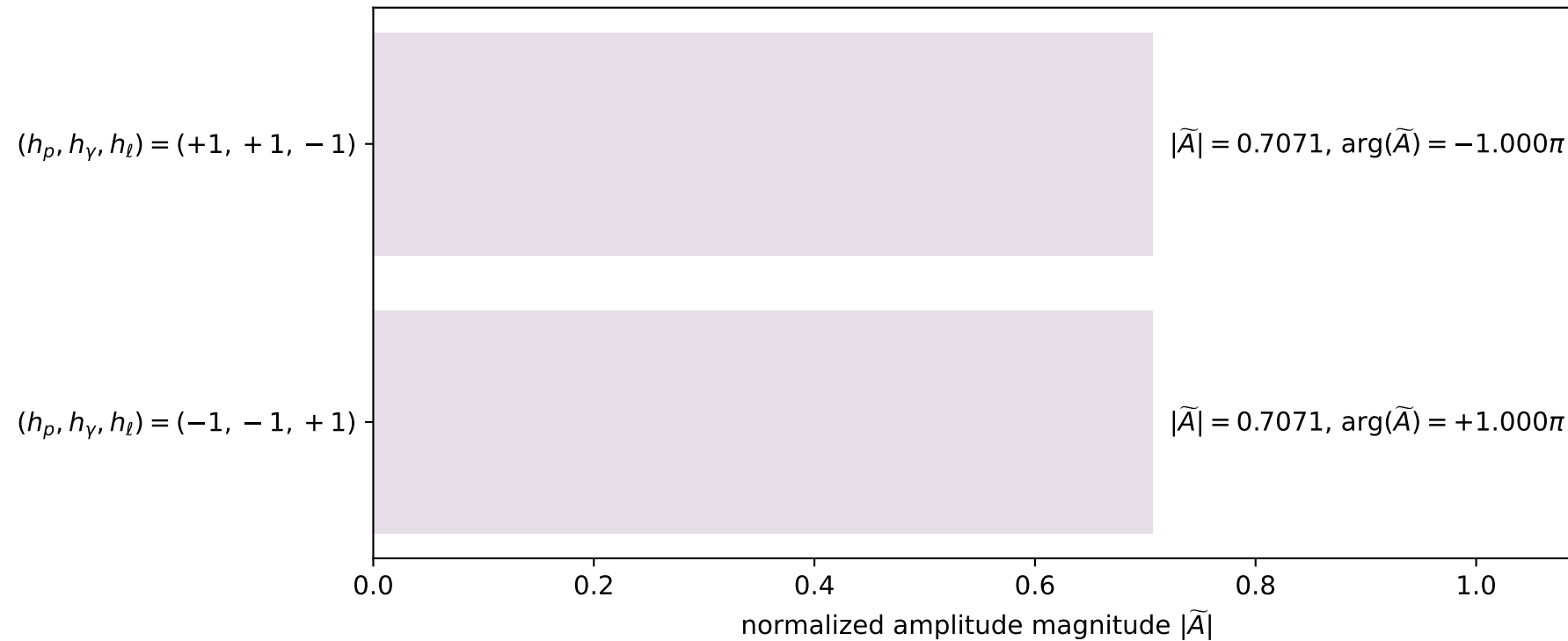


dghz_mixing_angles_polarization_02_local_minimum_14
 $d_{\text{GHZ}}=2.46768\text{e-}08$
 region: polarization_cluster_02_local_minimum_14
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0014
 lepton mixing angle: $\alpha_e=0.75308403$ rad
 incoming lepton: $0.729583|+\rangle +0.683892|-\rangle$
 proton mixing angle: $\alpha_p=2.3238894$ rad
 incoming proton: $-0.683899|+\rangle +0.729577|-\rangle$

$s=24.6277$, $\sqrt{s}=4.96263$, $|\vec{P}|=2.39267$, $|\vec{P}'|=2.39267$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.225183$, $\phi_\gamma=3.93532$
 $E_\gamma=0.176937$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.36678$
 $Q^2=21.2061$, $x_B=0.92352$, $t=-22.8995$, $W^2=2.63599$, $y=0.966916$

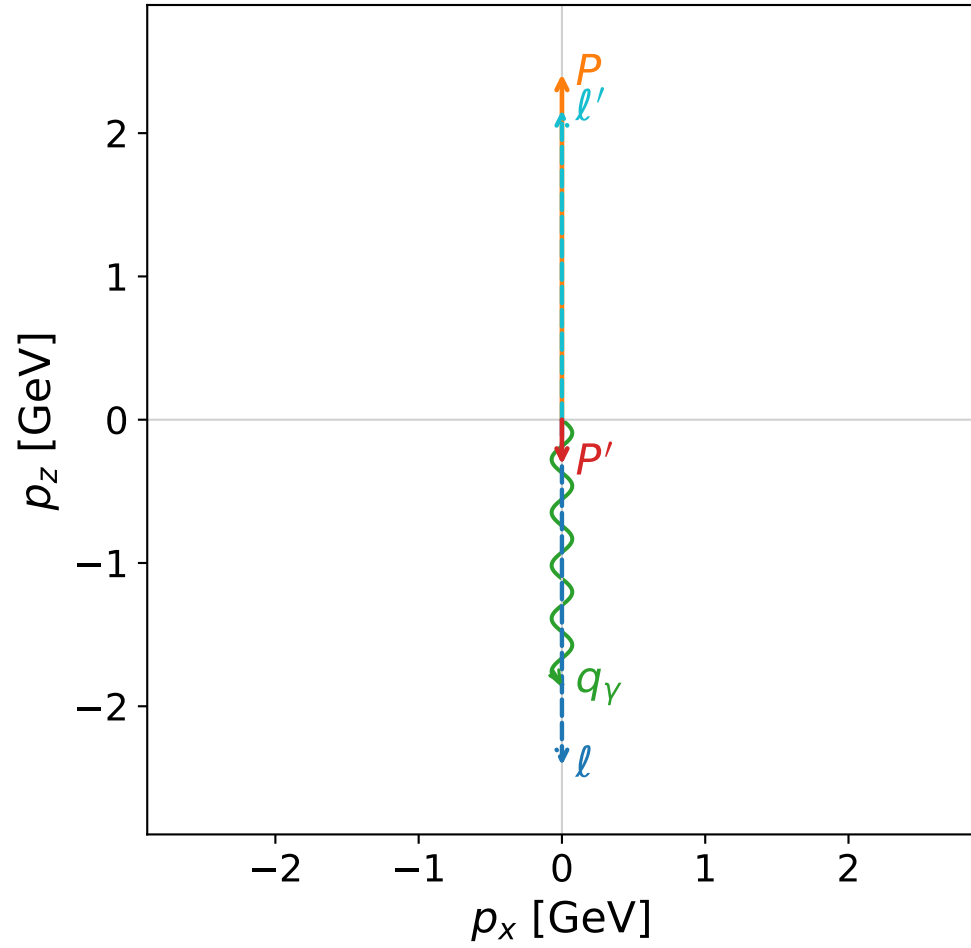
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3927$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3927$
 P $E= 2.5700$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3927$
 ℓ' $E= 2.2157$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2157$
 P' $E= 2.5700$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3927$
 q_γ $E= 0.1769$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1769$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

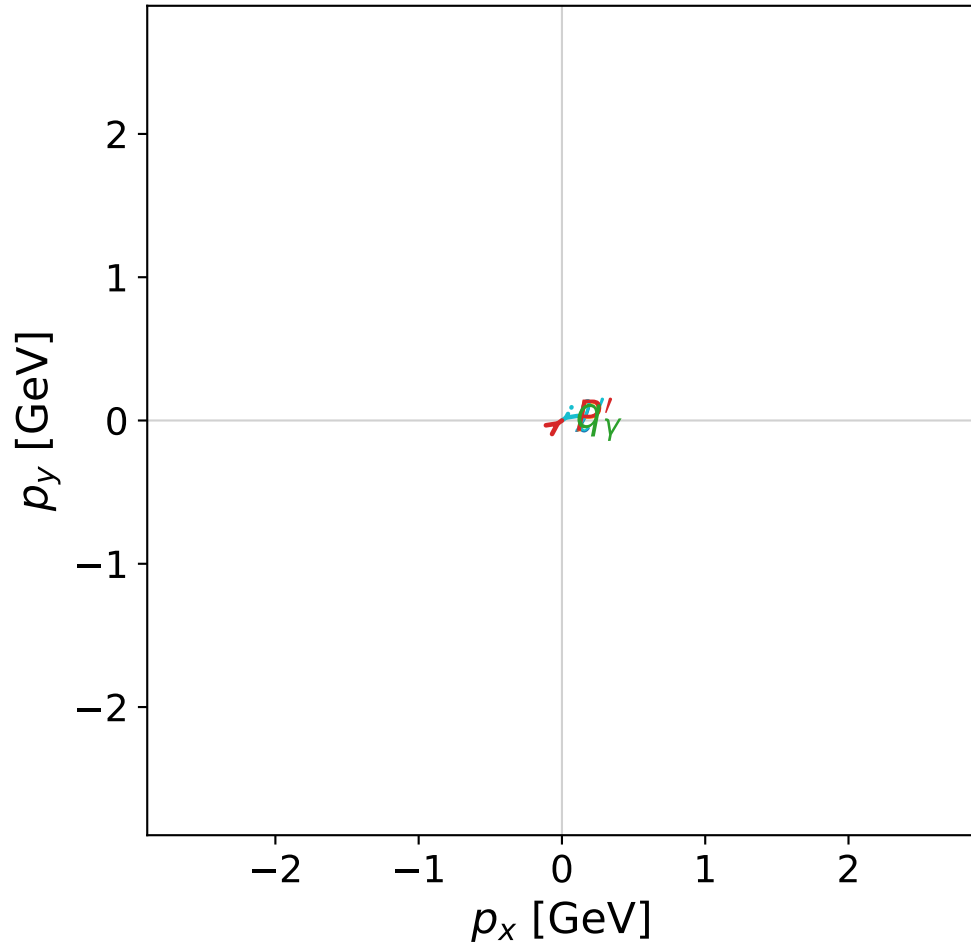


dghz_mixing_angles_polarization_02_local_minimum_21
 $d_{\text{GHZ}}=2.55996\text{e-}08$
 region: polarization_cluster_02_local_minimum_21
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0021
 lepton mixing angle: $\alpha_e=0.88921427$ rad
 incoming lepton: $0.630022|+\rangle + 0.776577|-\rangle$
 proton mixing angle: $\alpha_p=2.2523645$ rad
 incoming proton: $-0.630012|+\rangle + 0.776586|-\rangle$

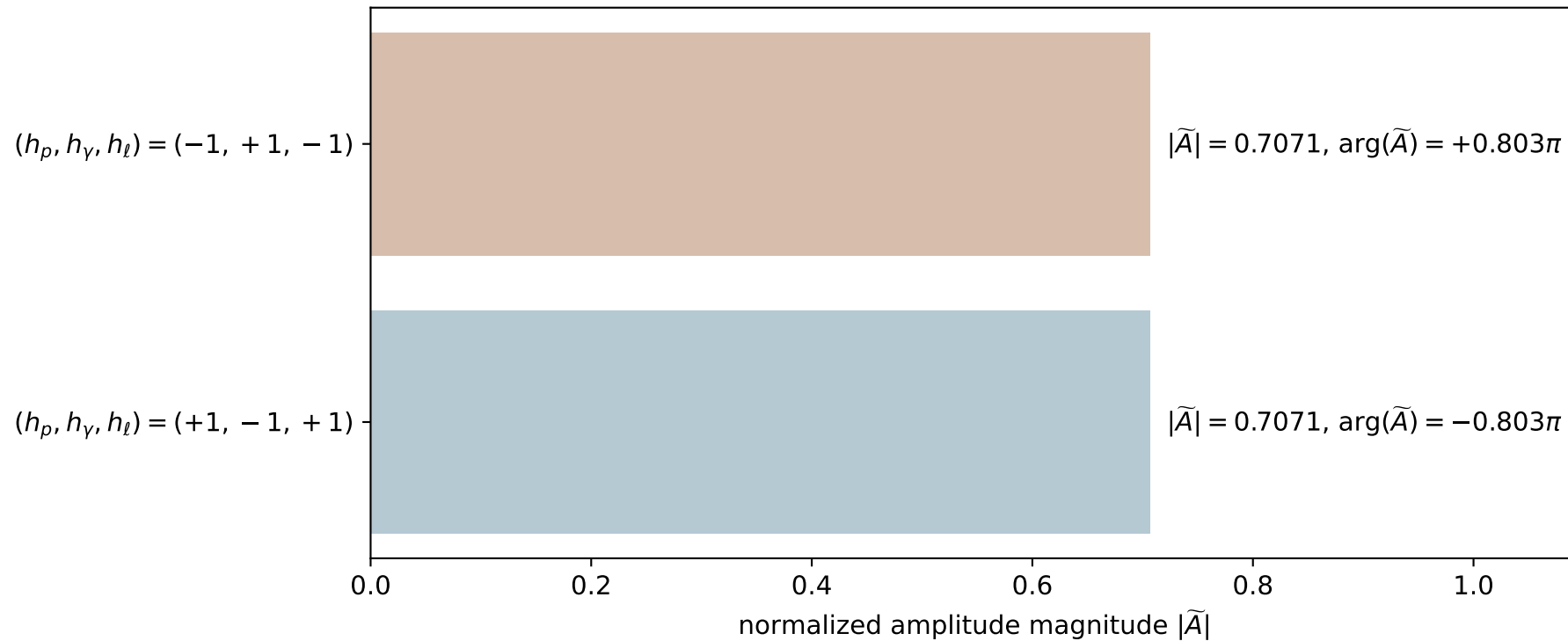
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.311681$
 $\theta_{P'}=3.14153$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.615032$, $\phi_\gamma=4.37445$
 $E_\gamma=1.84995$, $\theta_{\ell'}=8.59128\text{e-}06$, $\phi_{\ell'}=3.75662$
 $Q^2=20.8555$, $x_B=0.89281$, $t=-4.85994$, $W^2=3.38373$, $y=0.96846$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1616$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1616$
 P' $E= 0.9884$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3117$
 q_γ $E= 1.8499$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8499$

x--y plane

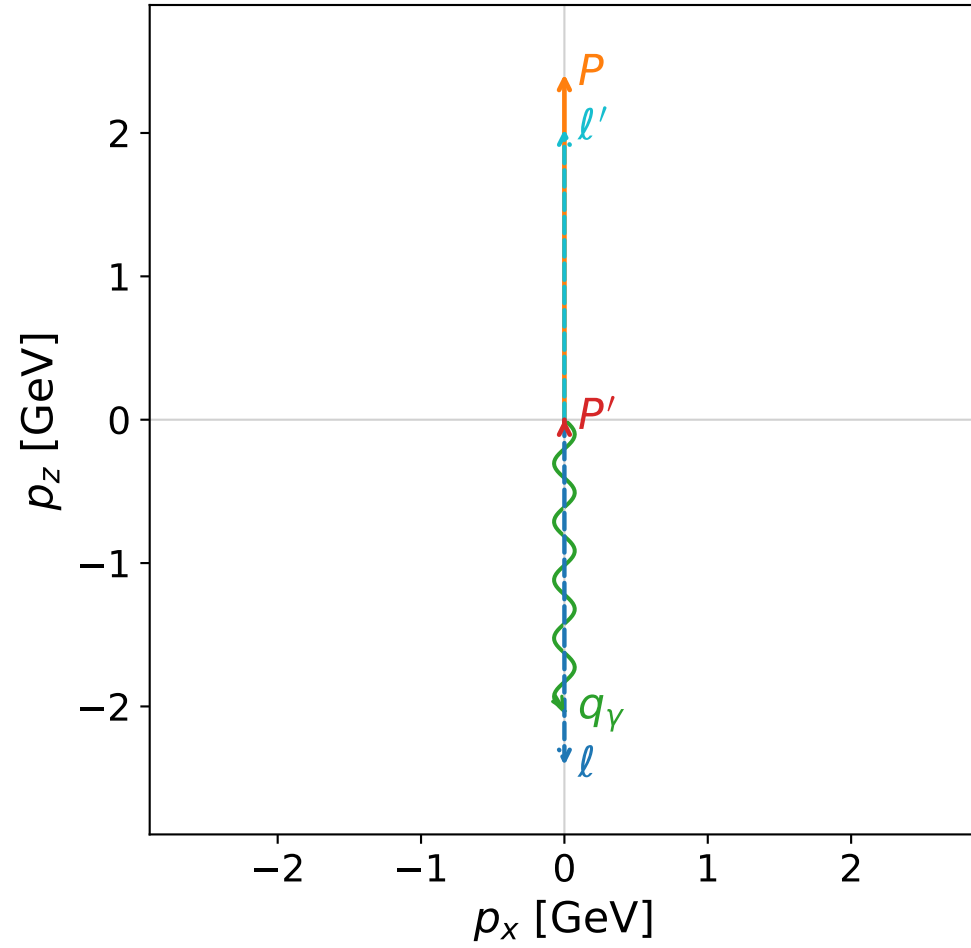


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

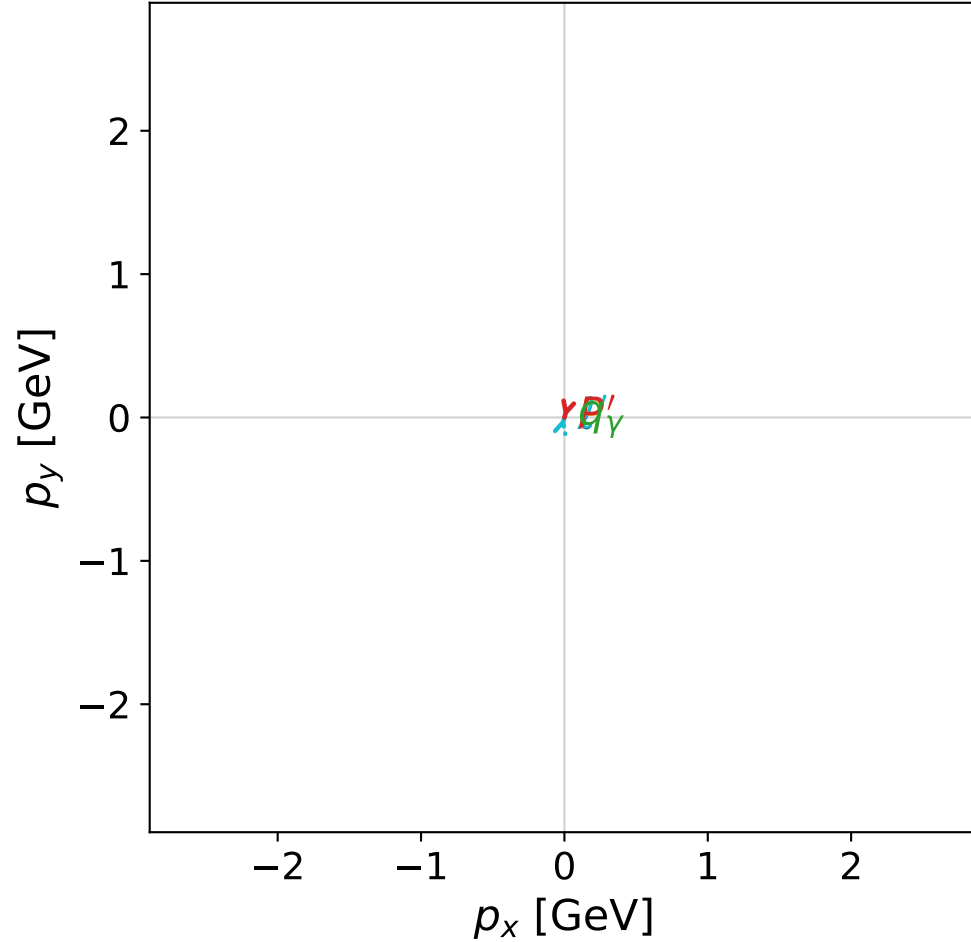


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



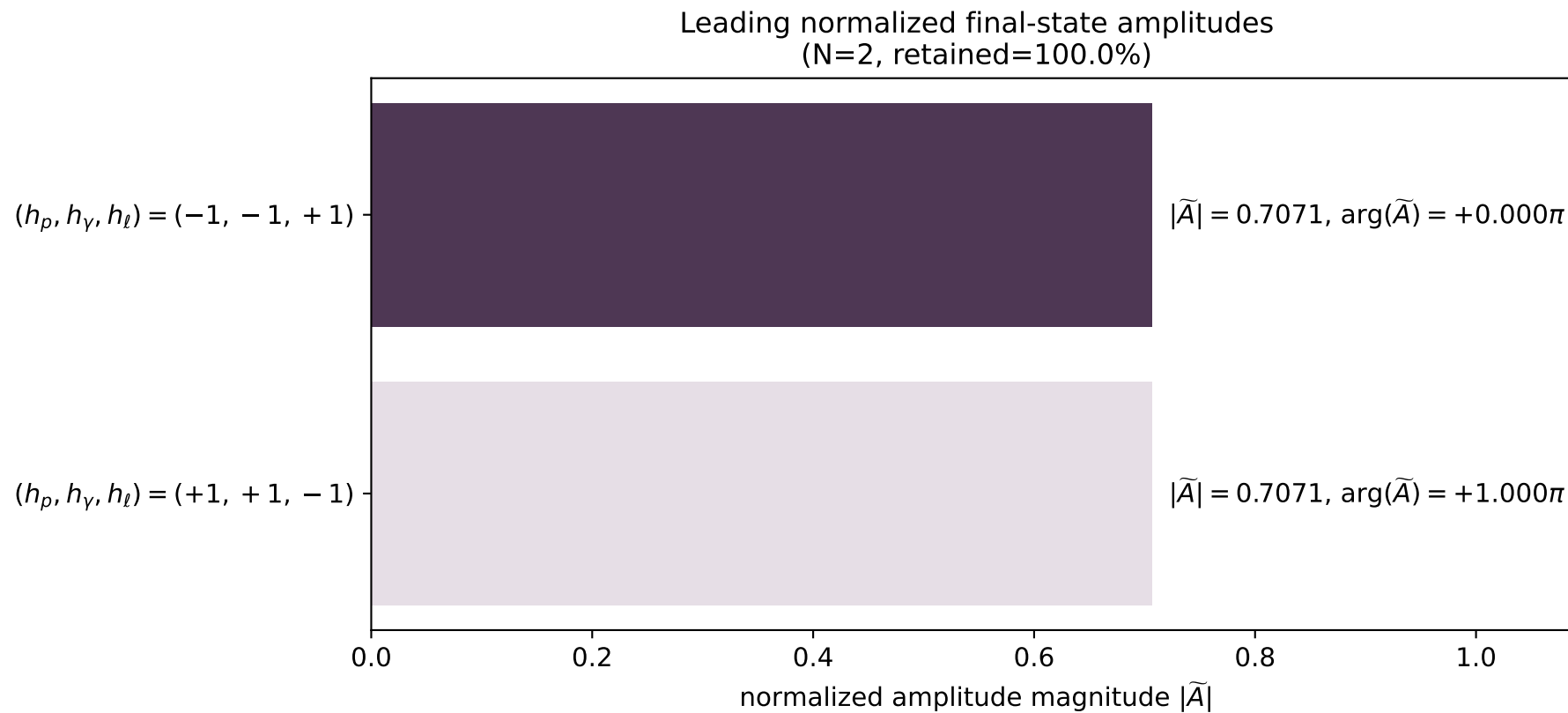
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_26
 $d_{\text{GHZ}}=2.59712\text{e-}08$
 region: polarization_cluster_02_local_minimum_26
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0026
 lepton mixing angle: $\alpha_e=0.77423168$ rad
 incoming lepton: $0.714958|+\rangle + 0.699167|-\rangle$
 proton mixing angle: $\alpha_p=2.3673692$ rad
 incoming proton: $-0.714964|+\rangle + 0.699161|-\rangle$

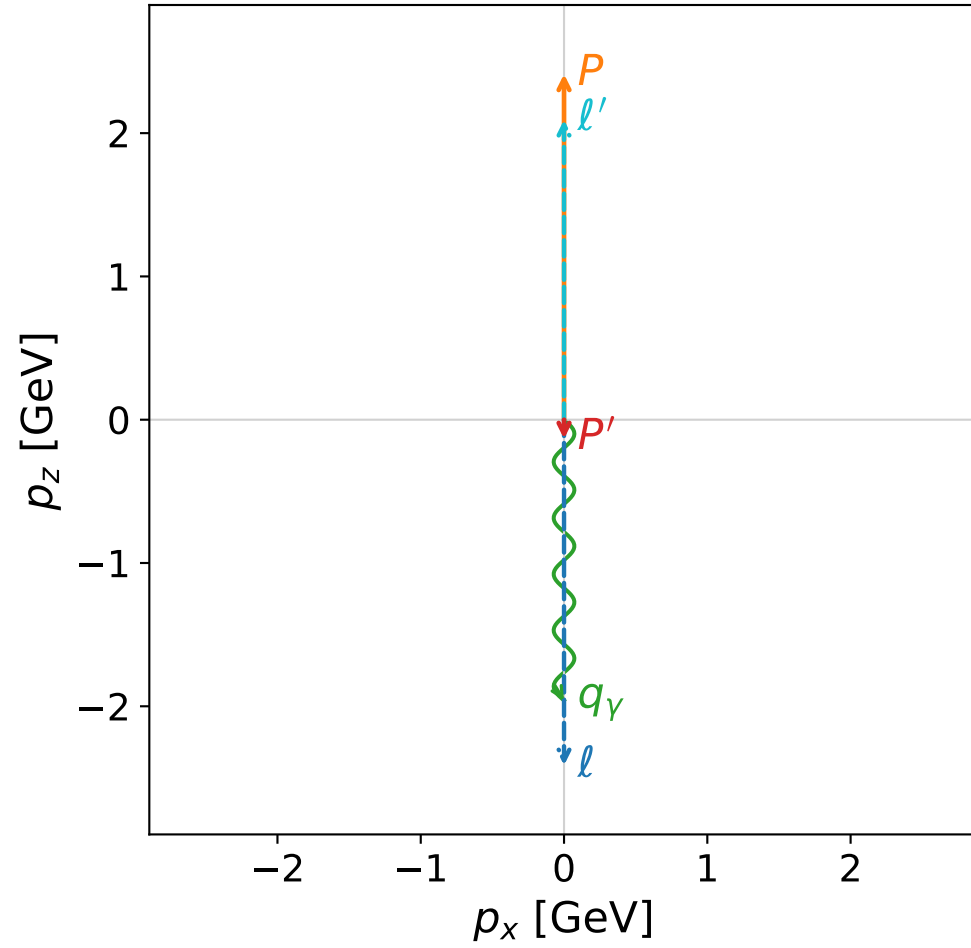
$s=24.9729$, $\sqrt{s}=4.99729$, $|\vec{P}|=2.41061$, $|\vec{P}'|=0.00645061$
 $\theta_{P'}=0.00979874$, $\theta_\gamma=3.14159$, $\phi_{P'}=4.43638$, $\phi_\gamma=6.38506\text{e-}05$
 $E_\gamma=2.03286$, $\theta_{\ell'}=3.11916\text{e-}05$, $\phi_{\ell'}=1.29479$
 $Q^2=19.5395$, $x_B=0.835755$, $t=-3.06193$, $W^2=4.7198$, $y=0.970383$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4106$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4106$
 P $E= 2.5867$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4106$
 ℓ' $E= 2.0264$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 2.0264$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 0.0065$
 q_γ $E= 2.0329$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0329$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

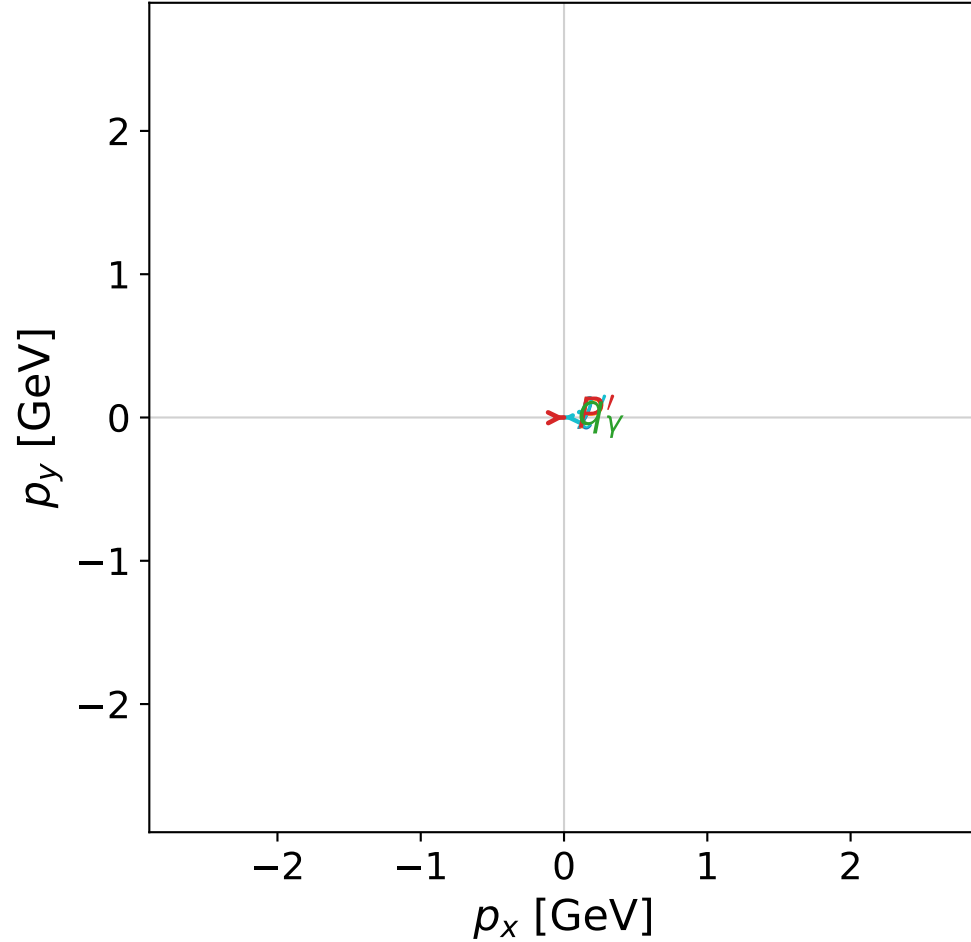


dghz_mixing_angles_polarization_02_local_minimum_31
 $d_{\text{GHZ}}=2.62572\text{e-}08$
 region: polarization_cluster_02_local_minimum_31
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0031
 lepton mixing angle: $\alpha_e=0.72294087$ rad
 incoming lepton: $0.749863|+\rangle + 0.661593|-\rangle$
 proton mixing angle: $\alpha_p=2.4186407$ rad
 incoming proton: $-0.749856|+\rangle + 0.661601|-\rangle$

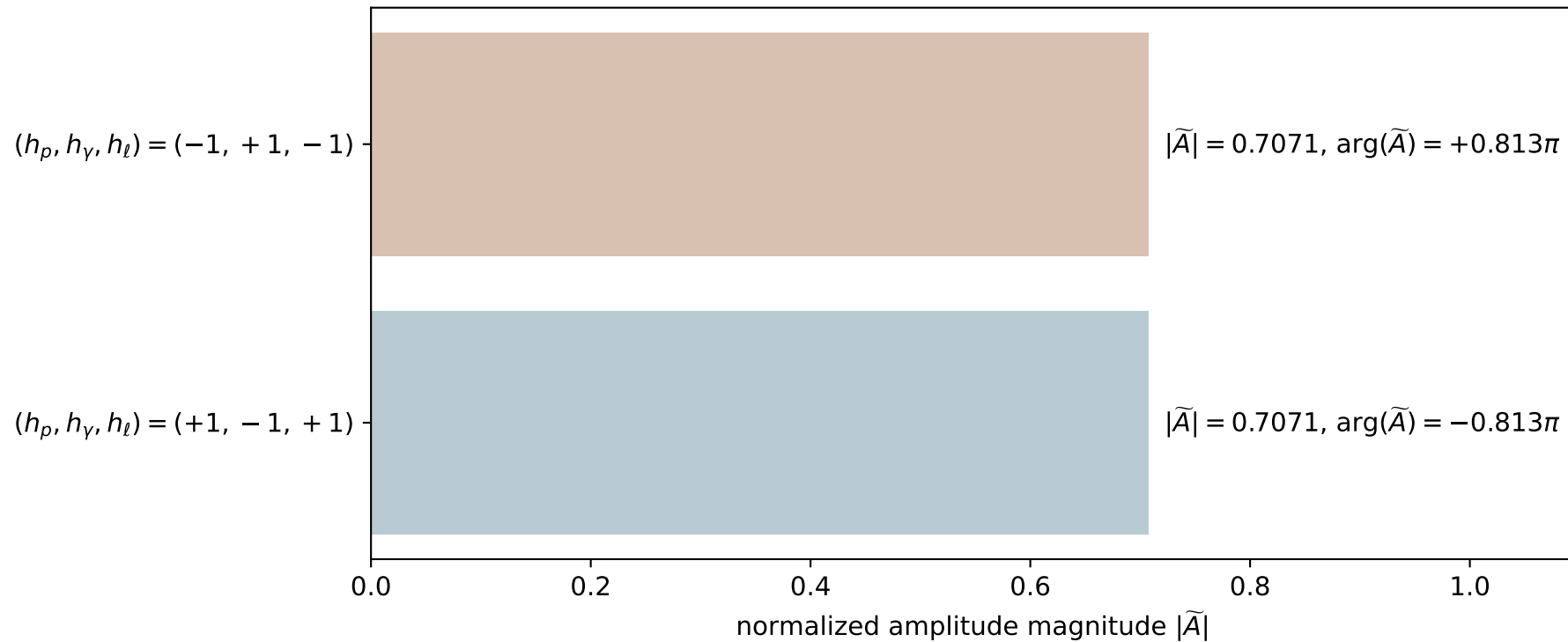
$s=24.9982$, $\sqrt{s}=4.99982$, $|\vec{P}|=2.41192$, $|\vec{P}'|=0.133473$
 $\theta_{P'}=3.14093$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0144773$, $\phi_\gamma=3.74298$
 $E_\gamma=1.95945$, $\theta_\ell=4.19601\text{e-}05$, $\phi_\ell=3.15607$
 $Q^2=20.1919$, $x_B=0.863574$, $t=-3.78797$, $W^2=4.06974$, $y=0.969459$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0929$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0929$
 P' $E= 0.9474$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.1335$
 q_γ $E= 1.9595$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9595$

x--y plane

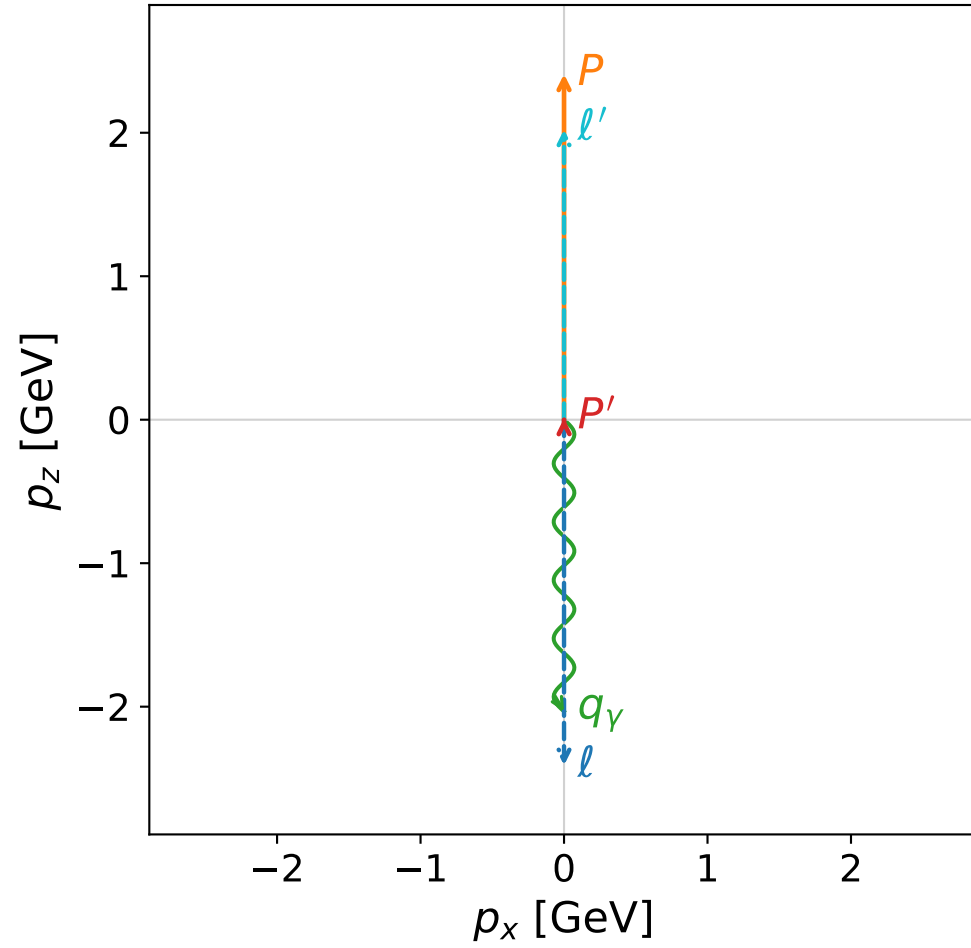


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

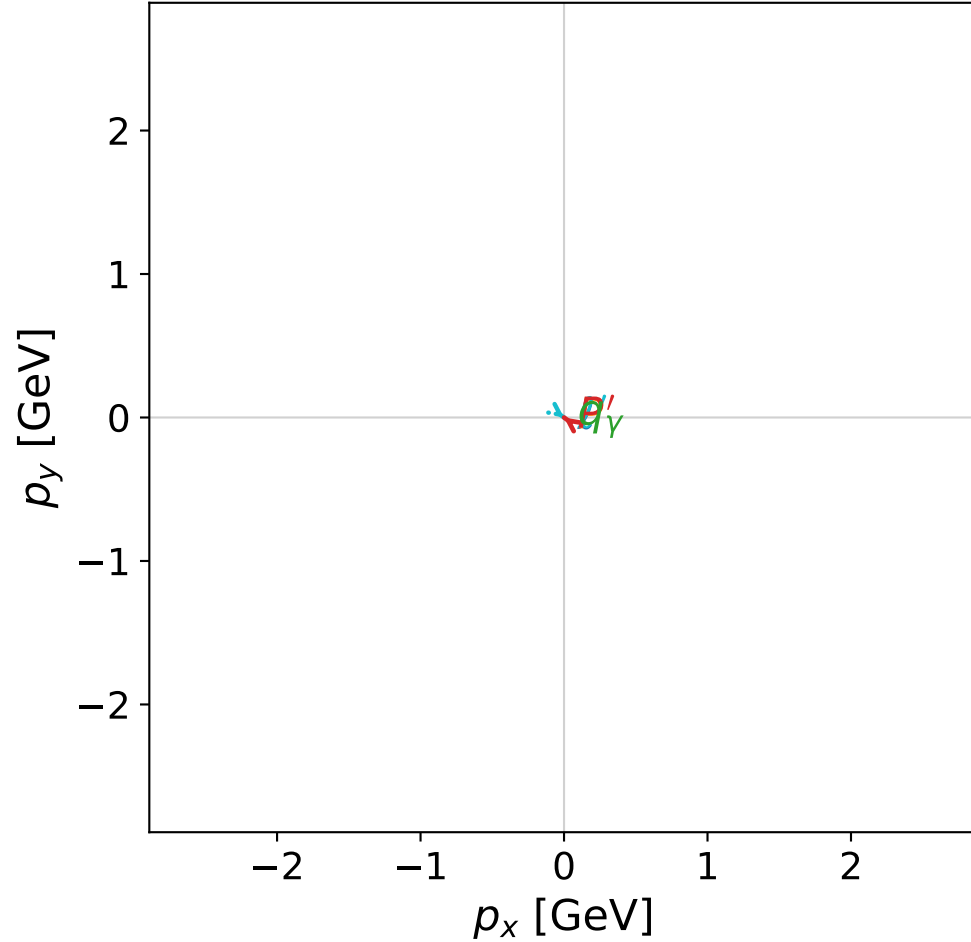


dghz_mixing_angles_polarization_02_local_minimum_33
 $d_{\{\rm GHZ\}}=2.63319\text{e-}08$
 region: polarization_cluster_02_local_minimum_33
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0033
 lepton mixing angle: $\alpha_e=0.79116621$ rad
 incoming lepton: $0.703016|+\rangle + 0.711174|-\rangle$
 proton mixing angle: $\alpha_p=2.3504057$ rad
 incoming proton: $-0.703002|+\rangle + 0.711188|-\rangle$

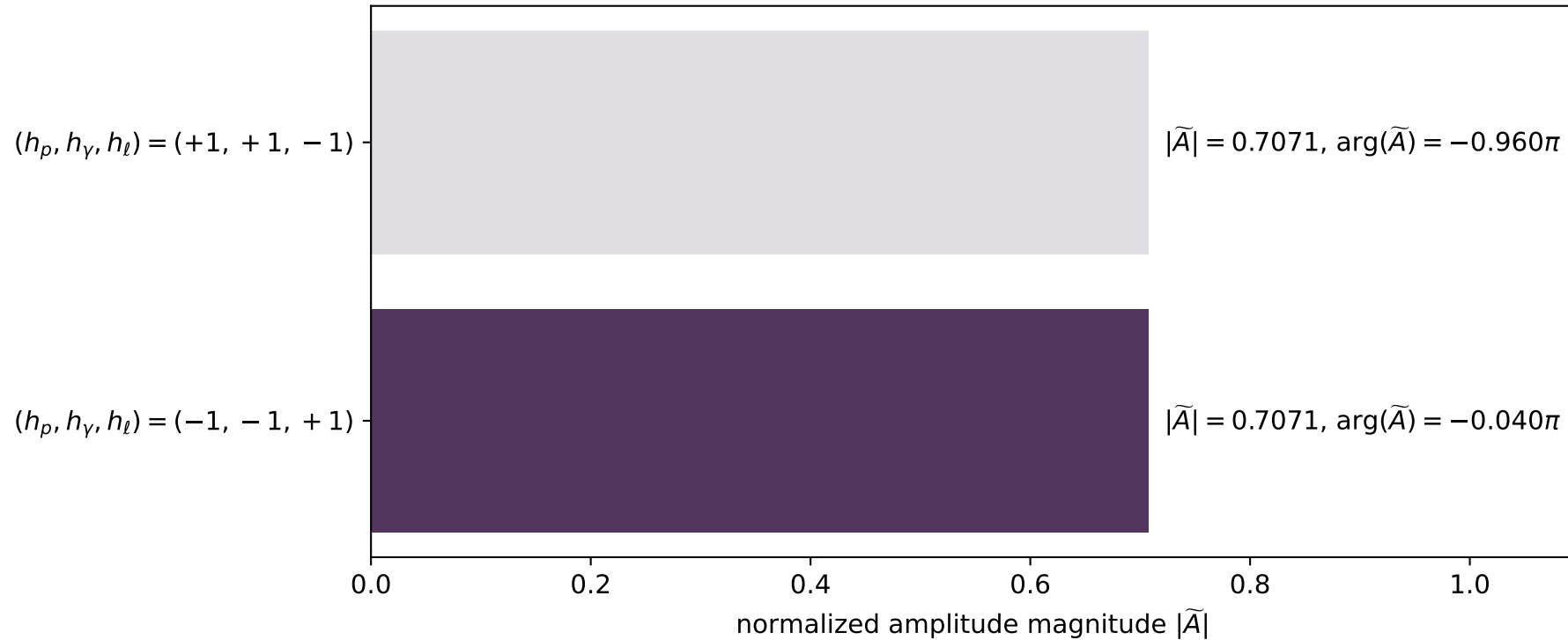
$s=24.934$, $\sqrt{s}=4.9934$, $|\vec{P}|=2.4086$, $|\vec{P}'|=0.0106855$
 $\theta_{P'}=0.0073264$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.51128$, $\phi_\gamma=6.15755$
 $E_\gamma=2.03301$, $\theta_{\ell'}=3.87108\text{e-}05$, $\phi_{\ell'}=5.65288$
 $Q^2=19.4839$, $x_B=0.834731$, $t=-3.03824$, $W^2=4.73747$, $y=0.970372$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4086$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4086$
 P $E= 2.5848$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4086$
 ℓ' $E= 2.0223$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 2.0223$
 P' $E= 0.9381$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 0.0107$
 q_γ $E= 2.0330$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0330$

x--y plane

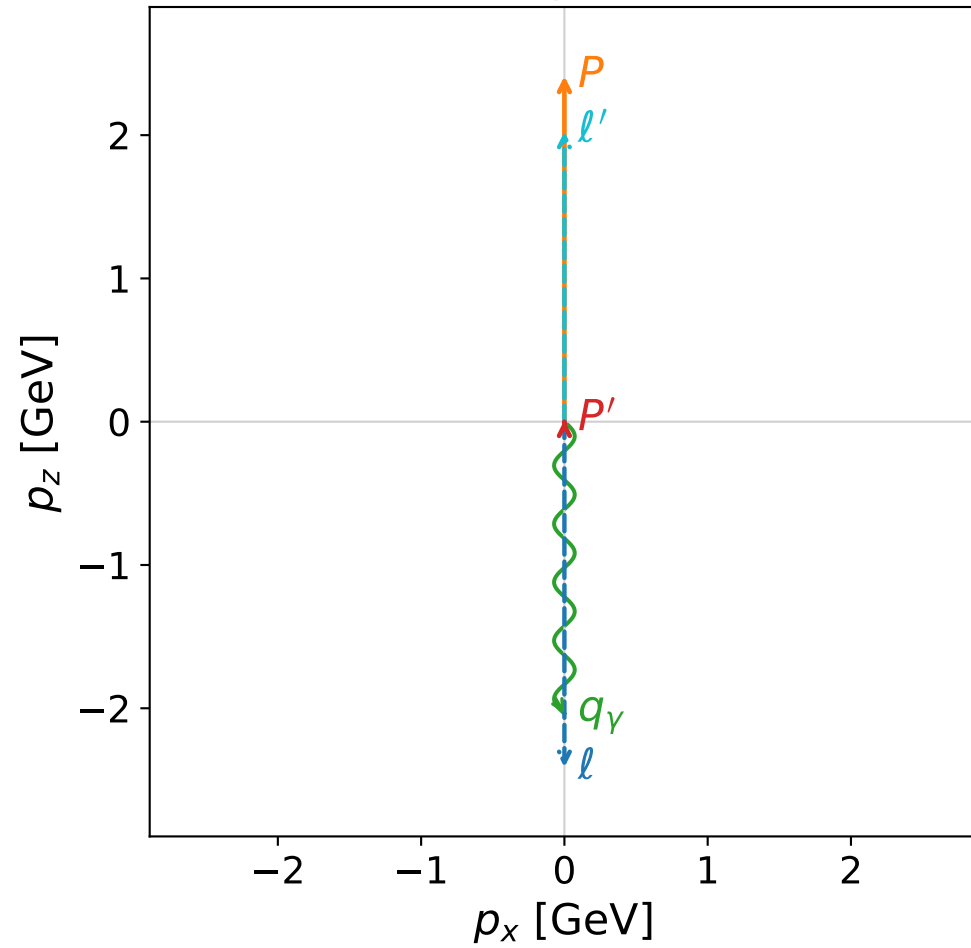


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

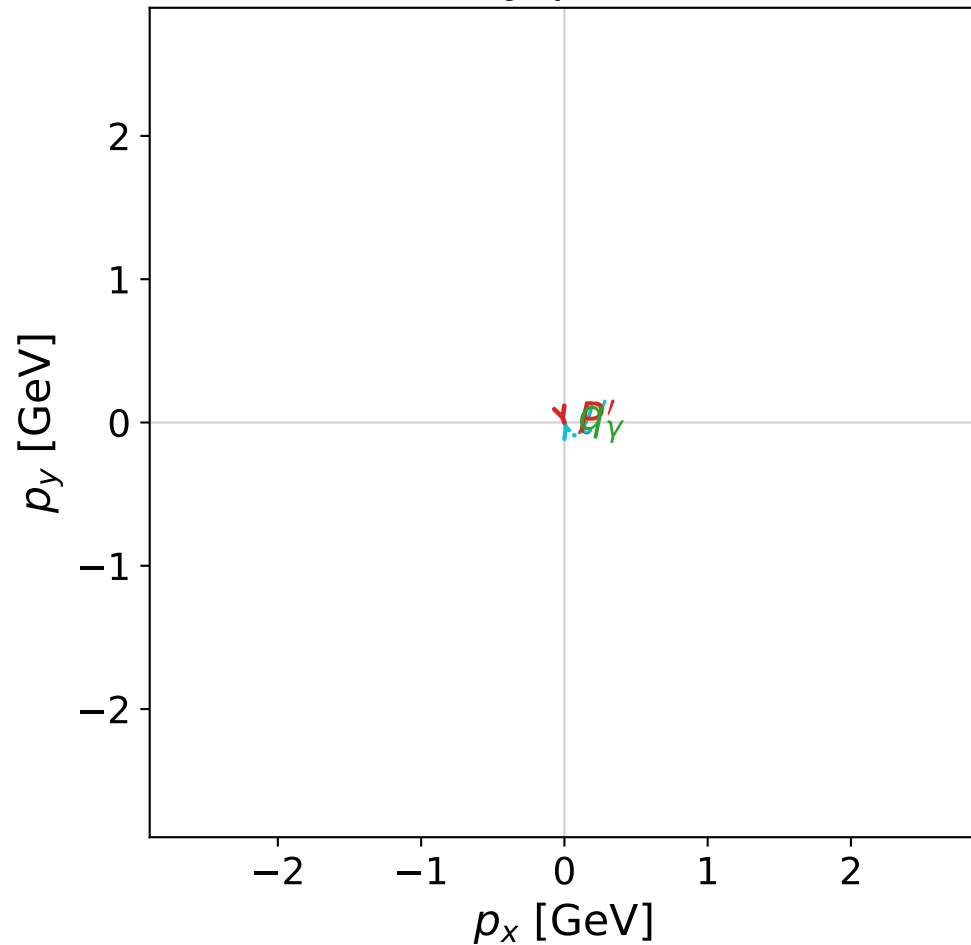


dghz_mixing_angles_polarization_02_local_minimum_34
 $d_{\text{GHZ}}=2.63343\text{e-}08$
 region: polarization_cluster_02_local_minimum_34
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0034
 lepton mixing angle: $\alpha_e=0.76007598$ rad
 incoming lepton: $0.724784|+\rangle + 0.688977|-\rangle$
 proton mixing angle: $\alpha_p=2.3815115$ rad
 incoming proton: $-0.72478|+\rangle + 0.68898|-\rangle$

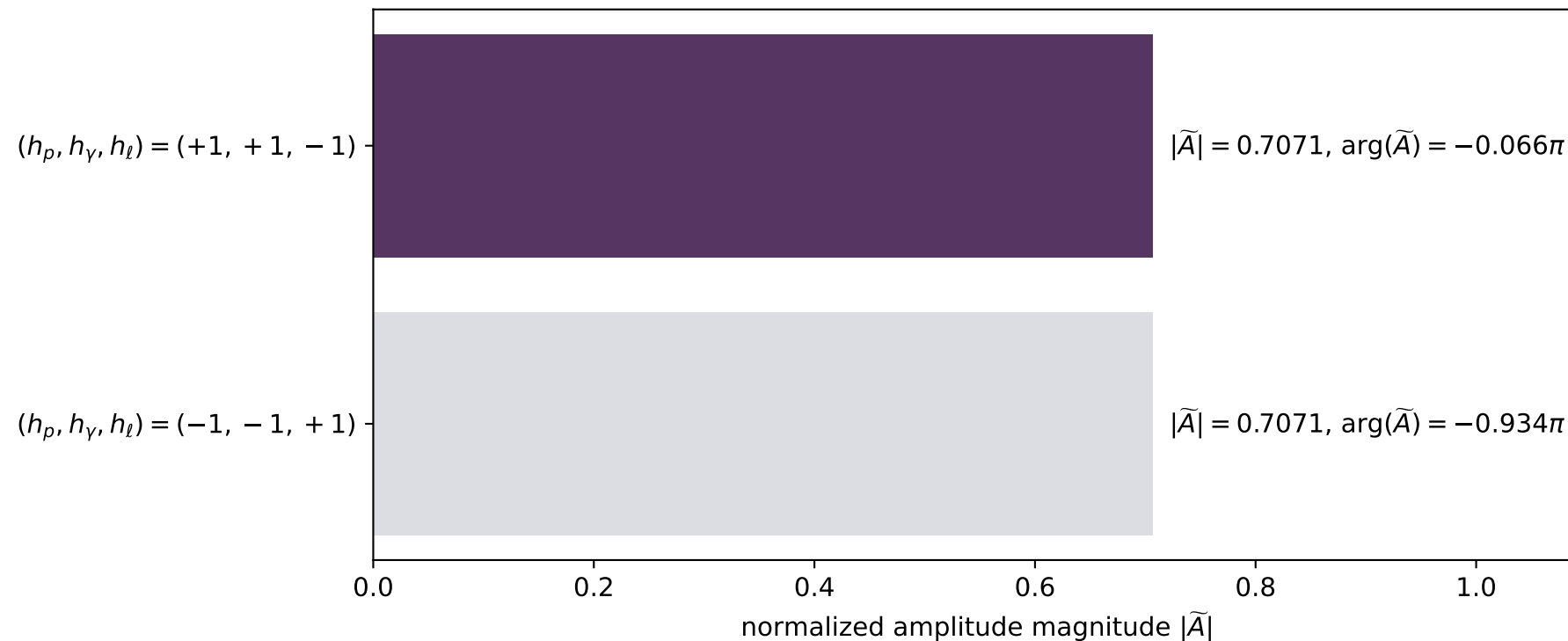
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0121954$
 $\theta_{P'}=0.00245437$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.04048$, $\phi_\gamma=3.34859$
 $E_\gamma=2.03706$, $\theta_{\ell'}=1.47823\text{e-}05$, $\phi_{\ell'}=1.89889$
 $Q^2=19.536$, $x_B=0.834603$, $t=-3.03695$, $W^2=4.75137$, $y=0.970455$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0249$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0249$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0122$
 q_γ $E= 2.0371$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0371$

x--y plane

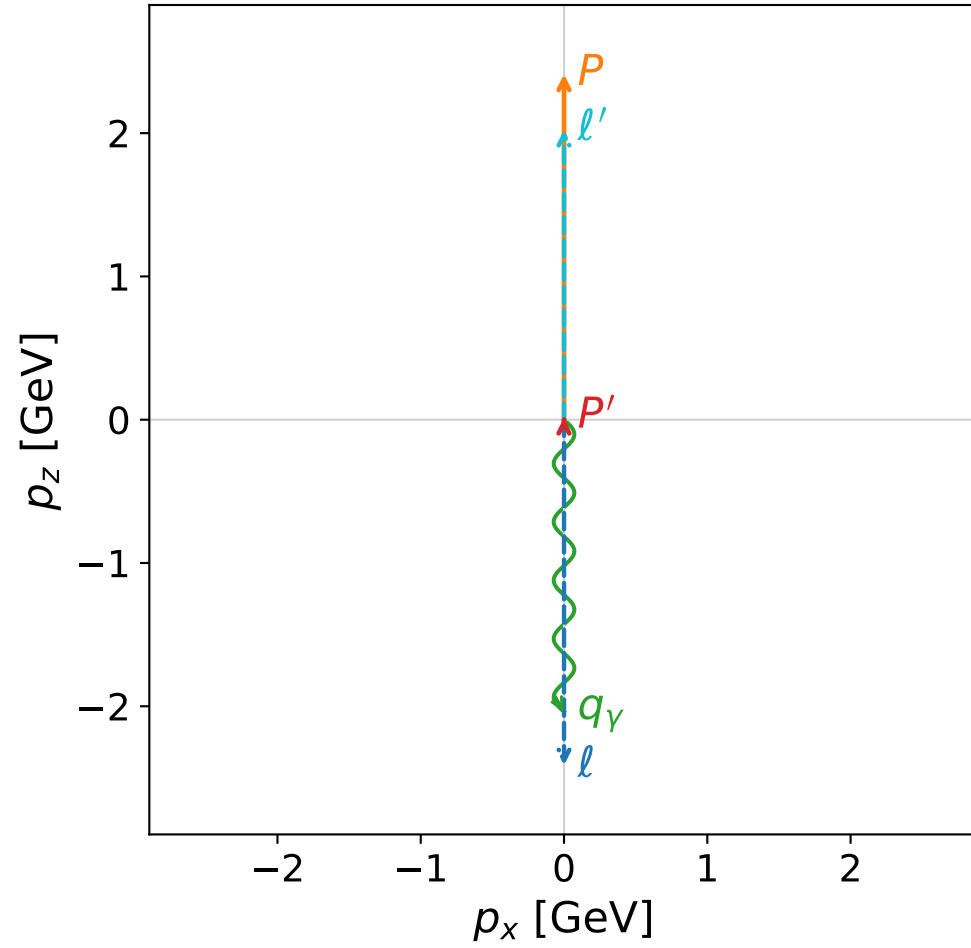


Leading normalized final-state amplitudes (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

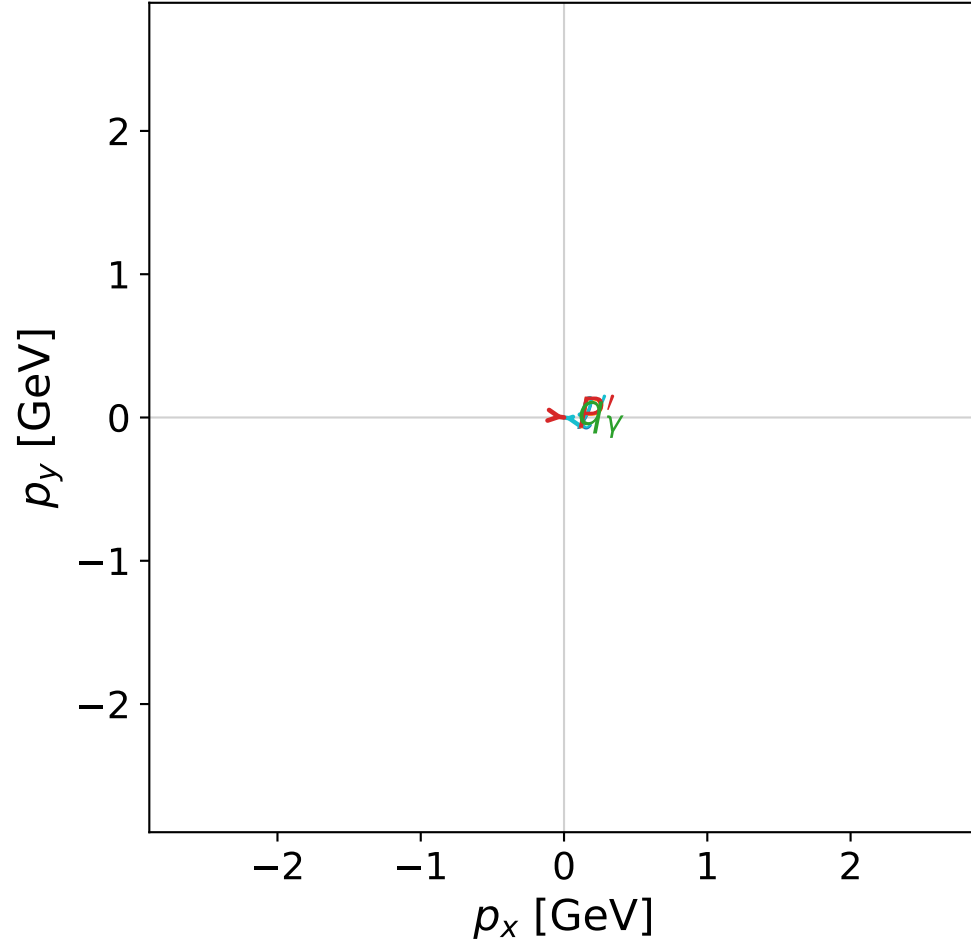


dghz_mixing_angles_polarization_02_local_minimum_36
 $d_{\{\rm GHZ\}}=2.63908\text{e-}08$
 region: polarization_cluster_02_local_minimum_36
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0036
 lepton mixing angle: $\alpha_e=0.80546213$ rad
 incoming lepton: $0.692778|+\rangle +0.721151|-\rangle$
 proton mixing angle: $\alpha_p=2.3361295$ rad
 incoming proton: $-0.692777|+\rangle +0.721152|-\rangle$

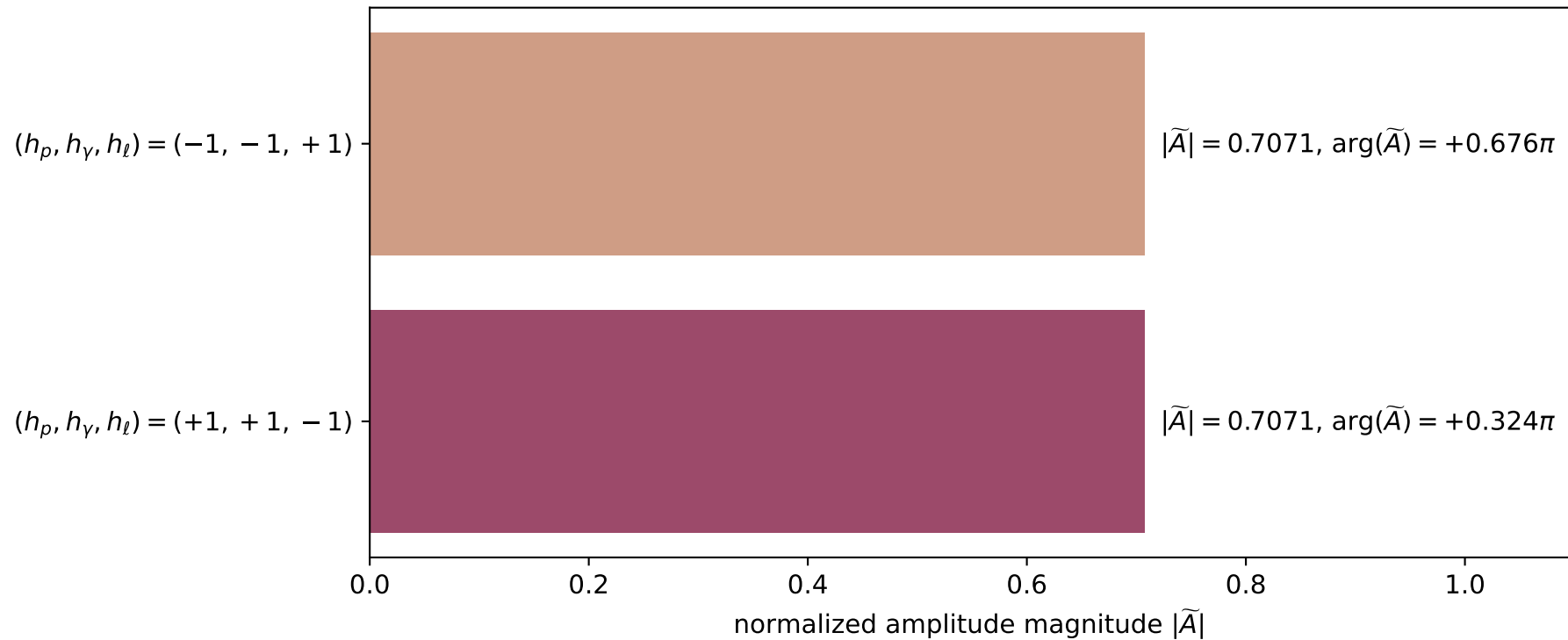
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0152919$
 $\theta_{p'}=0.00490874$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.14104$, $\phi_\gamma=2.12467$
 $E_\gamma=2.03858$, $\theta_{\ell'}=3.70997\text{e-}05$, $\phi_{\ell'}=2.99944$
 $Q^2=19.5208$, $x_B=0.833936$, $t=-3.02225$, $W^2=4.76708$, $y=0.970478$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0233$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0233$
 P' $E= 0.9381$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0153$
 q_γ $E= 2.0386$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0386$

x--y plane

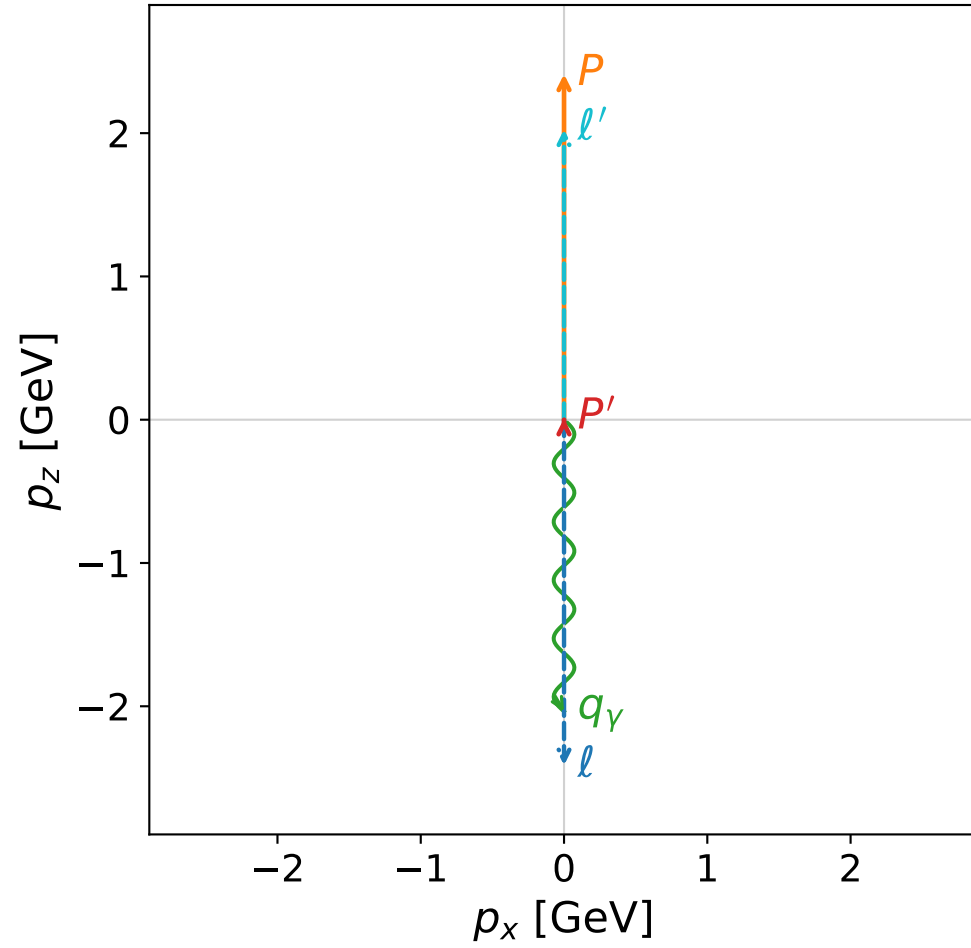


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

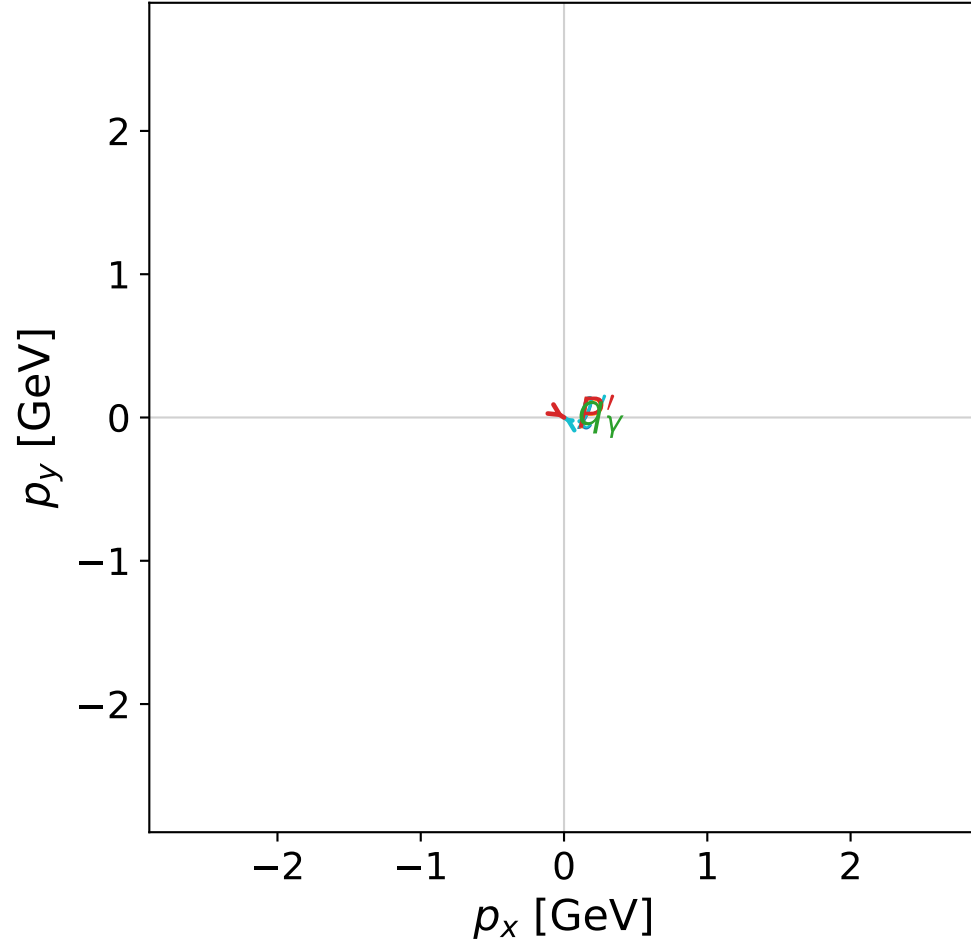


dghz_mixing_angles_polarization_02_local_minimum_37
 $d_{\{\rm GHZ\}}=2.64099\text{e-}08$
 region: polarization_cluster_02_local_minimum_37
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0037
 lepton mixing angle: $\alpha_e=0.81766347$ rad
 incoming lepton: $0.683928|+\rangle +0.72955|-\rangle$
 proton mixing angle: $\alpha_p=2.3239276$ rad
 incoming proton: $-0.683927|+\rangle +0.729551|-\rangle$

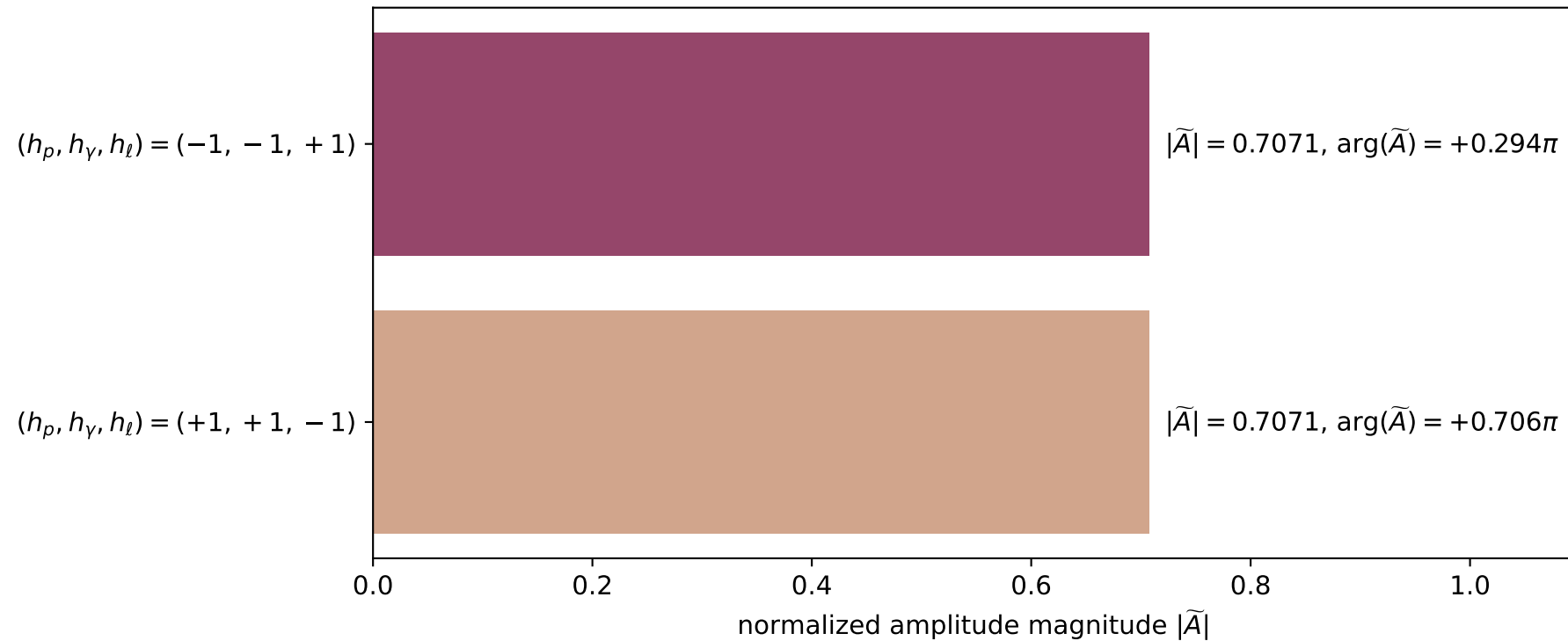
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0096556$
 $\theta_{P'}=0.00245437$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.7193$, $\phi_\gamma=0.92409$
 $E_\gamma=2.0358$, $\theta_{\ell'}=1.16963\text{e-}05$, $\phi_{\ell'}=2.5777$
 $Q^2=19.5484$, $x_B=0.835149$, $t=-3.04905$, $W^2=4.73853$, $y=0.970436$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0261$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0261$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0097$
 q_γ $E= 2.0358$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0358$

x--y plane

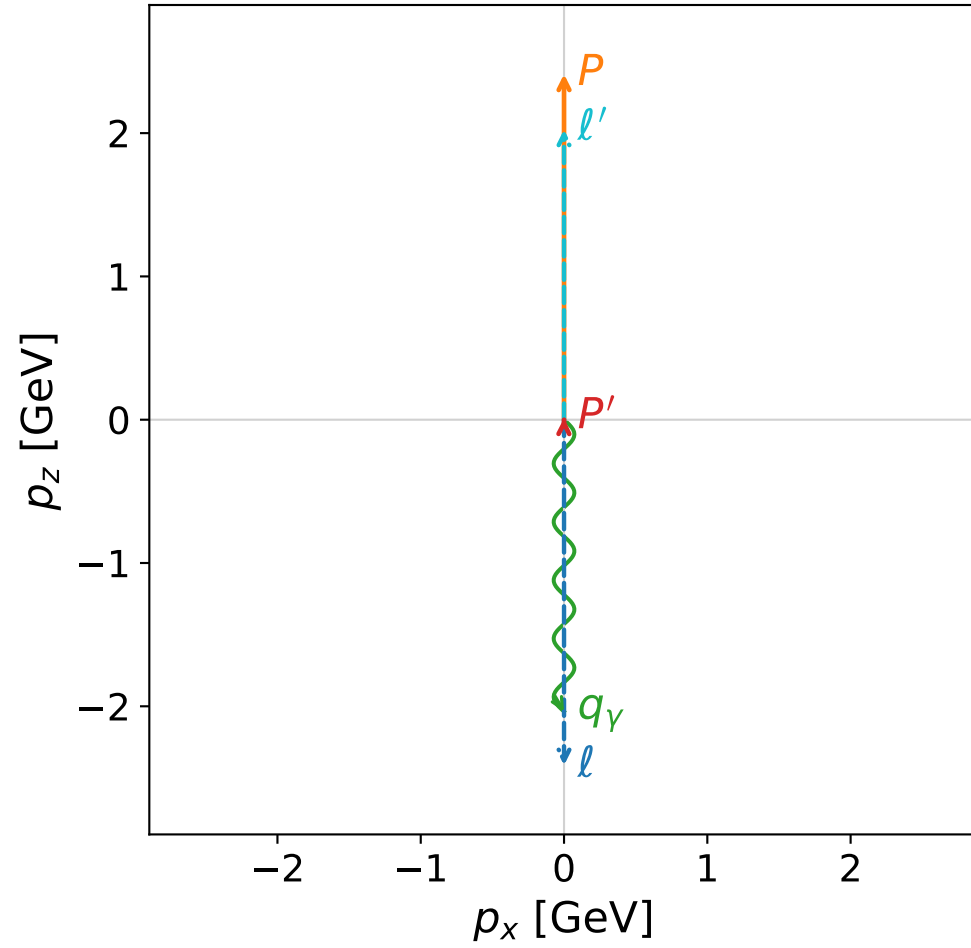


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

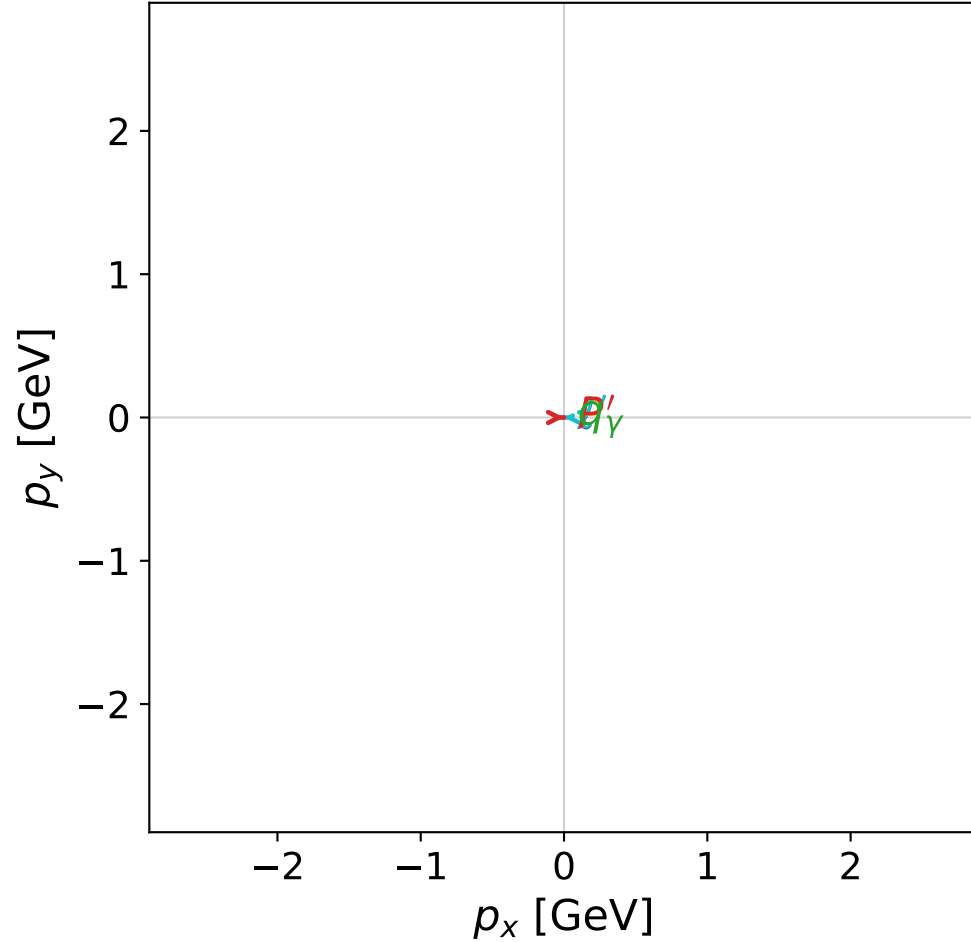


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



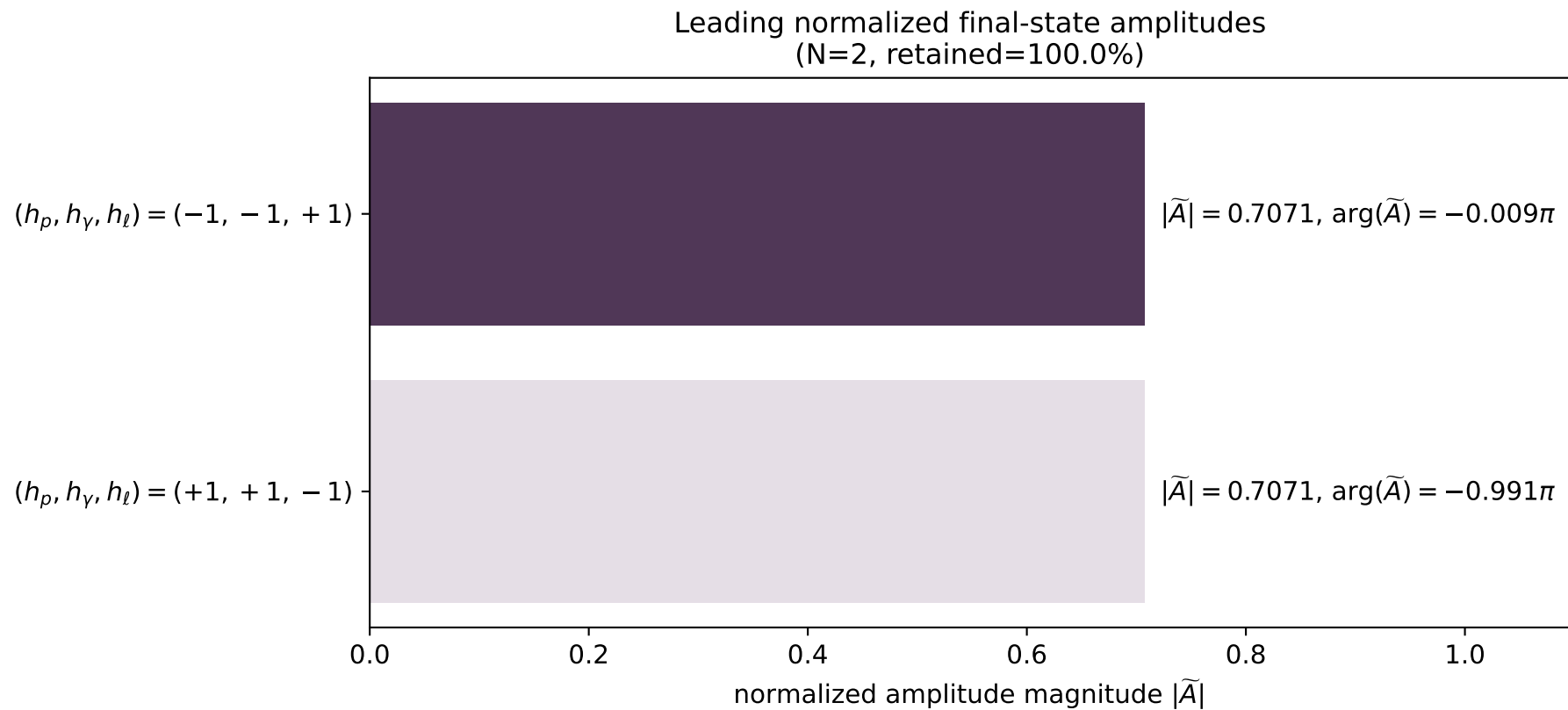
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_40
 $d_{\text{GHZ}}=2.64788\text{e-}08$
 region: polarization_cluster_02_local_minimum_40
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0040
 lepton mixing angle: $\alpha_e=0.76726665$ rad
 incoming lepton: $0.719811|+\rangle +0.69417|-\rangle$
 proton mixing angle: $\alpha_p=2.3743383$ rad
 incoming proton: $-0.719819|+\rangle +0.694161|-\rangle$

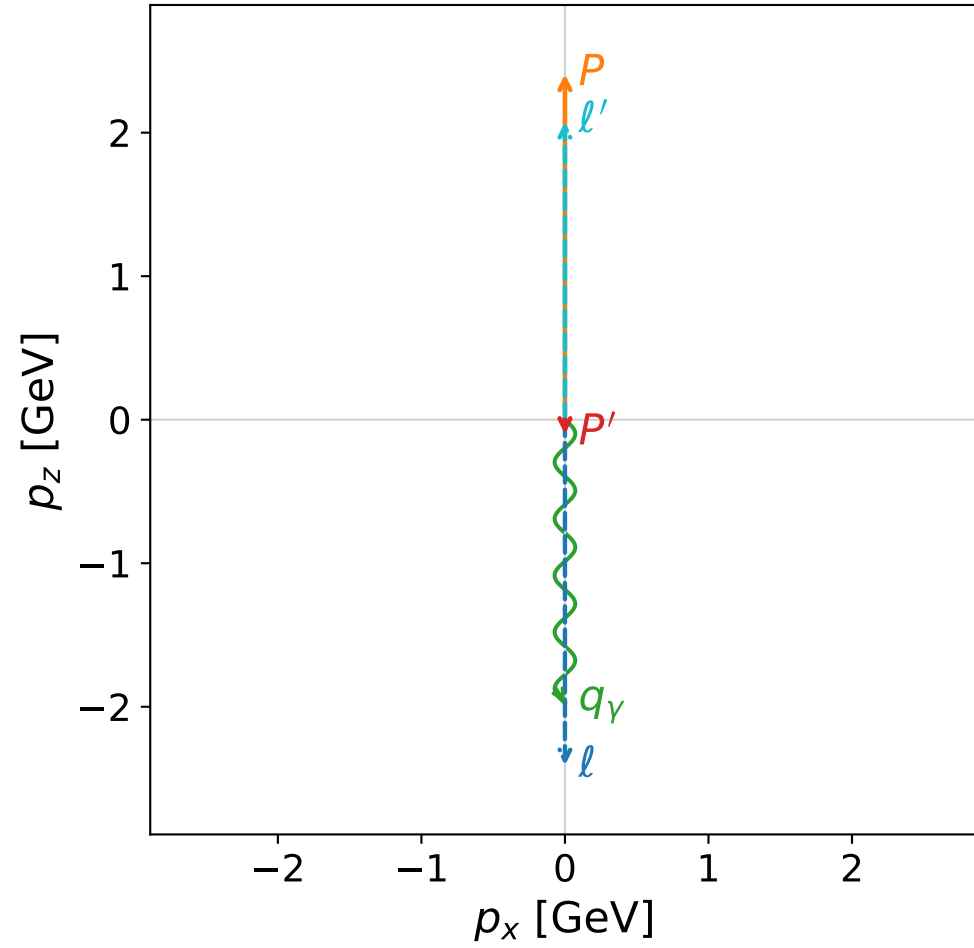
$s=24.9961$, $\sqrt{s}=4.99961$, $|\vec{P}|=2.41181$, $|\vec{P}'|=0.0116328$
 $\theta_{P'}=0.00959617$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.28284$, $\phi_\gamma=6.2538$
 $E_\gamma=2.03658$, $\theta_{\ell'}=5.51265\text{e-}05$, $\phi_{\ell'}=3.14125$
 $Q^2=19.5352$, $x_B=0.834713$, $t=-3.03928$, $W^2=4.74815$, $y=0.970447$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.0250$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0250$
 P' $E= 0.9381$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0116$
 q_γ $E= 2.0366$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0366$

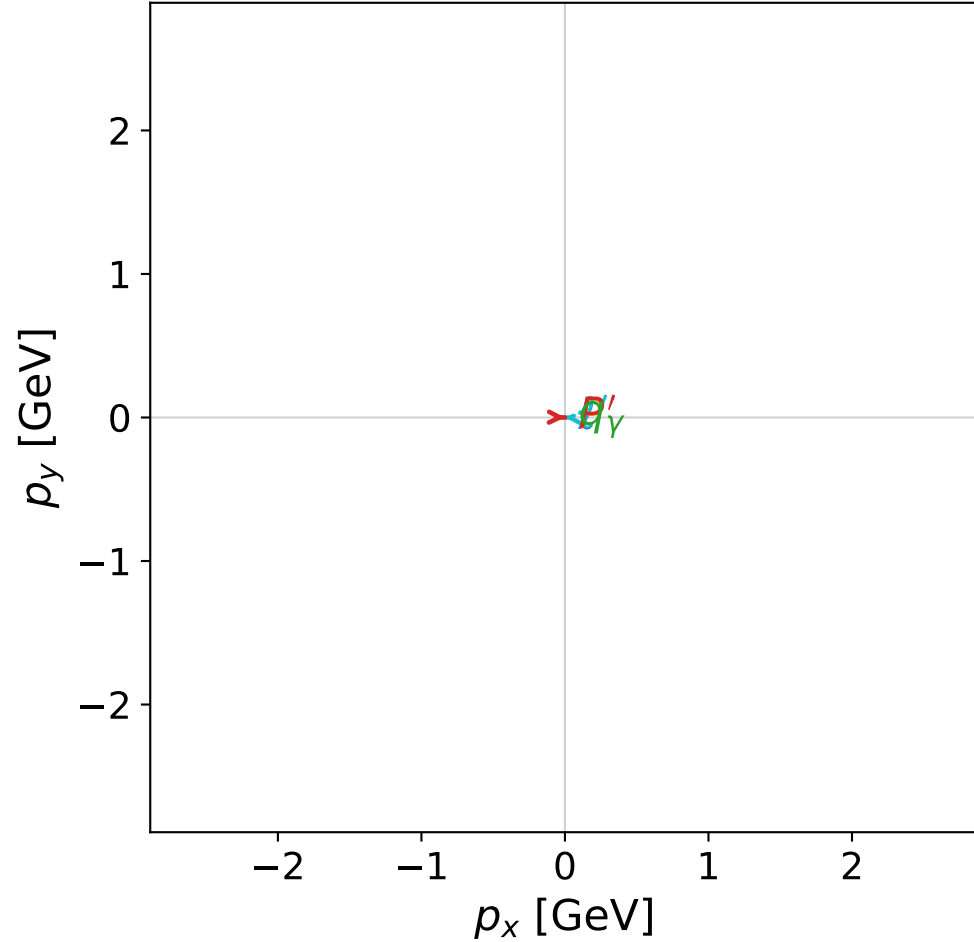


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

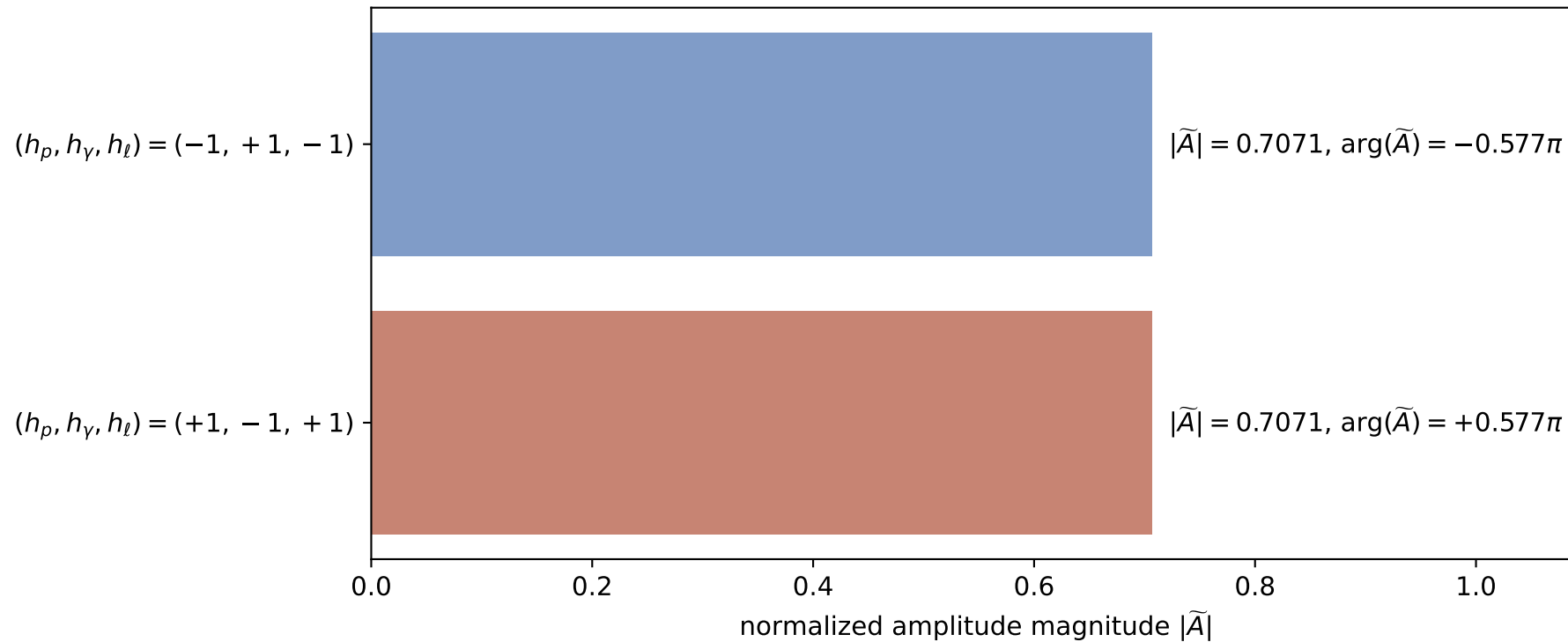


dghz_mixing_angles_polarization_02_local_minimum_41
 $d_{\text{GHZ}}=2.65018\text{e-}08$
 region: polarization_cluster_02_local_minimum_41
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0041
 lepton mixing angle: $\alpha_e=0.70648598$ rad
 incoming lepton: $0.760648|+\rangle + 0.649165|-\rangle$
 proton mixing angle: $\alpha_p=2.4350875$ rad
 incoming proton: $-0.760635|+\rangle + 0.649179|-\rangle$

$s=24.9206$, $\sqrt{s}=4.99205$, $|\vec{P}|=2.4079$, $|\vec{P}'|=0.101522$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.26287$, $\phi_\gamma=1.79274$
 $E_\gamma=1.97352$, $\theta_{\ell'}=3.12924\text{e-}07$, $\phi_{\ell'}=3.12128$
 $Q^2=19.986$, $x_B=0.857428$, $t=-3.6054$, $W^2=4.20309$, $y=0.969575$

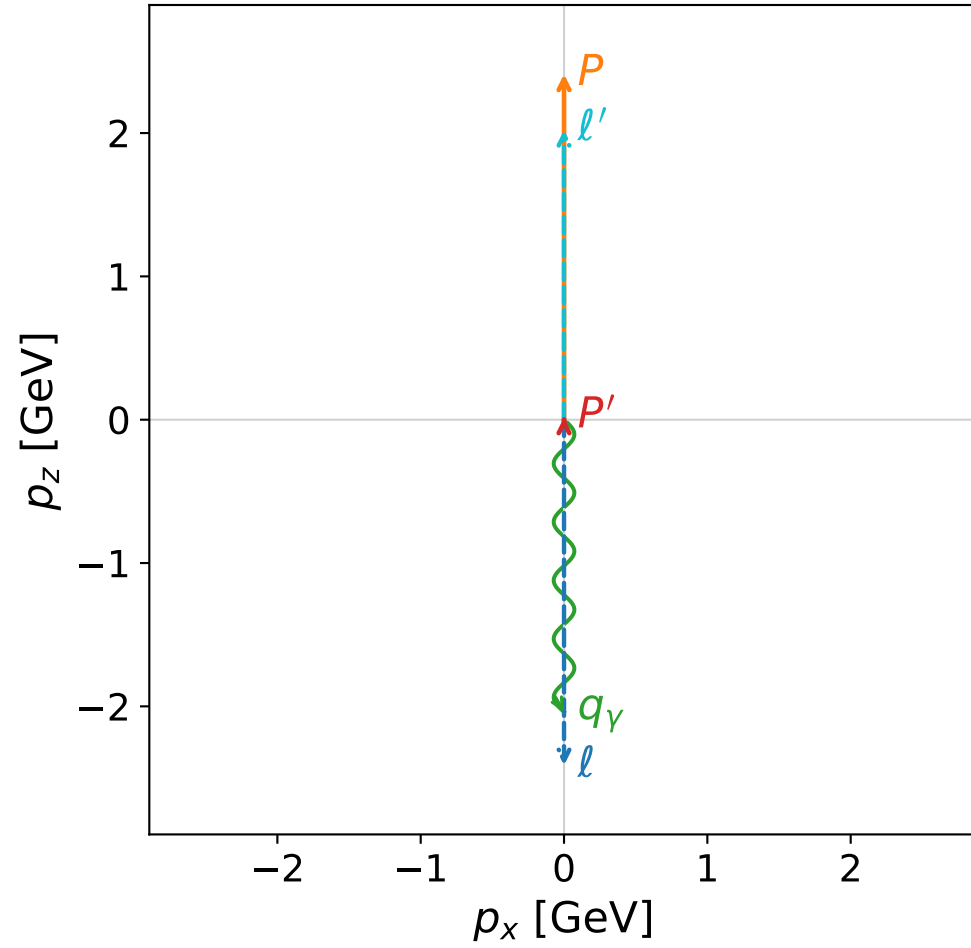
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4079$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4079$
 P $E= 2.5841$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4079$
 ℓ' $E= 2.0750$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0750$
 P' $E= 0.9435$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.1015$
 q_γ $E= 1.9735$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9735$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

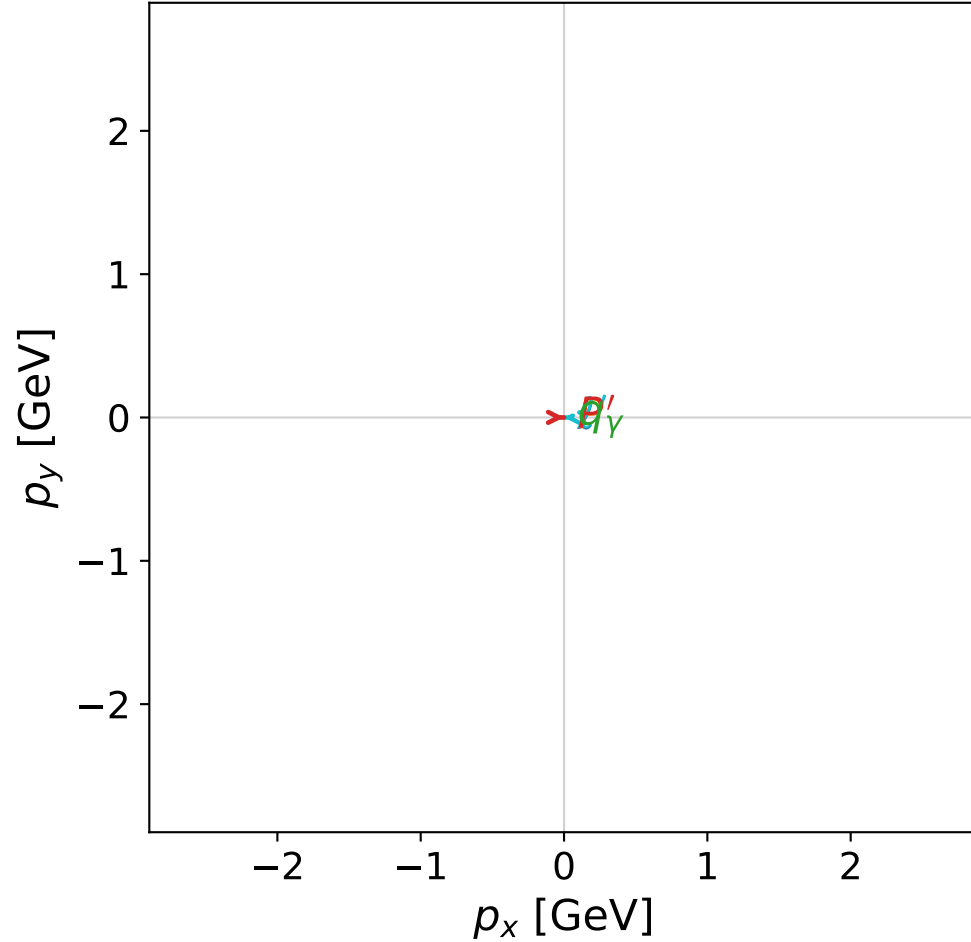


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



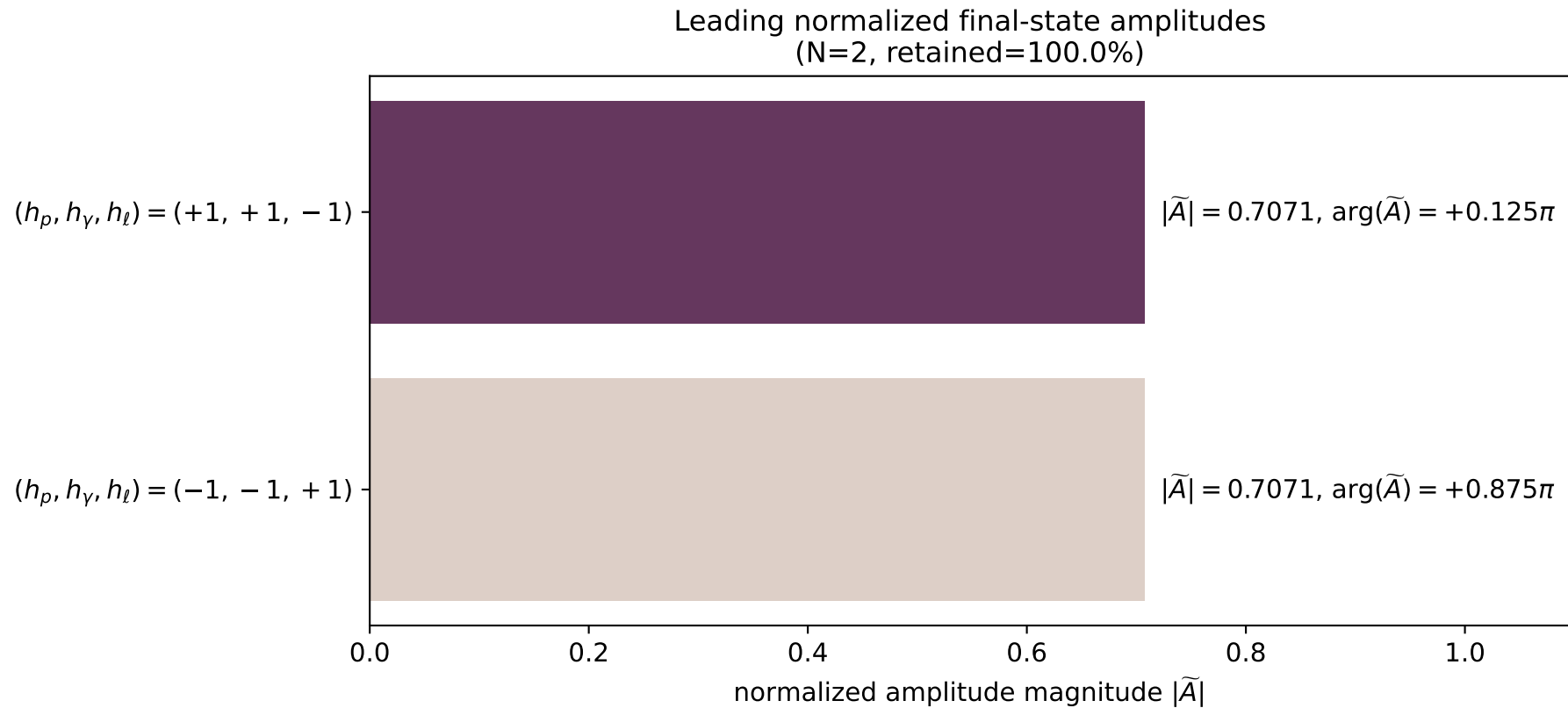
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_42
 $d_{\text{GHZ}}=2.65173\text{e-}08$
 region: polarization_cluster_02_local_minimum_42
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0042
 lepton mixing angle: $\alpha_e=0.7472291$ rad
 incoming lepton: $0.733575|+\rangle + 0.679609|-\rangle$
 proton mixing angle: $\alpha_p=2.394356$ rad
 incoming proton: $-0.73357|+\rangle + 0.679614|-\rangle$

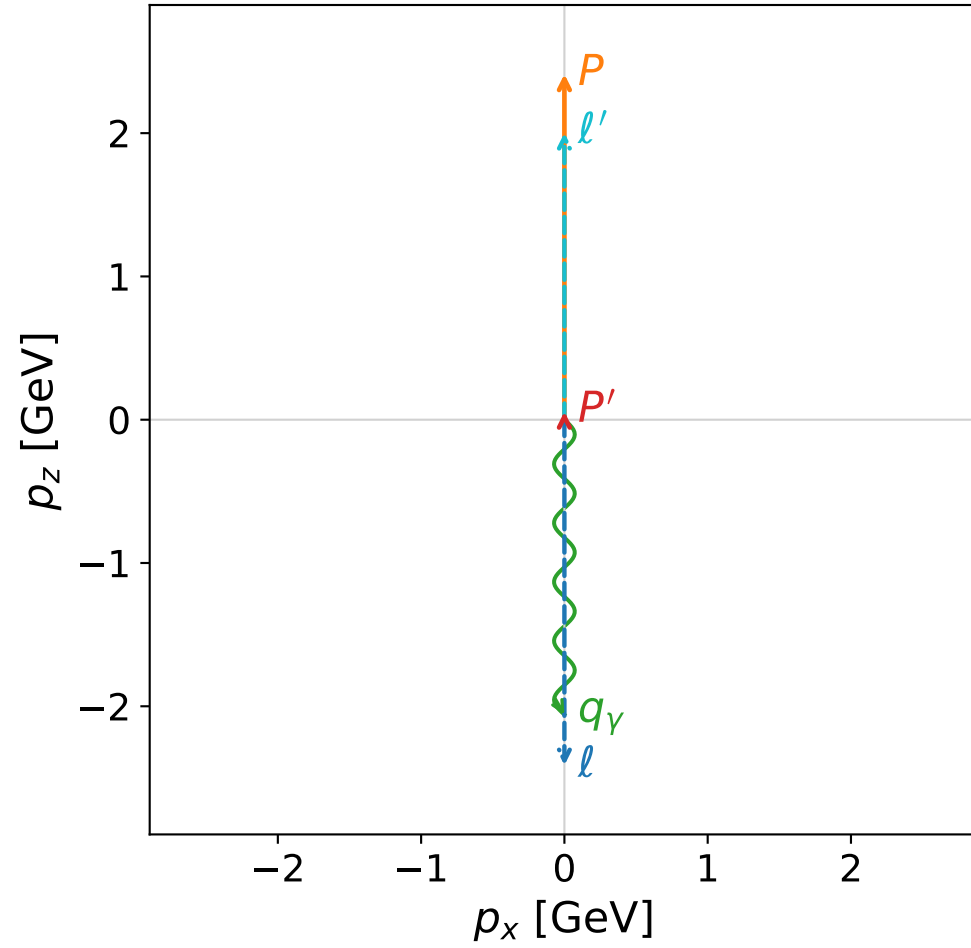
$s=24.9854$, $\sqrt{s}=4.99854$, $|\vec{P}|=2.41126$, $|\vec{P}'|=0.0171201$
 $\theta_{P'}=3.14159\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.27545$, $\phi_\gamma=2.74997$
 $E_\gamma=2.03875$, $\theta_{\ell'}=2.66143\text{e-}07$, $\phi_{\ell'}=3.13386$
 $Q^2=19.4987$, $x_B=0.833497$, $t=-3.01229$, $W^2=4.77498$, $y=0.970476$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 2.0216$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0216$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0171$
 q_γ $E= 2.0388$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0388$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

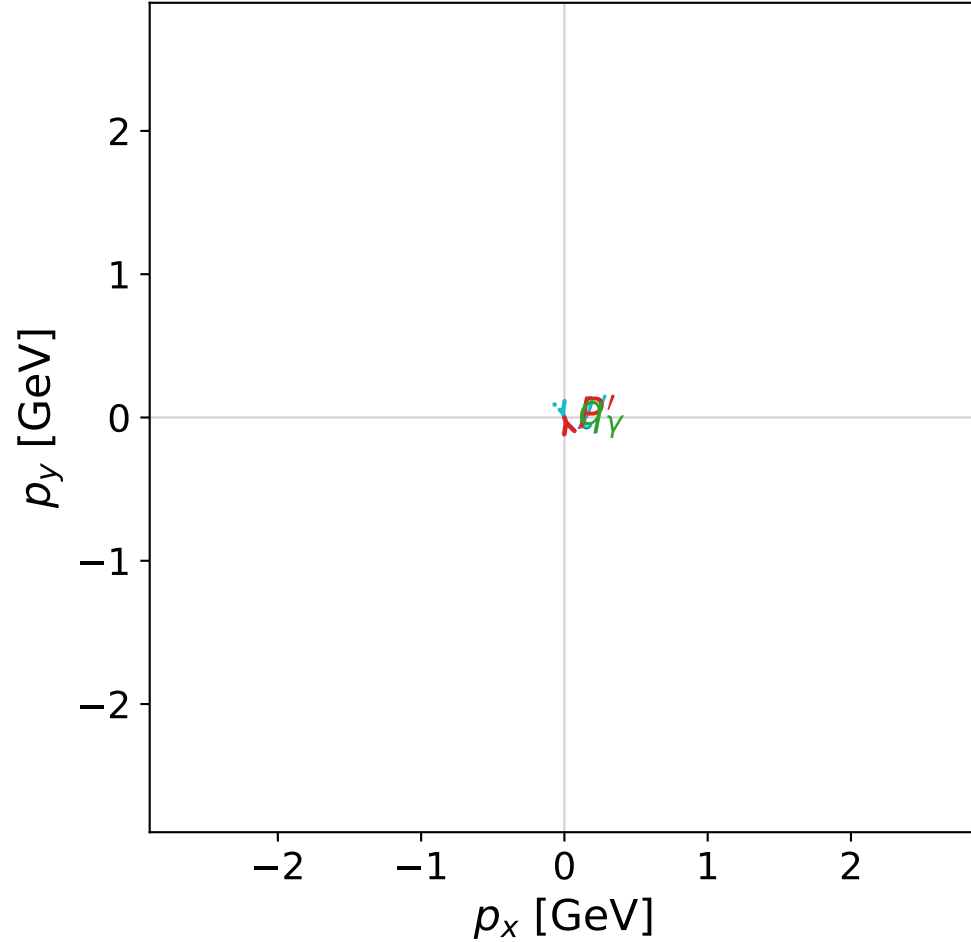


dghz_mixing_angles_polarization_02_local_minimum_49
 $d_{\text{GHZ}}=2.67844\text{e-}08$
 region: polarization_cluster_02_local_minimum_49
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0049
 lepton mixing angle: $\alpha_e=0.81607917$ rad
 incoming lepton: $0.685083|+\rangle + 0.728465|-\rangle$
 proton mixing angle: $\alpha_p=2.3255025$ rad
 incoming proton: $-0.685075|+\rangle + 0.728473|-\rangle$

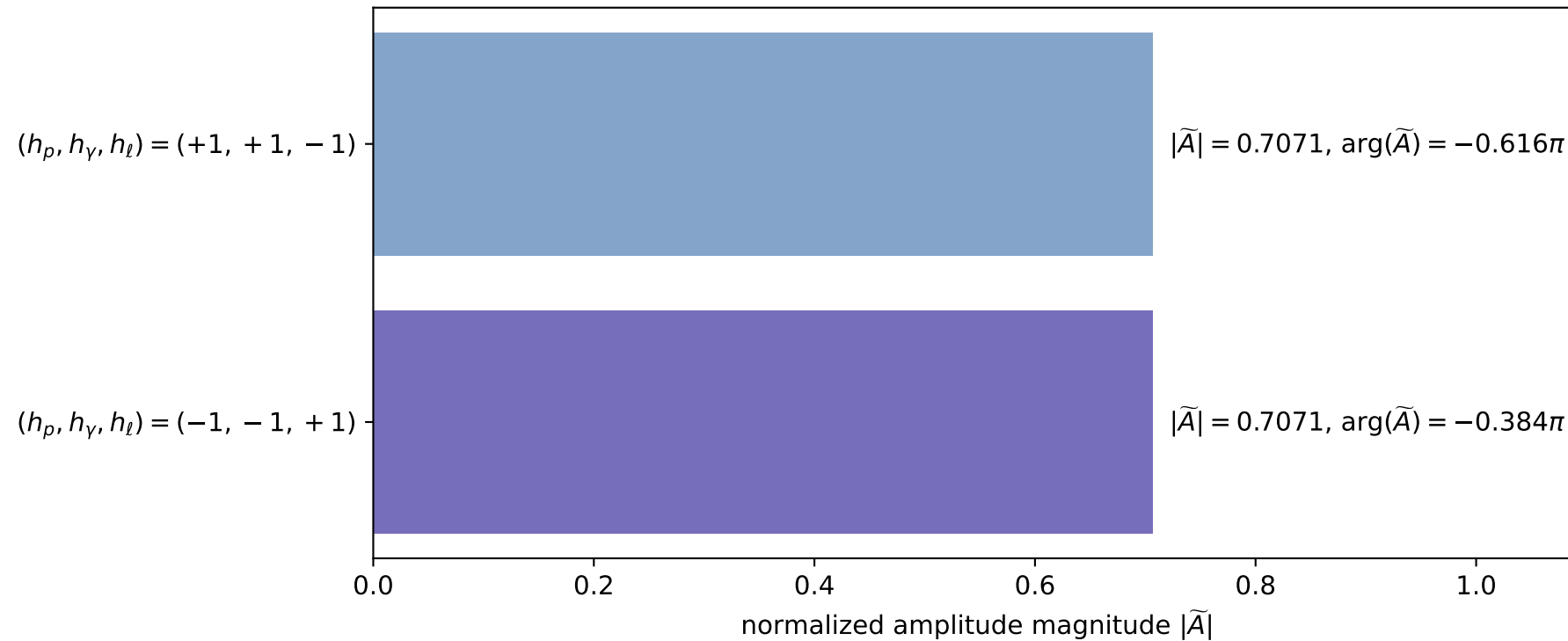
$s=24.9993$, $\sqrt{s}=4.99993$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.0563059$
 $\theta_{P'}=2.80434\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.88915$, $\phi_\gamma=5.07566$
 $E_\gamma=2.05828$, $\theta_{\ell'}=7.88636\text{e-}07$, $\phi_{\ell'}=5.03075$
 $Q^2=19.3148$, $x_B=0.824895$, $t=-2.83243$, $W^2=4.9799$, $y=0.970788$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0020$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0020$
 P' $E= 0.9397$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0563$
 q_γ $E= 2.0583$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0583$

x--y plane

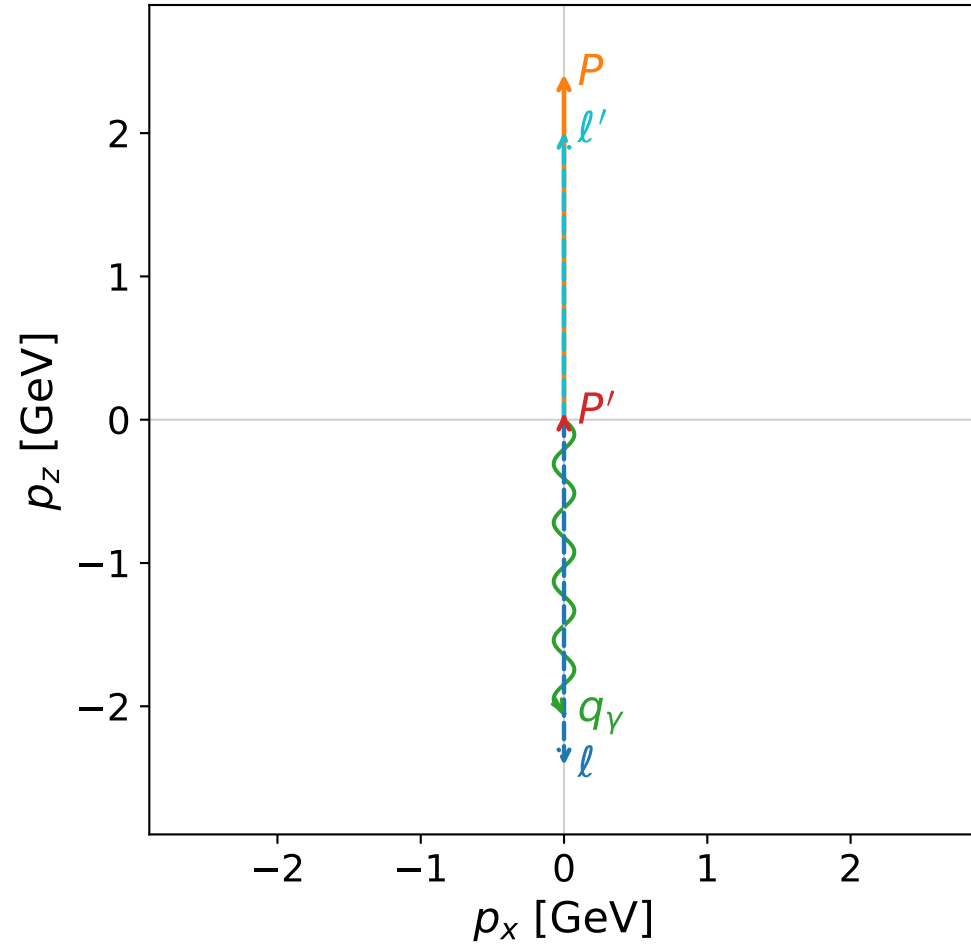


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

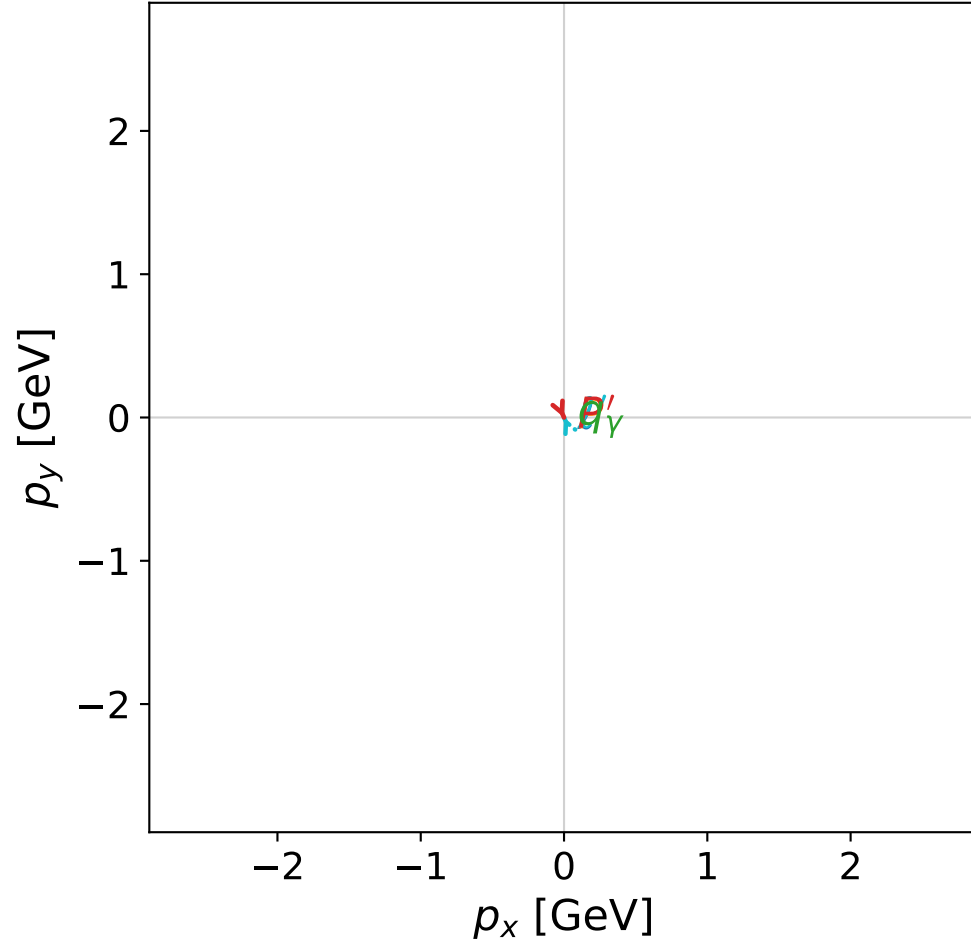


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



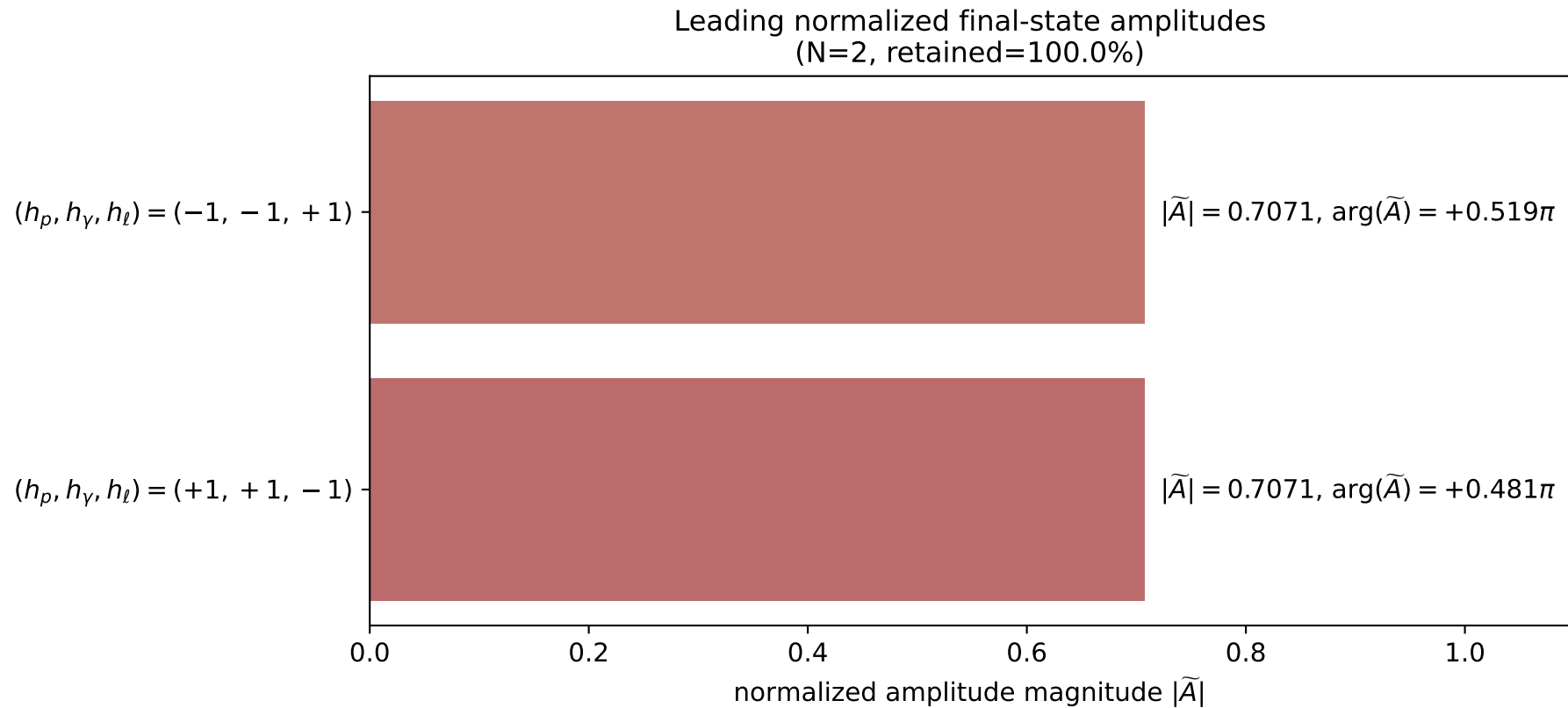
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_54
 $d_{\{\text{rm GHZ}\}}=2.70595\text{e-}08$
 region: polarization_cluster_02_local_minimum_54
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0054
 lepton mixing angle: $\alpha_e=0.85741662$ rad
 incoming lepton: $0.654393|+\rangle + 0.756155|-\rangle$
 proton mixing angle: $\alpha_p=2.2841843$ rad
 incoming proton: $-0.654399|+\rangle + 0.756149|-\rangle$

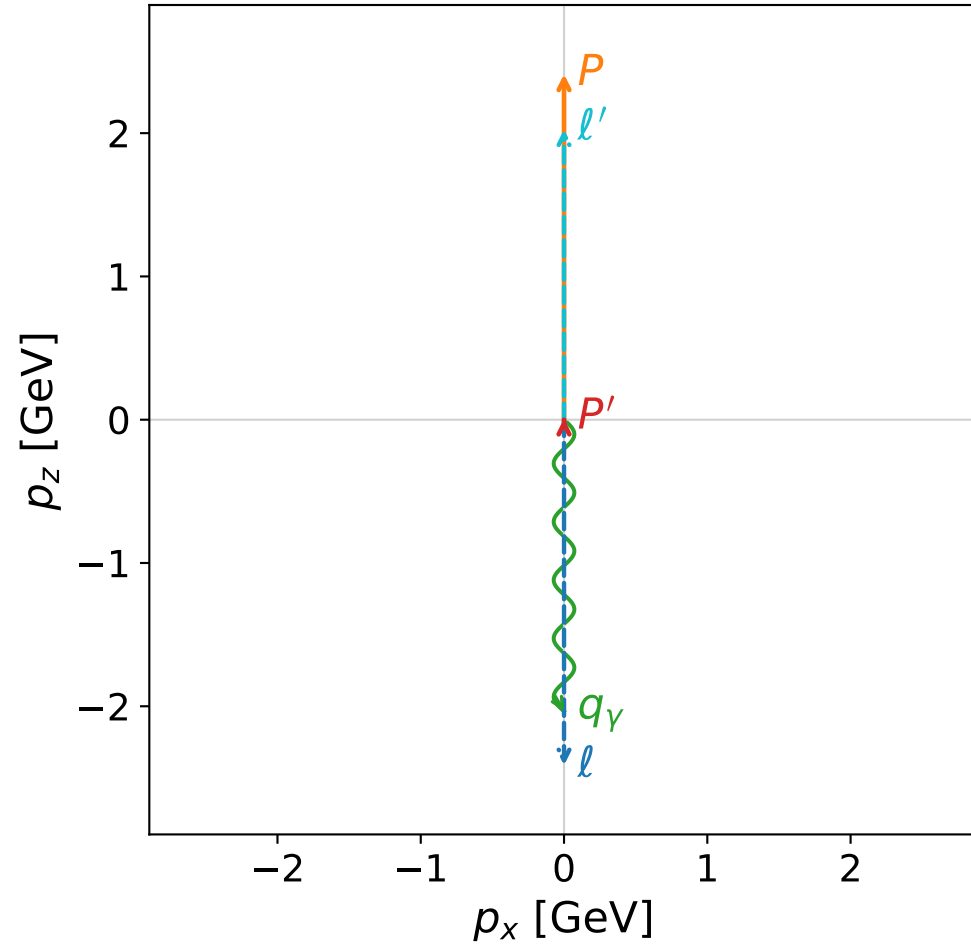
$s=24.9965$, $\sqrt{s}=4.99965$, $|\vec{P}|=2.41184$, $|\vec{P}'|=0.0441066$
 $\theta_{p'}=0.00122325$, $\theta_Y=3.14159$, $\phi_{p'}=5.12862$, $\phi_Y=1.63085$
 $E_Y=2.05236$, $\theta_{\ell'}=2.68658\text{e-}05$, $\phi_{\ell'}=1.98702$
 $Q^2=19.3743$, $x_B=0.827614$, $t=-2.88766$, $W^2=4.91537$, $y=0.970691$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.0083$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0083$
 P' $E= 0.9390$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0441$
 q_Y $E= 2.0524$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0524$

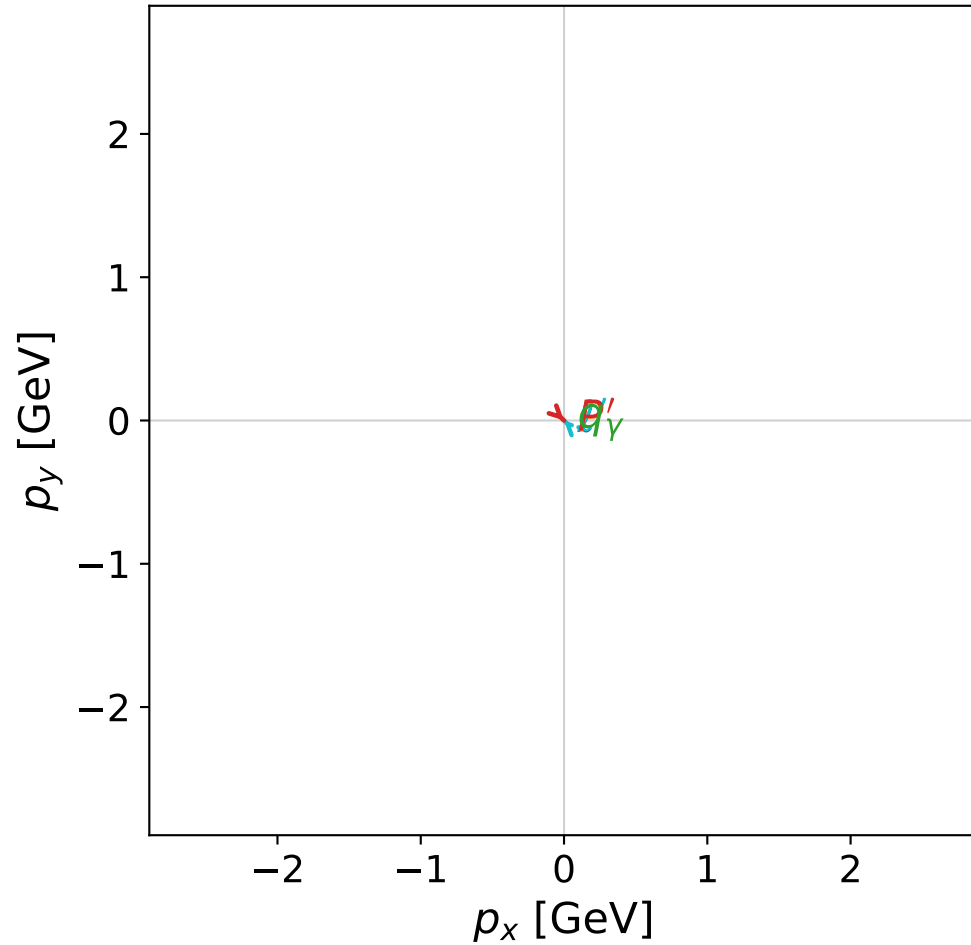


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

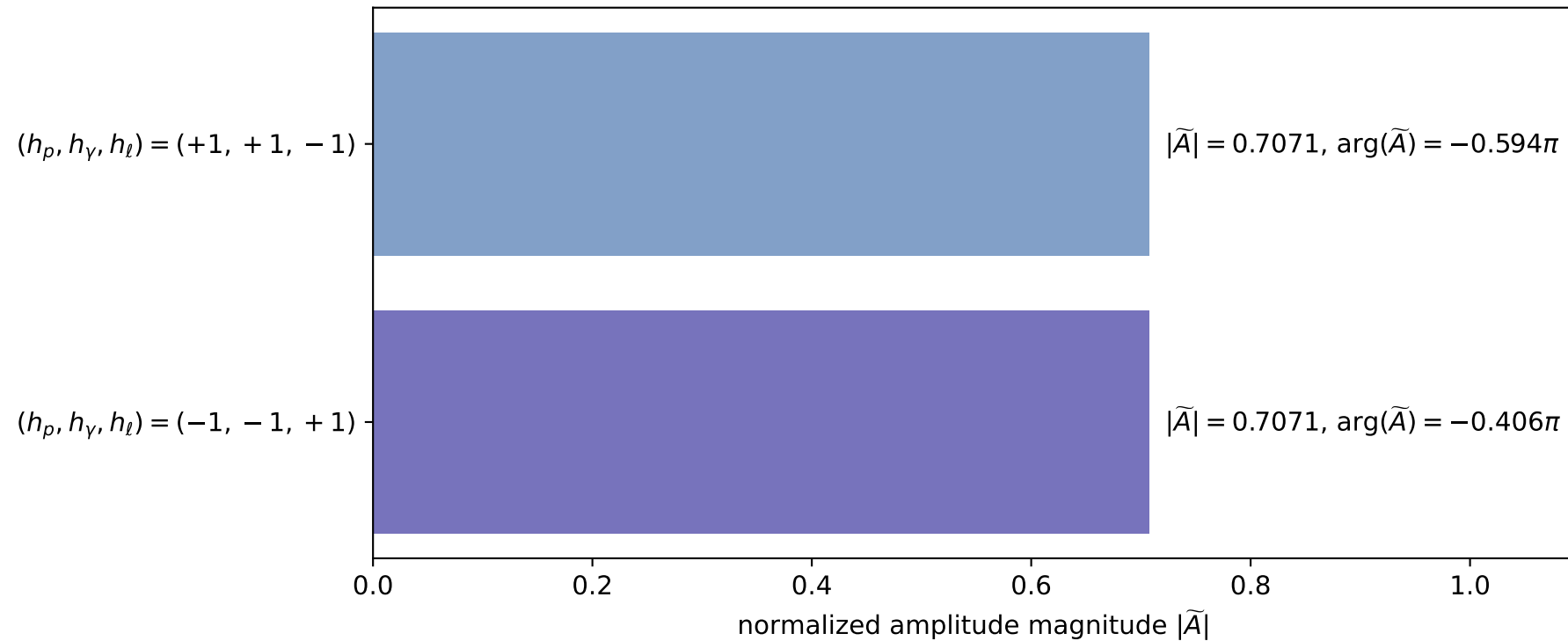


dghz_mixing_angles_polarization_02_local_minimum_62
 $d_{\text{GHZ}}=2.75303\text{e-}08$
 region: polarization_cluster_02_local_minimum_62
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0062
 lepton mixing angle: $\alpha_e=0.65340217$ rad
 incoming lepton: $0.79402|+\rangle + 0.607891|-\rangle$
 proton mixing angle: $\alpha_p=2.4881891$ rad
 incoming proton: $-0.794019|+\rangle + 0.607892|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00901078$
 $\theta_{P'}=0.000613592$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.51618$, $\phi_\gamma=5.00919$
 $E_\gamma=2.03548$, $\theta_{\ell'}=2.72834\text{e-}06$, $\phi_{\ell'}=2.37459$
 $Q^2=19.5515$, $x_B=0.835287$, $t=-3.05213$, $W^2=4.73527$, $y=0.970432$

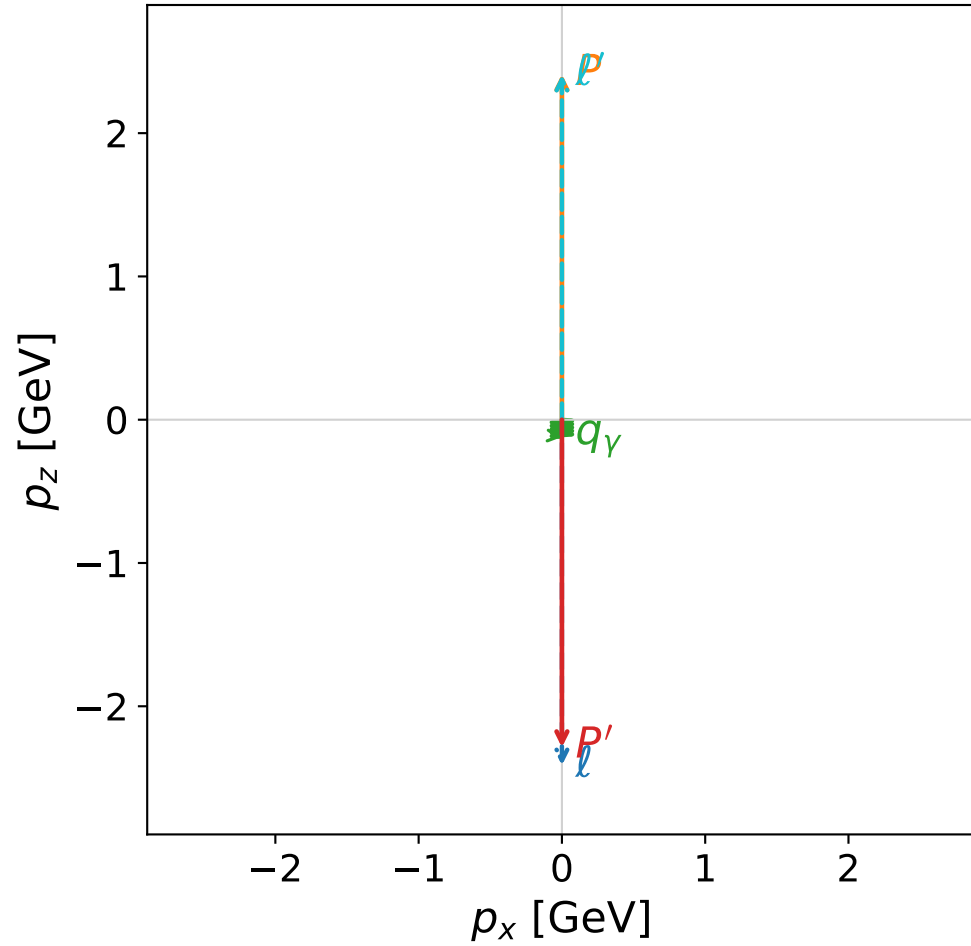
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0265$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0265$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0090$
 q_γ $E= 2.0355$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0355$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_02_local_minimum_77

$d_{\{\rm GHZ\}}=2.96256\text{e-}08$

region: polarization_cluster_02_local_minimum_77

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0077

lepton mixing angle: $\alpha_e=1.1396606$ rad

incoming lepton: $0.417903|+\rangle + 0.908492|-\rangle$

proton mixing angle: $\alpha_p=2.0019206$ rad

incoming proton: $-0.417892|+\rangle + 0.908497|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28693$

$\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=4.37912$, $\phi_\gamma=0.0937313$

$E_\gamma=0.120627$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.28657$

$Q^2=23.2282$, $x_B=0.998083$, $t=-22.0666$, $W^2=0.924449$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$

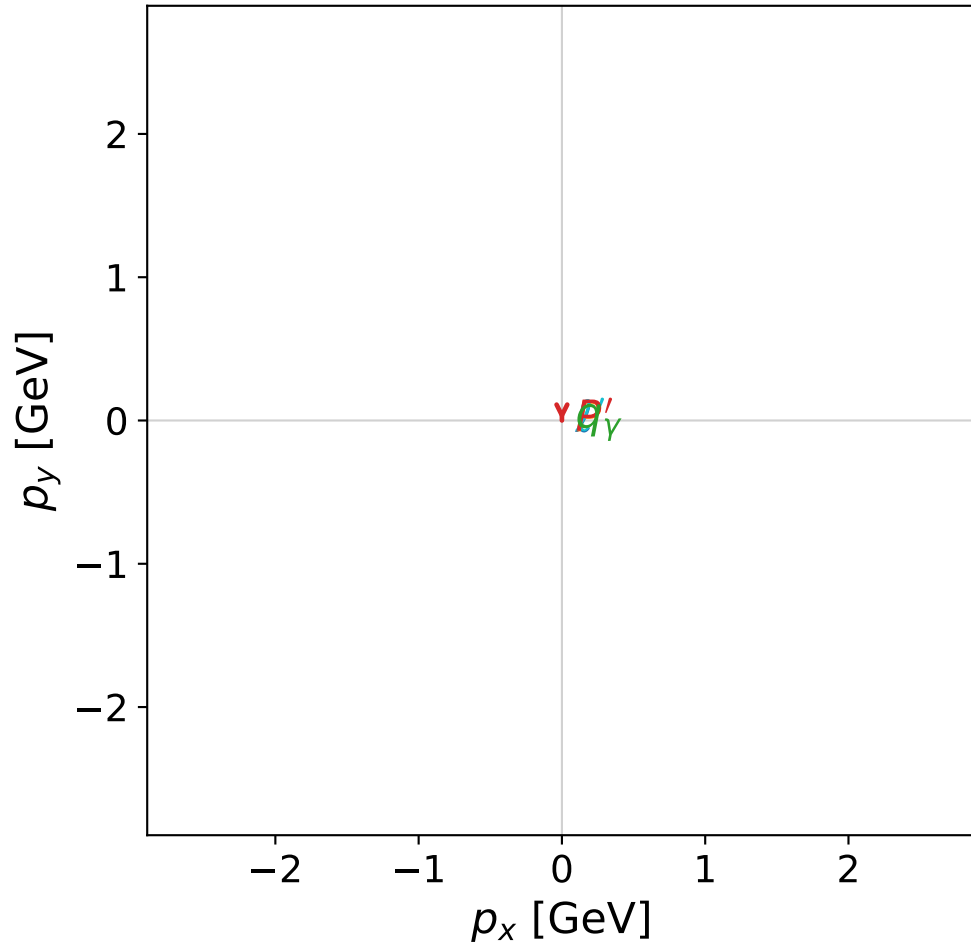
P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$

ℓ' $E= 2.4076$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4076$

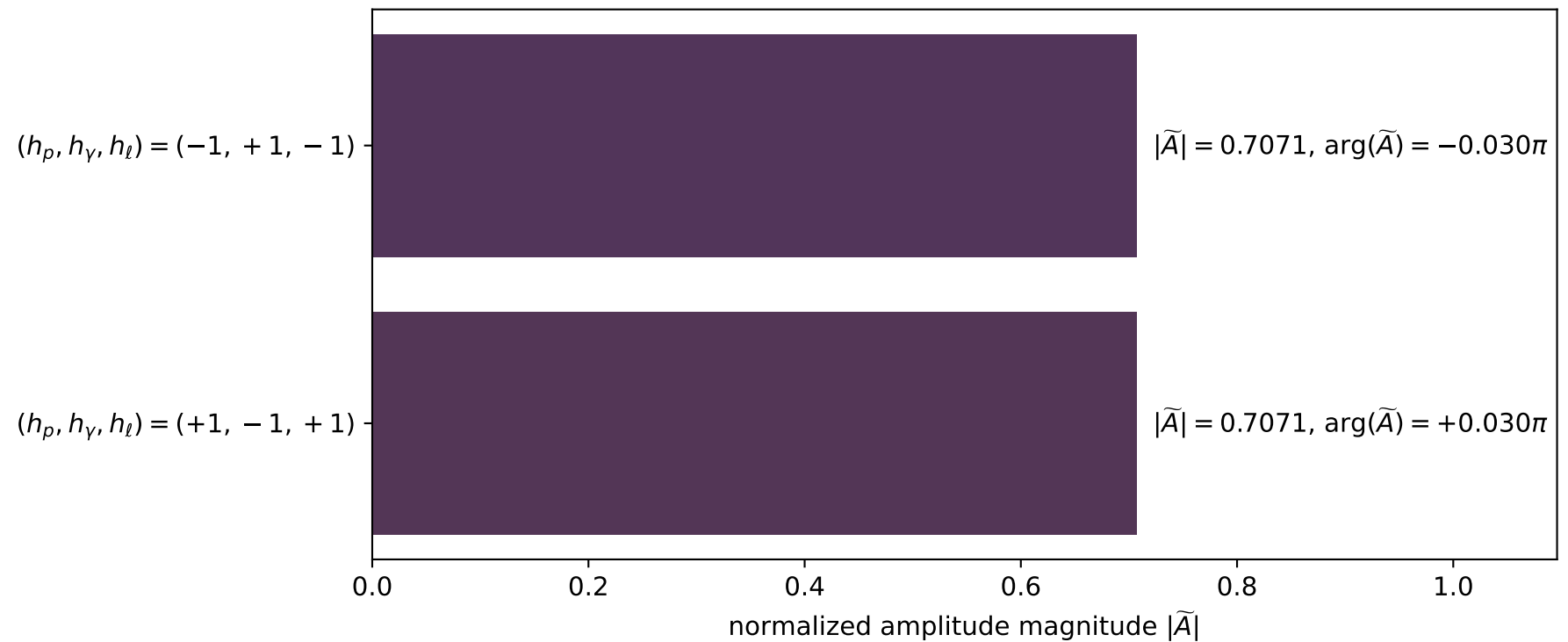
P' $E= 2.4718$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.2869$

q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1206$

x--y plane

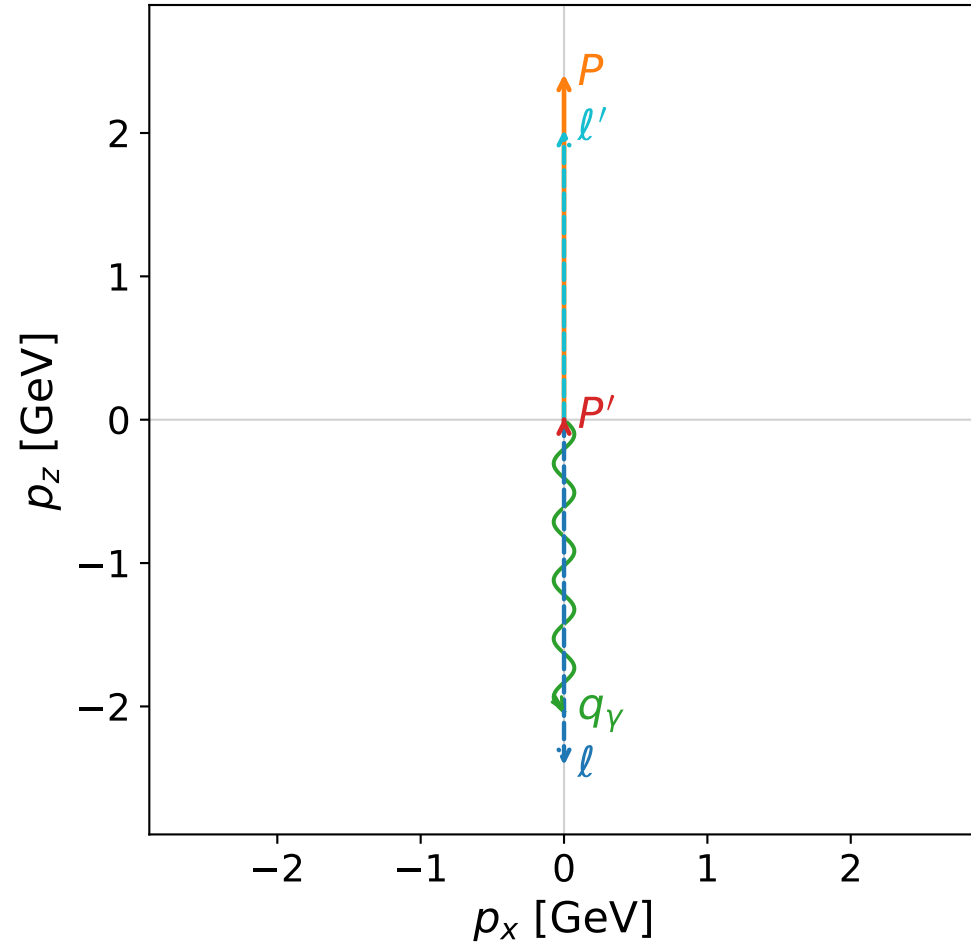


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

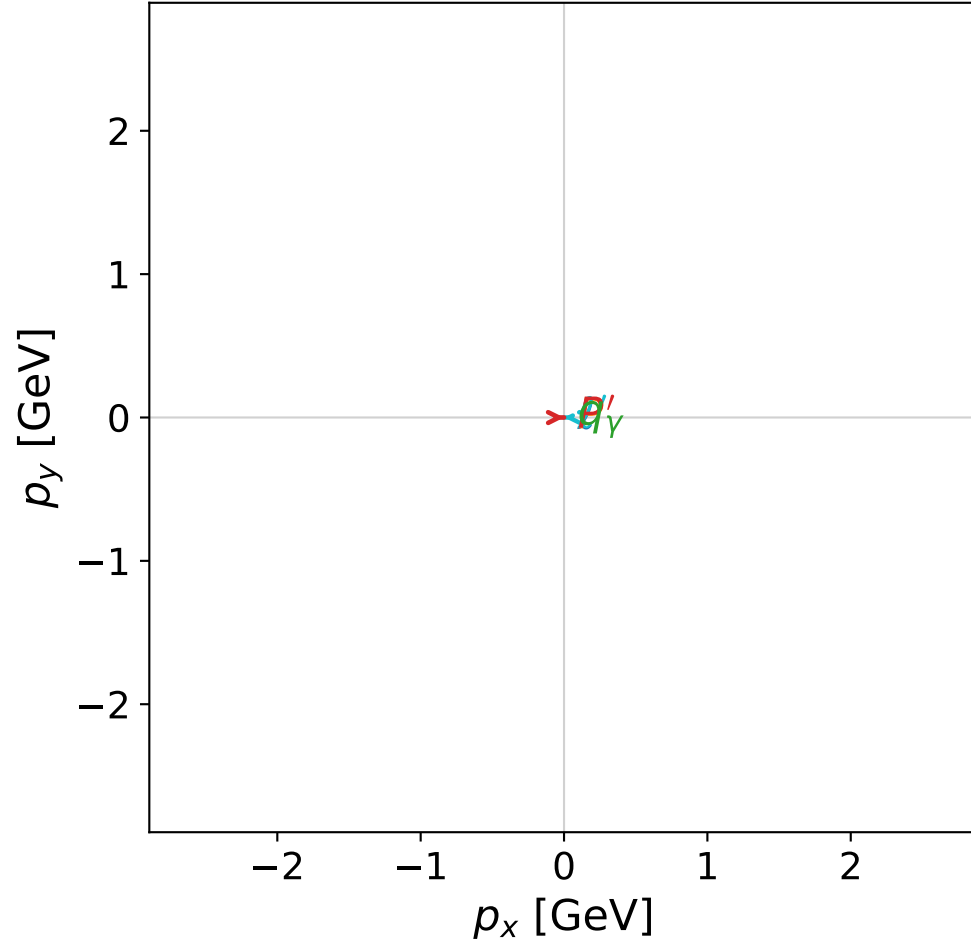


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



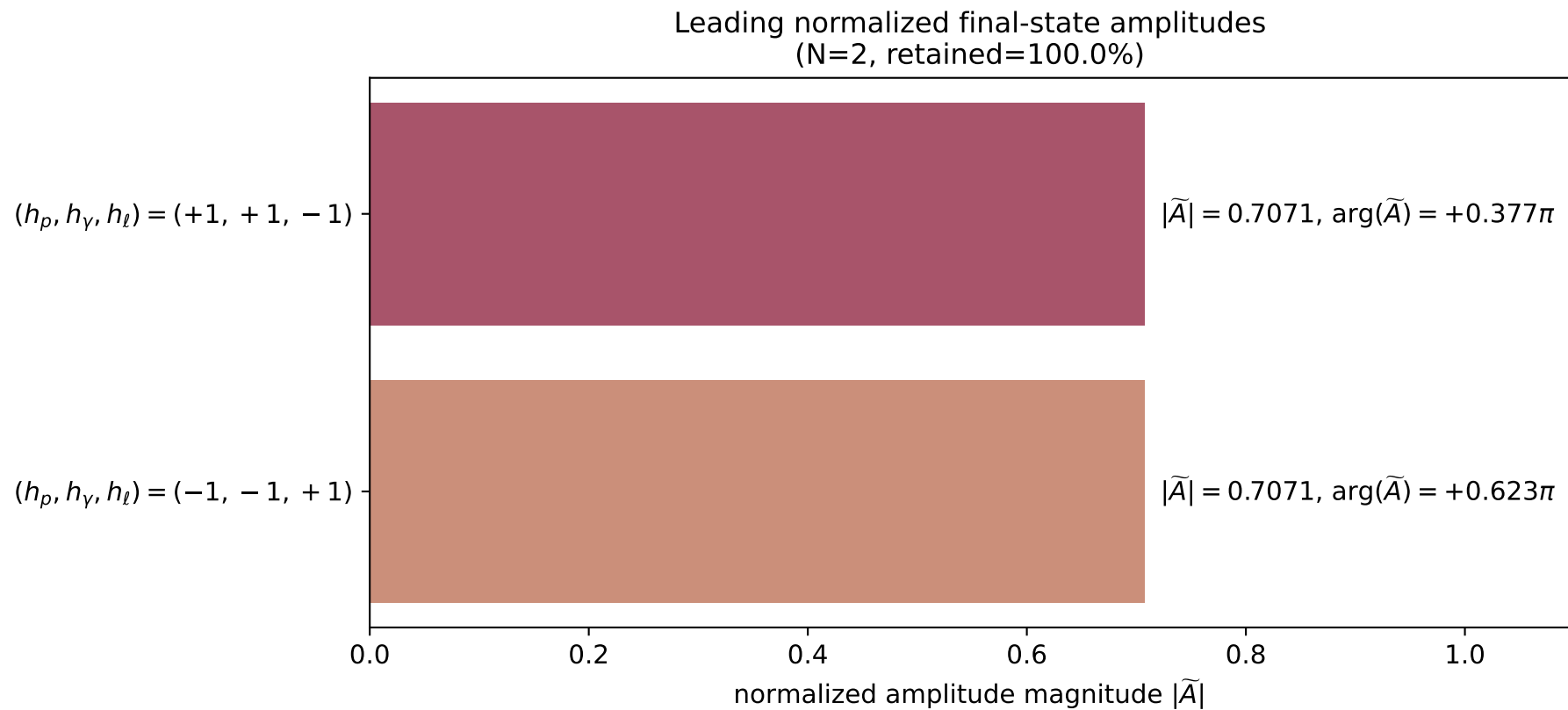
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_78
 $d_{\text{GHZ}}=2.97007\text{e-}08$
 region: polarization_cluster_02_local_minimum_78
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0078
 lepton mixing angle: $\alpha_e=0.55713074$ rad
 incoming lepton: $0.848776|+\rangle + 0.528753|-\rangle$
 proton mixing angle: $\alpha_p=2.5844424$ rad
 incoming proton: $-0.848765|+\rangle + 0.52877|-\rangle$

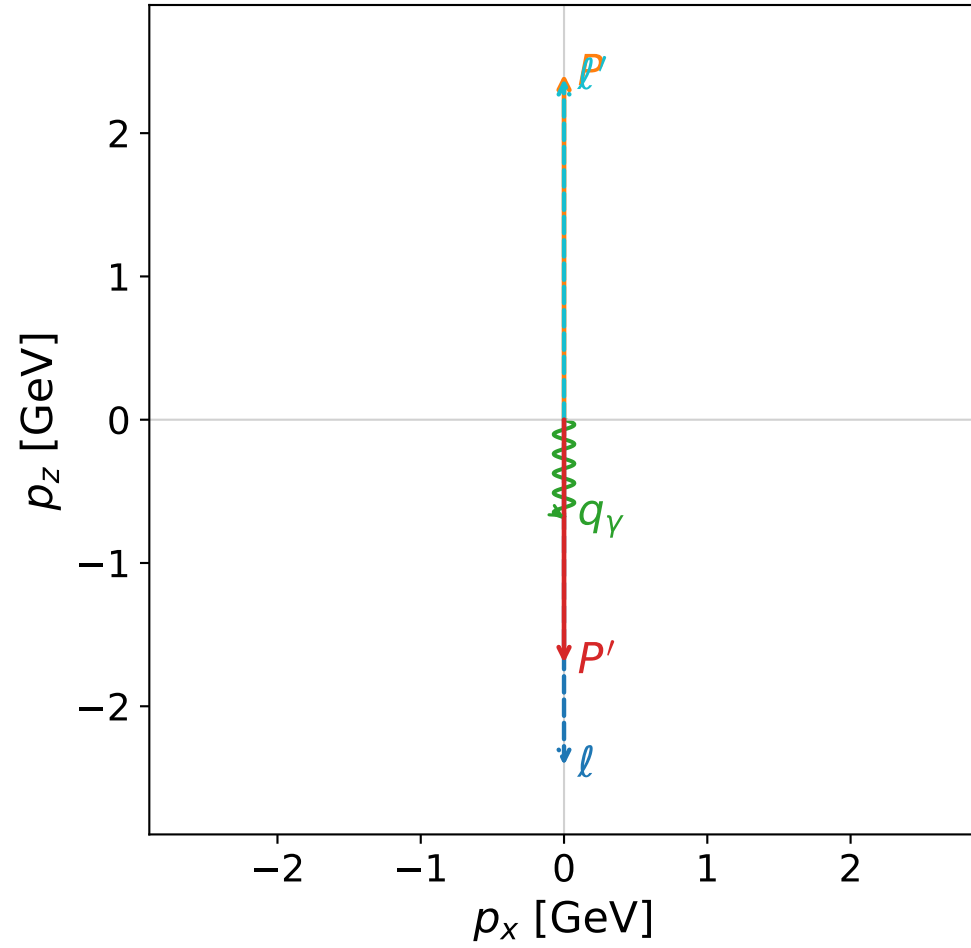
$s=24.9739$, $\sqrt{s}=4.99739$, $|\vec{P}|=2.41067$, $|\vec{P}'|=0.0137879$
 $\theta_{P'}=0.000837323$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.00351513$, $\phi_\gamma=1.9584$
 $E_\gamma=2.03654$, $\theta_{\ell'}=5.70749\text{e-}06$, $\phi_{\ell'}=3.14511$
 $Q^2=19.5047$, $x_B=0.834182$, $t=-3.02706$, $W^2=4.75696$, $y=0.970439$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4107$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4107$
 P $E= 2.5867$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4107$
 ℓ' $E= 2.0228$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0228$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0138$
 q_γ $E= 2.0365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0365$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

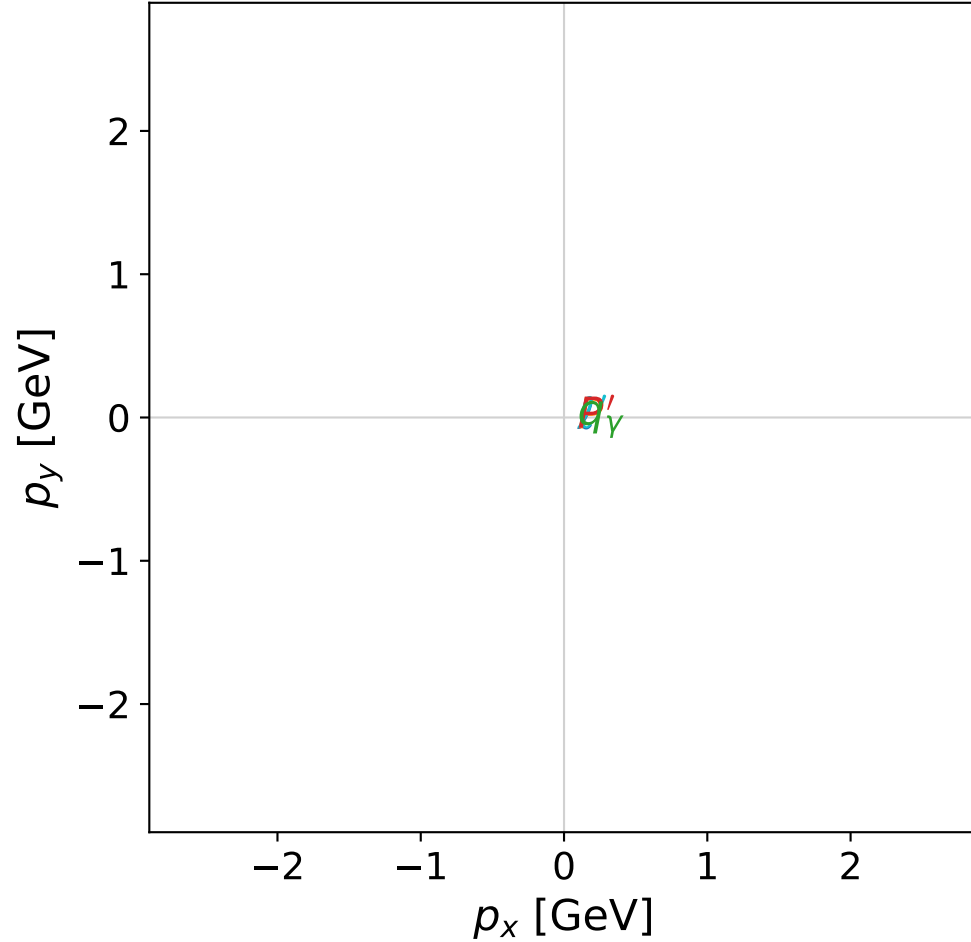


dghz_mixing_angles_polarization_02_local_minimum_86
 $d_{\{\text{rm GHZ}\}}=3.12213\text{e-}08$
 region: polarization_cluster_02_local_minimum_86
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0086
 lepton mixing angle: $\alpha_e=1.1624483$ rad
 incoming lepton: $0.397094|+\rangle + 0.917778|-\rangle$
 proton mixing angle: $\alpha_p=1.9791355$ rad
 incoming proton: $-0.397086|+\rangle + 0.917782|-\rangle$

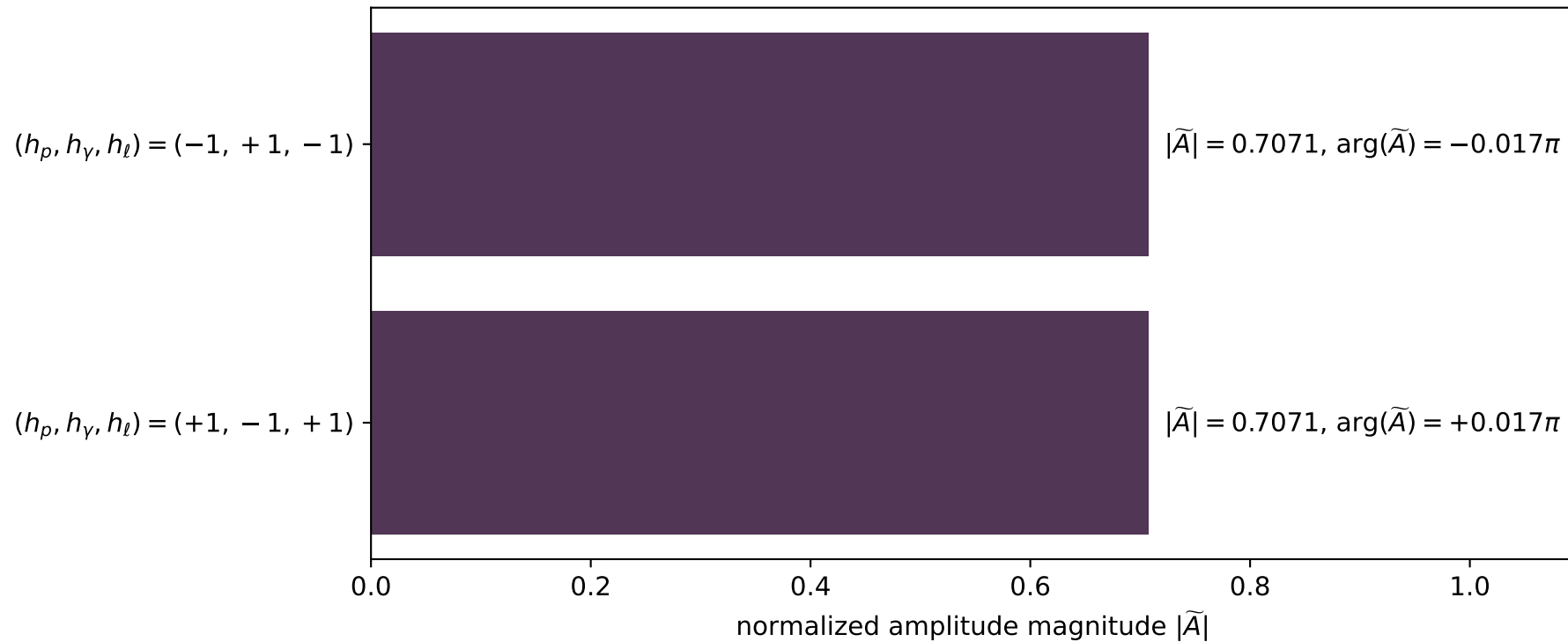
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.69806$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.694775$, $\phi_\gamma=0.0541782$
 $E_\gamma=0.681015$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.65694$
 $Q^2=22.9535$, $x_B=0.985852$, $t=-16.4727$, $W^2=1.20925$, $y=0.965287$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3791$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.3791$
 P' $E= 1.9399$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6981$
 q_γ $E= 0.6810$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6810$

x--y plane

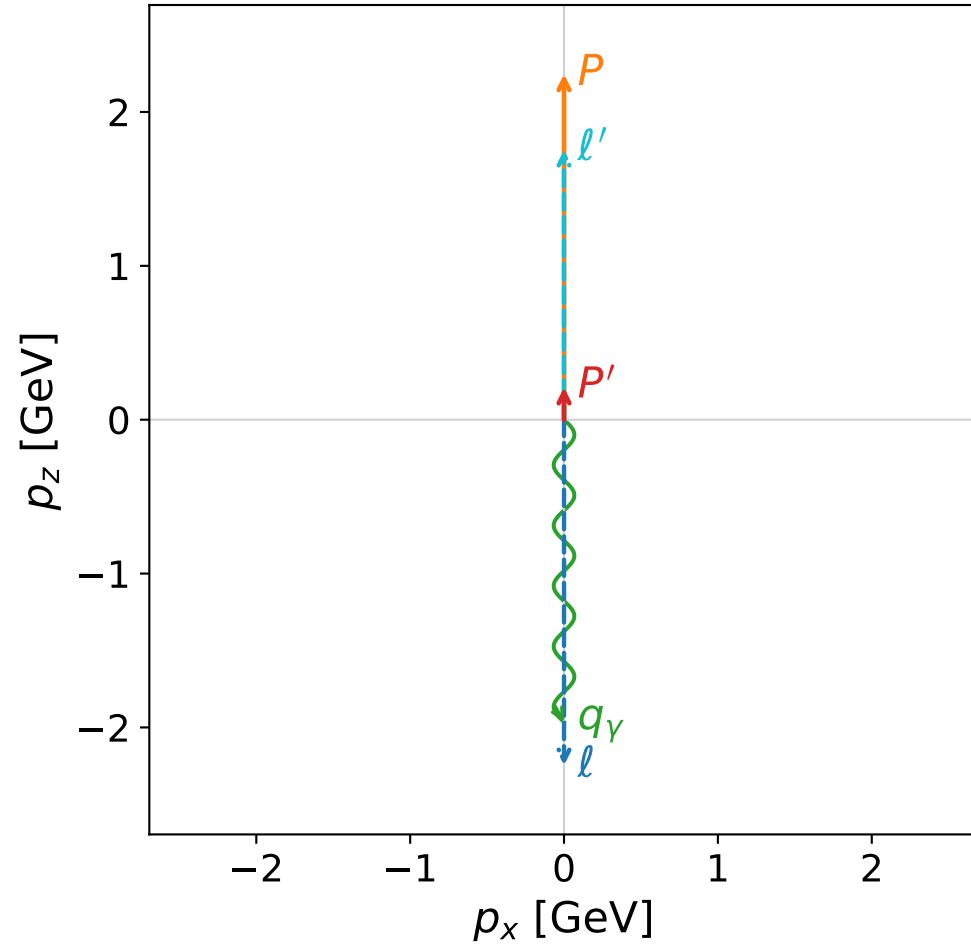


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

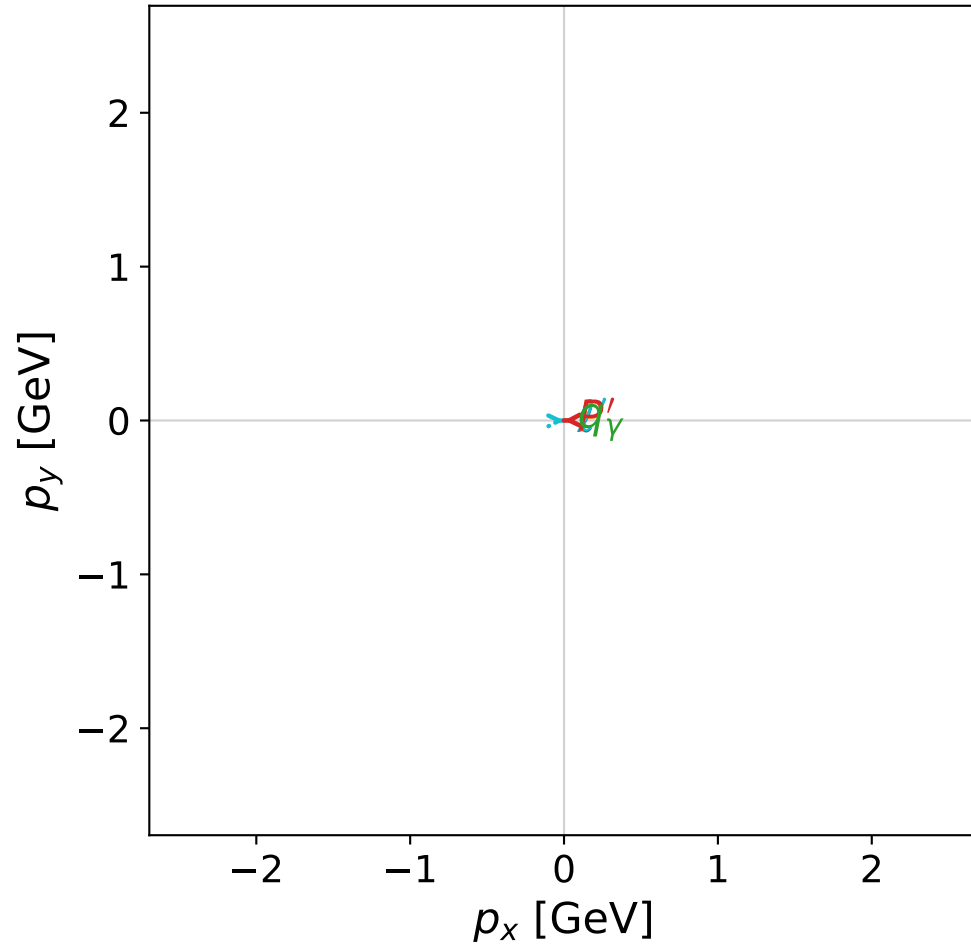


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



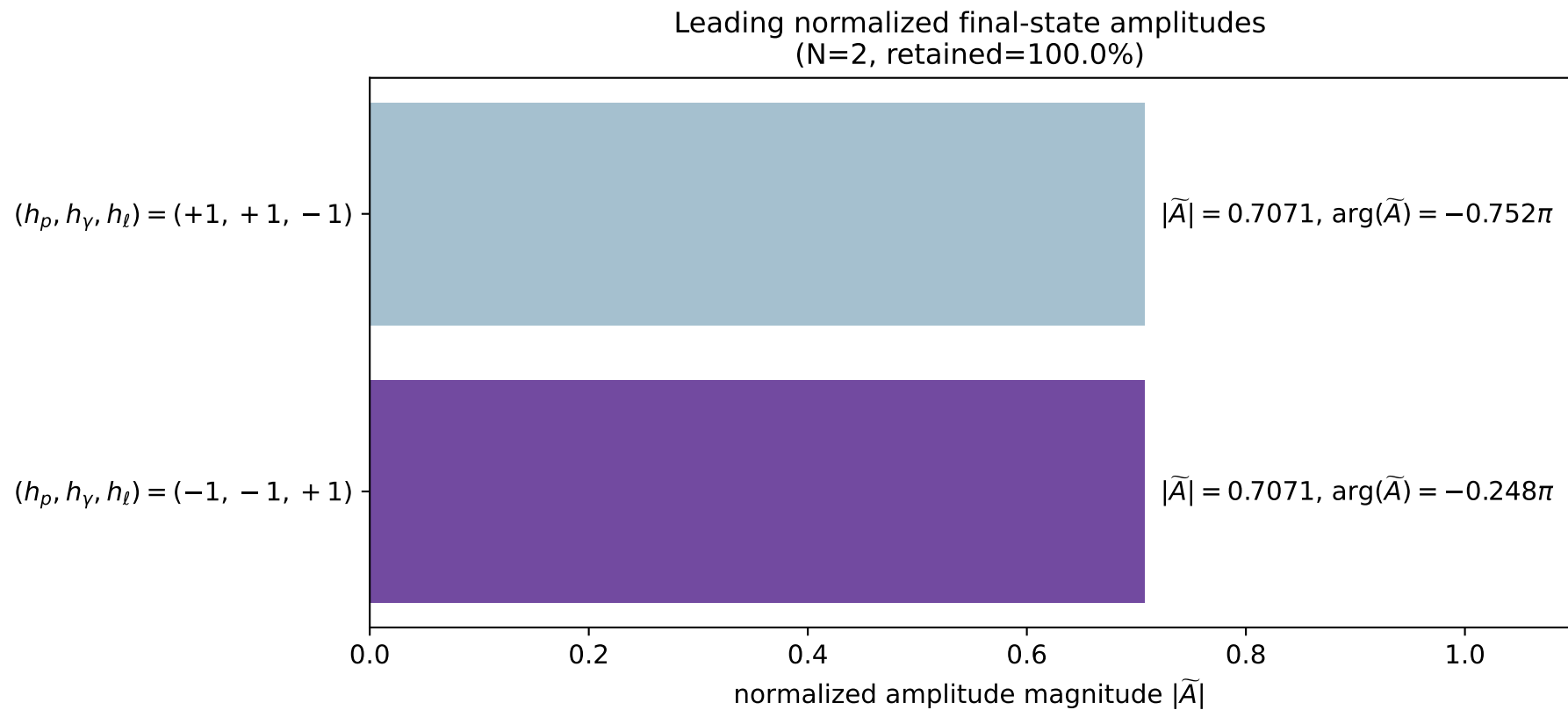
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_90
 $d_{\text{GHZ}}=3.23249\text{e-}08$
 region: polarization_cluster_02_local_minimum_90
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0090
 lepton mixing angle: $\alpha_e=0.78715457$ rad
 incoming lepton: $0.705864|+\rangle +0.708348|-\rangle$
 proton mixing angle: $\alpha_p=2.3544342$ rad
 incoming proton: $-0.705861|+\rangle +0.70835|-\rangle$

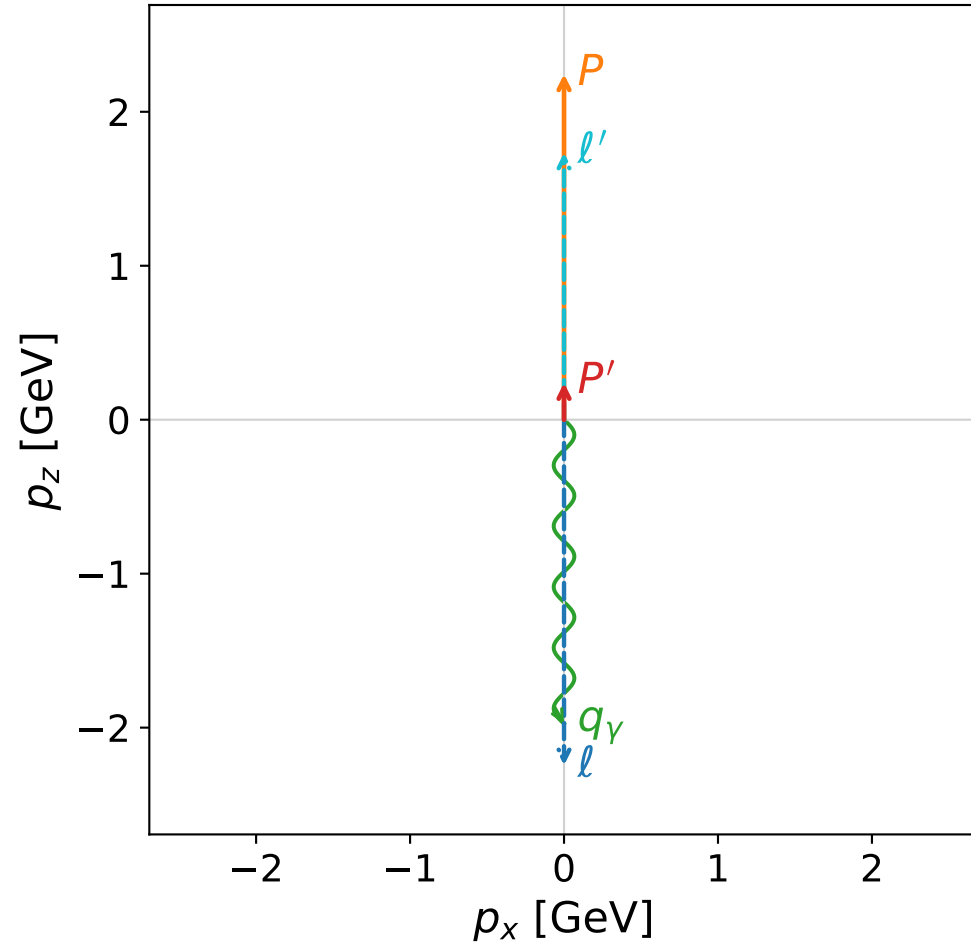
$s=21.9038$, $\sqrt{s}=4.68015$, $|\vec{P}|=2.24608$, $|\vec{P}'|=0.210445$
 $\theta_{P'}=6.7809\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.16268$, $\phi_\gamma=5.50499$
 $E_\gamma=1.96464$, $\theta_{\ell'}=8.13445\text{e-}07$, $\phi_{\ell'}=0.0210885$
 $Q^2=15.7602$, $x_B=0.773911$, $t=-1.97479$, $W^2=5.48402$, $y=0.968628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2461$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2461$
 P $E= 2.4341$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2461$
 ℓ' $E= 1.7542$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7542$
 P' $E= 0.9613$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2104$
 q_γ $E= 1.9646$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9646$

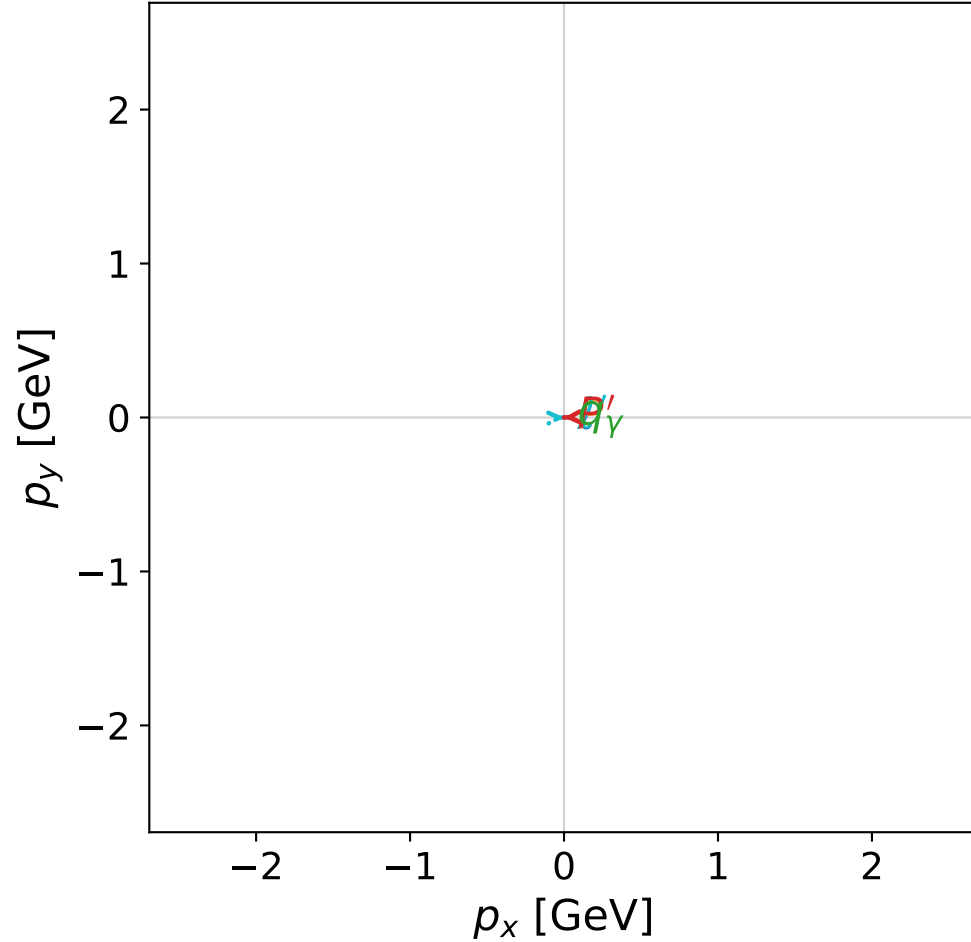


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



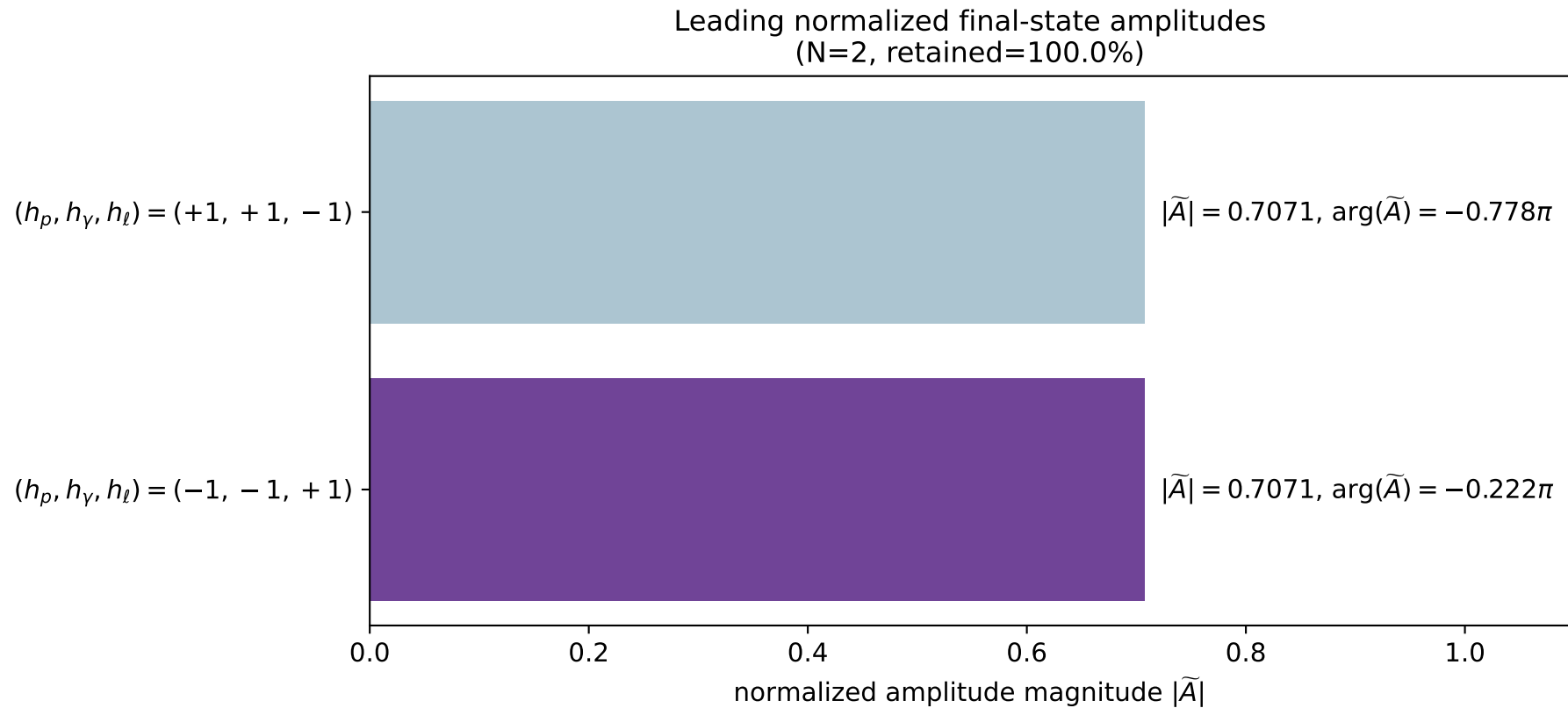
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_94
 $d_{\text{GHZ}}=3.33046\text{e-}08$
 region: polarization_cluster_02_local_minimum_94
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0094
 lepton mixing angle: $\alpha_e=0.79992921$ rad
 incoming lepton: $0.696757|+\rangle + 0.717307|-\rangle$
 proton mixing angle: $\alpha_p=2.3416514$ rad
 incoming proton: $-0.696749|+\rangle + 0.717315|-\rangle$

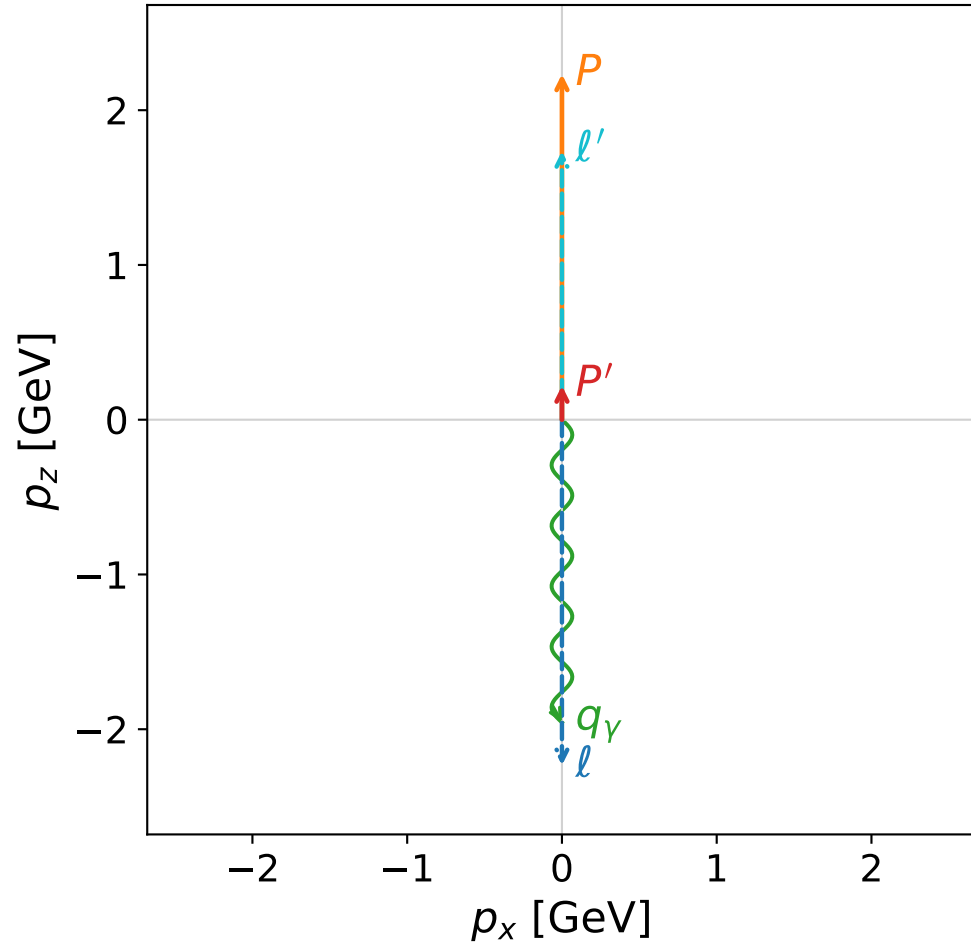
$s=21.8786$, $\sqrt{s}=4.67746$, $|\vec{P}|=2.24468$, $|\vec{P}'|=0.239827$
 $\theta_{P'}=0.000137432$, $\theta_Y=3.14159$, $\phi_{P'}=3.17695$, $\phi_Y=5.58436$
 $E_Y=1.97455$, $\theta_{\ell'}=1.9\text{e-}05$, $\phi_{\ell'}=0.0353563$
 $Q^2=15.5756$, $x_B=0.765531$, $t=-1.87435$, $W^2=5.65037$, $y=0.968921$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2447$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2447$
 P $E= 2.4328$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2447$
 ℓ' $E= 1.7347$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7347$
 P' $E= 0.9682$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2398$
 q_Y $E= 1.9746$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9746$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

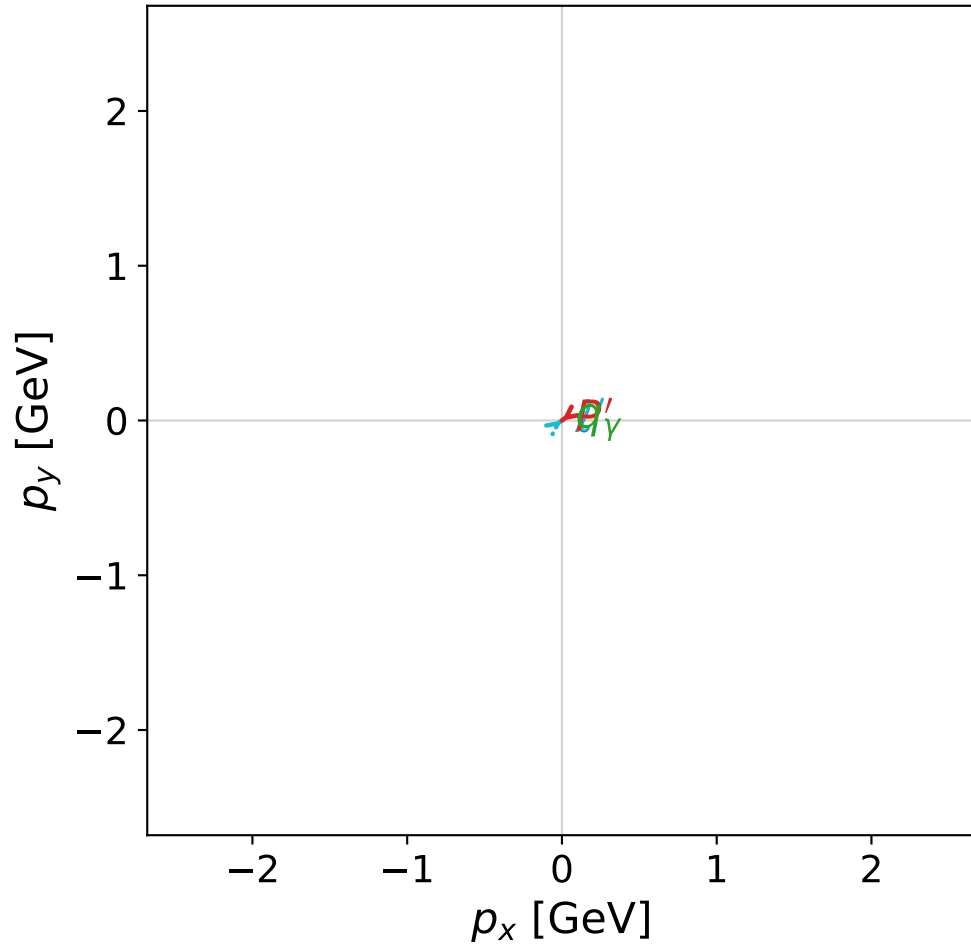


dghz_mixing_angles_polarization_02_local_minimum_96
 $d_{\text{GHZ}}=3.42638\text{e-}08$
 region: polarization_cluster_02_local_minimum_96
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0096
 lepton mixing angle: $\alpha_e=0.66918829$ rad
 incoming lepton: $0.784325|+\rangle +0.62035|-\rangle$
 proton mixing angle: $\alpha_p=2.4723833$ rad
 incoming proton: $-0.784312|+\rangle +0.620366|-\rangle$

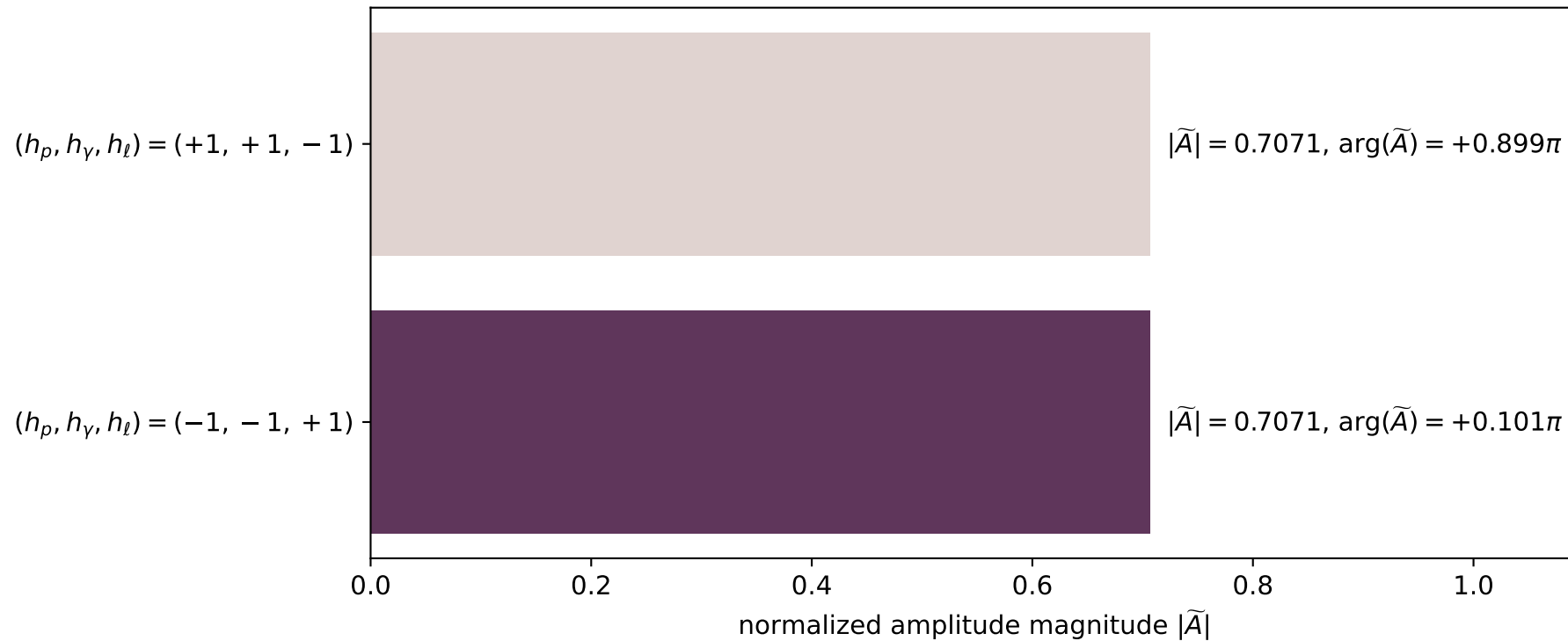
$s=21.6714$, $\sqrt{s}=4.65526$, $|\vec{P}|=2.23313$, $|\vec{P}'|=0.219926$
 $\theta_{P'}=3.14159\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.77898$, $\phi_\gamma=0.317472$
 $E_\gamma=1.95587$, $\theta_{l'}=3.98006\text{e-}06$, $\phi_{l'}=0.637385$
 $Q^2=15.5064$, $x_B=0.770105$, $t=-1.9252$, $W^2=5.50887$, $y=0.96844$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2331$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2331$
 P $E= 2.4221$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2331$
 l' $E= 1.7359$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7359$
 P' $E= 0.9634$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2199$
 q_γ $E= 1.9559$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9559$

x--y plane

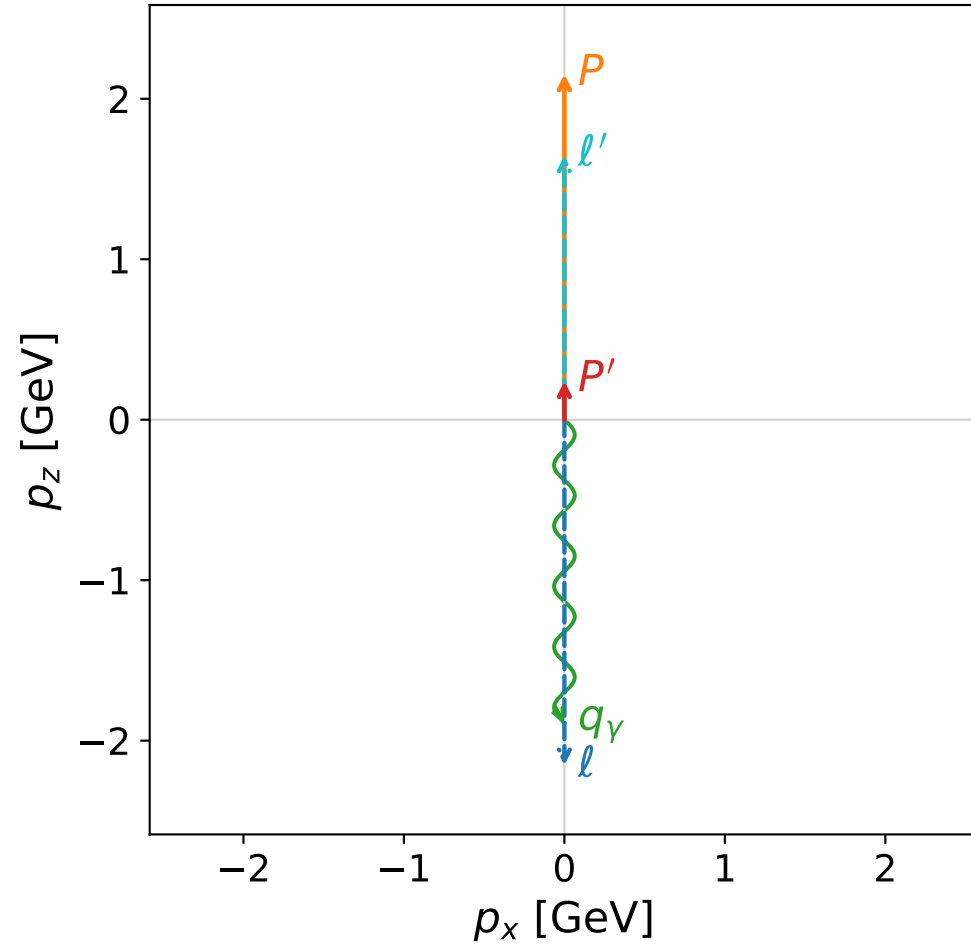


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

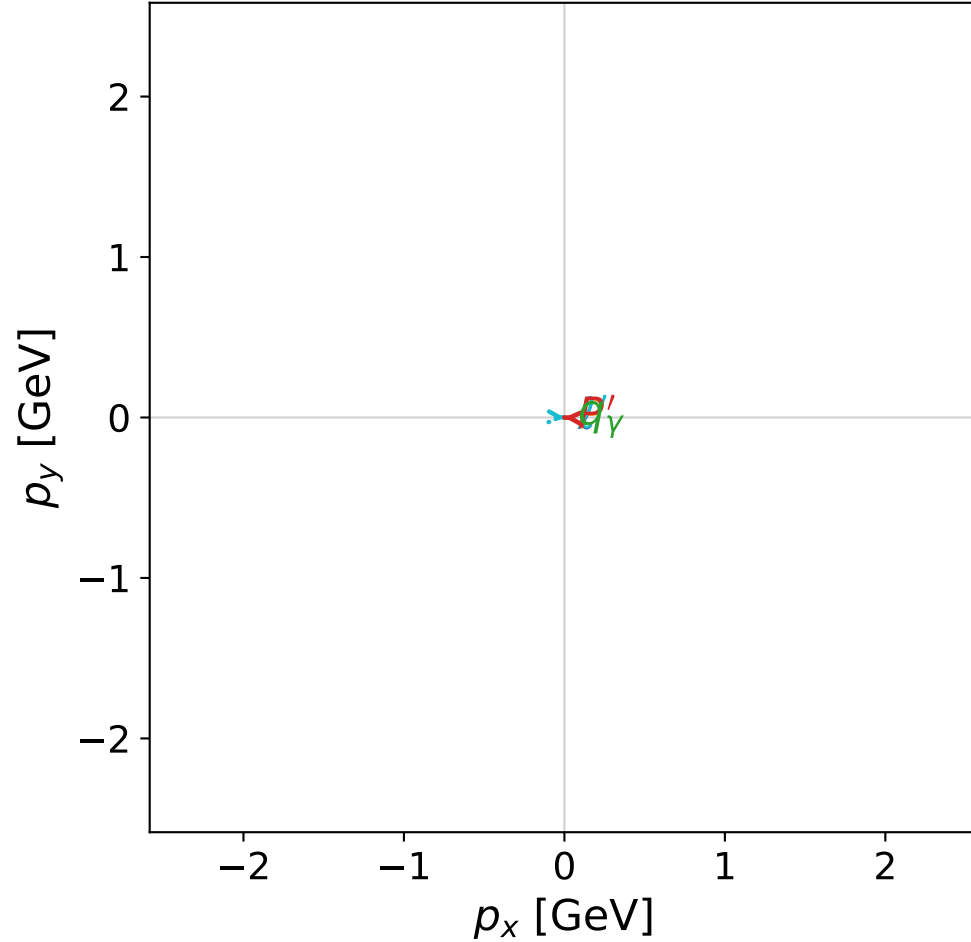


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



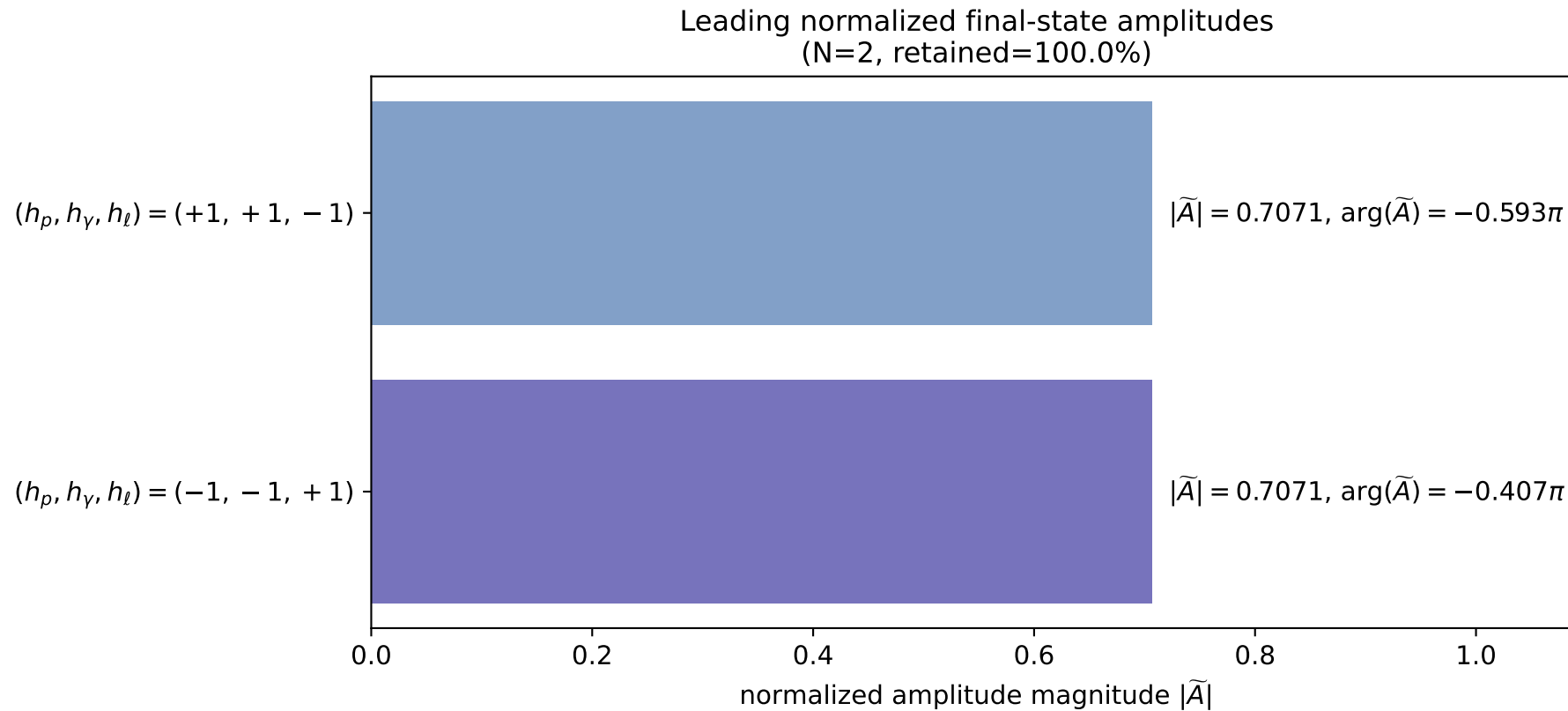
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_99
 $d_{\text{GHZ}}=3.58348\text{e-}08$
 region: polarization_cluster_02_local_minimum_99
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0099
 lepton mixing angle: $\alpha_e=0.78633215$ rad
 incoming lepton: $0.706446|+\rangle + 0.707767|-\rangle$
 proton mixing angle: $\alpha_p=2.3552466$ rad
 incoming proton: $-0.706436|+\rangle + 0.707777|-\rangle$

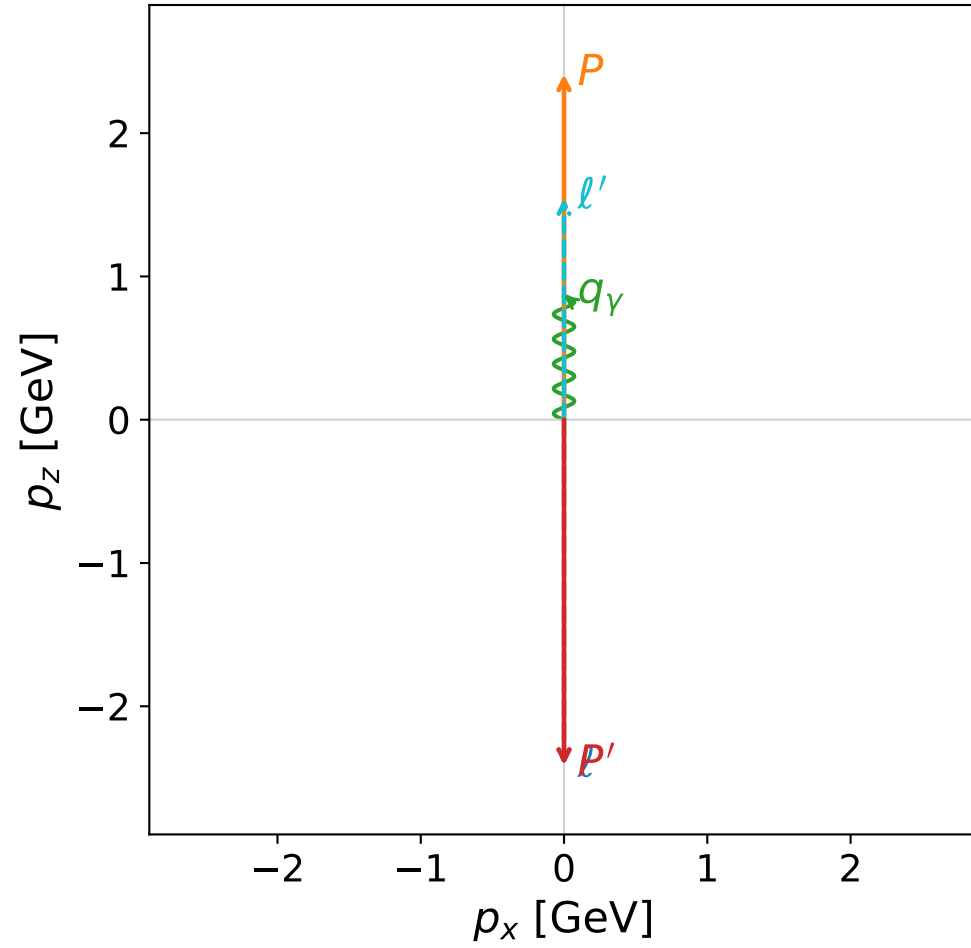
$s=20.2744$, $\sqrt{s}=4.5027$, $|\vec{P}|=2.15365$, $|\vec{P}'|=0.24118$
 $\theta_{P'}=6.93359\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.0985$, $\phi_\gamma=5.00556$
 $E_\gamma=1.88769$, $\theta_{l'}=1.01558\text{e-}06$, $\phi_{l'}=6.2401$
 $Q^2=14.184$, $x_B=0.756439$, $t=-1.75164$, $W^2=5.44687$, $y=0.966822$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1537$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1537$
 P $E= 2.3491$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1537$
 l' $E= 1.6465$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.6465$
 P' $E= 0.9685$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.2412$
 q_γ $E= 1.8877$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8877$

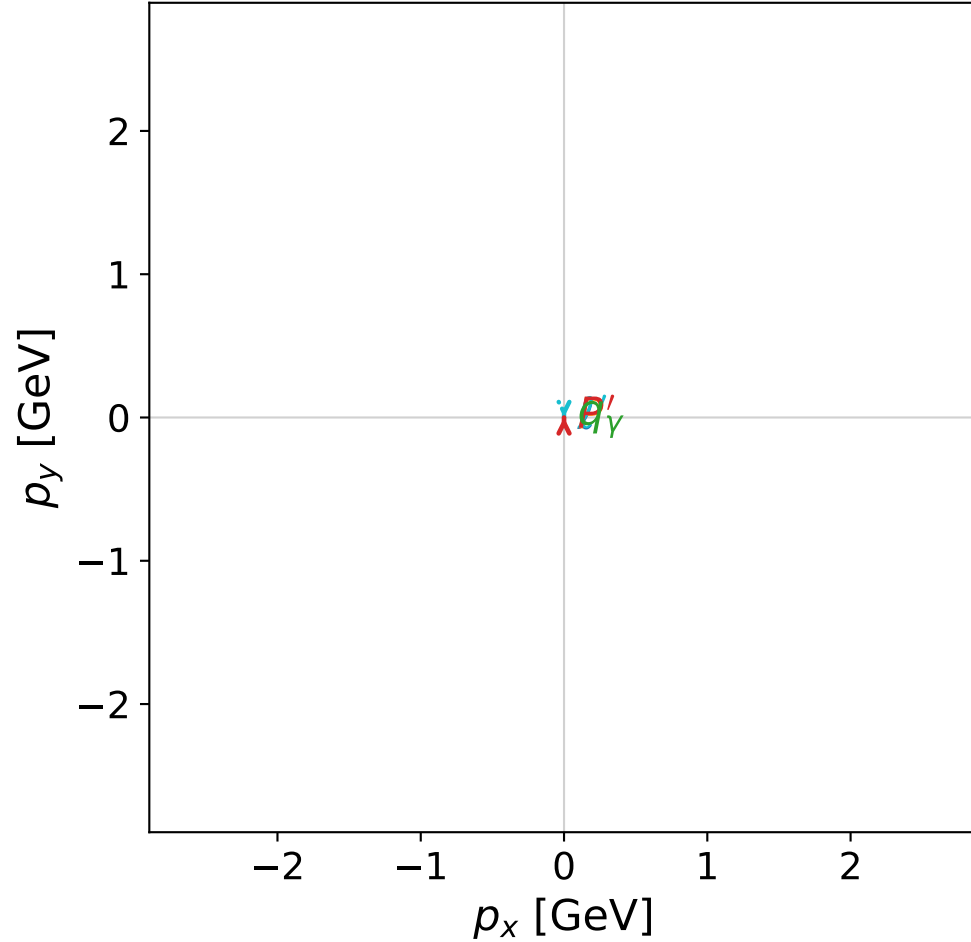


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

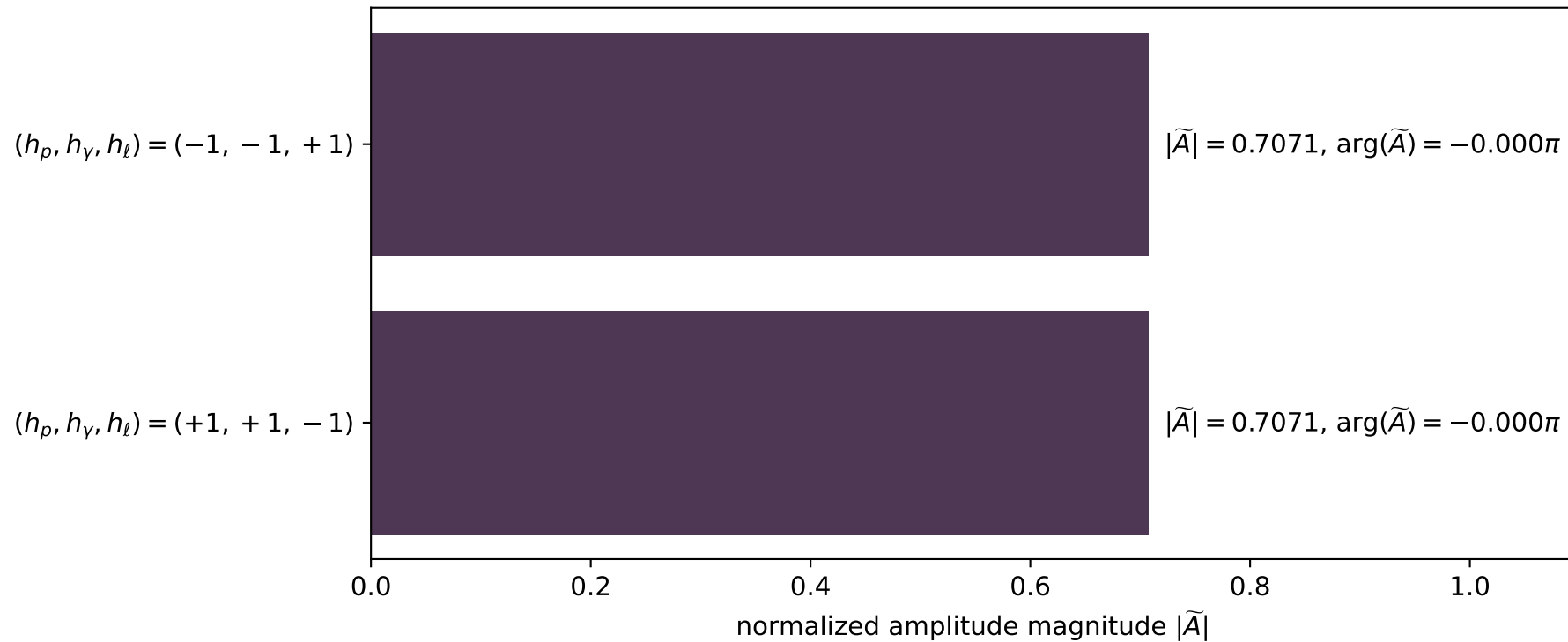


dghz_mixing_angles_polarization_02_local_minimum_100
 $d_{\{\text{rm GHZ}\}}=3.62681\text{e-}08$
 region: polarization_cluster_02_local_minimum_100
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0100
 lepton mixing angle: $\alpha_e=0.65339387$ rad
 incoming lepton: $0.794025|+\rangle + 0.607885|-\rangle$
 proton mixing angle: $\alpha_p=2.224186$ rad
 incoming proton: $-0.607881|+\rangle + 0.794028|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.52284$, $\phi_\gamma=1.07341$
 $E_\gamma=0.865987$, $\theta_{l'}=0$, $\phi_{l'}=4.66443$
 $Q^2=14.9162$, $x_B=0.632684$, $t=-23.2713$, $W^2=9.53971$, $y=0.977442$

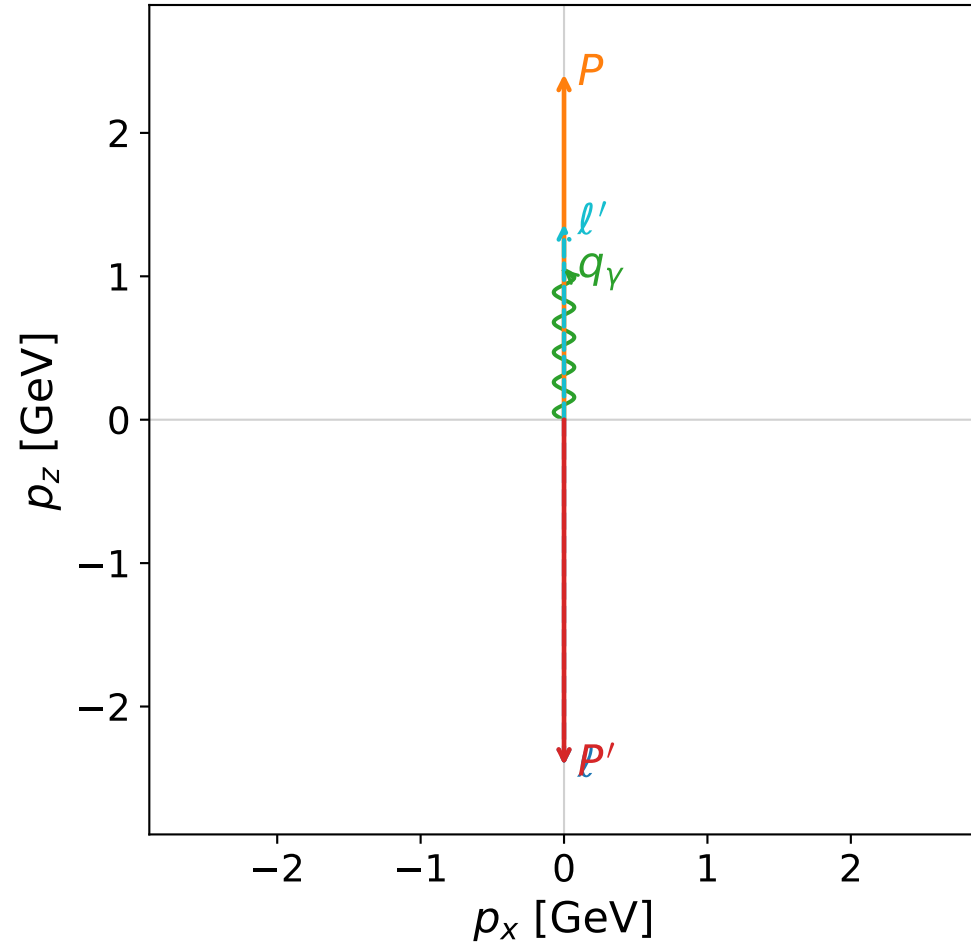
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.5460$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5460$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.8660$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8660$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

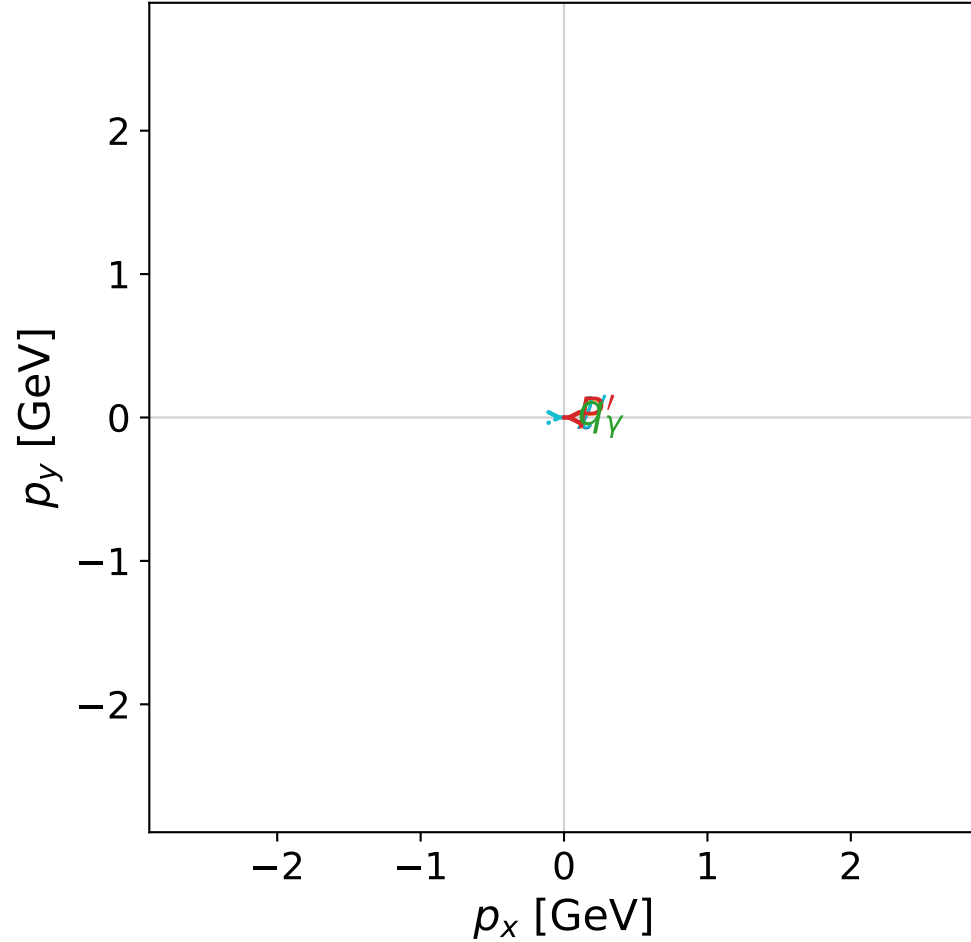


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

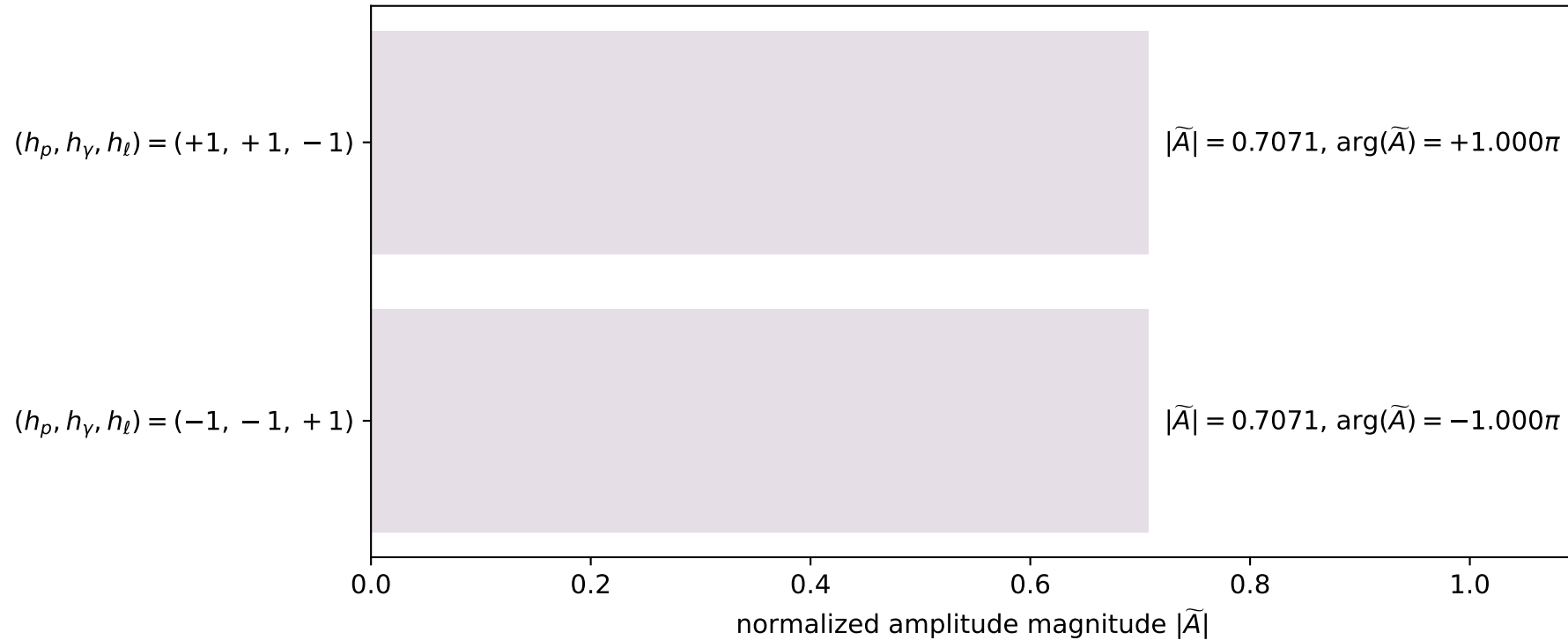


dghz_mixing_angles_polarization_02_local_minimum_110
 $d_{\text{GHZ}}=4.05095\text{e-}08$
 region: polarization_cluster_02_local_minimum_110
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0110
 lepton mixing angle: $\alpha_e=0.68998436$ rad
 incoming lepton: $0.771256|+\rangle + 0.636525|-\rangle$
 proton mixing angle: $\alpha_p=2.2608057$ rad
 incoming proton: $-0.636544|+\rangle + 0.77124|-\rangle$

$s=24.9589$, $\sqrt{s}=4.99589$, $|\vec{P}|=2.40989$, $|\vec{P}'|=2.40989$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.7054$, $\phi_\gamma=1.95022$
 $E_\gamma=1.04501$, $\theta_{l'}=0$, $\phi_{l'}=5.84699$
 $Q^2=13.1568$, $x_B=0.557533$, $t=-23.2303$, $W^2=11.3213$, $y=0.980035$

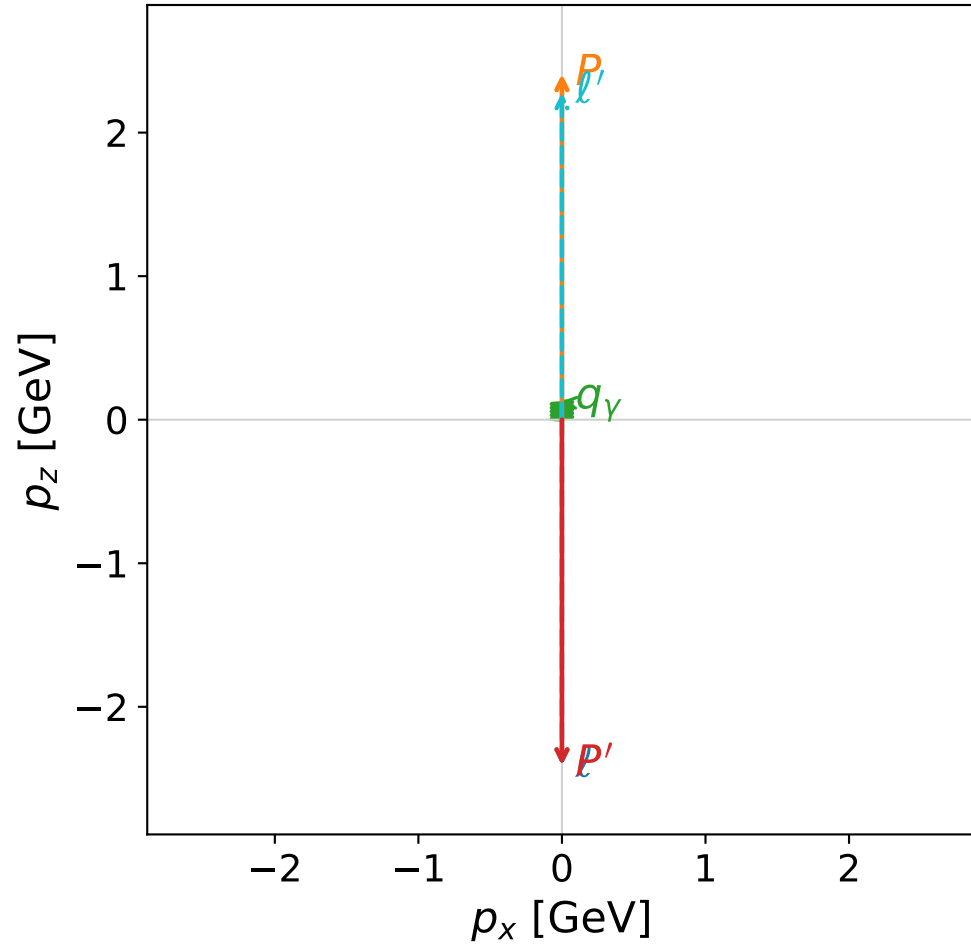
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4099$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4099$
 P $E= 2.5860$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4099$
 l' $E= 1.3649$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.3649$
 P' $E= 2.5860$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4099$
 q_γ $E= 1.0450$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.0450$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

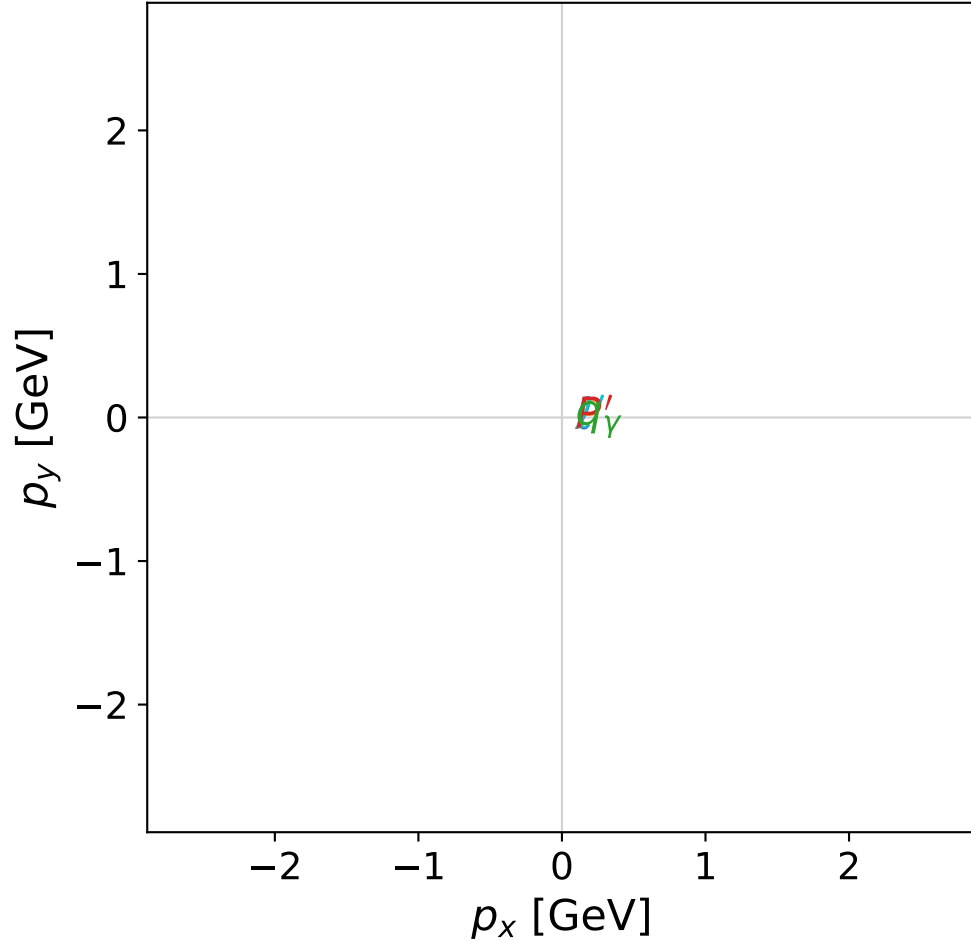


dghz_mixing_angles_polarization_02_local_minimum_116
 $d_{\text{GHZ}}=4.3521\text{e-}08$
 region: polarization_cluster_02_local_minimum_116
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0116
 lepton mixing angle: $\alpha_e=0.28922643$ rad
 incoming lepton: $0.958465|+\rangle + 0.285211|-\rangle$
 proton mixing angle: $\alpha_p=1.8600136$ rad
 incoming proton: $-0.285202|+\rangle + 0.958467|-\rangle$

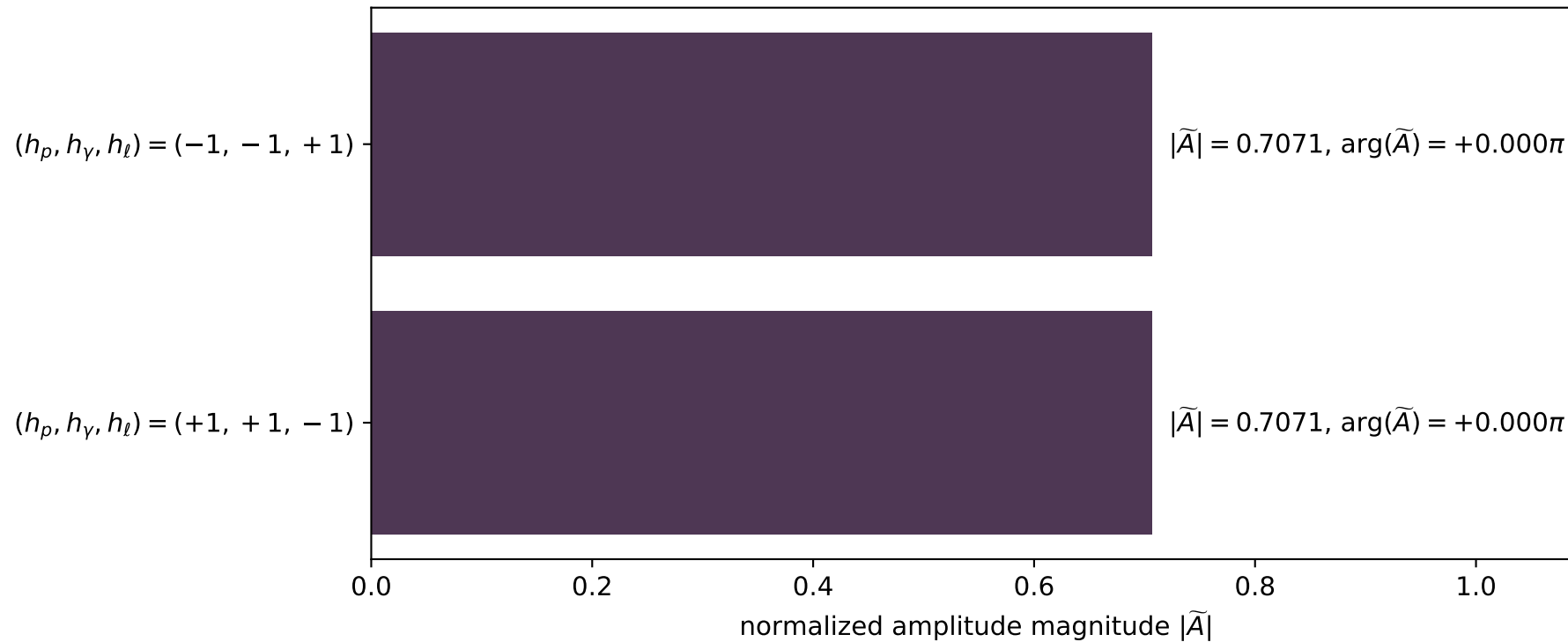
$s=24.913$, $\sqrt{s}=4.99129$, $|\vec{P}|=2.40751$, $|\vec{P}'|=2.40751$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.69158$, $\phi_\gamma=0.269509$
 $E_\gamma=0.128615$, $\theta_{l'}=0$, $\phi_{l'}=2.54999$
 $Q^2=21.9458$, $x_B=0.94473$, $t=-23.1844$, $W^2=2.16375$, $y=0.96657$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4075$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4075$
 P $E= 2.5838$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4075$
 ℓ' $E= 2.2789$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.2789$
 P' $E= 2.5838$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4075$
 q_γ $E= 0.1286$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1286$

x--y plane

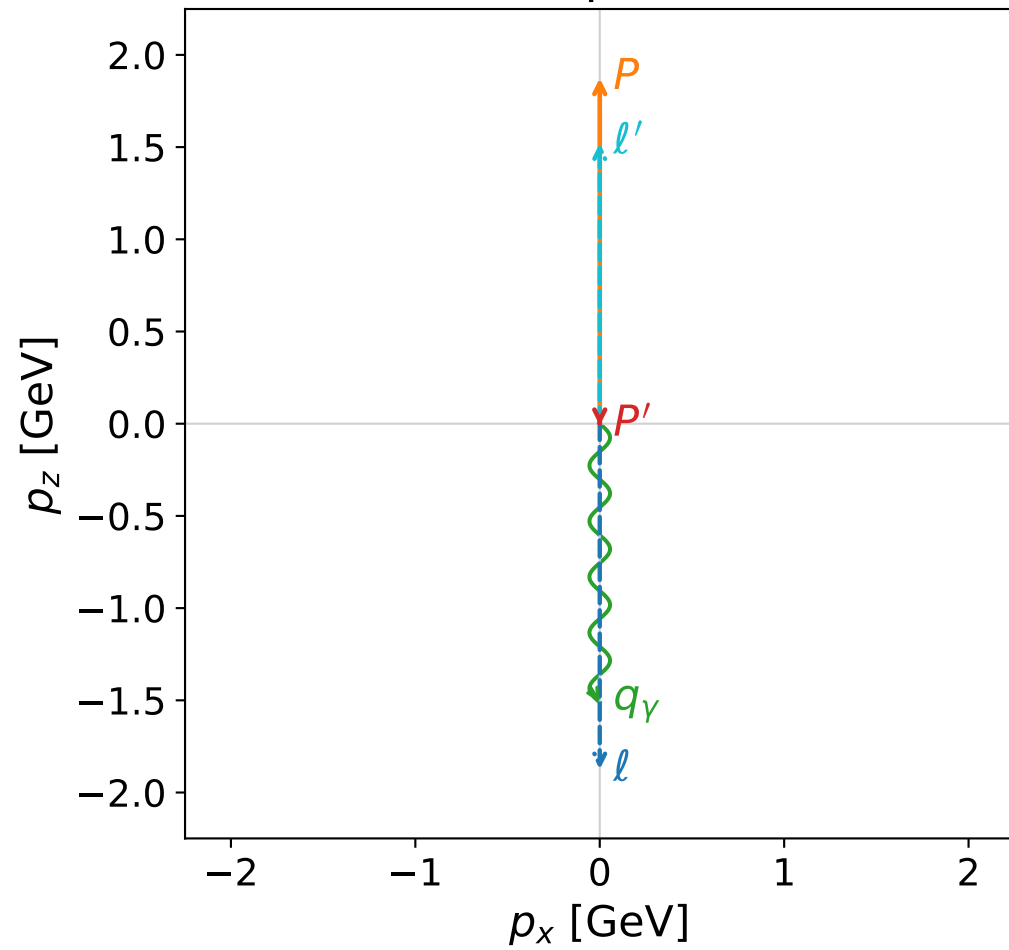


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

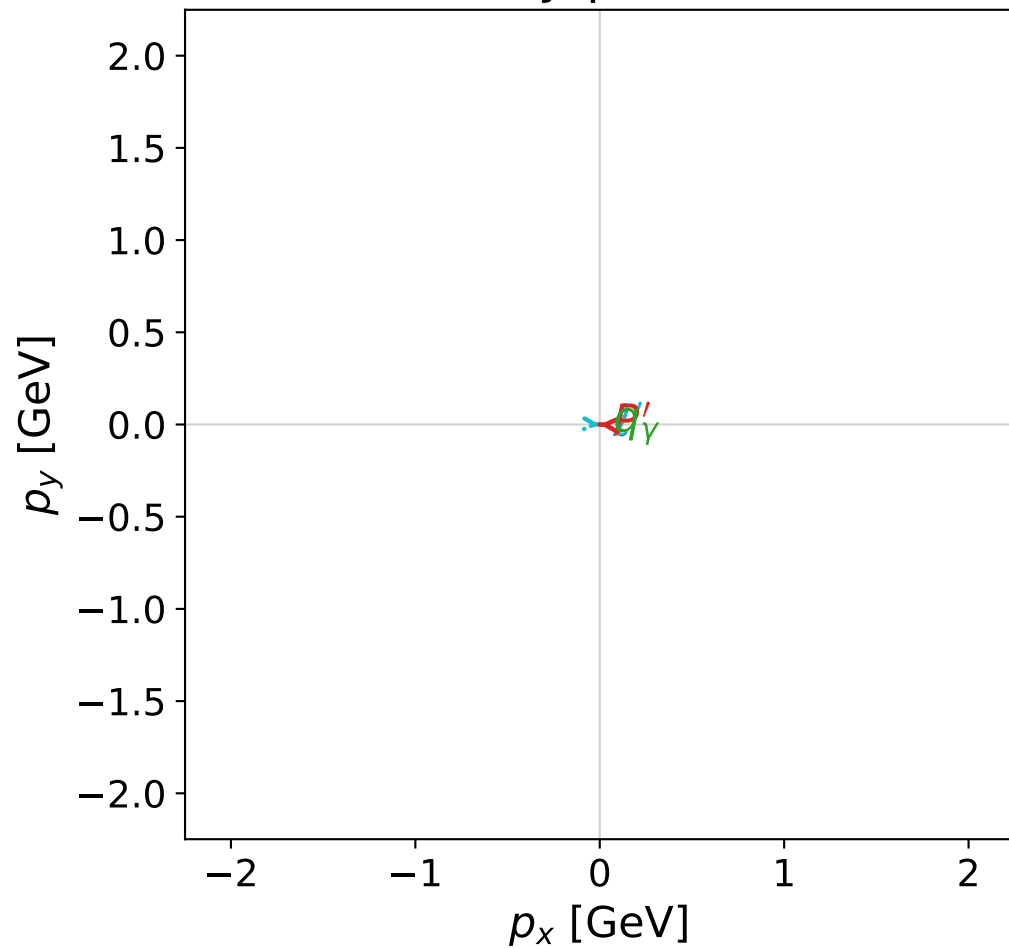


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



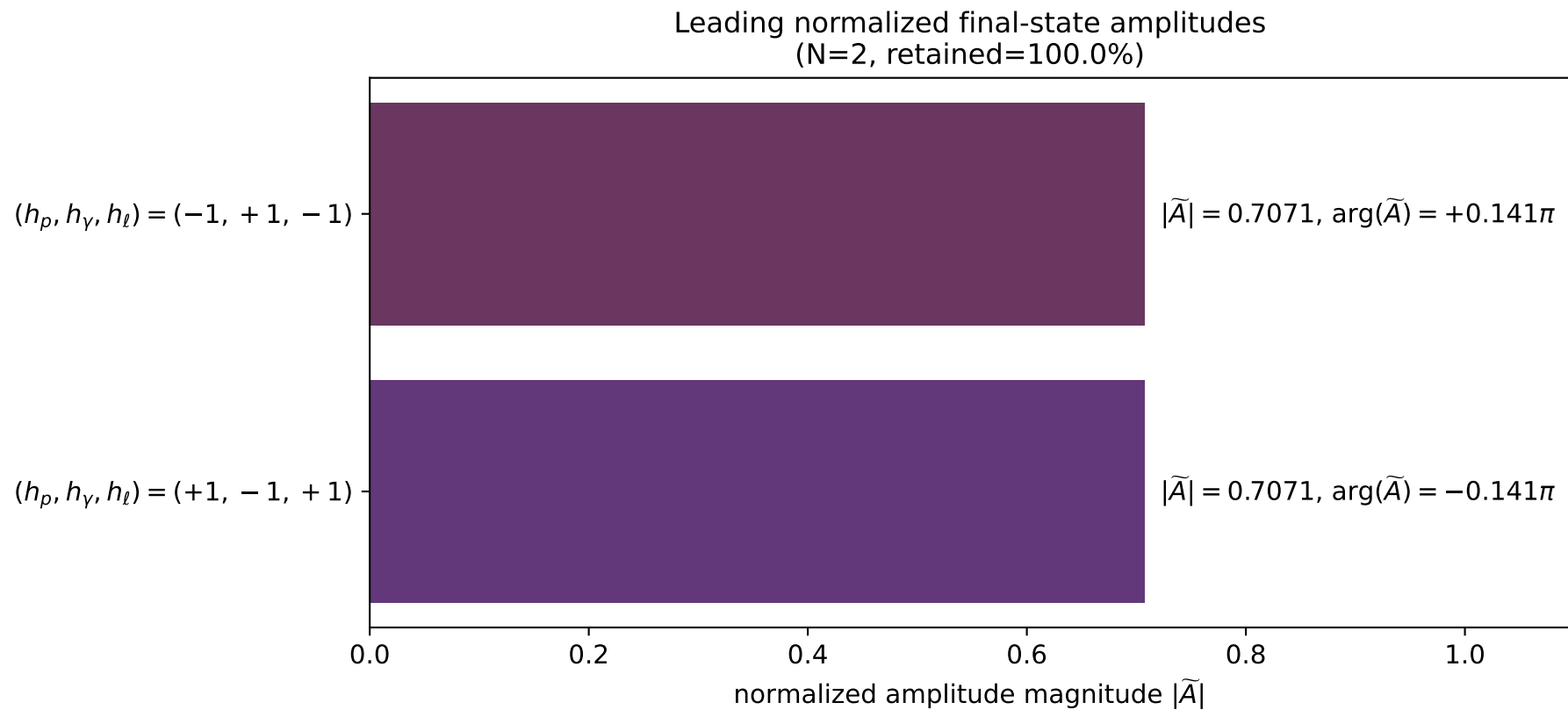
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_120
 $d_{\text{GHZ}}=4.59538\text{e-}08$
 region: polarization_cluster_02_local_minimum_120
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0120
 lepton mixing angle: $\alpha_e=0.74588136$ rad
 incoming lepton: $0.73449|+\rangle +0.678619|-\rangle$
 proton mixing angle: $\alpha_p=2.3956994$ rad
 incoming proton: $-0.734482|+\rangle +0.678628|-\rangle$

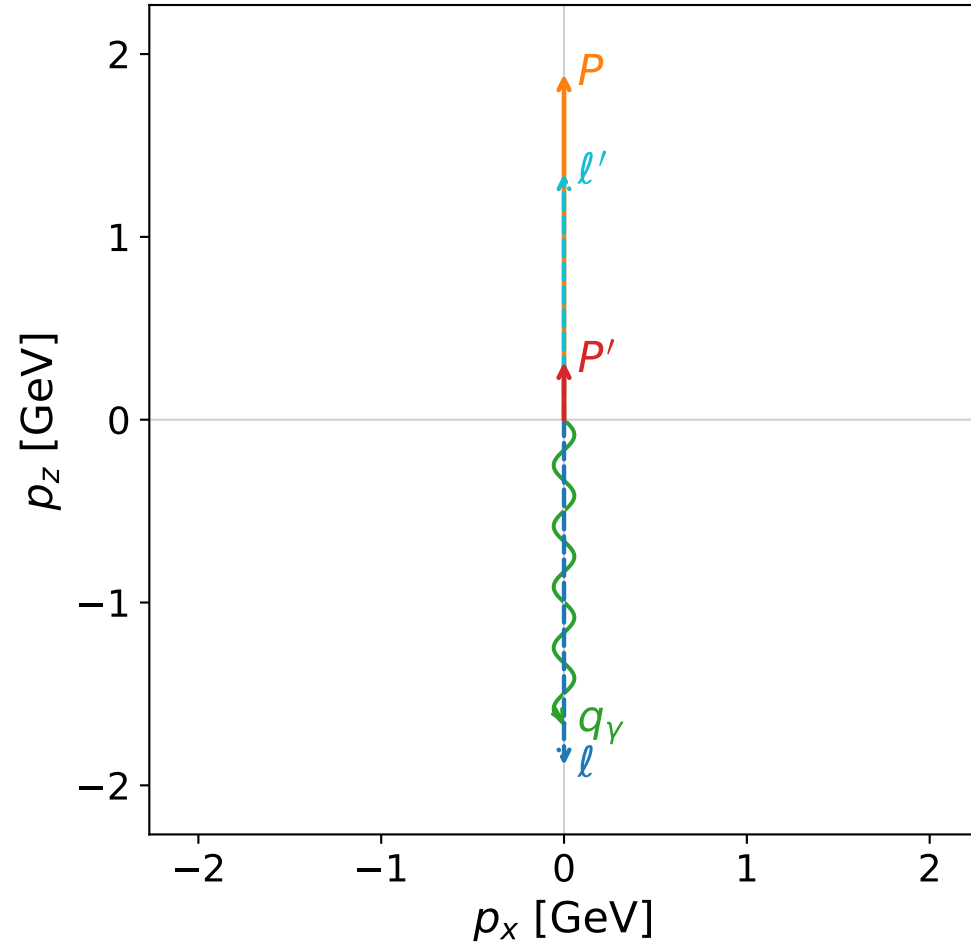
$s=15.7593$, $\sqrt{s}=3.9698$, $|\vec{P}|=1.87408$, $|\vec{P}'|=0.00992053$
 $\theta_{p'}=3.14111$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.09611$, $\phi_\gamma=2.65443$
 $E_\gamma=1.51091$, $\theta_{\ell'}=3.16449\text{e-}06$, $\phi_{\ell'}=6.23771$
 $Q^2=11.4006$, $x_B=0.802563$, $t=-2.20928$, $W^2=3.68449$, $y=0.954693$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8741$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8741$
 P $E= 2.0957$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8741$
 ℓ' $E= 1.5208$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.5208$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0099$
 q_γ $E= 1.5109$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.5109$

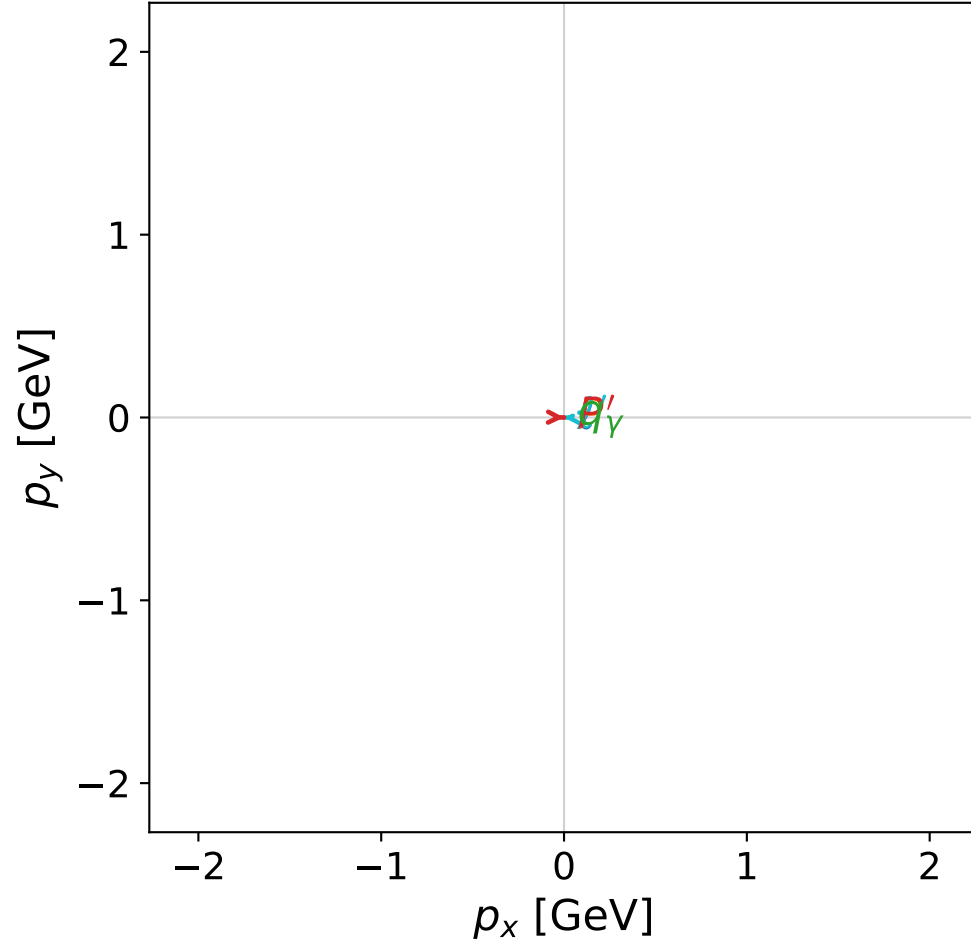


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



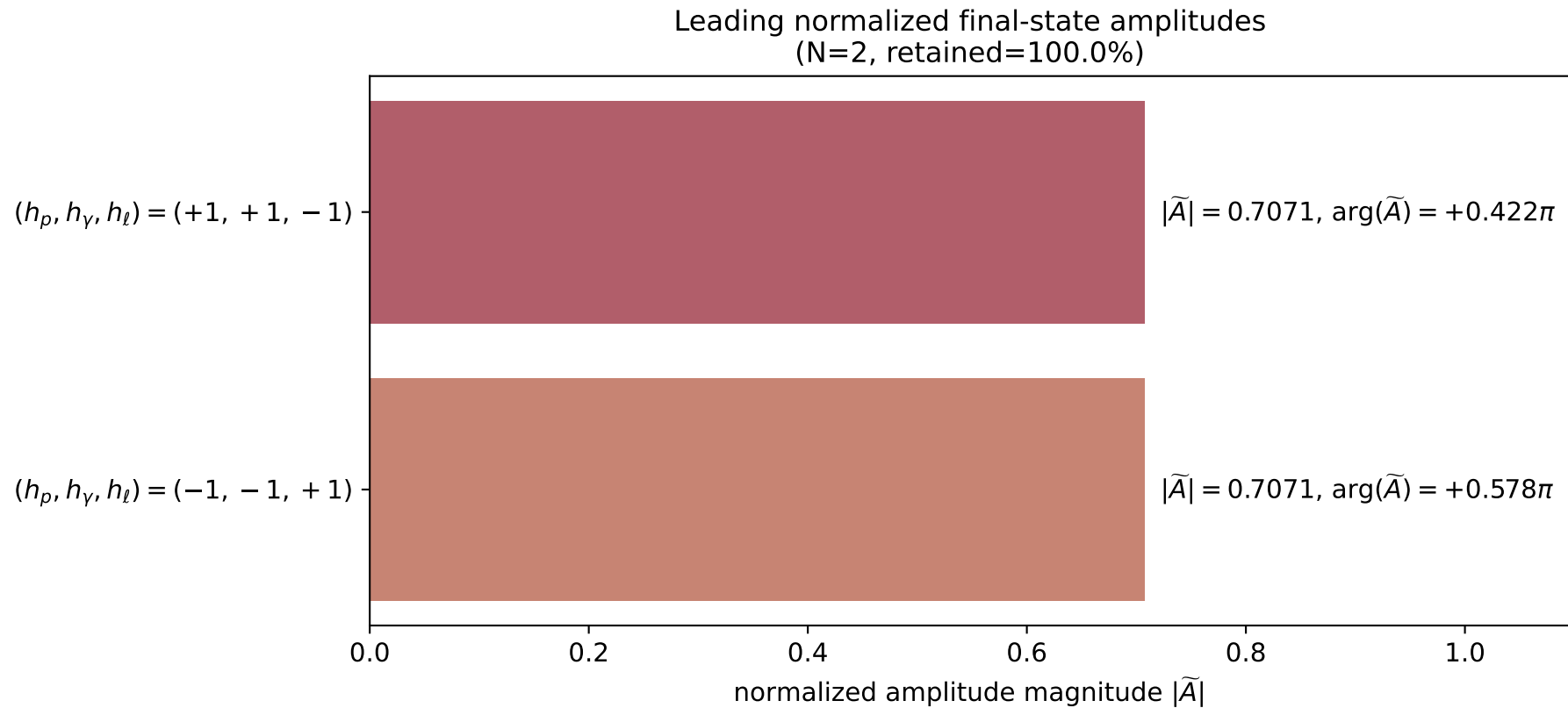
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_127
 $d_{\{\text{rm GHZ}\}}=5.10304\text{e-}08$
 region: polarization_cluster_02_local_minimum_127
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0127
 lepton mixing angle: $\alpha_e=0.78216053$ rad
 incoming lepton: $0.709392|+\rangle + 0.704814|-\rangle$
 proton mixing angle: $\alpha_p=2.3594272$ rad
 incoming proton: $-0.709389|+\rangle + 0.704817|-\rangle$

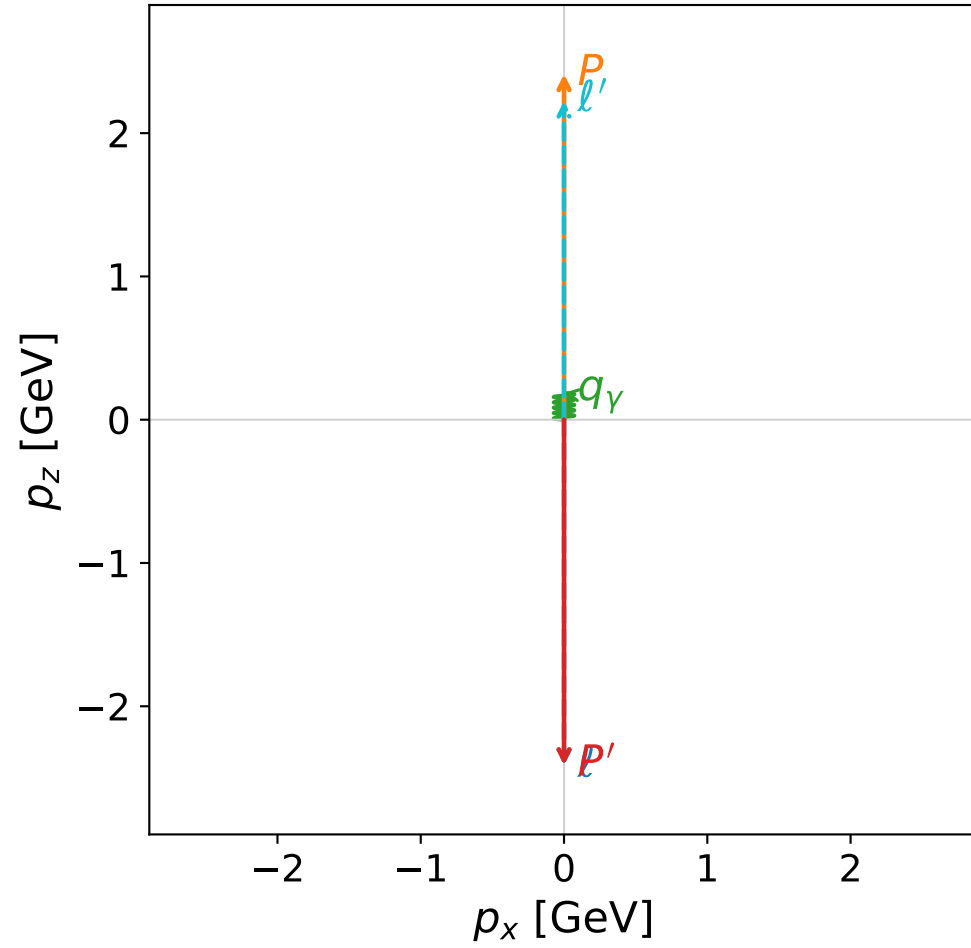
$s=16.0046$, $\sqrt{s}=4.00057$, $|\vec{P}|=1.89032$, $|\vec{P}'|=0.316348$
 $\theta_{p'}=9.29374\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.26205$, $\phi_\gamma=1.81607$
 $E_\gamma=1.6635$, $\theta_{l'}=2.18242\text{e-}05$, $\phi_{l'}=3.12046$
 $Q^2=10.1862$, $x_B=0.700944$, $t=-1.22222$, $W^2=5.22577$, $y=0.960822$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8903$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8903$
 P $E= 2.1102$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8903$
 l' $E= 1.3472$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.3472$
 P' $E= 0.9899$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3163$
 q_γ $E= 1.6635$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.6635$

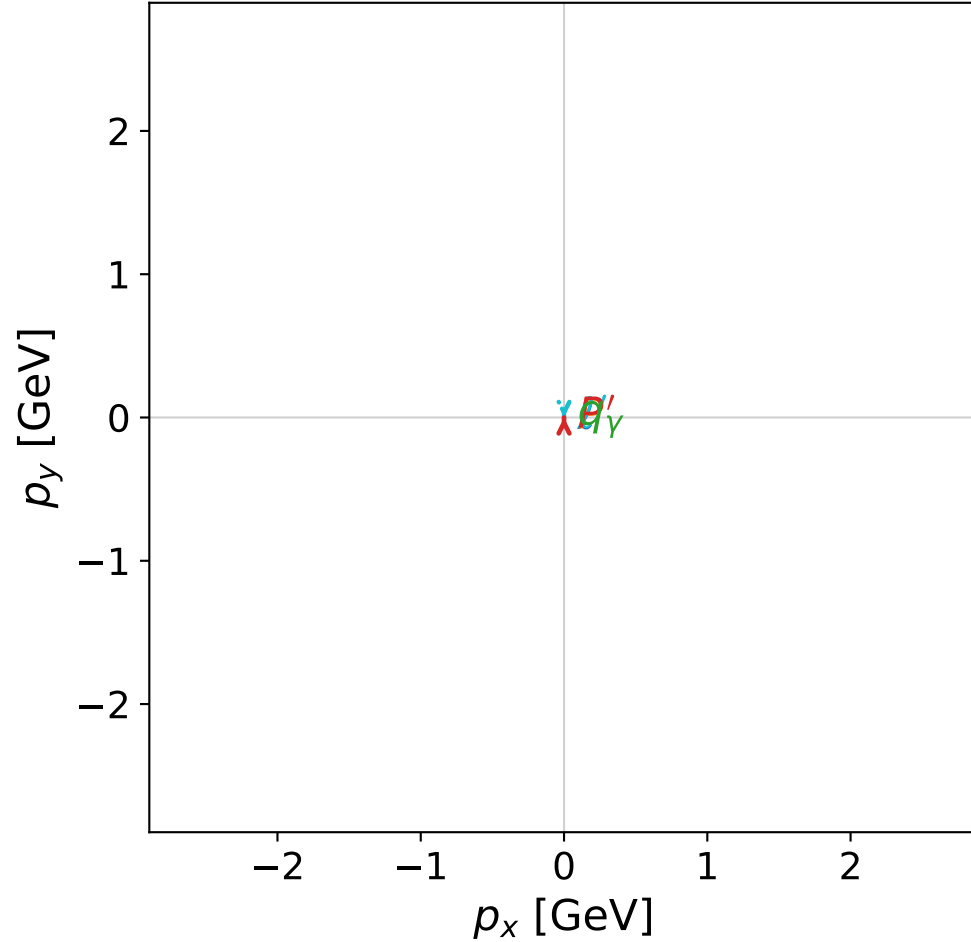


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

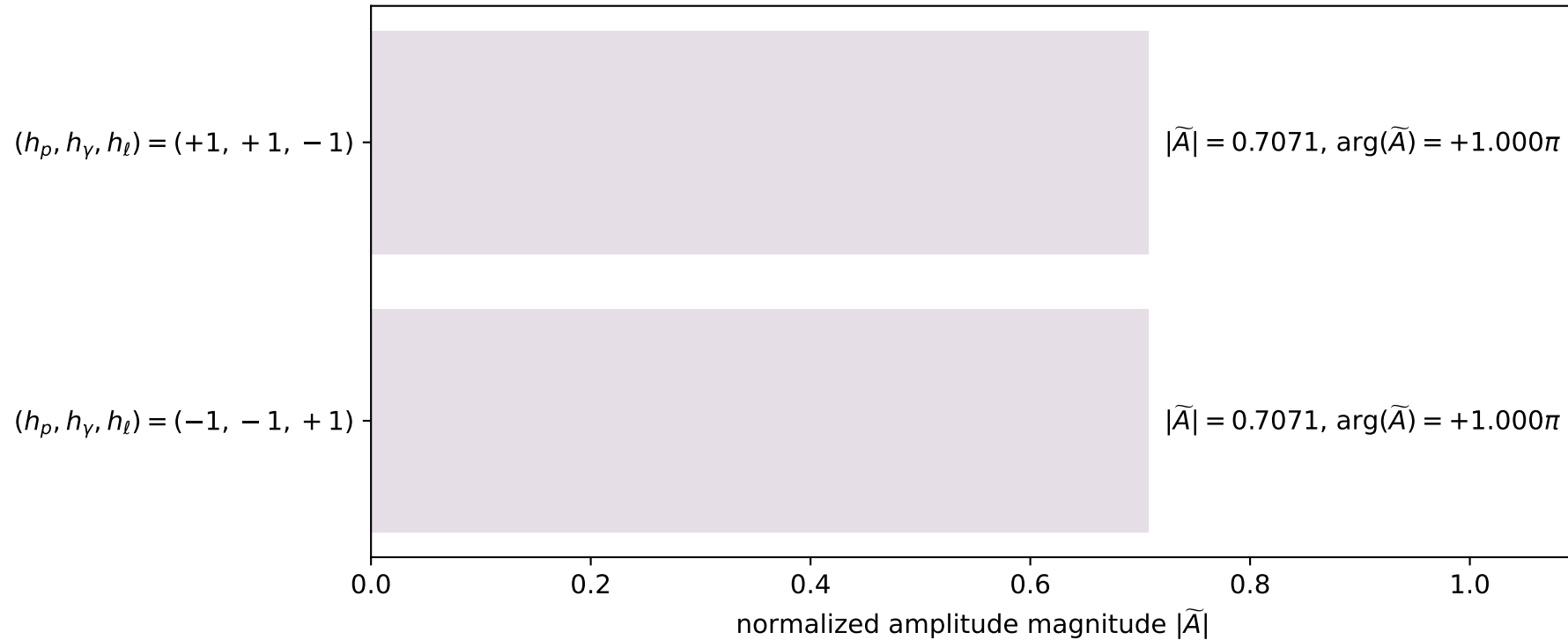


dghz_mixing_angles_polarization_02_local_minimum_136
 $d_{\{\rm GHZ\}}=6.21675\text{e-}08$
 region: polarization_cluster_02_local_minimum_136
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0136
 lepton mixing angle: $\alpha_e=0.20100578$ rad
 incoming lepton: $0.979866|+\rangle + 0.199655|-\rangle$
 proton mixing angle: $\alpha_p=1.7718087$ rad
 incoming proton: $-0.199661|+\rangle + 0.979865|-\rangle$

$s=24.9997$, $\sqrt{s}=4.99997$, $|\vec{P}|=2.412$, $|\vec{P}'|=2.412$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.40635$, $\phi_\gamma=3.24397$
 $E_\gamma=0.186808$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.54794$
 $Q^2=21.4686$, $x_B=0.919952$, $t=-23.2709$, $W^2=2.74791$, $y=0.967532$

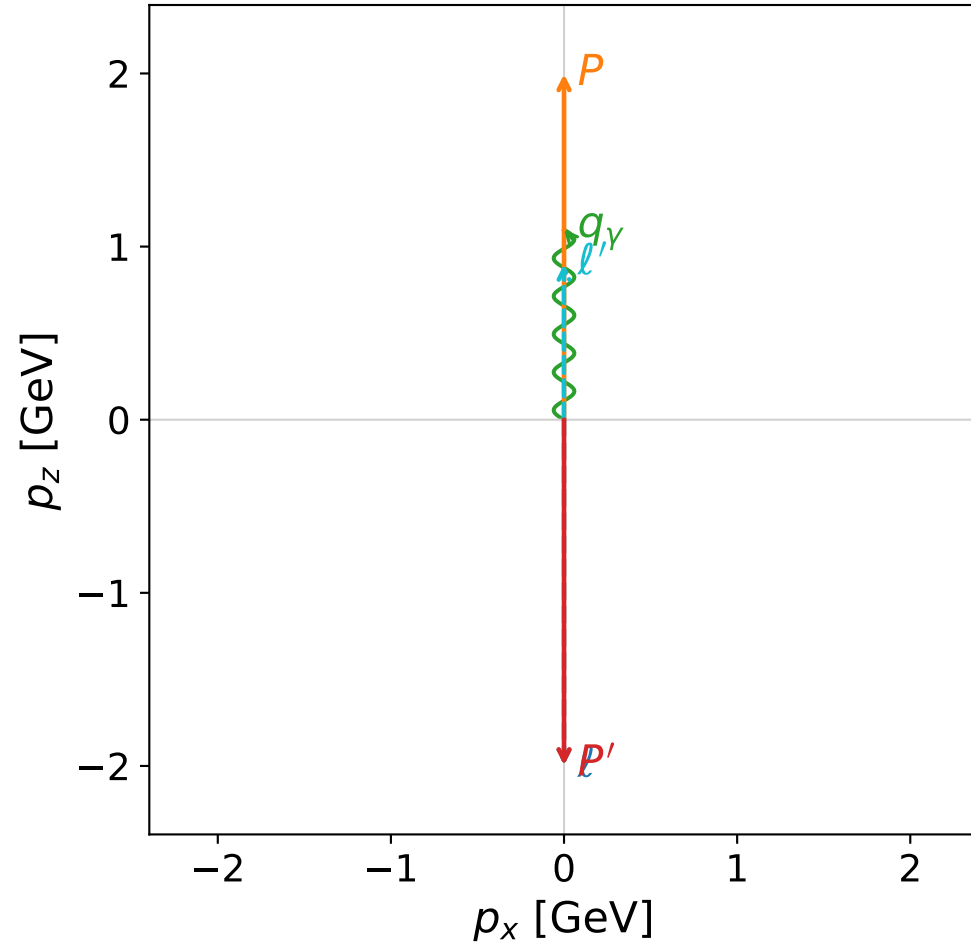
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2252$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2252$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1868$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1868$

Leading normalized final-state amplitudes (N=2, retained=100.0%)

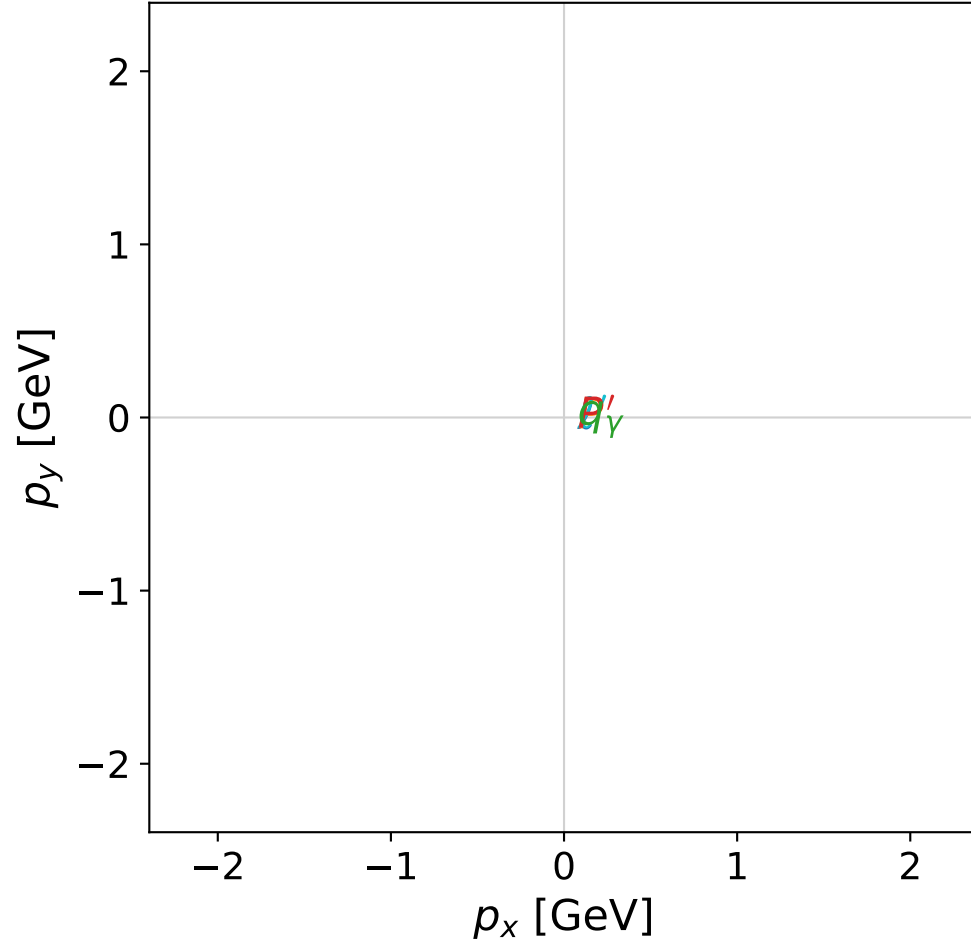


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



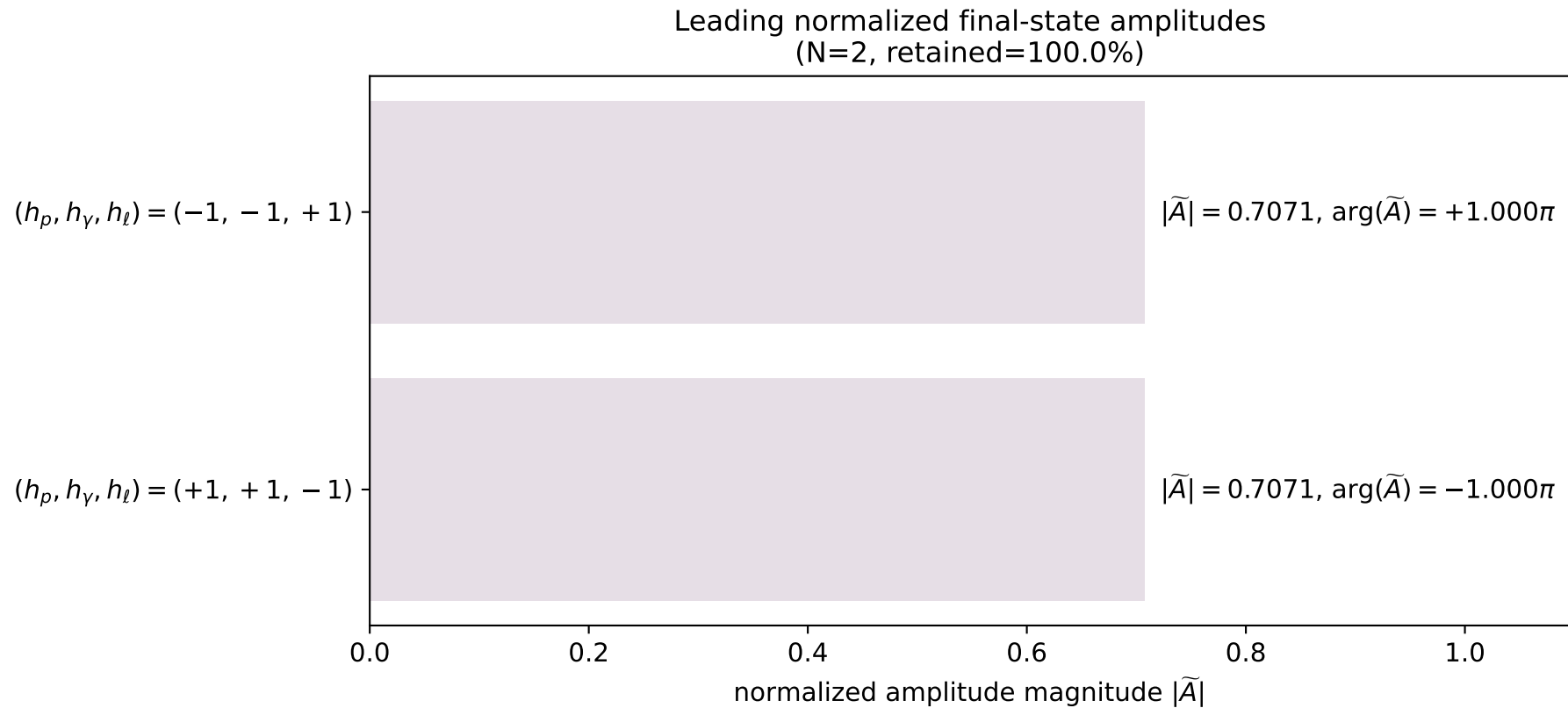
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_142
 $d_{\{\rm GHZ\}}=7.52972\text{e-}08$
 region: polarization_cluster_02_local_minimum_142
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0142
 lepton mixing angle: $\alpha_e=0.65625654$ rad
 incoming lepton: $0.792282|+\rangle + 0.610155|-\rangle$
 proton mixing angle: $\alpha_p=2.2270306$ rad
 incoming proton: $-0.610138|+\rangle + 0.792295|-\rangle$

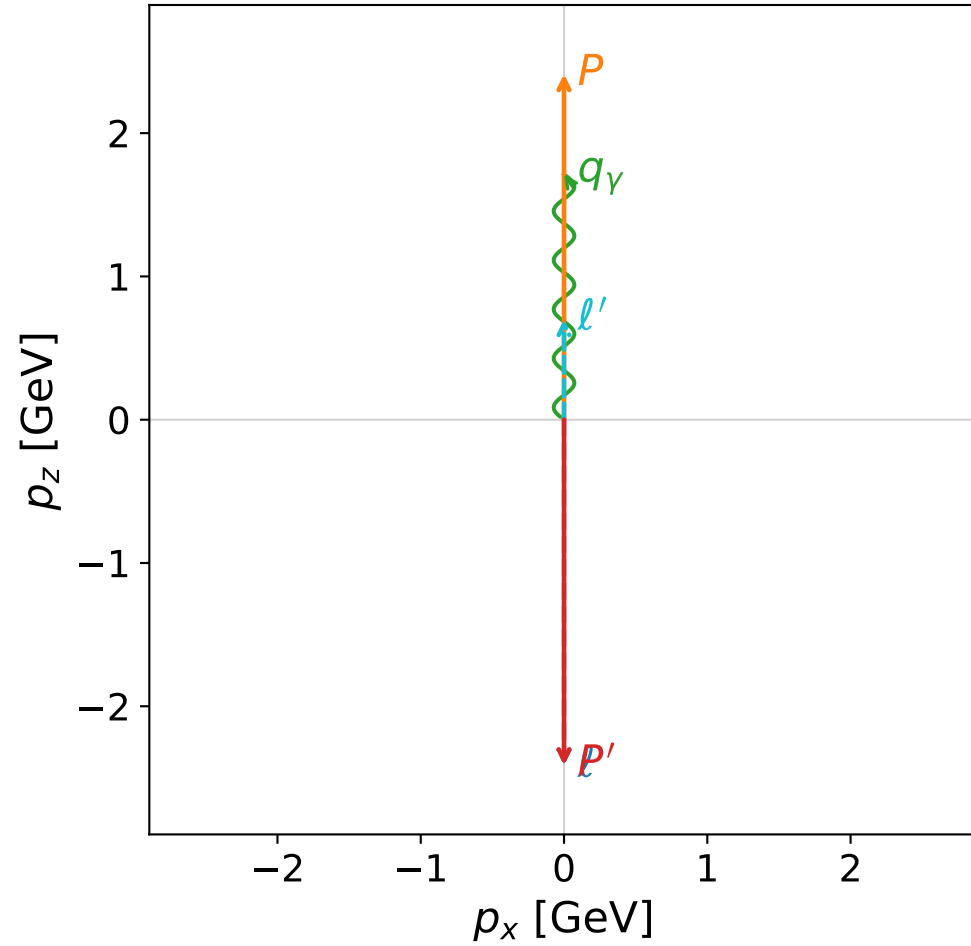
$s=17.6556$, $\sqrt{s}=4.20186$, $|\vec{P}|=1.99623$, $|\vec{P}'|=1.99623$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=6.06891$, $\phi_Y=4.38119$
 $E_Y=1.09748$, $\theta_{l'}=0$, $\phi_{l'}=2.92732$
 $Q^2=7.17649$, $x_B=0.437607$, $t=-15.9398$, $W^2=10.1027$, $y=0.977564$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.9962$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9962$
 P $E= 2.2056$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9962$
 l' $E= 0.8988$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.8988$
 P' $E= 2.2056$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9962$
 q_Y $E= 1.0975$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0975$

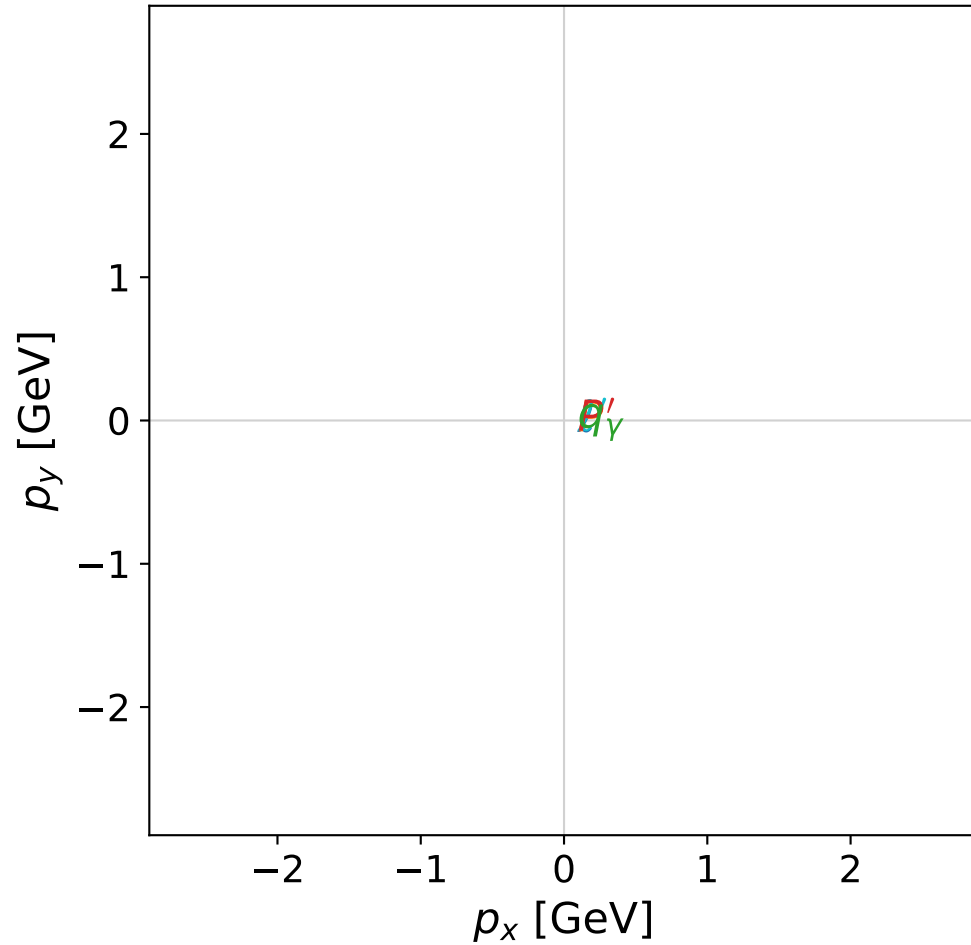


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

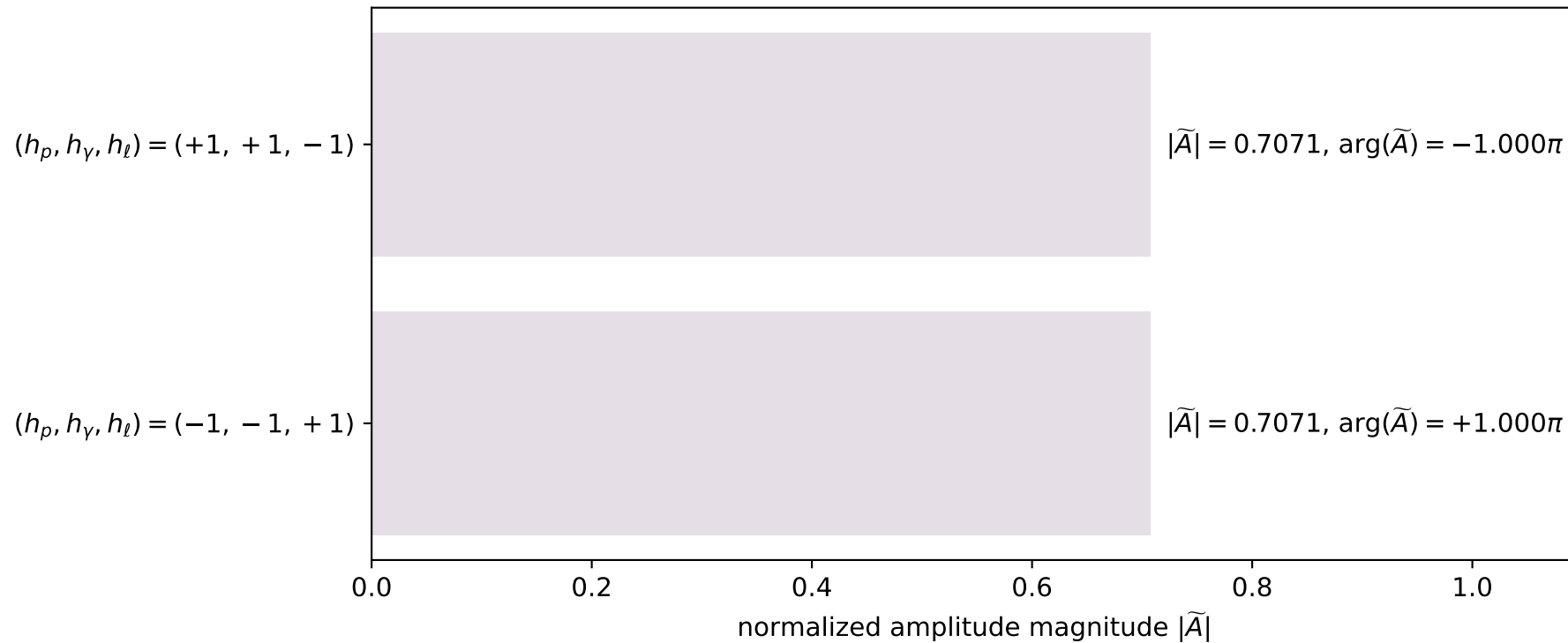


dghz_mixing_angles_polarization_02_local_minimum_144
 $d_{\text{GHZ}}=7.83262\text{e-}08$
 region: polarization_cluster_02_local_minimum_144
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0144
 lepton mixing angle: $\alpha_e=0.70738913$ rad
 incoming lepton: $0.760061|+\rangle +0.649852|-\rangle$
 proton mixing angle: $\alpha_p=2.2781902$ rad
 incoming proton: $-0.649855|+\rangle +0.760058|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.15414$, $\phi_\gamma=4.45412$
 $E_\gamma=1.71245$, $\theta_{l'}=0$, $\phi_{l'}=5.29573$
 $Q^2=6.74947$, $x_B=0.282713$, $t=-23.2713$, $W^2=18.0043$, $y=0.989793$

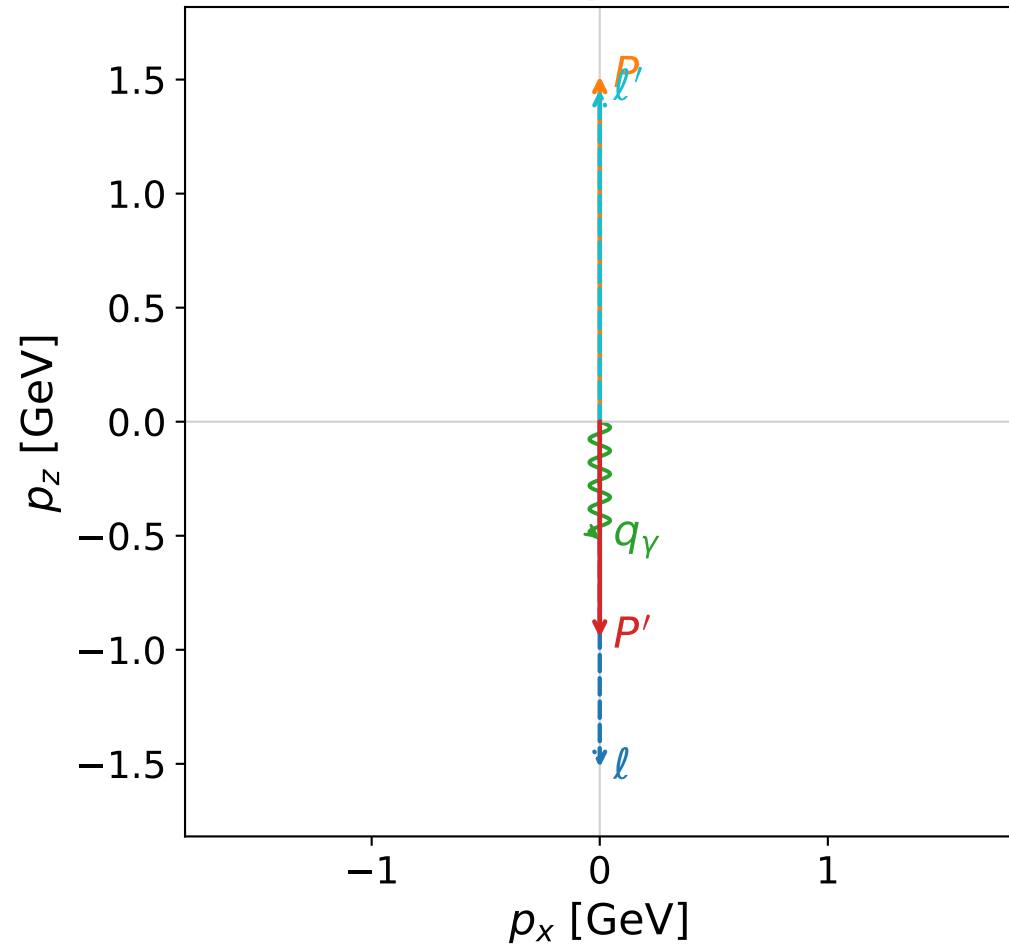
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 0.6996$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6996$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 1.7124$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.7124$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

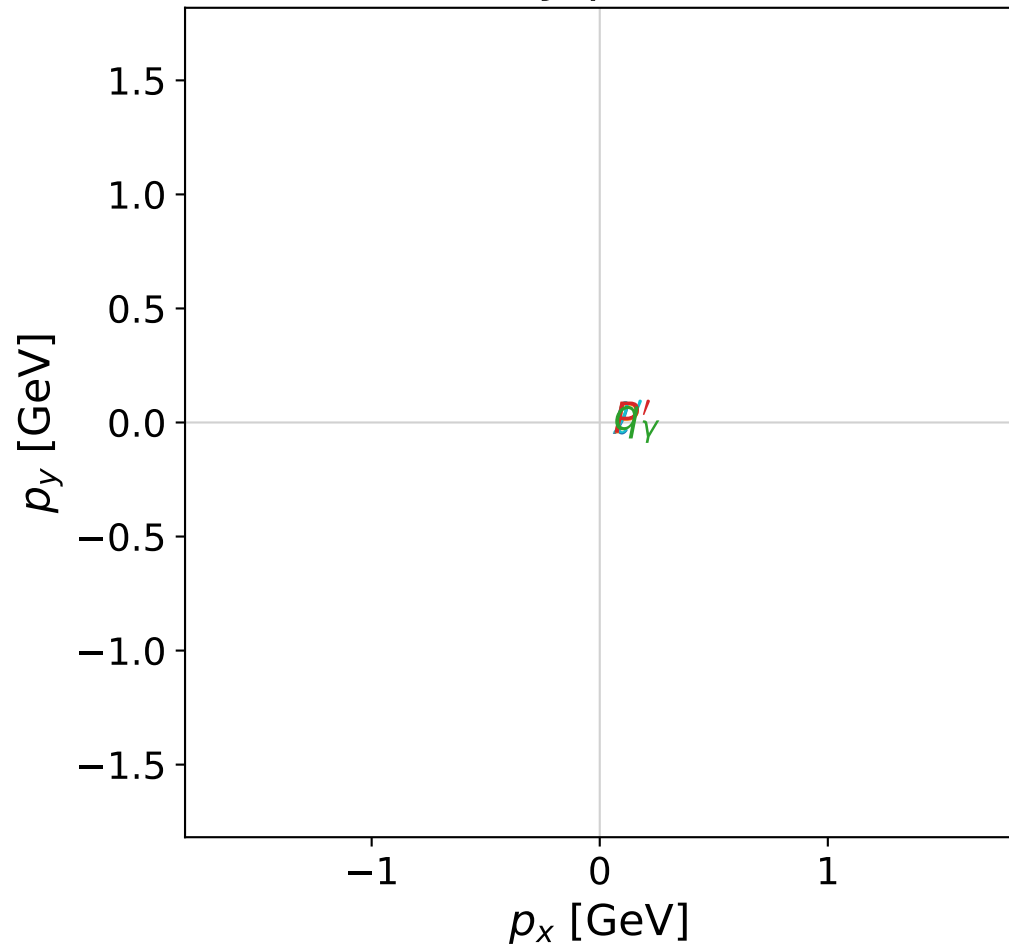


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



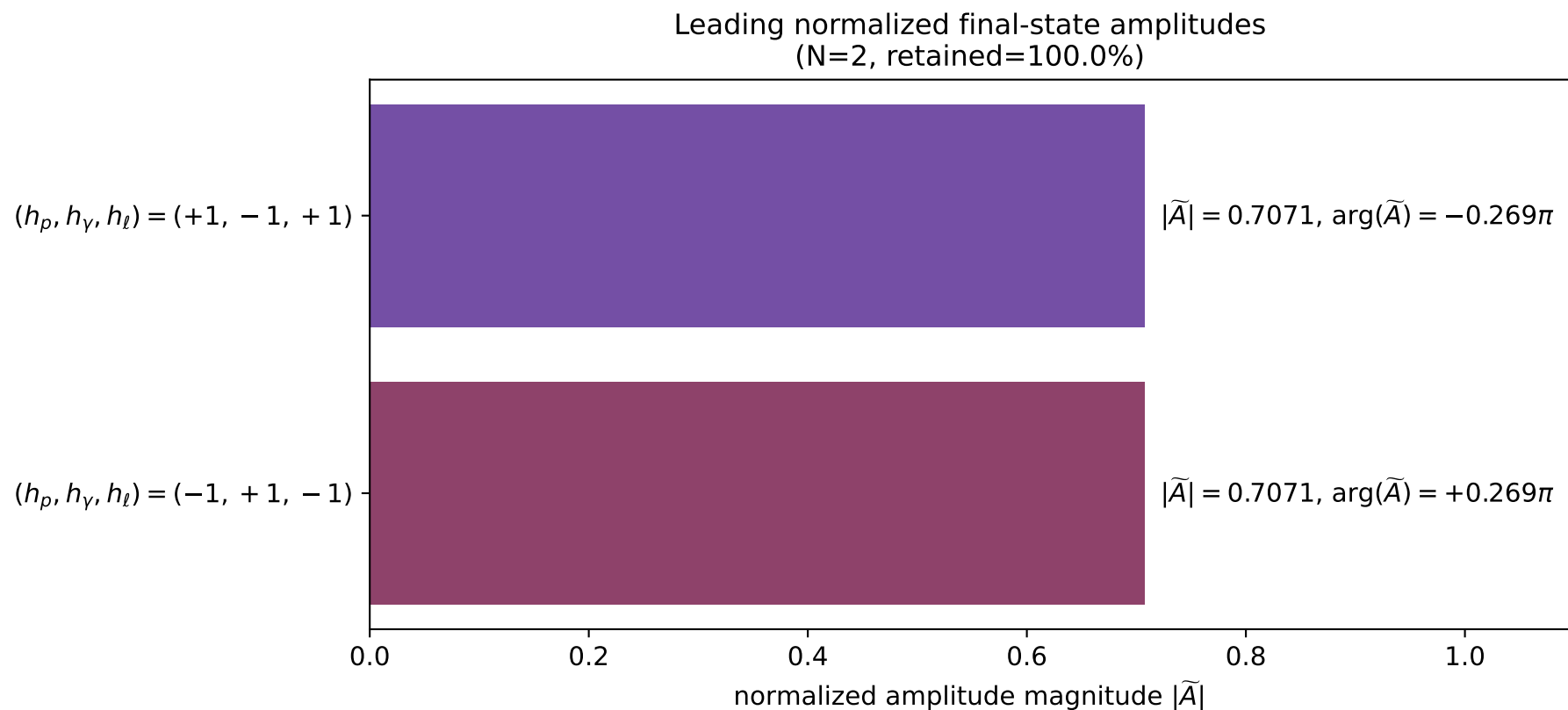
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_147
 $d_{\{\rm GHZ\}}=8.24447\text{e-}08$
 region: polarization_cluster_02_local_minimum_147
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0147
 lepton mixing angle: $\alpha_e=0.40049047$ rad
 incoming lepton: $0.92087|+\rangle + 0.38987|-\rangle$
 proton mixing angle: $\alpha_p=2.7411031$ rad
 incoming proton: $-0.92087|+\rangle + 0.389869|-\rangle$

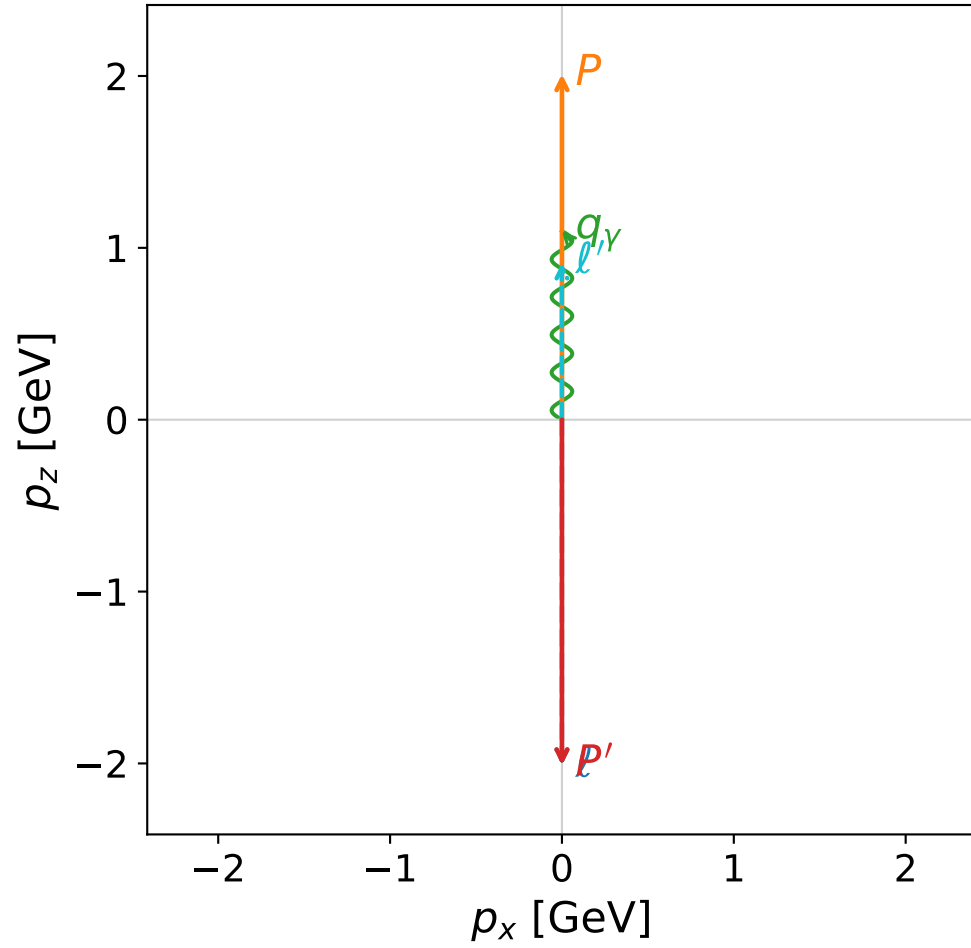
$s=10.8717$, $\sqrt{s}=3.29723$, $|\vec{P}|=1.51519$, $|\vec{P}'|=0.945903$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.07214$, $\phi_\gamma=5.43934$
 $E_\gamma=0.509598$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.96521$
 $Q^2=8.82147$, $x_B=0.957283$, $t=-5.85459$, $W^2=1.27349$, $y=0.922259$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5152$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5152$
 P $E= 1.7820$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5152$
 ℓ' $E= 1.4555$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.4555$
 P' $E= 1.3321$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.9459$
 q_γ $E= 0.5096$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.5096$

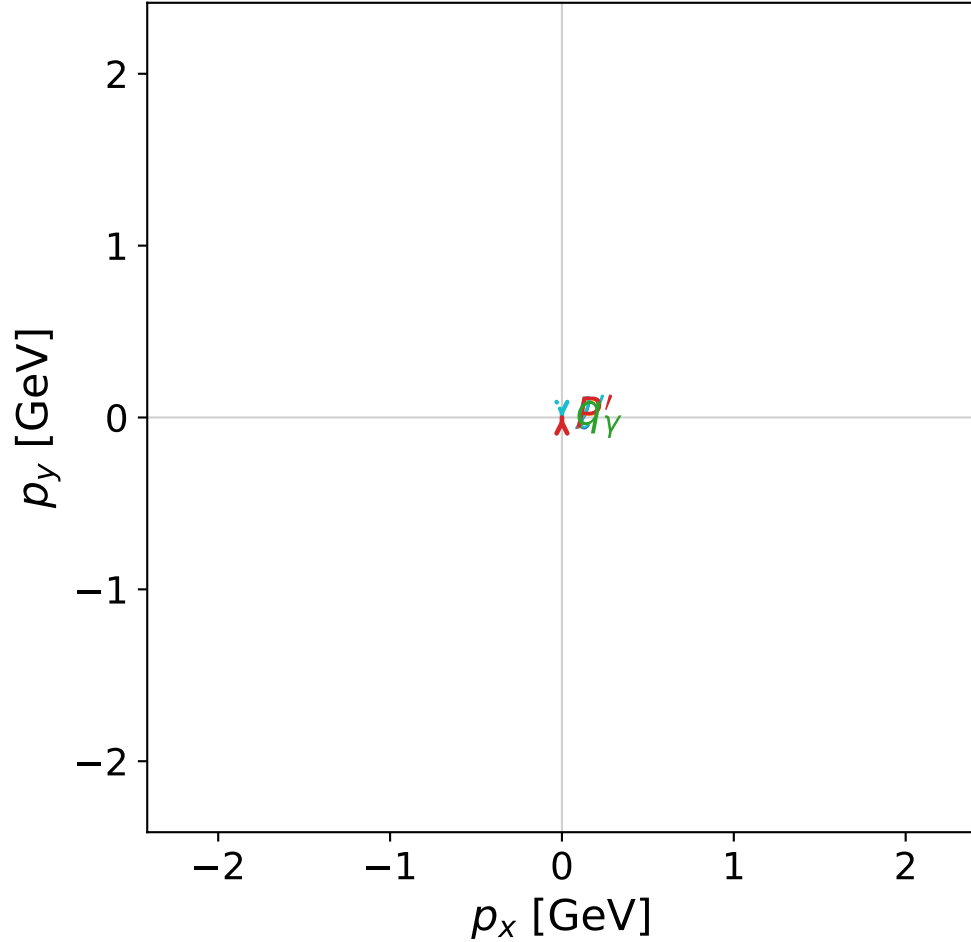


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



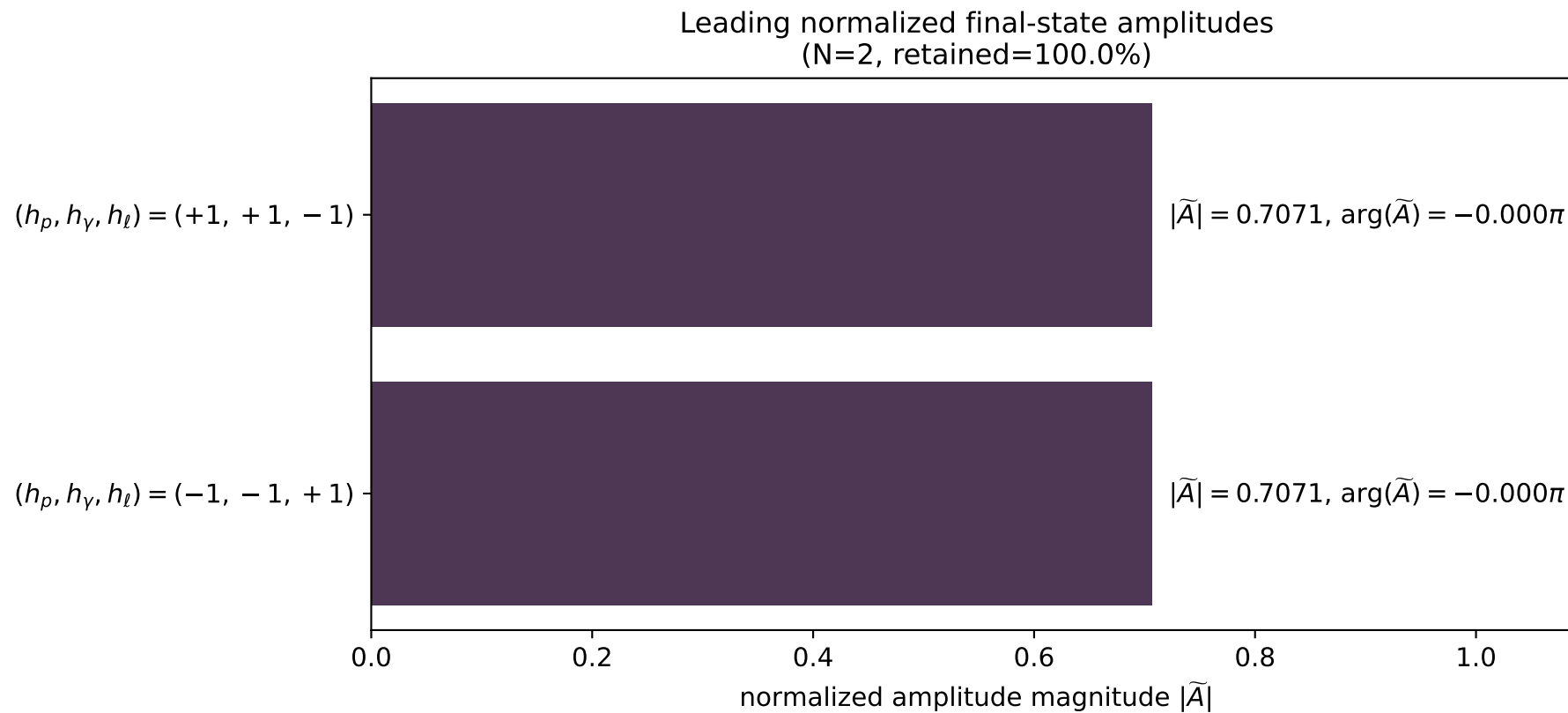
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_148
 $d_{\text{GHZ}}=8.27454\text{e-}08$
 region: polarization_cluster_02_local_minimum_148
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0148
 lepton mixing angle: $\alpha_e=1.0537095$ rad
 incoming lepton: $0.49435|+\rangle +0.869263|-\rangle$
 proton mixing angle: $\alpha_p=2.6245189$ rad
 incoming proton: $-0.869269|+\rangle +0.494339|-\rangle$

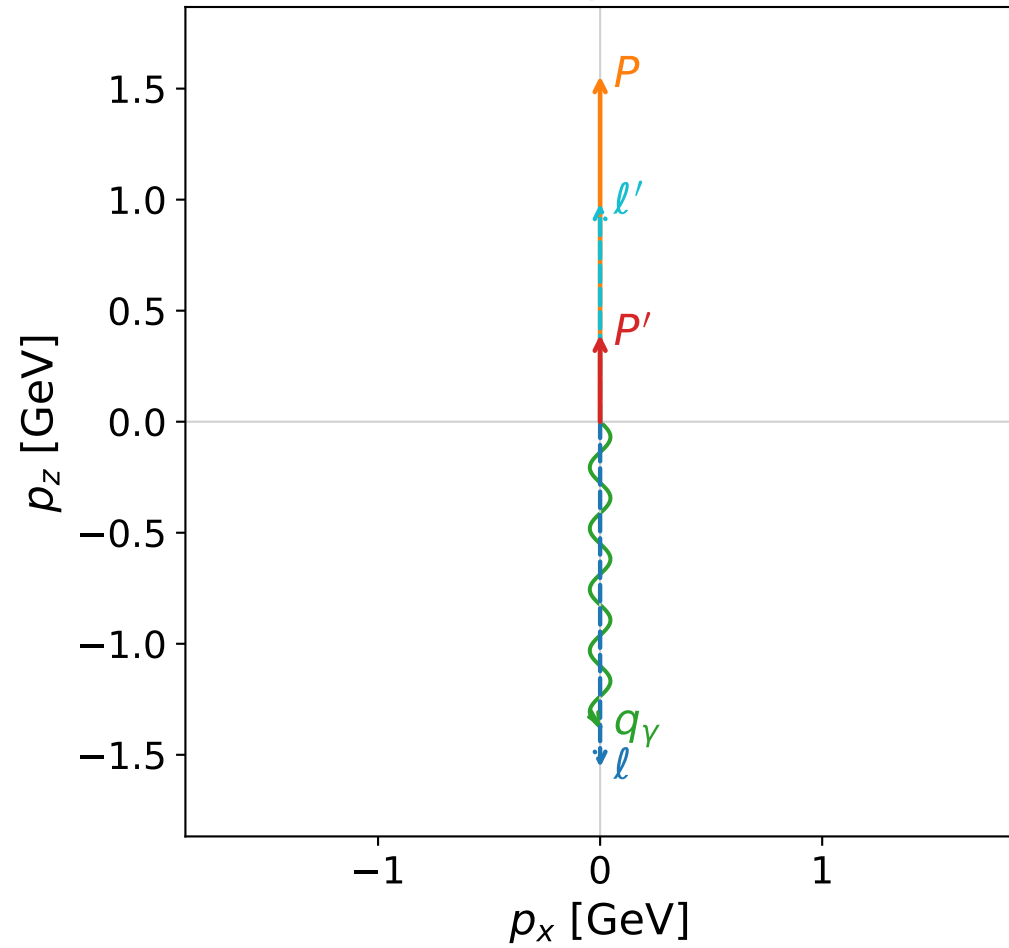
$s=17.8907$, $\sqrt{s}=4.22973$, $|\vec{P}|=2.01086$, $|\vec{P}'|=2.01086$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.82192$, $\phi_\gamma=5.84502$
 $E_\gamma=1.09781$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.96351$
 $Q^2=7.34406$, $x_B=0.44159$, $t=-16.1742$, $W^2=10.1667$, $y=0.97767$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0109$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0109$
 P $E= 2.2189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0109$
 ℓ' $E= 0.9130$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.9130$
 P' $E= 2.2189$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0109$
 q_γ $E= 1.0978$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0978$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

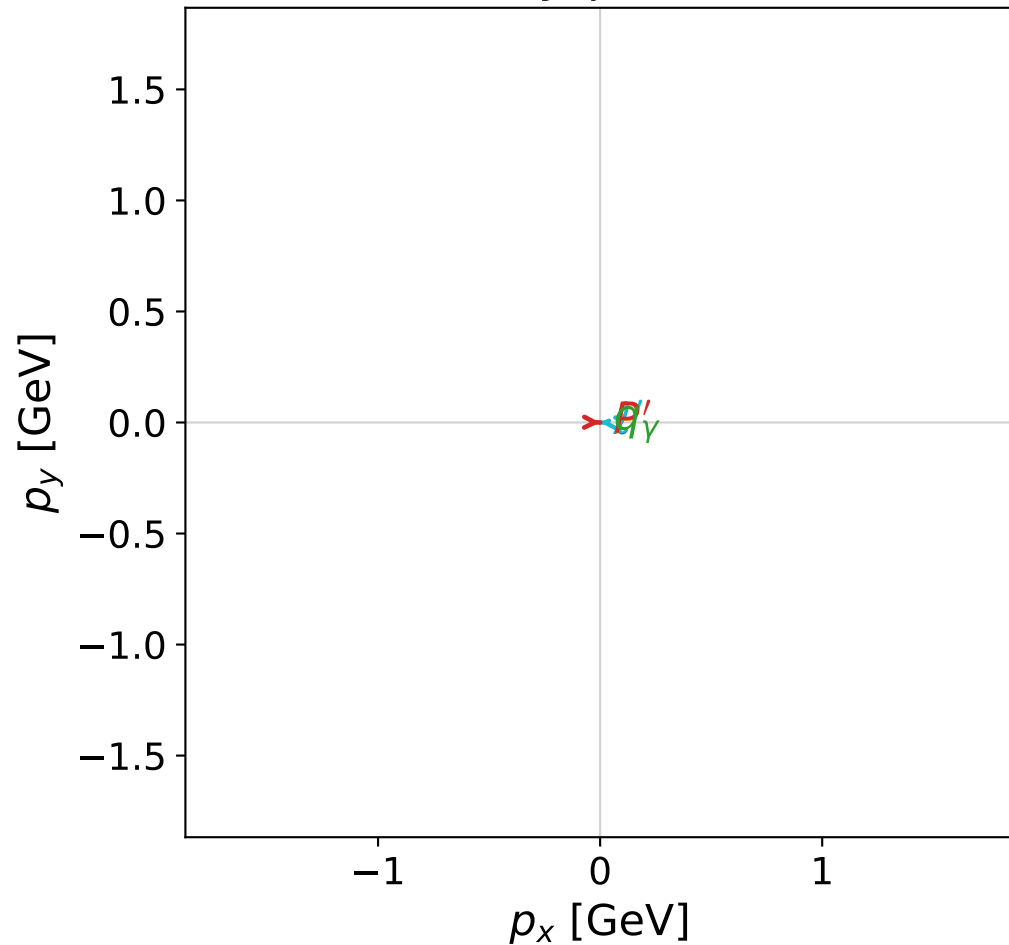


```
dghz_mixing_angles_polarization_02_local_minimum_150
d_{\rm GHZ}=8.40814e-08
region: polarization_cluster_02_local_minimum_150
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0150
lepton mixing angle: alpha_e=0.77196915 rad
incoming lepton: 0.716538|+> +0.697548|->
proton mixing angle: alpha_p=2.3696107 rad
incoming proton: -0.71653|+> +0.697557|->
```

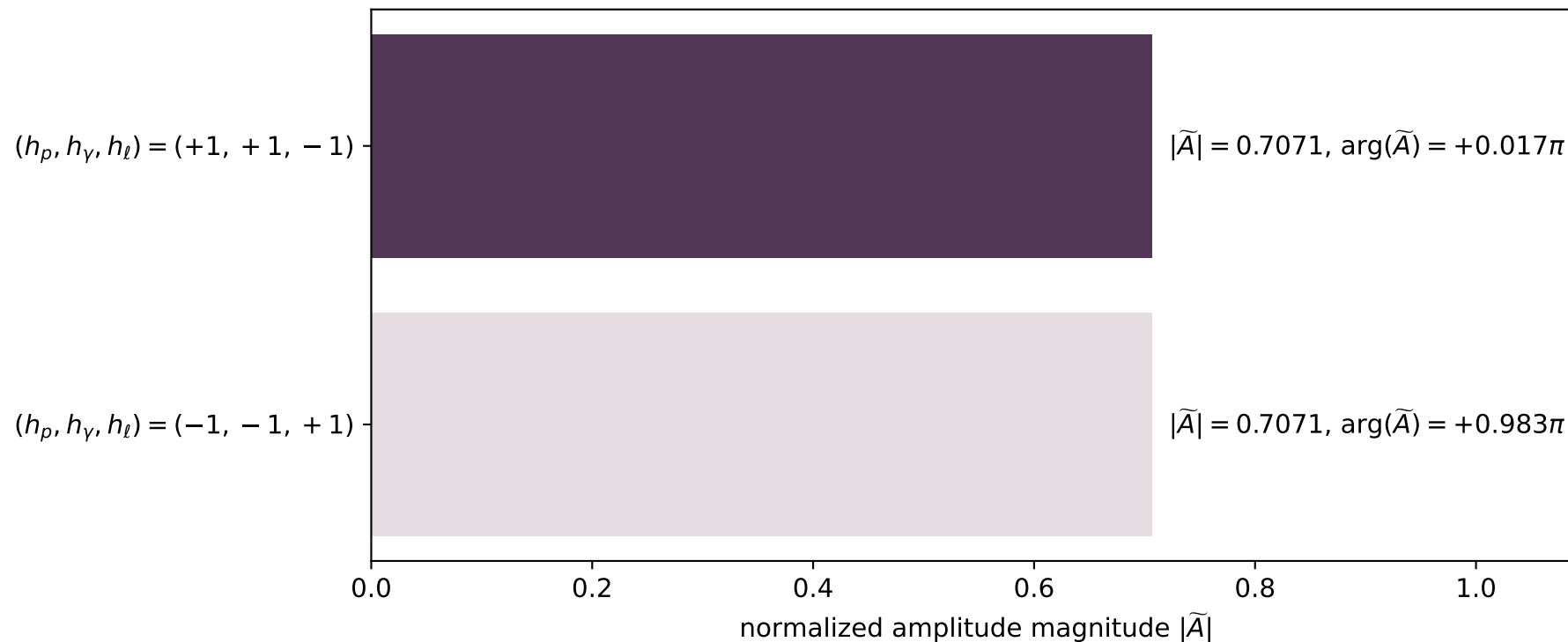
```
s=11.3797, sqrt(s)=3.37339, |P|=1.55628, |P'|=0.391465
theta_{p'}=7.31453e-06, theta_gamma=3.14159, phi_{p'}=6.26658, phi_gamma=3.08863
E_gamma=1.37422, theta_{l'}=2.91362e-06, phi_{l'}=3.12499
Q^2=6.1178, x_B=0.61256, t=-0.715694, W^2=4.74931, y=0.951176
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  1.5563 p_x=  0.0000 p_y=  0.0000 p_z= -1.5563
P      E=  1.8171 p_x=  0.0000 p_y=  0.0000 p_z=  1.5563
l'     E=  0.9828 p_x= -0.0000 p_y=  0.0000 p_z=  0.9828
P'     E=  1.0164 p_x=  0.0000 p_y= -0.0000 p_z=  0.3915
q_gamma E=  1.3742 p_x= -0.0000 p_y=  0.0000 p_z= -1.3742
```

x--y plane

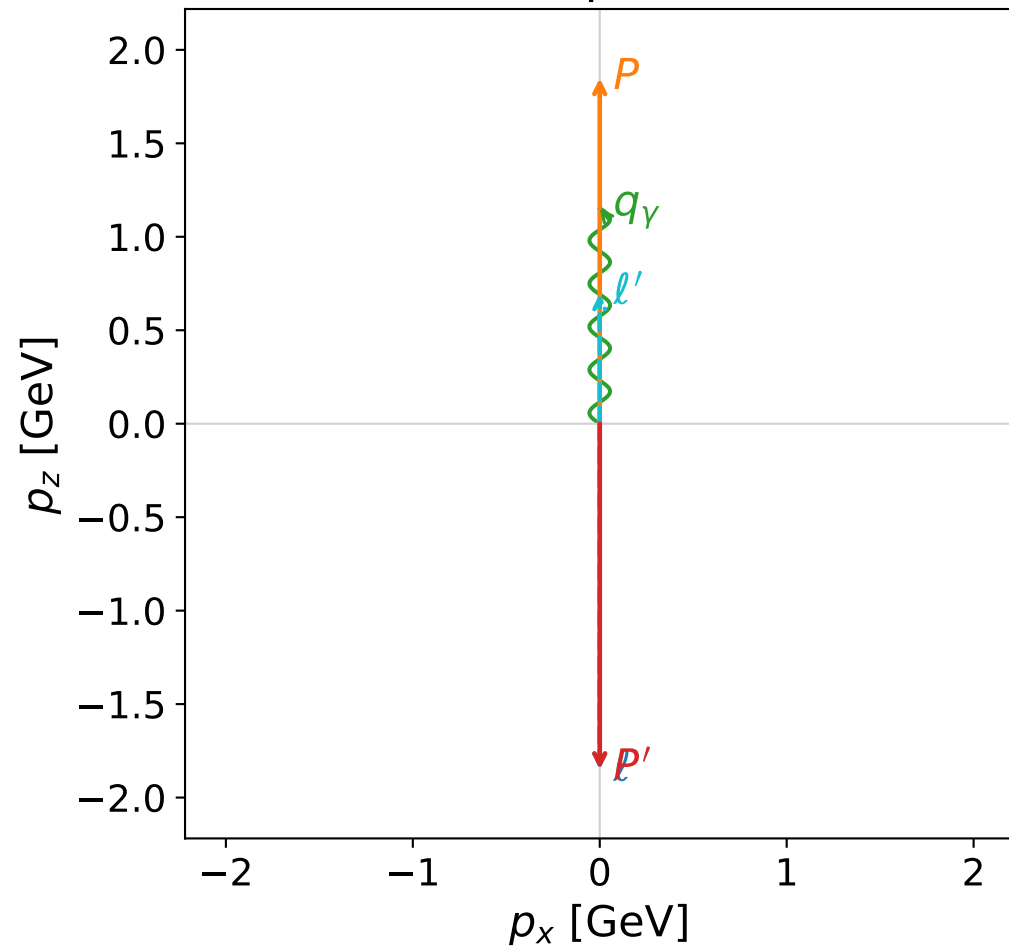


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

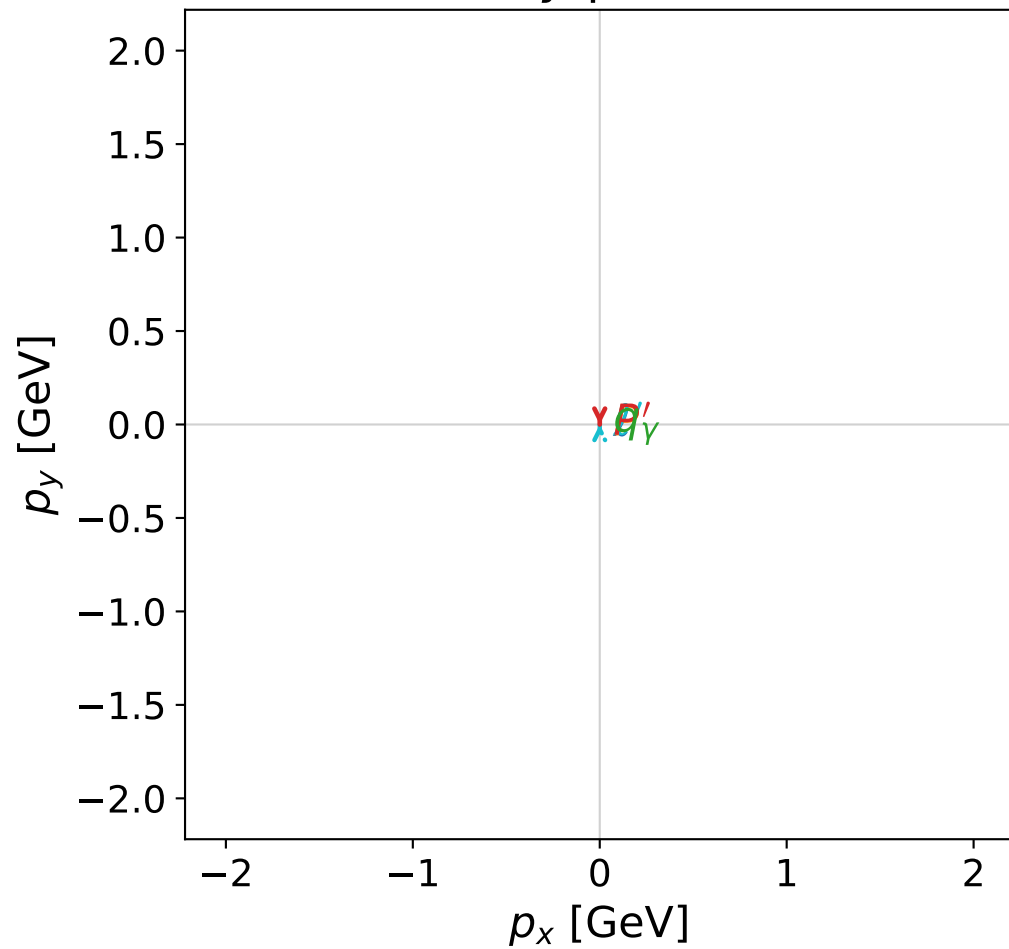


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

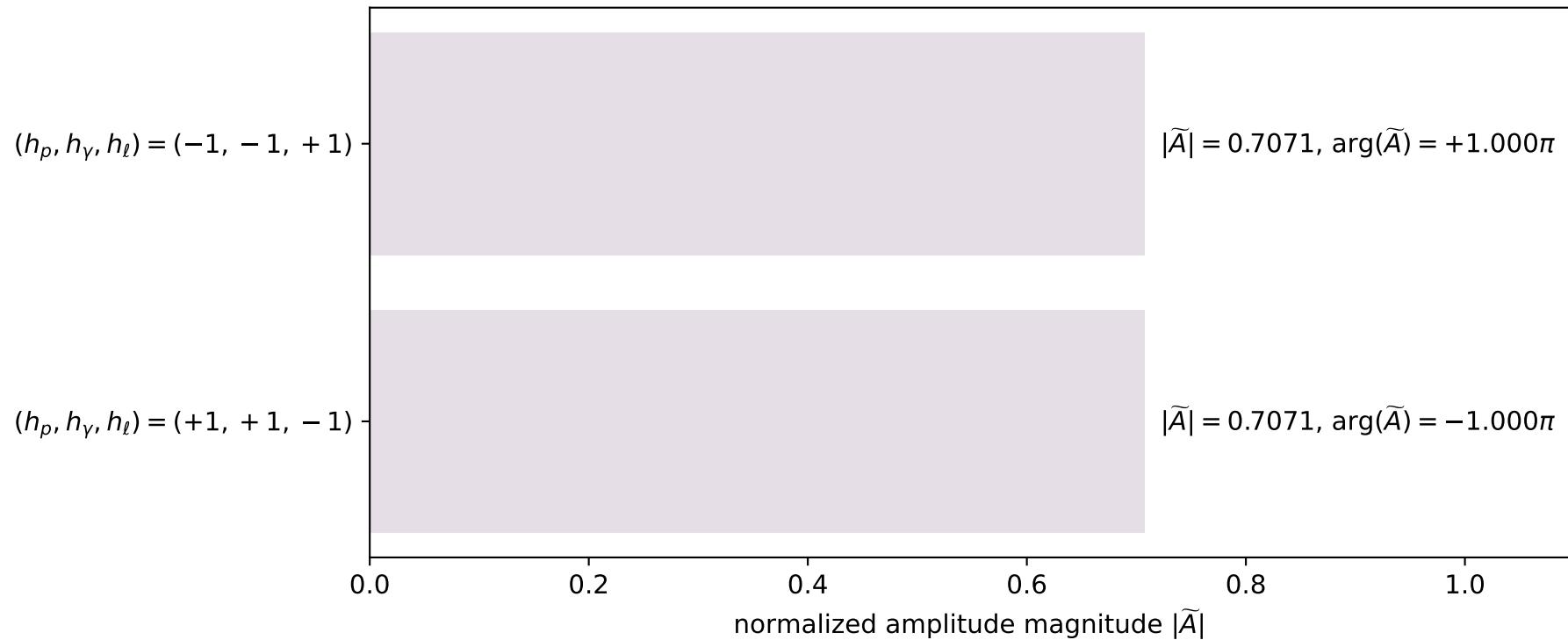


dghz_mixing_angles_polarization_02_local_minimum_157
 $d_{\{\rm GHZ\}}=1.01492\text{e-}07$
 region: polarization_cluster_02_local_minimum_157
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0157
 lepton mixing angle: $\alpha_e=0.78801275$ rad
 incoming lepton: $0.705256|+\rangle + 0.708953|-\rangle$
 proton mixing angle: $\alpha_p=2.3588075$ rad
 incoming proton: $-0.708952|+\rangle + 0.705257|-\rangle$

$s=15.3816$, $\sqrt{s}=3.92194$, $|\vec{P}|=1.8488$, $|\vec{P}'|=1.8488$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.11449$, $\phi_\gamma=3.59494$
 $E_\gamma=1.15299$, $\theta_{l'}=0$, $\phi_{l'}=1.97289$
 $Q^2=5.14568$, $x_B=0.362639$, $t=-13.6722$, $W^2=9.92371$, $y=0.978472$

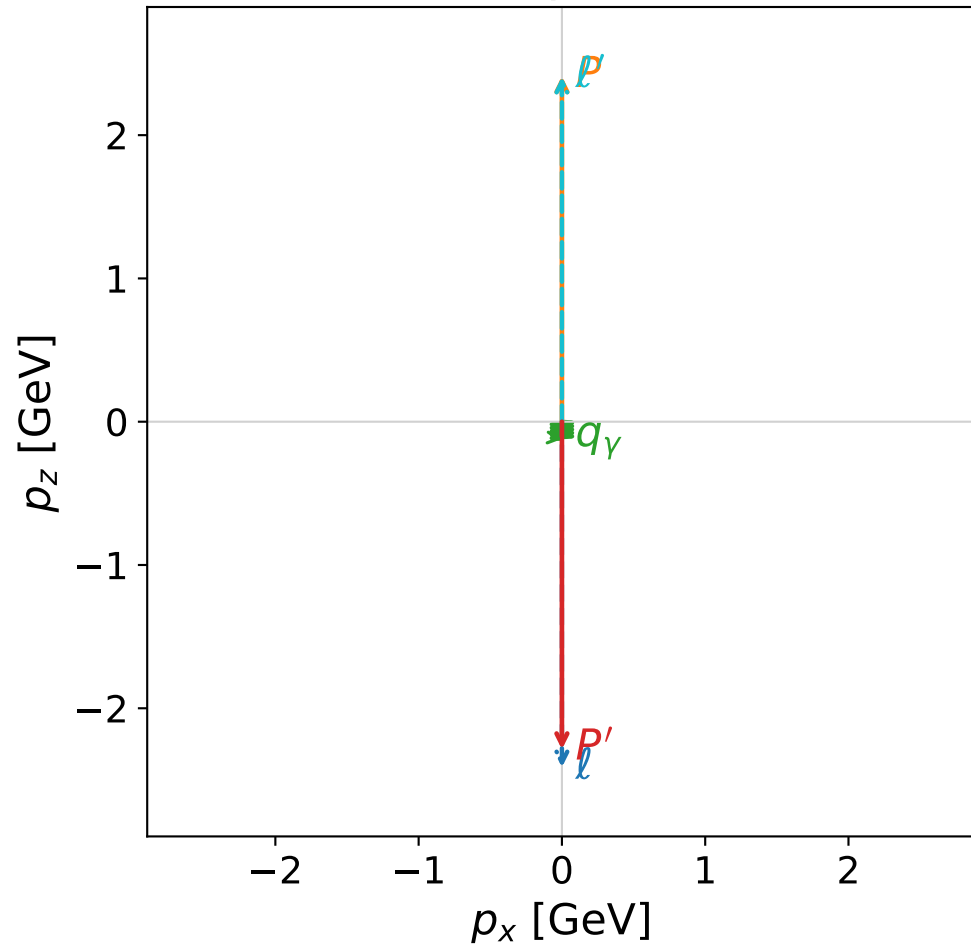
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8488$
 P $E= 2.0731$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8488$
 ℓ' $E= 0.6958$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6958$
 P' $E= 2.0731$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8488$
 q_γ $E= 1.1530$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.1530$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

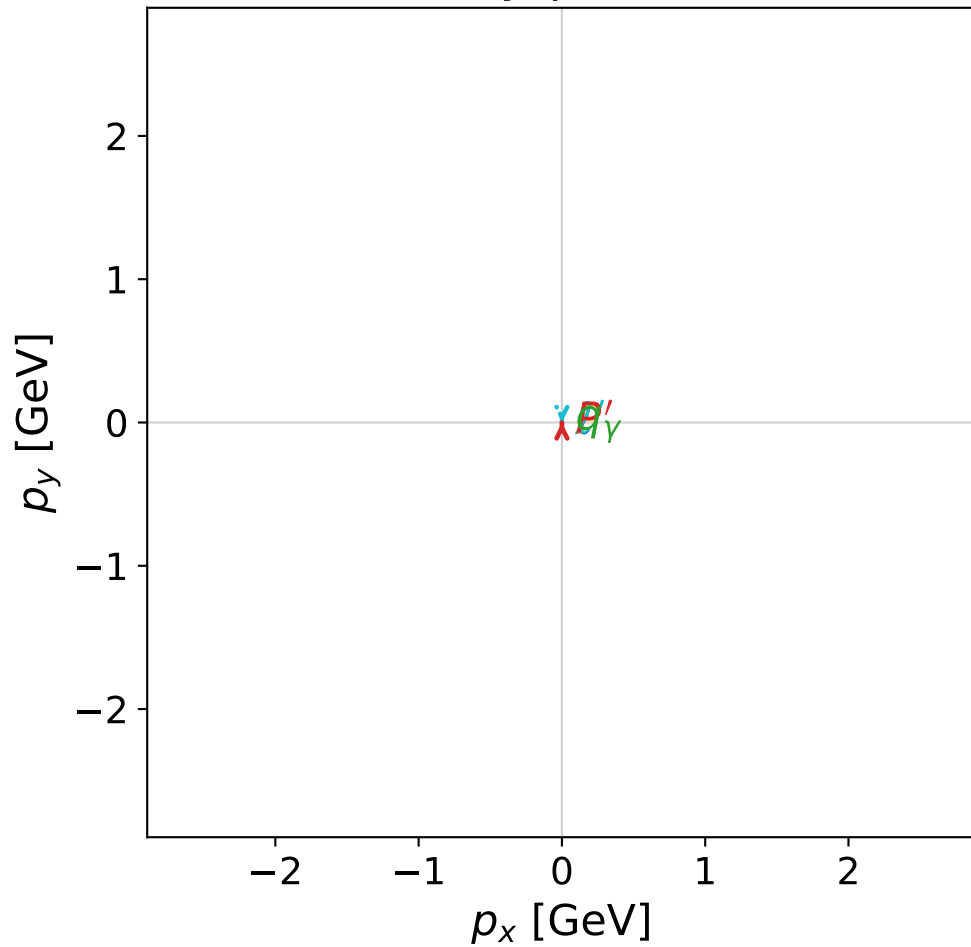


dghz_mixing_angles_polarization_02_local_minimum_159
 $d_{\{\rm GHZ\}}=1.07274\text{e-}07$
 region: polarization_cluster_02_local_minimum_159
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0159
 lepton mixing angle: $\alpha_e=0.10562157$ rad
 incoming lepton: $0.994427|+\rangle +0.105425|-\rangle$
 proton mixing angle: $\alpha_p=3.0359631$ rad
 incoming proton: $-0.994426|+\rangle +0.105433|-\rangle$

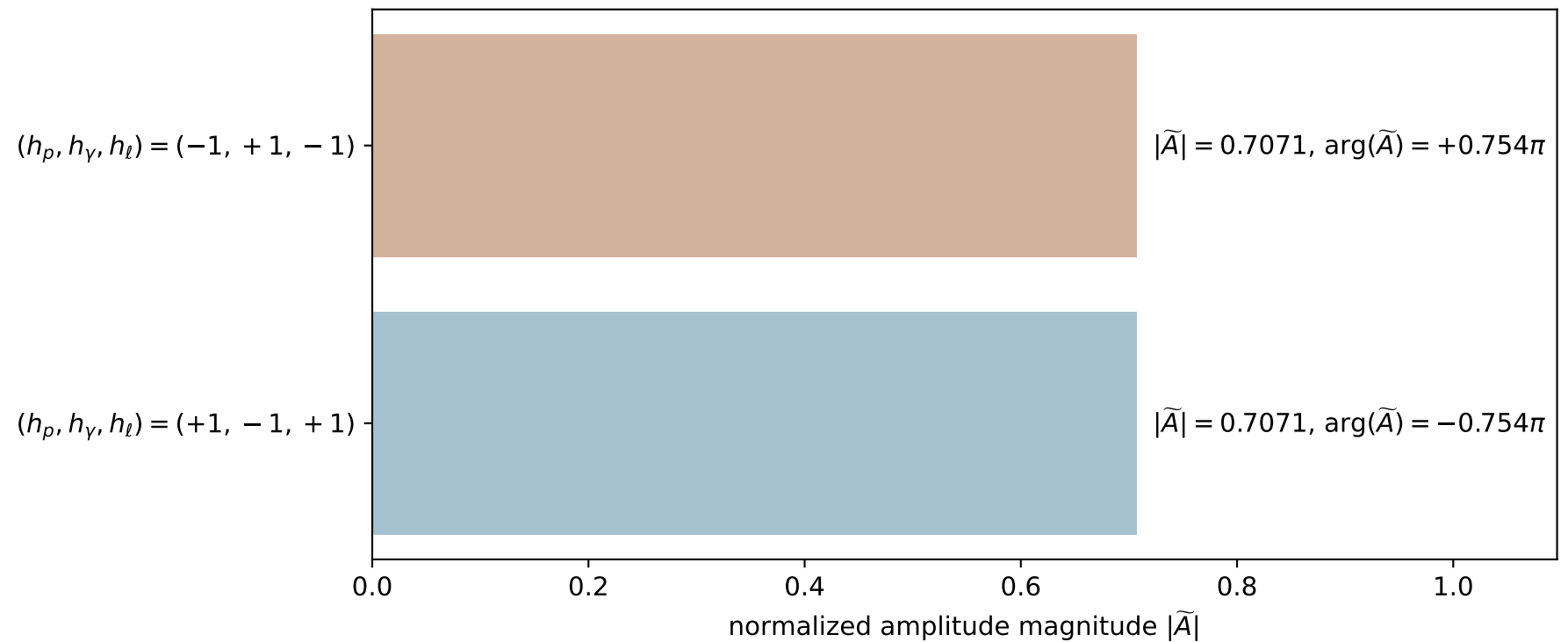
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28696$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.69535$, $\phi_\gamma=3.91457$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=4.88033$
 $Q^2=23.2283$, $x_B=0.998084$, $t=-22.0669$, $W^2=0.924437$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.4076$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.4076$
 P' $E= 2.4718$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2870$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1206$

x--y plane

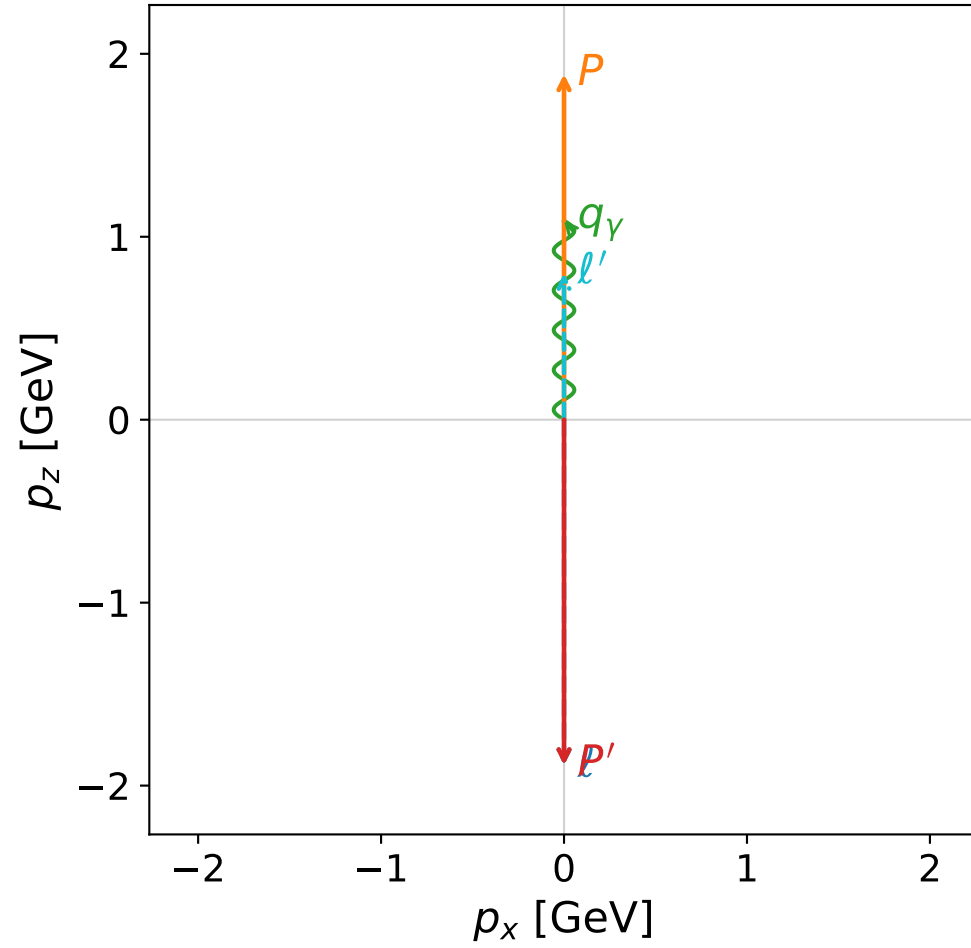


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

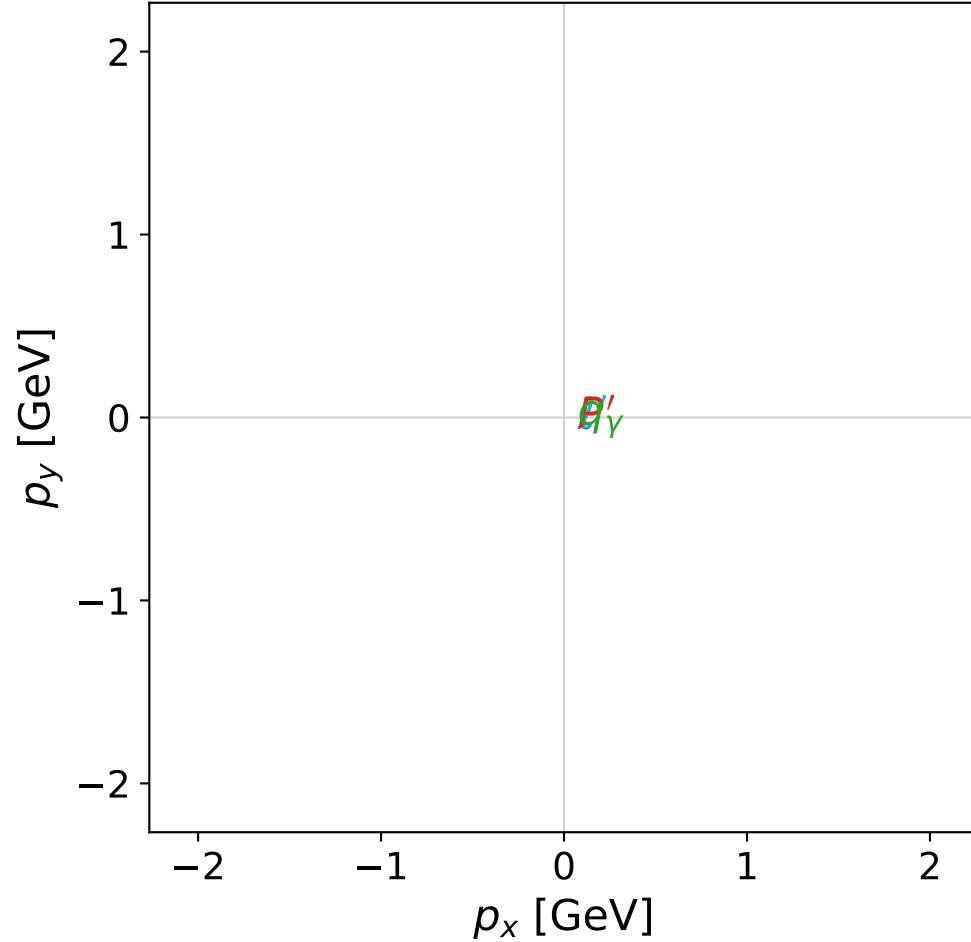


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

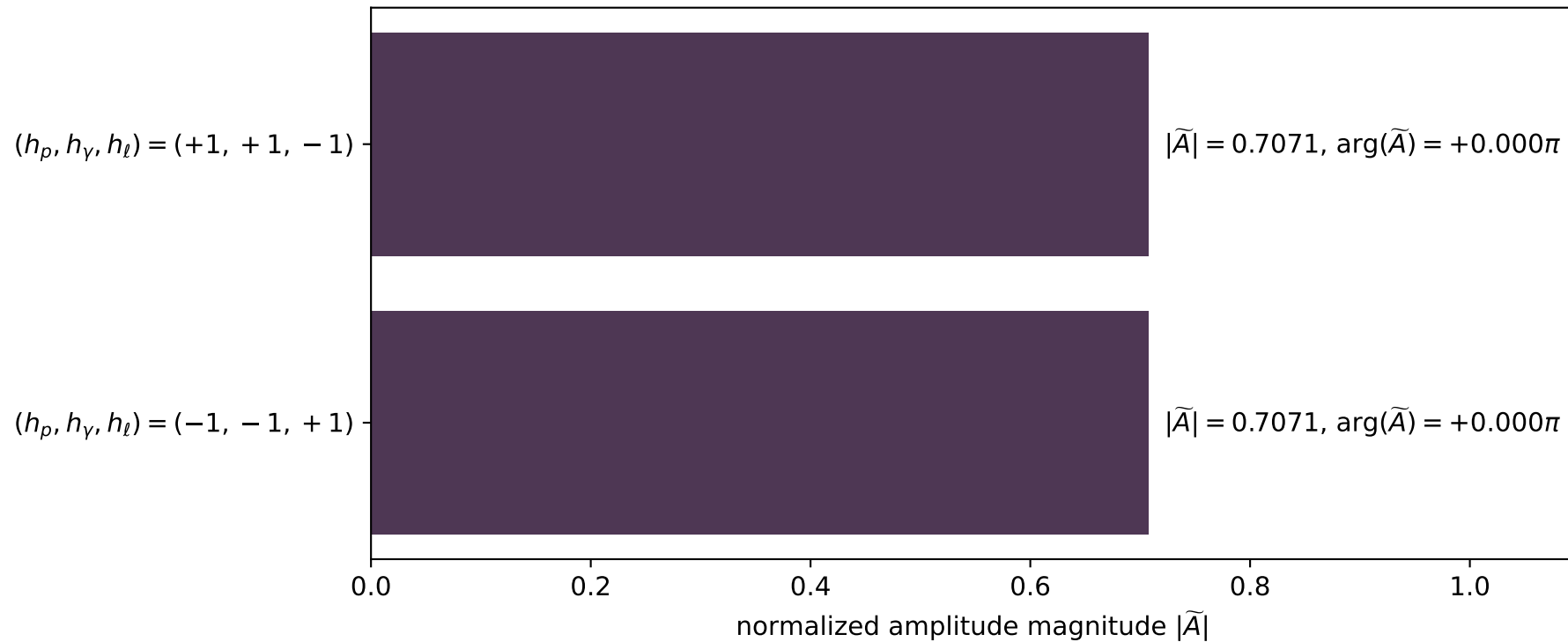


dghz_mixing_angles_polarization_02_local_minimum_160
 $d_{\{\text{rm GHZ}\}}=1.12987\text{e-}07$
 region: polarization_cluster_02_local_minimum_160
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0160
 lepton mixing angle: $\alpha_e=1.1362778$ rad
 incoming lepton: $0.420974|+\rangle + 0.907073|-\rangle$
 proton mixing angle: $\alpha_p=2.7070795$ rad
 incoming proton: $-0.907075|+\rangle + 0.420969|-\rangle$

$s=15.9848$, $\sqrt{s}=3.9981$, $|\vec{P}|=1.88901$, $|\vec{P}'|=1.88901$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.2262$, $\phi_\gamma=6.10955$
 $E_\gamma=1.08804$, $\theta_{l'}=0$, $\phi_{l'}=1.08461$
 $Q^2=6.05225$, $x_B=0.410256$, $t=-14.2735$, $W^2=9.57998$, $y=0.976661$

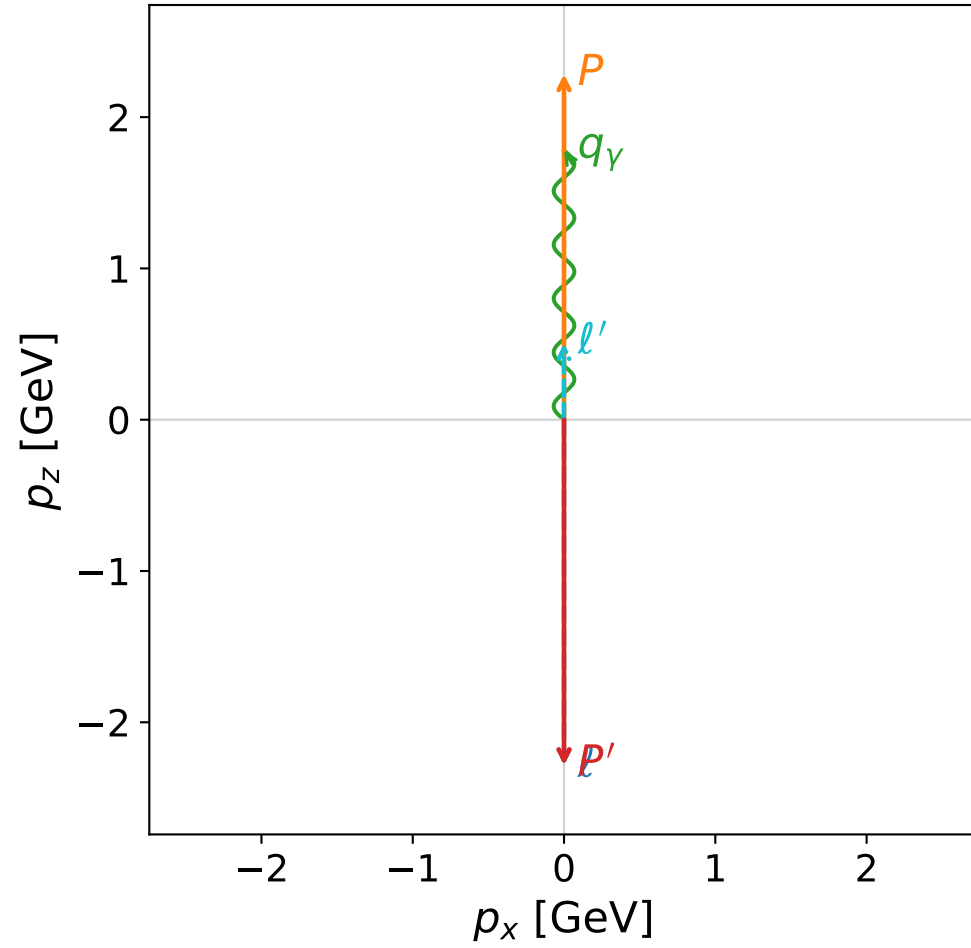
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8890$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8890$
 P $E= 2.1091$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8890$
 l' $E= 0.8010$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8010$
 P' $E= 2.1091$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8890$
 q_γ $E= 1.0880$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0880$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

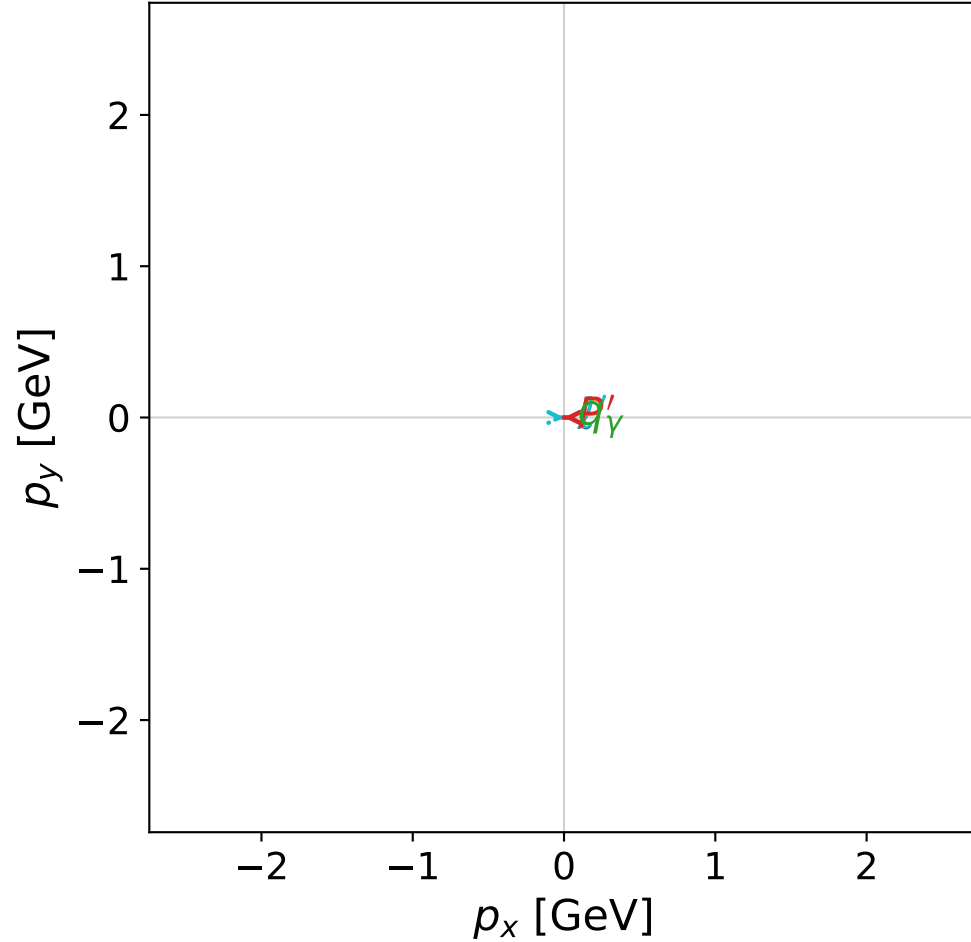


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



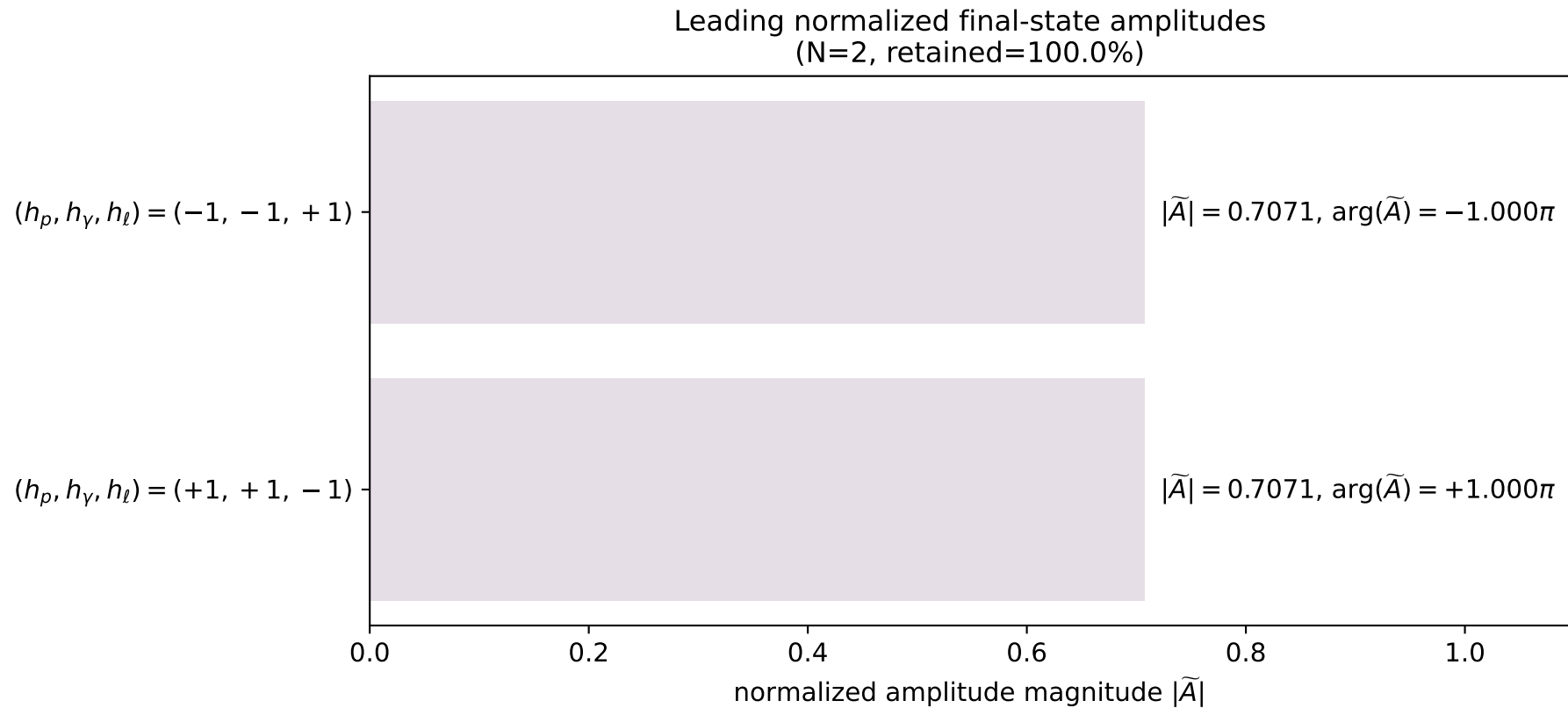
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_164
 $d_{\text{GHZ}}=1.19686\text{e-}07$
 region: polarization_cluster_02_local_minimum_164
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0164
 lepton mixing angle: $\alpha_e=0.61981444$ rad
 incoming lepton: $0.813986|+\rangle + 0.580884|-\rangle$
 proton mixing angle: $\alpha_p=2.1905907$ rad
 incoming proton: $-0.580868|+\rangle + 0.813998|-\rangle$

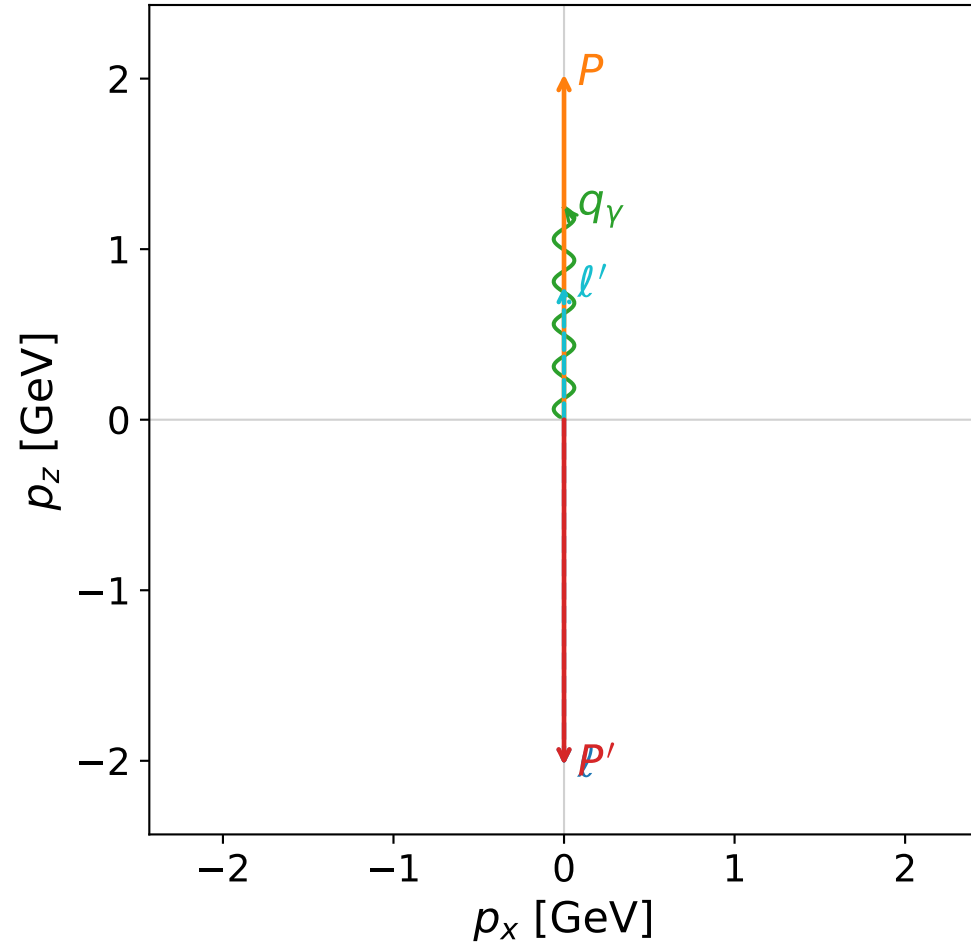
$s=22.5997$, $\sqrt{s}=4.75391$, $|\vec{P}|=2.28442$, $|\vec{P}'|=2.28442$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.04512$, $\phi_\gamma=2.44102$
 $E_\gamma=1.7794$, $\theta_{l'}=0$, $\phi_{l'}=6.18671$
 $Q^2=4.61466$, $x_B=0.214308$, $t=-20.8742$, $W^2=17.7981$, $y=0.991393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2844$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2844$
 P $E= 2.4695$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2844$
 l' $E= 0.5050$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.5050$
 P' $E= 2.4695$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2844$
 q_γ $E= 1.7794$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7794$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

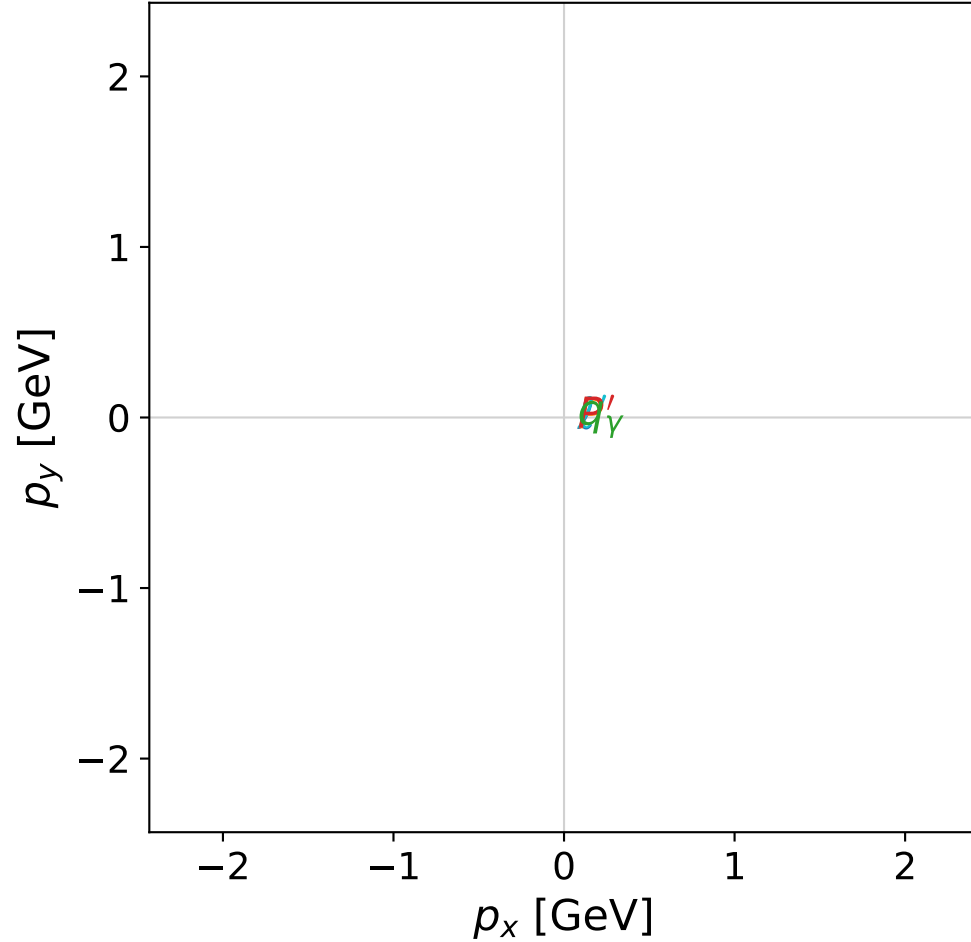


dghz_mixing_angles_polarization_02_local_minimum_167
 $d_{\{\text{rm GHZ}\}}=1.5068\text{e-}07$
 region: polarization_cluster_02_local_minimum_167
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0167
 lepton mixing angle: $\alpha_e=0.28996137$ rad
 incoming lepton: $0.958255|+\rangle + 0.285915|-\rangle$
 proton mixing angle: $\alpha_p=1.8607761$ rad
 incoming proton: $-0.285933|+\rangle + 0.95825|-\rangle$

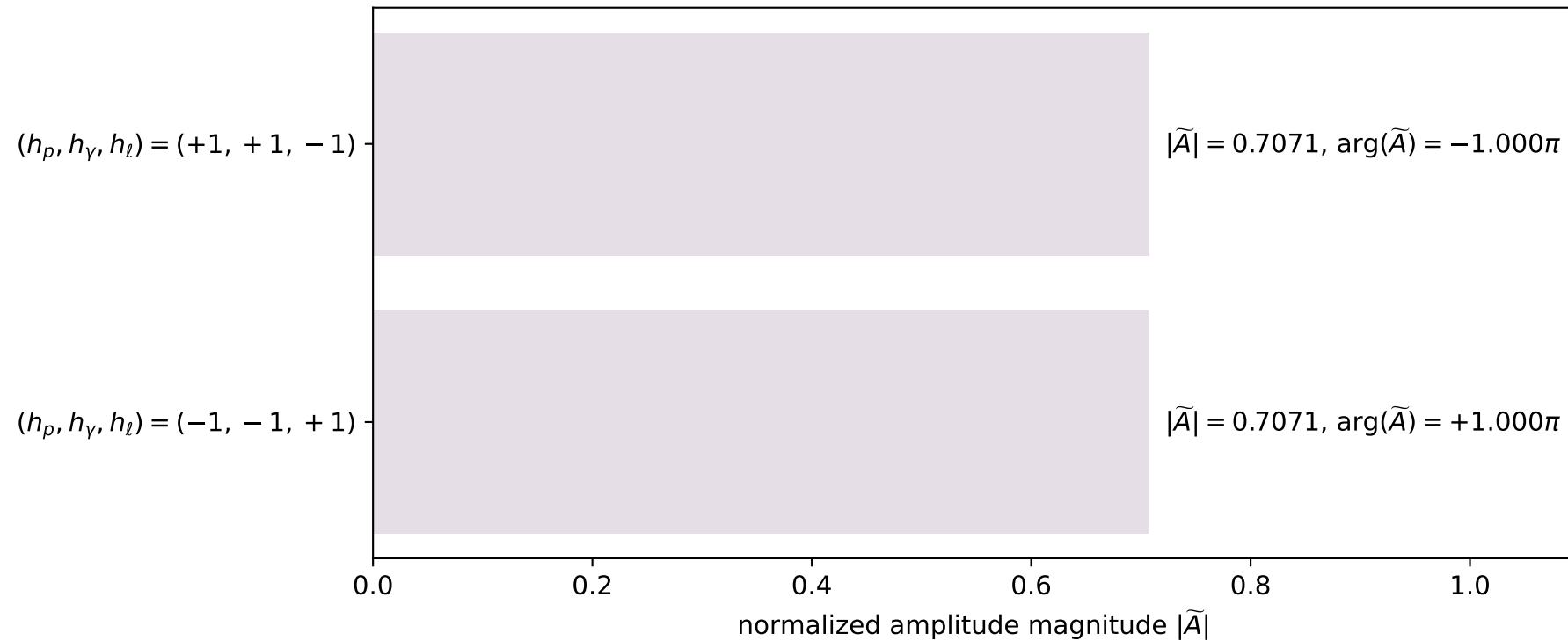
$s=18.1422$, $\sqrt{s}=4.25936$, $|\vec{P}|=2.0264$, $|\vec{P}'|=2.0264$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=0.950105$, $\phi_Y=4.60864$
 $E_Y=1.24571$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.0917$
 $Q^2=6.32793$, $x_B=0.373554$, $t=-16.4252$, $W^2=11.4917$, $y=0.981316$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0264$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0264$
 P $E= 2.2330$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0264$
 ℓ' $E= 0.7807$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7807$
 P' $E= 2.2330$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0264$
 q_Y $E= 1.2457$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.2457$

x--y plane

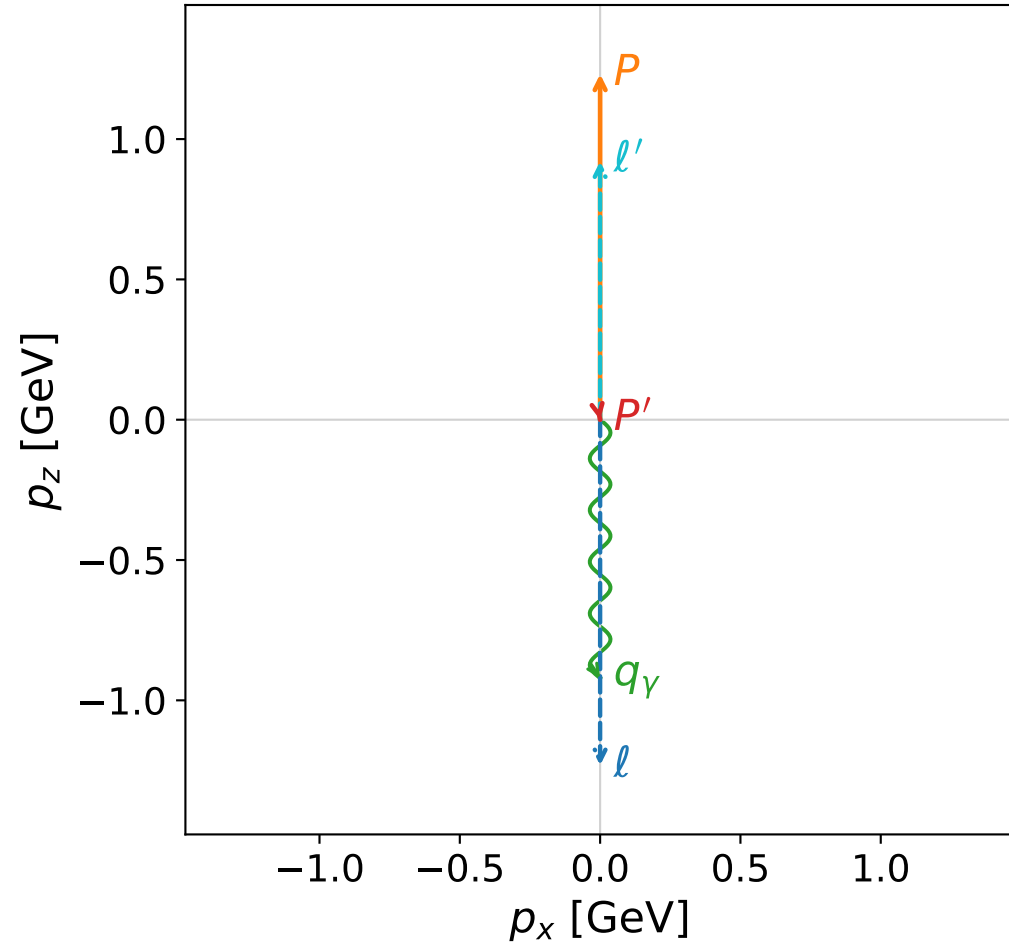


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

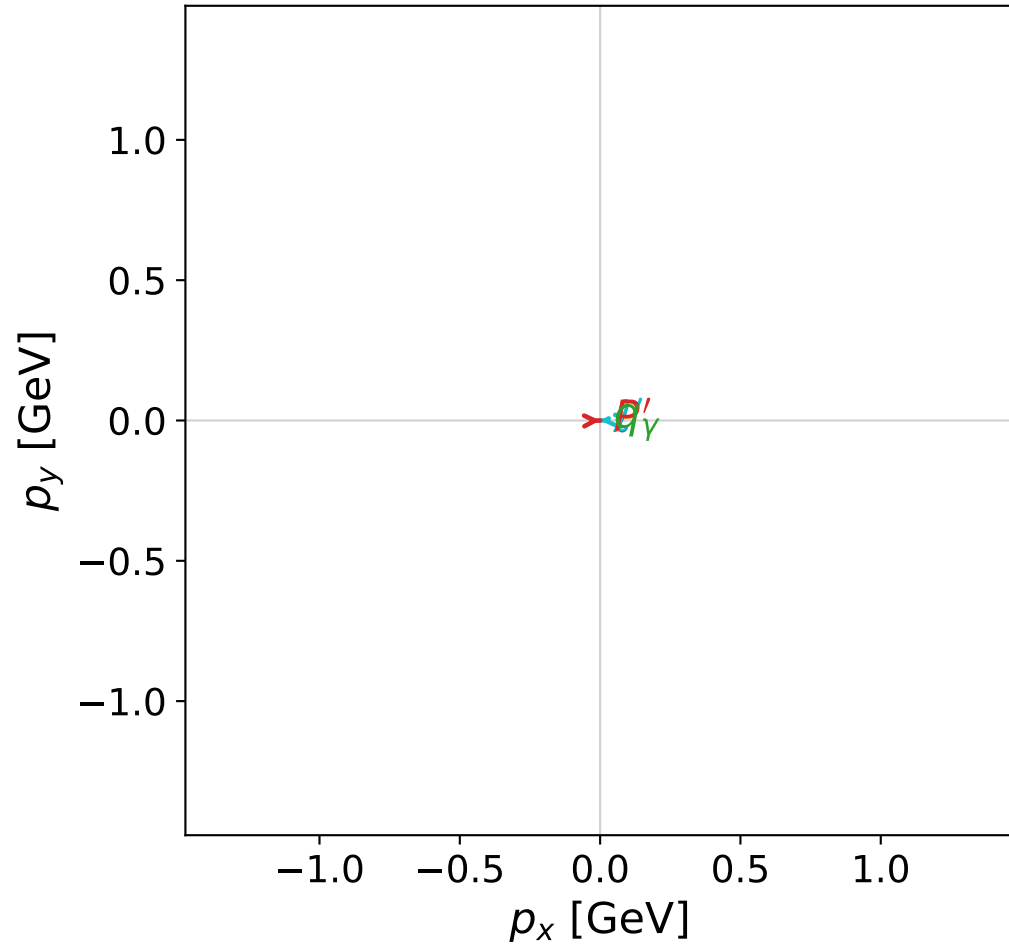


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

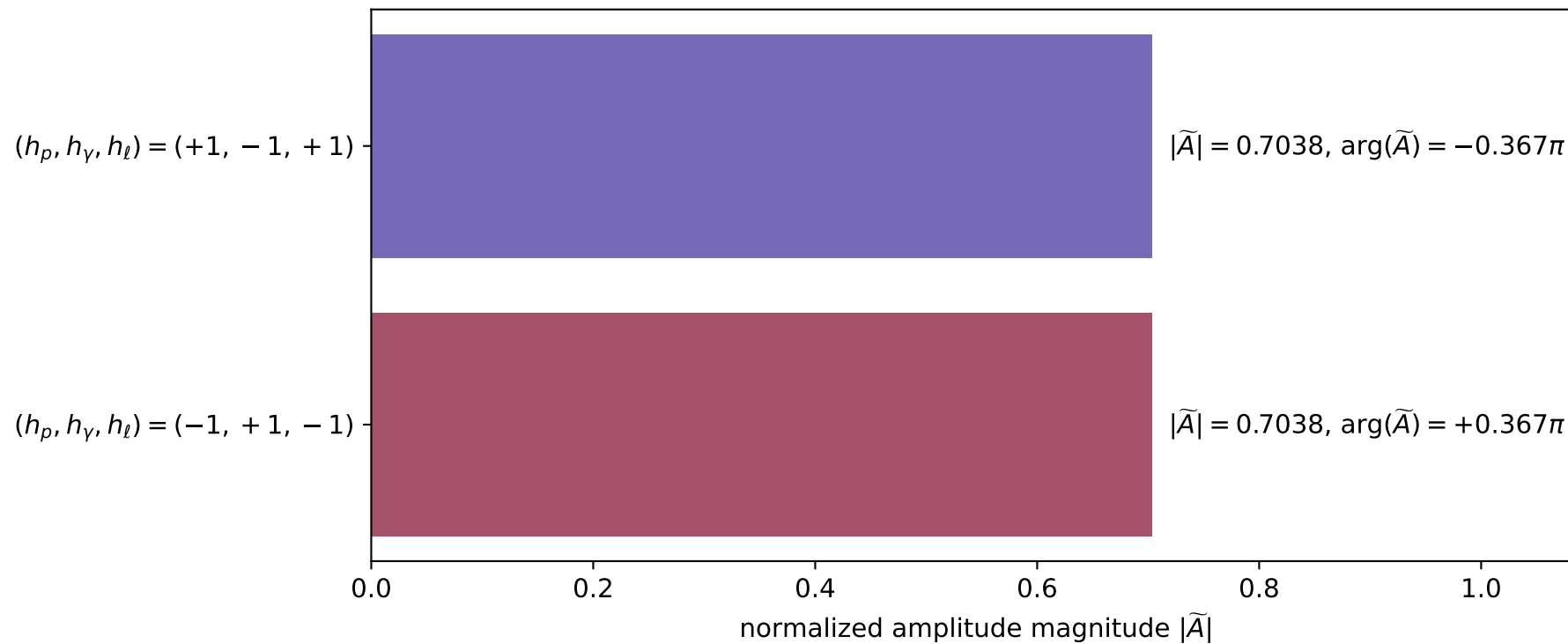


dghz_mixing_angles_polarization_02_local_minimum_168
 $d_{\text{GHZ}}=1.70137\text{e-}07$
 region: polarization_cluster_02_local_minimum_168
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0168
 lepton mixing angle: $\alpha_e=1.2036147$ rad
 incoming lepton: $0.358986|+\rangle + 0.933343|-\rangle$
 proton mixing angle: $\alpha_p=1.9380061$ rad
 incoming proton: $-0.359013|+\rangle + 0.933333|-\rangle$

$s=7.72597$, $\sqrt{s}=2.77956$, $|\vec{P}|=1.23151$, $|\vec{P}'|=0.000244016$
 $\theta_{P'}=2.94845$, $\theta_Y=3.14159$, $\phi_{P'}=0.027548$, $\phi_Y=5.15721$
 $E_Y=0.920662$, $\theta_{\ell'}=5.08603\text{e-}05$, $\phi_{\ell'}=3.16914$
 $Q^2=4.5364$, $x_B=0.724304$, $t=-1.14505$, $W^2=2.60656$, $y=0.914842$

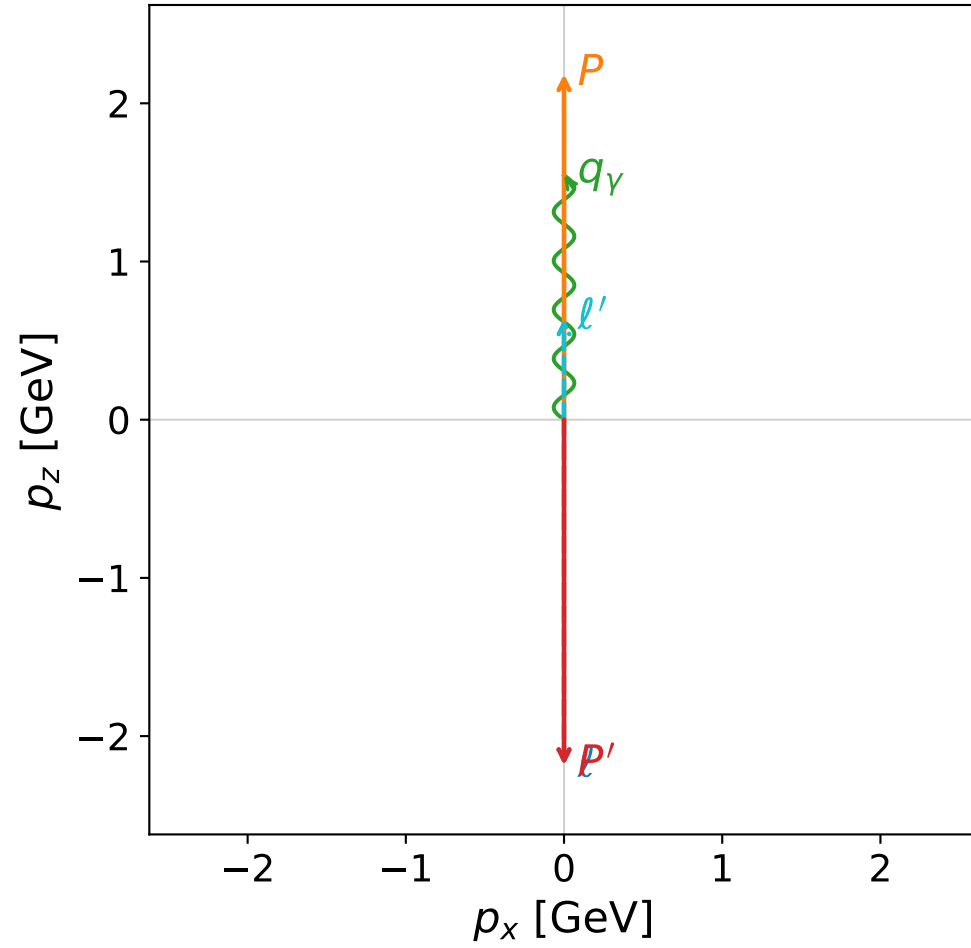
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2315$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2315$
 P $E= 1.5481$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2315$
 ℓ' $E= 0.9209$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.9209$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0002$
 q_Y $E= 0.9207$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.9207$

Leading normalized final-state amplitudes
(N=2, retained=99.1%)

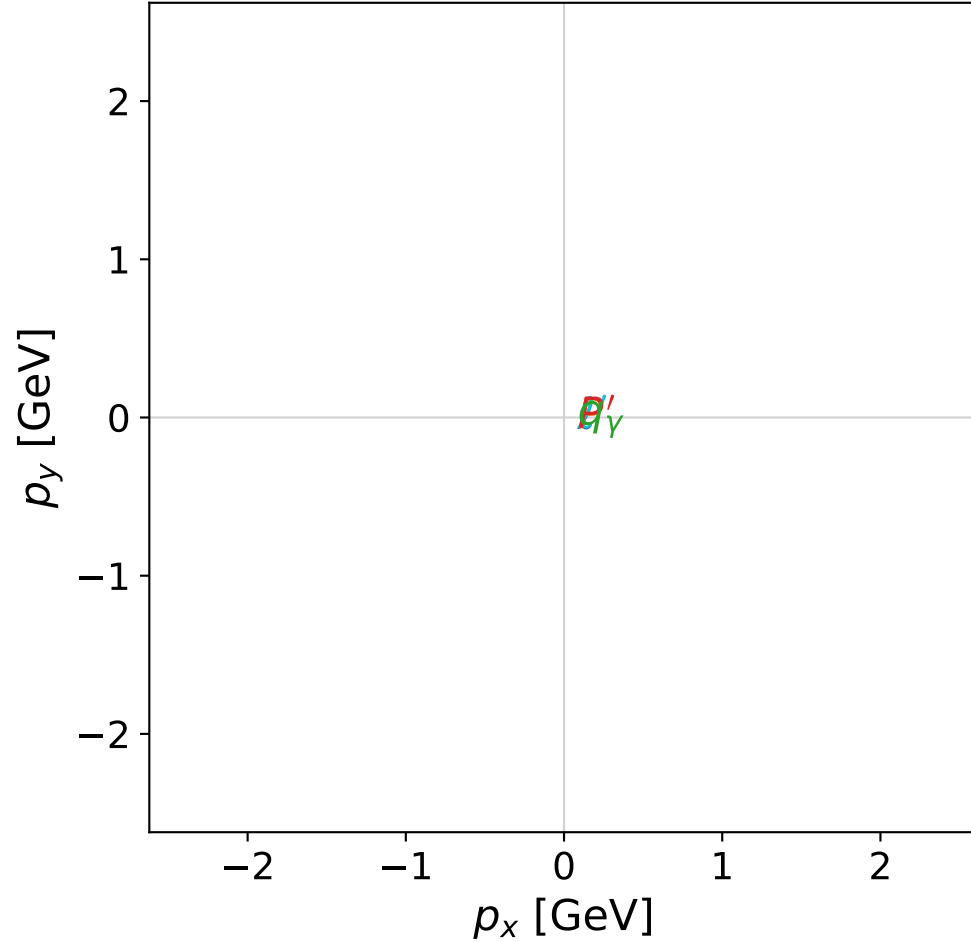


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



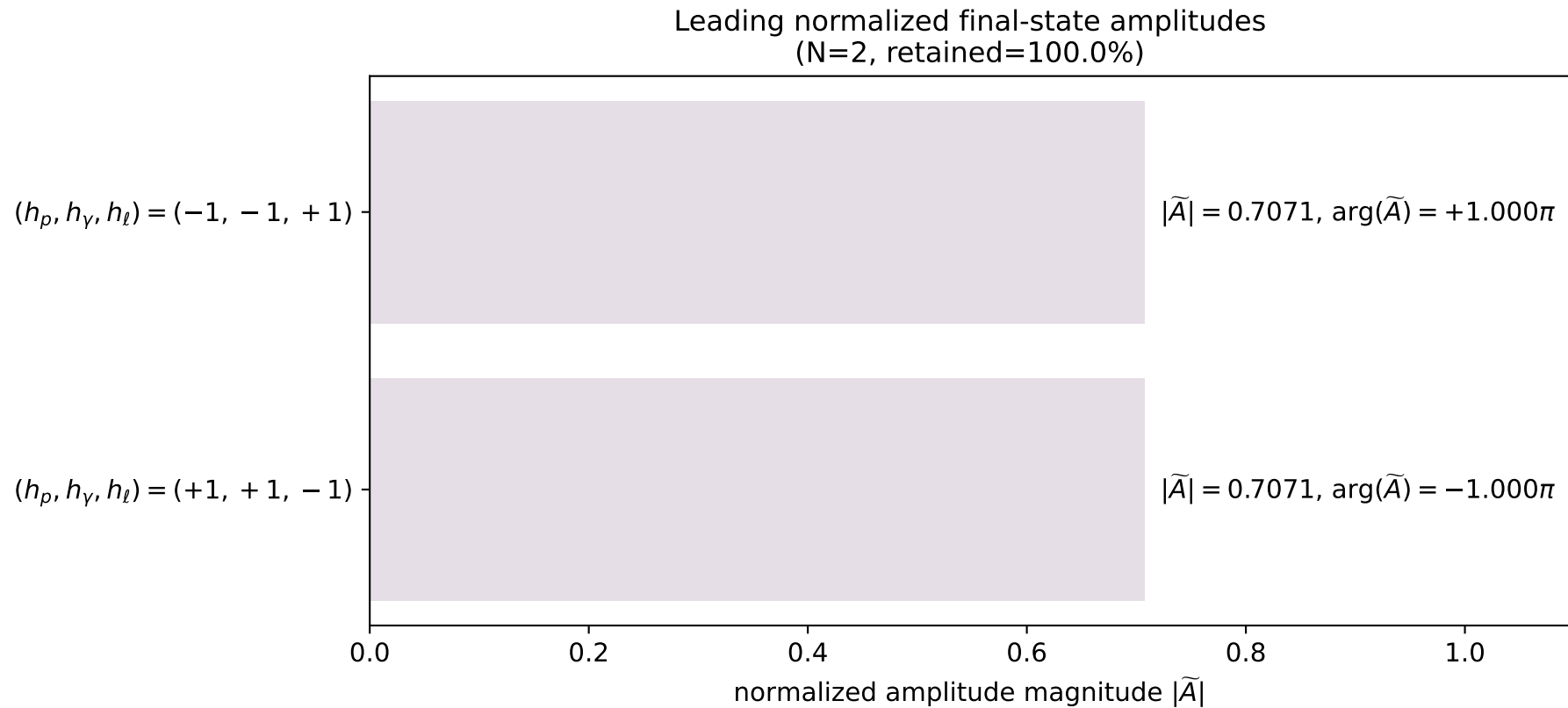
x--y plane



dghz_mixing_angles_polarization_02_local_minimum_171
 $d_{\{\rm GHZ\}}=2.04591\text{e-}07$
 region: polarization_cluster_02_local_minimum_171
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0171
 lepton mixing angle: $\alpha_e=0.23742392$ rad
 incoming lepton: $0.971947|+\rangle + 0.2352|-\rangle$
 proton mixing angle: $\alpha_p=1.8082114$ rad
 incoming proton: $-0.235191|+\rangle + 0.971949|-\rangle$

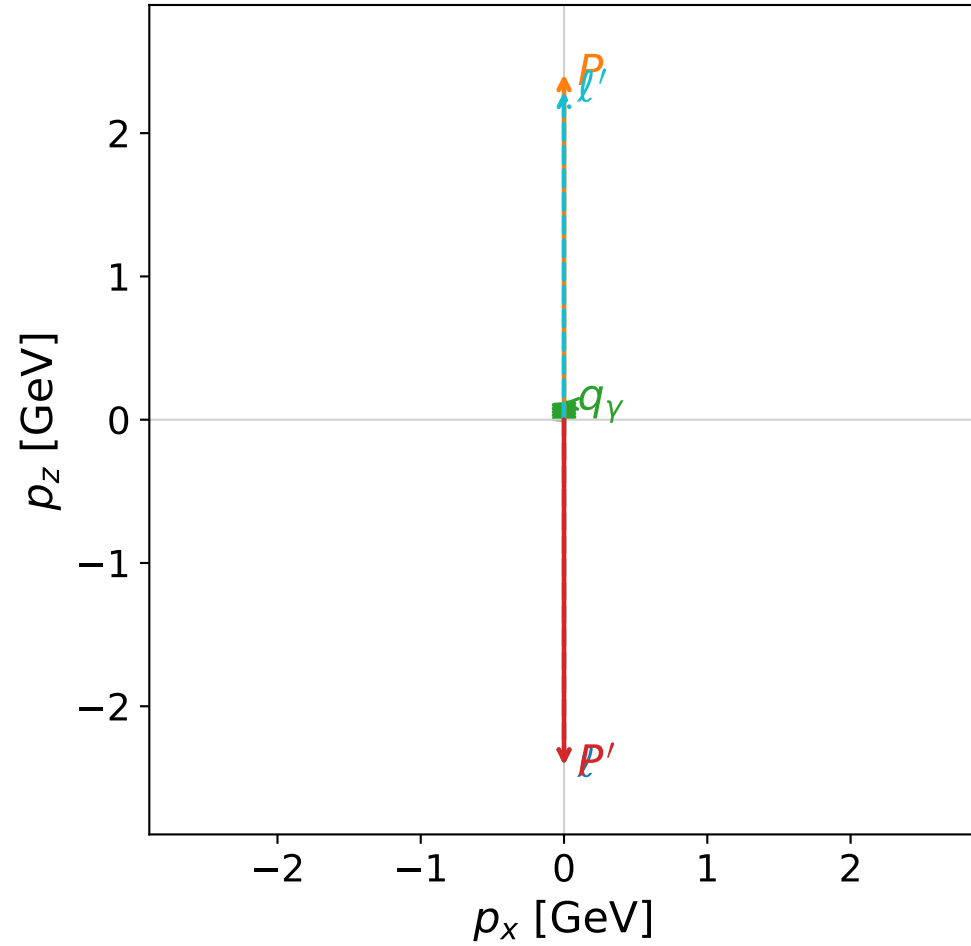
$s=20.8112$, $\sqrt{s}=4.56193$, $|\vec{P}|=2.18453$, $|\vec{P}'|=2.18453$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.23383$, $\phi_\gamma=4.47775$
 $E_\gamma=1.54558$, $\theta_{l'}=0$, $\phi_{l'}=5.37543$
 $Q^2=5.58324$, $x_B=0.283631$, $t=-19.0887$, $W^2=14.9815$, $y=0.987634$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1845$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1845$
 P $E= 2.3774$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1845$
 l' $E= 0.6390$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6390$
 P' $E= 2.3774$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1845$
 q_γ $E= 1.5456$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5456$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

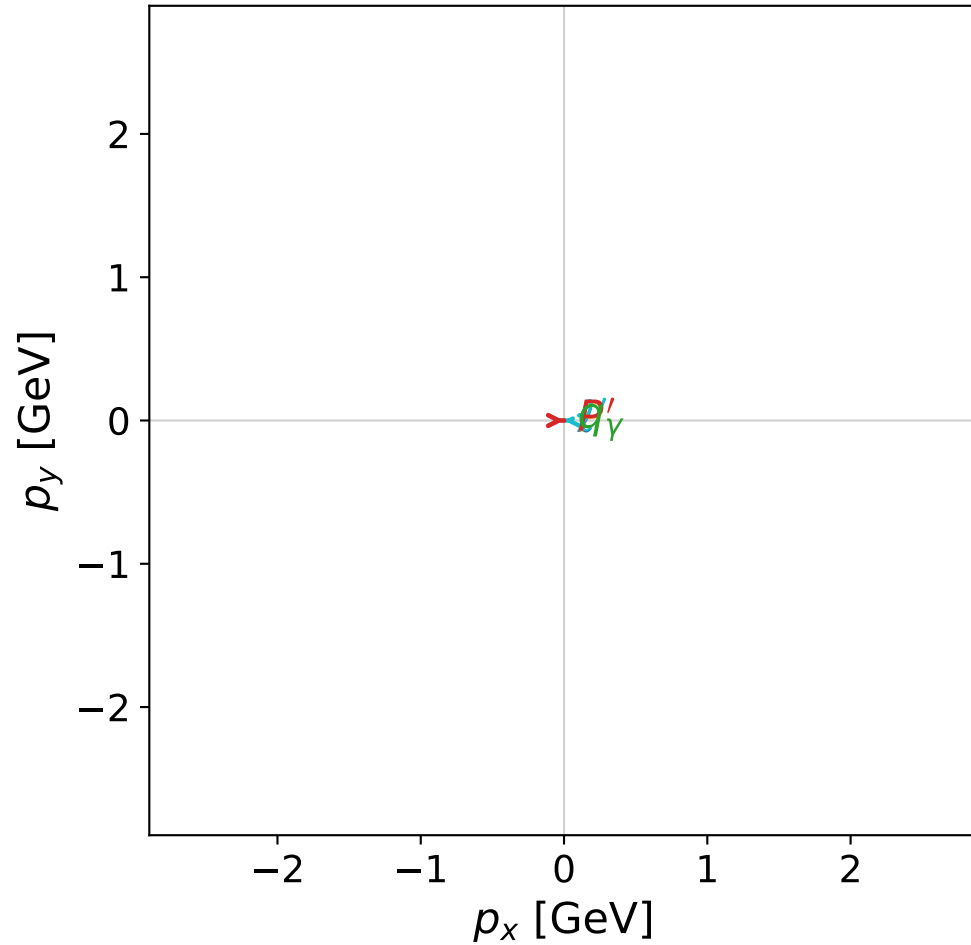


dghz_mixing_angles_polarization_02_local_minimum_175
 $d_{\{\text{rm GHZ}\}}=2.71276\text{e-}07$
 region: polarization_cluster_02_local_minimum_175
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0175
 lepton mixing angle: $\alpha_e=0.045198059$ rad
 incoming lepton: $0.998979|+\rangle +0.0451827|-\rangle$
 proton mixing angle: $\alpha_p=1.6160065$ rad
 incoming proton: $-0.0451948|+\rangle +0.998978|-\rangle$

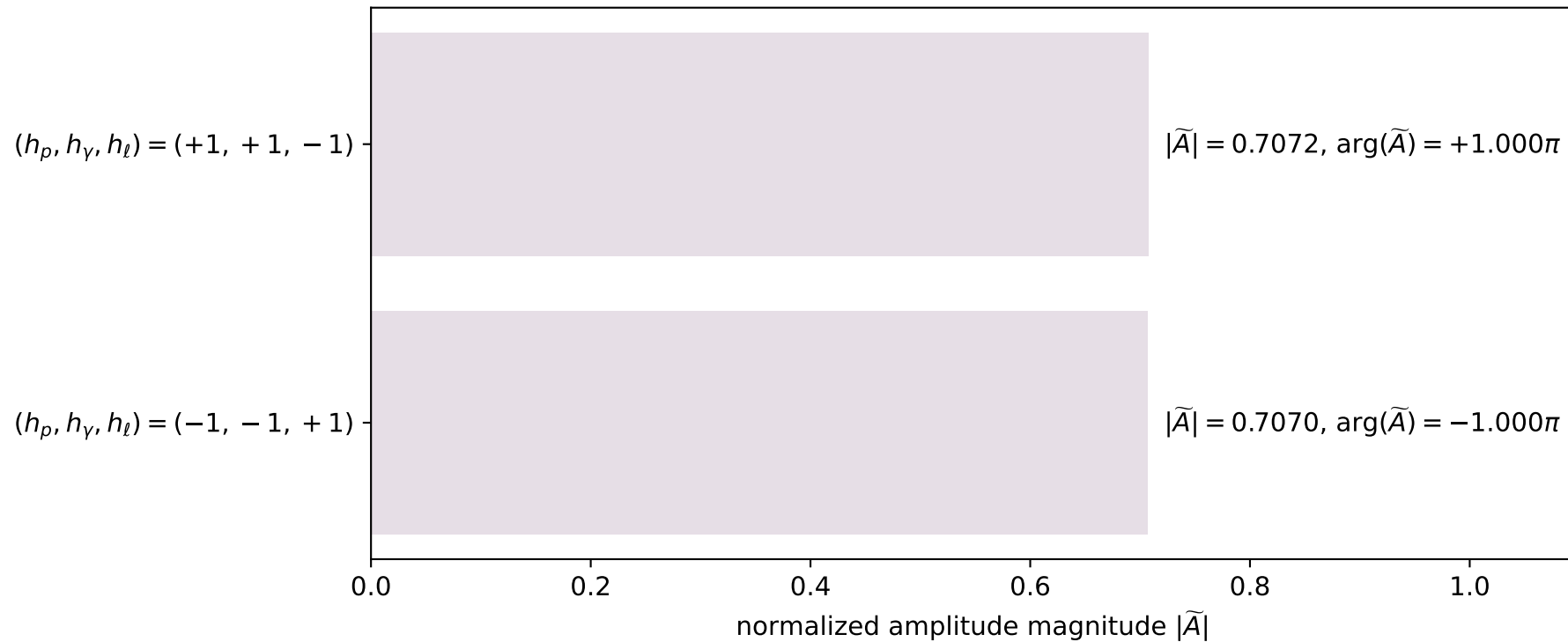
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.96908\text{e-}29$, $\phi_\gamma=1.572$
 $E_\gamma=0.120652$, $\theta_{l'}=0$, $\phi_{l'}=3.14159$
 $Q^2=22.1072$, $x_B=0.948249$, $t=-23.2713$, $W^2=2.08636$, $y=0.966567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1207$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.1207$

x--y plane

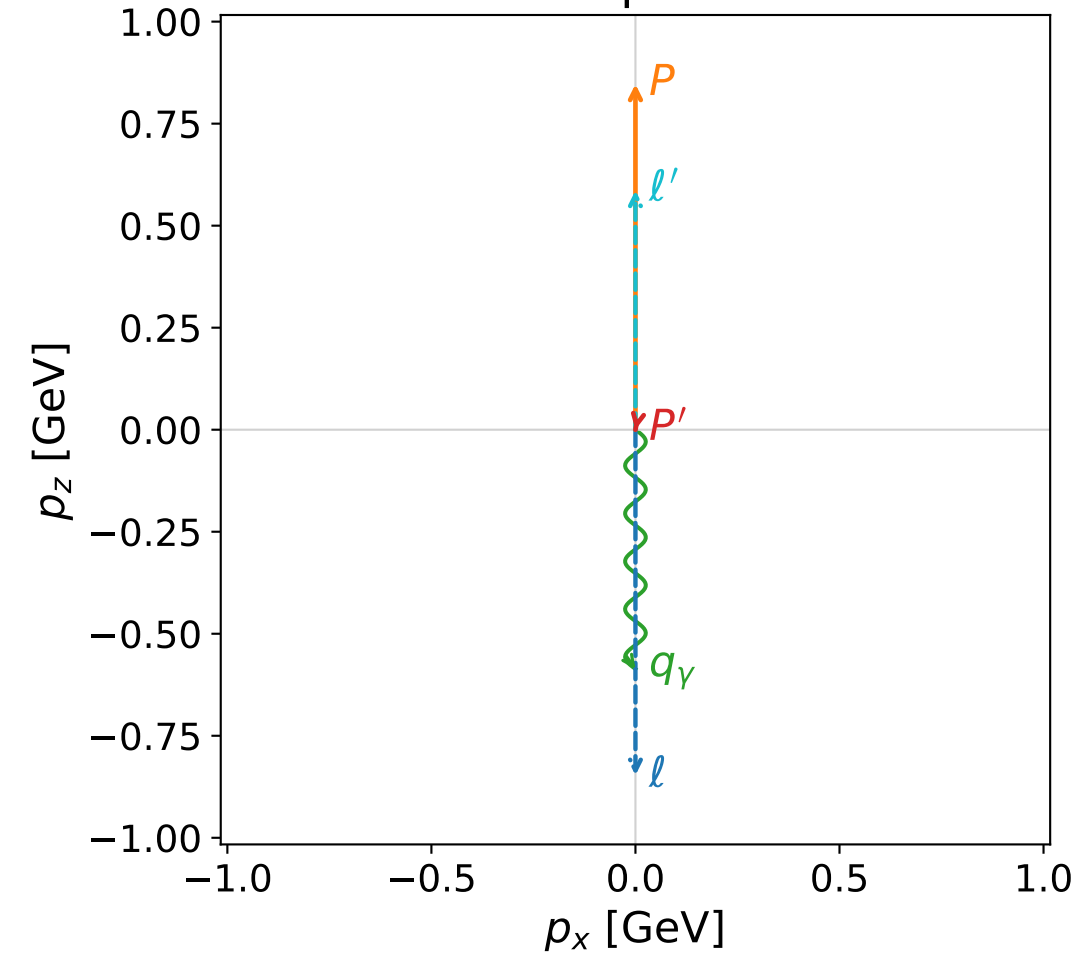


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

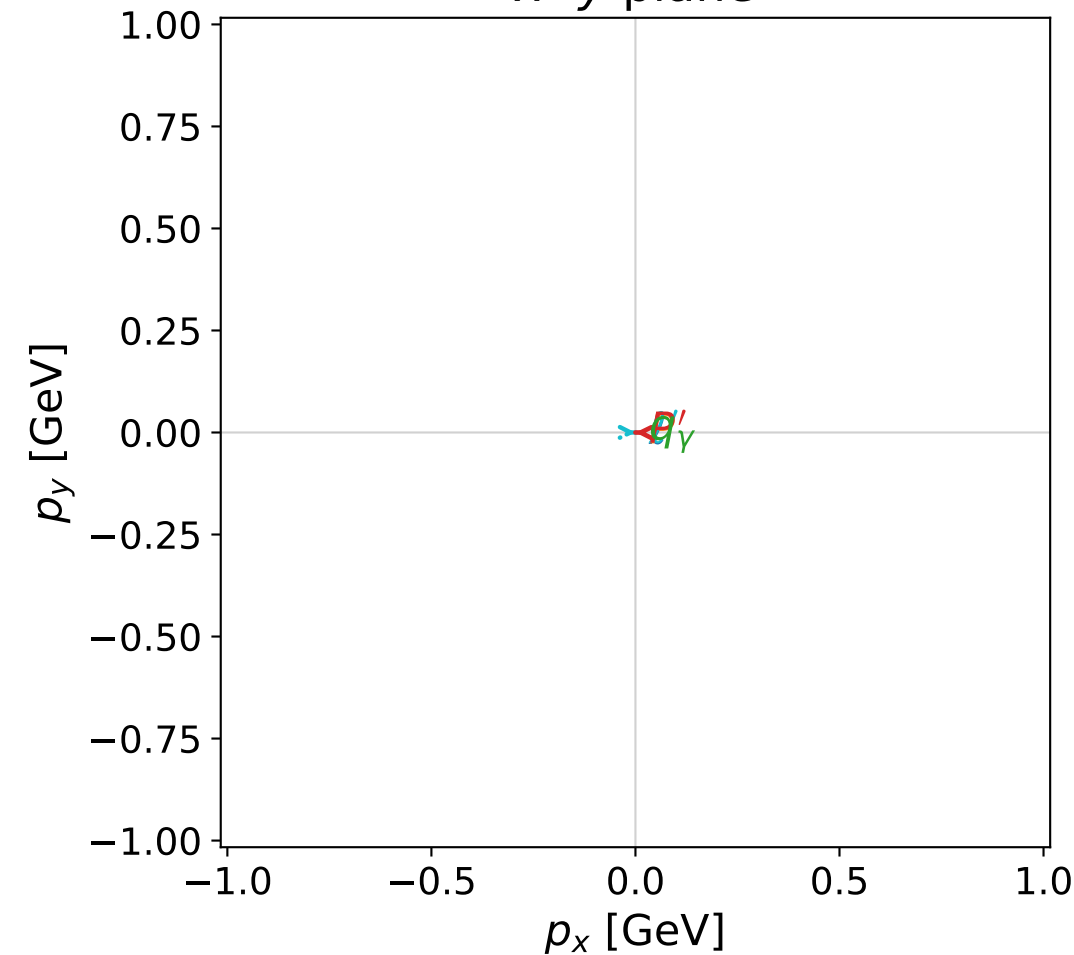


dghz_mixing_angles_polarization_02_local_minimum_182
 $d_{\{\rm GHZ\}}=6.61545\text{e-}07$
 region: polarization_cluster_02_local_minimum_182
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0182
 lepton mixing angle: $\alpha_e=0.2035518$ rad
 incoming lepton: $0.979355|+\rangle +0.202149|-\rangle$
 proton mixing angle: $\alpha_p=2.9380379$ rad
 incoming proton: $-0.979354|+\rangle +0.202152|-\rangle$

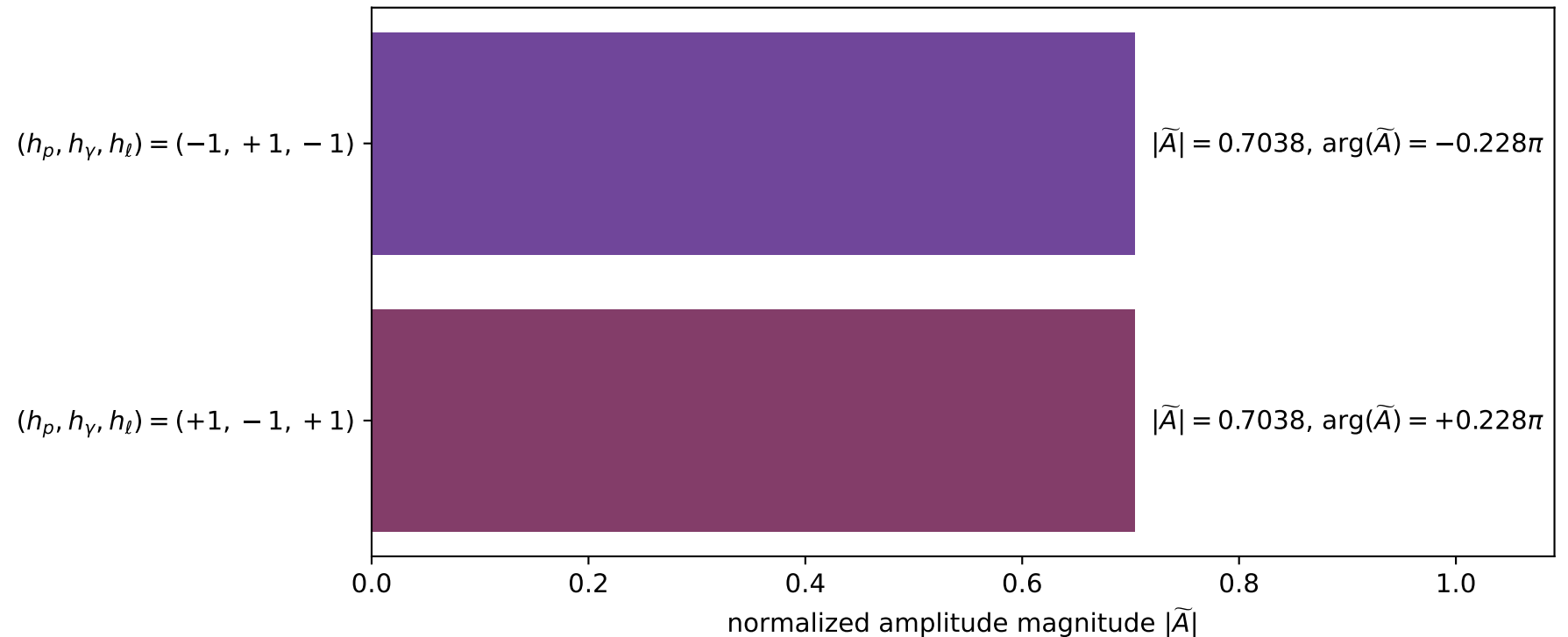
$s=4.45467$, $\sqrt{s}=2.11061$, $|\vec{P}|=0.846871$, $|\vec{P}'|=1.86603\text{e-}05$
 $\theta_{p'}=2.94793$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.13457$, $\phi_\gamma=3.85199$
 $E_\gamma=0.586295$, $\theta_{\ell'}=6.12523\text{e-}06$, $\phi_{\ell'}=6.27616$
 $Q^2=1.98613$, $x_B=0.643594$, $t=-0.611116$, $W^2=1.97971$, $y=0.863258$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.8469$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.8469$
 P $E= 1.2637$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8469$
 ℓ' $E= 0.5863$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.5863$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 0.5863$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.5863$

x--y plane

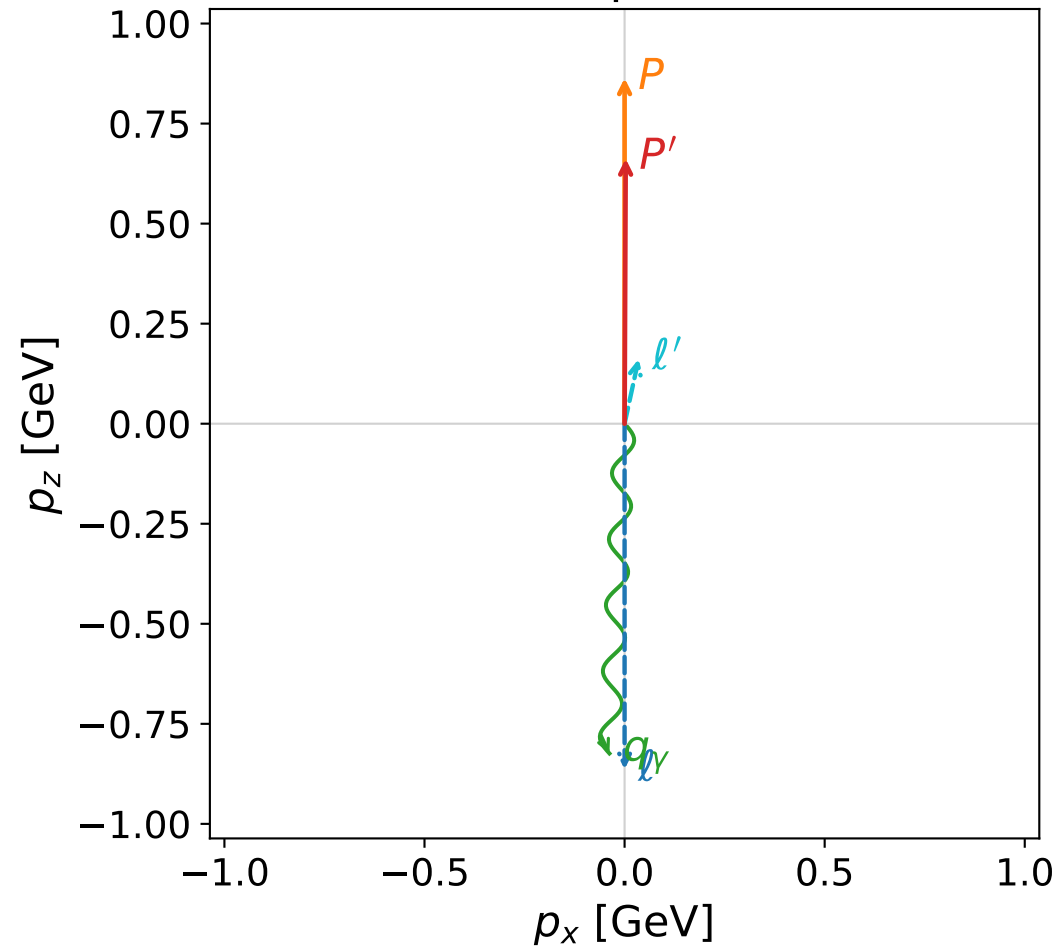


Leading normalized final-state amplitudes
 (N=2, retained=99.1%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_02_local_minimum_185

$d_{\{\rm GHZ\}}=3.76832\text{e-}06$

region: polarization_cluster_02_local_minimum_185

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0185

lepton mixing angle: $\alpha_e=1.2208886$ rad

incoming lepton: $0.342811|+\rangle +0.939404|-\rangle$

proton mixing angle: $\alpha_p=2.8239017$ rad

incoming proton: $-0.949959|+\rangle +0.312374|-\rangle$

$s=4.57376$, $\sqrt{s}=2.13864$, $|\vec{P}|=0.863616$, $|\vec{P}'|=0.662371$

$\theta_{P'}=0.00416983$, $\theta_Y=3.09573$, $\phi_{P'}=6.28319$, $\phi_Y=3.1416$

$E_Y=0.824908$, $\theta_{\ell'}=0.213519$, $\phi_{\ell'}=6.85947\text{e-}06$

$Q^2=0.564992$, $x_B=0.159094$, $t=-0.0244499$, $W^2=3.86616$, $y=0.961393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.8636$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.8636$

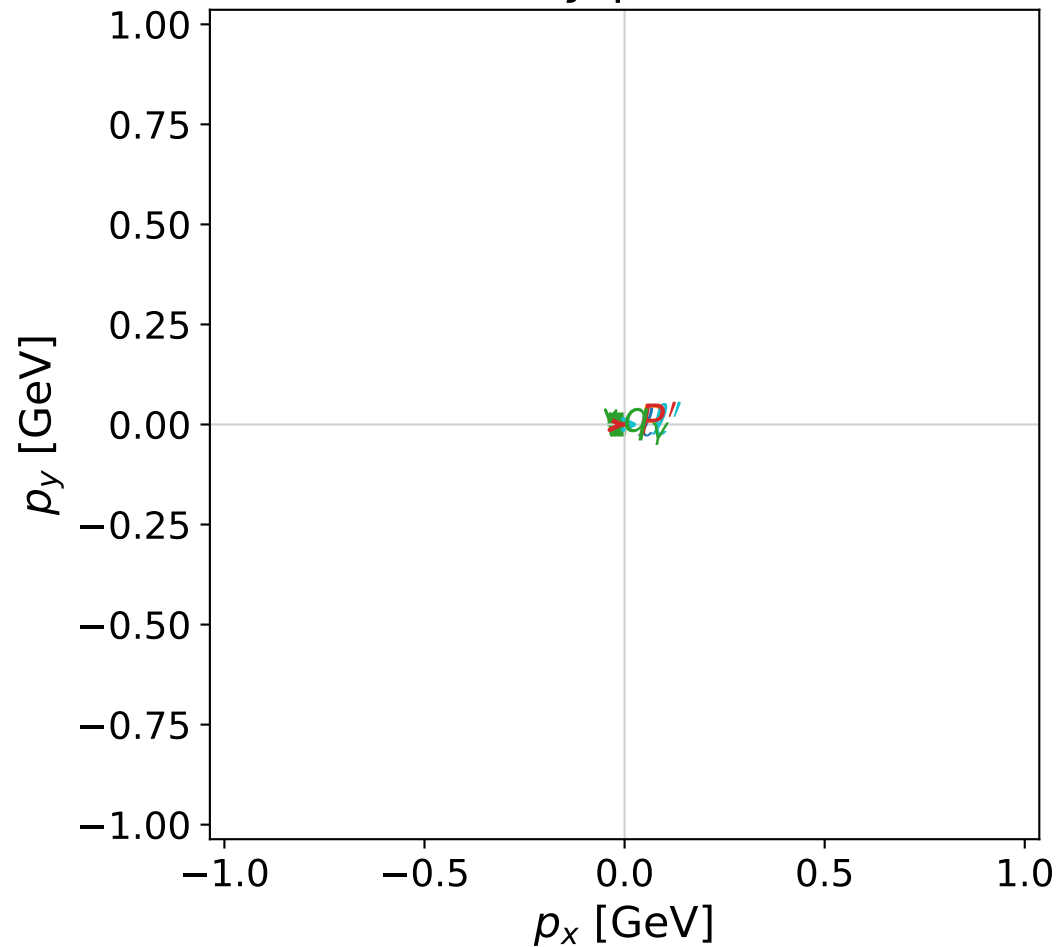
P $E= 1.2750$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8636$

ℓ' $E= 0.1654$ $p_x= 0.0351$ $p_y= 0.0000$ $p_z= 0.1617$

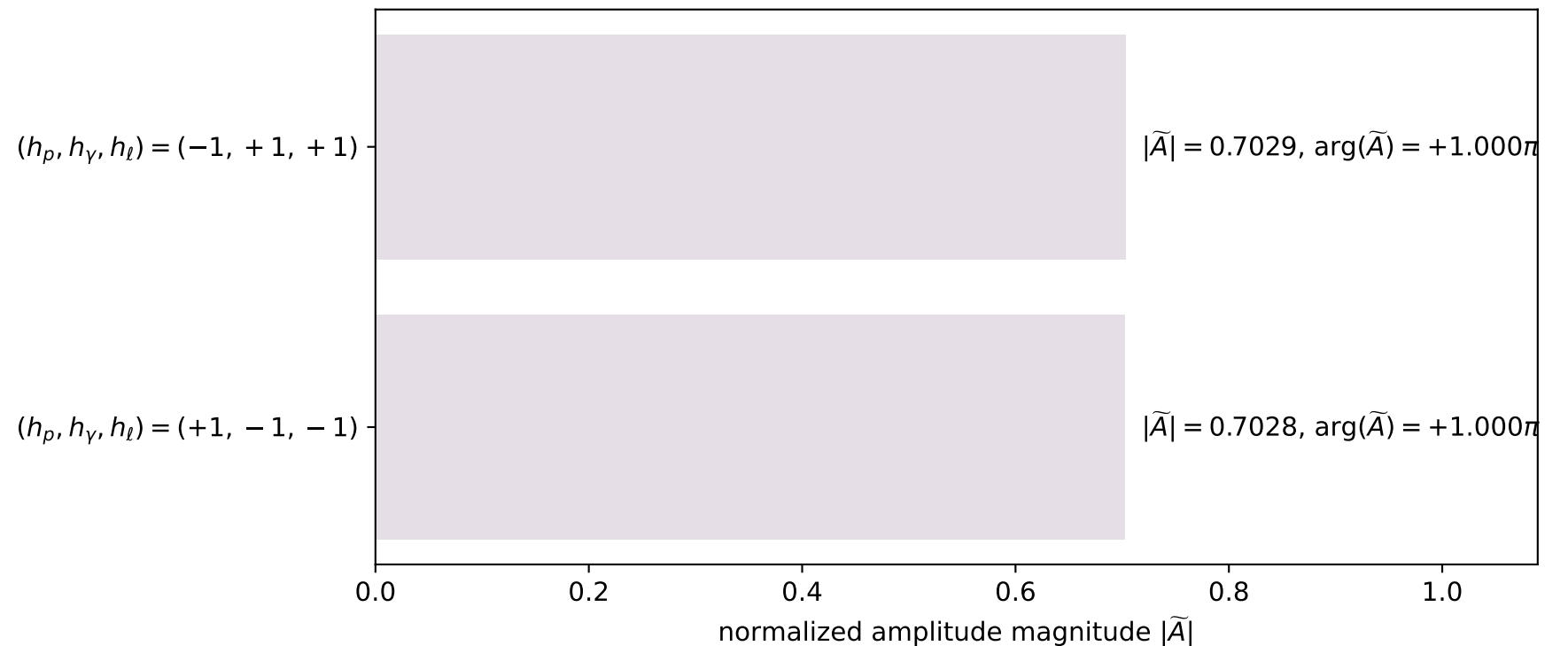
P' $E= 1.1483$ $p_x= 0.0028$ $p_y= -0.0000$ $p_z= 0.6624$

q_Y $E= 0.8249$ $p_x= -0.0378$ $p_y= -0.0000$ $p_z= -0.8240$

x--y plane

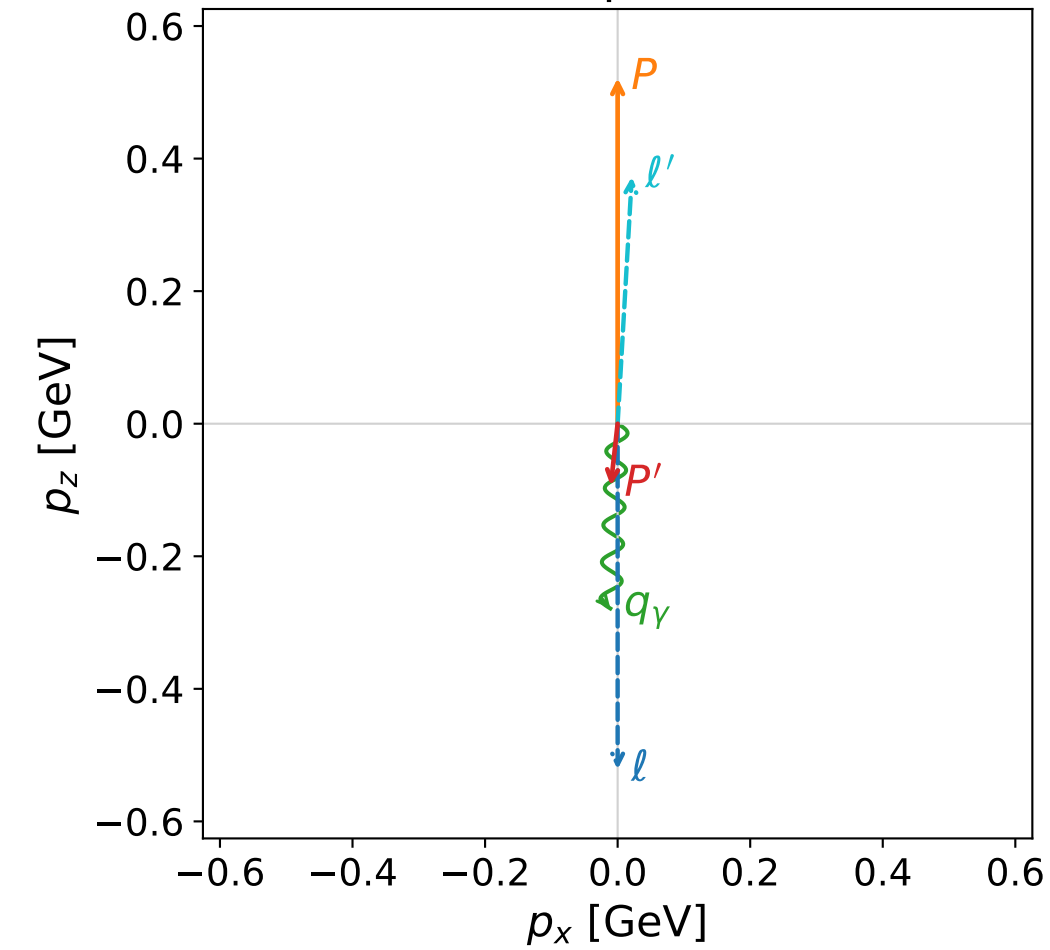


Leading normalized final-state amplitudes
(N=2, retained=98.8%)

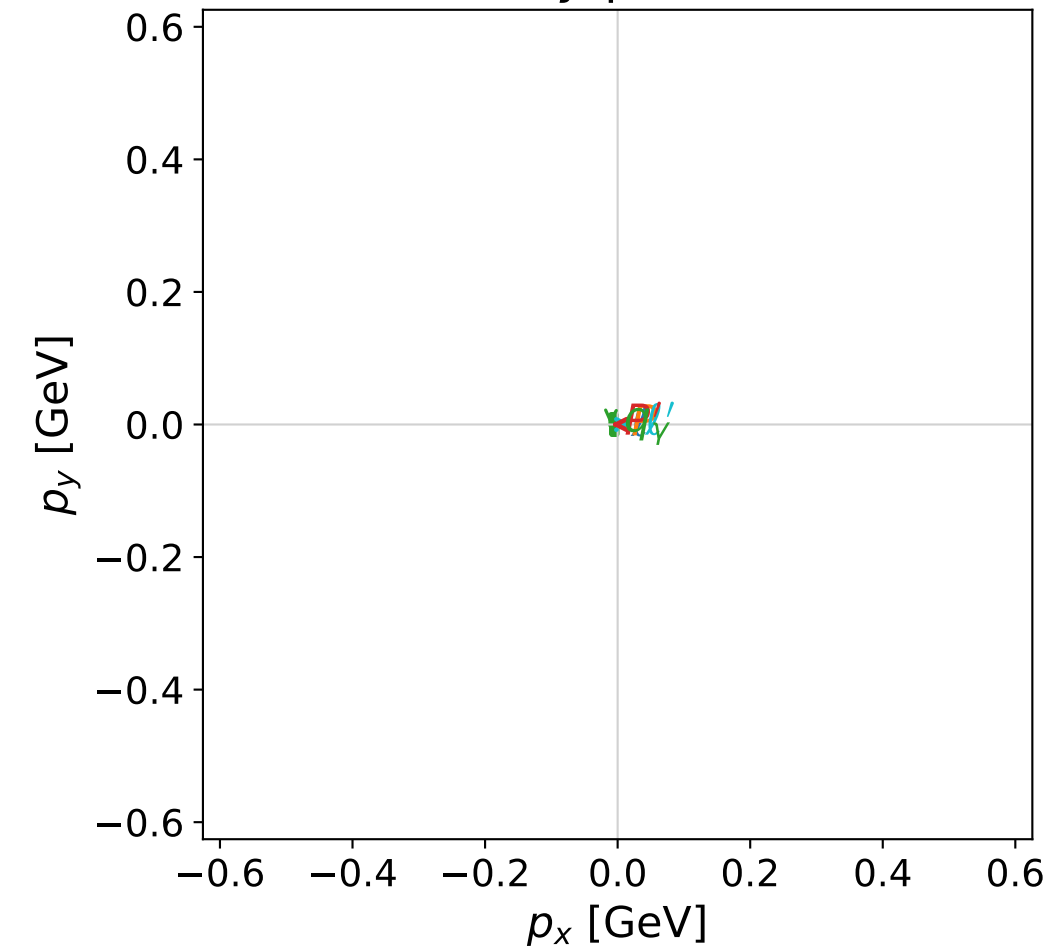


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

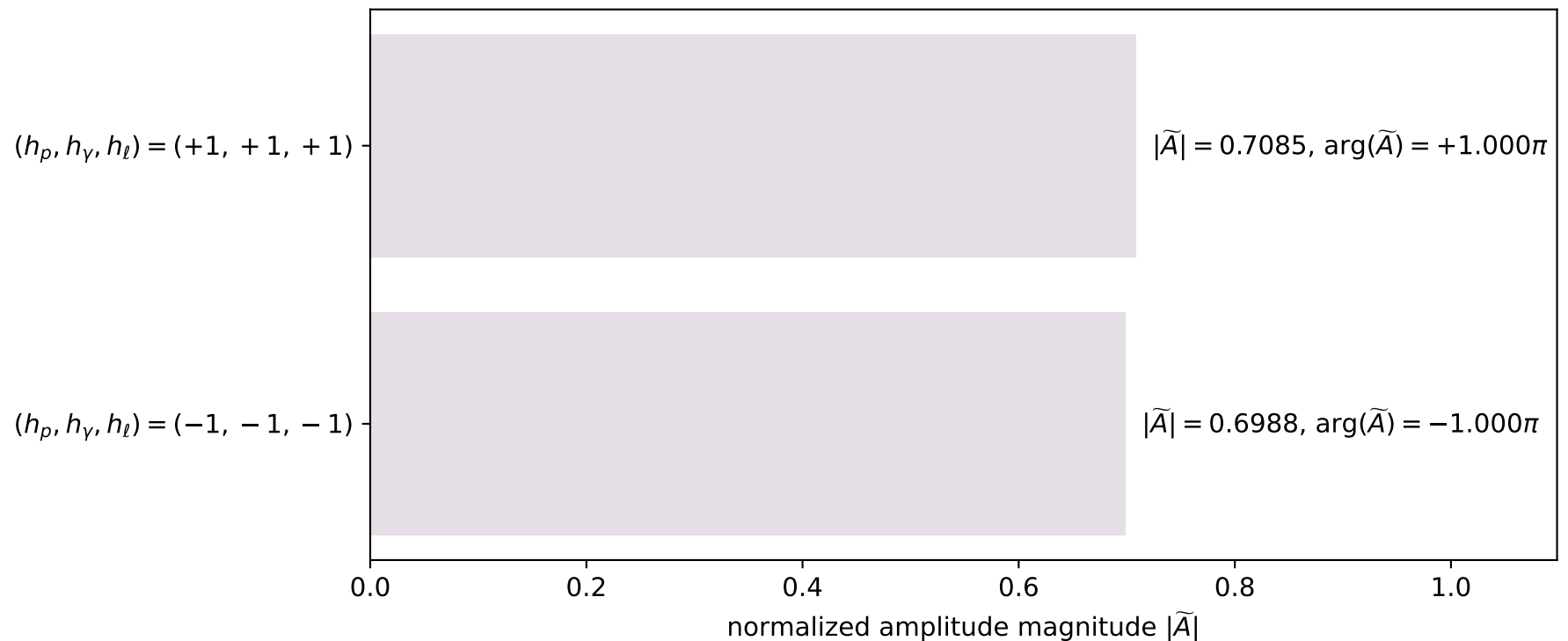


dghz_mixing_angles_polarization_02_local_minimum_286
 $d_{\text{GHZ}}=0.000253567$
 region: polarization_cluster_02_local_minimum_286
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0286
 lepton mixing angle: $\alpha_e=0.023844319$ rad
 incoming lepton: $0.999716|+\rangle +0.0238421|-\rangle$
 proton mixing angle: $\alpha_p=1.5427631$ rad
 incoming proton: $0.0280295|+\rangle +0.999607|-\rangle$

$s=2.54271$, $\sqrt{s}=1.59459$, $|\vec{P}|=0.52141$, $|\vec{P}'|=0.0940246$
 $\theta_{p'}=3.03139$, $\theta_\gamma=3.10212$, $\phi_{p'}=3.1416$, $\phi_\gamma=3.14162$
 $E_\gamma=0.27902$, $\theta_{\ell'}=0.0572926$, $\phi_{\ell'}=1.62633\text{e-}05$
 $Q^2=0.777031$, $x_B=0.621249$, $t=-0.36114$, $W^2=1.35357$, $y=0.752168$

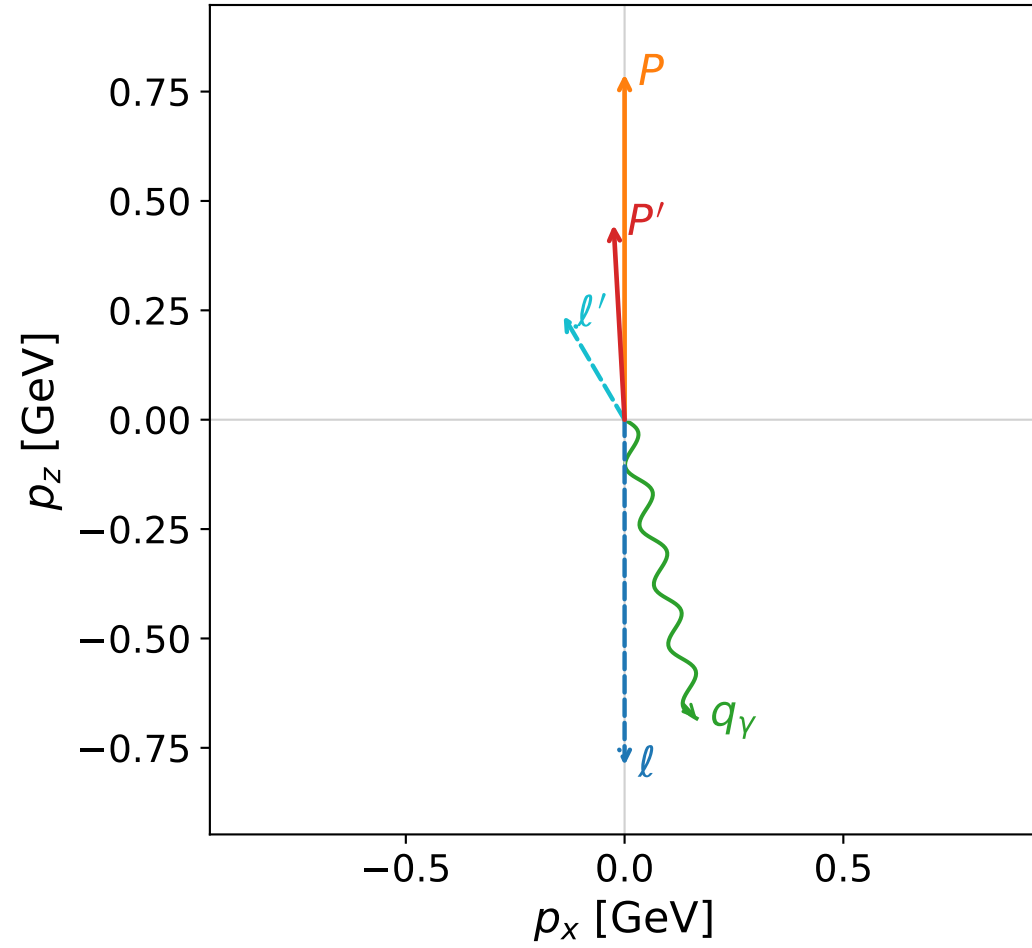
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5214$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5214$
 P $E= 1.0732$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5214$
 ℓ' $E= 0.3729$ $p_x= 0.0214$ $p_y= 0.0000$ $p_z= 0.3723$
 P' $E= 0.9427$ $p_x= -0.0103$ $p_y= -0.0000$ $p_z= -0.0935$
 q_γ $E= 0.2790$ $p_x= -0.0110$ $p_y= -0.0000$ $p_z= -0.2788$

Leading normalized final-state amplitudes
 (N=2, retained=99.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

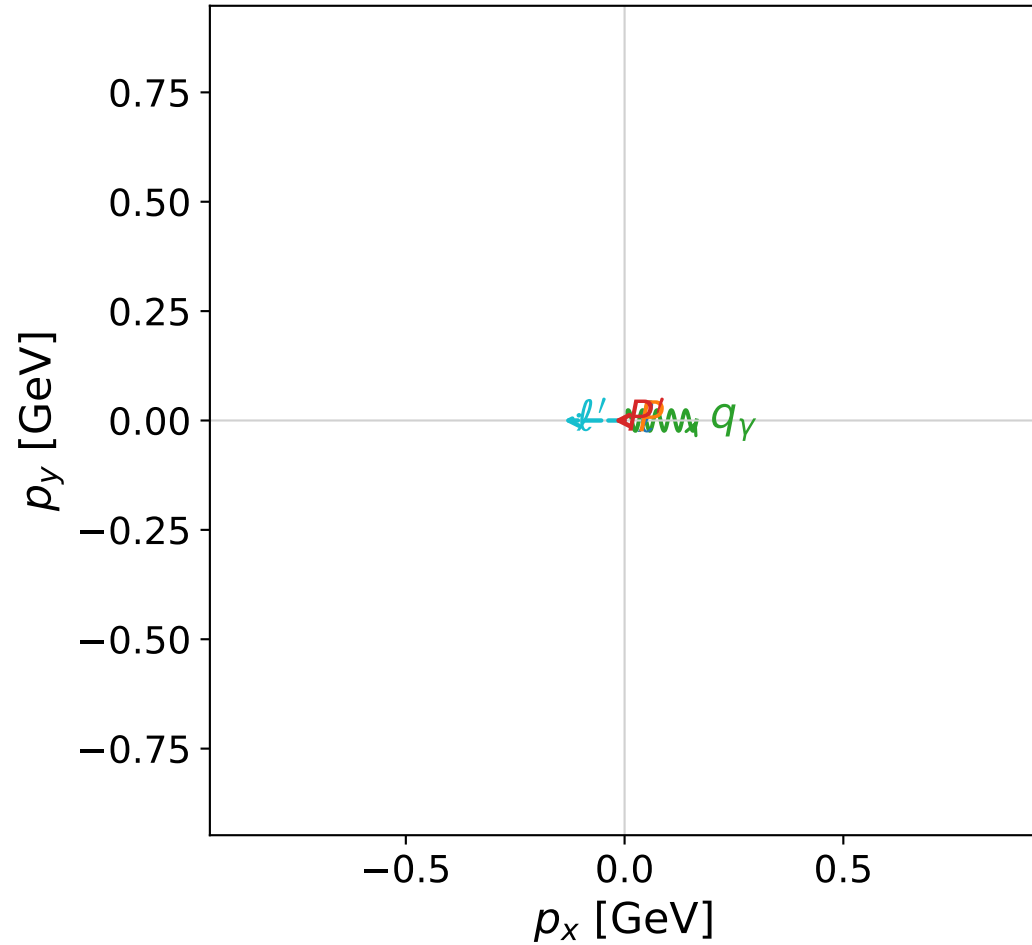


dghz_mixing_angles_polarization_02_local_minimum_296
 $d_{\text{GHZ}}=0.000302995$
 region: polarization_cluster_02_local_minimum_296
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0296
 lepton mixing angle: $\alpha_e=0.55513734$ rad
 incoming lepton: $0.849828|+\rangle +0.52706|-\rangle$
 proton mixing angle: $\alpha_p=2.048015$ rad
 incoming proton: $-0.45931|+\rangle +0.888276|-\rangle$

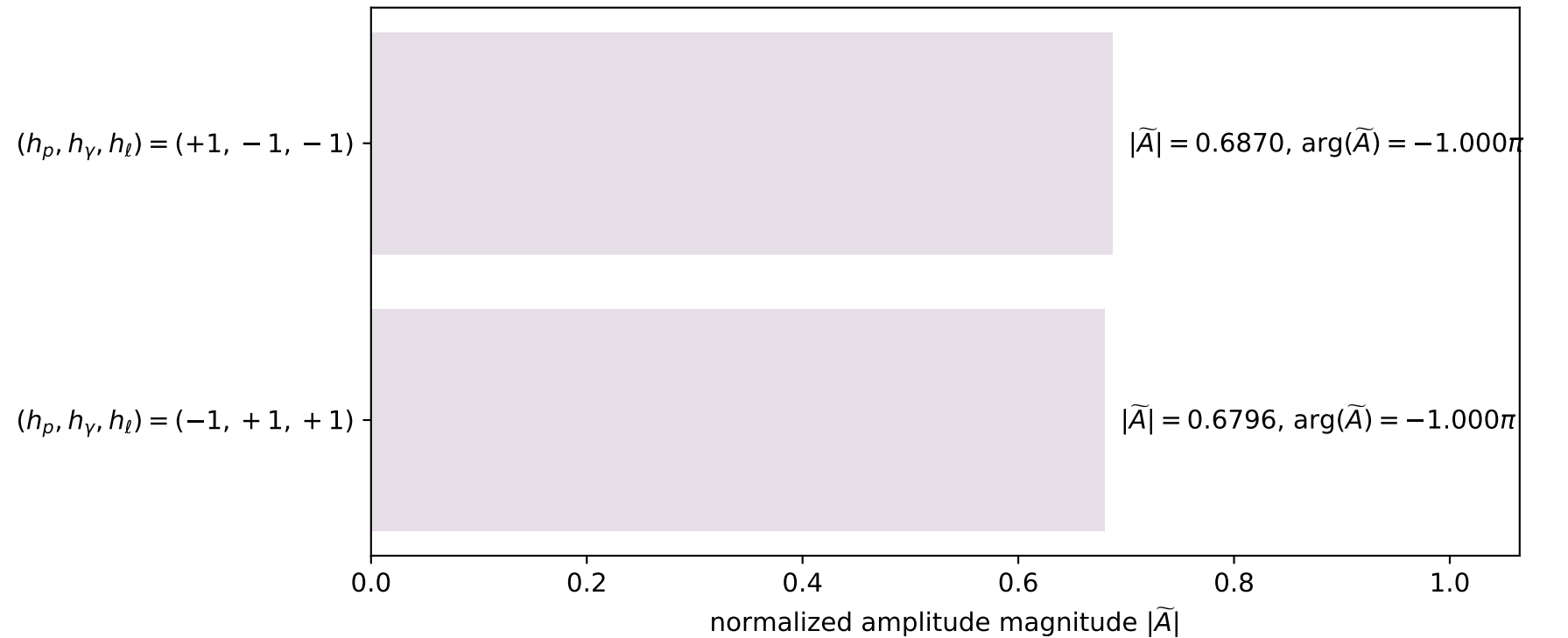
$s=4.06514$, $\sqrt{s}=2.01622$, $|\vec{P}|=0.789919$, $|\vec{P}'|=0.446256$
 $\theta_{P'}=0.0562449$, $\theta_\gamma=2.9051$, $\phi_{P'}=3.14149$, $\phi_\gamma=6.28313$
 $E_\gamma=0.702308$, $\theta_{l'}=0.531494$, $\phi_{l'}=3.14154$
 $Q^2=0.809472$, $x_B=0.280563$, $t=-0.0840407$, $W^2=2.95554$, $y=0.905777$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.7899$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7899$
 P $E= 1.2263$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7899$
 l' $E= 0.2752$ $p_x= -0.1395$ $p_y= 0.0000$ $p_z= 0.2372$
 P' $E= 1.0387$ $p_x= -0.0251$ $p_y= 0.0000$ $p_z= 0.4456$
 q_γ $E= 0.7023$ $p_x= 0.1645$ $p_y= -0.0000$ $p_z= -0.6828$

x--y plane

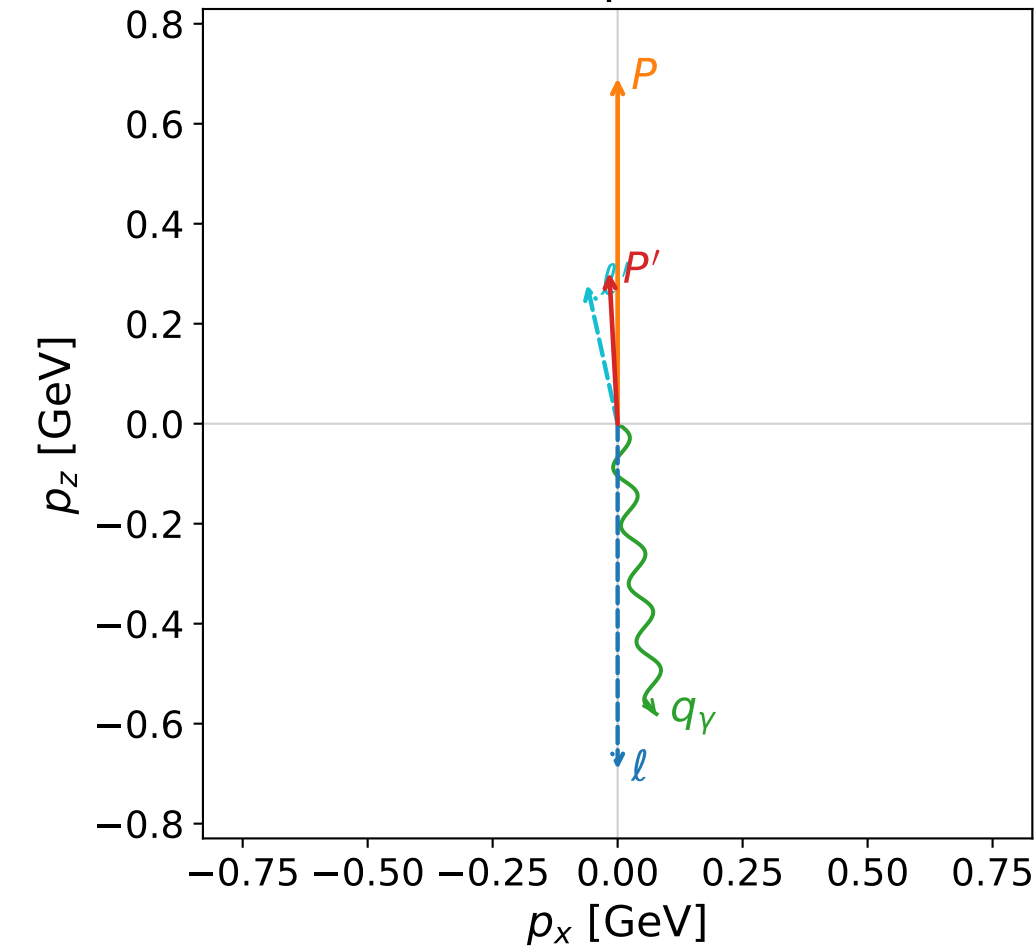


Leading normalized final-state amplitudes
 (N=2, retained=93.4%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

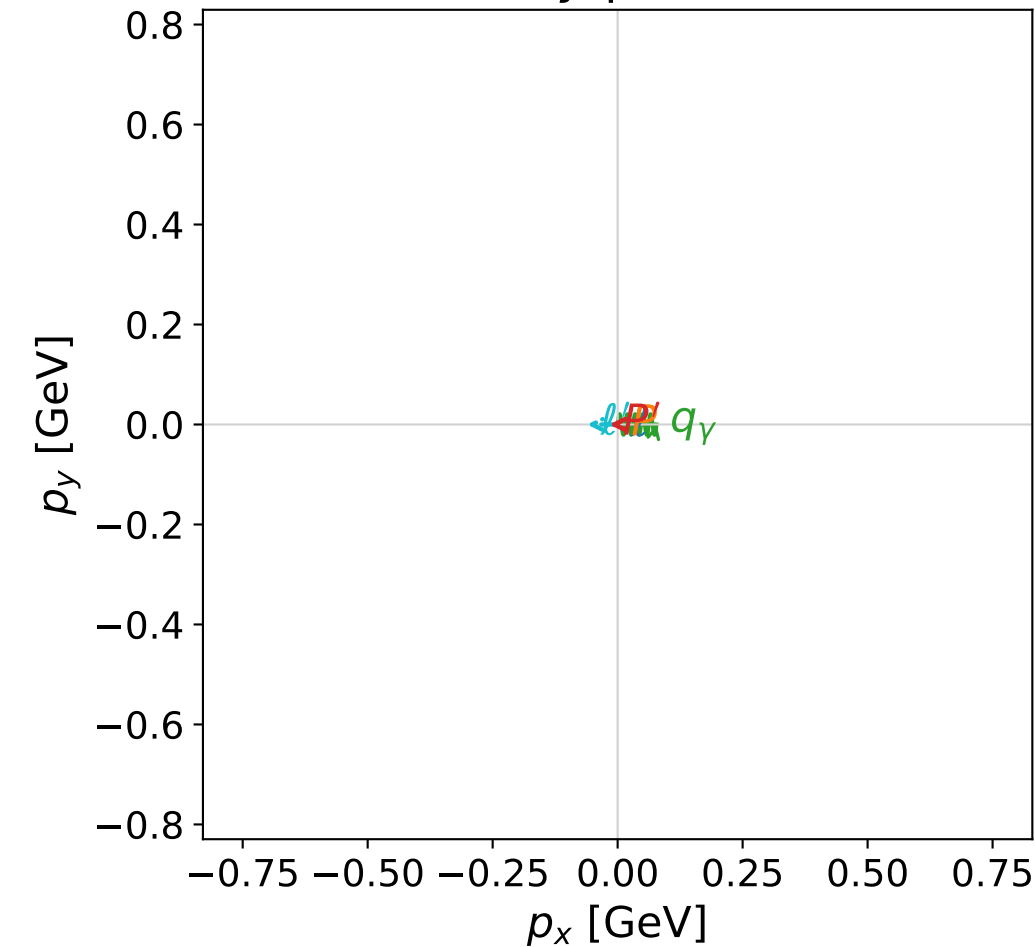


dghz_mixing_angles_polarization_02_local_minimum_308
 $d_{\text{GHZ}}=0.000352127$
 region: polarization_cluster_02_local_minimum_308
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0308
 lepton mixing angle: $\alpha_e=0.20616489$ rad
 incoming lepton: $0.978823|+\rangle +0.204708|-\rangle$
 proton mixing angle: $\alpha_p=1.7420637$ rad
 incoming proton: $-0.170431|+\rangle +0.98537|-\rangle$

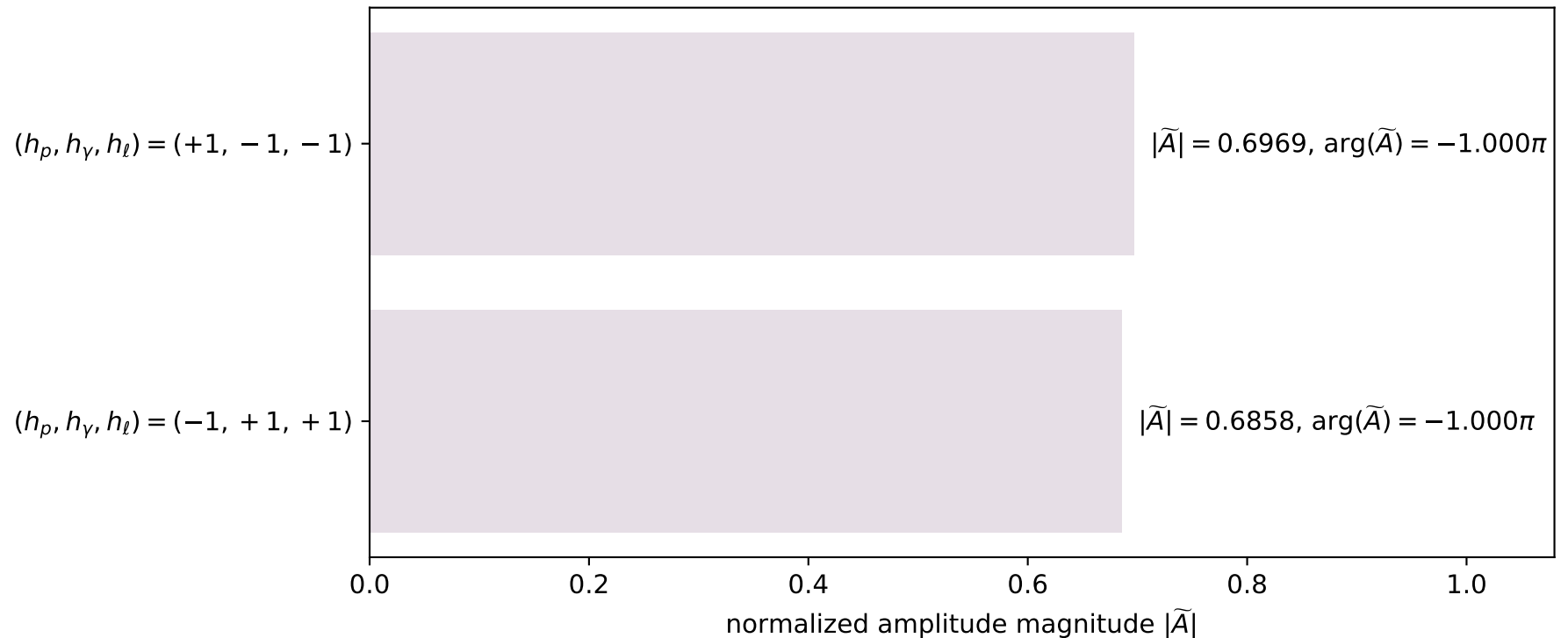
$s=3.44654$, $\sqrt{s}=1.85649$, $|\vec{P}|=0.691278$, $|\vec{P}'|=0.302862$
 $\theta_{p'}=0.0554148$, $\theta_\gamma=3.00846$, $\phi_{p'}=3.1414$, $\phi_\gamma=6.28312$
 $E_\gamma=0.58589$, $\theta_{l'}=0.215754$, $\phi_{l'}=3.14156$
 $Q^2=0.778686$, $x_B=0.340408$, $t=-0.11928$, $W^2=2.38866$, $y=0.891225$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.6913$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.6913$
 P $E=1.1652$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.6913$
 l' $E=0.2849$ $p_x=-0.0610$ $p_y=0.0000$ $p_z=0.2783$
 P' $E=0.9857$ $p_x=-0.0168$ $p_y=0.0000$ $p_z=0.3024$
 q_γ $E=0.5859$ $p_x=0.0778$ $p_y=-0.0000$ $p_z=-0.5807$

x--y plane

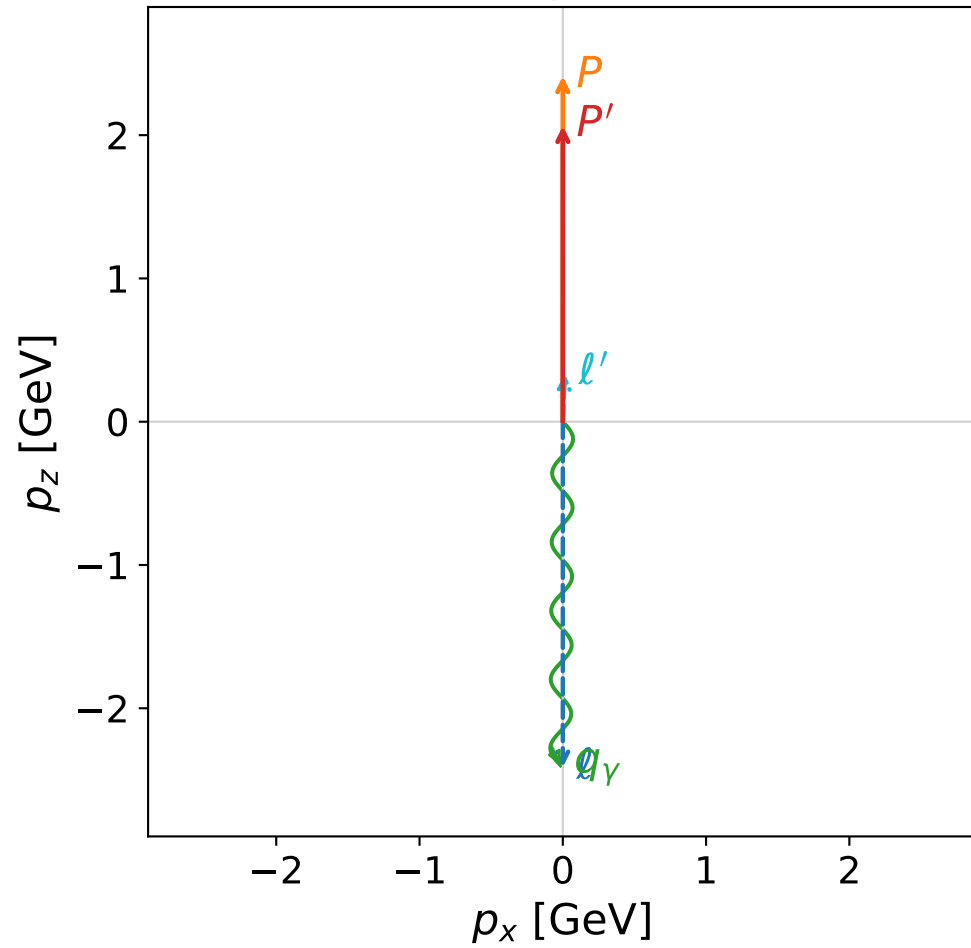


Leading normalized final-state amplitudes
 (N=2, retained=95.6%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

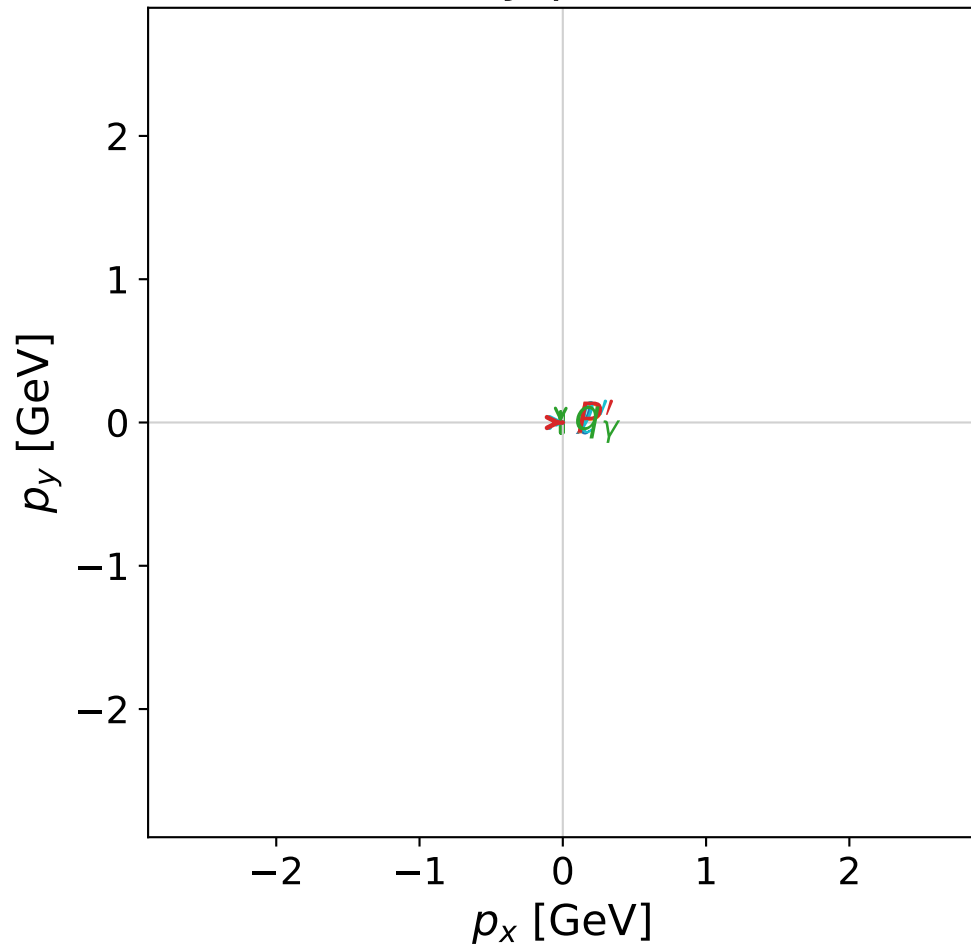


dghz_mixing_angles_polarization_02_local_minimum_323
 $d_{\{\rm GHZ\}}=0.000435487$
 region: polarization_cluster_02_local_minimum_323
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0323
 lepton mixing angle: $\alpha_e=0.69364314$ rad
 incoming lepton: $0.768922|+\rangle + 0.639343|-\rangle$
 proton mixing angle: $\alpha_p=2.2770928$ rad
 incoming proton: $-0.649021|+\rangle + 0.760771|-\rangle$

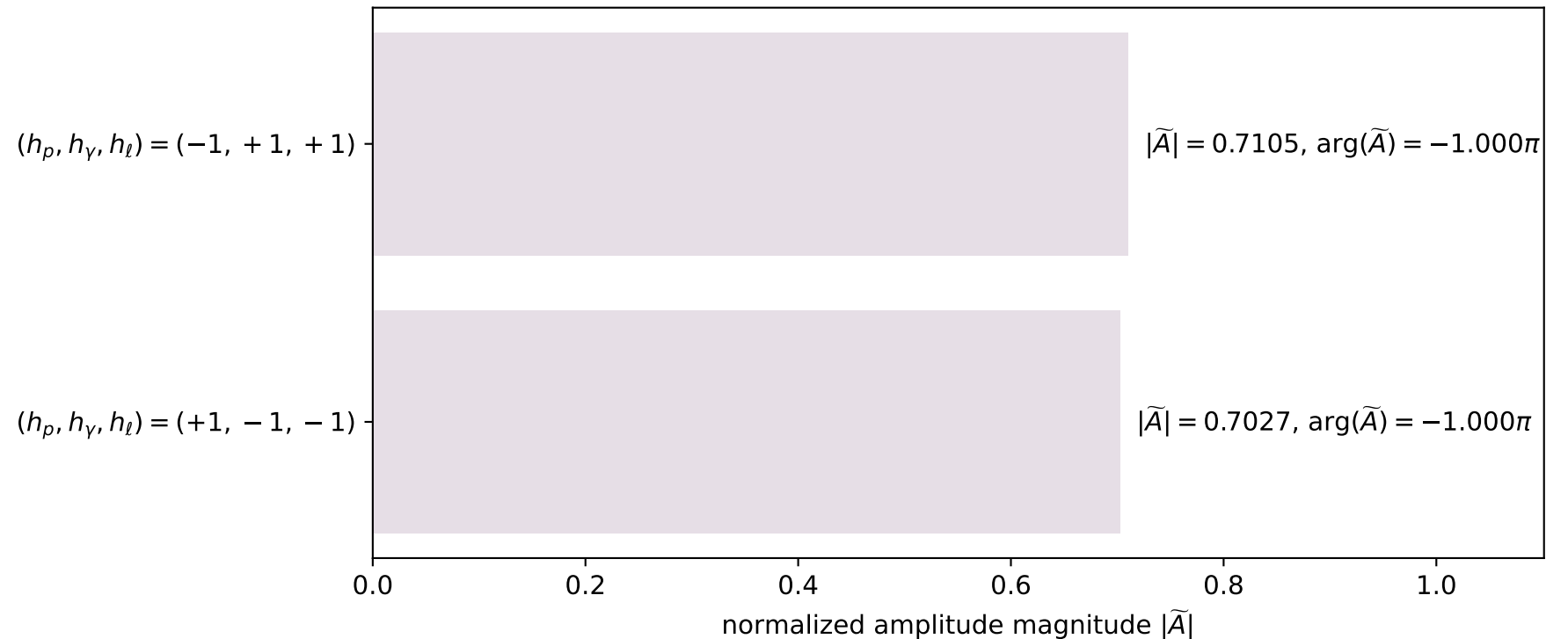
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.06443$
 $\theta_{p'}=0.000218156$, $\theta_\gamma=3.13499$, $\phi_{p'}=0.00267229$, $\phi_\gamma=3.14155$
 $E_\gamma=2.3983$, $\theta_{l'}=0.0460371$, $\phi_{l'}=6.28306$
 $Q^2=3.22234$, $x_B=0.134259$, $t=-0.0181278$, $W^2=21.6583$, $y=0.995053$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 0.3342$ $p_x= 0.0154$ $p_y= -0.0000$ $p_z= 0.3338$
 P' $E= 2.2675$ $p_x= 0.0005$ $p_y= 0.0000$ $p_z= 2.0644$
 q_γ $E= 2.3983$ $p_x= -0.0158$ $p_y= 0.0000$ $p_z= -2.3982$

x--y plane

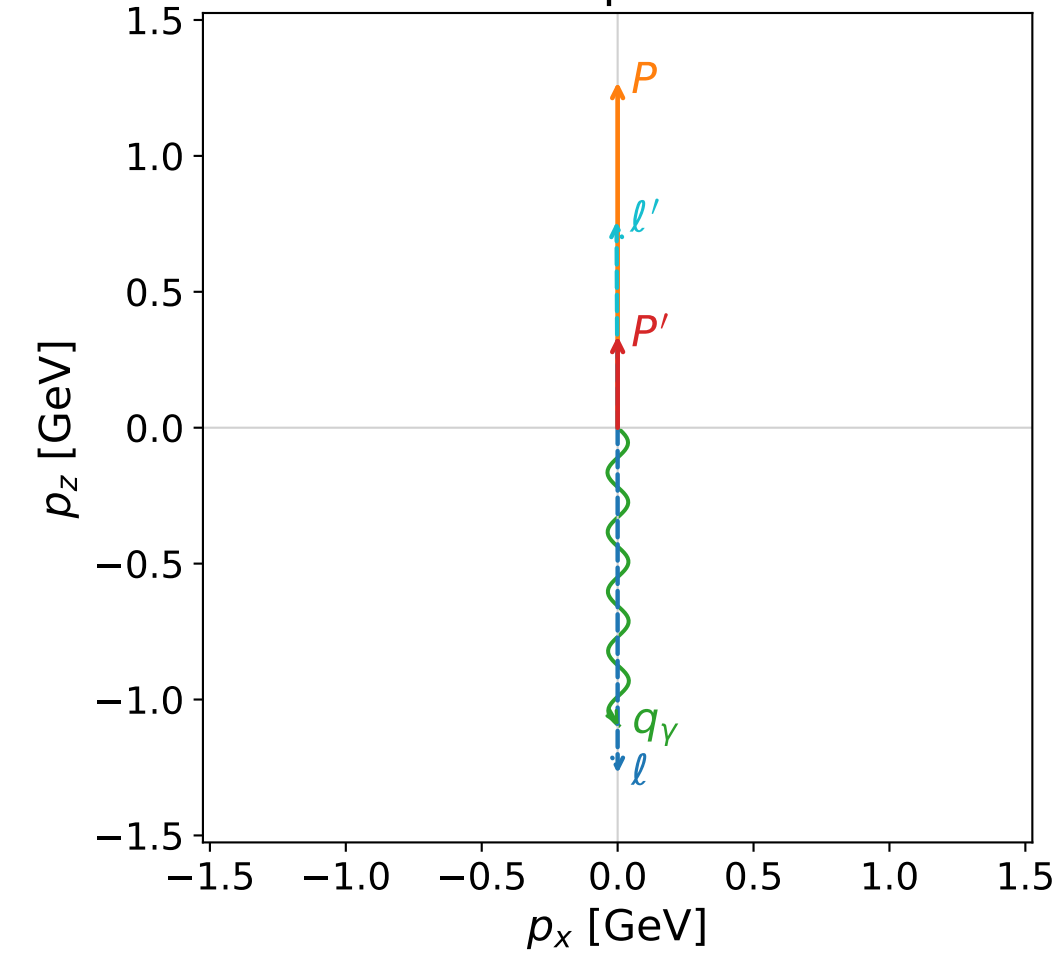


Leading normalized final-state amplitudes
(N=2, retained=99.9%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



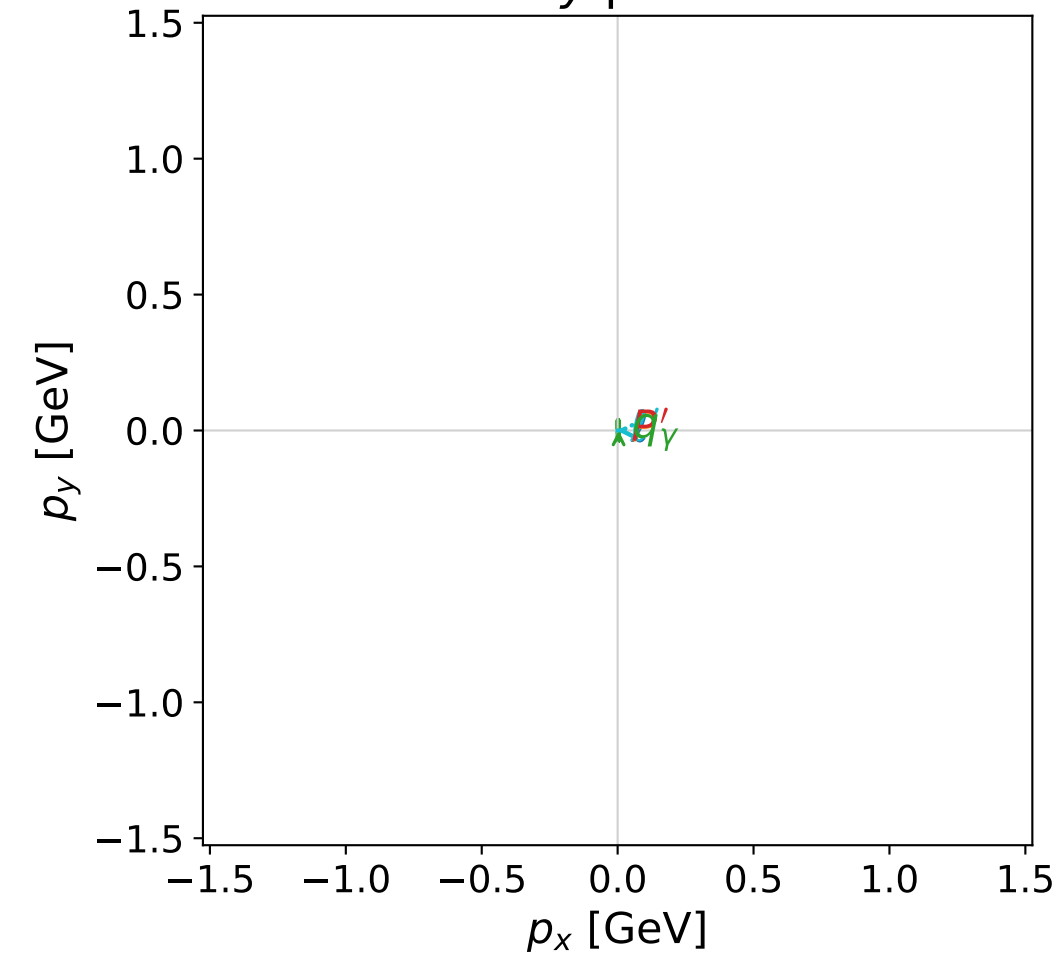
dghz_mixing_angles_polarization_02_local_minimum_339
 $d_{\text{GHZ}}=0.000560516$
 region: polarization_cluster_02_local_minimum_339
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0339
 lepton mixing angle: $\alpha_e=1.2628387$ rad
 incoming lepton: $0.303113|+\rangle + 0.952955|-\rangle$
 proton mixing angle: $\alpha_p=2.8313726$ rad
 incoming proton: $-0.952266|+\rangle + 0.305268|-\rangle$

$s=8.13048$, $\sqrt{s}=2.8514$, $|\vec{P}|=1.27142$, $|\vec{P}'|=0.337119$
 $\theta_{P'}=0$, $\theta_Y=3.13807$, $\phi_{P'}=5.79128$, $\phi_Y=2.46636e-05$
 $E_Y=1.09589$, $\theta_{\ell'}=0.005094$, $\phi_{\ell'}=3.14162$
 $Q^2=3.85883$, $x_B=0.568952$, $t=-0.532742$, $W^2=3.80336$, $y=0.935414$

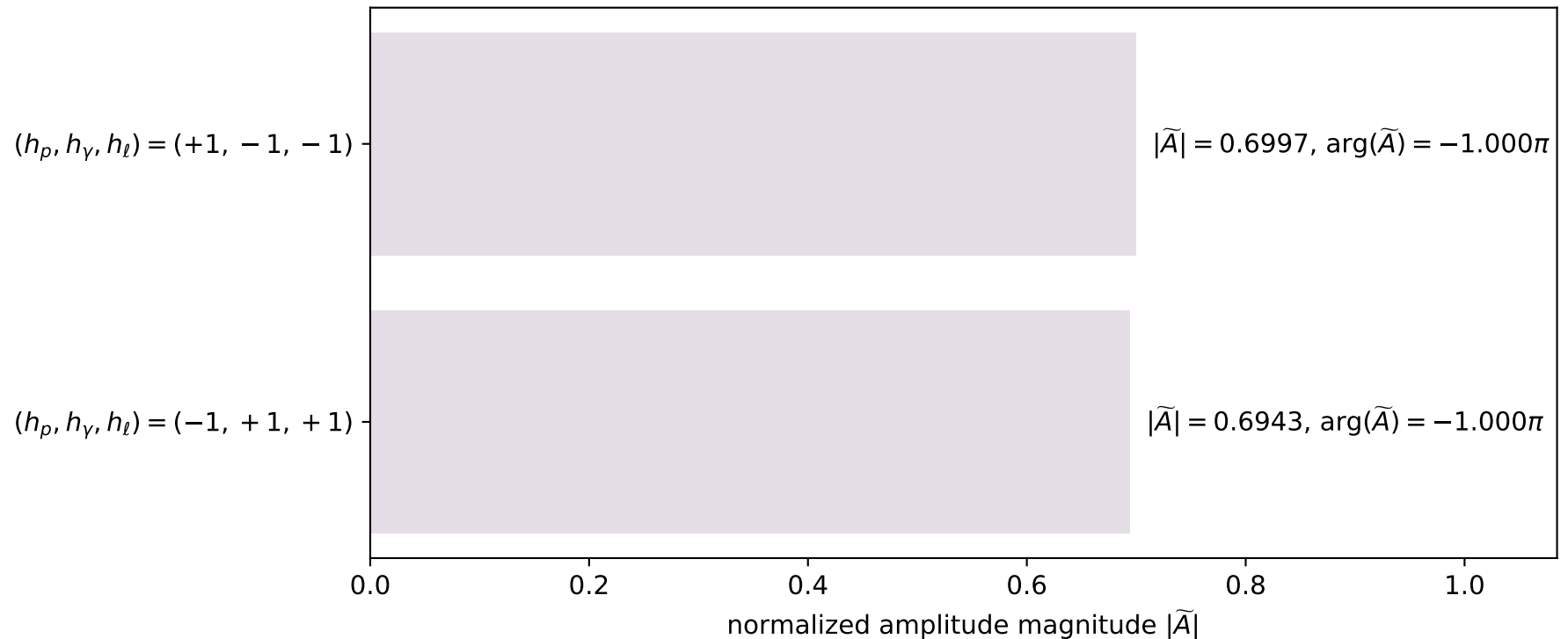
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	1.2714	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-1.2714
P	$E=$	1.5800	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	1.2714
ℓ'	$E=$	0.7588	$p_x=$	-0.0039	$p_y=$	-0.0000	$p_z=$	0.7588
P'	$E=$	0.9967	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	0.3371
q_Y	$E=$	1.0959	$p_x=$	0.0039	$p_y=$	0.0000	$p_z=$	-1.0959

x--y plane

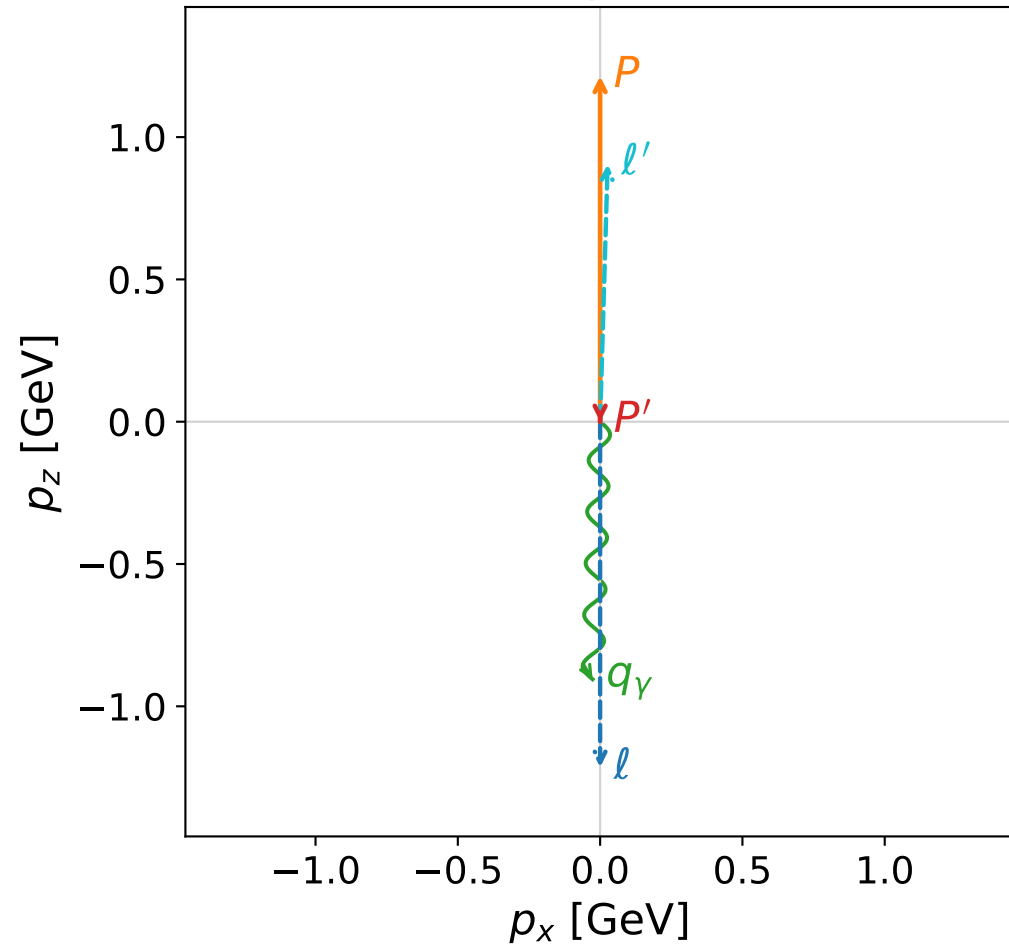


Leading normalized final-state amplitudes
(N=2, retained=97.2%)

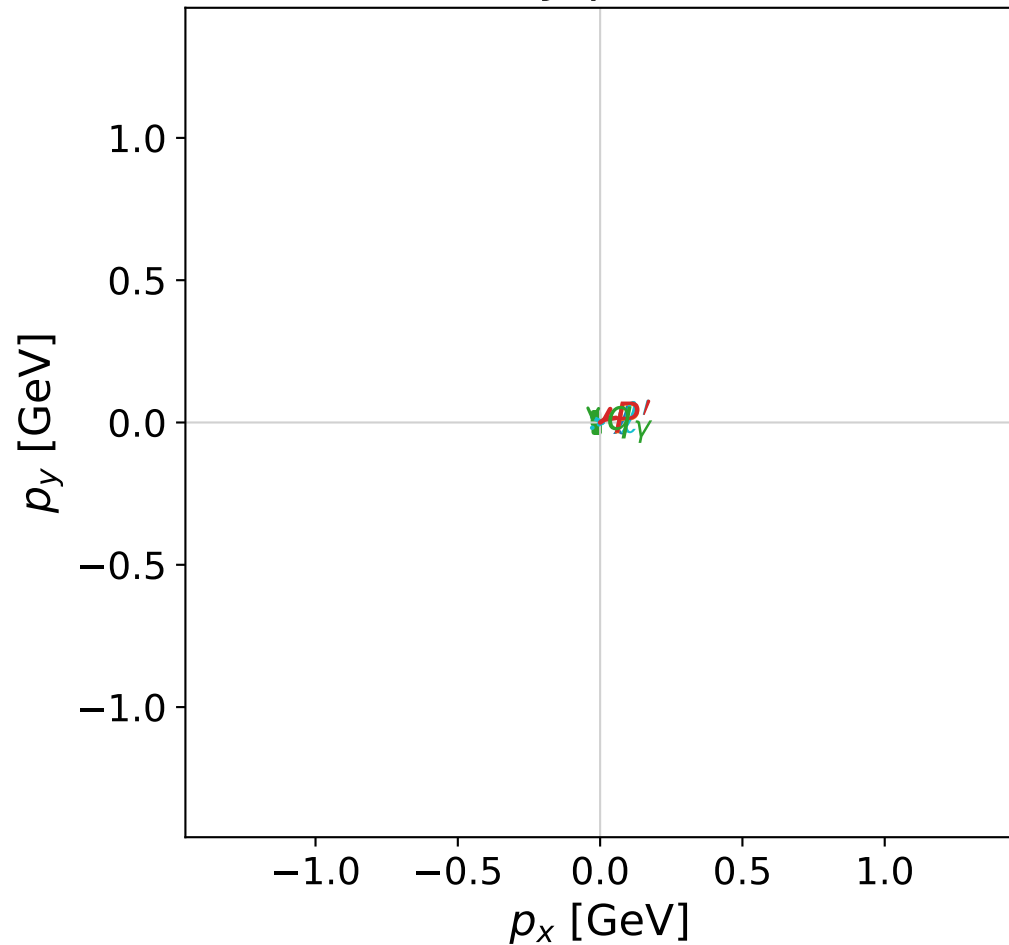


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

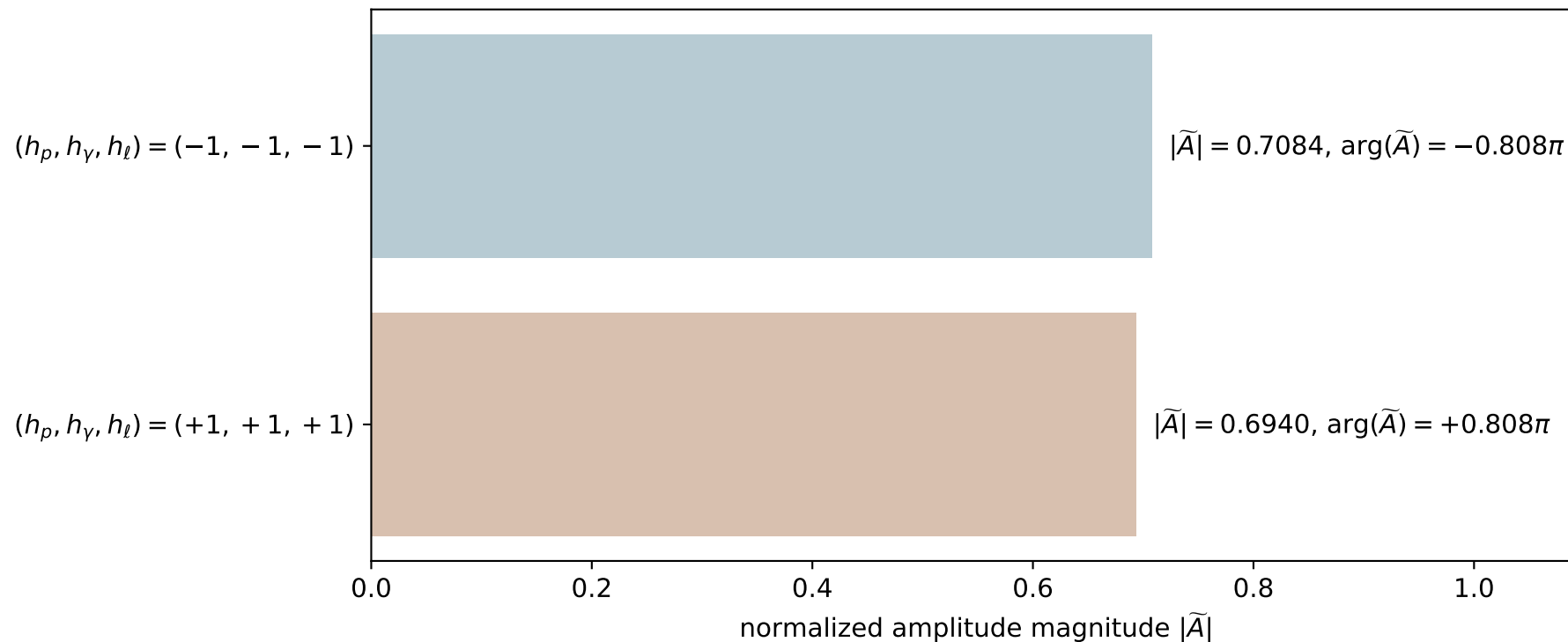


dghz_mixing_angles_polarization_02_local_minimum_354
 $d_{\text{GHZ}}=0.0010861$
 region: polarization_cluster_02_local_minimum_354
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0354
 lepton mixing angle: $\alpha_e=0.027969497$ rad
 incoming lepton: $0.999609|+\rangle + 0.0279659|-\rangle$
 proton mixing angle: $\alpha_p=1.5433927$ rad
 incoming proton: $0.0274002|+\rangle + 0.999625|-\rangle$

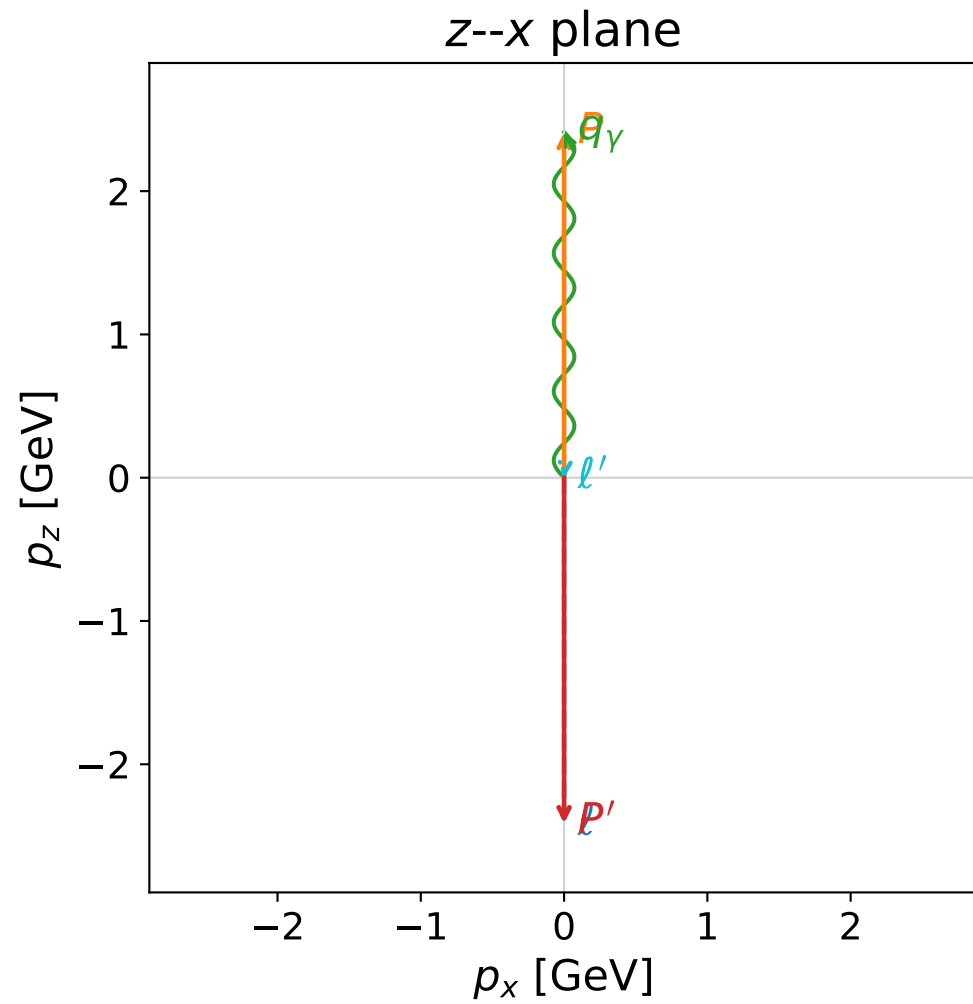
$s=7.55541$, $\sqrt{s}=2.74871$, $|\vec{P}|=1.21431$, $|\vec{P}'|=4.5944\text{e-}06$
 $\theta_{P'}=3.14154$, $\theta_\gamma=3.11242$, $\phi_{P'}=3.74475$, $\phi_\gamma=3.14156$
 $E_\gamma=0.905353$, $\theta_{\ell'}=0.0291745$, $\phi_{\ell'}=6.28315$
 $Q^2=4.3966$, $x_B=0.721341$, $t=-1.11886$, $W^2=2.57828$, $y=0.913036$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2143$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2143$
 P $E= 1.5344$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2143$
 ℓ' $E= 0.9054$ $p_x= 0.0264$ $p_y= -0.0000$ $p_z= 0.9050$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_γ $E= 0.9054$ $p_x= -0.0264$ $p_y= 0.0000$ $p_z= -0.9050$

Leading normalized final-state amplitudes
 (N=2, retained=98.3%)



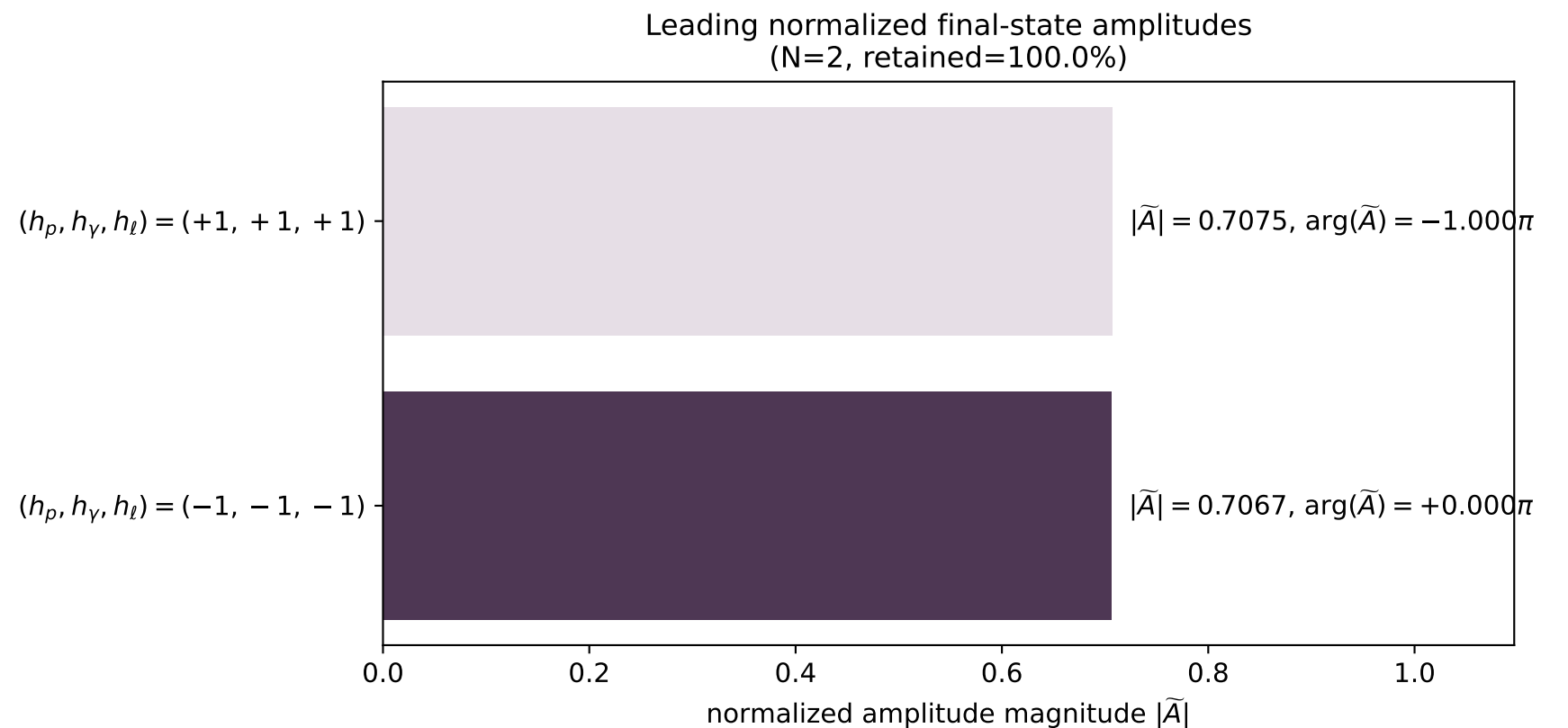
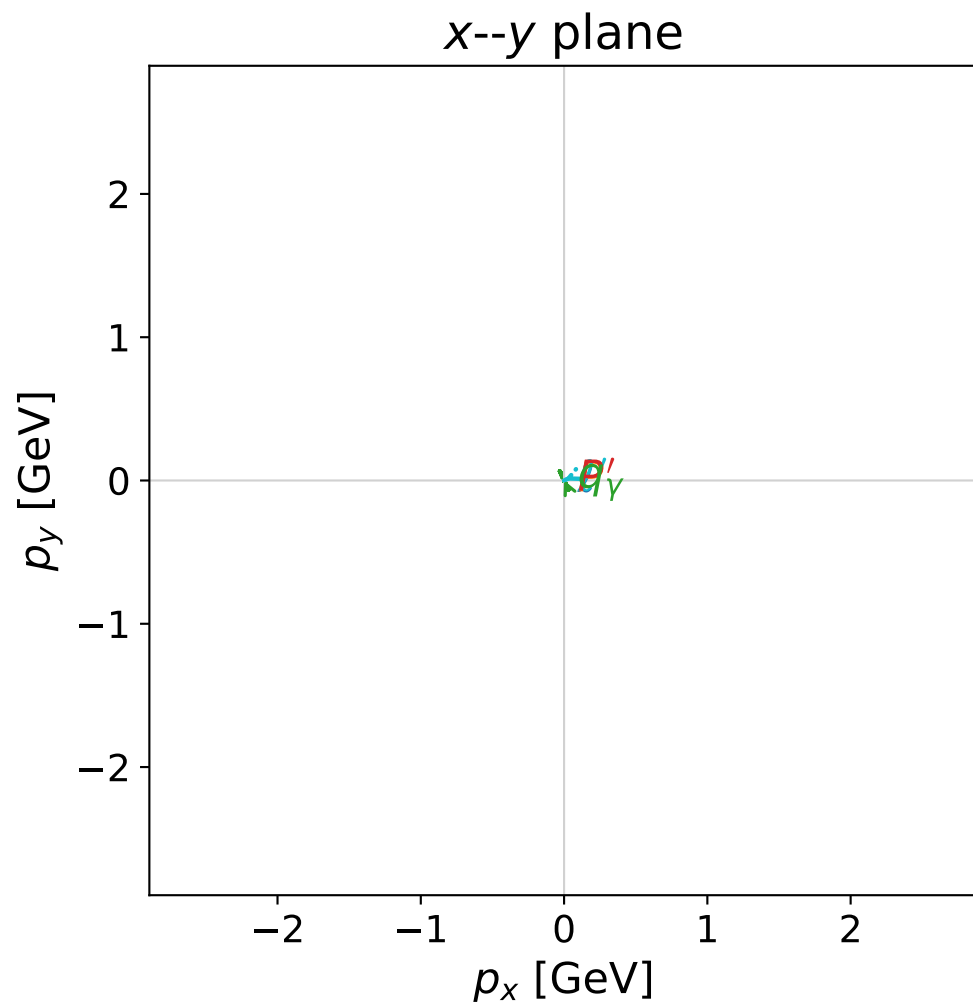
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



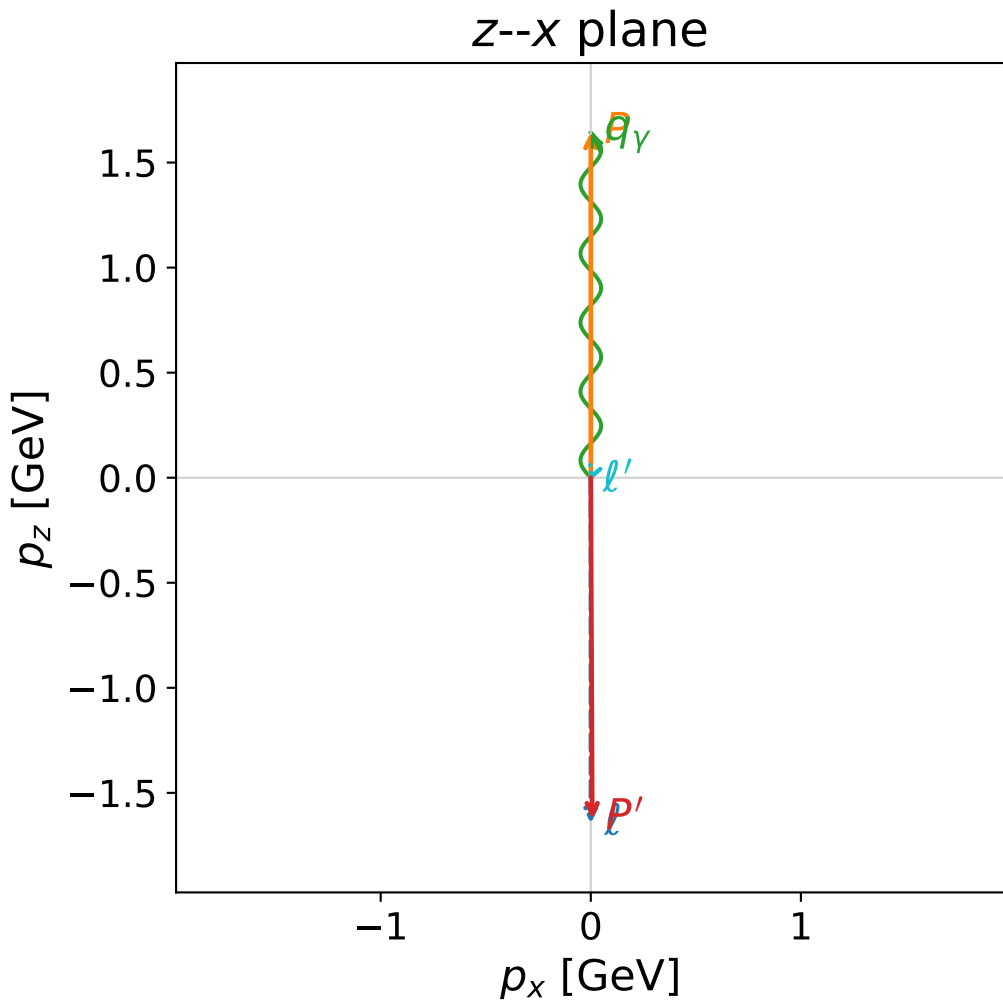
```
dghz_mixing_angles_polarization_02_local_minimum_355
d_{\rm GHZ}=0.00117646
region: polarization_cluster_02_local_minimum_355
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0355
lepton mixing angle: alpha_e=0.78572109 rad
incoming lepton: 0.706878|+> +0.707335|->
proton mixing angle: alpha_p=2.3570726 rad
incoming proton: -0.707727|+> +0.706486|->
```

```
s=25, sqrt(s)=5, |P|=2.41202, |P'|=2.41055
theta_p=3.14159, theta_gamma=4.76846e-05, phi_p=3.33831, phi_gamma=0.431012
E_gamma=2.41192, theta_l'=3.05787, phi_l'=3.5726
Q^2=0.000465873, x_B=1.9326e-05, t=-23.2571, W^2=24.9853, y=0.999411
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  2.4120 p_x=  0.0000 p_y=  0.0000 p_z= -2.4120
P      E=  2.5880 p_x=  0.0000 p_y=  0.0000 p_z=  2.4120
l'     E=  0.0015 p_x= -0.0001 p_y= -0.0000 p_z= -0.0014
P'     E=  2.5866 p_x= -0.0000 p_y= -0.0000 p_z= -2.4105
q_gamma E=  2.4119 p_x=  0.0001 p_y=  0.0000 p_z=  2.4119
```



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

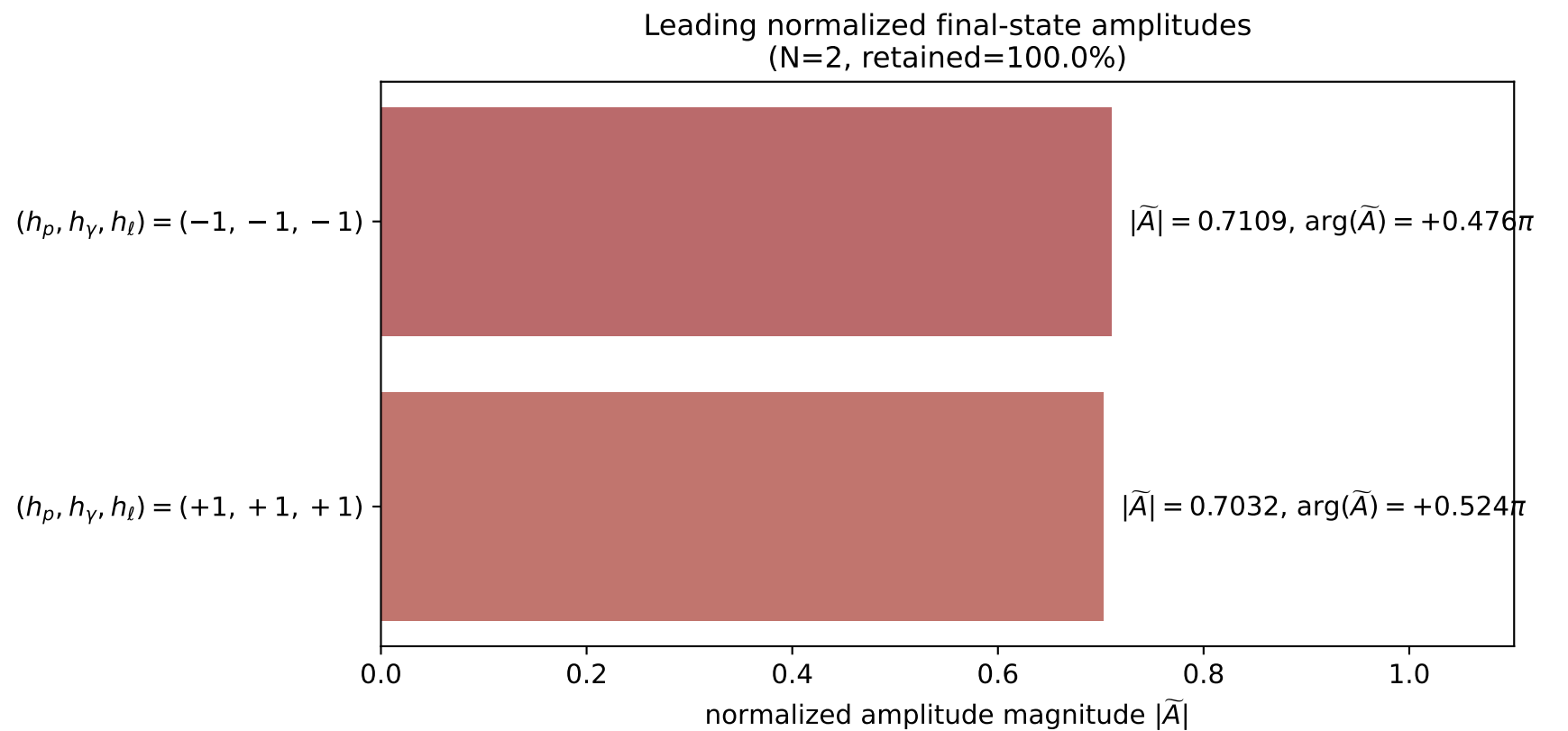
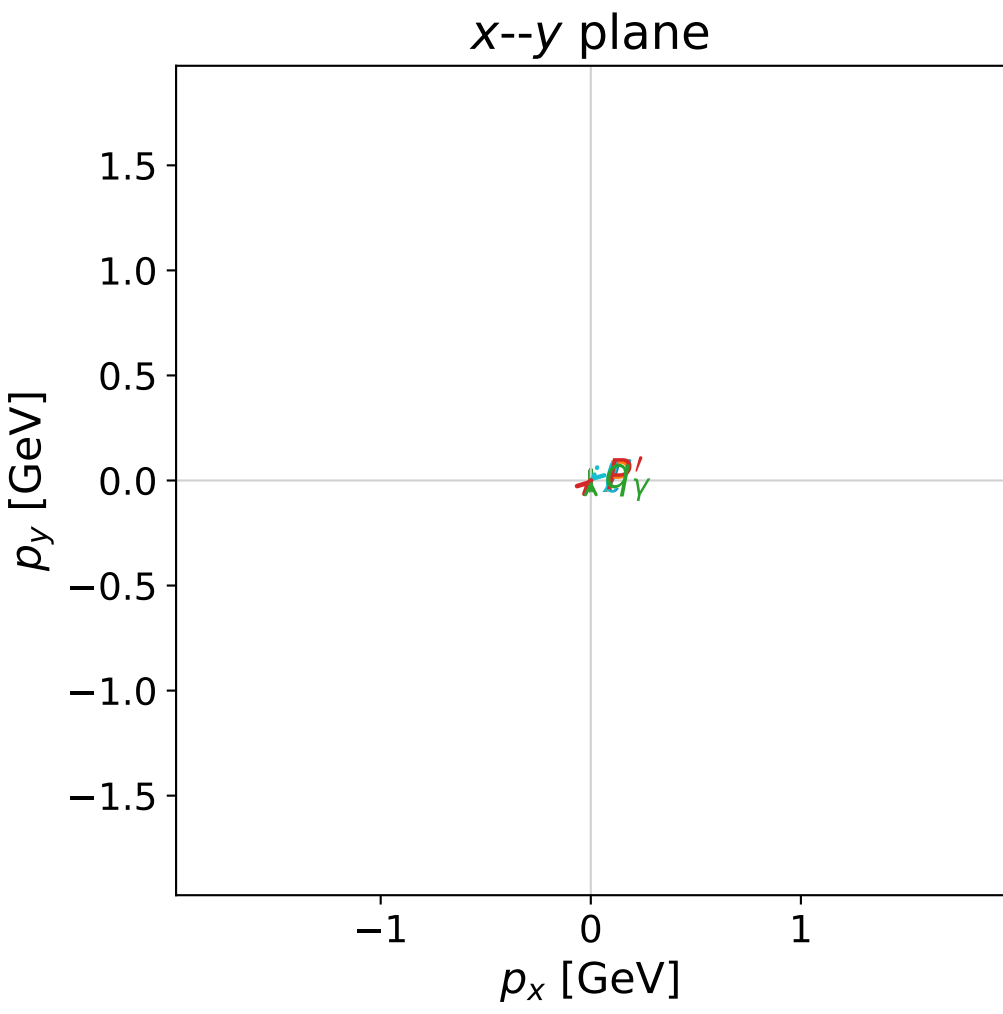


dghz_mixing_angles_polarization_02_local_minimum_370
 $d_{\text{GHZ}}=0.0225057$
 region: polarization_cluster_02_local_minimum_370
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0370
 lepton mixing angle: $\alpha_e=0.78532657$ rad
 incoming lepton: $0.707157|+\rangle + 0.707056|-\rangle$
 proton mixing angle: $\alpha_p=2.3500697$ rad
 incoming proton: $-0.702763|+\rangle + 0.711424|-\rangle$

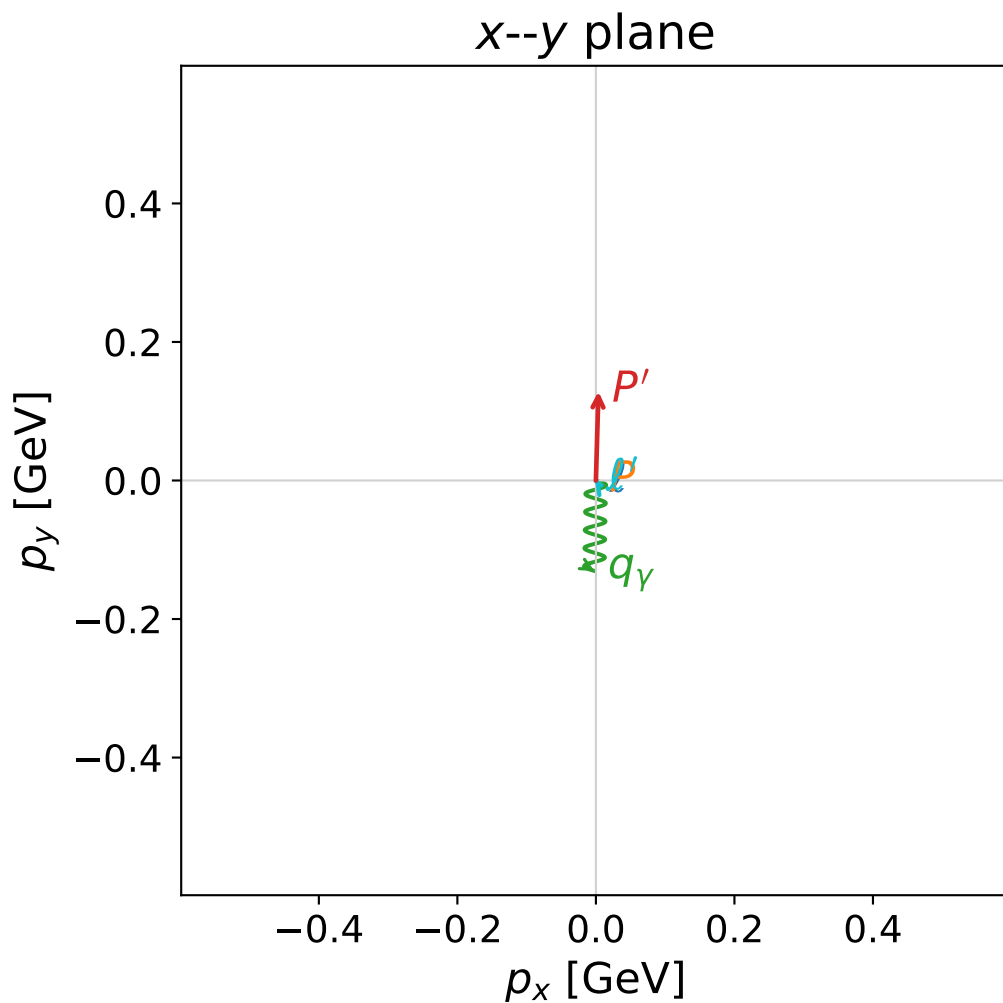
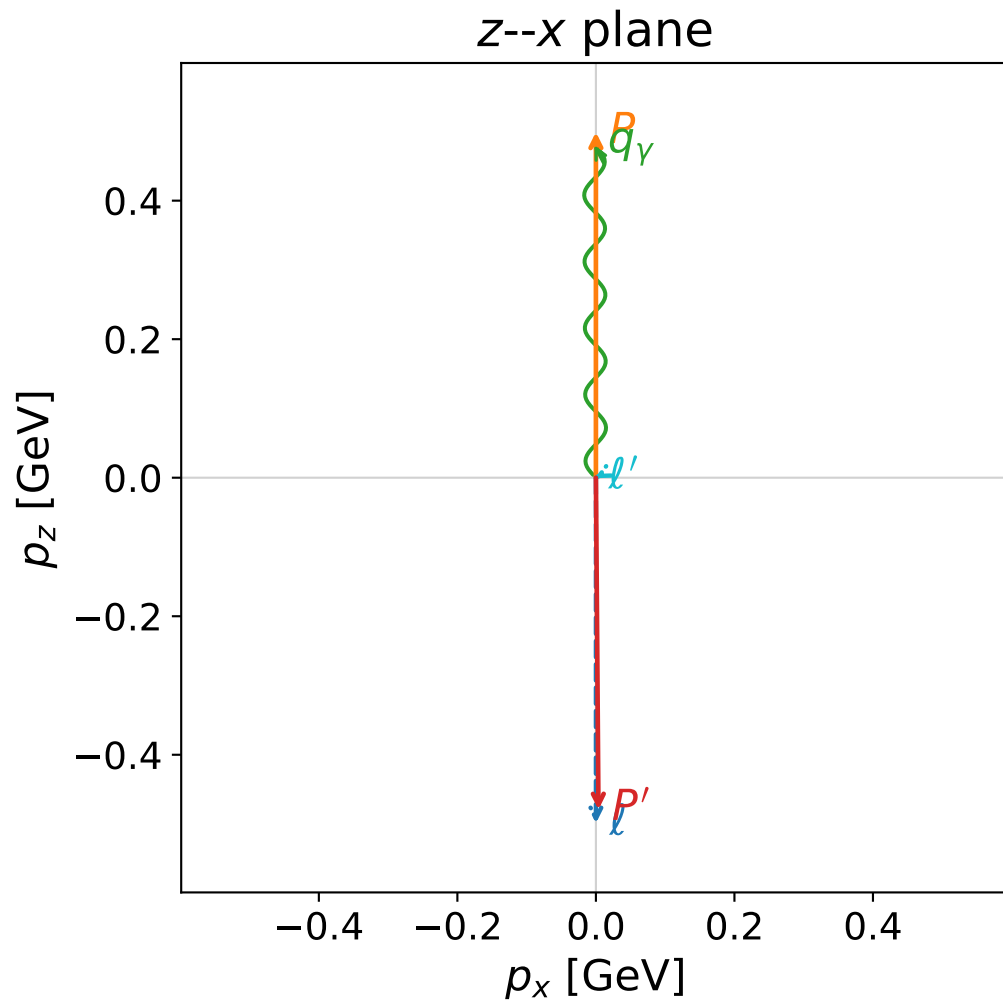
$s=12.52$, $\sqrt{s}=3.53836$, $|\vec{P}|=1.64485$, $|\vec{P}'|=1.62509$
 $\theta_{p'}=3.13567$, $\theta_\gamma=0.000106839$, $\phi_{p'}=0.753952$, $\phi_\gamma=0.000369628$
 $E_\gamma=1.64223$, $\theta_{l'}=2.62489$, $\phi_{l'}=3.88324$
 $Q^2=0.00850448$, $x_B=0.000738953$, $t=-10.6921$, $W^2=12.3802$, $y=0.988717$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	1.6449	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-1.6449
P	$E=$	1.8935	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	1.6449
ℓ'	$E=$	0.0198	$p_x=$	-0.0072	$p_y=$	-0.0066	$p_z=$	-0.0172
P'	$E=$	1.8764	$p_x=$	0.0070	$p_y=$	0.0066	$p_z=$	-1.6251
q_γ	$E=$	1.6422	$p_x=$	0.0002	$p_y=$	0.0000	$p_z=$	1.6422



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

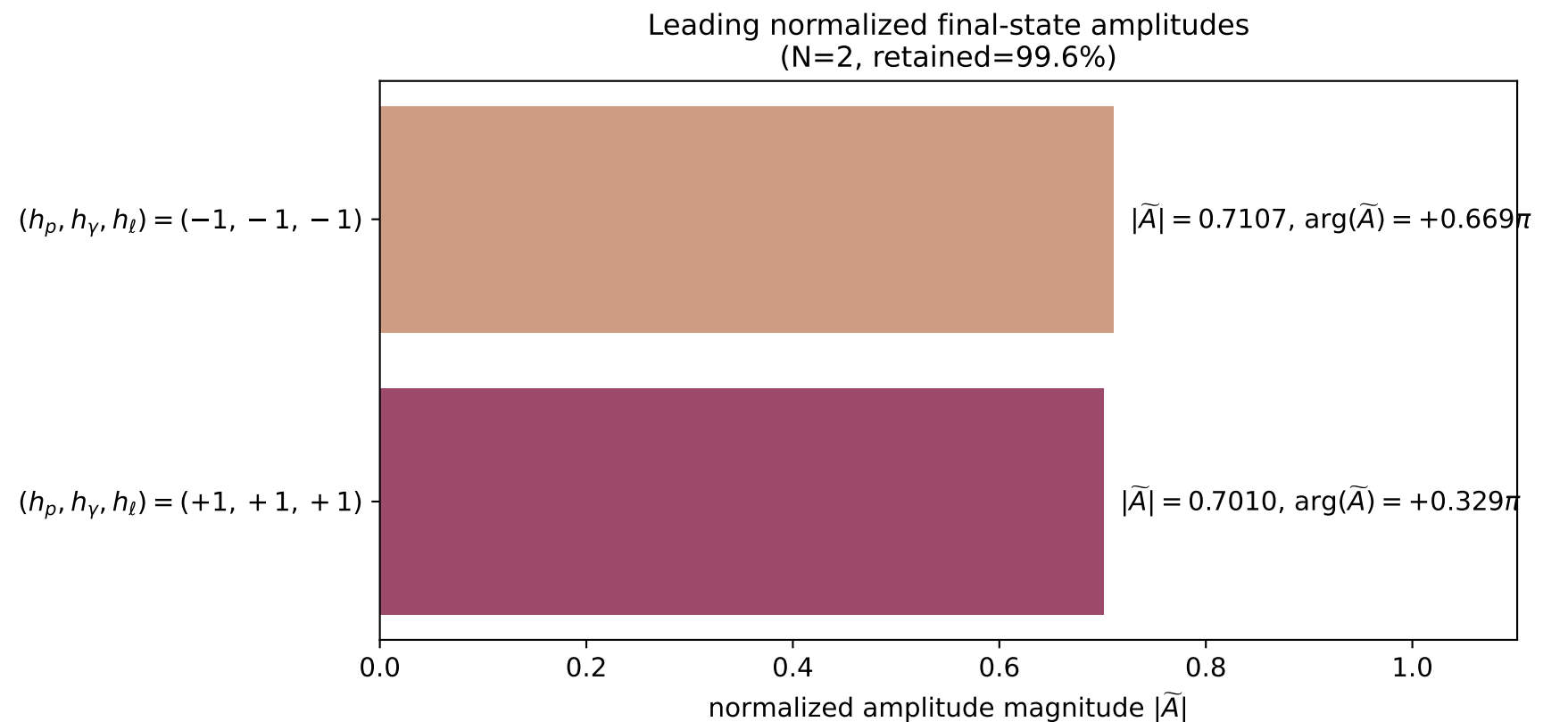


dghz_mixing_angles_polarization_02_local_minimum_371
 $d_{\{\rm GHZ\}}=0.0256597$
 region: polarization_cluster_02_local_minimum_371
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0371
 lepton mixing angle: $\alpha_e=0.78708844$ rad
 incoming lepton: $0.705911|+\rangle +0.708301|-\rangle$
 proton mixing angle: $\alpha_p=2.3508161$ rad
 incoming proton: $-0.703293|+\rangle +0.7109|-\rangle$

$s=2.43798$, $\sqrt{s}=1.5614$, $|\vec{P}|=0.498955$, $|\vec{P}'|=0.496309$
 $\theta_{p'}=2.87957$, $\theta_\gamma=0.26554$, $\phi_{p'}=1.54217$, $\phi_\gamma=4.69489$
 $E_\gamma=0.497557$, $\theta_{\ell'}=1.86425$, $\phi_{\ell'}=2.16912$
 $Q^2=0.0018846$, $x_B=0.00121447$, $t=-0.973645$, $W^2=2.42975$, $y=0.995924$

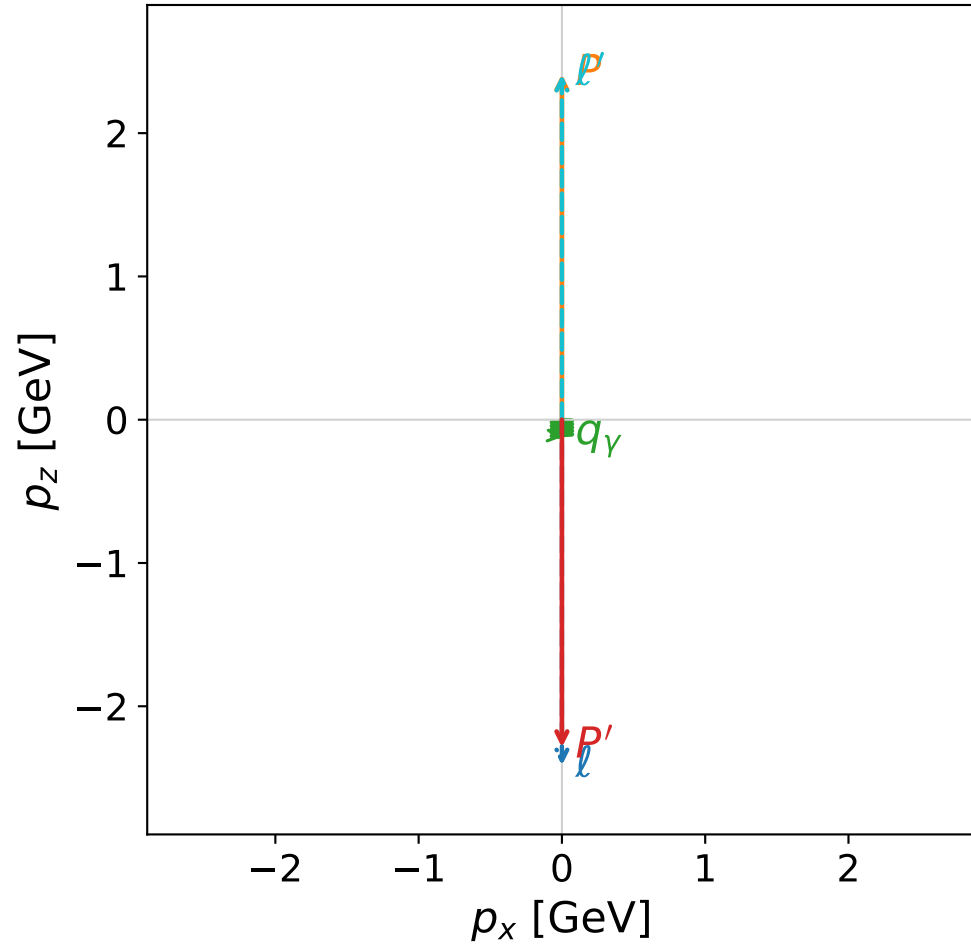
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.4990	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.4990
P	$E=$	1.0624	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.4990
ℓ'	$E=$	0.0026	$p_x=$	-0.0014	$p_y=$	0.0020	$p_z=$	-0.0007
P'	$E=$	1.0612	$p_x=$	0.0037	$p_y=$	0.1285	$p_z=$	-0.4794
q_γ	$E=$	0.4976	$p_x=$	-0.0023	$p_y=$	-0.1306	$p_z=$	0.4801



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

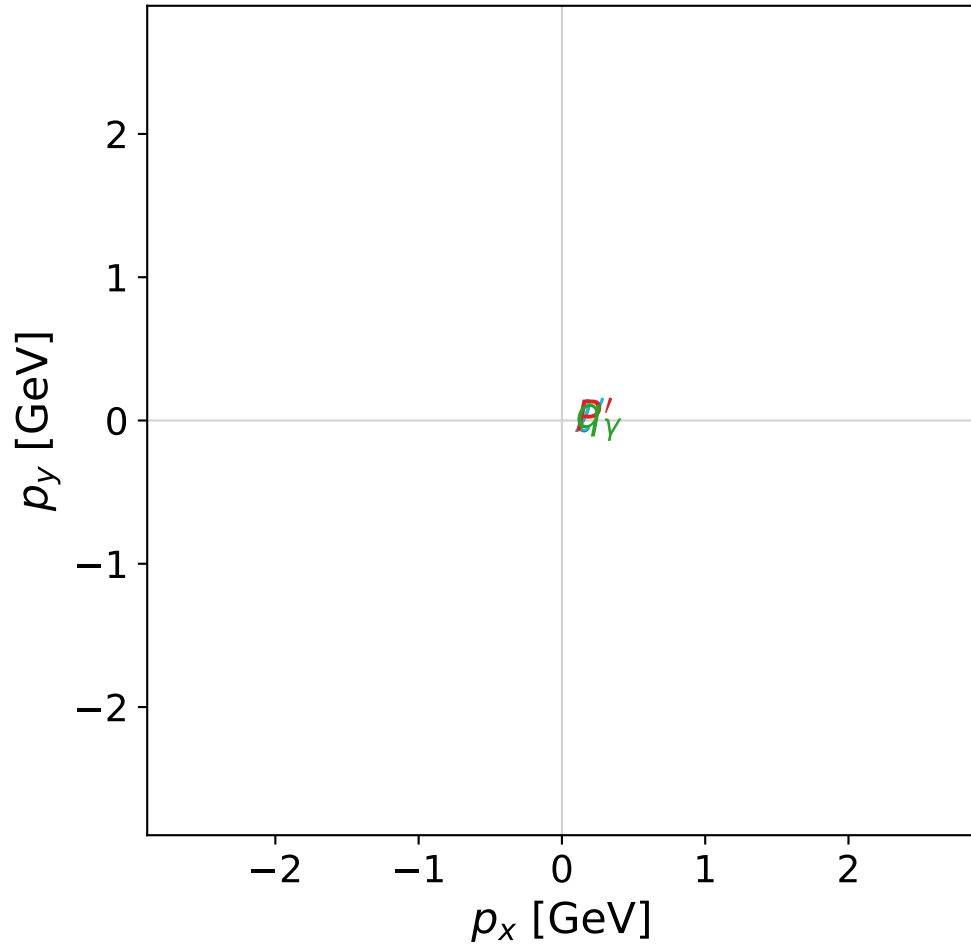


dghz_mixing_angles_polarization_03_local_minimum_1
 $d_{\{\rm GHZ\}}=2.28069\text{e-}08$
 region: polarization_cluster_03_local_minimum_1
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0001
 lepton mixing angle: $\alpha_e=2.4389177$ rad
 incoming lepton: $-0.763116|+\rangle +0.646261|-\rangle$
 proton mixing angle: $\alpha_p=0.70266188$ rad
 incoming proton: $0.763125|+\rangle +0.646251|-\rangle$

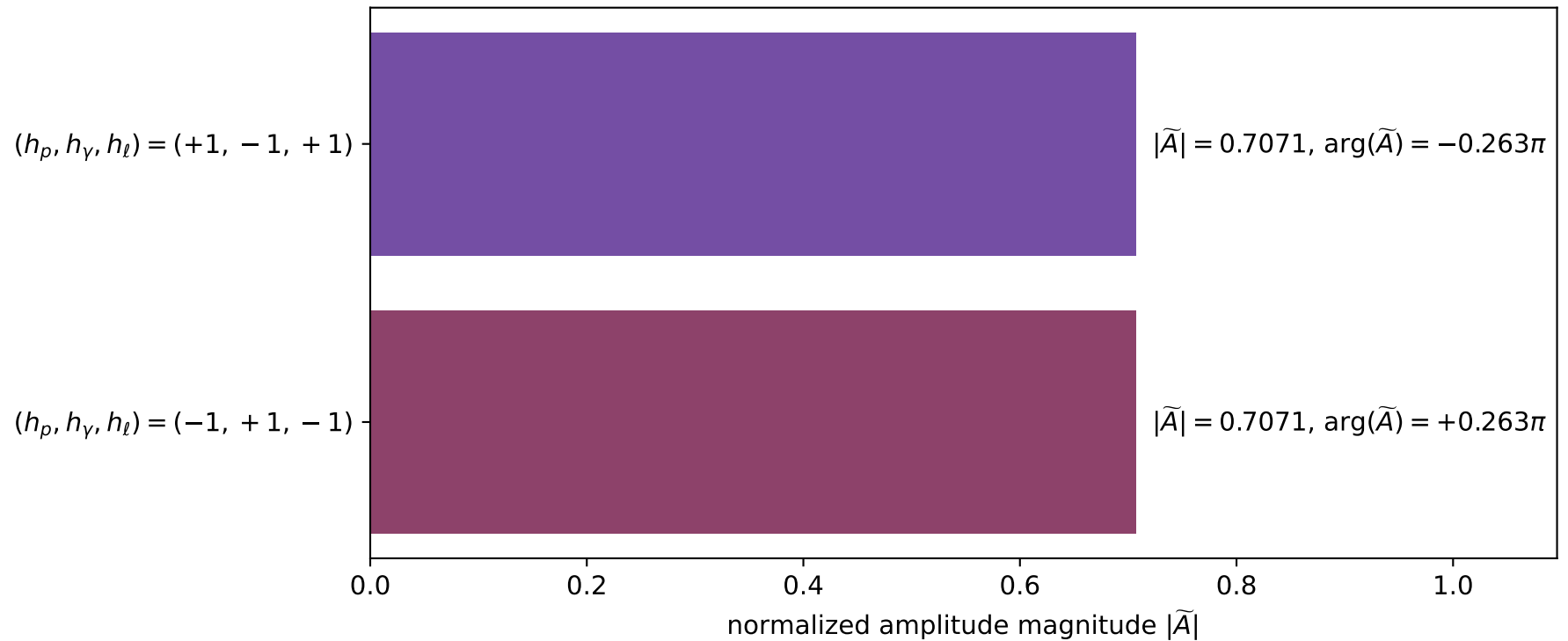
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28696$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=8.73394\text{e-}09$, $\phi_Y=2.31425$
 $E_Y=0.120598$, $\theta_{l'}=0$, $\phi_{l'}=3.18183$
 $Q^2=23.2283$, $x_B=0.998084$, $t=-22.0669$, $W^2=0.924438$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.4076$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.4076$
 P' $E= 2.4718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2870$
 q_Y $E= 0.1206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1206$

x--y plane

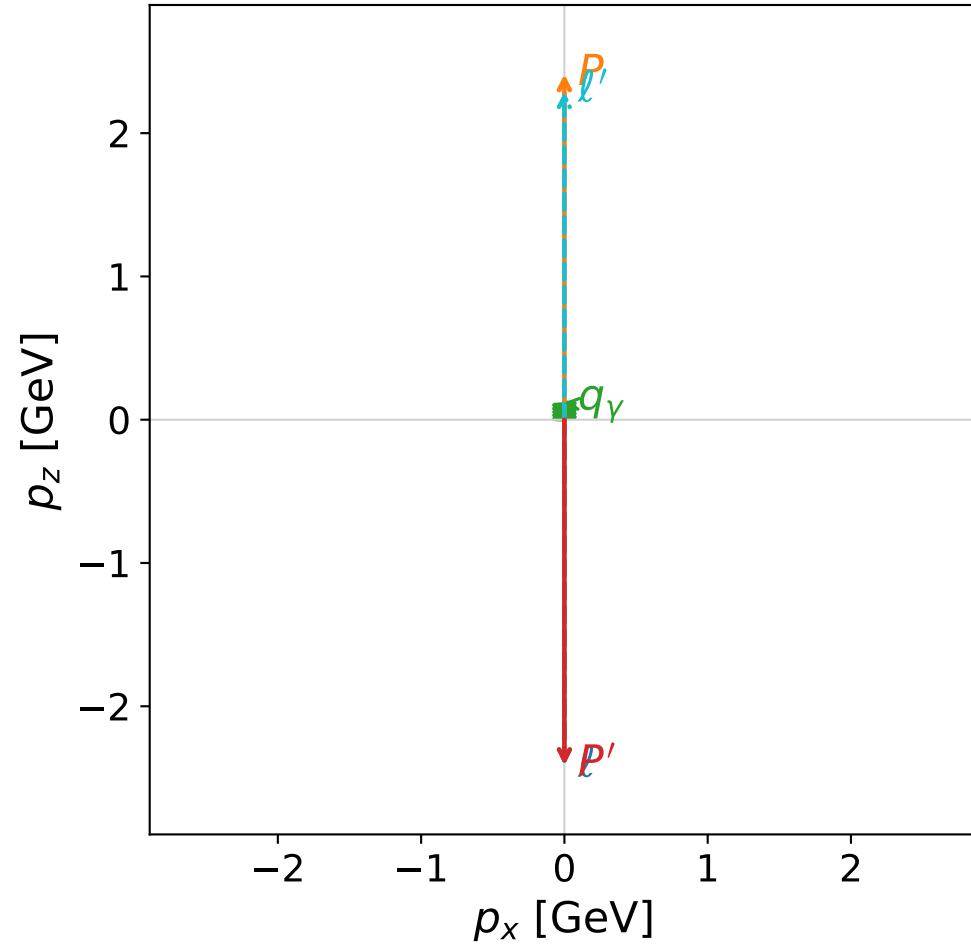


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

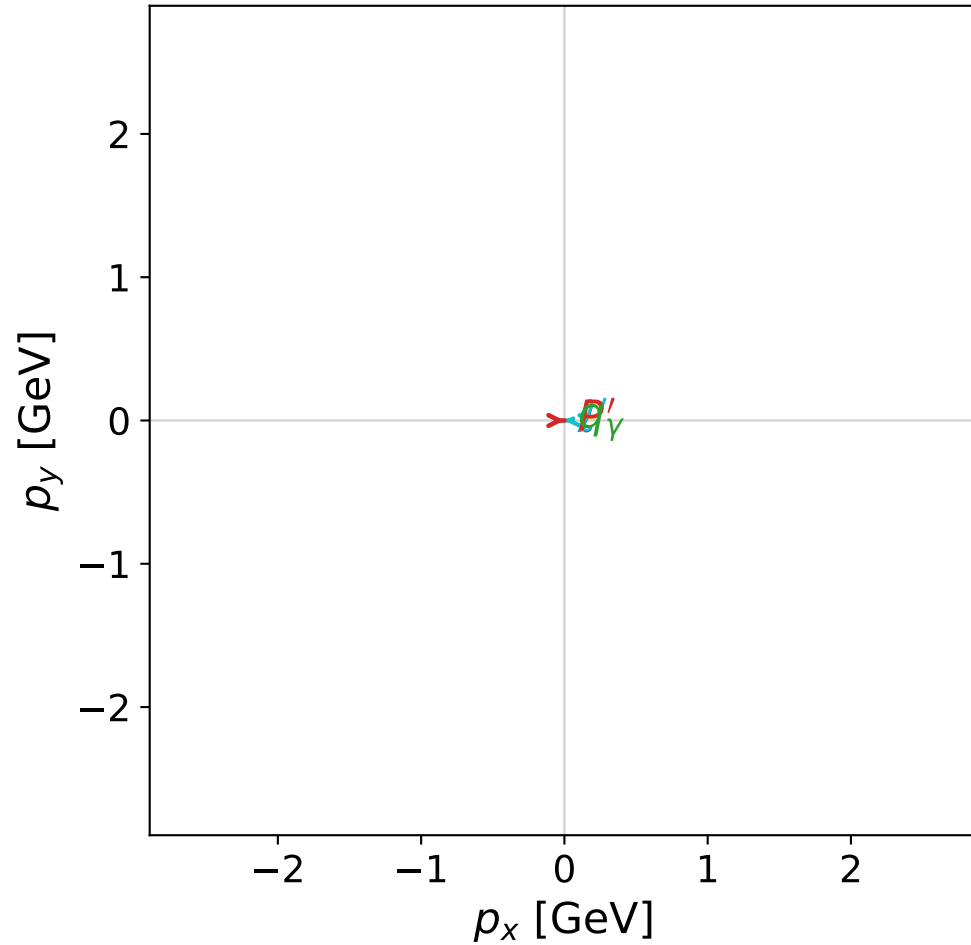


dghz_mixing_angles_polarization_03_local_minimum_5
 $d_{\text{GHZ}}=2.39007\text{e-}08$
 region: polarization_cluster_03_local_minimum_5
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0005
 lepton mixing angle: $\alpha_e=2.2809494$ rad
 incoming lepton: $-0.65195|+\rangle +0.758262|-\rangle$
 proton mixing angle: $\alpha_p=0.71016018$ rad
 incoming proton: $0.758257|+\rangle +0.651955|-\rangle$

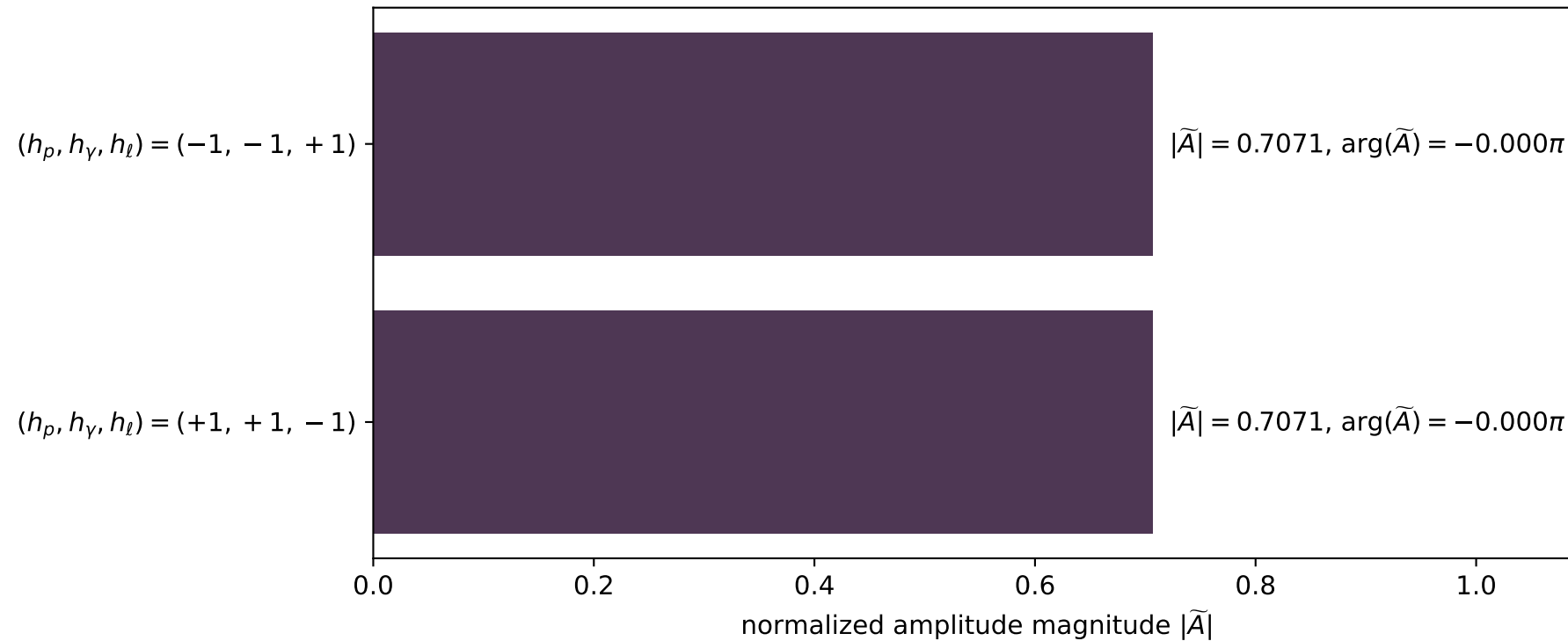
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=6.28319$, $\phi_\gamma=0$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

x--y plane

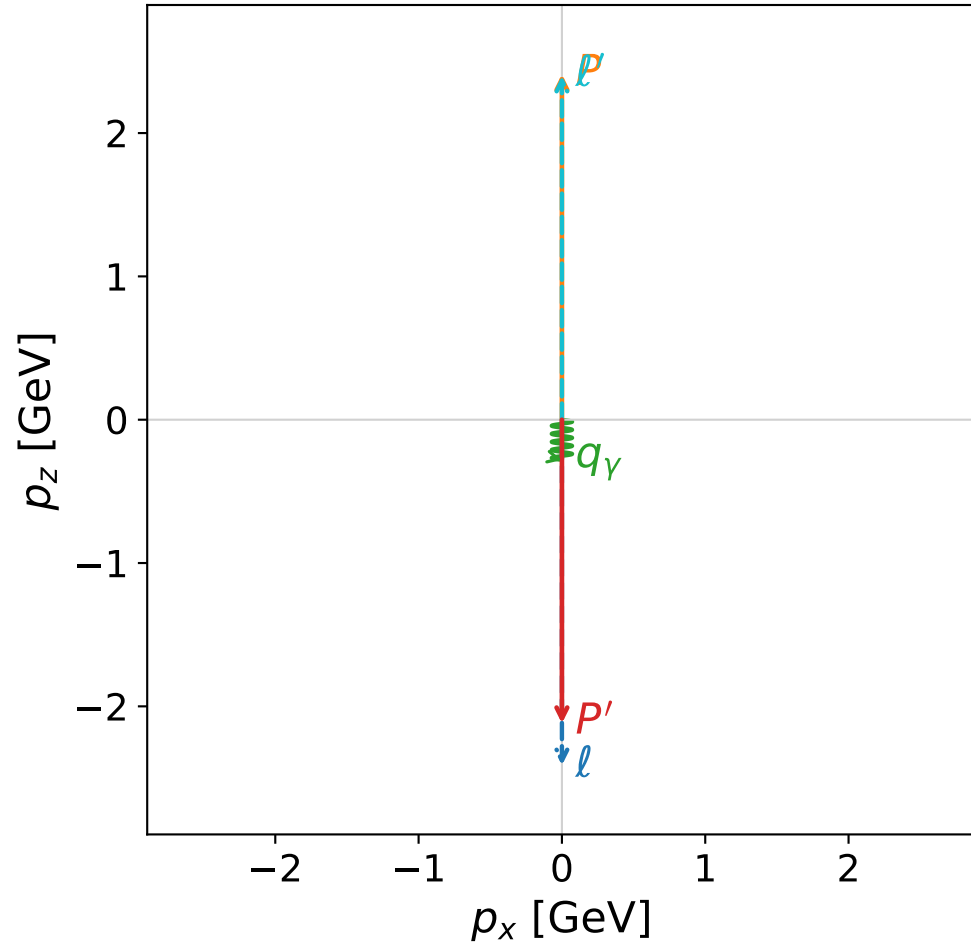


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

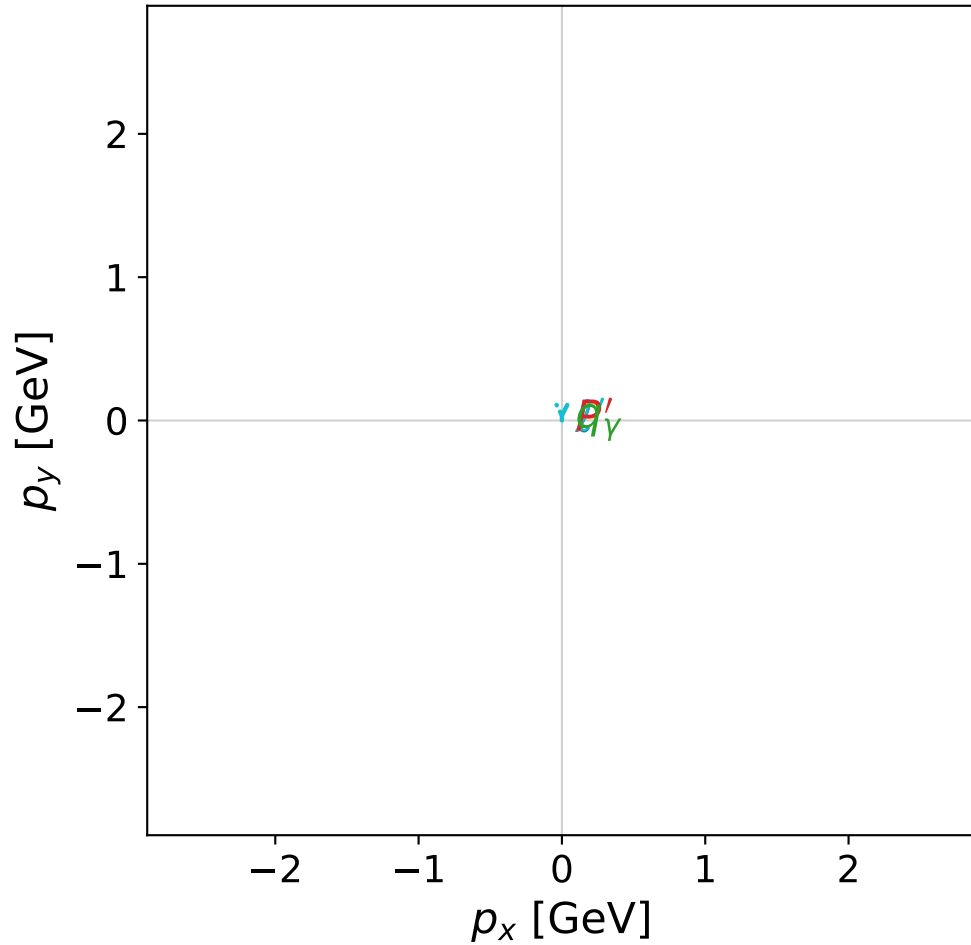


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_12

$d_{\{\text{rm GHZ}\}}=2.46147\text{e-}08$

region: polarization_cluster_03_local_minimum_12

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0012

lepton mixing angle: $\alpha_e=2.1502889$ rad

incoming lepton: $-0.547599|+\rangle +0.836741|-\rangle$

proton mixing angle: $\alpha_p=0.99129539$ rad

incoming proton: $0.547606|+\rangle +0.836736|-\rangle$

$s=24.9869$, $\sqrt{s}=4.99869$, $|\vec{P}|=2.41134$, $|\vec{P}'|=2.11911$

$\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.71126$, $\phi_\gamma=1.54366$

$E_\gamma=0.281071$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.83329$

$Q^2=23.1506$, $x_B=0.995208$, $t=-20.4521$, $W^2=0.991326$, $y=0.964951$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$

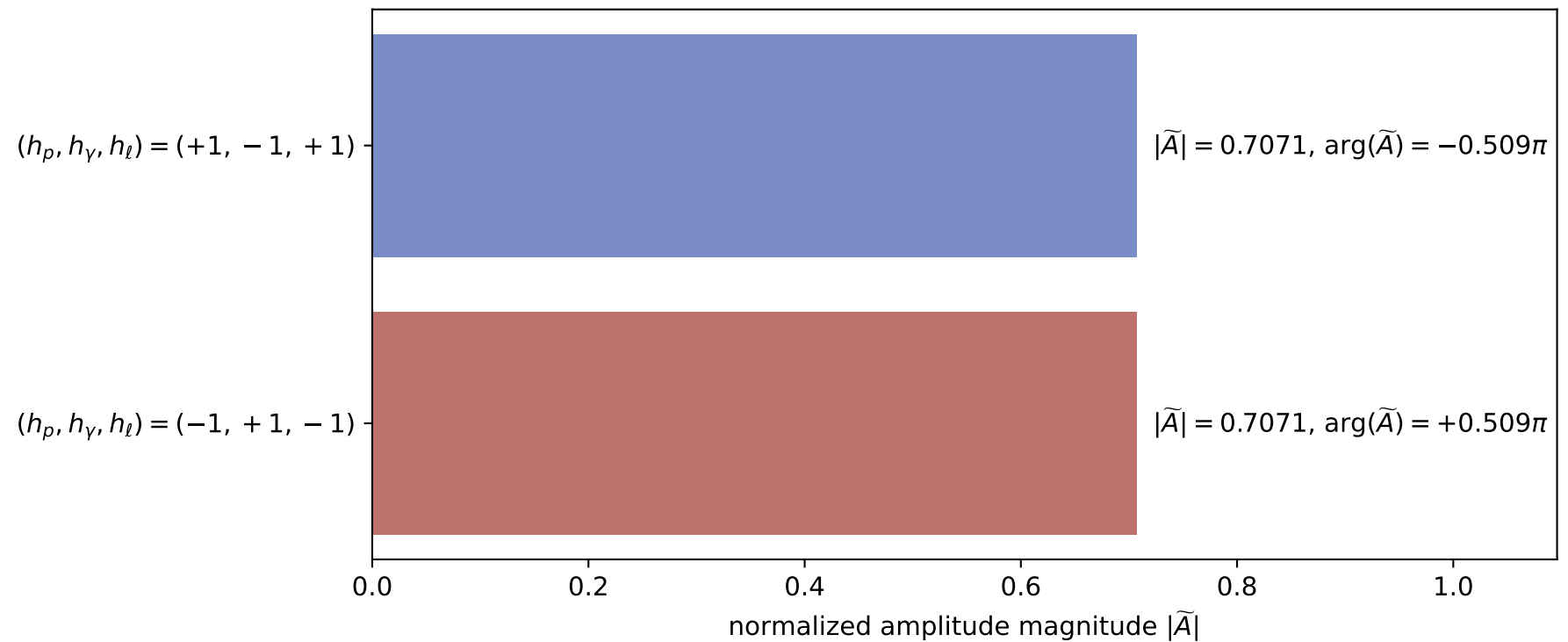
P $E= 2.5874$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$

ℓ' $E= 2.4002$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.4002$

P' $E= 2.3174$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1191$

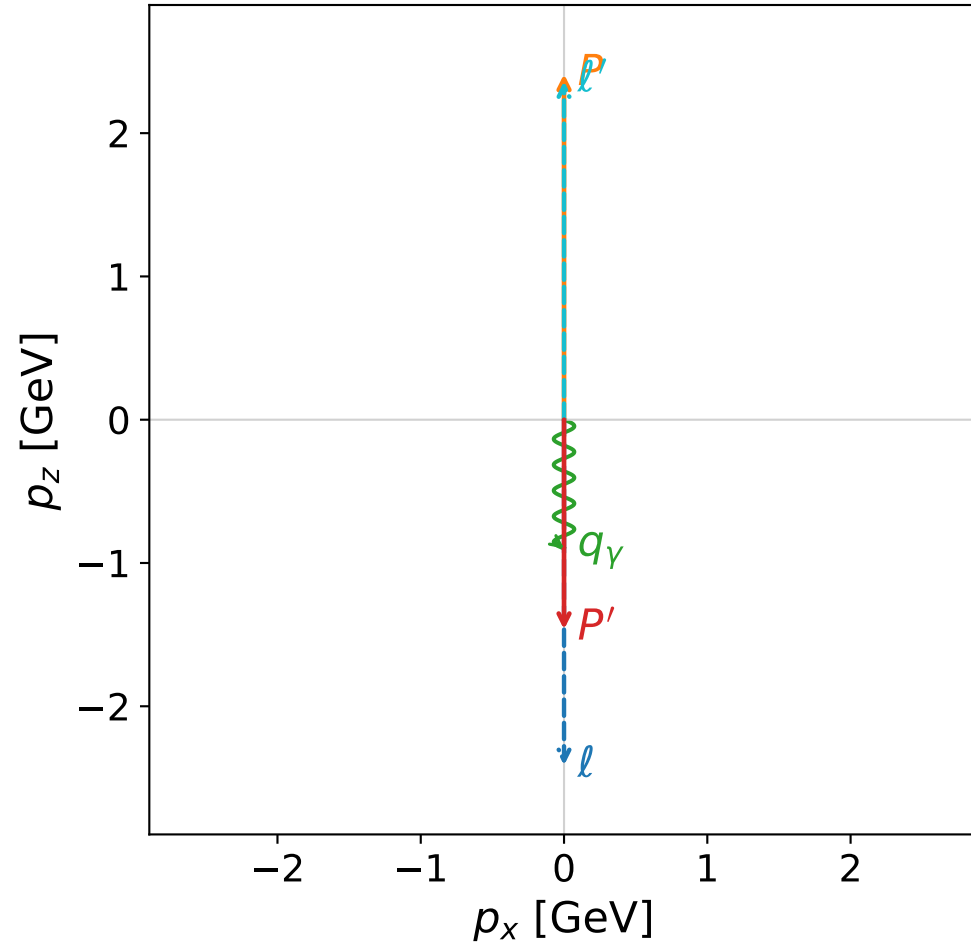
q_γ $E= 0.2811$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2811$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

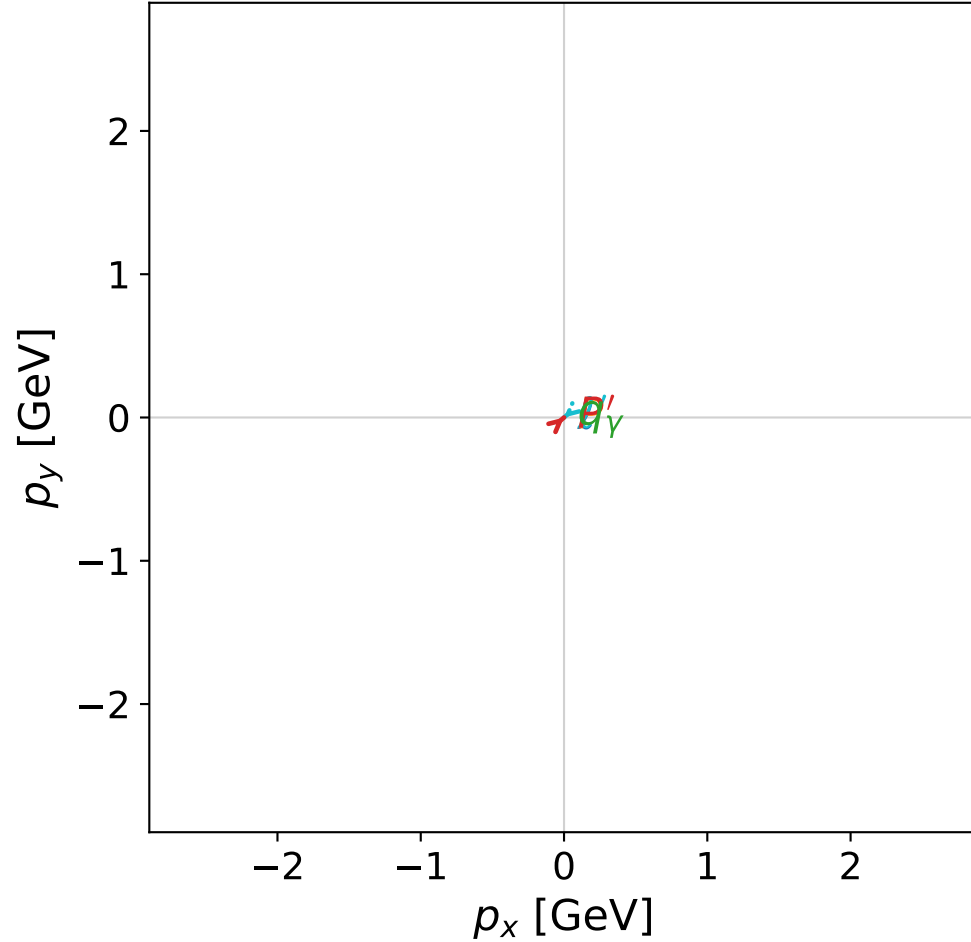


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



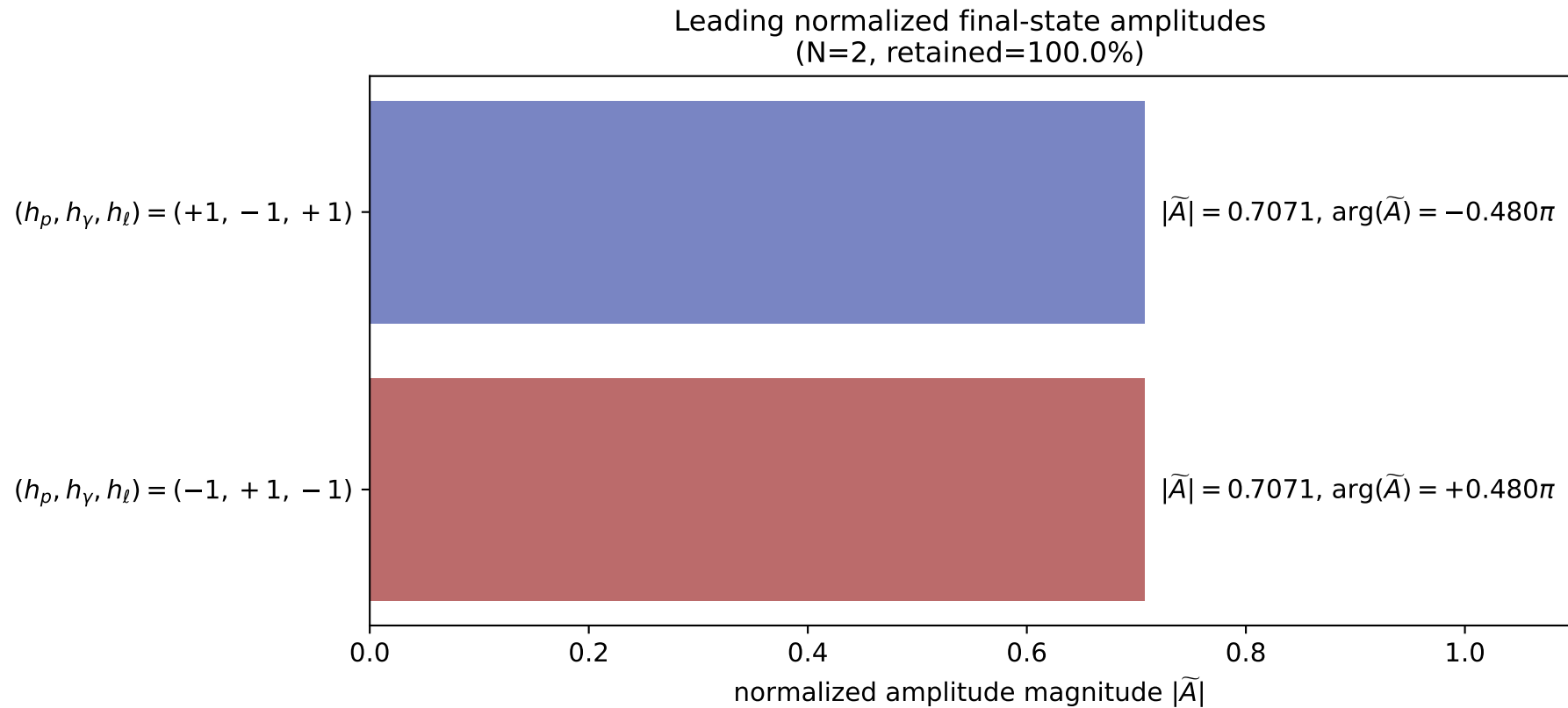
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_13
 $d_{\{\rm GHZ\}}=2.46222\text{e-}08$
 region: polarization_cluster_03_local_minimum_13
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0013
 lepton mixing angle: $\alpha_e=2.1691893$ rad
 incoming lepton: $-0.563315|+\rangle +0.826242|-\rangle$
 proton mixing angle: $\alpha_p=0.97239039$ rad
 incoming proton: $0.563326|+\rangle +0.826235|-\rangle$

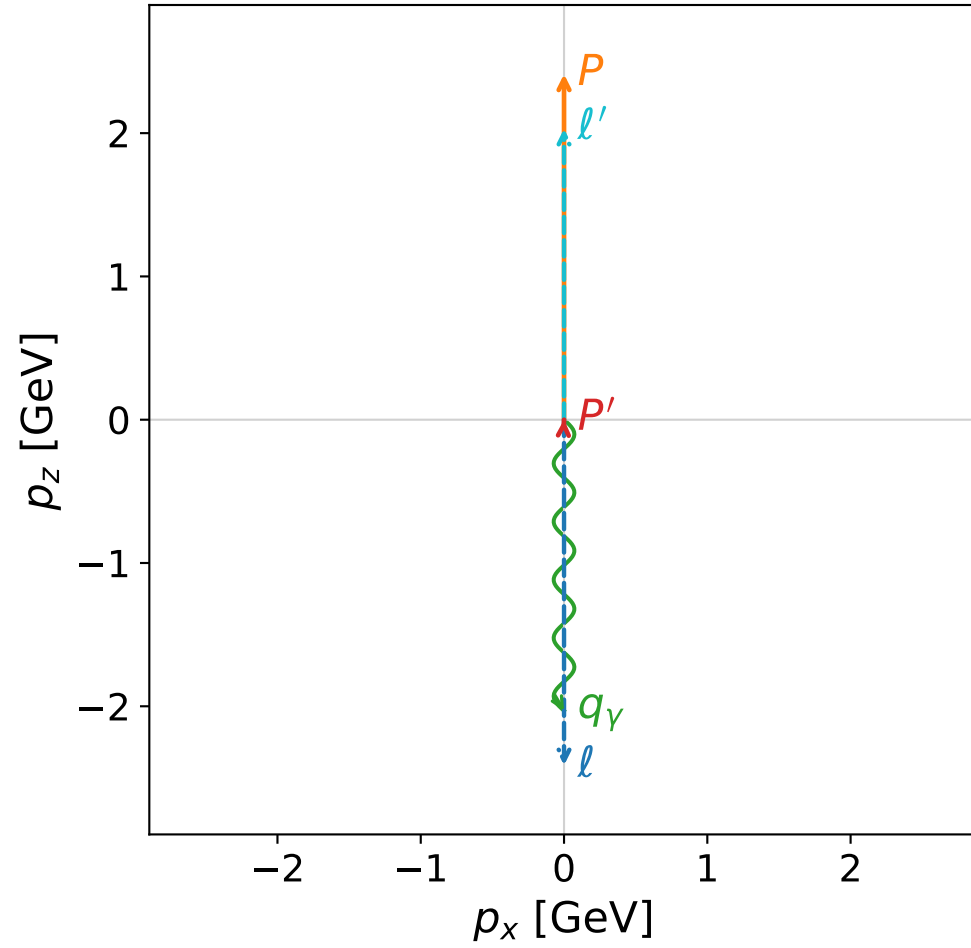
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.46356$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.711355$, $\phi_\gamma=2.34502$
 $E_\gamma=0.899044$, $\theta_{\ell'}=4.02607\text{e-}07$, $\phi_{\ell'}=3.85295$
 $Q^2=22.7946$, $x_B=0.978784$, $t=-14.2982$, $W^2=1.37394$, $y=0.965527$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3626$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.3626$
 P' $E= 1.7383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4636$
 q_γ $E= 0.8990$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.8990$

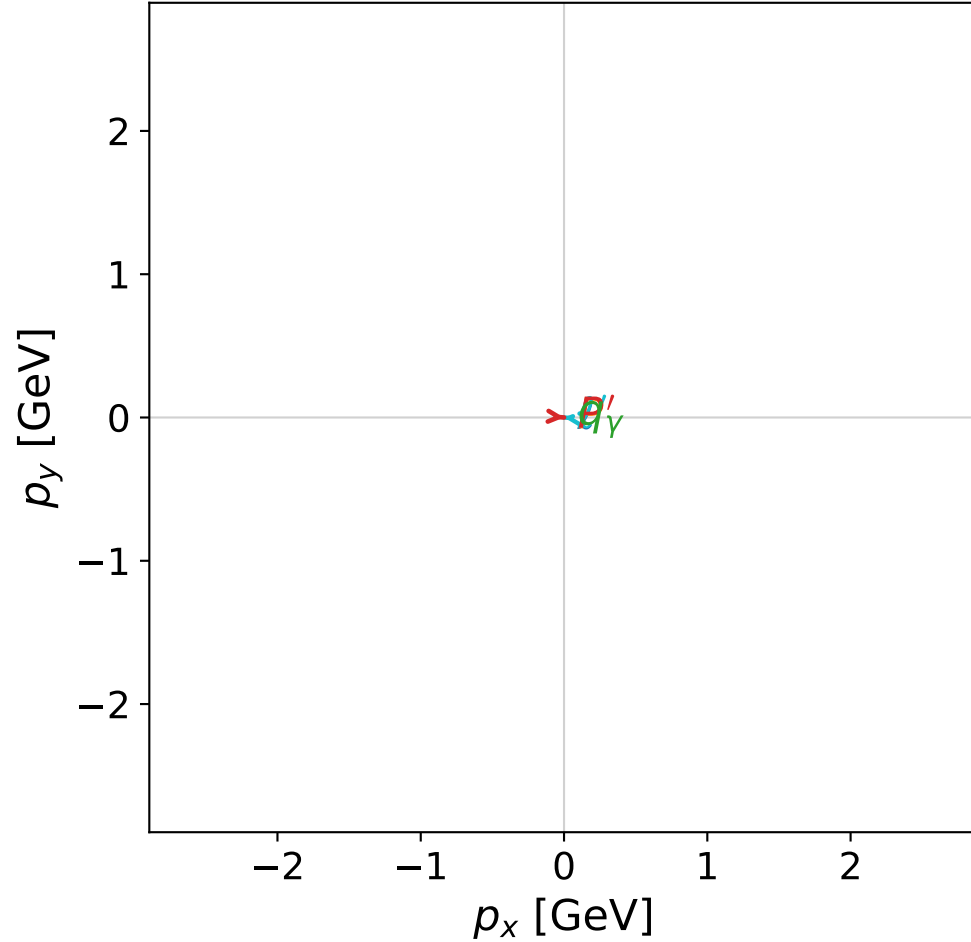


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



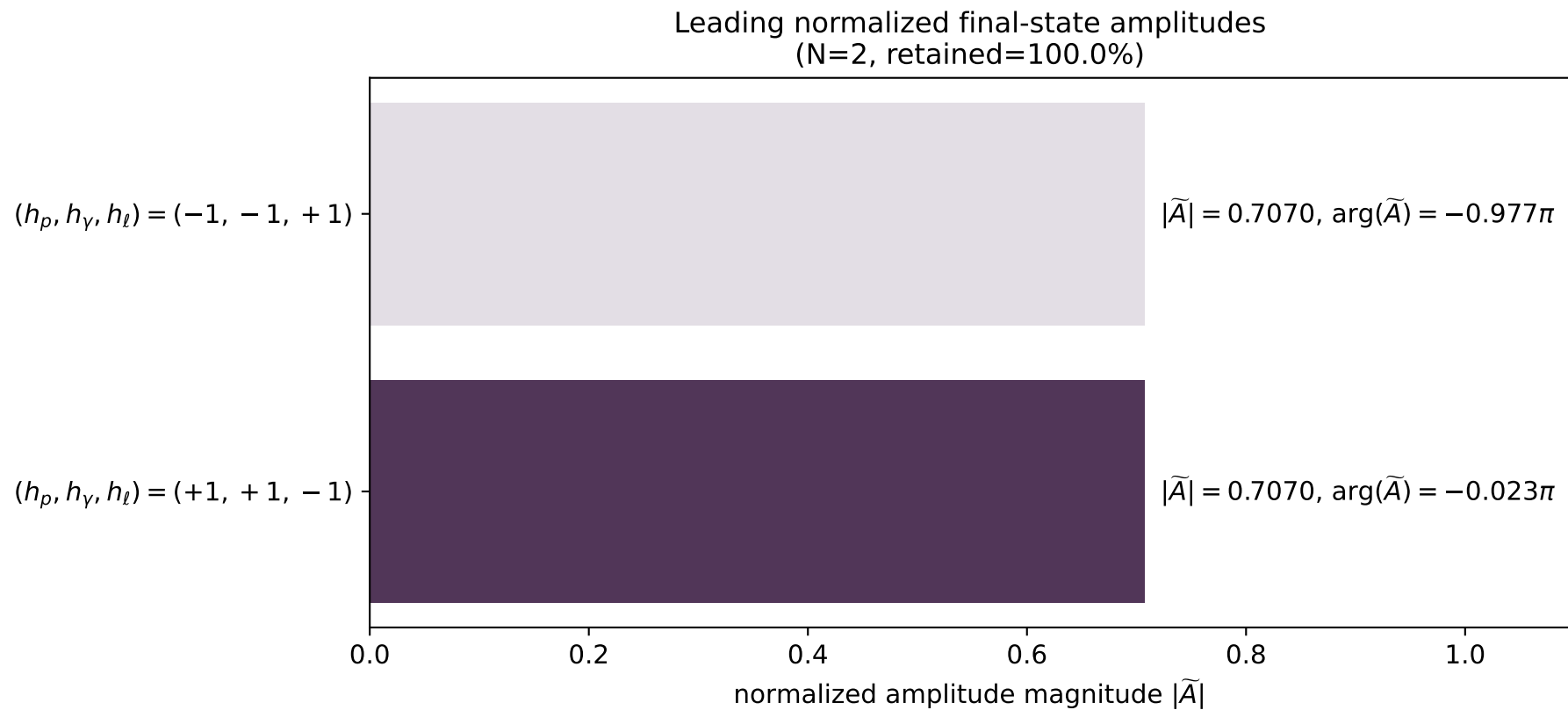
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_15
 $d_{\text{GHZ}}=2.47594\text{e-}08$
 region: polarization_cluster_03_local_minimum_15
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0015
 lepton mixing angle: $\alpha_e=2.4020754$ rad
 incoming lepton: $-0.738794|+\rangle +0.673931|-\rangle$
 proton mixing angle: $\alpha_p=0.73950355$ rad
 incoming proton: $0.738803|+\rangle +0.673921|-\rangle$

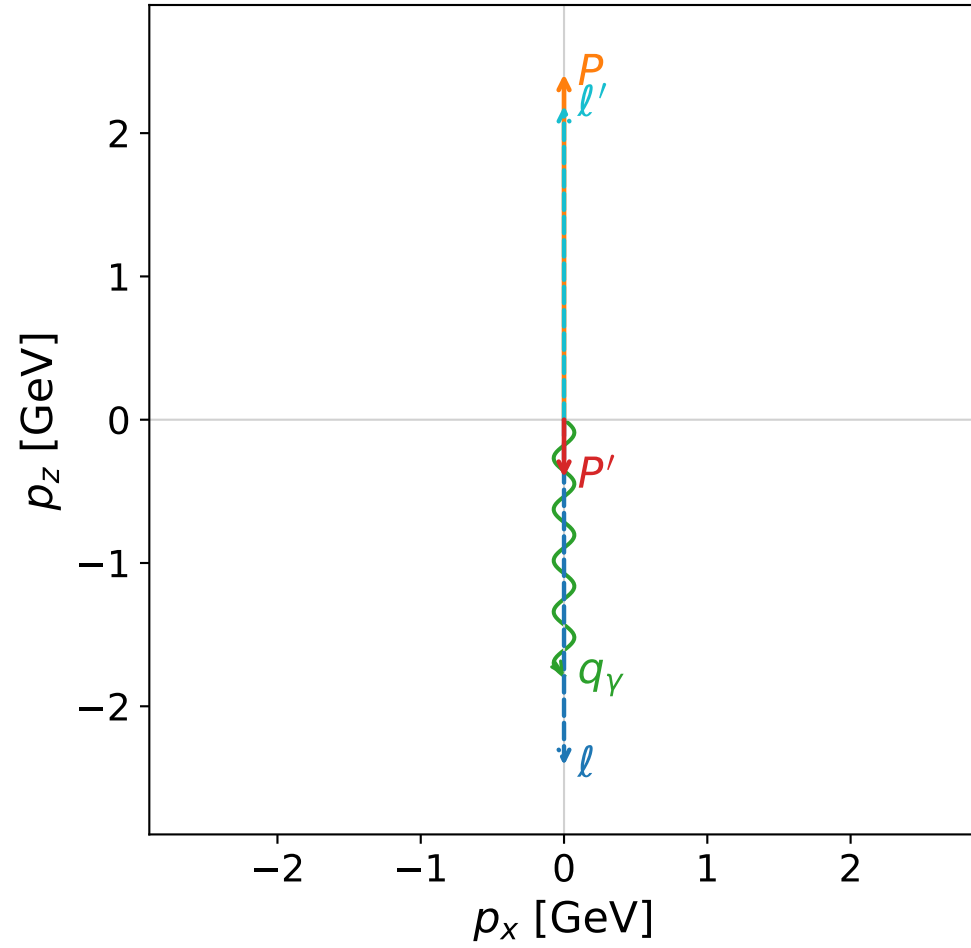
$s=24.9985$, $\sqrt{s}=4.99985$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.00056453$
 $\theta_{P'}=0.0393096$, $\theta_Y=3.14159$, $\phi_{P'}=6.20497$, $\phi_Y=0.0735511$
 $E_Y=2.03121$, $\theta_{l'}=1.09255\text{e-}05$, $\phi_{l'}=3.06338$
 $Q^2=19.5912$, $x_B=0.837086$, $t=-3.09252$, $W^2=4.69268$, $y=0.970368$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 l' $E= 2.0306$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0306$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0006$
 q_Y $E= 2.0312$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0312$

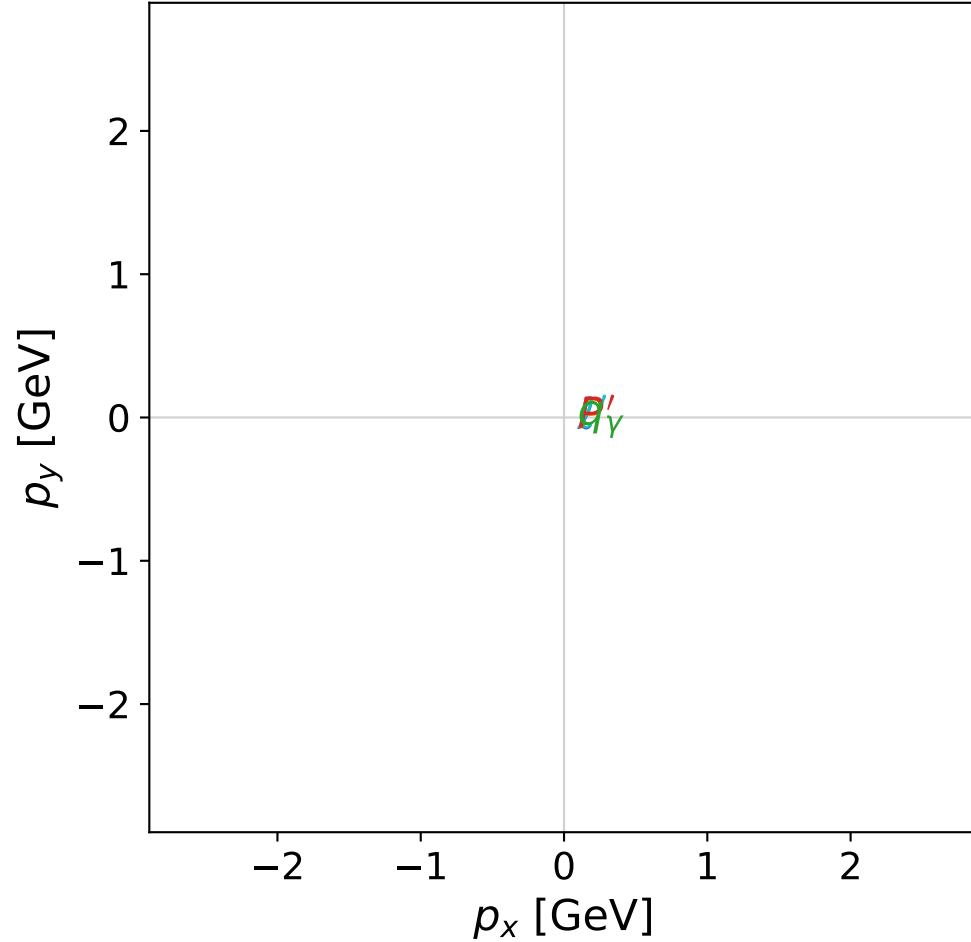


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



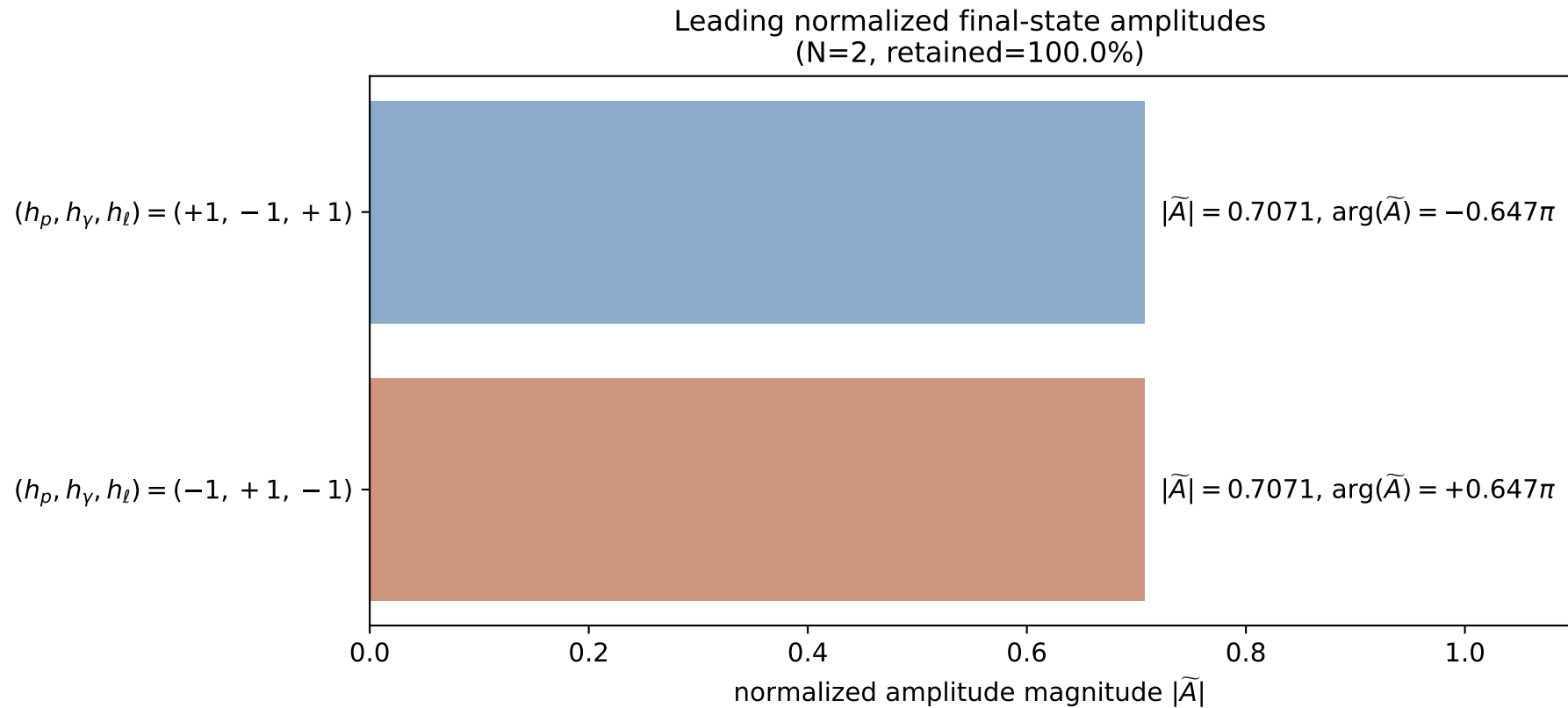
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_17
 $d_{\text{GHZ}}=2.4853\text{e-}08$
 region: polarization_cluster_03_local_minimum_17
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0017
 lepton mixing angle: $\alpha_e=2.4106629$ rad
 incoming lepton: $-0.744554|+\rangle +0.667562|-\rangle$
 proton mixing angle: $\alpha_p=0.73092649$ rad
 incoming proton: $0.744556|+\rangle +0.66756|-\rangle$

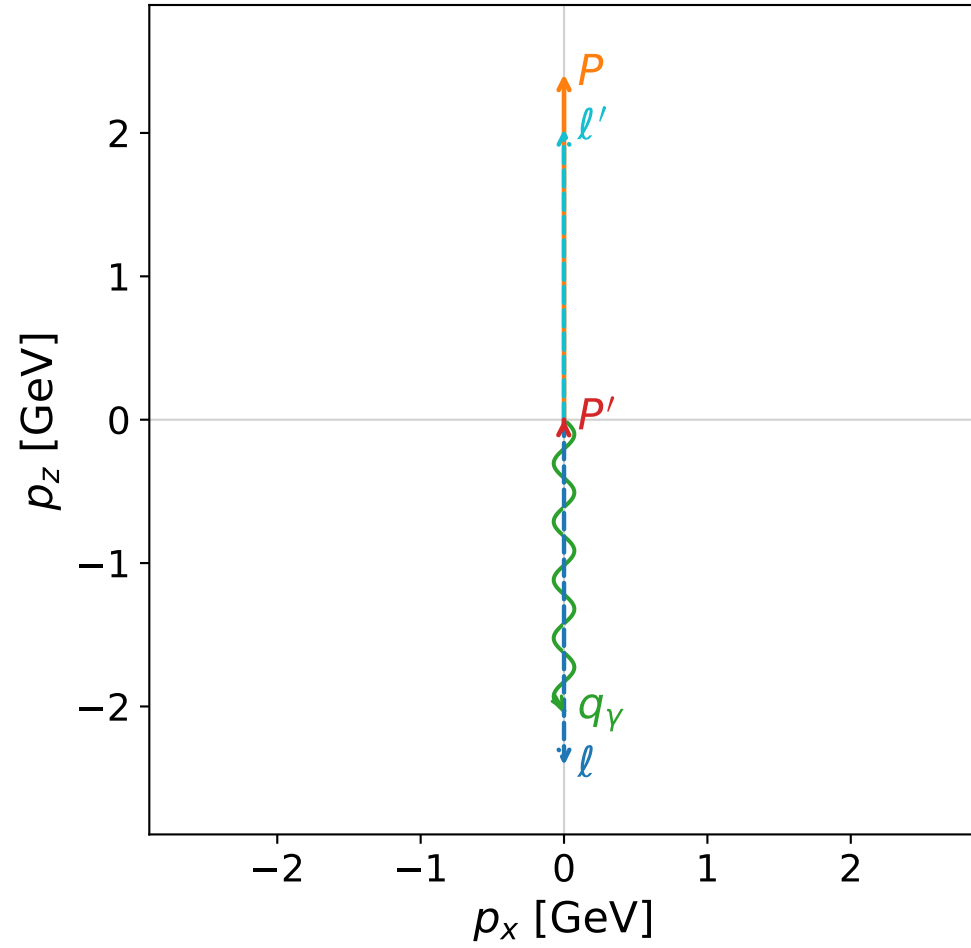
$s=24.9975$, $\sqrt{s}=4.99975$, $|\vec{P}|=2.41189$, $|\vec{P}'|=0.403477$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.09629$, $\phi_\gamma=1.10925$
 $E_\gamma=1.78759$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.47422$
 $Q^2=21.1384$, $x_B=0.905421$, $t=-5.47151$, $W^2=3.08794$, $y=0.968025$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.1911$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1911$
 P' $E= 1.0211$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.4035$
 q_γ $E= 1.7876$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7876$

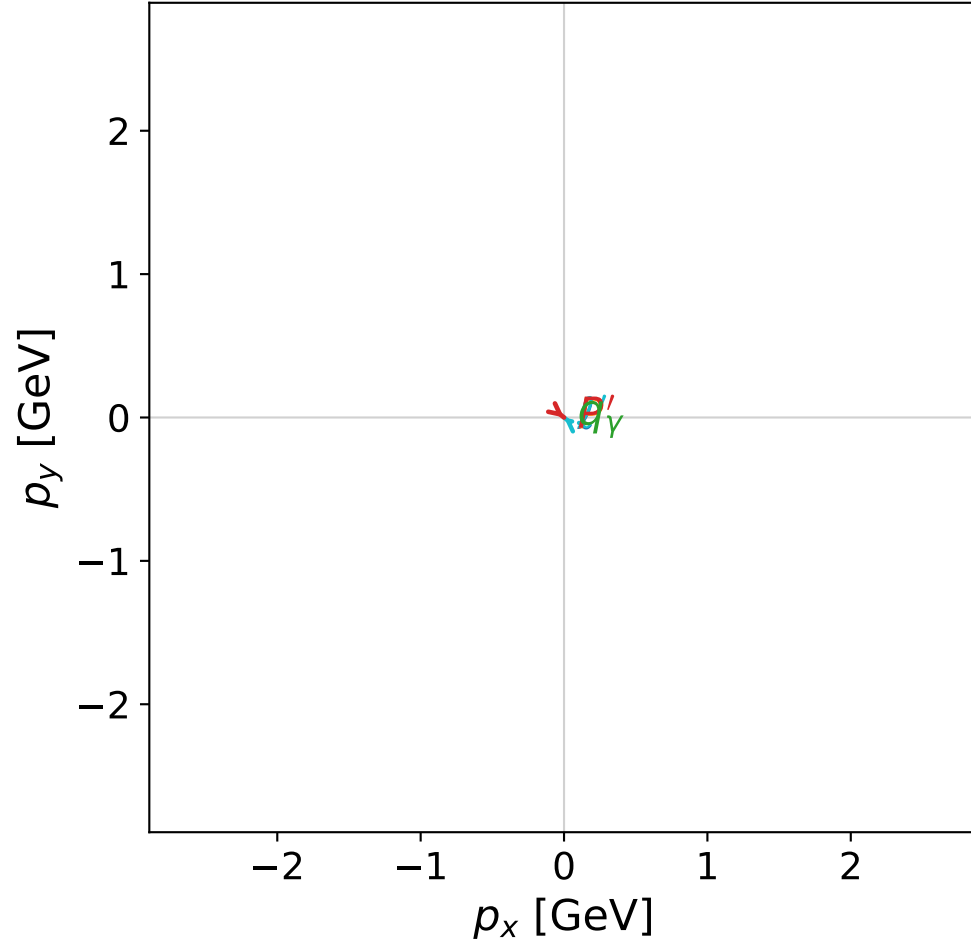


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



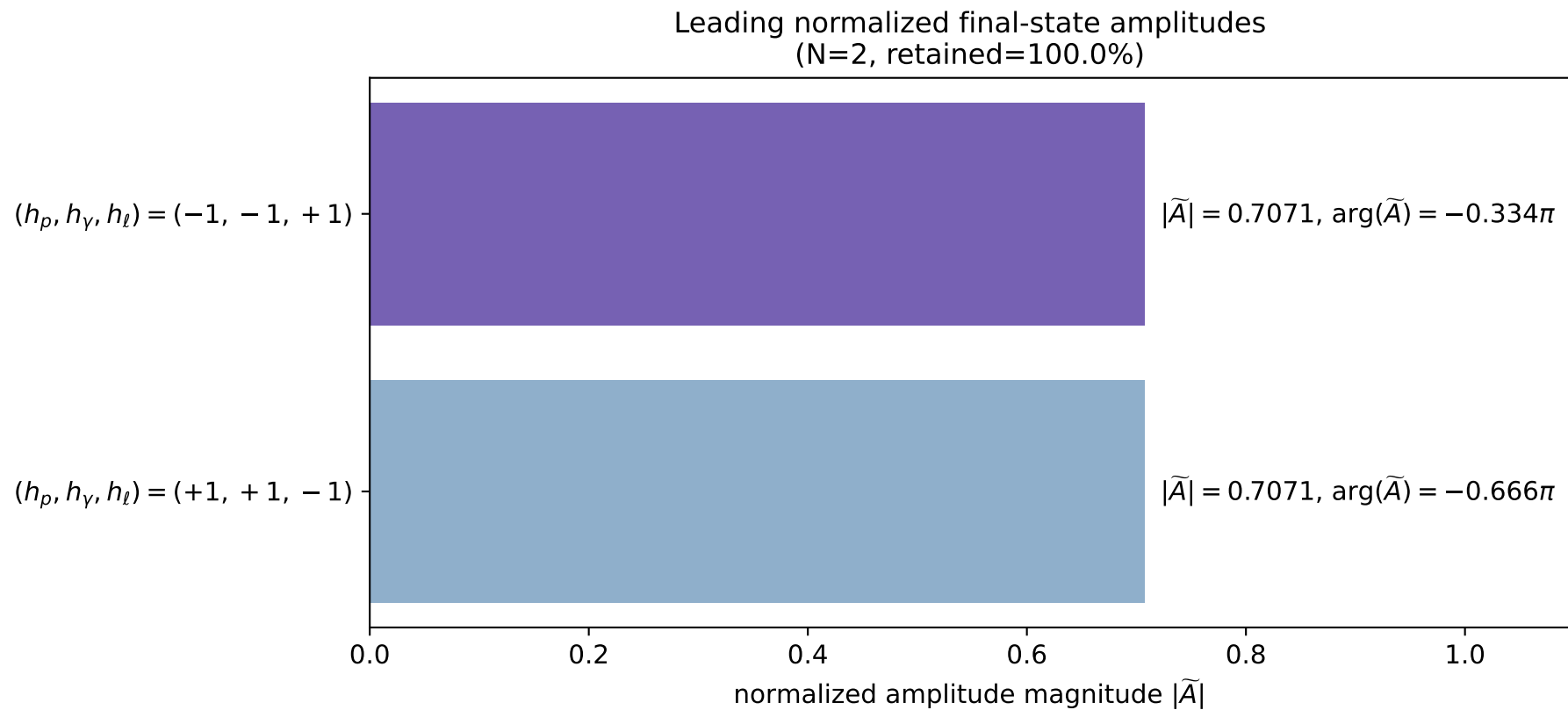
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_20
 $d_{\text{GHZ}}=2.54044\text{e-}08$
 region: polarization_cluster_03_local_minimum_20
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0020
 lepton mixing angle: $\alpha_e=2.3449091$ rad
 incoming lepton: $-0.699082|+\rangle +0.715042|-\rangle$
 proton mixing angle: $\alpha_p=0.79667666$ rad
 incoming proton: $0.699087|+\rangle +0.715037|-\rangle$

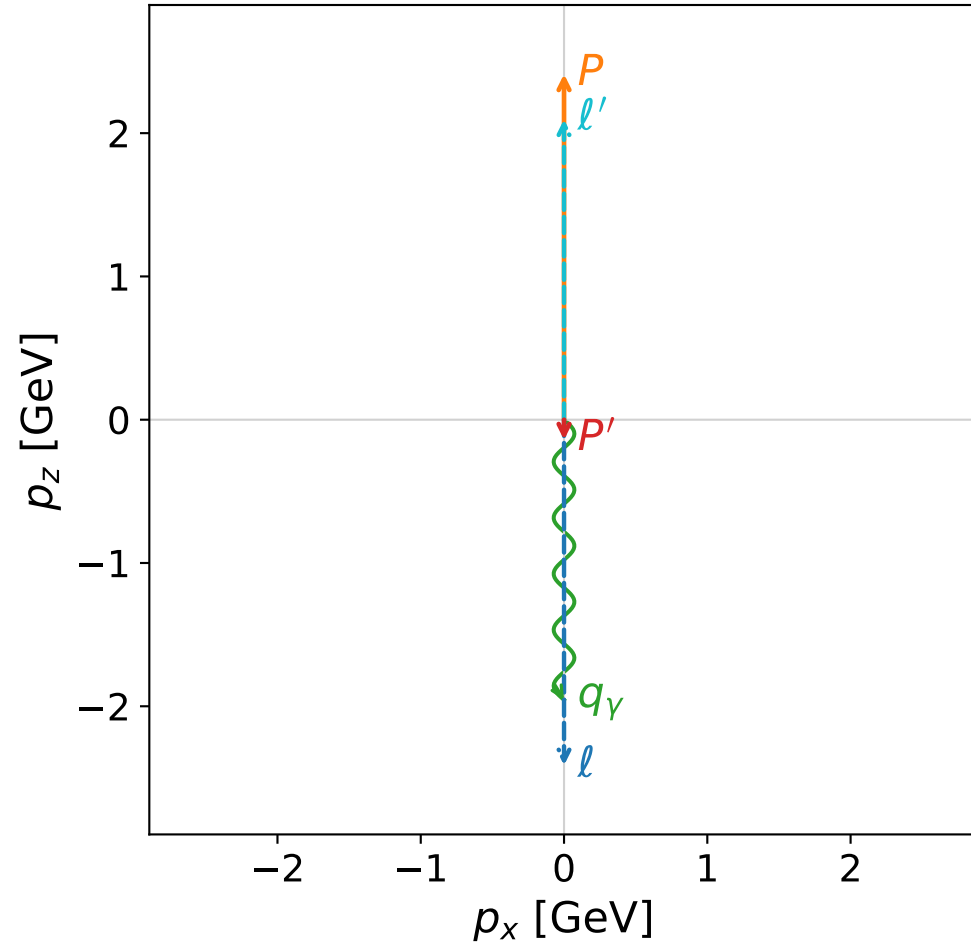
$s=24.9686$, $\sqrt{s}=4.99686$, $|\vec{P}|=2.41039$, $|\vec{P}'|=0.0036789$
 $\theta_{P'}=0.019635$, $\theta_Y=3.14159$, $\phi_{P'}=5.61047$, $\phi_Y=2.09314$
 $E_Y=2.03127$, $\theta_{l'}=3.56238\text{e-}05$, $\phi_{l'}=2.46887$
 $Q^2=19.5491$, $x_B=0.836335$, $t=-3.07484$, $W^2=4.70547$, $y=0.970358$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4104$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4104$
 P $E= 2.5865$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4104$
 l' $E= 2.0276$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0276$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0037$
 q_Y $E= 2.0313$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0313$

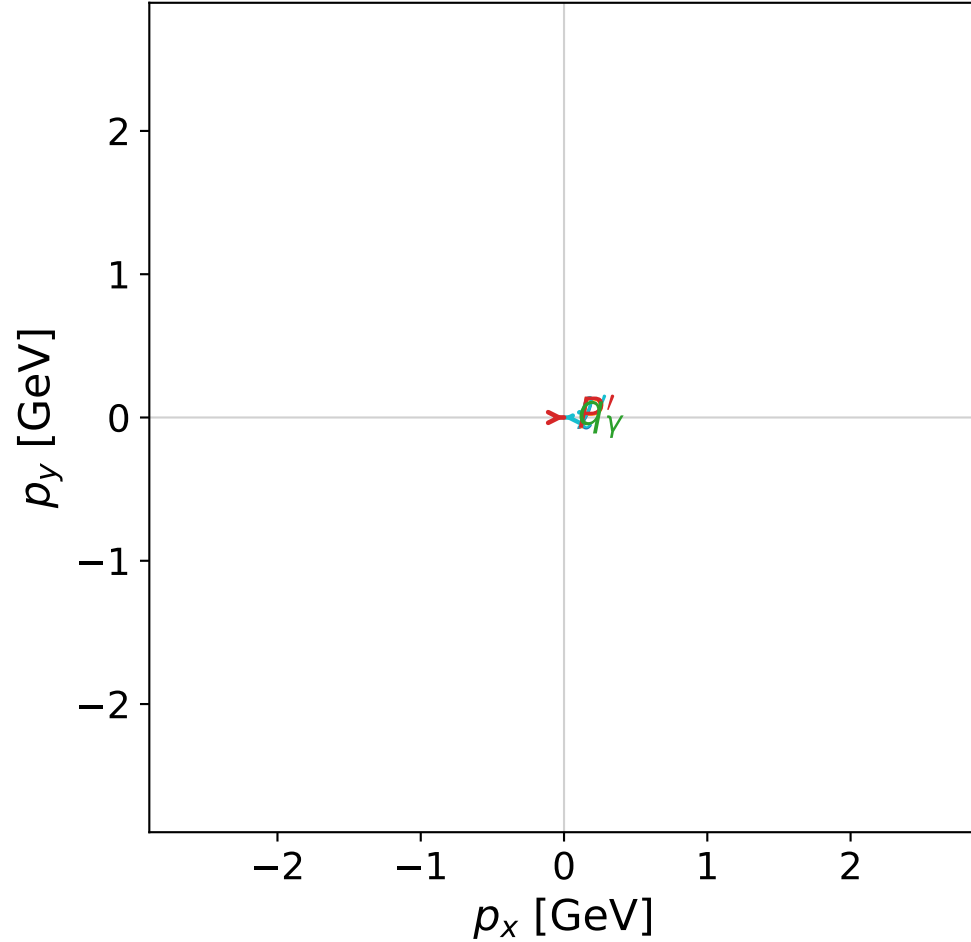


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



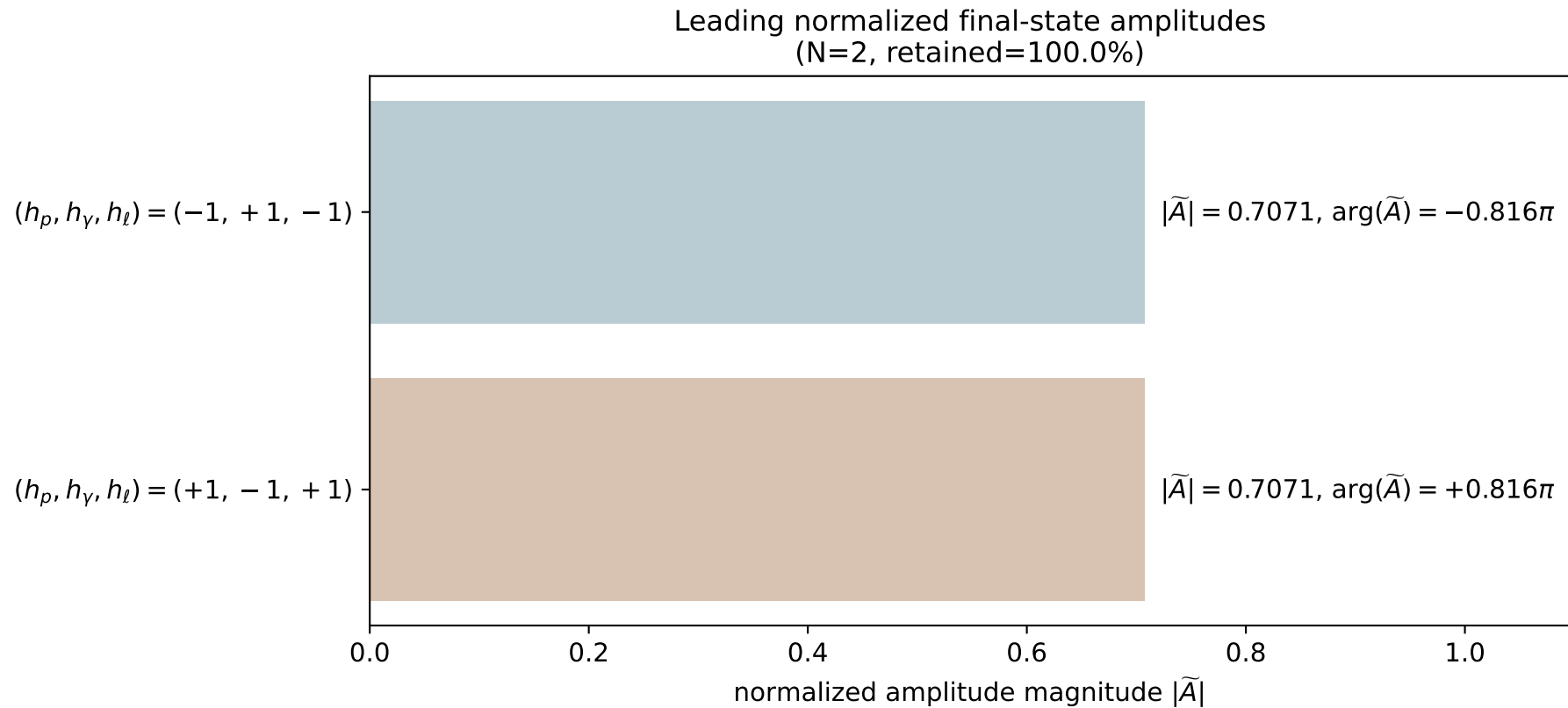
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_38
 $d_{\text{GHZ}}=2.64248\text{e-}08$
 region: polarization_cluster_03_local_minimum_38
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0038
 lepton mixing angle: $\alpha_e=2.4362483$ rad
 incoming lepton: $-0.761388|+\rangle +0.648296|-\rangle$
 proton mixing angle: $\alpha_p=0.70535267$ rad
 incoming proton: $0.761383|+\rangle +0.648302|-\rangle$

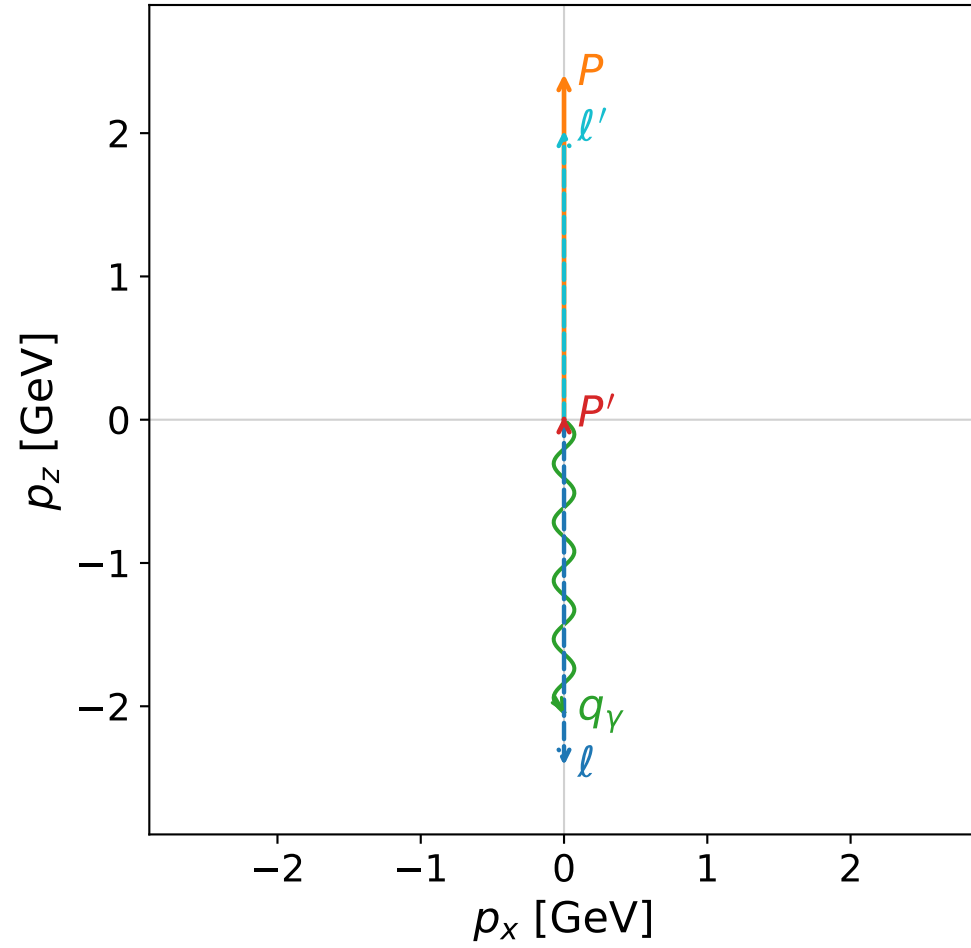
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.140649$
 $\theta_{p'}=3.14094$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.000316874$, $\phi_\gamma=5.70698$
 $E_\gamma=1.95543$, $\theta_{l'}=4.39858\text{e-}05$, $\phi_{l'}=3.14191$
 $Q^2=20.2231$, $x_B=0.864884$, $t=-3.82814$, $W^2=4.03919$, $y=0.969416$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0961$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0961$
 P' $E= 0.9485$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.1406$
 q_γ $E= 1.9554$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9554$

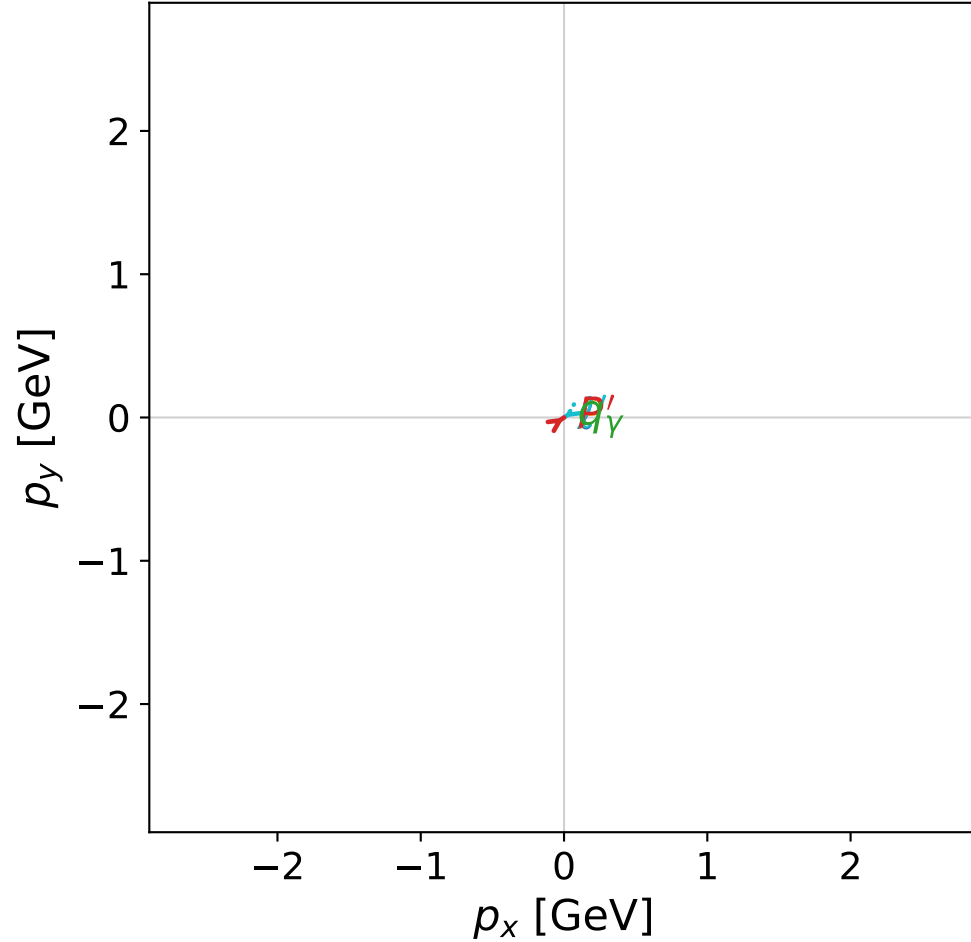


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



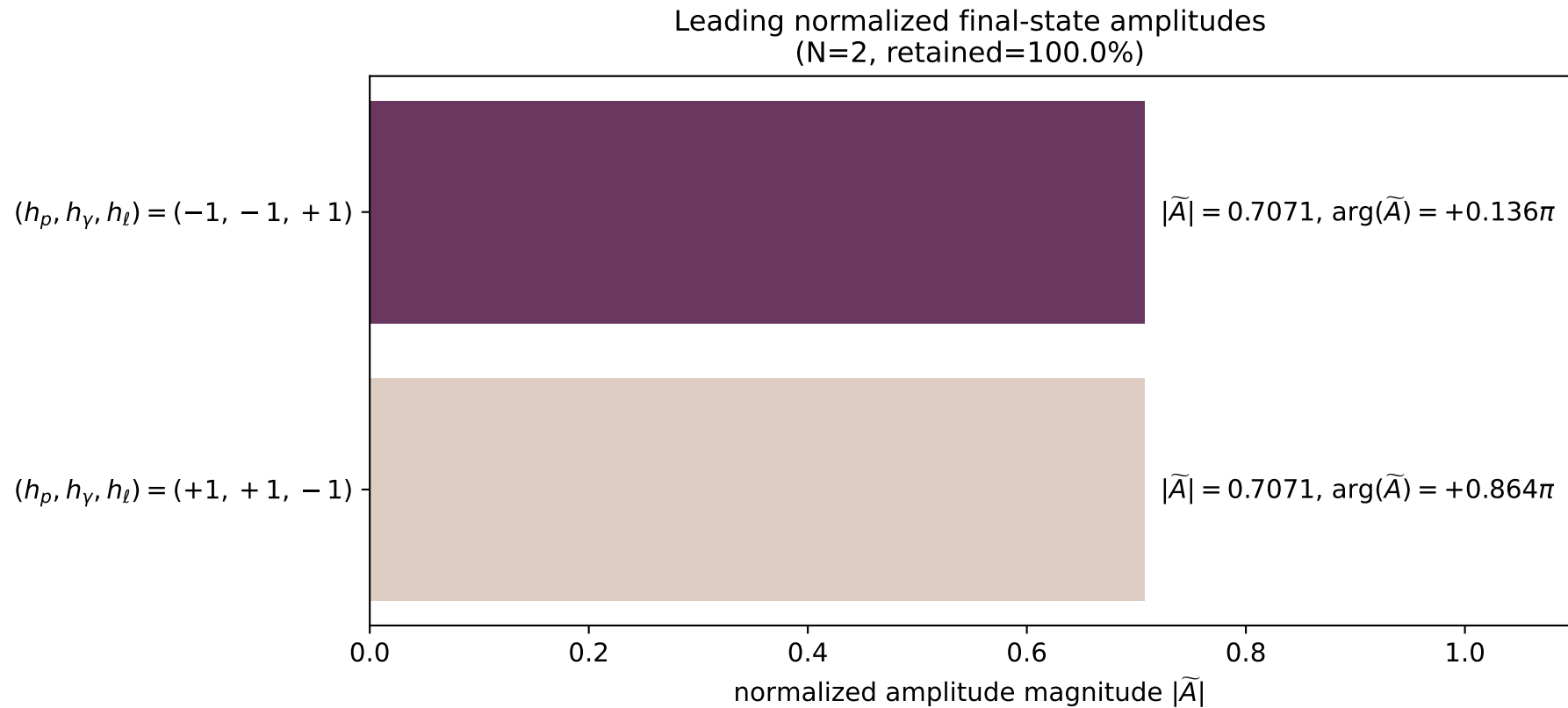
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_43
 $d_{\{\text{rm GHZ}\}}=2.6533\text{e-}08$
 region: polarization_cluster_03_local_minimum_43
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0043
 lepton mixing angle: $\alpha_e=2.3049352$ rad
 incoming lepton: $-0.669948|+\rangle +0.742408|-\rangle$
 proton mixing angle: $\alpha_p=0.83665018$ rad
 incoming proton: $0.669953|+\rangle +0.742403|-\rangle$

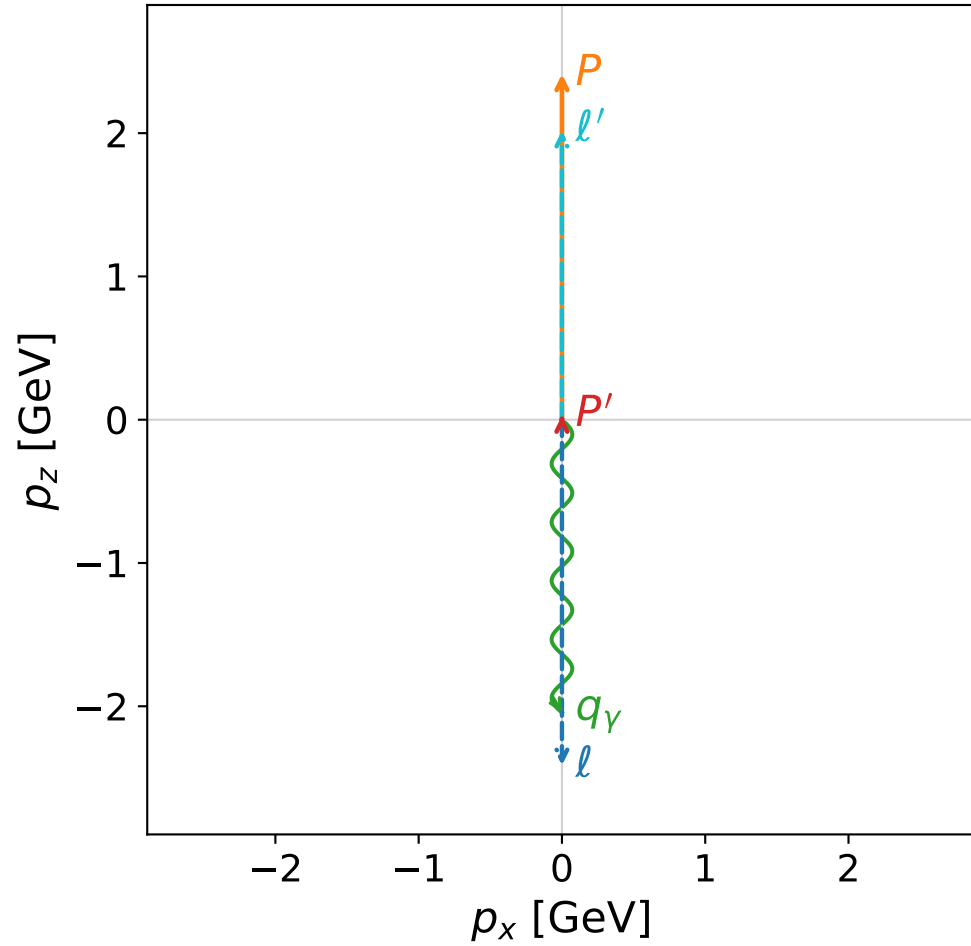
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.023742$
 $\theta_{p'}=3.55227\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.592674$, $\phi_\gamma=3.5688$
 $E_\gamma=2.04272$, $\theta_{\ell'}=4.1803\text{e-}07$, $\phi_{\ell'}=3.73427$
 $Q^2=19.4792$, $x_B=0.832104$, $t=-2.98239$, $W^2=4.81021$, $y=0.970541$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0190$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0190$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0237$
 q_γ $E= 2.0427$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0427$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

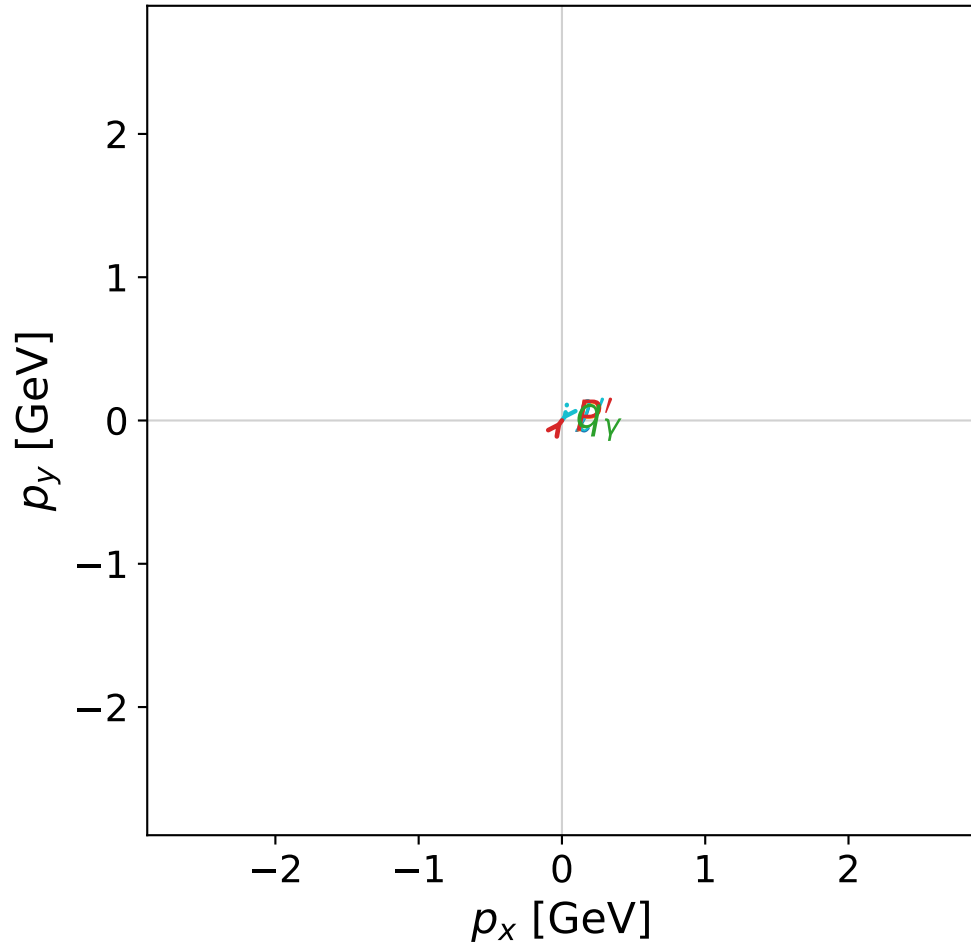


dghz_mixing_angles_polarization_03_local_minimum_45
 $d_{\text{GHZ}}=2.6597\text{e-}08$
 region: polarization_cluster_03_local_minimum_45
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0045
 lepton mixing angle: $\alpha_e=2.3367051$ rad
 incoming lepton: $-0.693192|+\rangle +0.720753|-\rangle$
 proton mixing angle: $\alpha_p=0.80488895$ rad
 incoming proton: $0.693191|+\rangle +0.720754|-\rangle$

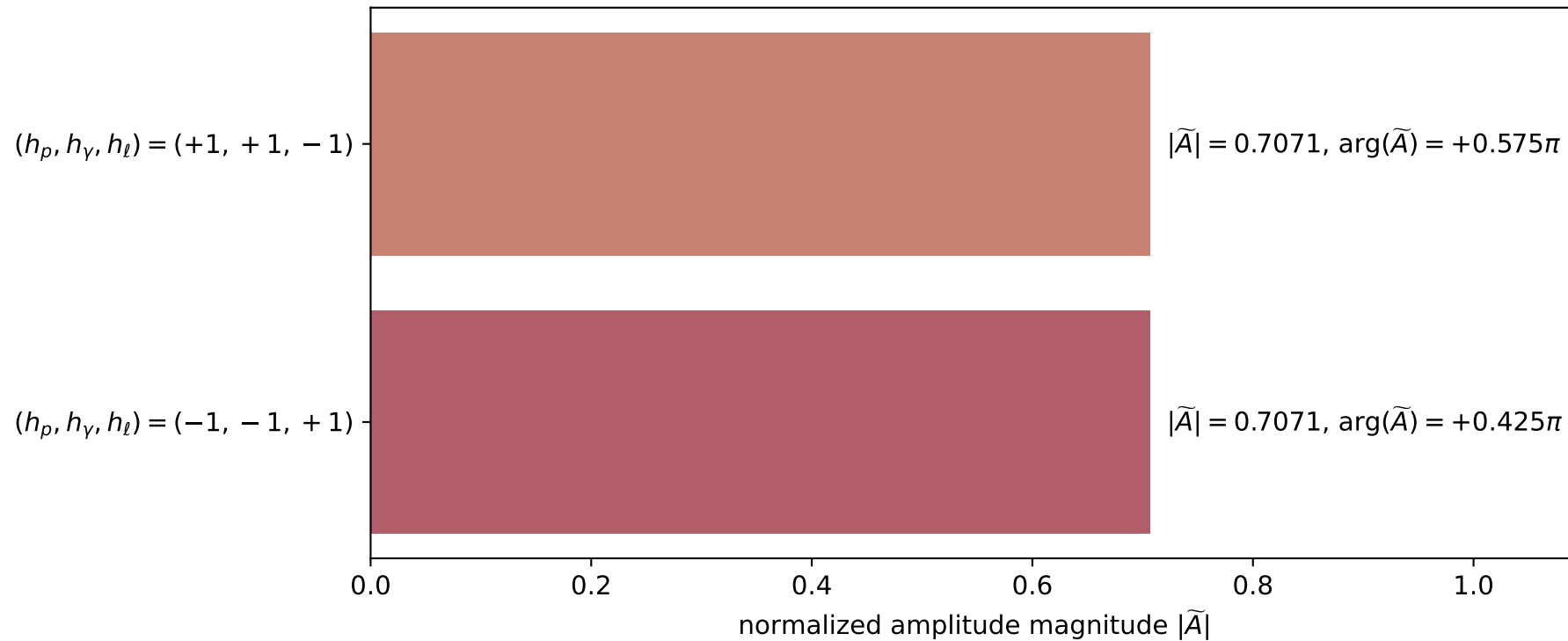
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0277887$
 $\theta_{P'}=0.00245437$, $\theta_Y=3.14159$, $\phi_{P'}=0.937225$, $\phi_Y=4.47598$
 $E_Y=2.04469$, $\theta_{\ell'}=3.38161\text{e-}05$, $\phi_{\ell'}=4.07882$
 $Q^2=19.4592$, $x_B=0.831222$, $t=-2.96345$, $W^2=4.831$, $y=0.970571$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0169$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 2.0169$
 P' $E= 0.9384$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0278$
 q_Y $E= 2.0447$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0447$

x--y plane

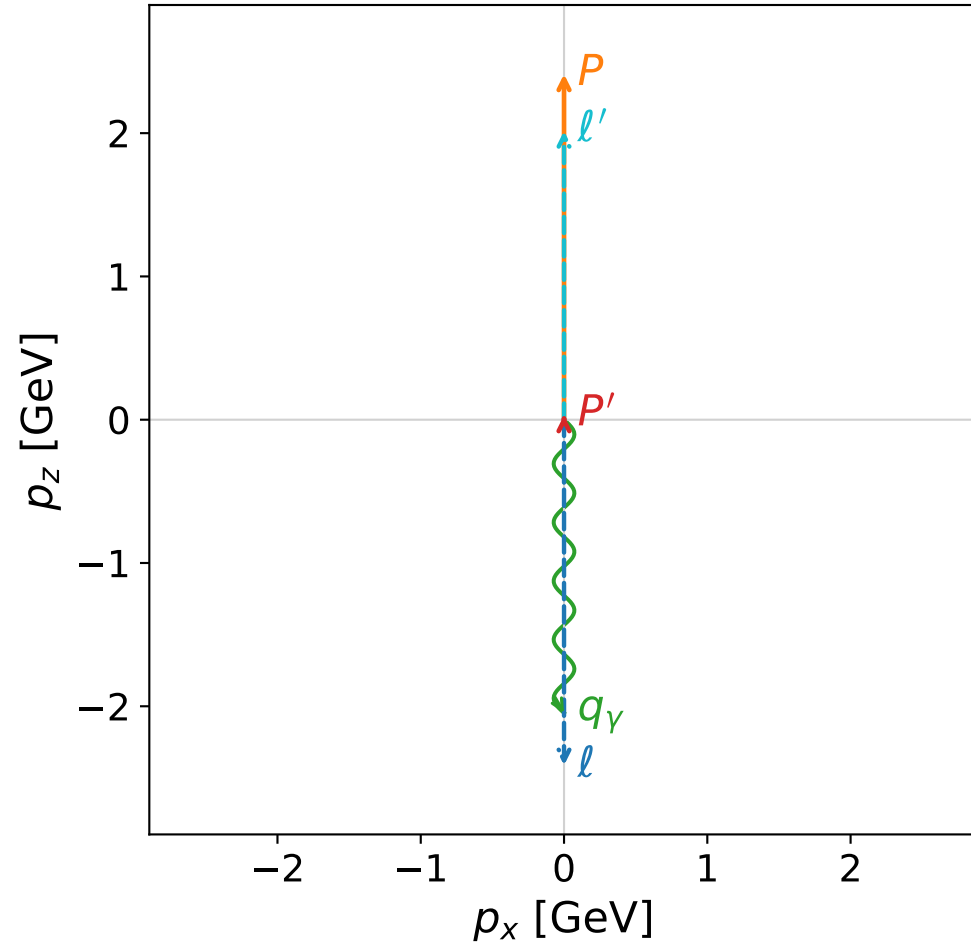


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

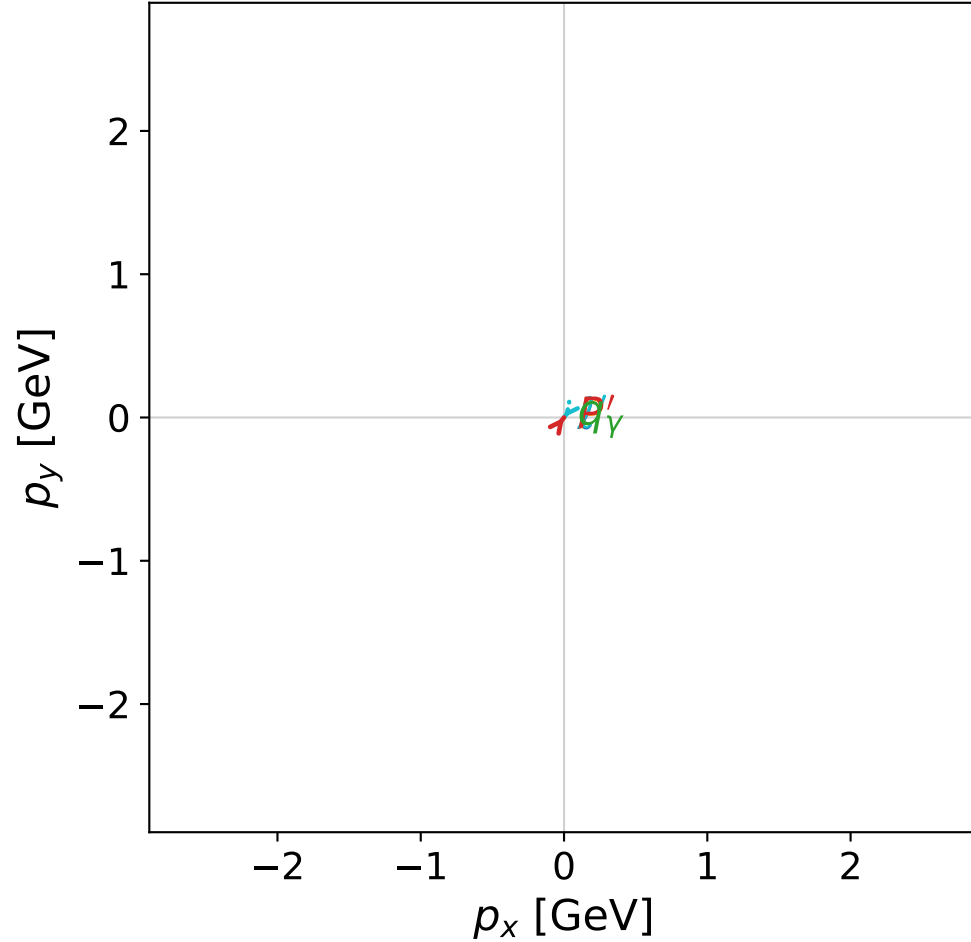


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



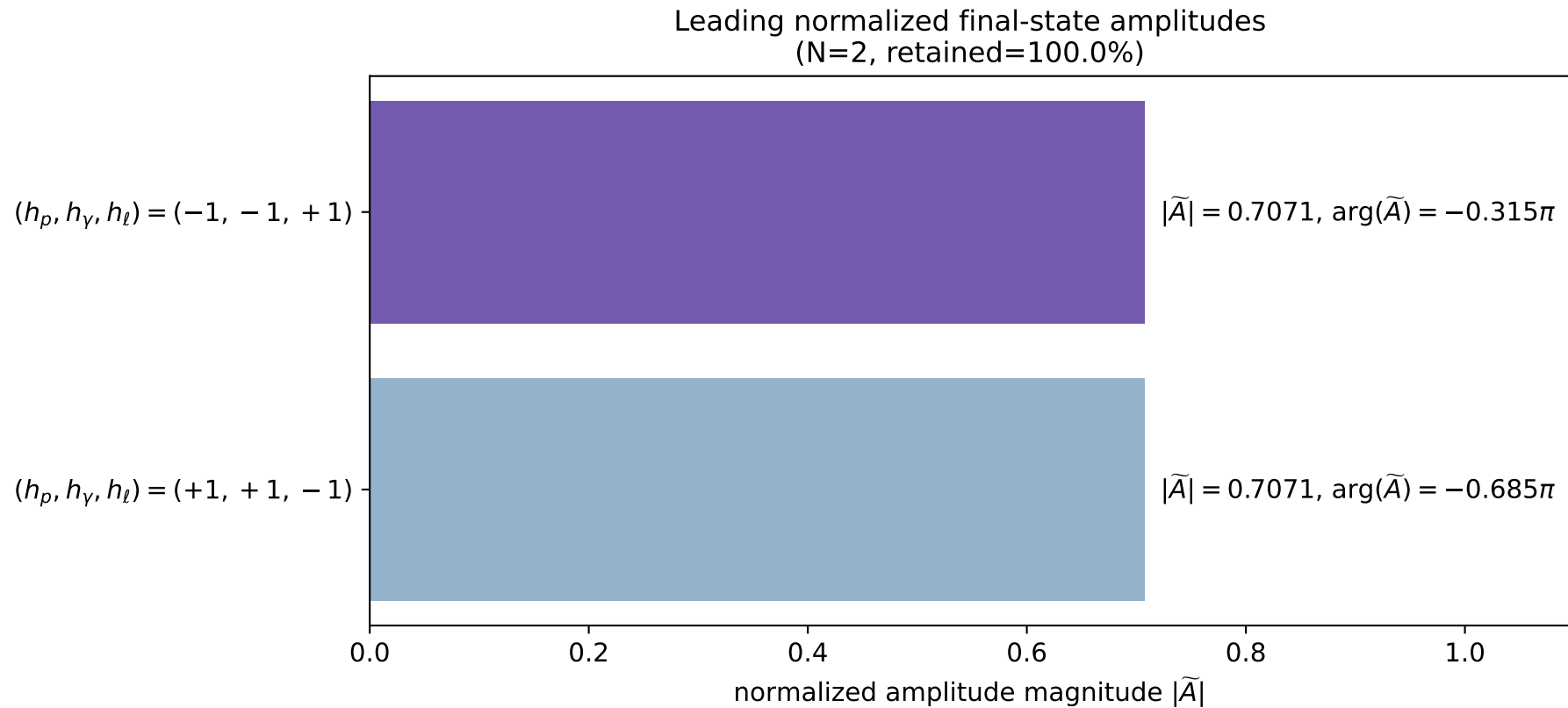
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_48
 $d_{\{\rm GHZ\}}=2.67266\text{e-}08$
 region: polarization_cluster_03_local_minimum_48
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0048
 lepton mixing angle: $\alpha_e=2.3692897$ rad
 incoming lepton: $-0.716306|+\rangle +0.697787|-\rangle$
 proton mixing angle: $\alpha_p=0.772294$ rad
 incoming proton: $0.716312|+\rangle +0.69778|-\rangle$

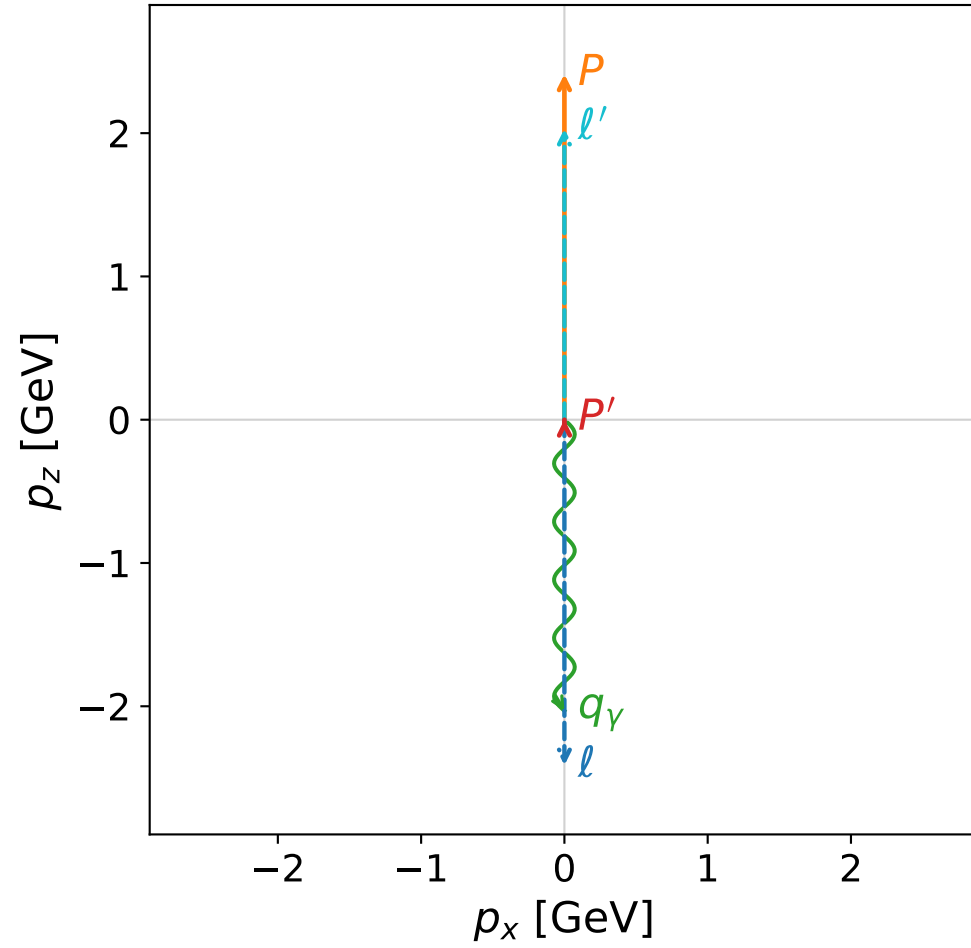
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0314813$
 $\theta_{p'}=0.000613592$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.92167$, $\phi_\gamma=2.15105$
 $E_\gamma=2.04648$, $\theta_{\ell'}=9.58645\text{e-}06$, $\phi_{\ell'}=4.06326$
 $Q^2=19.4408$, $x_B=0.830413$, $t=-2.94624$, $W^2=4.85005$, $y=0.970599$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0150$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0150$
 P' $E= 0.9385$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0315$
 q_γ $E= 2.0465$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0465$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

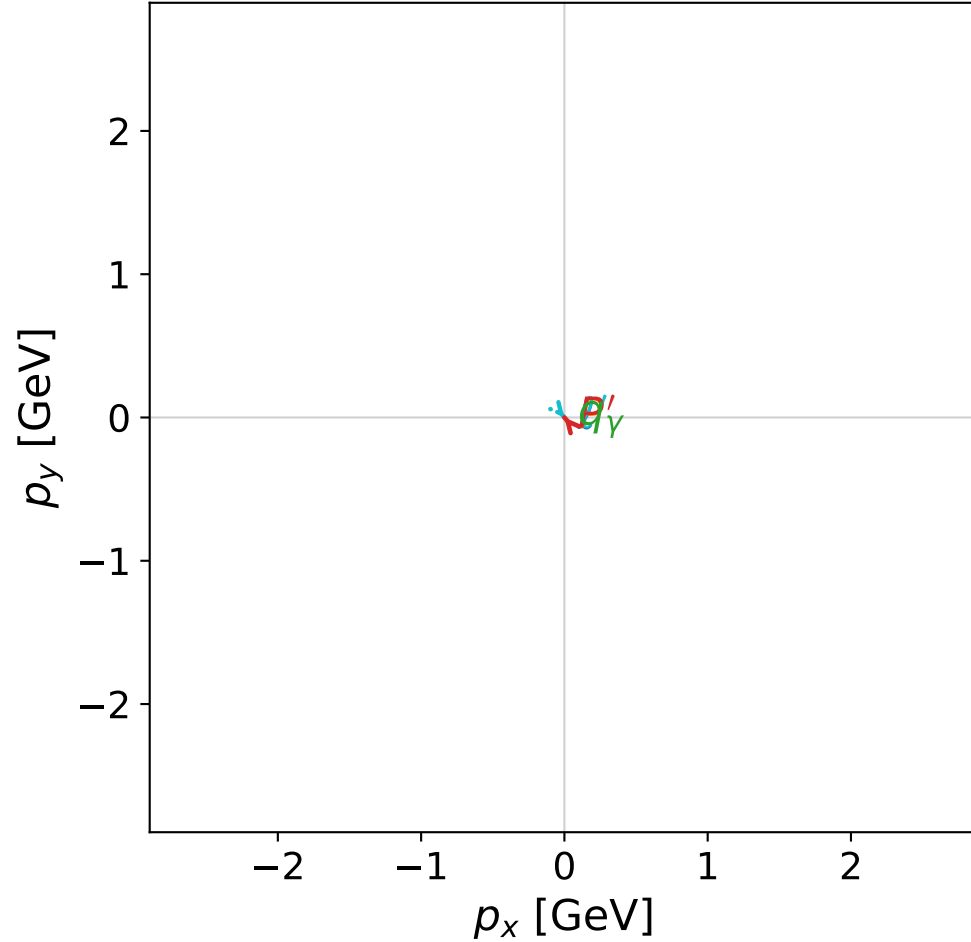


dghz_mixing_angles_polarization_03_local_minimum_51
 $d_{\text{GHZ}}=2.68417\text{e-}08$
 region: polarization_cluster_03_local_minimum_51
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0051
 lepton mixing angle: $\alpha_e=2.2220181$ rad
 incoming lepton: $-0.606159|+\rangle +0.795344|-\rangle$
 proton mixing angle: $\alpha_p=0.91958237$ rad
 incoming proton: $0.606152|+\rangle +0.795349|-\rangle$

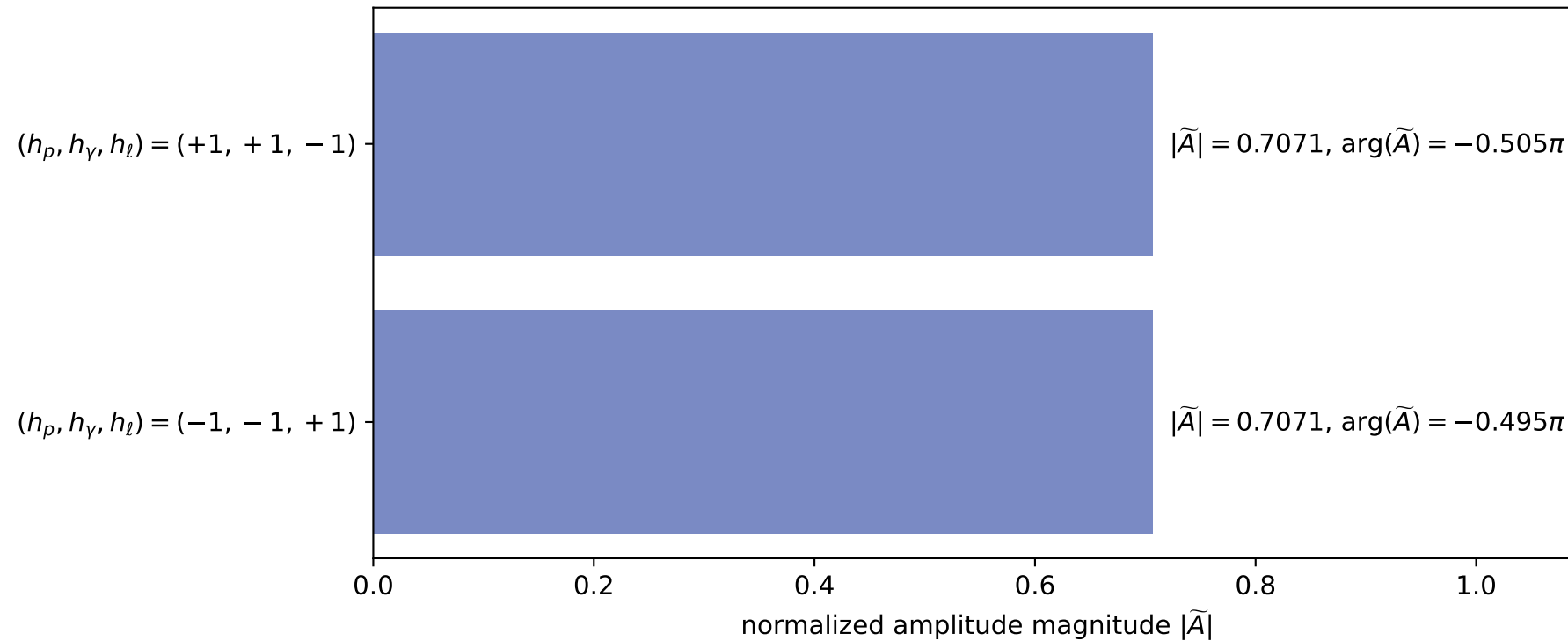
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.001359$
 $\theta_{P'}=0.00981748$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.27173$, $\phi_\gamma=1.5862$
 $E_\gamma=2.03168$, $\theta_{\ell'}=6.57126\text{e-}06$, $\phi_{\ell'}=5.41332$
 $Q^2=19.5887$, $x_B=0.836921$, $t=-3.08882$, $W^2=4.6968$, $y=0.970376$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0303$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0303$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0014$
 q_γ $E= 2.0317$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0317$

x--y plane

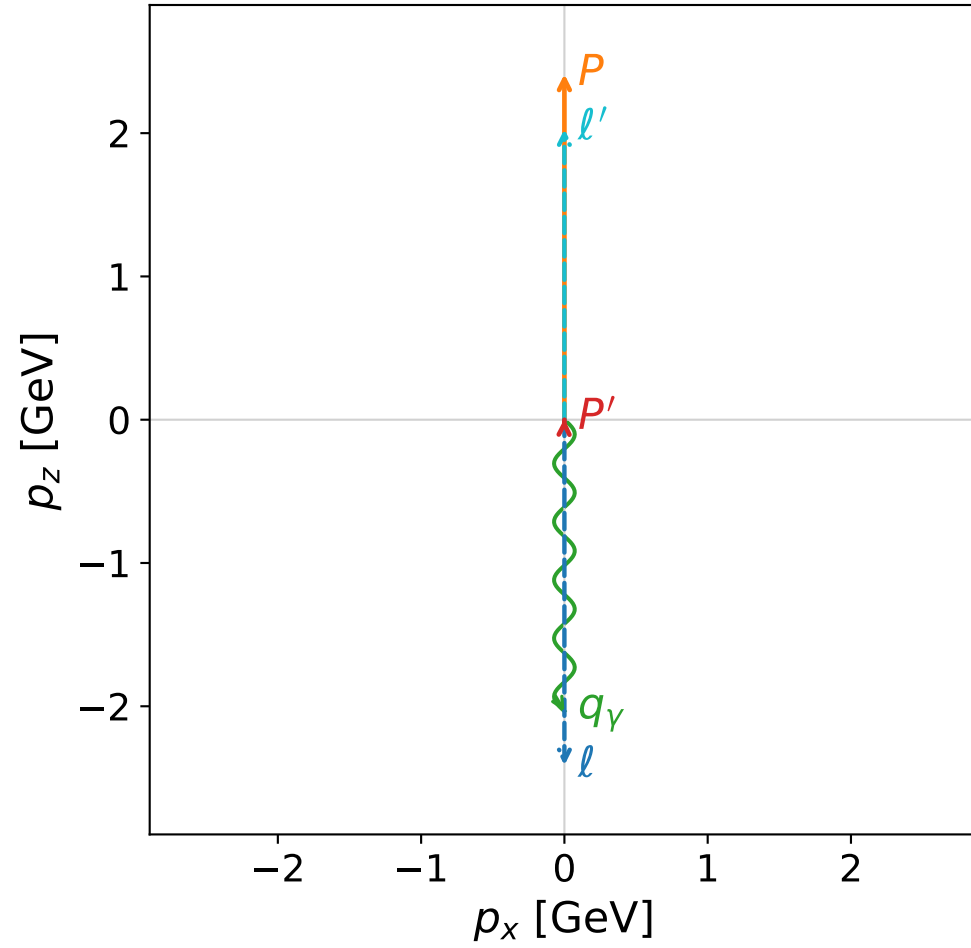


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

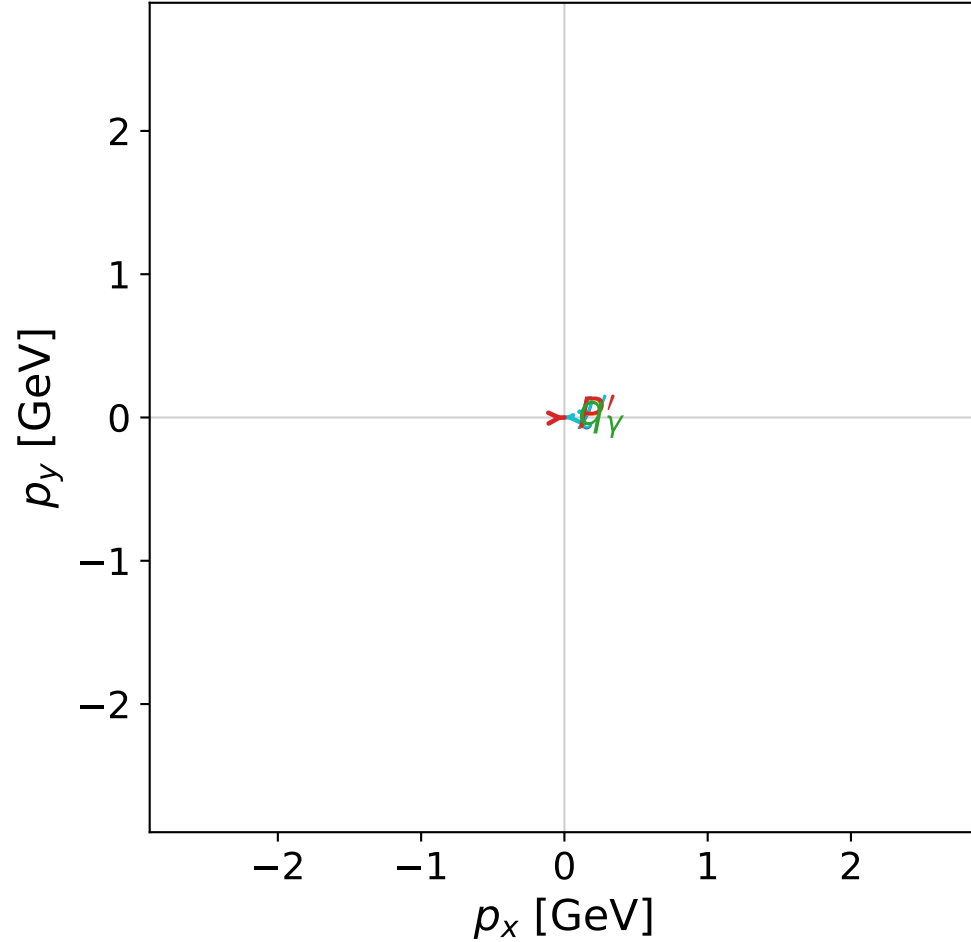


dghz_mixing_angles_polarization_03_local_minimum_52
 $d_{\text{GHZ}}=2.69014\text{e-}08$
 region: polarization_cluster_03_local_minimum_52
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0052
 lepton mixing angle: $\alpha_e=2.4223145$ rad
 incoming lepton: $-0.752281|+\rangle +0.658842|-\rangle$
 proton mixing angle: $\alpha_p=0.71927442$ rad
 incoming proton: $0.752284|+\rangle +0.658839|-\rangle$

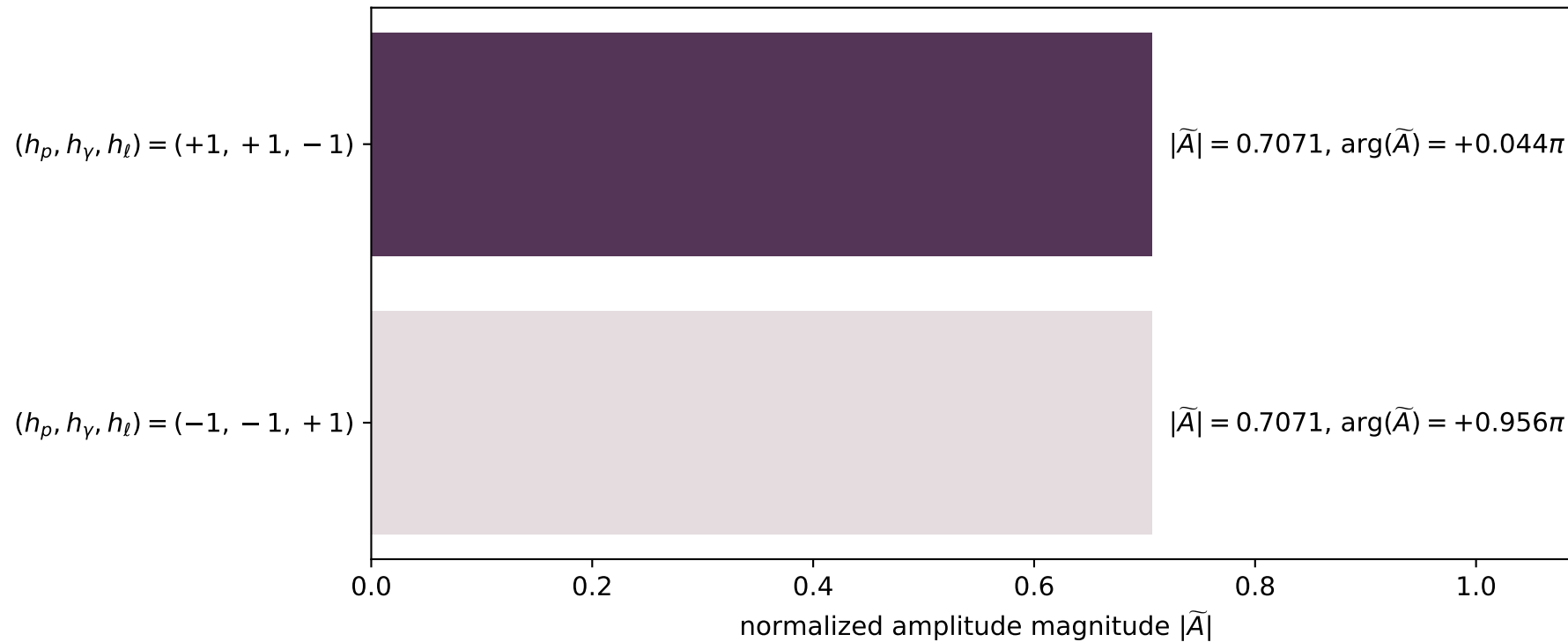
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00684209$
 $\theta_{P'}=0.0196328$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0386213$, $\phi_\gamma=6.14533$
 $E_\gamma=2.03441$, $\theta_{\ell'}=6.62473\text{e-}05$, $\phi_{\ell'}=3.18021$
 $Q^2=19.5621$, $x_B=0.835752$, $t=-3.0625$, $W^2=4.72433$, $y=0.970416$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0276$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0276$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0068$
 q_γ $E= 2.0344$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0344$

x--y plane

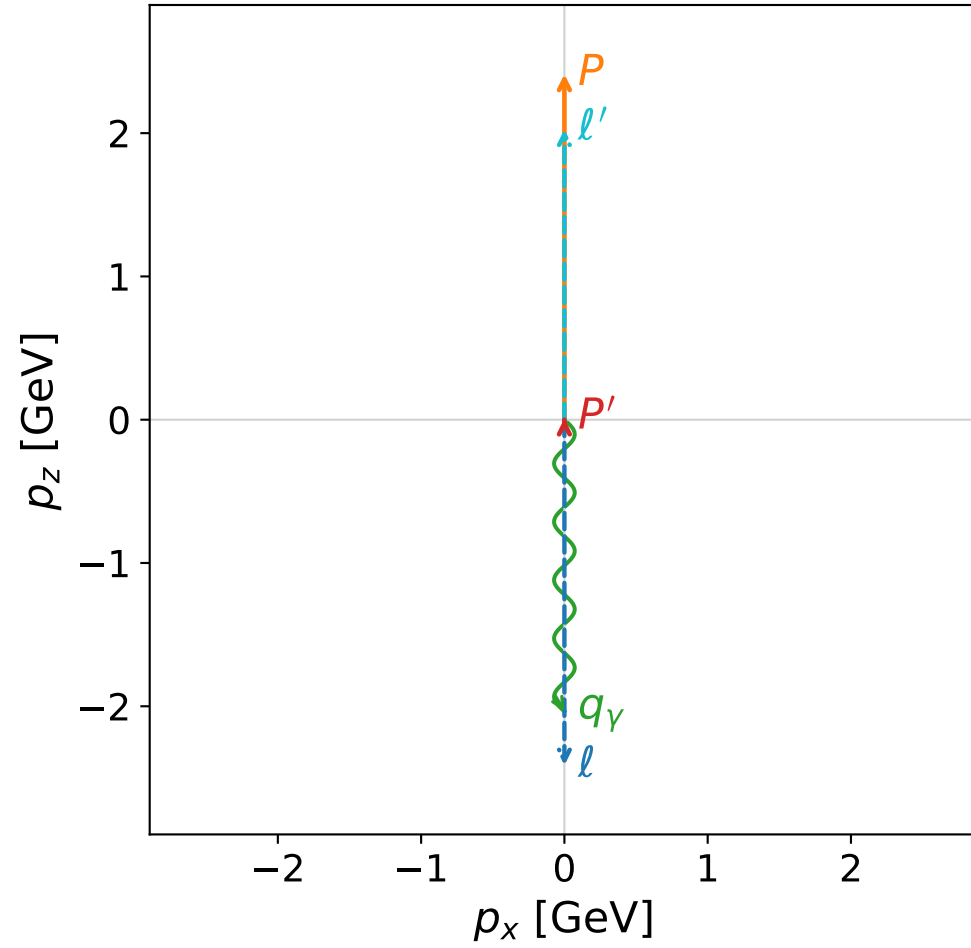


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

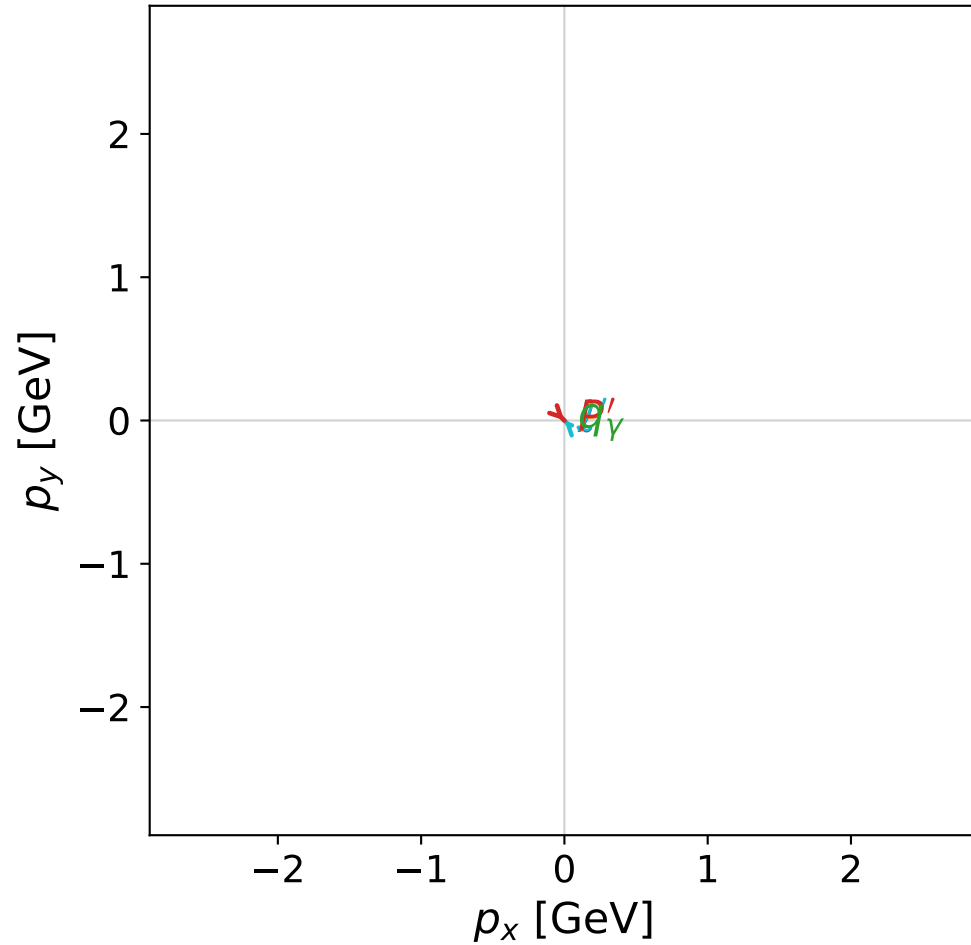


dghz_mixing_angles_polarization_03_local_minimum_53
 $d_{\text{GHZ}}=2.69649\text{e-}08$
 region: polarization_cluster_03_local_minimum_53
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0053
 lepton mixing angle: $\alpha_e=2.2615246$ rad
 incoming lepton: $-0.637099|+\rangle +0.770782|-\rangle$
 proton mixing angle: $\alpha_p=0.8800608$ rad
 incoming proton: $0.637104|+\rangle +0.770778|-\rangle$

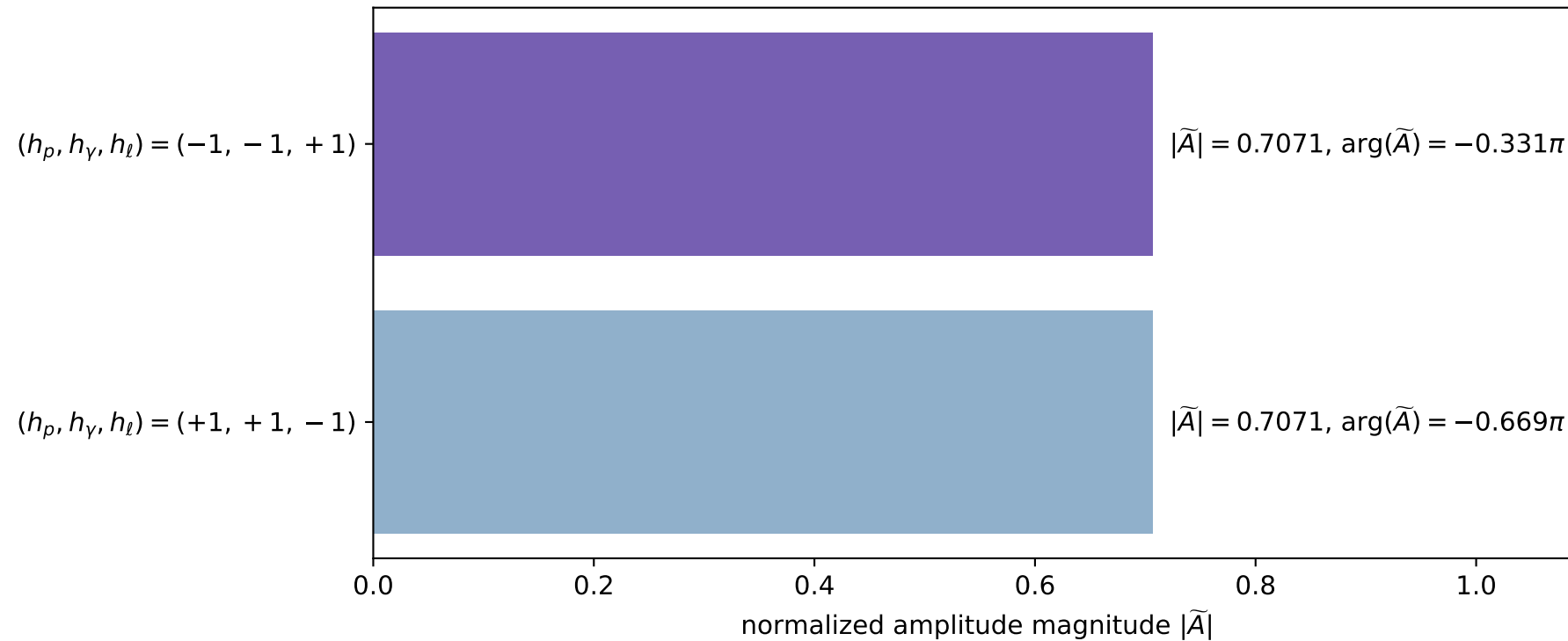
$s=24.9999$, $\sqrt{s}=4.99999$, $|\vec{P}|=2.41201$, $|\vec{P}'|=0.00908808$
 $\theta_{P'}=0.000472367$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.49656$, $\phi_\gamma=2.10193$
 $E_\gamma=2.03552$, $\theta_{\ell'}=2.11843\text{e-}06$, $\phi_{\ell'}=2.35497$
 $Q^2=19.5511$, $x_B=0.83527$, $t=-3.05175$, $W^2=4.73565$, $y=0.970432$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0264$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0264$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0091$
 q_γ $E= 2.0355$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0355$

x--y plane

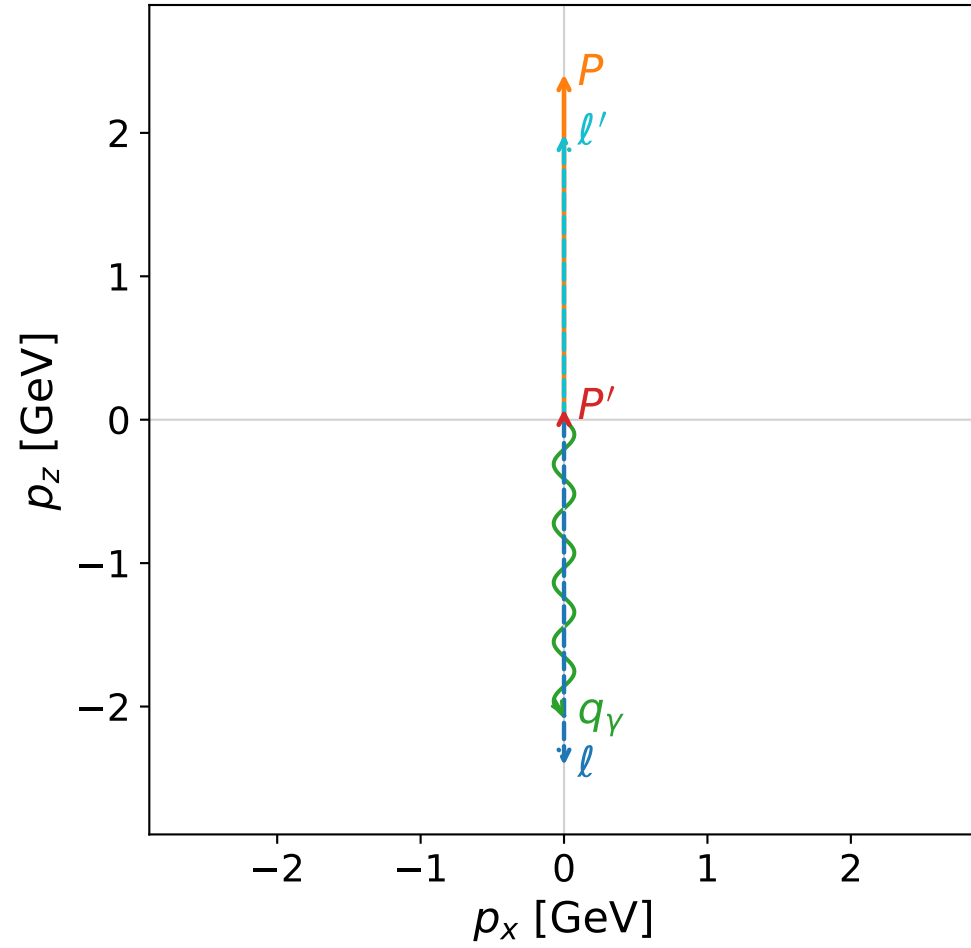


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

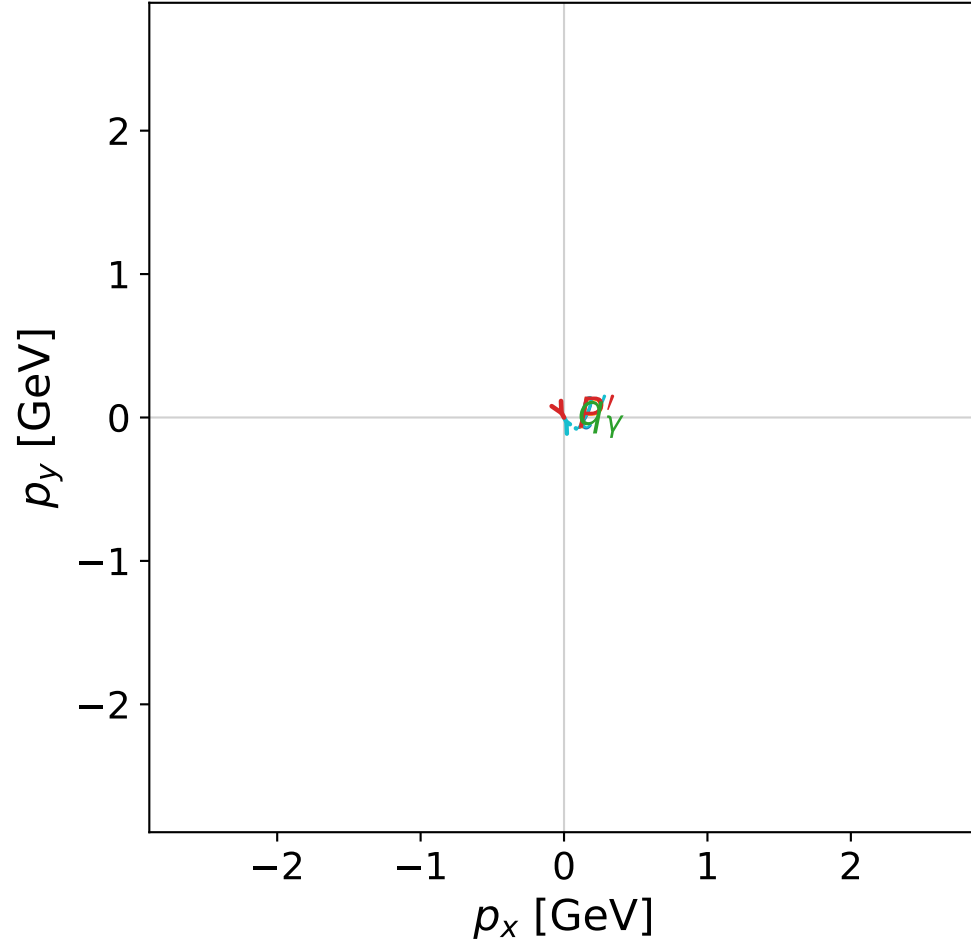


dghz_mixing_angles_polarization_03_local_minimum_55
 $d_{\{\rm GHZ\}}=2.70614\text{e-}08$
 region: polarization_cluster_03_local_minimum_55
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0055
 lepton mixing angle: $\alpha_e=2.360551$ rad
 incoming lepton: $-0.710181|+\rangle +0.70402|-\rangle$
 proton mixing angle: $\alpha_p=0.78104501$ rad
 incoming proton: $0.710178|+\rangle +0.704022|-\rangle$

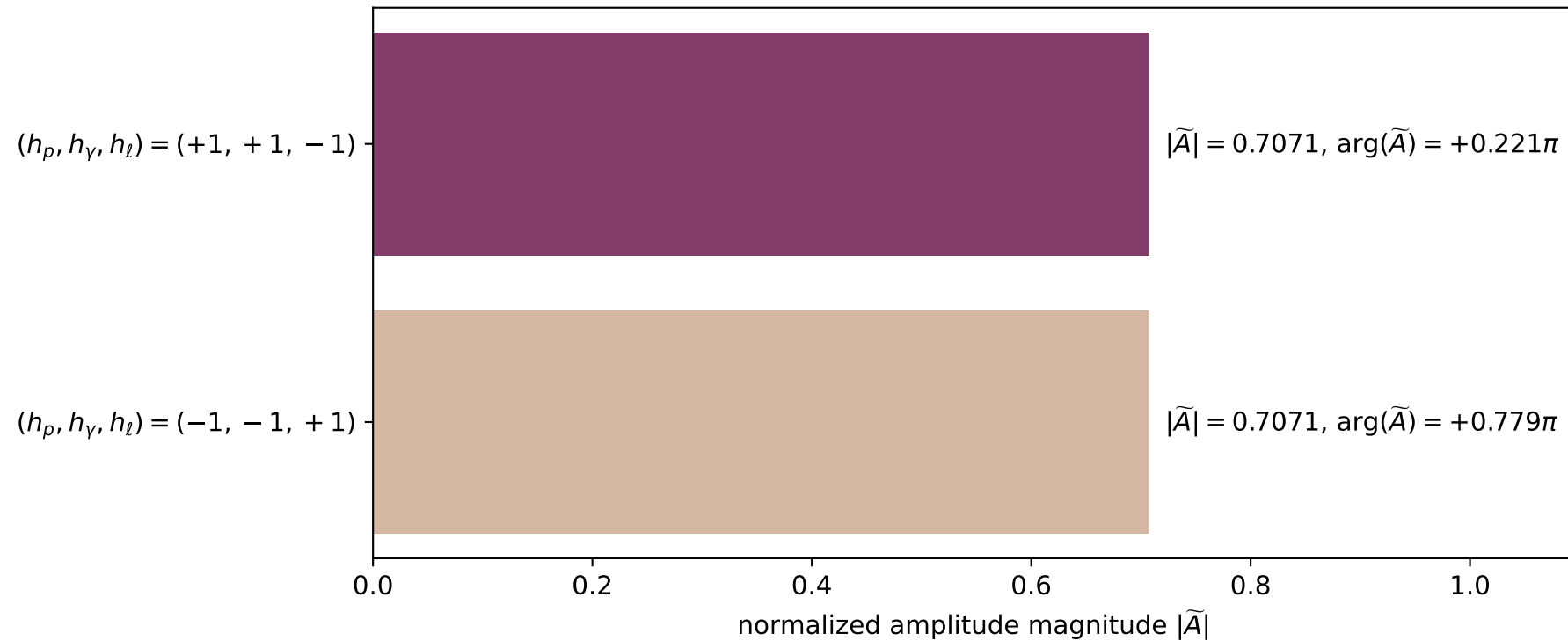
$s=24.9555$, $\sqrt{s}=4.99555$, $|\vec{P}|=2.40971$, $|\vec{P}'|=0.0757091$
 $\theta_{p'}=7.31897\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=5.21638$, $\phi_\gamma=5.58978$
 $E_\gamma=2.0651$, $\theta_{\ell'}=2.78532\text{e-}06$, $\phi_{\ell'}=2.07479$
 $Q^2=19.1755$, $x_B=0.820345$, $t=-2.74225$, $W^2=5.07927$, $y=0.970893$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4097$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4097$
 P $E= 2.5858$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4097$
 ℓ' $E= 1.9894$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9894$
 P' $E= 0.9411$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0757$
 q_γ $E= 2.0651$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0651$

x--y plane

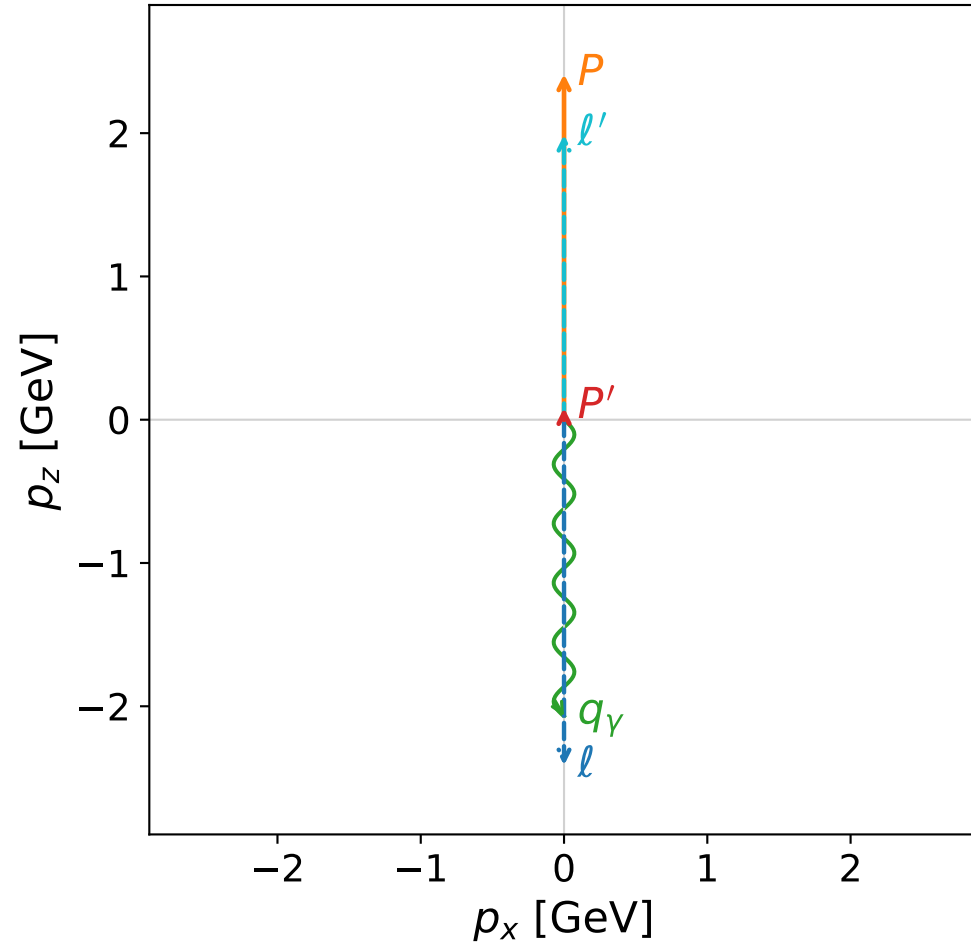


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

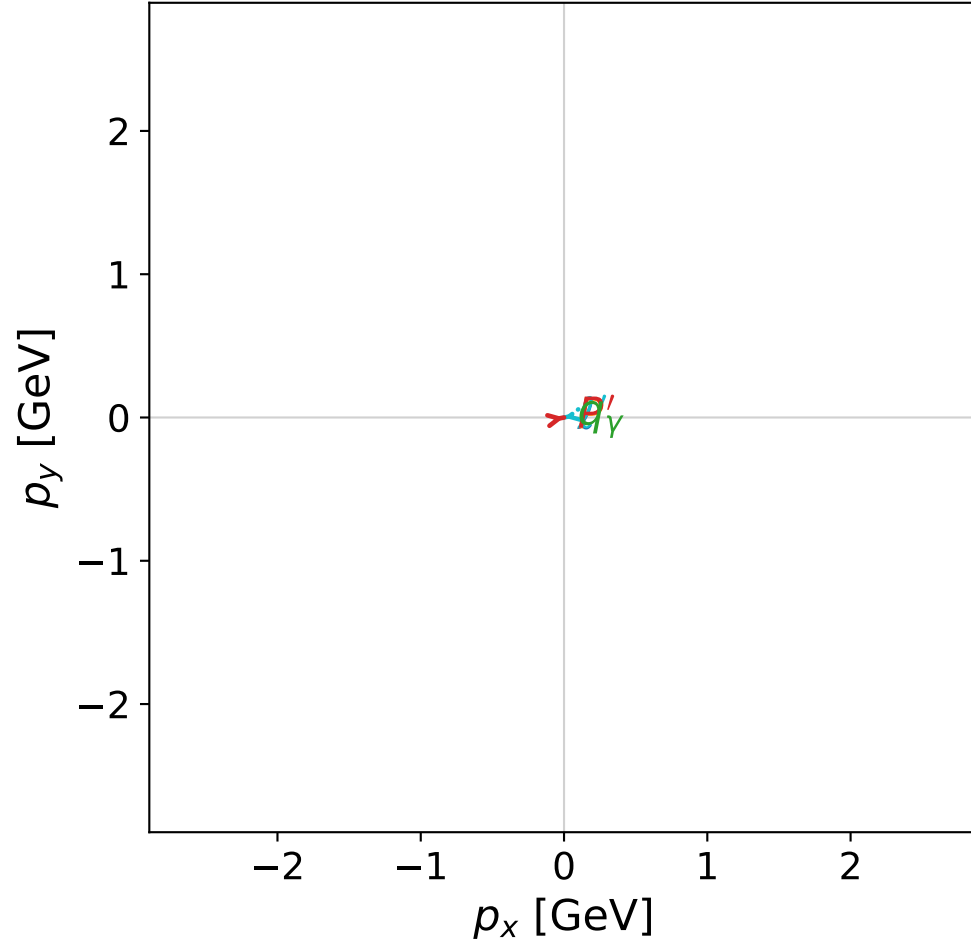


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



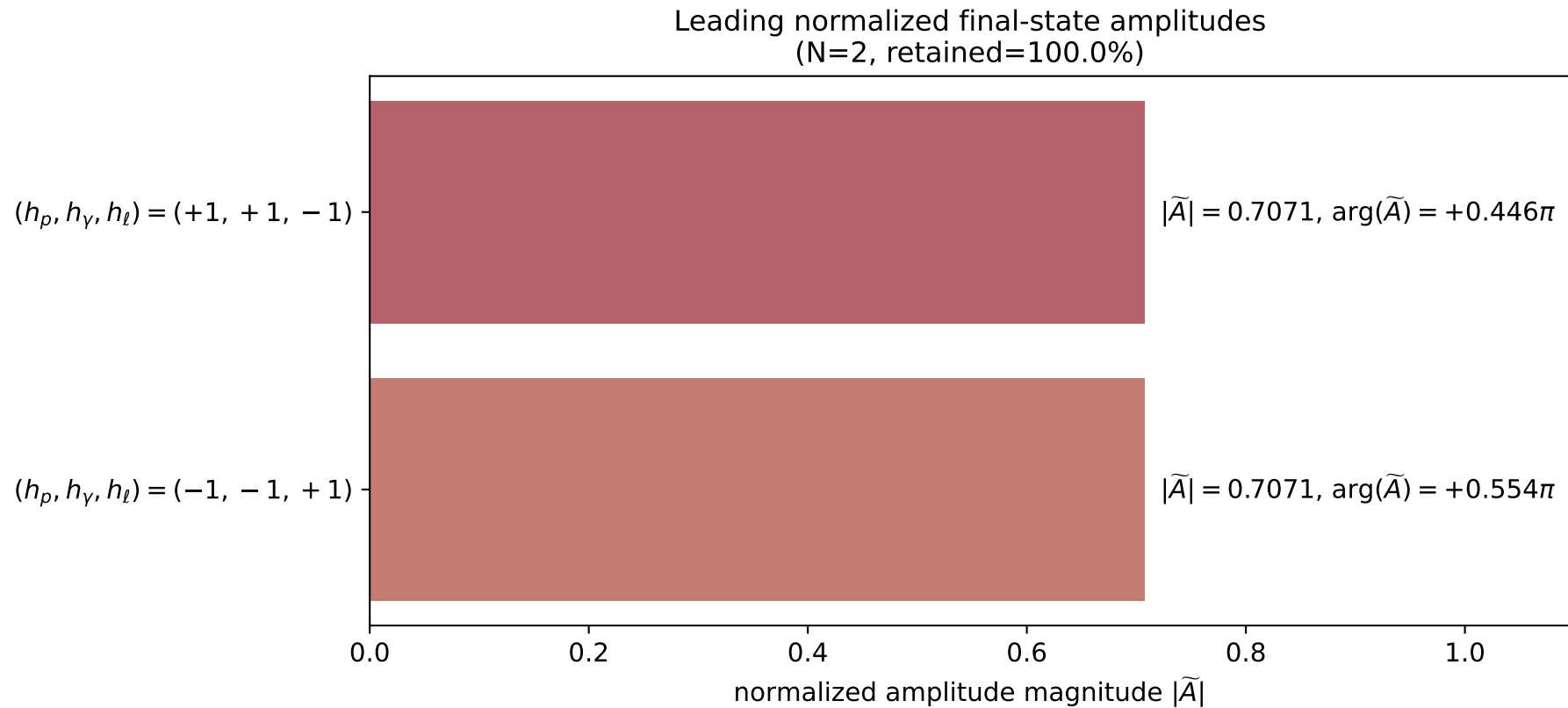
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_56
 $d_{\text{GHZ}}=2.70735\text{e-}08$
 region: polarization_cluster_03_local_minimum_56
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0056
 lepton mixing angle: $\alpha_e=2.3601707$ rad
 incoming lepton: $-0.709913|+\rangle +0.70429|-\rangle$
 proton mixing angle: $\alpha_p=0.78143038$ rad
 incoming proton: $0.709907|+\rangle +0.704296|-\rangle$

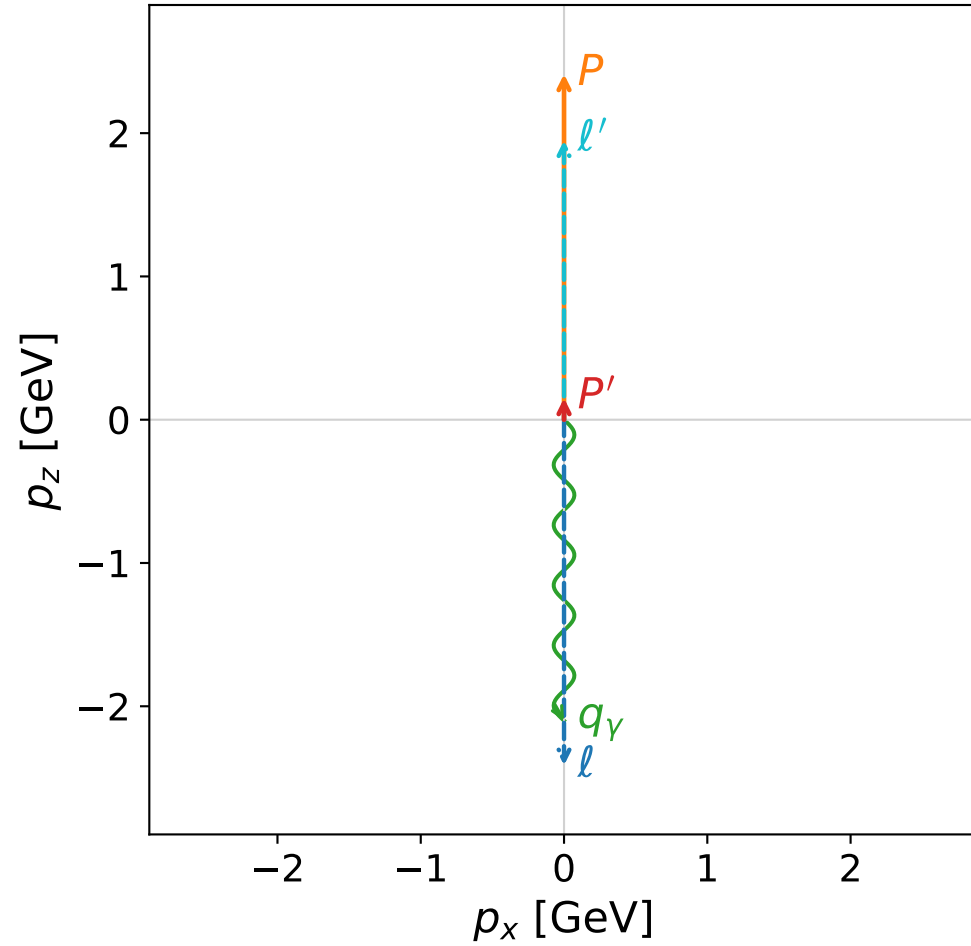
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0826338$
 $\theta_{P'}=7.6699\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.192921$, $\phi_\gamma=4.88356$
 $E_\gamma=2.0705$, $\theta_{\ell'}=3.18833\text{e-}06$, $\phi_{\ell'}=3.33451$
 $Q^2=19.1791$, $x_B=0.818899$, $t=-2.71555$, $W^2=5.12133$, $y=0.970995$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9879$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9879$
 P' $E= 0.9416$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0826$
 q_γ $E= 2.0705$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0705$

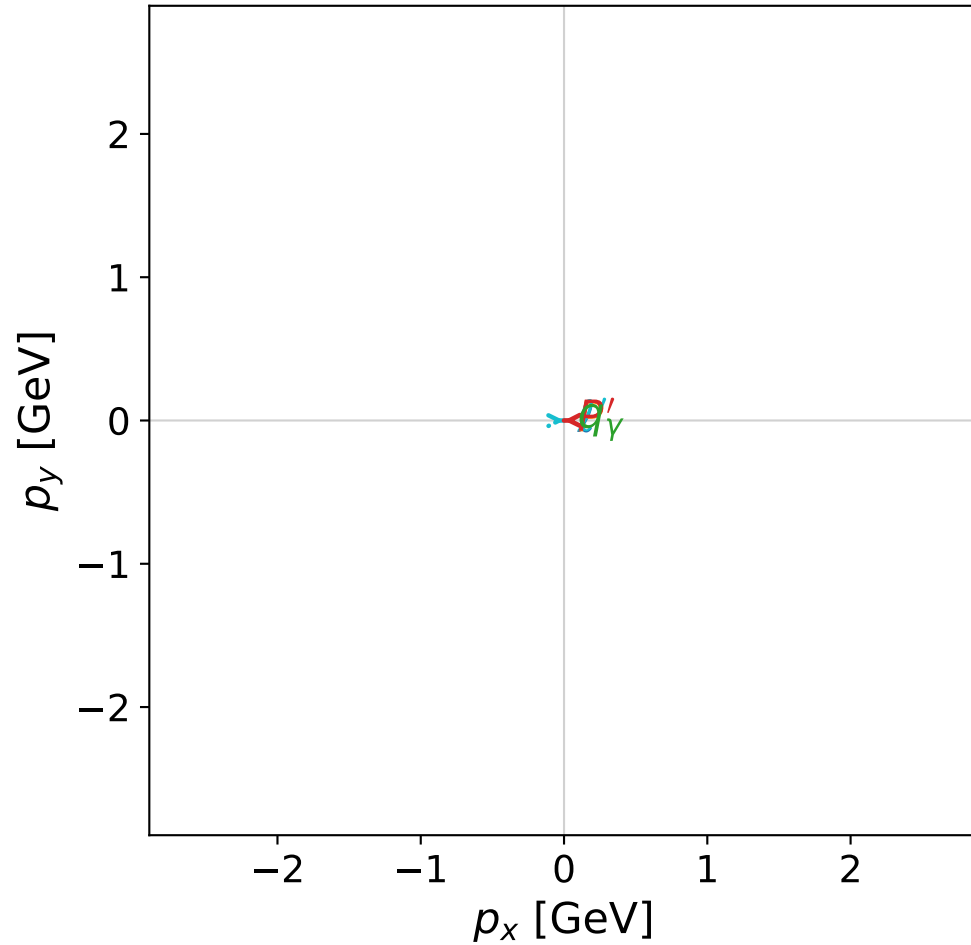


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



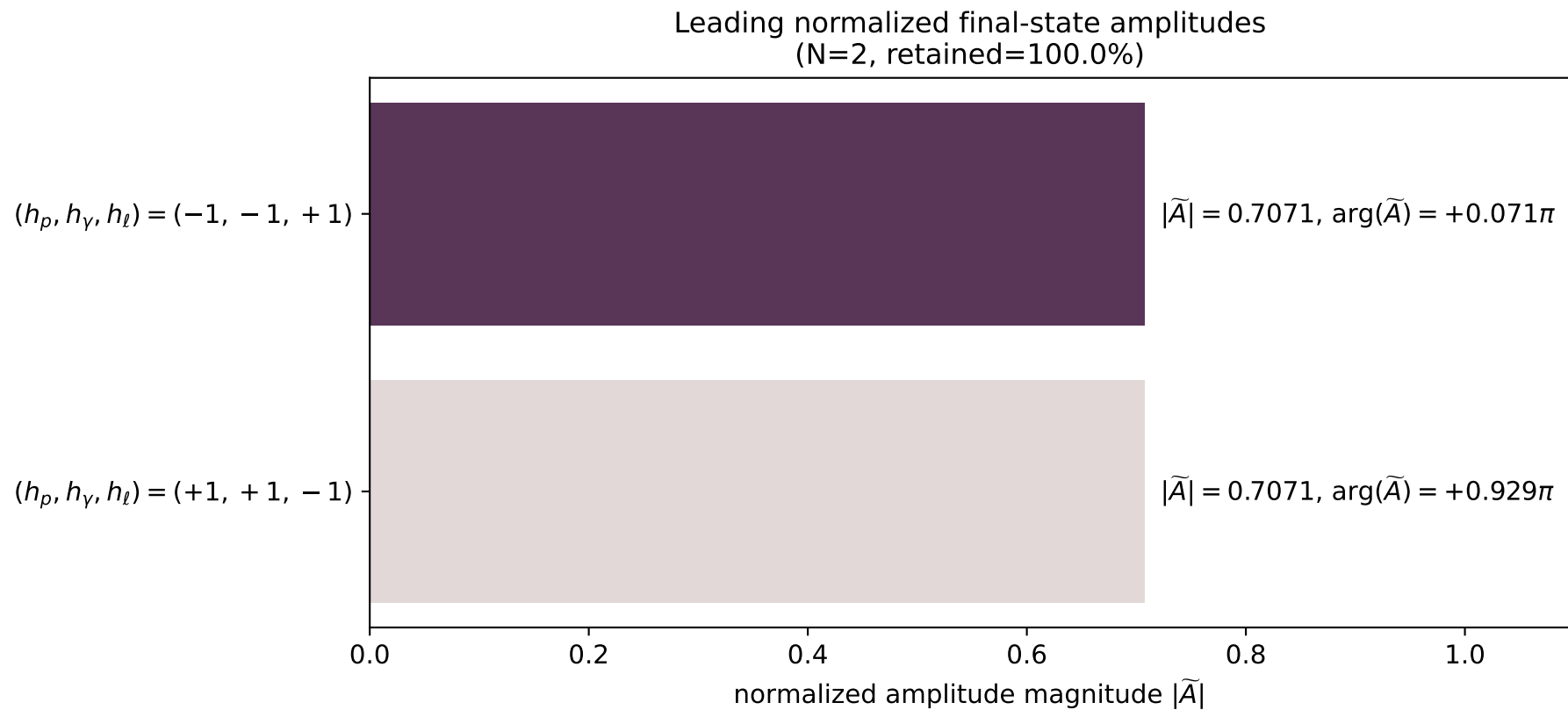
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_60
 $d_{\{\rm GHZ\}}=2.74796\text{e-}08$
 region: polarization_cluster_03_local_minimum_60
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0060
 lepton mixing angle: $\alpha_e=2.3243577$ rad
 incoming lepton: $-0.68424|+\rangle +0.729257|-\rangle$
 proton mixing angle: $\alpha_p=0.8172232$ rad
 incoming proton: $0.684249|+\rangle +0.729249|-\rangle$

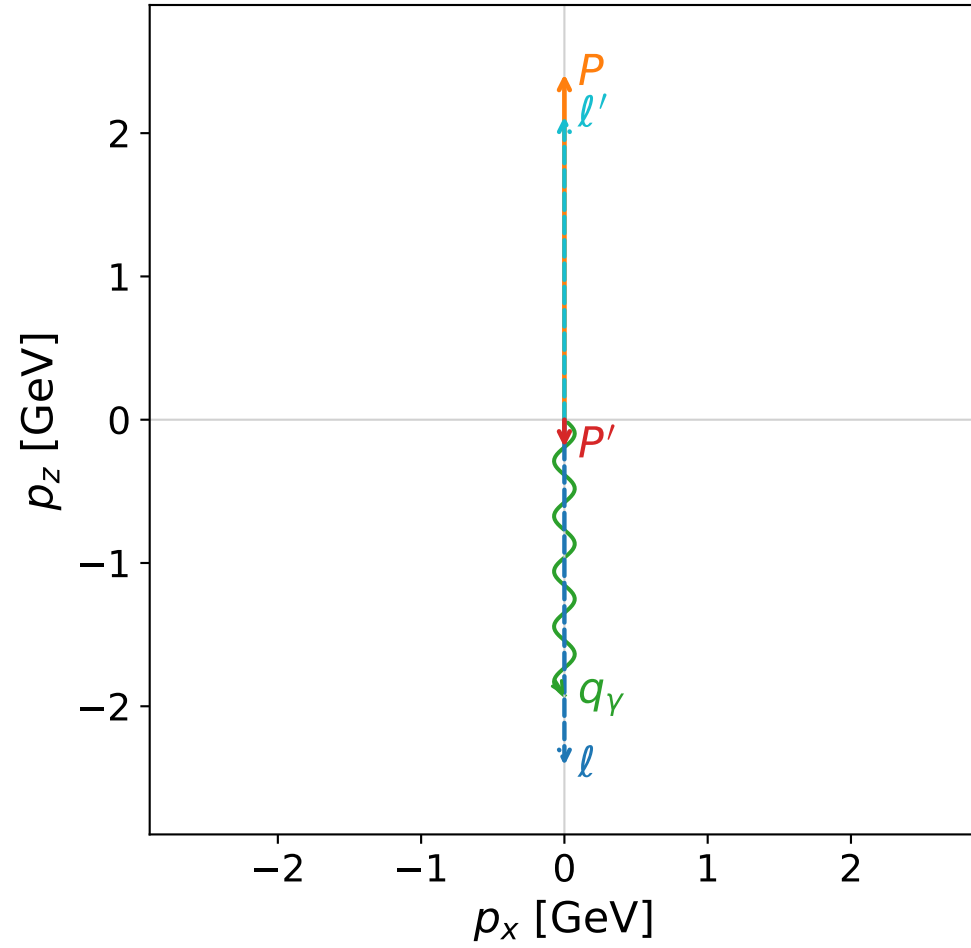
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.150619$
 $\theta_{p'}=5.43472\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.14933$, $\phi_\gamma=3.36397$
 $E_\gamma=2.1003$, $\theta_{\ell'}=4.19848\text{e-}06$, $\phi_{\ell'}=0.00773789$
 $Q^2=18.8107$, $x_B=0.802708$, $t=-2.43097$, $W^2=5.50317$, $y=0.971552$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9497$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9497$
 P' $E= 0.9500$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1506$
 q_γ $E= 2.1003$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1003$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

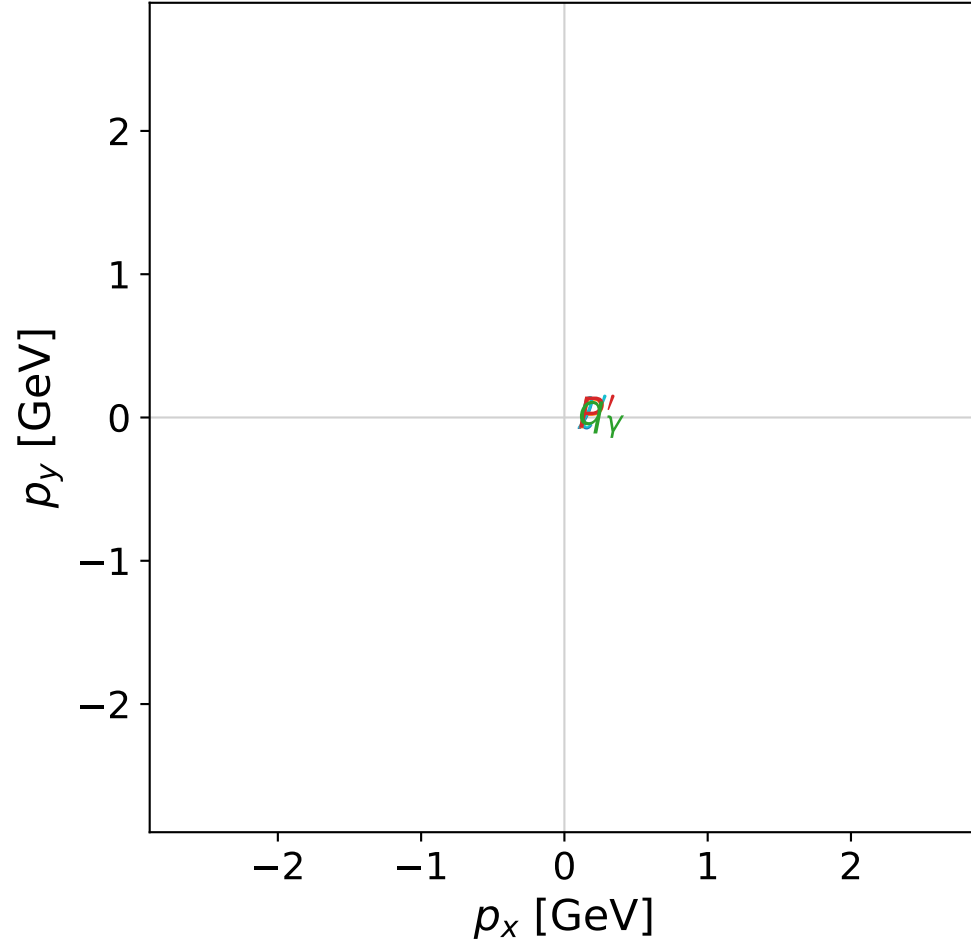


dghz_mixing_angles_polarization_03_local_minimum_66
 $d_{\text{GHZ}}=2.78662\text{e-}08$
 region: polarization_cluster_03_local_minimum_66
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0066
 lepton mixing angle: $\alpha_e=2.5600686$ rad
 incoming lepton: $-0.835626|+\rangle +0.549298|-\rangle$
 proton mixing angle: $\alpha_p=0.58153866$ rad
 incoming proton: $0.835618|+\rangle +0.54931|-\rangle$

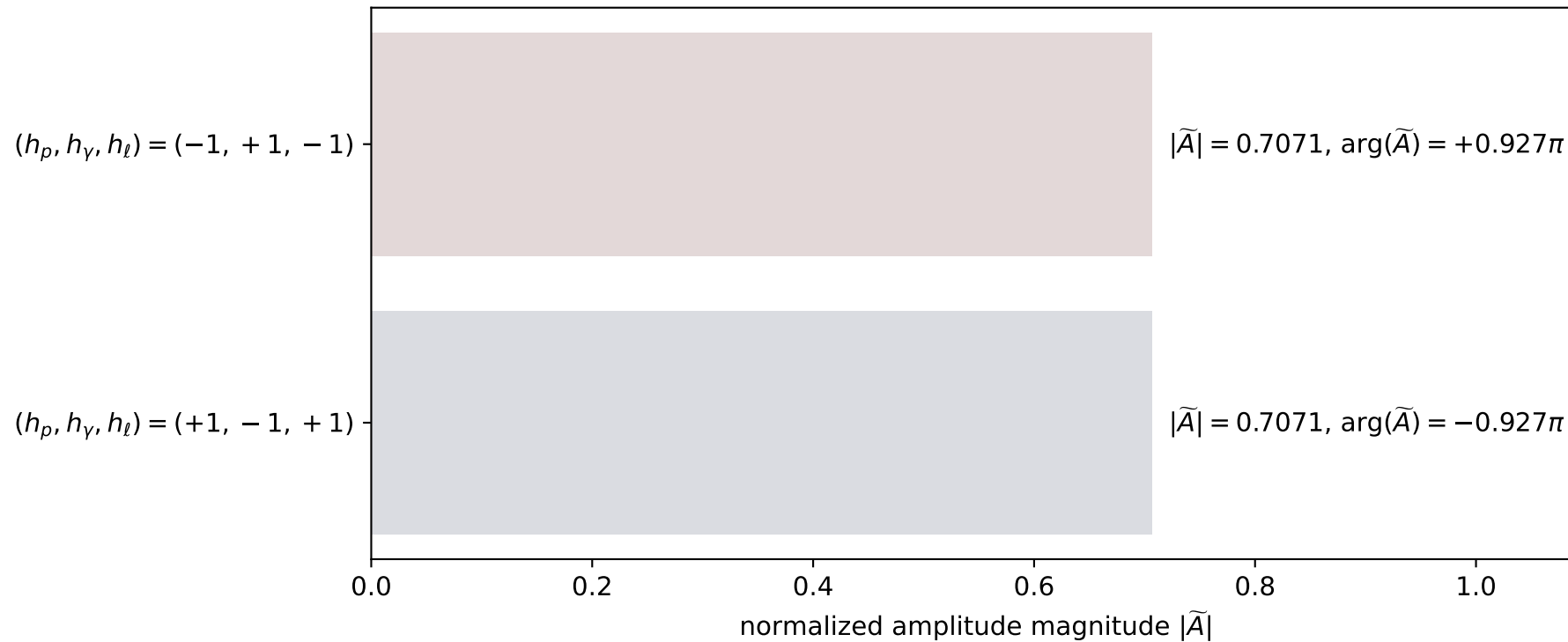
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.192212$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14281$, $\phi_\gamma=0.229158$
 $E_\gamma=1.92515$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.39574$
 $Q^2=20.4284$, $x_B=0.873944$, $t=-4.12349$, $W^2=3.8264$, $y=0.969106$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1174$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1174$
 P' $E= 0.9575$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1922$
 q_γ $E= 1.9251$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9251$

x--y plane

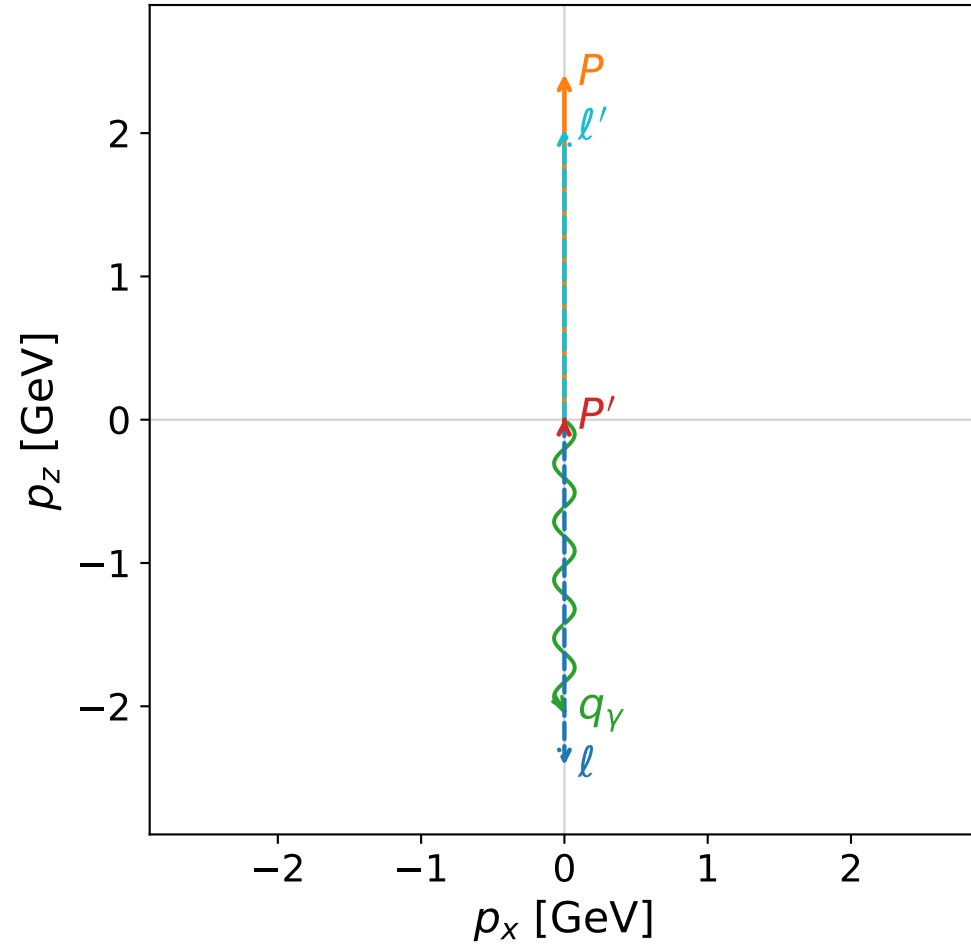


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

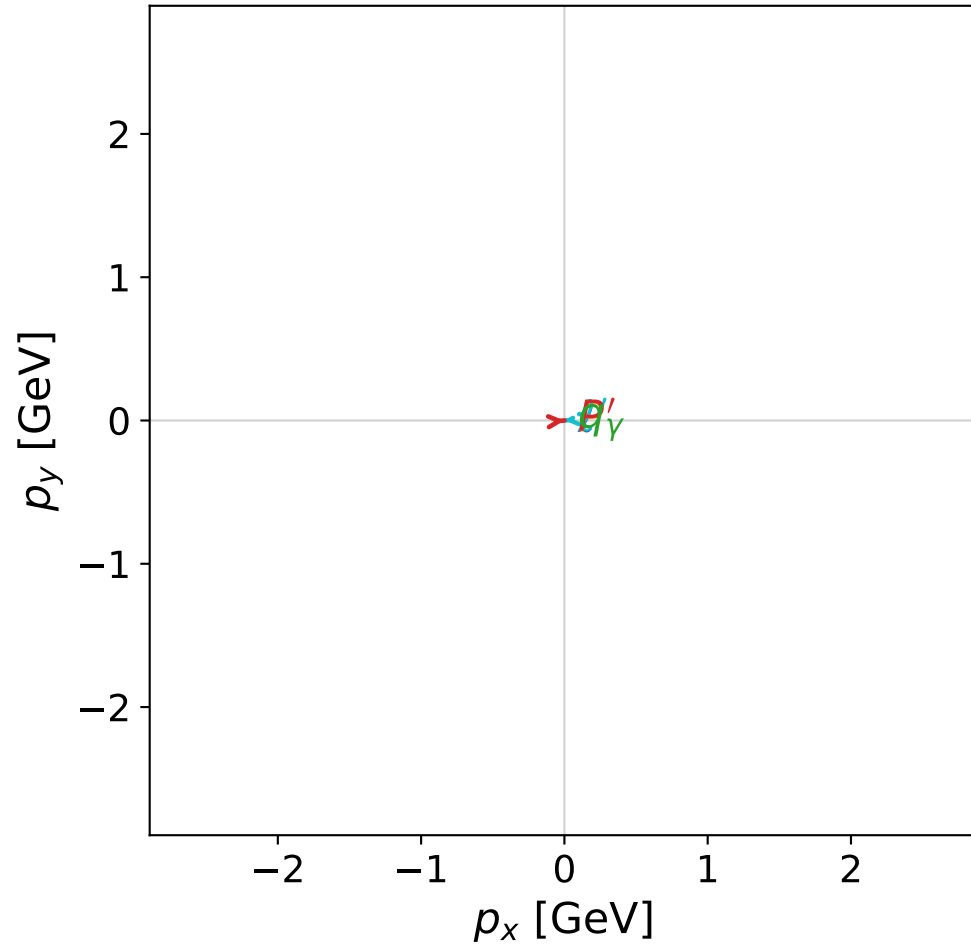


dghz_mixing_angles_polarization_03_local_minimum_67
 $d_{\text{GHZ}}=2.80689\text{e-}08$
 region: polarization_cluster_03_local_minimum_67
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0067
 lepton mixing angle: $\alpha_e=2.235145$ rad
 incoming lepton: $-0.616546|+\rangle + 0.787319|-\rangle$
 proton mixing angle: $\alpha_p=0.90642644$ rad
 incoming proton: $0.616563|+\rangle + 0.787305|-\rangle$

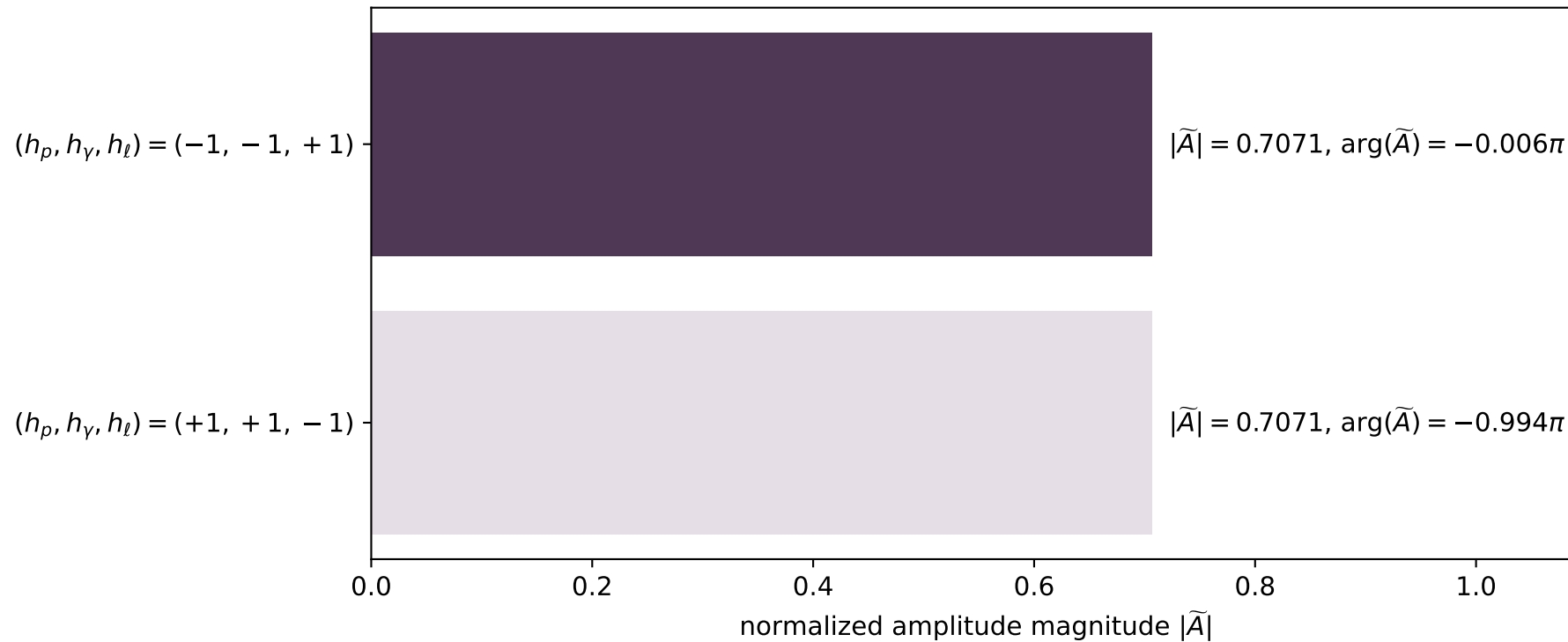
$s=24.9985$, $\sqrt{s}=4.99985$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.0078299$
 $\theta_{P'}=0.0184851$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0794118$, $\phi_\gamma=3.1225$
 $E_\gamma=2.03482$, $\theta_{\ell'}=7.14003\text{e-}05$, $\phi_{\ell'}=3.221$
 $Q^2=19.556$, $x_B=0.835536$, $t=-3.05764$, $W^2=4.72917$, $y=0.970421$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0270$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0270$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0078$
 q_γ $E= 2.0348$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0348$

x--y plane

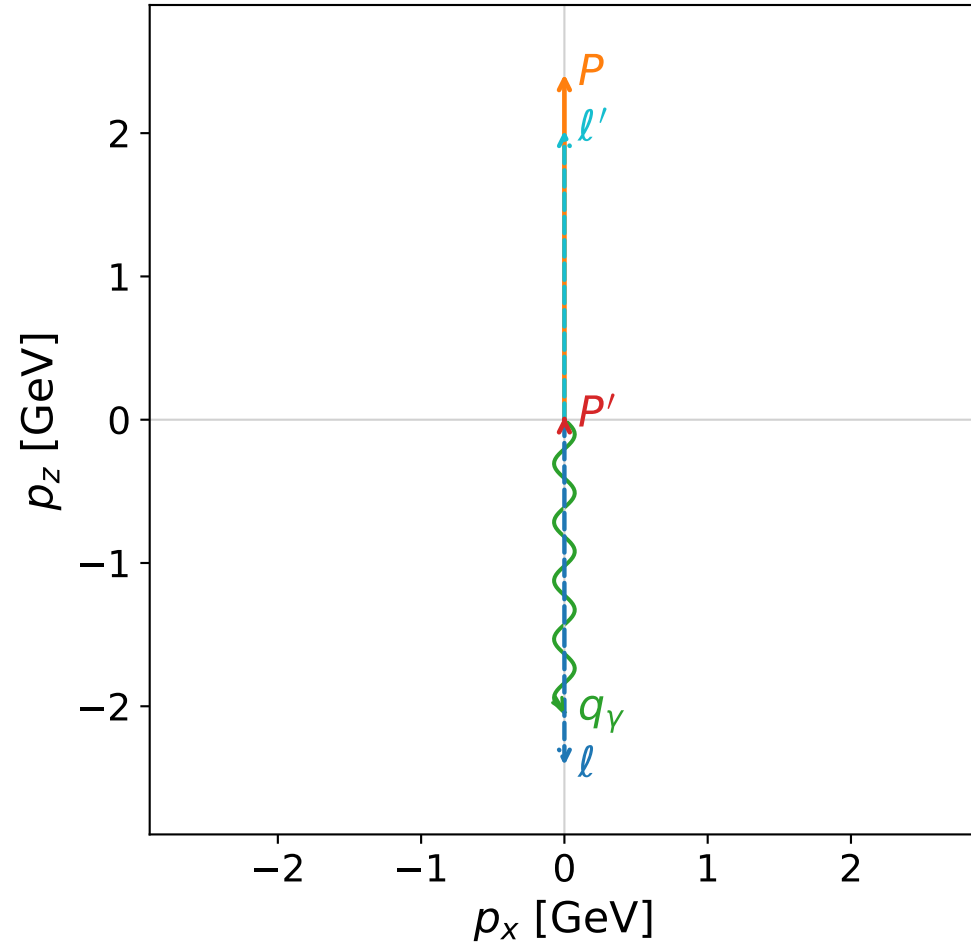


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

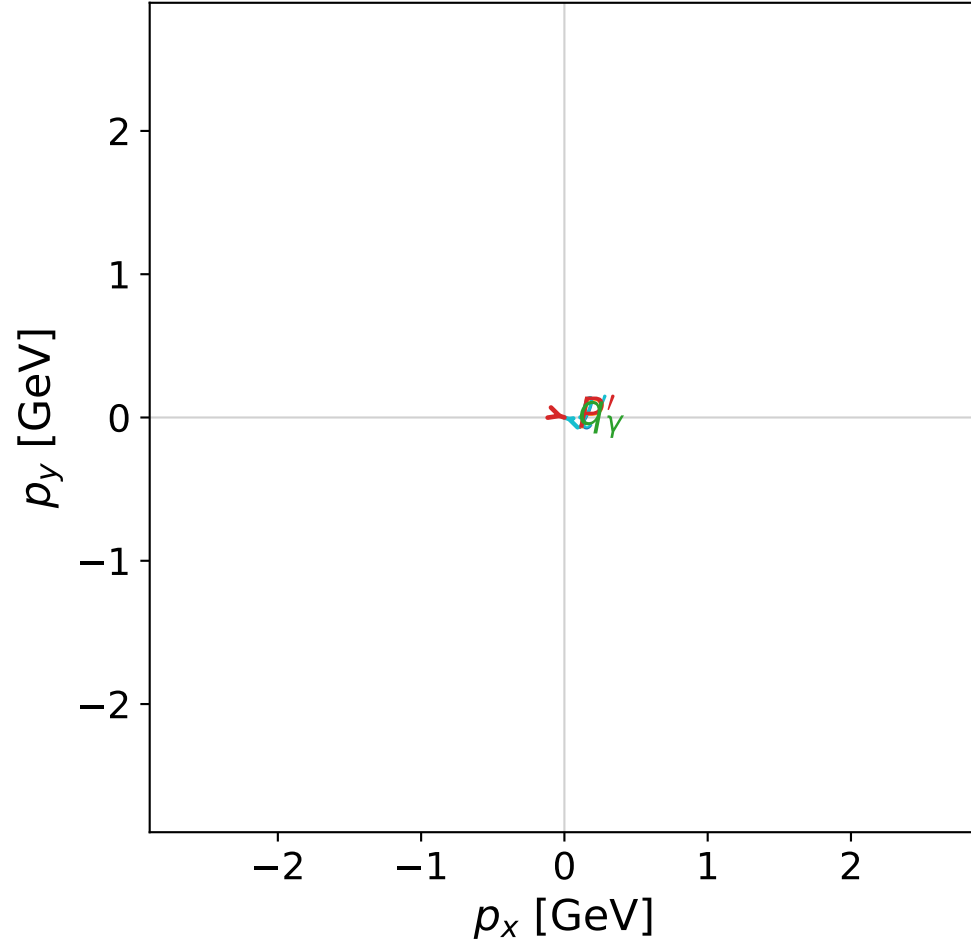


dghz_mixing_angles_polarization_03_local_minimum_73
 $d_{\text{GHZ}}=2.91539\text{e-}08$
 region: polarization_cluster_03_local_minimum_73
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0073
 lepton mixing angle: $\alpha_e=2.1495745$ rad
 incoming lepton: $-0.547002|+\rangle +0.837132|-\rangle$
 proton mixing angle: $\alpha_p=0.99200753$ rad
 incoming proton: $0.54701|+\rangle +0.837126|-\rangle$

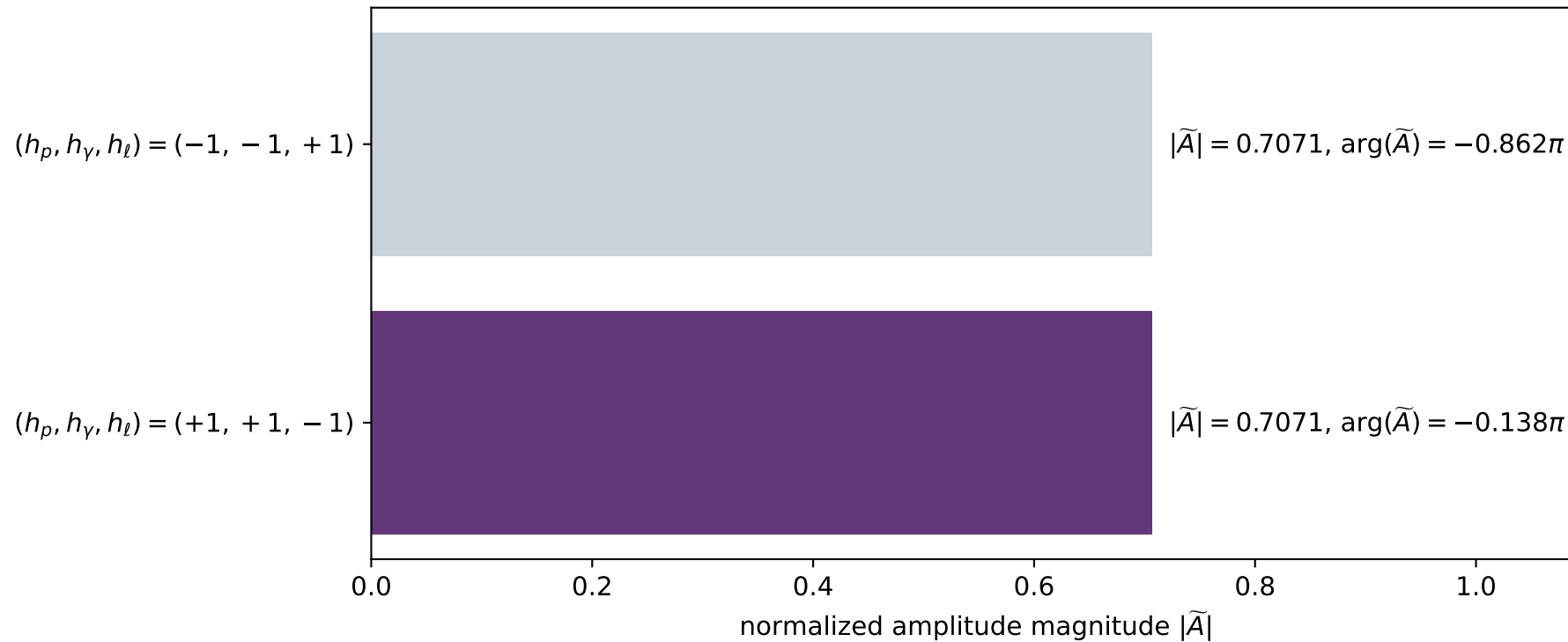
$s=24.9997$, $\sqrt{s}=4.99997$, $|\vec{P}|=2.412$, $|\vec{P}'|=0.0232778$
 $\theta_{P'}=0.000120512$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.96763$, $\phi_\gamma=0.433102$
 $E_\gamma=2.04248$, $\theta_{\ell'}=1.38933\text{e-}06$, $\phi_{\ell'}=2.82604$
 $Q^2=19.4813$, $x_B=0.832205$, $t=-2.98455$, $W^2=4.8078$, $y=0.970537$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0192$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0192$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0233$
 q_γ $E= 2.0425$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0425$

x--y plane

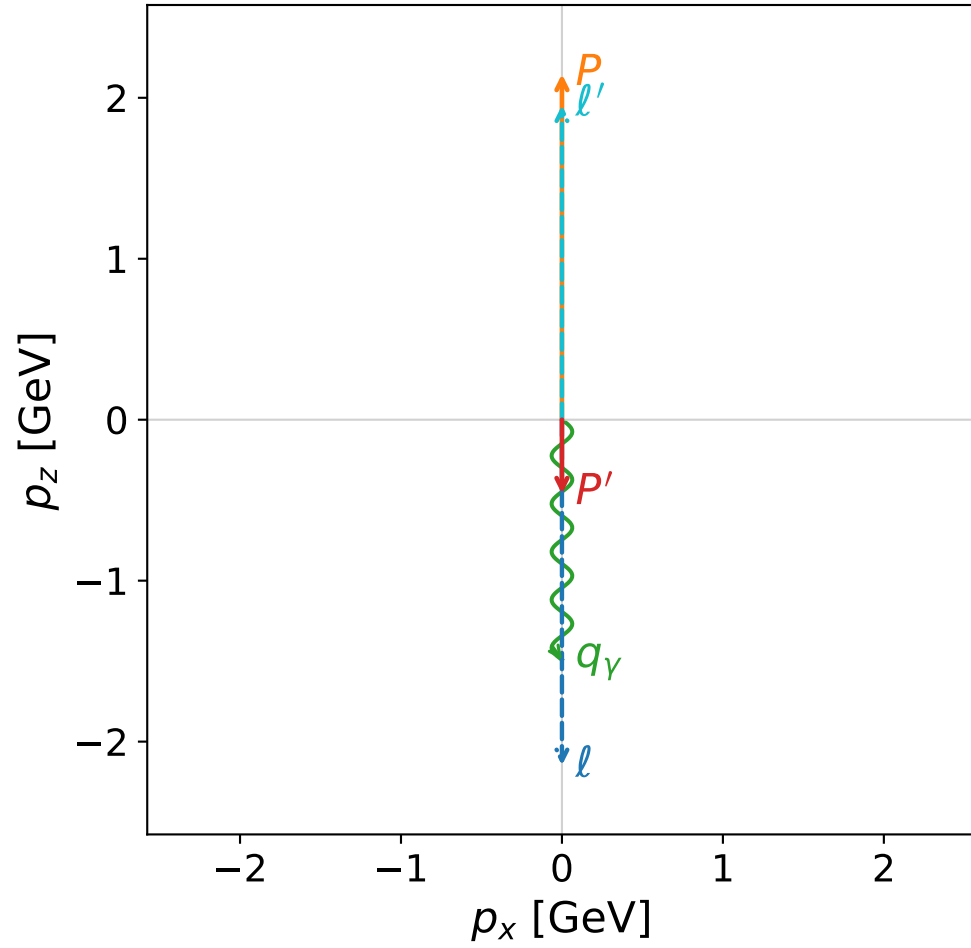


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

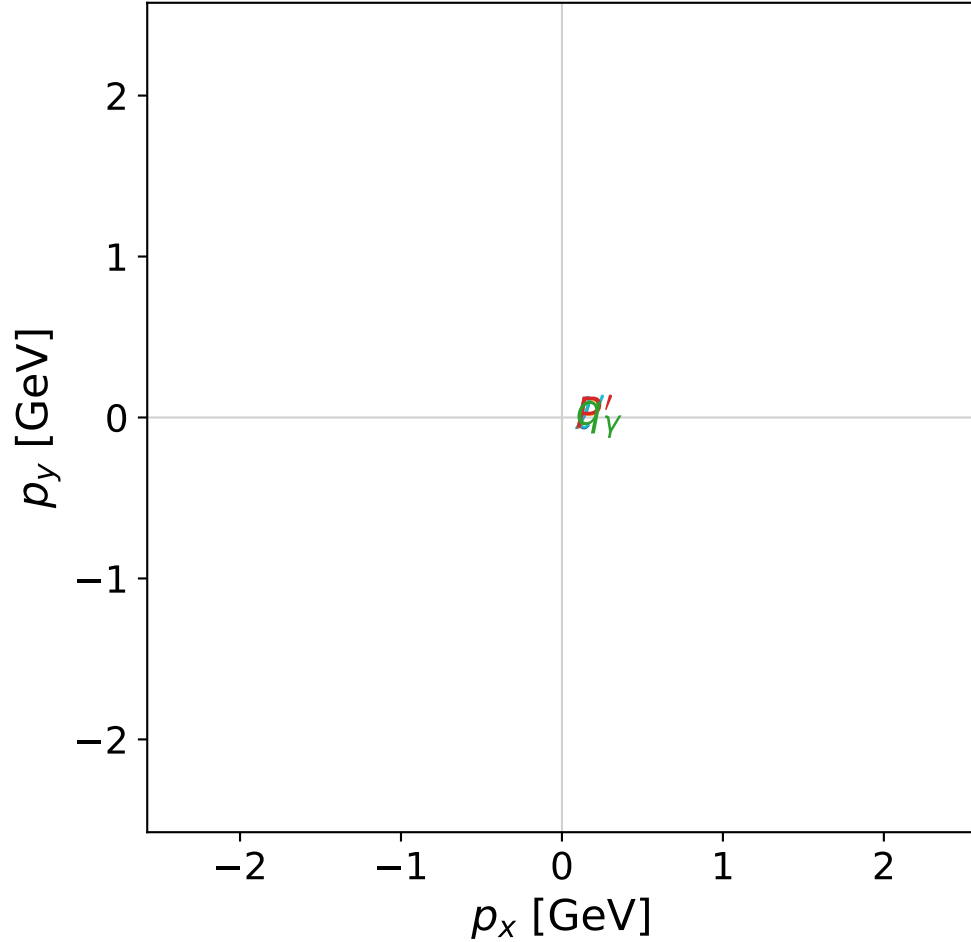


dghz_mixing_angles_polarization_03_local_minimum_87
 $d_{\text{GHZ}}=3.19716\text{e-}08$
 region: polarization_cluster_03_local_minimum_87
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0087
 lepton mixing angle: $\alpha_e=2.4687294$ rad
 incoming lepton: $-0.78204|+\rangle +0.623228|-\rangle$
 proton mixing angle: $\alpha_p=0.67288163$ rad
 incoming proton: $0.782029|+\rangle +0.623242|-\rangle$

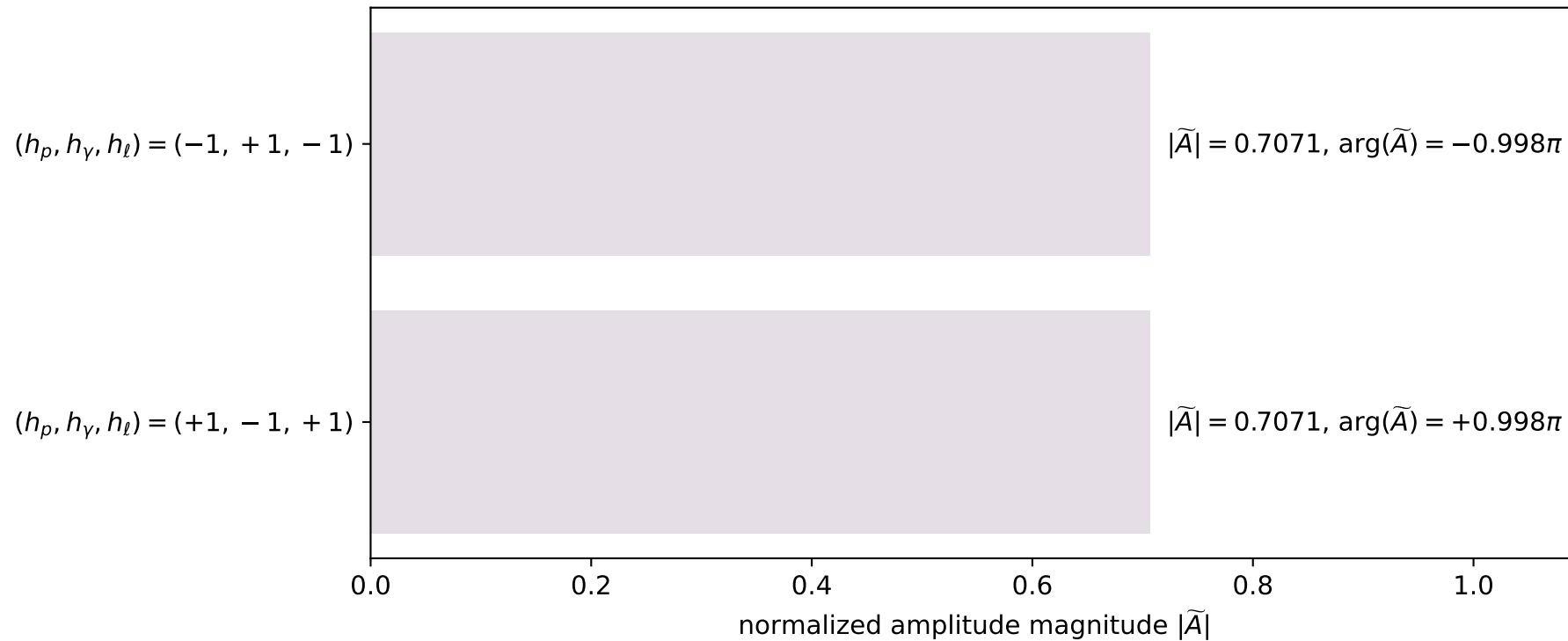
$s=20.1618$, $\sqrt{s}=4.49019$, $|\vec{P}|=2.14712$, $|\vec{P}'|=0.461231$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.72593$, $\phi_\gamma=6.27676$
 $E_\gamma=1.49185$, $\theta_{l'}=0$, $\phi_{l'}=2.90749$
 $Q^2=16.774$, $x_B=0.905891$, $t=-5.1192$, $W^2=2.62242$, $y=0.960305$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1471$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1471$
 P $E= 2.3431$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1471$
 l' $E= 1.9531$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9531$
 P' $E= 1.0453$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.4612$
 q_γ $E= 1.4918$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4918$

x--y plane

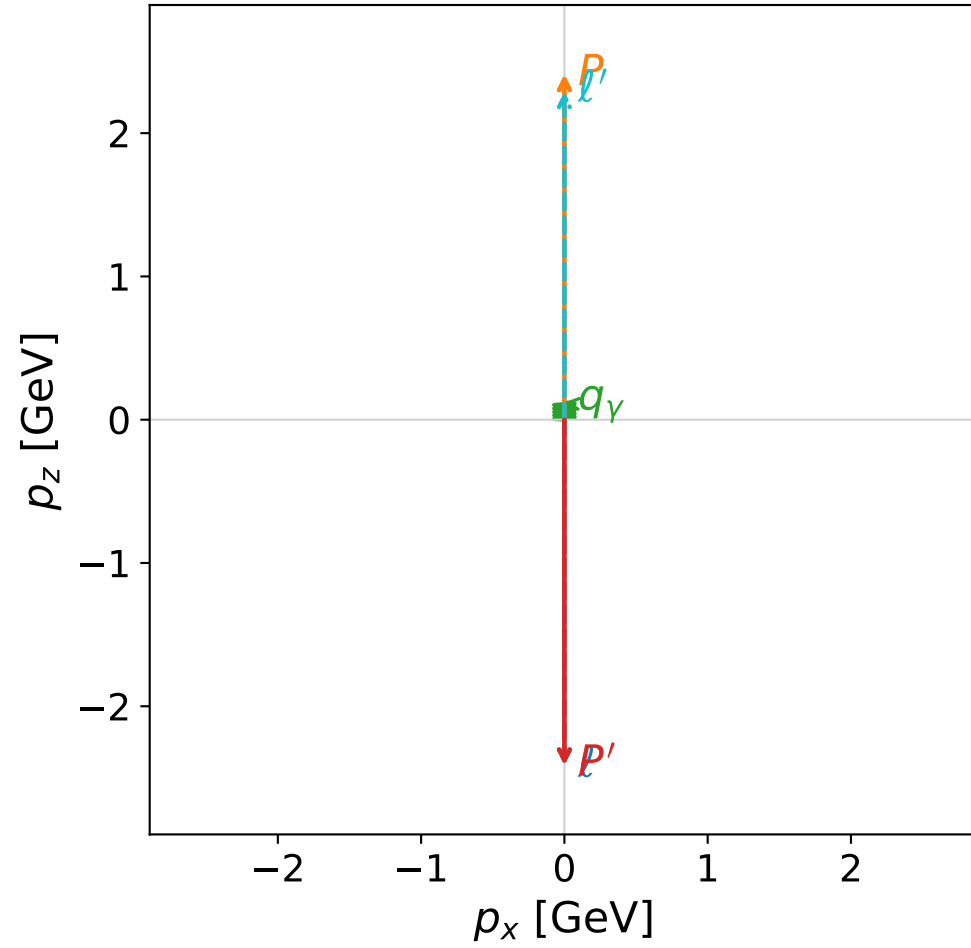


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

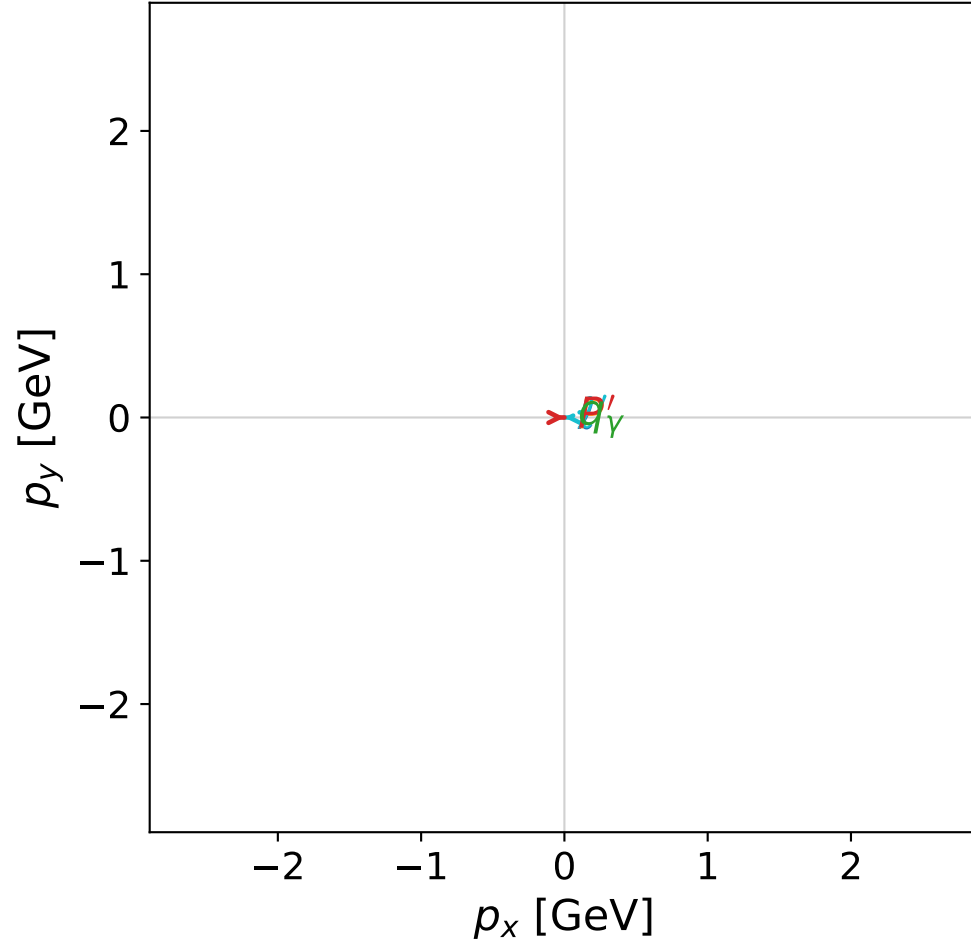


dghz_mixing_angles_polarization_03_local_minimum_89
 $d_{\text{GHZ}}=3.21673\text{e-}08$
 region: polarization_cluster_03_local_minimum_89
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0089
 lepton mixing angle: $\alpha_e=1.9834494$ rad
 incoming lepton: $-0.401041|+\rangle + 0.91606|- \rangle$
 proton mixing angle: $\alpha_p=0.41264296$ rad
 incoming proton: $0.916064|+\rangle + 0.401032|- \rangle$

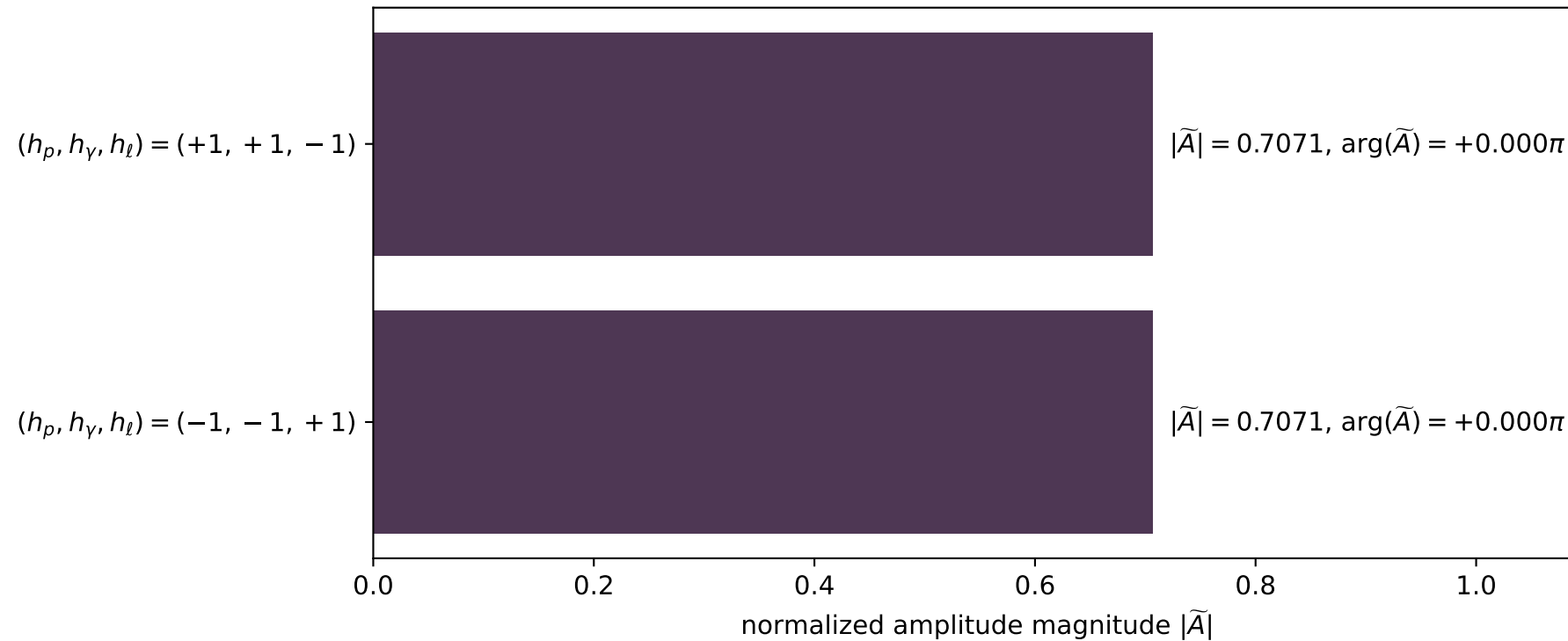
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0$, $\phi_\gamma=0$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

x--y plane

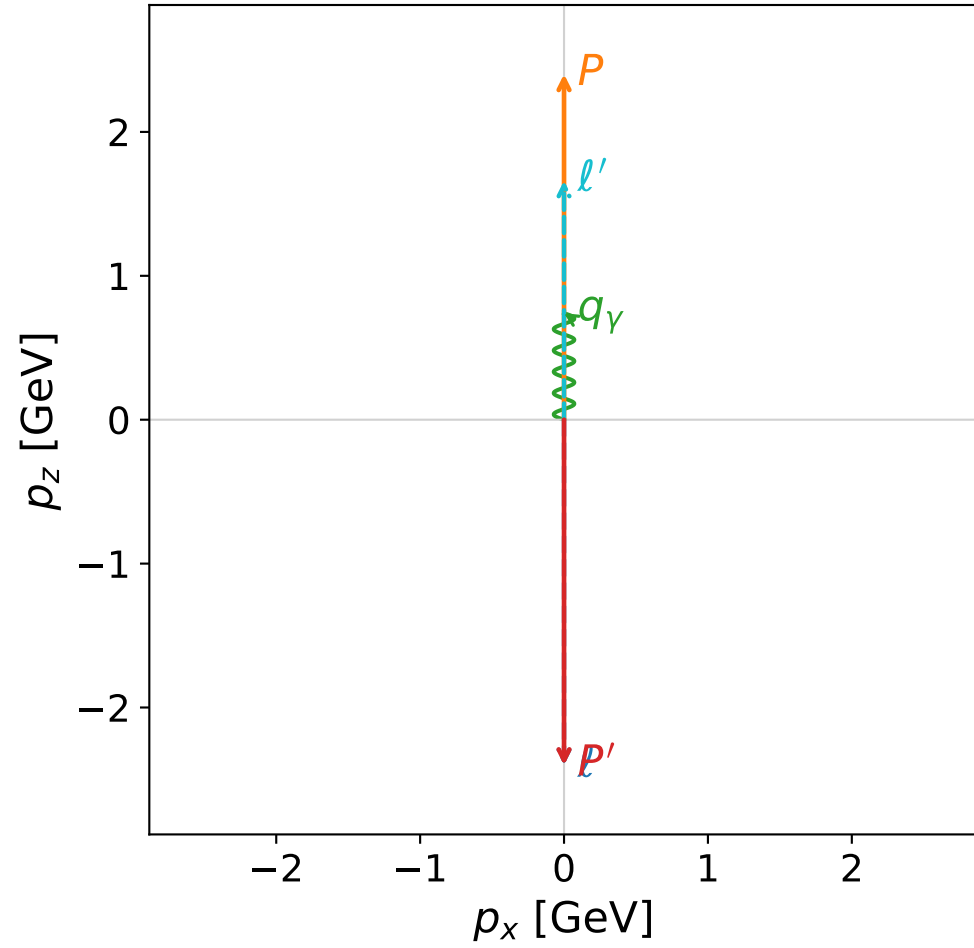


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

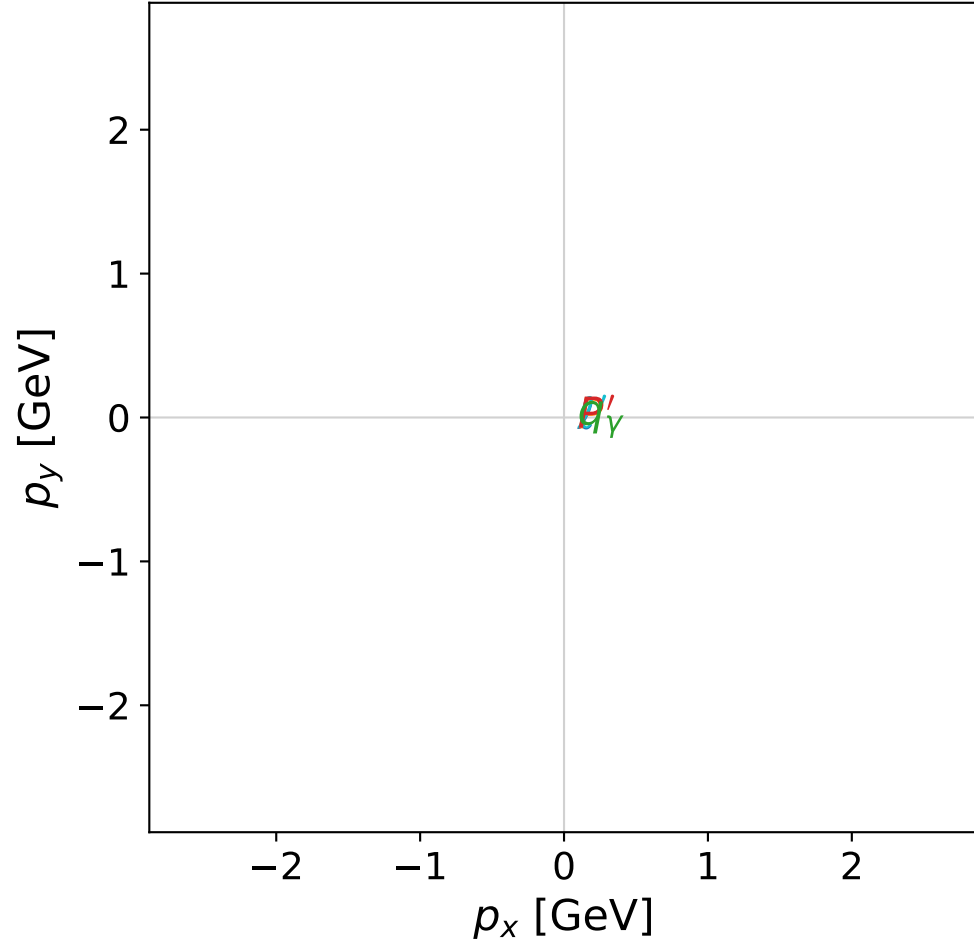


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



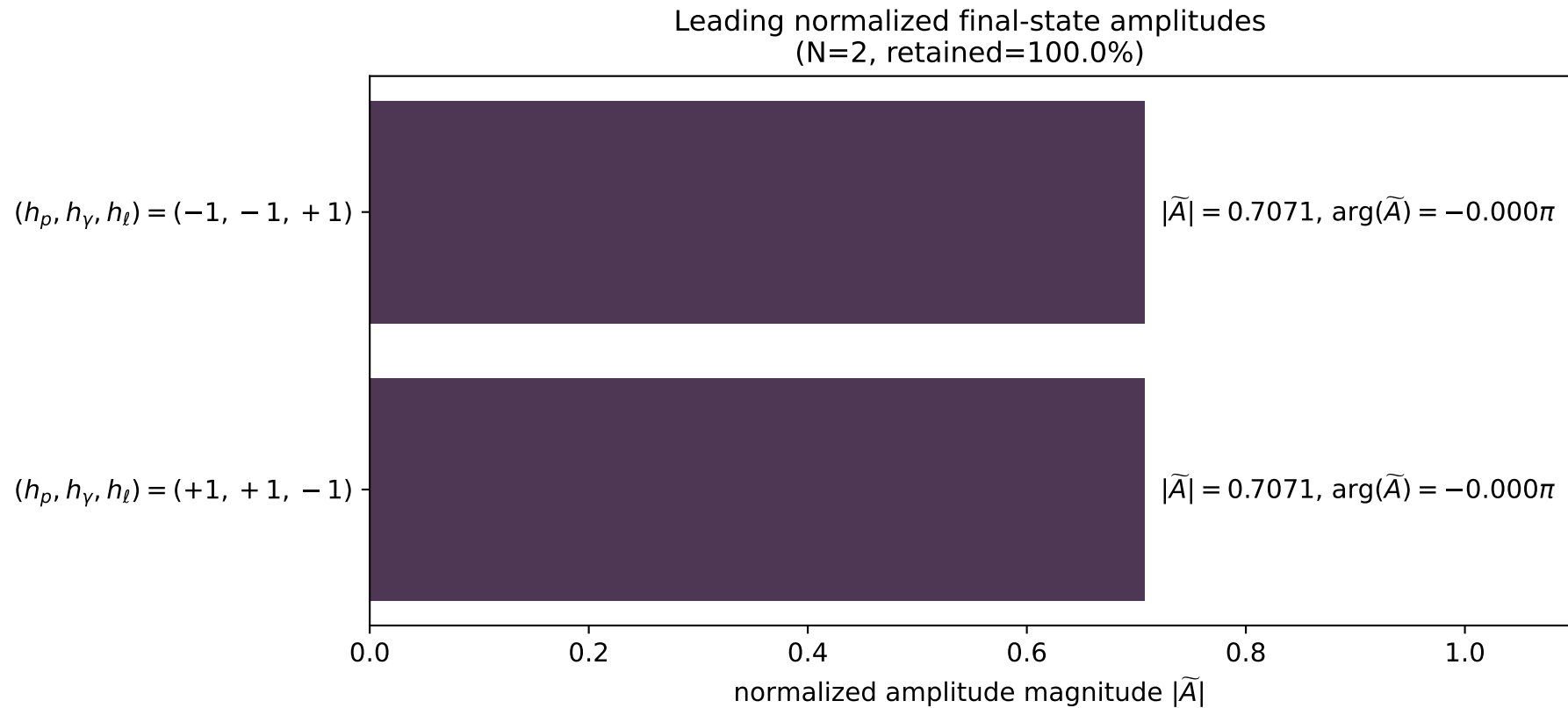
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_92
 $d_{\{\rm GHZ\}}=3.31027\text{e-}08$
 region: polarization_cluster_03_local_minimum_92
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0092
 lepton mixing angle: $\alpha_e=2.4281603$ rad
 incoming lepton: $-0.75612|+\rangle +0.654433|-\rangle$
 proton mixing angle: $\alpha_p=0.85738407$ rad
 incoming proton: $0.654418|+\rangle +0.756133|-\rangle$

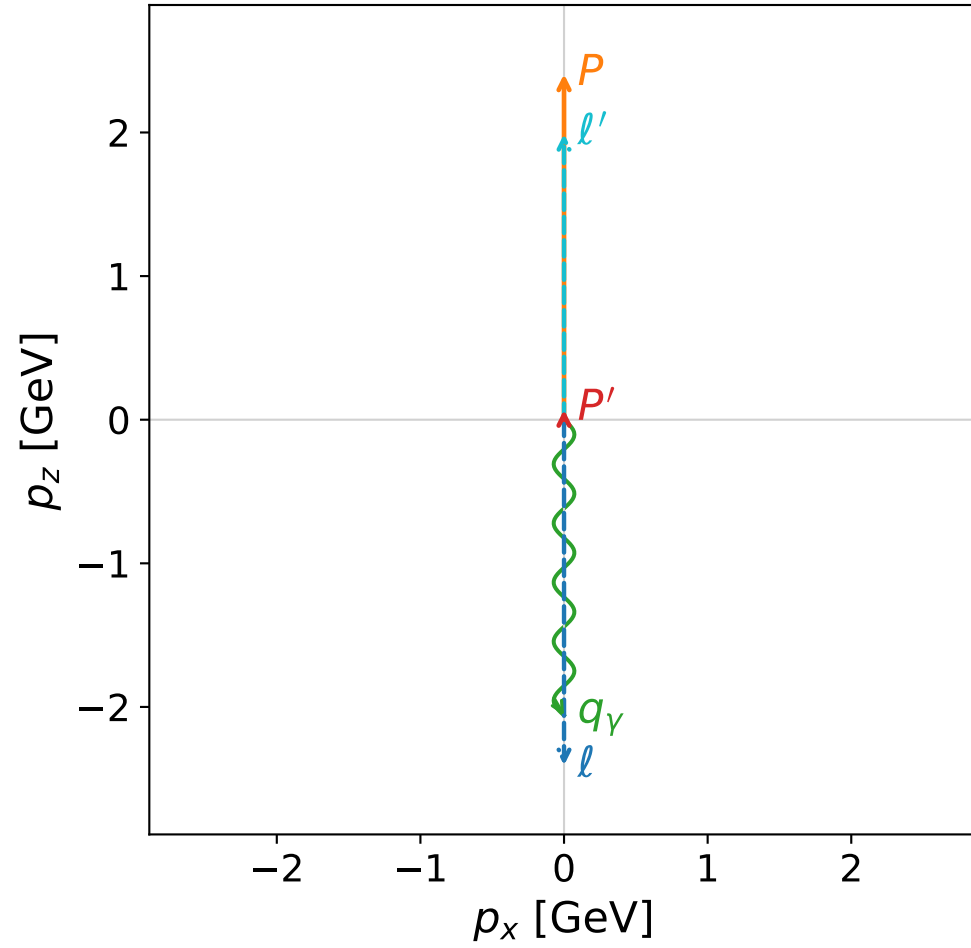
$s=24.8013$, $\sqrt{s}=4.98009$, $|\vec{P}|=2.40171$, $|\vec{P}'|=2.40171$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.23216$, $\phi_\gamma=5.8557$
 $E_\gamma=0.740299$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.09057$
 $Q^2=15.9609$, $x_B=0.684007$, $t=-23.0728$, $W^2=8.25336$, $y=0.975459$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4017$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4017$
 P $E= 2.5784$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4017$
 ℓ' $E= 1.6614$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.6614$
 P' $E= 2.5784$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4017$
 q_γ $E= 0.7403$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7403$

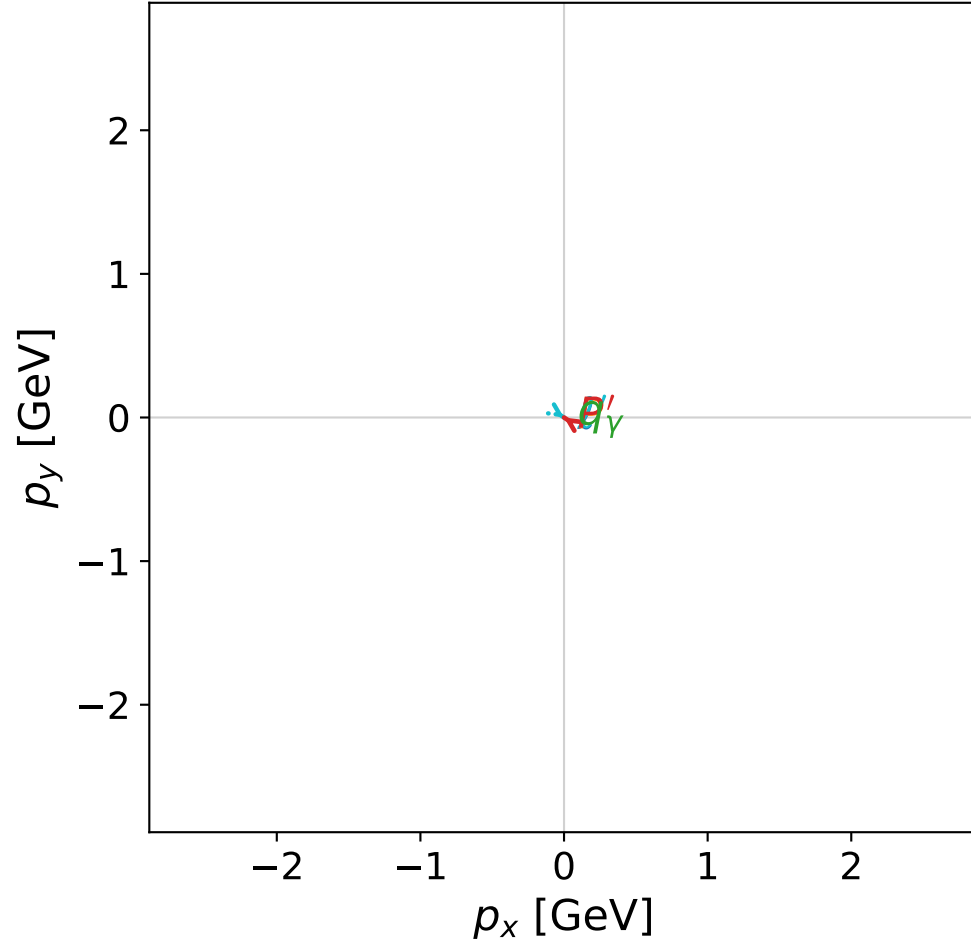


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



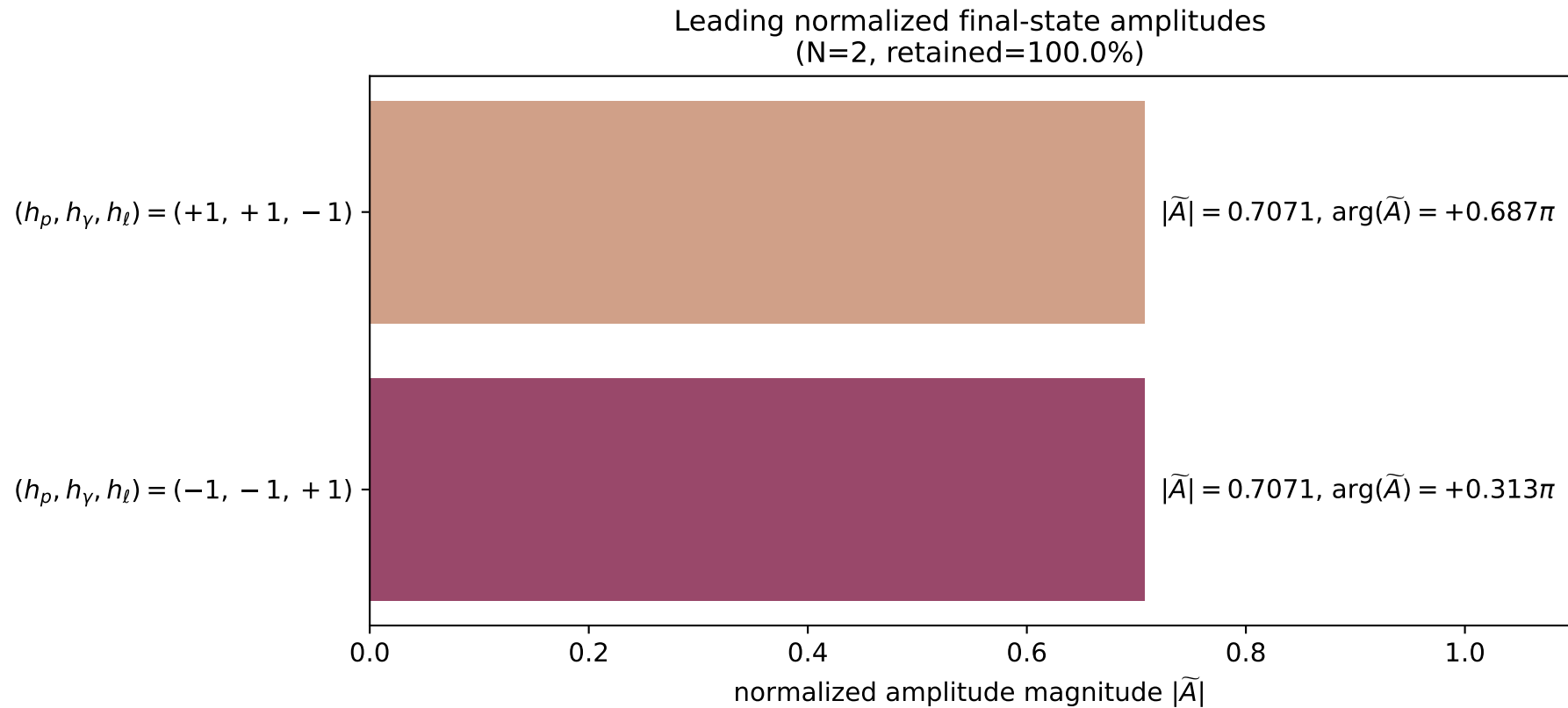
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_93
 $d_{\text{GHZ}}=3.31851\text{e-}08$
 region: polarization_cluster_03_local_minimum_93
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0093
 lepton mixing angle: $\alpha_e=2.6709187$ rad
 incoming lepton: $-0.891263|+\rangle +0.453487|-\rangle$
 proton mixing angle: $\alpha_p=0.47067803$ rad
 incoming proton: $0.891261|+\rangle +0.453491|-\rangle$

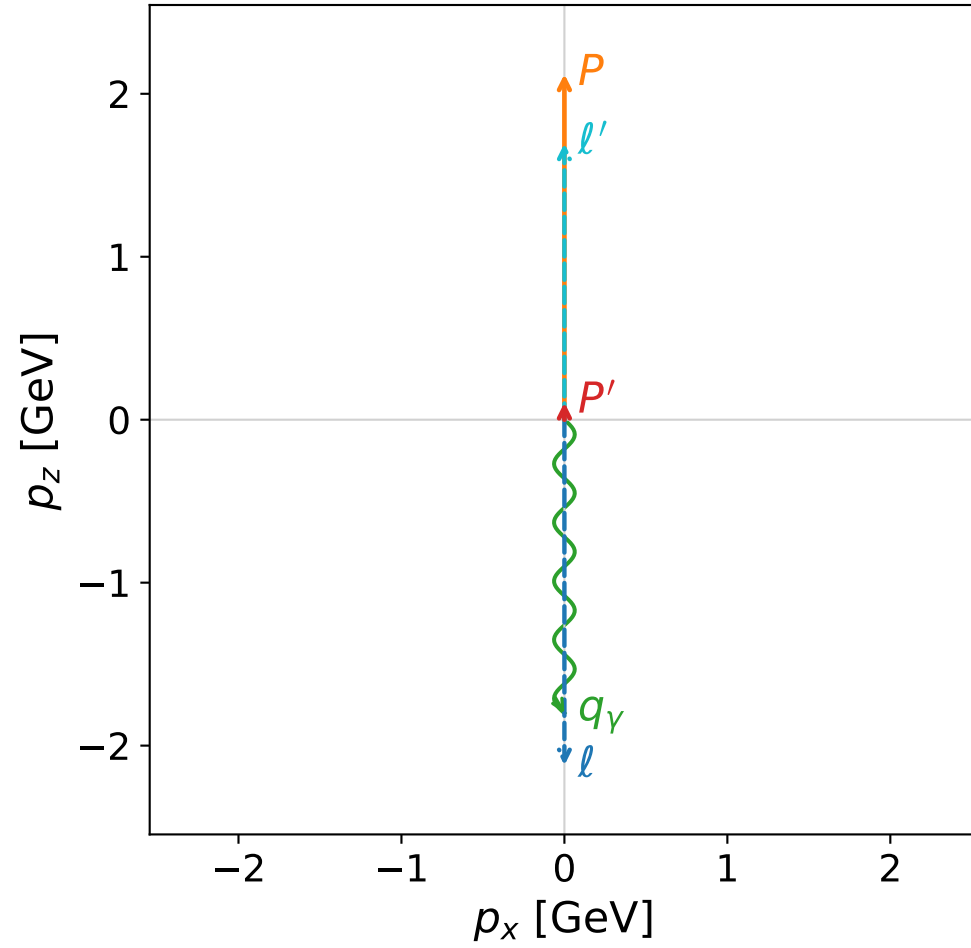
$s=24.8907$, $\sqrt{s}=4.98906$, $|\vec{P}|=2.40635$, $|\vec{P}'|=0.0697271$
 $\theta_{p'}=1.43494\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.55848$, $\phi_\gamma=4.12637$
 $E_\gamma=2.0591$, $\theta_{\ell'}=5.029\text{e-}07$, $\phi_{\ell'}=5.70007$
 $Q^2=19.1485$, $x_B=0.821501$, $t=-2.76326$, $W^2=5.04052$, $y=0.970777$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4064$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4064$
 P $E= 2.5827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4064$
 ℓ' $E= 1.9894$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9894$
 P' $E= 0.9406$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0697$
 q_γ $E= 2.0591$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0591$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

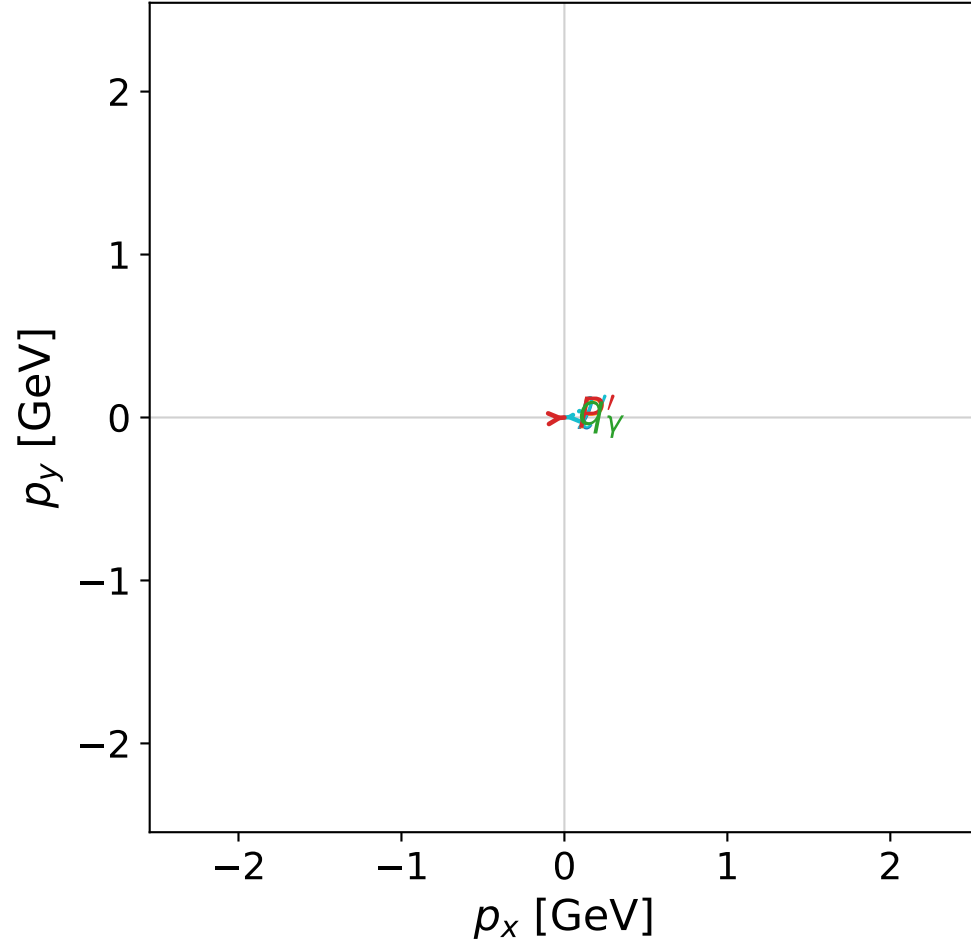


dghz_mixing_angles_polarization_03_local_minimum_97
 $d_{\text{GHZ}}=3.45768\text{e-}08$
 region: polarization_cluster_03_local_minimum_97
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0097
 lepton mixing angle: $\alpha_e=2.363027$ rad
 incoming lepton: $-0.711922|+\rangle +0.702259|-\rangle$
 proton mixing angle: $\alpha_p=0.77853959$ rad
 incoming proton: $0.71194|+\rangle +0.70224|-\rangle$

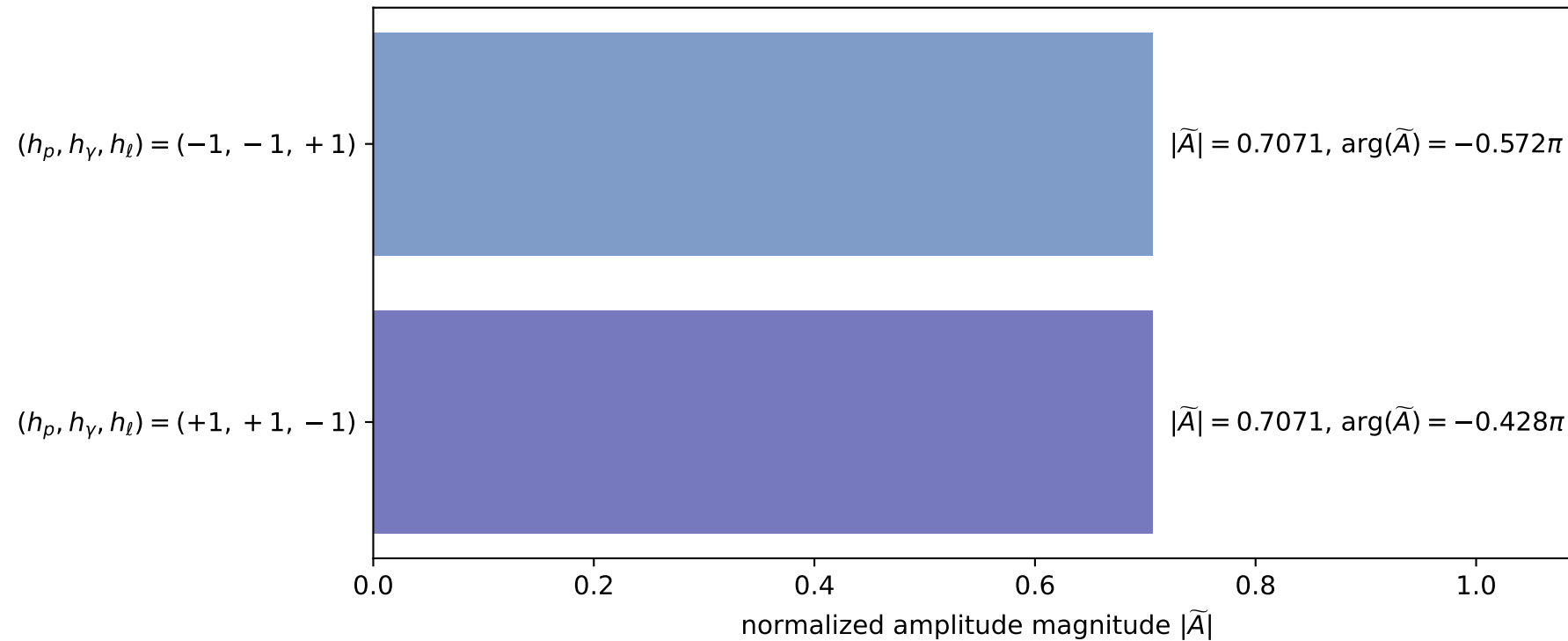
$s=19.711$, $\sqrt{s}=4.4397$, $|\vec{P}|=2.12076$, $|\vec{P}'|=0.105882$
 $\theta_{P'}=3.23647\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0751304$, $\phi_\gamma=1.34534$
 $E_\gamma=1.80081$, $\theta_{\ell'}=2.02129\text{e-}07$, $\phi_{\ell'}=3.21672$
 $Q^2=14.3782$, $x_B=0.79178$, $t=-2.16917$, $W^2=4.66097$, $y=0.964325$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1208$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1208$
 P $E= 2.3189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1208$
 ℓ' $E= 1.6949$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.6949$
 P' $E= 0.9440$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1059$
 q_γ $E= 1.8008$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8008$

x--y plane

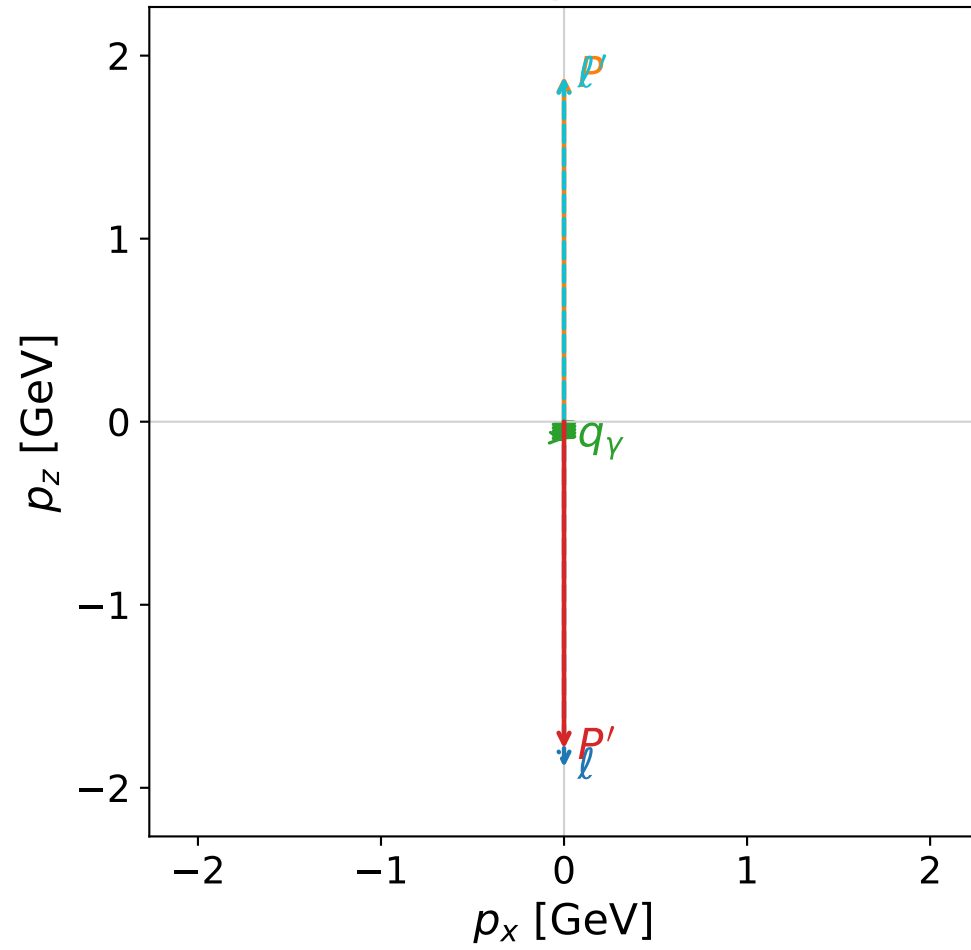


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

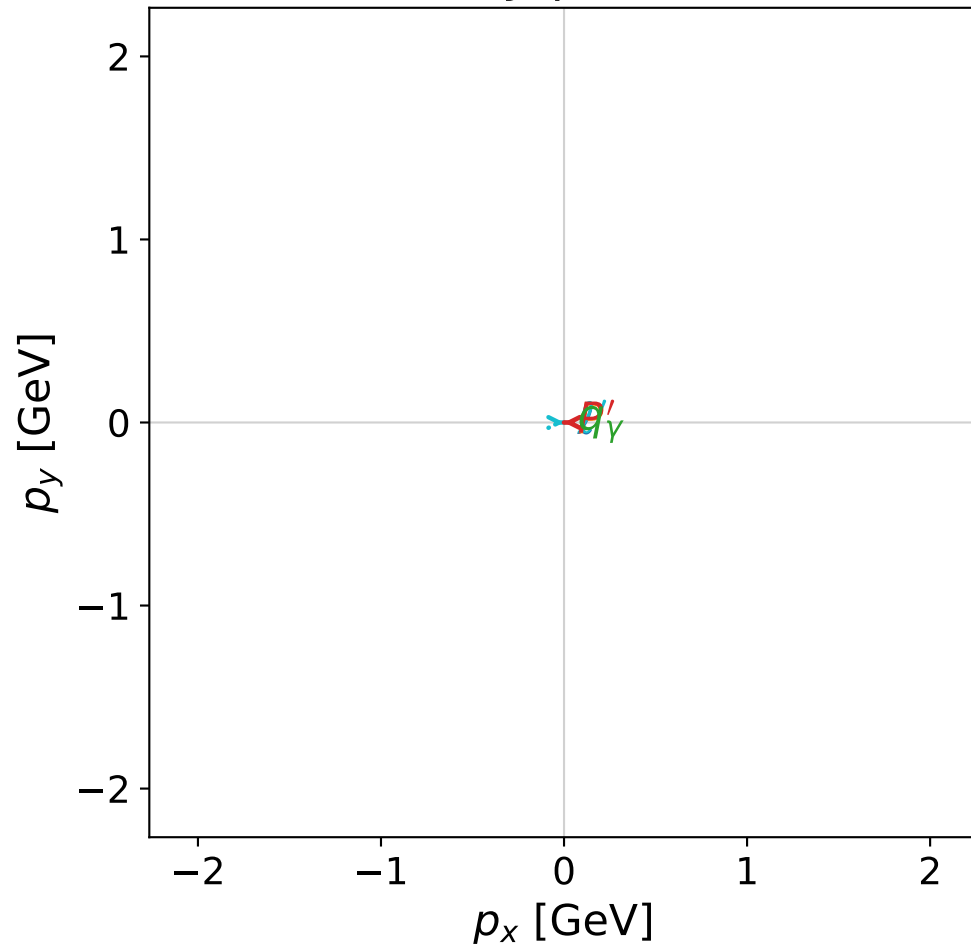


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



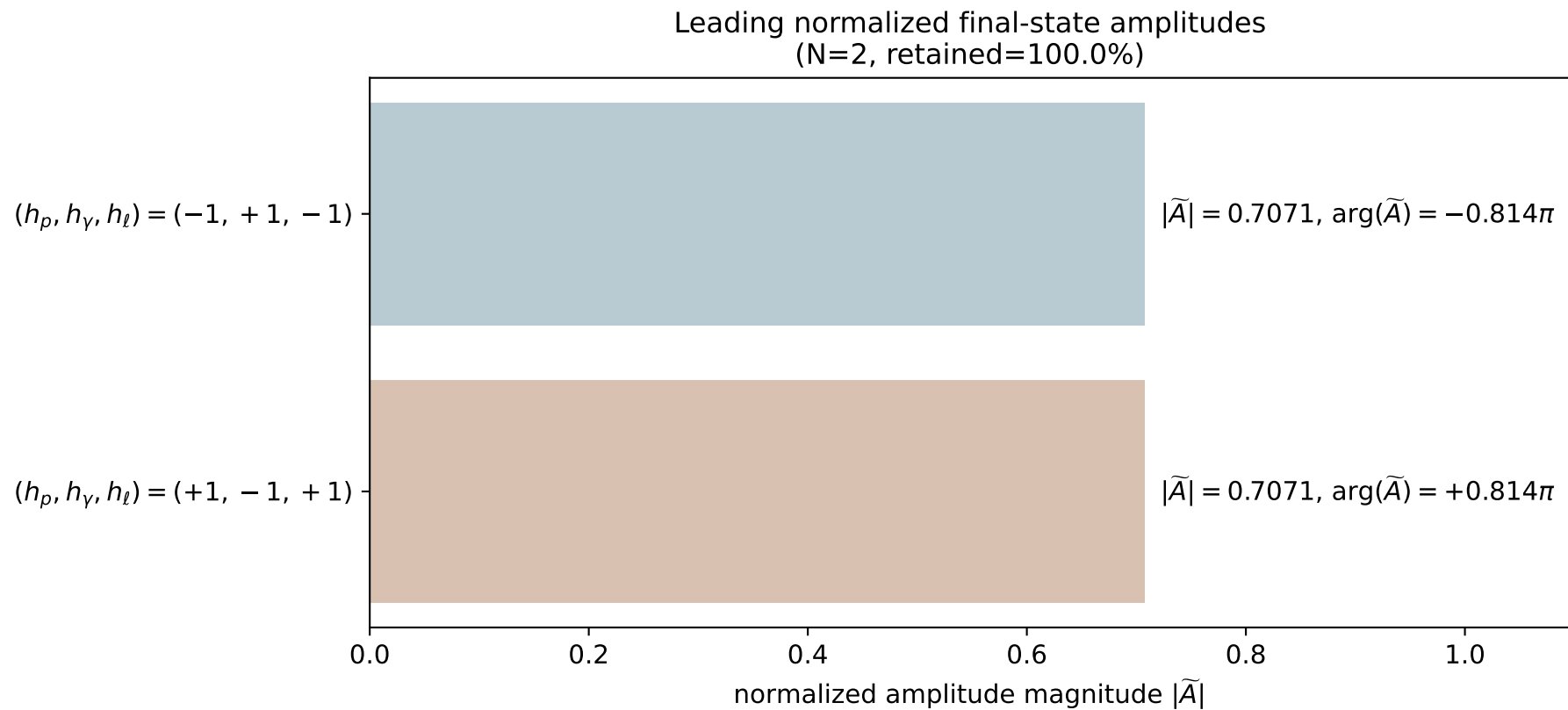
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_102
 $d_{\{\rm GHZ\}}=3.6745\text{e-}08$
 region: polarization_cluster_03_local_minimum_102
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0102
 lepton mixing angle: $\alpha_e=2.3754496$ rad
 incoming lepton: $-0.72059|+\rangle +0.693361|-\rangle$
 proton mixing angle: $\alpha_p=0.7661545$ rad
 incoming proton: $0.720582|+\rangle +0.693369|-\rangle$

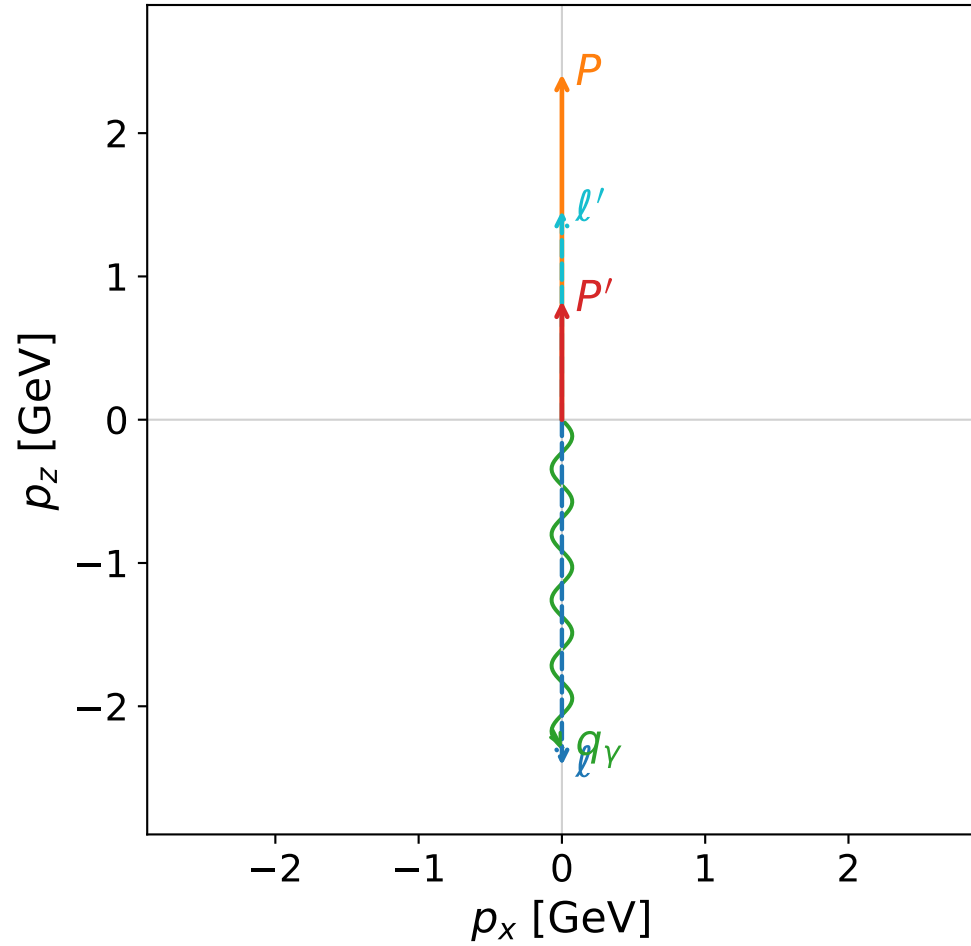
$s=15.9699$, $\sqrt{s}=3.99624$, $|\vec{P}|=1.88803$, $|\vec{P}'|=1.78706$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.12959$, $\phi_\gamma=5.69987$
 $E_\gamma=0.0954542$, $\theta_{l'}=0$, $\phi_{l'}=0.0182279$
 $Q^2=14.217$, $x_B=0.996905$, $t=-13.4982$, $W^2=0.923985$, $y=0.945067$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8880$
 P $E= 2.1082$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8880$
 l' $E= 1.8825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8825$
 P' $E= 2.0183$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.7871$
 q_γ $E= 0.0955$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0955$

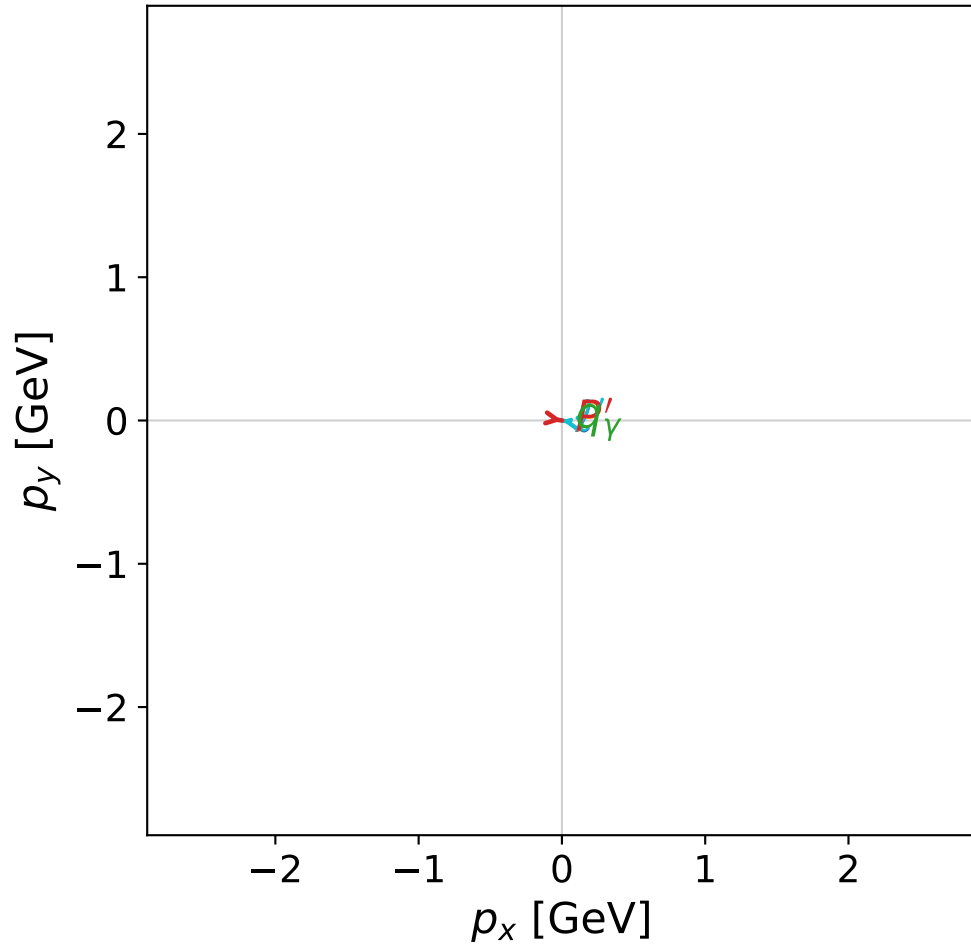


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



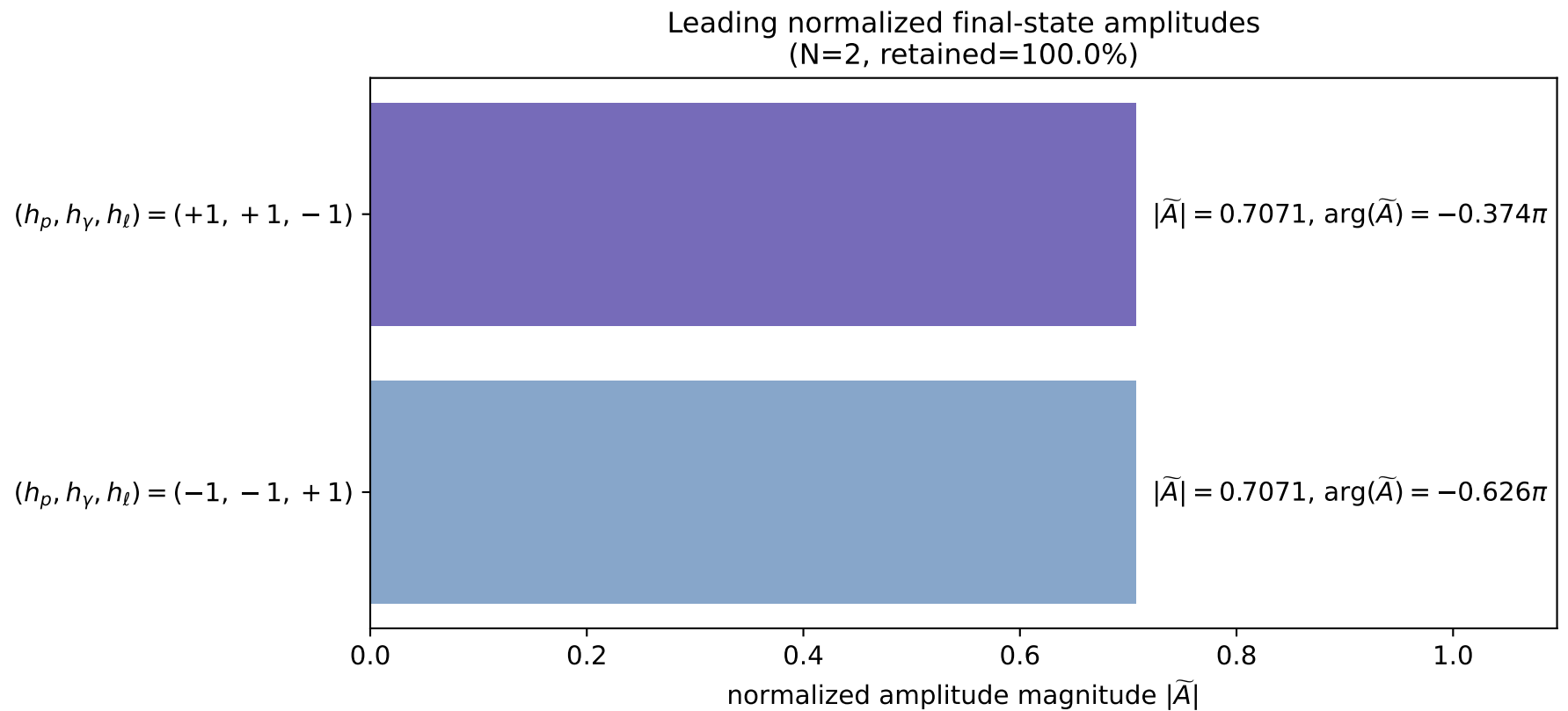
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_103
 $d_{\{\rm GHZ\}}=3.69193\text{e-}08$
 region: polarization_cluster_03_local_minimum_103
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0103
 lepton mixing angle: $\alpha_e=2.3646323$ rad
 incoming lepton: $-0.713048|+\rangle +0.701115|-\rangle$
 proton mixing angle: $\alpha_p=0.77696696$ rad
 incoming proton: $0.713043|+\rangle +0.70112|-\rangle$

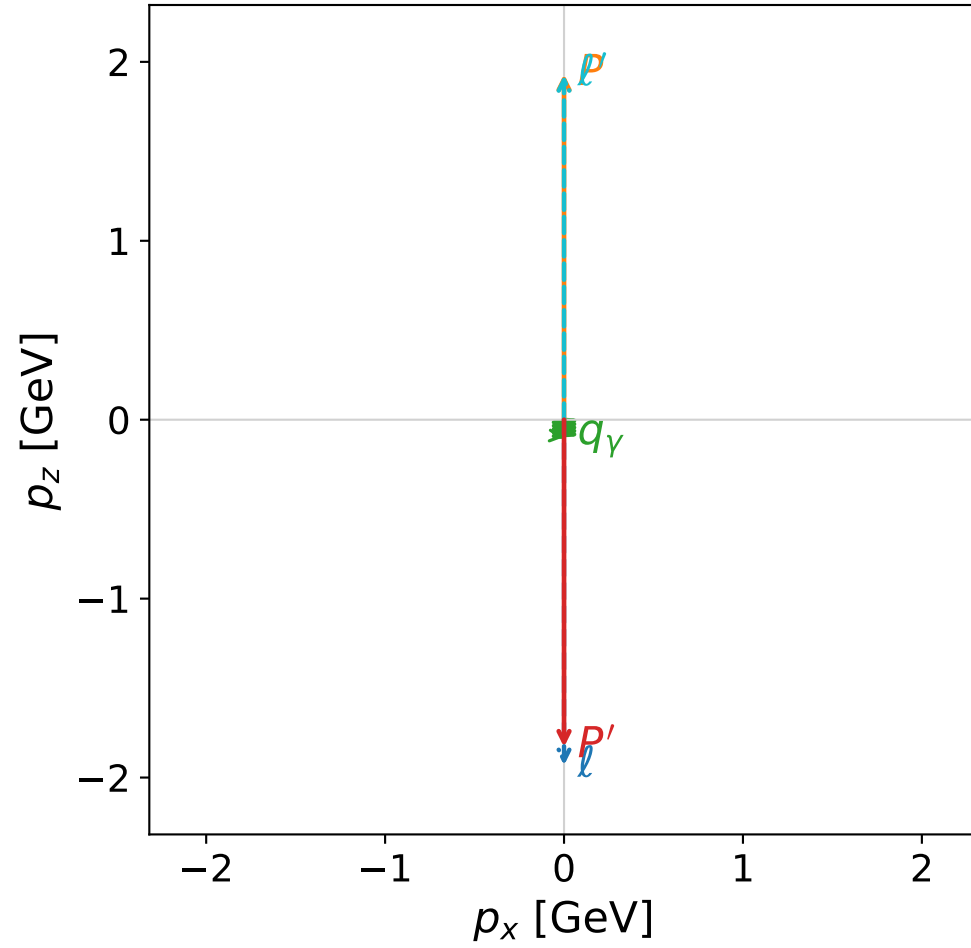
$s=24.9995$, $\sqrt{s}=4.99995$, $|\vec{P}|=2.41199$, $|\vec{P}'|=0.829186$
 $\theta_{p'}=3.04918\text{e-}05$, $\theta_Y=3.14159$, $\phi_{p'}=6.11999$, $\phi_Y=1.17541$
 $E_Y=2.28859$, $\theta_{\ell'}=1.73244\text{e-}05$, $\phi_{\ell'}=2.9784$
 $Q^2=14.0803$, $x_B=0.596469$, $t=-0.720361$, $W^2=10.4056$, $y=0.978705$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.4594$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.4594$
 P' $E= 1.2520$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8292$
 q_Y $E= 2.2886$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2886$

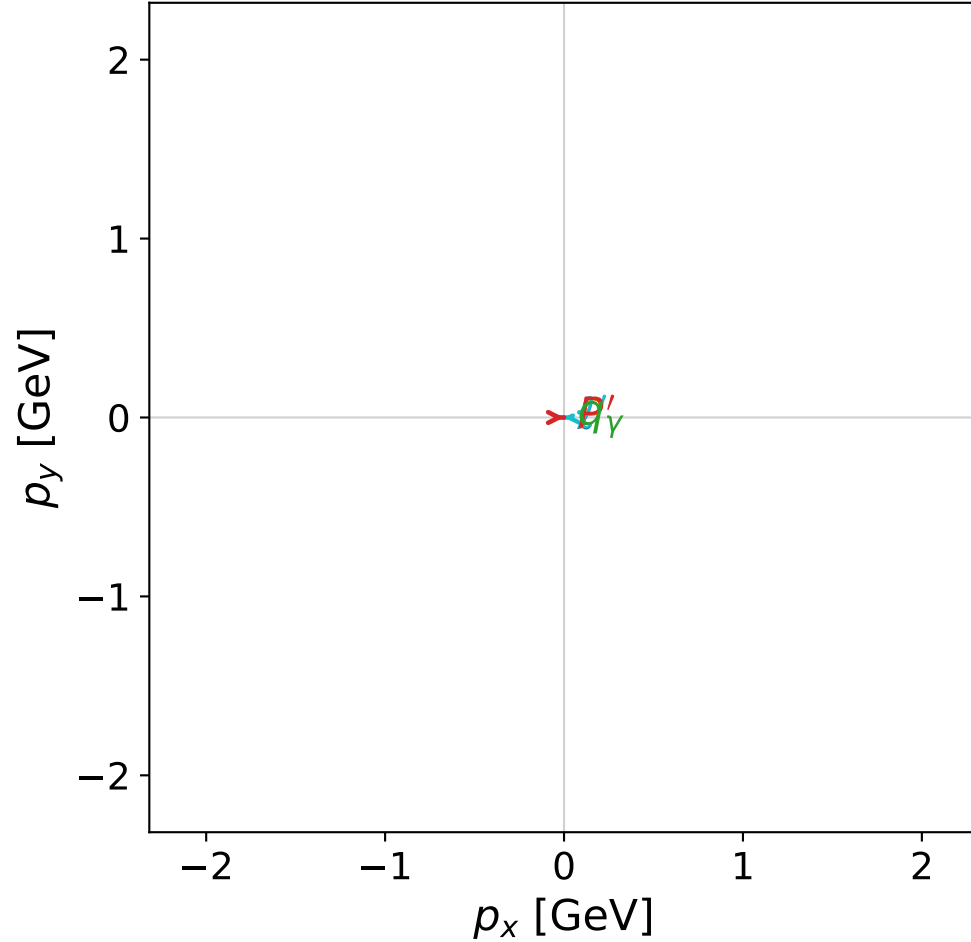


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



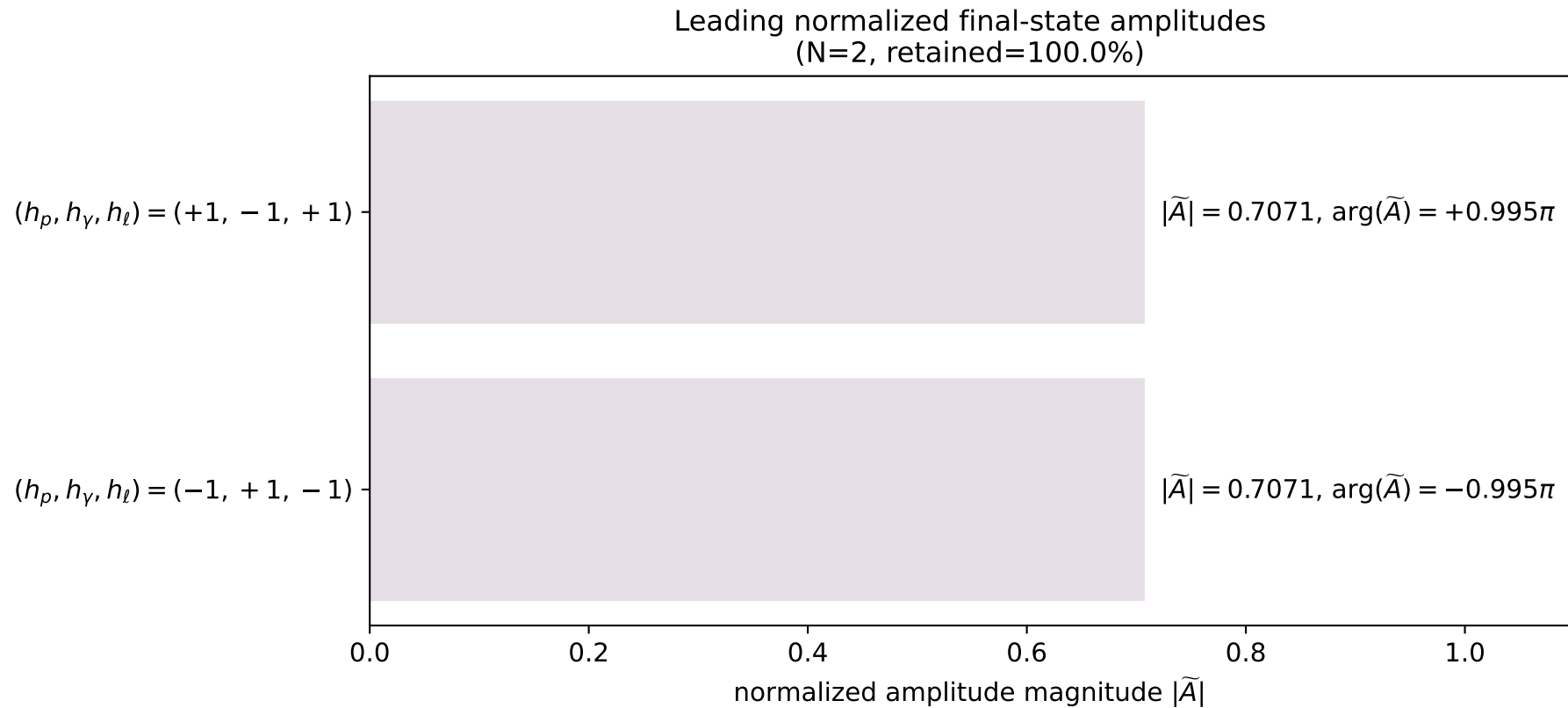
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_106
 $d_{\{\text{rm GHZ}\}}=3.85907\text{e-}08$
 region: polarization_cluster_03_local_minimum_106
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0106
 lepton mixing angle: $\alpha_e=2.1441764$ rad
 incoming lepton: $-0.542475|+\rangle +0.840072|-\rangle$
 proton mixing angle: $\alpha_p=0.99739265$ rad
 incoming proton: $0.542494|+\rangle +0.840059|-\rangle$

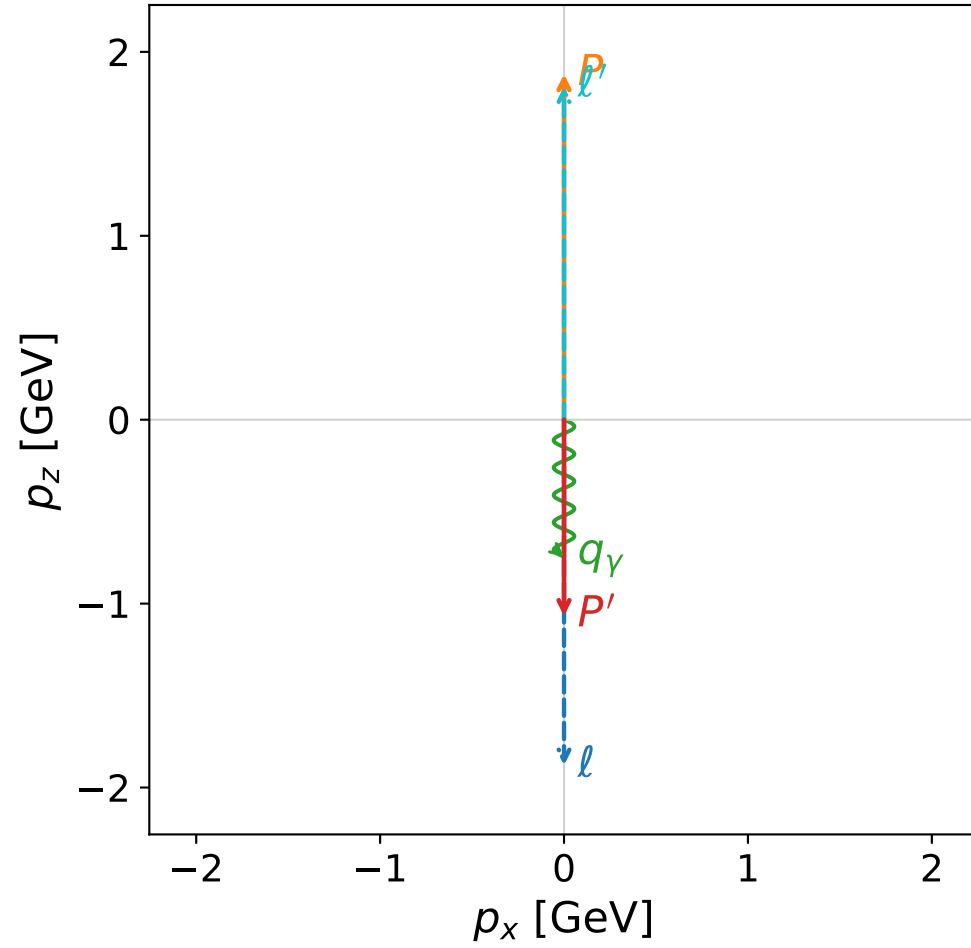
$s=16.6364$, $\sqrt{s}=4.07877$, $|\vec{P}|=1.93153$, $|\vec{P}'|=1.82957$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.0149987$, $\phi_\gamma=6.26819$
 $E_\gamma=0.0966003$, $\theta_{l'}=0$, $\phi_{l'}=3.15509$
 $Q^2=14.8818$, $x_B=0.997069$, $t=-14.1375$, $W^2=0.923592$, $y=0.94726$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.9315$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9315$
 P $E= 2.1472$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9315$
 l' $E= 1.9262$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9262$
 P' $E= 2.0560$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8296$
 q_γ $E= 0.0966$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0966$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

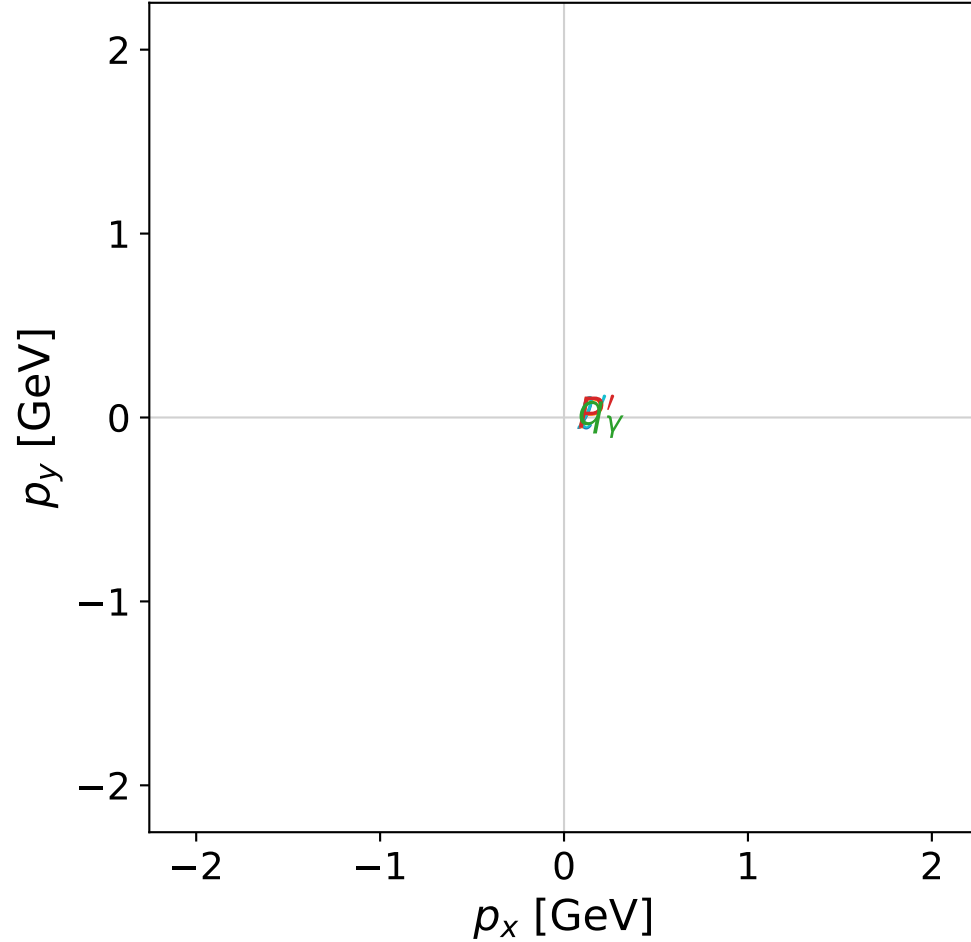


dghz_mixing_angles_polarization_03_local_minimum_107
 $d_{\{\rm GHZ\}}=3.86512\text{e-}08$
 region: polarization_cluster_03_local_minimum_107
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0107
 lepton mixing angle: $\alpha_e=2.4205789$ rad
 incoming lepton: $-0.751137|+\rangle +0.660146|-\rangle$
 proton mixing angle: $\alpha_p=0.72102631$ rad
 incoming proton: $0.751129|+\rangle +0.660156|-\rangle$

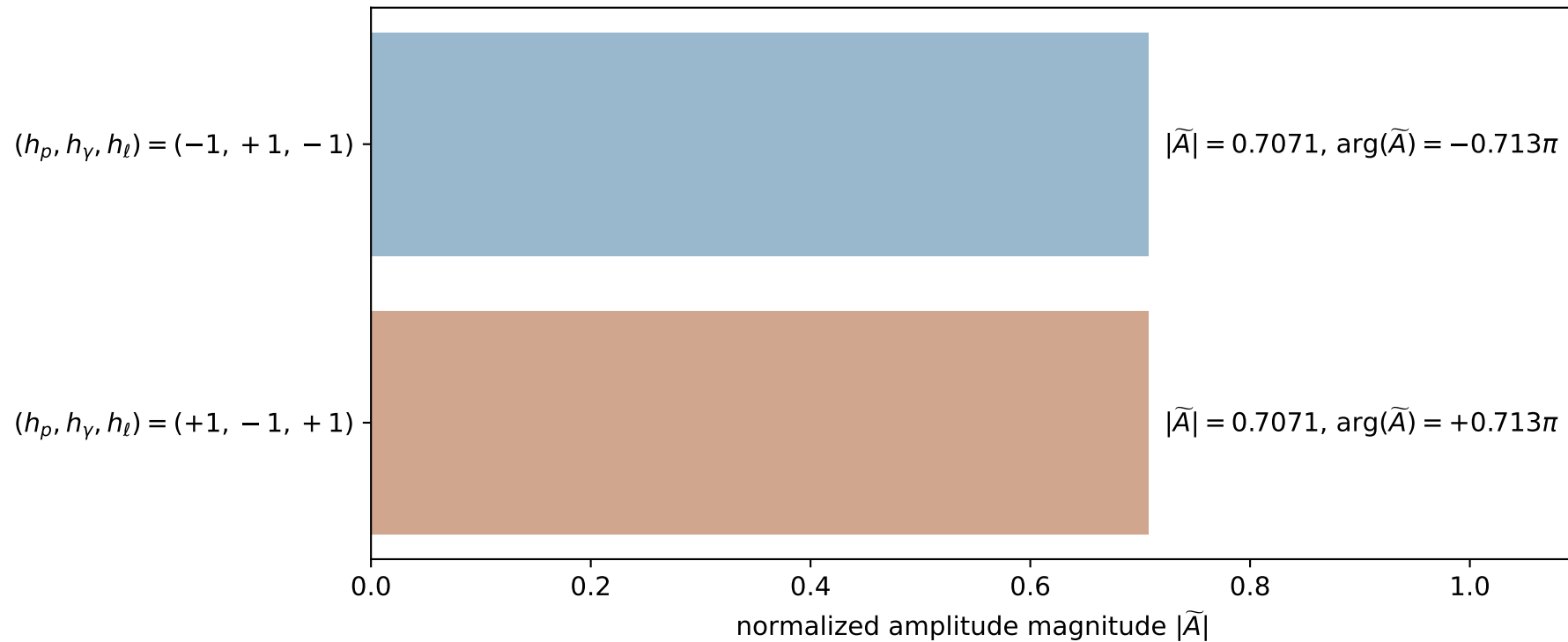
$s=15.8347$, $\sqrt{s}=3.97928$, $|\vec{P}|=1.87909$, $|\vec{P}'|=1.06808$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.05624$, $\phi_\gamma=5.38261$
 $E_\gamma=0.744851$, $\theta_{l'}=0$, $\phi_{l'}=3.47839$
 $Q^2=13.6266$, $x_B=0.962801$, $t=-8.22517$, $W^2=1.40632$, $y=0.946392$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8791$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8791$
 P $E= 2.1002$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8791$
 l' $E= 1.8129$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.8129$
 P' $E= 1.4215$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.0681$
 q_γ $E= 0.7449$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.7449$

x--y plane

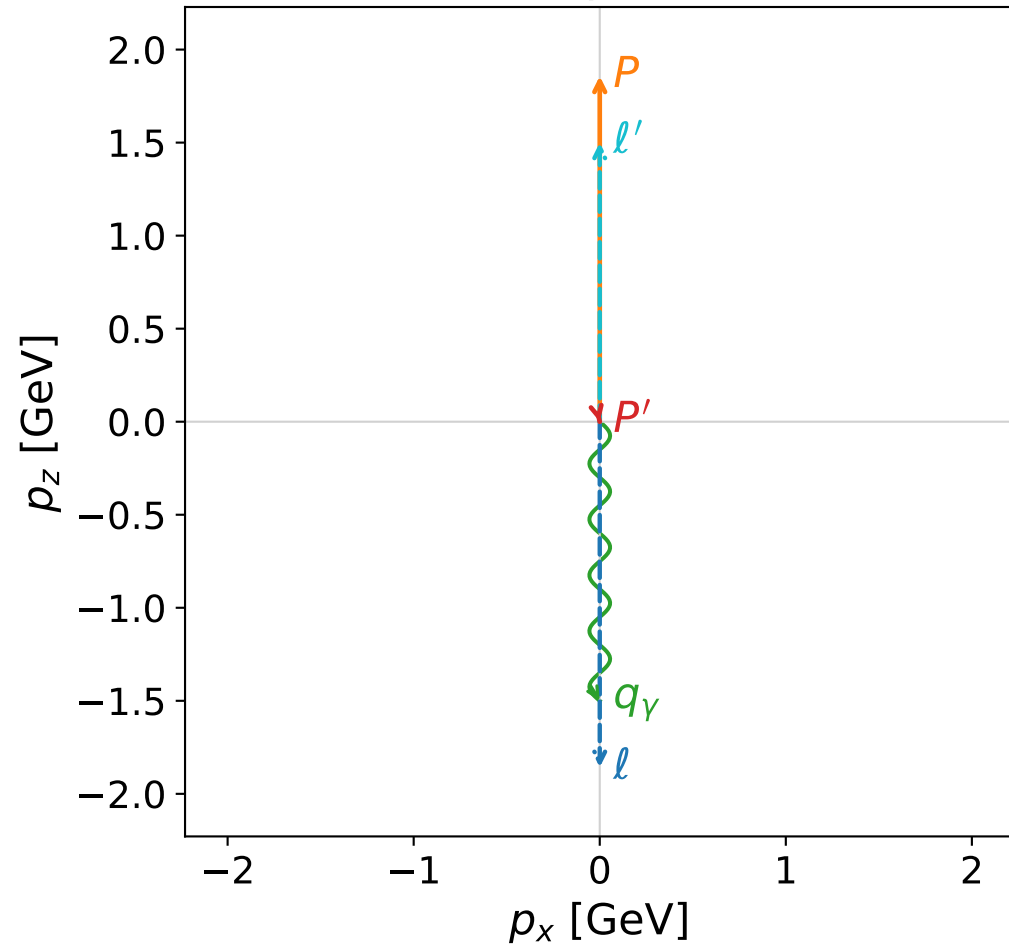


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

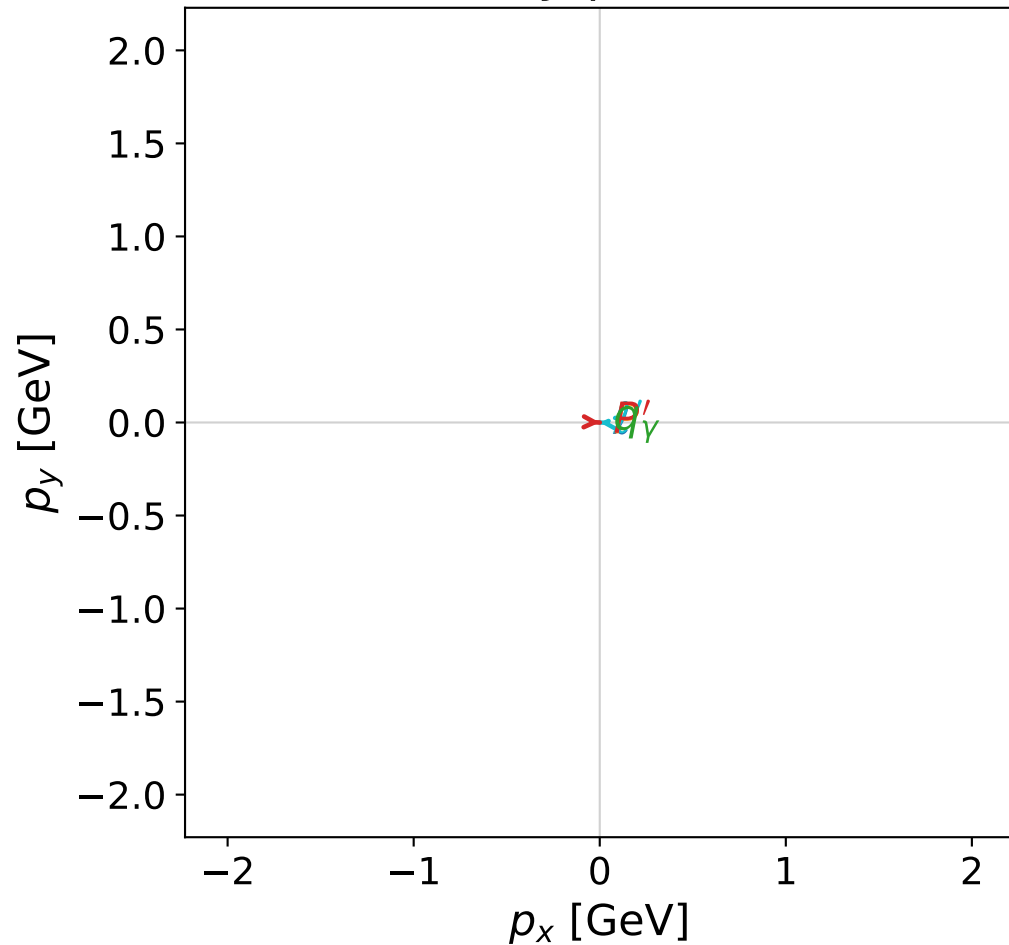


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

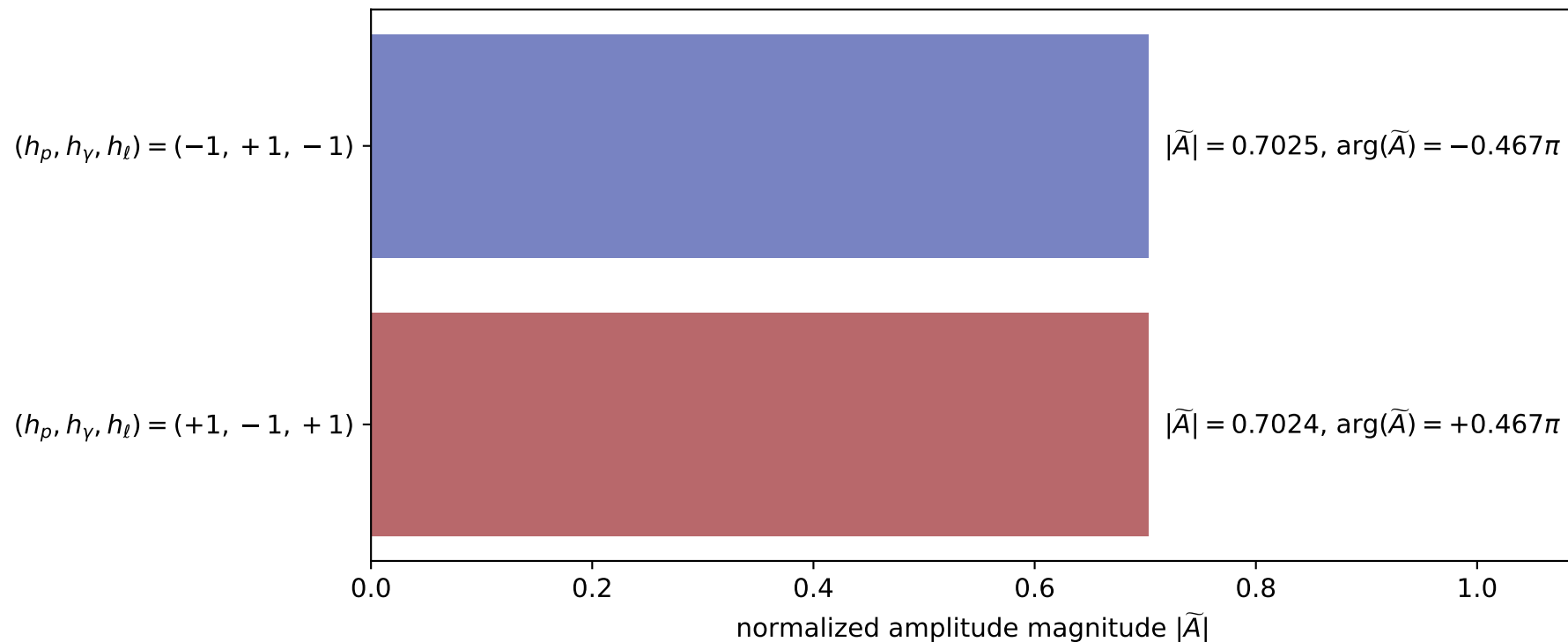


dghz_mixing_angles_polarization_03_local_minimum_118
 $d_{\text{GHZ}}=4.43159\text{e-}08$
 region: polarization_cluster_03_local_minimum_118
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0118
 lepton mixing angle: $\alpha_e=2.3993315$ rad
 incoming lepton: $-0.736942|+\rangle +0.675956|-\rangle$
 proton mixing angle: $\alpha_p=0.74228136$ rad
 incoming proton: $0.736928|+\rangle +0.675971|-\rangle$

$s=15.5089$, $\sqrt{s}=3.93814$, $|\vec{P}|=1.85736$, $|\vec{P}'|=0.000247174$
 $\theta_{p'}=2.91175$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.24252$, $\phi_\gamma=4.56954$
 $E_\gamma=1.49995$, $\theta_{\ell'}=3.7536\text{e-}05$, $\phi_{\ell'}=3.10092$
 $Q^2=11.1456$, $x_B=0.798465$, $t=-2.14474$, $W^2=3.69303$, $y=0.954178$

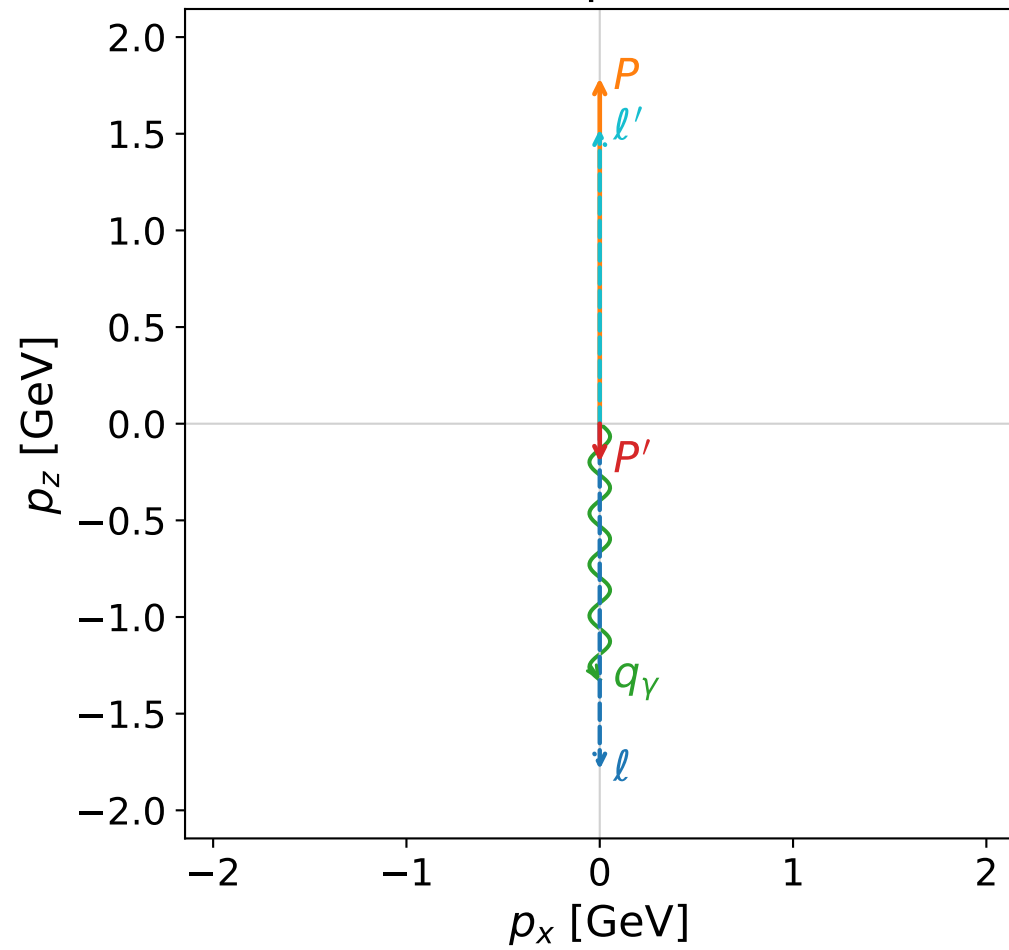
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8574$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8574$
 P $E= 2.0808$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8574$
 ℓ' $E= 1.5002$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 1.5002$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.0002$
 q_γ $E= 1.4999$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4999$

Leading normalized final-state amplitudes
 (N=2, retained=98.7%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

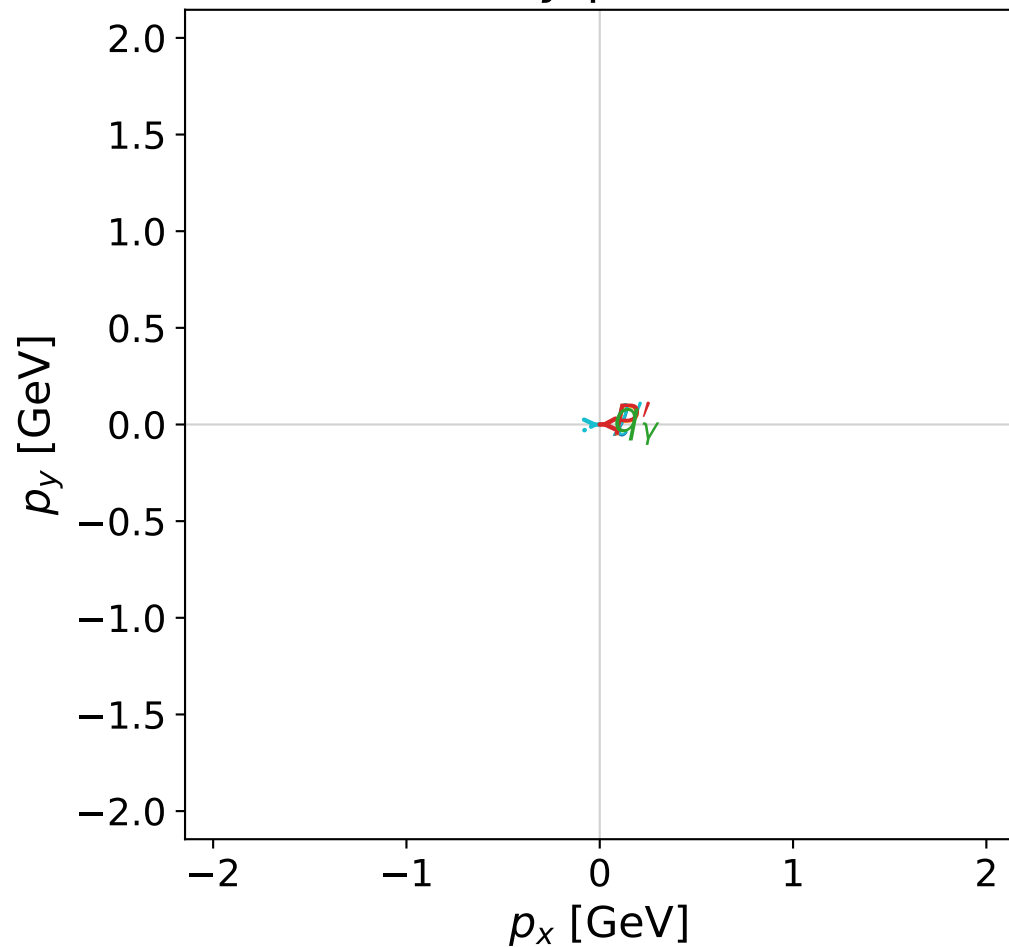


dghz_mixing_angles_polarization_03_local_minimum_124
 $d_{\text{GHZ}}=4.80438\text{e-}08$
 region: polarization_cluster_03_local_minimum_124
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0124
 lepton mixing angle: $\alpha_e=2.3690498$ rad
 incoming lepton: $-0.716138|+\rangle +0.697959|-\rangle$
 proton mixing angle: $\alpha_p=0.77252117$ rad
 incoming proton: $0.716153|+\rangle +0.697943|-\rangle$

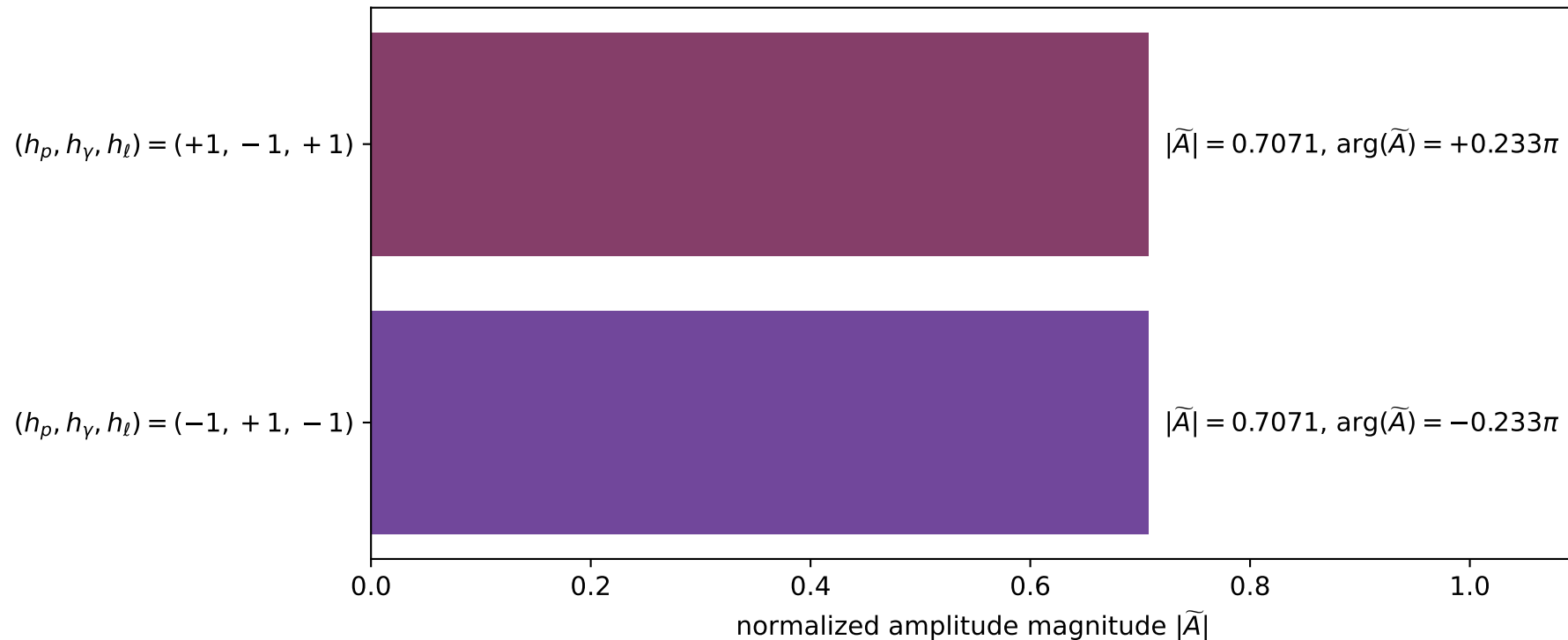
$s=14.4913$, $\sqrt{s}=3.80674$, $|\vec{P}|=1.78781$, $|\vec{P}'|=0.198316$
 $\theta_{p'}=3.14137$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.17027$, $\phi_\gamma=0.760181$
 $E_\gamma=1.32485$, $\theta_{\ell'}=2.88699\text{e-}05$, $\phi_{\ell'}=0.0286774$
 $Q^2=10.8925$, $x_B=0.843897$, $t=-2.82066$, $W^2=2.89472$, $y=0.948272$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7878$
 P $E= 2.0189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7878$
 ℓ' $E= 1.5232$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5232$
 P' $E= 0.9587$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1983$
 q_γ $E= 1.3248$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3248$

x--y plane

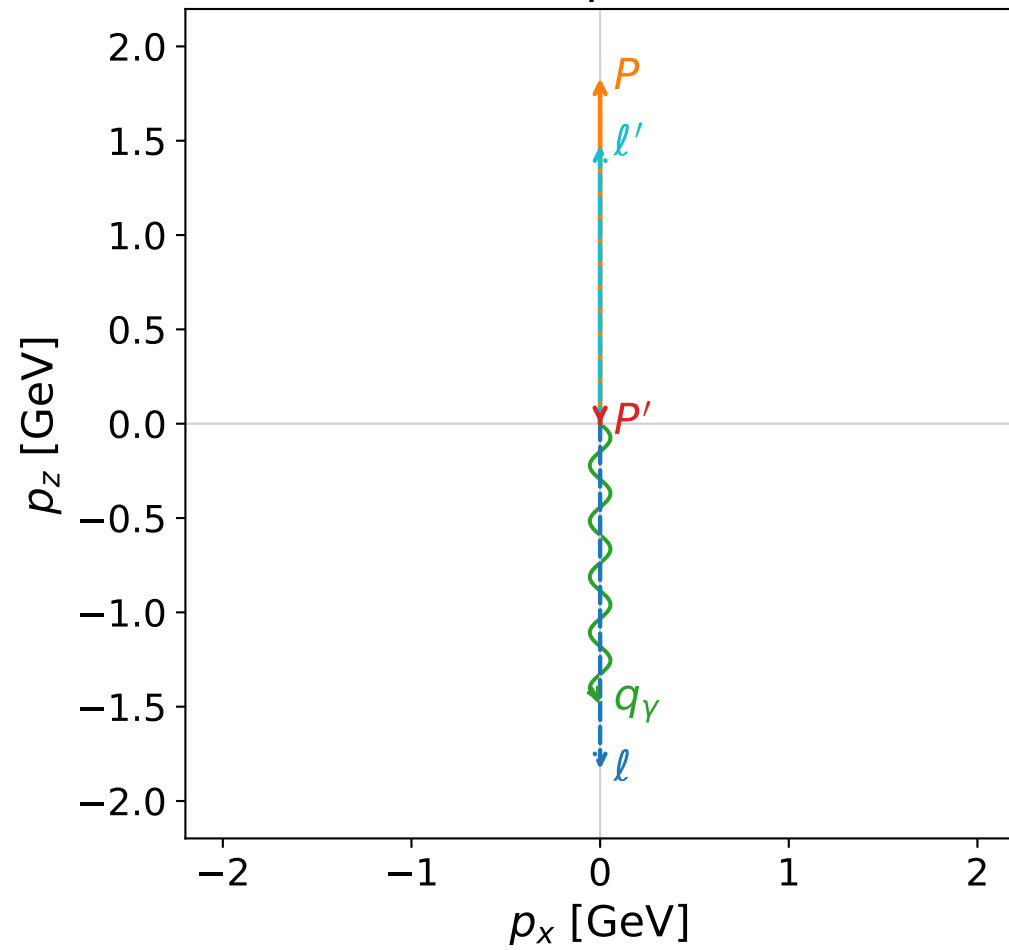


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

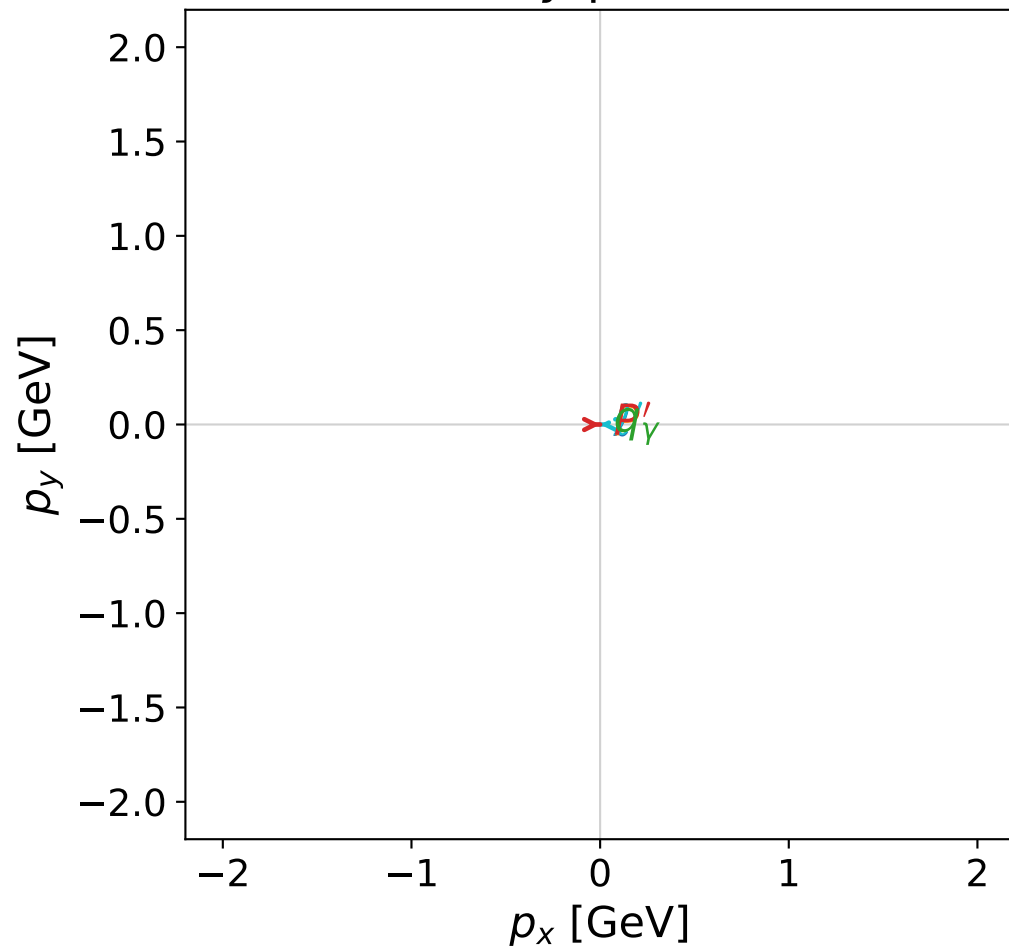


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



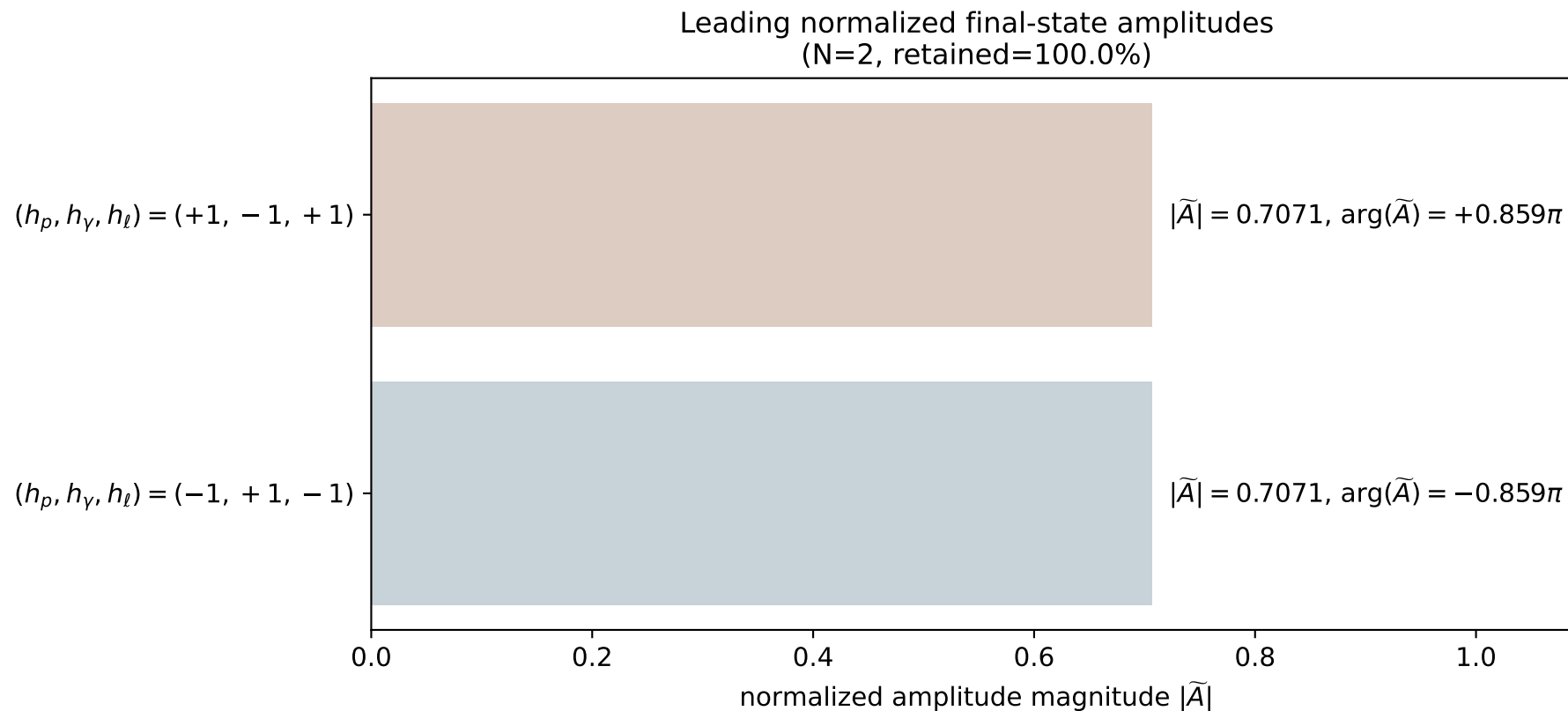
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_128
 $d_{\text{GHZ}}=5.13615\text{e-}08$
 region: polarization_cluster_03_local_minimum_128
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0128
 lepton mixing angle: $\alpha_e=2.5301823$ rad
 incoming lepton: $-0.818839|+\rangle +0.574023|-\rangle$
 proton mixing angle: $\alpha_p=0.61140686$ rad
 incoming proton: $0.818841|+\rangle +0.57402|-\rangle$

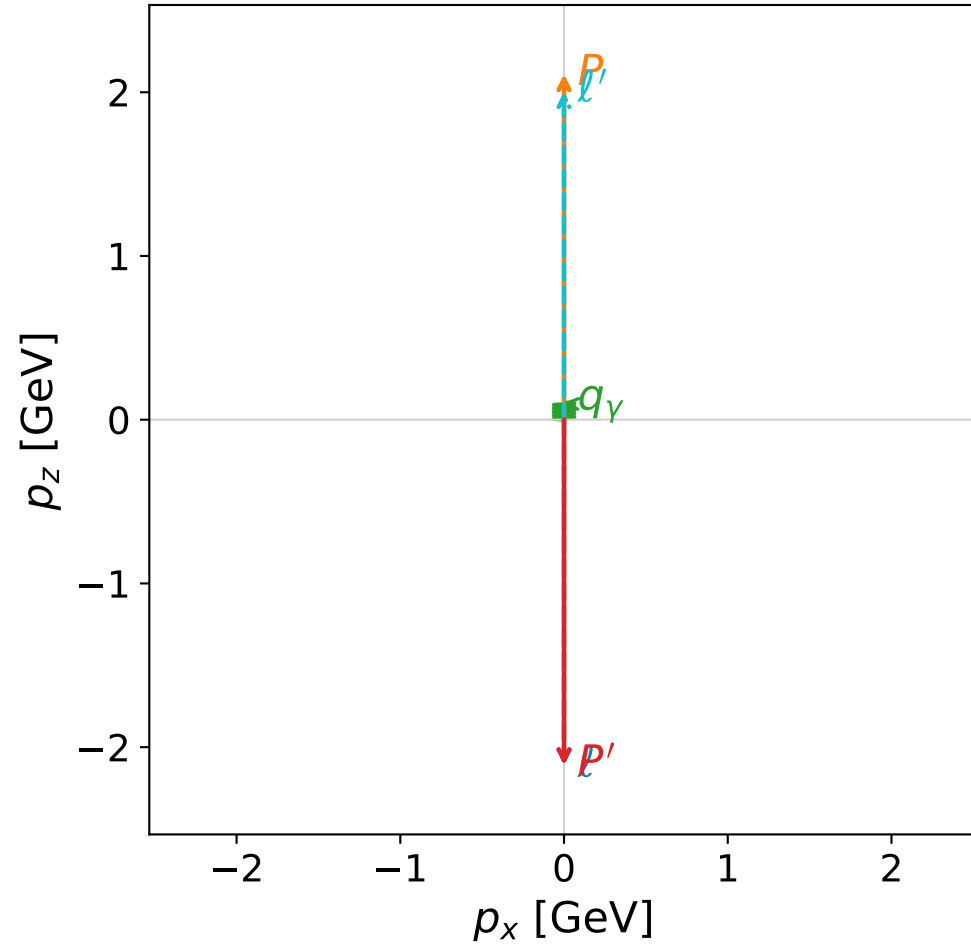
$s=15.1318$, $\sqrt{s}=3.88997$, $|\vec{P}|=1.83189$, $|\vec{P}'|=0.000225325$
 $\theta_{P'}=3.14145$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.00233824$, $\phi_\gamma=5.84287$
 $E_\gamma=1.47587$, $\theta_{\ell'}=1.49012\text{e-}08$, $\phi_{\ell'}=3.14393$
 $Q^2=10.8162$, $x_B=0.79623$, $t=-2.10208$, $W^2=3.64791$, $y=0.953148$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8319$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8319$
 P $E= 2.0581$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8319$
 ℓ' $E= 1.4761$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4761$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0002$
 q_γ $E= 1.4759$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4759$

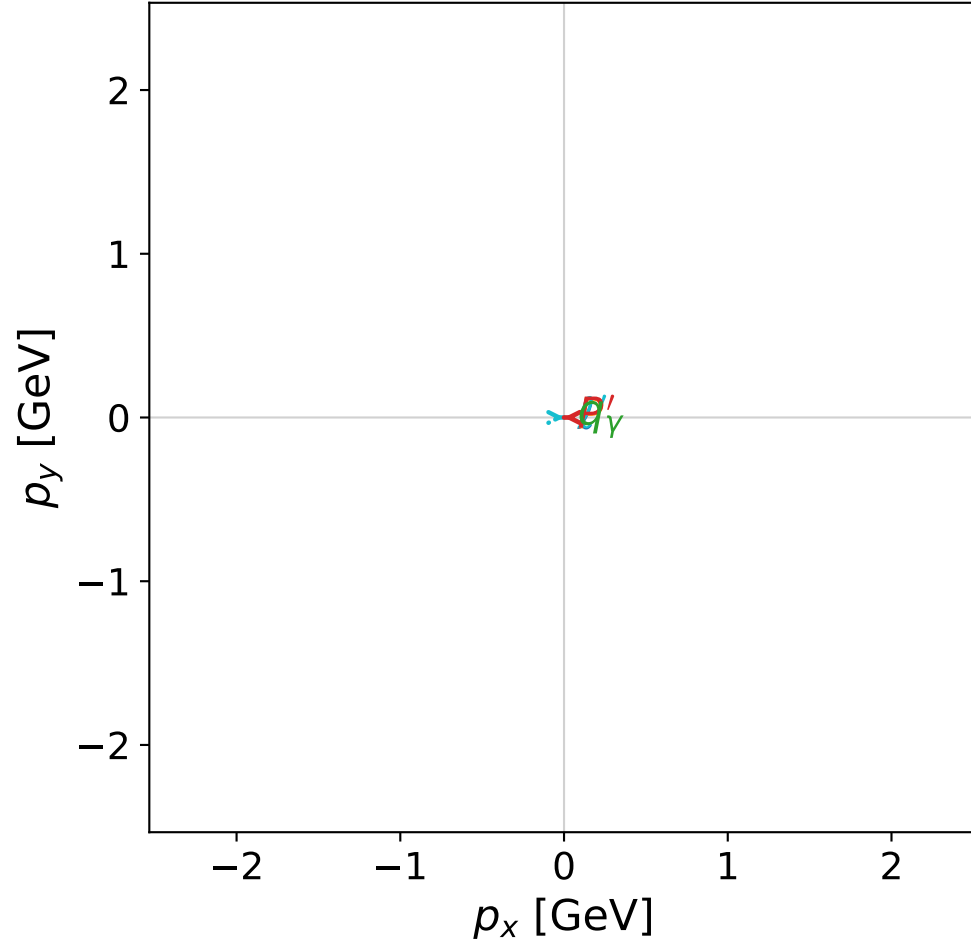


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



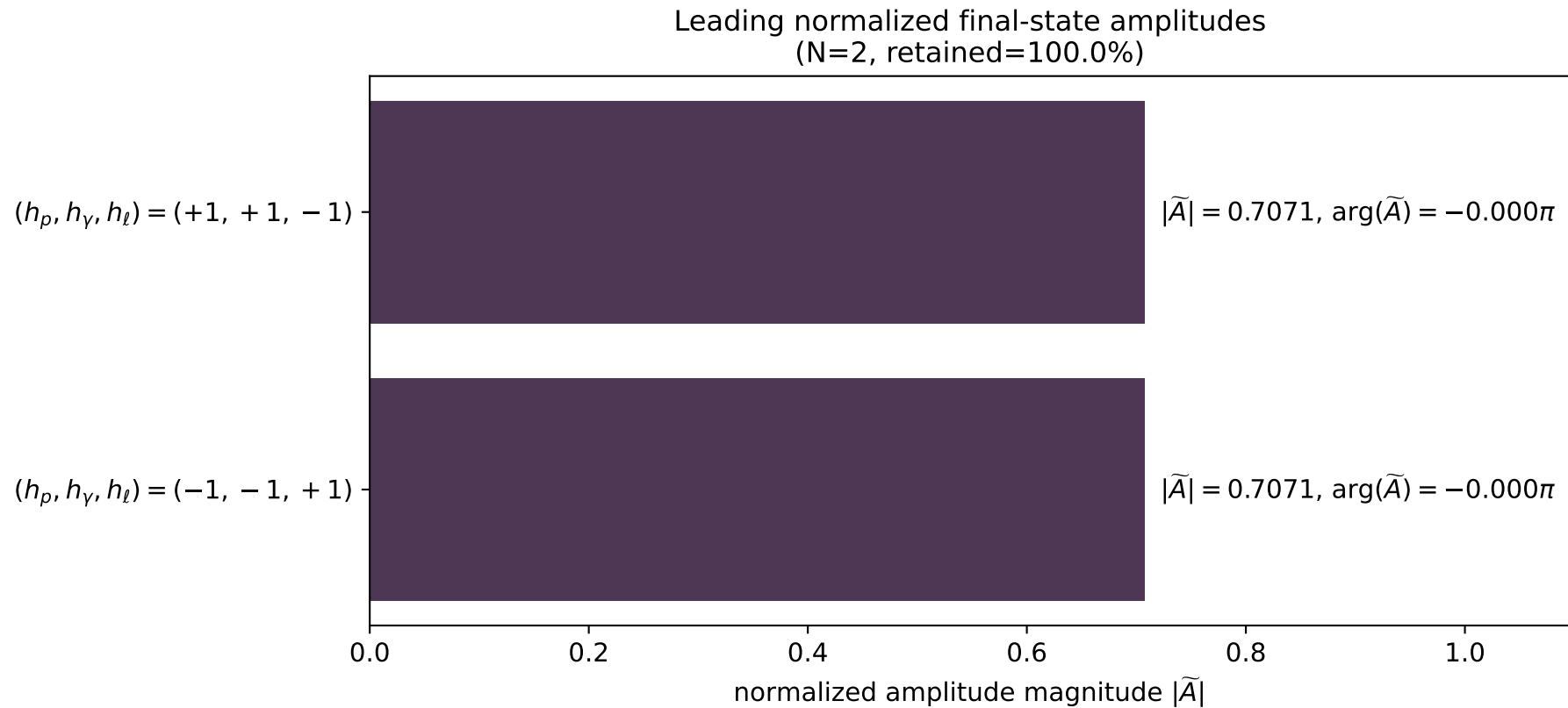
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_131
 $d_{\{\text{rm GHZ}\}}=5.31748\text{e-}08$
 region: polarization_cluster_03_local_minimum_131
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0131
 lepton mixing angle: $\alpha_e=1.8802171$ rad
 incoming lepton: $-0.304507|+\rangle +0.95251|-\rangle$
 proton mixing angle: $\alpha_p=0.30941263$ rad
 incoming proton: $0.952513|+\rangle +0.304499|-\rangle$

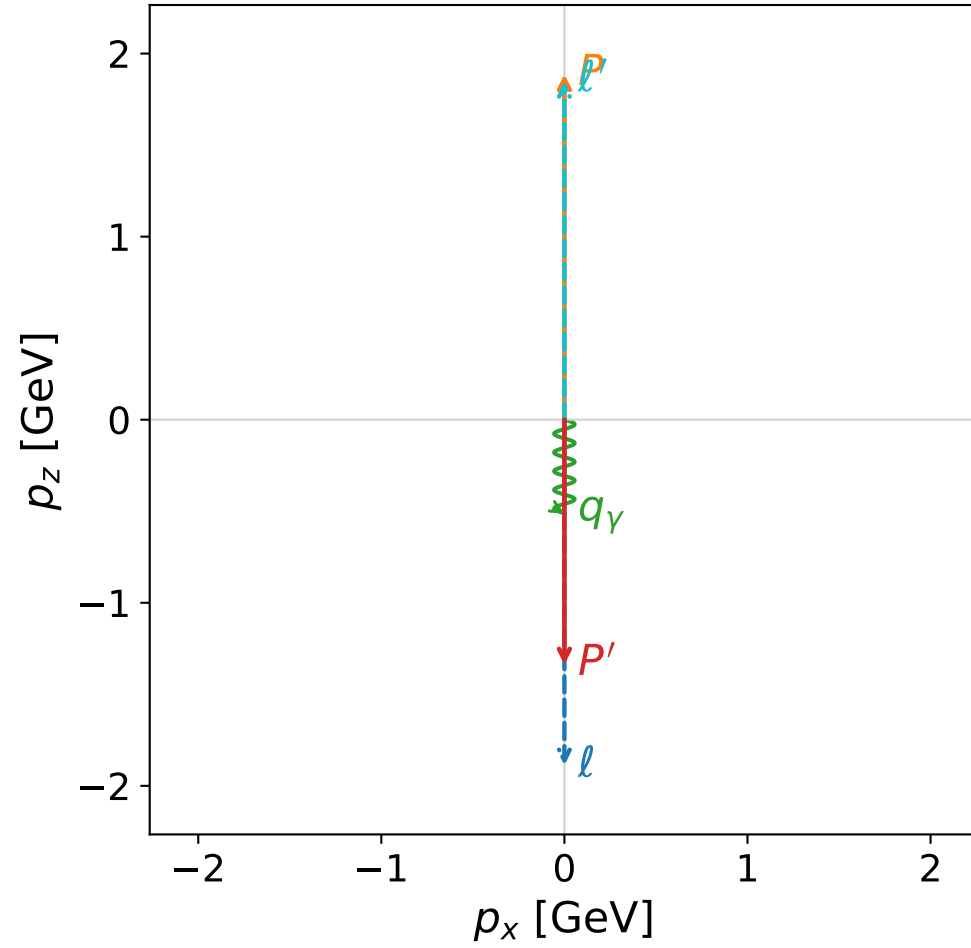
$s=19.5413$, $\sqrt{s}=4.42055$, $|\vec{P}|=2.11076$, $|\vec{P}'|=2.11076$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.40678$, $\phi_\gamma=1.45431$
 $E_\gamma=0.105533$, $\theta_{l'}=0$, $\phi_{l'}=0.265183$
 $Q^2=16.9302$, $x_B=0.947768$, $t=-17.8212$, $W^2=1.81287$, $y=0.957226$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1108$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1108$
 P $E= 2.3098$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1108$
 l' $E= 2.0052$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0052$
 P' $E= 2.3098$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1108$
 q_γ $E= 0.1055$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1055$

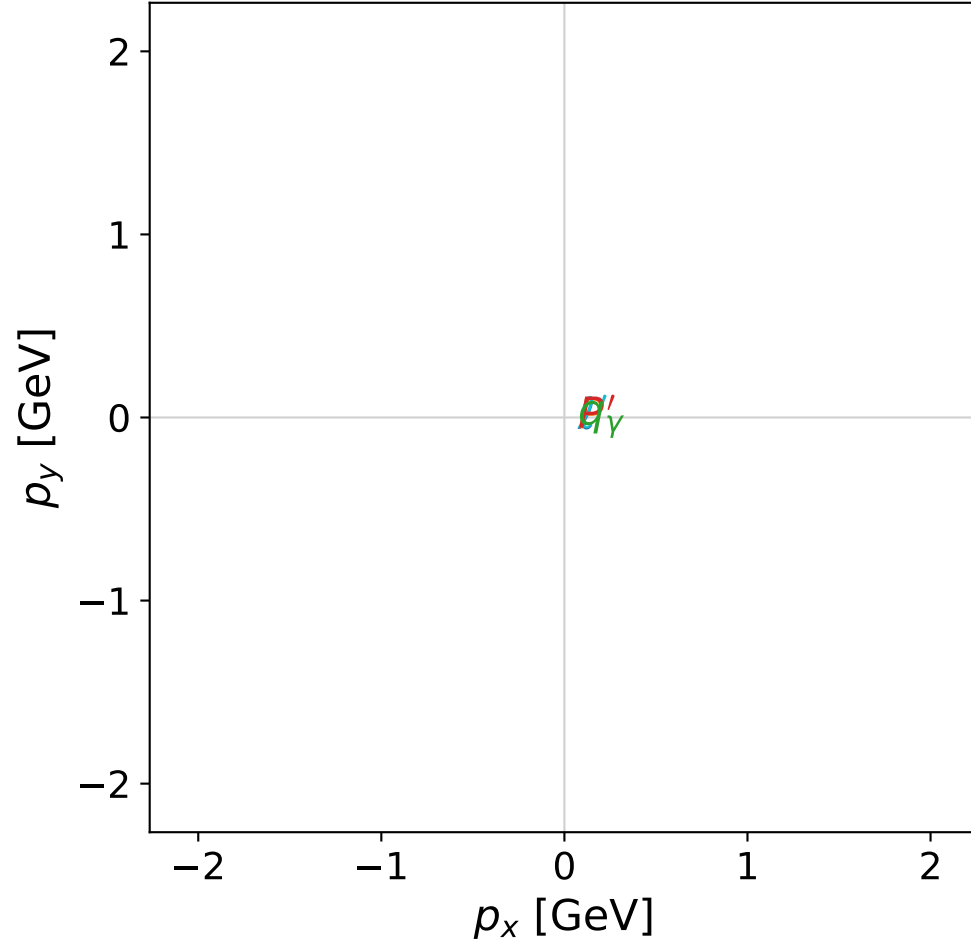


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



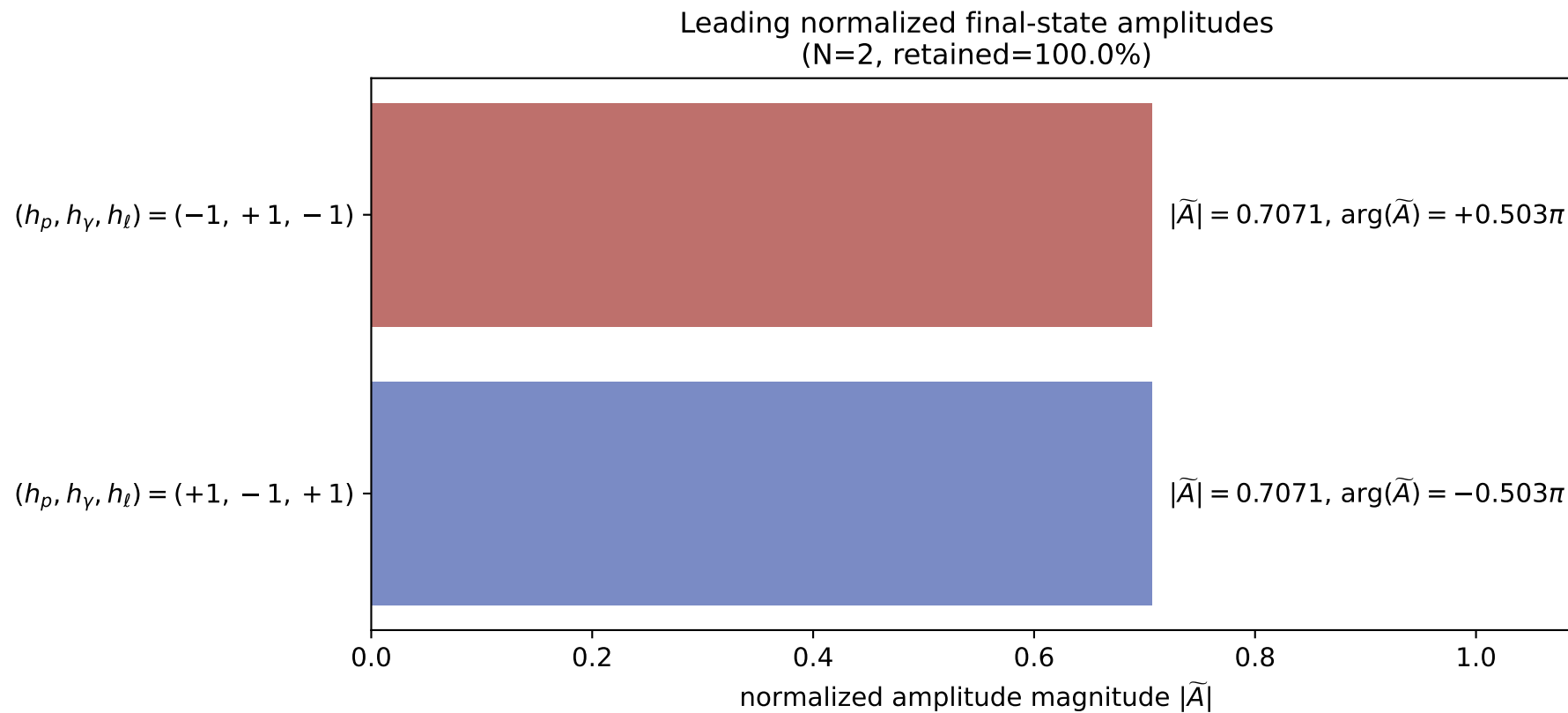
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_132
 $d_{\text{GHZ}}=5.32794\text{e-}08$
 region: polarization_cluster_03_local_minimum_132
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0132
 lepton mixing angle: $\alpha_e=2.7526505$ rad
 incoming lepton: $-0.925311|+\rangle +0.37921|-\rangle$
 proton mixing angle: $\alpha_p=0.38894583$ rad
 incoming proton: $0.925309|+\rangle +0.379213|-\rangle$

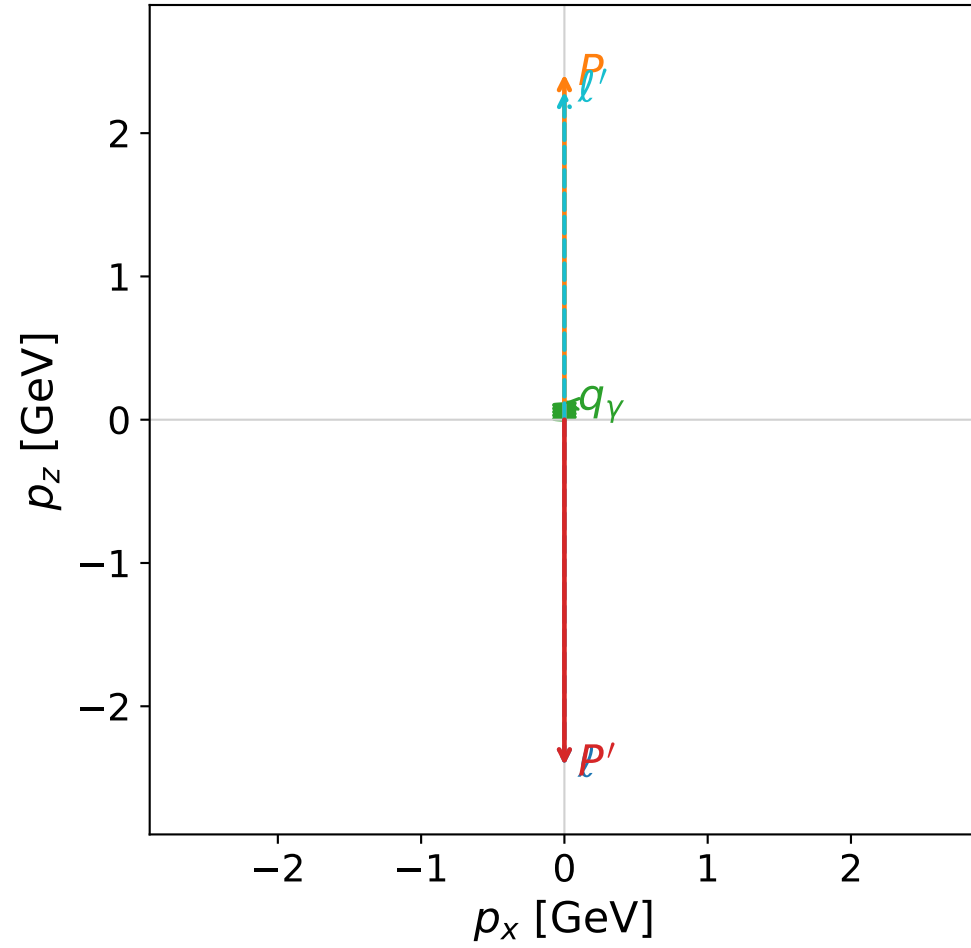
$s=15.965$, $\sqrt{s}=3.99562$, $|\vec{P}|=1.88771$, $|\vec{P}'|=1.33879$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=5.83465$, $\phi_\gamma=1.56232$
 $E_\gamma=0.511073$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.08424$
 $Q^2=13.968$, $x_B=0.978806$, $t=-10.1864$, $W^2=1.18229$, $y=0.945994$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=1.8877$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-1.8877$
 P $E=2.1079$ $p_x=0.0000$ $p_y=0.0000$ $p_z=1.8877$
 ℓ' $E=1.8499$ $p_x=-0.0000$ $p_y=0.0000$ $p_z=1.8499$
 P' $E=1.6347$ $p_x=0.0000$ $p_y=-0.0000$ $p_z=-1.3388$
 q_γ $E=0.5111$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.5111$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

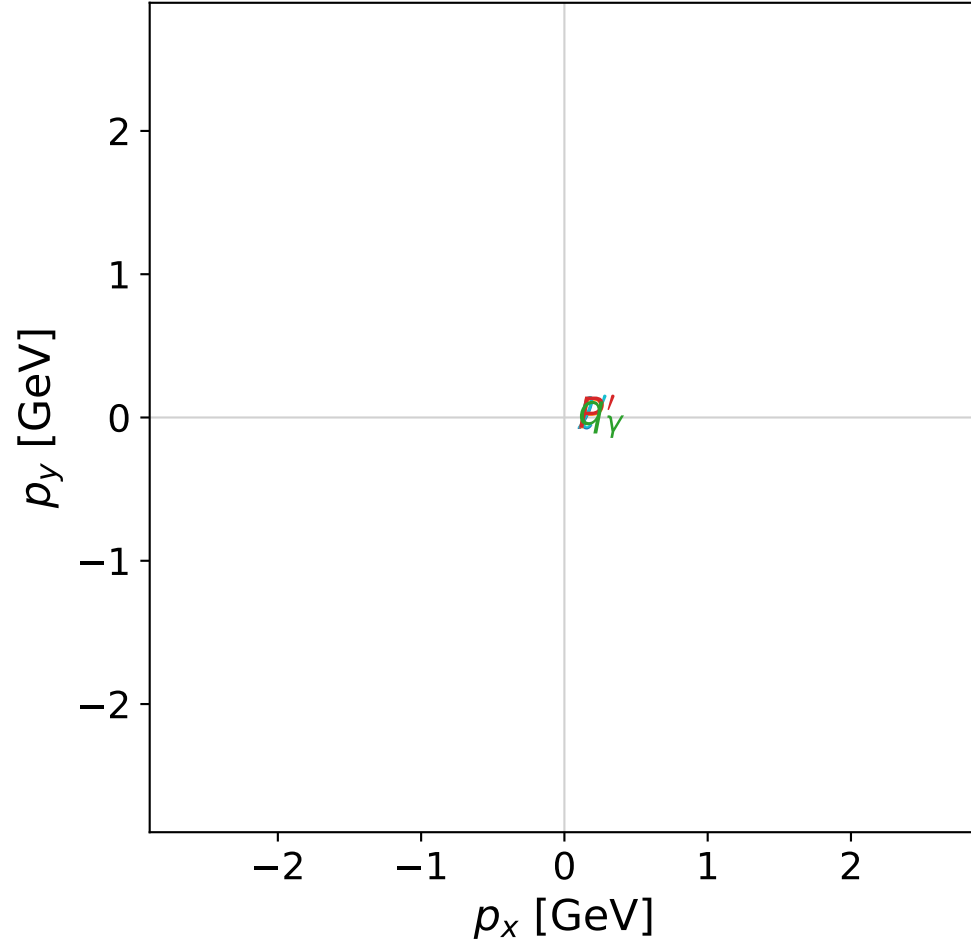


dghz_mixing_angles_polarization_03_local_minimum_134
 $d_{\text{GHZ}}=5.41283\text{e-}08$
 region: polarization_cluster_03_local_minimum_134
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0134
 lepton mixing angle: $\alpha_e=2.9154679$ rad
 incoming lepton: $-0.974543|+\rangle +0.224203|-\rangle$
 proton mixing angle: $\alpha_p=1.3446609$ rad
 incoming proton: $0.224213|+\rangle +0.97454|-\rangle$

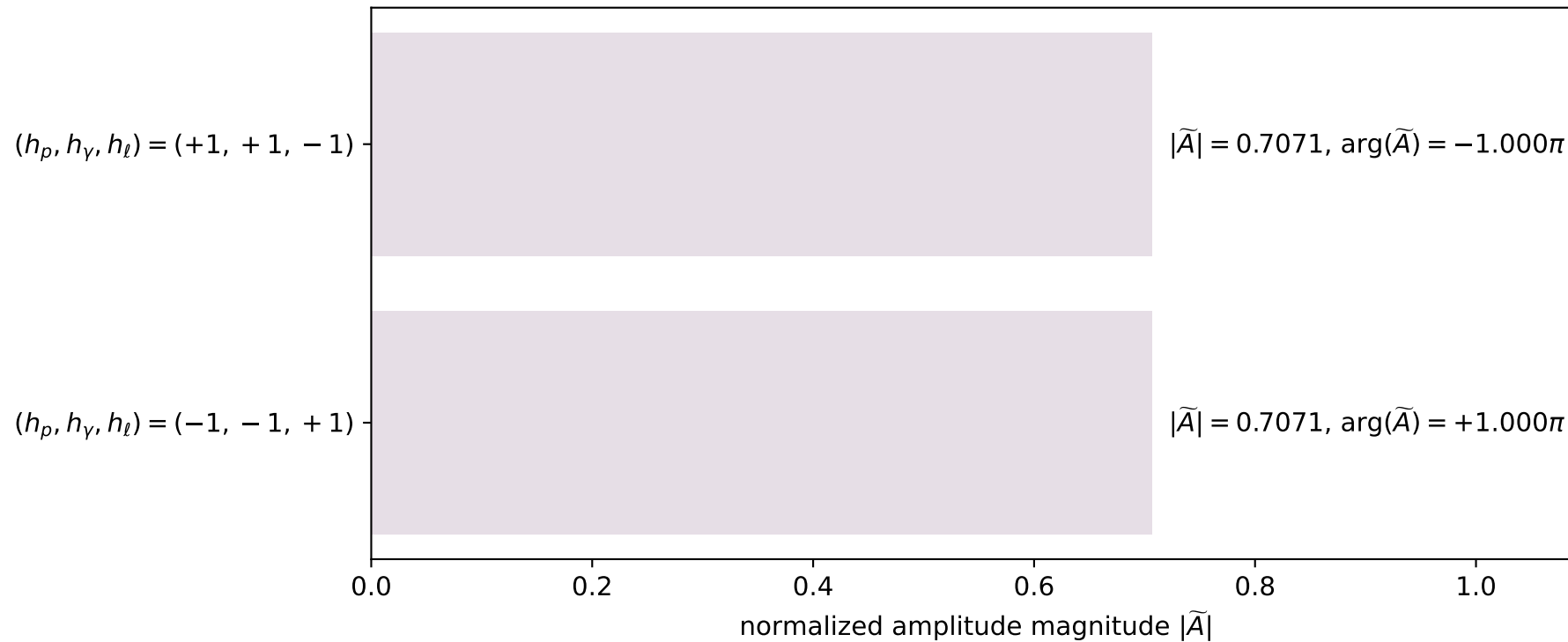
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.5701$, $\phi_\gamma=3.68393$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=5.71169$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1206$

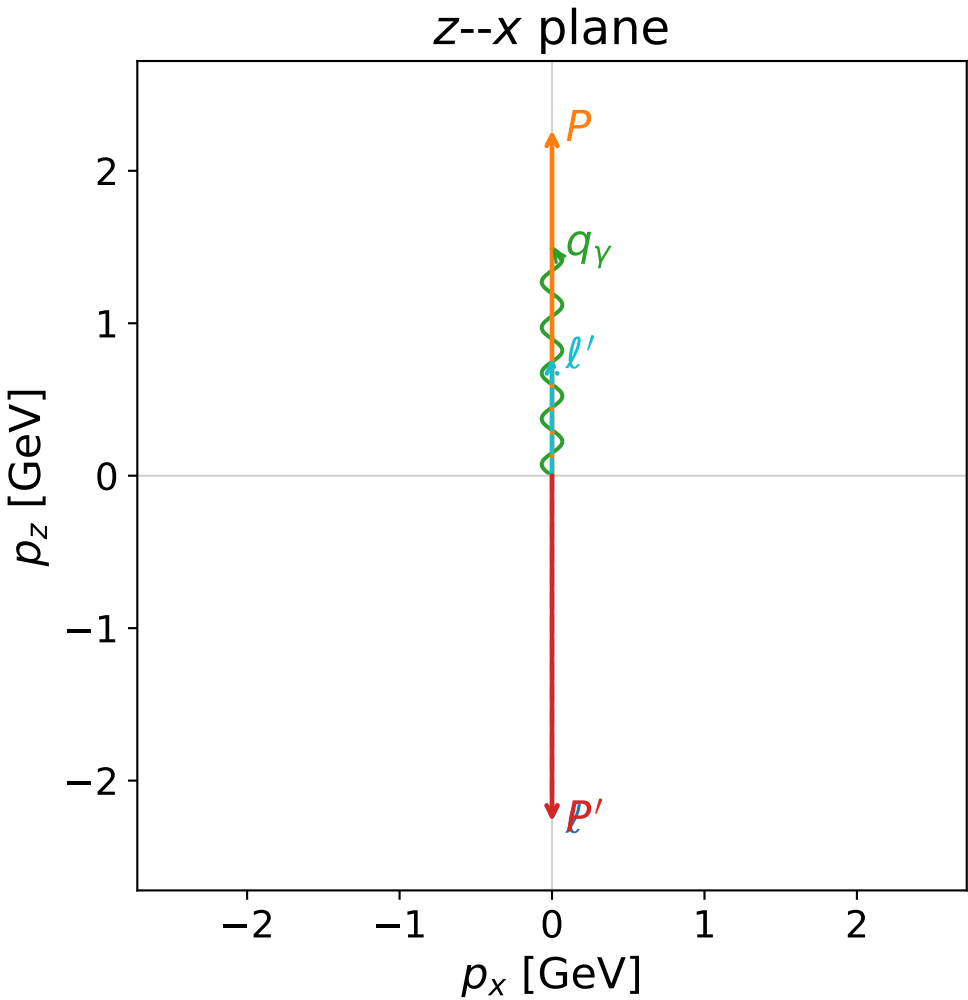
x--y plane



Leading normalized final-state amplitudes
(N=2, retained=100.0%)



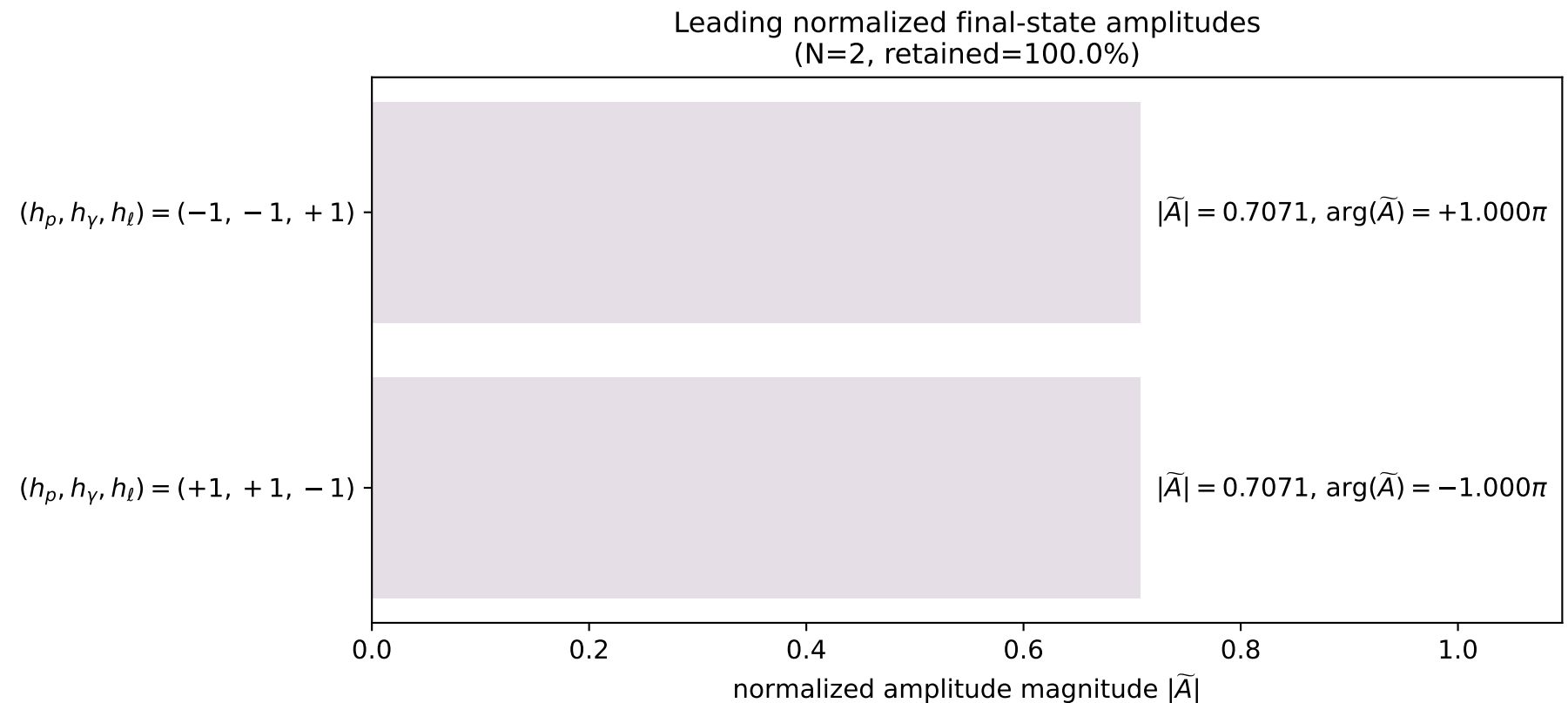
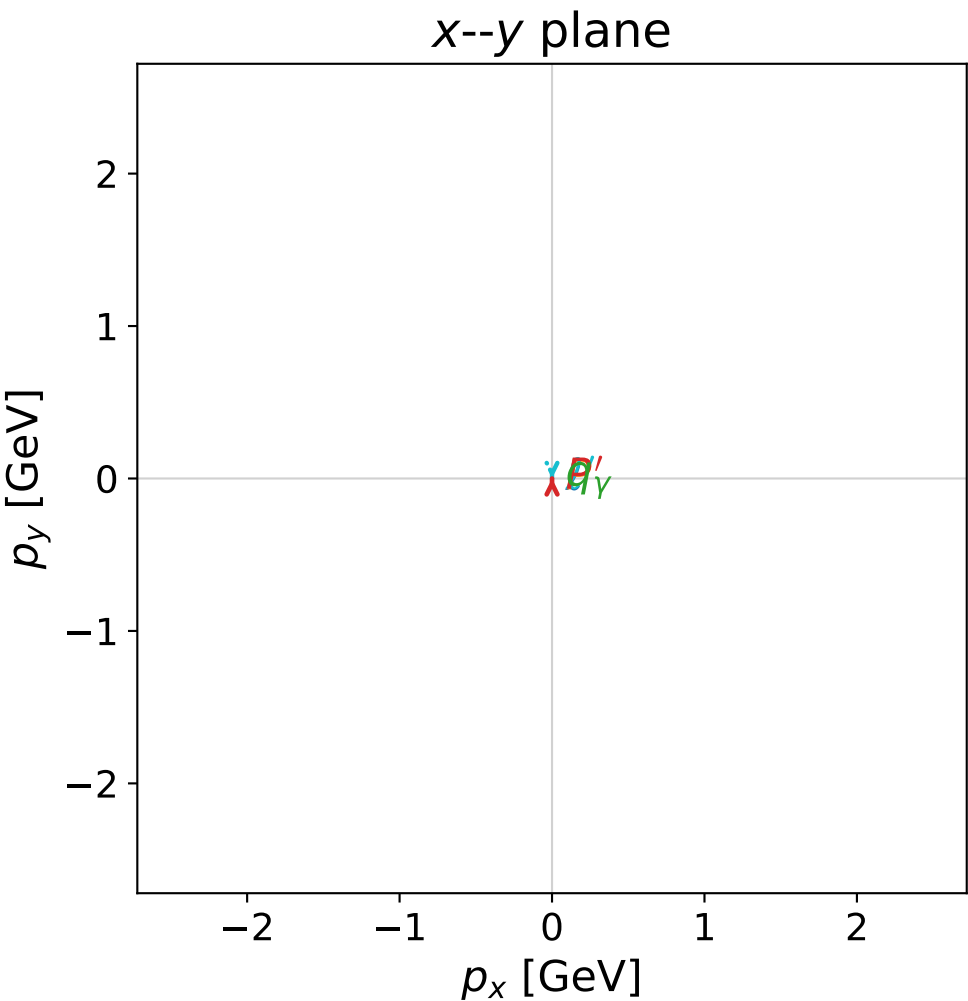
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



dghz_mixing_angles_polarization_03_local_minimum_143
d_{\rm GHZ}=7.74877e-08
region: polarization_cluster_03_local_minimum_143
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0143
lepton mixing angle: $\alpha_e=2.2187554$ rad
incoming lepton: $-0.60356|+\rangle +0.797317|-\rangle$
proton mixing angle: $\alpha_p=0.64796548$ rad
incoming proton: $0.797313|+\rangle +0.603566|-\rangle$

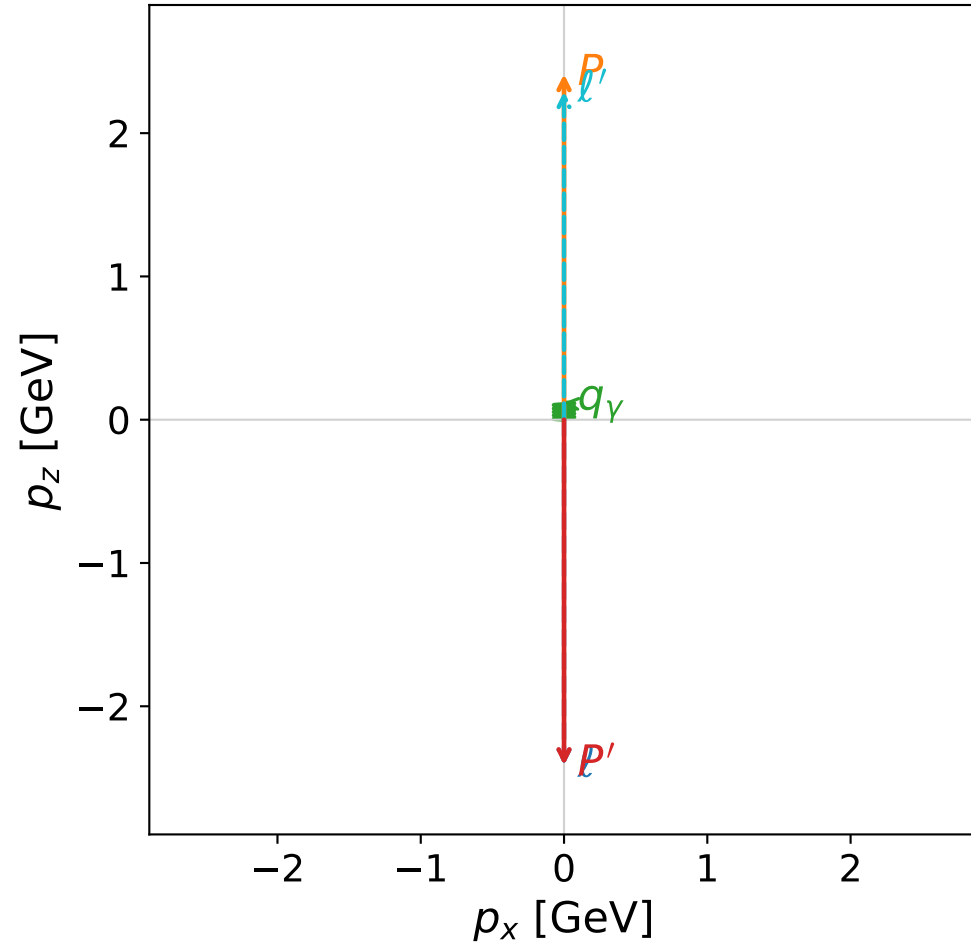
$s=22.2813$, $\sqrt{s}=4.7203$, $|\vec{P}|=2.26695$, $|\vec{P}'|=2.26695$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.65437$, $\phi_\gamma=3.92312$
 $E_\gamma=1.49471$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.79596$
 $Q^2=7.00254$, $x_B=0.331661$, $t=-20.5563$, $W^2=14.9908$, $y=0.986548$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2670$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2670$
 P $E= 2.4533$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2670$
 ℓ' $E= 0.7722$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7722$
 P' $E= 2.4533$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2670$
 q_γ $E= 1.4947$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4947$

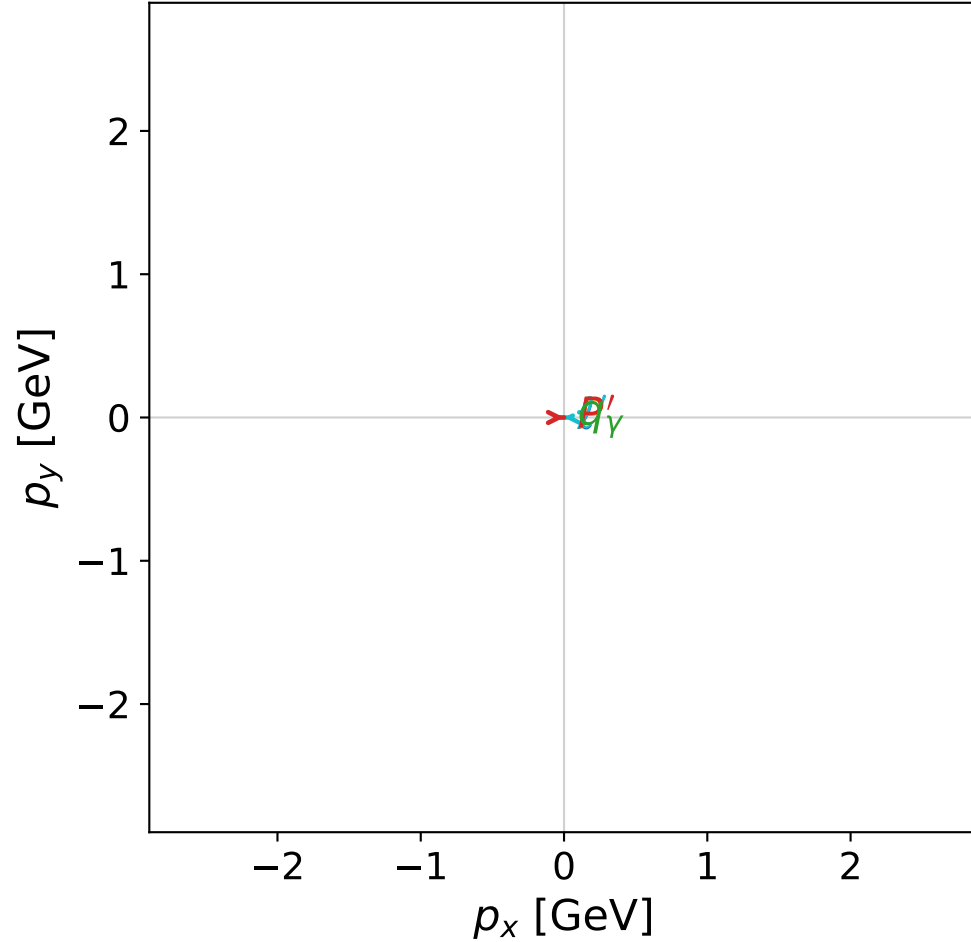


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

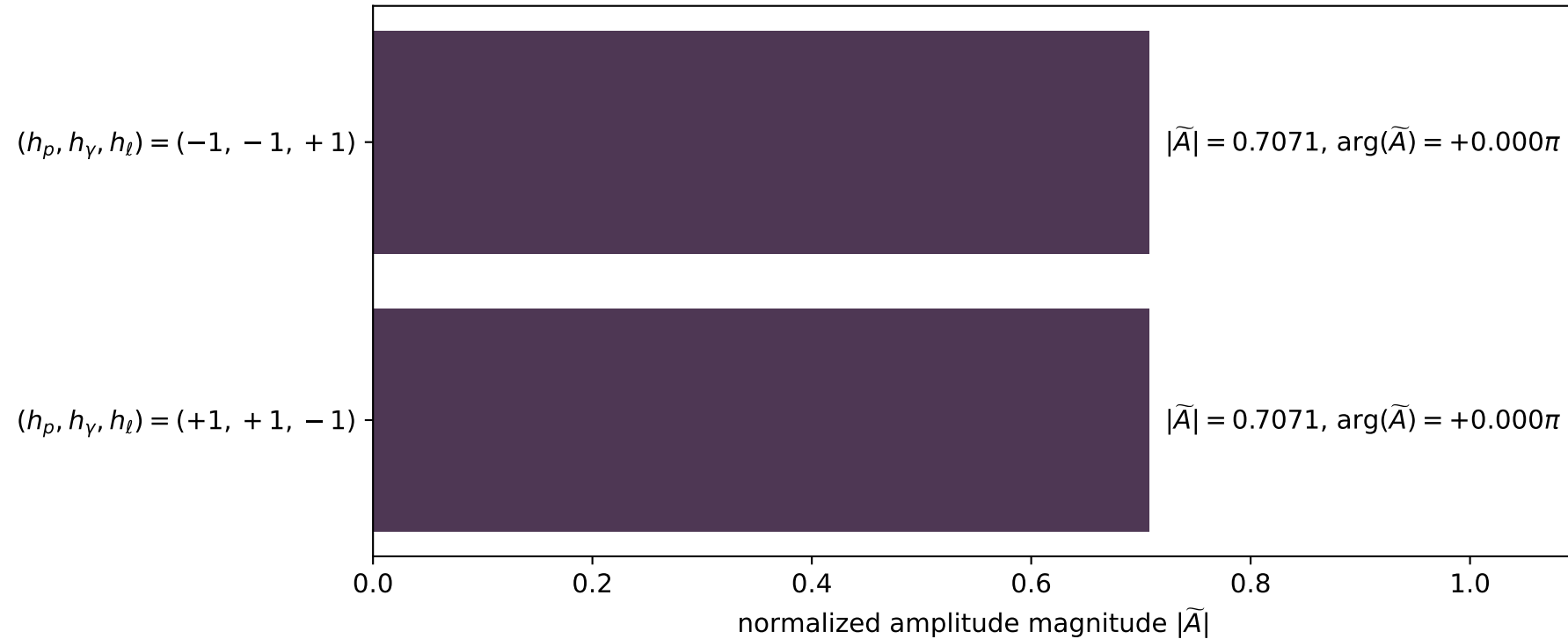


dghz_mixing_angles_polarization_03_local_minimum_154
 $d_{\{\text{rm GHZ}\}}=9.73504\text{e-}08$
 region: polarization_cluster_03_local_minimum_154
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0154
 lepton mixing angle: $\alpha_e=3.0190559$ rad
 incoming lepton: $-0.992502|+\rangle +0.12223|-\rangle$
 proton mixing angle: $\alpha_p=1.4482597$ rad
 incoming proton: $0.12223|+\rangle +0.992502|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.116076$, $\phi_\gamma=6.10382$
 $E_\gamma=0.120666$, $\theta_{l'}=0$, $\phi_{l'}=3.25767$
 $Q^2=22.1071$, $x_B=0.948242$, $t=-23.2713$, $W^2=2.08651$, $y=0.966567$

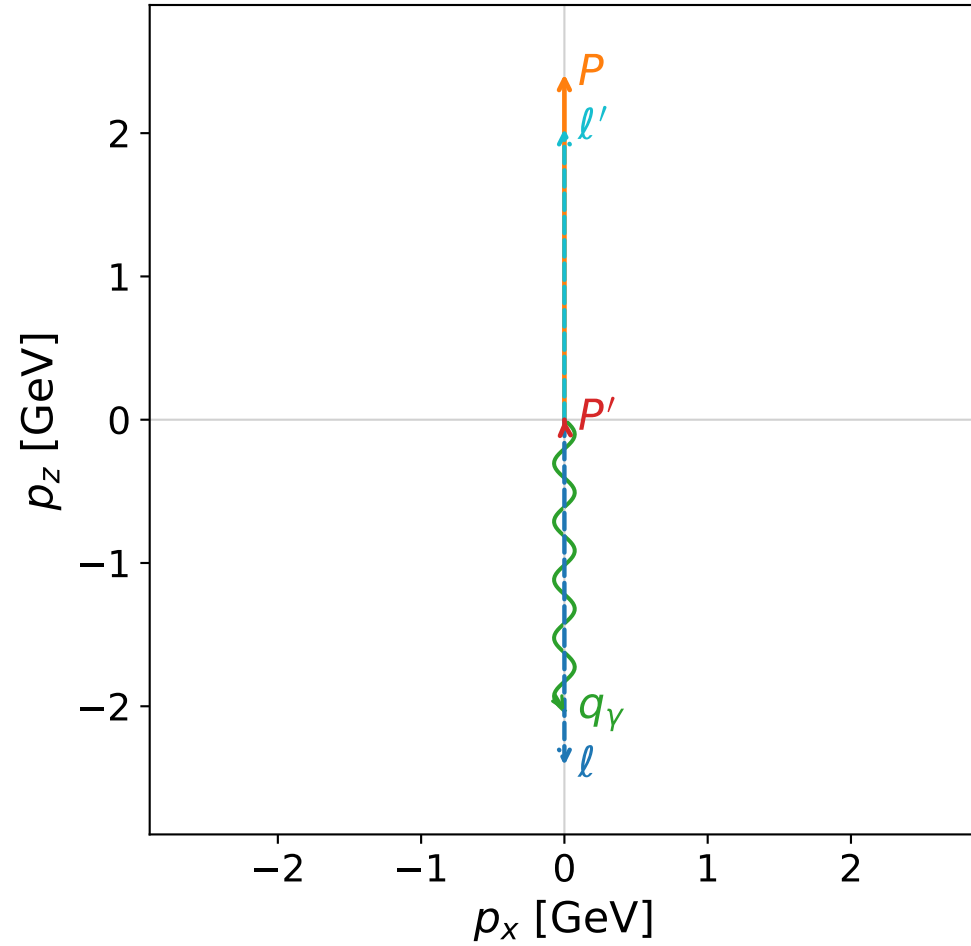
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.2913$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2913$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1207$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.1207$

Leading normalized final-state amplitudes (N=2, retained=100.0%)

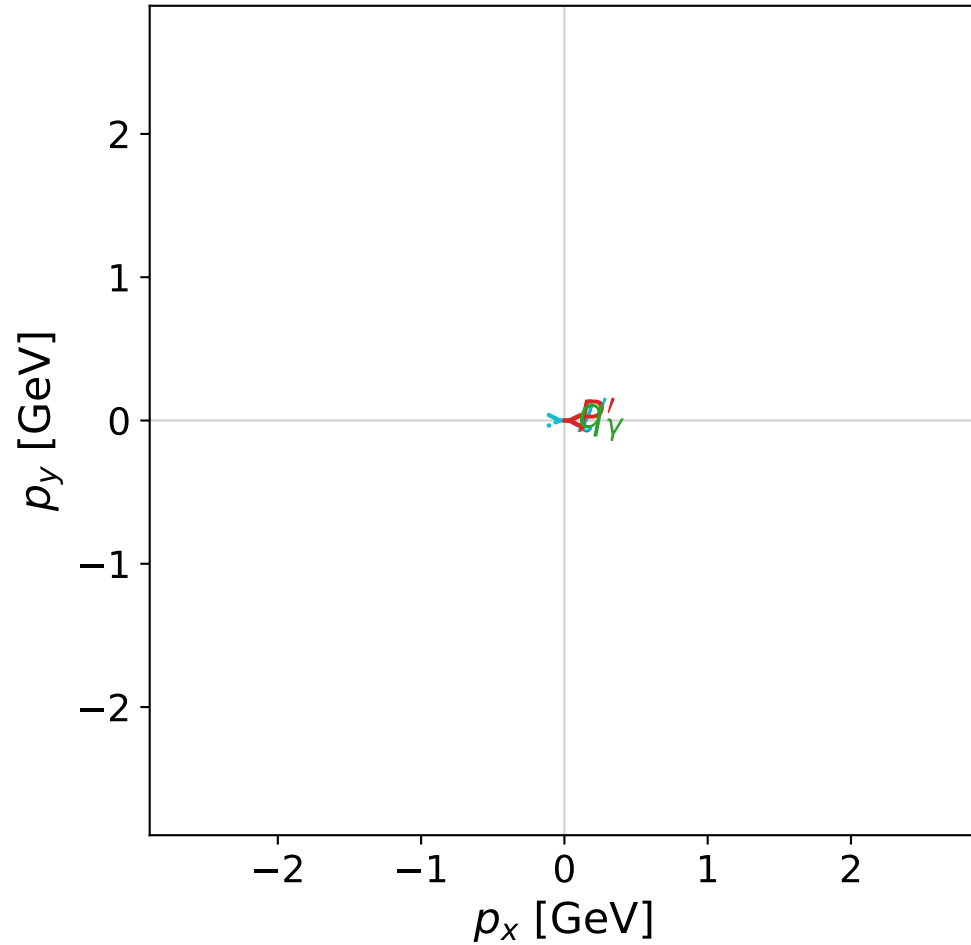


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

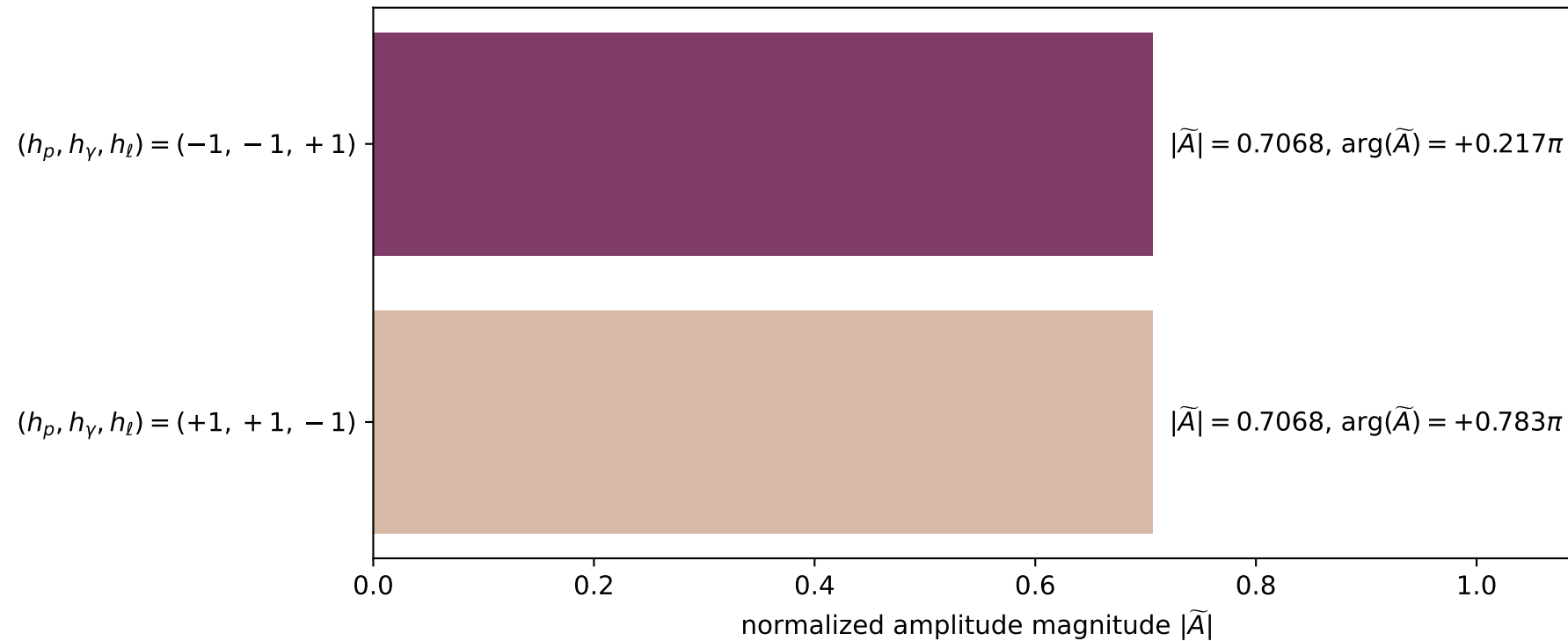


dghz_mixing_angles_polarization_03_local_minimum_156
 $d_{\text{GHZ}}=9.98211\text{e-}08$
 region: polarization_cluster_03_local_minimum_156
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0156
 lepton mixing angle: $\alpha_e=3.0075642$ rad
 incoming lepton: $-0.991032|+\rangle +0.133628|-\rangle$
 proton mixing angle: $\alpha_p=0.13401798$ rad
 incoming proton: $0.991033|+\rangle +0.133617|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.000632829$
 $\theta_{P'}=0.0587284$, $\theta_Y=3.14159$, $\phi_{P'}=3.12895$, $\phi_Y=3.82447$
 $E_Y=2.03132$, $\theta_{\ell'}=1.82912\text{e-}05$, $\phi_{\ell'}=6.27054$
 $Q^2=19.5922$, $x_B=0.837076$, $t=-3.09232$, $W^2=4.69316$, $y=0.97037$

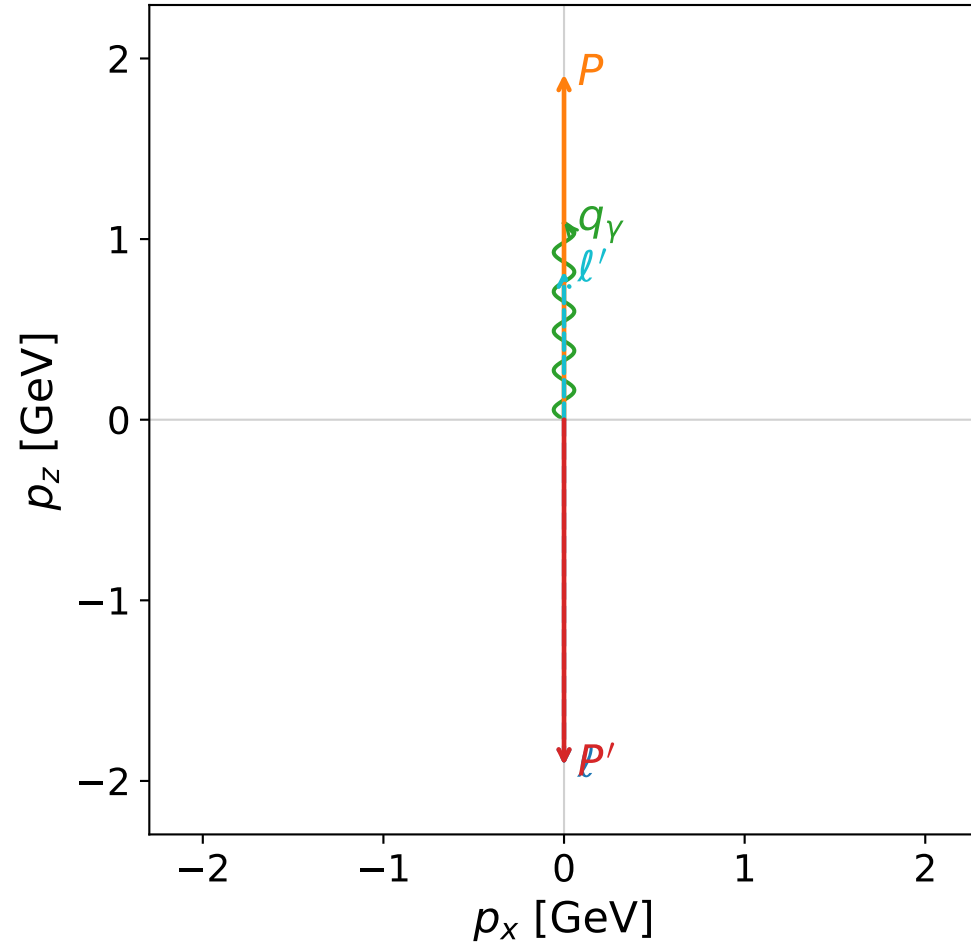
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0307$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0307$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0006$
 q_Y $E= 2.0313$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0313$

Leading normalized final-state amplitudes
 (N=2, retained=99.9%)

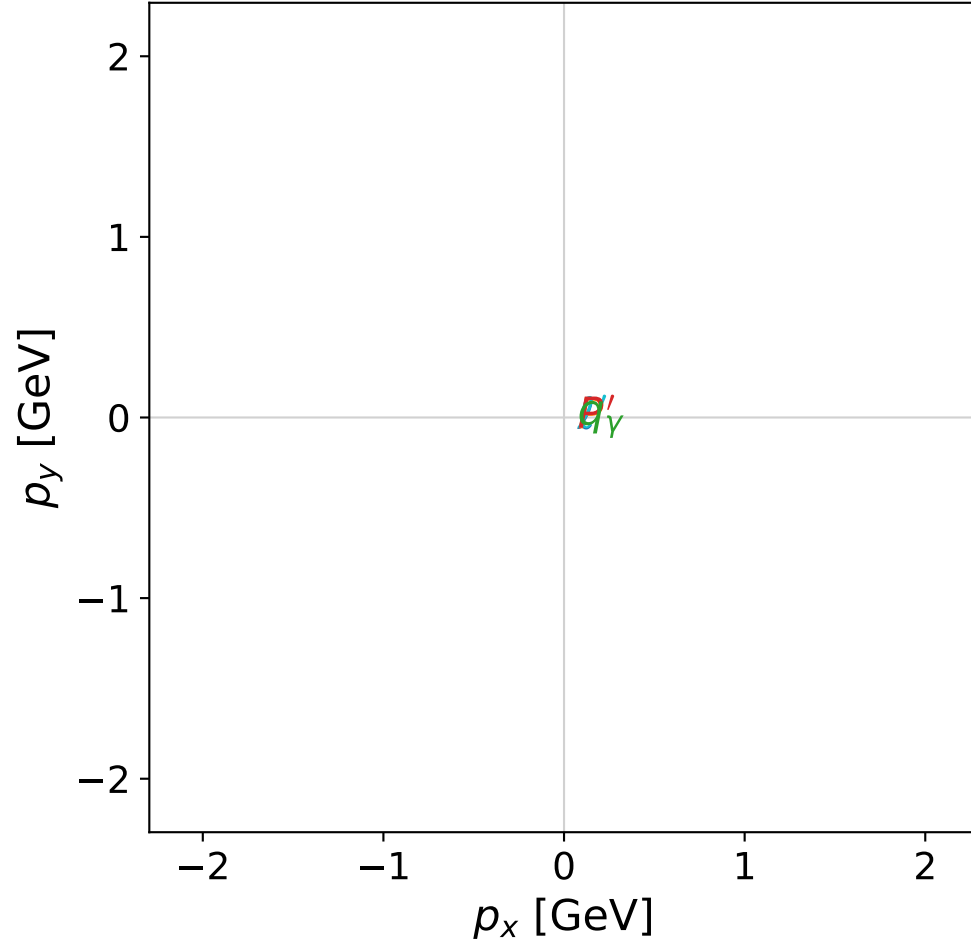


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



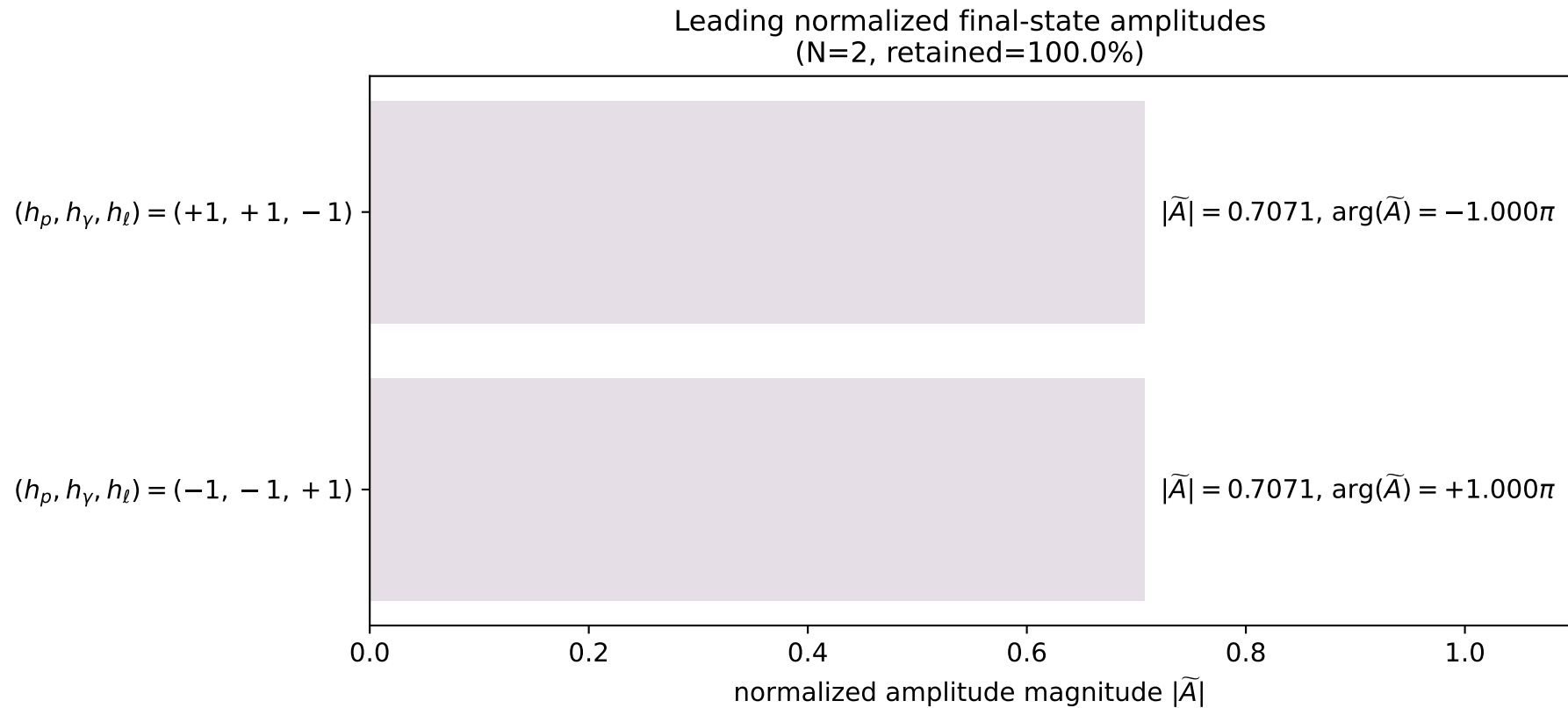
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_158
 $d_{\text{GHZ}}=1.03904\text{e-}07$
 region: polarization_cluster_03_local_minimum_158
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0158
 lepton mixing angle: $\alpha_e=2.0331734$ rad
 incoming lepton: $-0.446077|+\rangle +0.894995|-\rangle$
 proton mixing angle: $\alpha_p=0.46236846$ rad
 incoming proton: $0.894999|+\rangle +0.446069|-\rangle$

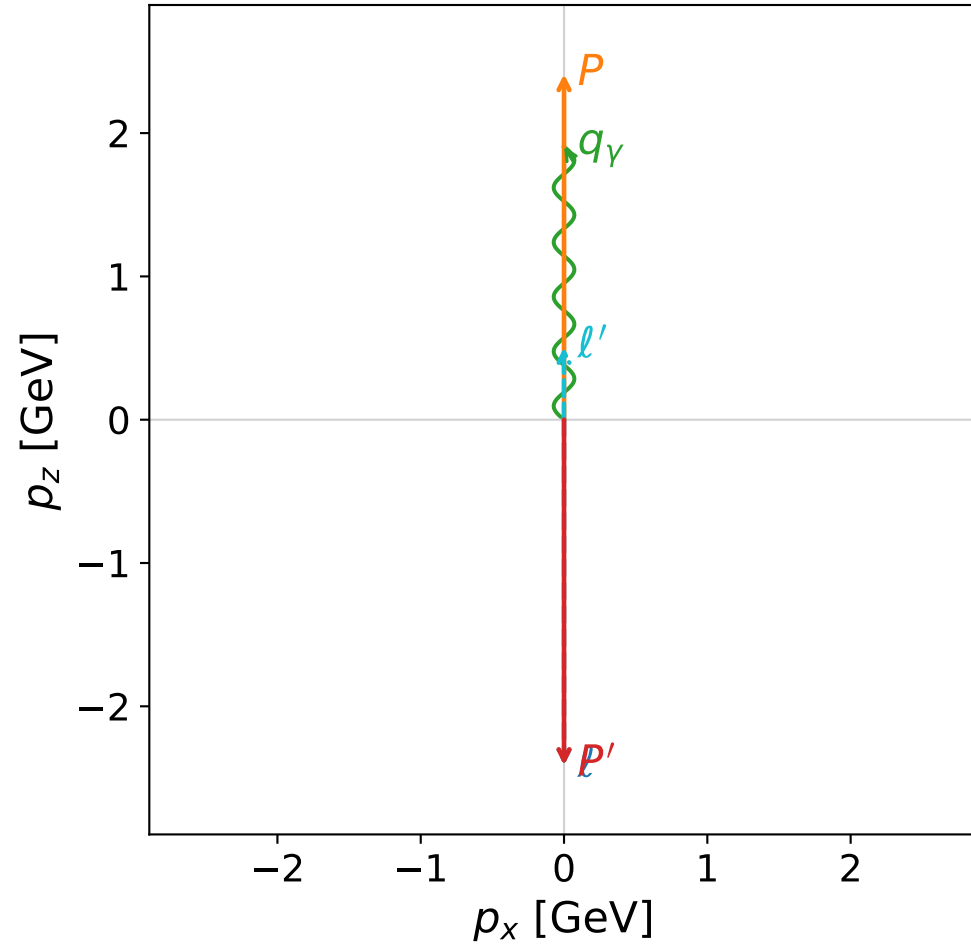
$s=16.3566$, $\sqrt{s}=4.04434$, $|\vec{P}|=1.91339$, $|\vec{P}'|=1.91339$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.40943$, $\phi_\gamma=3.29633$
 $E_\gamma=1.09092$, $\theta_{l'}=0$, $\phi_{l'}=5.55102$
 $Q^2=6.29483$, $x_B=0.416353$, $t=-14.6443$, $W^2=9.70397$, $y=0.976878$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.9134$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9134$
 P $E= 2.1309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9134$
 l' $E= 0.8225$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8225$
 P' $E= 2.1309$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9134$
 q_γ $E= 1.0909$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0909$

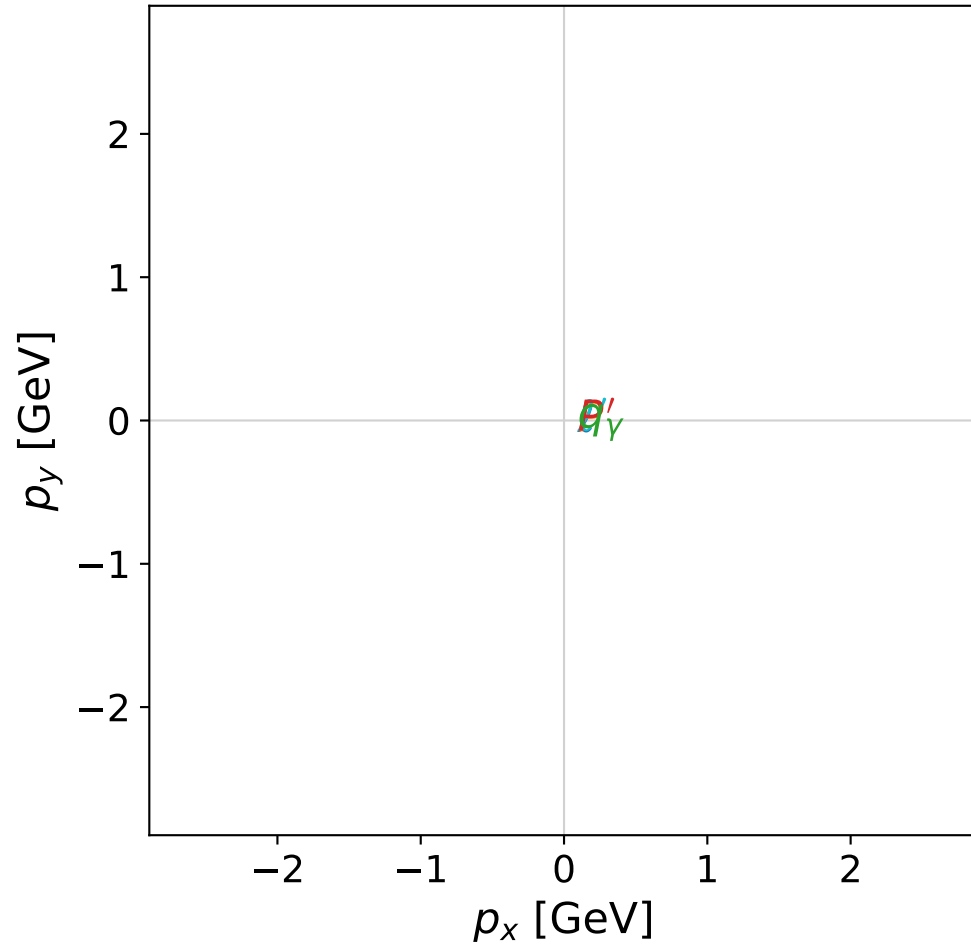


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

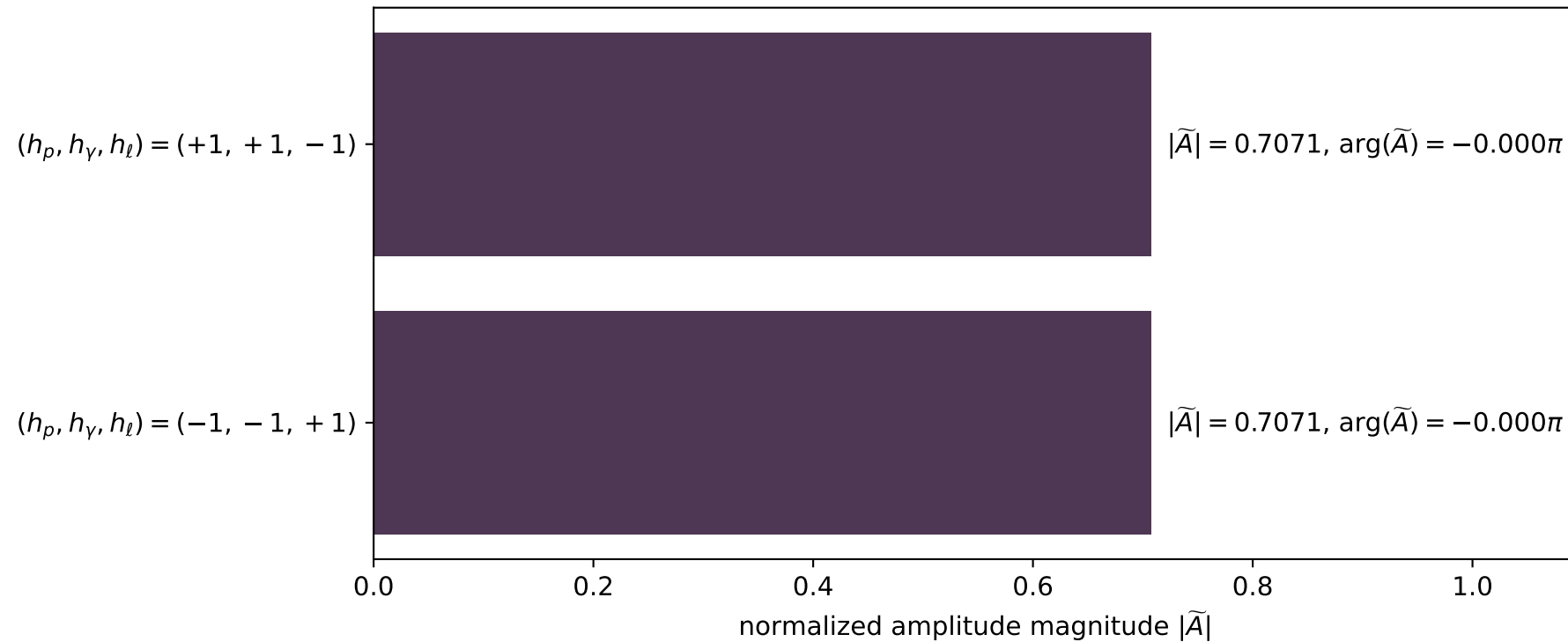


dghz_mixing_angles_polarization_03_local_minimum_161
 $d_{\{\text{rm GHZ}\}}=1.13122\text{e-}07$
 region: polarization_cluster_03_local_minimum_161
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0161
 lepton mixing angle: $\alpha_e=2.5224319$ rad
 incoming lepton: $-0.814366|+\rangle +0.580352|-\rangle$
 proton mixing angle: $\alpha_p=0.95162257$ rad
 incoming proton: $0.580362|+\rangle +0.814358|-\rangle$

$s=24.9928$, $\sqrt{s}=4.99928$, $|\vec{P}|=2.41164$, $|\vec{P}'|=2.41164$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=5.60969$, $\phi_Y=5.70666$
 $E_Y=1.90533$, $\theta_{l'}=0$, $\phi_{l'}=2.4681$
 $Q^2=4.88418$, $x_B=0.204062$, $t=-23.2641$, $W^2=19.9304$, $y=0.992609$

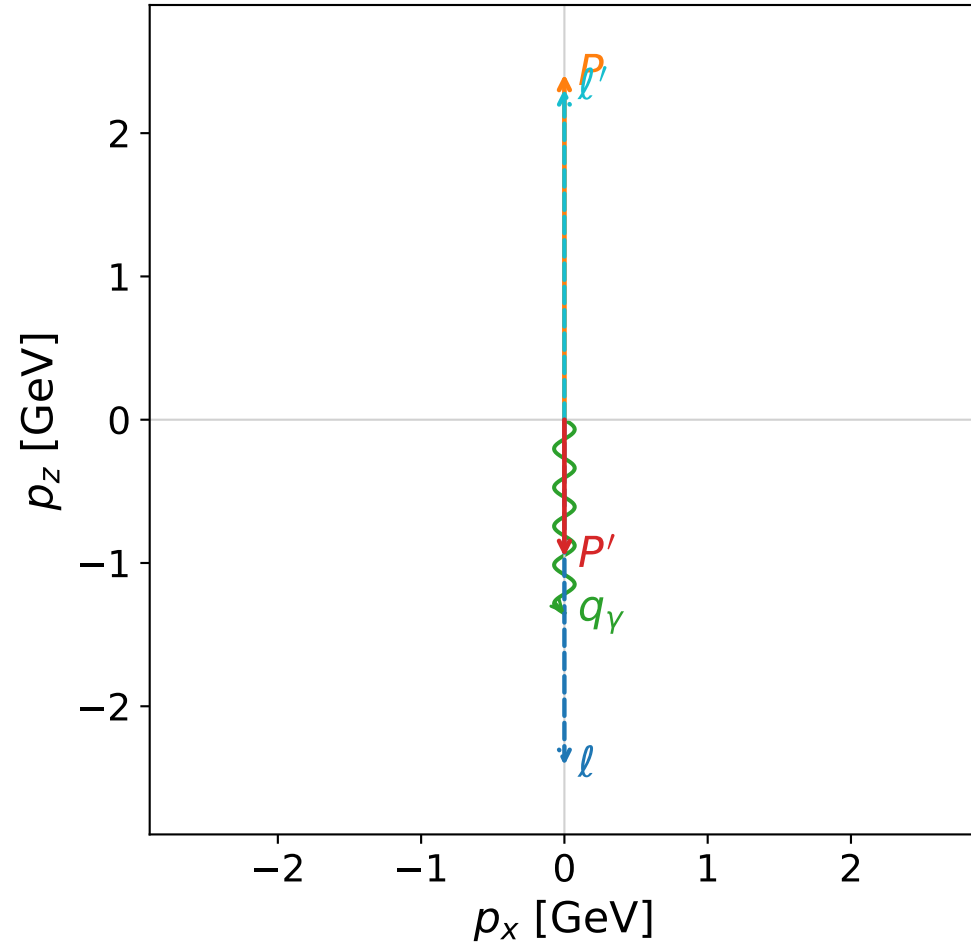
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4116$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4116$
 P $E= 2.5876$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4116$
 ℓ' $E= 0.5063$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.5063$
 P' $E= 2.5876$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4116$
 q_Y $E= 1.9053$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9053$

Leading normalized final-state amplitudes (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

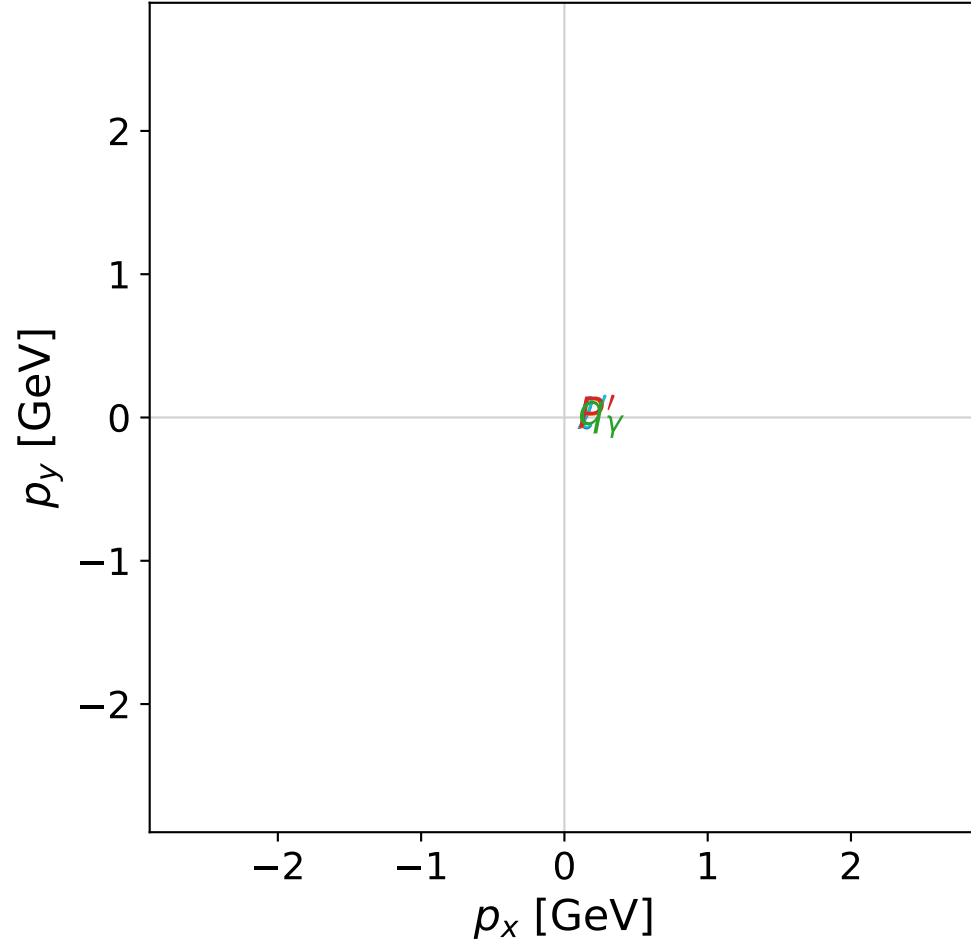


dghz_mixing_angles_polarization_03_local_minimum_174
 $d_{\text{GHZ}}=2.67083\text{e-}07$
 region: polarization_cluster_03_local_minimum_174
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0174
 lepton mixing angle: $\alpha_e=3.0976361$ rad
 incoming lepton: $-0.999034|+\rangle +0.0439424|-\rangle$
 proton mixing angle: $\alpha_p=0.043954198$ rad
 incoming proton: $0.999034|+\rangle +0.04394|-\rangle$

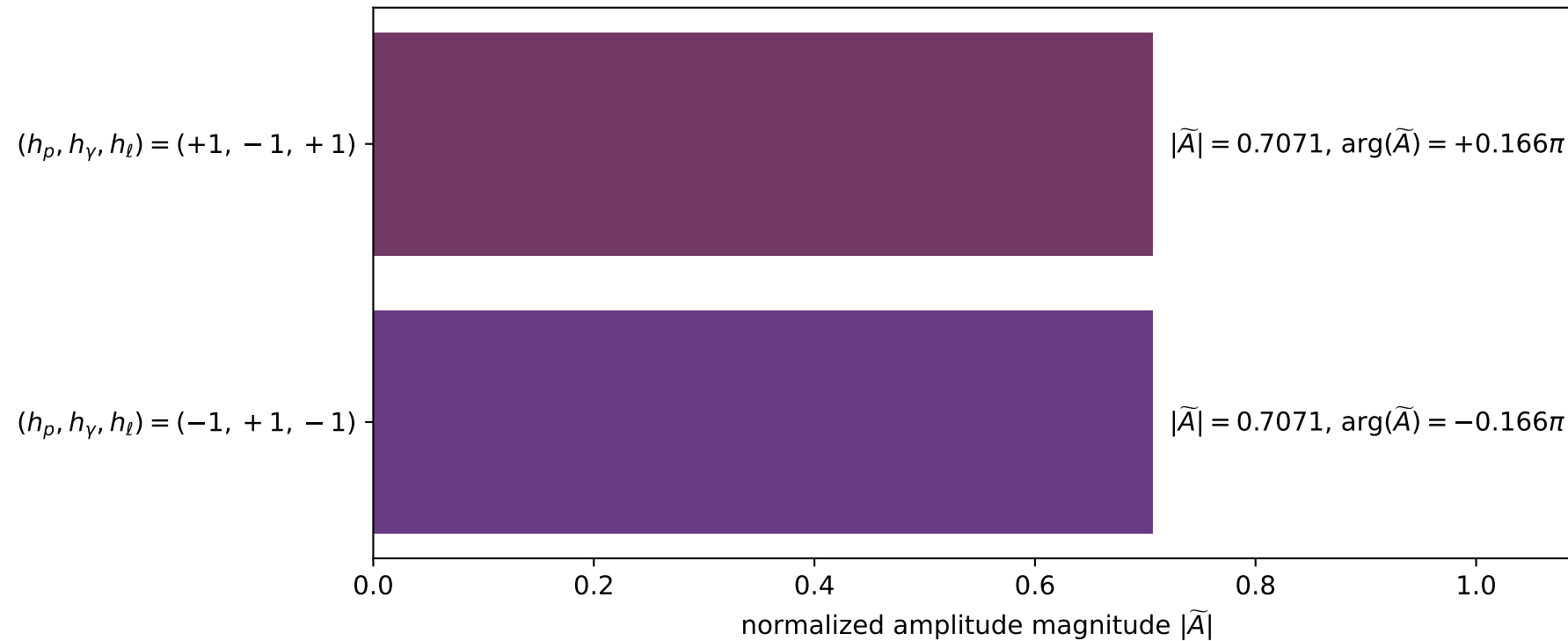
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.956774$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14456$, $\phi_\gamma=3.66323$
 $E_\gamma=1.35168$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.307656$
 $Q^2=22.2721$, $x_B=0.955566$, $t=-9.79096$, $W^2=1.9155$, $y=0.966317$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3085$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3085$
 P' $E= 1.3399$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.9568$
 q_γ $E= 1.3517$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.3517$

x--y plane

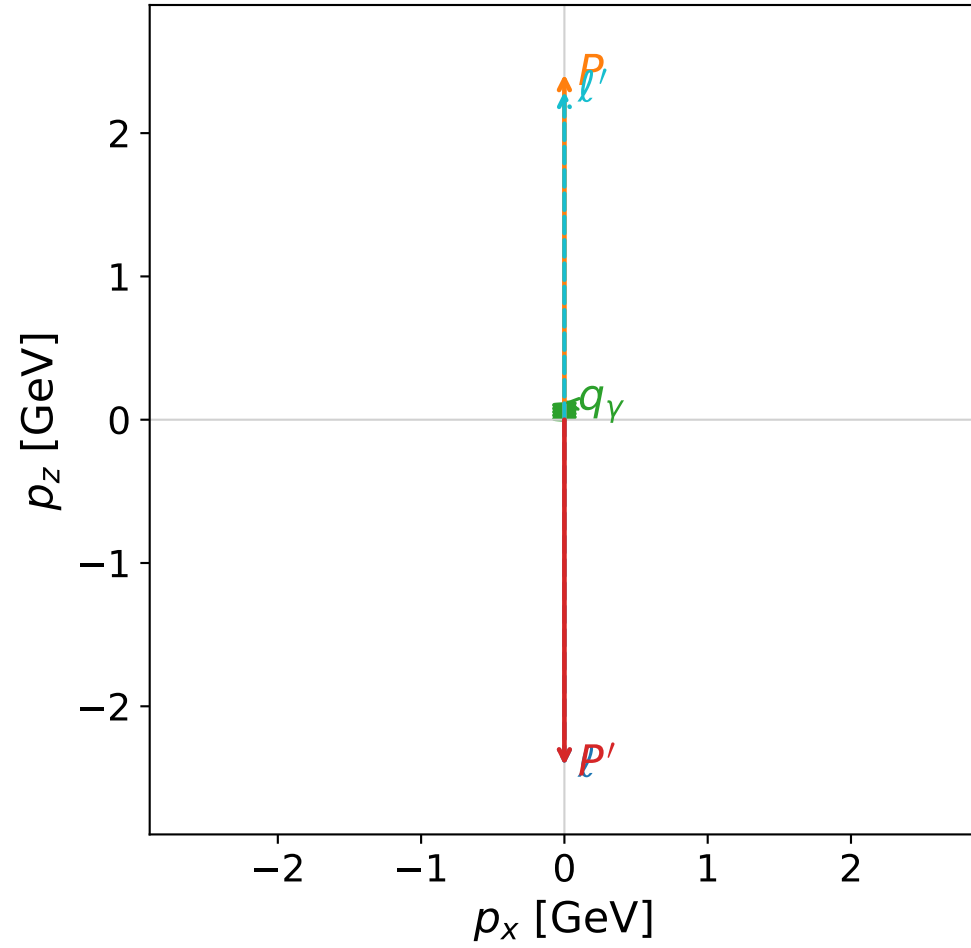


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

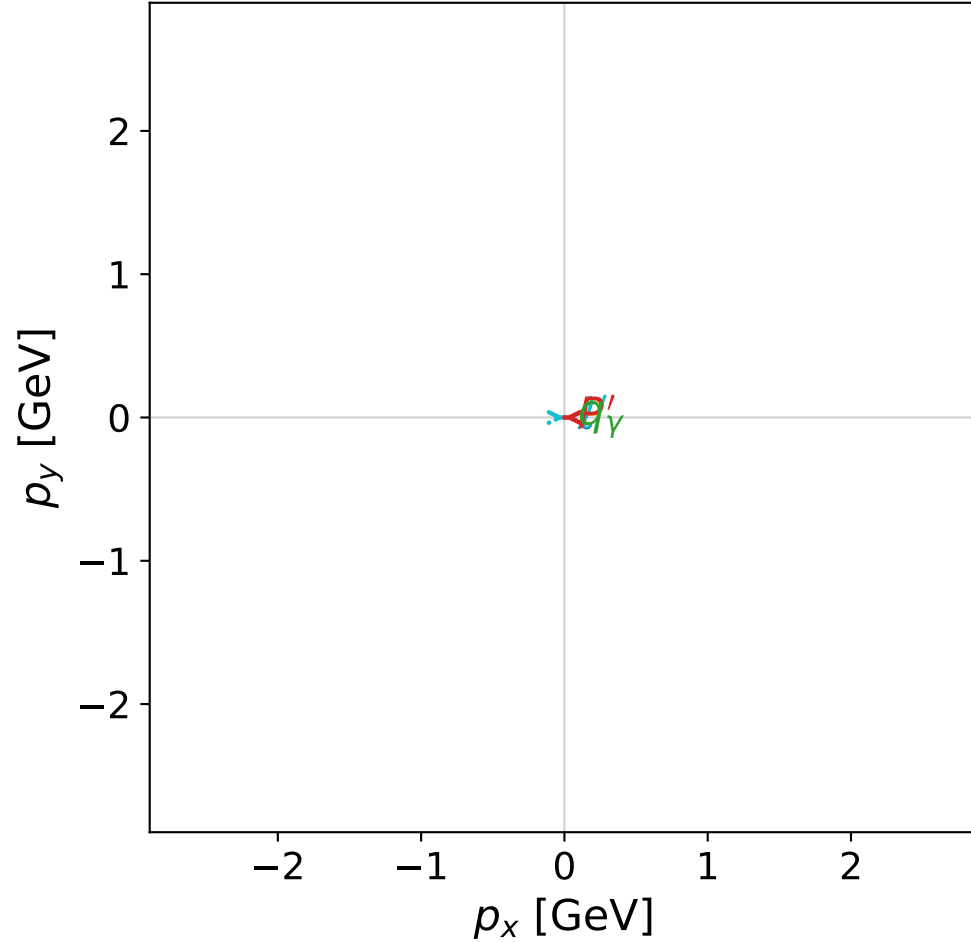


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

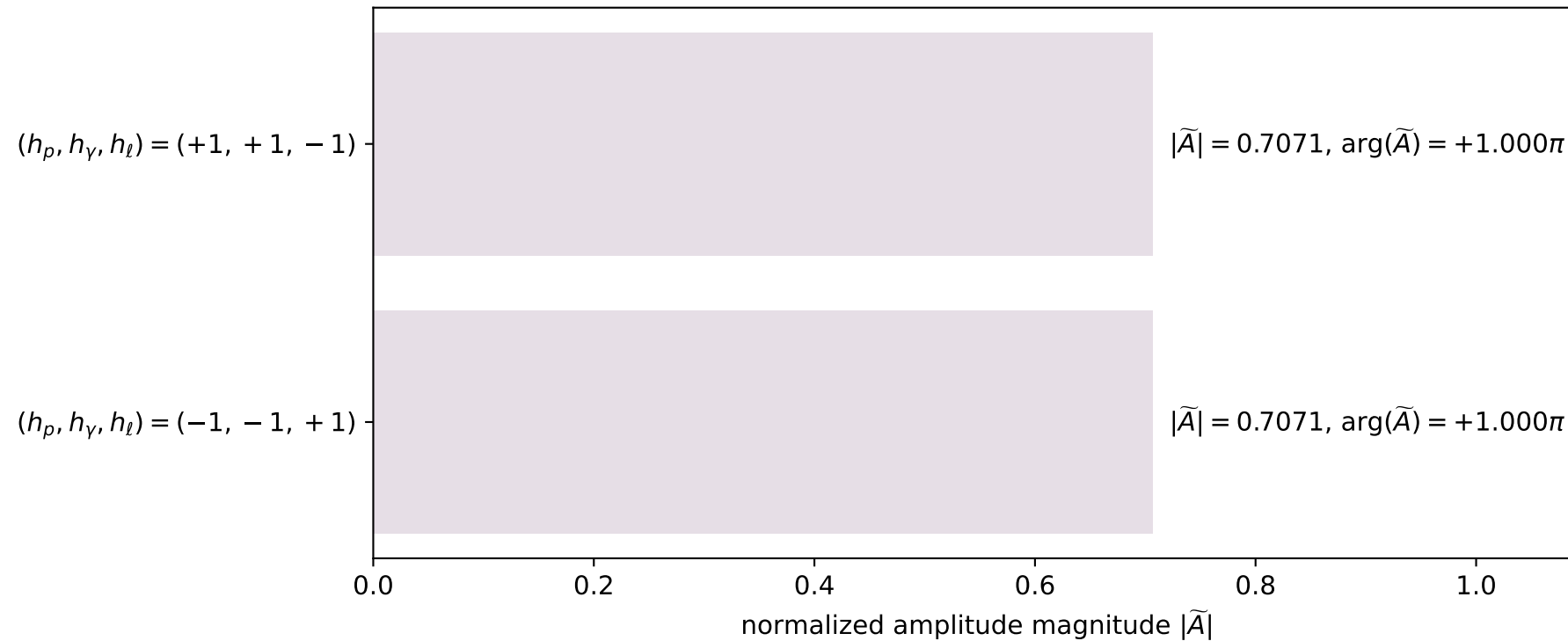


dghz_mixing_angles_polarization_03_local_minimum_178
 $d_{\text{GHZ}}=4.23054\text{e-}07$
 region: polarization_cluster_03_local_minimum_178
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0178
 lepton mixing angle: $\alpha_e=3.1136594$ rad
 incoming lepton: $-0.99961|+\rangle +0.0279296|-\rangle$
 proton mixing angle: $\alpha_p=1.5428613$ rad
 incoming proton: $0.0279314|+\rangle +0.99961|-\rangle$

$s=24.9961$, $\sqrt{s}=4.99961$, $|\vec{P}|=2.41181$, $|\vec{P}'|=2.41181$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.46255$, $\phi_\gamma=3.37801$
 $E_\gamma=0.120586$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.320954$
 $Q^2=22.104$, $x_B=0.948272$, $t=-23.2674$, $W^2=2.08561$, $y=0.966561$

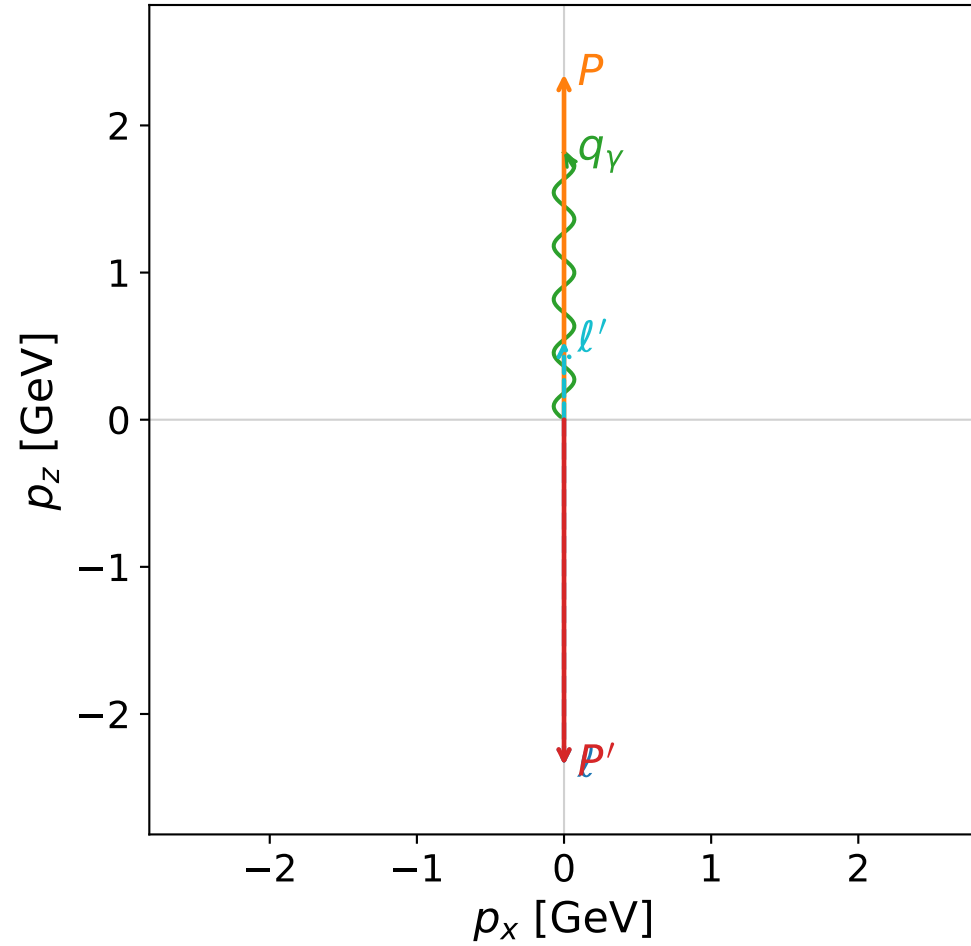
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.2912$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2912$
 P' $E= 2.5878$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4118$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1206$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

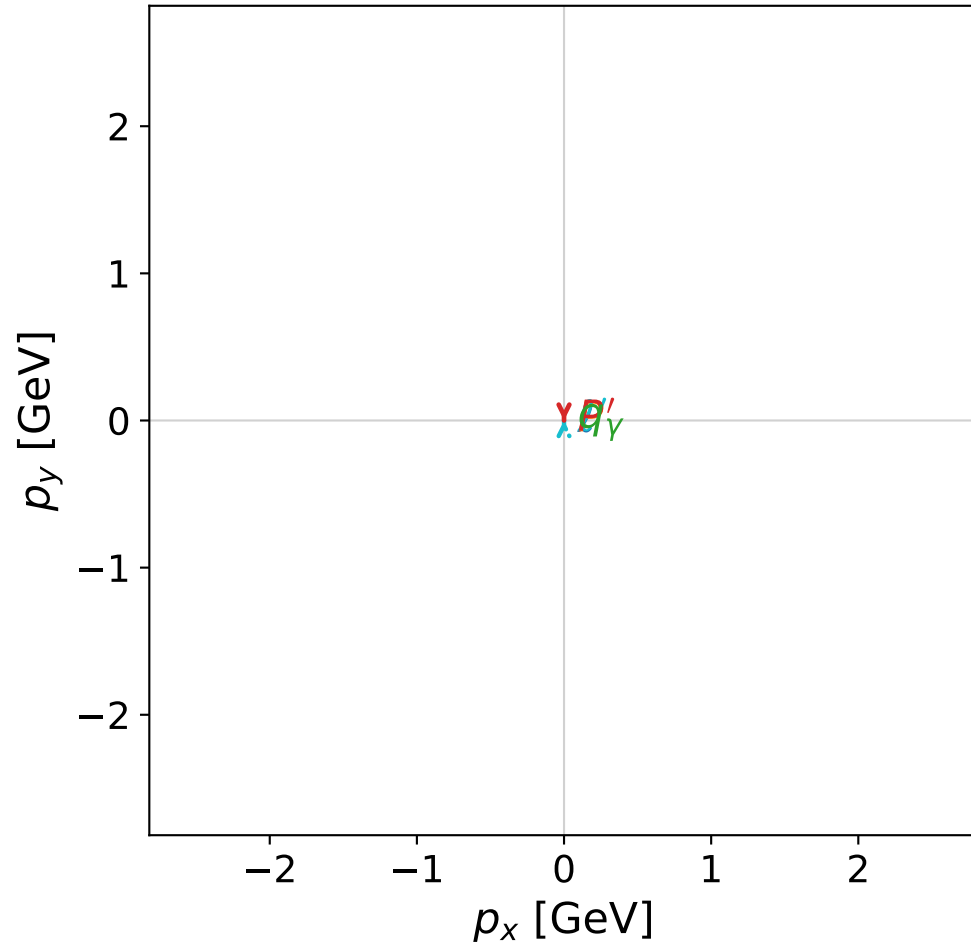


dghz_mixing_angles_polarization_03_local_minimum_181
 $d_{\{\text{rm GHZ}\}}=5.4582\text{e-}07$
 region: polarization_cluster_03_local_minimum_181
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0181
 lepton mixing angle: $\alpha_e=3.0452158$ rad
 incoming lepton: $-0.995359|+\rangle +0.0962277|-\rangle$
 proton mixing angle: $\alpha_p=1.4744086$ rad
 incoming proton: $0.0962386|+\rangle +0.995358|-\rangle$

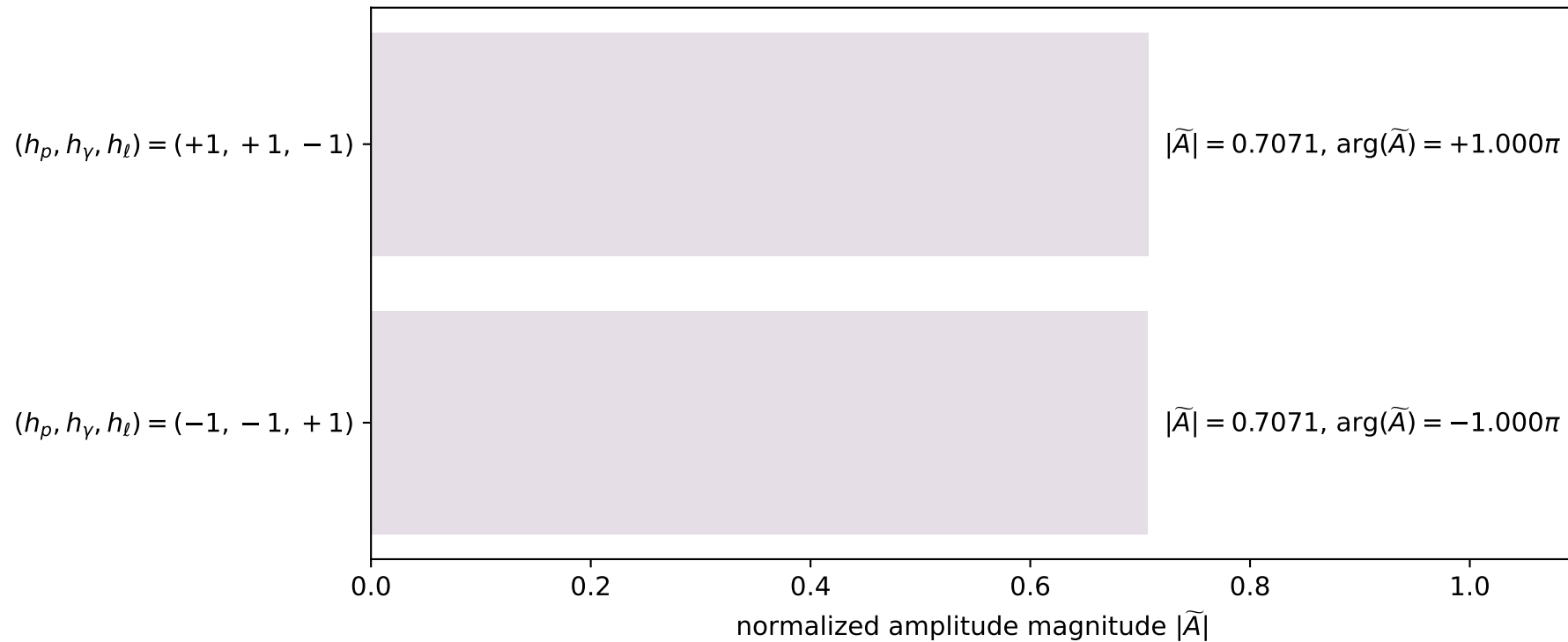
$s=23.7939$, $\sqrt{s}=4.8779$, $|\vec{P}|=2.34876$, $|\vec{P}'|=2.34876$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.56992$, $\phi_\gamma=2.75915$
 $E_\gamma=1.81698$, $\theta_{l'}=0$, $\phi_{l'}=1.42833$
 $Q^2=4.99613$, $x_B=0.219879$, $t=-22.0667$, $W^2=18.6059$, $y=0.991628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3488$
 P $E= 2.5291$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3488$
 l' $E= 0.5318$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5318$
 P' $E= 2.5291$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.3488$
 q_γ $E= 1.8170$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8170$

x--y plane

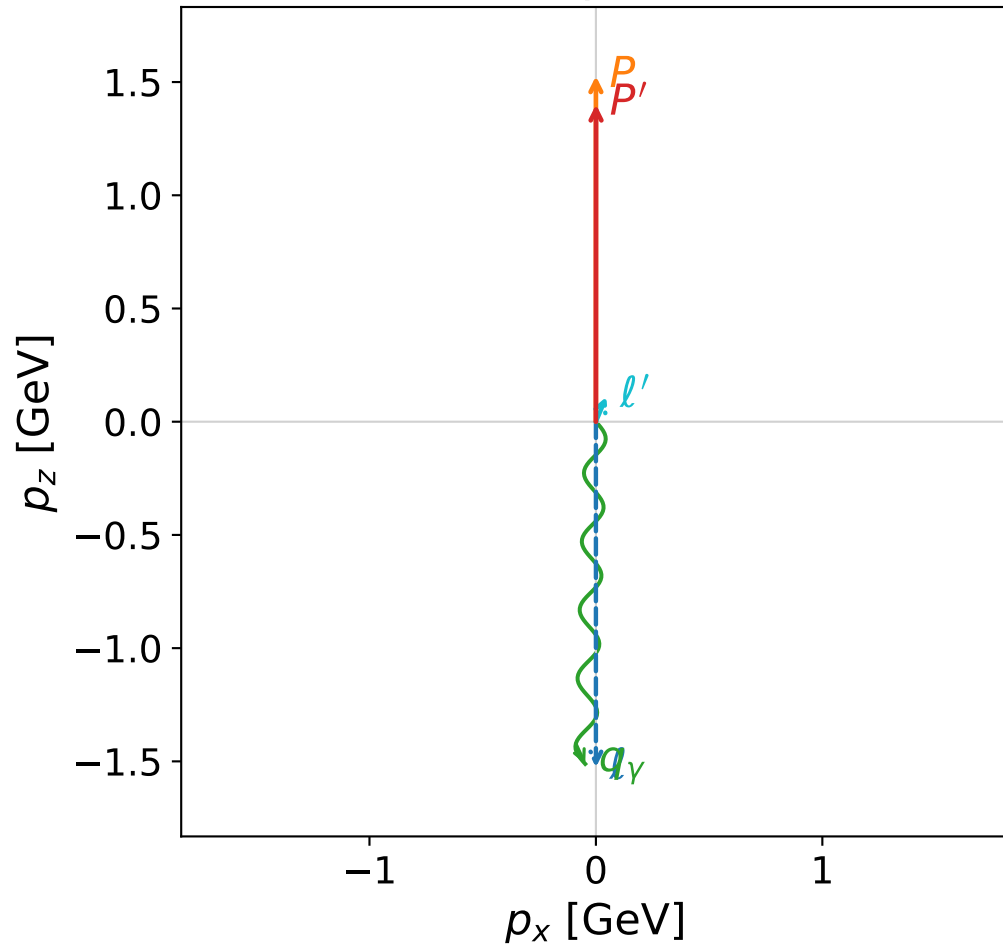


Leading normalized final-state amplitudes (N=2, retained=100.0%)

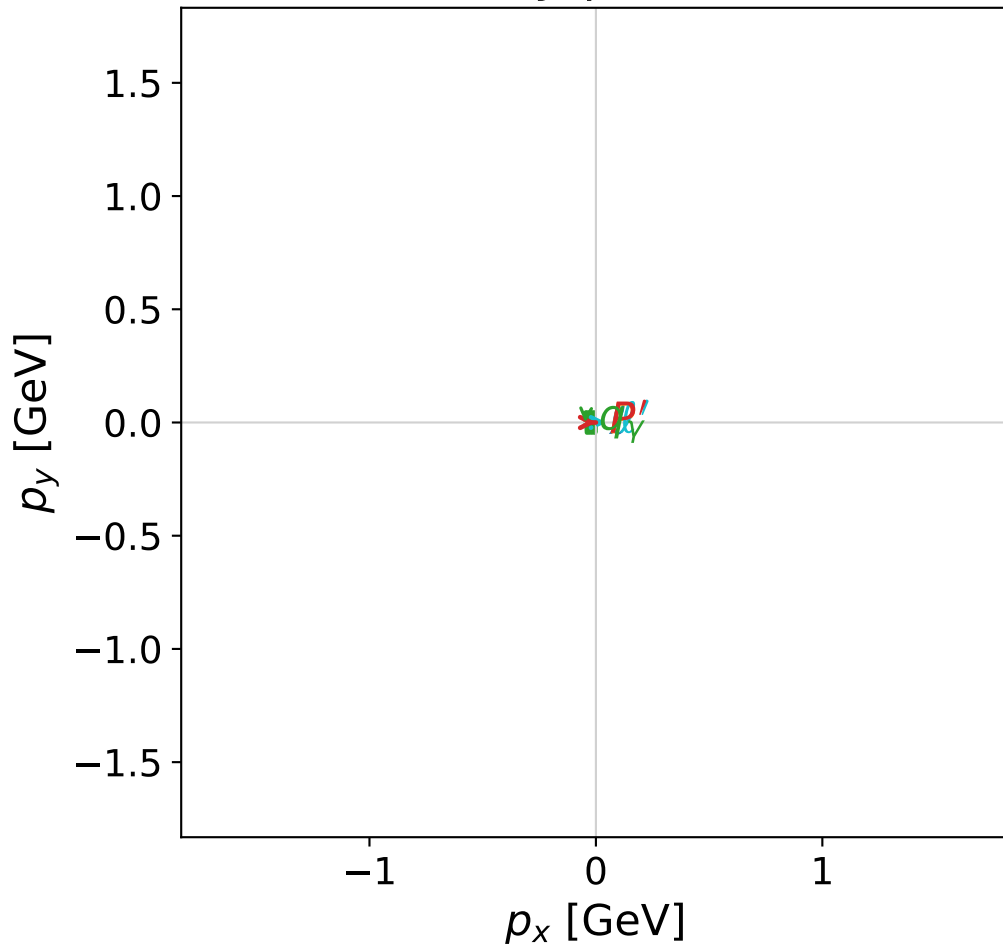


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



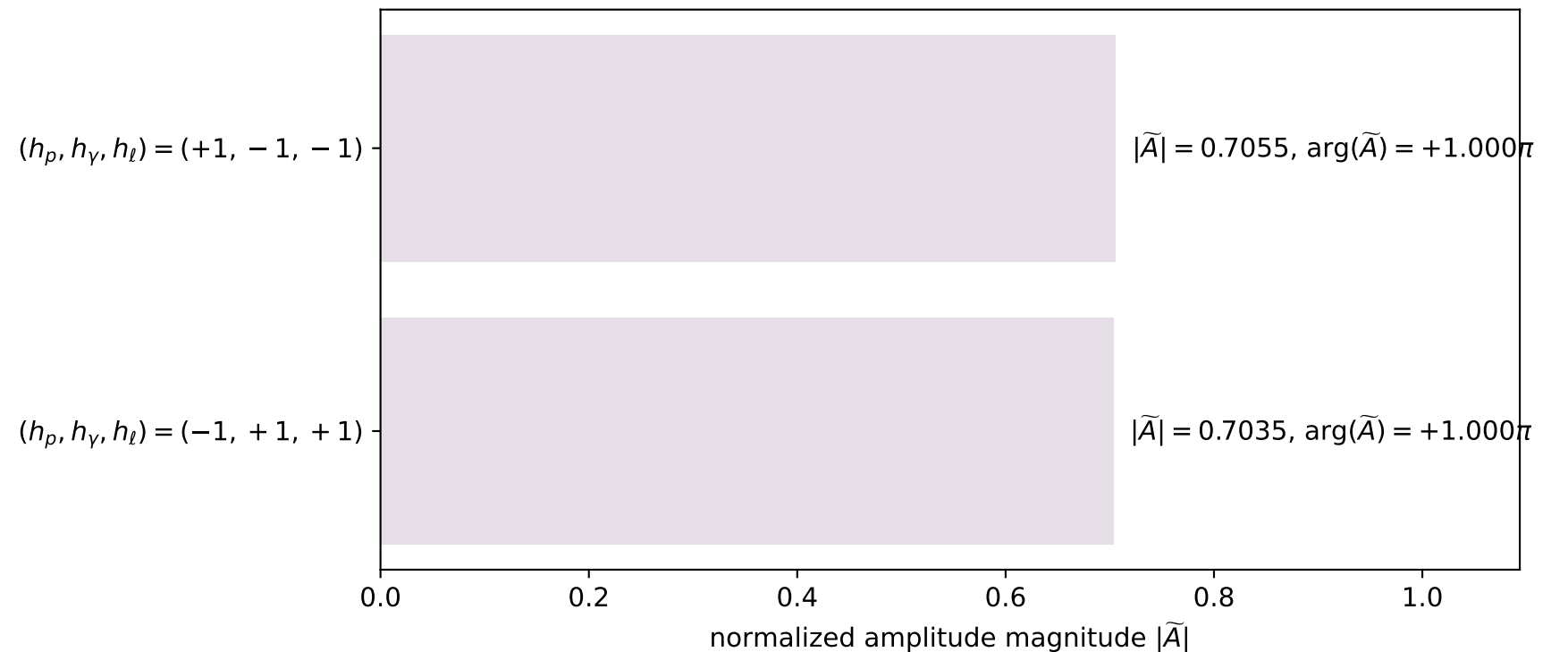
dghz_mixing_angles_polarization_03_local_minimum_188
 $d_{\{\text{rm GHZ}\}}=1.11719\text{e-}05$
 region: polarization_cluster_03_local_minimum_188
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0188
 lepton mixing angle: $\alpha_e=2.3764974$ rad
 incoming lepton: $-0.721316|+\rangle +0.692606|-\rangle$
 proton mixing angle: $\alpha_p=0.84717575$ rad
 incoming proton: $0.662102|+\rangle +0.749413|-\rangle$

$s=11.0066$, $\sqrt{s}=3.31762$, $|\vec{P}|=1.52621$, $|\vec{P}'|=1.40088$
 $\theta_{p'}=0.000417261$, $\theta_\gamma=3.11038$, $\phi_{p'}=6.35544\text{e-}05$, $\phi_\gamma=3.14165$
 $E_\gamma=1.51194$, $\theta_{\ell'}=0.399604$, $\phi_{\ell'}=6.18232\text{e-}05$
 $Q^2=0.702332$, $x_B=0.0699923$, $t=-0.00457805$, $W^2=10.2119$, $y=0.990883$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

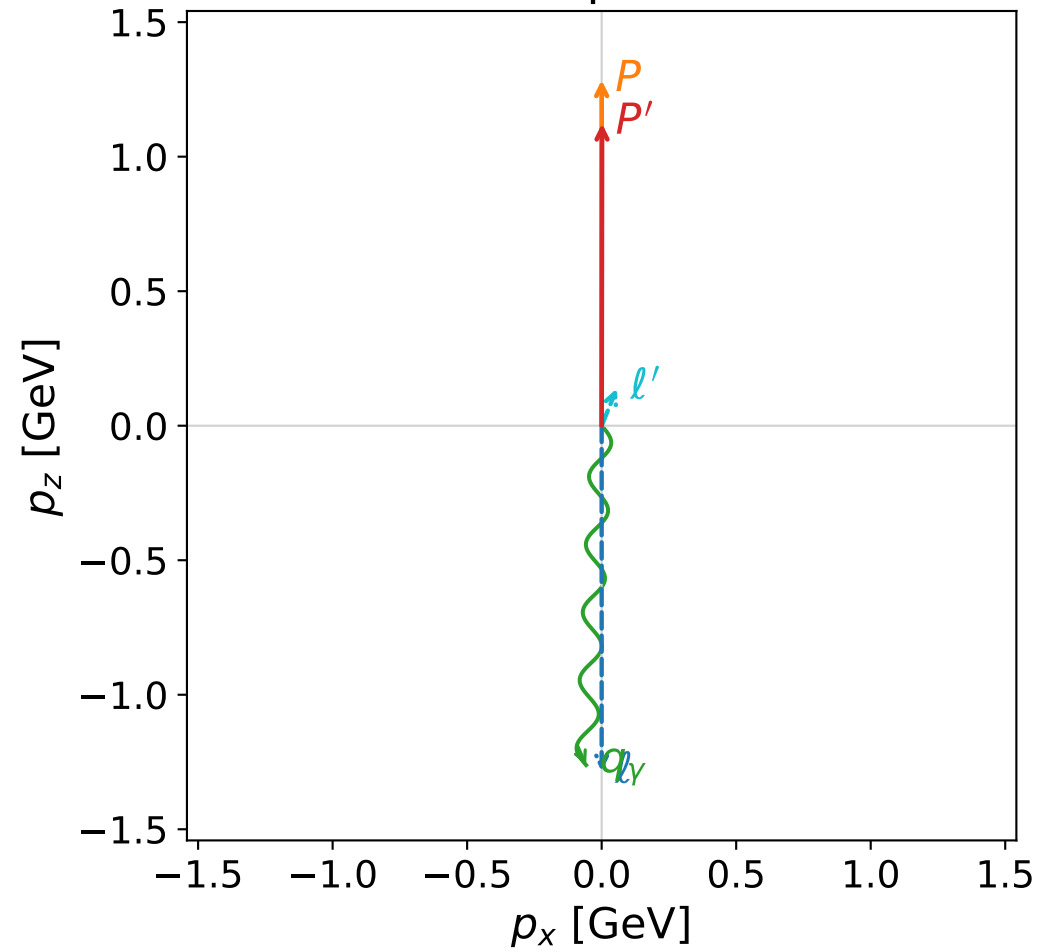
Particle	E	p_x	p_y	p_z
ℓ	1.5262	0.0000	0.0000	-1.5262
P	1.7914	0.0000	0.0000	1.5262
ℓ'	0.1198	0.0466	0.0000	0.1103
P'	1.6859	0.0006	0.0000	1.4009
q_γ	1.5119	-0.0472	-0.0000	-1.5112

Leading normalized final-state amplitudes
(N=2, retained=99.3%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

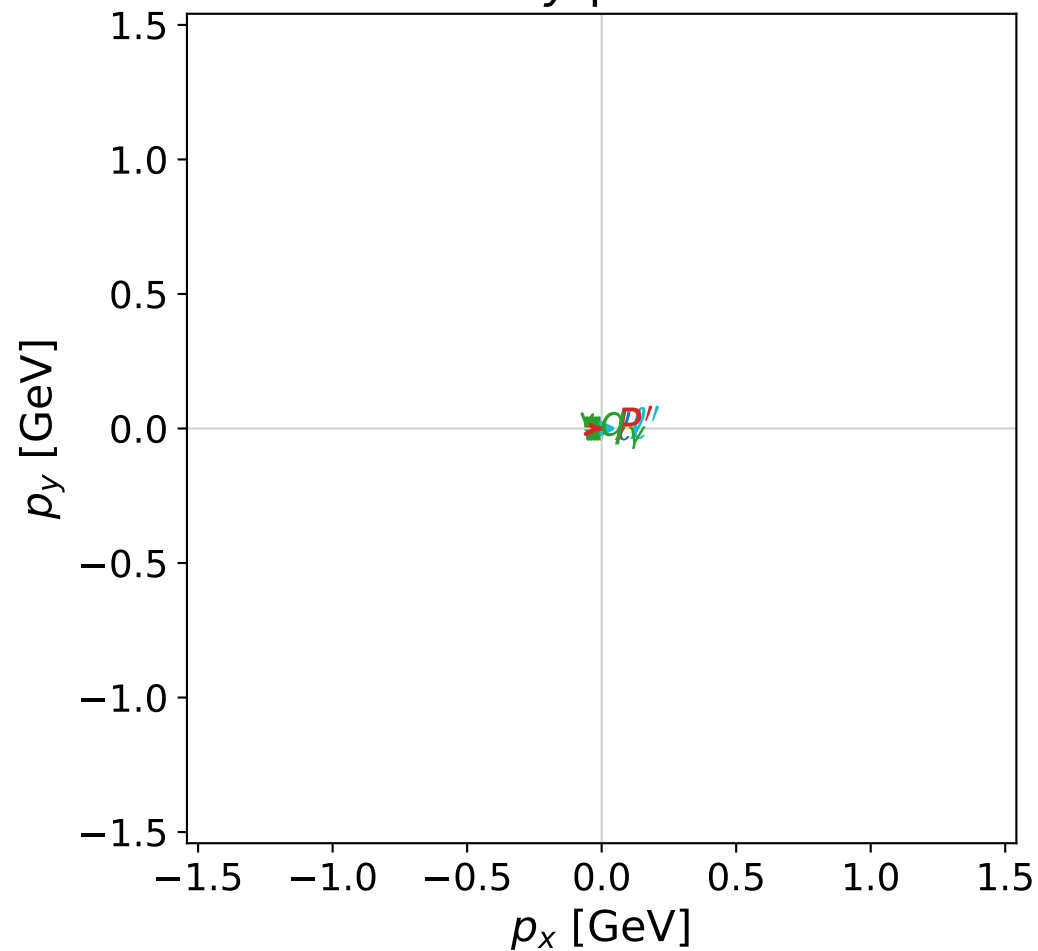


dghz_mixing_angles_polarization_03_local_minimum_189
 $d_{\{\rm GHZ\}}=1.25241\text{e-}05$
 region: polarization_cluster_03_local_minimum_189
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0189
 lepton mixing angle: $\alpha_e=2.4349155$ rad
 incoming lepton: $-0.760524|+\rangle +0.64931|-\rangle$
 proton mixing angle: $\alpha_p=0.90852471$ rad
 incoming proton: $0.61491|+\rangle +0.788597|-\rangle$

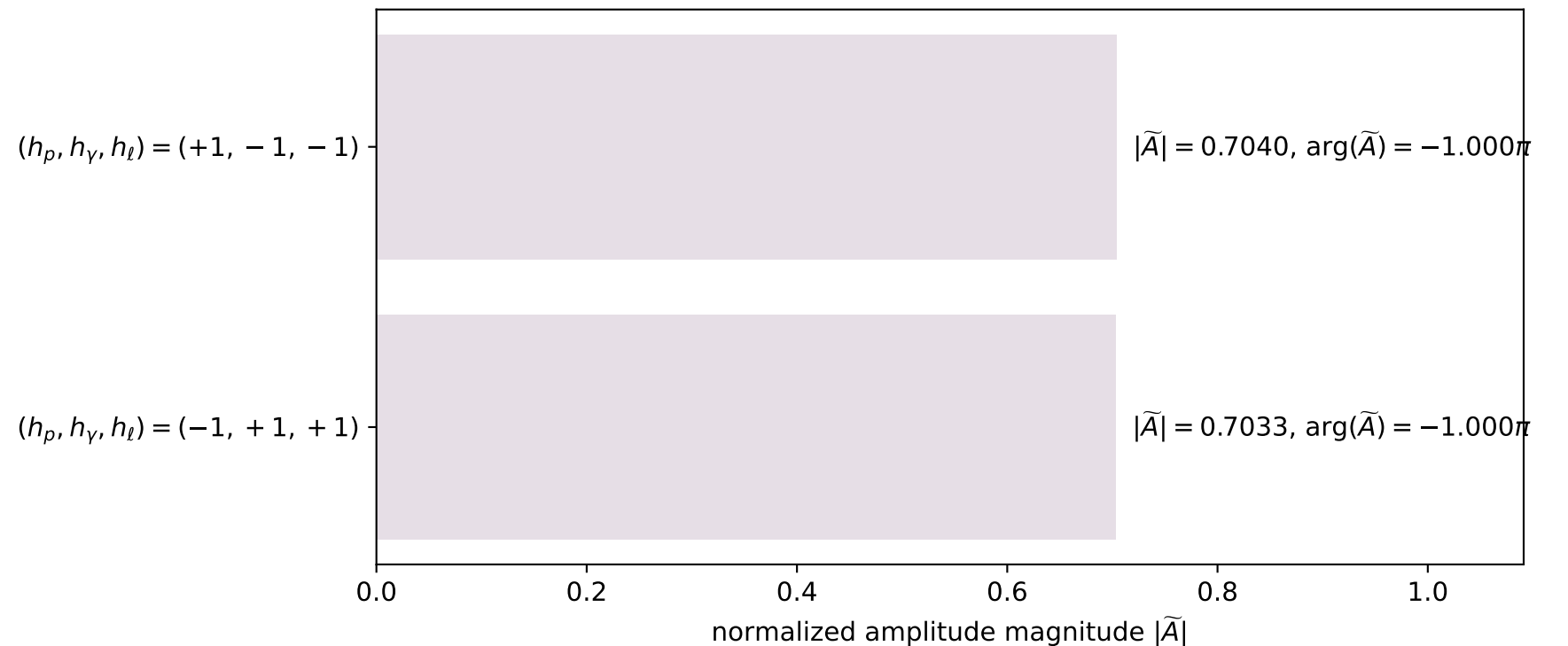
$s=8.26616$, $\sqrt{s}=2.87509$, $|\vec{P}|=1.28454$, $|\vec{P}'|=1.12322$
 $\theta_{p'}=0.00126895$, $\theta_Y=3.09582$, $\phi_{p'}=6.28313$, $\phi_Y=3.14157$
 $E_Y=1.2626$, $\theta_{l'}=0.38751$, $\phi_{l'}=6.28316$
 $Q^2=0.737755$, $x_B=0.101526$, $t=-0.00984887$, $W^2=7.40874$, $y=0.983798$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.2845$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2845$
 P $E= 1.5906$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2845$
 l' $E= 0.1491$ $p_x= 0.0563$ $p_y= -0.0000$ $p_z= 0.1381$
 P' $E= 1.4634$ $p_x= 0.0014$ $p_y= -0.0000$ $p_z= 1.1232$
 q_Y $E= 1.2626$ $p_x= -0.0578$ $p_y= 0.0000$ $p_z= -1.2613$

x--y plane

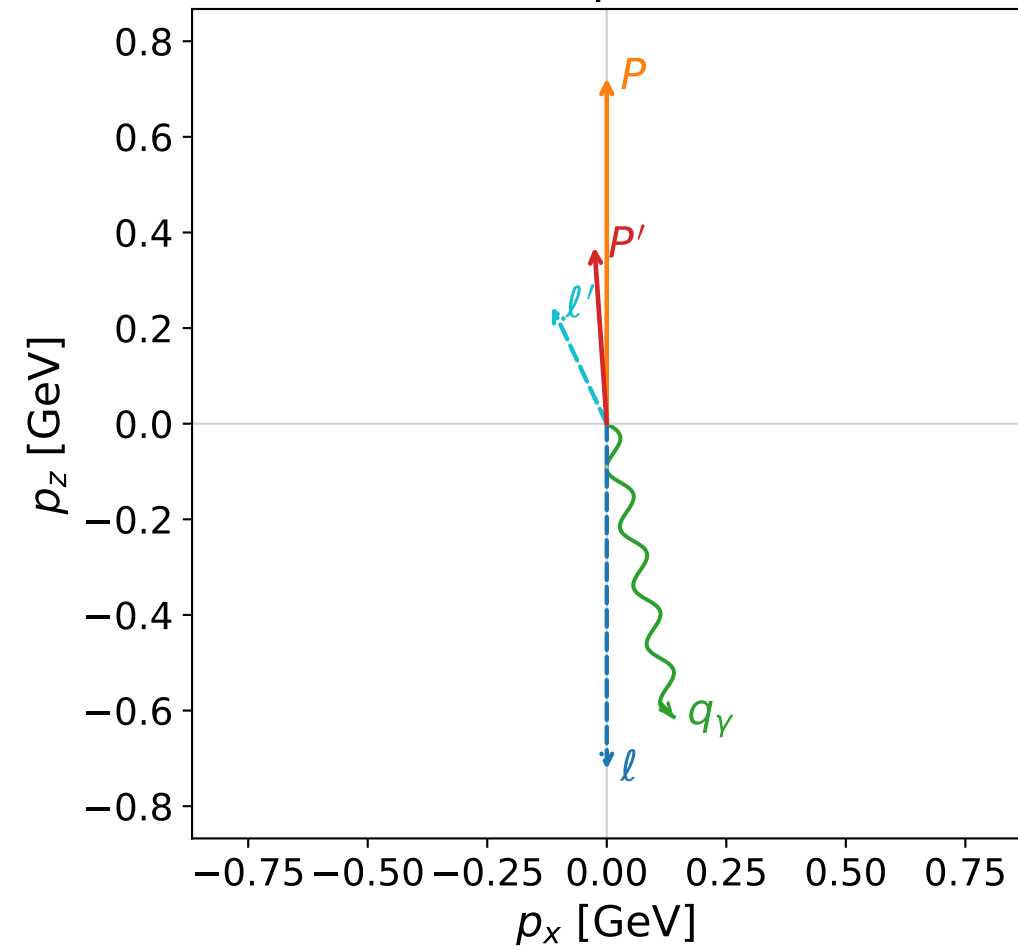


Leading normalized final-state amplitudes
(N=2, retained=99.0%)

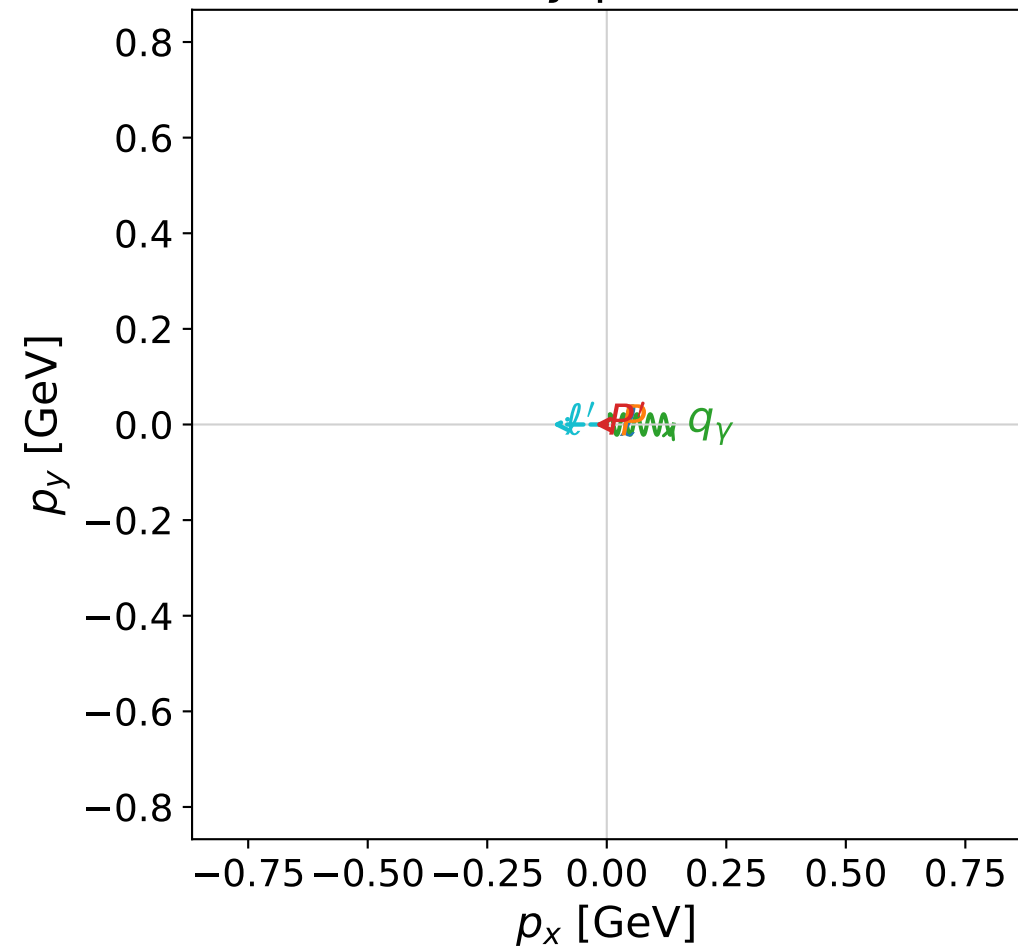


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_204

$d_{\{\text{rm GHZ}\}}=2.47799\text{e-}05$

region: polarization_cluster_03_local_minimum_204

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0204

lepton mixing angle: $\alpha_e=2.0175873$ rad

incoming lepton: $-0.432074|+\rangle +0.901838|-\rangle$

proton mixing angle: $\alpha_p=0.38602922$ rad

incoming proton: $0.926411|+\rangle +0.376513|-\rangle$

$s=3.63688$, $\sqrt{s}=1.90706$, $|\vec{P}|=0.72285$, $|\vec{P}'|=0.370003$

$\theta_{P'}=0.0696716$, $\theta_Y=2.91723$, $\phi_{P'}=3.14156$, $\phi_Y=0$

$E_Y=0.629105$, $\theta_{\ell'}=0.437411$, $\phi_{\ell'}=3.1416$

$Q^2=0.742871$, $x_B=0.300568$, $t=-0.0948674$, $W^2=2.60853$, $y=0.896454$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.7228$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7228$

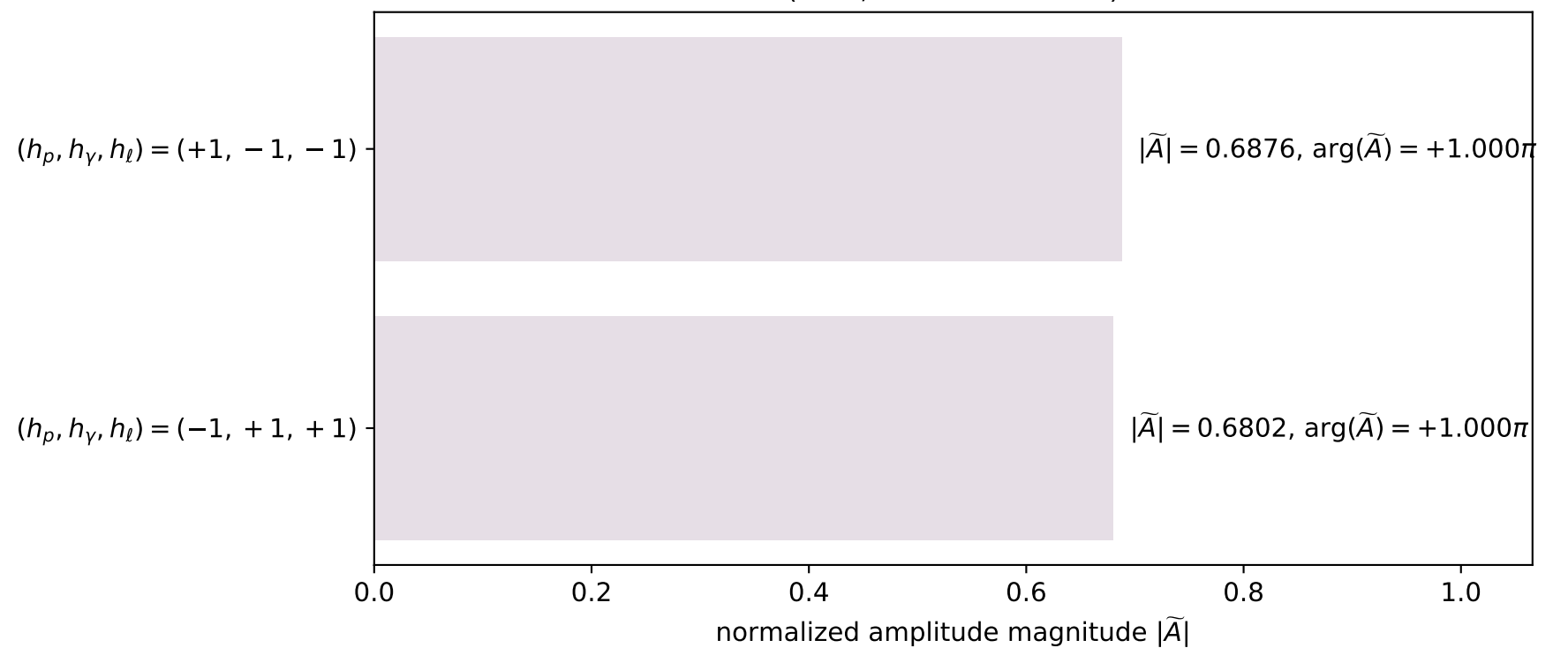
P $E= 1.1842$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7228$

ℓ' $E= 0.2696$ $p_x= -0.1142$ $p_y= -0.0000$ $p_z= 0.2442$

P' $E= 1.0083$ $p_x= -0.0258$ $p_y= 0.0000$ $p_z= 0.3691$

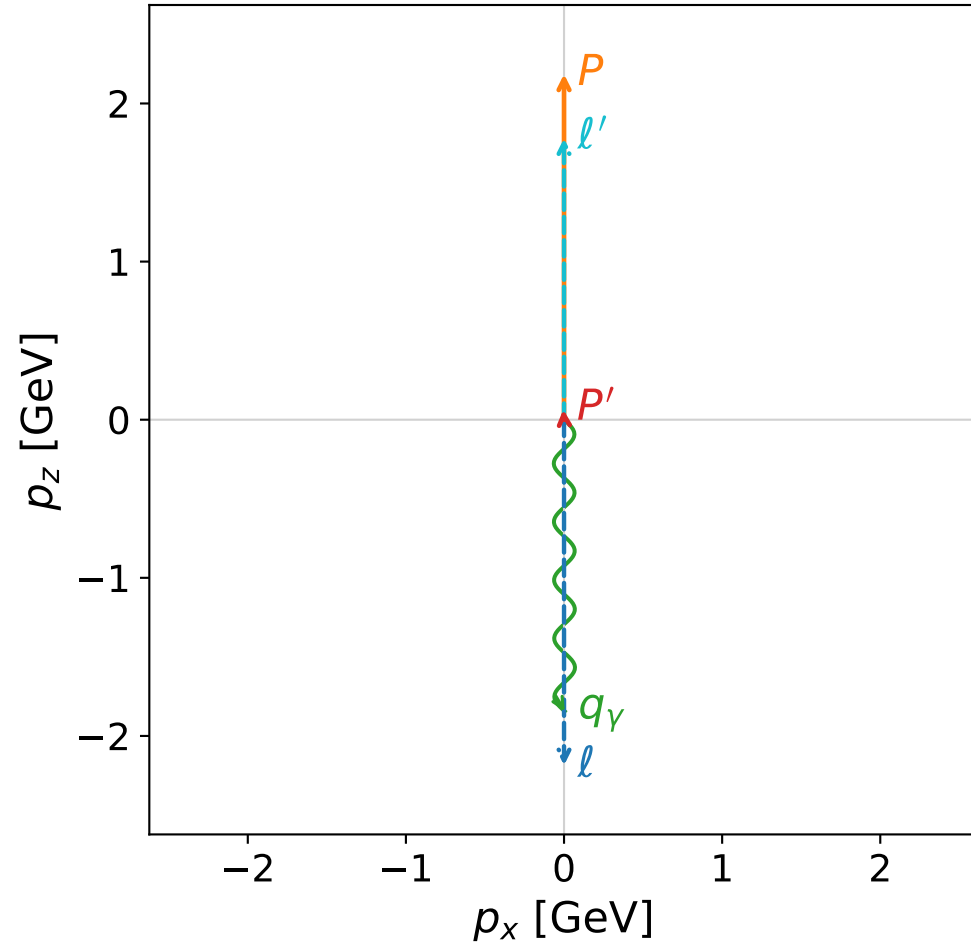
q_Y $E= 0.6291$ $p_x= 0.1400$ $p_y= 0.0000$ $p_z= -0.6133$

Leading normalized final-state amplitudes
(N=2, retained=93.6%)

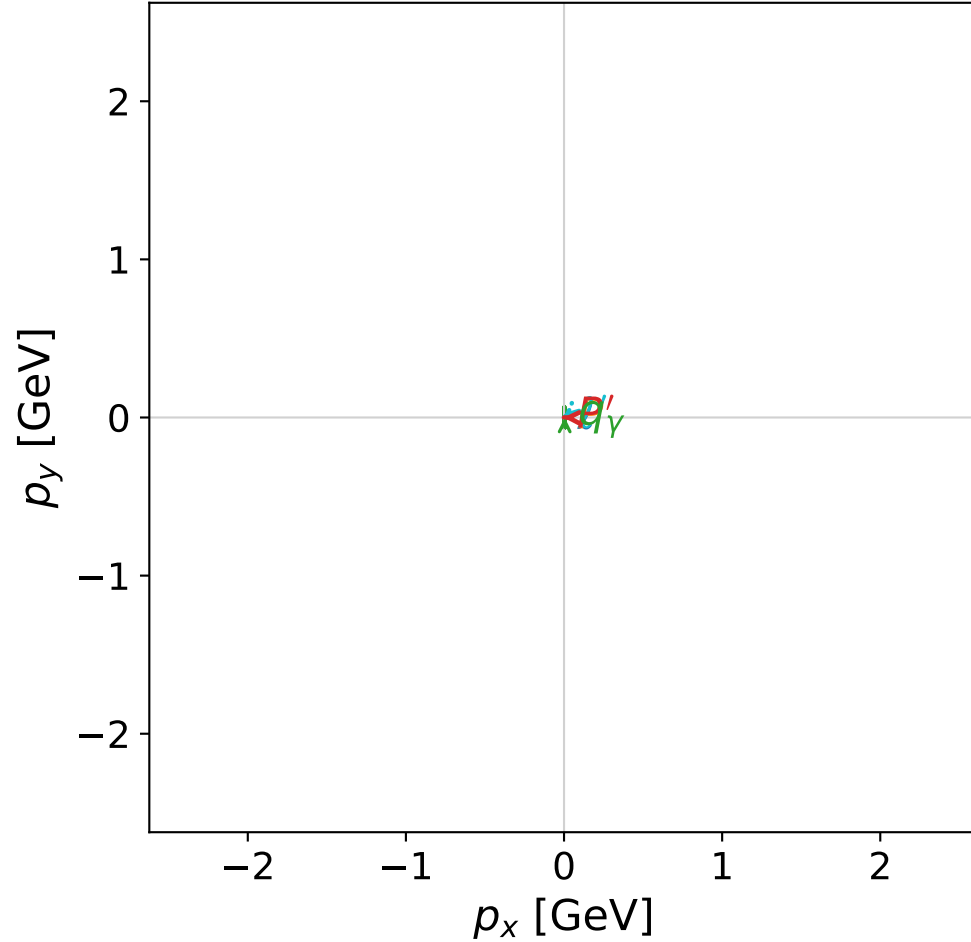


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

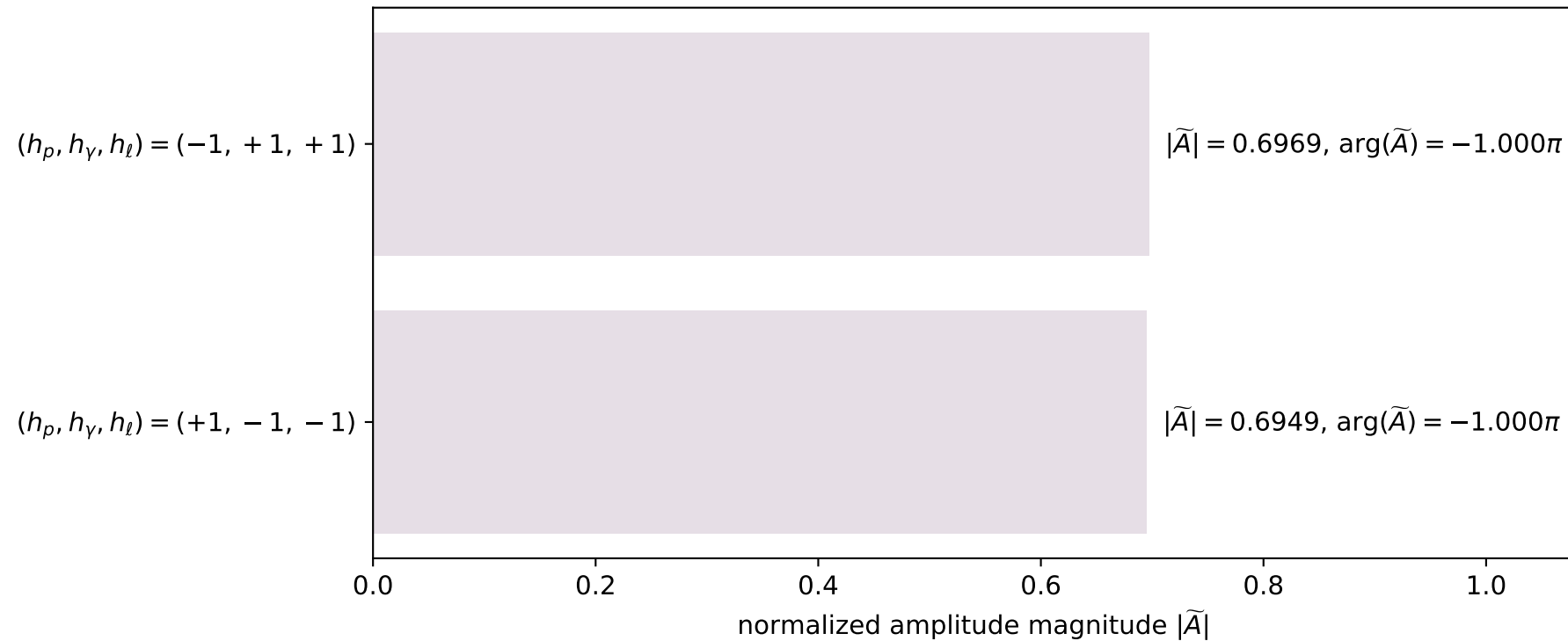


dghz_mixing_angles_polarization_03_local_minimum_248
 $d_{\{\text{rm GHZ}\}}=0.000127763$
 region: polarization_cluster_03_local_minimum_248
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0248
 lepton mixing angle: $\alpha_e=2.9049644$ rad
 incoming lepton: $-0.972134|+\rangle +0.234426|-\rangle$
 proton mixing angle: $\alpha_p=1.3339106$ rad
 incoming proton: $0.234676|+\rangle +0.972074|-\rangle$

$s=20.8306$, $\sqrt{s}=4.56405$, $|\vec{P}|=2.18564$, $|\vec{P}'|=0.064417$
 $\theta_{P'}=0.0689908$, $\theta_\gamma=3.13913$, $\phi_{P'}=3.12116$, $\phi_\gamma=0$
 $E_\gamma=1.84406$, $\theta_{\ell'}=7.53344\text{e-}05$, $\phi_{\ell'}=3.88476$
 $Q^2=15.5599$, $x_B=0.807696$, $t=-2.43181$, $W^2=4.58449$, $y=0.965605$

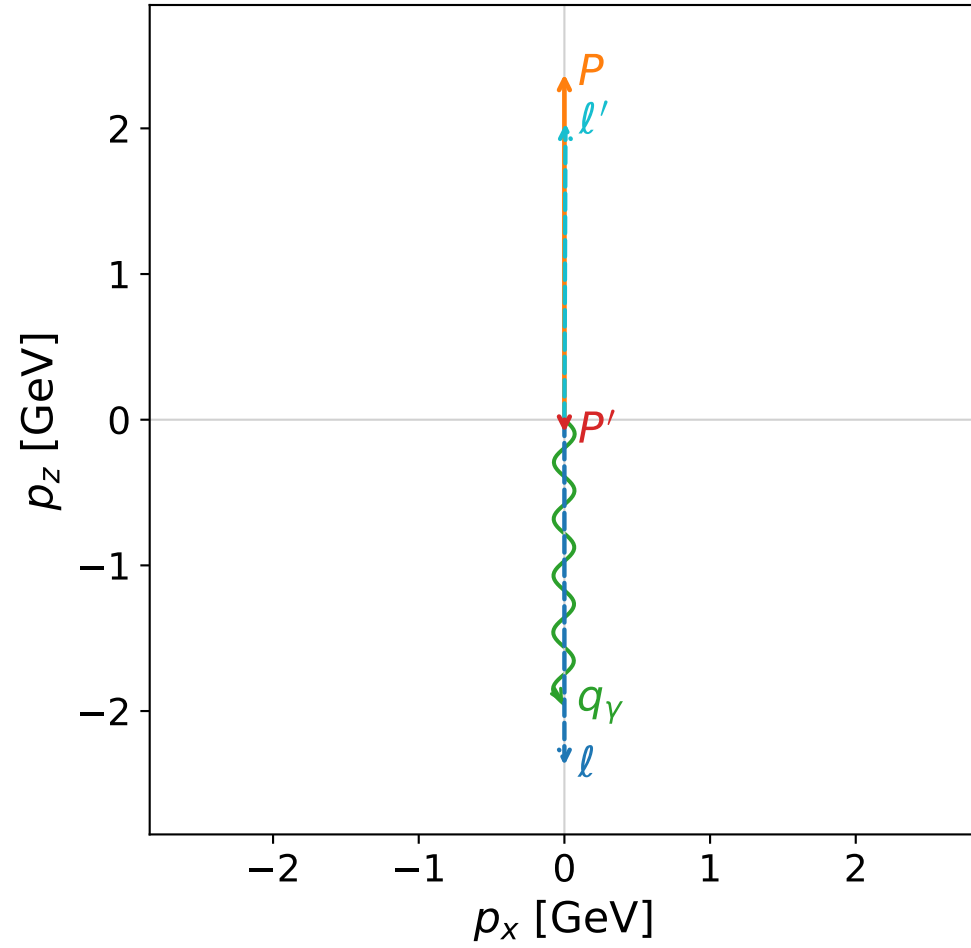
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1856$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1856$
 P $E= 2.3784$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1856$
 ℓ' $E= 1.7798$ $p_x= -0.0001$ $p_y= -0.0001$ $p_z= 1.7798$
 P' $E= 0.9402$ $p_x= -0.0044$ $p_y= 0.0001$ $p_z= 0.0643$
 q_γ $E= 1.8441$ $p_x= 0.0045$ $p_y= 0.0000$ $p_z= -1.8441$

Leading normalized final-state amplitudes
(N=2, retained=96.9%)

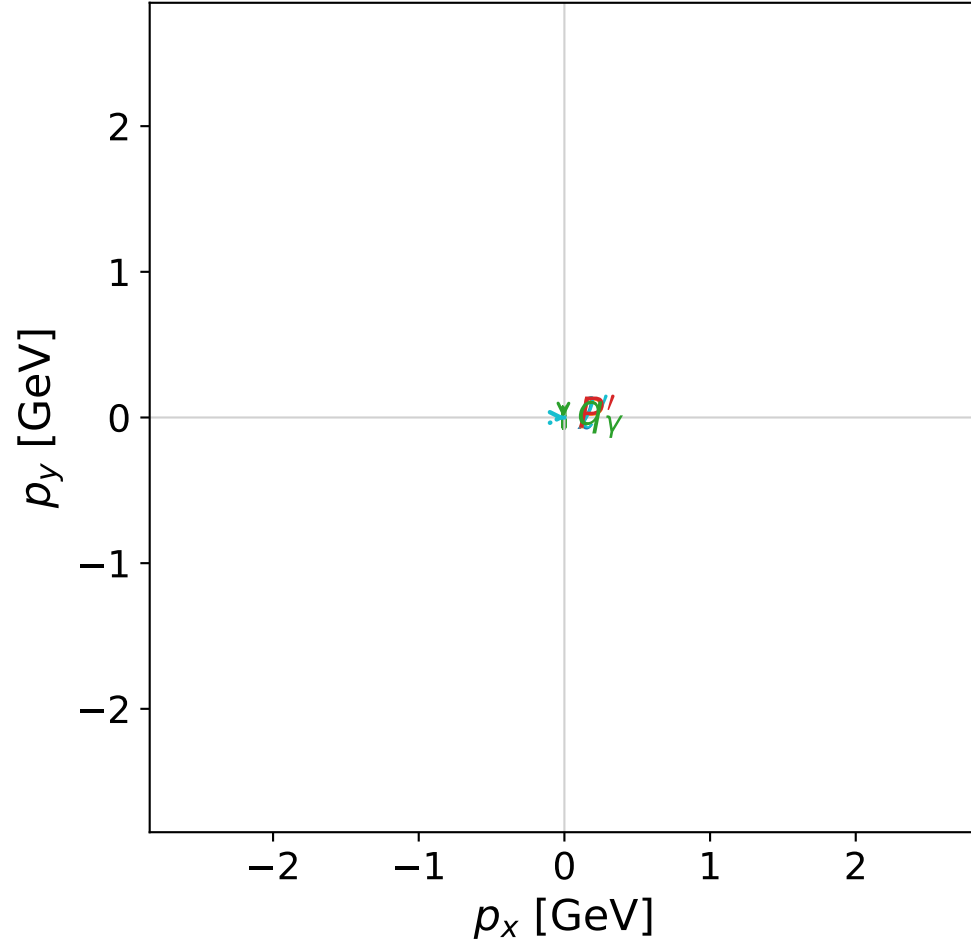


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

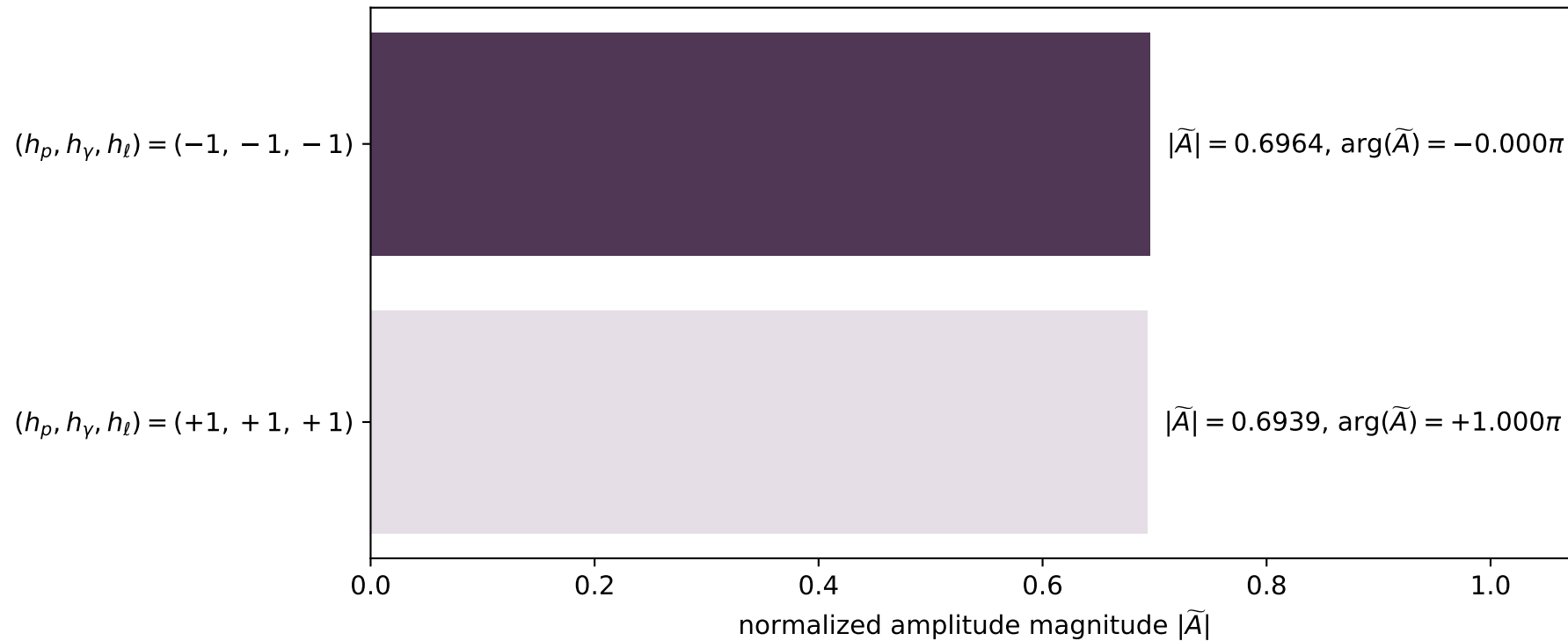


dghz_mixing_angles_polarization_03_local_minimum_262
 $d_{\text{GHZ}}=0.000177709$
 region: polarization_cluster_03_local_minimum_262
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0262
 lepton mixing angle: $\alpha_e=1.6990512$ rad
 incoming lepton: $-0.127904|+\rangle + 0.991787|-\rangle$
 proton mixing angle: $\alpha_p=0.12870773$ rad
 incoming proton: $0.991729|+\rangle + 0.128353|-\rangle$

$s=24.2413$, $\sqrt{s}=4.92355$, $|\vec{P}|=2.37242$, $|\vec{P}'|=0.0876034$
 $\theta_{P'}=3.14159$, $\theta_Y=3.13785$, $\phi_{P'}=3.00433$, $\phi_Y=3.14156$
 $E_Y=1.94693$, $\theta_{l'}=0.00358498$, $\phi_{l'}=6.28316$
 $Q^2=19.307$, $x_B=0.853001$, $t=-3.46271$, $W^2=4.20707$, $y=0.968871$

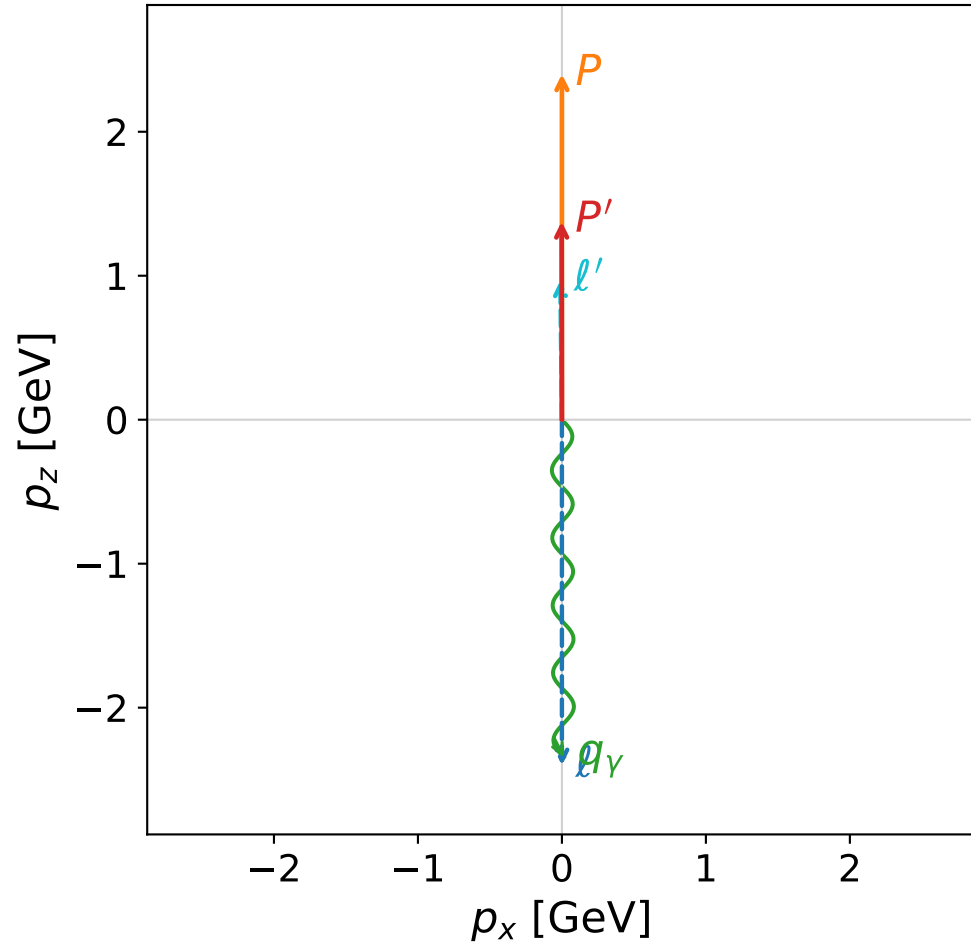
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3724$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3724$
 P $E= 2.5511$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3724$
 l' $E= 2.0345$ $p_x= 0.0073$ $p_y= -0.0000$ $p_z= 2.0345$
 P' $E= 0.9421$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0876$
 q_Y $E= 1.9469$ $p_x= -0.0073$ $p_y= 0.0000$ $p_z= -1.9469$

Leading normalized final-state amplitudes
 (N=2, retained=96.6%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

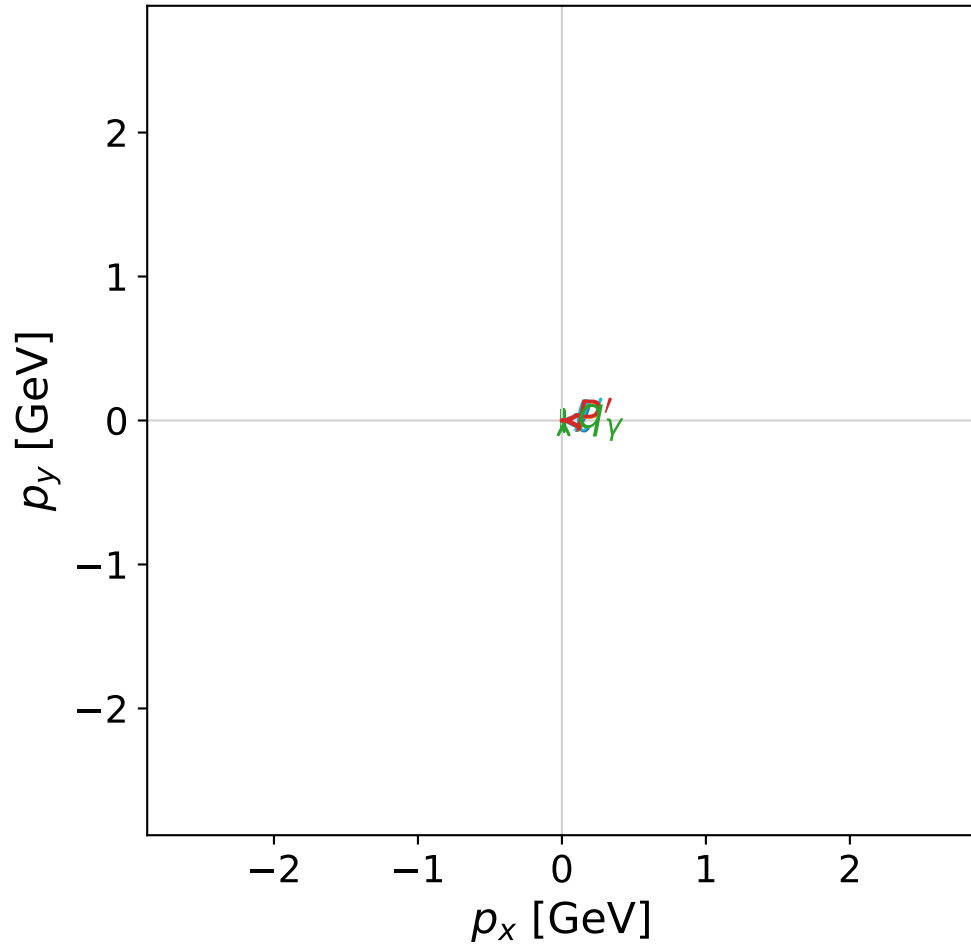


dghz_mixing_angles_polarization_03_local_minimum_267
 $d_{\text{GHZ}}=0.000191782$
 region: polarization_cluster_03_local_minimum_267
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0267
 lepton mixing angle: $\alpha_e=2.647731$ rad
 incoming lepton: $-0.880509|+\rangle +0.47403|-\rangle$
 proton mixing angle: $\alpha_p=1.0728653$ rad
 incoming proton: $0.477609|+\rangle +0.878573|-\rangle$

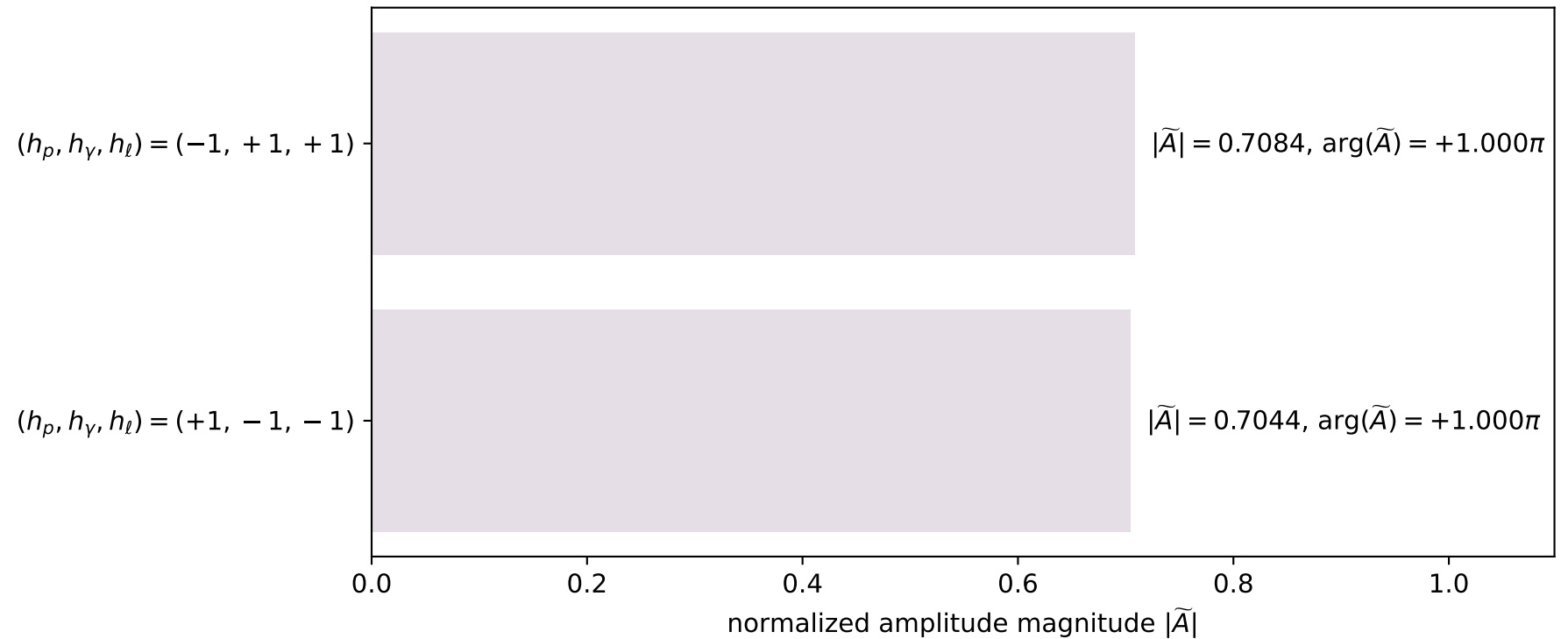
$s=24.7744$, $\sqrt{s}=4.97739$, $|\vec{P}|=2.40031$, $|\vec{P}'|=1.37624$
 $\theta_{P'}=0.00145035$, $\theta_\gamma=3.13587$, $\phi_{P'}=3.13934$, $\phi_\gamma=0$
 $E_\gamma=2.34405$, $\theta_{l'}=0.0117929$, $\phi_{l'}=3.14199$
 $Q^2=9.29214$, $x_B=0.394536$, $t=-0.217746$, $W^2=15.1398$, $y=0.985667$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4003$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4003$
 P $E= 2.5771$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4003$
 l' $E= 0.9678$ $p_x= -0.0114$ $p_y= -0.0000$ $p_z= 0.9678$
 P' $E= 1.6655$ $p_x= -0.0020$ $p_y= 0.0000$ $p_z= 1.3762$
 q_γ $E= 2.3441$ $p_x= 0.0134$ $p_y= 0.0000$ $p_z= -2.3440$

x--y plane

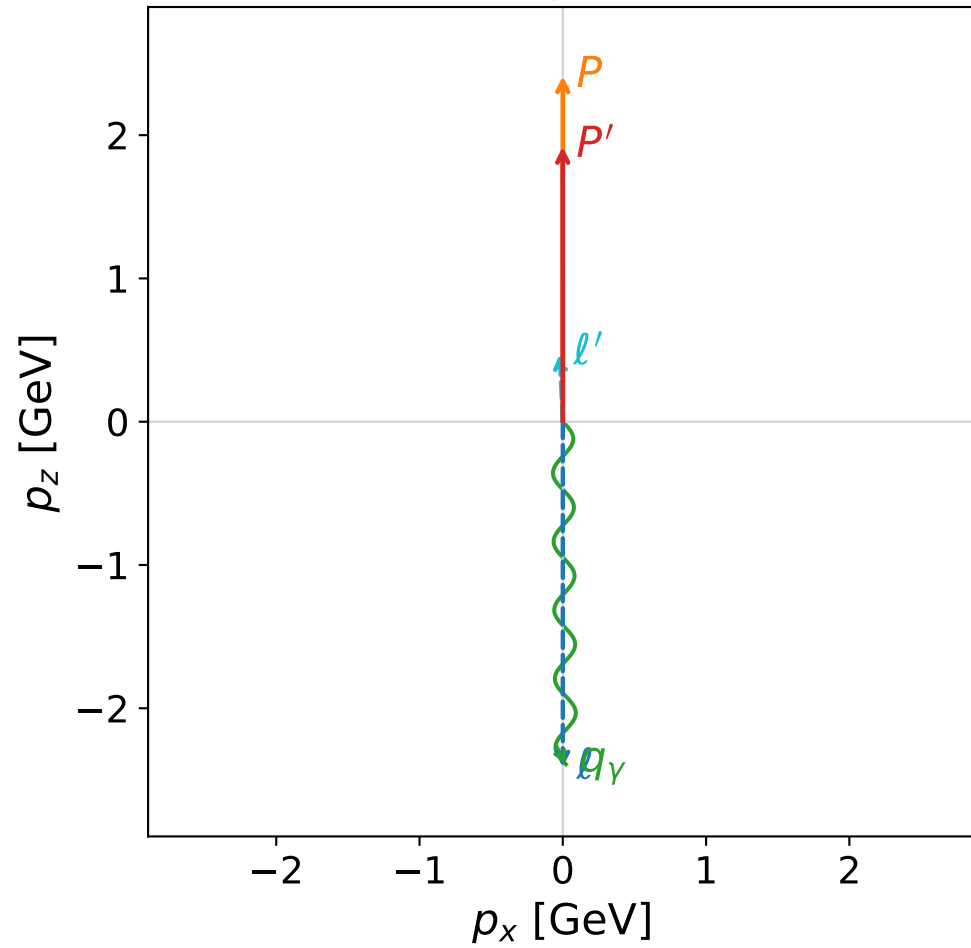


Leading normalized final-state amplitudes
(N=2, retained=99.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_03_local_minimum_316

$d_{\text{GHZ}}=0.00040247$

region: polarization_cluster_03_local_minimum_316

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0316

lepton mixing angle: $\alpha_e=2.555944$ rad

incoming lepton: $-0.833354|+\rangle + 0.55274|-\rangle$

proton mixing angle: $\alpha_p=0.97424173$ rad

incoming proton: $0.561796|+\rangle + 0.827276|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.91934$

$\theta_{p'}=0.000445521$, $\theta_\gamma=3.13237$, $\phi_{p'}=3.14125$, $\phi_\gamma=6.28316$

$E_\gamma=2.39134$, $\theta_{l'}=0.0448925$, $\phi_{l'}=3.14158$

$Q^2=4.55525$, $x_B=0.190186$, $t=-0.0386979$, $W^2=20.2762$, $y=0.993012$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$

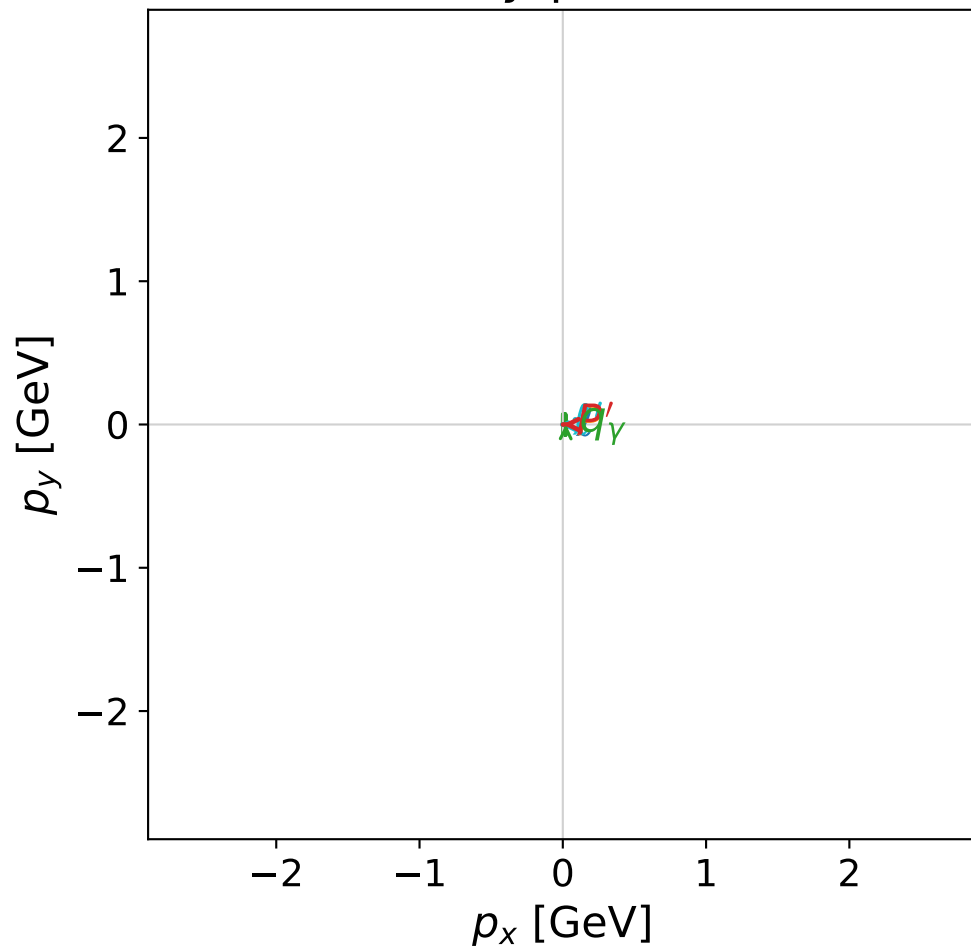
P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$

l' $E= 0.4724$ $p_x= -0.0212$ $p_y= 0.0000$ $p_z= 0.4719$

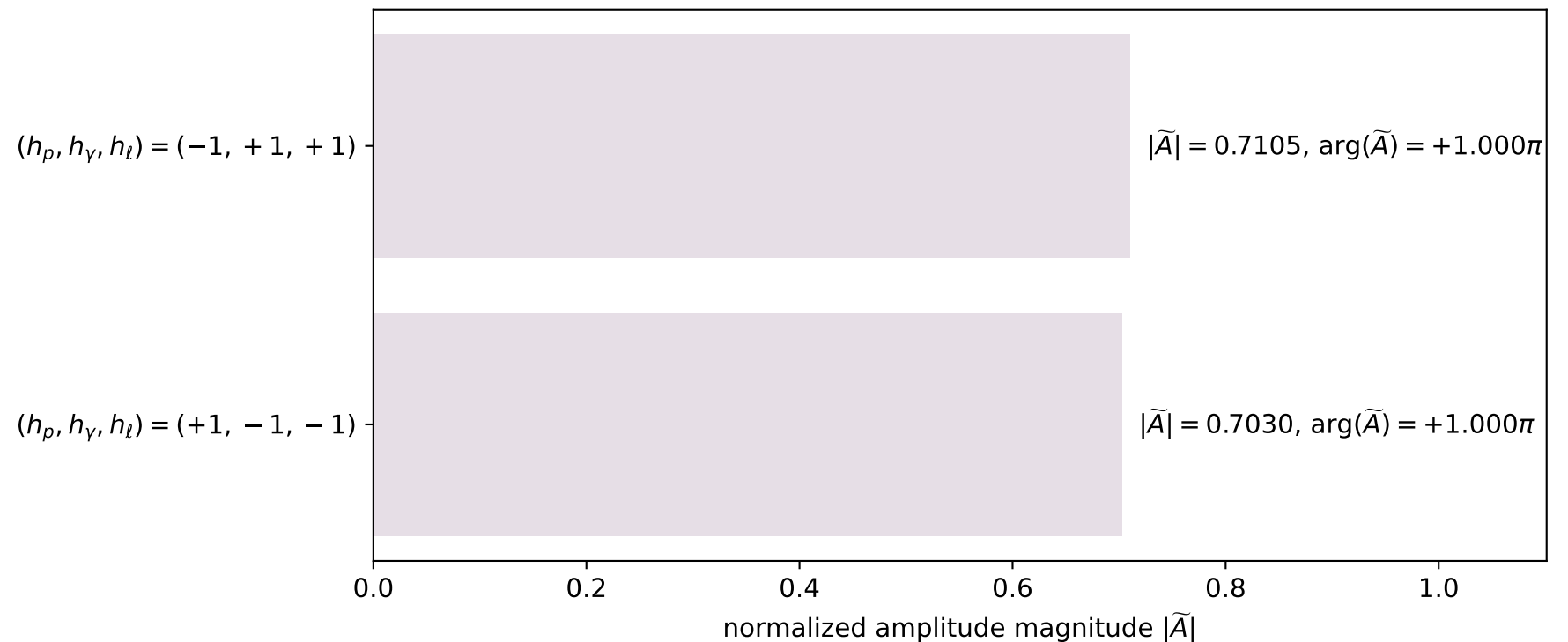
P' $E= 2.1363$ $p_x= -0.0009$ $p_y= 0.0000$ $p_z= 1.9193$

q_γ $E= 2.3913$ $p_x= 0.0221$ $p_y= -0.0000$ $p_z= -2.3912$

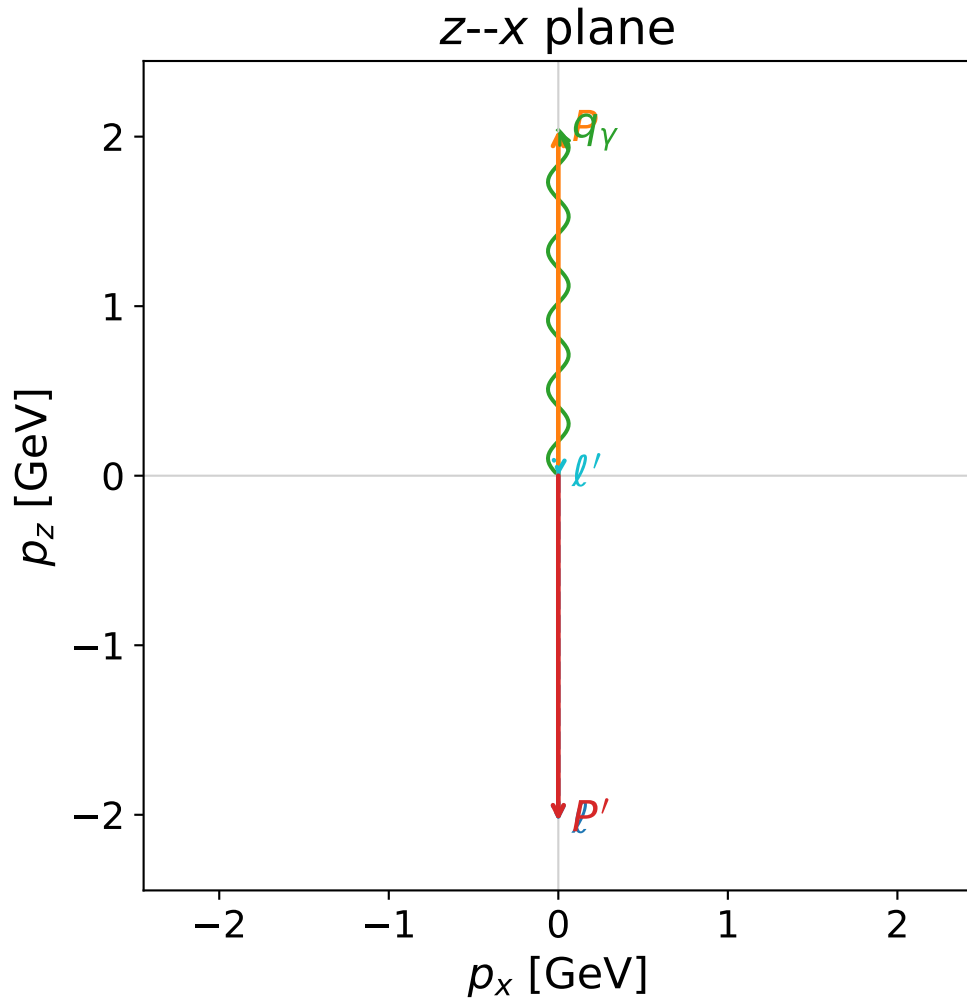
x--y plane



Leading normalized final-state amplitudes
(N=2, retained=99.9%)



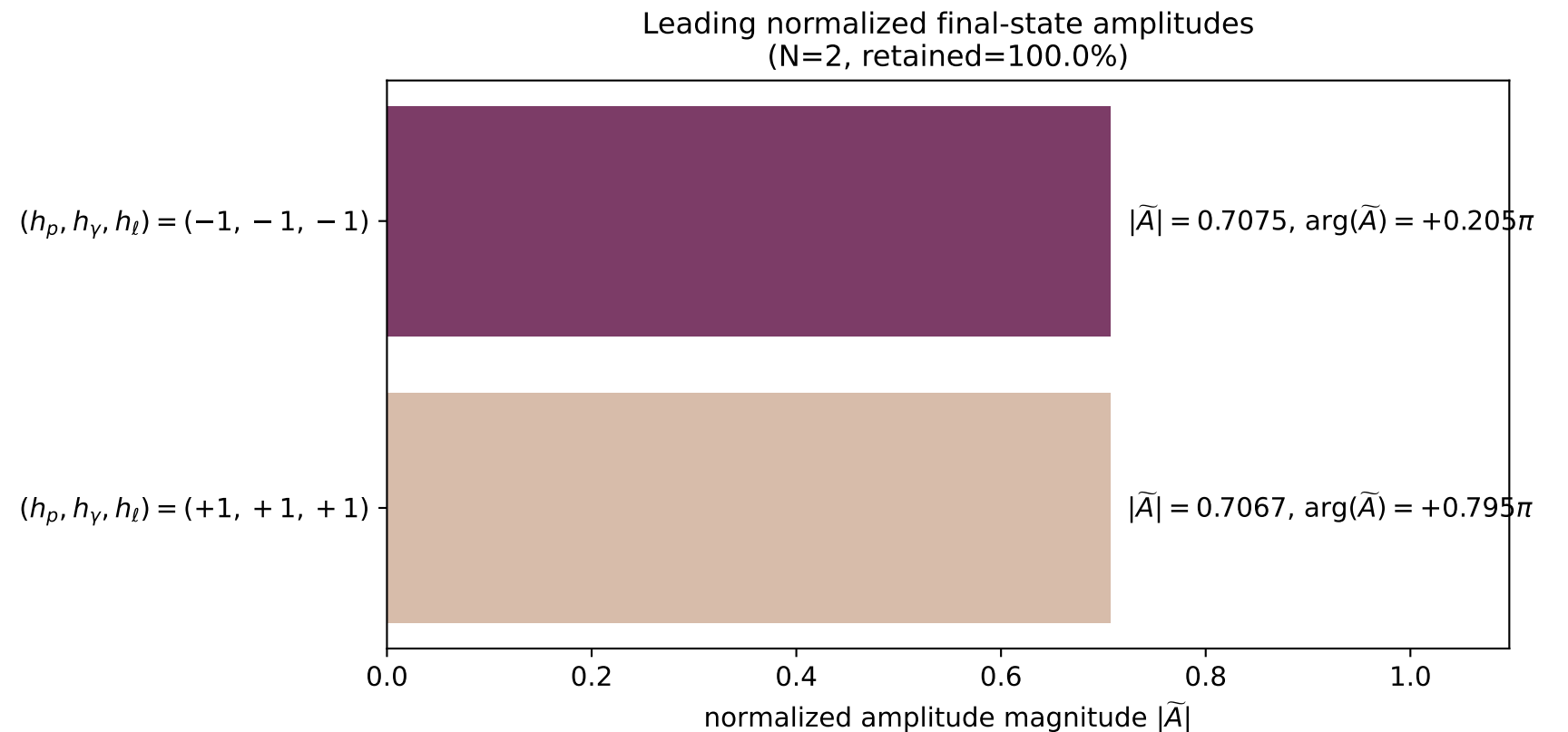
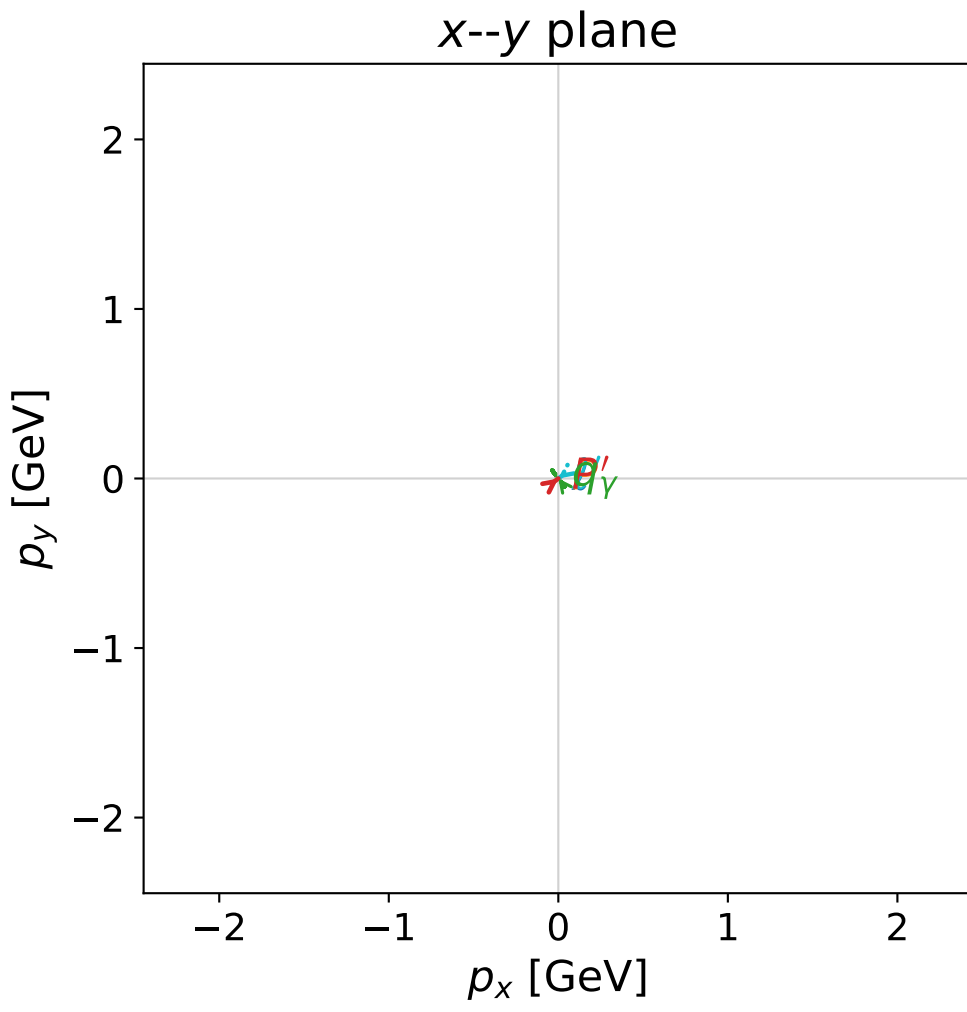
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



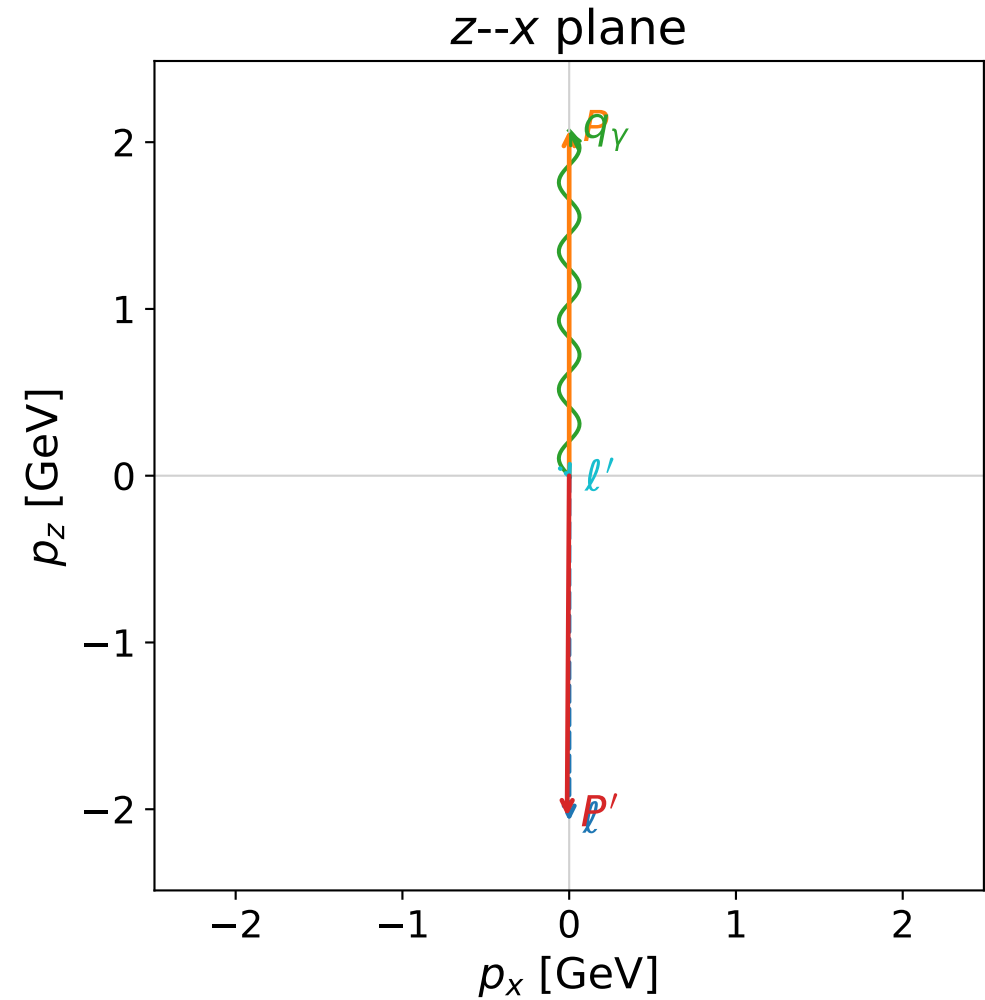
```
dghz_mixing_angles_polarization_03_local_minimum_357
d_{\rm GHZ}=0.00125916
region: polarization_cluster_03_local_minimum_357
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0357
lepton mixing angle: alpha_e=2.3565227 rad
incoming lepton: -0.707339|+> +0.706875|->
proton mixing angle: alpha_p=0.78632898 rad
incoming proton: 0.706448|+> +0.707765|->
```

```
s=18.3383, sqrt(s)=4.28232, |P|=2.03843, |P'|=2.03709
theta_p=3.14158, theta_gamma=2.79619e-05, phi_p=0.644969, phi_gamma=0.64563
E_gamma=2.03832, theta_l'=3.06993, phi_l'=3.78699
Q^2=0.000426632, x_B=2.44525e-05, t=-16.6098, W^2=18.3268, y=0.999369
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  2.0384 p_x=  0.0000 p_y=  0.0000 p_z= -2.0384
P      E=  2.2439 p_x=  0.0000 p_y=  0.0000 p_z=  2.0384
l'     E=  0.0013 p_x= -0.0001 p_y= -0.0001 p_z= -0.0012
P'     E=  2.2427 p_x=  0.0000 p_y=  0.0000 p_z= -2.0371
q_gamma E=  2.0383 p_x=  0.0000 p_y=  0.0000 p_z=  2.0383
```



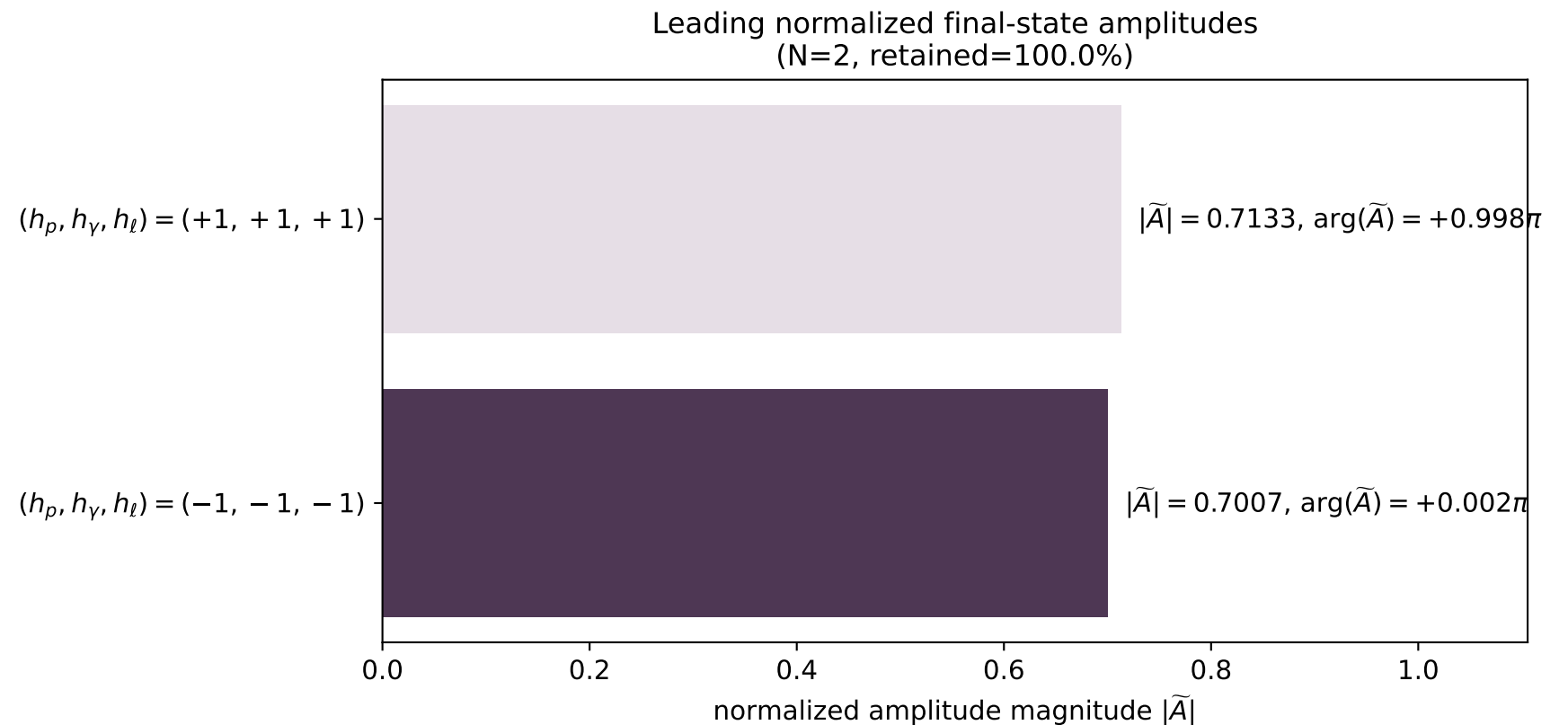
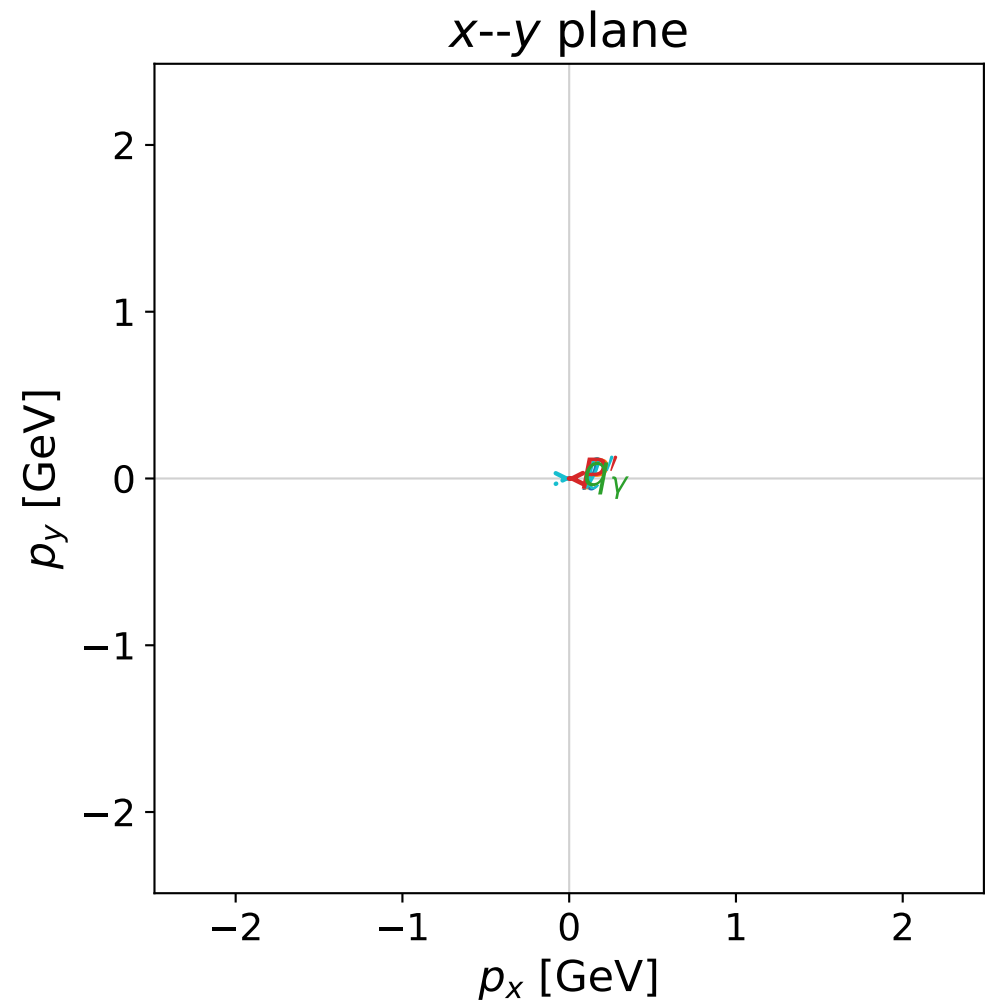
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



```
dghz_mixing_angles_polarization_03_local_minimum_372
d_{\rm GHZ}=0.0286865
region: polarization_cluster_03_local_minimum_372
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0372
lepton mixing angle: alpha_e=2.3560927 rad
incoming lepton: -0.707035|+> +0.707179|->
proton mixing angle: alpha_p=0.77718399 rad
incoming proton: 0.712891|+> +0.701275|->
```

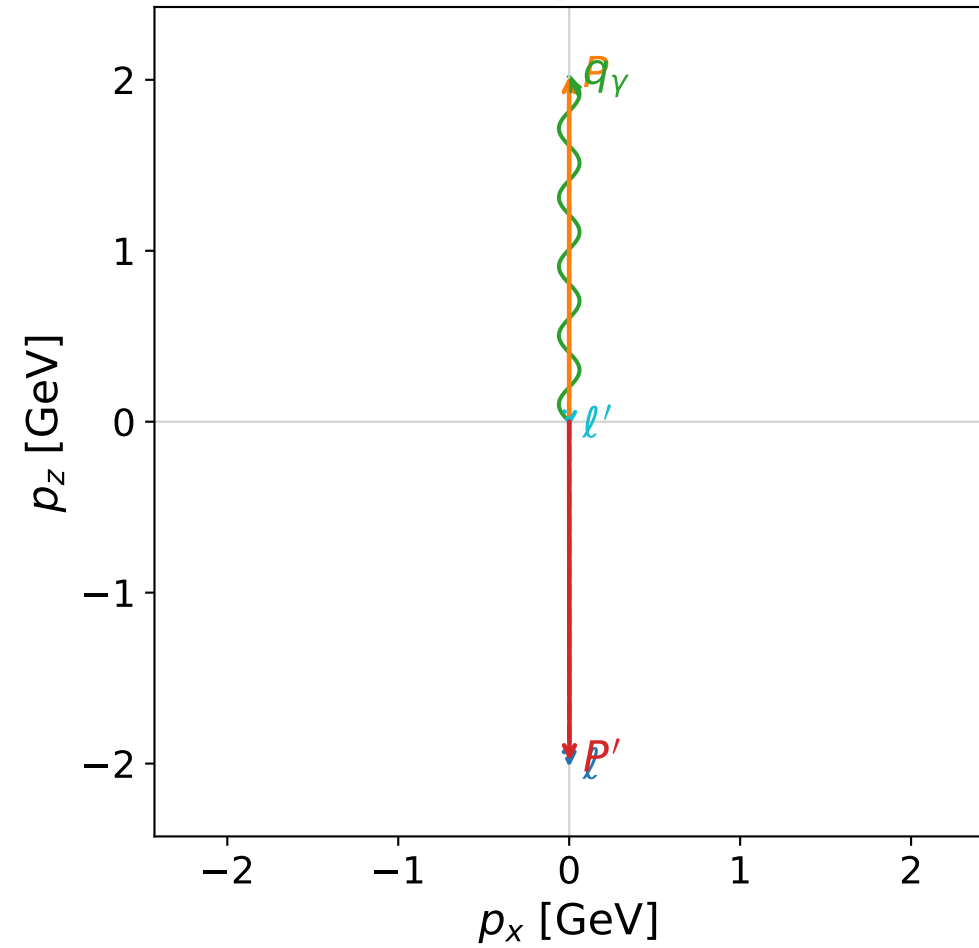
```
s=18.8971,  $\sqrt{s}$ =4.34708,  $|\vec{P}|$ =2.07234,  $|\vec{P}'|$ =2.04126
 $\theta_{p'}$ =3.13533,  $\theta_\gamma$ =0,  $\phi_{p'}$ =3.14439,  $\phi_\gamma$ =1.40154
 $E_\gamma$ =2.06954,  $\theta_{\ell'}$ =2.71784,  $\phi_{\ell'}$ =0.00279457
 $Q^2$ =0.0114083,  $x_B$ =0.000642414,  $t$ =-16.9207,  $W^2$ =18.6269,  $y$ =0.985637
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
 $\ell$       E=  2.0723 p_x=  0.0000 p_y=  0.0000 p_z= -2.0723
P         E=  2.2747 p_x=  0.0000 p_y=  0.0000 p_z=  2.0723
 $\ell'$     E=  0.0311 p_x=  0.0128 p_y=  0.0000 p_z= -0.0283
P'        E=  2.2465 p_x= -0.0128 p_y= -0.0000 p_z= -2.0412
 $q_\gamma$  E=  2.0695 p_x=  0.0000 p_y=  0.0000 p_z=  2.0695
```

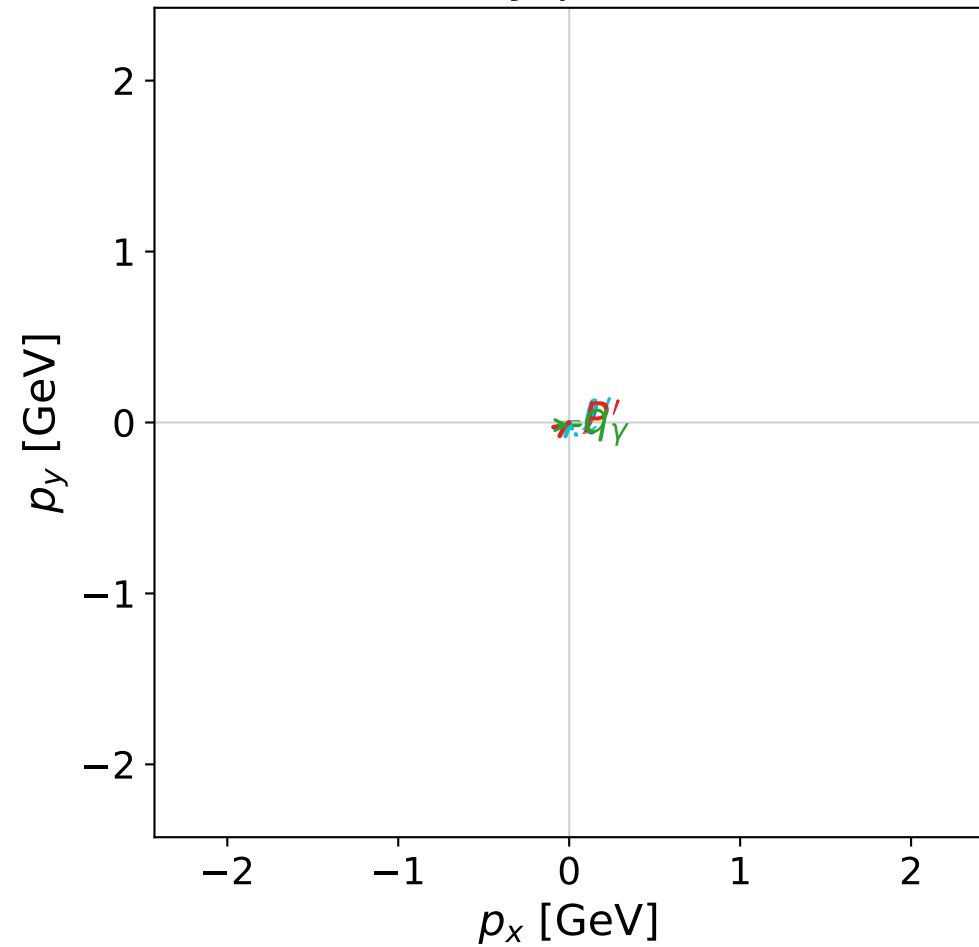


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



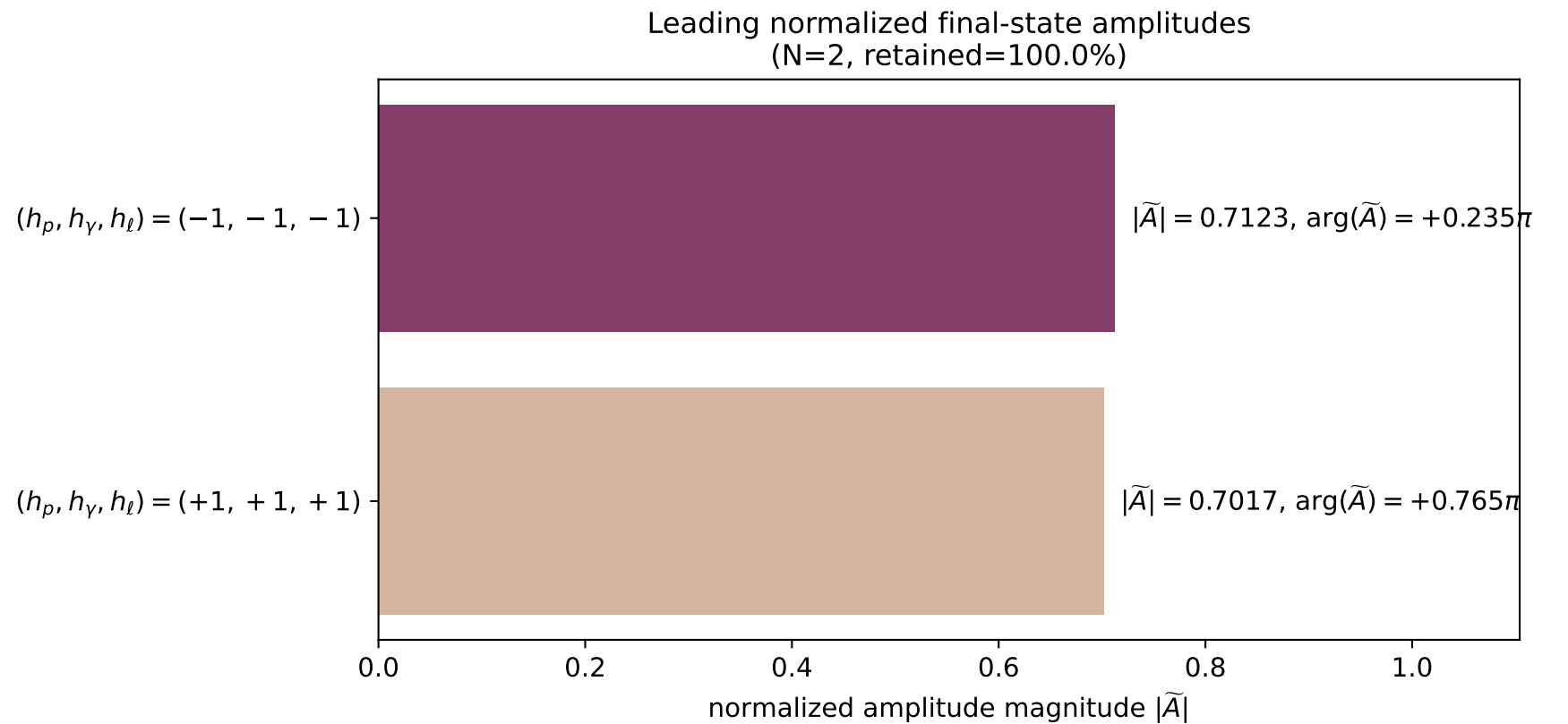
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_374
 $d_{\text{GHZ}}=0.0335652$
 region: polarization_cluster_03_local_minimum_374
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0374
 lepton mixing angle: $\alpha_e=2.356188$ rad
 incoming lepton: $-0.707102|+\rangle +0.707111|-\rangle$
 proton mixing angle: $\alpha_p=0.79277071$ rad
 incoming proton: $0.701874|+\rangle +0.712301|-\rangle$

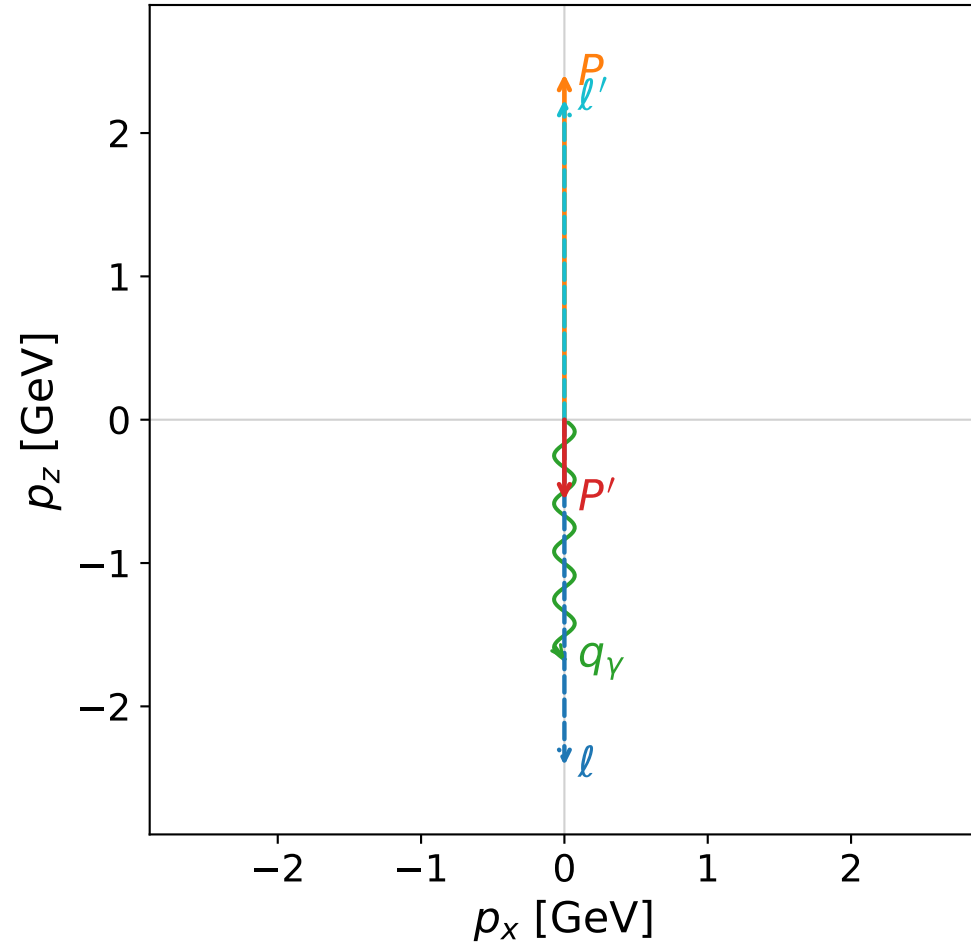
$s=18.0658$, $\sqrt{s}=4.25039$, $|\vec{P}|=2.02169$, $|\vec{P}'|=1.98608$
 $\theta_{p'}=3.14052$, $\theta_\gamma=0.00808957$, $\phi_{p'}=0.622432$, $\phi_\gamma=4.72104$
 $E_\gamma=2.01834$, $\theta_{l'}=2.70049$, $\phi_{l'}=1.69454$
 $Q^2=0.0137918$, $x_B=0.000816221$, $t=-16.0612$, $W^2=17.7632$, $y=0.983193$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0217$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0217$
 P $E= 2.2287$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0217$
 l' $E= 0.0356$ $p_x= -0.0019$ $p_y= 0.0151$ $p_z= -0.0322$
 P' $E= 2.1964$ $p_x= 0.0017$ $p_y= 0.0012$ $p_z= -1.9861$
 q_γ $E= 2.0183$ $p_x= 0.0001$ $p_y= -0.0163$ $p_z= 2.0183$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_8

$d_{\text{GHZ}}=2.4276\text{e-}08$

region: polarization_cluster_04_local_minimum_8

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0008

lepton mixing angle: $\alpha_e=2.332037$ rad

incoming lepton: $-0.68982|+\rangle +0.723981|-\rangle$

proton mixing angle: $\alpha_p=2.3320298$ rad

incoming proton: $-0.689815|+\rangle +0.723986|-\rangle$

$s=24.9835$, $\sqrt{s}=4.99835$, $|\vec{P}|=2.41116$, $|\vec{P}'|=0.561095$

$\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=4.05473$, $\phi_\gamma=1.2574$

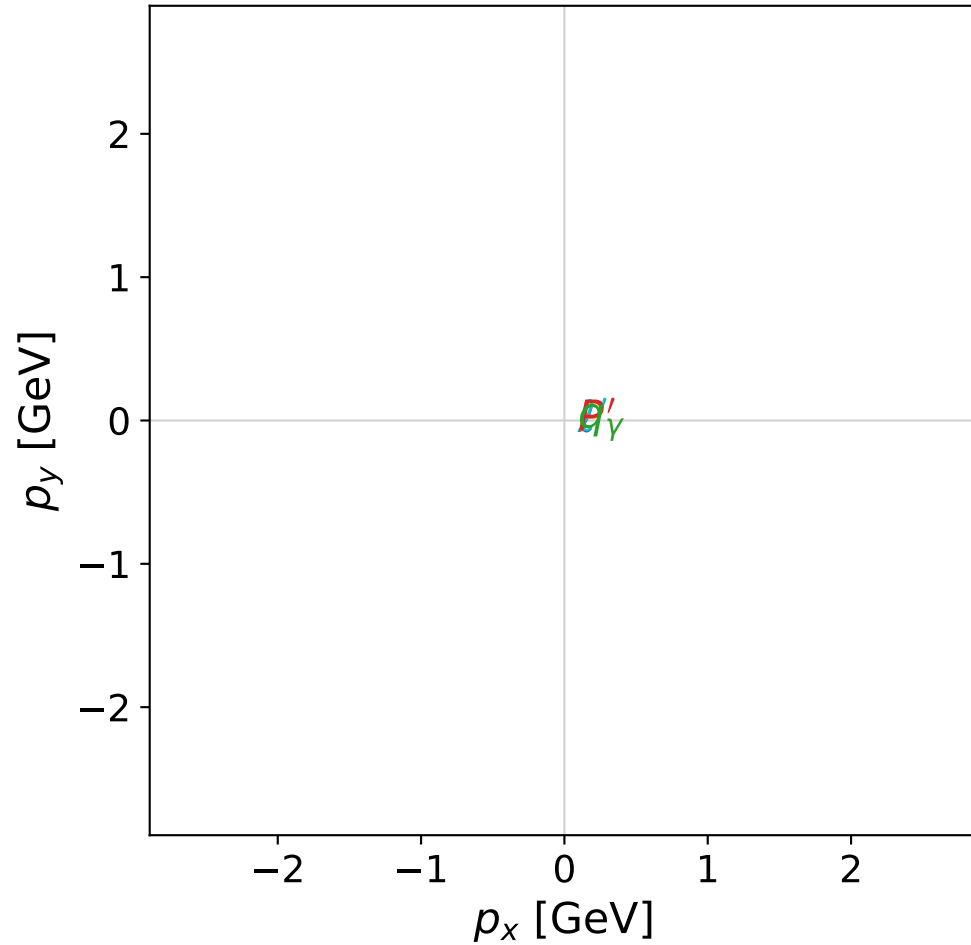
$E_\gamma=1.67212$, $\theta_{l'}=0$, $\phi_{l'}=4.56305$

$Q^2=21.5386$, $x_B=0.923711$, $t=-6.60174$, $W^2=2.6587$, $y=0.967382$

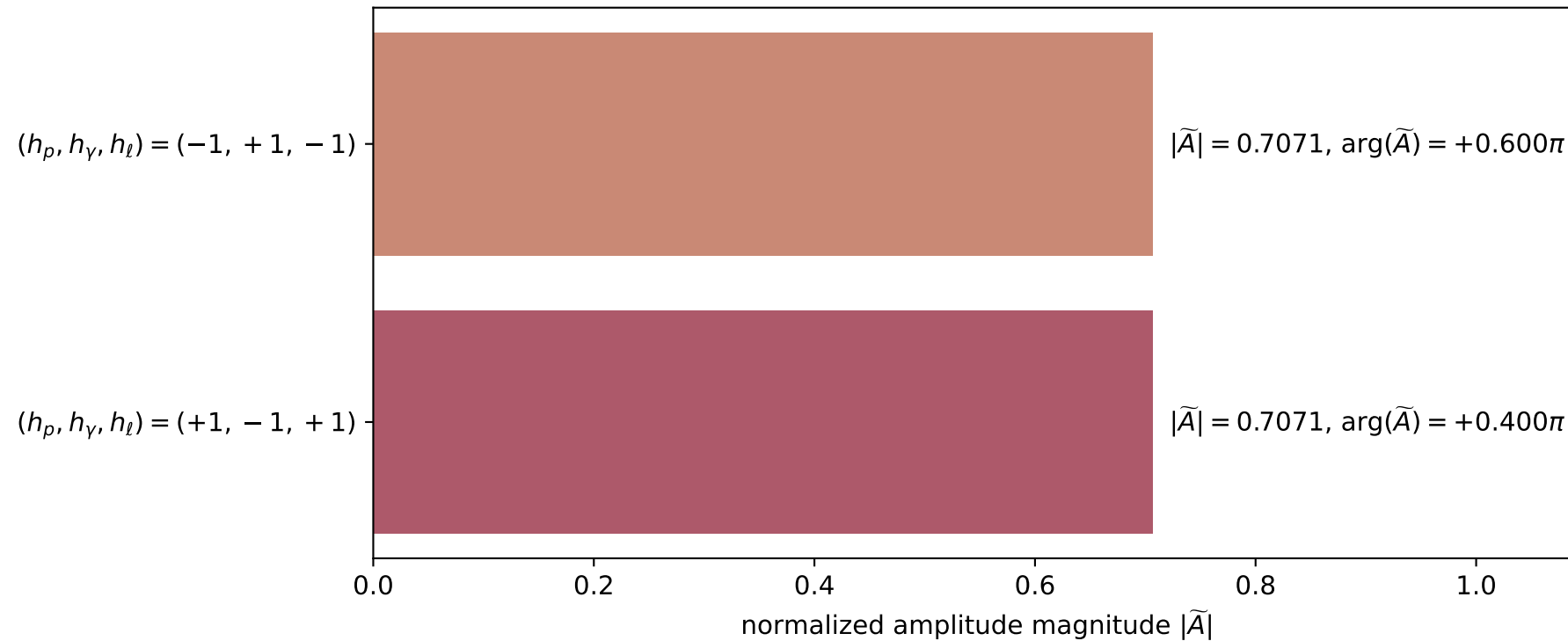
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
l	2.4112	0.0000	0.0000	-2.4112
P	2.5872	0.0000	0.0000	2.4112
l'	2.2332	-0.0000	-0.0000	2.2332
P'	1.0930	-0.0000	-0.0000	-0.5611
q_γ	1.6721	0.0000	0.0000	-1.6721

x--y plane

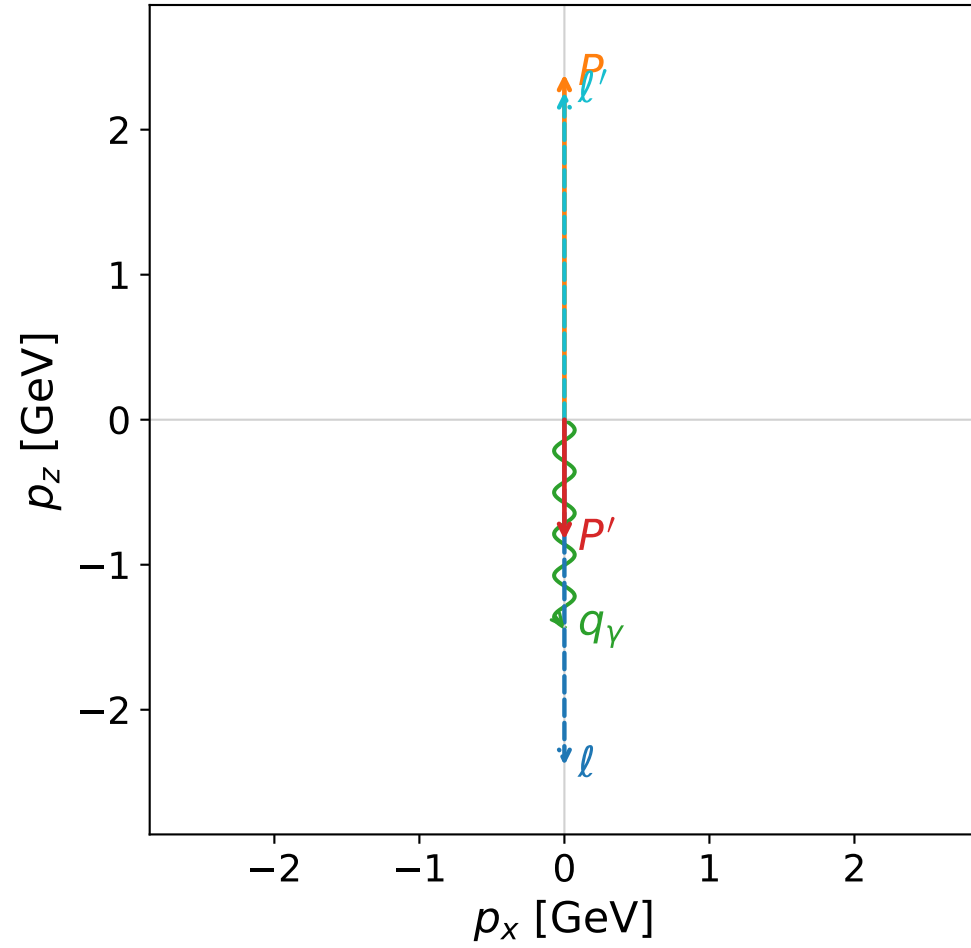


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_9

$d_{\text{GHZ}}=2.43084\text{e-}08$

region: polarization_cluster_04_local_minimum_9

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0009

lepton mixing angle: $\alpha_e=2.380064$ rad

incoming lepton: $-0.723782|+\rangle + 0.690029|-\rangle$

proton mixing angle: $\alpha_p=2.3800831$ rad

incoming proton: $-0.723795|+\rangle + 0.690015|-\rangle$

$s=24.4448$, $\sqrt{s}=4.94417$, $|\vec{P}|=2.38311$, $|\vec{P}'|=0.827879$

$\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14928$, $\phi_\gamma=4.99248$

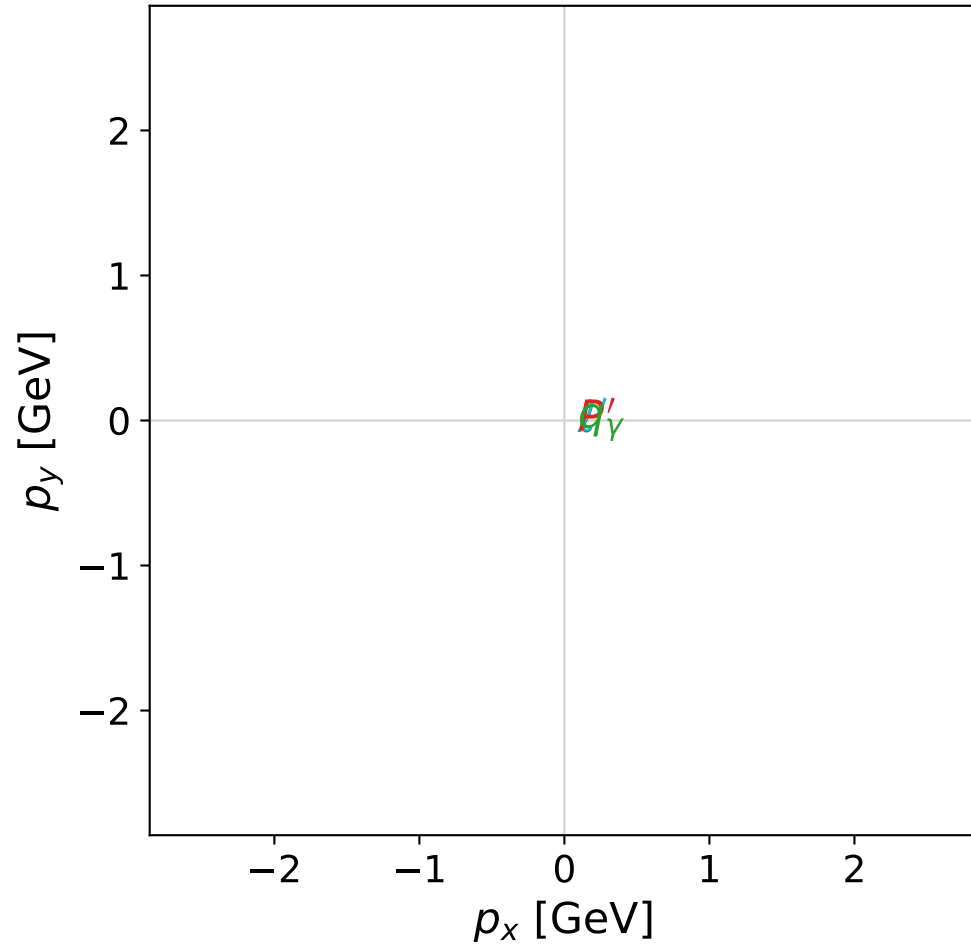
$E_\gamma=1.4326$, $\theta_{l'}=0$, $\phi_{l'}=1.26818$

$Q^2=21.5479$, $x_B=0.946724$, $t=-8.5944$, $W^2=2.09243$, $y=0.965859$

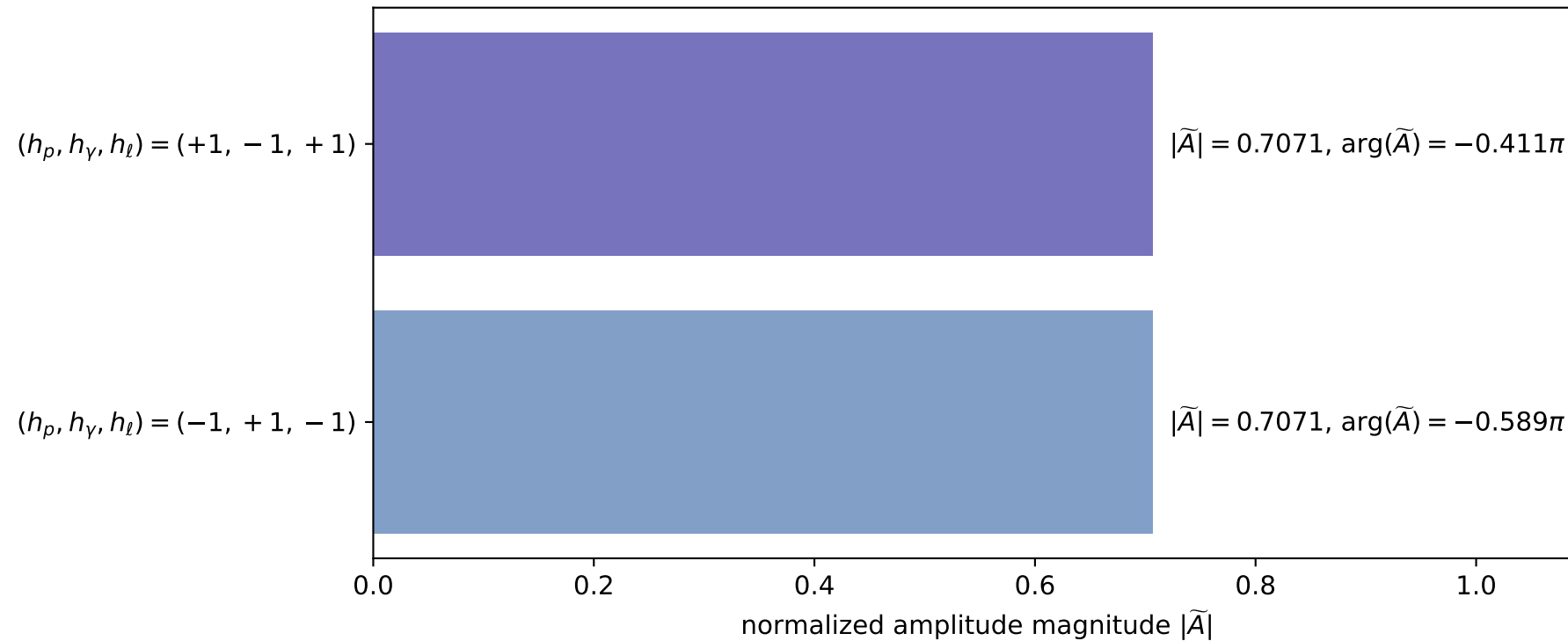
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l	$E=$	2.3831	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-2.3831
P	$E=$	2.5611	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	2.3831
l'	$E=$	2.2605	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	2.2605
P'	$E=$	1.2511	$p_x=$	-0.0000	$p_y=$	-0.0000	$p_z=$	-0.8279
q_γ	$E=$	1.4326	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	-1.4326

x--y plane

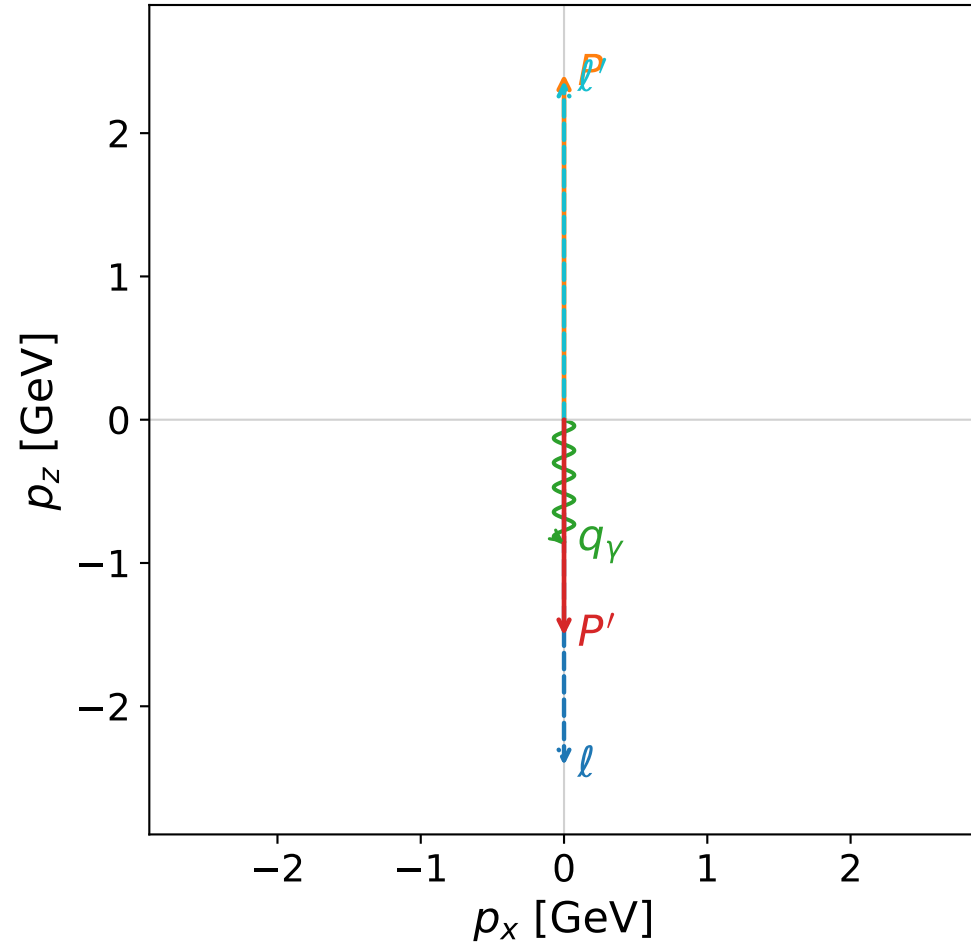


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

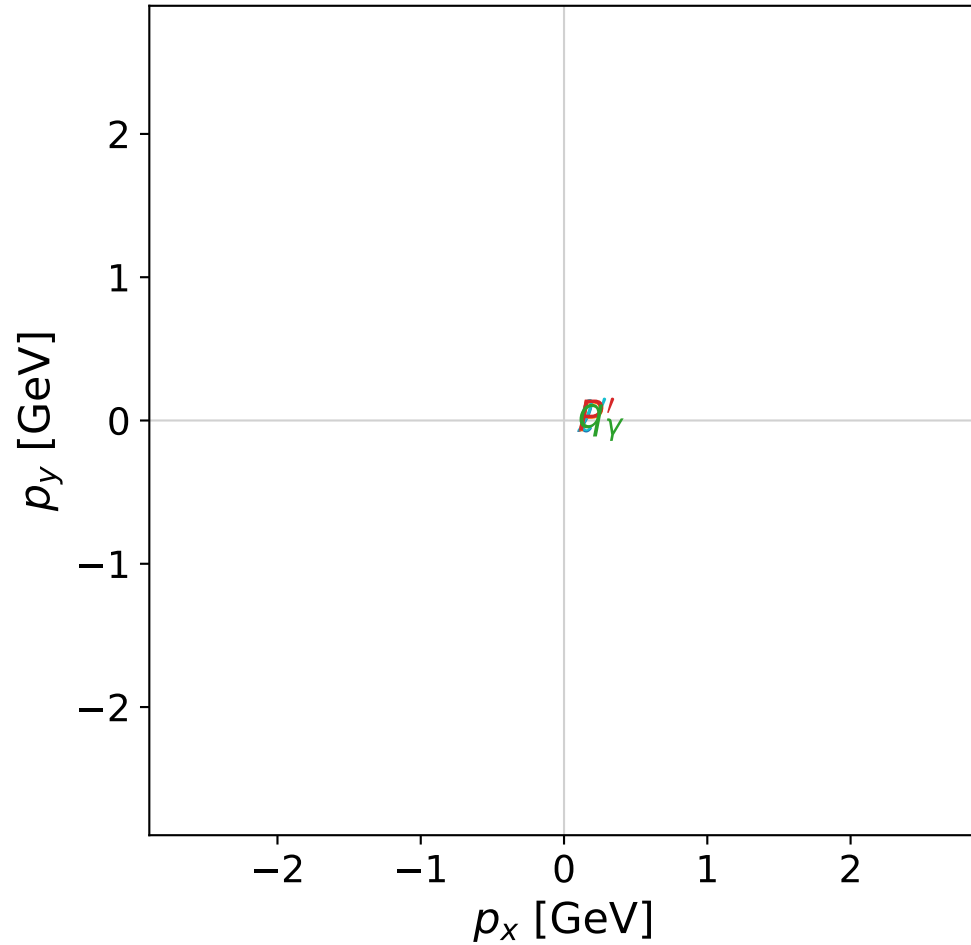


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



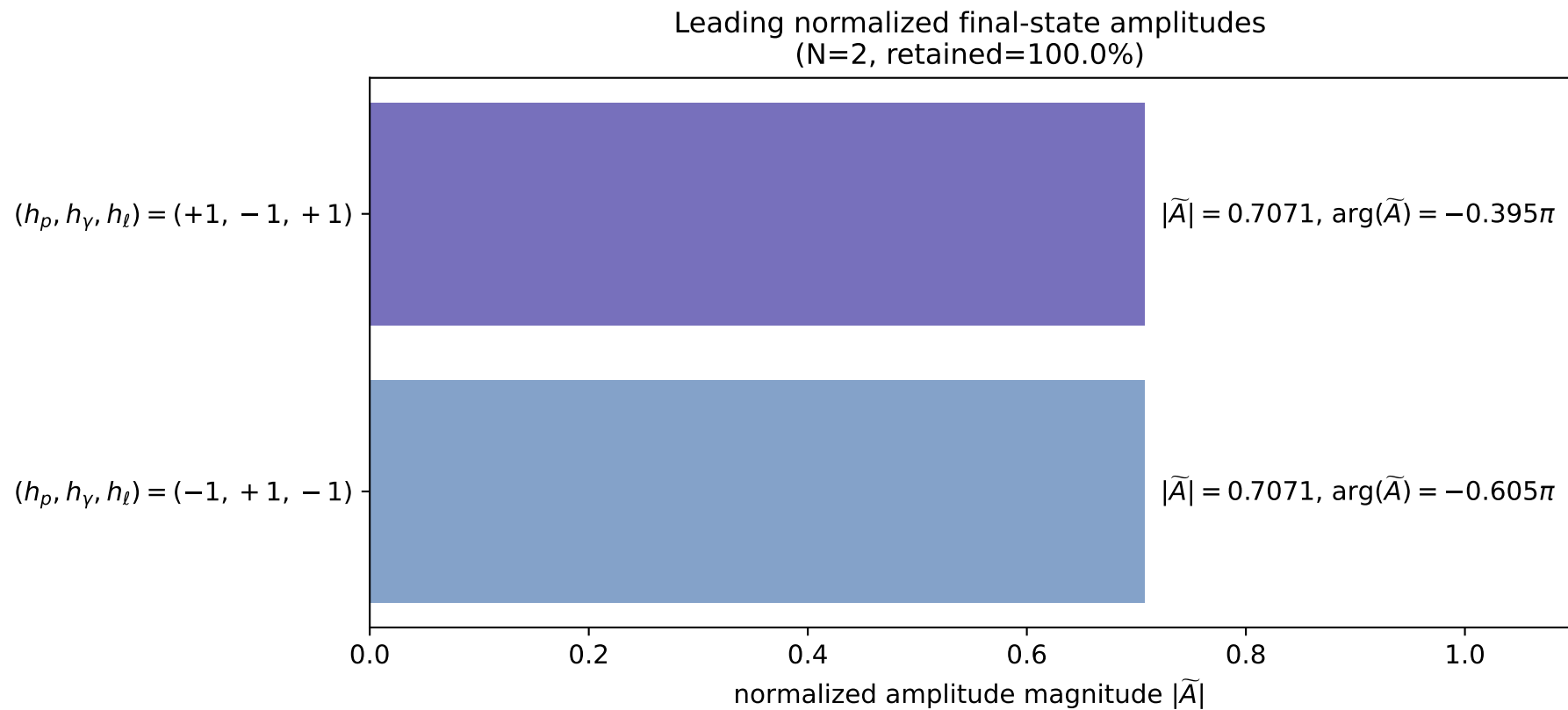
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_18
 $d_{\{\rm GHZ\}}=2.51612\text{e-}08$
 region: polarization_cluster_04_local_minimum_18
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0018
 lepton mixing angle: $\alpha_e=2.5703729$ rad
 incoming lepton: $-0.841242|+\rangle +0.540659|-\rangle$
 proton mixing angle: $\alpha_p=2.5703847$ rad
 incoming proton: $-0.841248|+\rangle +0.540649|-\rangle$

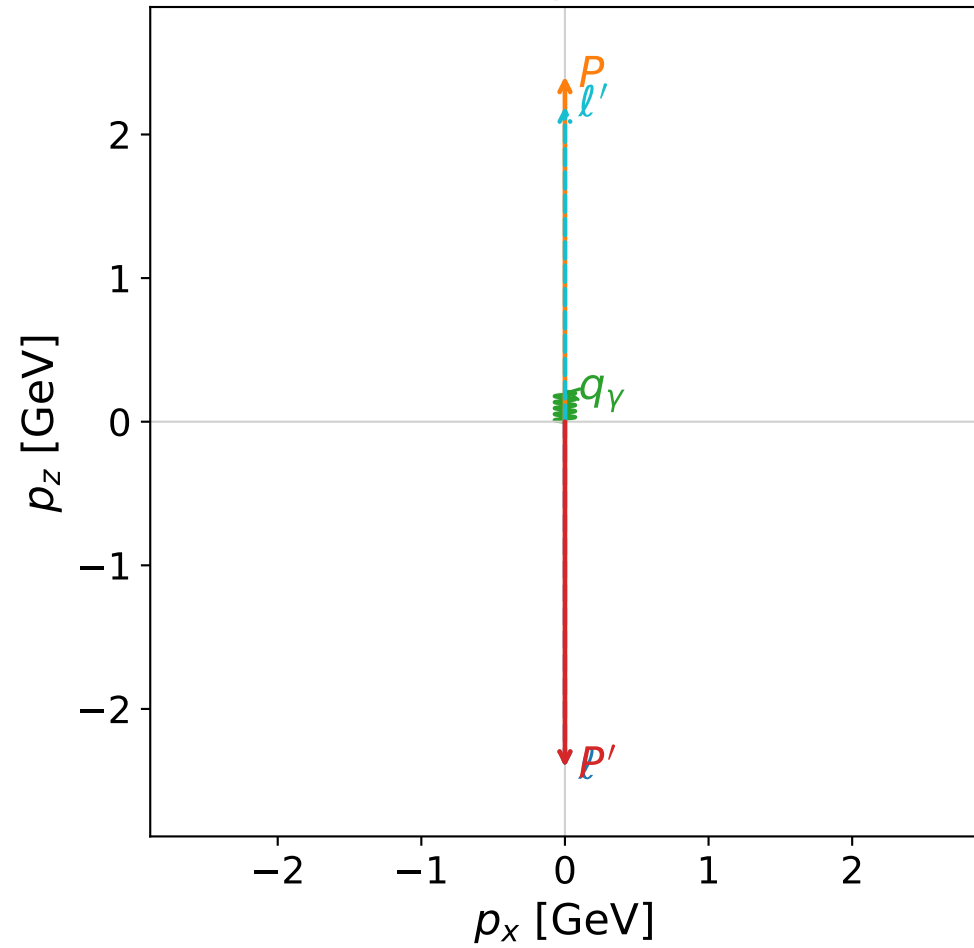
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.50685$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.529804$, $\phi_\gamma=5.04178$
 $E_\gamma=0.859098$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.10903$
 $Q^2=22.8268$, $x_B=0.980219$, $t=-14.6965$, $W^2=1.34049$, $y=0.965478$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3660$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.3660$
 P' $E= 1.7750$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5069$
 q_γ $E= 0.8591$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.8591$

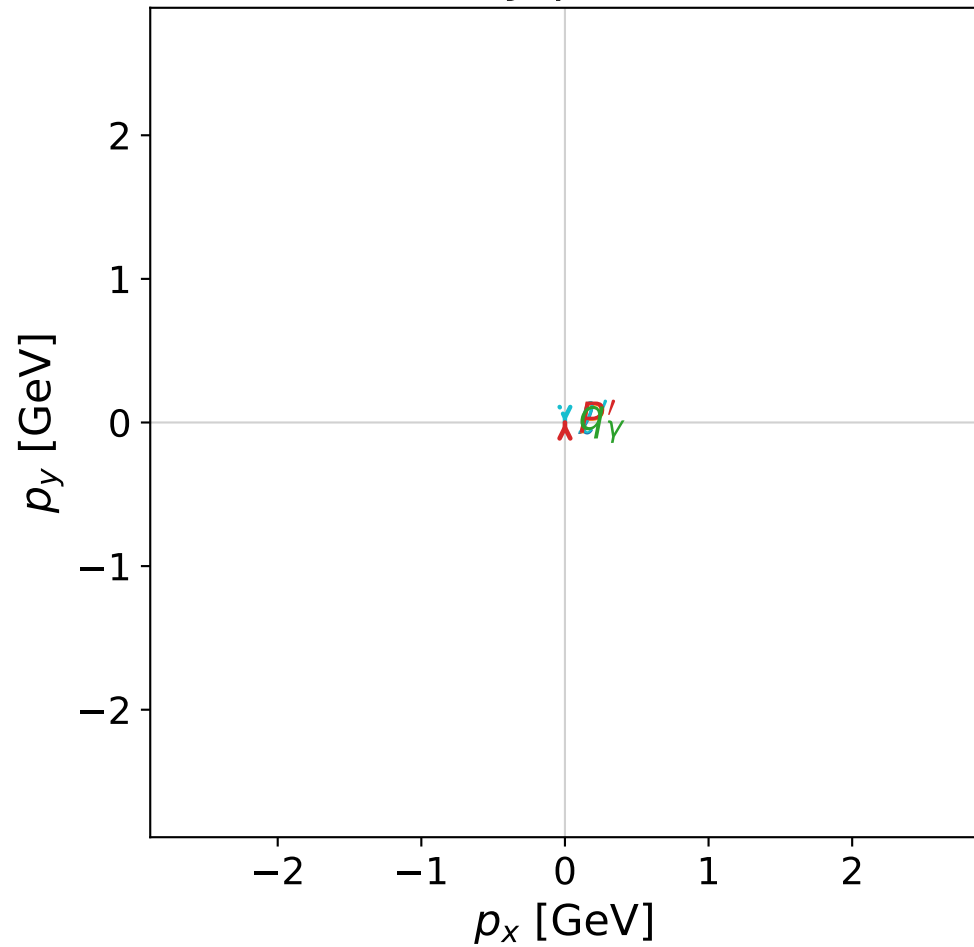


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

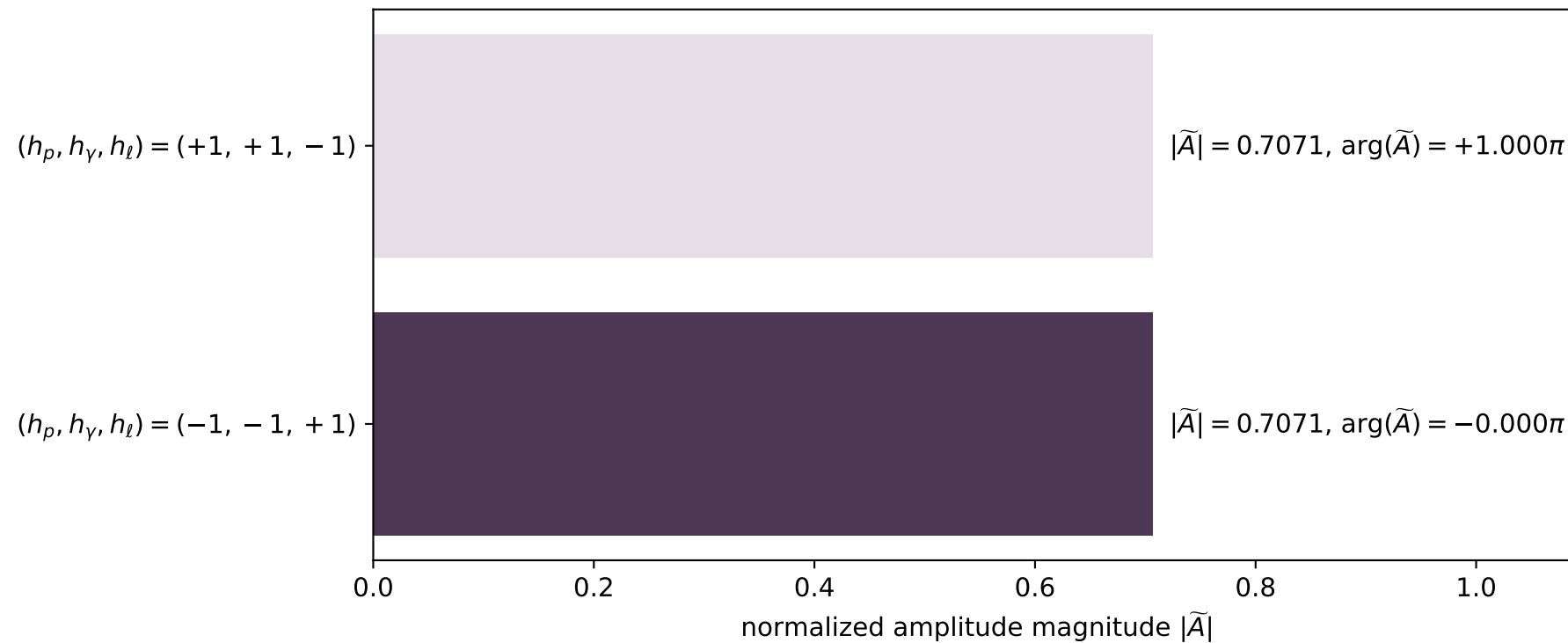


dghz_mixing_angles_polarization_04_local_minimum_22
 $d_{\text{GHZ}}=2.56605\text{e-}08$
 region: polarization_cluster_04_local_minimum_22
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0022
 lepton mixing angle: $\alpha_e=2.4937434$ rad
 incoming lepton: $-0.797384|+\rangle +0.603473|-\rangle$
 proton mixing angle: $\alpha_p=2.218653$ rad
 incoming proton: $-0.603479|+\rangle +0.797379|-\rangle$

$s=24.8915$, $\sqrt{s}=4.98914$, $|\vec{P}|=2.4064$, $|\vec{P}'|=2.4064$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.87557$, $\phi_\gamma=0.0606964$
 $E_\gamma=0.208957$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.01716$
 $Q^2=21.1516$, $x_B=0.91027$, $t=-23.163$, $W^2=2.96488$, $y=0.967722$

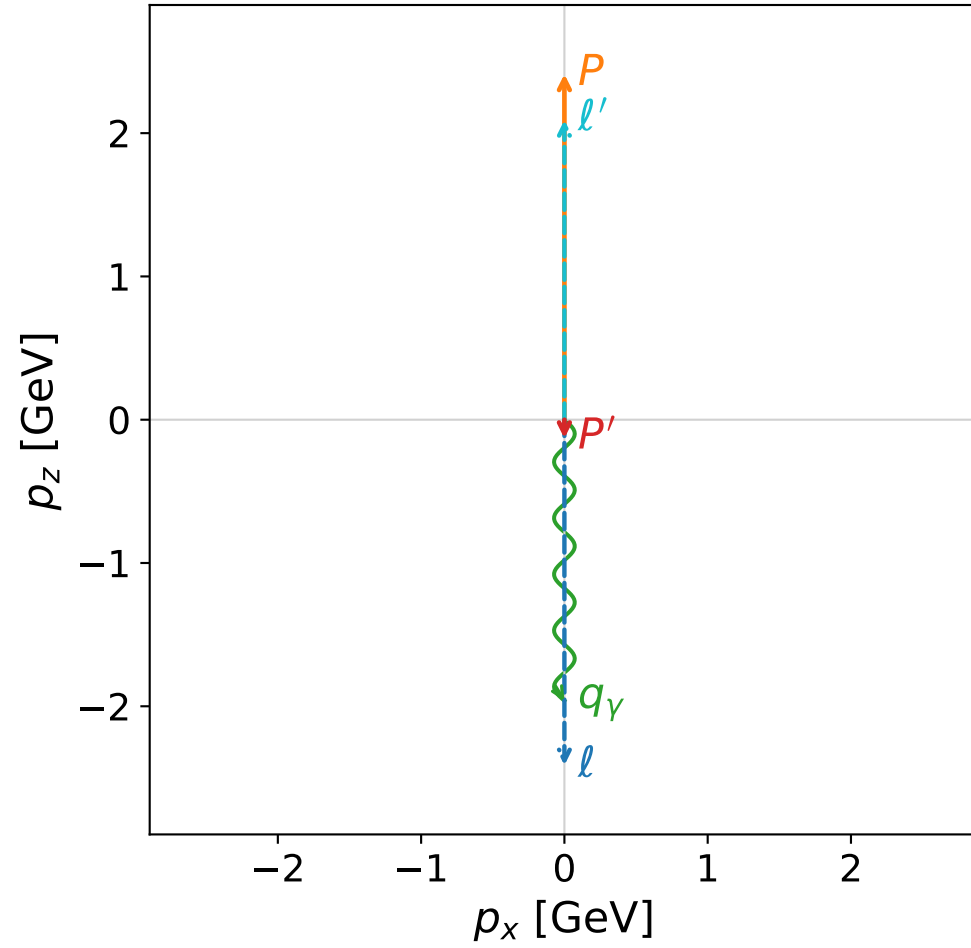
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	2.4064	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-2.4064
P	$E=$	2.5827	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	2.4064
ℓ'	$E=$	2.1974	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	2.1974
P'	$E=$	2.5827	$p_x=$	-0.0000	$p_y=$	0.0000	$p_z=$	-2.4064
q_γ	$E=$	0.2090	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.2090

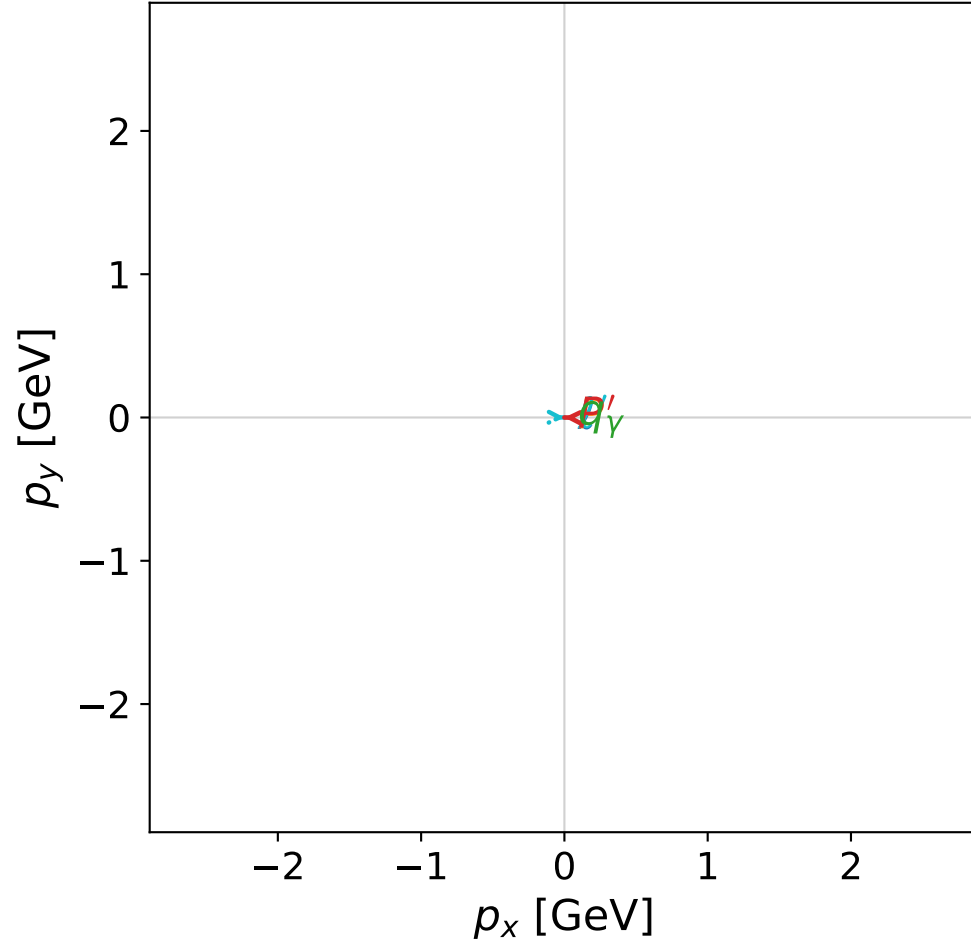
Leading normalized final-state amplitudes
(N=2, retained=100.0%)

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



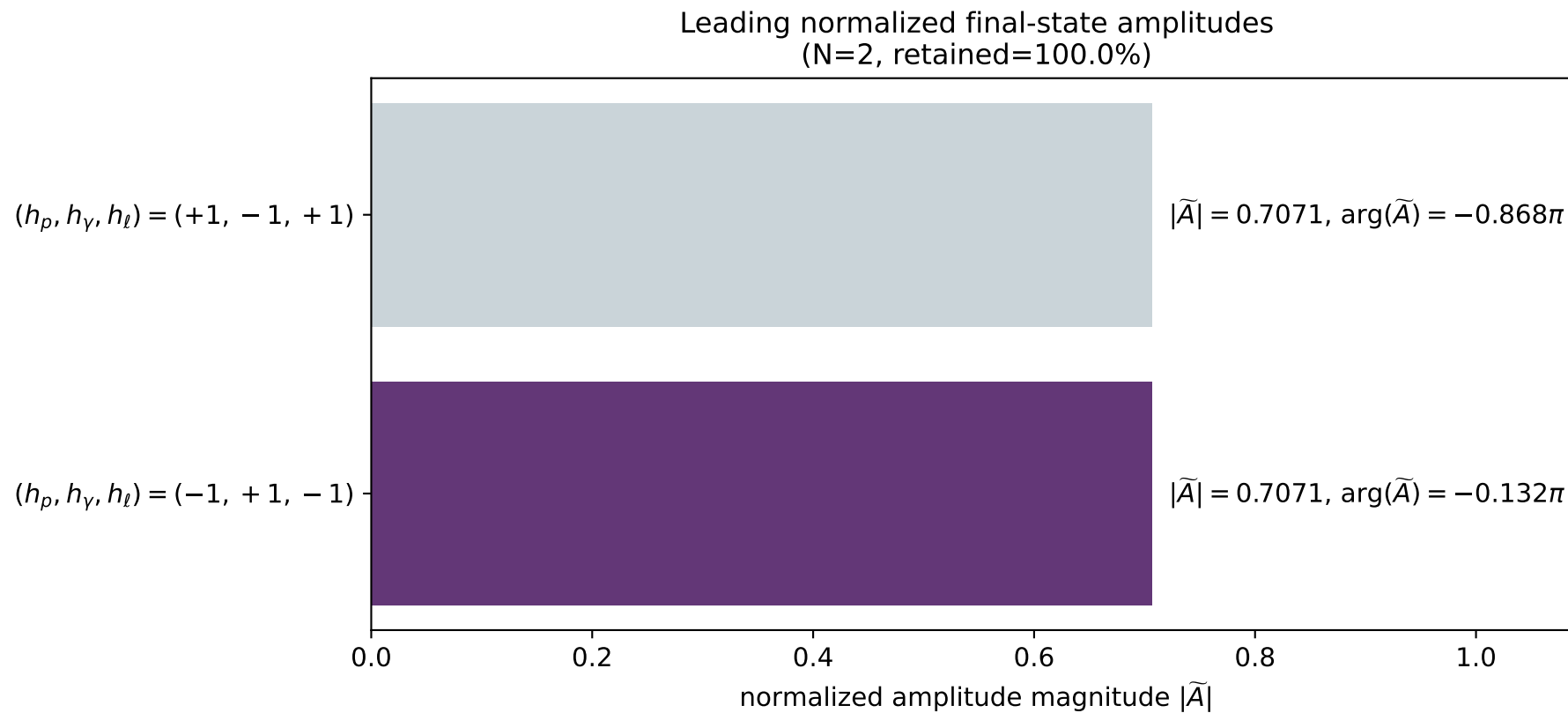
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_29
 $d_{\text{GHZ}}=2.60368\text{e-}08$
 region: polarization_cluster_04_local_minimum_29
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0029
 lepton mixing angle: $\alpha_e=2.3616845$ rad
 incoming lepton: $-0.710978|+\rangle +0.703214|-\rangle$
 proton mixing angle: $\alpha_p=2.3617041$ rad
 incoming proton: $-0.710992|+\rangle +0.7032|-\rangle$

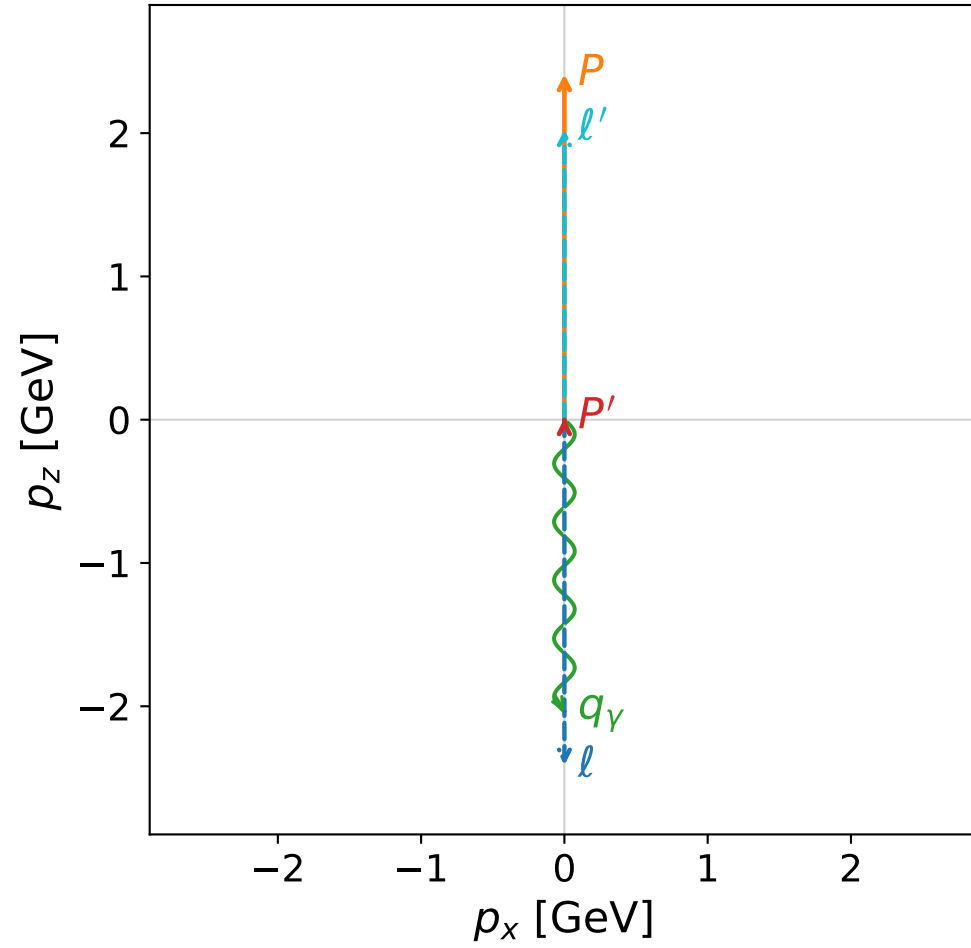
$s=24.9994$, $\sqrt{s}=4.99994$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.13022$
 $\theta_{P'}=3.14095$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.1261$, $\phi_\gamma=0.397731$
 $E_\gamma=1.96136$, $\theta_{l'}=3.98045\text{e-}05$, $\phi_{l'}=6.2677$
 $Q^2=20.1795$, $x_B=0.86298$, $t=-3.77005$, $W^2=4.08383$, $y=0.969481$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0916$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 2.0916$
 P' $E= 0.9470$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= -0.1302$
 q_γ $E= 1.9614$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9614$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

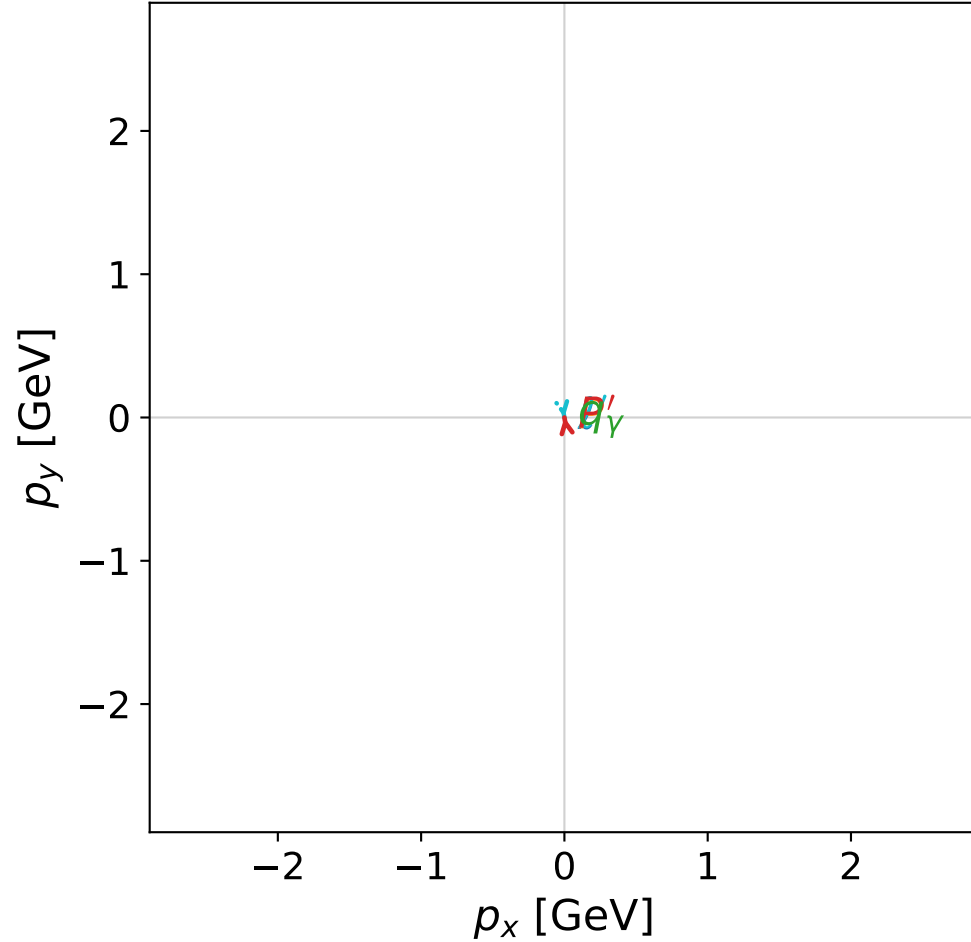


dghz_mixing_angles_polarization_04_local_minimum_30
 $d_{\text{GHZ}}=2.62358\text{e-}08$
 region: polarization_cluster_04_local_minimum_30
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0030
 lepton mixing angle: $\alpha_e=2.3547844$ rad
 incoming lepton: $-0.706109|+\rangle +0.708103|-\rangle$
 proton mixing angle: $\alpha_p=2.3547913$ rad
 incoming proton: $-0.706114|+\rangle +0.708098|-\rangle$

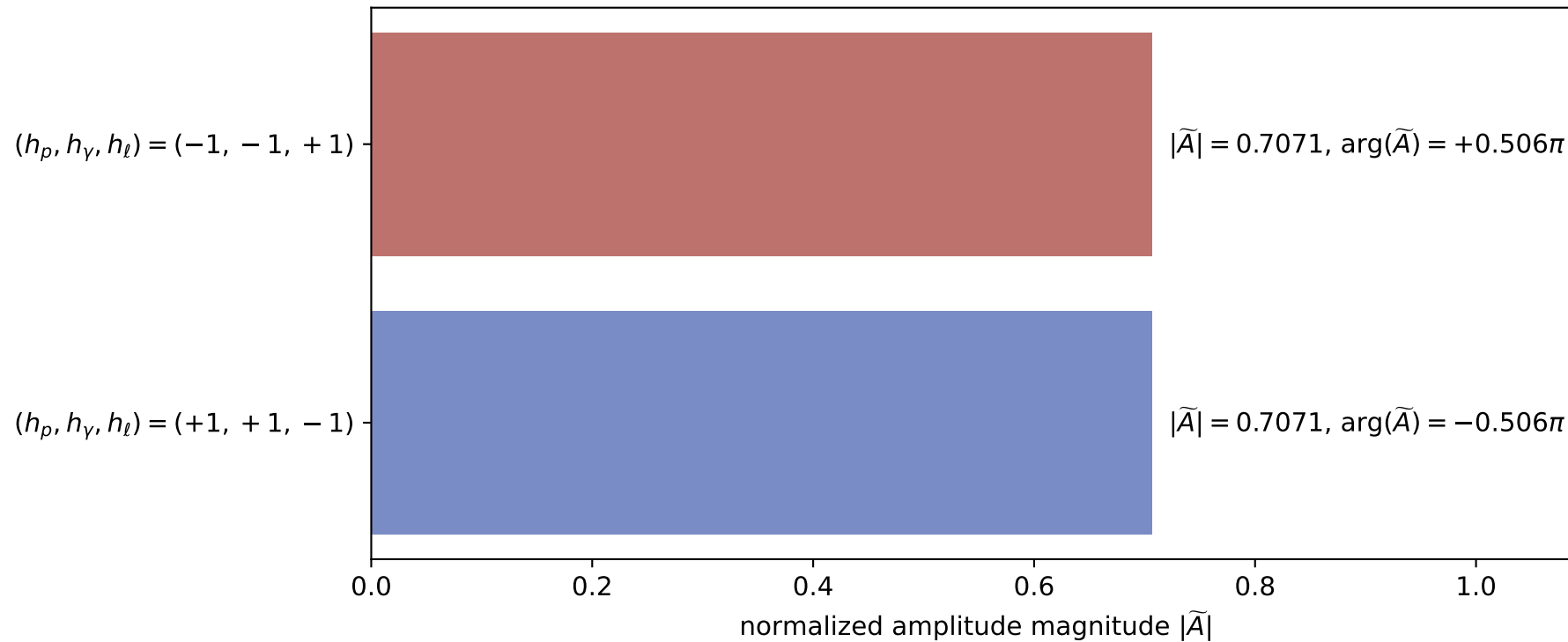
$s=24.9986$, $\sqrt{s}=4.99986$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.0112972$
 $\theta_{P'}=0.0085095$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.74098$, $\phi_\gamma=1.5897$
 $E_\gamma=2.03655$, $\theta_{\ell'}=4.7467\text{e-}05$, $\phi_{\ell'}=4.88257$
 $Q^2=19.5391$, $x_B=0.834792$, $t=-3.0411$, $W^2=4.74669$, $y=0.970447$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0252$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= 2.0252$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0001$ $p_z= 0.0113$
 q_γ $E= 2.0365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0365$

x--y plane

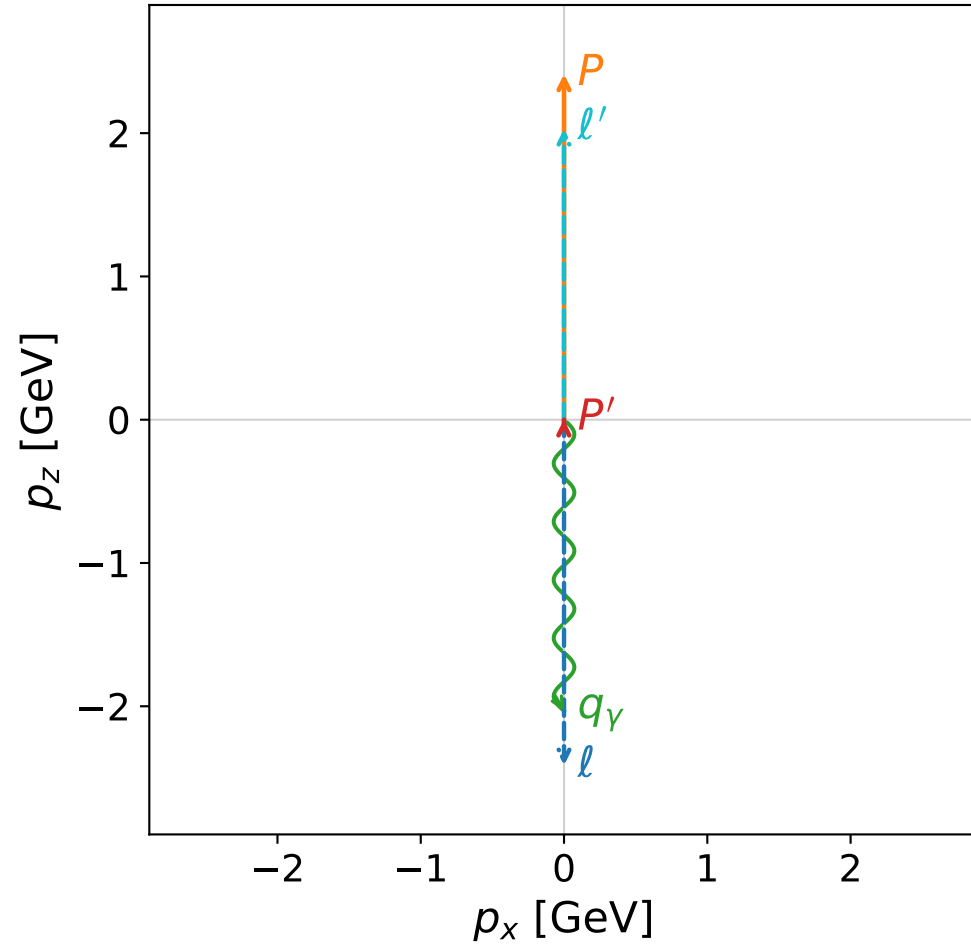


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

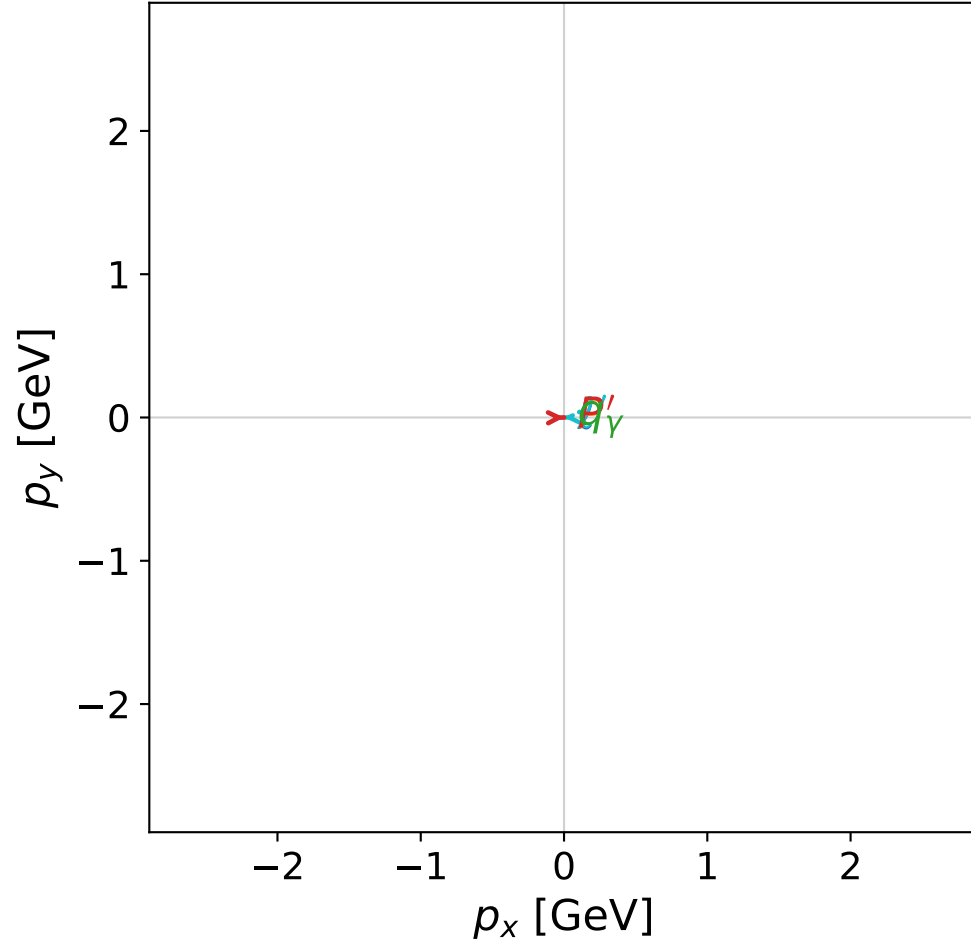


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



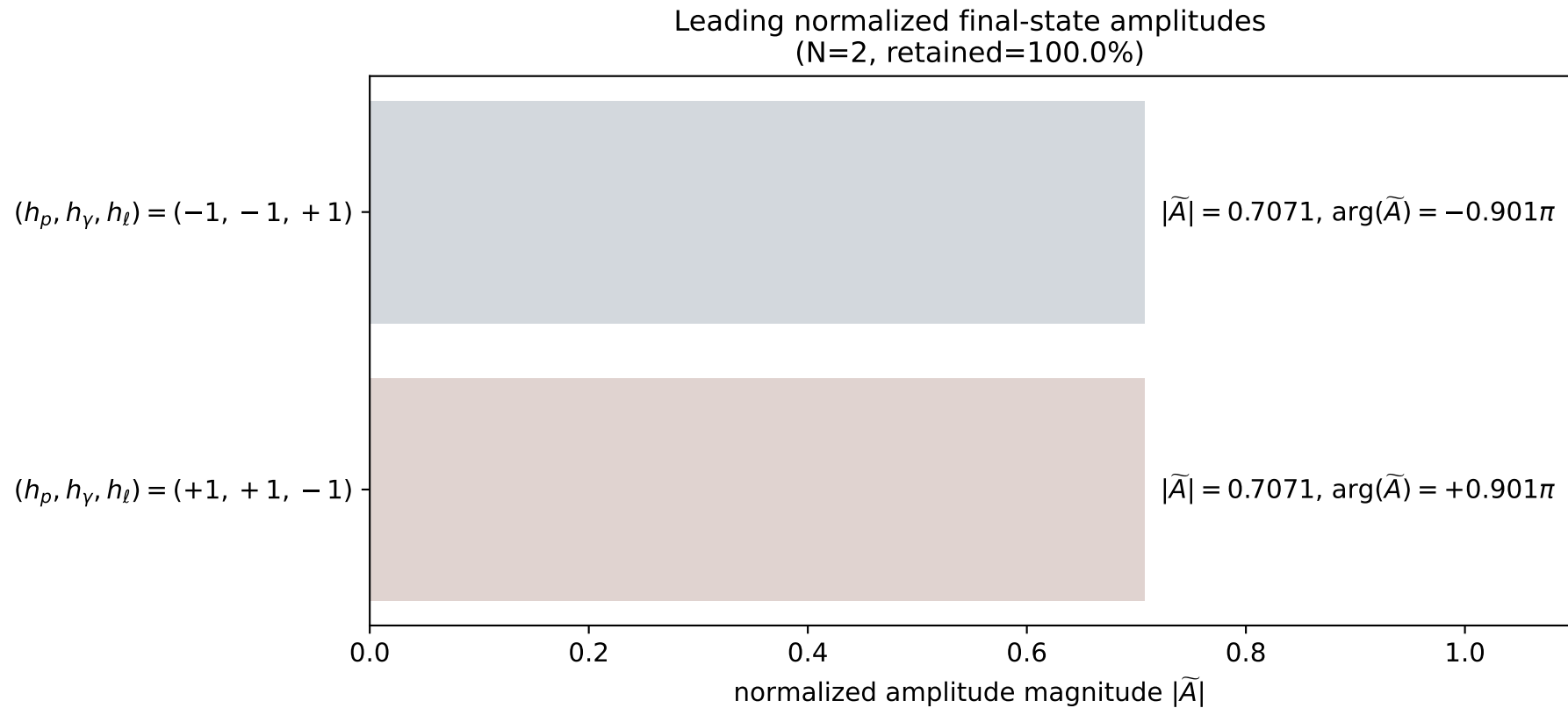
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_39
 $d_{\{\rm GHZ\}}=2.64421\text{e-}08$
 region: polarization_cluster_04_local_minimum_39
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0039
 lepton mixing angle: $\alpha_e=2.4732875$ rad
 incoming lepton: $-0.784873|+\rangle +0.619657|-\rangle$
 proton mixing angle: $\alpha_p=2.4732933$ rad
 incoming proton: $-0.784877|+\rangle +0.619652|-\rangle$

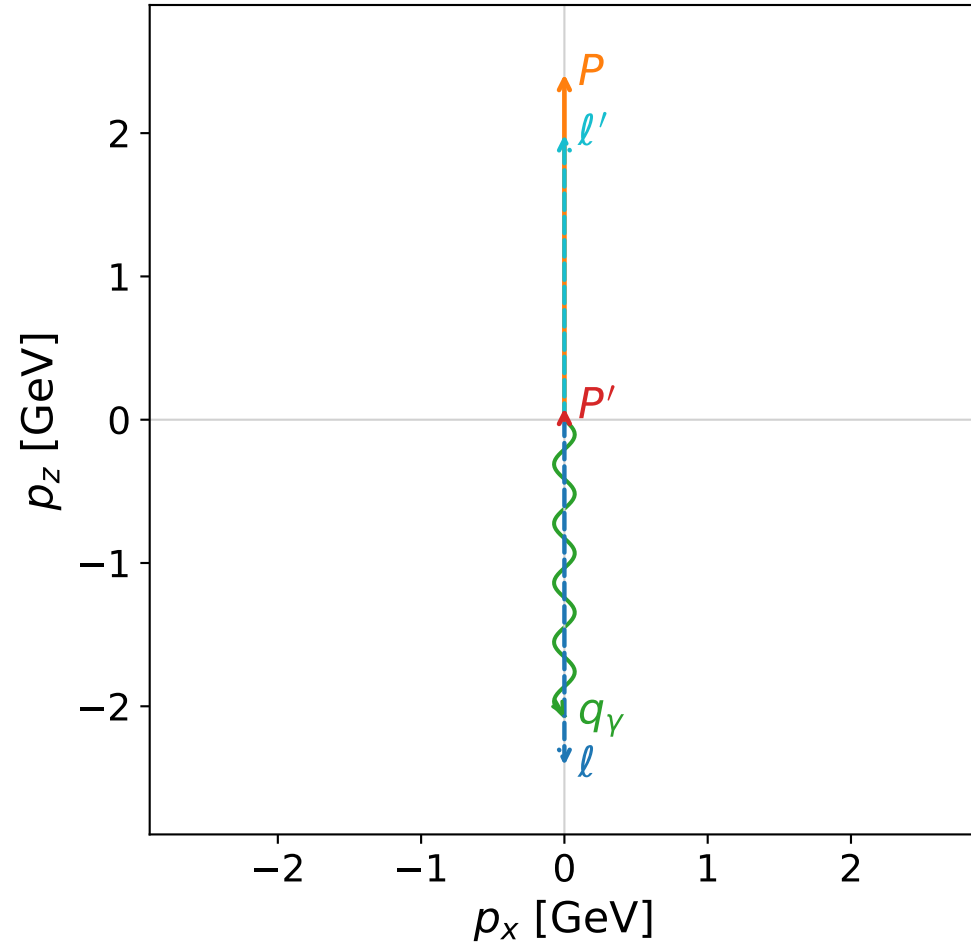
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00214662$
 $\theta_{p'}=0.00981748$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.019635$, $\phi_\gamma=3.45408$
 $E_\gamma=2.03207$, $\theta_{l'}=1.03817\text{e-}05$, $\phi_{l'}=3.16123$
 $Q^2=19.5848$, $x_B=0.836754$, $t=-3.08503$, $W^2=4.70074$, $y=0.970381$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0299$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0299$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0021$
 q_γ $E= 2.0321$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0321$

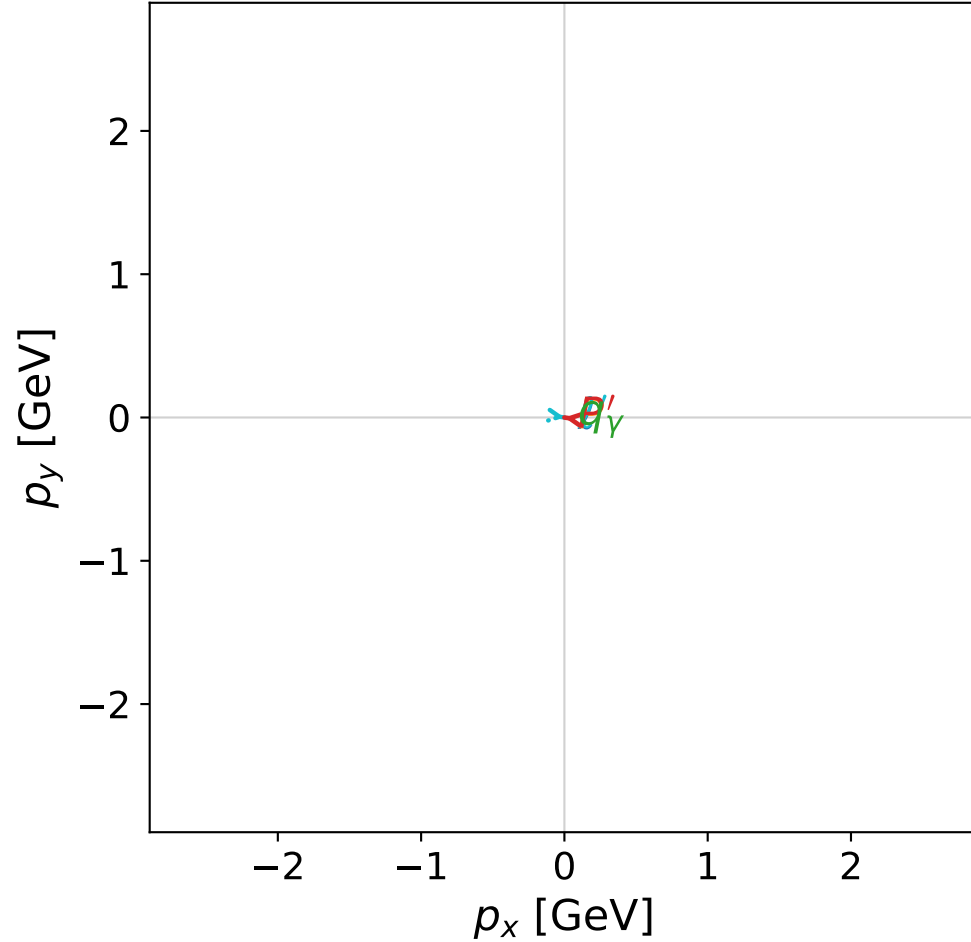


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

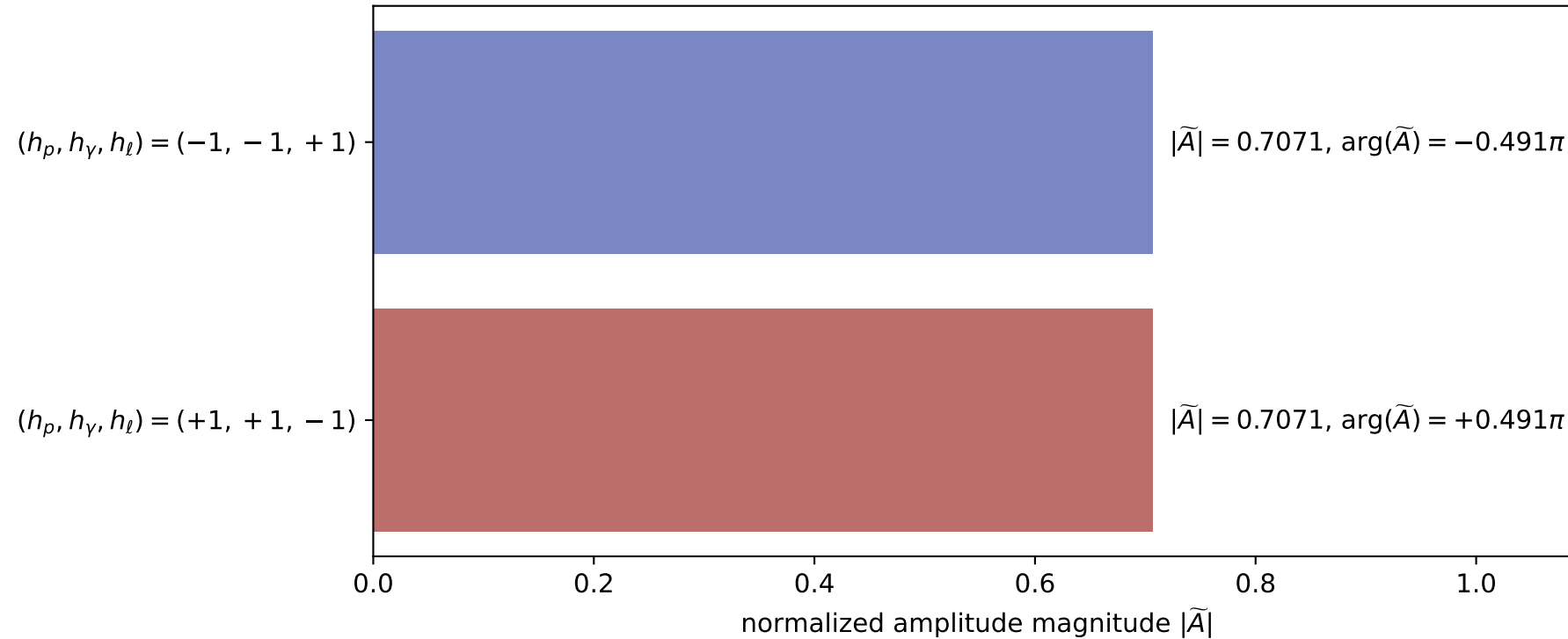


dghz_mixing_angles_polarization_04_local_minimum_50
 $d_{\text{GHZ}}=2.68327\text{e-}08$
 region: polarization_cluster_04_local_minimum_50
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0050
 lepton mixing angle: $\alpha_e=2.2455924$ rad
 incoming lepton: $-0.624738|+\rangle + 0.780834|-\rangle$
 proton mixing angle: $\alpha_p=2.2455942$ rad
 incoming proton: $-0.624739|+\rangle + 0.780833|-\rangle$

$s=24.9974$, $\sqrt{s}=4.99974$, $|\vec{P}|=2.41188$, $|\vec{P}'|=0.0814227$
 $\theta_{P'}=7.88291\text{e-}06$, $\theta_Y=3.14159$, $\phi_{P'}=2.99389$, $\phi_Y=4.74182$
 $E_Y=2.06982$, $\theta_{\ell'}=3.23049\text{e-}07$, $\phi_{\ell'}=6.13548$
 $Q^2=19.183$, $x_B=0.819169$, $t=-2.72062$, $W^2=5.11448$, $y=0.970983$

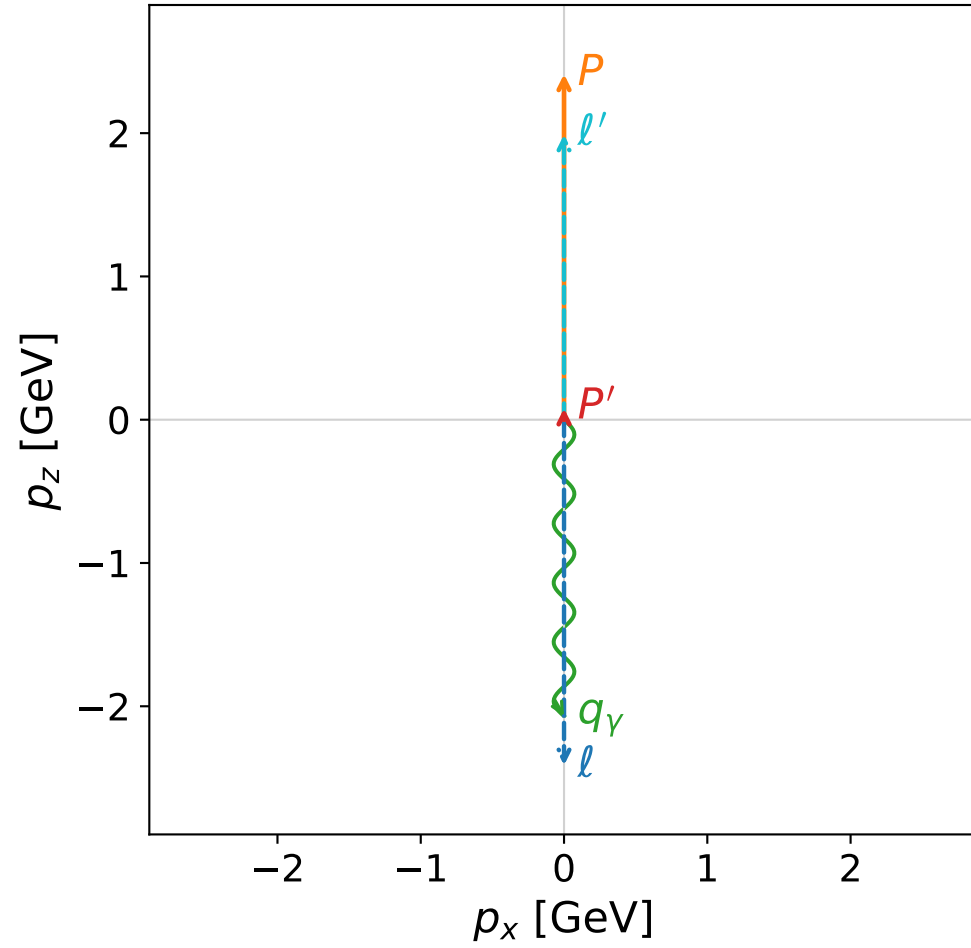
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 1.9884$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9884$
 P' $E= 0.9415$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0814$
 q_Y $E= 2.0698$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0698$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

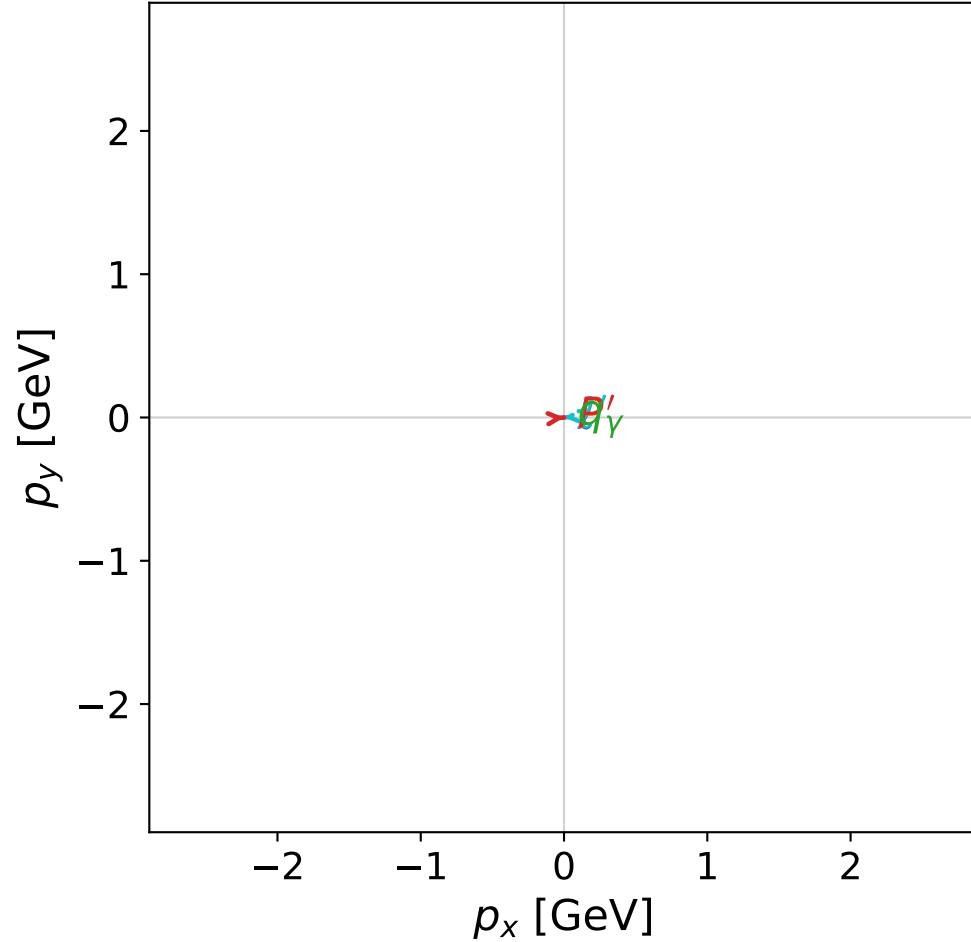


dghz_mixing_angles_polarization_04_local_minimum_59
 $d_{\{\rm GHZ\}}=2.72599\text{e-}08$
 region: polarization_cluster_04_local_minimum_59
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0059
 lepton mixing angle: $\alpha_e=2.4105855$ rad
 incoming lepton: $-0.744502|+\rangle +0.66762|-\rangle$
 proton mixing angle: $\alpha_p=2.4105852$ rad
 incoming proton: $-0.744502|+\rangle +0.66762|-\rangle$

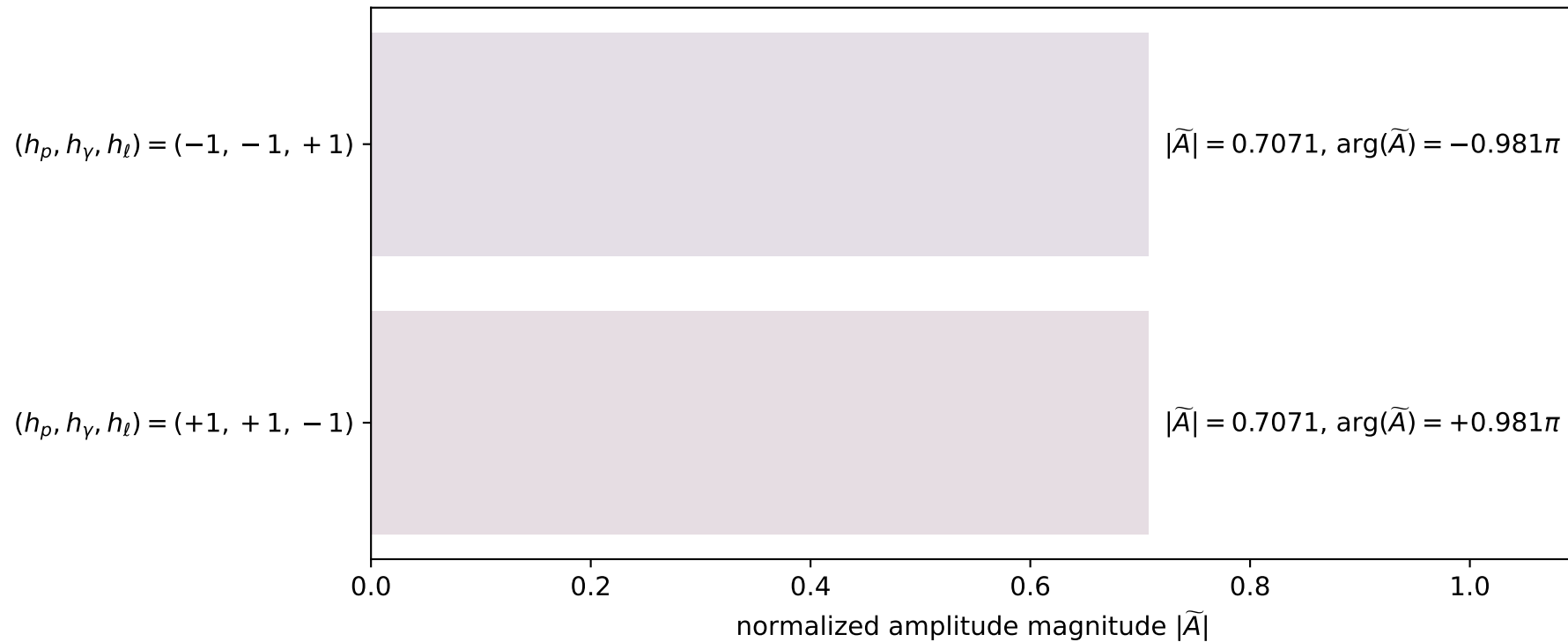
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0792932$
 $\theta_{P'}=0.00083419$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0772728$, $\phi_\gamma=3.20129$
 $E_\gamma=2.06897$, $\theta_{l'}=3.32443\text{e-}05$, $\phi_{l'}=3.21887$
 $Q^2=19.1966$, $x_B=0.819668$, $t=-2.73017$, $W^2=5.10319$, $y=0.970969$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 1.9897$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 1.9897$
 P' $E= 0.9413$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0793$
 q_γ $E= 2.0690$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0690$

x--y plane

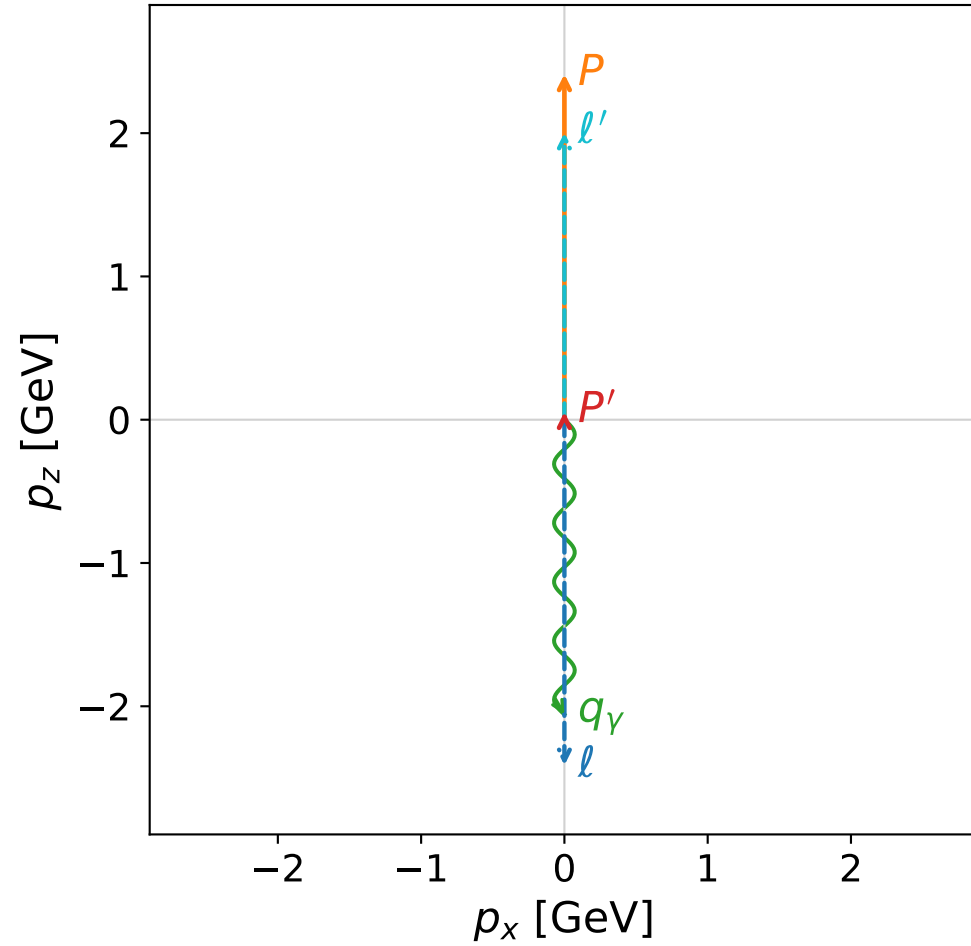


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

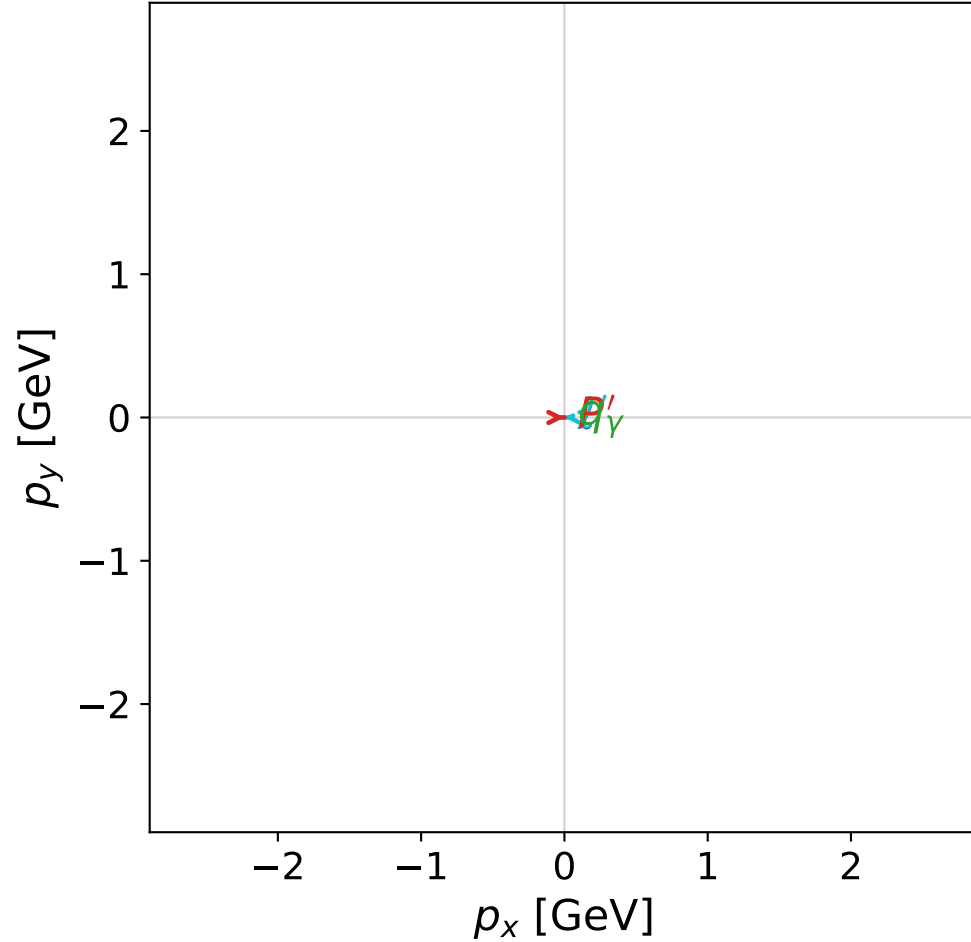


dghz_mixing_angles_polarization_04_local_minimum_63
 $d_{\text{GHZ}}=2.75718\text{e-}08$
 region: polarization_cluster_04_local_minimum_63
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0063
 lepton mixing angle: $\alpha_e=2.4612083$ rad
 incoming lepton: $-0.777331|+\rangle +0.629092|-\rangle$
 proton mixing angle: $\alpha_p=2.461193$ rad
 incoming proton: $-0.777321|+\rangle +0.629104|-\rangle$

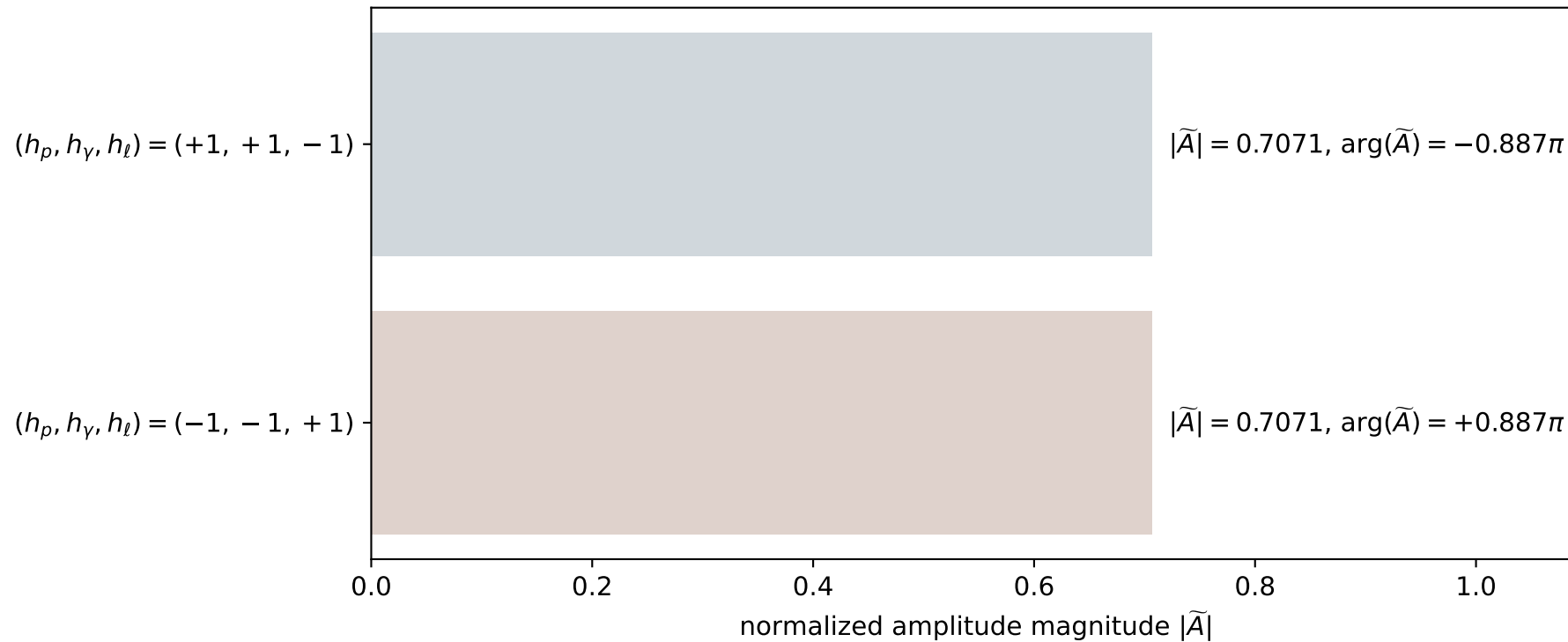
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0542678$
 $\theta_{P'}=0.00132352$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.28313$, $\phi_\gamma=2.78563$
 $E_\gamma=2.05735$, $\theta_{\ell'}=3.58569\text{e-}05$, $\phi_{\ell'}=3.14154$
 $Q^2=19.3259$, $x_B=0.825355$, $t=-2.8417$, $W^2=4.96918$, $y=0.970773$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0031$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0031$
 P' $E= 0.9396$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0543$
 q_γ $E= 2.0573$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0573$

x--y plane

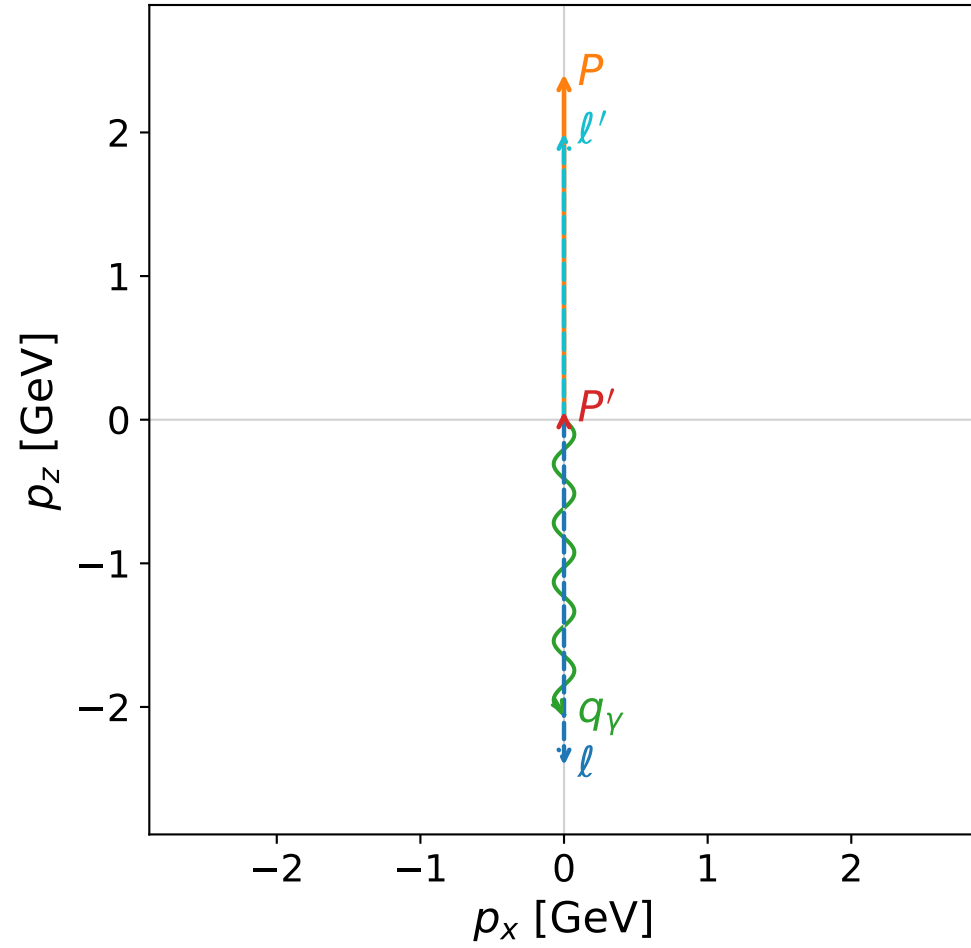


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

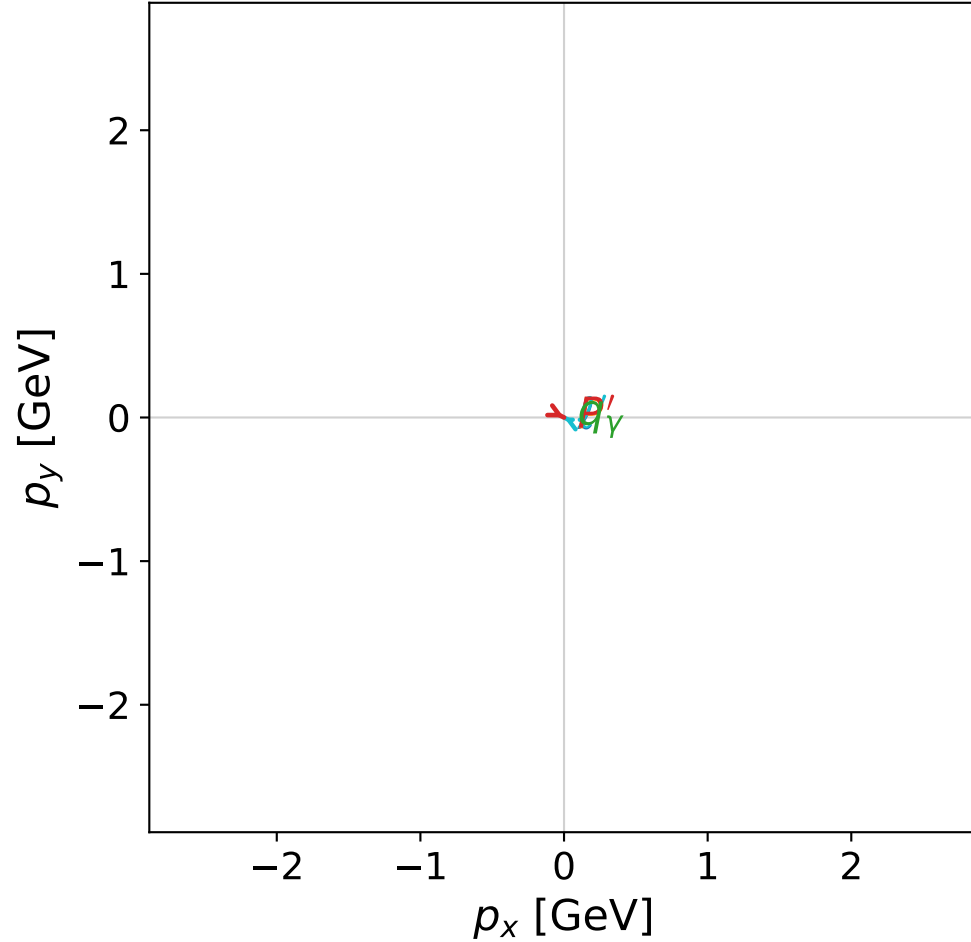


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



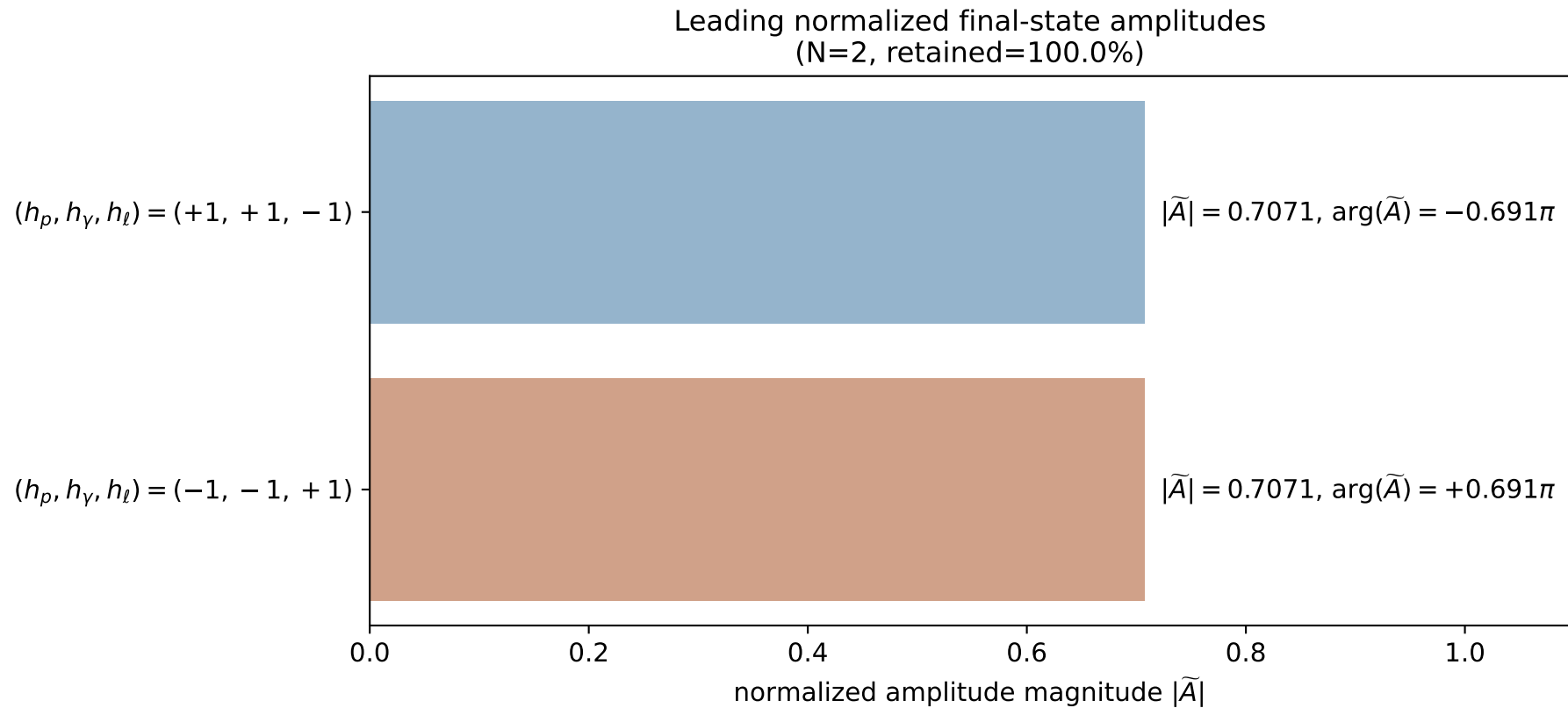
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_64
 $d_{\{\rm GHZ\}}=2.76858\text{e-}08$
 region: polarization_cluster_04_local_minimum_64
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0064
 lepton mixing angle: $\alpha_e=2.2443831$ rad
 incoming lepton: $-0.623793|+\rangle +0.781589|-\rangle$
 proton mixing angle: $\alpha_p=2.2443657$ rad
 incoming proton: $-0.62378|+\rangle +0.7816|-\rangle$

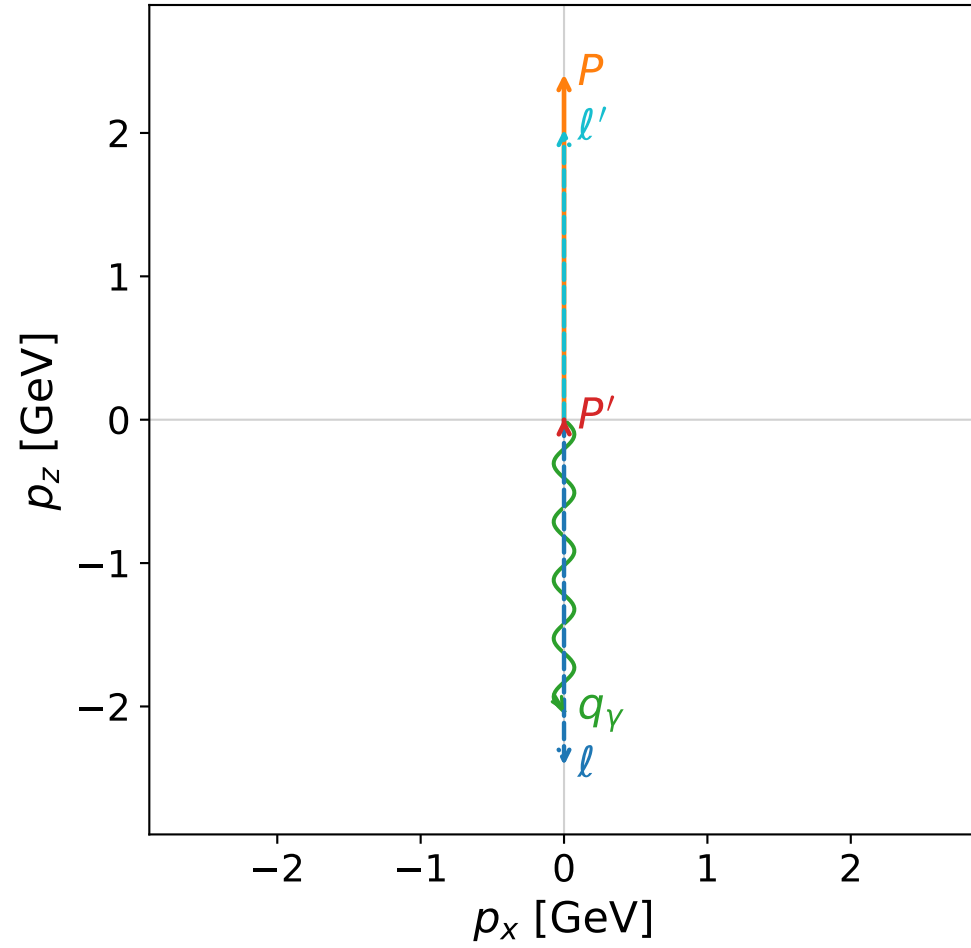
$s=24.887$, $\sqrt{s}=4.98869$, $|\vec{P}|=2.40616$, $|\vec{P}'|=0.0592441$
 $\theta_{P'}=7.76676\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.81389$, $\phi_\gamma=2.17126$
 $E_\gamma=2.05403$, $\theta_{\ell'}=2.30665\text{e-}06$, $\phi_{\ell'}=2.67229$
 $Q^2=19.1991$, $x_B=0.823871$, $t=-2.80969$, $W^2=4.98427$, $y=0.970691$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4062$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4062$
 P $E= 2.5825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4062$
 ℓ' $E= 1.9948$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9948$
 P' $E= 0.9399$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0592$
 q_γ $E= 2.0540$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0540$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

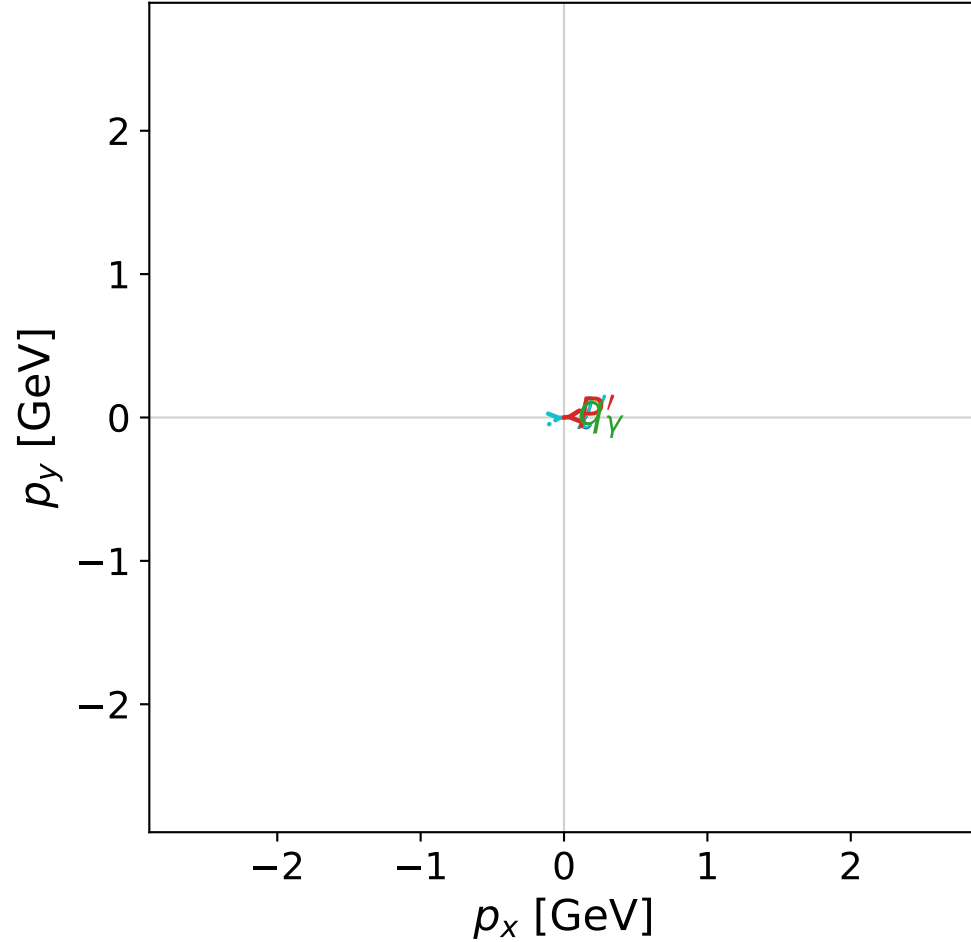


dghz_mixing_angles_polarization_04_local_minimum_74
 $d_{\{\rm GHZ\}}=2.93202\text{e-}08$
 region: polarization_cluster_04_local_minimum_74
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0074
 lepton mixing angle: $\alpha_e=2.1227555$ rad
 incoming lepton: $-0.524356|+\rangle +0.851499|-\rangle$
 proton mixing angle: $\alpha_p=2.12275$ rad
 incoming proton: $-0.524352|+\rangle +0.851502|-\rangle$

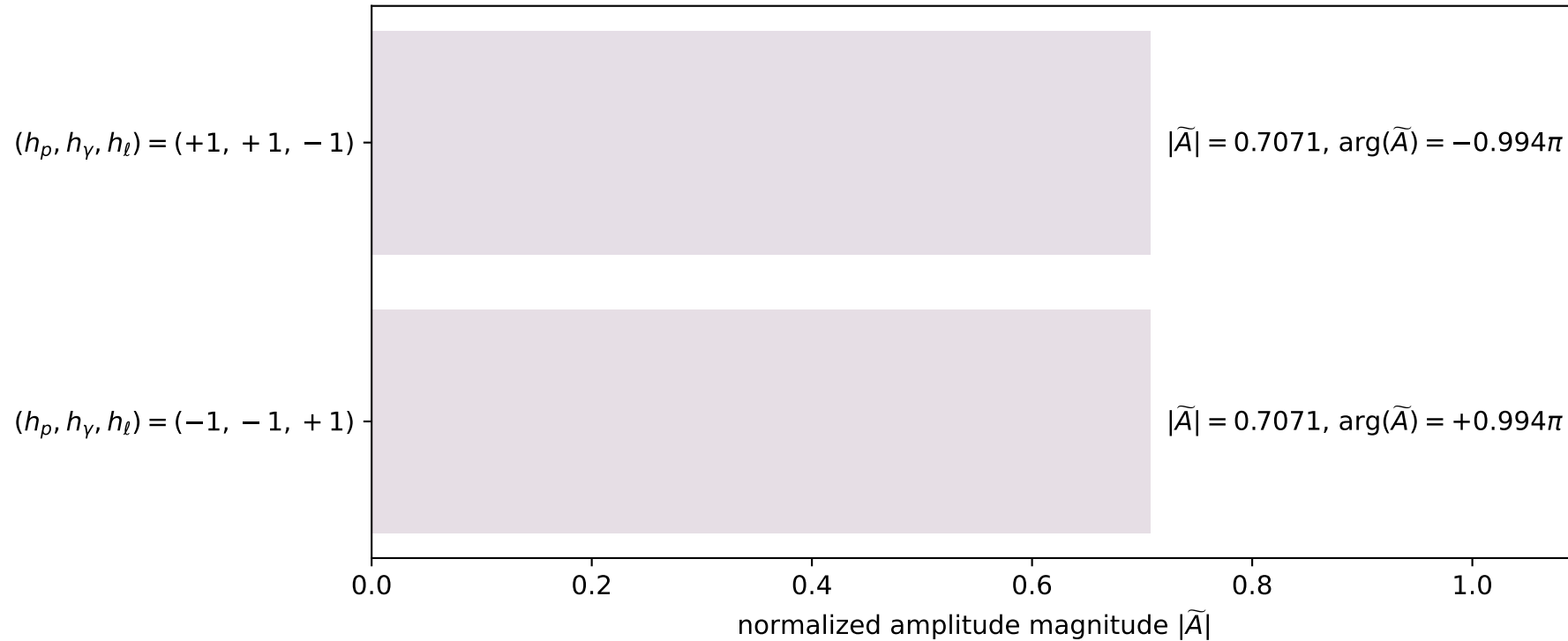
$s=24.9688$, $\sqrt{s}=4.99688$, $|\vec{P}|=2.4104$, $|\vec{P}'|=0.00957823$
 $\theta_{P'}=0.00965936$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.2353$, $\phi_\gamma=3.12295$
 $E_\gamma=2.0342$, $\theta_{\ell'}=4.56964\text{e-}05$, $\phi_{\ell'}=0.0937025$
 $Q^2=19.5206$, $x_B=0.835073$, $t=-3.04662$, $W^2=4.73517$, $y=0.970402$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4104$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4104$
 P $E= 2.5865$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4104$
 ℓ' $E= 2.0246$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.0246$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 0.0096$
 q_γ $E= 2.0342$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0342$

x--y plane

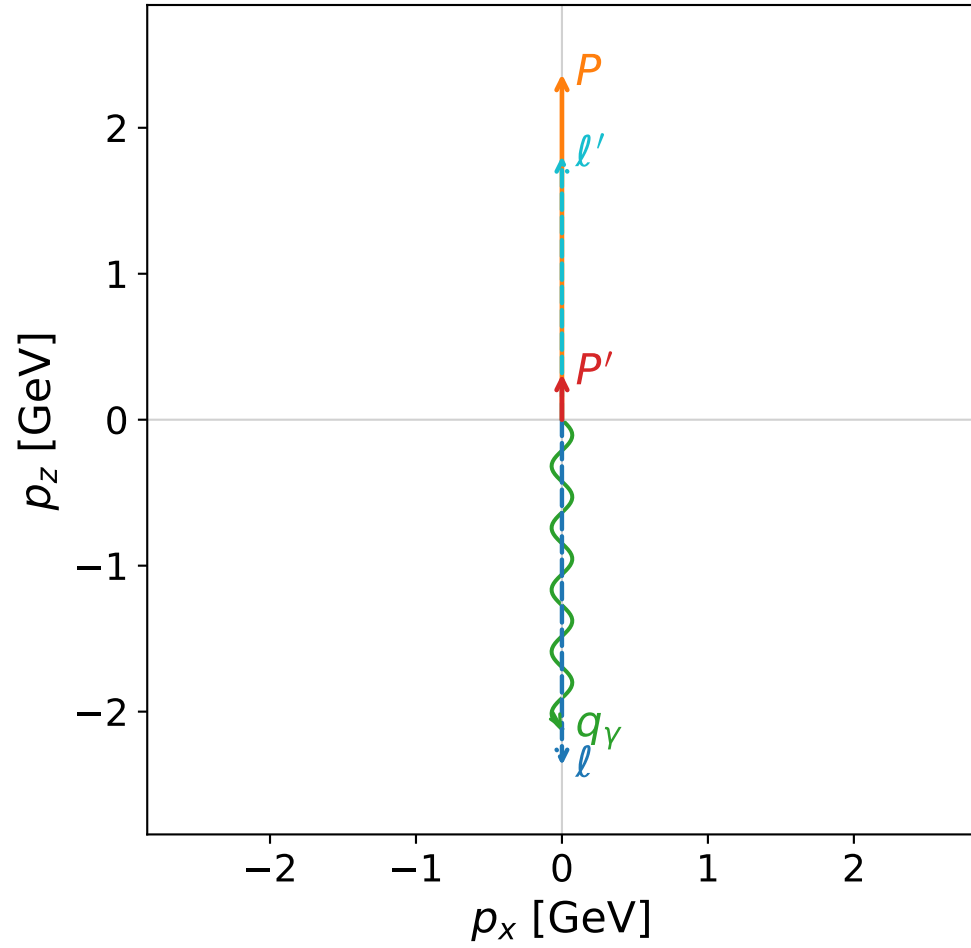


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

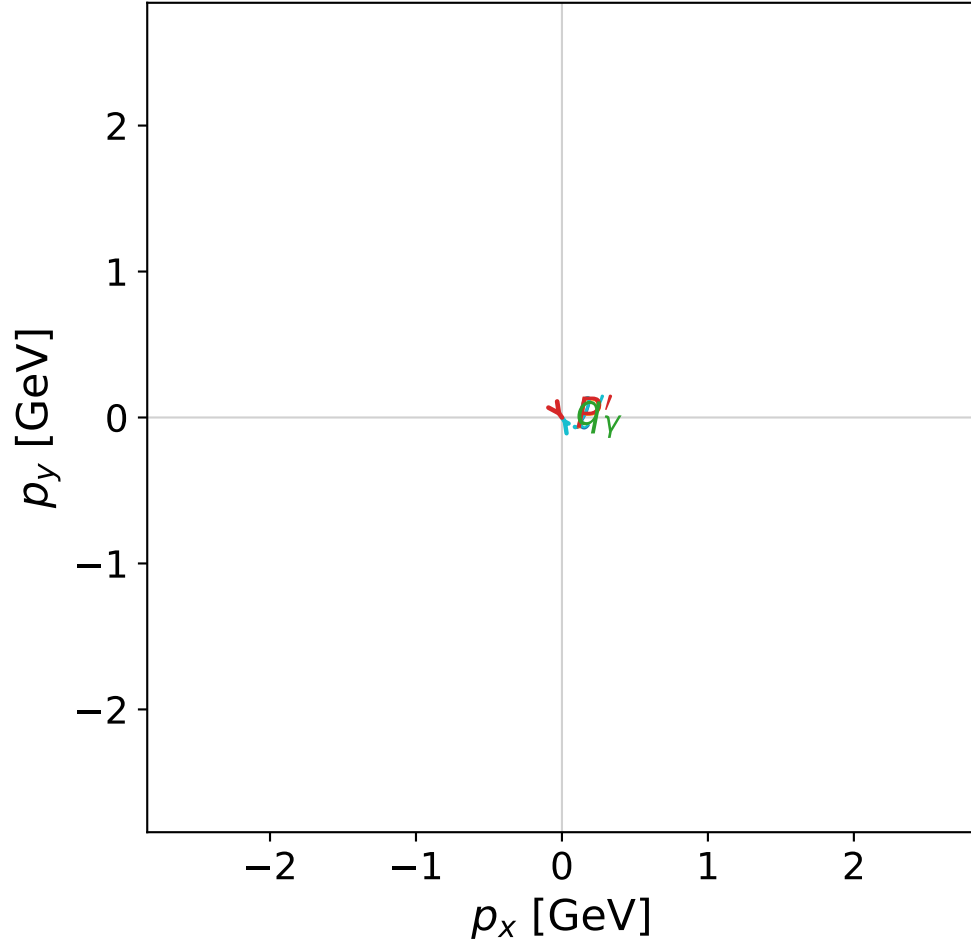


dghz_mixing_angles_polarization_04_local_minimum_80
 $d_{\text{GHZ}}=3.02153\text{e-}08$
 region: polarization_cluster_04_local_minimum_80
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0080
 lepton mixing angle: $\alpha_e=2.3404957$ rad
 incoming lepton: $-0.695919|+\rangle + 0.71812|-\rangle$
 proton mixing angle: $\alpha_p=2.3404627$ rad
 incoming proton: $-0.695896|+\rangle + 0.718143|-\rangle$

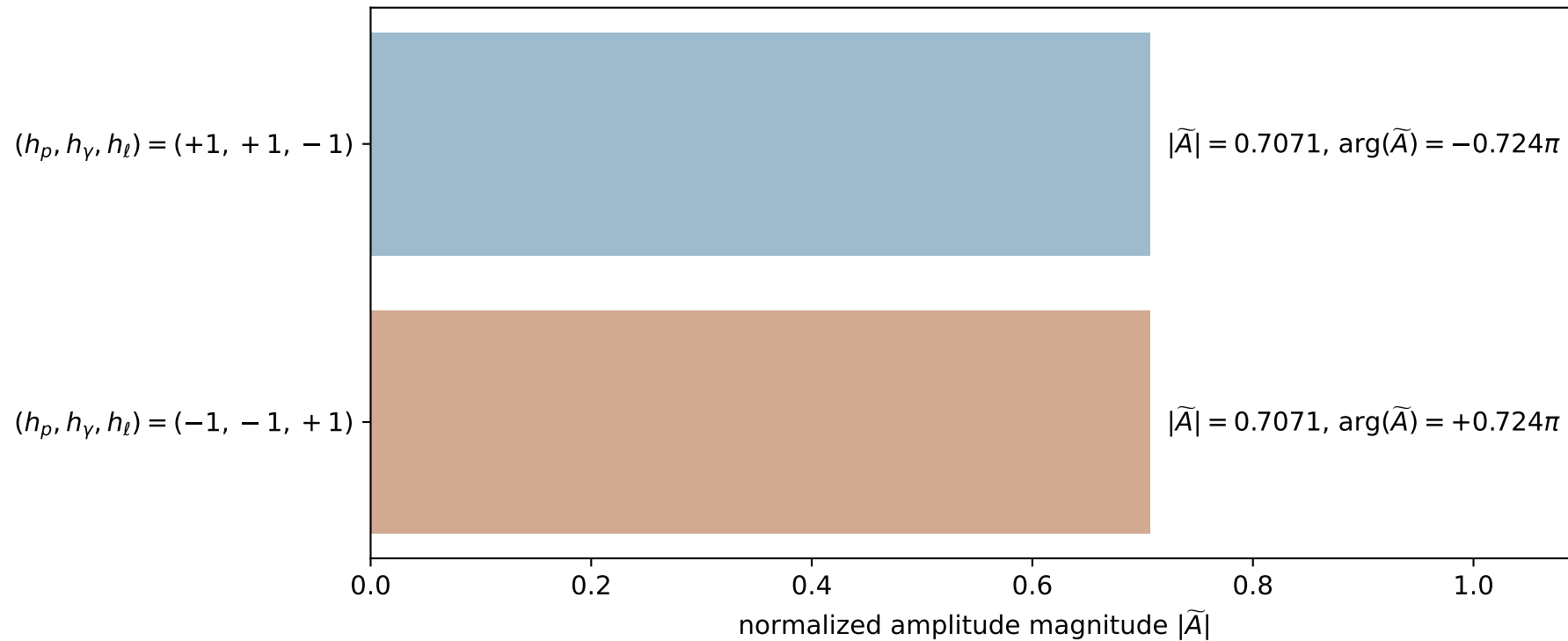
$s=24.1511$, $\sqrt{s}=4.91438$, $|\vec{P}|=2.36767$, $|\vec{P}'|=0.31225$
 $\theta_{P'}=9.71376\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.3268$, $\phi_\gamma=2.27603$
 $E_\gamma=2.11901$, $\theta_{l'}=1.67875\text{e-}06$, $\phi_{l'}=2.18521$
 $Q^2=17.1113$, $x_B=0.756322$, $t=-1.79708$, $W^2=6.39291$, $y=0.9722$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3677$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3677$
 P $E= 2.5467$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3677$
 l' $E= 1.8068$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8068$
 P' $E= 0.9886$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3123$
 q_γ $E= 2.1190$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1190$

x--y plane

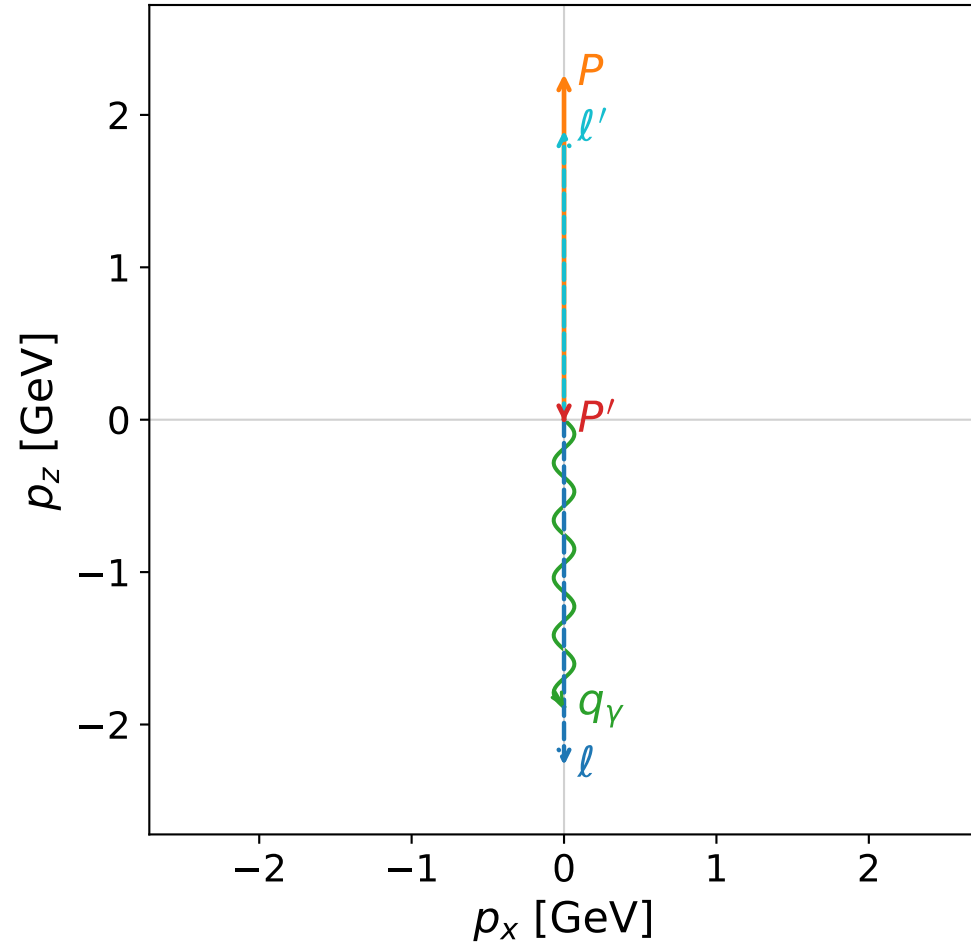


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

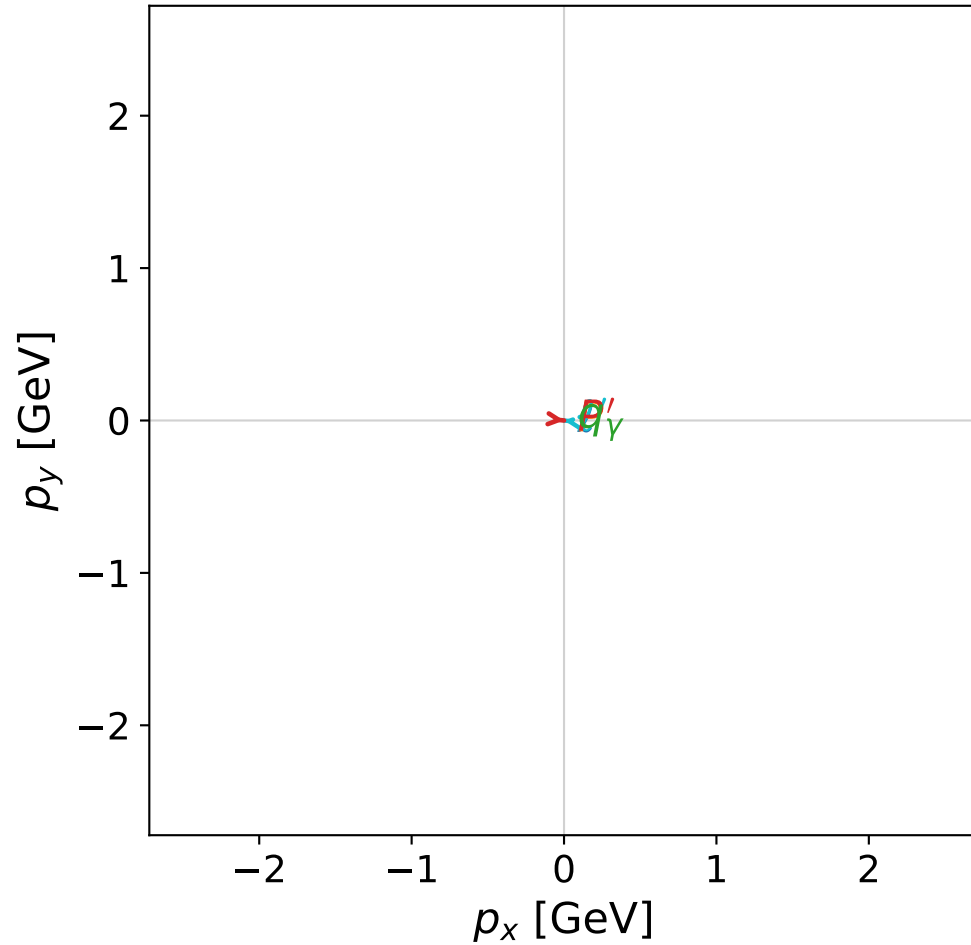


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



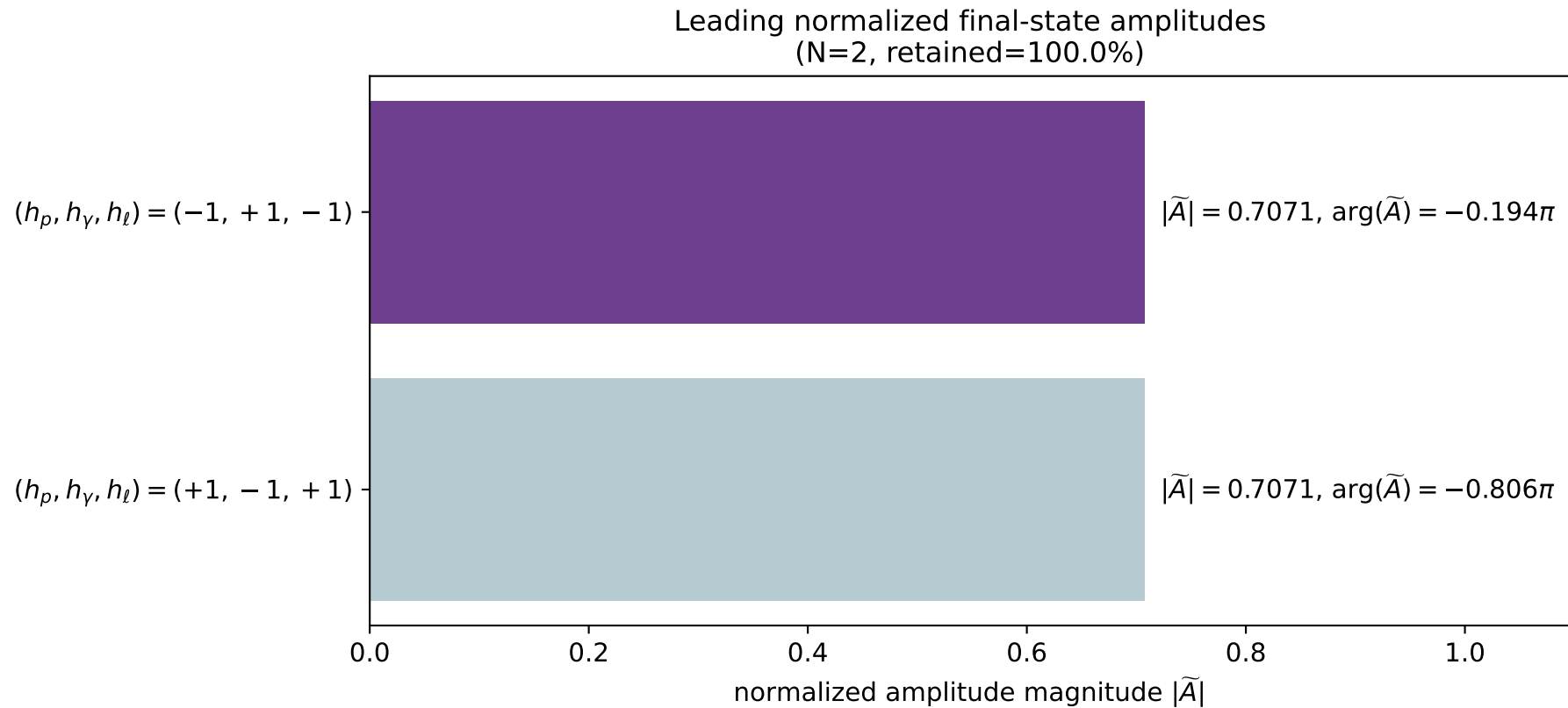
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_81
 $d_{\{\rm GHZ\}}=3.03827\text{e-}08$
 region: polarization_cluster_04_local_minimum_81
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0081
 lepton mixing angle: $\alpha_e=2.3663814$ rad
 incoming lepton: $-0.714273|+\rangle +0.699867|-\rangle$
 proton mixing angle: $\alpha_p=2.3663611$ rad
 incoming proton: $-0.714259|+\rangle +0.699881|-\rangle$

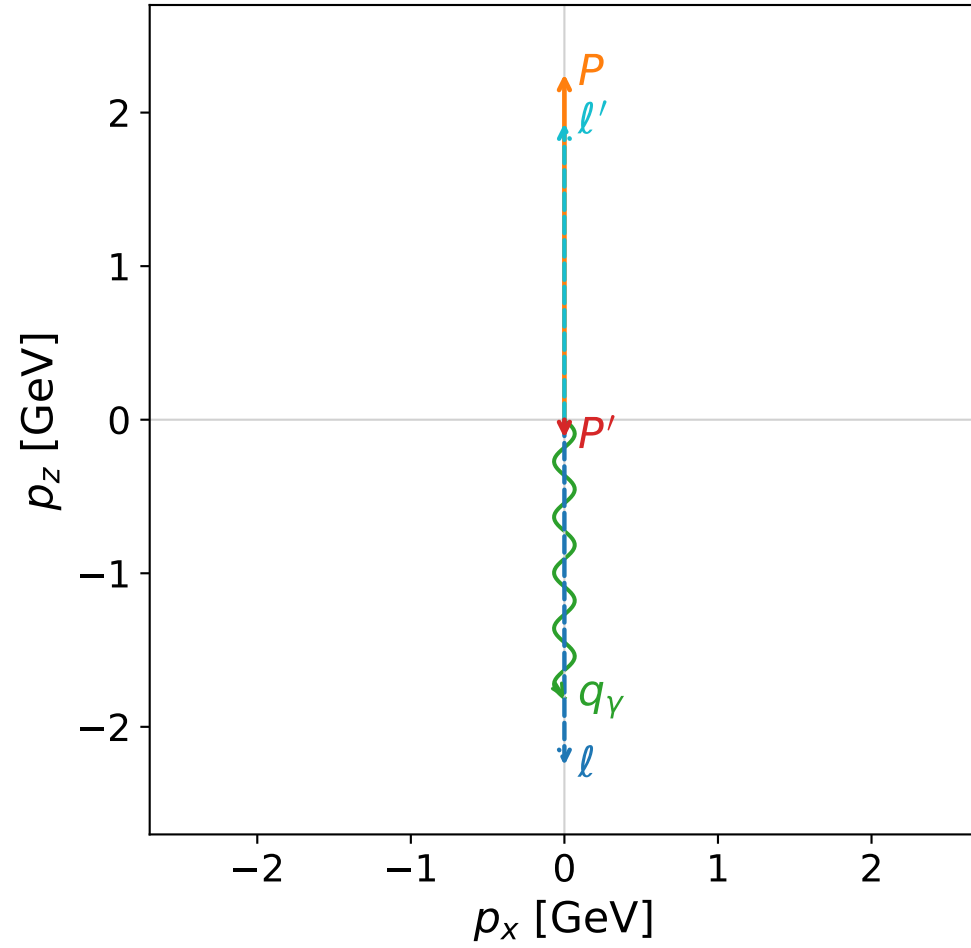
$s=22.294$, $\sqrt{s}=4.72165$, $|\vec{P}|=2.26765$, $|\vec{P}'|=0.0124921$
 $\theta_{p'}=3.14148$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.17498$, $\phi_\gamma=3.64321$
 $E_\gamma=1.88554$, $\theta_{\ell'}=7.48922\text{e-}07$, $\phi_{\ell'}=3.03338$
 $Q^2=17.2163$, $x_B=0.831433$, $t=-2.90107$, $W^2=4.37032$, $y=0.966967$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2677$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2677$
 P $E= 2.4540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2677$
 ℓ' $E= 1.8980$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8980$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0125$
 q_γ $E= 1.8855$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8855$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

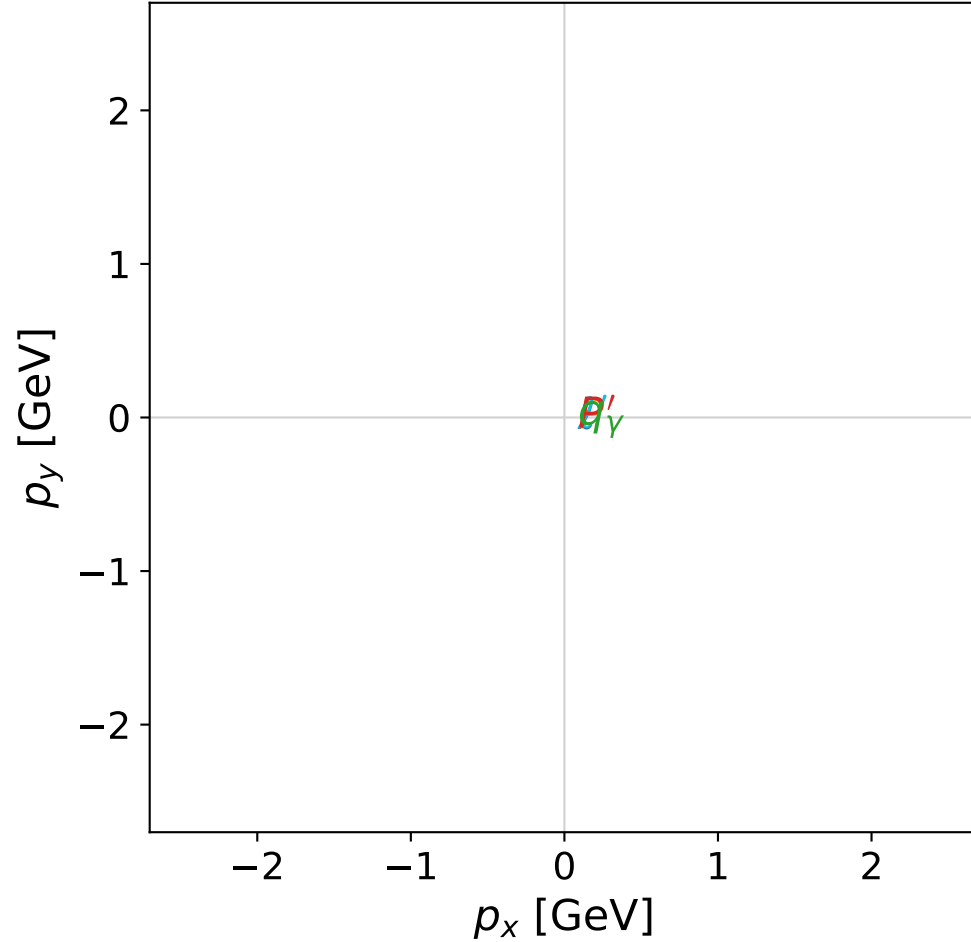


dghz_mixing_angles_polarization_04_local_minimum_82
 $d_{\text{GHZ}}=3.04333\text{e-}08$
 region: polarization_cluster_04_local_minimum_82
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0082
 lepton mixing angle: $\alpha_e=2.2763558$ rad
 incoming lepton: $-0.64846|+\rangle +0.761249|-\rangle$
 proton mixing angle: $\alpha_p=2.2763635$ rad
 incoming proton: $-0.648466|+\rangle +0.761244|-\rangle$

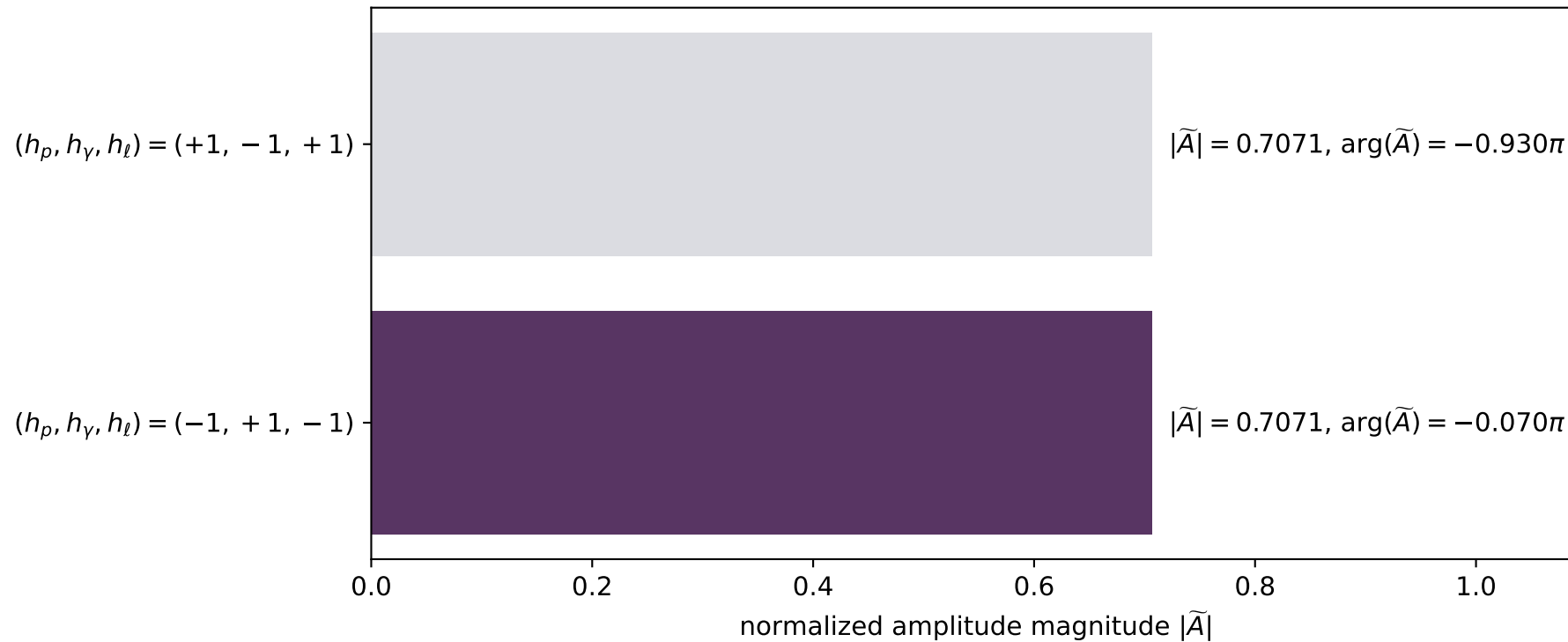
$s=21.9839$, $\sqrt{s}=4.68869$, $|\vec{P}|=2.25052$, $|\vec{P}'|=0.118469$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=3.11906$, $\phi_Y=3.36218$
 $E_Y=1.81239$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.205792$
 $Q^2=17.3817$, $x_B=0.852909$, $t=-3.3839$, $W^2=3.87747$, $y=0.965662$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2505$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2505$
 P $E= 2.4382$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2505$
 ℓ' $E= 1.9309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9309$
 P' $E= 0.9455$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1185$
 q_Y $E= 1.8124$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8124$

x--y plane

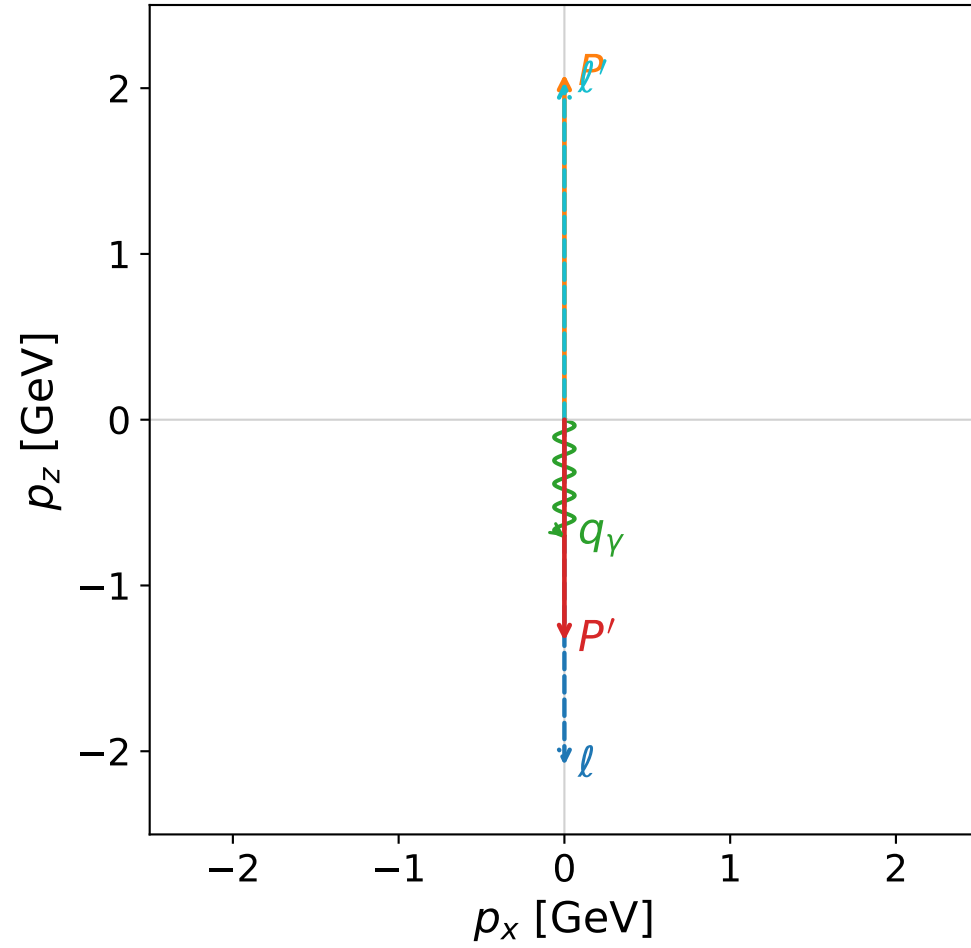


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

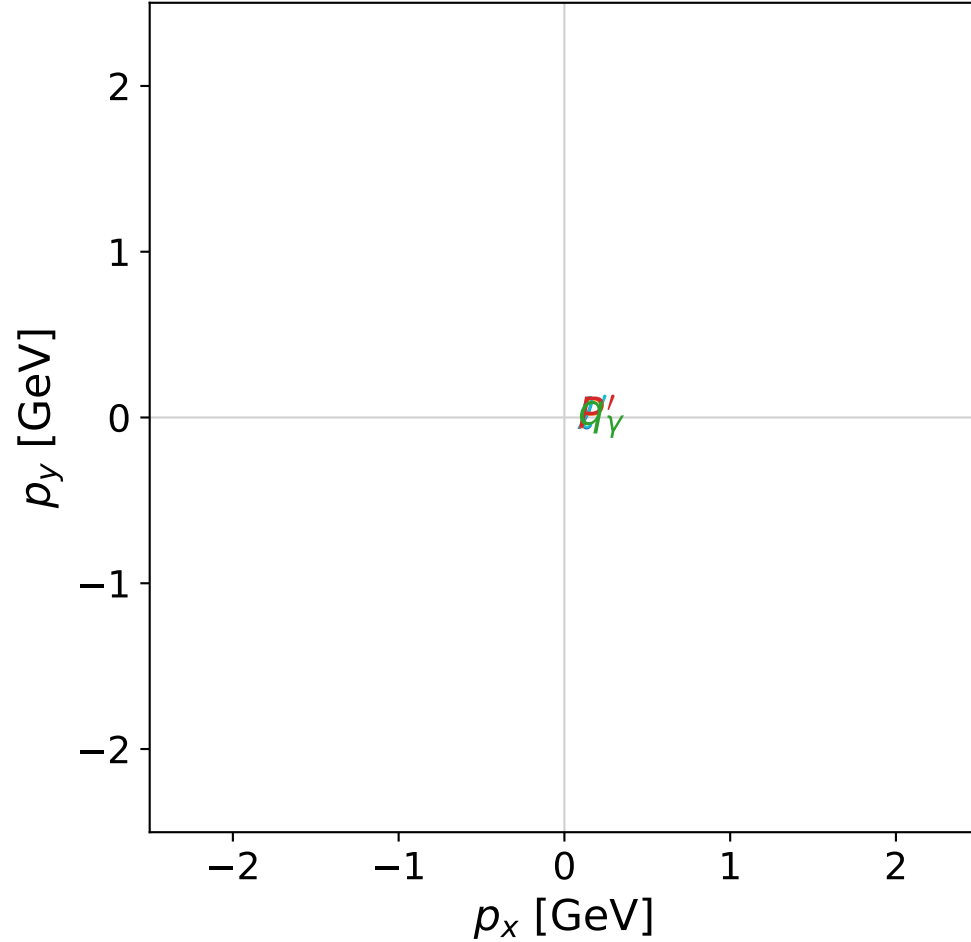


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



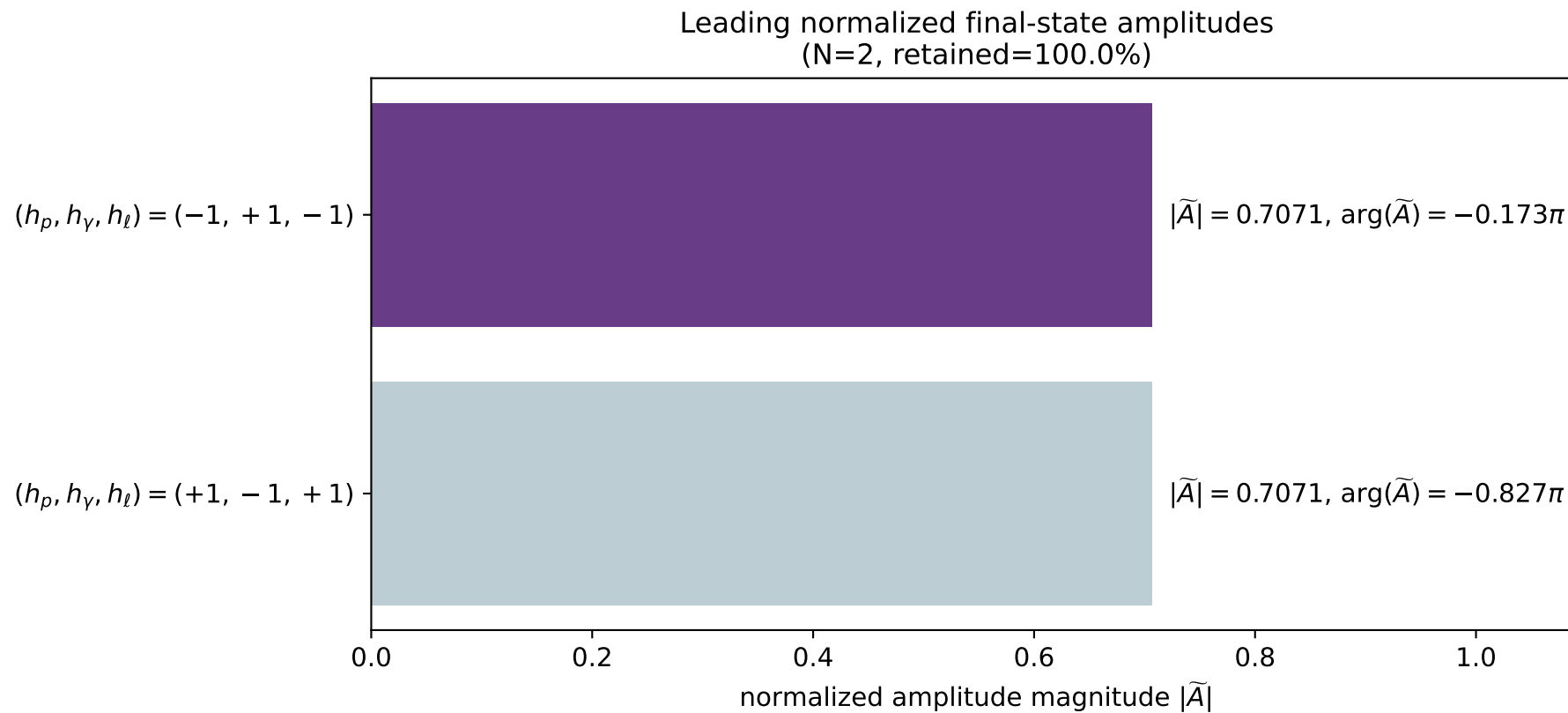
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_84
 $d_{\text{GHZ}}=3.10329\text{e-}08$
 region: polarization_cluster_04_local_minimum_84
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0084
 lepton mixing angle: $\alpha_e=2.2903682$ rad
 incoming lepton: $-0.659063|+\rangle +0.752088|-\rangle$
 proton mixing angle: $\alpha_p=2.2903511$ rad
 incoming proton: $-0.65905|+\rangle +0.752099|-\rangle$

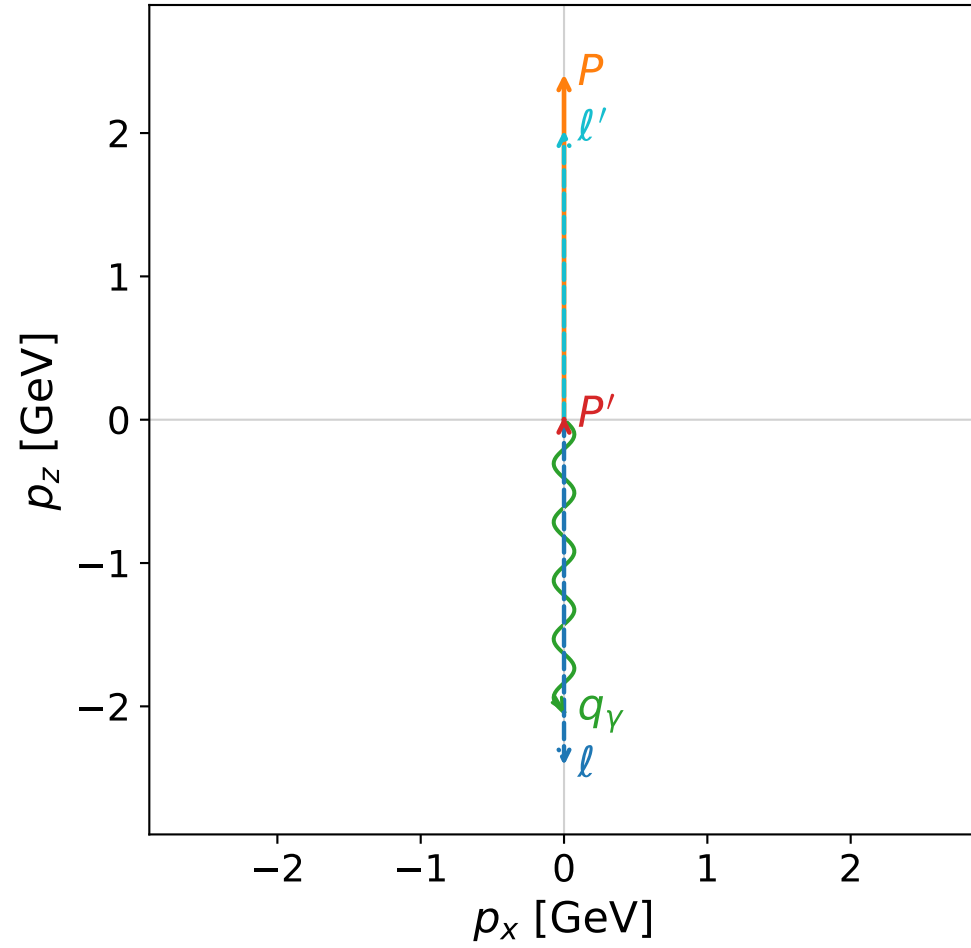
$s=19.1051$, $\sqrt{s}=4.37093$, $|\vec{P}|=2.08482$, $|\vec{P}'|=1.33443$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0100816$, $\phi_\gamma=3.68413$
 $E_\gamma=0.702689$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.69661$
 $Q^2=16.9881$, $x_B=0.976044$, $t=-11.2623$, $W^2=1.2968$, $y=0.955001$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0848$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0848$
 P $E= 2.2861$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0848$
 ℓ' $E= 2.0371$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0371$
 P' $E= 1.6311$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3344$
 q_γ $E= 0.7027$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.7027$

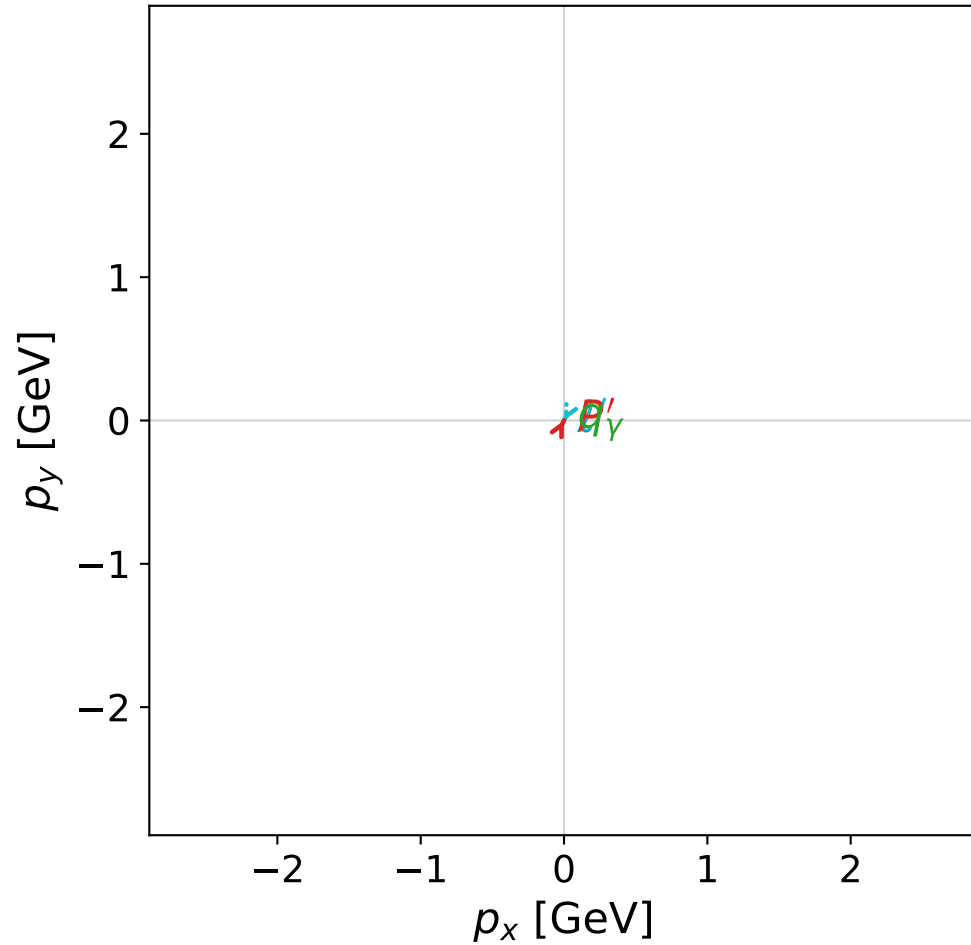


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



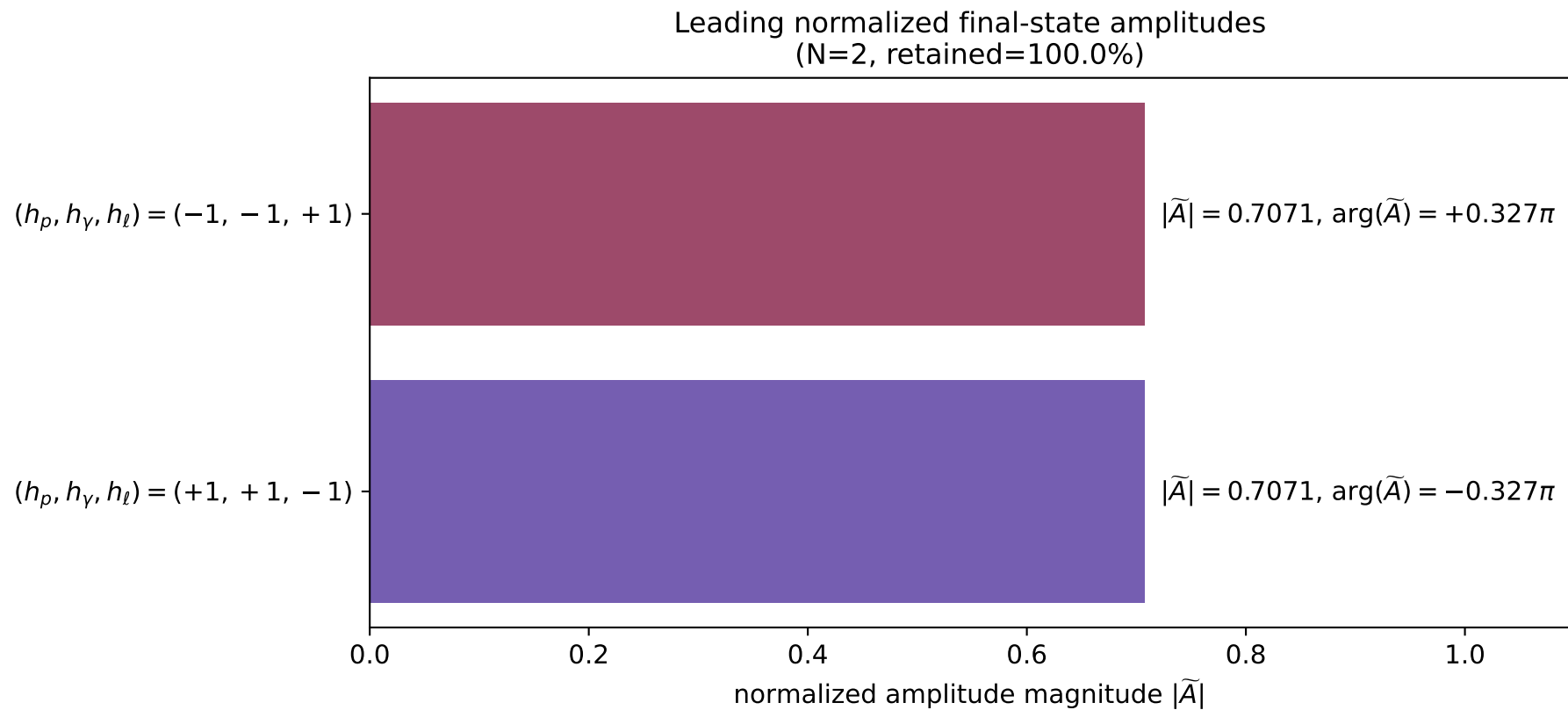
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_85
 $d_{\text{GHZ}}=3.10962\text{e-}08$
 region: polarization_cluster_04_local_minimum_85
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0085
 lepton mixing angle: $\alpha_e=2.1008763$ rad
 incoming lepton: $-0.505602|+\rangle +0.862767|-\rangle$
 proton mixing angle: $\alpha_p=2.1008833$ rad
 incoming proton: $-0.505608|+\rangle +0.862763|-\rangle$

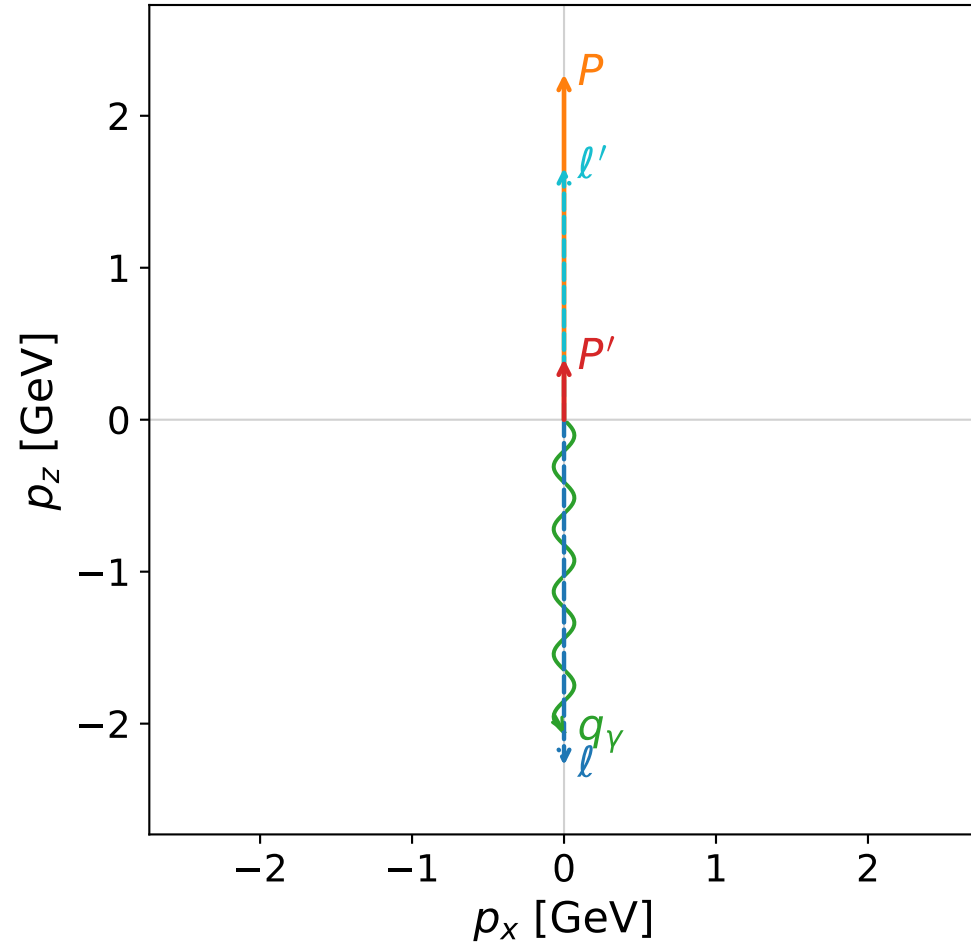
$s=24.9853$, $\sqrt{s}=4.99853$, $|\vec{P}|=2.41125$, $|\vec{P}'|=0.0216116$
 $\theta_{P'}=0.00498868$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.09327$, $\phi_\gamma=1.0281$
 $E_\gamma=2.04095$, $\theta_{\ell'}=5.33904\text{e-}05$, $\phi_{\ell'}=4.23487$
 $Q^2=19.4765$, $x_B=0.832523$, $t=-2.99111$, $W^2=4.79789$, $y=0.970509$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 2.0193$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 2.0193$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0216$
 q_γ $E= 2.0409$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0409$

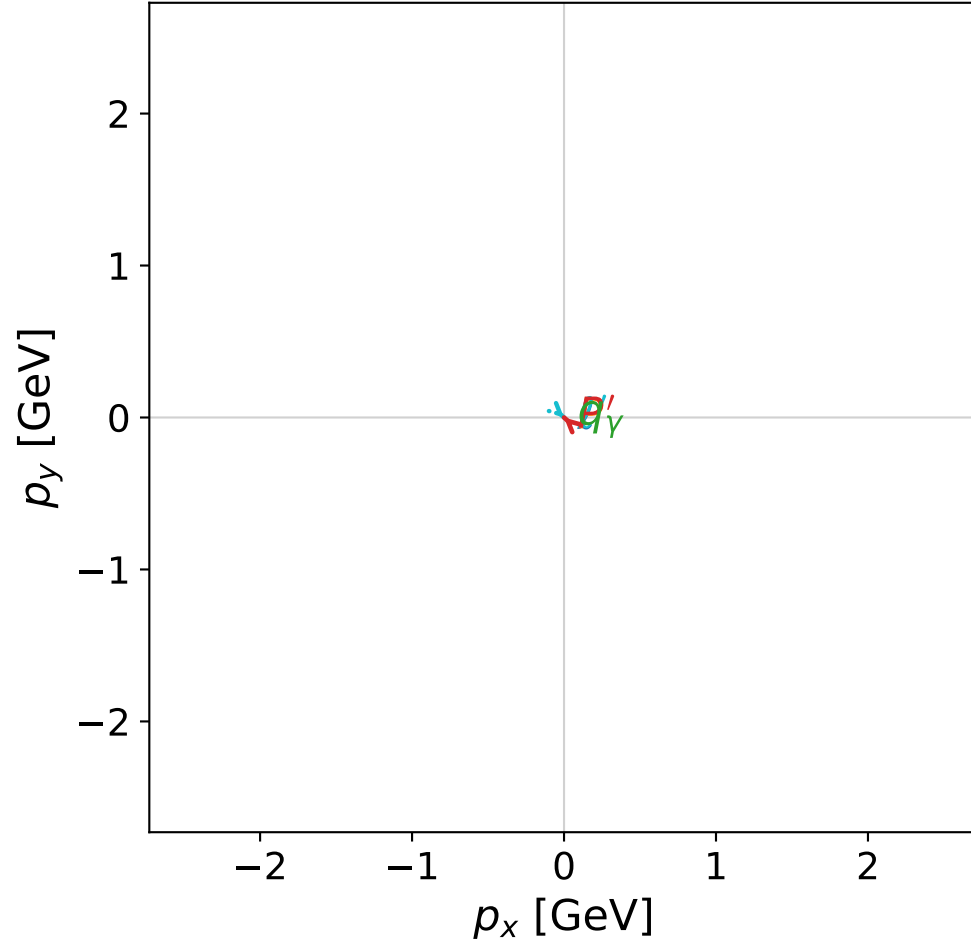


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



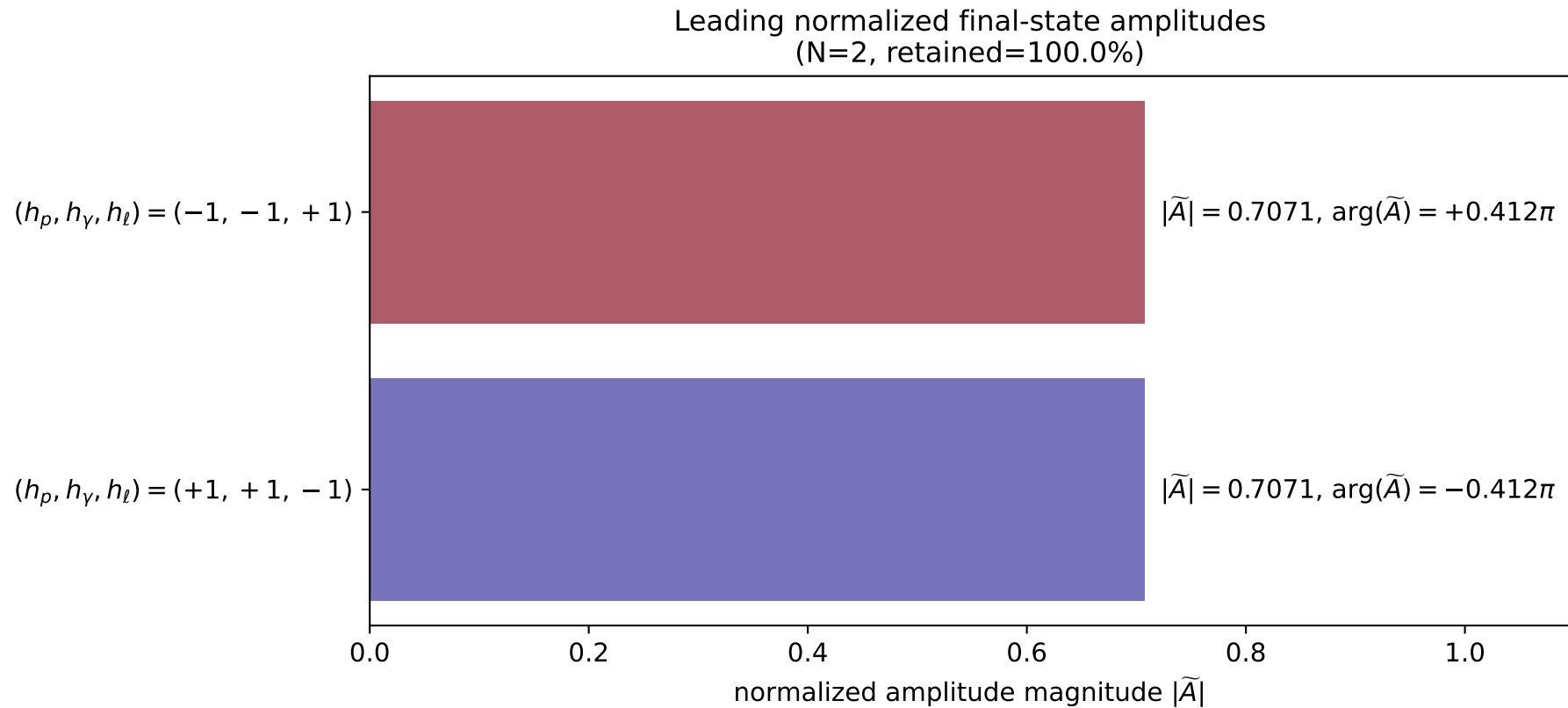
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_88
 $d_{\text{GHZ}}=3.20878\text{e-}08$
 region: polarization_cluster_04_local_minimum_88
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0088
 lepton mixing angle: $\alpha_e=2.2717872$ rad
 incoming lepton: $-0.644975|+\rangle +0.764203|-\rangle$
 proton mixing angle: $\alpha_p=2.2717877$ rad
 incoming proton: $-0.644976|+\rangle +0.764203|-\rangle$

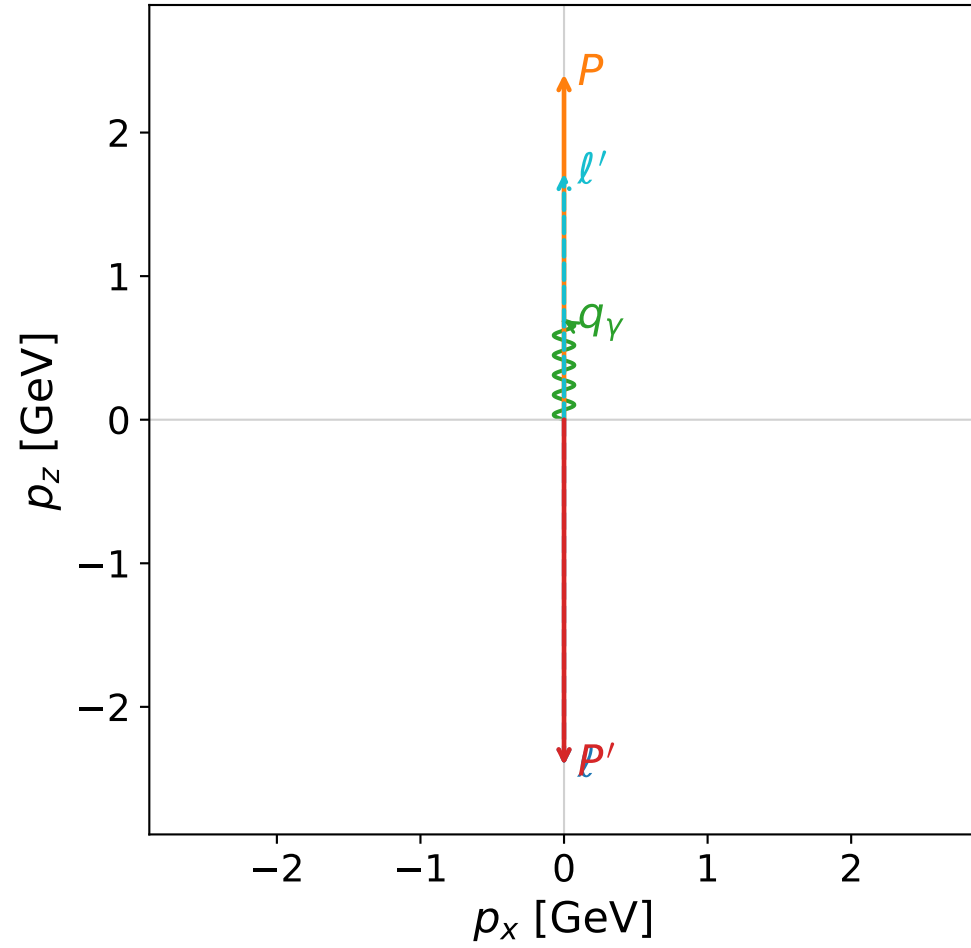
$s=22.4118$, $\sqrt{s}=4.73411$, $|\vec{P}|=2.27413$, $|\vec{P}'|=0.400283$
 $\theta_{p'}=1.00102\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.40749$, $\phi_\gamma=1.29355$
 $E_\gamma=2.05728$, $\theta_{\ell'}=2.41656\text{e-}07$, $\phi_{\ell'}=5.54909$
 $Q^2=15.0729$, $x_B=0.720637$, $t=-1.43729$, $W^2=6.72301$, $y=0.971395$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2741$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2741$
 P $E= 2.4600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2741$
 ℓ' $E= 1.6570$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.6570$
 P' $E= 1.0198$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.4003$
 q_γ $E= 2.0573$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0573$

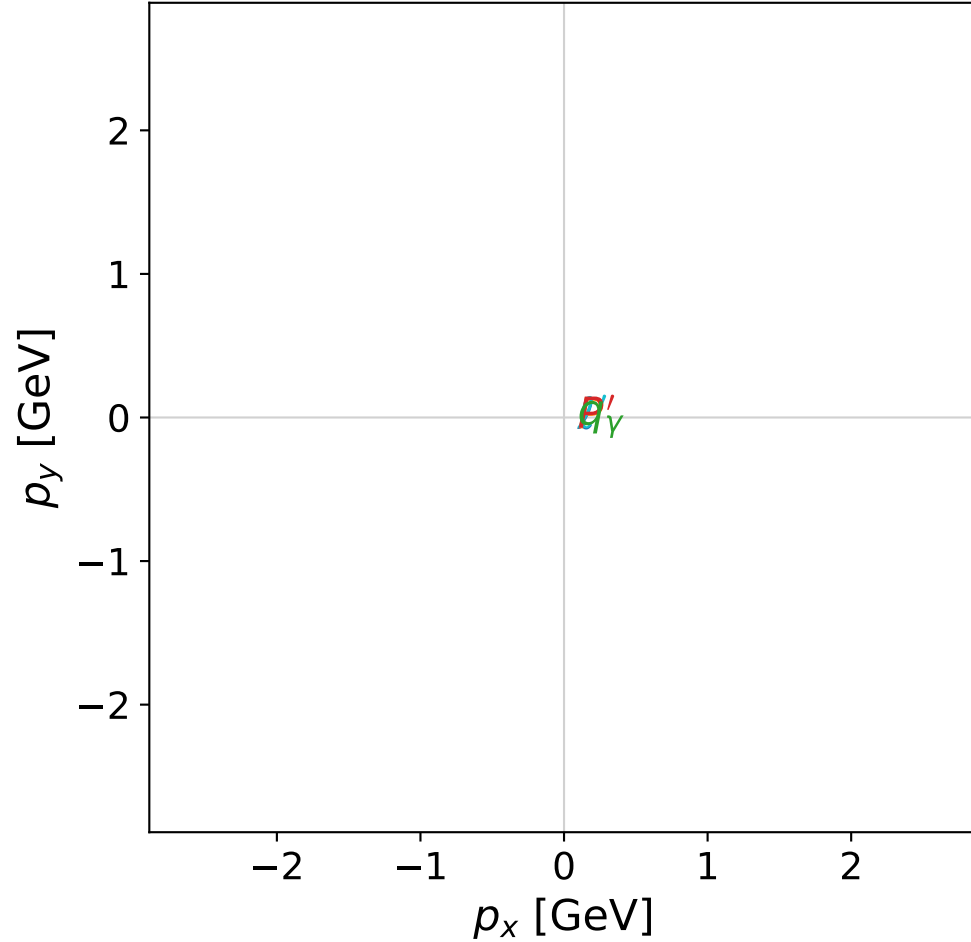


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



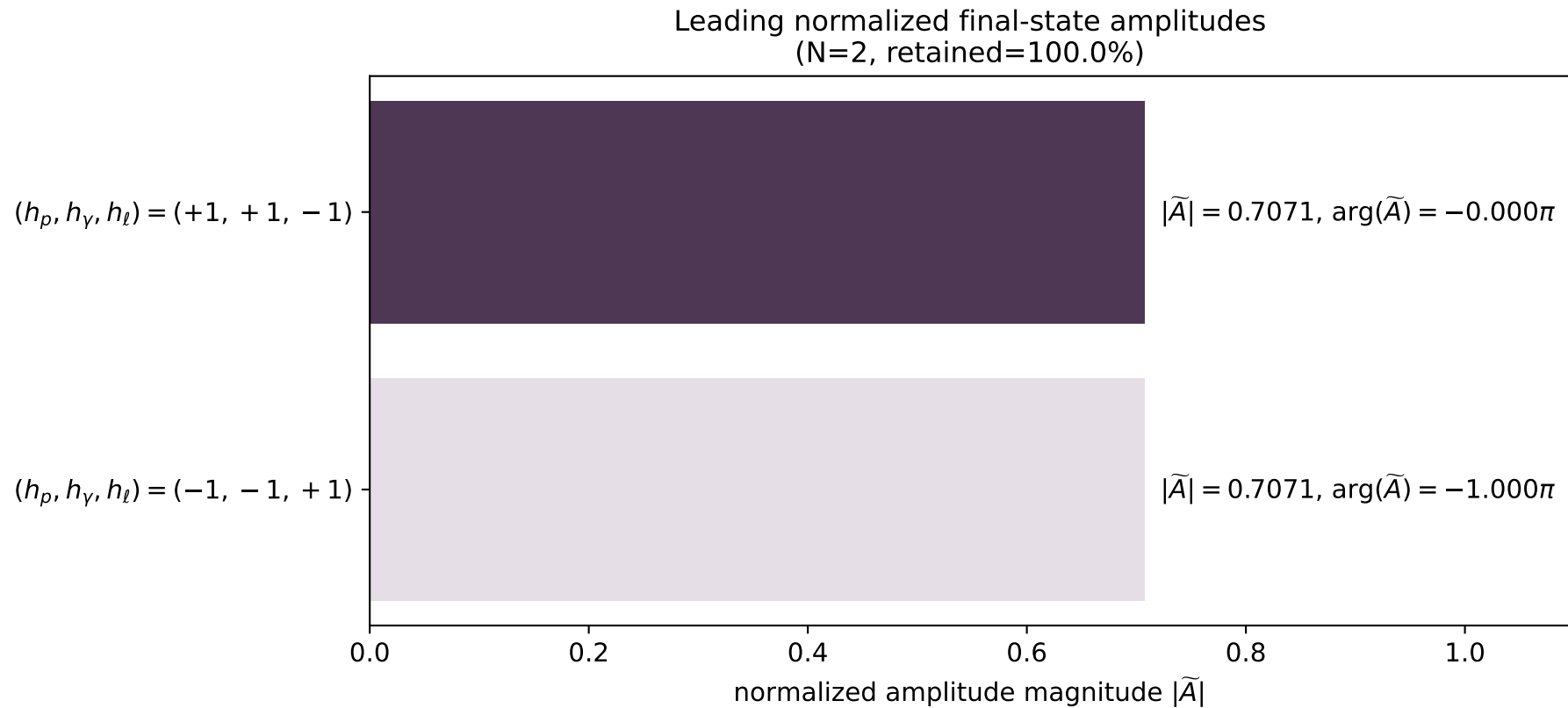
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_91
 $d_{\{\text{rm GHZ}\}}=3.29073\text{e-}08$
 region: polarization_cluster_04_local_minimum_91
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0091
 lepton mixing angle: $\alpha_e=2.4970427$ rad
 incoming lepton: $-0.79937|+\rangle +0.600839|-\rangle$
 proton mixing angle: $\alpha_p=2.2153473$ rad
 incoming proton: $-0.60084|+\rangle +0.79937|-\rangle$

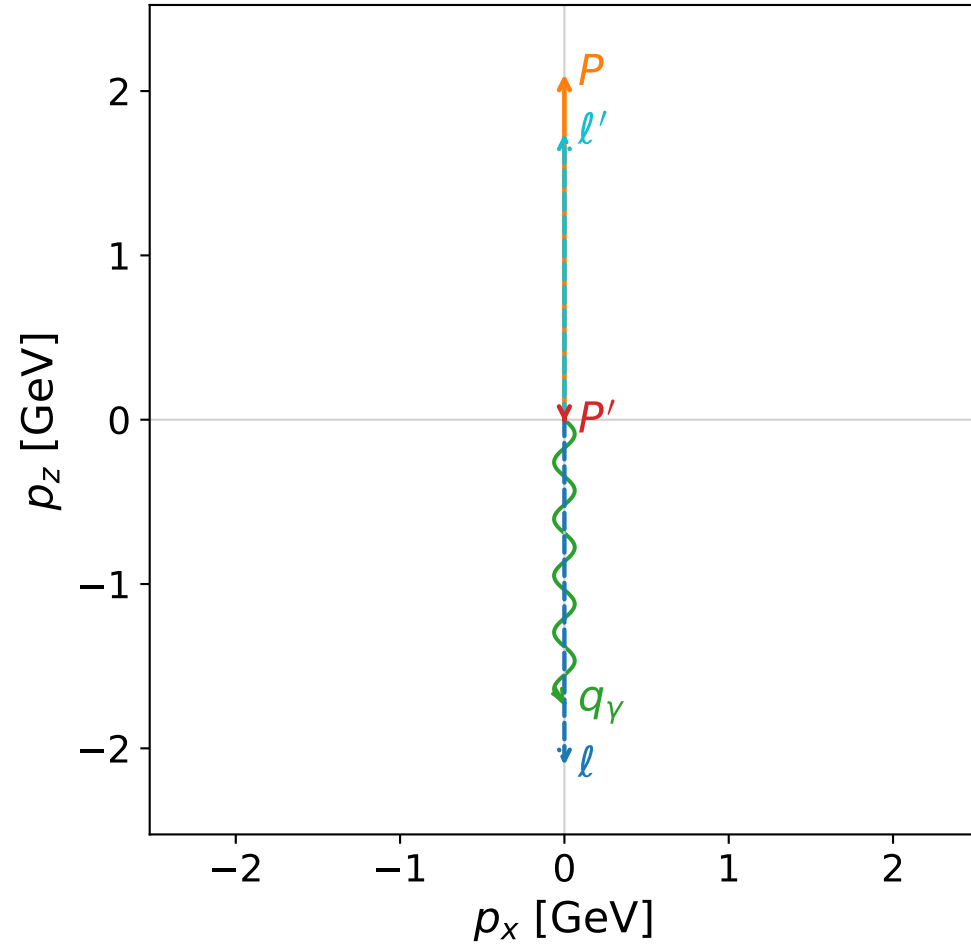
$s=24.9047$, $\sqrt{s}=4.99046$, $|\vec{P}|=2.40708$, $|\vec{P}'|=2.40708$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=4.19237$, $\phi_Y=1.989$
 $E_Y=0.691167$, $\theta_{l'}=0$, $\phi_{l'}=1.05077$
 $Q^2=16.5213$, $x_B=0.705442$, $t=-23.1761$, $W^2=7.77833$, $y=0.974816$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4071$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4071$
 P $E= 2.5834$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4071$
 l' $E= 1.7159$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7159$
 P' $E= 2.5834$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4071$
 q_Y $E= 0.6912$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6912$

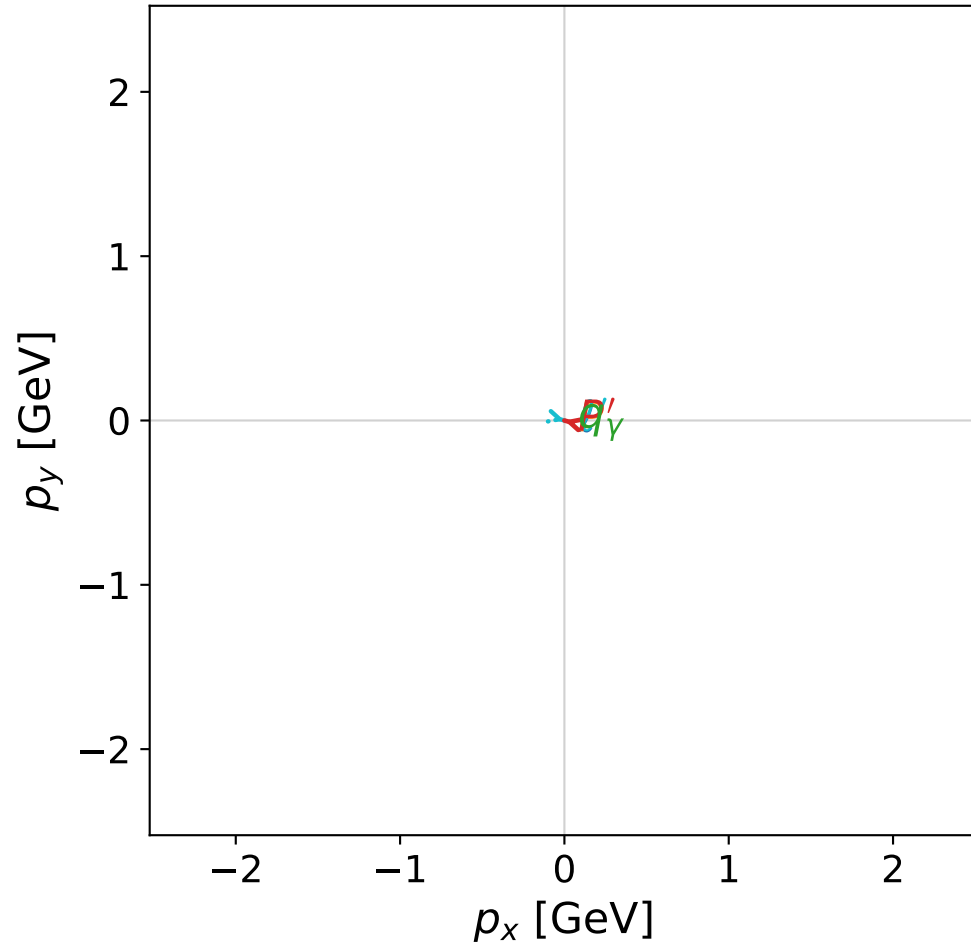


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



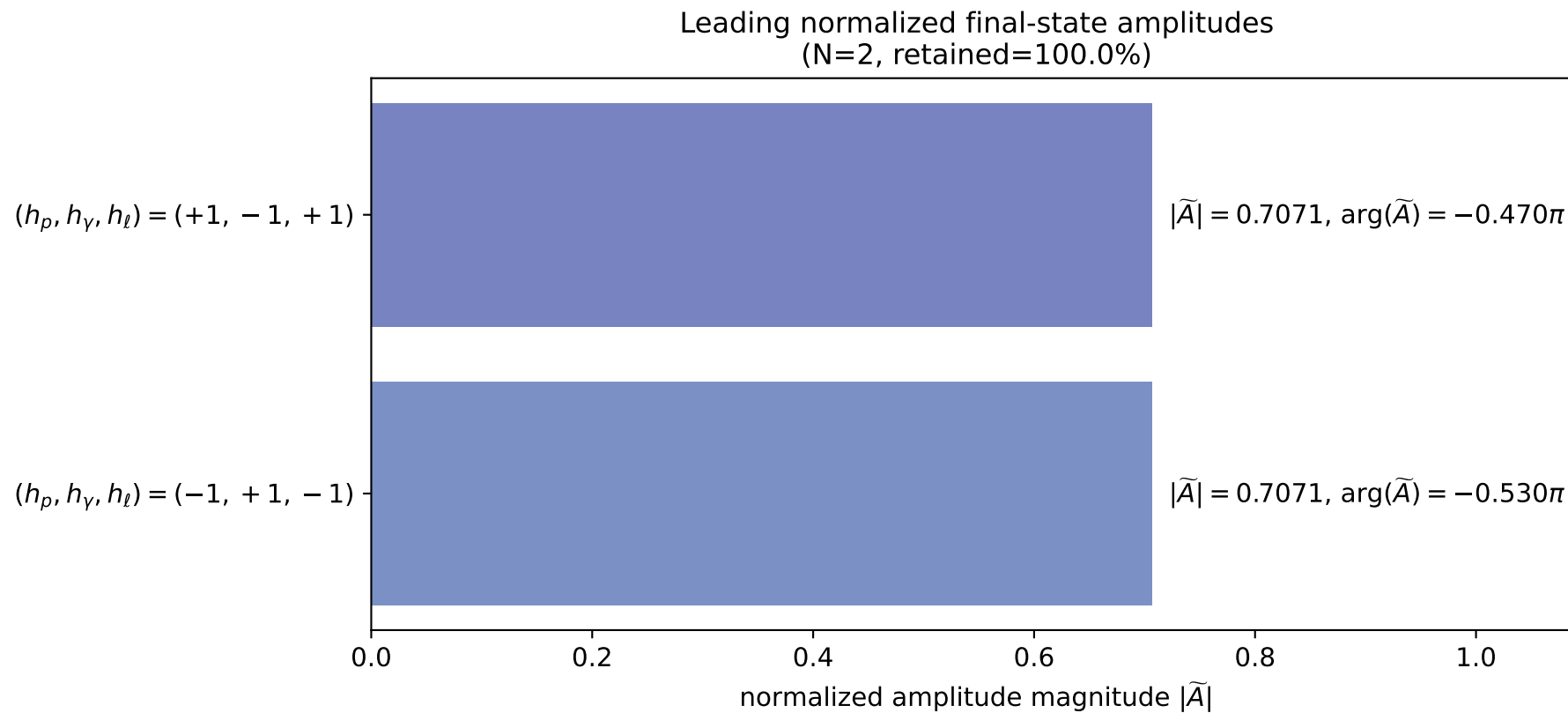
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_98
 $d_{\text{GHZ}}=3.56394\text{e-}08$
 region: polarization_cluster_04_local_minimum_98
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0098
 lepton mixing angle: $\alpha_e=2.3614211$ rad
 incoming lepton: $-0.710793|+\rangle +0.703401|-\rangle$
 proton mixing angle: $\alpha_p=2.3614305$ rad
 incoming proton: $-0.7108|+\rangle +0.703395|-\rangle$

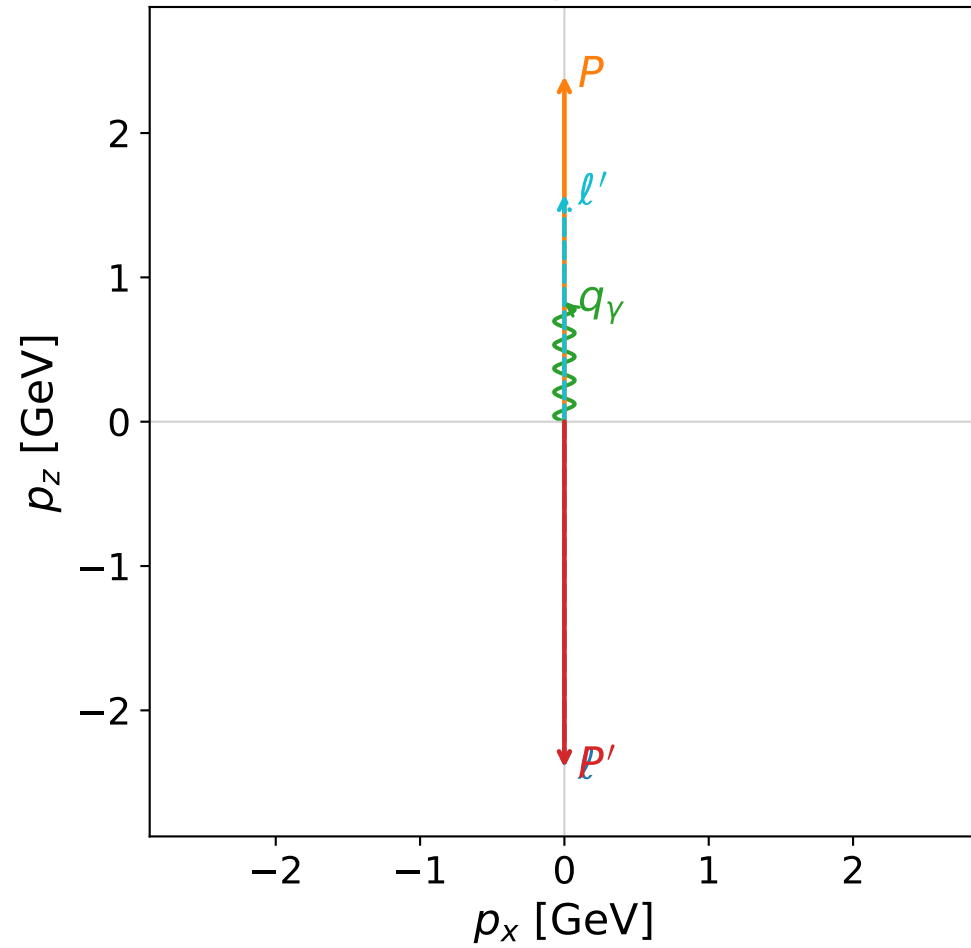
$s=19.4131$, $\sqrt{s}=4.40603$, $|\vec{P}|=2.10317$, $|\vec{P}'|=0.0160914$
 $\theta_{P'}=3.14148$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.86856$, $\phi_\gamma=1.39128$
 $E_\gamma=1.7259$, $\theta_{\ell'}=1.00525\text{e-}06$, $\phi_{\ell'}=6.01016$
 $Q^2=14.6548$, $x_B=0.821572$, $t=-2.6288$, $W^2=4.06257$, $y=0.962461$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1032$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1032$
 P $E= 2.3029$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1032$
 ℓ' $E= 1.7420$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7420$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0161$
 q_γ $E= 1.7259$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7259$

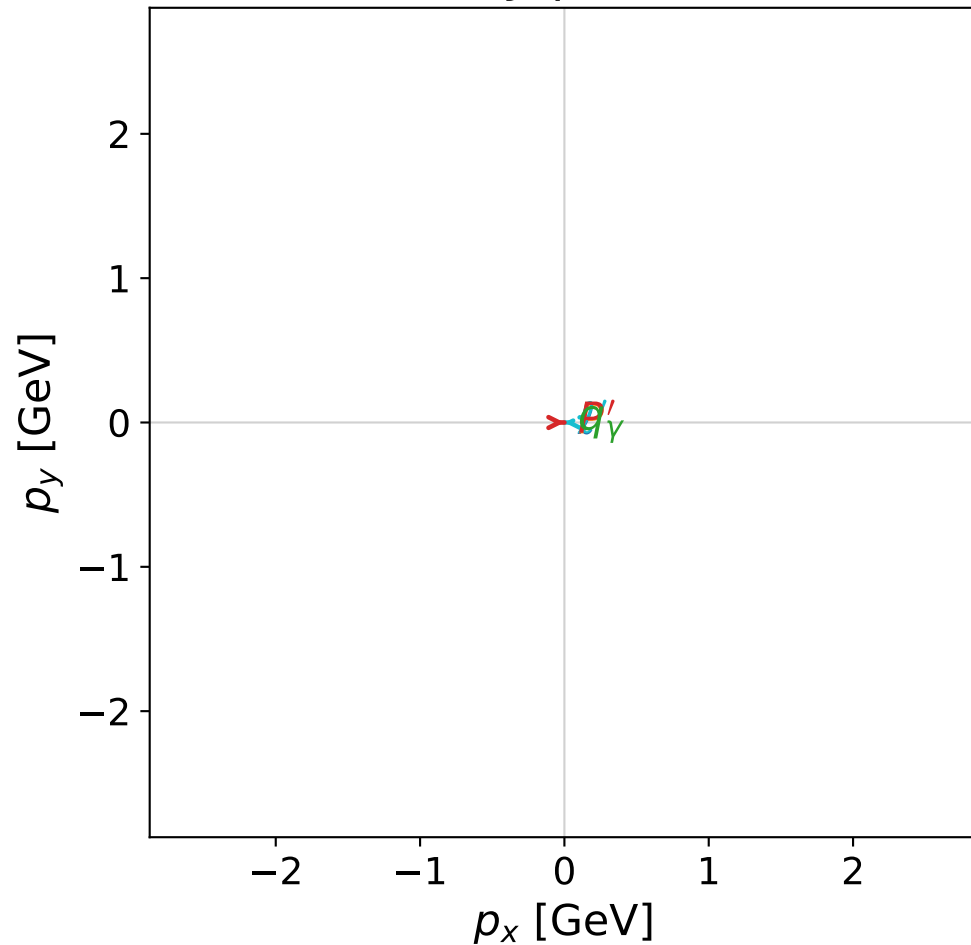


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

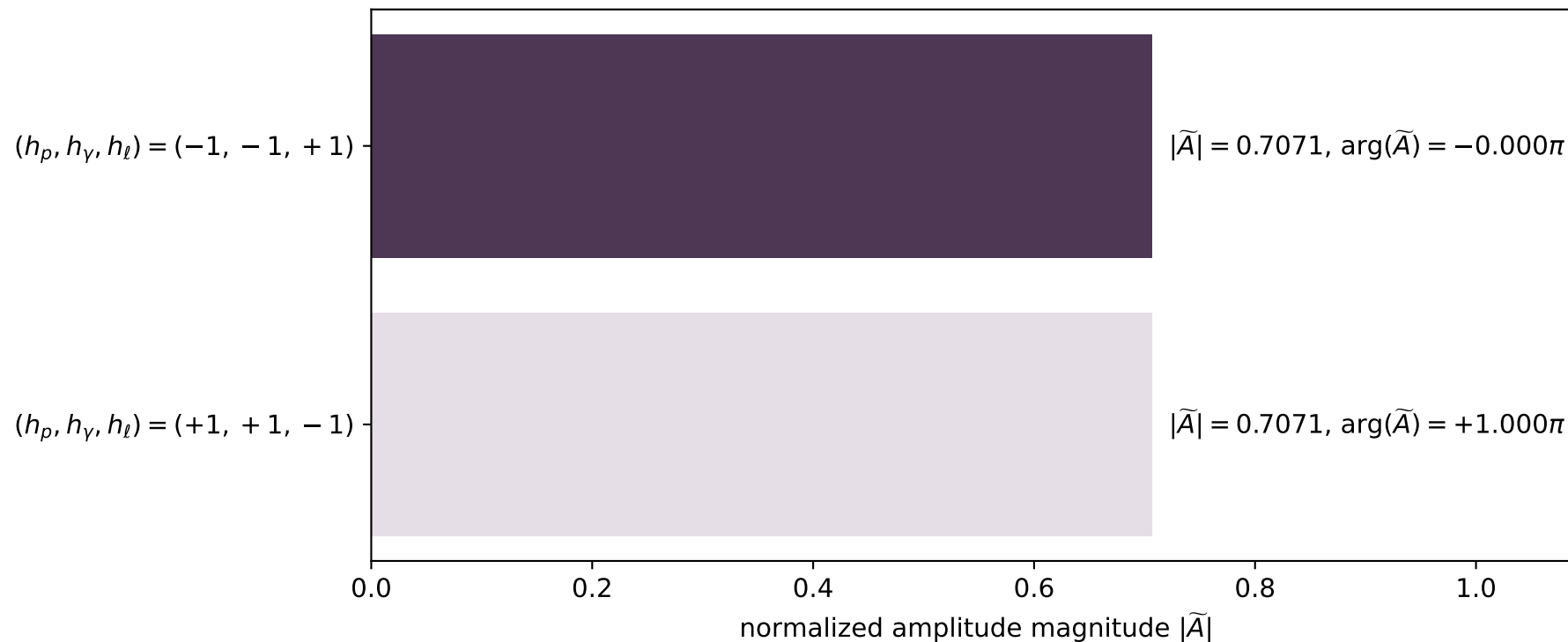


dghz_mixing_angles_polarization_04_local_minimum_101
 $d_{\text{GHZ}}=3.64201\text{e-}08$
 region: polarization_cluster_04_local_minimum_101
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0101
 lepton mixing angle: $\alpha_e=2.19789$ rad
 incoming lepton: $-0.586794|+\rangle +0.809736|-\rangle$
 proton mixing angle: $\alpha_p=2.514492$ rad
 incoming proton: $-0.809732|+\rangle +0.5868|-\rangle$

$s=24.6624$, $\sqrt{s}=4.96613$, $|\vec{P}|=2.39448$, $|\vec{P}'|=2.39448$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=0.290949$, $\phi_Y=6.12241$
 $E_Y=0.819028$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.43254$
 $Q^2=15.0896$, $x_B=0.64973$, $t=-22.9341$, $W^2=9.01465$, $y=0.976527$

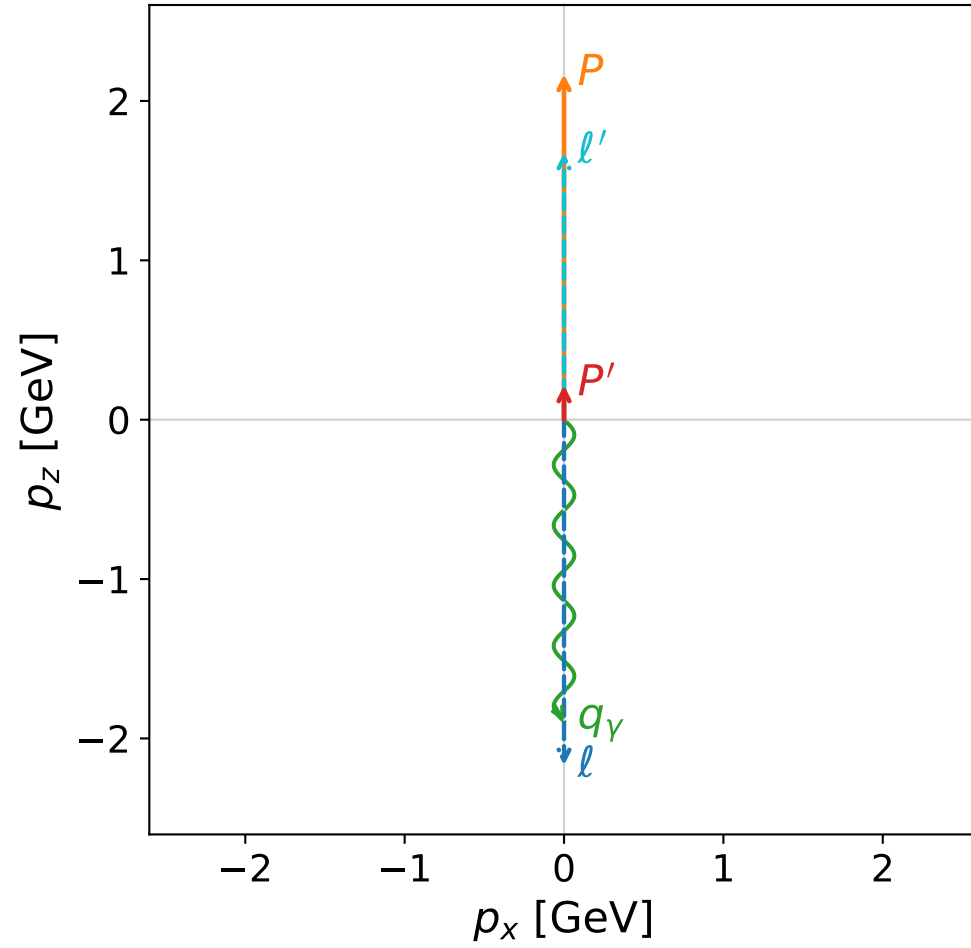
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3945$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3945$
 P $E= 2.5716$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3945$
 ℓ' $E= 1.5755$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5755$
 P' $E= 2.5716$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3945$
 q_Y $E= 0.8190$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8190$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

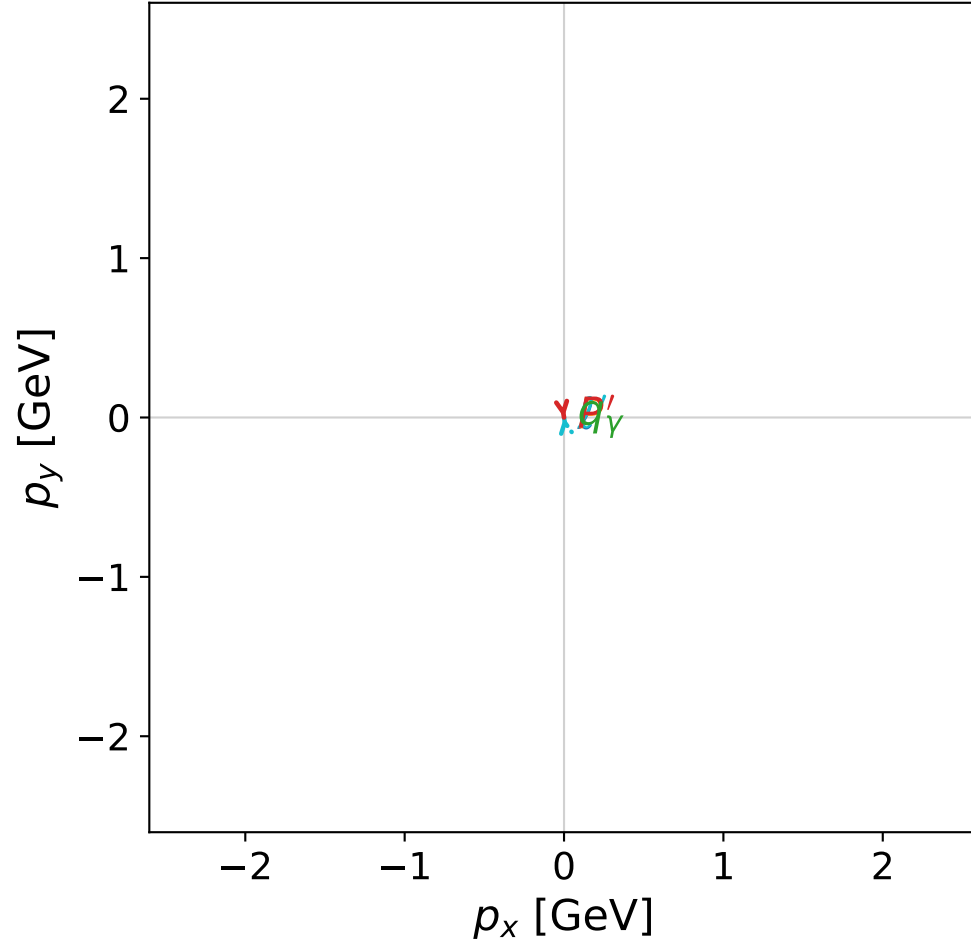


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



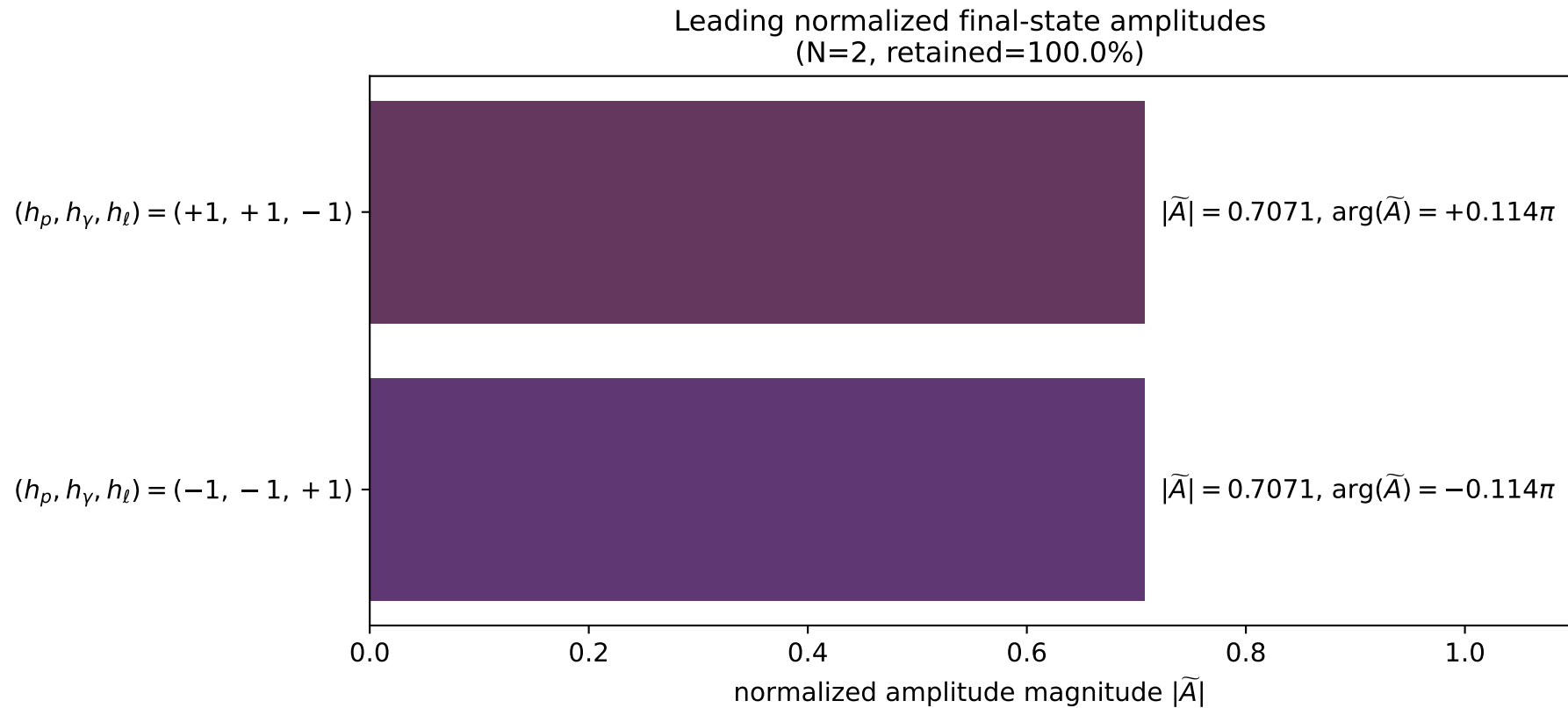
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_105
 $d_{\text{GHZ}}=3.82625\text{e-}08$
 region: polarization_cluster_04_local_minimum_105
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0105
 lepton mixing angle: $\alpha_e=2.5411507$ rad
 incoming lepton: $-0.825086|+\rangle +0.565007|-\rangle$
 proton mixing angle: $\alpha_p=2.5411495$ rad
 incoming proton: $-0.825085|+\rangle +0.565008|-\rangle$

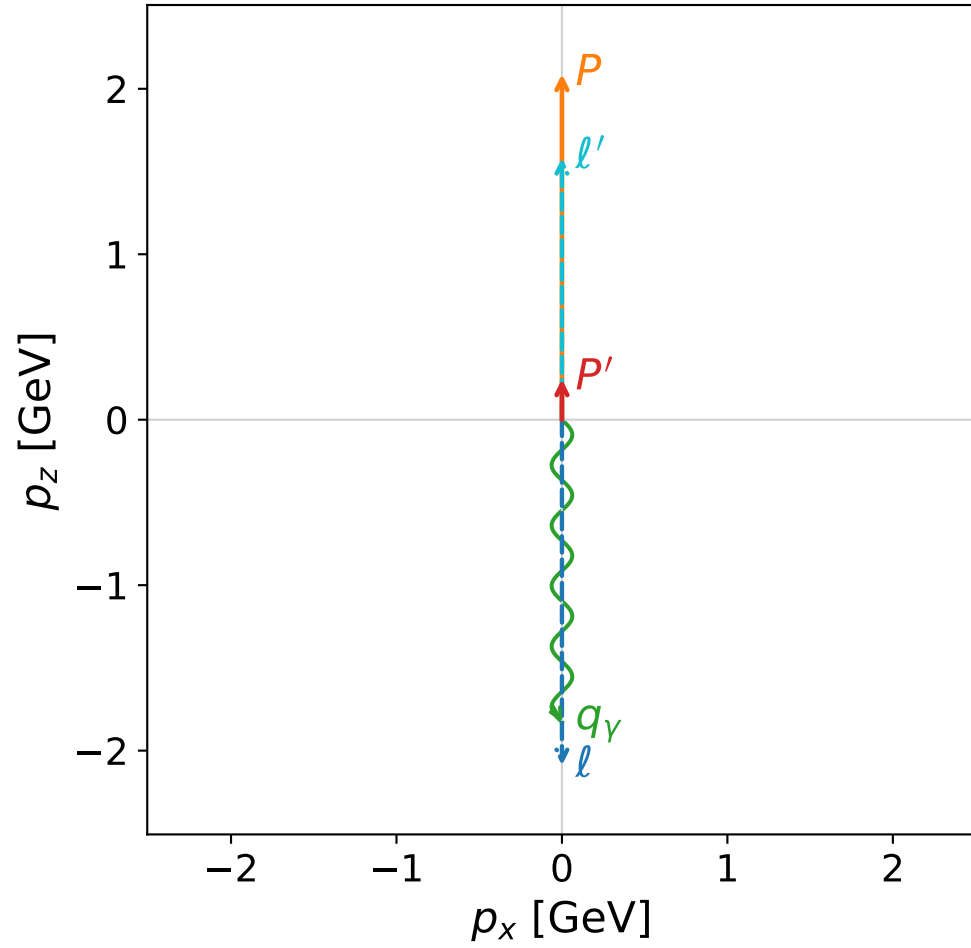
$s=20.5321$, $\sqrt{s}=4.53124$, $|\vec{P}|=2.16853$, $|\vec{P}'|=0.216611$
 $\theta_{P'}=7.6699\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=4.86682$, $\phi_Y=5.92443$
 $E_Y=1.89258$, $\theta_{l'}=9.91298\text{e-}06$, $\phi_{l'}=1.72522$
 $Q^2=14.5376$, $x_B=0.765079$, $t=-1.84994$, $W^2=5.34368$, $y=0.966881$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1685$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1685$
 P $E= 2.3627$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1685$
 l' $E= 1.6760$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.6760$
 P' $E= 0.9627$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.2166$
 q_Y $E= 1.8926$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8926$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

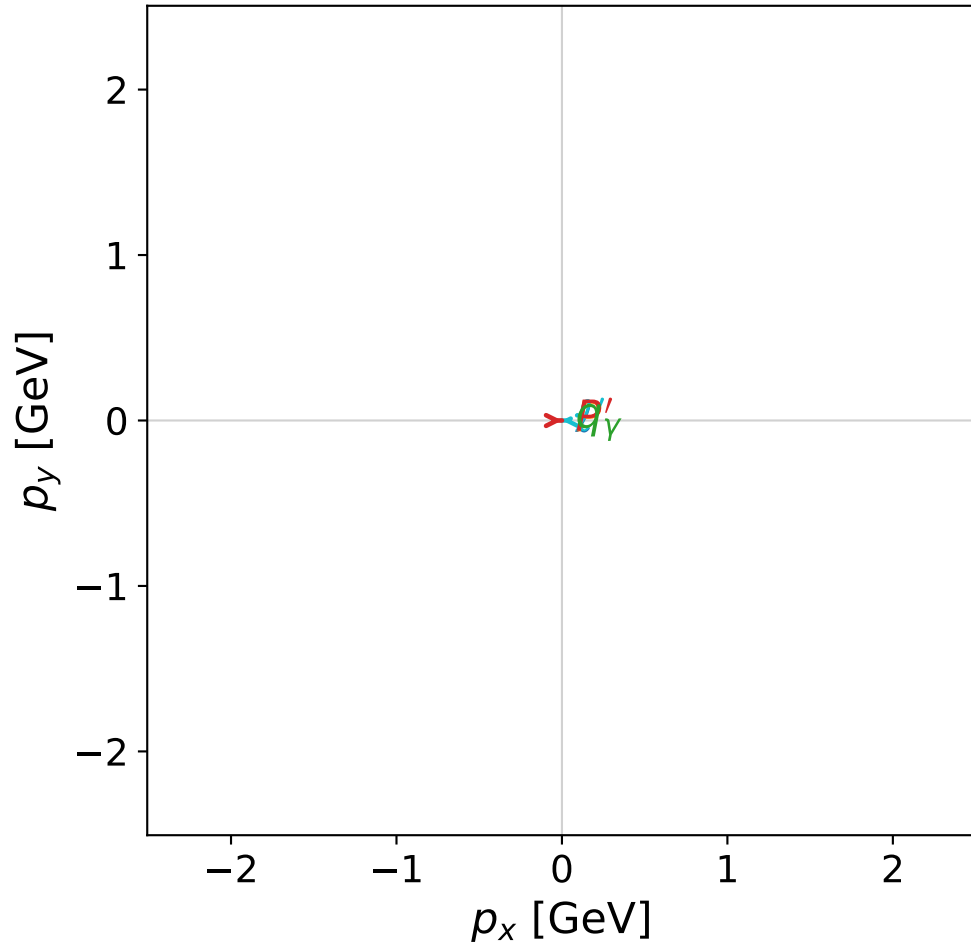


dghz_mixing_angles_polarization_04_local_minimum_108
 $d_{\text{GHZ}}=3.93085\text{e-}08$
 region: polarization_cluster_04_local_minimum_108
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0108
 lepton mixing angle: $\alpha_e=2.3726047$ rad
 incoming lepton: $-0.718615|+\rangle + 0.695408|-\rangle$
 proton mixing angle: $\alpha_p=2.3726093$ rad
 incoming proton: $-0.718618|+\rangle + 0.695405|-\rangle$

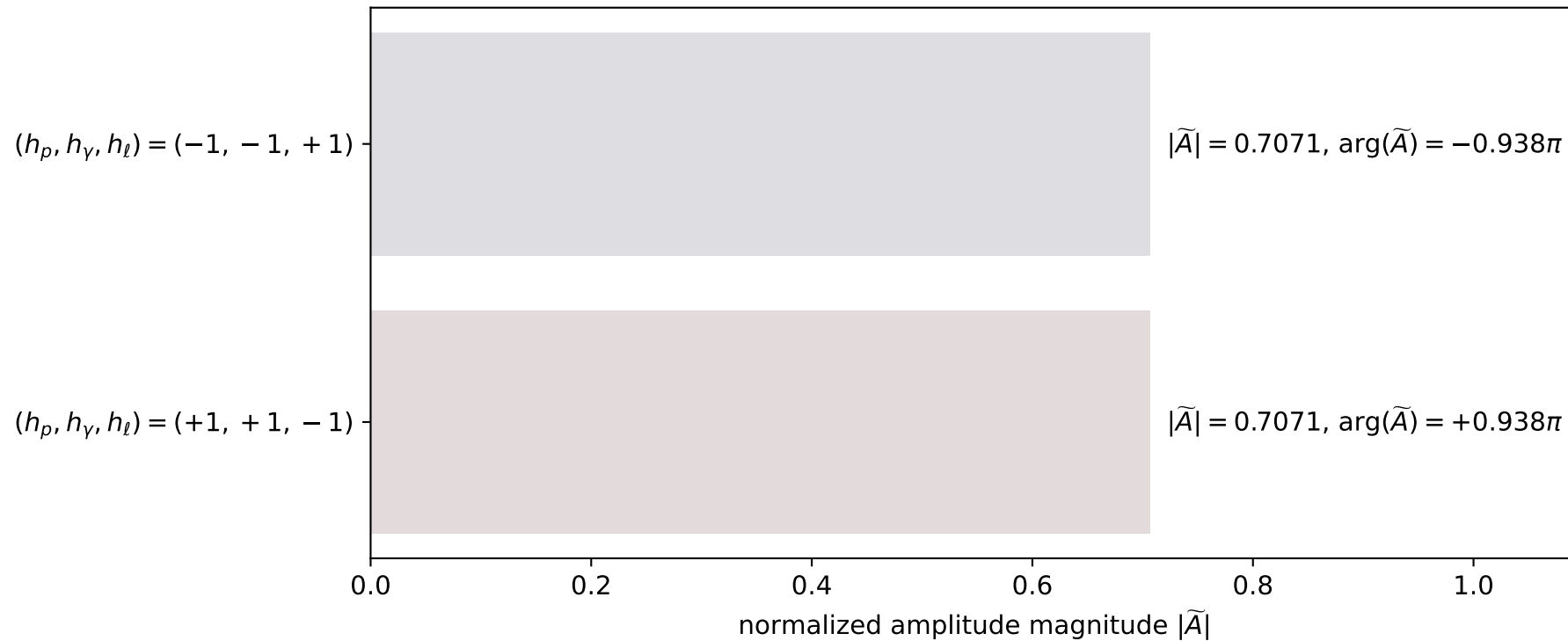
$s=19.1697$, $\sqrt{s}=4.37832$, $|\vec{P}|=2.08868$, $|\vec{P}'|=0.24491$
 $\theta_{P'}=0.000152337$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.00207501$, $\phi_\gamma=3.33601$
 $E_\gamma=1.82689$, $\theta_{l'}=2.35835\text{e-}05$, $\phi_{l'}=3.14367$
 $Q^2=13.217$, $x_B=0.74867$, $t=-1.65659$, $W^2=5.31684$, $y=0.965237$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0887$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0887$
 P $E= 2.2896$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0887$
 l' $E= 1.5820$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5820$
 P' $E= 0.9694$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2449$
 q_γ $E= 1.8269$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8269$

x--y plane

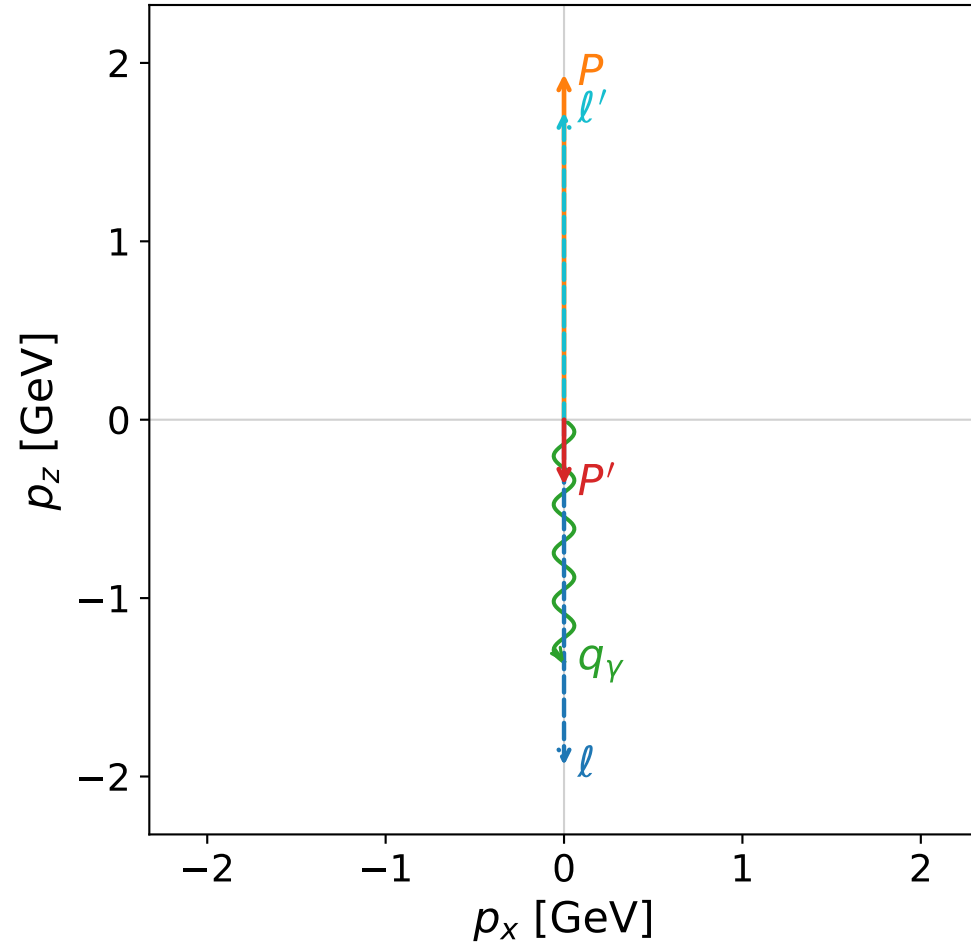


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

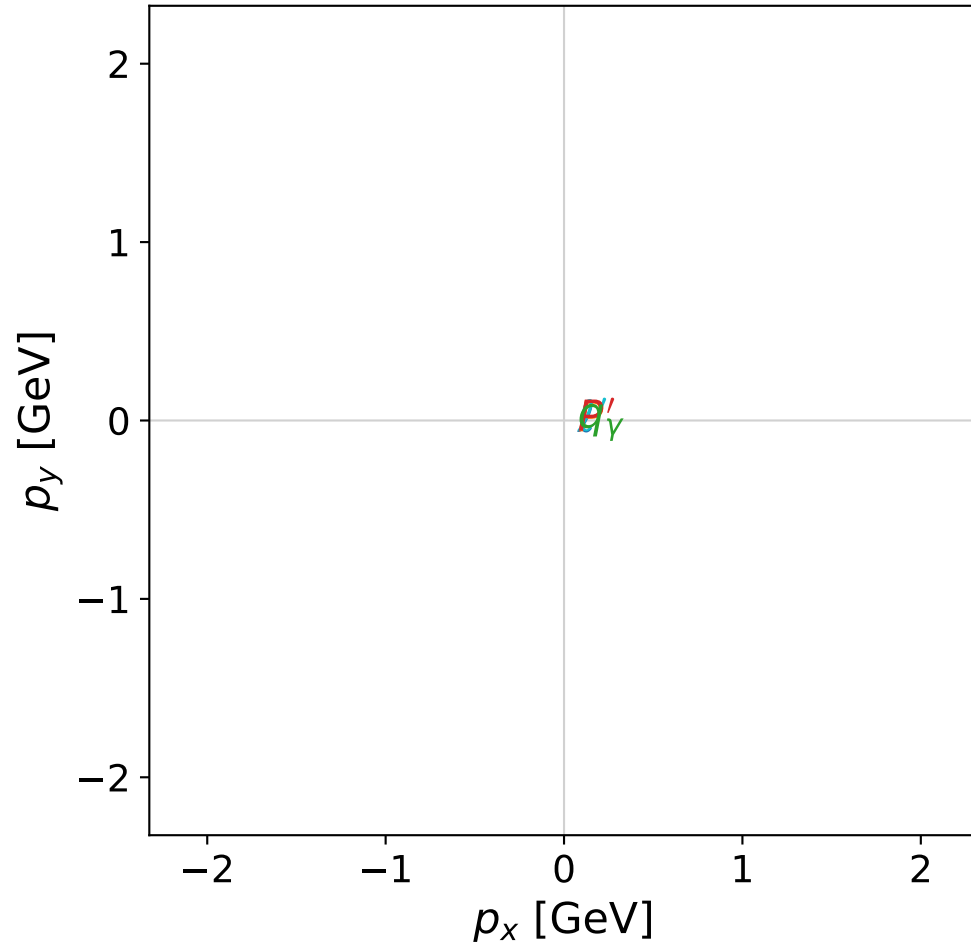


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



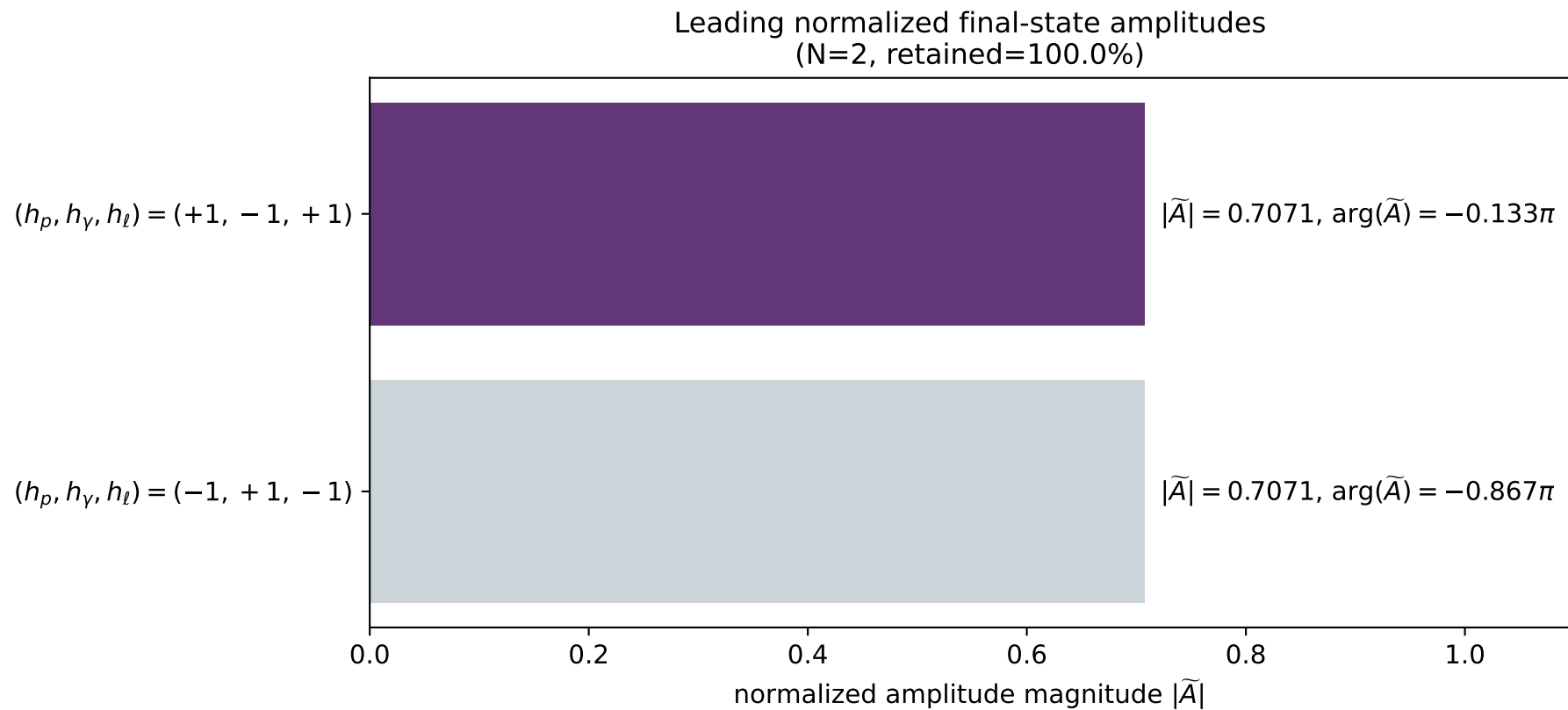
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_109
 $d_{\{\text{rm GHZ}\}}=3.93746\text{e-}08$
 region: polarization_cluster_04_local_minimum_109
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0109
 lepton mixing angle: $\alpha_e=2.4147598$ rad
 incoming lepton: $-0.747283|+\rangle +0.664506|-\rangle$
 proton mixing angle: $\alpha_p=2.4147771$ rad
 incoming proton: $-0.747294|+\rangle +0.664493|-\rangle$

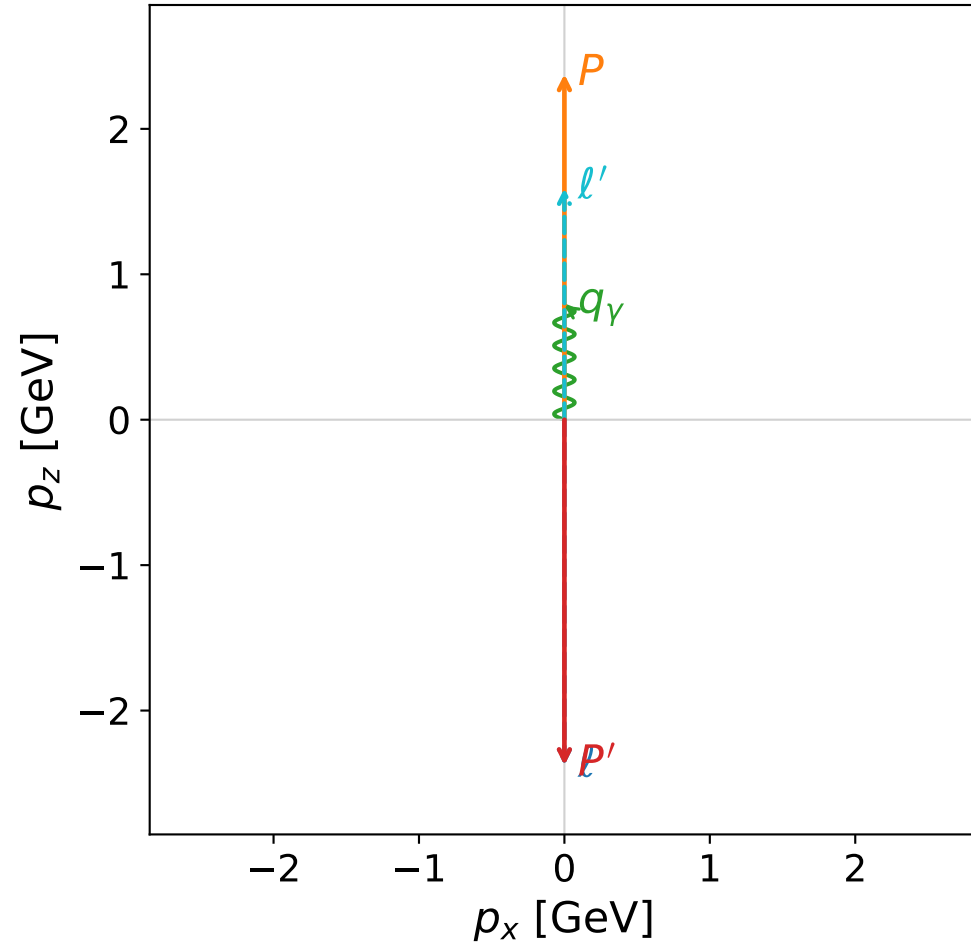
$s=16.7248$, $\sqrt{s}=4.0896$, $|\vec{P}|=1.93723$, $|\vec{P}'|=0.365737$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.18745$, $\phi_\gamma=5.86465$
 $E_\gamma=1.35854$, $\theta_{l'}=0$, $\phi_{l'}=2.56558$
 $Q^2=13.3613$, $x_B=0.884674$, $t=-3.99128$, $W^2=2.62161$, $y=0.953176$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9372$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9372$
 P $E= 2.1524$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9372$
 ℓ' $E= 1.7243$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7243$
 P' $E= 1.0068$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.3657$
 q_γ $E= 1.3585$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.3585$

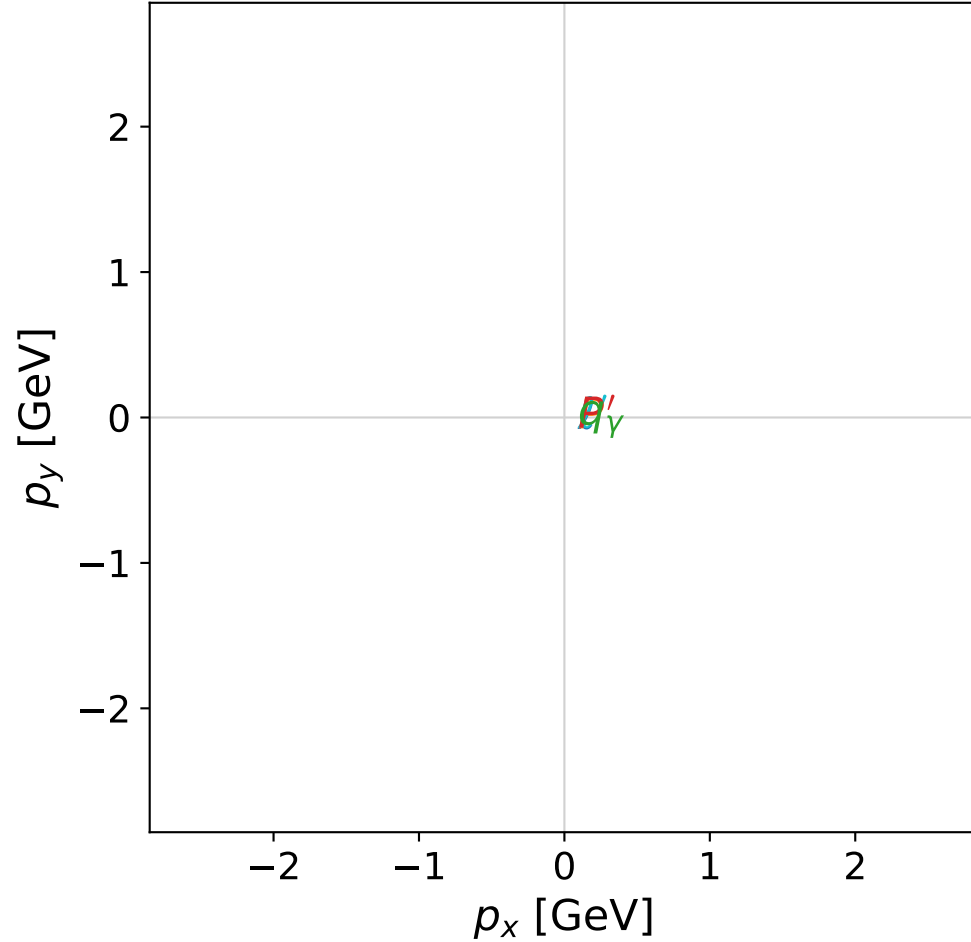


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



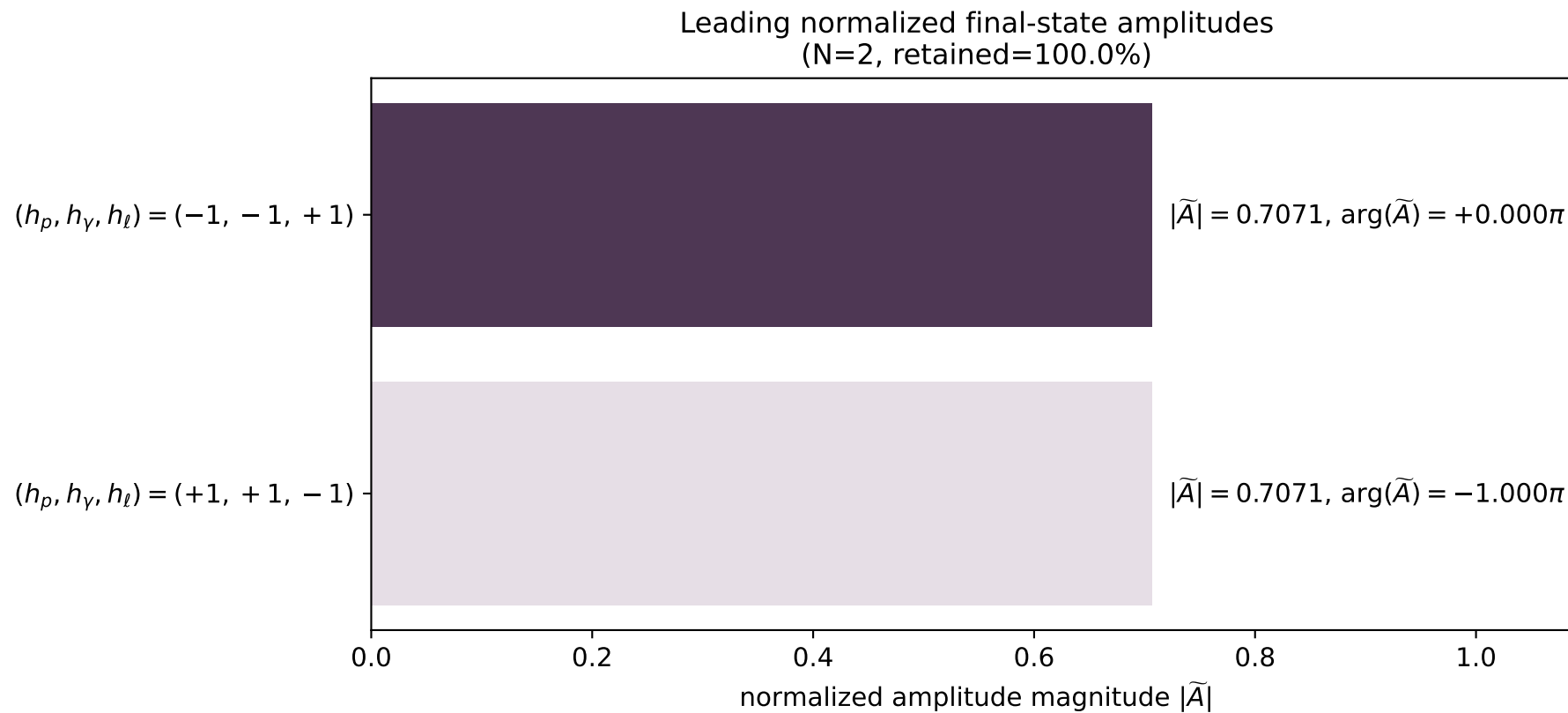
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_117
 $d_{\text{GHZ}}=4.36167\text{e-}08$
 region: polarization_cluster_04_local_minimum_117
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0117
 lepton mixing angle: $\alpha_e=2.0273398$ rad
 incoming lepton: $-0.440848|+\rangle +0.897582|-\rangle$
 proton mixing angle: $\alpha_p=2.6850445$ rad
 incoming proton: $-0.89758|+\rangle +0.440852|-\rangle$

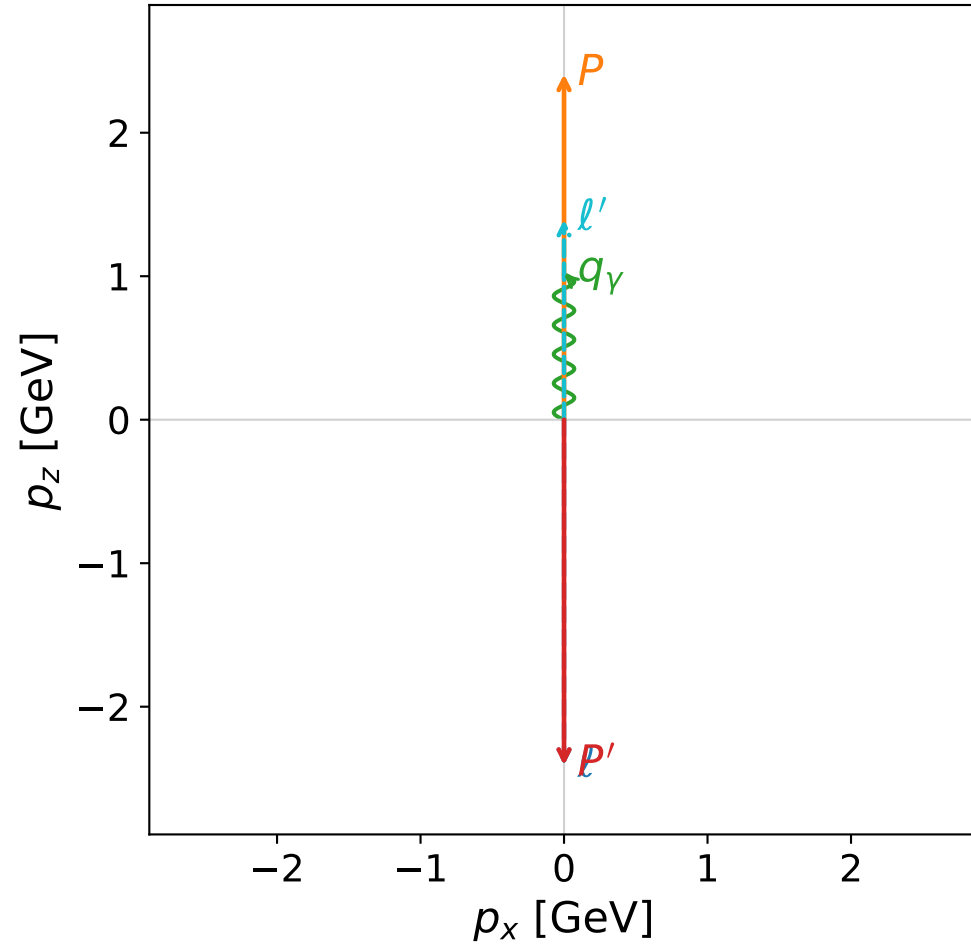
$s=24.3182$, $\sqrt{s}=4.93135$, $|\vec{P}|=2.37646$, $|\vec{P}'|=2.37646$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.60836$, $\phi_\gamma=5.61305$
 $E_\gamma=0.784824$, $\theta_{l'}=0$, $\phi_{l'}=0.466771$
 $Q^2=15.1299$, $x_B=0.66155$, $t=-22.5903$, $W^2=8.62032$, $y=0.975768$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3765$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3765$
 P $E= 2.5549$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3765$
 ℓ' $E= 1.5916$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5916$
 P' $E= 2.5549$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.3765$
 q_γ $E= 0.7848$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7848$

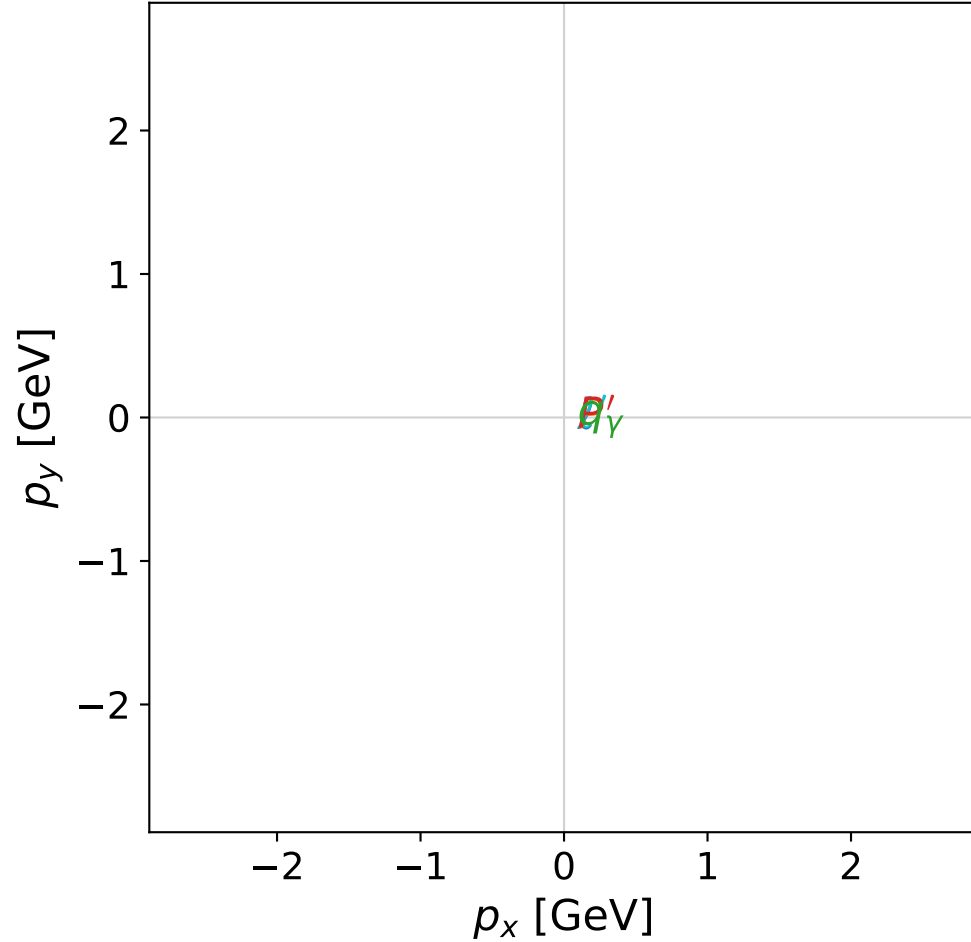


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



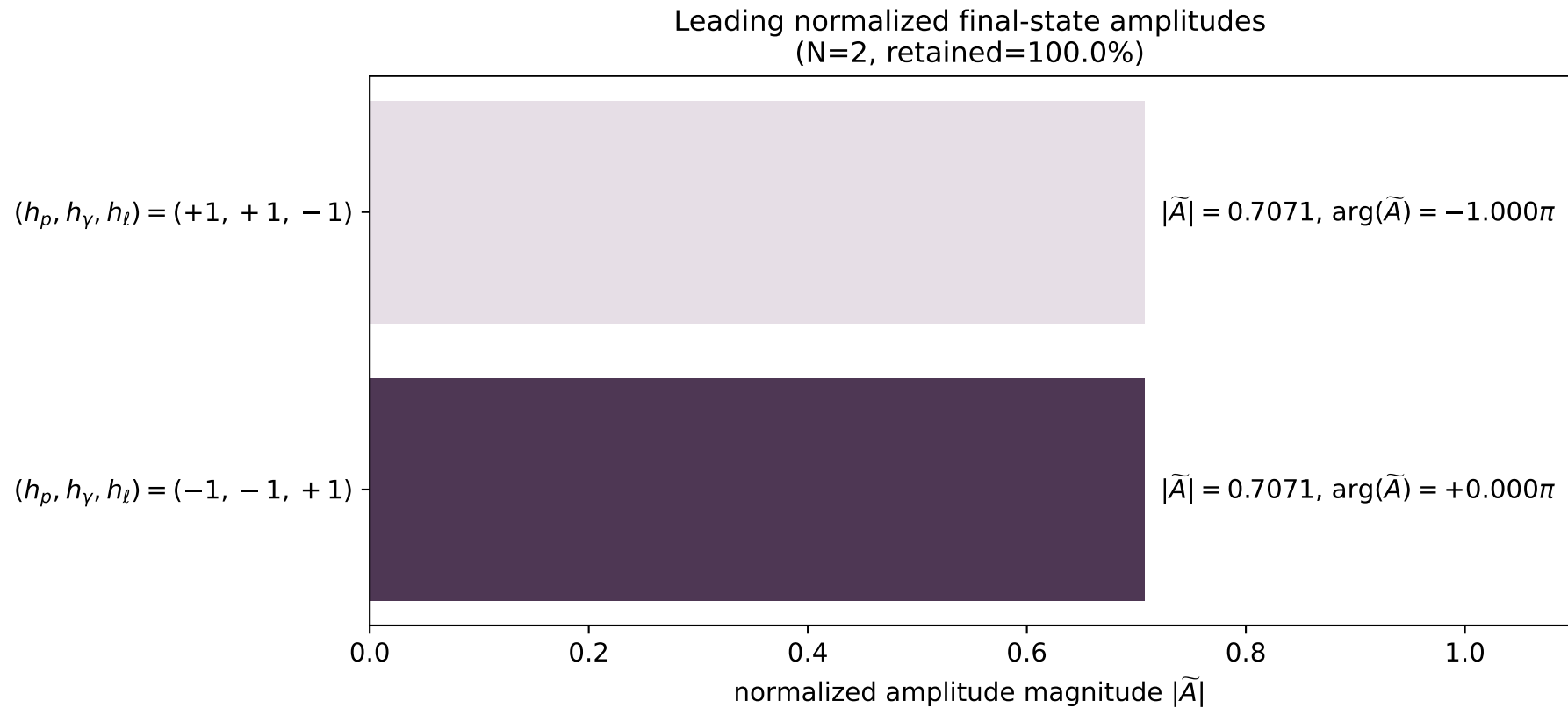
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_119
 $d_{\{\rm GHZ\}}=4.46361\text{e-}08$
 region: polarization_cluster_04_local_minimum_119
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0119
 lepton mixing angle: $\alpha_e=2.099717$ rad
 incoming lepton: $-0.504602|+\rangle +0.863352|-\rangle$
 proton mixing angle: $\alpha_p=2.61268$ rad
 incoming proton: $-0.863356|+\rangle +0.504595|-\rangle$

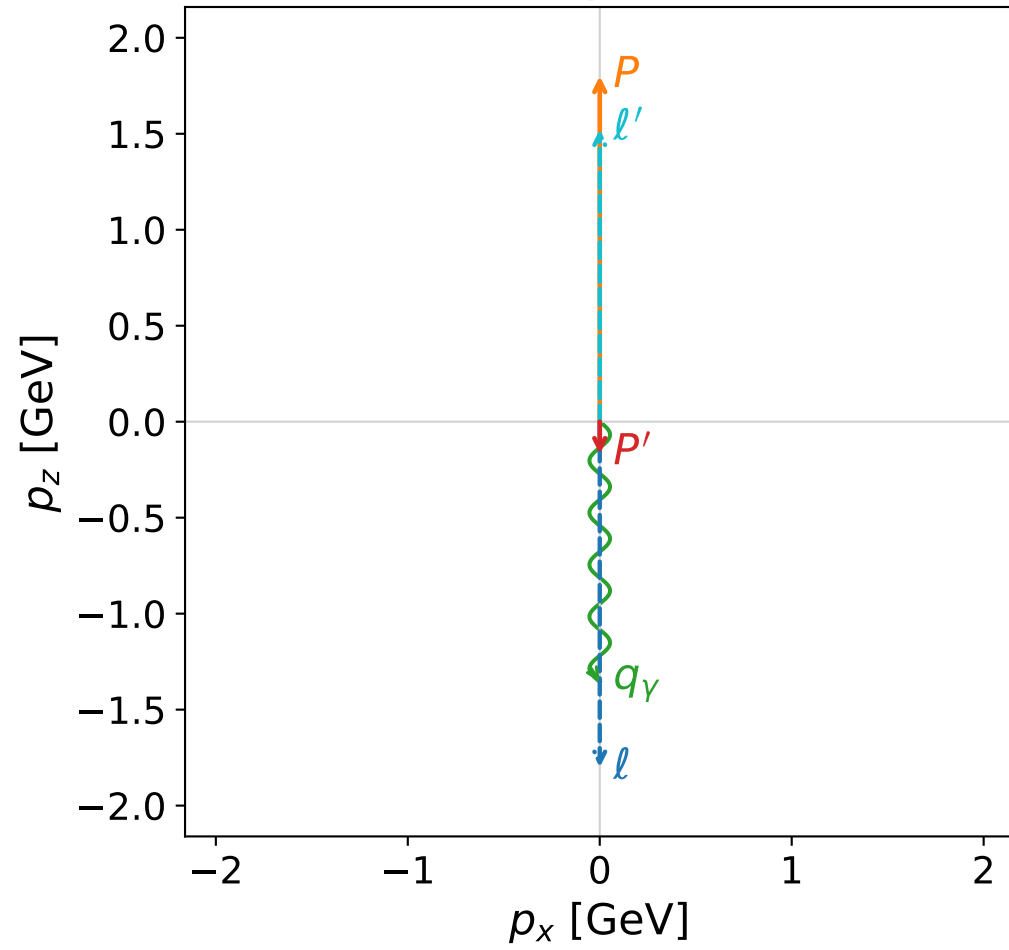
$s=24.93$, $\sqrt{s}=4.993$, $|\vec{P}|=2.40839$, $|\vec{P}'|=2.40839$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.04535$, $\phi_\gamma=0.354409$
 $E_\gamma=1.01446$, $\theta_{l'}=0$, $\phi_{l'}=0.903761$
 $Q^2=13.4285$, $x_B=0.569998$, $t=-23.2014$, $W^2=11.0102$, $y=0.979573$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4084$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4084$
 P $E= 2.5846$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4084$
 l' $E= 1.3939$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3939$
 P' $E= 2.5846$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4084$
 q_γ $E= 1.0145$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0145$

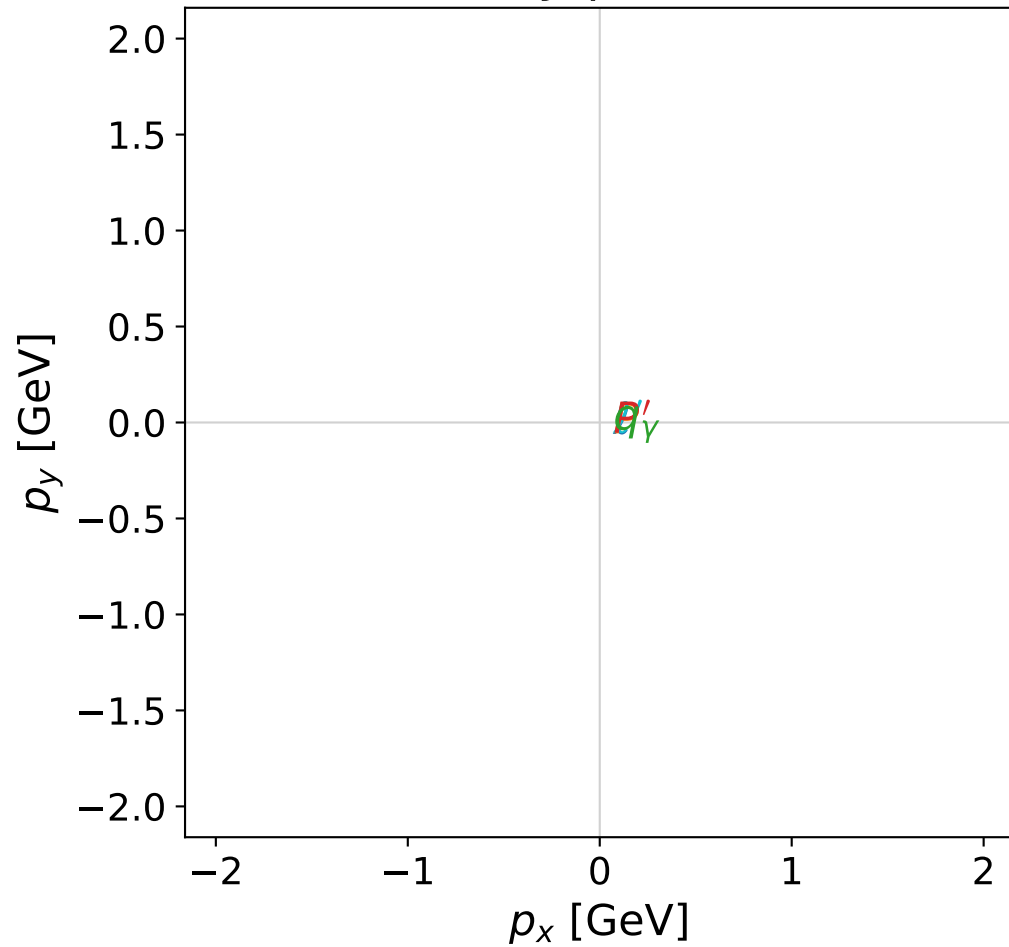


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



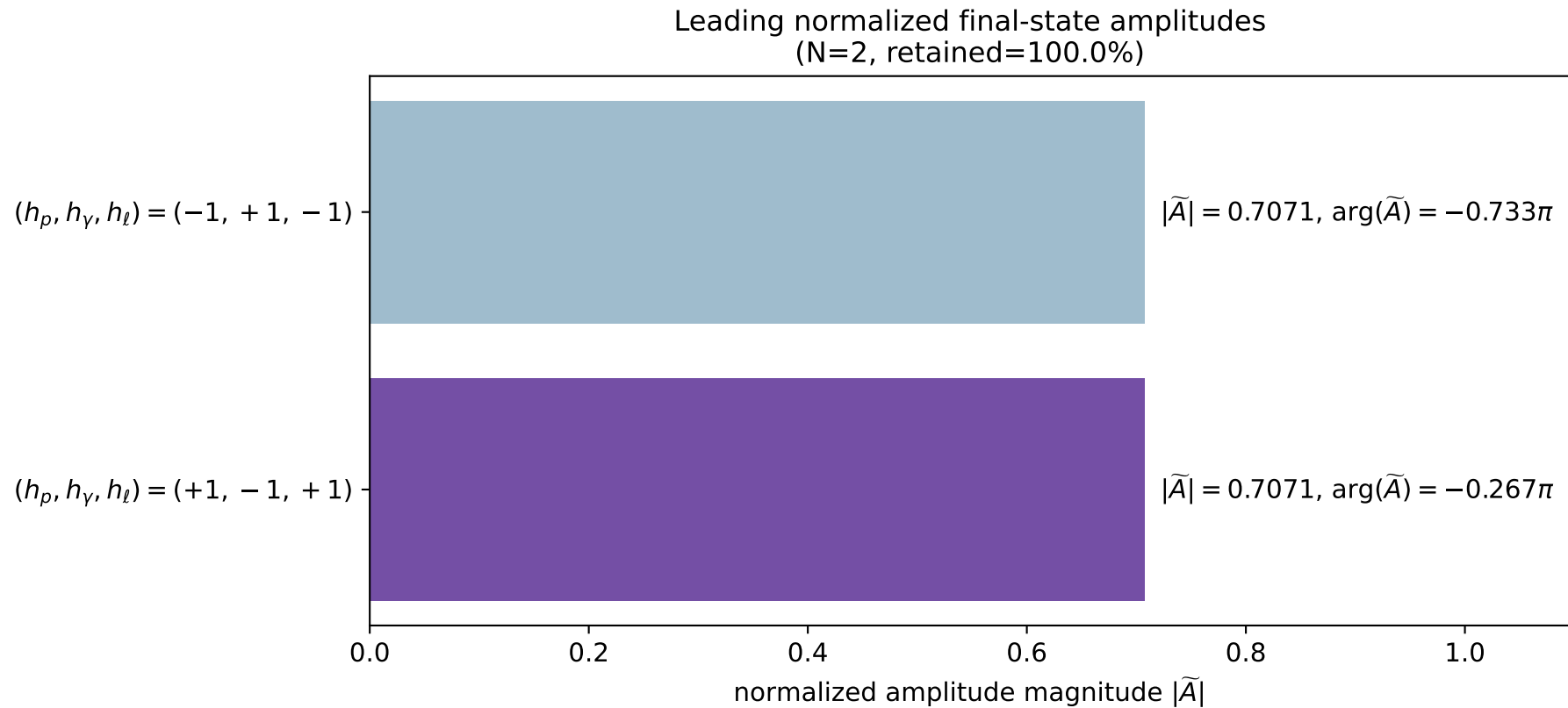
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_122
 $d_{\text{GHZ}}=4.77096\text{e-}08$
 region: polarization_cluster_04_local_minimum_122
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0122
 lepton mixing angle: $\alpha_e=2.3265008$ rad
 incoming lepton: $-0.685802|+\rangle +0.727789|-\rangle$
 proton mixing angle: $\alpha_p=2.3264837$ rad
 incoming proton: $-0.685789|+\rangle +0.7278|-\rangle$

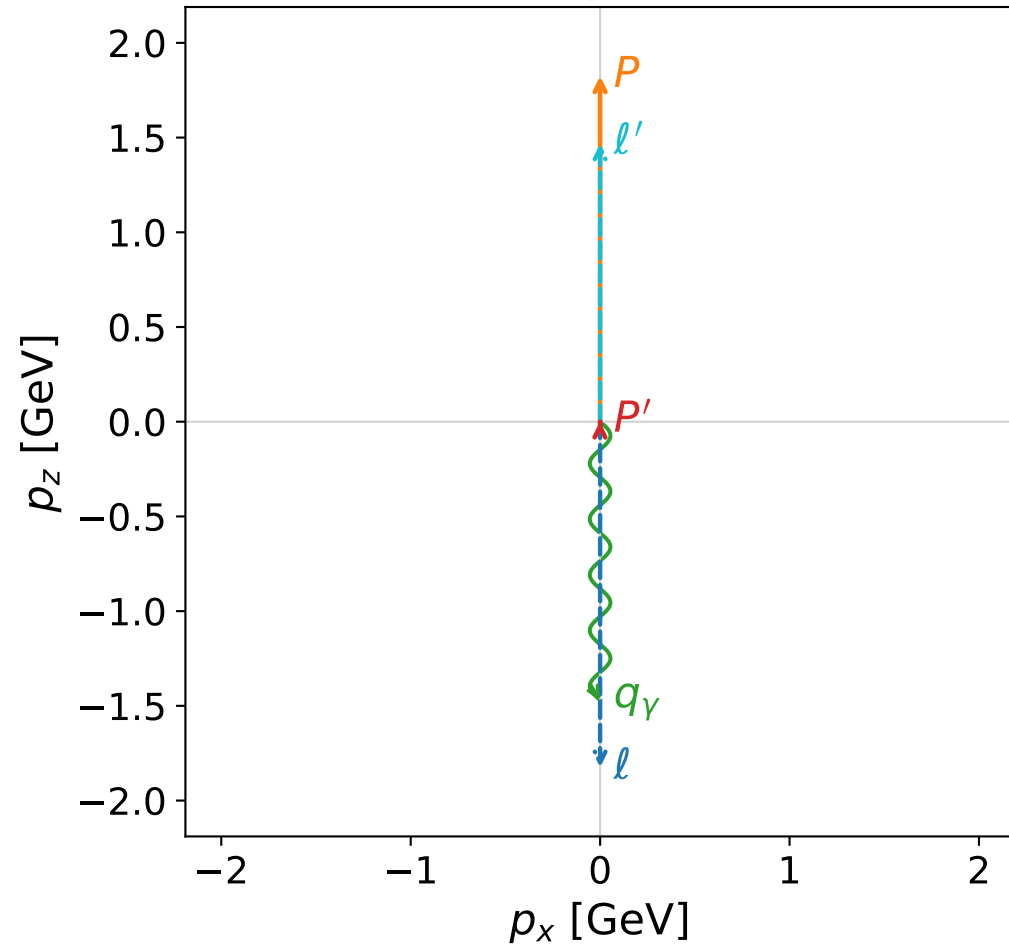
$s=14.673$, $\sqrt{s}=3.83054$, $|\vec{P}|=1.80042$, $|\vec{P}'|=0.16869$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.10573$, $\phi_\gamma=5.44421$
 $E_\gamma=1.3544$, $\theta_{l'}=0$, $\phi_{l'}=2.20483$
 $Q^2=10.9688$, $x_B=0.837731$, $t=-2.71733$, $W^2=3.00452$, $y=0.949273$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8004$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8004$
 P $E= 2.0301$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8004$
 ℓ' $E= 1.5231$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5231$
 P' $E= 0.9530$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1687$
 q_γ $E= 1.3544$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.3544$

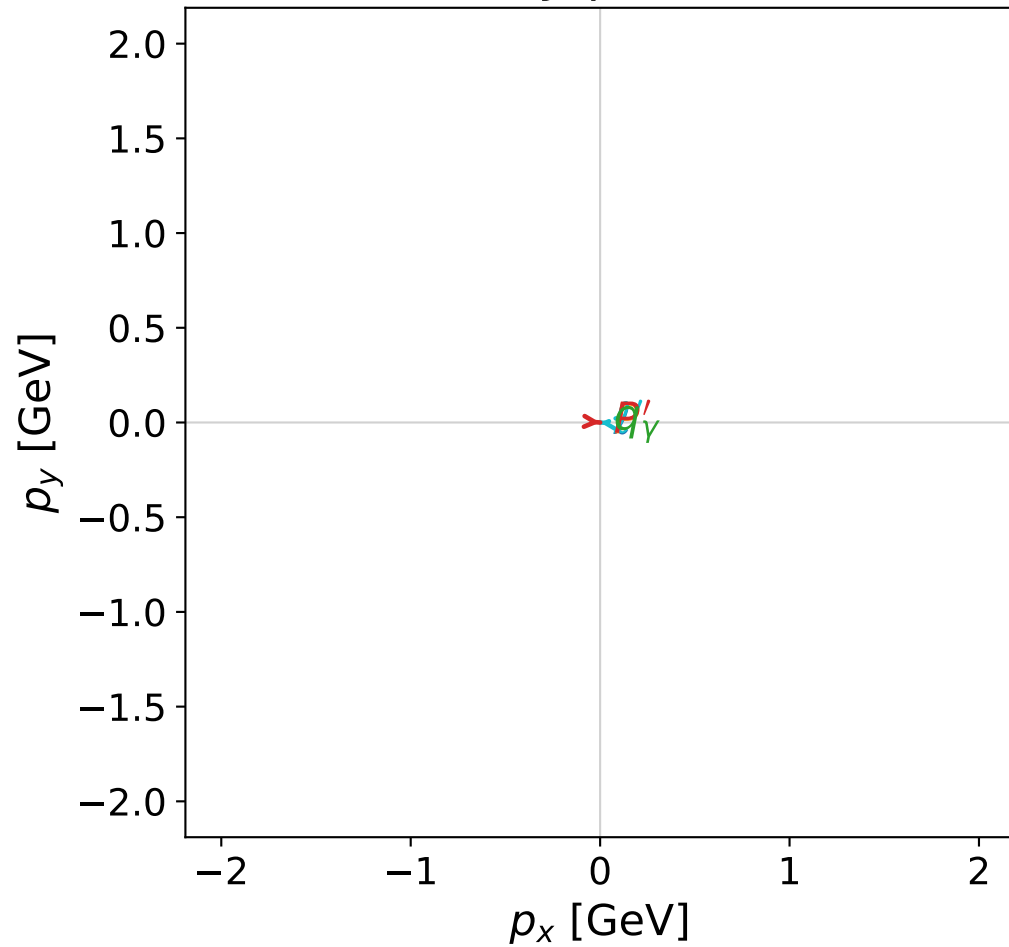


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



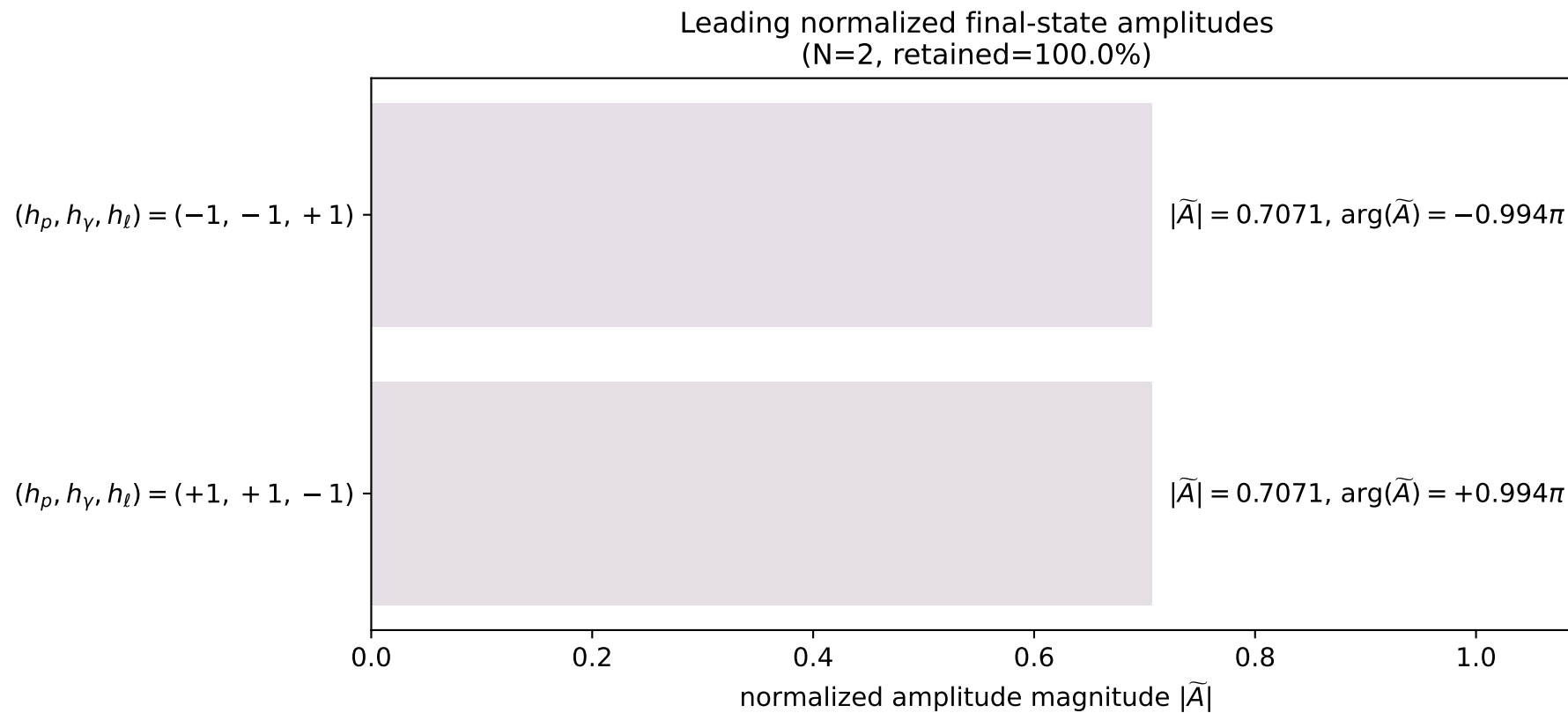
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_123
 $d_{\text{GHZ}}=4.77825\text{e-}08$
 region: polarization_cluster_04_local_minimum_123
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0123
 lepton mixing angle: $\alpha_e=2.3820098$ rad
 incoming lepton: $-0.725123|+\rangle + 0.688619|-\rangle$
 proton mixing angle: $\alpha_p=2.3820131$ rad
 incoming proton: $-0.725126|+\rangle + 0.688617|-\rangle$

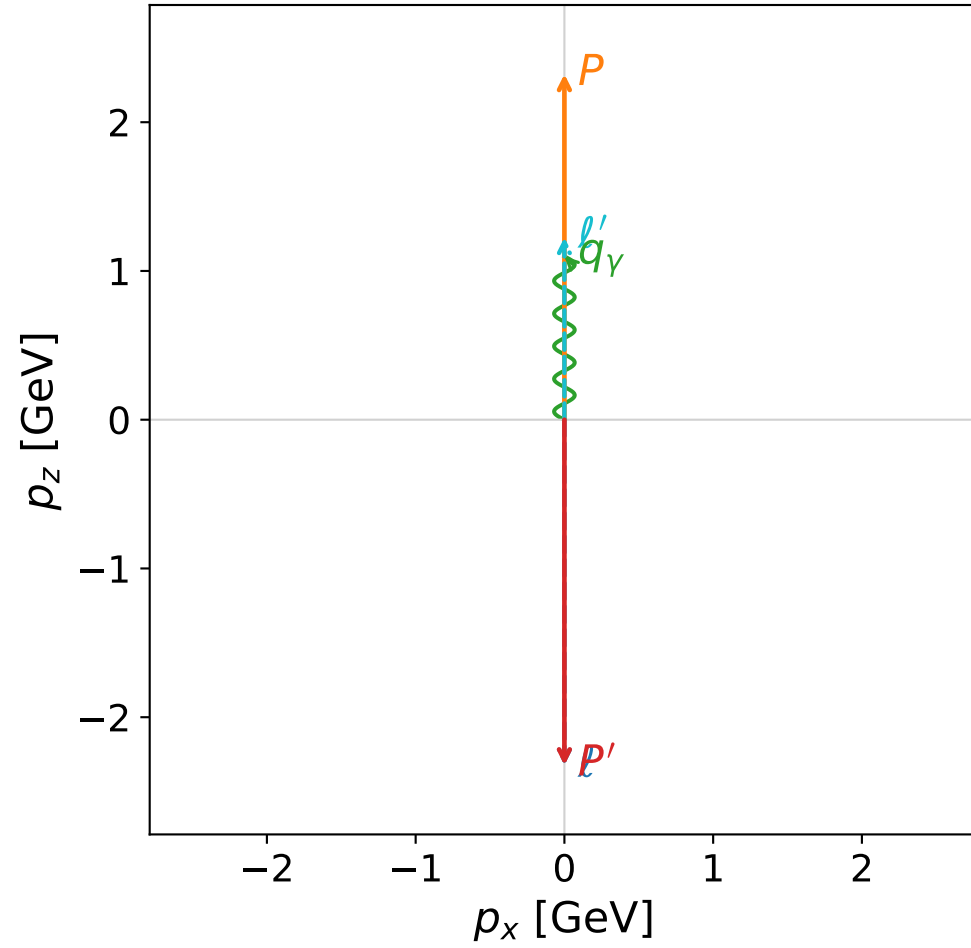
$s=15.0226$, $\sqrt{s}=3.8759$, $|\vec{P}|=1.82445$, $|\vec{P}'|=0.00112548$
 $\theta_{P'}=0.0124896$, $\theta_Y=3.14159$, $\phi_{P'}=6.21424$, $\phi_Y=3.15921$
 $E_Y=1.46951$, $\theta_{\ell'}=9.57262\text{e-}06$, $\phi_{\ell'}=3.07264$
 $Q^2=10.716$, $x_B=0.795185$, $t=-2.08473$, $W^2=3.63996$, $y=0.952862$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8244$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8244$
 P $E= 2.0515$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8244$
 ℓ' $E= 1.4684$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.4684$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0011$
 q_Y $E= 1.4695$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4695$

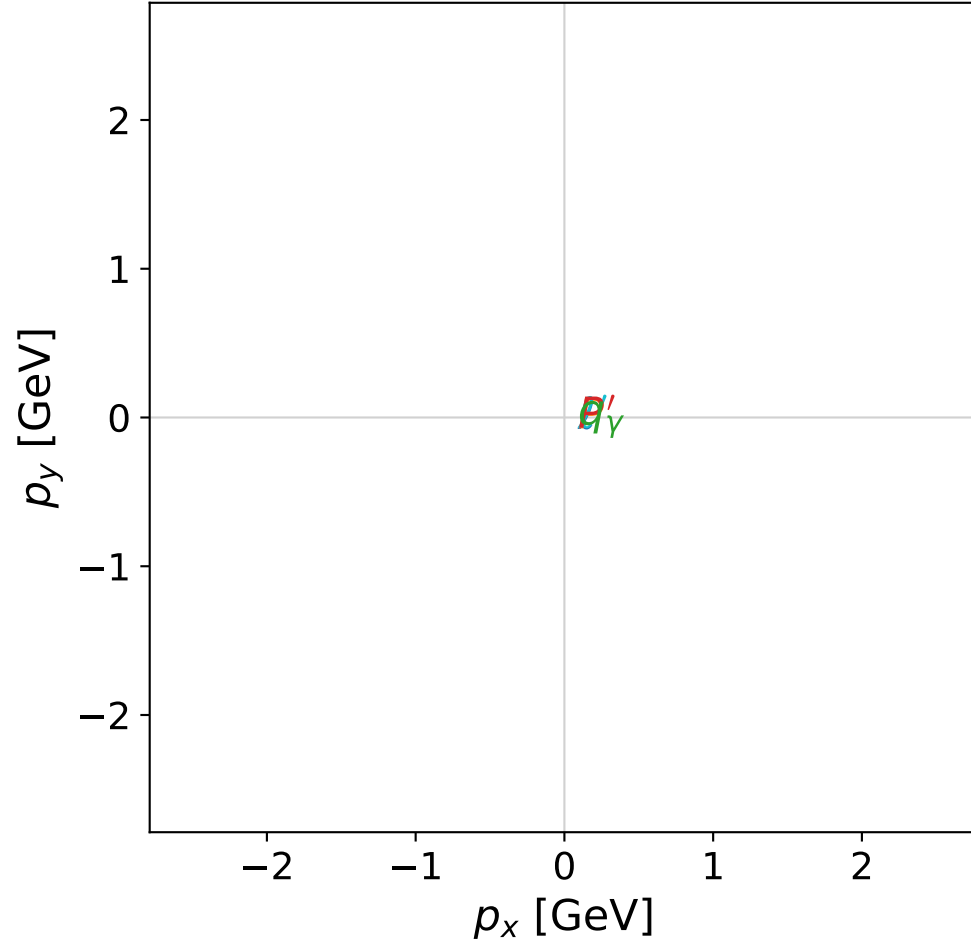


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



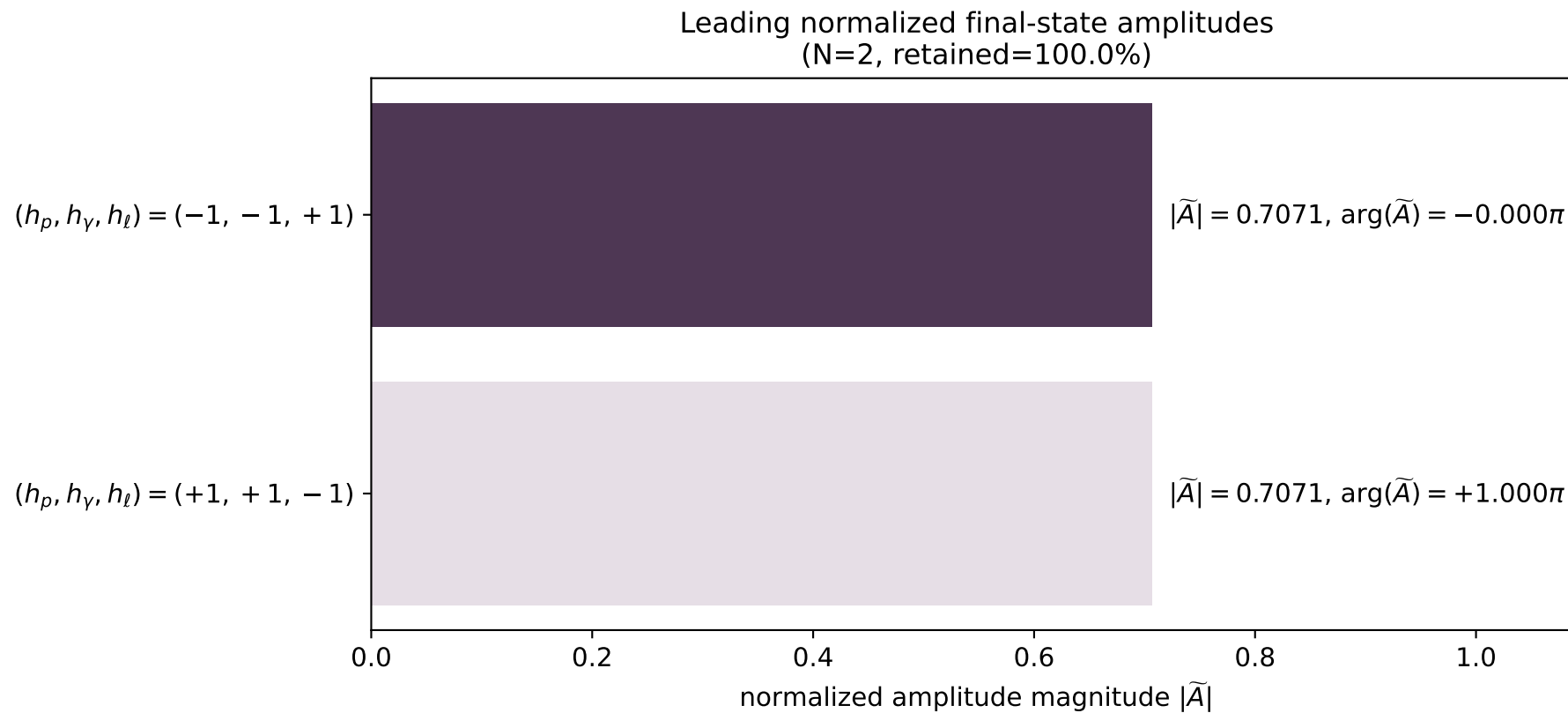
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_125
 $d_{\text{GHZ}}=4.83377\text{e-}08$
 region: polarization_cluster_04_local_minimum_125
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0125
 lepton mixing angle: $\alpha_e=2.1954113$ rad
 incoming lepton: $-0.584785|+\rangle +0.811188|-\rangle$
 proton mixing angle: $\alpha_p=2.5169716$ rad
 incoming proton: $-0.811185|+\rangle +0.58479|-\rangle$

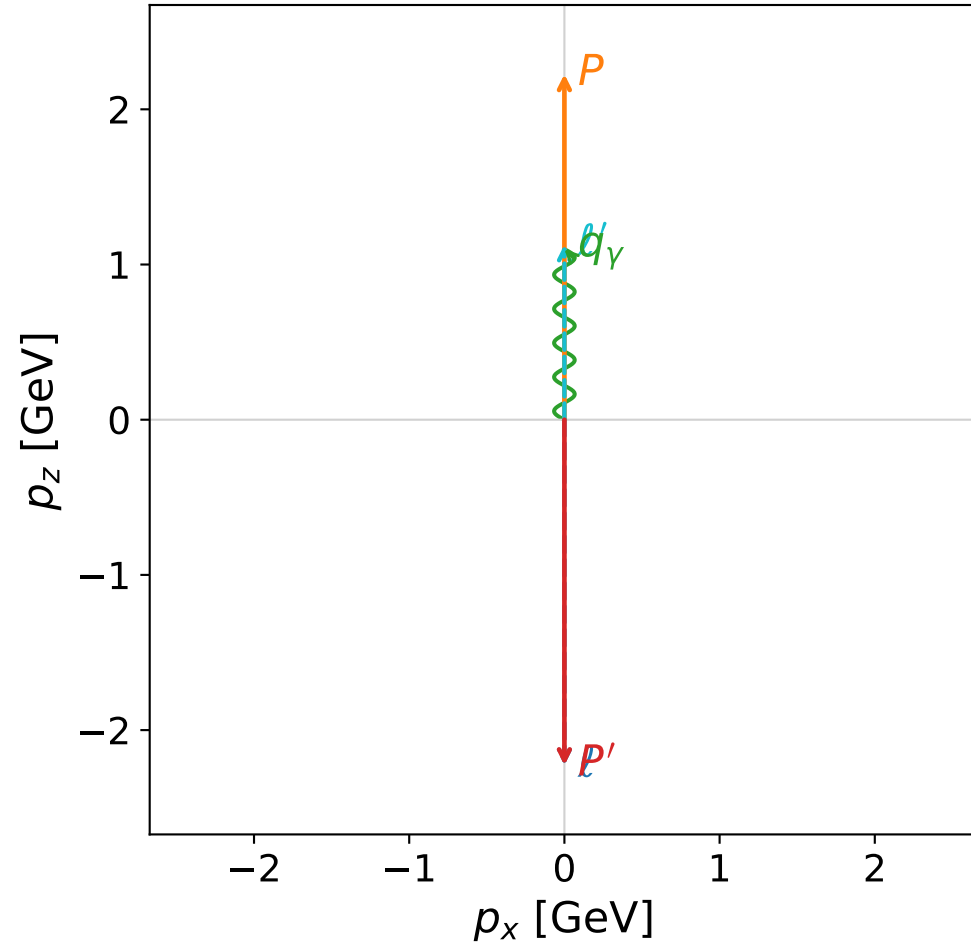
$s=23.3175$, $\sqrt{s}=4.82882$, $|\vec{P}|=2.32331$, $|\vec{P}'|=2.32331$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.30408$, $\phi_\gamma=5.61598$
 $E_\gamma=1.09792$, $\theta_{l'}=0$, $\phi_{l'}=5.44568$
 $Q^2=11.3878$, $x_B=0.517838$, $t=-21.591$, $W^2=11.4831$, $y=0.980098$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3233$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3233$
 P $E= 2.5055$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3233$
 ℓ' $E= 1.2254$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.2254$
 P' $E= 2.5055$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.3233$
 q_γ $E= 1.0979$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0979$

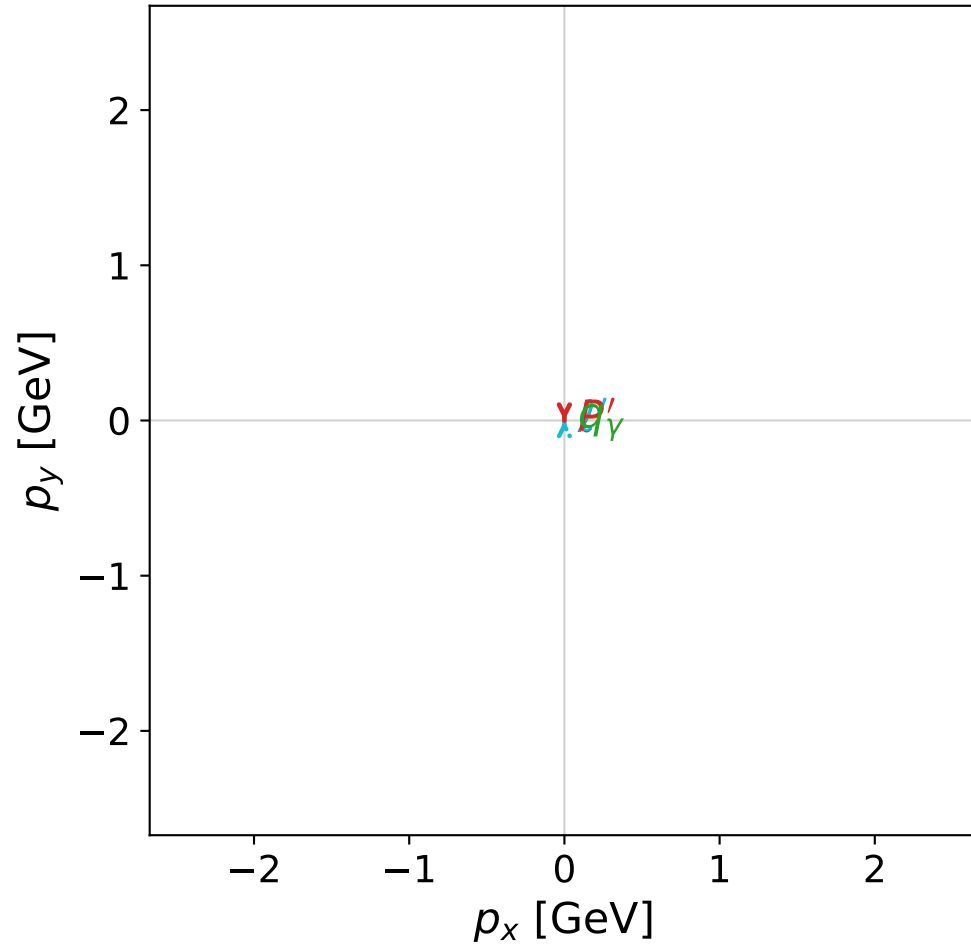


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

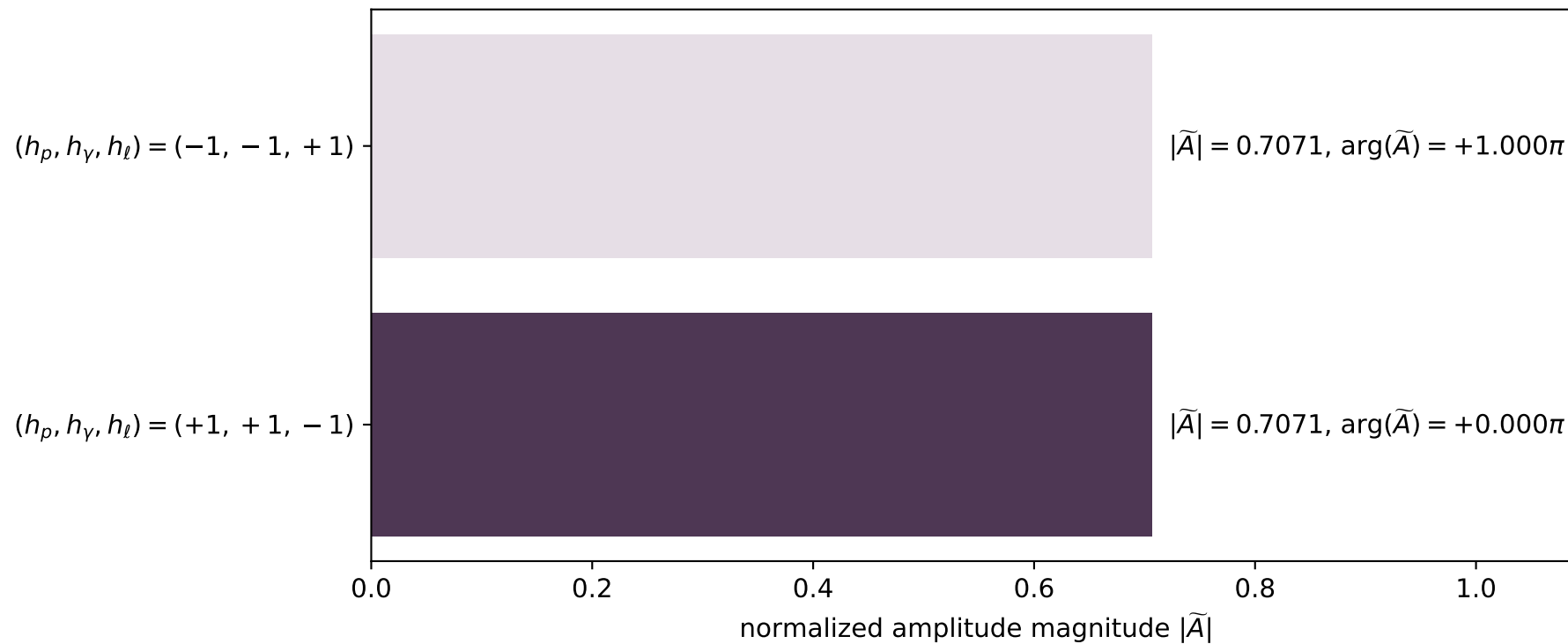


dghz_mixing_angles_polarization_04_local_minimum_133
 $d_{\text{GHZ}}=5.36187\text{e-}08$
 region: polarization_cluster_04_local_minimum_133
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0133
 lepton mixing angle: $\alpha_e=2.2324916$ rad
 incoming lepton: $-0.614455|+\rangle +0.788952|-\rangle$
 proton mixing angle: $\alpha_p=2.4798937$ rad
 incoming proton: $-0.788949|+\rangle +0.614458|-\rangle$

$s=21.5586$, $\sqrt{s}=4.64312$, $|\vec{P}|=2.22681$, $|\vec{P}'|=2.22681$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.93115$, $\phi_\gamma=3.73044$
 $E_\gamma=1.09901$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.78956$
 $Q^2=10.0456$, $x_B=0.496047$, $t=-19.8348$, $W^2=11.0856$, $y=0.97933$

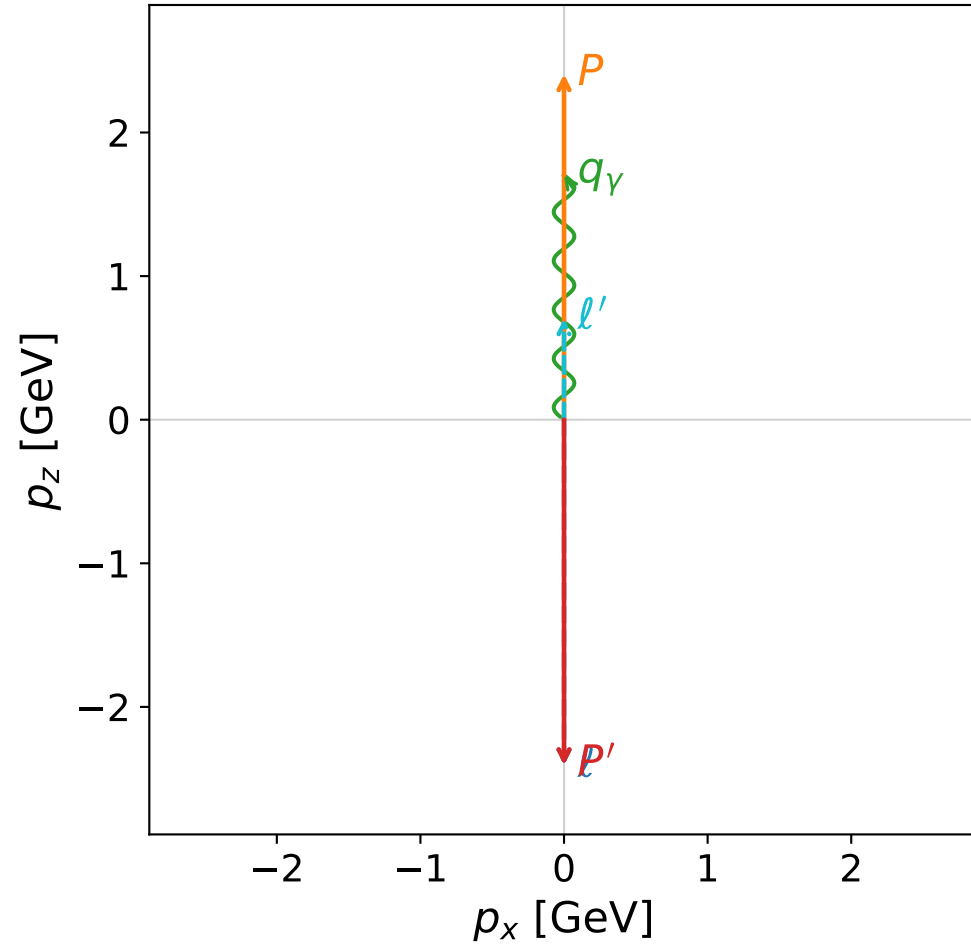
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2268$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2268$
 P $E= 2.4163$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2268$
 ℓ' $E= 1.1278$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.1278$
 P' $E= 2.4163$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.2268$
 q_γ $E= 1.0990$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0990$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

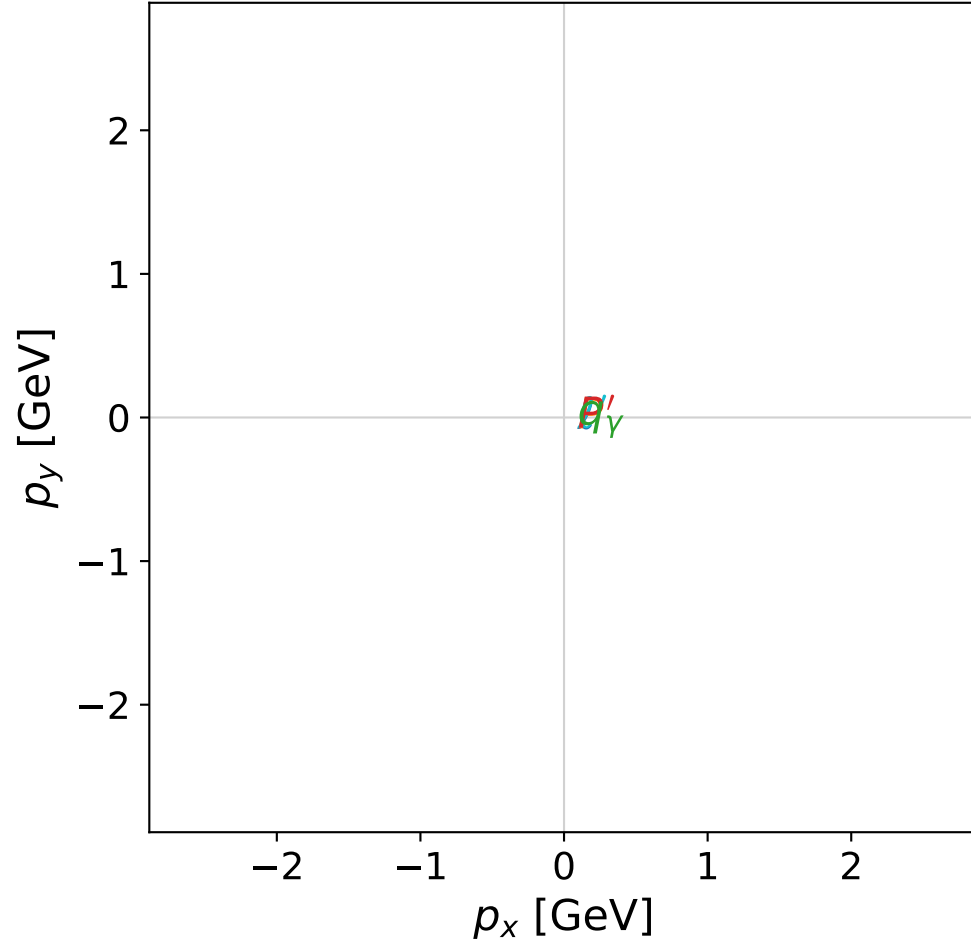


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

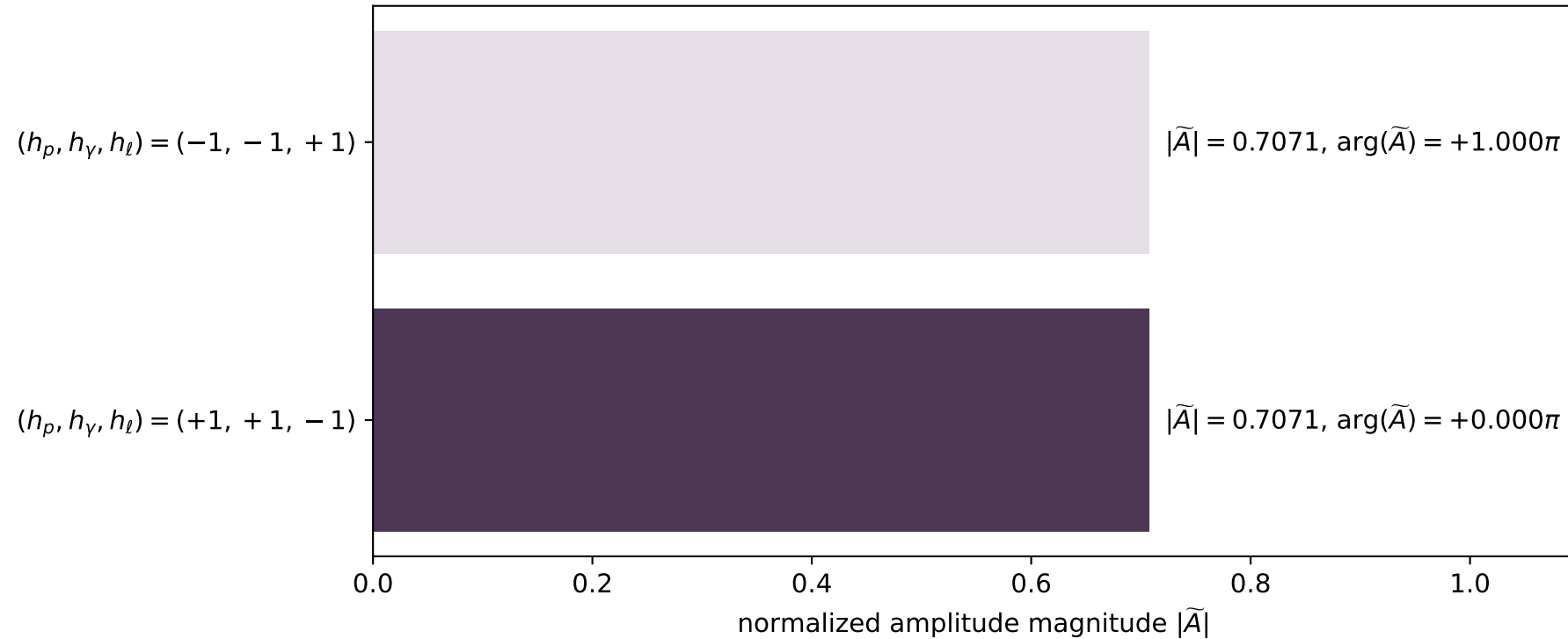


dghz_mixing_angles_polarization_04_local_minimum_145
 $d_{\{\rm GHZ\}}=7.88665\text{e-}08$
 region: polarization_cluster_04_local_minimum_145
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0145
 lepton mixing angle: $\alpha_e=2.4645339$ rad
 incoming lepton: $-0.779419|+\rangle +0.626503|-\rangle$
 proton mixing angle: $\alpha_p=2.2478416$ rad
 incoming proton: $-0.626493|+\rangle +0.779427|-\rangle$

$s=24.8957$, $\sqrt{s}=4.98956$, $|\vec{P}|=2.40661$, $|\vec{P}'|=2.40661$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=2.62423$, $\phi_Y=3.19394$
 $E_Y=1.70223$, $\theta_{l'}=0$, $\phi_{l'}=5.76582$
 $Q^2=6.78069$, $x_B=0.285293$, $t=-23.1671$, $W^2=17.8666$, $y=0.989656$

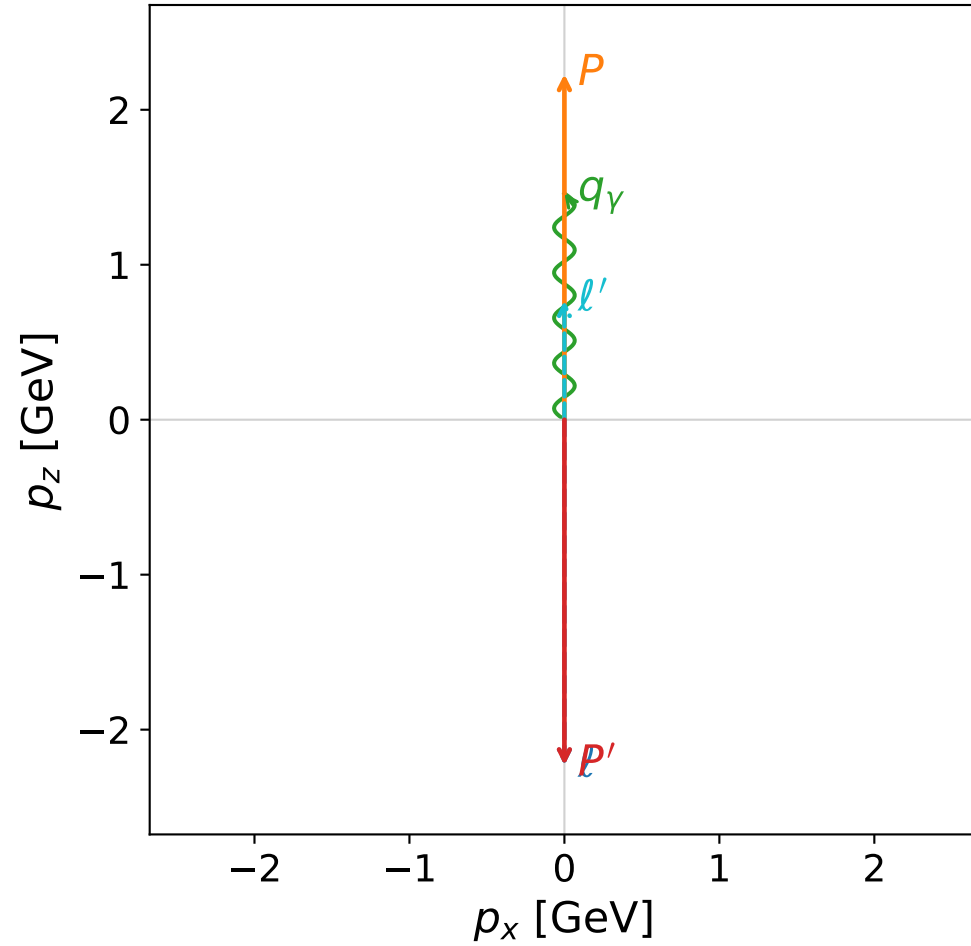
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4066$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4066$
 P $E= 2.5829$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4066$
 l' $E= 0.7044$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7044$
 P' $E= 2.5829$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4066$
 q_Y $E= 1.7022$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.7022$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

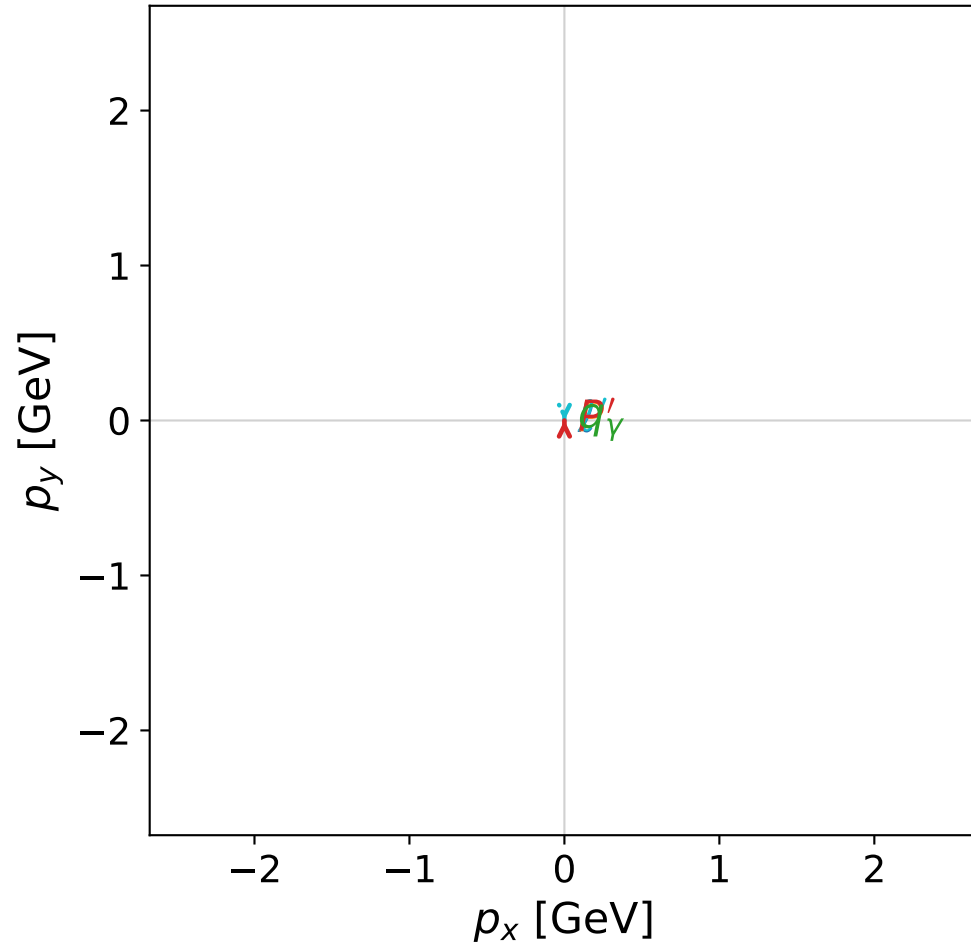


dghz_mixing_angles_polarization_04_local_minimum_153
 $d_{\text{GHZ}}=9.35668\text{e-}08$
 region: polarization_cluster_04_local_minimum_153
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0153
 lepton mixing angle: $\alpha_e=2.6674747$ rad
 incoming lepton: $-0.889696|+\rangle +0.456554|-\rangle$
 proton mixing angle: $\alpha_p=2.0449223$ rad
 incoming proton: $-0.456561|+\rangle +0.889692|-\rangle$

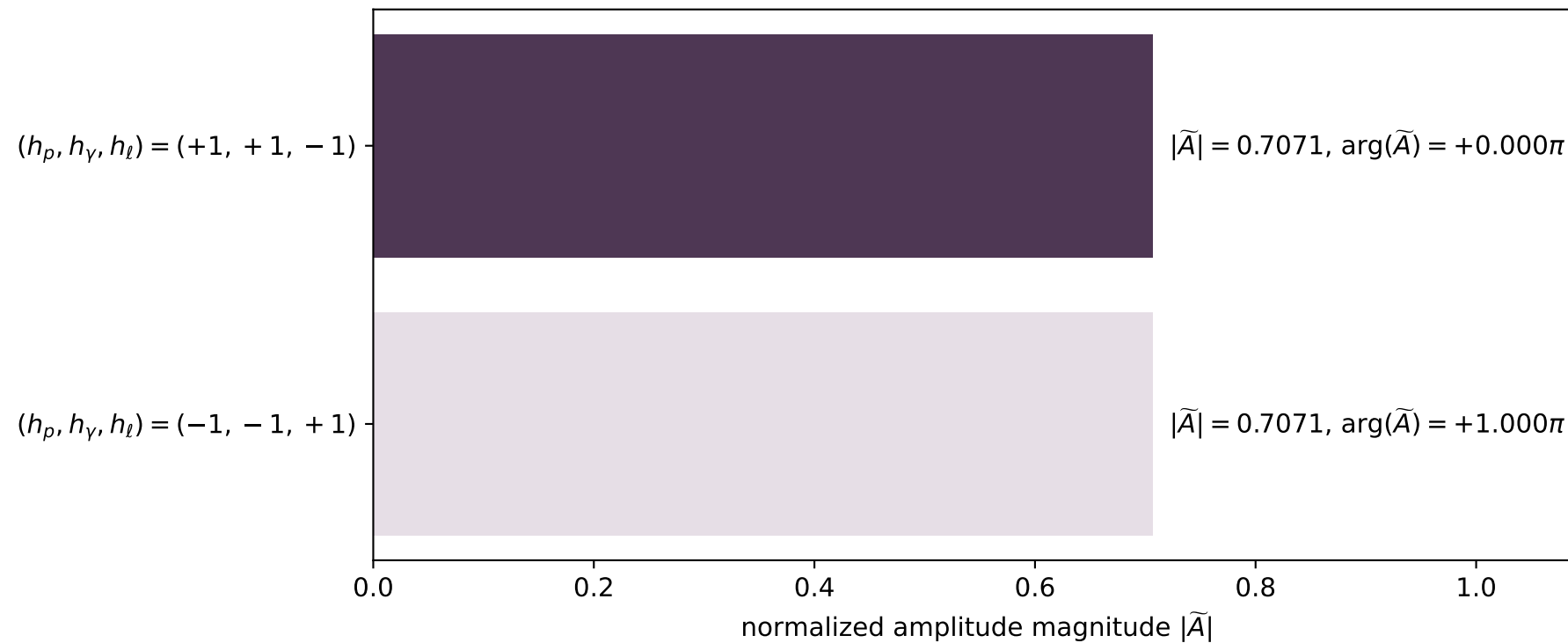
$s=21.6182$, $\sqrt{s}=4.64954$, $|\vec{P}|=2.23015$, $|\vec{P}'|=2.23015$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.53792$, $\phi_\gamma=3.63958$
 $E_\gamma=1.45997$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.67951$
 $Q^2=6.8705$, $x_B=0.336017$, $t=-19.8944$, $W^2=14.4563$, $y=0.985945$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2302$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2302$
 P $E= 2.4194$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2302$
 ℓ' $E= 0.7702$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7702$
 P' $E= 2.4194$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2302$
 q_γ $E= 1.4600$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4600$

x--y plane

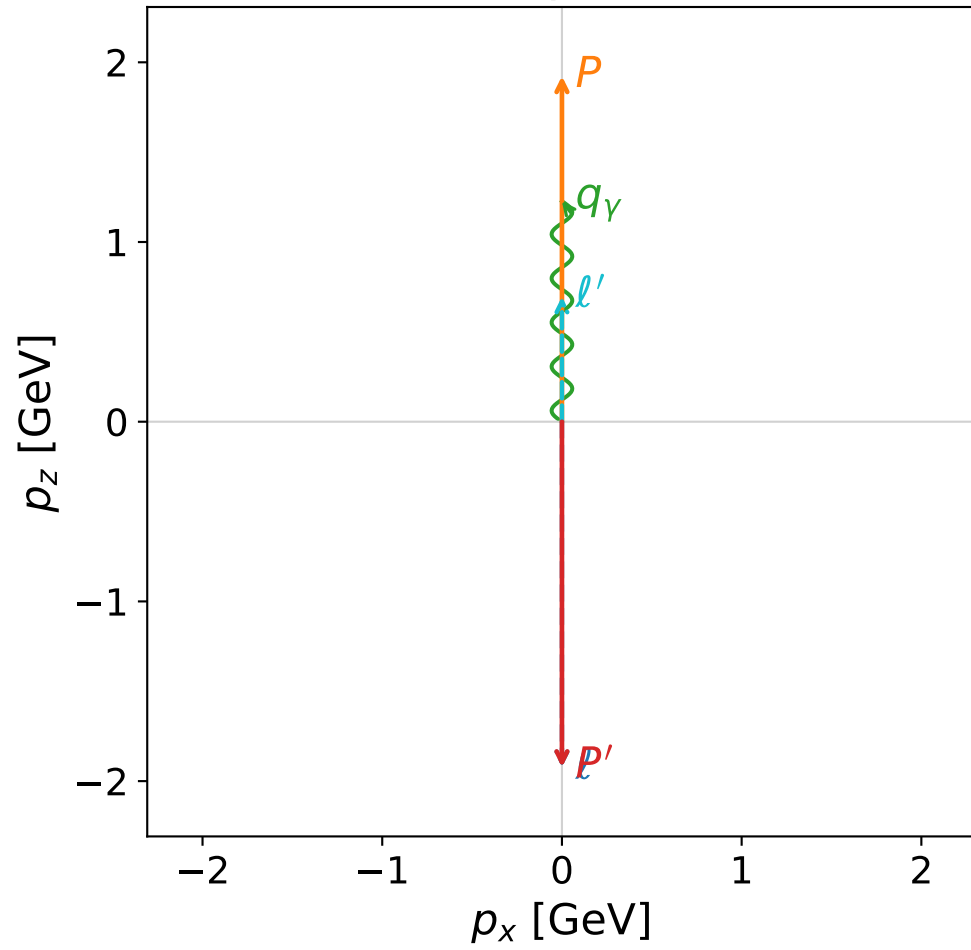


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

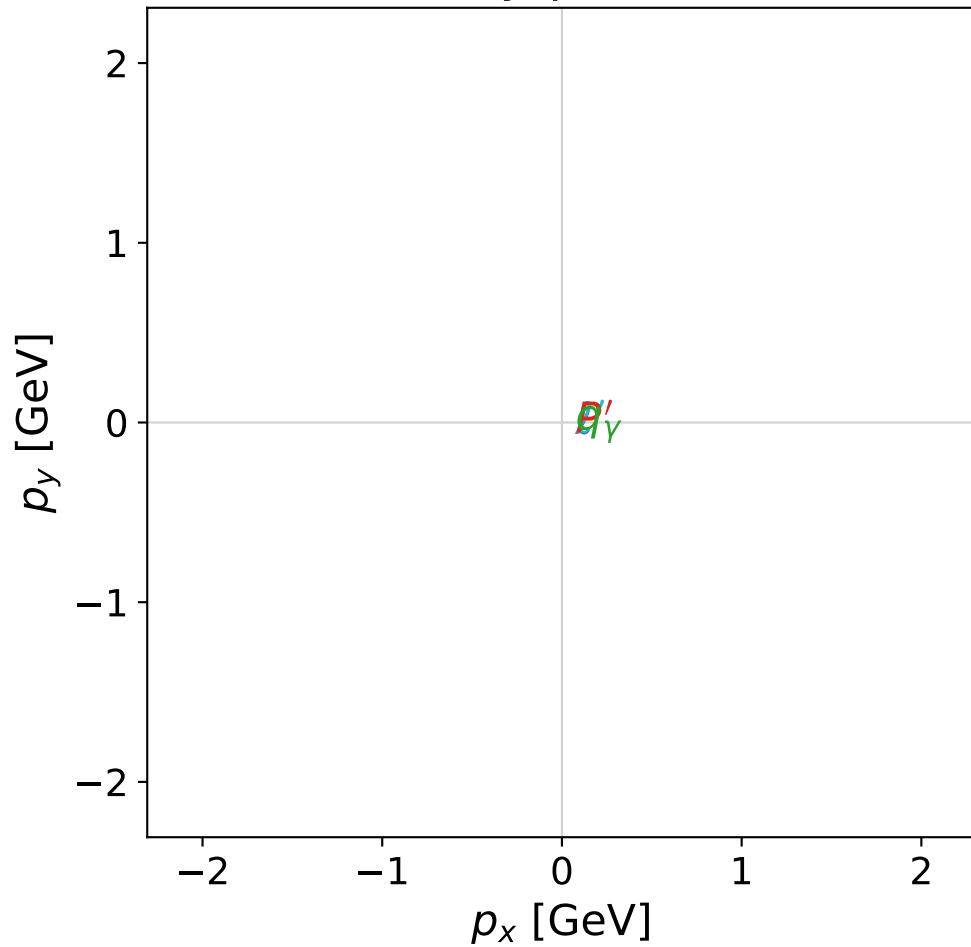


dghz_mixing_angles_polarization_04_local_minimum_155
 $d_{\text{GHZ}}=9.76885\text{e-}08$
 region: polarization_cluster_04_local_minimum_155
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0155
 lepton mixing angle: $\alpha_e=2.379922$ rad
 incoming lepton: $-0.723684|+\rangle + 0.690131|-\rangle$
 proton mixing angle: $\alpha_p=2.3324913$ rad
 incoming proton: $-0.690149|+\rangle + 0.723667|-\rangle$

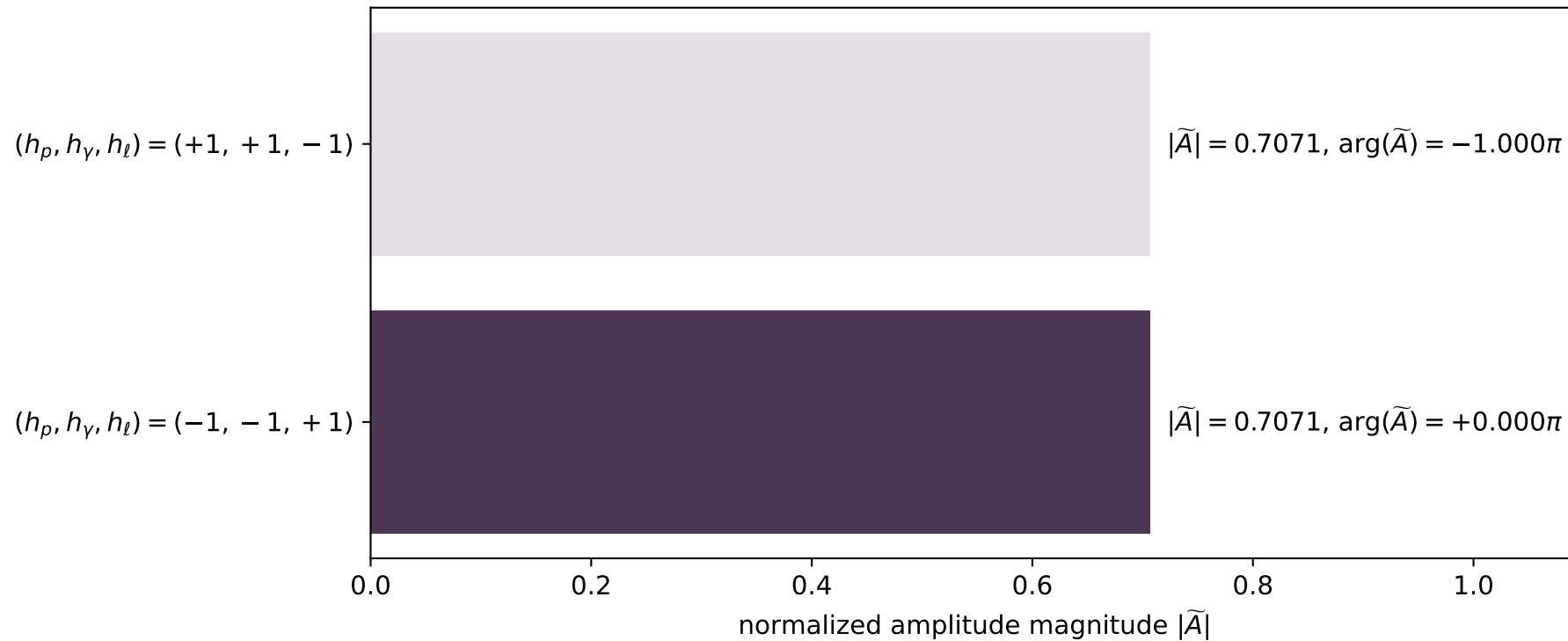
$s=16.5062$, $\sqrt{s}=4.06278$, $|\vec{P}|=1.92311$, $|\vec{P}'|=1.92311$
 $\theta_{P'}=3.14159$, $\theta_\gamma=0$, $\phi_{P'}=4.19286$, $\phi_\gamma=0.182475$
 $E_\gamma=1.22715$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.05127$
 $Q^2=5.35358$, $x_B=0.34934$, $t=-14.7934$, $W^2=10.8511$, $y=0.98071$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9231$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9231$
 P $E= 2.1397$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9231$
 ℓ' $E= 0.6960$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6960$
 P' $E= 2.1397$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9231$
 q_γ $E= 1.2272$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2272$

x--y plane

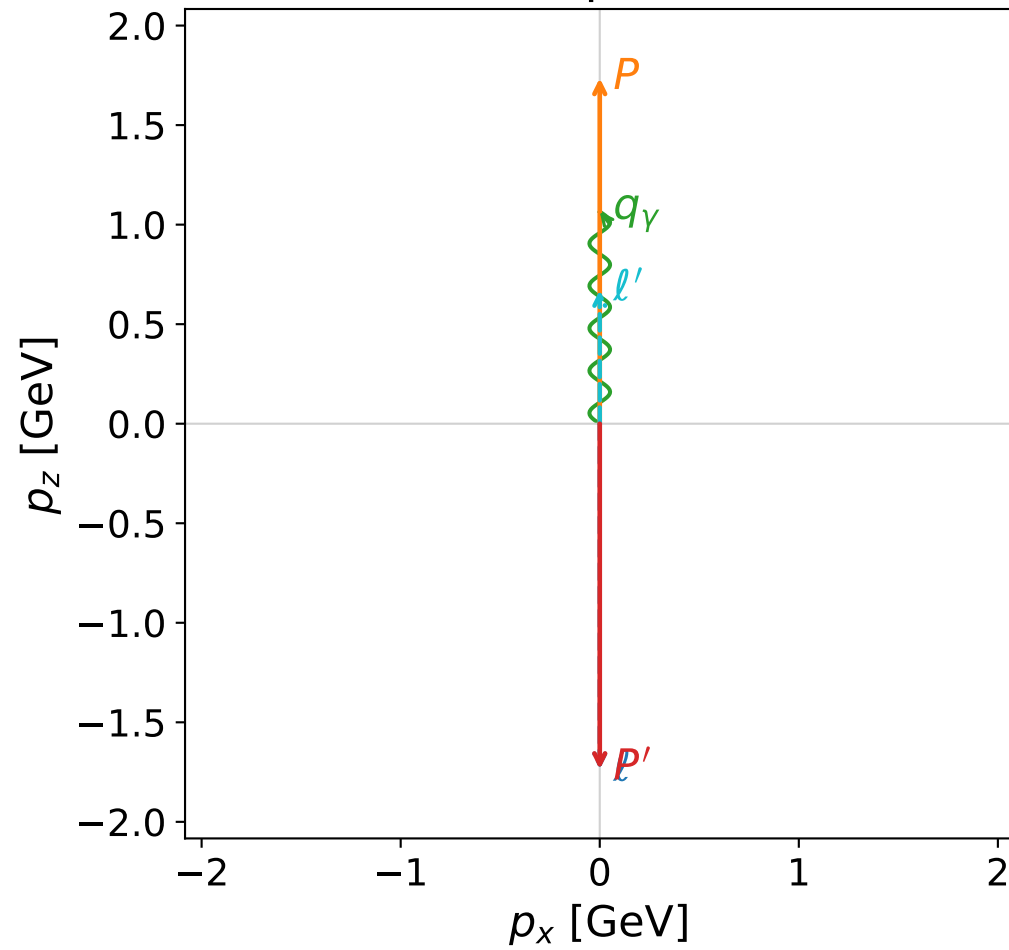


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

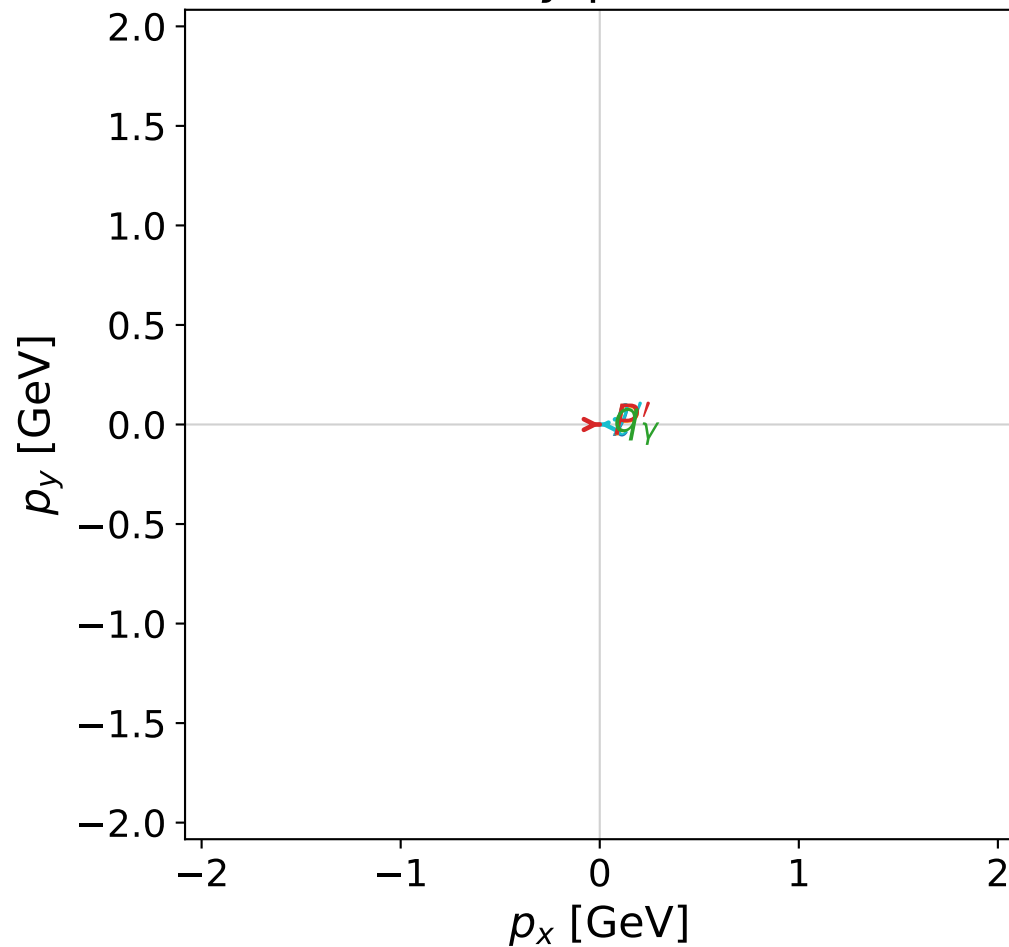


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

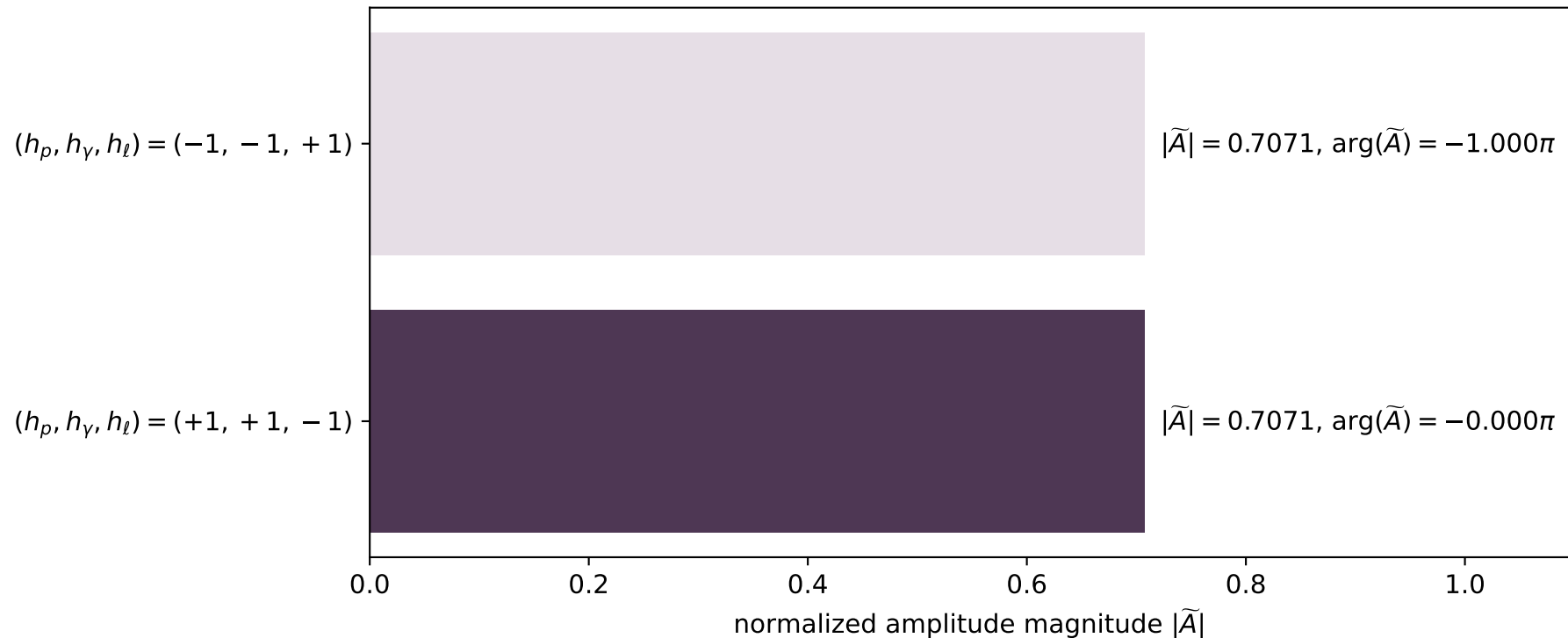


dghz_mixing_angles_polarization_04_local_minimum_162
 $d_{\{\rm GHZ\}}=1.14455\text{e-}07$
 region: polarization_cluster_04_local_minimum_162
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0162
 lepton mixing angle: $\alpha_e=2.254464$ rad
 incoming lepton: $-0.631641|+\rangle +0.775261|-\rangle$
 proton mixing angle: $\alpha_p=2.4579197$ rad
 incoming proton: $-0.775258|+\rangle +0.631645|-\rangle$

$s=13.7588$, $\sqrt{s}=3.70928$, $|\vec{P}|=1.73604$, $|\vec{P}'|=1.73604$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.143001$, $\phi_\gamma=2.52234$
 $E_\gamma=1.06513$, $\theta_{l'}=0$, $\phi_{l'}=3.28459$
 $Q^2=4.65894$, $x_B=0.370915$, $t=-12.0554$, $W^2=8.78157$, $y=0.975287$

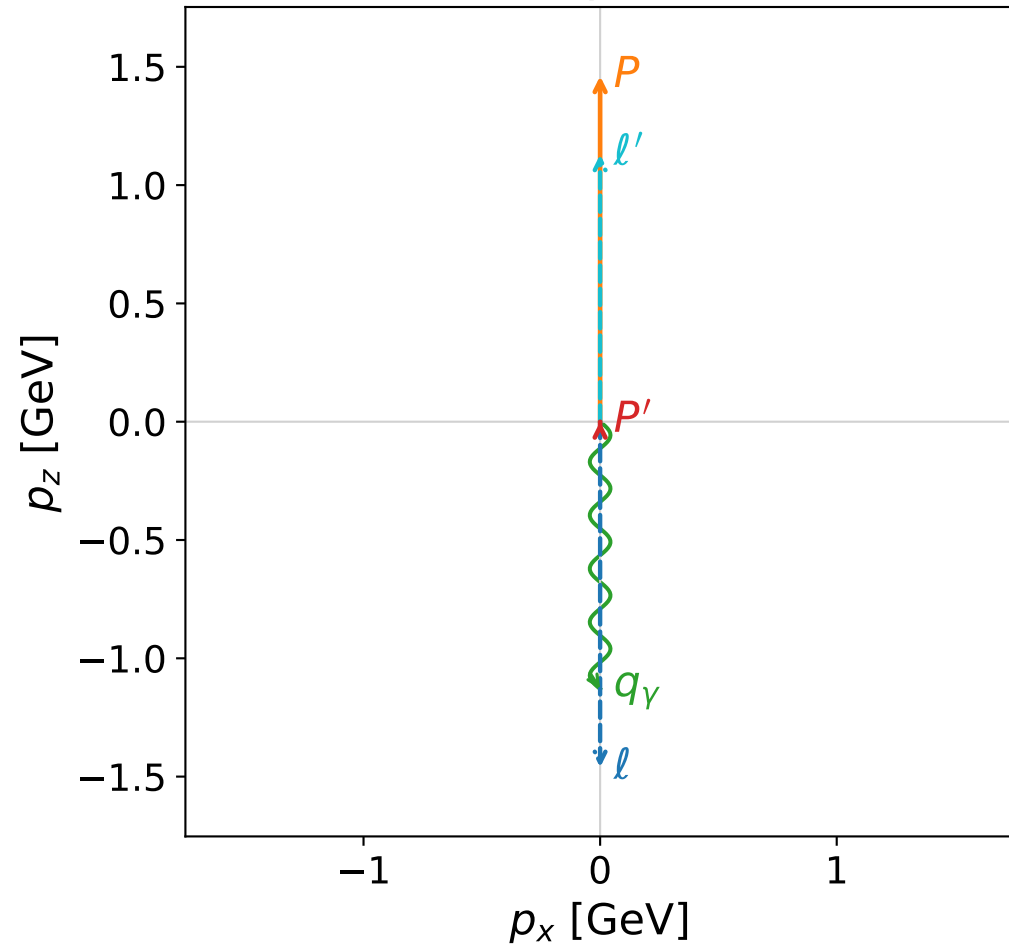
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7360$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7360$
 P $E= 1.9732$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7360$
 ℓ' $E= 0.6709$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.6709$
 P' $E= 1.9732$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7360$
 q_γ $E= 1.0651$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.0651$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

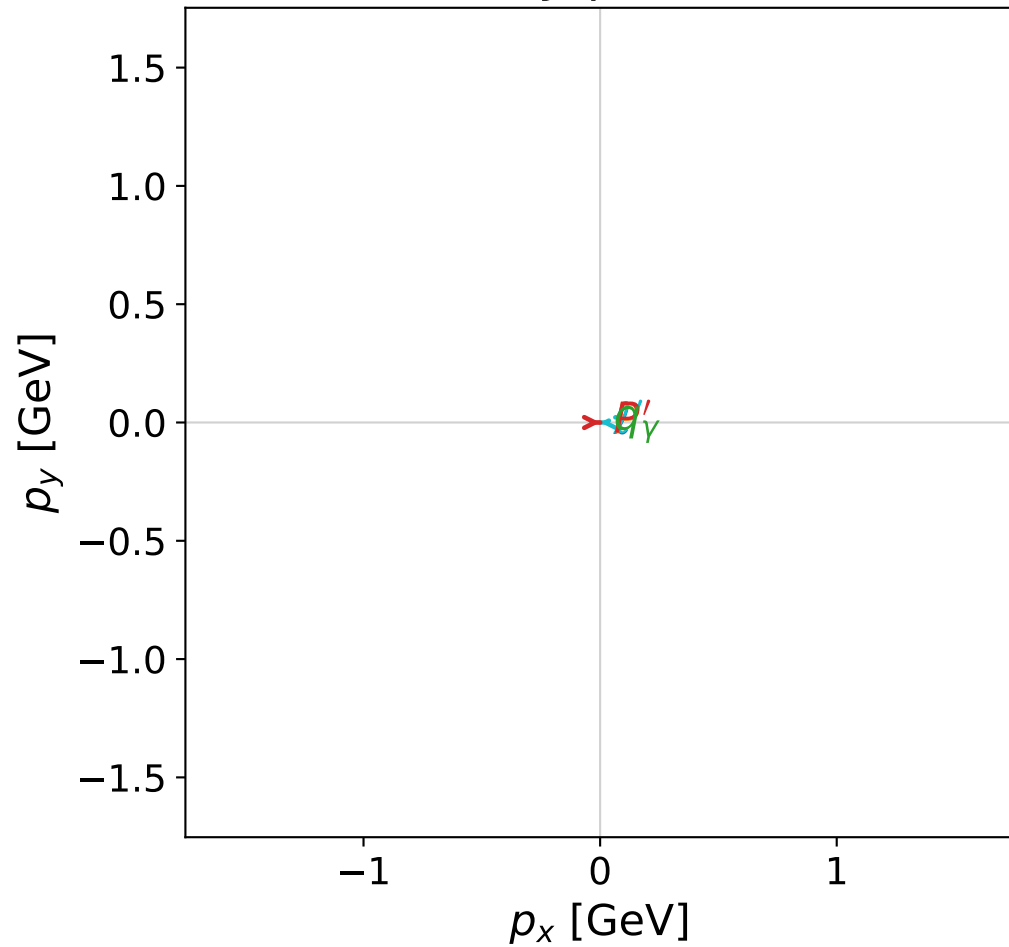


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

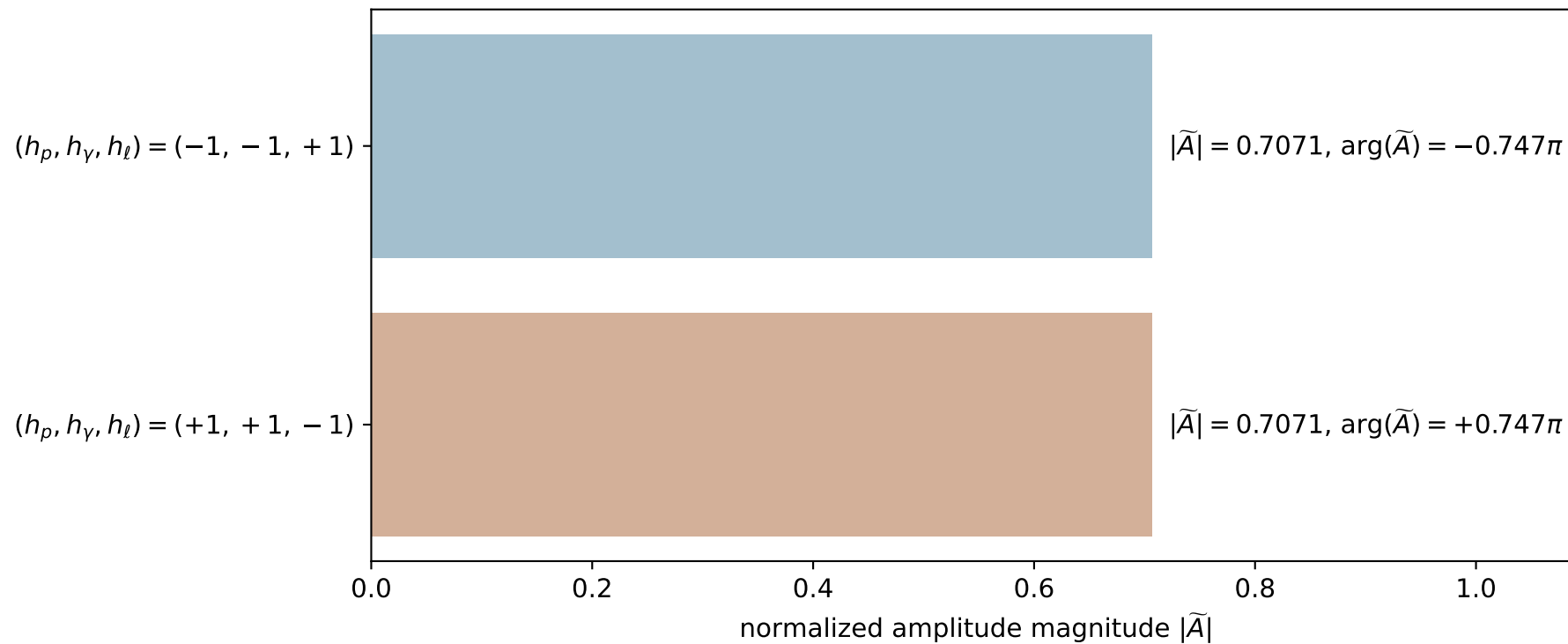


dghz_mixing_angles_polarization_04_local_minimum_166
 $d_{\text{GHZ}}=1.3556\text{e-}07$
 region: polarization_cluster_04_local_minimum_166
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0166
 lepton mixing angle: $\alpha_e=1.8819764$ rad
 incoming lepton: $-0.306182|+\rangle + 0.951973|-\rangle$
 proton mixing angle: $\alpha_p=1.8819874$ rad
 incoming proton: $-0.306193|+\rangle + 0.95197|-\rangle$

$s=10.2184$, $\sqrt{s}=3.19662$, $|\vec{P}|=1.46069$, $|\vec{P}'|=0.000294288$
 $\theta_{P'}=0.000613329$, $\theta_Y=3.14159$, $\phi_{P'}=0.00132073$, $\phi_Y=3.93698$
 $E_Y=1.12946$, $\theta_{\ell'}=1.59797\text{e-}07$, $\phi_{\ell'}=3.14291$
 $Q^2=6.59744$, $x_B=0.75685$, $t=-1.49606$, $W^2=2.99937$, $y=0.933439$

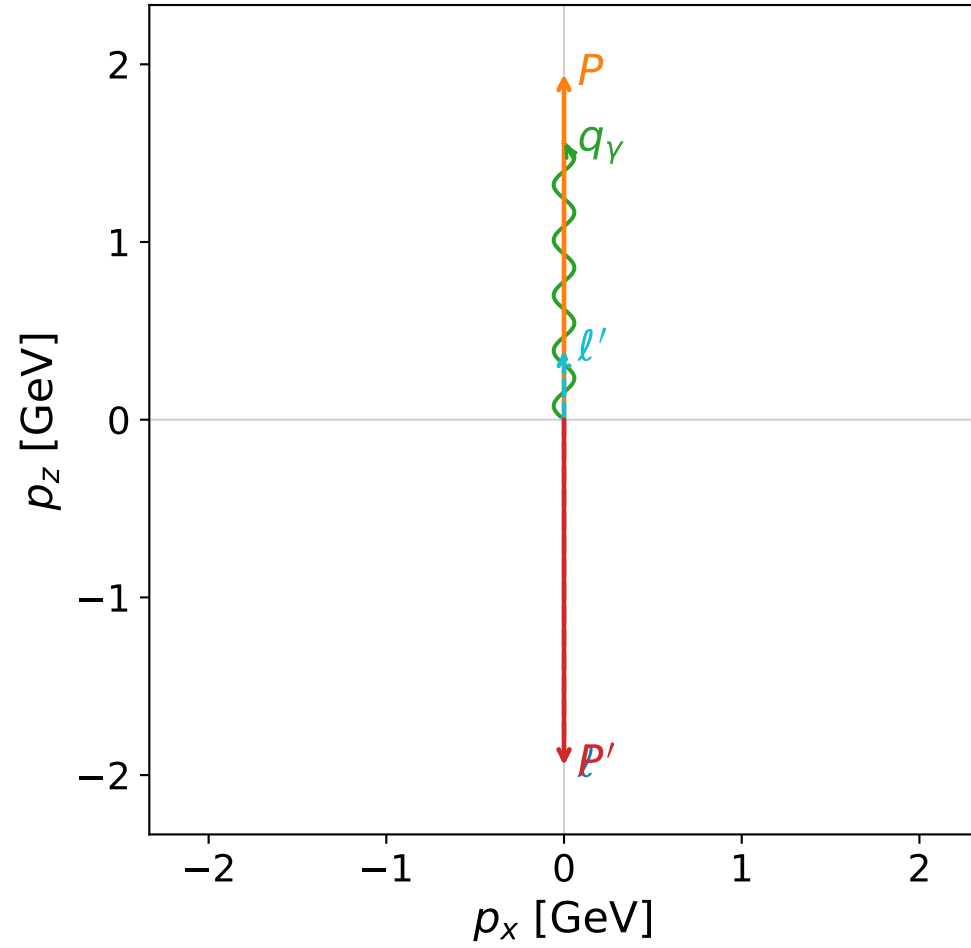
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.4607$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4607$
 P $E= 1.7359$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4607$
 ℓ' $E= 1.1292$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.1292$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0003$
 q_Y $E= 1.1295$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.1295$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

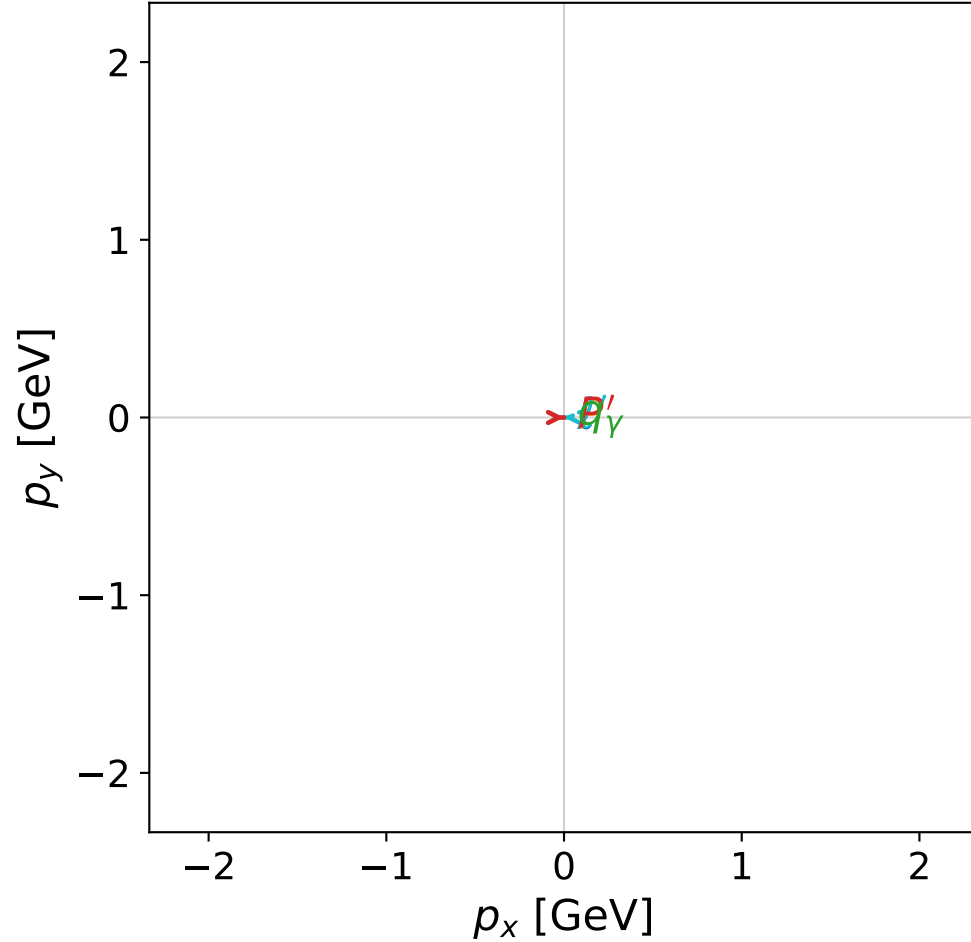


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



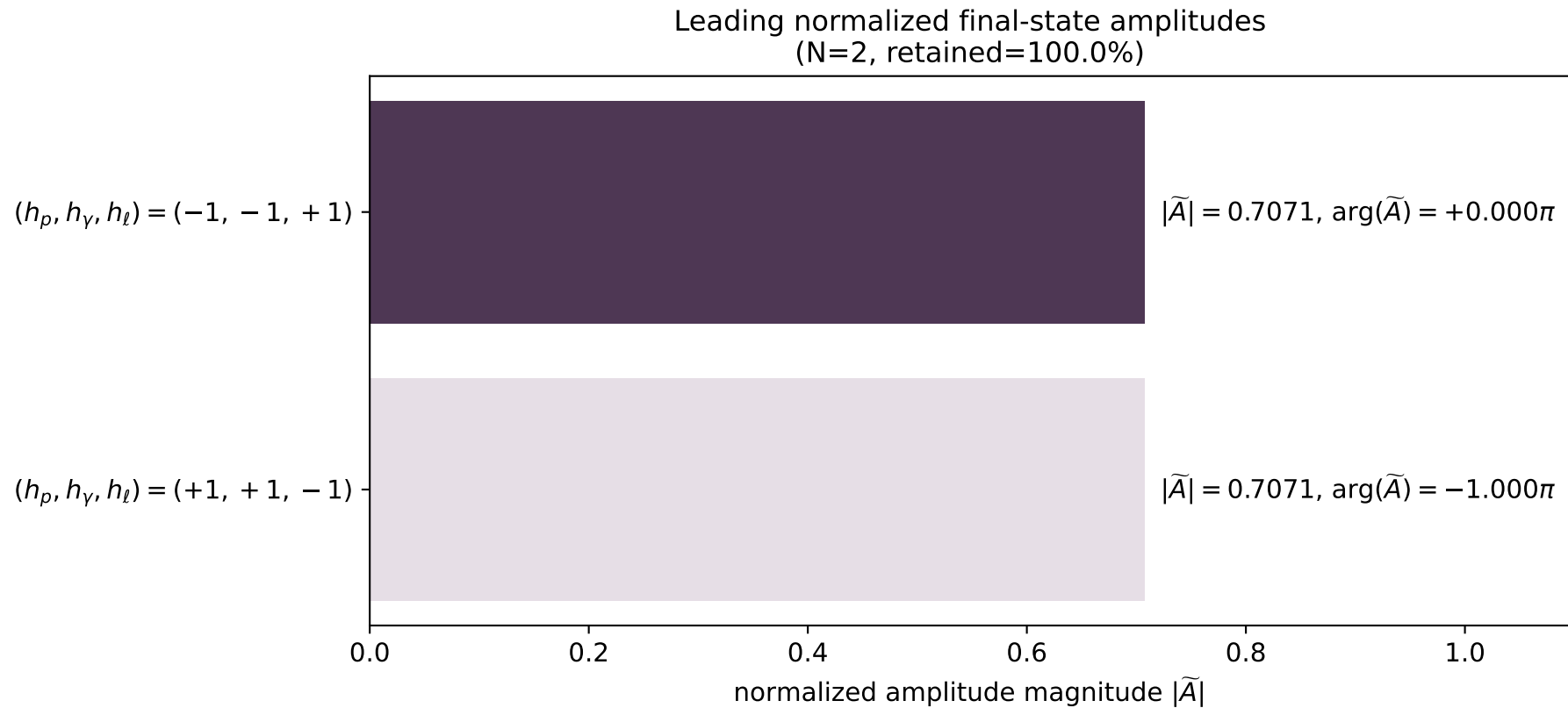
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_169
 $d_{\{\text{rm GHZ}\}}=1.74025\text{e-}07$
 region: polarization_cluster_04_local_minimum_169
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0169
 lepton mixing angle: $\alpha_e=2.2827776$ rad
 incoming lepton: $-0.653335|+\rangle +0.757069|-\rangle$
 proton mixing angle: $\alpha_p=2.4296097$ rad
 incoming proton: $-0.757068|+\rangle +0.653336|-\rangle$

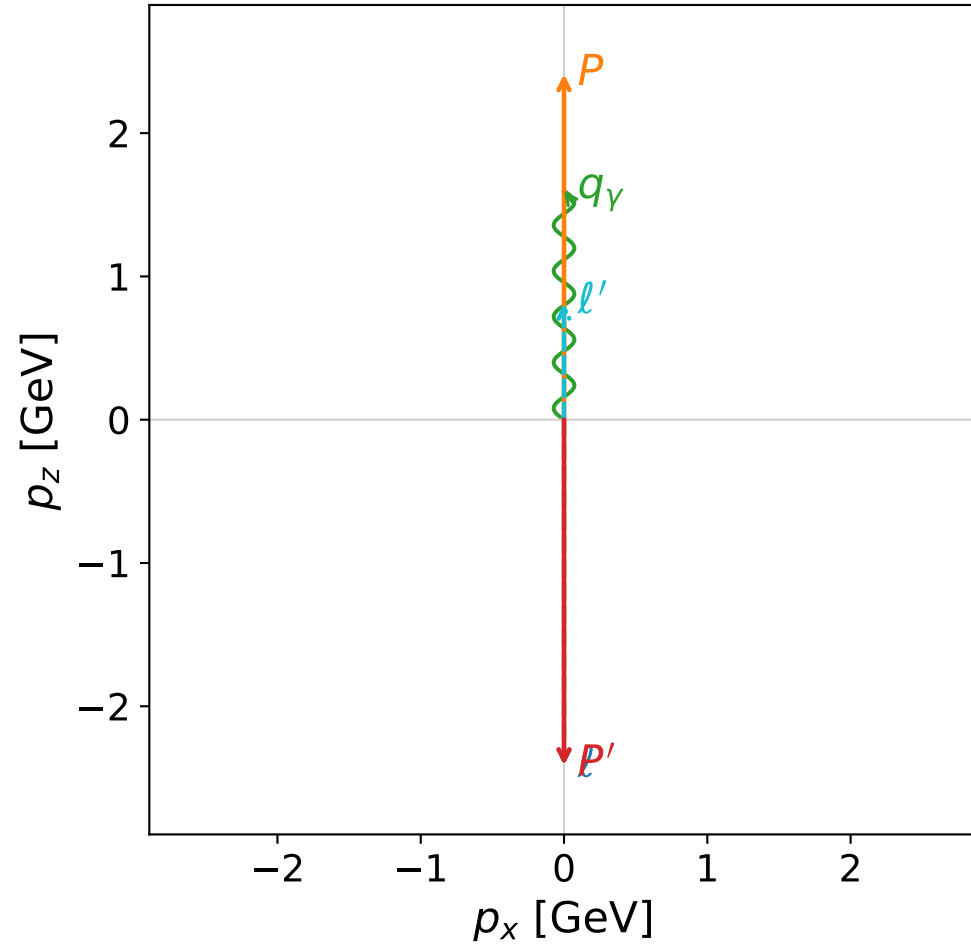
$s=16.843$, $\sqrt{s}=4.10403$, $|\vec{P}|=1.94482$, $|\vec{P}'|=1.94482$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=6.13063$, $\phi_Y=0.266108$
 $E_Y=1.55486$, $\theta_{l'}=0$, $\phi_{l'}=2.98904$
 $Q^2=3.03359$, $x_B=0.192048$, $t=-15.1293$, $W^2=13.6422$, $y=0.989526$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9448$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9448$
 P $E= 2.1592$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9448$
 ℓ' $E= 0.3900$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3900$
 P' $E= 2.1592$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9448$
 q_Y $E= 1.5549$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5549$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

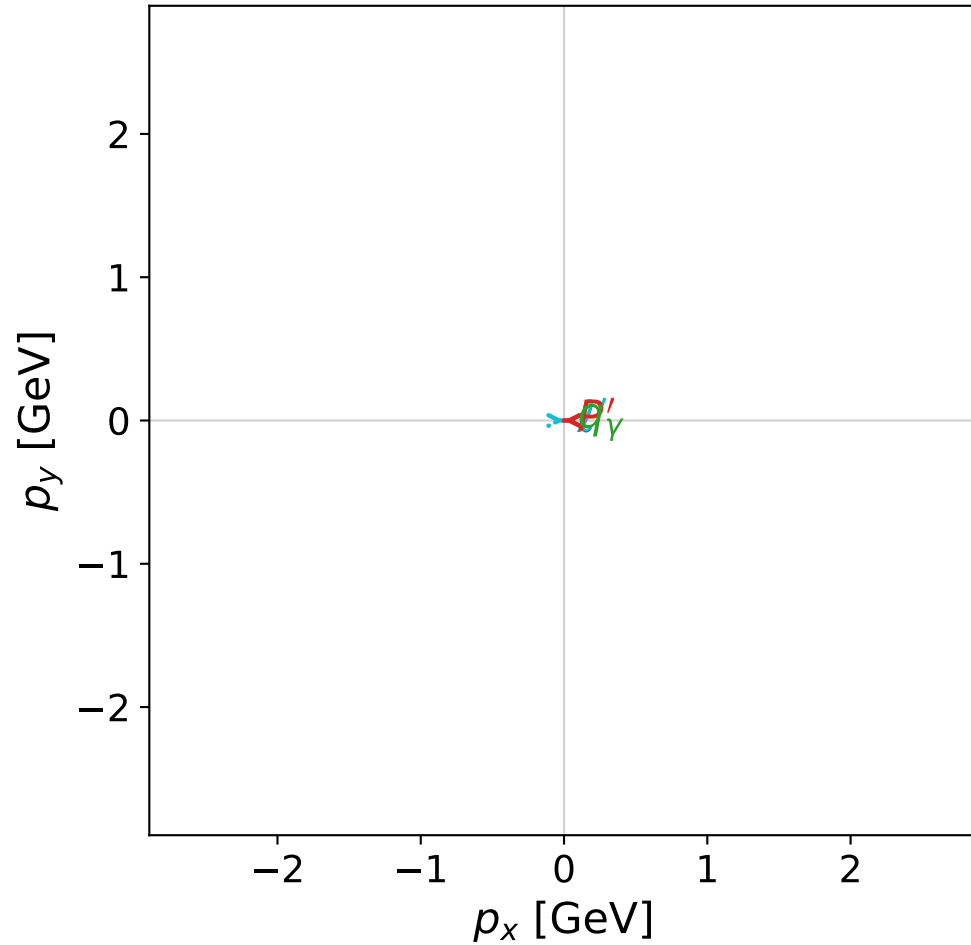


dghz_mixing_angles_polarization_04_local_minimum_176
 $d_{\{\rm GHZ\}}=3.49411\text{e-}07$
 region: polarization_cluster_04_local_minimum_176
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0176
 lepton mixing angle: $\alpha_e=3.0459519$ rad
 incoming lepton: $-0.99543|+\rangle +0.095495|-\rangle$
 proton mixing angle: $\alpha_p=1.6664366$ rad
 incoming proton: $-0.0954945|+\rangle +0.99543|-\rangle$

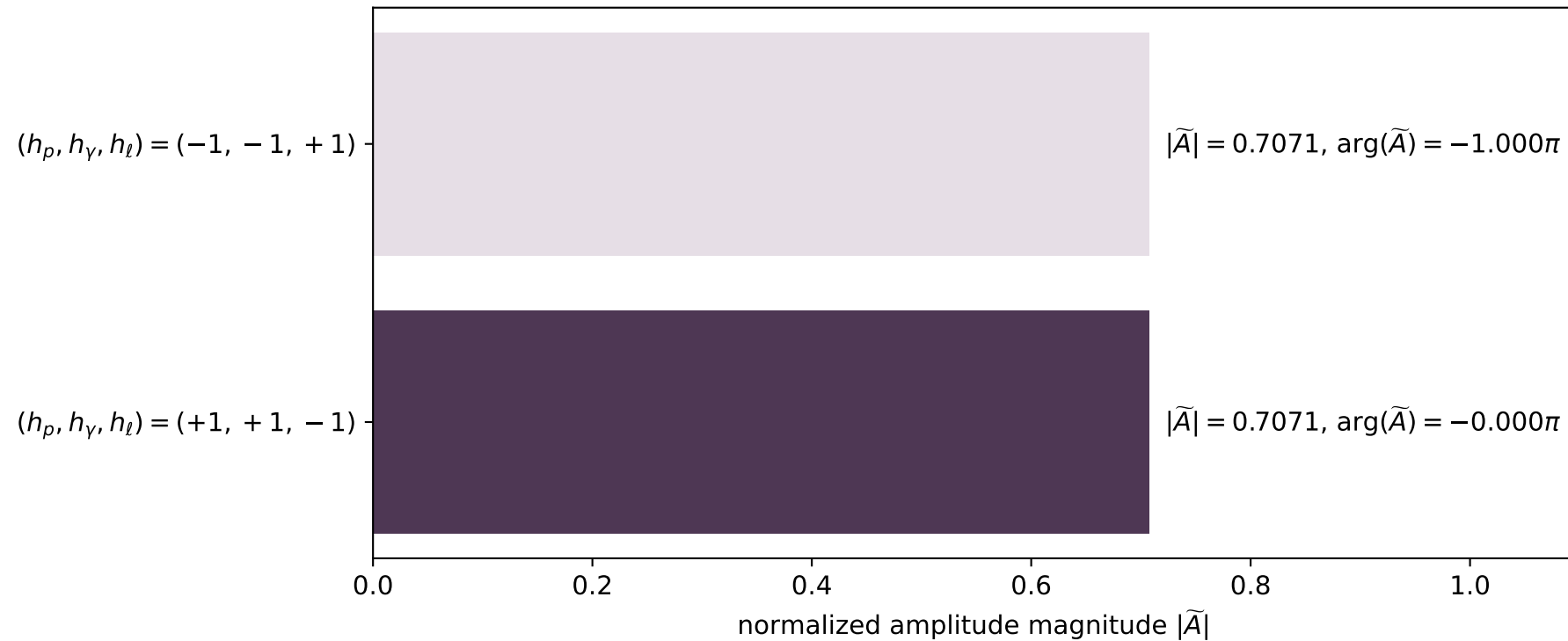
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.3165$, $\phi_\gamma=2.32512$
 $E_\gamma=1.59717$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.17491$
 $Q^2=7.86165$, $x_B=0.329859$, $t=-23.2713$, $W^2=16.8516$, $y=0.988111$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 0.8148$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8148$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4120$
 q_γ $E= 1.5972$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5972$

x--y plane

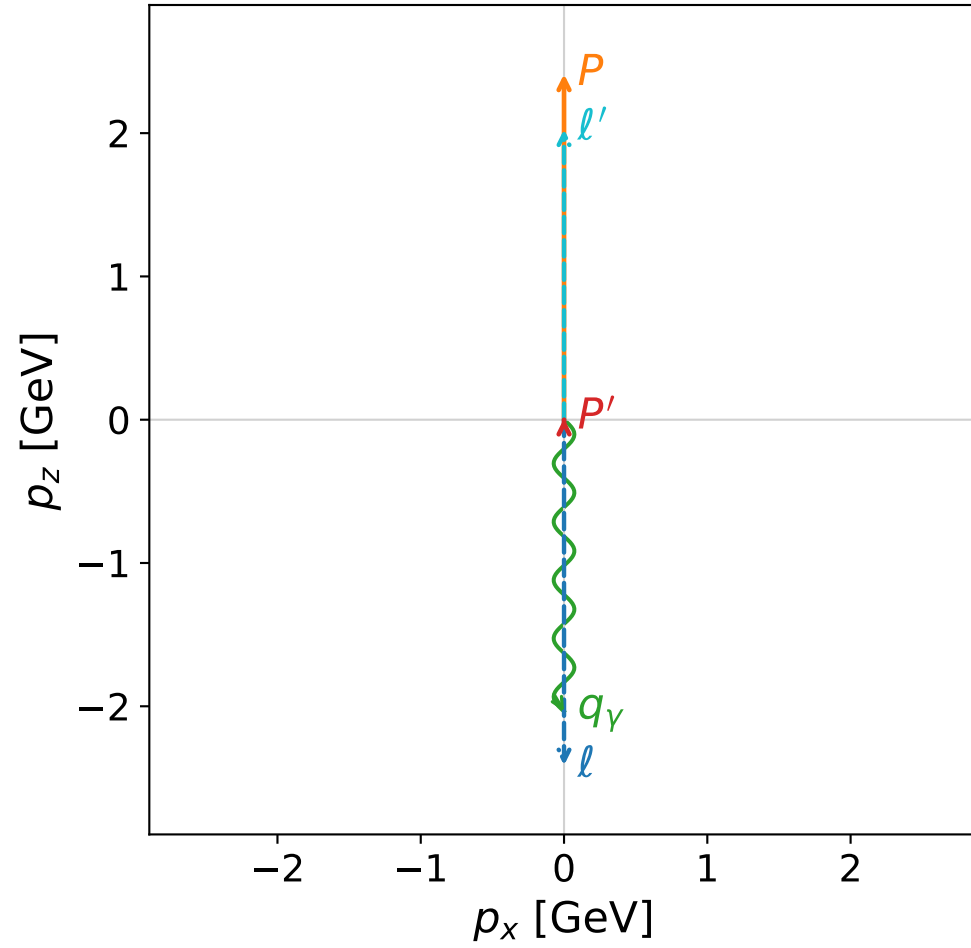


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

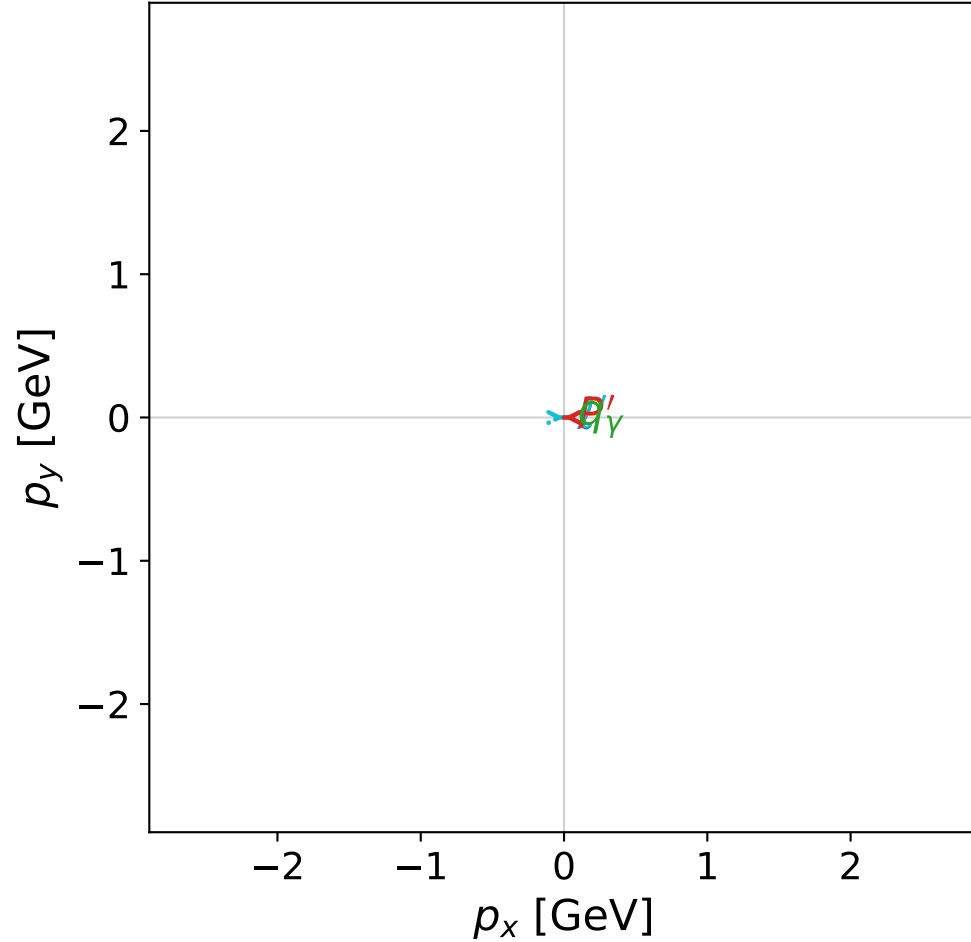


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



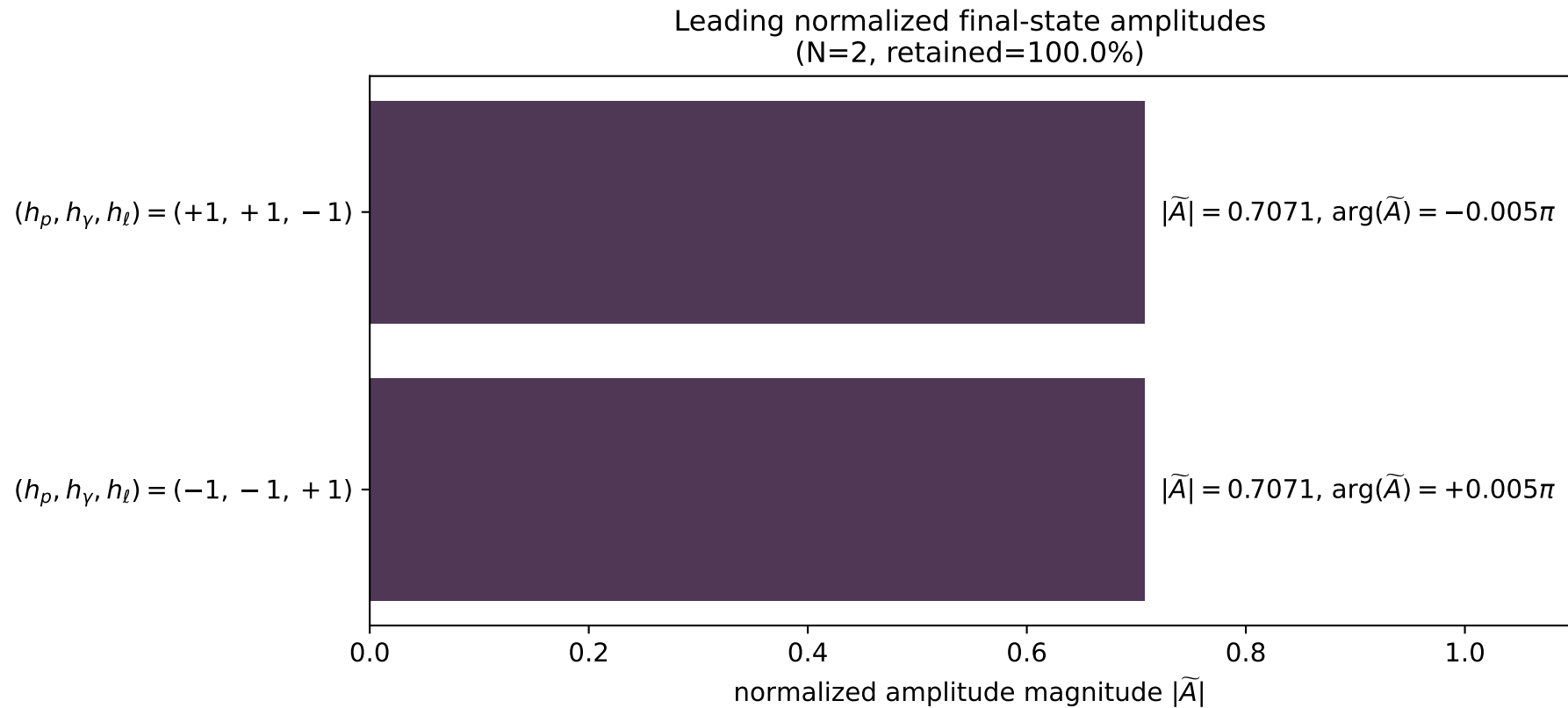
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_177
 $d_{\{\rm GHZ\}}=4.14871\text{e-}07$
 region: polarization_cluster_04_local_minimum_177
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0177
 lepton mixing angle: $\alpha_e=3.1093625$ rad
 incoming lepton: $-0.999481|+\rangle +0.0322245|-\rangle$
 proton mixing angle: $\alpha_p=3.1093621$ rad
 incoming proton: $-0.999481|+\rangle +0.0322249|-\rangle$

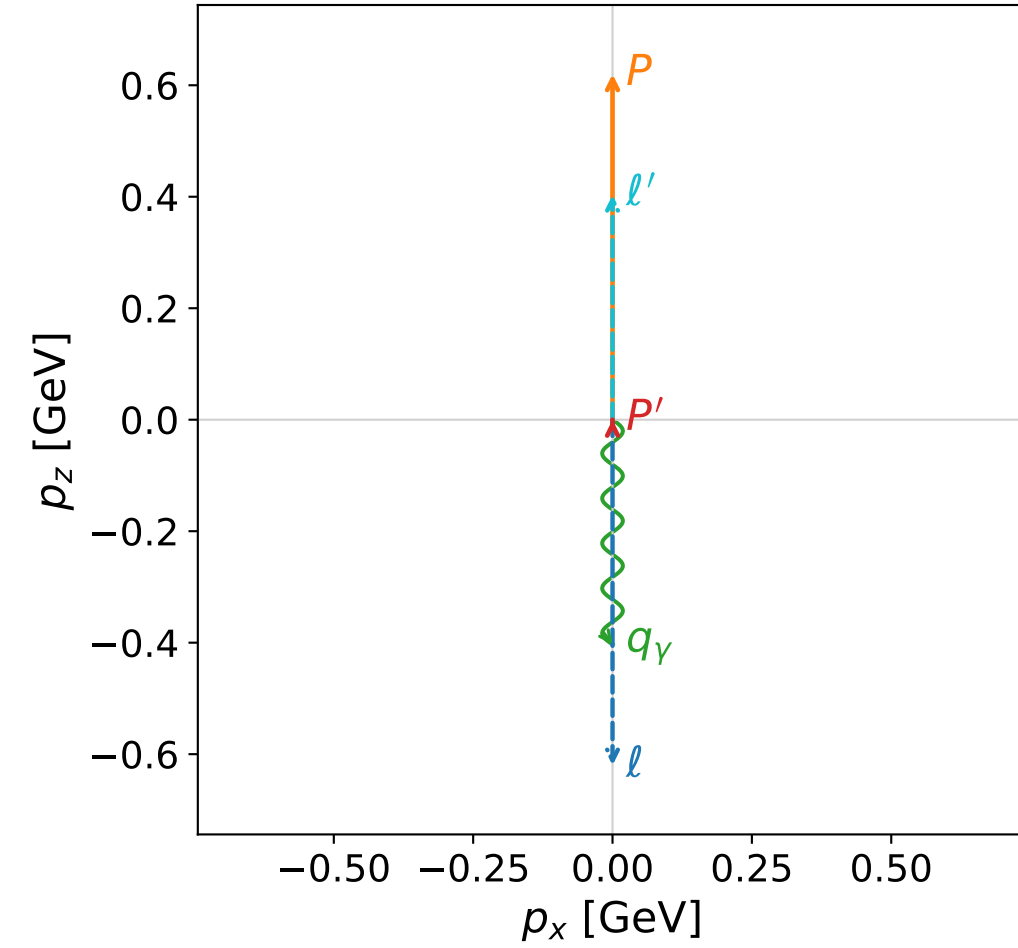
$s=24.9987$, $\sqrt{s}=4.99987$, $|\vec{P}|=2.41195$, $|\vec{P}'|=0.00931426$
 $\theta_{P'}=0.00233638$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14277$, $\phi_\gamma=0.0168325$
 $E_\gamma=2.03557$, $\theta_{\ell'}=1.07398\text{e-}05$, $\phi_{\ell'}=0.00117805$
 $Q^2=19.5489$, $x_B=0.835218$, $t=-3.05056$, $W^2=4.73668$, $y=0.970433$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0263$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0263$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0093$
 q_γ $E= 2.0356$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0356$

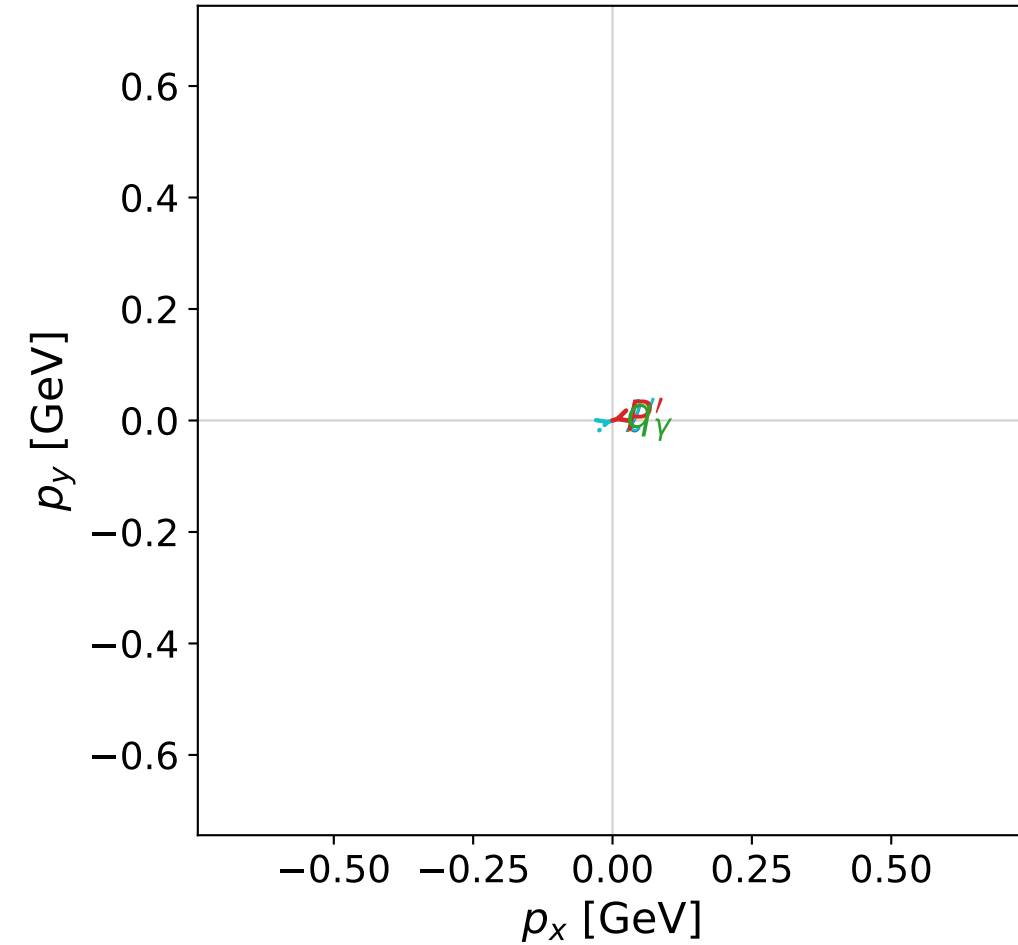


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

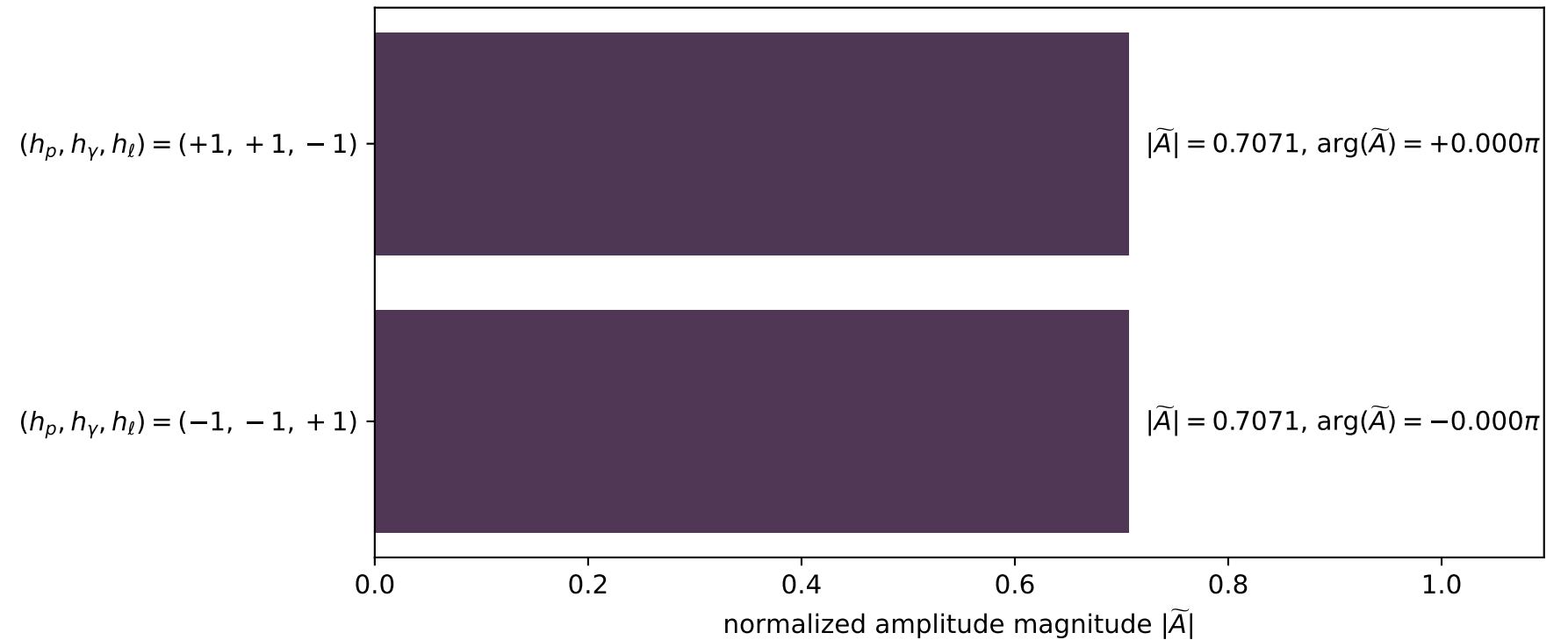


dghz_mixing_angles_polarization_04_local_minimum_183
 $d_{\text{GHZ}}=1.00228\text{e-}06$
 region: polarization_cluster_04_local_minimum_183
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0183
 lepton mixing angle: $\alpha_e=1.8447186$ rad
 incoming lepton: $-0.27051|+\rangle +0.962717|-\rangle$
 proton mixing angle: $\alpha_p=1.8447094$ rad
 incoming proton: $-0.270501|+\rangle +0.96272|-\rangle$

$s=3.04272$, $\sqrt{s}=1.74434$, $|\vec{P}|=0.61997$, $|\vec{P}'|=1.00646\text{e-}06$
 $\theta_{P'}=0.00076699$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.44967$, $\phi_\gamma=6.28171$
 $E_\gamma=0.40317$, $\theta_{l'}=0$, $\phi_{l'}=0.308074$
 $Q^2=0.999811$, $x_B=0.569317$, $t=-0.349628$, $W^2=1.63619$, $y=0.811955$

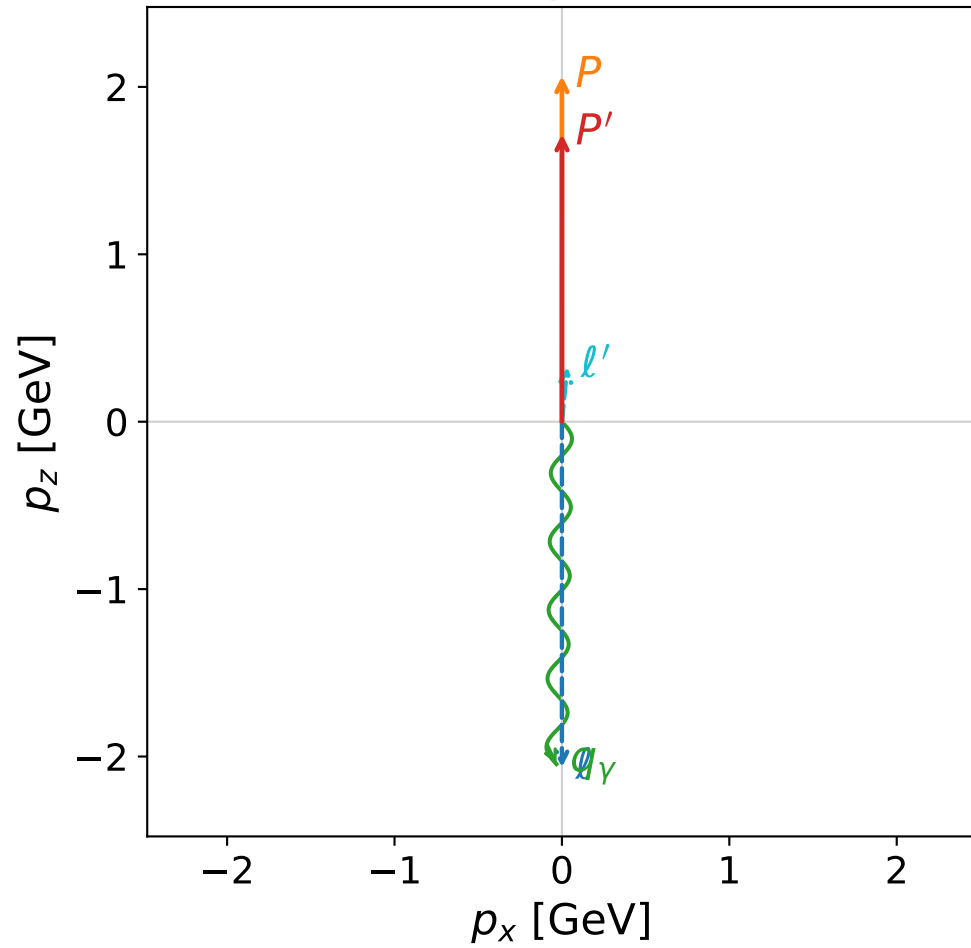
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.6200$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6200$
 P $E= 1.1244$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6200$
 l' $E= 0.4032$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4032$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_γ $E= 0.4032$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.4032$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

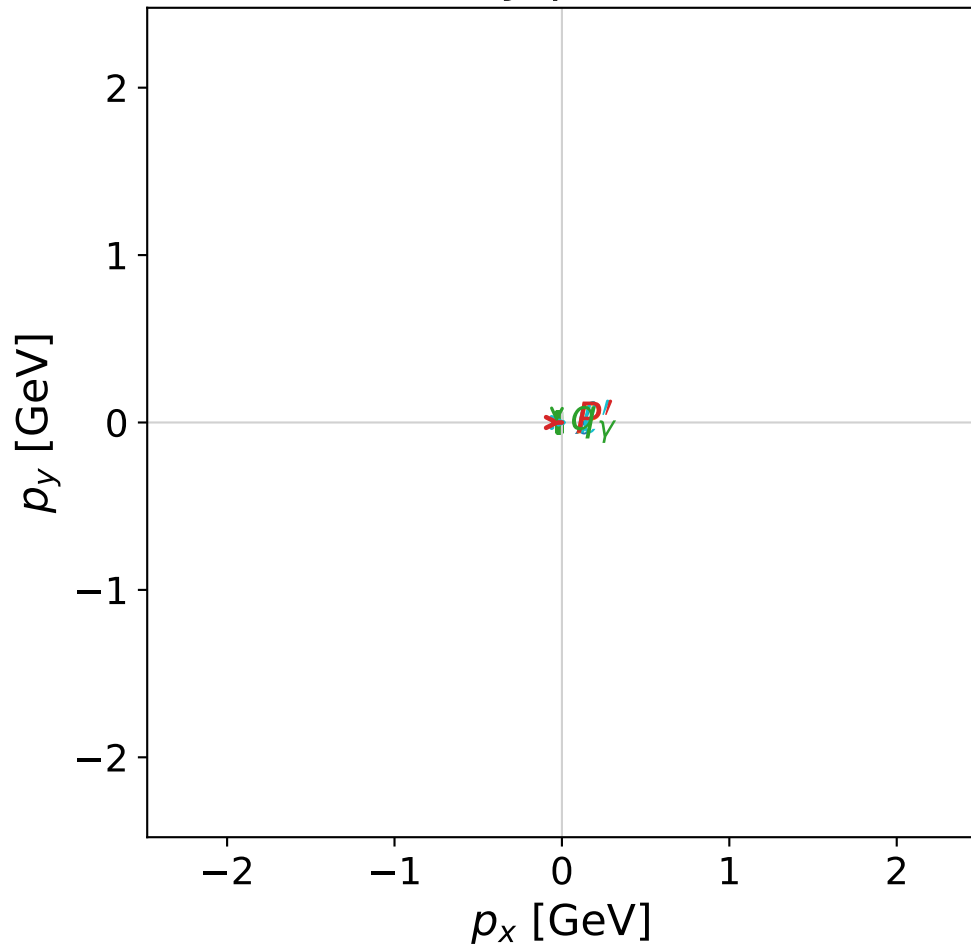


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



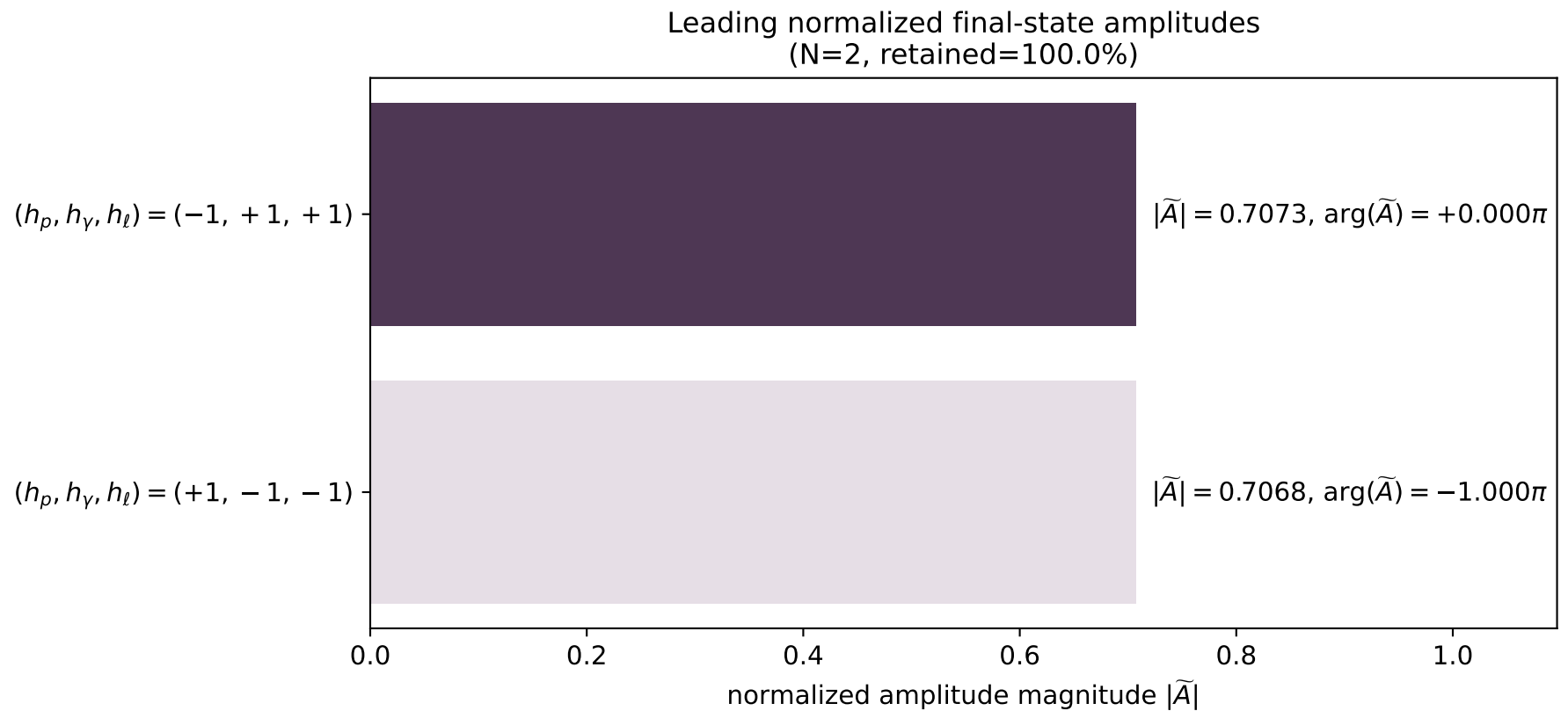
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_184
 $d_{\text{GHZ}}=2.62168\text{e-}06$
 region: polarization_cluster_04_local_minimum_184
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0184
 lepton mixing angle: $\alpha_e=2.1744032$ rad
 incoming lepton: $-0.567616|+\rangle +0.823294|-\rangle$
 proton mixing angle: $\alpha_p=2.5444445$ rad
 incoming proton: $-0.826943|+\rangle +0.562286|-\rangle$

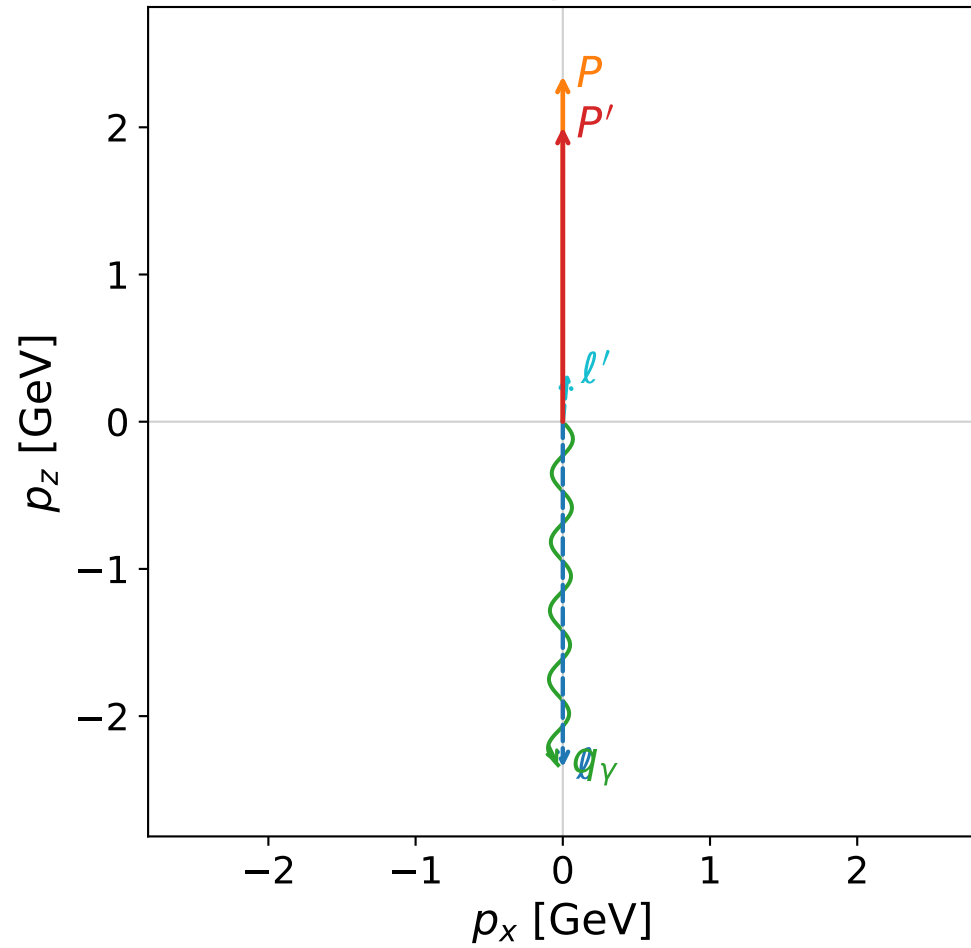
$s=18.765$, $\sqrt{s}=4.33186$, $|\vec{P}|=2.06438$, $|\vec{P}'|=1.71704$
 $\theta_{p'}=0.000299242$, $\theta_Y=3.12573$, $\phi_{p'}=8.42046\text{e-}05$, $\phi_Y=3.14159$
 $E_Y=2.04553$, $\theta_{l'}=0.0969549$, $\phi_{l'}=6.28318$
 $Q^2=2.71682$, $x_B=0.153105$, $t=-0.023959$, $W^2=15.9079$, $y=0.992152$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0644$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0644$
 P $E= 2.2675$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0644$
 l' $E= 0.3298$ $p_x= 0.0319$ $p_y= -0.0000$ $p_z= 0.3282$
 P' $E= 1.9565$ $p_x= 0.0005$ $p_y= 0.0000$ $p_z= 1.7170$
 q_Y $E= 2.0455$ $p_x= -0.0324$ $p_y= 0.0000$ $p_z= -2.0453$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_186

$d_{\{\rm GHZ\}}=7.14153\text{e-}06$

region: polarization_cluster_04_local_minimum_186

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0186

lepton mixing angle: $\alpha_e=2.2229182$ rad

incoming lepton: $-0.606874|+\rangle +0.794798|-\rangle$

proton mixing angle: $\alpha_p=2.4957371$ rad

incoming proton: $-0.798585|+\rangle +0.601882|-\rangle$

$s=23.7724$, $\sqrt{s}=4.8757$, $|\vec{P}|=2.34762$, $|\vec{P}'|=2.00059$

$\theta_{p'}=0.000172458$, $\theta_\gamma=3.12791$, $\phi_{p'}=0.000436125$, $\phi_\gamma=3.14159$

$E_\gamma=2.33272$, $\theta_{l'}=0.0948293$, $\phi_{l'}=6.28318$

$Q^2=3.12386$, $x_B=0.137221$, $t=-0.0189857$, $W^2=20.5212$, $y=0.994436$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.3476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3476$

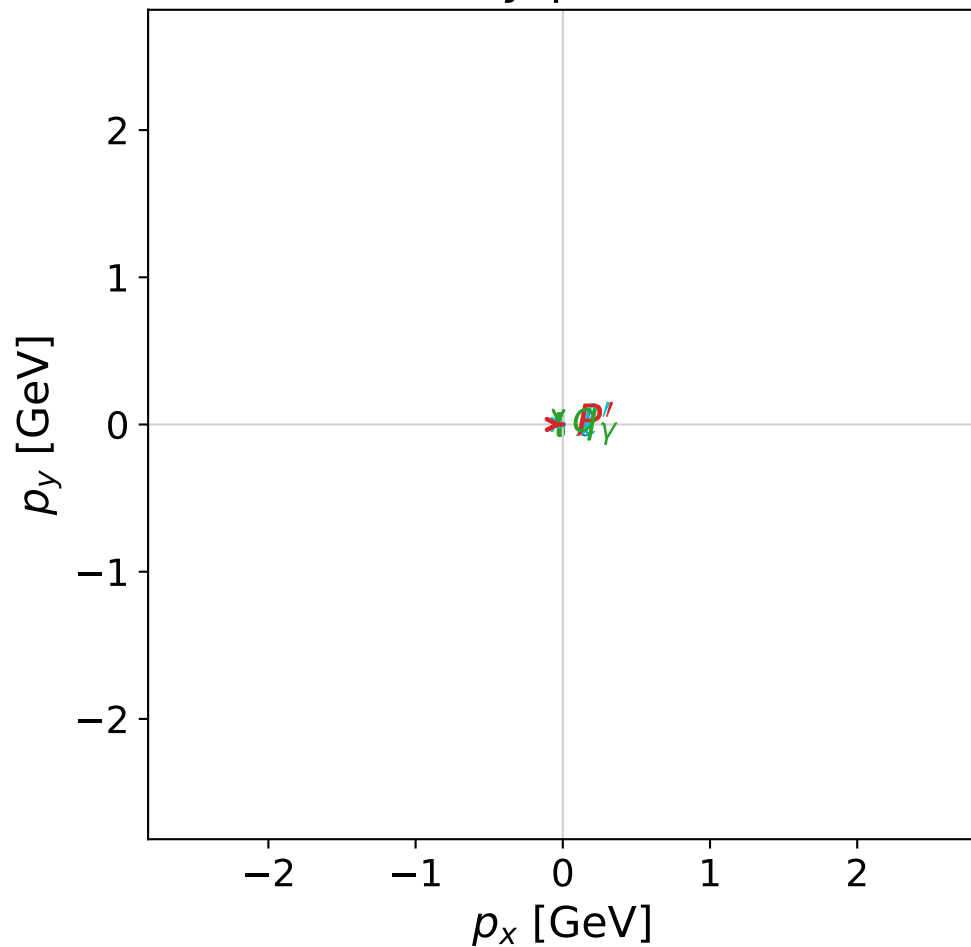
P $E= 2.5281$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3476$

l' $E= 0.3334$ $p_x= 0.0316$ $p_y= -0.0000$ $p_z= 0.3319$

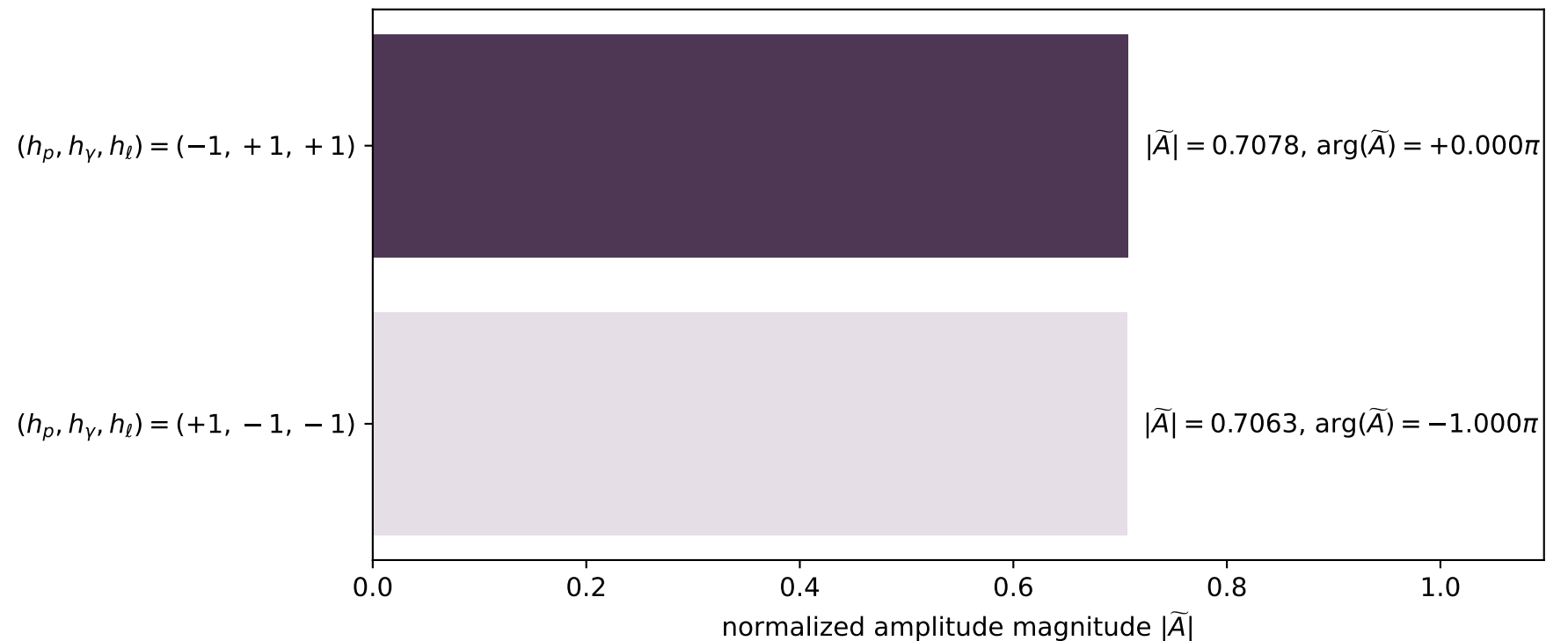
P' $E= 2.2096$ $p_x= 0.0003$ $p_y= 0.0000$ $p_z= 2.0006$

q_γ $E= 2.3327$ $p_x= -0.0319$ $p_y= 0.0000$ $p_z= -2.3325$

x--y plane

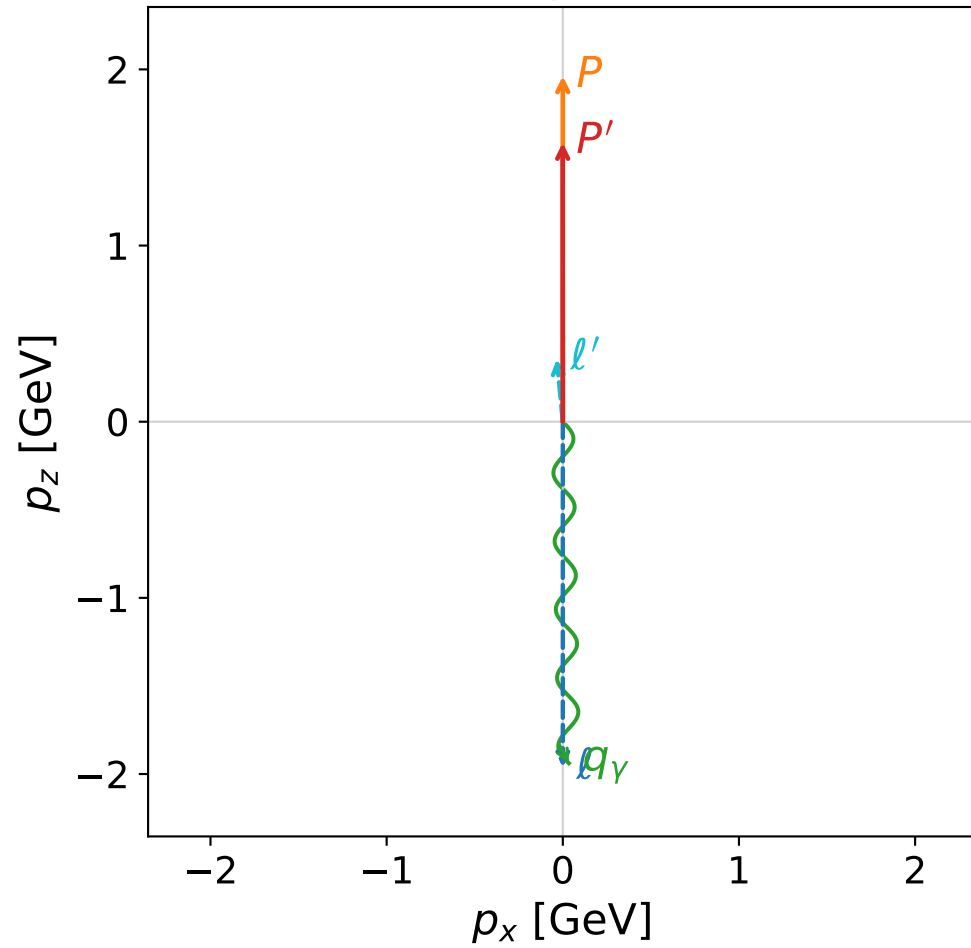


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_205

$d_{\{\rm GHZ\}}=2.5832\text{e-}05$

region: polarization_cluster_04_local_minimum_205

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0205

lepton mixing angle: $\alpha_e=2.5877528$ rad

incoming lepton: $-0.850511|+\rangle +0.525957|-\rangle$

proton mixing angle: $\alpha_p=2.116265$ rad

incoming proton: $-0.518819|+\rangle +0.854884|-\rangle$

$s=17.1102$, $\sqrt{s}=4.13645$, $|\vec{P}|=1.96187$, $|\vec{P}'|=1.58344$

$\theta_{p'}=0.000451651$, $\theta_\gamma=3.1237$, $\phi_{p'}=3.14162$, $\phi_\gamma=6.28317$

$E_\gamma=1.93908$, $\theta_{l'}=0.0953256$, $\phi_{l'}=3.14157$

$Q^2=2.79482$, $x_B=0.173893$, $t=-0.0315471$, $W^2=14.1571$, $y=0.990252$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 1.9619$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9619$

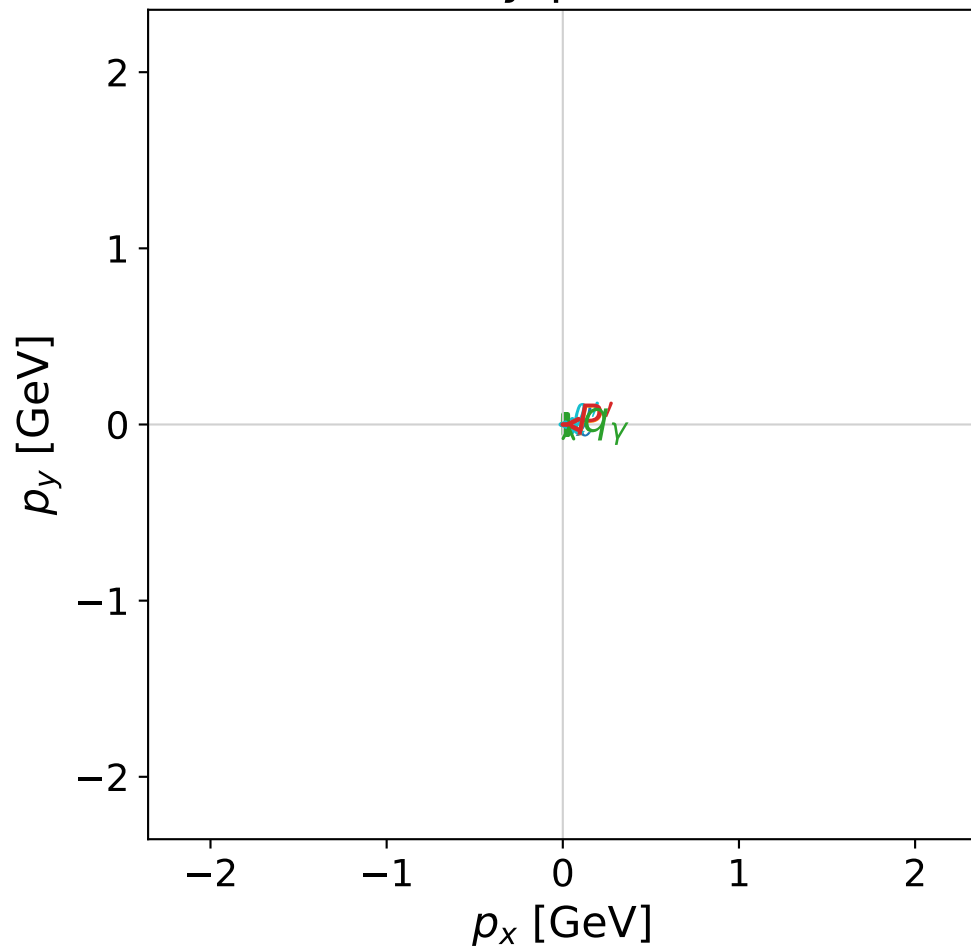
P $E= 2.1746$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9619$

l' $E= 0.3570$ $p_x= -0.0340$ $p_y= 0.0000$ $p_z= 0.3553$

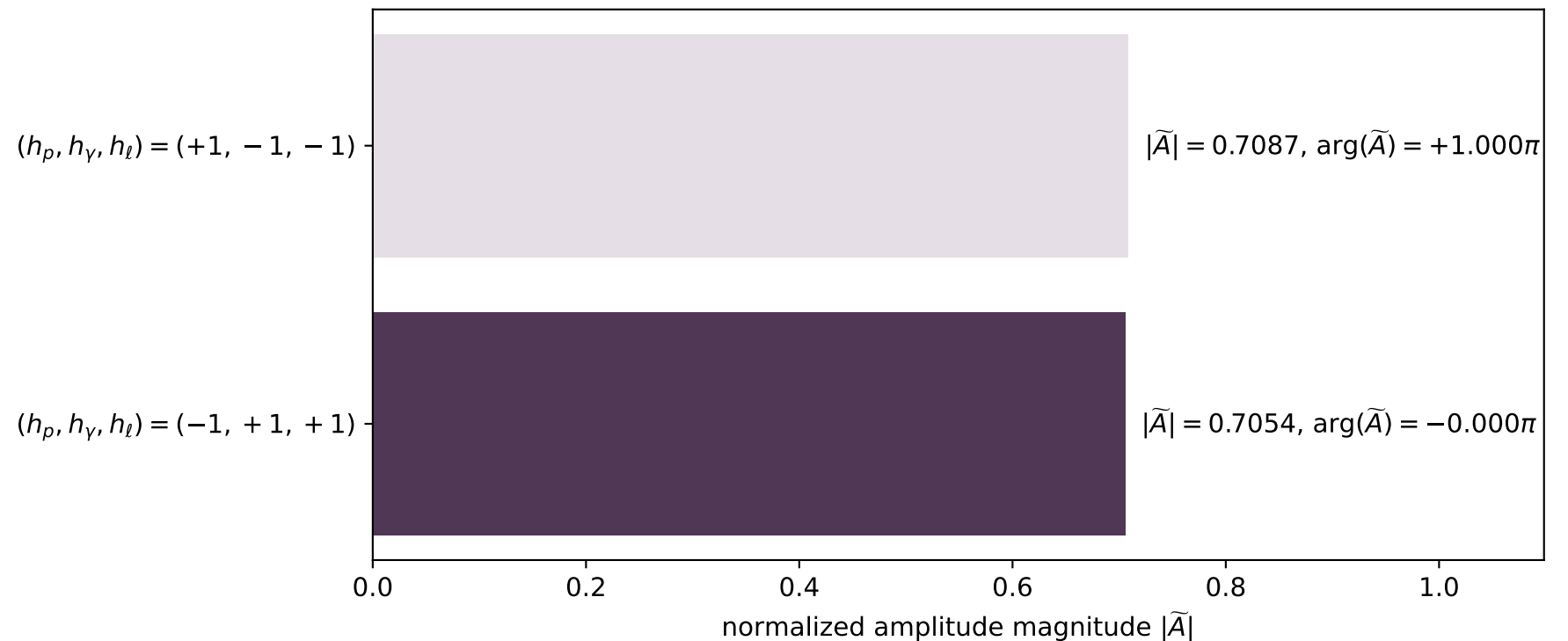
P' $E= 1.8404$ $p_x= -0.0007$ $p_y= -0.0000$ $p_z= 1.5834$

q_γ $E= 1.9391$ $p_x= 0.0347$ $p_y= -0.0000$ $p_z= -1.9388$

x--y plane

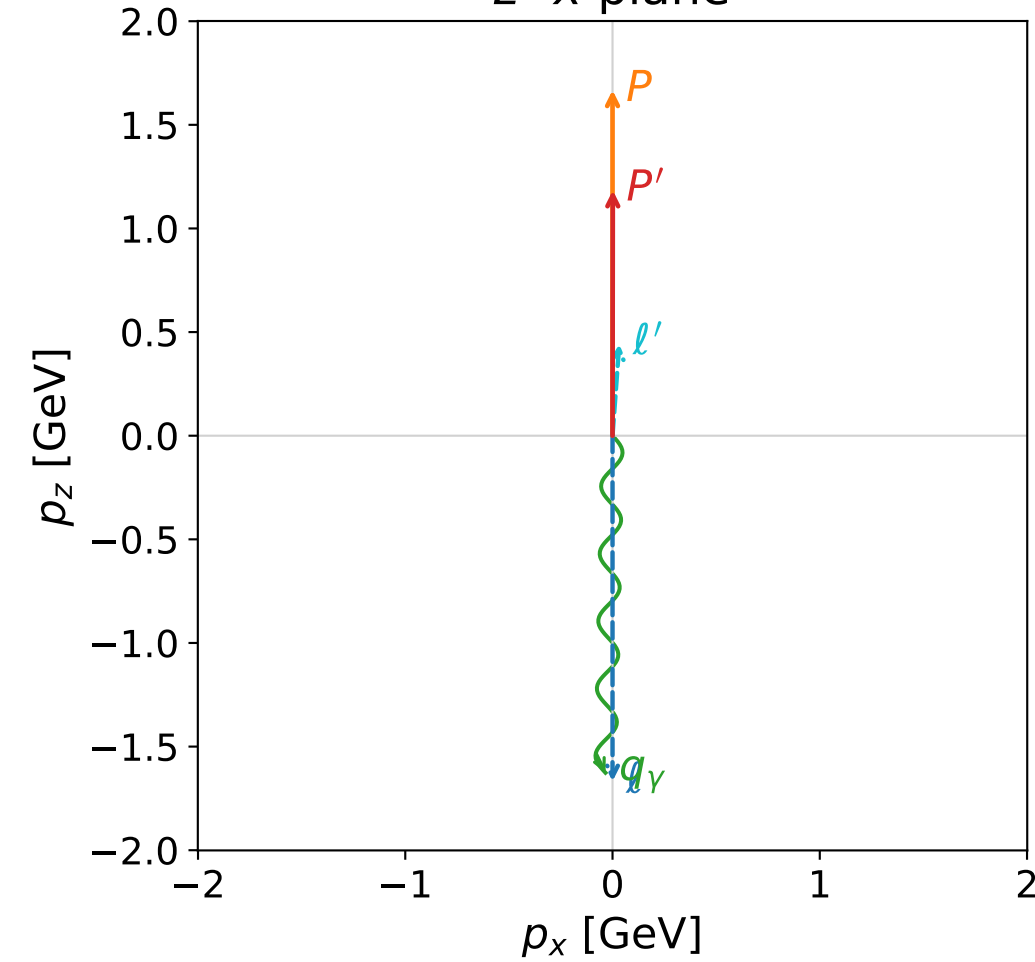


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

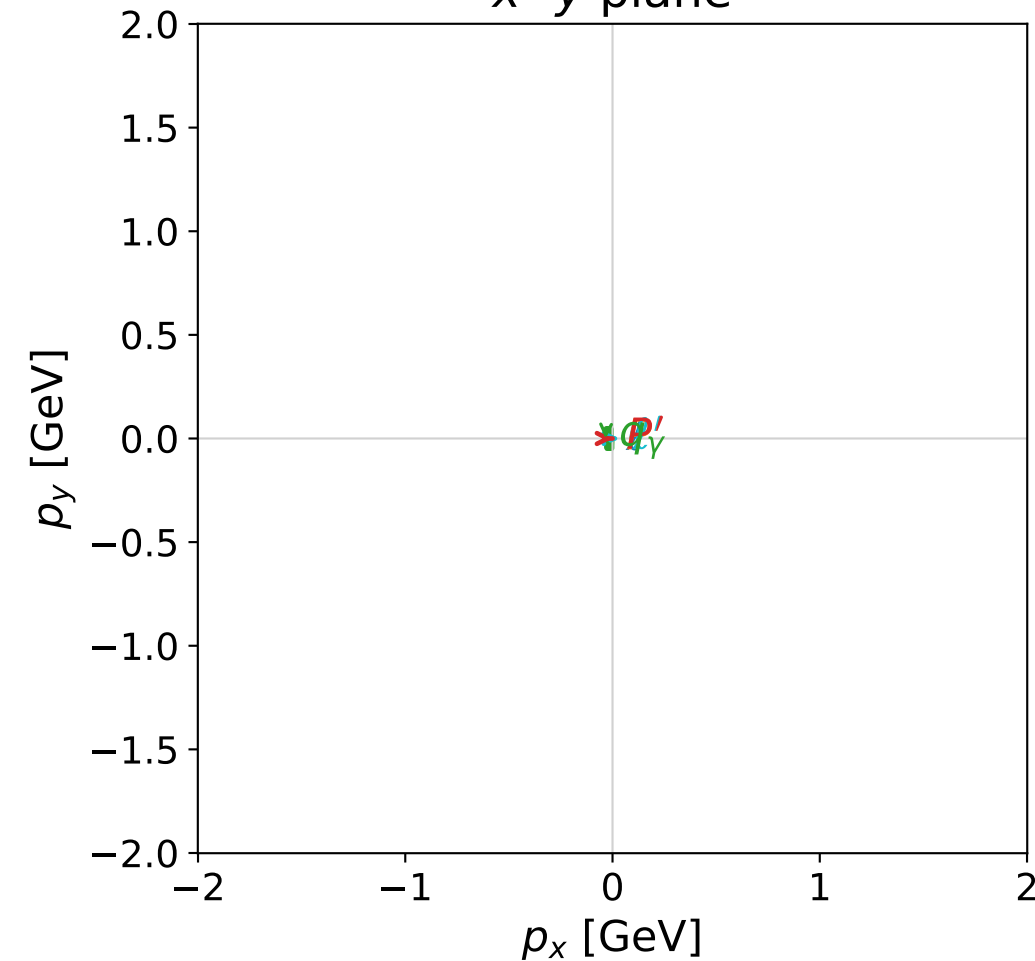


dghz_mixing_angles_polarization_04_local_minimum_210
 $d_{\{\rm GHZ\}}=2.82327\text{e-}05$
 region: polarization_cluster_04_local_minimum_210
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0210
 lepton mixing angle: $\alpha_e=1.938937$ rad
 incoming lepton: $-0.359881|+\rangle +0.932998|-\rangle$
 proton mixing angle: $\alpha_p=2.7798744$ rad
 incoming proton: $-0.93529|+\rangle +0.353882|-\rangle$

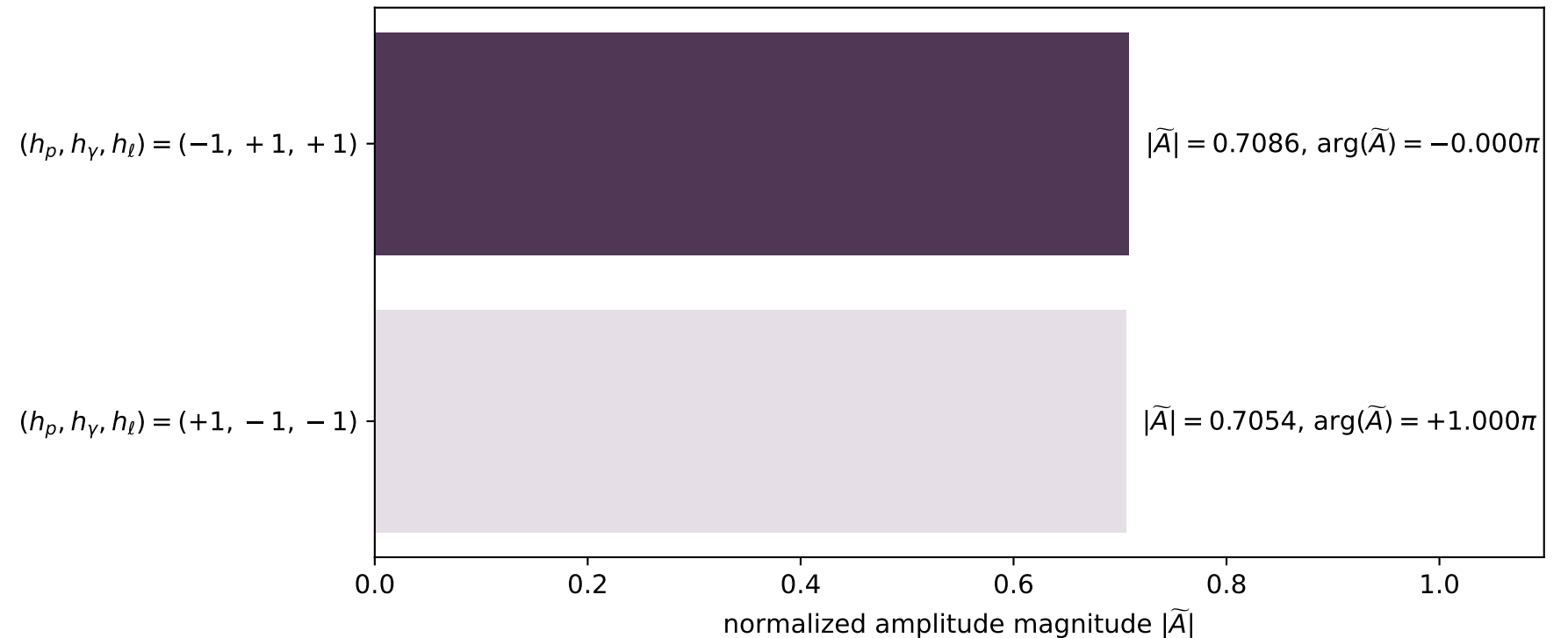
$s=12.8213$, $\sqrt{s}=3.58068$, $|\vec{P}|=1.66748$, $|\vec{P}'|=1.1839$
 $\theta_{p'}=0.00120427$, $\theta_\gamma=3.12072$, $\phi_{p'}=6.28227$, $\phi_\gamma=3.14164$
 $E_\gamma=1.62664$, $\theta_{l'}=0.0733953$, $\phi_{l'}=8.56722\text{e-}05$
 $Q^2=2.95466$, $x_B=0.252116$, $t=-0.0716446$, $W^2=9.64462$, $y=0.981411$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.6675$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6675$
 P $E= 1.9132$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6675$
 ℓ' $E= 0.4436$ $p_x= 0.0325$ $p_y= 0.0000$ $p_z= 0.4424$
 P' $E= 1.5105$ $p_x= 0.0014$ $p_y= -0.0000$ $p_z= 1.1839$
 q_γ $E= 1.6266$ $p_x= -0.0340$ $p_y= -0.0000$ $p_z= -1.6263$

x--y plane

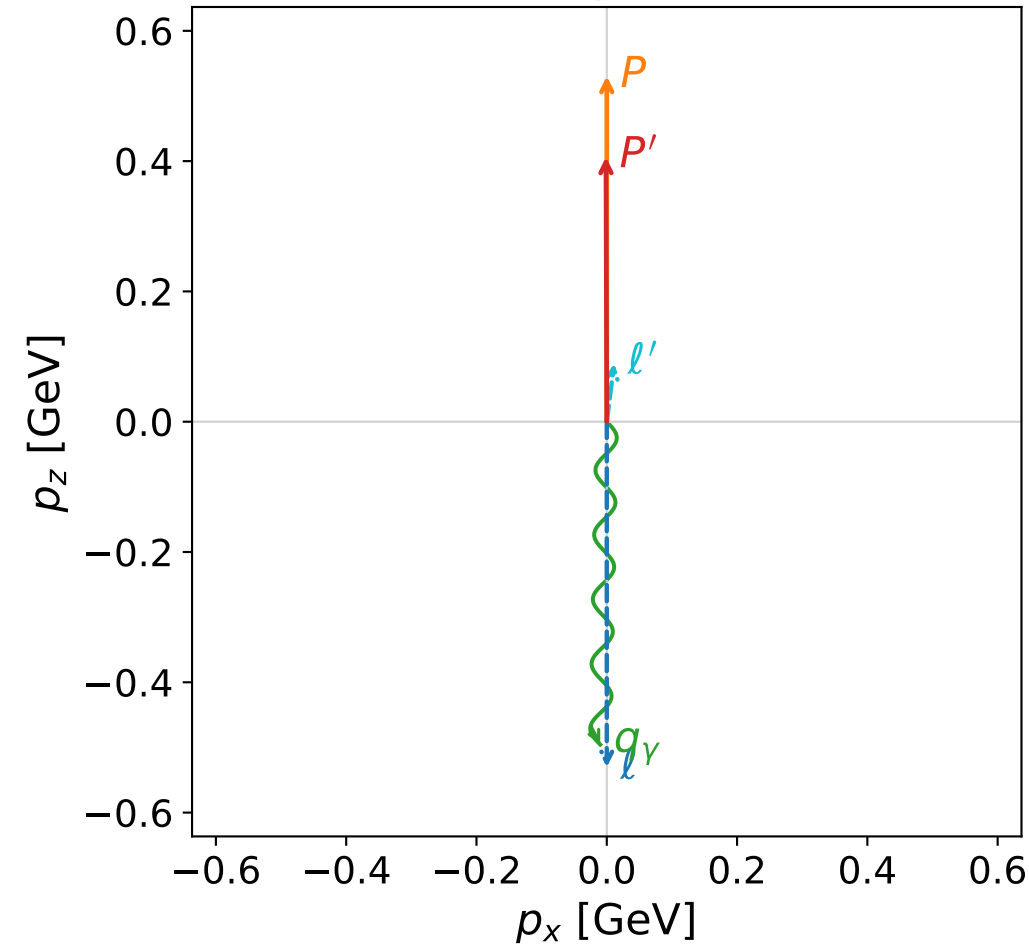


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_221

$d_{\{\text{rm GHZ}\}}=4.54196\text{e-}05$

region: polarization_cluster_04_local_minimum_221

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0221

lepton mixing angle: $\alpha_e=1.989916$ rad

incoming lepton: $-0.406956|+\rangle +0.913448|-\rangle$

proton mixing angle: $\alpha_p=2.6885858$ rad

incoming proton: $-0.899135|+\rangle +0.437671|-\rangle$

$s=2.58602$, $\sqrt{s}=1.60811$, $|\vec{P}|=0.53049$, $|\vec{P}'|=0.406256$

$\theta_{p'}=0.00318369$, $\theta_\gamma=3.12141$, $\phi_{p'}=3.14161$, $\phi_\gamma=3.14159$

$E_\gamma=0.495777$, $\theta_{l'}=0.125661$, $\phi_{l'}=6.28318$

$Q^2=0.190506$, $x_B=0.118564$, $t=-0.0123647$, $W^2=2.29612$, $y=0.941749$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 0.5305$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5305$

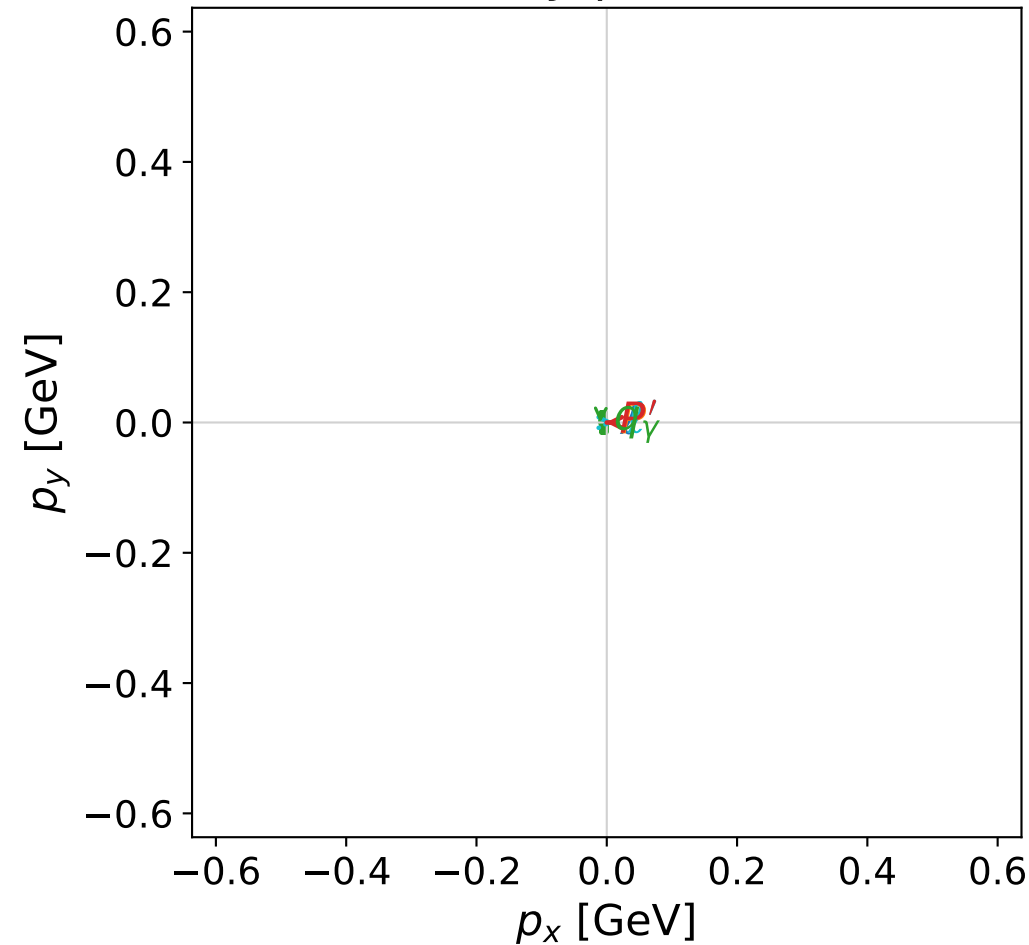
P $E= 1.0776$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5305$

l' $E= 0.0901$ $p_x= 0.0113$ $p_y= -0.0000$ $p_z= 0.0894$

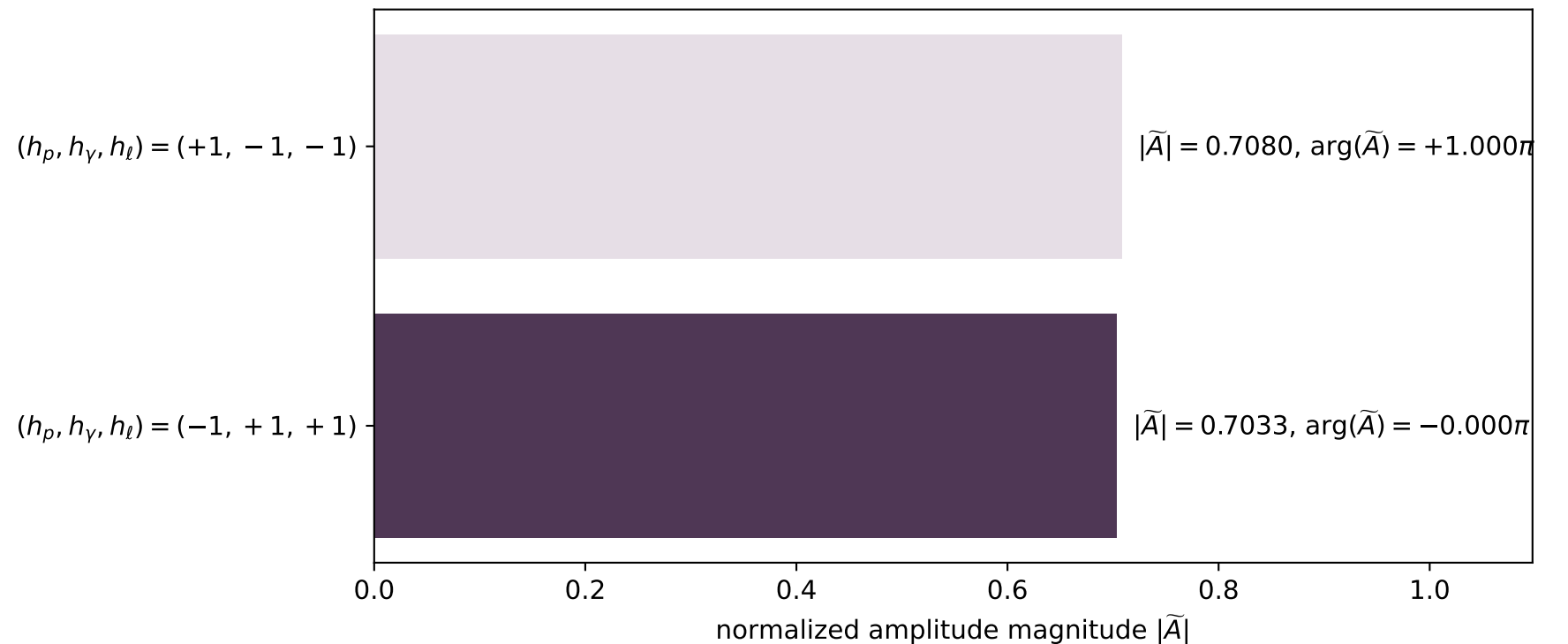
P' $E= 1.0222$ $p_x= -0.0013$ $p_y= -0.0000$ $p_z= 0.4063$

q_γ $E= 0.4958$ $p_x= -0.0100$ $p_y= 0.0000$ $p_z= -0.4957$

x--y plane

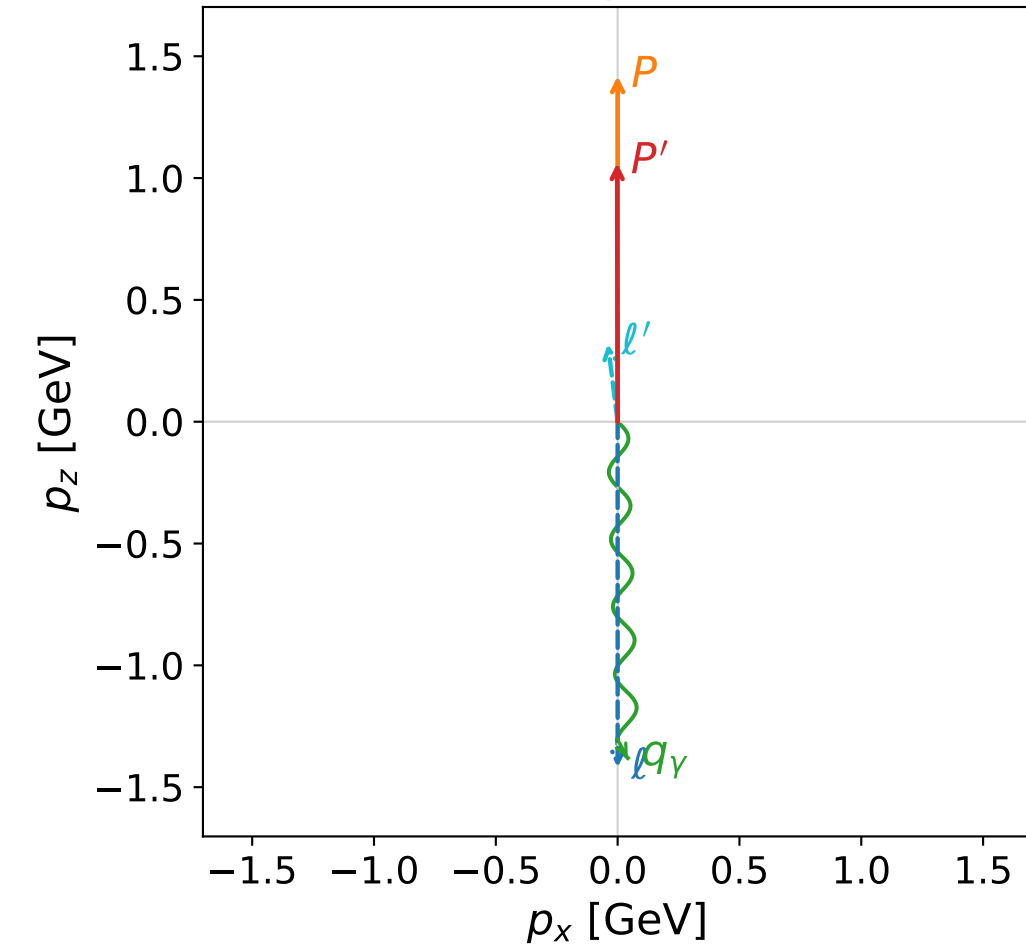


Leading normalized final-state amplitudes
(N=2, retained=99.6%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

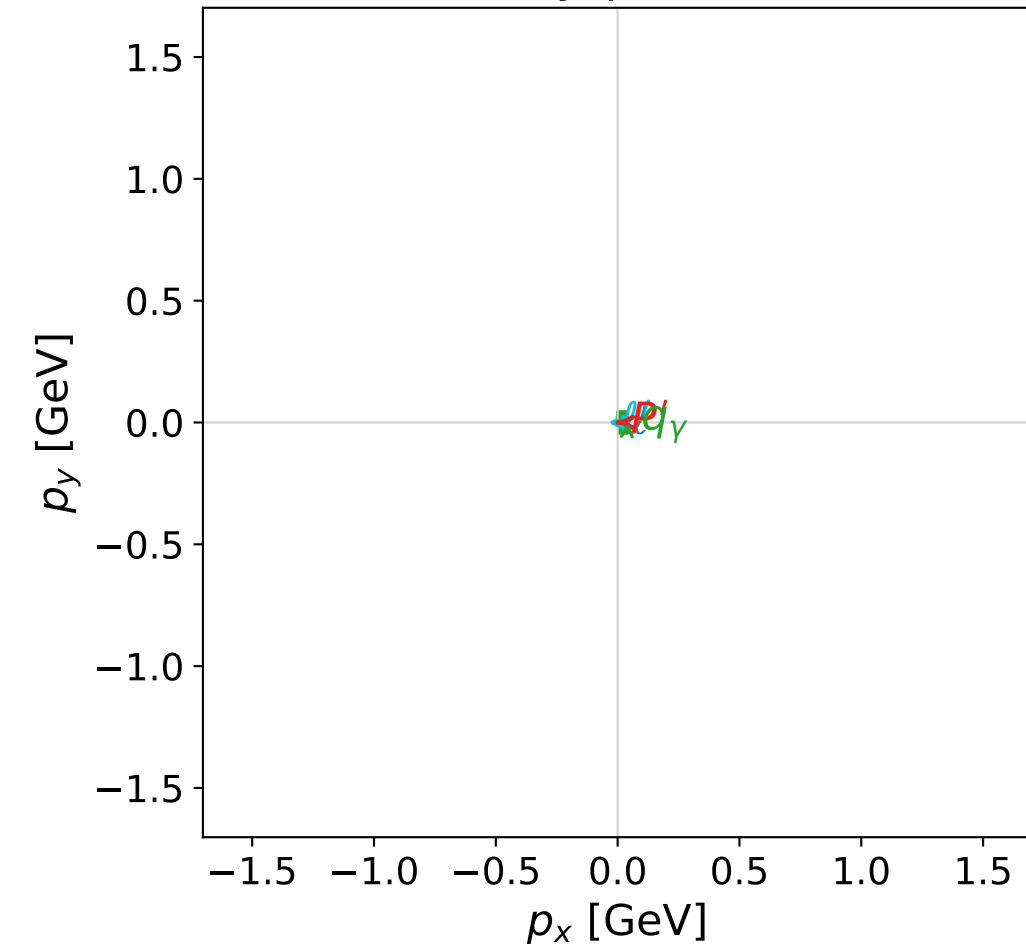


dghz_mixing_angles_polarization_04_local_minimum_228
 $d_{\text{GHZ}}=5.43438\text{e-}05$
 region: polarization_cluster_04_local_minimum_228
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0228
 lepton mixing angle: $\alpha_e=2.7339948$ rad
 incoming lepton: $-0.918076|+\rangle +0.396405|-\rangle$
 proton mixing angle: $\alpha_p=1.9650585$ rad
 incoming proton: $-0.384127|+\rangle +0.92328|-\rangle$

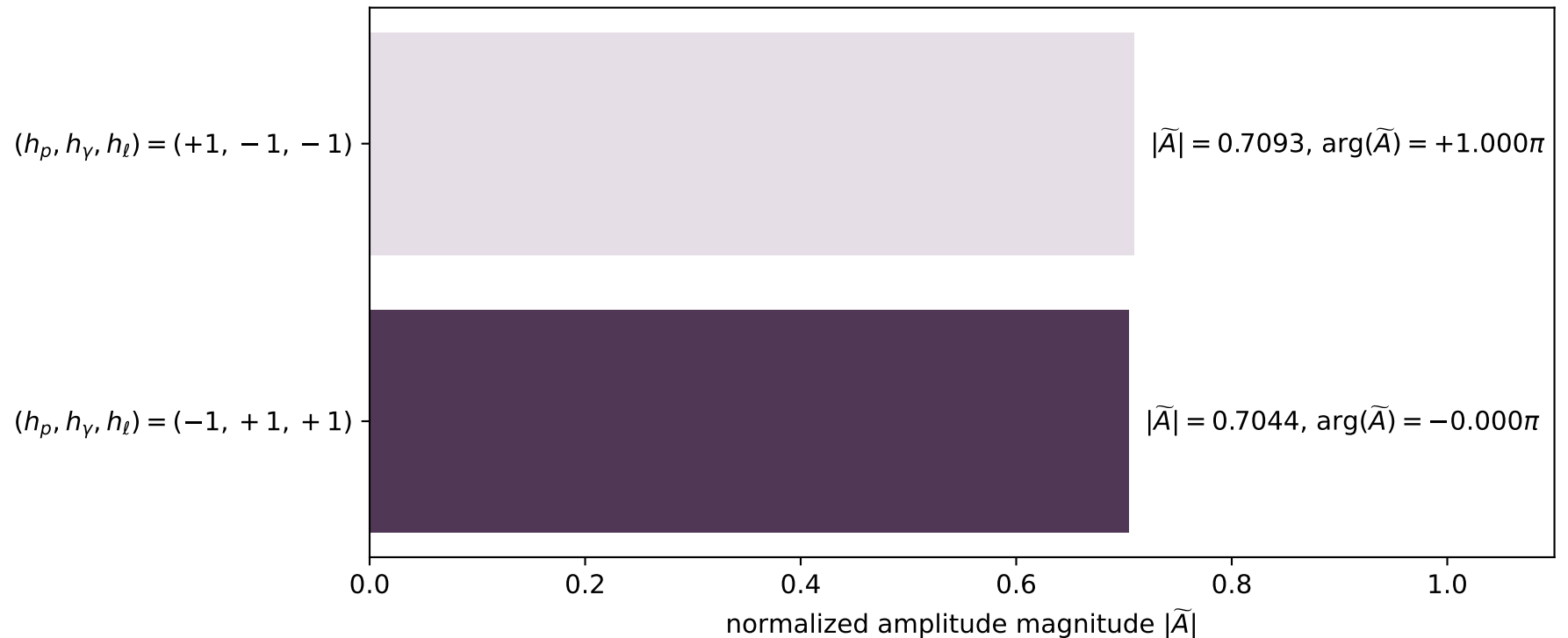
$s=9.72826$, $\sqrt{s}=3.11902$, $|\vec{P}|=1.41846$, $|\vec{P}'|=1.06154$
 $\theta_{P'}=0.00179591$, $\theta_\gamma=3.11125$, $\phi_{P'}=3.1417$, $\phi_\gamma=2.99541\text{e-}05$
 $E_\gamma=1.38105$, $\theta_{l'}=0.124754$, $\phi_{l'}=3.14162$
 $Q^2=1.81636$, $x_B=0.209741$, $t=-0.0467601$, $W^2=7.7235$, $y=0.978708$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.4185$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4185$
 P $E= 1.7006$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4185$
 l' $E= 0.3214$ $p_x= -0.0400$ $p_y= -0.0000$ $p_z= 0.3189$
 P' $E= 1.4166$ $p_x= -0.0019$ $p_y= -0.0000$ $p_z= 1.0615$
 q_γ $E= 1.3811$ $p_x= 0.0419$ $p_y= 0.0000$ $p_z= -1.3804$

x--y plane

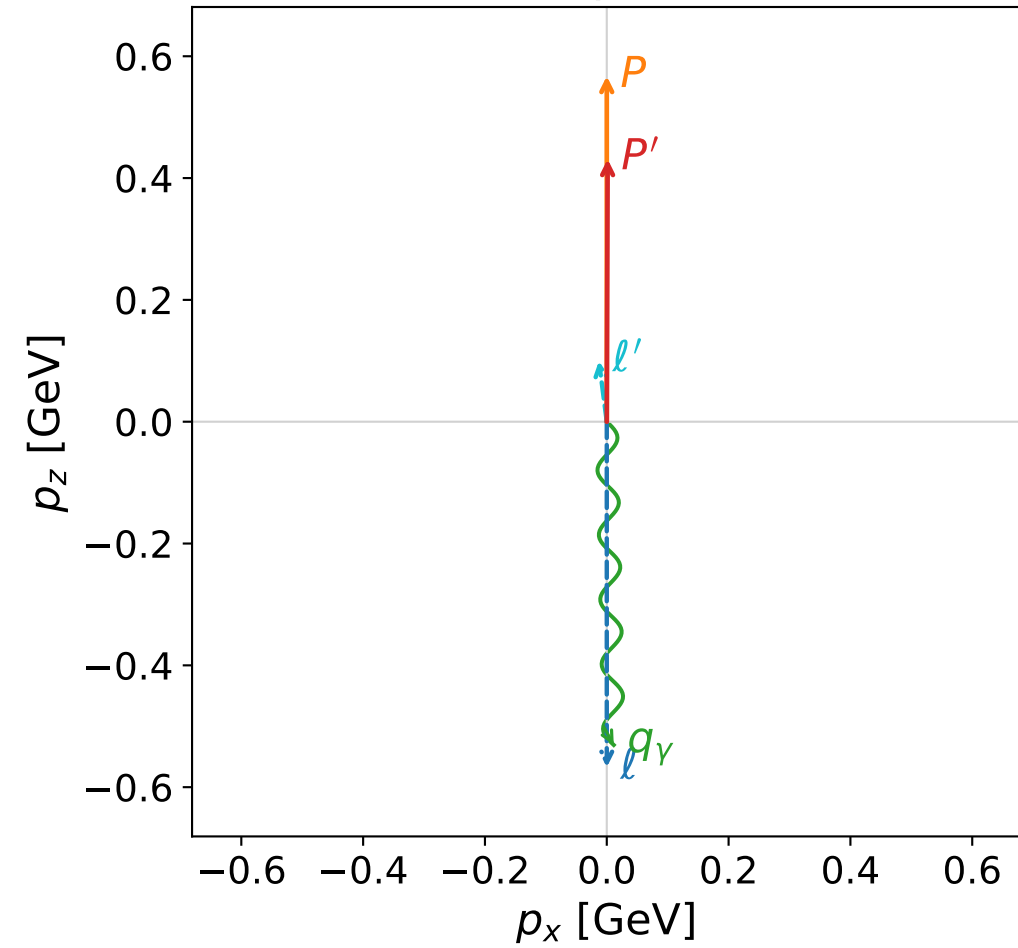


Leading normalized final-state amplitudes
(N=2, retained=99.9%)

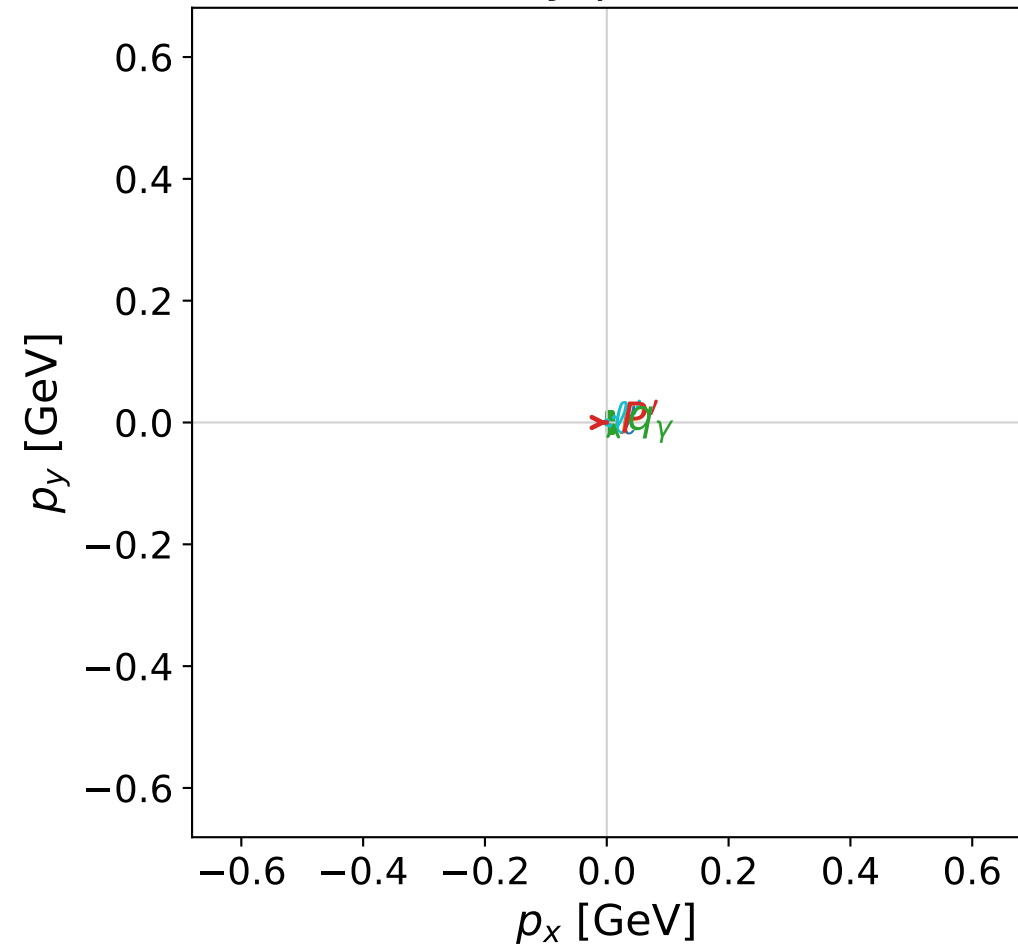


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_235

$d_{\{\rm GHZ\}}=6.62947\text{e-}05$

region: polarization_cluster_04_local_minimum_235

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0235

lepton mixing angle: $\alpha_e=2.7359514$ rad

incoming lepton: $-0.91885|+\rangle +0.394608|-\rangle$

proton mixing angle: $\alpha_p=2.0031027$ rad

incoming proton: $-0.418966|+\rangle +0.908002|-\rangle$

$s=2.76767$, $\sqrt{s}=1.66363$, $|\vec{P}|=0.567381$, $|\vec{P}'|=0.431117$

$\theta_{p'}=0.00334598$, $\theta_Y=3.12$, $\phi_{p'}=4.16802\text{e-}06$, $\phi_Y=6.28316$

$E_Y=0.530853$, $\theta_{\ell'}=0.128826$, $\phi_{\ell'}=3.14157$

$Q^2=0.227024$, $x_B=0.127496$, $t=-0.0144849$, $W^2=2.43345$, $y=0.943218$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.5674$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5674$

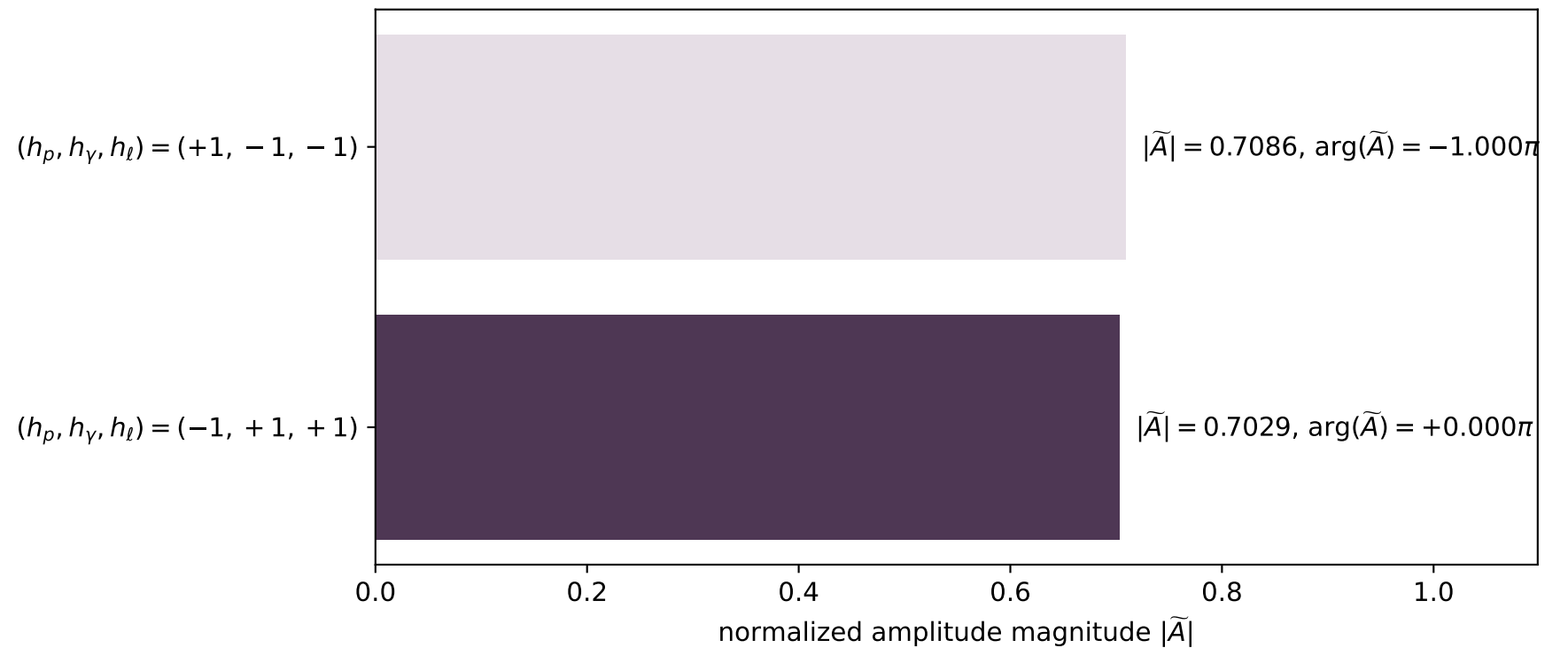
P $E= 1.0963$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5674$

ℓ' $E= 0.1004$ $p_x= -0.0129$ $p_y= 0.0000$ $p_z= 0.0996$

P' $E= 1.0323$ $p_x= 0.0014$ $p_y= 0.0000$ $p_z= 0.4311$

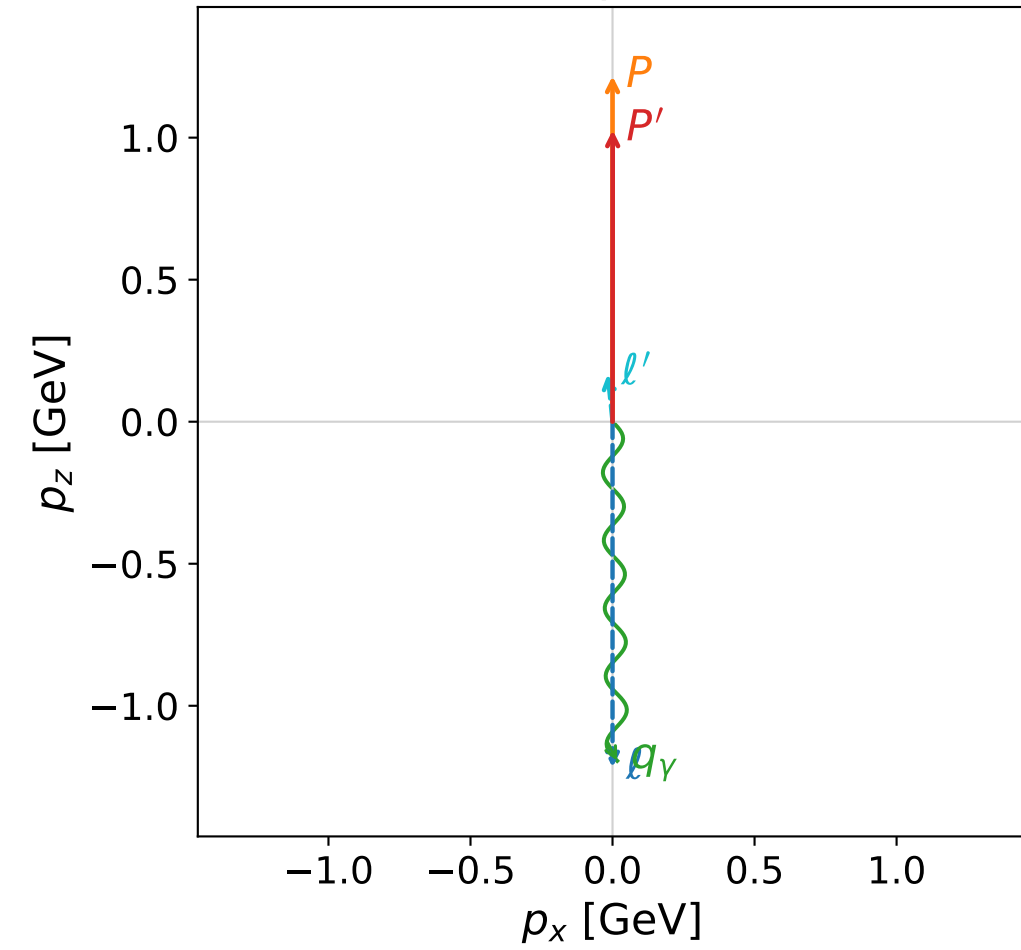
q_Y $E= 0.5309$ $p_x= 0.0115$ $p_y= -0.0000$ $p_z= -0.5307$

Leading normalized final-state amplitudes
(N=2, retained=99.6%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_238

$d_{\{\rm GHZ\}}=7.24626\text{e-}05$

region: polarization_cluster_04_local_minimum_238

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0238

lepton mixing angle: $\alpha_e=2.5683975$ rad

incoming lepton: $-0.840172|+\rangle + 0.542319|-\rangle$

proton mixing angle: $\alpha_p=2.1590186$ rad

incoming proton: $-0.554883|+\rangle + 0.831928|-\rangle$

$s=7.57815$, $\sqrt{s}=2.75284$, $|\vec{P}|=1.21662$, $|\vec{P}'|=1.02635$

$\theta_{P'}=0.000568325$, $\theta_\gamma=3.12781$, $\phi_{P'}=0.000173411$, $\phi_\gamma=8.59629\text{e-}06$

$E_\gamma=1.19402$, $\theta_{\ell'}=0.101329$, $\phi_{\ell'}=3.14161$

$Q^2=0.817507$, $x_B=0.12408$, $t=-0.0149387$, $W^2=6.65088$, $y=0.983613$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 1.2166$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2166$

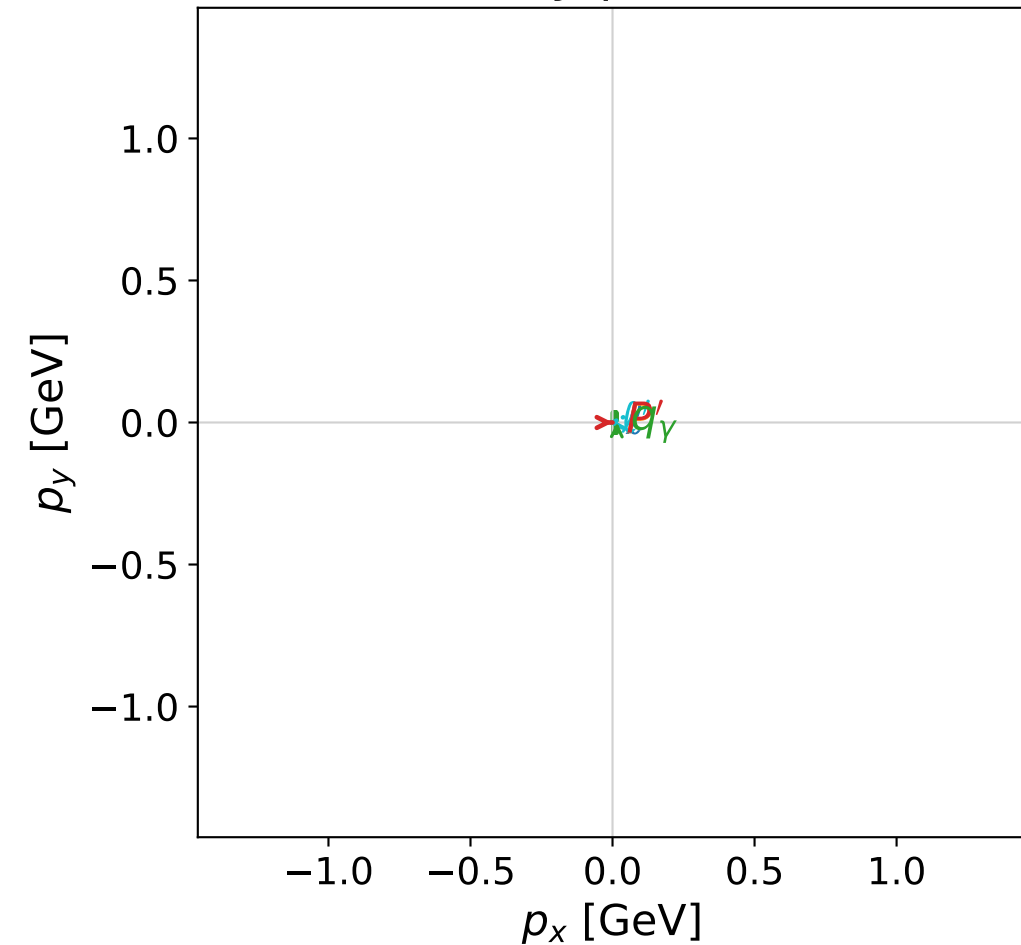
P $E= 1.5362$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2166$

ℓ' $E= 0.1684$ $p_x= -0.0170$ $p_y= -0.0000$ $p_z= 0.1676$

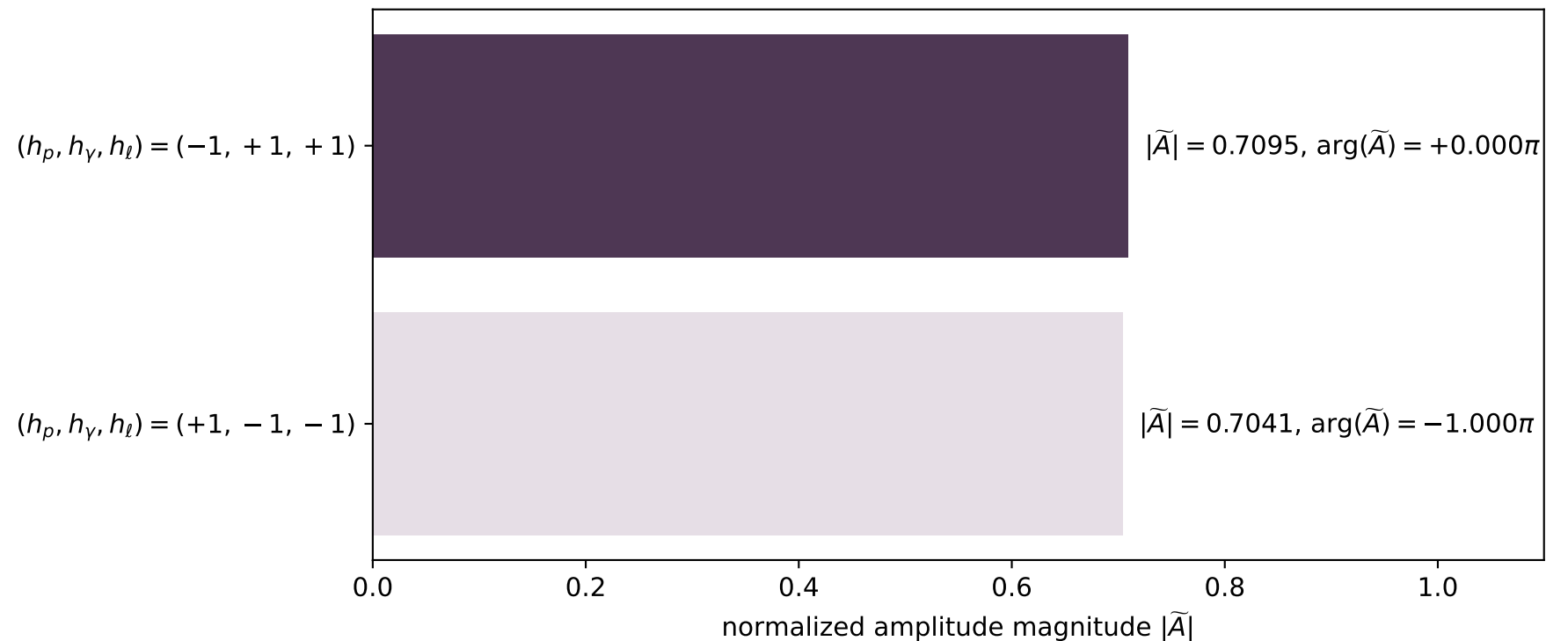
P' $E= 1.3904$ $p_x= 0.0006$ $p_y= 0.0000$ $p_z= 1.0263$

q_γ $E= 1.1940$ $p_x= 0.0165$ $p_y= 0.0000$ $p_z= -1.1939$

x--y plane

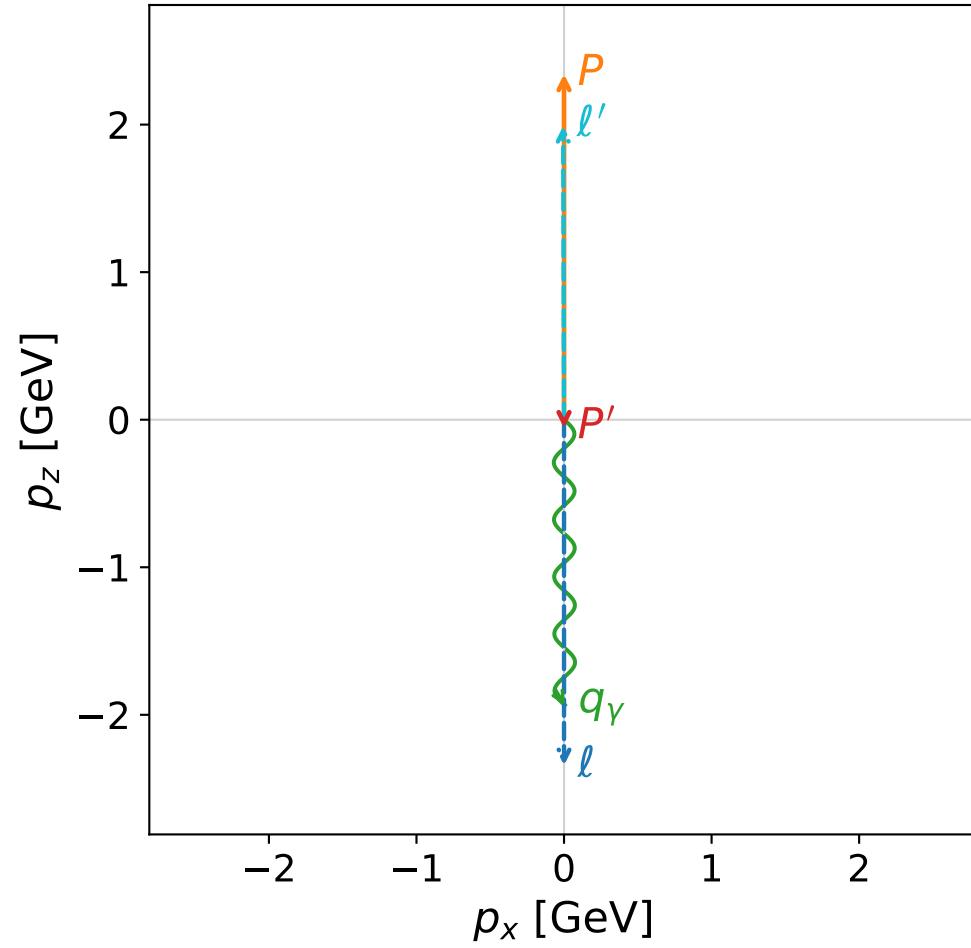


Leading normalized final-state amplitudes
(N=2, retained=99.9%)

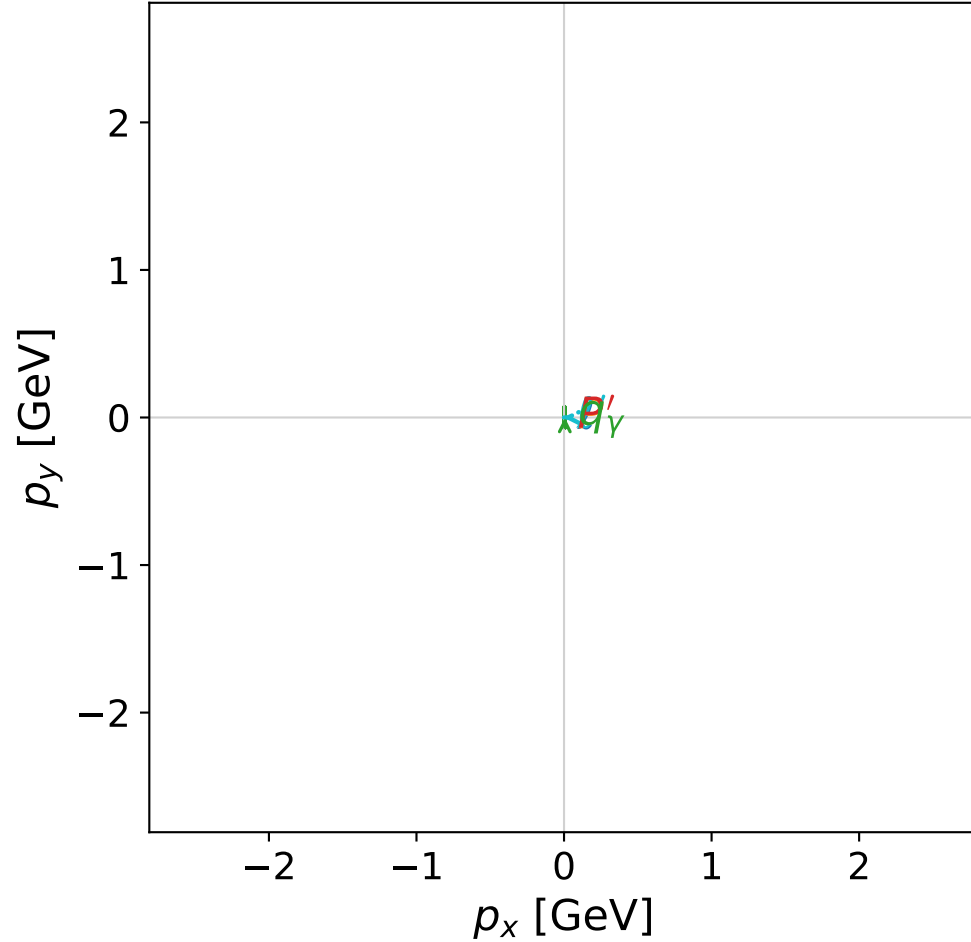


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



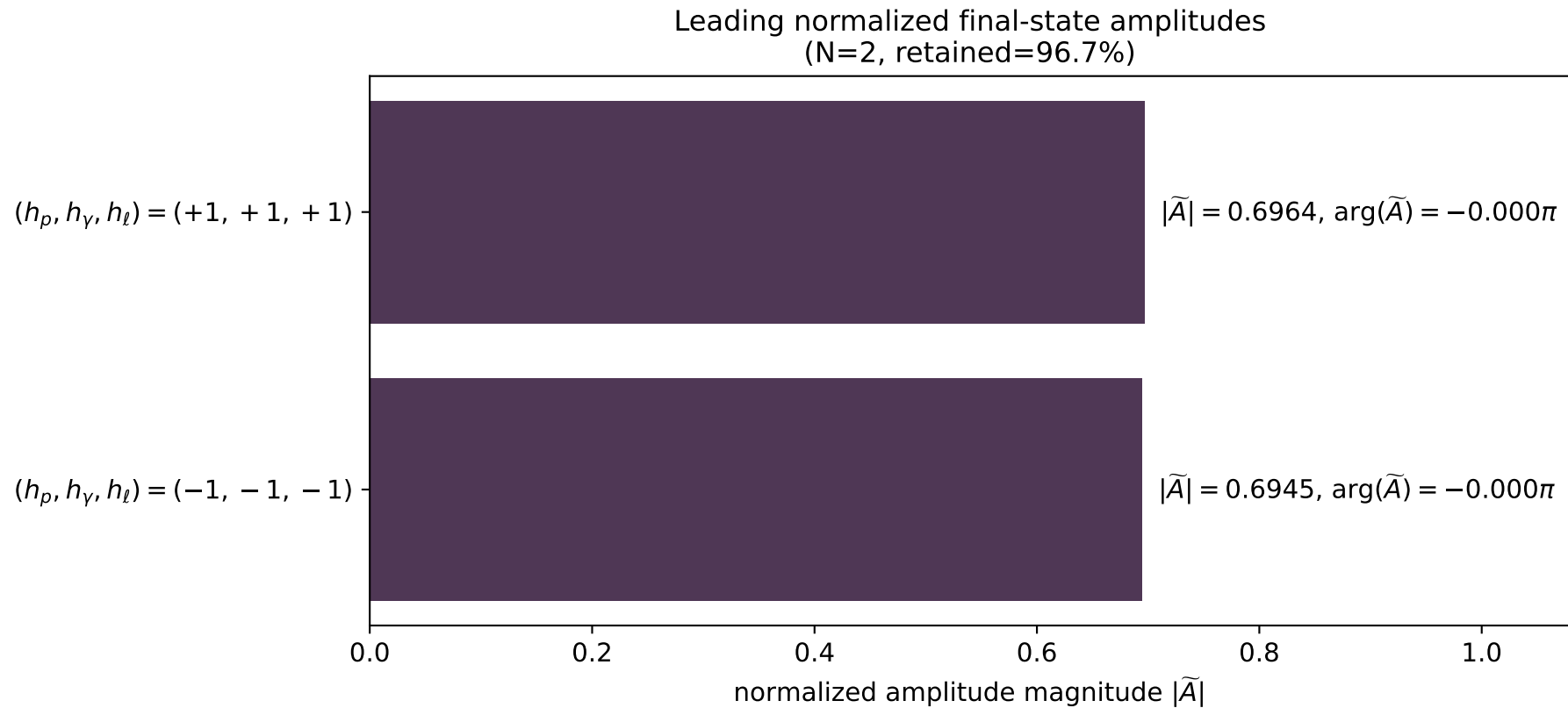
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_245
 $d_{\{\text{rm GHZ}\}}=0.000119118$
 region: polarization_cluster_04_local_minimum_245
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0245
 lepton mixing angle: $\alpha_e=2.9733431$ rad
 incoming lepton: $-0.985879|+\rangle +0.167457|-\rangle$
 proton mixing angle: $\alpha_p=1.7395034$ rad
 incoming proton: $-0.167908|+\rangle +0.985803|-\rangle$

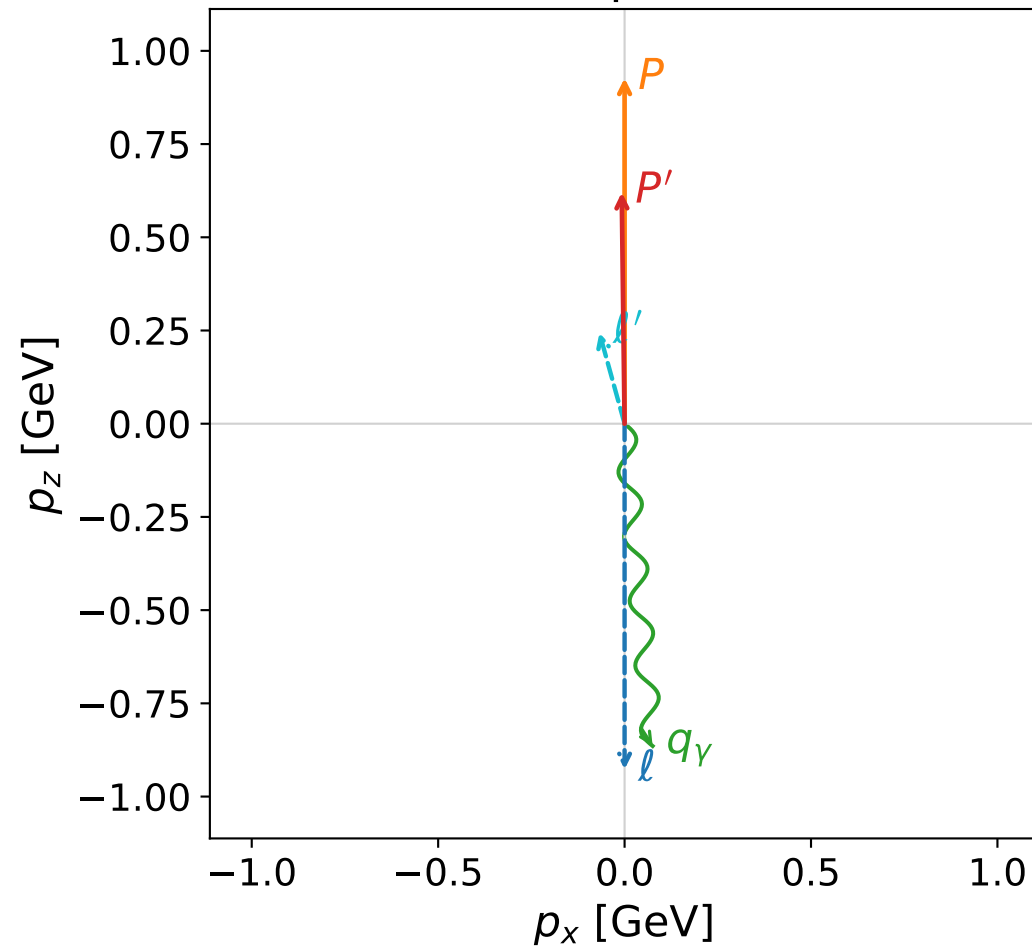
$s=23.6705$, $\sqrt{s}=4.86523$, $|\vec{P}|=2.3422$, $|\vec{P}'|=0.0567288$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.13863$, $\phi_{p'}=2.24178$, $\phi_\gamma=4.57737\text{e-}07$
 $E_\gamma=1.9344$, $\theta_{\ell'}=0.00287975$, $\phi_{\ell'}=3.14159$
 $Q^2=18.6544$, $x_B=0.845219$, $t=-3.24792$, $W^2=4.29593$, $y=0.968399$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3422$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3422$
 P $E= 2.5230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3422$
 ℓ' $E= 1.9911$ $p_x= -0.0057$ $p_y= -0.0000$ $p_z= 1.9911$
 P' $E= 0.9397$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0567$
 q_γ $E= 1.9344$ $p_x= 0.0057$ $p_y= 0.0000$ $p_z= -1.9344$

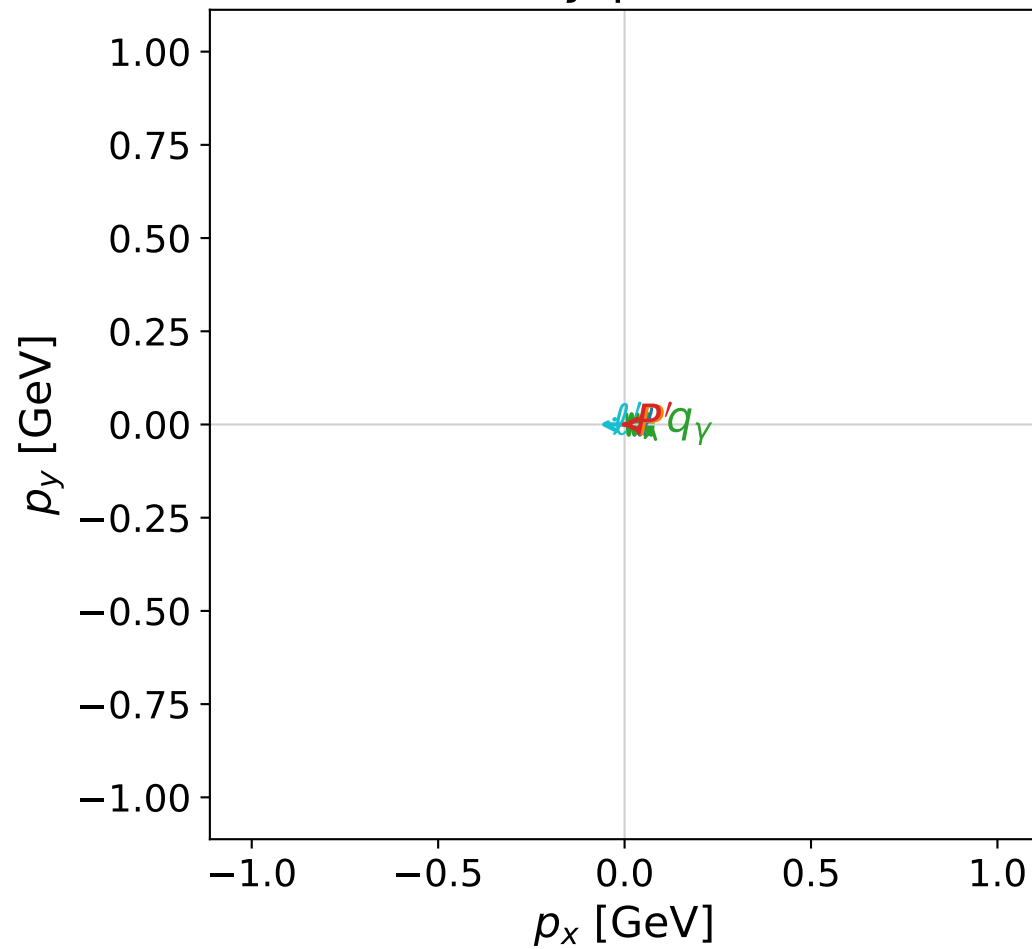


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

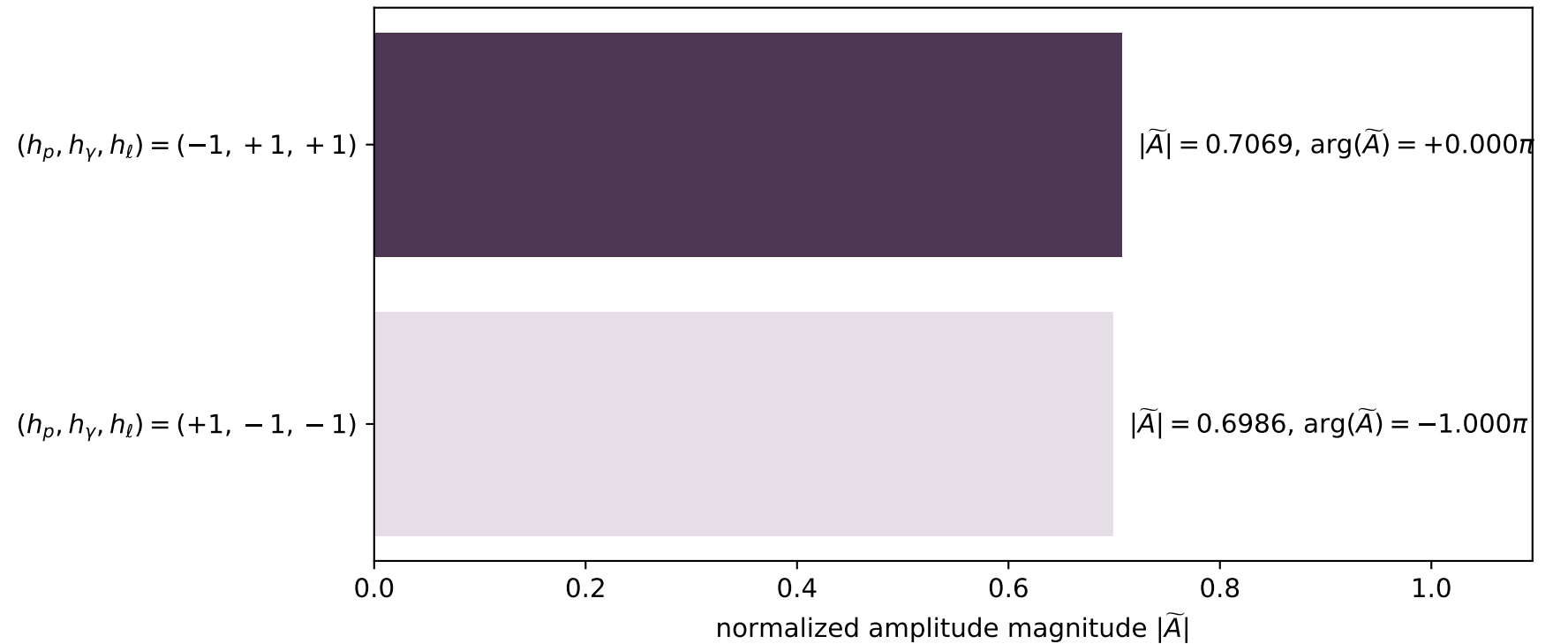


dghz_mixing_angles_polarization_04_local_minimum_251
 $d_{\text{GHZ}}=0.000133889$
 region: polarization_cluster_04_local_minimum_251
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0251
 lepton mixing angle: $\alpha_e=2.7320295$ rad
 incoming lepton: $-0.917295|+\rangle +0.398209|-\rangle$
 proton mixing angle: $\alpha_p=1.949413$ rad
 incoming proton: $-0.369635|+\rangle +0.929177|-\rangle$

$s=5.04265$, $\sqrt{s}=2.24559$, $|\vec{P}|=0.926887$, $|\vec{P}'|=0.620376$
 $\theta_{p'}=0.0123959$, $\theta_\gamma=3.05471$, $\phi_{p'}=3.14155$, $\phi_\gamma=6.28317$
 $E_\gamma=0.867702$, $\theta_{l'}=0.270165$, $\phi_{l'}=3.14158$
 $Q^2=0.922051$, $x_B=0.233591$, $t=-0.0563611$, $W^2=3.90509$, $y=0.948228$

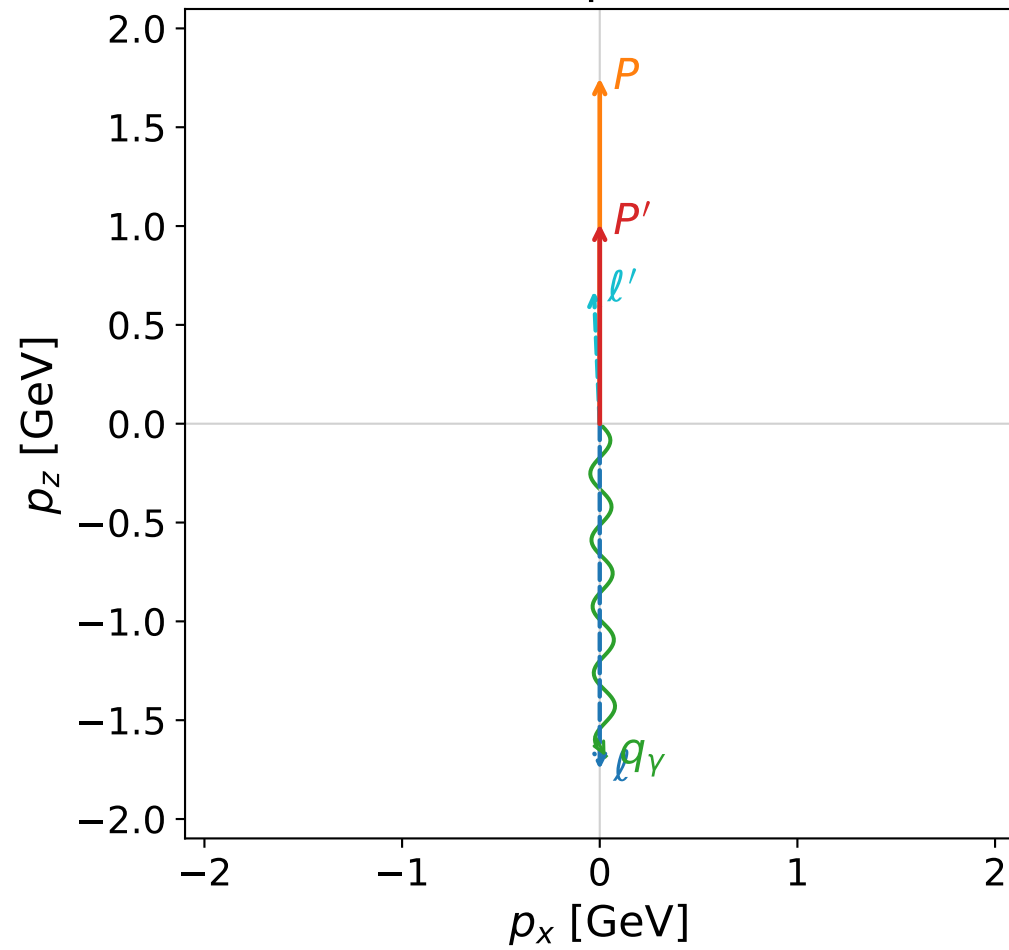
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.9269$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9269$
 P $E= 1.3187$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9269$
 l' $E= 0.2533$ $p_x= -0.0676$ $p_y= 0.0000$ $p_z= 0.2441$
 P' $E= 1.1246$ $p_x= -0.0077$ $p_y= 0.0000$ $p_z= 0.6203$
 q_γ $E= 0.8677$ $p_x= 0.0753$ $p_y= -0.0000$ $p_z= -0.8644$

Leading normalized final-state amplitudes
(N=2, retained=98.8%)

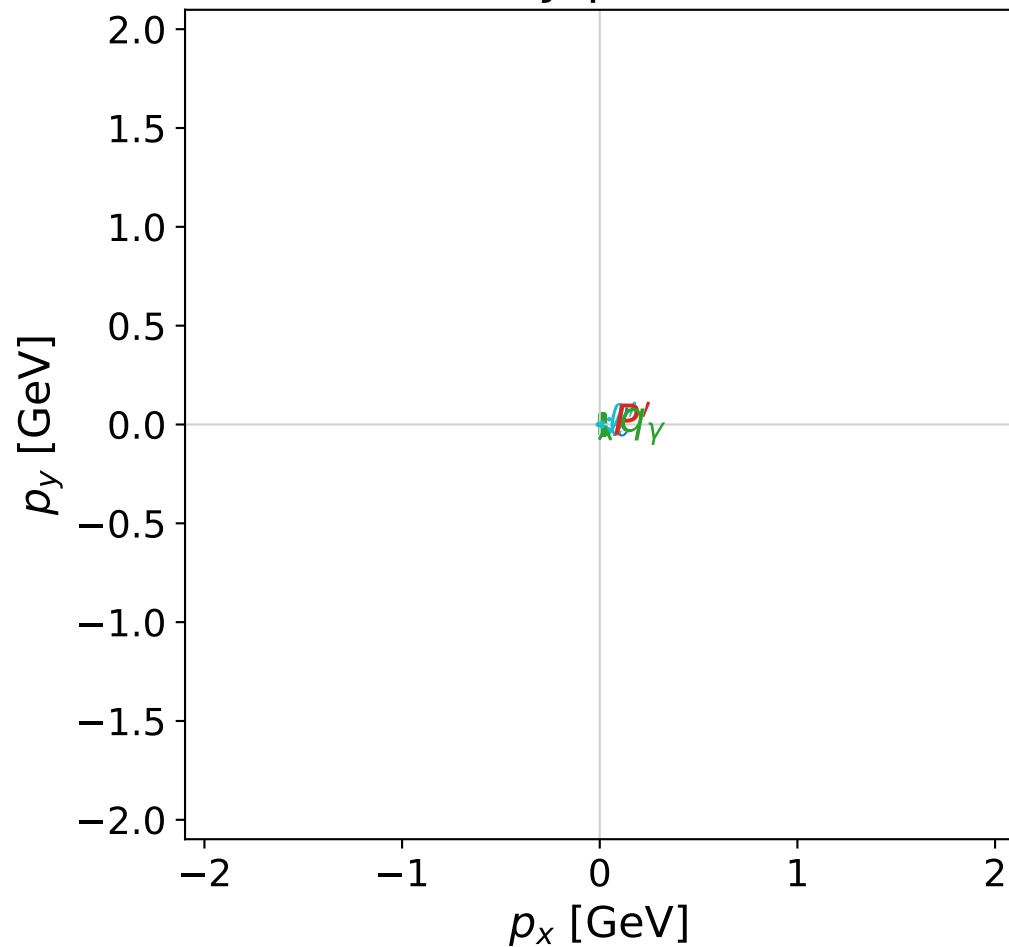


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



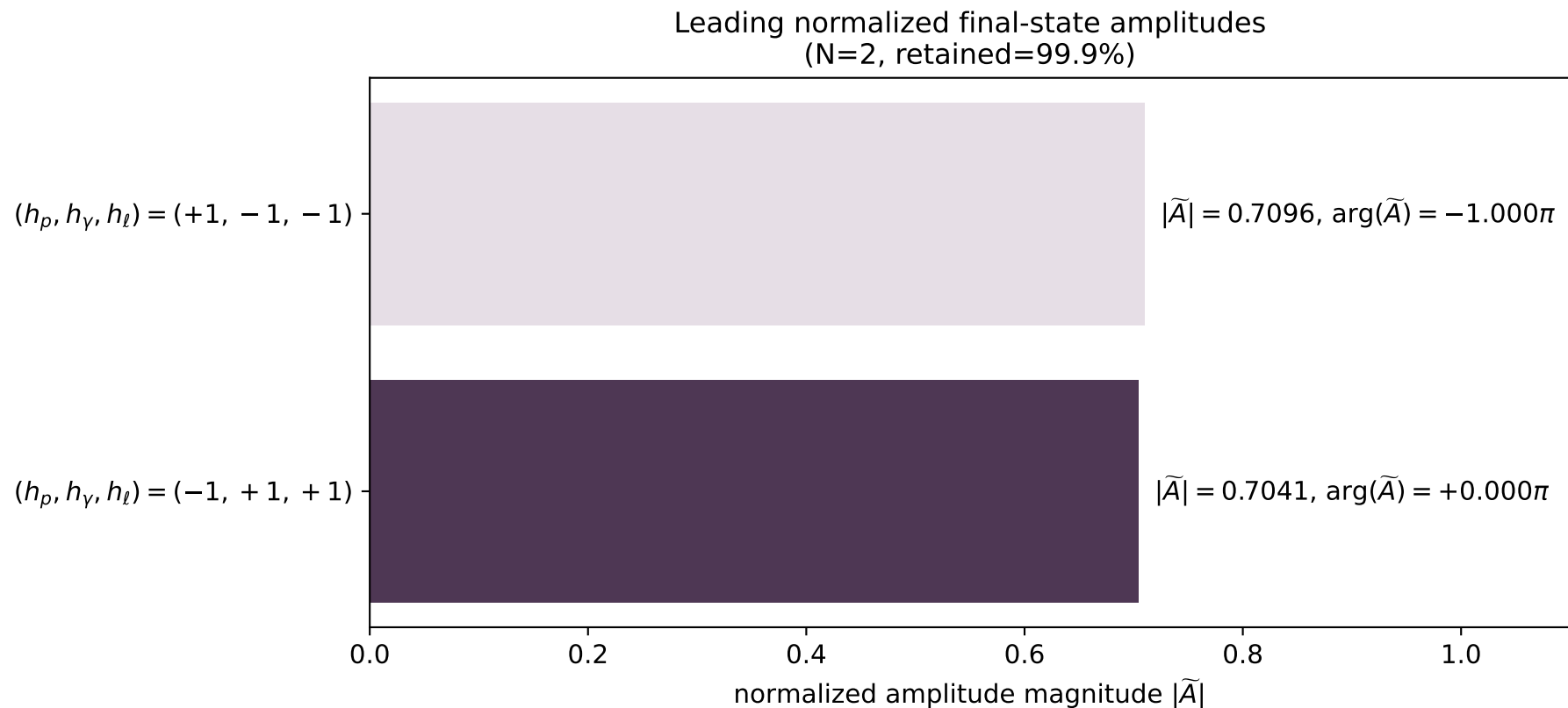
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_252
 $d_{\text{GHZ}}=0.000137569$
 region: polarization_cluster_04_local_minimum_252
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0252
 lepton mixing angle: $\alpha_e=2.8965137$ rad
 incoming lepton: $-0.970118|+\rangle +0.242633|-\rangle$
 proton mixing angle: $\alpha_p=1.8140654$ rad
 incoming proton: $-0.240877|+\rangle +0.970556|-\rangle$

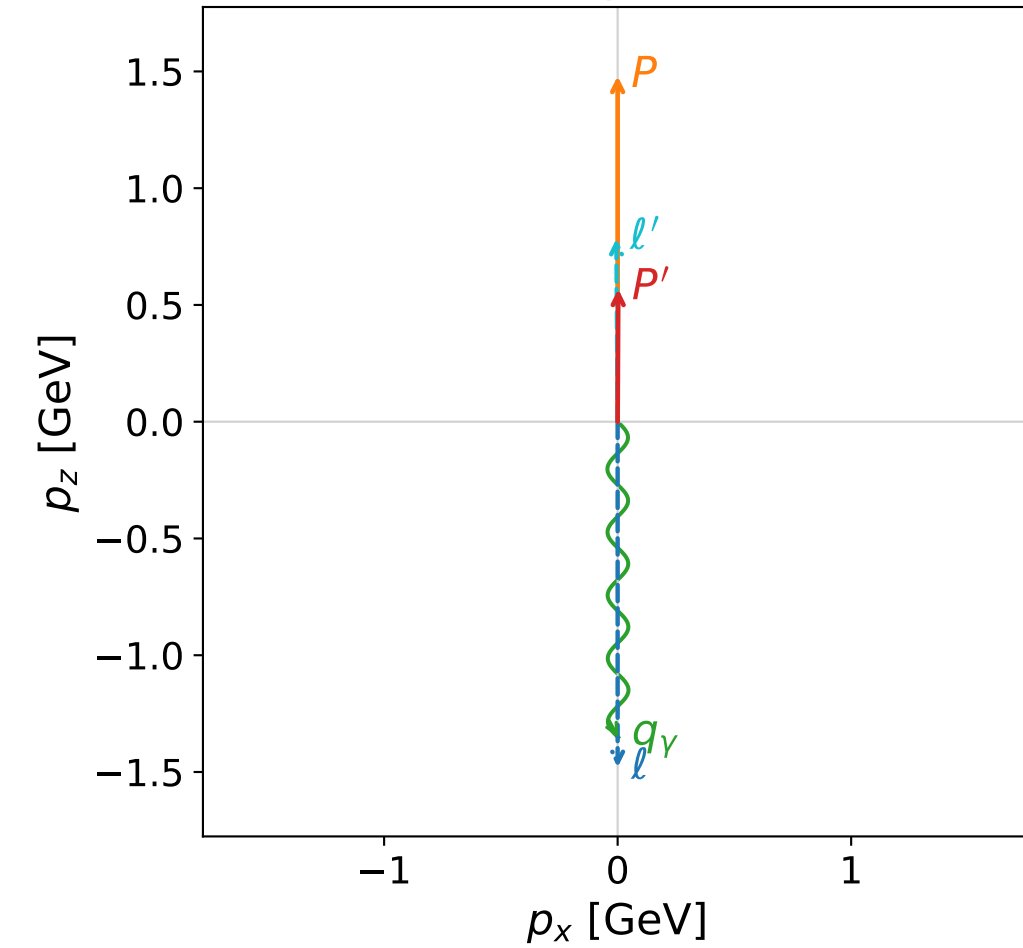
$s=13.9325$, $\sqrt{s}=3.73262$, $|\vec{P}|=1.74845$, $|\vec{P}'|=1.01096$
 $\theta_{P'}=0$, $\theta_\gamma=3.12412$, $\phi_{P'}=5.03716$, $\phi_\gamma=6.28317$
 $E_\gamma=1.68205$, $\theta_{l'}=0.0437779$, $\phi_{l'}=3.14158$
 $Q^2=4.69394$, $x_B=0.368621$, $t=-0.177771$, $W^2=8.9197$, $y=0.975575$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7485$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7485$
 P $E= 1.9842$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7485$
 l' $E= 0.6715$ $p_x= -0.0294$ $p_y= 0.0000$ $p_z= 0.6708$
 P' $E= 1.3791$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0110$
 q_γ $E= 1.6821$ $p_x= 0.0294$ $p_y= -0.0000$ $p_z= -1.6818$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

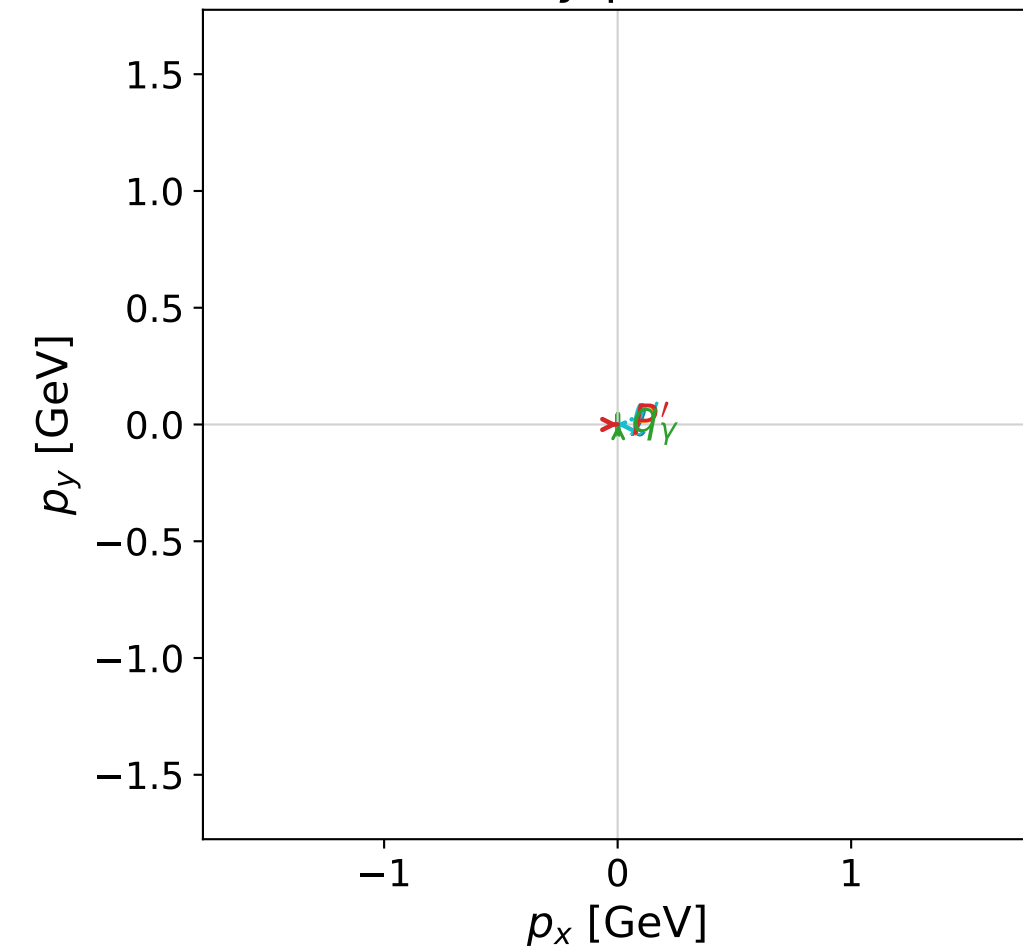


dghz_mixing_angles_polarization_04_local_minimum_293
 $d_{\text{GHZ}}=0.000295565$
 region: polarization_cluster_04_local_minimum_293
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0293
 lepton mixing angle: $\alpha_e=2.6023939$ rad
 incoming lepton: $-0.85812|+\rangle +0.513449|-\rangle$
 proton mixing angle: $\alpha_p=2.1130396$ rad
 incoming proton: $-0.516059|+\rangle +0.856553|-\rangle$

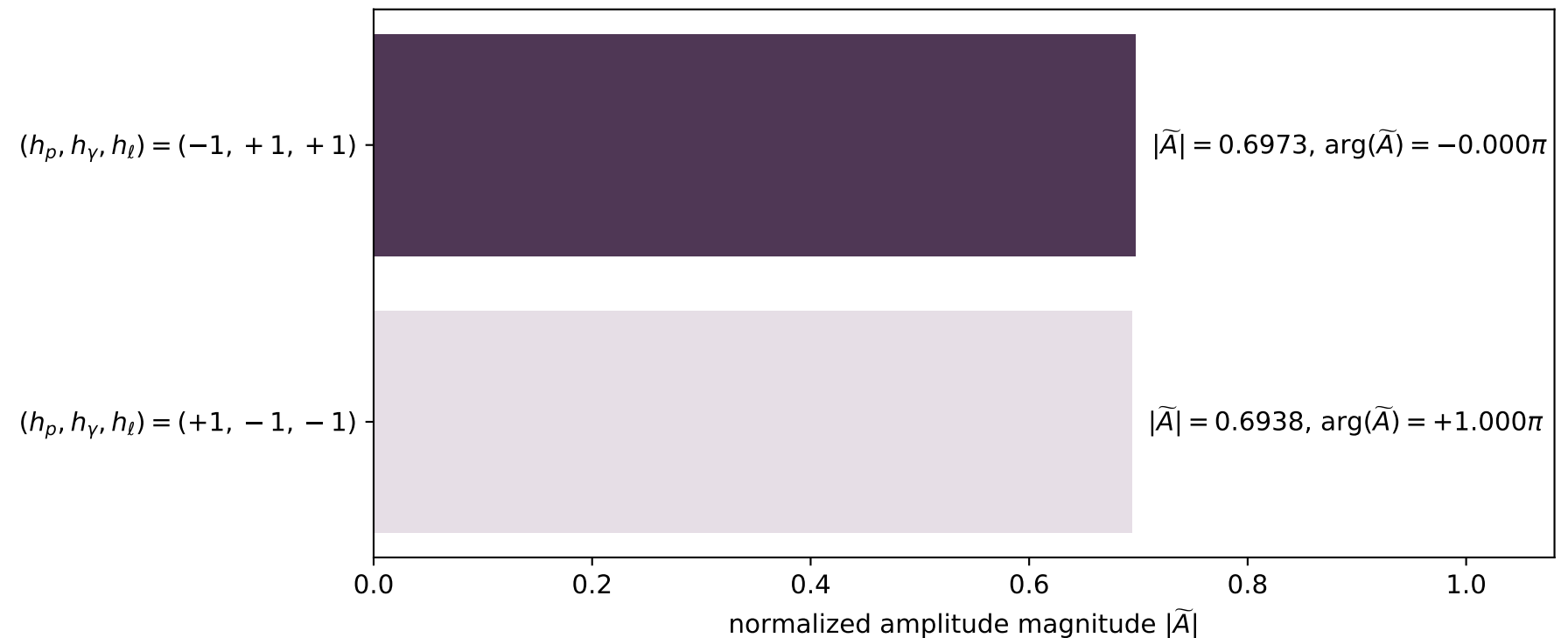
$s=10.4463$, $\sqrt{s}=3.23207$, $|\vec{P}|=1.47992$, $|\vec{P}'|=0.568077$
 $\theta_{P'}=0.00483204$, $\theta_Y=3.13964$, $\phi_{P'}=6.28121$, $\phi_Y=0$
 $E_Y=1.35176$, $\theta_{\ell'}=0.00687461$, $\phi_{\ell'}=3.14059$
 $Q^2=4.63923$, $x_B=0.507591$, $t=-0.401758$, $W^2=5.38032$, $y=0.955392$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.4799$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4799$
 P $E= 1.7521$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4799$
 ℓ' $E= 0.7837$ $p_x= -0.0054$ $p_y= 0.0000$ $p_z= 0.7837$
 P' $E= 1.0966$ $p_x= 0.0027$ $p_y= -0.0000$ $p_z= 0.5681$
 q_Y $E= 1.3518$ $p_x= 0.0026$ $p_y= 0.0000$ $p_z= -1.3518$

x--y plane

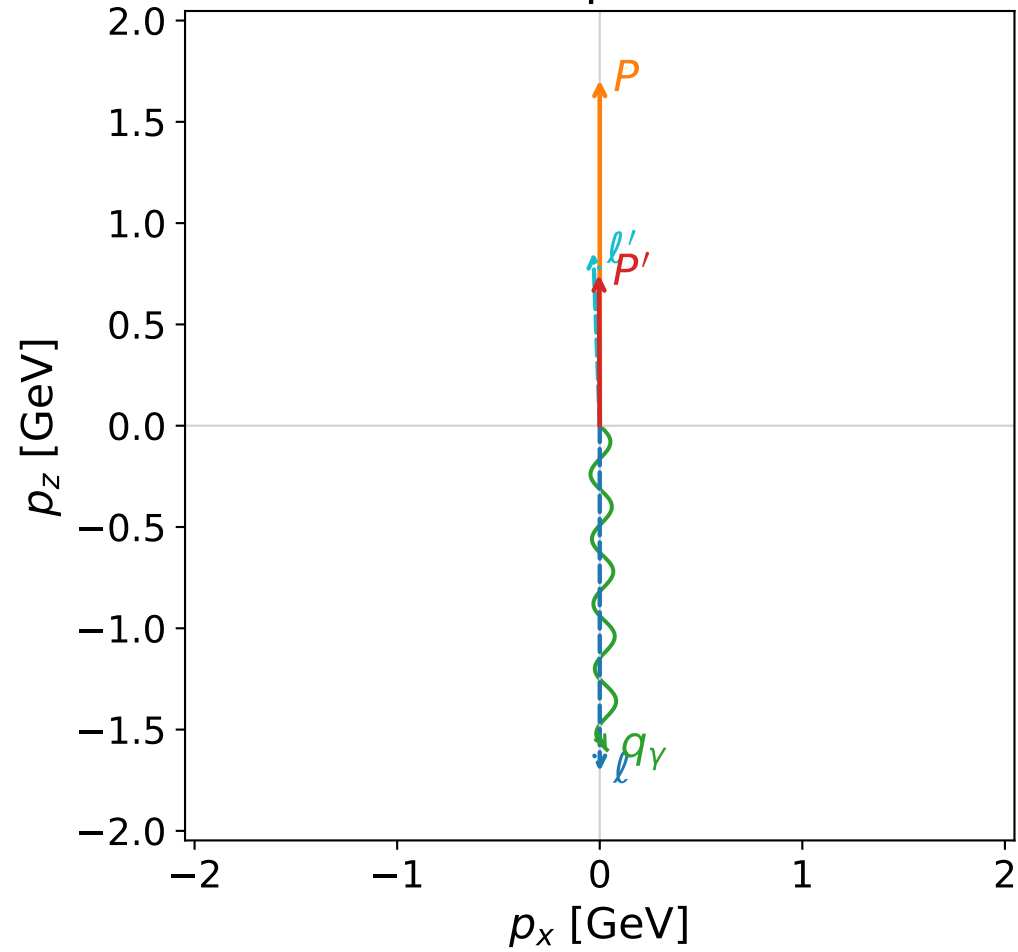


Leading normalized final-state amplitudes
 (N=2, retained=96.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

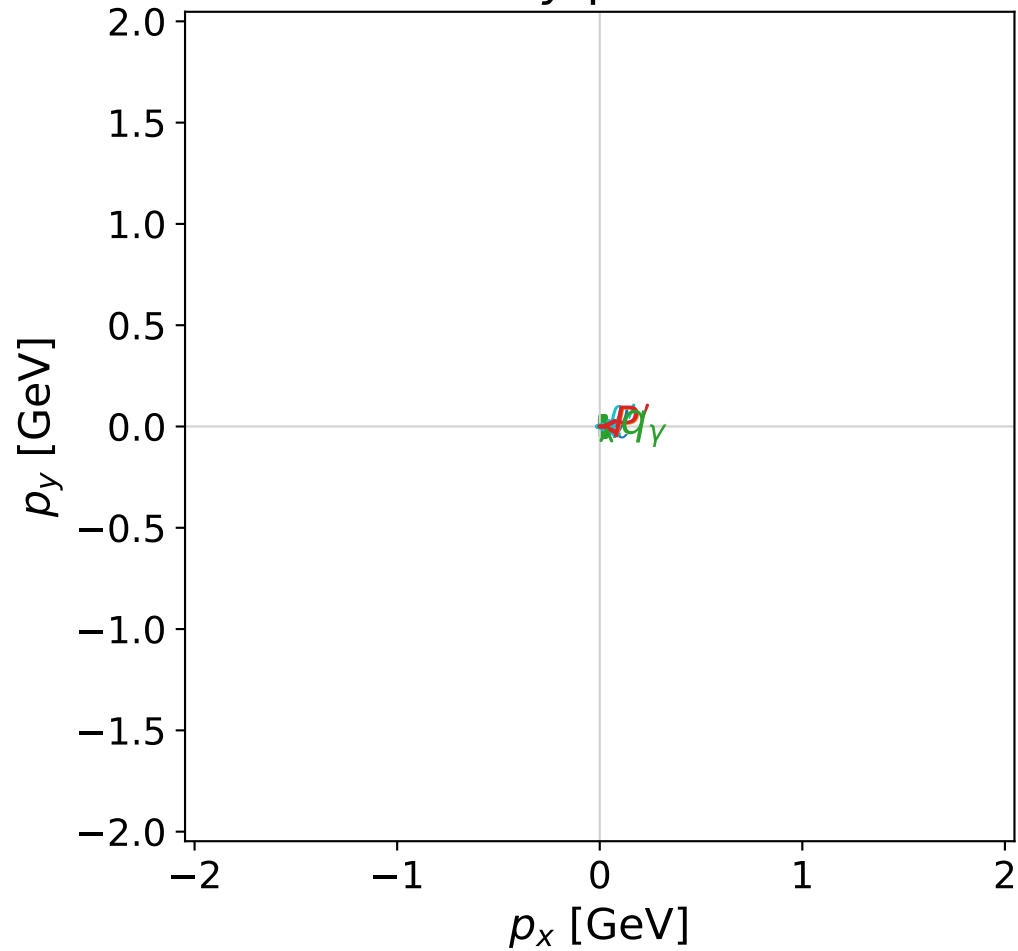


dghz_mixing_angles_polarization_04_local_minimum_297
 $d_{\{\rm GHZ\}}=0.000306617$
 region: polarization_cluster_04_local_minimum_297
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0297
 lepton mixing angle: $\alpha_e=3.0107596$ rad
 incoming lepton: $-0.991454|+\rangle +0.13046|-\rangle$
 proton mixing angle: $\alpha_p=1.6989407$ rad
 incoming proton: $-0.127794|+\rangle +0.991801|-\rangle$

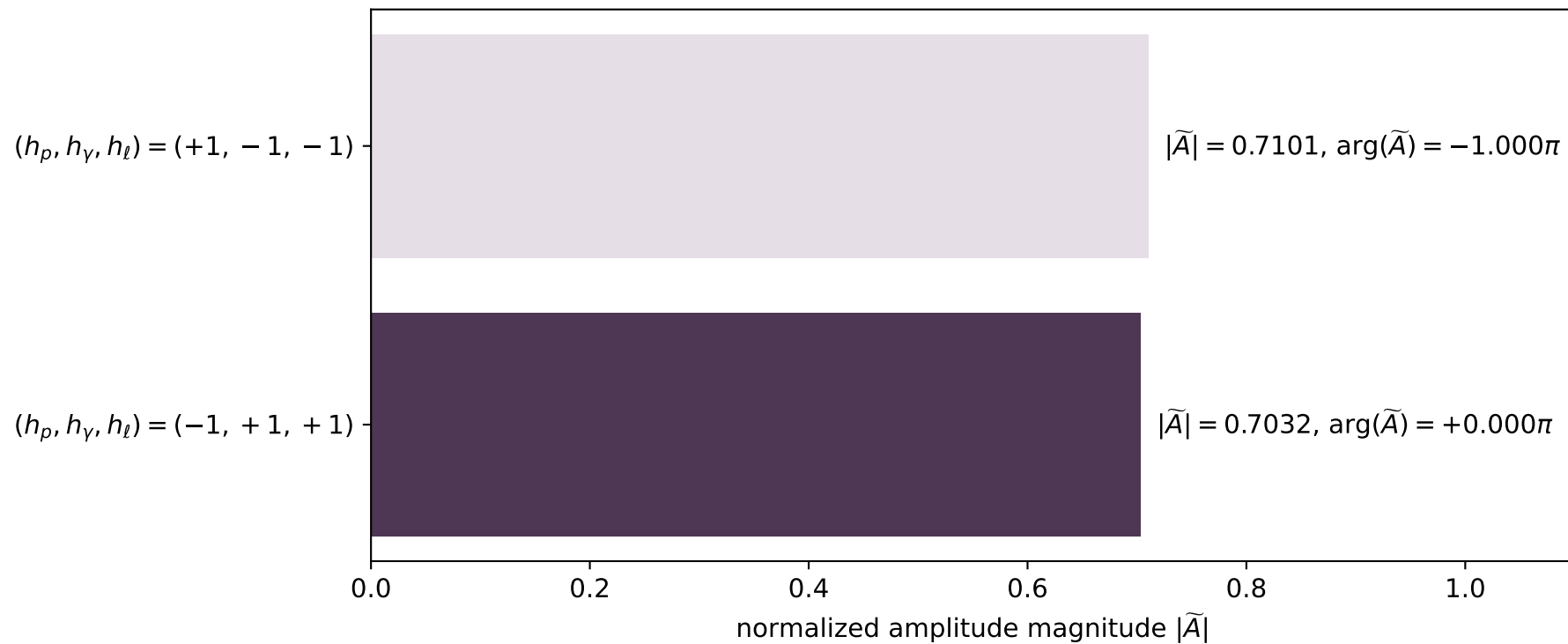
$s=13.3426$, $\sqrt{s}=3.65276$, $|\vec{P}|=1.70594$, $|\vec{P}'|=0.745662$
 $\theta_{p'}=0.00520095$, $\theta_\gamma=3.11943$, $\phi_{p'}=3.14107$, $\phi_\gamma=0$
 $E_\gamma=1.59997$, $\theta_{l'}=0.0369583$, $\phi_{l'}=3.14166$
 $Q^2=5.82901$, $x_B=0.483769$, $t=-0.361859$, $W^2=7.09999$, $y=0.96681$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7059$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7059$
 P $E= 1.9468$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7059$
 l' $E= 0.8545$ $p_x= -0.0316$ $p_y= -0.0000$ $p_z= 0.8539$
 P' $E= 1.1983$ $p_x= -0.0039$ $p_y= 0.0000$ $p_z= 0.7457$
 q_γ $E= 1.6000$ $p_x= 0.0355$ $p_y= 0.0000$ $p_z= -1.5996$

x--y plane

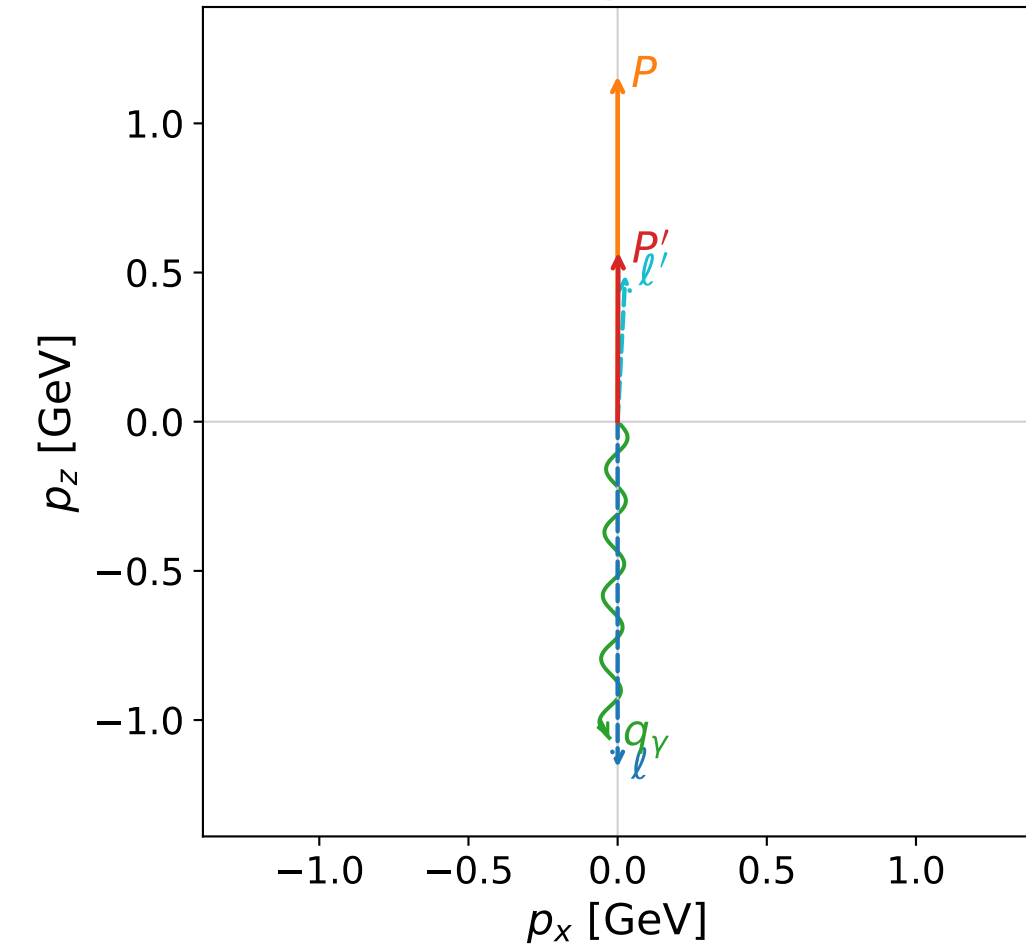


Leading normalized final-state amplitudes
(N=2, retained=99.9%)

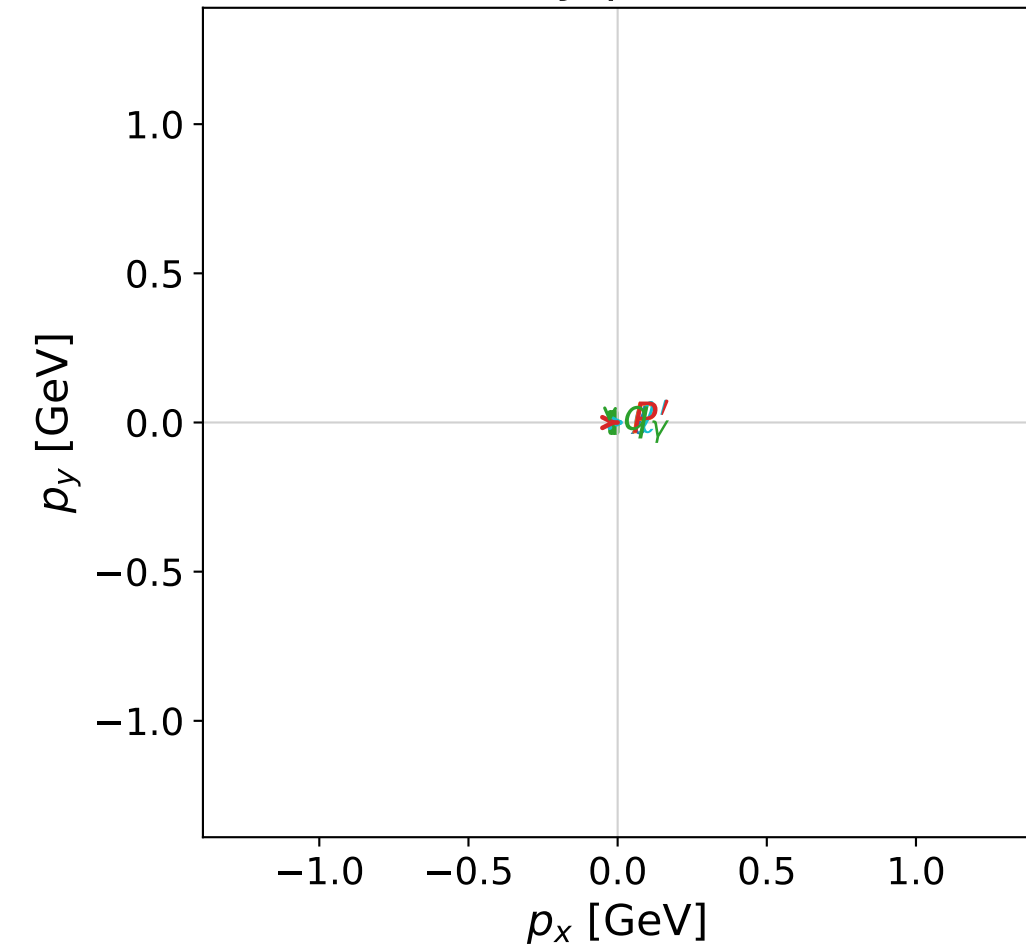


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



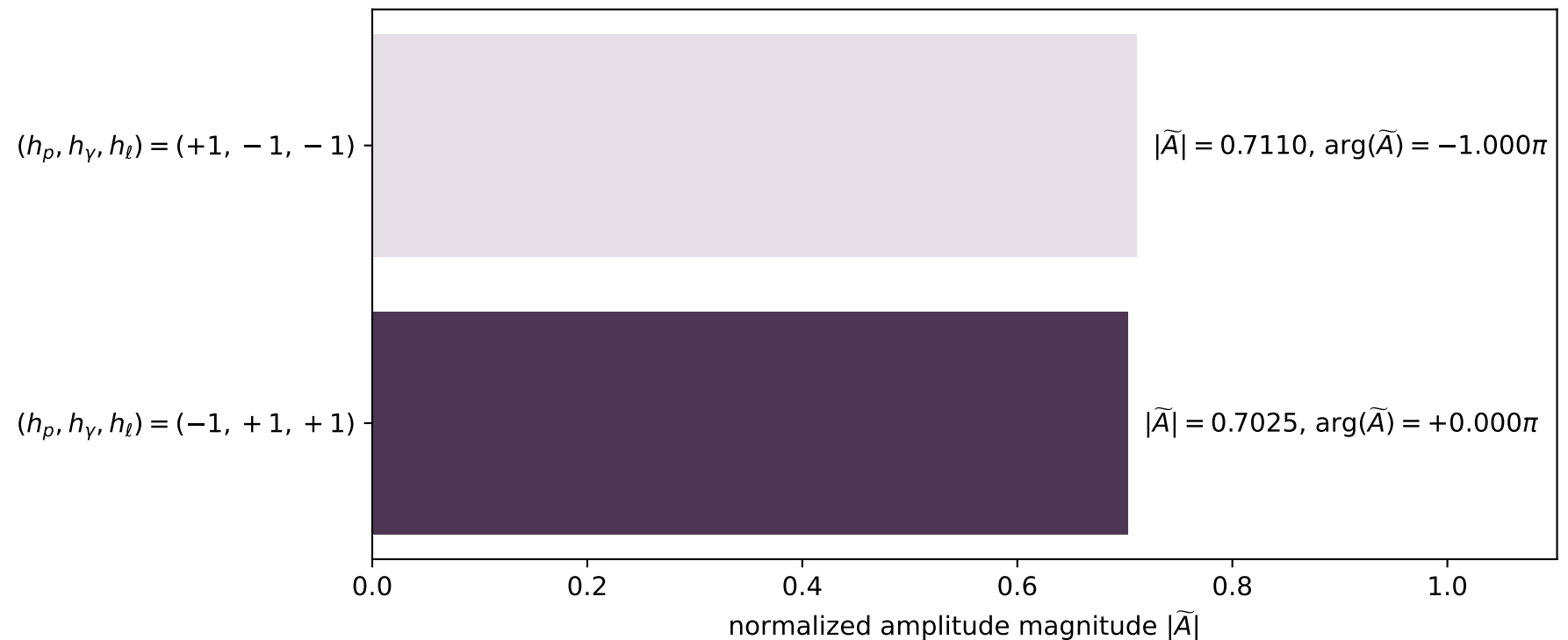
dghz_mixing_angles_polarization_04_local_minimum_319
 $d_{\text{GHZ}}=0.000411475$
 region: polarization_cluster_04_local_minimum_319
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0319
 lepton mixing angle: $\alpha_e=1.704901$ rad
 incoming lepton: $-0.133703|+> +0.991021|->$
 proton mixing angle: $\alpha_p=3.0080191$ rad
 incoming proton: $-0.991092|+> +0.133177|->$

$s=7.01694$, $\sqrt{s}=2.64895$, $|\vec{P}|=1.1584$, $|\vec{P}'|=0.567777$
 $\theta_{p'}=0.00378308$, $\theta_Y=3.11539$, $\phi_{p'}=0.00339324$, $\phi_Y=3.14154$
 $E_Y=1.05998$, $\theta_{l'}=0.0520539$, $\phi_{l'}=6.28284$
 $Q^2=2.28056$, $x_B=0.392633$, $t=-0.193536$, $W^2=4.40765$, $y=0.946437$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

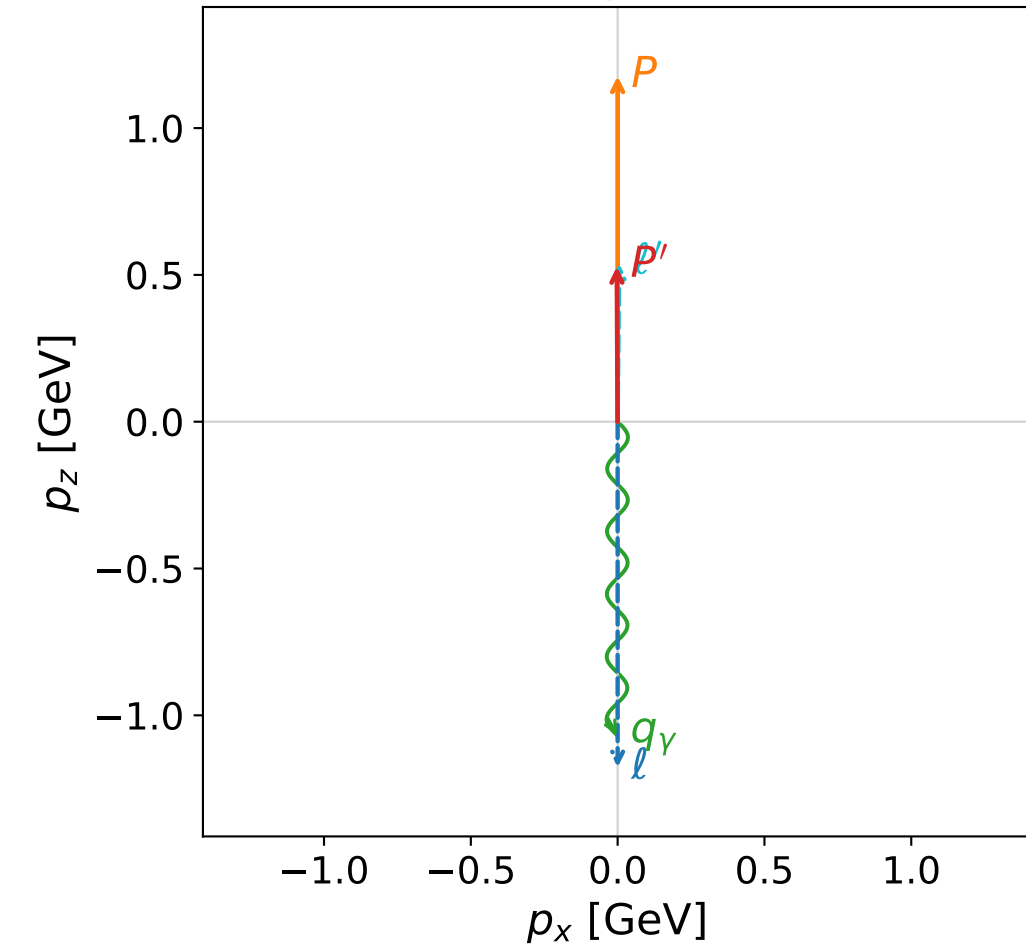
ℓ	$E=$	1.1584	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-1.1584
P	$E=$	1.4905	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	1.1584
ℓ'	$E=$	0.4925	$p_x=$	0.0256	$p_y=$	-0.0000	$p_z=$	0.4918
P'	$E=$	1.0965	$p_x=$	0.0021	$p_y=$	0.0000	$p_z=$	0.5678
q_Y	$E=$	1.0600	$p_x=$	-0.0278	$p_y=$	0.0000	$p_z=$	-1.0596

Leading normalized final-state amplitudes
 (N=2, retained=99.9%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



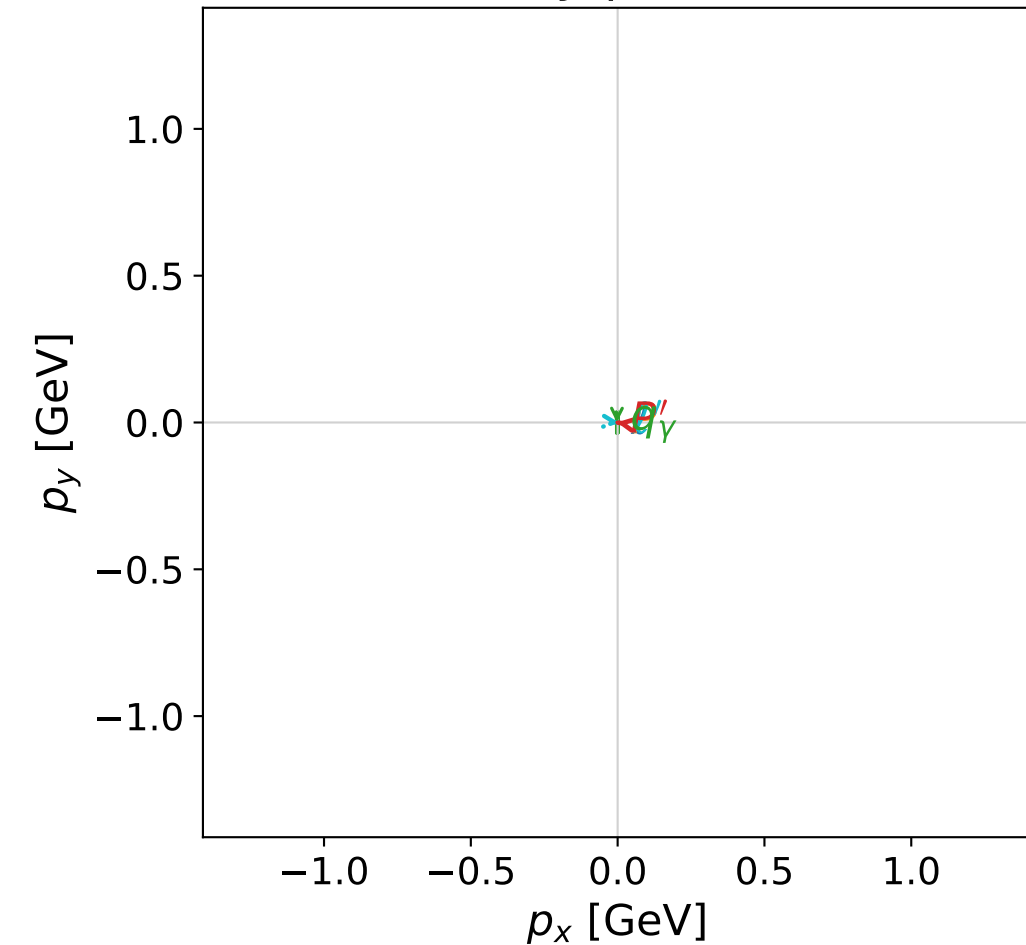
dghz_mixing_angles_polarization_04_local_minimum_336
 $d_{\text{GHZ}}=0.000558565$
 region: polarization_cluster_04_local_minimum_336
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0336
 lepton mixing angle: $\alpha_e=2.0949563$ rad
 incoming lepton: $-0.500486|+\rangle +0.865745|-\rangle$
 proton mixing angle: $\alpha_p=2.6121434$ rad
 incoming proton: $-0.863085|+\rangle +0.505058|-\rangle$

$s=7.19362$, $\sqrt{s}=2.68209$, $|\vec{P}|=1.17702$, $|\vec{P}'|=0.528084$
 $\theta_{p'}=0.00510742$, $\theta_\gamma=3.13917$, $\phi_{p'}=2.97399$, $\phi_\gamma=3.14159$
 $E_\gamma=1.06685$, $\theta_{l'}=0.00977172$, $\phi_{l'}=6.19762$
 $Q^2=2.53666$, $x_B=0.425599$, $t=-0.237414$, $W^2=4.30339$, $y=0.944002$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	1.1770	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-1.1770
P	$E=$	1.5051	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	1.1770
ℓ'	$E=$	0.5388	$p_x=$	0.0052	$p_y=$	-0.0004	$p_z=$	0.5388
P'	$E=$	1.0764	$p_x=$	-0.0027	$p_y=$	0.0004	$p_z=$	0.5281
q_γ	$E=$	1.0669	$p_x=$	-0.0026	$p_y=$	-0.0000	$p_z=$	-1.0669

x--y plane



Leading normalized final-state amplitudes
 (N=2, retained=96.6%)

